FRM PART II • 2026 • EXAM 25 NOVEMBER 2026

Credit Risk

Notes + Question Workbook

P1 • P2 • P3, in order of importance

111 learning objectives from 22 readings, banded by the Master LO Map v3. P4 is excluded. Built from Vardeez's notes, the GARP 2026 Credit Risk book, the SchweserNotes 2026 book, the Tier A and B–D Recall Workbooks and the Credit Risk Question Bank.

How to use this book

- The book runs **P1, then P2, then P3**. Inside each part, LOs follow syllabus order. The syllabus index at the front lists all 111 LOs in GARP order with their band and page.
- Each LO is one block: **focus box** (why it sits in its band, Vardeez's pointer and the reported exam shapes), **notes** limited to that focus, a **formula card** or comparison table, **exam traps**, then **6 to 10 practice questions**. **Solutions** sit at the end of each block, so attempt first.
- LO wording is verbatim from the official GARP 2026 learning objectives, as listed in the Master LO Map.
- Type tags: CALC numerical questions expected; CONCEPT statement questions (only the reported workbook calculations appear); BOTH a mix set by the reported shapes and Vardeez's focus.
- Question sources, in priority order: Tier workbook (August 2026 recall and mapped shapes), GARP-tagged bank questions (practice exams, GARP book, mocks), other bank questions, and *New* questions written in GARP style from the readings where an LO was short. Every number has been recomputed.
- Boxes headed *Correction to the source notes* mark errors found in Vardeez, GARP, Schweser or a bank key, with the correct value used throughout. Figures are drawn for black-and-white printing.

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Syllabus index (GARP order)

Every LO in this book, in the order of the official 2026 learning objectives. P4 LOs are not in this book.

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CR-18 • Central Clearing

CR-18 f P3 CONCEPT 1098

CR-19 • Future Value and Exposure

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CR-21 b P2 CONCEPT 753
CR-21 c P2 CONCEPT 761
CR-21 d P2 CALC 771
CR-21 e P2 CONCEPT 781
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CR-22 h P3 CONCEPT 1120
CR-22 i P2 CONCEPT 818
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CR-23 • An Introduction to Securitisation

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CR-23 h P1 CALC 471

CREDIT RISK • PART

P1

Most important

Tier A recall topics and the concepts Vardeez flags in Very Important chapters. 41 LOs from 11 readings.

CR-2 • Governance • Vardeez: NOT SO IMP

CR-2 a

P1 CONCEPT

Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control.

Workbook: A22 Three Lines of Defence and Operational Risk Governance • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A22): the August 2026 recall reports two questions on the three lines of defence, and Map #29 lists "three lines of Operational Risk Management". The same model is tested under ORR-2 c, so a question can arrive in either book's vocabulary.
Vardeez: the reading is rated NOT SO IMP ("not much expected here"), but his table carries the one distinction that matters: *own* versus *oversee* versus *independently assure*, and which function sits in each line.

Reported exam shapes

- **A22·S1** Three lines of operational risk management: who does what, with every distractor moving an activity across a line.

What the questions look like: statement questions only. You get four functions or four role assignments; three of them move one activity (risk ownership, policy setting, sign-off, approval, independent assurance) into the wrong line or the wrong person. No calculations.

1. Why governance decides credit losses

GARP's starting point (Bouteille and Coogan-Pushner, Ch.2): one bad judgment on one transaction can lose a lot of money, but it rarely bankrupts a firm. Firms fail when **portfolios of toxic transactions** are built, and in the absence of fraud that happens through a poor risk management framework and a corporate governance failure. Investigations of massive losses often find that every procedure was respected: the rules did not work, staff were not skilled enough, origination lacked oversight, incentives rewarded the wrong goals or the approval process was flawed.

- **Prevention beats cure.** No portfolio management or mitigation strategy compensates for deficient risk taking, and no price, however good, compensates for a bad transaction.
- **Incentives push volume.** Most incentive systems reward top-line growth, so originators push for volume, margins and expected returns and present their deals in the most favourable light. The risk manager's job is to **control quality**.
- **The answer.** To balance the stakeholders and define risk management responsibilities clearly, most organizations use the framework the Institute of Internal Auditors (IIA) set out in its 2013 position paper, *The Three Lines of Defense in Effective Risk Management and Control*.

2. The three lines of defense

Line	Who (GARP)	What they do
First	Business owners (originators, business units, the managers who run them)	**Own and manage** risk: take it, manage it day to day, follow the guidelines.
Second	Enterprise risk management (under the CRO), compliance, legal	**Monitor and oversee** risk: establish policies and procedures and serve as management oversight of the first line.
Third	Internal audit, external auditors, special audit committees	**Independent assurance** over the risk management and risk monitoring done by the first and second lines.

Three points carry most questions:

- **Ownership stays in the first line.** The second line sets the rules and challenges, but it does not take over the risk. A risk function that "owns and manages" a business's risks has collapsed two lines into one.
- **The second line is still management.** It is independent of the *business* (not in a profit center, not paid on profitability), but it is part of the management structure: its oversight is "management oversight". Compliance and legal sit here, not in the third line.
- **Only the third line is independent of management.** It gives assurance over *both* other lines. GARP puts external auditors in the third line too, even though they are outside the firm. Oversight runs one way: second over first, third over both. The first line never oversees the second, and the second never oversees audit.

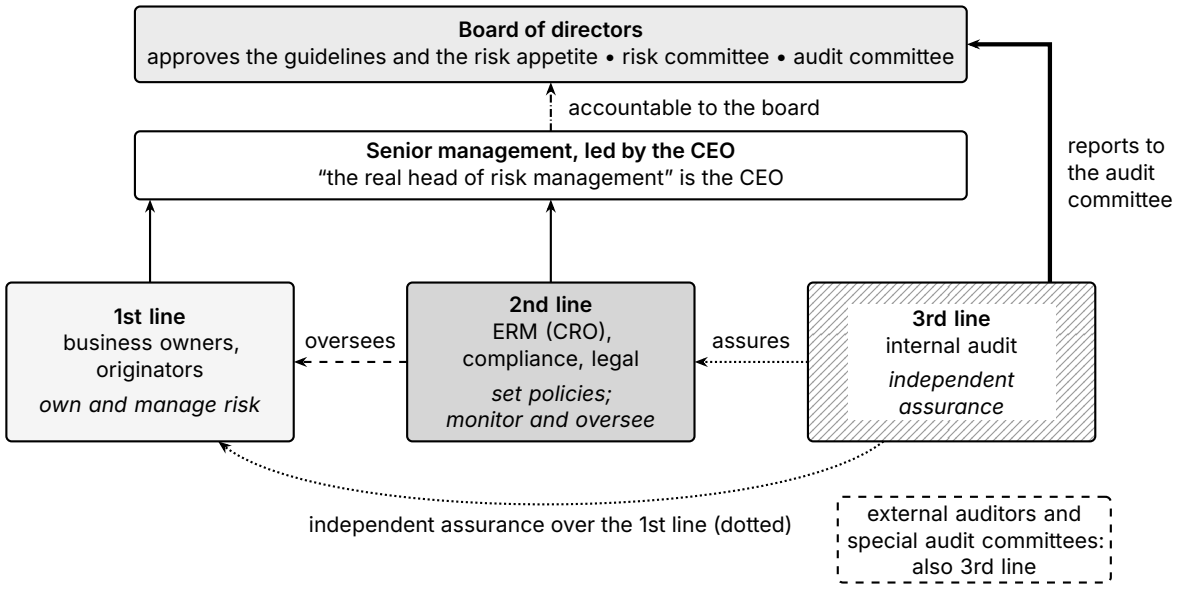


Figure 1. The three lines of defense under the board. Solid arrows are reporting lines (the thick one: internal audit reports to the board's audit committee); the dashed arrow is the 2nd line's oversight of the 1st line; dotted arrows are the 3rd line's independent assurance over both other lines; the dash-dot arrow is management's accountability to the board. Fills: light grey 1st line, darker grey 2nd line, hatched 3rd line.

3. Responsibilities above and beside the lines

The LO asks for "risk management responsibilities in an organization", not just the three boxes. GARP and the notes assign them as follows (CR-2 c gives the detail).

- **Board of directors.** Approves the credit guidelines (sponsored by the CRO or CFO). For listed firms the board's audit committee reviews risk management policies; most financial institutions also have a board risk committee that oversees, reviews and approves them. Nobody may take more risk than the board has accepted.
- **CEO.** Responsible for ensuring adequate governance guidelines are in place (Vardeez, Schweser). GARP's final words go further: "the real head of risk management is the CEO", who has the *absolute* responsibility to make the whole organization respect risk management principles, must set the example with his direct reports, and must involve risk management at the right time in strategic decisions such as mergers and acquisitions.
- **CRO.** Owns the guidelines (drafts them, seeks their approval, promulgates and maintains them). Every risk management employee's chain of command ends at the CRO, not at a business head. The CRO usually reports to the CEO and has privileged access to the board's risk and audit committees.
- **Risk management unit.** Not a profit center and **no approval authority**: it plays an advisory role, because business managers are accountable for the top and bottom lines of their units. Guidelines state where its recommendation is required. A veto right is rare; a risk manager who strongly disagrees attaches a written memo to the committee package and presents the view in person.
- **Originators and business heads.** Own the transactions and their risk, and are accountable for their units' results; approval authority is delegated to them and to committees according to transaction risk (CR-2 c).

4. The operational risk version (ORR-2 c and Workbook A22.1)

The August questions were set in operational risk language (GARP ORR Ch.2, summarised in Workbook A22). The logic is the same, with more detail on who signs what.

- **Line 1, risk owners** (BCBS: "business unit management"). Risk is managed where it is generated: the head of IT owns IT risk. They identify and assess inherent risks, set and test controls, report residual risks, and give the **final sign-off** on their risk and control self-assessment (RCSA).
- **Line 1.5, risk champions.** Specialists inside each business who record events and losses, map risks and controls and follow up action plans. Ownership of operational risk must *never* be assumed by, or passed to, them.
- **Line 2, the corporate operational risk function (CORF).** Develops policies, standards and training, forms an independent view and challenges the business, with evidence of that challenge. "The second line owns the methodology; the first line owns the risks": it may run the RCSA workshop but does not steer the answers or sign them off. In operational risk it needs delegated authority to veto decisions that breach regulation or board risk appetite; in GARP's credit chapter a veto right for risk managers is rare.
- **Line 3, internal audit.** Independent of the risk process, including of the risk function. It may take account of others' work after evaluating that function, but it never relies *exclusively* on risk, compliance or finance testing and always examines its own sample. It reports in a solid line to a non-executive, usually the audit committee chair, never to the CRO.
- **Hybrid functions** (finance, HR, legal, IT, security in smaller firms). BCBS asks them to document and distinguish their first-line and second-line roles. *The role defines the line, not the department's*

name: IT security encrypting data and partitioning the network is line 1; verifying ICT controls in business operations and enforcing security policies is line 2. Legal drafting contracts is line 1; legal overseeing legal activity in the business is line 2.

	First line	Second line	Third line
Who	Business owners, originators	ERM (CRO), compliance, legal	Internal audit, external auditors, special audit committees
Core verb	Own and manage	Monitor and oversee	Independently assure
Typical act	Takes the risk; signs off its own RCSA	Sets policies and methods; challenges	Audits lines 1 and 2 with its own sample
Independent of	Nobody	The business (not management)	Management
Reports to	Business head, CEO	CRO, then CEO; access to board committees	Audit committee of the board
Never	Hands ownership to the 2nd line or to risk champions	Owns the risk or audits itself	Relies exclusively on others' testing; reports to the CRO

Exam traps

- **Ownership moved to the second line.** "Risk management owns and manages the risk" or "takes over day-to-day management to centralise it". The second line oversees; the business owns.
- **Compliance or legal placed in the third line.** They are independent of the business but they are management oversight, so second line. External auditors, though outside the firm, are third line.
- **Reversed oversight.** The first line "overseeing" the second, or the second line "overseeing" internal audit.
- **Sign-off moved.** The second line facilitating an RCSA is fine; the second line signing it off is not. Risk champions never take ownership.
- **Absolute claims on audit.** Internal audit may reduce scrutiny of a function it has evaluated, but never relies on it exclusively, and never reports to the CRO.
- **Department name instead of role.** A hybrid function sits in line 1 or line 2 activity by activity.
- **Approval authority given to risk management.** It advises and recommends; approval sits with the business hierarchy and committees. The CEO, not the CRO, carries the ultimate responsibility.

Practice questions: CR-2 a

Question CR-2 a.1

Bank Q22 • Mock 1 Q17 (rebuilt)

A consultant is mapping a bank's functions to the three lines of defense framework described by GARP (the IIA's 2013 position paper). Which of the following parties belongs to the third line of defense?

A. Business line staff who originate credit transactions.

B. The credit risk management department.

C. External auditors.

D. The compliance function.

Question CR-2 a.2

Bank Q23 • Schweser Module Quiz 20.1 Q1

To ensure an adequate second line of defense, organizations should:

A. include the compliance and legal functions in risk oversight.

B. look primarily to business owners because they own and manage the risks.

C. ensure that the first line of defense has proper oversight of the second line.

D. ensure that the audit function provides an independent risk management role.

Question CR-2 a.3

Bank Q20 • AnalystPrep Q.5791 (rebuilt)

A bank is launching a new consumer lending product and must assign risk management responsibilities under the three lines of defense framework. Which assignment is most appropriate?

A. The business owns and manages the risks; risk management, compliance and legal set policies and oversee the business; auditors provide independent assurance.

B. Enterprise risk management owns and manages the risks; the business monitors the risk function; internal audit sets the policies and procedures.

C. The business owns and manages the risks and also provides the independent assurance over its own controls; the risk function only advises.

D. Internal and external auditors own and manage the risks and report to the business; compliance and legal provide the independent assurance.

Question CR-2 a.4

Bank Q24 • AP Mock 3 Q36

Which of the following is most likely a role and responsibility of the second line of defense for the overall risk management of a firm under risk governance?

A. Making decisions for managing risks.

B. Establishing risk management methods and measurement methods.

C. Overseeing the risk management activities of the third line of defense.

D. Reporting independently to the board of directors.

Question CR-2 a.5

Bank Q21 • AnalystPrep Q.5792 (rebuilt)

A retailer is implementing a new inventory system that carries data security and supply chain risks. The CRO heads the firm's second line of defense. Which action best fits the CRO's responsibility under the three lines of defense?

A. Take over day-to-day management of the system's risks from the business and IT owners, so that responsibility is centralized.

B. Delegate all risk management for the system to the IT department and rely on its reports without challenge.

C. Perform independent audits of the system implementation and report the results directly to the audit committee.

D. Set policies and risk measurement standards and oversee the business and IT owners, who continue to manage the risks.

Question CR-2 a.6

Bank Q27 • Mock 2 Q33 (rebuilt) • re-filed from CR-2 b

A bank's credit guidelines require input from risk management on transactions above certain size and rating thresholds. Which statement best describes the responsibility of the risk management function in the credit approval process?

A. It approves or vetoes every transaction, since it is the second line of defense and independent of the business.

B. It advises the approvers without holding approval authority, and its recommendation is required where the guidelines say so.

C. It approves the transactions below the business unit head's limit, so that originators are not slowed down.

D. It originates transactions jointly with the business, so that risk is built into each structure from the start.

Question CR-2 a.7

New • built from GARP Ch.2, Final Words, and 2.4 Oversight

A bank has just appointed an independent CRO who heads the second line of defense. At the next board meeting the new CEO says that, from now on, risk management "is the CRO's job". Which response from the chair of the board risk committee is most consistent with GARP's description of risk management responsibilities?

A. Agreed: the CRO is now accountable for the bank's risk profile, and the CEO's part is limited to approving the guidelines.

B. Agreed: the CEO should stay out of risk decisions, so that the independence of the second line from management is kept.

C. Not quite: the CEO is the real head of risk management, must make the whole bank respect its principles and must involve risk in strategy.

D. Not quite: the CEO should personally approve every large credit transaction, because the CEO is the bank's first line of defense.

Question CR-2 a.8 Workbook A22.1 • S1 • Aug-2026 • Map #29

The head of operational risk at a Canadian bank is reviewing a draft responsibilities matrix ahead of a supervisory visit. The bank uses the three lines of defence model, and its IT security, legal and finance departments are hybrid functions. Which of the following entries in the draft is consistent with the three lines of defence model for operational risk governance?

A. A second-line risk manager facilitates each business unit's risk and control self-assessment and gives the final sign-off on its results, as the second line owns the methodology.

B. The payments department's risk champions take over ownership of the department's operational risks, so that the head of payments can concentrate on running the business.

C. After evaluating the corporate operational risk function as effective, internal audit relies exclusively on that function's control testing when auditing the payments department.

D. In IT security, encrypting data and partitioning the network are documented as first-line roles, while verifying ICT controls and enforcing security policies are documented as second-line roles.

Solutions • CR-2 a

Solution CR-2 a.1

Answer: C

Bank Q22 • Mock 1 Q17

Step 1 – Place each party in its line

First line: business owners and originators. Second line: enterprise risk management, compliance and legal. Third line: internal audit, external auditors and special audit committees.
⇒ Originators (A) are first line; credit risk management (B) and compliance (D) are second line.

Step 2 – Test for independent assurance

The third line is the one that provides independent assurance over the first two. External auditors give an independent opinion and GARP lists them in the third line.
⇒ C is the only third-line party.

Interpretation

External auditors sit outside the firm, which tempts candidates to leave them out of the framework altogether. GARP's version of the 2013 IIA paper includes them, together with internal audit and special audit committees.

Why the other options are wrong

A is incorrect. Originators own and manage the risk they take: first line.
B is incorrect. Credit risk management sets policies and oversees the business: second line.
D is incorrect. Compliance is independent of the business but is management oversight, so second line, not independent assurance.

LO: CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. • **Reference:** Bouteille and Coogan-Pushner, The Handbook of Credit Risk Management 2e, Ch.2 (GARP CR Ch.2), introduction

Solution CR-2 a.2

Answer: A

Bank Q23 • Schweser Module Quiz 20.1 Q1

Step 1 – Who makes up the second line

Enterprise risk management, compliance and legal. They establish policies and procedures and oversee the first line.
⇒ Including compliance and legal in risk oversight builds the second line as GARP defines it.

Step 2 – Check the direction of oversight in the other options

Oversight runs from the second line to the first, and assurance from the third line to both.

Interpretation

The second line is defined by its members and by what they do to the first line. Each wrong option describes another line or reverses an arrow.

Why the other options are wrong

B is incorrect. Business owners are the first line; leaning on them is not a second line.
C is incorrect. Reversed: the second line oversees the first, not the other way round.
D is incorrect. Independent assurance is the third line (audit), not the second.

LO: CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. • **Reference:** GARP CR Ch.2, introduction; SchweserNotes 2026 Book 2, Reading 20, LO 20.a

Solution CR-2 a.3 — Answer: A

Bank Q20 • AnalystPrep Q.5791

Step 1 – First line

The business that launches the product takes and manages its risks.

Step 2 – Second line

Risk management, compliance and legal set the policies and procedures and oversee the business.

Step 3 – Third line

Internal and external auditors give independent assurance over the first two lines.
⇒ Only A gets all three assignments right.

Interpretation

The assignment follows the verbs: own, oversee, assure. Each wrong option collapses two lines into one or hands an activity to the wrong function.

Why the other options are wrong

B is incorrect. Reverses the first two lines (the risk function never owns the business's risk) and gives policy setting to audit.
C is incorrect. Removes independence: a business cannot assure its own controls, and the second line oversees rather than merely advises.
D is incorrect. Gives ownership to auditors, who must stay independent, and assurance to compliance and legal, which are second line.

LO: CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. • **Reference:** GARP CR Ch.2, introduction

Solution CR-2 a.4 — Answer: B

Bank Q24 • AP Mock 3 Q36

Step 1 – What the second line does

It establishes policies, procedures and the methods used to identify and measure risk, and it monitors and oversees the first line.
⇒ Establishing risk management and measurement methods is a second-line role.

Step 2 – Assign the other options

Making risk decisions is first line. Reporting independently to the board (in practice its audit committee) is the mark of the third line.

Interpretation

The CRO does have privileged access to the board's risk and audit committees, but that is access, not the independent assurance role that defines internal audit. Read "independently" as the third-line signal.

Why the other options are wrong

A is incorrect. The business makes and manages risk decisions: first line.
C is incorrect. Reversed direction: the third line assures the second, so the second line does not oversee audit.
D is incorrect. Independent reporting to the board is internal audit's hallmark (third line).

LO: CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. • **Reference:** GARP CR Ch.2, introduction and Oversight (Independence)

Solution CR-2 a.5 — Answer: D

Bank Q21 • AnalystPrep Q.5792

Step 1 – The CRO's line

The CRO leads enterprise risk management, the core of the second line, alongside compliance and legal.

Step 2 – Second-line duties

Set policies and measurement standards, monitor and oversee the risks taken by the first line.
⇒ Ownership stays with the business and IT owners who generate the risks.

Step 3 – Exclude the other lines

Running the risks is first line; independent audit is third line.

Interpretation

The CRO's value is independent challenge of the owners, not doing their job for them. Taking over the risks would leave no one overseeing the people managing them.

Why the other options are wrong

A is incorrect. Takes ownership away from the first line; the second line oversees rather than manages.
B is incorrect. Passing everything to IT with no challenge abandons the oversight role.
C is incorrect. Independent audit is the third line; the CRO cannot assure work he oversees.

LO: CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. • **Reference:** GARP CR Ch.2, introduction

Solution CR-2 a.6 — Answer: B

Bank Q27 • Mock 2 Q33 • re-filed from CR-2 b

Step 1 – Who holds approval authority

Business managers are accountable for the top and bottom lines of their units, so approval authority sits with the business hierarchy and the committees to which the board's authority is delegated.

Step 2 – What risk management does

It is not a profit center and has no approval authority; it plays an advisory role. Guidelines state the criteria under which its input or recommendation is required, and that recommendation must then be obtained.
⇒ B describes the role.

Step 3 – What happens on disagreement

A veto right is rare. A risk manager who strongly disagrees attaches a written memo to the committee package and presents the opinion in person.

Interpretation

Being the second line makes risk management independent of the business, not the decision maker. Its influence comes through required recommendations, dissent on the record and the CRO's access to the CEO and the board committees.

Why the other options are wrong

A is incorrect. Overstates the role: risk managers rarely hold a veto, and they do not approve.
C is incorrect. Gives risk management approval authority, which it does not have.
D is incorrect. Confuses risk management with origination; joint origination would also compromise its independence.

LO: CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense frame-

work for effective risk management and control. CR-2 b: Explain the processes that lead to risk taking including credit origination, credit risk assessment, and credit approval processes. • **Reference:** GARP CR Ch.2, Skills

Solution CR-2 a.7 — Answer: C

New • built from GARP Ch.2, Final Words

Step 1 – GARP's statement on the CEO

"The real head of risk management is the chief executive officer." The CEO has the absolute responsibility to make the whole organization respect risk management principles.

Step 2 – What that requires in practice

The CEO and his direct reports support the principles and behave as if they were directly in charge of risk management; senior management must set the example; in strategic decisions such as mergers and acquisitions the CEO involves the risk management department at the right time.
⇒ Appointing a CRO adds an independent second line; it does not move the CEO's responsibility.

Interpretation

The CRO reports to the CEO precisely because the CEO remains responsible. Vardeez and Schweser put it more narrowly (the CEO ensures adequate governance guidelines are in place), which points the same way.

Why the other options are wrong

A is incorrect. The board, not the CEO, approves the guidelines, and the CEO's responsibility is not limited or transferred to the CRO.
B is incorrect. Independence of the second line is from the business, not from management; the CEO must involve risk management, not avoid it.
D is incorrect. Approval is delegated by transaction risk to individuals and committees; the CEO is not the first line and does not approve every large deal.

LO: CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. • **Reference:** GARP CR Ch.2, Final Words and Oversight (Independence)

Solution CR-2 a.8 — Answer: D

Workbook A22.1 • S1 • Aug-2026

Step 1 – Who owns the risk

The first line ("business unit management") owns the risks it generates and must give the final sign-off on its risk and control assessment. The second line may run the RCSA workshop, then collect and challenge the answers, while staying independent of the content.
⇒ "The second line owns the methodology; the first line owns the risks." Sign-off is a first-line act.

Step 2 – Where risk champions sit

Risk champions (line 1.5) record events, map risks and controls and follow up action plans inside the business. Firms must ensure ownership of operational risk is not assumed by, or passed to, them.

Step 3 – How far internal audit may rely on others

IIA guidance: internal audit may take account of others' work after evaluating that function, but in no circumstances relies exclusively on risk, compliance or finance; it always examines its own sample.

Step 4 – Hybrid functions

BCBS requires a hybrid function to document and distinguish its first and second line responsibilities, emphasising the independence of the second line. Table 2.4 gives exactly this split for IT security: encrypting data, partitioning and monitoring the network (line 1) versus verifying the implementation of ICT controls in business operations and enforcing security policies (line 2).

Interpretation

D is correct. The matrix entry places each IT security activity in the line its role defines, which is what the supervisor will test: the line is set by the role, not by the department's name. The fix for the other entries is to return sign-off and ownership to the first line and restore internal audit's own sampling.

Why the other options are wrong

A is incorrect. Role misassignment: the final sign-off on the risk and control assessment belongs to the first line; the second line owns only the methodology and the challenge.

B is incorrect. Sibling swap: substitutes the risk champions (line 1.5) for the risk owners (line 1); ownership must stay with the business managers.

C is incorrect. Absolute claim: a positive evaluation can justify less intense scrutiny, never exclusive reliance; internal audit must always test its own sample.

LO: ORR-2 c: Describe the "three lines of defense" model for operational risk governance and compare roles and responsibilities for each line of defense. CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. • **Reference:** GARP ORR Ch.2, Section 2.3 d) and Table 2.4; Bouteille and Coogan-Pushner Ch.2

CR-4 • Capital Structure in Banks • Vardeez: VERY IMPORTANT

CR-4 d

P1 CALC

Define and calculate unexpected loss (UL).

Workbook: A1 Unexpected Loss, UL Contribution and Economic Capital • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A1 is the map's number one topic (10 of 16 sources). The recall notes name "UL for a single component in a portfolio (numerical)", and the August 2026 sitting asked for a UL contribution built on it.
Vardeez: "Important reading: unexpected loss calculations as well as risk contributions can show up." Pointer for this LO: the unexpected loss calculation.

Reported exam shapes

- **A1·S1** Standalone UL of one loan from EA, PD, LR, σ_{LR} and the binomial σ_{PD}.
- **A1·S5** Intermediate-result trap: the single-asset UL is planted as a distractor when the UL contribution is asked.

What the questions look like: mostly numerical. You get a loan's exposure (often next to a larger commitment you must not use), PD, loss rate, the standard deviation of the loss rate, and either σ_{PD} or the words "default is a Bernoulli event". Concept questions test that UL is a standard deviation (not the mean, not the variance) and that it cannot be priced into the loan rate.

Worked elsewhere: A1.2 (standalone UL, then the UL contribution) at Question CR-4 f.7; A1.3 (portfolio UL into economic capital) under CR-4 g. This block carries the shared background for CR-4; the later blocks recap it.

1. From expected loss to economic capital: the chain in this reading

Credit risk is the risk of loss from any non-payment or rescheduling of promised payments (default events) or from unexpected *credit migration* (a change in the borrower's credit quality). Schroeck's bottom-up approach turns that risk into a capital number in four steps, and each LO of the reading is one link:

Step	Quantity	Formula	LO
1	Expected loss	$\mathrm{EL} = \mathrm{EA} \times \mathrm{PD} \times \mathrm{LR}$	CR-4 c
2	Standalone unexpected loss	$\mathrm{UL} = \mathrm{EA}\sqrt{\mathrm{PD}\,\sigma_{LR}^2 + \mathrm{LR}^2\sigma_{PD}^2}$	CR-4 d, e
3	UL contribution in the portfolio	$\mathrm{ULC}_i = \mathrm{UL}_i \sum_j \rho_{ij}\mathrm{UL}_j/\mathrm{UL}_P$	CR-4 f
4	Economic capital	$\mathrm{EC}_P = \mathrm{UL}_P \times \mathrm{CM}$, $\mathrm{EC}_i = \mathrm{ULC}_i \times \mathrm{CM}$	CR-4 g

The three inputs (CR-4 b)

- **Probability of default (PD)**, also called expected default frequency (EDF): the probability that the borrower defaults before the end of the horizon, by convention one year. It is borrower-specific and tied to the rating.
- **Exposure amount (EA)**: the expected exposure at the time of default, including all promised interest. Amortisation schedules and credit lines (limit against utilisation) change it, so it can differ a lot from the amount at origination.
- **Loss rate (LR)**: the fraction of EA lost if default occurs, after collateral sales and collection costs.

Also called severity, loss given default (LGD) or loss in the event of default (LIED). LR $= 1 -$ RR, with RR the recovery rate; LR is largely a function of collateral.

Notation. This reading writes EA and LR; CR-3 and CR-5 write EAD and LGD for the same quantities. Questions use both.

Expected loss is a cost, unexpected loss is the risk

EL is the long-run average loss, an actuarial cost of doing business that is reserved for (loan loss reserves, "standard risk costs") and recovered through loan pricing. In GARP's words, "by definition, EL does not itself constitute risk": if losses always equalled EL there would be no reason to hold capital. Risk is the *variation* of losses around EL. That variation is the unexpected loss, it cannot be anticipated or priced into the loan rate, and it is what economic capital is held against.

2. Standalone unexpected loss

UL is the standard deviation of credit losses at the horizon H: $\text{UL}_H \equiv \sqrt{\text{var}(L_H)}$. "Standalone" means the loan is looked at on its own, ignoring diversification (that comes in CR-4 f).

Where the formula comes from

Write the loss as $L = \text{EA} \times D \times \widetilde{LR}$, where the default indicator D is 1 with probability PD and 0 otherwise, and the realised loss rate $\widetilde{LR}$ has mean LR and standard deviation σ_{LR}. Take EA as given and D independent of $\widetilde{LR}$ (default is a binary event, so PD is independent of EA and LR; EA and LR are treated as independent too). Then

$$\text{E}[L] = \text{EA} \cdot \text{PD} \cdot \text{LR}, \qquad \text{E}[L^2] = \text{EA}^2 \cdot \text{PD} \cdot (\sigma_{LR}^2 + \text{LR}^2)$$

$$\text{var}(L) = \text{EA}^2 \left[\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\,\text{PD}(1-\text{PD})\right] = \text{EA}^2 \left[\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\,\sigma_{PD}^2\right]$$

using $\sigma_{PD}^2 = \text{PD}(1-\text{PD})$ for the Bernoulli default indicator (CR-4 e). Taking the square root gives GARP's eq. 4.2:

$$\boxed{\text{UL} = \text{EA} \times \sqrt{\text{PD} \cdot \sigma_{LR}^2 + \text{LR}^2 \cdot \sigma_{PD}^2}}$$

- **Two sources of uncertainty.** $\text{PD}\,\sigma_{LR}^2$ is severity uncertainty (it only matters if default happens, so it is weighted by PD); $\text{LR}^2\sigma_{PD}^2$ is default uncertainty. Note the asymmetry: PD enters the first term unsquared, LR enters the second squared.
- **No uncertainty, no UL.** If both variances were zero, UL would be zero and EL would be known with certainty: no credit risk.
- **Loss rate known** ($\sigma_{LR} = 0$): $\text{UL} = \text{EA} \times \text{LR} \times \sigma_{PD} = \text{EA} \times \text{LR} \times \sqrt{\text{PD}(1-\text{PD})}$.
- **UL is a fraction of EA.** Each term under the root is below one, so the root is the UL per unit of exposure.
- **Horizon.** One year by convention, and all volatilities are annualised so that business units can be compared. A loan's remaining maturity does not enter.

3. Worked examples

XYZ Bank (Vardeez and Schweser). Commitment \$2,000,000, of which \$1,800,000 is outstanding; PD 1%, LR 40%, $\sigma_{PD} = 10\%$, $\sigma_{LR} = 30\%$.

$$\text{EL} = 1{,}800{,}000 \times 0.01 \times 0.40 = \$7{,}200$$

$$\begin{aligned}\text{UL} &= 1{,}800{,}000\sqrt{0.01(0.30)^2 + (0.40)^2(0.10)^2} = 1{,}800{,}000\sqrt{0.0009 + 0.0016} \\ &= 1{,}800{,}000 \times 0.05 = \$90{,}000\end{aligned}$$

UL is 5% of the exposure and 12.5 times EL. The exposure is the \$1.8 million drawn, not the \$2 million commitment. The 10% σ_{PD} is the binomial value rounded: $\sqrt{0.01 \times 0.99} = 9.95\%$ would give \$89,712.

Bigger Bank, asset A. EA \$8,250,000, PD 0.5%, LR 50%, $\sigma_{PD} = 7.05\%$ $(= \sqrt{0.005 \times 0.995})$, $\sigma_{LR} = 25\%$:

$$\begin{aligned}\text{UL}_A &= 8{,}250{,}000\sqrt{0.005(0.25)^2 + (0.50)^2(0.0705)^2} = 8{,}250{,}000\sqrt{0.0003125 + 0.0012426} \\ &= 8{,}250{,}000 \times 0.039434 = \$325{,}333\end{aligned}$$

against $\text{EL}_A = 8{,}250{,}000 \times 0.005 \times 0.50 = \$20{,}625$: UL is about 15.8 times EL. In both examples the default term dominates (64% of the variance for XYZ, 80% for asset A).

UL is not proportional to EL

With the loss rate known, $\text{UL}/\text{EL} = \sigma_{PD}/\text{PD} = \sqrt{(1 - \text{PD})/\text{PD}}$:

PD	0.5%	1%	2%	5%	20%	50%
UL/EL	14.11	9.95	7.00	4.36	2.00	1.00

Low-PD loans carry far more risk per unit of expected loss, which is why UL "typically exceeds" EL for bank loans. It is not a rule: above a 50% PD the UL falls below the EL, and both are zero when PD is zero (Question CR-4 d.2).

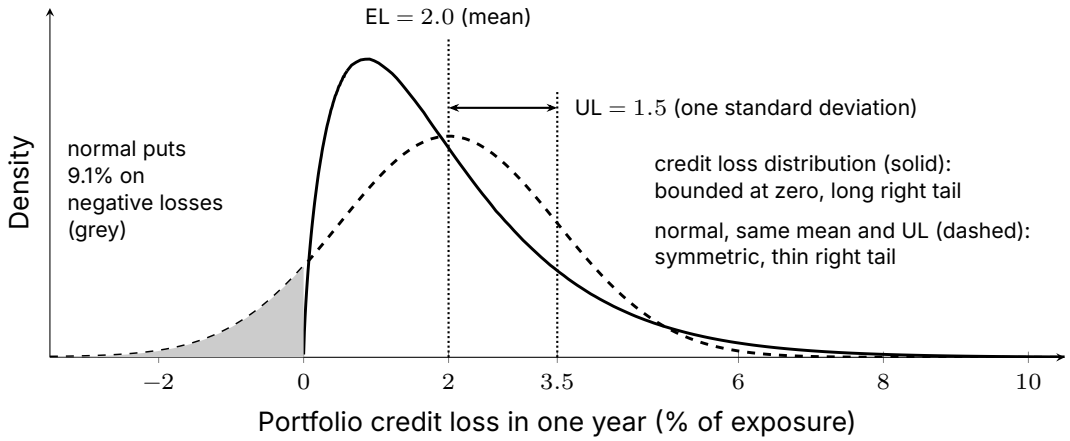


Figure 1. A credit loss distribution with EL $= 2\%$ and UL $= 1.5\%$ of exposure (a beta distribution, solid) against a normal curve with the same two moments (dashed). EL is the mean, not the most likely outcome: the peak is near 0.9% and 60% of years lose less than EL. UL measures the spread around EL; the capital question is how far into the right tail to go (CR-4 g).

4. Which inputs go into the formula

- **Exposure: drawn, not committed.** Use the outstanding amount (XYZ: \$1.8 million of \$2 million). If the stem gives a drawdown on default for the unused line, add it: EA = drawn + drawdown fraction × undrawn. A \$100,000 line, 80% drawn, 70% drawdown: EA = $80{,}000 + 0.70 \times 20{,}000 = 94{,}000$, and the same EA goes into both EL and UL (Question CR-4 d.5).
- **σ_{PD}: given or binomial.** If the stem gives σ_{PD}, use it as given. If it says default is Bernoulli or binomial, compute $\sqrt{\text{PD}(1-\text{PD})}$ (CR-4 e). AP Mock 3 Q59 gives $\sigma_{PD} = 4\%$ for a 1% PD, which is not the binomial 9.95%; the key uses the 4% (Question CR-4 d.4).
- **Loss rate, not recovery rate.** A 60% recovery is a 40% loss rate. Putting RR where LR belongs is the commonest wrong-input twin.
- **Standard deviations, not variances.** The squares go *inside* the root: σ_{LR}^2 and σ_{PD}^2. Adding the two square roots (a "sum of standard deviations") overstates UL.
- **Red herrings.** Remaining maturity, the commitment, the EL itself when it is quoted, and portfolio data when only the standalone UL is asked.

Standalone UL against UL contribution (shape S5)

Inside a portfolio a loan adds only its **UL contribution**, which is smaller than its standalone UL because default correlations are below one. For a large pool of similar loans $\text{ULC}_i \approx \text{UL}_i\sqrt{\rho}$ (CR-4 f). When the question asks for the contribution, the standalone UL you computed on the way is the planted wrong answer (Question CR-4 d.6). Economic capital for one loan is $\text{ULC}_i \times \text{CM}$, not $\text{UL}_i \times \text{CM}$ (CR-4 g).

> **Correction to the source notes.** Vardeez's CR-4 c text says that "the difference between the actual loss and the expected loss is called the unexpected loss" and that "the sum EL + UL is the total potential loss". In GARP (Schroeck, eq. 4.2) UL is the *standard deviation* of credit losses around EL, a one-standard-deviation measure of spread, not a realised difference. EL + UL is not a loss bound either: in Figure 1, losses exceed EL + UL in about 15% of years. The capital cushion is $\text{CM} \times \text{UL}_P$ measured from EL, with CM around 7 at 99.97% (CR-4 g). Adding EL to UL is the planted wrong answer C of Question CR-4 d.3.

Formula card: CR-4 d

Expected loss (CR-4 c)	$\text{EL} = \text{EA} \times \text{PD} \times \text{LR}, \quad \text{LR} = 1 - \text{RR}$
Standalone UL	$\text{UL} = \text{EA}\sqrt{\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\,\sigma_{PD}^2}$
Binomial default variance	$\sigma_{PD}^2 = \text{PD}(1-\text{PD})$ (use a given σ_{PD} if the stem supplies one)
Known loss rate, $\sigma_{LR} = 0$	$\text{UL} = \text{EA} \times \text{LR} \times \sqrt{\text{PD}(1-\text{PD})}, \quad \text{UL/EL} = \sqrt{(1-\text{PD})/\text{PD}}$
Exposure on a credit line	EA = drawn + drawdown fraction × undrawn (never the full commitment by default)
Contribution (CR-4 f)	$\text{ULC}_i \approx \text{UL}_i\sqrt{\rho}$ for a large pool of similar loans

	Expected loss (EL)	Unexpected loss (UL)
Statistic	Mean of the loss distribution	Standard deviation of the loss distribution
Inputs	EA, PD, LR	EA, PD, LR, σ_{PD}, σ_{LR}
Covered by	Loan loss reserves and pricing	Economic capital, through CM
Priced in the loan rate?	Yes, a foreseeable cost	No, it cannot be anticipated
Portfolio	Additive: $\text{EL}_P = \sum \text{EL}_i$	Not additive: depends on correlations
Constitutes risk?	No (GARP)	Yes: risk is the variation around EL

Exam traps

- **Commitment as exposure.** Using the full facility instead of the drawn amount (plus any stated drawdown on default).
- **Dropped term.** Keeping only PD σ_{LR}^2 or only $\text{LR}^2\sigma_{PD}^2$. With a known loss rate the first term is zero by design; otherwise both belong.
- **Wrong input twin.** Recovery rate in place of the loss rate; PD or PD^2 in place of $\sigma_{PD}^2 = \text{PD}(1-\text{PD})$; σ_{PD} where σ_{PD}^2 belongs.
- **Variance reported as UL.** UL is the square root; a number in squared currency is not UL.
- **Sum of roots.** $\sqrt{a}+\sqrt{b}$ instead of $\sqrt{a+b}$ overstates UL.
- **EL added to UL.** Capital covers UL (times CM), not EL + UL; EL is reserved and priced.
- **Standalone UL when the contribution is asked (S5).** The intermediate result is always among the options.
- **Inconsistent exposure.** Computing EL on one exposure and UL on another when the stem defines a drawdown.

Practice questions: CR-4 d

Question CR-4 d.1 Bank Q111 • AnalystPrep Q.5877 (rebuilt)

A credit risk trainer at a commercial bank asks a new analyst to describe the standalone unexpected loss (UL) of a loan in the standard bottom-up credit risk framework. Which of the following descriptions is correct?

A. The mean of the credit loss distribution, equal to exposure times PD times loss rate.

B. The standard deviation of actual credit losses around the expected loss.

C. The variance of credit losses around the expected loss.

D. The part of credit losses that can be priced into the loan's interest rate.

Question CR-4 d.2

Bank Q114 • Schweser Module Quiz 22.1 Q2

A bank's credit committee is reviewing the expected loss (EL) and the unexpected loss (UL) reported for its corporate loans. Which of the following statements about EL and UL is correct?

A. EL always exceeds UL.

B. UL always exceeds EL.

C. EL and UL are parameterised by exactly the same set of variables.

D. EL is directly related to exposure.

Question CR-4 d.3

Workbook A1.1 • S1 • Map #1

A junior credit analyst at a Singapore commercial bank is computing the standalone risk of a single term loan for the bank's economic capital report. The data are:

- Exposure amount (EA): SGD 20 million
- One-year PD: 2.0%; default is a Bernoulli event
- Expected loss rate (LR): 40%; standard deviation of the loss rate: 25%
- EA and LR are independent of each other and of default

Which of the following is closest to the standalone unexpected loss (UL) of the loan?

A. SGD 1.12 million

B. SGD 1.32 million

C. SGD 1.48 million

D. SGD 1.82 million

Question CR-4 d.4

Bank Q108 • AP Mock 3 Q59 (rebuilt)

American Banking Group approved a loan to Hazard Corp five years ago with a total commitment of USD 120 million. Today the outstanding balance is USD 80 million. The credit risk department reports:

Parameter	Value
Probability of default over the next year	1%
Loss rate if the customer defaults	25%
Standard deviation of the probability of default	4%
Standard deviation of the loss rate	15%

Which of the following is closest to the unexpected loss for this loan today?

A. USD 0.800 million

B. USD 1.200 million

C. USD 1.442 million

D. USD 2.163 million

Question CR-4 d.5

Bank Q110 • AnalystPrep Q.3683

A bank has booked a credit line with a total commitment of USD 100,000, of which 80% is currently drawn. The default probability over the next year is 2% and the loss given default (LGD) is 40%. The standard deviation of LGD is 30% and the standard deviation of the probability of default is 14%. The bank expects 70% of the undrawn amount to be drawn before a default. Which of the following is closest to the expected and unexpected losses for the bank?

A. Expected loss USD 640; unexpected loss USD 5,621

B. Expected loss USD 640; unexpected loss USD 6,604

C. Expected loss USD 752; unexpected loss USD 5,621

D. Expected loss USD 752; unexpected loss USD 6,604

Question CR-4 d.6

Bank Q124 • GARP PE1 Q25 • Mock 3 Q55 (rebuilt) • re-filed from CR-4 f

An analyst in the credit risk department of a bank is estimating the unexpected loss contributions (ULC) of individual loans to overall credit portfolio risk. The portfolio consists of many loans that have approximately the same characteristics and size. Information about each loan:

Exposure amount	CNY 3,000,000
Constant pairwise default correlation	0.25
Annual probability of default (PD)	4%
Variance of PD, PD$(1 - \text{PD})$	0.0384
Annual loss rate (LR)	15%
Standard deviation of LR	9%

Which of the following is the correct estimate of the ULC of a single loan to the overall credit portfolio risk?

A. CNY 18,000

B. CNY 25,851

C. CNY 51,701

D. CNY 103,402

Solutions • CR-4 d

Solution CR-4 d.1 — Answer: B

Bank Q111 • AnalystPrep Q.5877

Step 1 – Place EL and UL on the same distribution

EL is the *mean* of the credit loss distribution, EA $\times$ PD $\times$ LR. It is predictable, so it is reserved for and priced.

Step 2 – Identify the dispersion measure

GARP: "UL, in statistical terms, is the standard deviation of credit losses, that is, the standard deviation of actual credit losses around the expected loss average."

$\Rightarrow$ UL is a standard deviation, in currency units, of the same distribution whose mean is EL.

Interpretation

UL is the input to everything that follows in the reading: the UL contribution (CR-4 f) and economic capital (CR-4 g). Because it cannot be anticipated, it needs a capital cushion rather than a price.

Why the other options are wrong

A is incorrect. This is the expected loss, the mean of the distribution, not its dispersion.

C is incorrect. The variance is in squared currency; UL is its square root.

D is incorrect. Polarity flip: EL is the part that is priced. ULs "cannot be adequately priced for in a loan's interest rate", which is why they need economic capital.

LO: CR-4 d: Define and calculate unexpected loss (UL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Unexpected Losses (UL-Standalone)

Solution CR-4 d.2 — Answer: D

Bank Q114 • Schweser Module Quiz 22.1 Q2

Step 1 – EL and exposure

EL $=$ EA $\times$ PD $\times$ LR is linear in EA: double the exposure and EL doubles.

$\Rightarrow$ D is correct.

Step 2 – Test the "always" statements

With the loss rate known, UL/EL $= \sqrt{(1-\text{PD})/\text{PD}}$: 9.95 at a 1% PD, 1.00 at 50%, and 0.50 at 80%. At PD $= 0$ both are zero.

$\Rightarrow$ UL usually exceeds EL for bank loans, but neither ordering holds always.

Step 3 – Compare the inputs

UL also needs σ_{LR} and σ_{PD}; EL does not.

Interpretation

The table in Section 3 is the quick way to see this: the lower the PD, the larger UL is relative to EL.

Why the other options are wrong

A is incorrect. For typical bank loans UL is several times EL (12.5 times for XYZ Bank).

B is incorrect. Absolute qualifier: UL falls below EL once PD exceeds 50% (with a known loss rate), and both are zero when PD is zero.

C is incorrect. UL needs the two standard deviations as well as EA, PD and LR.

LO: CR-4 d: Define and calculate unexpected loss (UL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.1 and 4.2; SchweserNotes 2026 Book 2, Reading 22

Solution CR-4 d.3 — Answer: B

Workbook A1.1 • S1 • Map #1

Step 1 – Variance of the default indicator

$$\sigma_{PD}^2 = \text{PD}(1-\text{PD}) = 0.02 \times 0.98 = 0.0196, \qquad \sigma_{PD} = 0.14$$

$\Rightarrow$ Default is a 0/1 event with a 2% chance of a 1; its standard deviation is 14 percentage points.

Step 2 – Loss-rate uncertainty term

$$\text{PD} \cdot \sigma_{LR}^2 = 0.02 \times 0.25^2 = 0.00125$$

Step 3 – Default uncertainty term

$$\text{LR}^2 \cdot \sigma_{PD}^2 = 0.40^2 \times 0.0196 = 0.003136$$

$\Rightarrow$ Default uncertainty dominates: about 72% of the loss variance per unit of EA comes from not knowing whether default happens.

Step 4 – Standalone UL

$$\text{UL} = 20 \times \sqrt{0.00125 + 0.003136} = 20 \times 0.06623 = \text{SGD } 1.32 \text{ million}$$

Interpretation

EL is only $0.02 \times 20 \times 0.40 = $ SGD 0.16 million, which is priced into the loan. The UL of SGD 1.32 million is the one-standard-deviation surprise around that average; it is the input to the loan's UL contribution and hence to its economic capital, after diversification.

Why the other options are wrong

A is incorrect. Ignored loss-rate volatility (dropped the $\text{PD} \cdot \sigma_{LR}^2$ term): $20 \times 0.40 \times 0.14 = 1.12$.
C is incorrect. Added EL to UL, treating the capital cushion as covering expected loss too: $1.32 + 0.16 = 1.48$.
D is incorrect. Used the recovery rate (60%) instead of the loss rate in the default term: $20 \times \sqrt{0.00125 + 0.36 \times 0.0196} = 1.82$.

LO: CR-4 d: Define and calculate unexpected loss (UL). CR-4 e: Estimate the variance of default probability assuming a binomial distribution. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Unexpected Losses (UL-Standalone), eqs. 4.2 and 4.3

Solution CR-4 d.4 — Answer: C

Bank Q108 • AP Mock 3 Q59

Step 1 – Exposure today

EA = USD 80 million, the outstanding balance. The USD 120 million commitment was the original limit.

Step 2 – The two variance terms

$$\text{PD}\,\sigma_{LR}^2 = 0.01 \times 0.15^2 = 0.000225, \qquad \text{LR}^2\sigma_{PD}^2 = 0.25^2 \times 0.04^2 = 0.0001$$

$\Rightarrow$ The stem supplies $\sigma_{PD} = 4\%$, so it is used as given.

Step 3 – Standalone UL

$$\text{UL} = 80 \times \sqrt{0.000325} = 80 \times 0.018028 = \text{USD } 1.442 \text{ million}$$

Interpretation

A binomial σ_{PD} for a 1% PD would be 9.95%, giving USD 2.32 million; the question fixes σ_{PD} at 4%, and the stem's number wins. The commitment is the planted input.

Why the other options are wrong

A is incorrect. Dropped the PD σ_{LR}^2 term: $80 \times \sqrt{0.0001} = 0.800$.
B is incorrect. Dropped the LR$^2\sigma_{PD}^2$ term: $80 \times \sqrt{0.000225} = 1.200$.
D is incorrect. Used the USD 120 million commitment as the exposure: $120 \times 0.018028 = 2.163$.

LO: CR-4 d: Define and calculate unexpected loss (UL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eq. 4.2; SchweserNotes 2026 Book 2, Reading 22, XYZ Bank example

Solution CR-4 d.5

Answer: D

Bank Q110 • AnalystPrep Q.3683

Step 1 – Exposure at default

$$\text{EA} = 80{,}000 + 0.70 \times 20{,}000 = 94{,}000$$

$\Rightarrow$ The drawn USD 80,000 plus the 70% of the undrawn USD 20,000 expected to be drawn before default.

Step 2 – Expected loss

$$\text{EL} = 94{,}000 \times 0.02 \times 0.40 = \text{USD } 752$$

Step 3 – Unexpected loss

$$\begin{aligned}\text{UL} &= 94{,}000\sqrt{0.02 \times 0.30^2 + 0.40^2 \times 0.14^2} = 94{,}000\sqrt{0.0018 + 0.003136} \\ &= 94{,}000 \times 0.070257 = \text{USD } 6{,}604\end{aligned}$$

Interpretation

GARP defines EA as the expected exposure *at the time of default*, and credit lines (limit against utilisation) change it. Both EL and UL are built on the same EA; UL is almost nine times EL.

Why the other options are wrong

A is incorrect. Ignored the drawdown and used the drawn USD 80,000 for both: EL 640 and UL $80{,}000 \times 0.070257 = 5{,}621$.
B is incorrect. Inconsistent exposure: EL on 80,000 (640) but UL on 94,000 (6,604).
C is incorrect. Inconsistent exposure the other way: EL on 94,000 (752) but UL on 80,000 (5,621).

LO: CR-4 d: Define and calculate unexpected loss (UL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Expected Losses (exposure amount) and eq. 4.2

Solution CR-4 d.6

Answer: C

Bank Q124 • GARP PE1 Q25 • Mock 3 Q55

Step 1 – Standalone UL of one loan

$$\begin{aligned}\text{UL} &= 3{,}000{,}000\sqrt{0.04 \times 0.09^2 + 0.15^2 \times 0.0384} = 3{,}000{,}000\sqrt{0.000324 + 0.000864} \\ &= 3{,}000{,}000 \times 0.034467 = \text{CNY } 103{,}402\end{aligned}$$

$\Rightarrow$ This is the intermediate result, not the answer.

Step 2 – Contribution in a large pool of similar loans

$$\text{ULC} = \text{UL} \times \sqrt{\rho} = 103{,}402 \times \sqrt{0.25} = 103{,}402 \times 0.5 = \text{CNY } 51{,}701$$

$\Rightarrow$ Inside the portfolio each loan keeps only half of its standalone risk.

Interpretation

This is shape S5: the stem walks you through the standalone UL, and GARP lists it as option D. The question asks for the contribution, so one more step is needed (CR-4 f derives the $\sqrt{\rho}$ shortcut).

Why the other options are wrong

A is incorrect. Computed the expected loss, $3{,}000{,}000 \times 0.04 \times 0.15 = 18{,}000$, which is not a risk measure.

B is incorrect. Multiplied by $\rho = 0.25$ instead of $\sqrt{\rho} = 0.5$: $103{,}402 \times 0.25 = 25{,}851$.

D is incorrect. Stopped at the standalone UL (intermediate result): 103,402.

LO: CR-4 d: Define and calculate unexpected loss (UL). CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.2, 4.14 and 4.15

CR-4 • Capital Structure in Banks • Vardeez: VERY IMPORTANT

CR-4 e

P1 CALC

Estimate the variance of default probability assuming a binomial distribution.

Workbook: A1 Unexpected Loss, UL Contribution and Economic Capital • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: the binomial σ_{PD} is the second input under the root of every A1 unexpected loss (shape S1), so a slip here breaks the UL, the UL contribution and the economic capital that follow.

Vardeez: no separate pointer; Vardeez's note for the reading flags unexpected loss calculations, and this LO is the piece of them that the binomial assumption supplies.

Reported exam shape

- **A1·S1** Standalone UL from EA, PD, LR, σ_{LR} and the binomial σ_{PD}: here the focus is on getting $\sigma_{PD} = \sqrt{\text{PD}(1-\text{PD})}$ right.

What the questions look like: usually the words "default is a Bernoulli (binomial) event" inside a UL stem. Standalone versions ask for the variance against the standard deviation against PD^2, which of two loans has the larger variance, or the variance of the number of defaults in a pool.

Worked elsewhere: A1.1 (binomial σ_{PD} into a standalone UL) at Question CR-4 d.3; A1.2 (the same with a known loss rate, then the UL contribution) under CR-4 f. Background on EL and UL: CR-4 d, Sections 1 and 2.

1. Default as a Bernoulli variable

Over the one-year horizon a borrower either defaults or it does not. Write the default indicator as $D = 1$ with probability PD and $D = 0$ with probability $1 - \text{PD}$: a Bernoulli variable, which is a binomial with one trial, $B(1; \text{PD})$. Since $D^2 = D$,

$$\text{E}[D] = \text{PD}, \qquad \text{E}[D^2] = \text{PD}, \qquad \boxed{\sigma_{PD}^2 = \text{E}[D^2] - \text{E}[D]^2 = \text{PD}(1-\text{PD})}$$

$$\sigma_{PD} = \sqrt{\text{PD}(1-\text{PD})}$$

This is GARP's eq. 4.3. "Variance of the default probability" in the LO wording means the variance of this 0/1 default event; the PD is its mean.

PD	0.5%	1%	2%	4%	10%	50%
$\sigma_{PD}^2 = \text{PD}(1-\text{PD})$	0.004975	0.0099	0.0196	0.0384	0.09	0.25
σ_{PD}	7.05%	9.95%	14.0%	19.6%	30.0%	50.0%
σ_{PD}/PD	14.1	9.95	7.00	4.90	3.00	1.00

- **Small PD: variance close to PD.** For PD of a few per cent, $\text{PD}(1-\text{PD}) \approx \text{PD}$, so $\sigma_{PD} \approx \sqrt{\text{PD}}$. The standard deviation is many times the PD itself: a 1% PD has a 9.95% standard deviation.
- **Maximum at 50%.** The variance peaks at 0.25 when $\text{PD} = 50\%$, the point of greatest uncertainty about the outcome.
- **Symmetric about 50%.** PD 10% and PD 90% have the same variance, 0.09. A PD of 80% (variance 0.16) is *more* uncertain than one of 10% (Question CR-4 e.2).

- **The numbers in the examples are binomial.** Bigger Bank's $\sigma_{PD} = 7.05\%$ for a 0.5% PD and A1.1's 14% for a 2% PD are exactly $\sqrt{\text{PD}(1-\text{PD})}$; XYZ Bank's 10% for a 1% PD is 9.95% rounded.

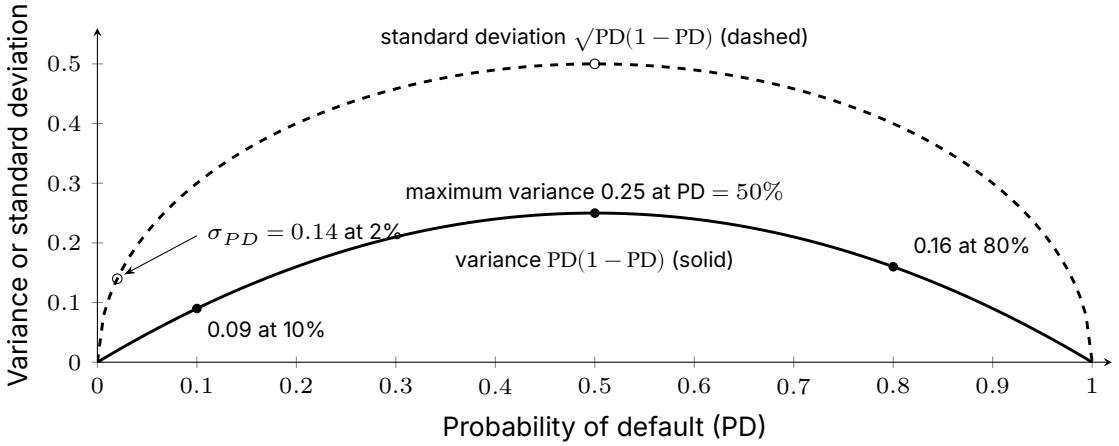


Figure 2. Variance (solid) and standard deviation (dashed) of the Bernoulli default indicator. Both are symmetric about 50%, where the outcome is least predictable. For the PDs of bank loans the standard deviation rises steeply: it is already 0.14 at a 2% PD (open circle). Filled dots: the two loans of Question CR-4 e.2 and the maximum.

2. From one loan to a pool: the binomial with n trials

If a pool holds n loans with the same PD and defaults are *independent*, the number of defaults N is binomial, $B(n;\text{PD})$, and variances add:

$$\text{E}[N] = n\,\text{PD}, \qquad \text{var}(N) = n\,\text{PD}(1-\text{PD}), \qquad \text{var}\left(\tfrac{N}{n}\right) = \frac{\text{PD}(1-\text{PD})}{n}$$

Example (Question CR-4 e.3): 500 loans with a 1.5% PD give an expected 7.5 defaults, a variance of $500 \times 0.015 \times 0.985 = 7.3875$ and a standard deviation of 2.72 defaults. The default *rate* has a standard deviation of only $\sqrt{0.014775/500} = 0.54\%$: with independence, size diversifies the risk away.

With default correlation ρ between every pair, the covariance terms come back:

$$\text{var}(N) = n\,\text{PD}(1-\text{PD})\big[1 + (n-1)\rho\big]$$

This is CR-4 f's identical-loan formula $\text{UL}_P^2 = \text{UL}^2(n + \rho(n^2 - n))$ with $\text{EA} = \text{LR} = 1$, so that each loan's UL is just σ_{PD}. The same 500 loans with $\rho = 0.02$ have $\text{var}(N) = 7.3875 \times 10.98 = 81.1$, a standard deviation of 9.0 defaults instead of 2.7. Even a small correlation dominates the pool's risk; this is the undiversifiable part that CR-4 f measures.

3. Where the default volatility goes: the default term of UL

σ_{PD}^2 is the second term under the root of the standalone UL (CR-4 d):

$$\text{UL} = \text{EA}\sqrt{\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\,\text{PD}(1-\text{PD})} \quad \xrightarrow{\sigma_{LR}=0} \quad \text{UL} = \text{EA} \times \text{LR} \times \sqrt{\text{PD}(1-\text{PD})}$$

- **UL grows with the square root of PD, EL linearly.** With a known loss rate, raising PD from 1% to 4% multiplies EL by 4 but UL by only $\sqrt{0.0384/0.0099} = 1.97$ (Question CR-4 e.5). A downgrade raises expected loss faster than unexpected loss.

- **Given or computed.** If the stem states σ_{PD}, use it; if it states that default is Bernoulli or binomial, compute it. A stem can give a value that is not binomial (AP Mock 3 Q59: 4% for a 1% PD, Question CR-4 d.4).
- **Reverse-solving.** A reported σ_{PD} implies a PD through $\text{PD}^2 - \text{PD} + \sigma_{PD}^2 = 0$:

$$\text{PD} = \frac{1 - \sqrt{1 - 4\sigma_{PD}^2}}{2}$$

taking the smaller root (the larger one, $1 - \text{PD}$, has the same variance). $\sigma_{PD} = 21.79\%$ gives $\text{PD} = (1 - \sqrt{0.81008})/2 = 5.00\%$ (Question CR-4 e.6).

4. The other variance: the loss rate

The default event has a natural distribution; the loss rate does not. GARP: σ_{LR}^2 is hard to measure for lack of historical data, so a distribution must be assumed, and each choice gives a different variance:

- **Binomial (all or nothing)** *overstates* the variance: either all of the exposure is lost or nothing. A 40% mean loss rate treated this way would have $\sigma_{LR} = \sqrt{0.4 \times 0.6} = 49\%$, far above the 25% to 30% used in the examples.
- **Uniform** assumes every loss between 0% and 100% is equally likely ($\sigma = 1/\sqrt{12} = 28.9\%$ around a fixed 50% mean).
- **Normal** is "the most reasonable assumption, because of the lack of better knowledge", shaped so that some borrowers recover almost everything and losing everything is very unlikely.

For convenience the volatility of LR is held constant over time. In short: the binomial is right for the default event and wrong for the loss rate.

Formula card: CR-4 e

Bernoulli default variance	$\sigma_{PD}^2 = \text{PD}(1 - \text{PD})$, $\sigma_{PD} = \sqrt{\text{PD}(1 - \text{PD})}$ (GARP eq. 4.3)
Shape	maximum 0.25 at $\text{PD} = 50\%$; symmetric: PD and $1 - \text{PD}$ give the same variance
Pool of n, independent	$\text{E}[N] = n\,\text{PD}$, $\text{var}(N) = n\,\text{PD}(1 - \text{PD})$
Pool of n, correlation ρ	$\text{var}(N) = n\,\text{PD}(1 - \text{PD})\,[1 + (n - 1)\rho]$
Into UL	$\text{UL} = \text{EA}\sqrt{\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\,\text{PD}(1 - \text{PD})}$; known LR: $\text{EA} \cdot \text{LR} \cdot \sigma_{PD}$
Reverse-solve	$\text{PD} = (1 - \sqrt{1 - 4\sigma_{PD}^2})/2$ (smaller root)

	One loan	Pool, independent	Pool, correlated
Distribution	Bernoulli, $B(1; \text{PD})$	Binomial, $B(n; \text{PD})$	Not binomial: fatter right tail
Variance	$\text{PD}(1 - \text{PD})$	$n\,\text{PD}(1 - \text{PD})$	$n\,\text{PD}(1-\text{PD})[1+(n-1)\rho]$
Default rate variance	$\text{PD}(1 - \text{PD})$	Falls as $1/n$	Tends to $\rho\,\text{PD}(1 - \text{PD})$
Used in	Standalone UL (CR-4 d)	Diversification intuition	Portfolio UL (CR-4 f)

Exam traps

- **PD^2 for the variance.** $0.02^2 = 0.0004$, not 0.0196: the missing term is $-\text{PD}^2$, not $\times\text{PD}$.
- **PD for the variance.** Close for small PD (0.0200 against 0.0196) but not equal, and it fails badly for large PDs.
- **Variance and standard deviation swapped.** The UL formula needs σ_{PD}^2 under the root; feeding σ_{PD} (0.14) in its place, or quoting 0.14 when the variance is asked, are both on the option list.
- **Low PD read as high uncertainty.** The variance rises with PD up to 50%; 80% is more uncertain than 10%.
- **Mean reported as a variance.** For a pool, n PD is the expected number of defaults; the variance carries the extra $(1 - \text{PD})$.
- **Wrong root.** Reverse-solving gives PD and $1 - \text{PD}$; a borrower with $\sigma_{PD} = 21.8\%$ has a 5% PD, not 95%.
- **Binomial loss rate.** The binomial overstates the variance of LR; GARP prefers the normal.

Practice questions: CR-4 e

Question CR-4 e.1

Bank Q116 • New question written for this bank (GARP practice-exam style)

A credit analyst at a bank models the default of an individual corporate borrower as a two-state event over a one-year horizon. The borrower has a 2% probability of default. Assuming default follows a binomial (Bernoulli) distribution, which of the following is the variance of the default probability?

A. 0.0004

B. 0.0196

C. 0.0200

D. 0.1400

Question CR-4 e.2

Bank Q117 • New question written for this bank (GARP practice-exam style)

A credit officer compares two loans that each follow a two-state default model over one year. Loan X has a probability of default of 10% and Loan Y has a probability of default of 80%. Assuming binomial default, which of the following statements about the variance of the default probability is correct?

A. Loan X has the higher variance, 0.09, because its probability of default is closer to zero and therefore more certain to be a non-default.

B. Loan Y has the higher variance, 0.16, compared with 0.09 for Loan X.

C. Loan Y has the higher variance, 0.64, because the variance equals the squared probability of default.

D. The two loans have equal variance, because both are two-state outcomes.

Question CR-4 e.3

Bank Q115 • AnalystPrep Q.6186 (rebuilt)

A portfolio manager analyses a subset of a loan portfolio made up of 500 loan exposures. Each loan has a one-year probability of default of 1.5%, and defaults are assumed independent, so that each loan is a two-state (default or no default) outcome. Assuming a binomial distribution, which of the following is the variance of the number of defaults in the subset?

A. 0.1125

B. 2.72

C. 7.3875

D. 7.5

Question CR-4 e.4

Bank Q112 • BT P2.T6.506.2 (rebuilt) • re-filed from CR-4 d

A bank's credit exposure to a customer consists of the following:

Exposure amount (EA)	USD 50.0 million
Probability of default (PD)	2.0%
Loss rate (LR)	50.0%
Standard deviation of the loss rate	40.0%
Expected loss	USD 0.5 million

Assume default is a binomial (Bernoulli) event. Which of the following is nearest to the exposure's unexpected loss (UL)?

A. USD 2.83 million

B. USD 2.87 million

C. USD 4.50 million

D. USD 6.33 million

Question CR-4 e.5

New • built from GARP Ch.4, eqs. 4.2 and 4.3 • shape A1·S1

A credit risk analyst at a UK bank is updating the standalone risk of a corporate loan after a downgrade. The bank's model treats the loss rate as known (standard deviation of the loss rate zero) and models default as a Bernoulli event.

	Before downgrade	After downgrade
Exposure amount	GBP 12 million	GBP 12 million
One-year PD	1%	4%
Loss rate	45%	45%
Remaining maturity	4 years	4 years

By which of the following factors is the loan's standalone unexpected loss closest to rising?

A. 1.97

B. 2.00

C. 3.88

D. 4.00

Question CR-4 e.6 New • built from GARP Ch.4, eq. 4.3 • shape A1·S1

A model validator at a European bank is reviewing the inputs of the internal credit portfolio model. For one corporate borrower the model reports a standard deviation of the default probability of 21.79%, and the model documentation states that default is a binomial (Bernoulli) event. The validator wants to check this against the borrower's one-year PD. Which of the following PDs is consistent with the reported figure?

A. 4.75%

B. 5.00%

C. 21.79%

D. 95.00%

Solutions • CR-4 e

Solution CR-4 e.1 — Answer: B

Bank Q116 • New question written for this bank

Step 1 – Bernoulli variance

$$\sigma^2_{PD} = \text{PD}(1-\text{PD}) = 0.02 \times 0.98 = 0.0196$$

$\Rightarrow$ The standard deviation is $\sqrt{0.0196} = 0.14$, seven times the PD.

Interpretation

This is the number that enters the default term of UL as $\text{LR}^2 \times 0.0196$. The other options are the three classic slips around it.

Why the other options are wrong

A is incorrect. Squared the PD: $0.02^2 = 0.0004$, dropping the $(1-\text{PD})$ structure.
C is incorrect. Reported the PD itself, 0.0200, ignoring the factor $(1-\text{PD})$.
D is incorrect. Reported the standard deviation, $\sqrt{0.0196} = 0.1400$, when the variance is asked.

LO: CR-4 e: Estimate the variance of default probability assuming a binomial distribution. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eq. 4.3

Solution CR-4 e.2 — Answer: B

Bank Q117 • New question written for this bank

Step 1 – Variance of each loan

$$\text{X: } 0.10 \times 0.90 = 0.09 \qquad \text{Y: } 0.80 \times 0.20 = 0.16$$

Step 2 – Compare

$0.16 > 0.09$: Loan Y is the more uncertain, because 80% is closer than 10% to the 50% point where $\text{PD}(1-\text{PD})$ peaks.
$\Rightarrow$ Loan Y has the higher variance (Figure 2).

Interpretation

Variance measures how unpredictable the outcome is, not how bad it is. A loan with a 90% PD would have the same 0.09 variance as Loan X: default is nearly certain, so the outcome is again quite predictable.

Why the other options are wrong

A is incorrect. Right number for X, wrong conclusion: a PD near zero does mean a predictable outcome, which is a *low* variance.
C is incorrect. Used $\text{PD}^2 = 0.64$ instead of $\text{PD}(1-\text{PD}) = 0.16$.
D is incorrect. Two states do not equalise the variance; it depends on PD through $\text{PD}(1-\text{PD})$.

LO: CR-4 e: Estimate the variance of default probability assuming a binomial distribution. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eq. 4.3

Solution CR-4 e.3 — Answer: C

Bank Q115 • AnalystPrep Q.6186

Step 1 – One loan

Each default indicator is Bernoulli with variance $0.015 \times 0.985 = 0.014775$.

Step 2 – Add the variances of independent loans

$$\text{var}(N) = n\,\text{PD}(1-\text{PD}) = 500 \times 0.015 \times 0.985 = 7.3875$$

$\Rightarrow$ The expected number of defaults is 7.5, with a standard deviation of $\sqrt{7.3875} = 2.72$.

Interpretation

Independence lets the variances add with no covariance terms. With even a small default correlation the variance would be several times larger (Section 2: 81.1 at $\rho = 0.02$).

Why the other options are wrong

A is incorrect. Used $n\,\text{PD}^2 = 500 \times 0.015^2 = 0.1125$.
B is incorrect. Reported the standard deviation, $\sqrt{7.3875} = 2.72$, instead of the variance.
D is incorrect. Reported the mean number of defaults, $n\,\text{PD} = 7.5$, forgetting the $(1 - \text{PD})$ factor.

LO: CR-4 e: Estimate the variance of default probability assuming a binomial distribution. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eq. 4.3 and footnote 12 (binomial $B(1;p)$)

Solution CR-4 e.4

Answer: C

Bank Q112 • BT P2.T6.506.2

Step 1 – Binomial variance of default

$$\sigma^2_{PD} = 0.02 \times 0.98 = 0.0196$$

Step 2 – The two terms under the root

$$\text{PD}\,\sigma^2_{LR} = 0.02 \times 0.40^2 = 0.0032, \qquad \text{LR}^2\sigma^2_{PD} = 0.50^2 \times 0.0196 = 0.0049$$

Step 3 – Unexpected loss

$$\text{UL} = 50 \times \sqrt{0.0032 + 0.0049} = 50 \times \sqrt{0.0081} = 50 \times 0.09 = \text{USD } 4.50 \text{ million}$$

$\Rightarrow$ UL is 9% of exposure and nine times the USD 0.5 million EL.

Interpretation

The EL in the stem is a check, not an input. The binomial assumption is what turns a 2% PD into a 14% standard deviation, and that default term supplies 60% of the variance here.

Why the other options are wrong

A is incorrect. Dropped the default term and kept only $\text{PD}\,\sigma^2_{LR}$: $50\sqrt{0.0032} = 2.83$.
B is incorrect. Used $\sigma^2_{PD} = \text{PD}^2 = 0.0004$: $50\sqrt{0.0032 + 0.25 \times 0.0004} = 2.87$.
D is incorrect. Added the two square roots instead of taking the root of the sum: $50 \times (0.0566 + 0.0700) = 6.33$.

LO: CR-4 e: Estimate the variance of default probability assuming a binomial distribution. CR-4 d: Define and calculate unexpected loss (UL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.2 and 4.3

Solution CR-4 e.5

Answer: A

New • built from GARP Ch.4, eqs. 4.2 and 4.3

Step 1 – UL with a known loss rate

$$\text{UL} = \text{EA} \times \text{LR} \times \sqrt{\text{PD}(1-\text{PD})}$$

$\Rightarrow$ EA and LR are unchanged, so the ratio depends only on σ_{PD}.

Step 2 – Binomial standard deviations

$$\text{Before: } \sqrt{0.01 \times 0.99} = 0.09950 \qquad \text{After: } \sqrt{0.04 \times 0.96} = 0.19596$$

Step 3 – Ratio

$$\frac{0.19596}{0.09950} = 1.97 \qquad (\text{UL : GBP } 0.537 \text{ million} \rightarrow \text{GBP } 1.058 \text{ million})$$

Interpretation

EL quadruples (GBP 0.054 million to 0.216 million) but UL roughly doubles: UL grows with the square root of PD. The maturity is a red herring; the model works on a one-year horizon.

Why the other options are wrong

B is incorrect. Approximated σ_{PD} by $\sqrt{\text{PD}}$, dropping $(1 - \text{PD})$: $\sqrt{0.04/0.01} = 2.00$.
C is incorrect. Took the ratio of the variances and forgot the square root: $0.0384/0.0099 = 3.88$.
D is incorrect. Scaled UL in proportion to PD, as EL scales: 4.00.

LO: CR-4 e: Estimate the variance of default probability assuming a binomial distribution. CR-4 d: Define and calculate unexpected loss (UL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.2 and 4.3

Solution CR-4 e.6 **Answer: B**

New • built from GARP Ch.4, eq. 4.3

Step 1 – Square the standard deviation

$$\sigma_{PD}^2 = 0.2179^2 = 0.047480$$

Step 2 – Solve $\text{PD}(1 - \text{PD}) = 0.047480$

$$\text{PD}^2 - \text{PD} + 0.047480 = 0 \quad \Rightarrow \quad \text{PD} = \frac{1 \pm \sqrt{1 - 4(0.047480)}}{2} = \frac{1 \pm 0.90004}{2}$$

Step 3 – Choose the root

The two roots are 5.00% and 95.00%; they share the same variance. For a performing corporate borrower the PD is the smaller root.
$\Rightarrow$ PD $= 5.00\%$. Check: $\sqrt{0.05 \times 0.95} = 0.2179$.

Interpretation

A validator can test the inputs this way: if the borrower's rating implies a PD far from 5%, the model's default volatility is inconsistent with its own binomial assumption.

Why the other options are wrong

A is incorrect. Read the variance as the PD (PD $\approx \sigma_{PD}^2 = 4.75\%$), the small-PD approximation, which drops the $(1 - \text{PD})$ factor.
C is incorrect. Read the standard deviation as the PD.
D is incorrect. Took the larger root, $1 - \text{PD} = 95.00\%$.

LO: CR-4 e: Estimate the variance of default probability assuming a binomial distribution. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eq. 4.3

CR-4 • Capital Structure in Banks • Vardeez: VERY IMPORTANT

CR-4 f

P1 CALC

Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size.

Workbook: A1 Unexpected Loss, UL Contribution and Economic Capital • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic and an August 2026 recall: "UL contribution, straightforward numerical (identical loans)". The map's recall notes add "Unexpected loss contribution: all loans with same value".
Vardeez: UL contributions (risk contributions) "can show up".

Reported exam shapes

- **A1·S2** UL contribution with all loans of the same value, ULC = UL $\times \sqrt{\rho}$, plain form.
- **A1·S3** Portfolio UL for n identical loans, $\mathrm{UL}_P = \mathrm{UL}\sqrt{n + \rho(n^2 - n)}$, forward and reverse-solved.
- **A1·S5** Intermediate-result trap: the single-asset UL planted when the ULC is asked.

What the questions look like: numerical. Identical-loan pools (the August shape) with the $\sqrt{\rho}$ shortcut or the exact n-loan formula; two- and three-asset portfolios where you combine standalone ULs with a correlation and split the result into contributions. Concept questions ask why ULs do not add and what the identical-loan shortcut hides.

Worked elsewhere: A1.3 (portfolio UL into economic capital) at Question CR-4 g.8; GARP PE1 Q25 (standalone UL then ULC) as Question CR-4 d.6. Standalone UL: CR-4 d.

1. Portfolio EL adds, portfolio UL does not

Recap from CR-4 d: each loan has an expected loss $\mathrm{EL}_i = \mathrm{EA}_i\,\mathrm{PD}_i\,\mathrm{LR}_i$ and a standalone unexpected loss UL_i, the standard deviation of its losses. At portfolio level (GARP eqs. 4.4 and 4.5):

$$\boxed{\mathrm{EL}_P = \sum_i \mathrm{EA}_i\,\mathrm{PD}_i\,\mathrm{LR}_i} \qquad \boxed{\mathrm{UL}_P = \sqrt{\sum_i \sum_j \rho_{ij}\,\mathrm{UL}_i\,\mathrm{UL}_j}}$$

ρ_{ij} is the correlation that default or a credit migration (in the same direction) of assets i and j occurs over the same horizon, with $\rho_{ii} = 1$. EL is linear, so it simply adds. UL is a standard deviation, so it adds only if every $\rho_{ij} = 1$; that sum $\sum \mathrm{UL}_i$ is the upper bound, and in practice UL_P is far below it. For two assets:

$$\mathrm{UL}_P = \sqrt{\mathrm{UL}_1^2 + \mathrm{UL}_2^2 + 2\rho\,\mathrm{UL}_1\mathrm{UL}_2}$$

Bigger Bank (Vardeez and Schweser): asset A has $\mathrm{UL}_A = \$325{,}333$ and $\mathrm{EL}_A = \$20{,}625$ (CR-4 d, Section 3); asset B is the XYZ loan, $\mathrm{UL}_B = \$90{,}000$ and $\mathrm{EL}_B = \$7{,}200$.

Correlation ρ	0	0.1	0.3	1
EL_P	\$27,825	\$27,825	\$27,825	\$27,825
UL_P	\$337,552	\$346,118	\$362,642	\$415,333

Correlation moves UL_P and leaves EL_P untouched. At $\rho = 1$ the portfolio UL is the plain sum $325{,}333 + 90{,}000$.

2. Risk contribution (UL contribution)

Only an asset's contribution to portfolio risk matters in a portfolio. GARP gets it in two steps (eqs. 4.6 and 4.9):

$$\text{ULMC}_i = \frac{\partial\,\text{UL}_P}{\partial\,\text{UL}_i} = \frac{\sum_j \rho_{ij}\,\text{UL}_j}{\text{UL}_P} \qquad \boxed{\text{ULC}_i = \text{ULMC}_i \times \text{UL}_i = \frac{\text{UL}_i \sum_j \rho_{ij}\,\text{UL}_j}{\text{UL}_P}}$$

- **The parts add to the whole** (eq. 4.10): $\sum_i \text{ULC}_i = \text{UL}_P$. This is what lets capital be allocated loan by loan in CR-4 g.
- **The derivative is with respect to** UL_i, the loan's standalone UL in currency, not with respect to a portfolio weight. The marginal contribution depends on the UL weights of the loans, not on the size of the portfolio.
- **Drivers.** A loan's contribution rises with its EL inputs (PD, LR, EA all enter UL_i) and with its correlation to the rest of the portfolio.

Two assets:

$$\text{RC}_1 = \text{UL}_1\,\frac{\text{UL}_1 + \rho\,\text{UL}_2}{\text{UL}_P} \qquad \text{RC}_2 = \text{UL}_2\,\frac{\text{UL}_2 + \rho\,\text{UL}_1}{\text{UL}_P} \qquad \text{RC}_1 + \text{RC}_2 = \text{UL}_P$$

Bigger Bank at $\rho = 0.3$:

$$\text{RC}_A = 325{,}333 \times \frac{325{,}333 + 0.3(90{,}000)}{362{,}642} = \$316{,}084$$

$$\text{RC}_B = 90{,}000 \times \frac{90{,}000 + 0.3(325{,}333)}{362{,}642} = \$46{,}558$$

and $316{,}084 + 46{,}558 = 362{,}642 = \text{UL}_P$. Asset A keeps 97% of its standalone UL; asset B keeps only 52%. The large position *is* most of the portfolio, so it diversifies little; the small one gets most of the diversification benefit.

> **Correction to the source notes.** SchweserNotes (LO 22.f) says that the risk contribution "isolates the incremental risk of adding asset i to the existing portfolio". The ULC is the *marginal* contribution times the standalone UL, an allocation whose parts sum to UL_P. The incremental (with-minus-without) UL is a different number: for Bigger Bank it is $362{,}642 - 90{,}000 = \$272{,}642$ for asset A (against $\text{RC}_A = \$316{,}084$) and \$37,309 for asset B (against \$46,558), and the two increments sum to \$309,952, not to UL_P. GARP (footnote 50) mentions the with-and-without calculation only as an alternative way of measuring a marginal risk contribution.

3. Identical loans: the August shape

Assume n loans of about the same size ($1/n$ each) and characteristics, so every $\text{UL}_i = \text{UL}$, and one constant pairwise correlation ρ. The double sum has n variance terms and $n^2 - n$ covariance terms (each pair counted twice), so (eqs. 4.13 to 4.15):

$$\boxed{\text{UL}_P = \text{UL}\sqrt{n + \rho\,(n^2 - n)}} \qquad \text{ULC}_i = \frac{\text{UL}_P}{n} = \text{UL}\sqrt{\frac{1}{n} + \rho\Big(1 - \frac{1}{n}\Big)}$$

$$\xrightarrow{\;n \text{ large}\;} \qquad \boxed{\text{ULC}_i = \text{UL}\sqrt{\rho}}$$

- **Diversifiable and undiversifiable risk.** The $1/n$ term is firm-specific risk; it vanishes in a large portfolio as the good and bad surprises cancel. The ρ term is the undiversifiable (systematic) residual that every loan carries however large the book. ρ here is the (weighted) average correlation between the loans (eq. 4.16).
- $\sqrt{\rho}$ **outside,** ρ **inside.** The contribution tends to $\mathrm{UL}\sqrt{\rho}$, but inside UL_P the correlation enters unrooted.
- **Reverse-solving** for the implied correlation (shape S3, Question CR-4 f.9):

$$\rho = \frac{(\mathrm{UL}_P/\mathrm{UL})^2 - n}{n^2 - n}$$

- **Small** n**: use the exact form.** Two identical loans with $\rho = 0.28$: $\mathrm{ULC} = \mathrm{UL}\sqrt{0.5 + 0.28 \times 0.5} = 0.80\,\mathrm{UL}$, while the shortcut gives $\sqrt{0.28} = 0.53\,\mathrm{UL}$ (Question CR-4 f.5).

Worked example. 250 similar loans, each with $\mathrm{UL} = \$0.12$ million, $\rho = 0.10$:

$$\mathrm{UL}_P = 0.12\sqrt{250 + 0.10 \times 62{,}250} = 0.12\sqrt{6{,}475} = 0.12 \times 80.47 = \$9.66 \text{ million}$$

$\mathrm{ULC}_i = 9.656/250 = \$38{,}624$ against the shortcut $0.12\sqrt{0.10} = \$37{,}947$. The undiversified sum is $250 \times 0.12 = \$30$ million: the book behaves like 80 standalone loans, so diversification removes about two thirds of the risk.

$\mathrm{ULC}_i/\mathrm{UL}_i$	$n = 1$	$n = 2$	$n = 10$	$n = 100$	$n = 1{,}000$	large n: $\sqrt{\rho}$
$\rho = 0.05$	1.0000	0.7246	0.3808	0.2439	0.2257	0.2236
$\rho = 0.28$	1.0000	0.8000	0.5933	0.5359	0.5298	0.5292

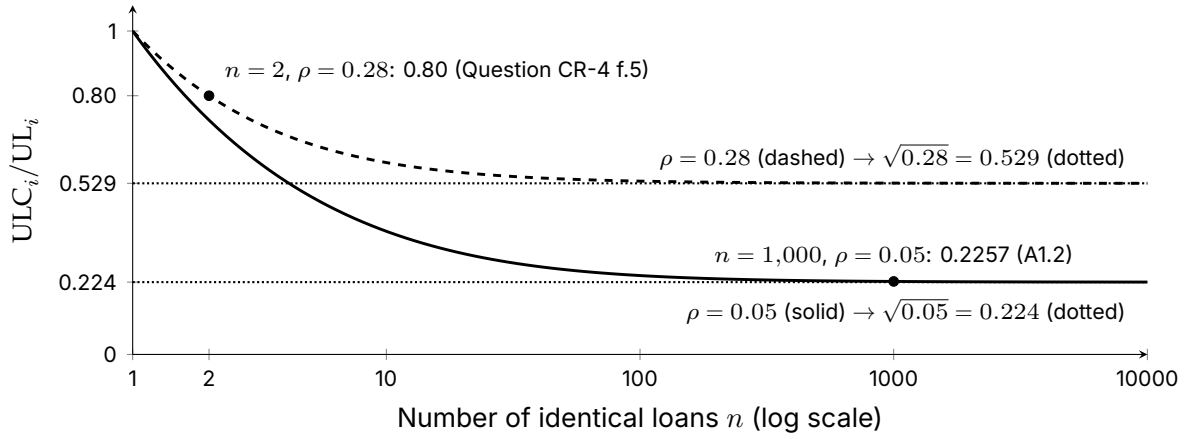


Figure 3. Each loan's UL contribution as a share of its standalone UL, $\sqrt{1/n + \rho(1 - 1/n)}$, for $\rho = 0.28$ (dashed) and $\rho = 0.05$ (solid). The contribution falls fast with the first loans and then levels off at $\sqrt{\rho}$ (dotted lines): the firm-specific part diversifies away, the correlated part never does. The shortcut $\mathrm{UL}\sqrt{\rho}$ is accurate from a few hundred loans, badly wrong for two.

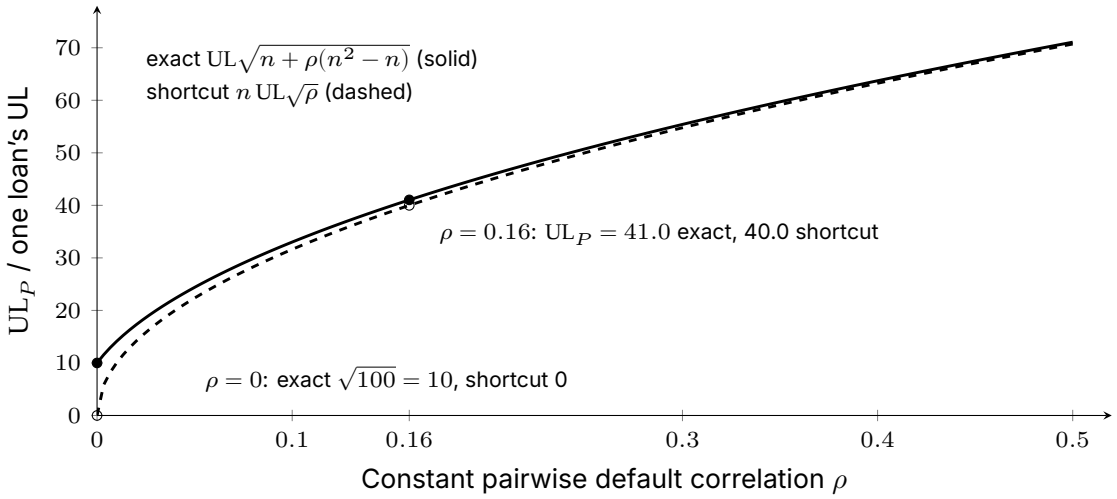


Figure 4. Portfolio UL of 100 identical loans against the default correlation (solid), with the large-n shortcut $n\,\mathrm{UL}\sqrt{\rho}$ (dashed). Even a small correlation lifts UL_P far above the independent case of 10; the curve then flattens, because the square root compresses it, and reaches the undiversified $n \times \mathrm{UL} = 100$ only at $\rho = 1$. The shortcut misses the firm-specific part, which matters only when ρ is close to zero. At $\rho = 0.16$ (Question CR-4 f.10) each loan contributes 41.0% of its standalone UL.

4. Why banks do not run the full formula

- **Too many correlations.** Eq. 4.5 needs $n(n-1)/2$ pairwise default correlations: 190 for 20 assets, 4,950 for 100. Real books hold thousands of credits, and default correlations are very difficult, if not impossible, to observe (they can be estimated from asset correlations).
- **The shortcut hides concentrations.** The identical-loan equation ignores different loan sizes and different correlations (industry, geography), so it cannot reveal the concentrations that lending limits are meant to prevent. It can be decomposed by segment so that industry or single-name correlations enter.
- **Correlation is concentration.** Higher correlation means defaults spill over (for example through industry effects), so high correlation = high concentration risk.
- **Default correlations are small but positive**, so the diversification benefits in credit portfolios are considerable.
- **Numerical methods.** The analytical approach is cumbersome and prone to estimation errors, so banks use numerical procedures such as Monte Carlo simulation.

Formula card: CR-4 f

Portfolio EL	$\mathrm{EL}_P = \sum_i \mathrm{EA}_i\,\mathrm{PD}_i\,\mathrm{LR}_i$ (additive; correlation irrelevant)
Portfolio UL	$\mathrm{UL}_P = \sqrt{\sum_i \sum_j \rho_{ij}\mathrm{UL}_i\mathrm{UL}_j}$; two assets: $\sqrt{\mathrm{UL}_1^2 + \mathrm{UL}_2^2 + 2\rho\,\mathrm{UL}_1\mathrm{UL}_2}$
Risk contribution	$\mathrm{ULC}_i = \mathrm{UL}_i \sum_j \rho_{ij}\mathrm{UL}_j/\mathrm{UL}_P$, $\sum_i \mathrm{ULC}_i = \mathrm{UL}_P$
Two assets	$\mathrm{RC}_1 = \mathrm{UL}_1(\mathrm{UL}_1 + \rho\,\mathrm{UL}_2)/\mathrm{UL}_P$
n identical loans	$\mathrm{UL}_P = \mathrm{UL}\sqrt{n + \rho(n^2 - n)}$, $\mathrm{ULC}_i = \mathrm{UL}_P/n = \mathrm{UL}\sqrt{1/n + \rho(1 - 1/n)}$
Large n	$\mathrm{ULC}_i = \mathrm{UL}\sqrt{\rho}$
Implied correlation	$\rho = [(\mathrm{UL}_P/\mathrm{UL})^2 - n]/(n^2 - n)$
Number of pairs	$n(n-1)/2$ distinct correlations

	General (eq. 4.5)	Two assets	n identical loans
Inputs	Every UL_i and all $n(n-1)/2$ correlations	UL_1, UL_2, one ρ	One UL, one ρ, n
Contribution	$\mathrm{UL}_i \sum_j \rho_{ij}\mathrm{UL}_j/\mathrm{UL}_P$	$\mathrm{UL}_1(\mathrm{UL}_1 + \rho\mathrm{UL}_2)/\mathrm{UL}_P$	$\mathrm{UL}_P/n \to \mathrm{UL}\sqrt{\rho}$
Shows concentration?	Yes, in principle	Yes: unequal shares	No: sizes and correlations are averaged away
Practical?	No: too many correlations to observe	Exam examples	Yes, the standard shortcut

Exam traps

- **Standalone UL when the contribution is asked (S5).** The intermediate result is always an option; for a pool the contribution is a fraction of it.
- **Summing ULs.** $\sum \mathrm{UL}_i$ is the $\rho = 1$ case. Only EL adds.
- **ρ against $\sqrt{\rho}$.** UL $\times\ \rho$ or UL $\times\ \rho^2$ for the contribution; $\sqrt{\rho}$ inside $\sqrt{n + \rho(n^2 - n)}$.
- **Missing factor 2.** The two-asset cross-term is $2\rho\,\mathrm{UL}_1\mathrm{UL}_2$; in the reverse solve, $n^2 - n$ (not $n(n-1)/2$) multiplies ρ.
- **Shortcut at small n.** $\mathrm{UL}\sqrt{\rho}$ is a large-n limit; for two or a dozen loans use UL_P/n.
- **Correlation dropped from the contribution.** $\mathrm{UL}_i^2/\mathrm{UL}_P$ is the $\rho = 0$ allocation; the numerator needs every $\rho_{ij}\mathrm{UL}_j$.
- **Pro-rata allocation.** Splitting UL_P in proportion to standalone ULs sums correctly but is not the risk contribution.
- **Swapped positions.** Attaching each UL to the wrong loan reverses the allocation.
- **Volatility as capital.** UL_P and ULC_i still need $\times$ CM to become economic capital (CR-4 g).

Practice questions: CR-4 f

Question CR-4 f.1

Bank Q118 • AP Mock 3 Q50

An analyst has calculated the expected loss for each bond in a corporate portfolio and must now estimate unexpected losses. The bonds have varying probabilities of default and loss given default, with associated volatilities. Why can the analyst not simply add up the unexpected losses of the individual bonds to estimate the total unexpected loss of the portfolio?

A. Because the unexpected loss is calculated from the average of the expected losses, and the aggregation of averages does not give the unexpected loss.

B. Because adding the individual unexpected losses assumes perfect correlation among the bonds, which is rarely the case in practice.

C. Because the bond portfolio is subject to market risk, which needs to be calculated separately from the credit risk of the individual bonds.

D. Because the unexpected loss calculation requires recovery rates, which have not been accounted for individually for each bond.

Question CR-4 f.2

Bank Q126 • Schweser Module Quiz 22.1 Q5 (rebuilt)

Bigger Bank has two loans outstanding with the features below. Assume the correlation between the two assets' losses is 0.2 and that $\text{UL}_i = \text{EA}_i\sqrt{\text{PD}_i\,\sigma^2_{LR,i} + \text{LR}_i^2\,\sigma^2_{PD,i}}$.

Item	Asset 1	Asset 2
Exposure amount (EA)	USD 3,000,000	USD 2,000,000
Probability of default (PD)	2%	4%
Loss rate (LR)	50%	40%
Standard deviation of PD	14%	20%
Standard deviation of LR	25%	30%

Which of the following is closest to the unexpected loss of the portfolio (UL_P)?

A. USD 308,788

B. USD 323,667

C. USD 337,892

D. USD 435,266

Question CR-4 f.3

Bank Q122 • BT P2.T6.921.3 (rebuilt, key corrected)

A credit portfolio contains two positions. The default correlation between them is 0.18, and the portfolio's unexpected loss is about USD 20.0 million.

	Position 1	Position 2
Exposure	USD 300.0 million	USD 100.0 million
Probability of default (PD), binomial	2.0%	6.0%
Loss given default (LGD)	20.0%	60.0%
Standard deviation of LGD	10.0%	30.0%
Standalone unexpected loss	USD 9.41 million	USD 16.03 million

Which of the following is nearest to the respective risk contributions (unexpected loss contributions) of Position 1 and Position 2?

A. USD 5.8 million and USD 14.2 million

B. USD 7.4 million and USD 12.6 million

C. USD 9.4 million and USD 16.0 million

D. USD 14.2 million and USD 5.8 million

Question CR-4 f.4

Bank Q131 • BT P2.T6.922.1 (rebuilt) • re-filed from CR-4 g

A credit portfolio manager attributes economic capital to three loans, each with an exposure of USD 10.0 million:

	Loan 1	Loan 2	Loan 3
Probability of default	2.0%	4.0%	6.0%
Standalone unexpected loss (USD)	597,000	840,000	1,023,500

The pairwise default correlation is 0.40 for every pair, so the portfolio's unexpected loss is USD 1,920,250. The risk contribution of Loan 1 is $597{,}000 \times [597{,}000 + 0.40(840{,}000) + 0.40(1{,}023{,}500)]/1{,}920{,}250 =$ USD 417,348, and with a capital multiplier of 5.50 its economic capital is about USD 2.30 million. Which of the following is nearest to the economic capital of Loan 2 and Loan 3, respectively?

A. USD 0.65 million and USD 0.85 million

B. USD 2.02 million and USD 3.00 million

C. USD 3.58 million and USD 4.69 million

D. USD 4.62 million and USD 5.63 million

Question CR-4 f.5

Bank Q120 • BT P2.T6.506.3 (distractors rebuilt)

A bank has extended two loans to customers in the same industry. Both loans have an exposure amount (EA) of USD 50.0 million, a default probability (PD) of 2.0%, a loss rate (LR) of 50.0% and a standard deviation of the loss rate of 60.0%, so that each loan has an expected loss of USD 500,000 and an unexpected loss of USD 5.5 million. The credit portfolio consists of these two loans, and the default correlation between them is 28.0%. Which of the following is nearest to the risk contribution of each loan to the portfolio's unexpected loss?

A. USD 2.91 million

B. USD 4.40 million

C. USD 5.50 million

D. USD 8.80 million

Question CR-4 f.6

Bank Q125 • GARP PE2 Q38 • Mock 4 Q39

A credit risk manager at a bank is estimating the unexpected loss (UL) of the bank's portfolio of loans and the UL contributions of the individual loans to the overall portfolio UL. The portfolio consists of a large number of loans and the manager assumes that the loans have approximately the same characteristics and size, with a constant pairwise default correlation of 0.32. Assuming the UL of each loan is USD 10,500, what is the approximate UL contribution of each loan to the portfolio UL?

A. USD 0

B. USD 1,075

C. USD 3,360

D. USD 5,940

Question CR-4 f.7

Workbook A1.2 • S1, S2, S5 • Aug-2026 • Map #1

A credit risk analyst at a Canadian bank is attributing portfolio risk to individual loans in a book of 1,000 small-business loans. Every loan has the same exposure and the same internal rating:

- Exposure amount per loan: CAD 5 million
- One-year PD: 1.0%; default is a Bernoulli event
- Loss rate: 50%, known with certainty
- Pairwise default correlation between any two loans: 0.05
- Weighted average remaining maturity: 3.2 years

Which of the following is closest to the unexpected loss contribution (ULC) of one loan to the portfolio's unexpected loss?

A. CAD 5,500

B. CAD 56,000

C. CAD 118,000

D. CAD 249,000

Question CR-4 f.8

Bank Q127 • GARP PE1 Q36

A senior risk manager at a financial institution is presenting to a group of junior analysts on the capital requirements for the firm's banking book. The manager demonstrates the calculation of economic capital, at the firm's chosen confidence level of 99.9%, for a portfolio of credit assets that have the same size and similar characteristics. The manager uses the following information:

Number of credit assets in the portfolio	12
Capital multiplier at the 99.9% confidence level	7.3
Unexpected loss of each credit asset	JPY 85 million
Default correlation between any pair of credit assets	0.3

What would the manager be correct to estimate as the economic capital for credit risk for this portfolio?

A. JPY 2.234 billion

B. JPY 4.078 billion

C. JPY 4.457 billion

D. JPY 7.446 billion

Question CR-4 f.9

Workbook A1.4 • S3 • Map #1 • options reordered

An internal model validator at a UK building society is backing out the default correlation implied by the credit portfolio model for a homogeneous pool of 200 equal-sized loans. The model reports:

- Standalone UL of each loan: GBP 0.30 million
- Portfolio UL: GBP 10.0 million

Using the exact identical-loan formula for portfolio UL, which of the following is closest to the implied pairwise default correlation?

A. 0.0229

B. 0.0278

C. 0.0458

D. 0.1667

Question CR-4 f.10 Bank Q123 • BT P2.T6.922.3 (rebuilt)

Consider a USD 20.0 million portfolio of 100 loans. In its general form, the portfolio's unexpected loss is $\mathrm{UL}_P = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n}\mathrm{UL}_i\,\mathrm{UL}_j\,\rho_{ij}}$. However, each loan in this portfolio has approximately the same characteristics and size (about USD 200,000), and for modelling purposes the pairwise correlation is set constant, $\rho_{ij} = 0.16$ for all $i \neq j$. These assumptions greatly simplify the calculation of the portfolio's unexpected loss and each loan's contribution to it. Which of the following statements is true?

A. A practical problem with the general form (specifying the full correlation matrix) is that default correlations are very difficult to observe.

B. Under the simplifying assumptions, each loan's unexpected loss contribution (ULC) is conveniently 16.0% of its standalone unexpected loss.

C. If the portfolio's unexpected loss were estimated by specifying each pairwise correlation ρ_{ij}, 100! (about 9.3×10^{157}) correlation pairs would be required.

D. When estimating the portfolio's unexpected loss and its contributions, banks prefer these analytical approaches over numerical procedures, because the latter are cumbersome and prone to estimation errors.

Solutions • CR-4 f

Solution CR-4 f.1

Answer: B

Bank Q118 • AP Mock 3 Q50

Step 1 – EL adds

$\text{EL}_P = \sum \text{EL}_i$ because EL is a mean and means are linear.

Step 2 – UL is a standard deviation

$\text{UL}_P = \sqrt{\sum_i \sum_j \rho_{ij} \text{UL}_i \text{UL}_j}$. Only if every $\rho_{ij} = 1$ does this collapse to $\sum \text{UL}_i$.

$\Rightarrow$ Adding the ULs assumes perfect correlation and so overstates portfolio risk.

Interpretation

Default correlations are small but positive, which is why UL_P is usually far below the sum of the standalone ULs (Bigger Bank: \$362,642 at $\rho = 0.3$ against \$415,333 summed).

Why the other options are wrong

A is incorrect. UL is not built from an average of ELs; it uses PD, LR and their volatilities. EL being linear is exactly why EL, unlike UL, can be added.

C is incorrect. Market risk is a separate issue; the reason ULs do not add is the correlation structure of credit losses.

D is incorrect. Recovery is already inside each bond's loss rate, and so inside each UL.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Unexpected Loss Contribution, eqs. 4.4 and 4.5

Solution CR-4 f.2

Answer: C

Bank Q126 • Schweser Module Quiz 22.1 Q5

Step 1 – Standalone ULs

$$\text{UL}_1 = 3{,}000{,}000\sqrt{0.02(0.25)^2 + 0.5^2(0.14)^2} = 3{,}000{,}000\sqrt{0.00615} = \text{USD } 235{,}266$$

$$\text{UL}_2 = 2{,}000{,}000\sqrt{0.04(0.30)^2 + 0.4^2(0.20)^2} = 2{,}000{,}000\sqrt{0.0036 + 0.0064} = \text{USD } 200{,}000$$

Step 2 – Combine with the correlation

$$\begin{aligned}\text{UL}_P &= \sqrt{235{,}266^2 + 200{,}000^2 + 2(0.2)(235{,}266)(200{,}000)} \\ &= \sqrt{1.1417 \times 10^{11}} = \text{USD } 337{,}892\end{aligned}$$

$\Rightarrow$ Between the zero-correlation value (308,788) and the perfect-correlation sum (435,266), as it must be for $0 < \rho < 1$.

Interpretation

Use the σ_{PD} values as given: asset 2's 20% is a rounded binomial figure ($\sqrt{0.04 \times 0.96} = 19.6\%$). EL does not enter UL_P at all.

Why the other options are wrong

A is incorrect. Set the correlation to zero, dropping the cross-term: $\sqrt{235{,}266^2 + 200{,}000^2} = 308{,}788$.

B is incorrect. Left out the factor 2 on the cross-term: $\sqrt{235{,}266^2 + 200{,}000^2 + 0.2(235{,}266)(200{,}000)} = 323{,}667$.

D is incorrect. Added the standalone ULs, the $\rho = 1$ case: $235{,}266 + 200{,}000 = 435{,}266$.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.2 and 4.5; SchweserNotes 2026 Book 2, Reading 22

Solution CR-4 f.3 Answer: A

Bank Q122 • BT P2.T6.921.3

Step 1 – Check the standalone ULs

$$\text{UL}_1 = 300\sqrt{0.02(0.10)^2 + 0.20^2(0.02 \times 0.98)} = 300 \times 0.031369 = 9.41$$

$$\text{UL}_2 = 100\sqrt{0.06(0.30)^2 + 0.60^2(0.06 \times 0.94)} = 100 \times 0.160325 = 16.03$$

$\Rightarrow$ The smaller position (USD 100 million) is the riskier one: higher PD and higher LGD.

Step 2 – Portfolio UL

$$\text{UL}_P = \sqrt{9.41^2 + 16.03^2 + 2(0.18)(9.41)(16.03)} = \sqrt{399.8} = 20.0$$

Step 3 – Risk contributions

$$\text{RC}_1 = 9.41 \times \frac{9.41 + 0.18(16.03)}{20.0} = 5.8 \qquad \text{RC}_2 = 16.03 \times \frac{16.03 + 0.18(9.41)}{20.0} = 14.2$$

$\Rightarrow 5.8 + 14.2 = 20.0 = \text{UL}_P$.

Interpretation

Each position keeps less than its standalone UL (62% and 89%), and the riskier position takes the larger share of portfolio risk even though its exposure is a third of the other's. The bank's original stem printed the two standalone ULs in the reverse order, which made its key (14.2 and 5.8) attach each contribution to the wrong position.

Why the other options are wrong

B is incorrect. Split UL_P in proportion to standalone UL: $20.0 \times 9.41/25.44 = 7.4$ and $20.0 \times 16.03/25.44 = 12.6$. It sums to UL_P but ignores how each position co-moves with the other.

C is incorrect. Reported the standalone ULs (9.4 and 16.0), the intermediate result; they sum to 25.4, more than UL_P.

D is incorrect. Swapped the positions: 14.2 belongs to Position 2, the USD 100 million loan.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.5, 4.9 and 4.10

Solution CR-4 f.4 Answer: C

Bank Q131 • BT P2.T6.922.1

Step 1 – Risk contribution of Loan 2

$$\text{RC}_2 = 840{,}000 \times \frac{840{,}000 + 0.40(597{,}000) + 0.40(1{,}023{,}500)}{1{,}920{,}250}$$

$$= 840{,}000 \times \frac{1{,}488{,}200}{1{,}920{,}250} = 651{,}003$$

Step 2 – Risk contribution of Loan 3

$$\text{RC}_3 = 1{,}023{,}500 \times \frac{1{,}023{,}500 + 0.40(597{,}000) + 0.40(840{,}000)}{1{,}920{,}250}$$
$$= 1{,}023{,}500 \times \frac{1{,}598{,}300}{1{,}920{,}250} = 851{,}899$$

$\Rightarrow$ Check: $417{,}348 + 651{,}003 + 851{,}899 = 1{,}920{,}250 = \text{UL}_P$.

Step 3 – Economic capital

$$\text{EC}_2 = 651{,}003 \times 5.50 = \text{USD } 3.58 \text{ million} \qquad \text{EC}_3 = 851{,}899 \times 5.50 = \text{USD } 4.69 \text{ million}$$

Interpretation

The three loans' capital adds to $\text{UL}_P \times \text{CM} = 1{,}920{,}250 \times 5.50 = \text{USD } 10.56$ million, the portfolio's economic capital. Capital follows the contribution, not the standalone UL (CR-4 g).

Why the other options are wrong

A is incorrect. Stopped at the risk contributions (0.65 and 0.85) and forgot the capital multiplier.
B is incorrect. Dropped the correlation terms from the numerator, using $\text{UL}_i^2/\text{UL}_P \times 5.50$: 2.02 and 3.00.
D is incorrect. Applied the multiplier to the standalone ULs, ignoring diversification: $840{,}000 \times 5.50 = 4.62$ and $1{,}023{,}500 \times 5.50 = 5.63$.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. CR-4 g: Describe how economic capital is derived. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.9, 4.10 and 4.18

Solution CR-4 f.5 **Answer: B**

Bank Q120 • BT P2.T6.506.3

Step 1 – Portfolio UL

$$\text{UL}_P = \sqrt{5.5^2 + 5.5^2 + 2(0.28)(5.5)(5.5)} = 5.5\sqrt{2.56} = 5.5 \times 1.6 = \text{USD } 8.80 \text{ million}$$

Step 2 – Risk contribution

$$\text{RC}_1 = 5.5 \times \frac{5.5 + 0.28(5.5)}{8.80} = 5.5 \times \frac{7.04}{8.80} = \text{USD } 4.40 \text{ million}$$

$\Rightarrow$ The loans are identical, so each takes half of UL_P; the exact n-loan form gives the same: $5.5\sqrt{1/2 + 0.28/2} = 4.40$.

Interpretation

With only two loans the firm-specific risk is far from diversified away: each keeps 80% of its standalone UL (Figure 3). The large-n shortcut belongs to big pools.

Why the other options are wrong

A is incorrect. Applied the large-n shortcut to a two-loan portfolio: $5.5\sqrt{0.28} = 2.91$; the two contributions would sum to 5.82, not 8.80.
C is incorrect. Reported the standalone UL, 5.50 (the $\rho = 1$ answer).
D is incorrect. Reported the portfolio UL, 8.80, instead of one loan's share.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.9 and 4.14

Solution CR-4 f.6 — Answer: D

Bank Q125 • GARP PE2 Q38 • Mock 4 Q39

Step 1 – Large homogeneous pool

Many loans of similar size and characteristics with one correlation: $\text{ULC}_i \approx \text{UL}_i\sqrt{\rho}$.

Step 2 – Apply

$$\text{ULC} = 10{,}500 \times \sqrt{0.32} = 10{,}500 \times 0.5657 = \text{USD } 5{,}940$$

Interpretation

This is shape S2 in its plain form. Each loan keeps about 57% of its standalone UL: the systematic part that no amount of diversification removes when $\rho = 0.32$.

Why the other options are wrong

A is incorrect. Assumed all risk diversifies away; that needs $\rho = 0$.
B is incorrect. Used ρ^2: $10{,}500 \times 0.1024 = 1{,}075$.
C is incorrect. Used ρ instead of $\sqrt{\rho}$: $10{,}500 \times 0.32 = 3{,}360$.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eq. 4.15

Solution CR-4 f.7 — Answer: B

Workbook A1.2 • S1, S2, S5 • Aug-2026

Step 1 – Standard deviation of the default indicator

$$\sigma_{PD} = \sqrt{\text{PD}(1-\text{PD})} = \sqrt{0.01 \times 0.99} = \sqrt{0.0099} = 0.09950$$

Step 2 – Standalone UL per loan ($\sigma_{LR} = 0$)

$$\text{UL}_i = \text{EA} \times \text{LR} \times \sigma_{PD} = 5 \times 0.50 \times 0.09950 = \text{CAD } 0.2487 \text{ million}$$

$\Rightarrow$ On its own, each loan has a loss standard deviation of about CAD 248,700.

Step 3 – UL contribution for identical loans

$$\text{ULC}_i \approx \text{UL}_i\sqrt{\rho} = 0.2487 \times \sqrt{0.05} = 0.2487 \times 0.2236 = \text{CAD } 0.0556 \text{ million}$$

$\Rightarrow$ With 1,000 loans the exact form gives the same answer: $\text{UL}_i\sqrt{1/n + \rho(1-1/n)} = 0.2487 \times 0.2257 = 0.0561$. Either way, about CAD 56,000.

Interpretation

Inside the portfolio each loan contributes only about 23% of its standalone UL, because default correlation (0.05) is small but positive. The 1,000 ULCs add back to $\text{UL}_P \approx$ CAD 56.1 million. The 3.2-year maturity is irrelevant: the model uses a one-year horizon.

Why the other options are wrong

A is incorrect. Forgot the square root of the binomial variance: $\text{UL}_i = 2.5 \times 0.0099 = 0.02475$, then $0.02475 \times 0.2236 = 0.0055$.
C is incorrect. Correlation misuse: put $\sqrt{\rho}$ instead of ρ inside the portfolio formula, $0.2487 \times \sqrt{1{,}000 + 0.2236 \times 999{,}000}/1{,}000 = 0.118$.

D is incorrect. Reported the standalone UL of one loan (the intermediate result GARP plants when ULC is asked): 0.249.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. CR-4 e: Estimate the variance of default probability assuming a binomial distribution. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Unexpected Loss Contribution (ULC), eqs. 4.14 and 4.15

Solution CR-4 f.8

Answer: C

Bank Q127 • GARP PE1 Q36

Step 1 – Portfolio UL for 12 identical assets

$$\begin{aligned}\text{UL}_P &= \text{UL}\sqrt{n+\rho(n^2-n)} = 85\sqrt{12+0.3\times 132} = 85\sqrt{51.6}\\ &= 85\times 7.1833 = \text{JPY } 610.6 \text{ million}\end{aligned}$$

$\Rightarrow$ Twelve assets behave like 7.2 standalone assets.

Step 2 – Economic capital

$$\text{EC}_P = \text{UL}_P \times \text{CM} = 610.6 \times 7.3 = \text{JPY } 4{,}457 \text{ million} = \text{JPY } 4.457 \text{ billion}$$

Interpretation

With only 12 assets the exact formula is required: the shortcut $n\,\text{UL}\sqrt{\rho}\times\text{CM}$ drops the $1/n$ (firm-specific) part and understates capital. Each asset is attributed $\text{UL}_P/12\times 7.3 = \text{JPY } 371$ million.

Why the other options are wrong

A is incorrect. Treated the correlation as a plain scalar on the summed ULs: $12\times 0.3\times 85\times 7.3 = 2.234$ billion.

B is incorrect. Used the large-n shortcut at $n = 12$: $12\times 85\times\sqrt{0.3}\times 7.3 = 4.078$ billion, dropping the own-variance terms.

D is incorrect. Summed the ULs, the no-diversification case: $12\times 85\times 7.3 = 7.446$ billion.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. CR-4 g: Describe how economic capital is derived. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.13 and 4.17

Solution CR-4 f.9

Answer: A

Workbook A1.4 • S3 • Map #1

Step 1 – Formula

$$\text{UL}_P = \text{UL}_i\sqrt{n+\rho(n^2-n)} \quad\Rightarrow\quad \rho = \frac{(\text{UL}_P/\text{UL}_i)^2 - n}{n^2-n}$$

Step 2 – Squared ratio

$$\left(\frac{10.0}{0.30}\right)^2 = 33.33^2 = 1{,}111.1$$

$\Rightarrow$ The portfolio behaves like $\sqrt{1{,}111} = 33.3$ standalone loans out of 200.

Step 3 – Remove the variance terms

$$1{,}111.1 - 200 = 911.1$$

$\Rightarrow$ This is the part of portfolio variance coming from the $n^2 - n = 39{,}800$ covariance terms.

Step 4 – Implied correlation

$$\rho = \frac{911.1}{39{,}800} = 0.0229$$

Interpretation

An implied default correlation of about 2.3% is small but positive, as Schroeck expects. The validator would compare it with correlations estimated from observable asset correlations for the segment. (The workbook prints the options in descending order; they are reordered here, so the key moves from D to A.)

Why the other options are wrong

B is incorrect. Used the large-n approximation ULC $=$ UL$\sqrt{\rho}$: $\rho = \left(10/(200 \times 0.30)\right)^2 = 0.0278$, which ignores the $1/n$ variance term.
C is incorrect. Counted each of the $n(n-1)/2$ distinct pairs once, without the factor 2: $\rho = 911.1/19{,}900 = 0.0458$.
D is incorrect. Forgot to square: ULC/UL $= 0.05/0.30 = 0.1667$ is (approximately) $\sqrt{\rho}$, not ρ.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Unexpected Loss Contribution (ULC), eqs. 4.13 to 4.16

Solution CR-4 f.10 **Answer: A**

Bank Q123 • BT P2.T6.922.3

Step 1 – Count the correlations

Distinct pairs: $n(n-1)/2 = 100 \times 99/2 = 4{,}950$, not 100!.
$\Rightarrow$ C is false.

Step 2 – Contribution under the simplifying assumptions

$\text{ULC}_i/\text{UL}_i = \sqrt{1/n + \rho(1 - 1/n)} = \sqrt{0.01 + 0.16 \times 0.99} = 41.0\%$, close to $\sqrt{0.16} = 40\%$.
$\Rightarrow$ B confuses ρ with $\sqrt{\rho}$.

Step 3 – Analytical against numerical

GARP: the analytical approach is cumbersome and prone to estimation errors, so banks now use numerical procedures (Monte Carlo).
$\Rightarrow$ D reverses this. A restates GARP: default correlations are "very difficult, if not impossible, to observe".

Interpretation

The constant-ρ, equal-size assumptions make the problem tractable, at the price of hiding concentrations by industry, geography and single name (Section 4).

Why the other options are wrong

B is incorrect. Each loan contributes about 41% of its UL ($\sqrt{0.16} = 40\%$ for large n), not 16%.
C is incorrect. The number of distinct pairs is $n(n-1)/2 = 4{,}950$.
D is incorrect. Reversed: numerical procedures are used precisely because the analytical approach is cumbersome and error-prone.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.5 and 4.13 to 4.16 and the insights that follow them

CR-4 • Capital Structure in Banks • Vardeez: VERY IMPORTANT

CR-4 g

P1 CONCEPT

Describe how economic capital is derived.

Workbook: A1 Unexpected Loss, UL Contribution and Economic Capital • Tier A • August 2026 recall D3 Regulatory vs Economic Capital and Capital Adequacy • Tier D

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A1, the map's number one): one source reports a candidate who got economic capital wrong by omitting the capital multiplier. D3 (Tier D) adds the regulatory-versus-economic distinction.
Vardeez: the reading is flagged for UL and risk-contribution calculations, not for economic capital, so this LO is taken as CONCEPT ("describe").

Reported exam shapes

- **A1·S4** Economic capital EC $=$ UL$_P \times$ CM: the omitted capital multiplier.
- **D3·S2** Unexpected loss into economic capital: EC$_P =$ UL$_P \times$ CM, and ULC$_i \times$ CM at loan level.

What the questions look like: mostly statements. How economic capital responds to the confidence level, to correlation and to the riskiness of the assets; that it is measured from EL and covers UL only; that loan-level capital follows the UL contribution; how it differs from regulatory capital.

Kept outside the TYPE rule (flagged): Question CR-4 g.8 is the numerical workbook question A1.3 (kept, as every workbook question is), and Question CR-4 g.6 is a GARP-practice numerical (Mock 2 Q30) whose arithmetic is a single subtraction, EC $=$ VaR $-$ EL.

Worked elsewhere: D3.1 (risk-weighted assets and the capital adequacy ratio, with an economic-capital red herring) under CR-5 c; the three-loan capital attribution (ULC$_i \times$ CM) as Question CR-4 f.4 and GARP PE1 Q36 (n identical loans, UL$_P \times$ CM) as Question CR-4 f.8.

1. What economic capital is

GARP Box 4.1: economic capital is "an estimate of the overall capital reserve needed to guarantee the solvency of a bank for a given confidence level". Recap from CR-4 d: expected loss is a cost, covered by loan loss reserves and pricing; unexpected loss is the risk. Economic capital is the cushion for *unexpected* losses, the excess capital beyond the credit reserves.

Two drivers.

- **The confidence level**, set to match the bank's target credit rating. A bank aiming for a very high rating must survive a very large loss, so it picks a very high confidence level (99.97% for a AA target). A higher confidence level means a higher capital multiplier (CM), so more of the loss distribution is covered and EC rises. A more aggressive bank targets a lower rating, a lower CM and less capital.
- **The riskiness of the assets.** Riskier assets (higher PD or LR, more correlated) have a higher UL, so at the same confidence level they need more EC. Lower default correlation lowers UL$_P$ and the capital with it.

Holding less capital frees it to earn returns elsewhere, so economic capital is where the bank sets its balance between risk and return.

2. The derivation in three steps

Schroeck translates UL into capital in three steps:

1. **Standalone UL** of each loan, $\text{UL}_i = \text{EA}\sqrt{\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\sigma_{PD}^2}$ (CR-4 d, e).
2. **UL contribution** of each loan to portfolio UL, $\text{ULC}_i = \text{UL}_i \sum_j \rho_{ij}\text{UL}_j/\text{UL}_P$, with $\sum_i \text{ULC}_i = \text{UL}_P$ (CR-4 f).
3. **Economic capital**: the distance between the expected loss and the loss at the chosen confidence level, expressed as a multiple of UL.

GARP: "the amount of economic capital needed is the distance between the expected outcome and the unexpected (negative) outcome at a certain confidence level". Knowing the shape of the loss distribution, EL_P and UL_P, that distance is written as a multiple of UL_P, the **capital multiplier** (eqs. 4.17 and 4.18):

$$\boxed{\text{EC}_P = \text{UL}_P \times \text{CM}} \qquad \boxed{\text{EC}_i = \text{ULC}_i \times \text{CM}}$$

- **Loan-level capital follows the contribution.** Because the ULC_i add up to UL_P, the loan capitals add up to EC_P: capital for one transaction "is directly proportional to its contribution to the overall portfolio credit risk", not to its standalone UL.
- **Measured from EL.** EC starts at EL_P, not at zero loss. The loss at the confidence level is $\text{EL}_P + \text{CM} \times \text{UL}_P$; equivalently, EC is that credit VaR minus EL (Question CR-4 g.6: $230 - 130 = 100$).
- **CM is not a normal quantile.** CM solves $\Pr\big[(X - \text{EL}_P)/\text{UL}_P \geq \text{CM}\big] = 1 - \alpha$ on the assumed loss distribution; GARP takes the inverse of a fitted beta distribution at the confidence level. It depends on the credit quality of the portfolio and on the confidence level. At 99.97% GARP puts CM between 7.0 and 7.5, "far higher than the capital multiples for the normally distributed events in market risk" (the normal multiple at 99.97% is 3.43), because credit losses are highly skewed.
- **UL alone is not capital.** UL_P and ULC_i are one-standard-deviation measures. Reporting them as capital is the omitted-multiplier error of shape S4.

3. Why the distribution matters

Only the tail matters for capital, so the crucial task is the choice of loss distribution. Credit losses are not normal: the upside is capped at receiving the promised payments, while very large losses are rare but possible. GARP's practical choice is the **beta distribution** on $[0, 1]$ (losses from 0% to 100%), which is flexible enough to be symmetric or skewed and is pinned down by two numbers, EL_P and UL_P. Fitting it exactly to the tail is hard, so the analytical beta fit is combined with Monte Carlo simulation (the detail belongs to CR-4 h).

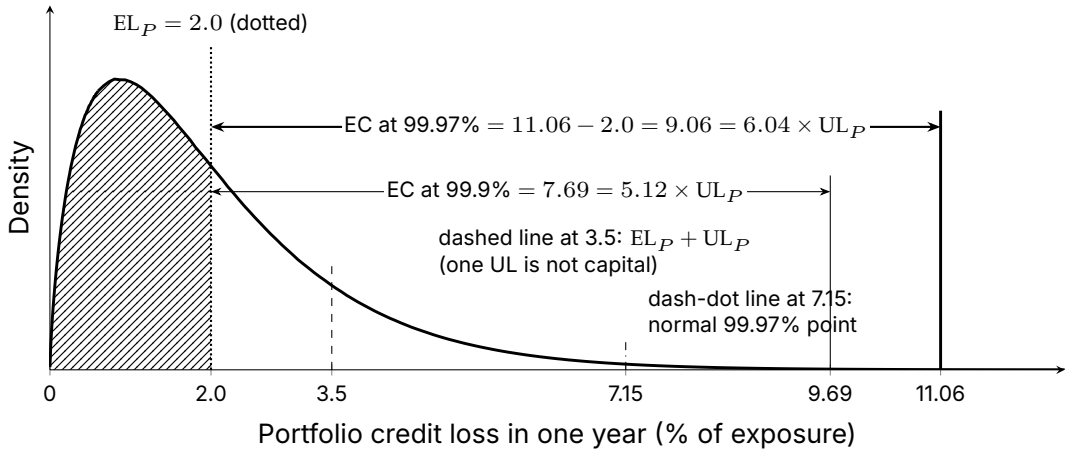


Figure 5. The loss distribution of Figure 1 (beta, $\text{EL}_P = 2\%$, $\text{UL}_P = 1.5\%$). The hatched area up to EL is the part covered by reserves and pricing. Economic capital is the distance from EL_P to the loss at the confidence level: 6.04 ULs at 99.97% (thick line) and 5.12 ULs at 99.9% (thin line). Raising the confidence level pushes the line right and raises CM; EL does not move. A normal curve with the same two moments would put the 99.97% point at only 3.43 ULs (dash-dot) and understate capital. For this fairly mild shape CM is 6.0; GARP quotes 7.0 to 7.5 for typical credit portfolios at 99.97%.

4. Economic capital against regulatory capital (D3)

Schroeck contrasts the economic approach with "the regulatory approach that assigns roughly 8% equity capital to credits on a standalone basis": the economic approach differentiates capital by borrower quality and reflects diversification in the portfolio. In the ORR reading (ORR-18 a), regulatory capital is "calculated according to a set of industrywide rules and formulas and sets only a minimum required level of capital adequacy"; economic (risk) capital is the bank's own estimate of the capital its risks require.

- Economic capital need not equal or exceed regulatory capital, and it does not absorb expected losses (Question CR-4 g.3).
- Neither replaces the other: the internal number can sit in the regulatory ratio only as a red herring (D3.1 at Question CR-5 c.9).

Correction to the source notes. Vardeez's CR-4 g text says that "the difference between the unexpected outcome and the confidence level is represented as a multiple of the portfolio's unexpected loss". GARP measures the distance between the *expected* outcome (EL_P) and the unexpected (negative) outcome at the confidence level, and it is that distance which is written as CM × UL_P. The first sentence of the same paragraph in Vardeez has it right.

	Expected loss	Economic capital	Regulatory capital
What it covers	The average loss, a cost of lending	Unexpected loss up to the confidence level	A supervisory minimum
How it is set	EA × PD × LR, loan by loan	$UL_P \times CM$; loan level $ULC_i \times CM$	Industrywide rules and formulas (roughly 8% standalone in Schroeck's comparison)
Diversification	None needed: EL adds	Yes, through the ULCs	Largely standalone
Confidence level	Not involved	Tied to the target rating (99.97% for AA)	Set by the regulator
Held as	Loan loss reserves and pricing	Capital	Capital

Exam traps

- **Omitted capital multiplier (S4).** UL_P or ULC_i reported as capital.
- **Normal multiple for credit.** 3.43 (99.97%) or 3.09 (99.9%) in place of the credit CM; credit losses are skewed, so CM is far larger.
- **EL inside EC.** Adding EL to $UL_P \times CM$ double counts what reserves cover; subtracting EL from $UL_P \times CM$ removes it twice, since CM is already measured from EL.
- **Standalone UL at loan level.** Loan capital is $ULC_i \times CM$, not $UL_i \times CM$.
- **Polarity of the target rating.** Higher rating, higher confidence level, higher CM, more capital.
- **Correlation ignored.** Two banks with the same EL and exposure need different capital if their default correlations differ.
- **Regulatory and economic swapped.** "Industrywide rules" and "minimum" describe regulatory capital.

Practice questions: CR-4 g

Question CR-4 g.1

Bank Q129 • AnalystPrep Q.6191

A bank is estimating its economic capital for credit risk. Its credit risk models estimate expected and unexpected losses over a one-year period, and the bank wants its capital reserves to reflect the credit risk in its portfolio. How should the bank derive economic capital from the expected and unexpected losses produced by its models?

A. By setting aside capital reserves equal to the sum of expected and unexpected losses.

B. By setting aside capital reserves equal to the unexpected losses, assuming no diversification benefit.

C. By estimating the excess capital reserves needed, subject to a predetermined confidence level, to cover unexpected losses.

D. By allocating resources to manage expected losses and disregarding unexpected losses.

Question CR-4 g.2 Bank Q121 • BT P2.T6.920.2 • re-filed from CR-4 f

Consider a credit portfolio that includes many loans. To derive economic capital (EC) for credit risk, a bank quantifies four measures: expected losses (EL), unexpected losses (UL), unexpected loss contributions (ULC) and economic capital (EC). Each of the following descriptions of these measures is accurate EXCEPT:

A. Expected losses can be viewed as payments to an insurance pool, do not by themselves constitute risk, and are reimbursed through adequate loan pricing.

B. Unexpected loss is the standard deviation of credit losses around the expected loss average.

C. The unexpected loss contribution is the first partial derivative of the portfolio's unexpected loss with respect to the position's weight in the portfolio.

D. Economic capital for credit risk is the difference between the expected outcome and the unexpected, negative outcome at a certain confidence level.

Question CR-4 g.3 Bank Q134 • BT P2.T7.1.1

A bank's risk committee is reviewing the following statements about economic capital (EC):

I. Economic capital must equal or exceed regulatory capital.

II. Economic capital should exceed equity capital.

III. Economic capital should equal the discounted stream of future excess returns on capital.

IV. Economic capital absorbs expected losses (EL) and should equal the firm's reserves.

V. Economic capital absorbs unexpected losses (UL) up to a certain confidence level and over a specified time horizon.

Which of the above statements is (are) true?

A. I only

B. V only

C. I, II and III

D. All are true

Question CR-4 g.4

Bank Q89 • GARP PE2 Q50 • Mock 4 Q51 • re-filed from CR-4 a

A risk analyst at a credit ratings agency is evaluating the economic capital for credit risk of two competing regional banks, Bank ABC and Bank XYZ. The two banks have the same credit asset exposure, duration of credit exposure, credit rating and expected loss. The average pairwise default correlation between the credit assets of Bank ABC is lower than that of Bank XYZ, and the two banks assess their risk appetite at the same predetermined confidence level. Which of the following statements would be correct?

A. If the confidence level for both banks is increased, the level of economic capital needed will increase for Bank ABC and for Bank XYZ.

B. If the confidence level for both banks is decreased, the level of economic capital needed will increase for Bank ABC but will decrease for Bank XYZ.

C. If the confidence level for both banks is increased, the level of economic capital needed will decrease for Bank ABC but will increase for Bank XYZ.

D. If the confidence level for both banks is kept unchanged, the level of economic capital needed for Bank ABC and for Bank XYZ will be equal.

Question CR-4 g.5

Bank Q137 • BT P2.T6.507.2 • re-filed from CR-4 h

Schroeck notes that the beta distribution, defined between 0 and c (here $c = 1$ for loss rates of 0% to 100%), is extremely flexible in the shapes it can accommodate. Which of the following statements about the beta distribution in deriving economic capital for credit risk is true?

A. It is flexible because it has four parameters, one for each moment, allowing precise calibration of kurtosis.

B. The most convenient and sufficiently realistic calibration sets the shape parameters α and β equal to each other.

C. It is characterised by the portfolio's expected loss, EL_P, and unexpected loss, UL_P; the challenge is fitting the tail, which depends on the ratio $\mathrm{EL}_P/\mathrm{UL}_P$.

D. Even when accurately fitted it gives no way to determine the capital multiplier, which therefore requires a different distribution.

Question CR-4 g.6

Bank Q132 • Mock 2 Q30 • numerical, kept and flagged

Using a confidence level of 99.9% over a one-year horizon, a bank under the IRB framework has an estimated credit VaR of USD 230 million. The expected loss on the portfolio is USD 130 million. A risk analyst is preparing a note for the risk committee. Which of the following is correct?

A. Economic capital is USD 100 million; if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would increase.

B. Economic capital is USD 100 million; if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would decrease, because a higher confidence level implies a more conservative expected loss.

C. Economic capital is USD 230 million; if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would increase.

D. Economic capital is USD 360 million; if the bank raised its confidence level from 99.9% to 99.97%, the required economic capital would remain unchanged.

Question CR-4 g.7

Workbook D3.2 • S1, S2 • Map #61

An internal auditor at an Australian bank is reviewing a draft board paper that compares the bank's regulatory capital for credit risk with the economic capital produced by its internal credit portfolio model. The bank targets an AA rating and calibrates its model at the 99.97% confidence level. Which of the following statements is correct for the paper to include?

A. A bank targeting a higher credit rating can choose a lower confidence level, which lowers the capital multiplier applied to unexpected loss and so reduces the economic capital it must hold.

B. Economic capital is calculated with industrywide rules and formulas and sets only a minimum level of capital adequacy, whereas regulatory capital reflects the bank's own estimate of its risk.

C. Economic capital for credit risk is the capital needed to absorb the expected loss plus the unexpected loss up to the confidence level, so it is measured from zero loss rather than from EL.

D. Economic capital at the loan level equals the loan's unexpected loss contribution times the capital multiplier, which reflects diversification, unlike a regulatory approach assigning roughly 8% capital to each credit standalone.

Question CR-4 g.8

Workbook A1.3 • S3, S4 • Map #1 • numerical, kept and flagged

A senior risk analyst at an Australian bank targeting a AA rating is preparing the economic capital figures for a portfolio of 400 near-identical mid-market loans. The data are:

- Each loan: EA AUD 2.5 million; PD 1.5%; LR 45%; standard deviation of LR 20%
- Average pairwise default correlation: 0.04
- Capital multiplier of the credit loss distribution at the 99.97% confidence level: 7.0
- The market-risk desk's normal-distribution multiple at 99.97%: 3.43

Which of the following is closest to the economic capital for the portfolio and the economic capital attributed to one loan?

A. Portfolio AUD 12.3 million; one loan AUD 0.031 million

B. Portfolio AUD 42.3 million; one loan AUD 0.106 million

C. Portfolio AUD 79.6 million; one loan AUD 0.199 million

D. Portfolio AUD 86.4 million; one loan AUD 0.216 million

Solutions • CR-4 g

Solution CR-4 g.1 — Answer: C

Bank Q129 • AnalystPrep Q.6191

Step 1 – Who covers what

Expected losses are covered by credit reserves and loan pricing; economic capital is for unexpected losses.

Step 2 – How much

The capital is the excess over EL needed to stay solvent at a predetermined confidence level, $\text{UL}_P \times \text{CM}$.

$\Rightarrow$ C describes exactly this.

Interpretation

Schweser: "the excess capital needed to match the bank's estimate of unexpected loss is referred to as economic capital". Both elements matter: unexpected loss only, and a confidence level.

Why the other options are wrong

A is incorrect. Adds EL, which is already reserved and priced; and UL without a confidence level is not a capital figure.

B is incorrect. No confidence-level scaling (CM), and ignoring diversification overstates the UL that capital must cover.

D is incorrect. Ignores unexpected loss, the very thing economic capital buffers.

LO: CR-4 g: Describe how economic capital is derived. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Box 4.1 and Economic Capital for Credit Risk; SchweserNotes 2026 Book 2, Reading 22

Solution CR-4 g.2 — Answer: C

Bank Q121 • BT P2.T6.920.2

Step 1 – Check EL and UL

A and B restate GARP: EL is "payments to an insurance pool", "does not itself constitute risk" and is reimbursed through pricing; UL is the standard deviation of losses around EL.

Step 2 – Check ULC

GARP takes the first partial derivative of UL_P with respect to UL_i, the loan's standalone UL in currency, and multiplies by UL_i: $\text{ULC}_i = (\partial \text{UL}_P / \partial \text{UL}_i)\,\text{UL}_i$. It explicitly ignores portfolio weights.

$\Rightarrow$ C is the inaccurate statement.

Step 3 – Check EC

D is GARP's definition: the distance between the expected outcome and the unexpected outcome at the confidence level.

Interpretation

The four measures are the chain of this reading: EL (cost), UL (risk), ULC (allocation), EC (capital).

Why the other options are wrong

A is incorrect. Accurate: EL is the foreseeable cost of lending.

B is incorrect. Accurate: UL is a standard deviation around EL.

D is incorrect. Accurate: EC is measured from EL to the confidence level.

LO: CR-4 g: Describe how economic capital is derived. CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:**

Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.1, 4.2, 4.6 and 4.9, footnote 42; Economic Capital for Credit Risk

Solution CR-4 g.3 — Answer: B

Bank Q134 • BT P2.T7.1.1

Step 1 – Expected against unexpected

IV is false: EL is absorbed by reserves and pricing. V is the definition of economic capital.

Step 2 – The other claims

I is false: economic capital is the bank's own estimate and can sit below or above the regulatory minimum. II has no basis. III describes a valuation (excess-return) concept, not capital for credit risk.

$\Rightarrow$ Only V is true.

Interpretation

Economic capital is defined by three things: unexpected loss, a confidence level and a horizon (usually one year).

Why the other options are wrong

A is incorrect. I is false: nothing requires internal capital to exceed the regulatory figure.

C is incorrect. I, II and III are all false.

D is incorrect. I to IV are false; IV confuses reserves for EL with capital for UL.

LO: CR-4 g: Describe how economic capital is derived. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Box 4.1 and Economic Capital for Credit Risk

Solution CR-4 g.4 — Answer: A

Bank Q89 • GARP PE2 Q50 • Mock 4 Q51

Step 1 – Write capital as CM times UL

$EC = CM \times UL_P$. Same confidence level means the same CM for both banks.

Step 2 – Correlation drives UL

Same exposure, rating and EL, but Bank XYZ has the higher default correlation, so less diversification and a higher UL_P. At the same confidence level, $EC_{XYZ} > EC_{ABC}$.

$\Rightarrow$ D is false.

Step 3 – Confidence level drives CM

Raising the confidence level raises CM, so EC rises for both banks; lowering it reduces EC for both.

$\Rightarrow$ A is correct.

Interpretation

The two drivers act separately: correlation sets the size of UL, the confidence level sets the multiple applied to it. Neither bank can move against the other when the confidence level changes.

Why the other options are wrong

B is incorrect. A lower confidence level lowers CM and capital for both banks.

C is incorrect. A higher confidence level raises CM for both; lower correlation keeps ABC's capital below XYZ's but does not make it fall.

D is incorrect. Ignores correlation: XYZ's higher correlation means a higher UL_P and more capital at the same confidence level.

LO: CR-4 g: Describe how economic capital is derived. CR-4 a: Evaluate a bank's economic capital relative to its level of credit risk. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Box 4.1; eqs. 4.5 and 4.17

Solution CR-4 g.5 — Answer: C

Bank Q137 • BT P2.T6.507.2

Step 1 – Parameters

The beta distribution has two shape parameters, and its form "is fully characterized by two parameters: EL_P and UL_P".
$\Rightarrow$ A is false.

Step 2 – Where it is weak

"The difficulty is fitting the beta distribution exactly to the tail", and the tail of a fitted beta depends on the ratio EL_P/UL_P; tail fitting is therefore combined with Monte Carlo simulation.
$\Rightarrow$ C is true.

Step 3 – From the distribution to CM

"Taking the inverse of the beta function at the chosen confidence level, we can determine CM."
$\Rightarrow$ D is false.

Interpretation

This is the link between the loss distribution and the capital multiplier: CM is read off the fitted (skewed) distribution, which is why it is so much larger than a normal multiple.

Why the other options are wrong

A is incorrect. Two shape parameters, set through EL_P and UL_P, not four.
B is incorrect. Equal shape parameters make the beta symmetric, which contradicts the skew of credit losses.
D is incorrect. The fitted beta is exactly what CM is taken from.

LO: CR-4 g: Describe how economic capital is derived. CR-4 h: Explain how the credit loss distribution is modeled. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Economic Capital for Credit Risk, eqs. 4.19 to 4.22 and footnote 58

Solution CR-4 g.6 — Answer: A

Bank Q132 • Mock 2 Q30

Step 1 – Economic capital

EC is the distance from EL to the loss at the confidence level: $230 - 130 = \text{USD } 100$ million.

Step 2 – Effect of a higher confidence level

The 99.97% loss quantile sits further into the tail than the 99.9% one, while EL does not change. EC rises.

Interpretation

The credit VaR of USD 230 million is $\text{EL}_P + \text{CM} \times \text{UL}_P$; the USD 100 million of capital is the $\text{CM} \times \text{UL}_P$ part. The bank's EL is covered by reserves and pricing.

Why the other options are wrong

B is incorrect. Direction wrong: EL is a mean and does not depend on the confidence level; the quantile, and with it EC, rises.
C is incorrect. Did not subtract EL: VaR itself is not economic capital.
D is incorrect. Added EL to VaR (360) and claimed capital is insensitive to the confidence level.

LO: CR-4 g: Describe how economic capital is derived. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Box 4.1 and Economic Capital for Credit Risk

Solution CR-4 g.7 — Answer: D

Workbook D3.2 • S1, S2 • Map #61

Why D is correct

Schroeck derives $\text{EC}_i = \text{ULC}_i \times \text{CM}$ (eq. 4.18): "the required economic capital at the single credit transaction level is directly proportional to its contribution to the overall portfolio credit risk". Footnote 62 contrasts this with "the regulatory approach that assigns roughly 8% equity capital to credits on a standalone basis", whereas the economic approach differentiates capital by borrower quality and reflects the benefits of diversification.

$\Rightarrow$ This is the map's discrimination: economic capital is the bank's own UL-based estimate; regulatory capital is a formula-driven minimum.

Why the other options are wrong

A is incorrect. Polarity flip. Box 4.1: a bank targeting a very high rating "must choose a very high confidence level (e.g., 99.97%), which corresponds to a higher capital multiplier", so economic capital rises. A lower CM belongs to a more aggressive bank targeting a lower rating.

B is incorrect. Sibling swap. Crouhy assigns "industrywide rules and formulas" and "only a minimum required level of capital adequacy" to regulatory capital; risk (economic) capital is intended to capture the economic realities of the risks the firm runs.

C is incorrect. Scope error. EL is "the foreseeable cost of doing business", covered by pricing and loan loss reserves, and "does not itself constitute risk". Economic capital is "the distance between the expected outcome and the unexpected (negative) outcome at a certain confidence level", that is, $\text{UL}_P \times \text{CM}$ measured from EL.

LO: CR-4 g: Describe how economic capital is derived. CR-4 a: Evaluate a bank's economic capital relative to its level of credit risk. ORR-18 a: Define, compare, and contrast risk capital, economic capital, and regulatory capital, and explain methods and motivations for using economic capital approaches to allocate risk capital. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Box 4.1, eqs. 4.17 and 4.18, footnote 62; Crouhy, Galai and Mark, Ch.17, 18.1

Solution CR-4 g.8 — Answer: D

Workbook A1.3 • S3, S4 • Map #1

Step 1 – Standalone UL

$$\text{PD} \cdot \sigma_{LR}^2 = 0.015 \times 0.04 = 0.000600, \qquad \text{LR}^2 \cdot \sigma_{PD}^2 = 0.2025 \times 0.014775 = 0.002992$$

$$\text{UL}_i = 2.5 \times \sqrt{0.003592} = \text{AUD } 0.1498 \text{ million}$$

Step 2 – Portfolio UL

$$\begin{aligned}\text{UL}_P &= \text{UL}_i \sqrt{n + \rho(n^2 - n)} = 0.1498 \times \sqrt{400 + 0.04 \times 159{,}600} \\ &= 0.1498 \times 82.37 = \text{AUD } 12.34 \text{ million}\end{aligned}$$

$\Rightarrow$ The 400 loans behave like 82 standalone loans: diversification removes most of the undiversified $400 \times 0.1498 = 59.9$.

Step 3 – UL contribution of one loan

$$\text{ULC}_i = \frac{\text{UL}_P}{n} = \frac{12.34}{400} = \text{AUD } 0.03085 \text{ million}$$

Step 4 – Apply the capital multiplier

$$\text{EC}_P = \text{UL}_P \times \text{CM} = 12.34 \times 7.0 = \text{AUD } 86.4 \text{ million}$$

$$\text{EC}_i = \text{ULC}_i \times \text{CM} = 0.03085 \times 7.0 = \text{AUD } 0.216 \text{ million}$$

$\Rightarrow$ Each loan carries AUD 216,000 of capital, about 8.6% of its exposure; the 400 attributions add back exactly to AUD 86.4 million.

Interpretation

CM converts a one-standard-deviation measure into the distance from EL_P to the 99.97% point of a skewed credit loss distribution. Portfolio EL ($400 \times 2.5 \times 0.015 \times 0.45 = \text{AUD } 6.75$ million) is covered by reserves and pricing, not by this capital.

Why the other options are wrong

A is incorrect. Omitted the capital multiplier: reported $\text{UL}_P = 12.3$ and $\text{ULC}_i = 0.031$ as if volatility were capital.

B is incorrect. Wrong input twin: used the normal multiple 3.43 instead of the credit CM, $12.34 \times 3.43 = 42.3$ and $0.03085 \times 3.43 = 0.106$. Schroeck: the credit CM is far higher than normal-distribution multiples because credit losses are skewed.

C is incorrect. Subtracted EL from the capital a second time: $86.4 - 6.75 = 79.6$ and $0.216 - 0.0169 = 0.199$. CM is already measured from EL_P.

LO: CR-4 g: Describe how economic capital is derived. CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Economic Capital for Credit Risk, eqs. 4.13, 4.17 and 4.18

CR-5 • Introduction to Credit Risk Modeling and Assessment • Vardeez: VERY IMPORTANT

CR-5 d

P1 CONCEPT

Describe the judgmental approaches, empirical models, and financial models to predict default.

Workbook: A10 Altman Z-Score and Linear Discriminant Analysis • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: Altman's Z-score (A10, Map #15, 6/16) came up in the August 2026 sitting, and the map also reports it as "calculate z score (LDA)", which places Altman's discriminant function inside this LO's empirical models.

Vardeez: CR-5 is a VERY IMPORTANT chapter and "LO 23.d is Imp". Know the three approaches side by side: data source, method, inputs, which borrowers each can cover, strengths and weaknesses.

Reported exam shapes

- **A10·S4** "Calculate z score (LDA)": Z as Altman's linear discriminant function, a fixed-weight score whose cut-offs assign a firm to a group.

What the questions look like: conceptual. A lender is described (rich loan history, no history, listed borrowers, credit derivatives) and you pick the approach, or four statements test inputs, applicability (consumer versus corporate), weaknesses, and structural versus reduced form. The LO is CONCEPT, so the Z-score is tested here as a method; its arithmetic belongs to CR-9 b.

Worked elsewhere: A10.1 (Z from filings), A10.2 (the discriminant score and its group) and A10.3 (reverse-solving EBIT) under CR-9 b.

1. Three ways to predict default

GARP's Chapter 5 (Doumpos et al.) sorts the approaches for assessing the likelihood of default into three schemes. Their main differences are the **data they need** and their **scope**: which borrowers they can be used for. One line to hold: judgmental means experts, empirical means history, financial means theory plus market prices.

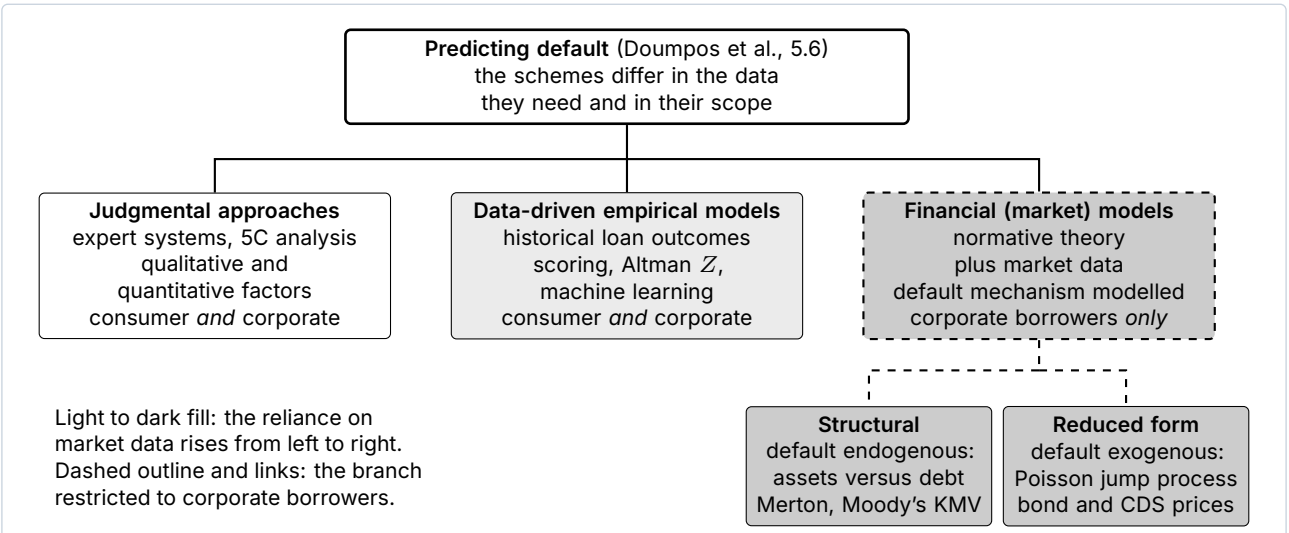


Figure 1. The three schemes of GARP 5.6. Only the financial branch (dashed) is limited to corporate debt; it splits into structural models (CR-5 e, CR-5 f) and reduced-form models (the hazard-rate approach of CR-9).

2. Judgmental approaches

The oldest and simplest scheme, also called **qualitative approaches** or **expert systems**: the assessment rests solely on the expert judgment of credit analysts about the borrower's fundamental qualitative and quantitative characteristics. The best-known framework is **5C analysis**: **character** (the borrower's personality), **capacity** (ability to repay from income after expenses and other debts), **capital** (the borrower's own capital at risk), **collateral** (guarantees) and **conditions** (the business environment and the loan's terms). Inputs run from financial statements and a business plan to the sector and economic outlook, so the approach serves **consumer and corporate** loans.

Limitations (GARP). Because the process rests on expertise rather than theory or data, it is very difficult to (a) **validate** the quality of the results, (b) **update** the process to changes in the environment, and (c) **examine scenarios** that affect one borrower or the portfolio. Transparency and consistency suffer when protocols are unclear. **Where it still earns its place:** wherever history is too thin for a model, such as **project finance**, shipping, start-ups and SMEs, and complex cases where analysts read unstructured information.

3. Data-driven empirical models

Models fitted to **historical data** on loans accepted, rejected, paid as agreed and in default, from internal databases, rating agencies and data providers. They apply to **corporate and consumer** loans wherever such history exists. The inputs are the kind of information a judgmental analyst uses; the difference is that a fitted model, not an expert, turns them into an explicit relationship between default and the risk factors. The roots are the bankruptcy models of the late 1960s (Altman, 1968); today machine learning is a technique *within* this scheme, not a fourth approach, and governance, news analytics and real-time market data have joined the financial ratios.

Advantages (GARP): transparency and consistency; structure and predictive power can be validated against actual data; hypotheses, scenarios and stress tests can be examined; timely updating; customisation to large firms, SMEs, sectors, cards and mortgages. **Weaknesses:** reliance on historical data that may not describe future risks, above all in crises; incomplete data; backward-looking and static inputs (financial statements reflect only the current status) that are not updated in real time.

Altman's Z-score as a linear discriminant function (shape S4)

Altman (1968) used a statistical technique called **discriminant analysis** to predict default from five accounting ratios. For publicly traded manufacturing companies (Hull eq. 9.1, GARP Ch.9):

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

X_1 working capital/total assets; X_2 retained earnings/total assets; X_3 EBIT/total assets; X_4 market value of equity/book value of total liabilities; X_5 sales/total assets. The weights were estimated on 66 listed manufacturers, 33 that failed within one year and 33 that did not, and the model proved fairly accurate out of sample.

"Linear discriminant function" describes three features. **Linear, fixed weights:** one score, the same weights for every firm. **Weights from data:** chosen so the score separates failed from non-failed firms in the sample, with no theory of default behind them, which is why Z is an *empirical* model. **Cut-offs assign a group:** above 3.0 unlikely to default; 2.7 to 3.0 "on alert"; 1.8 to 2.7 a good chance of default; below 1.8 the probability of "a financial embarrassment" is very high (Figure 2). Every weight is positive, so a higher Z is safer. In a scoring stem, "LDA" means linear discriminant analysis; in an operational-risk stem it means the loss distribution approach (frequency and severity combined into an annual loss distribution). They share only the acronym.

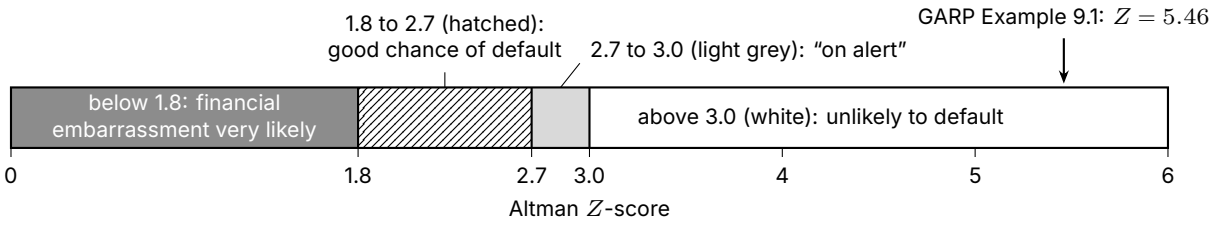


Figure 2. Altman's cut-offs turn the discriminant score into a group (GARP Ch.9, 9.1.2). The bands are dark grey, hatched, light grey and white from left to right. Example 9.1 (working capital 170,000, total assets 670,000, EBIT 60,000, sales 2,200,000, market equity 380,000, liabilities 240,000, retained earnings 300,000) scores 5.46, not in danger in the near future.

4. Financial (market) models

In contrast to the data-driven models, financial models are mostly based on **theory**: they take a **normative** approach founded on economic and financial principles and describe the *mechanism* of default. They are called **market models** because they rely on financial market data, so they are focused on **corporate debt**. Two types:

- **Structural models**: default is **endogenous**, linked to the firm's structural characteristics such as the value of its assets and its debt. Merton's option-pricing model (CR-5 e) and its best-known implementation, Moody's KMV (CR-5 f).
- **Reduced-form models**: default is a **random event** that may happen at any time, driven by a shift in an **exogenous** random variable, usually a Poisson jump process. The main inputs are market data on bonds and credit derivatives (Jarrow and Turnbull; Duffie and Singleton). The hazard rates of CR-9 are this family.

What the evidence says (GARP 5.8). Mixed: some studies find structural models beat empirical ones (Hillegeist et al., 2004), others that they add little (Agarwal and Taffler, 2008; Campbell et al., 2008); Chen and Hill (2013) find an empirical z-score model tracks agency ratings more closely than two structural models.

	Judgmental	Empirical	Financial
Data source	Experts (credit officers, committees)	Historical data: internal databases, agencies, providers	Market data on equity, bonds and credit derivatives
Method	Expert judgment of qualitative and quantitative factors	Models fitted to past loan outcomes	Normative economic and financial theory
Inputs	5Cs: character, capacity, capital, collateral, conditions	Borrower characteristics, loan status, external risk factors	Market prices; asset value and debt (structural); spreads (reduced form)
Borrowers	Consumer and corporate	Consumer and corporate	Corporate only
Strengths	Works without history; unique loans (project finance); insight on complex cases	Transparent, consistent, validated, scenario and stress tests, customisable	Theory of the default mechanism; market-based, forward-looking inputs
Weaknesses	Hard to validate, update and stress; transparency and consistency	Backward-looking, static, incomplete data; accounting inputs lag	Corporate debt only; model assumptions (CR-5 e); mixed evidence
Examples	5C analysis, expert systems	Credit scoring, Altman Z, machine learning	Merton, Moody's KMV; Poisson-jump hazard models

Correction to the source notes. Vardeez's three-approach table lists "may not capture market changes quickly" as a weakness of the financial models. GARP states no such weakness: these are market models, and GARP Ch.9 notes that market-based PDs (Moody's KMV, Kamakura) respond faster than ratings. GARP's stated limitation is scope (corporate debt only). Slowness is GARP's criticism of the other two schemes, so the table's "real-time updates" strength for empirical models is also dropped. The table above is corrected.

Exam traps

- **Applicability swap.** Only the financial models are restricted to corporate borrowers. Empirical models use data too, but historical loan data, so they cover consumers as well.
- **"Market data" versus "historical data".** Market prices drive the financial models; past loan outcomes drive the empirical models; experts drive the judgmental approach.
- **Structural versus reduced form.** Asset value against debt (endogenous) is structural; an exogenous Poisson jump calibrated to bond and CDS prices is reduced form. Questions swap the two definitions.
- **Right weakness, wrong approach.** "Hard to validate, update, run scenarios" is judgmental; "backward-looking, may fail in crises" is empirical. "Empirical models cannot be validated" reverses an advantage.
- **Z-score traps.** High Z is safe (all weights positive); "LDA" here is linear discriminant analysis, not the loss distribution approach; the weights come from data, not option pricing (distance to default is Merton's output, CR-5 e).

Practice questions: CR-5 d

Question CR-5 d.1

Bank Q184 • AnalystPrep Q.5973

A financial institution is restructuring its credit risk assessment process and has decided to use judgmental approaches for certain credit portfolios. Which of the following best describes a fundamental aspect of judgmental approaches in credit risk assessment?

A. They rely heavily on real-time market data to assess the creditworthiness of borrowers.

B. They require extensive use of credit scoring models based on empirical data.

C. They emphasize quantitative assessment using historical financial ratios and trends.

D. They involve qualitative analysis, focusing on aspects such as the borrower's character and capacity.

Question CR-5 d.2

Bank Q183 • Schweser Module Quiz 23.2 Q1 (rebuilt)

A financial analyst and a risk manager are discussing ways to derive the probability of default for a bank's loan portfolio. The analyst prefers models that rely on the perspectives of internal experts. The manager acknowledges the value of internal experts but prefers a model that is data driven and can assess loans for both consumer and corporate customers. Which types of approaches are the analyst and the manager, respectively, describing?

A. Judgmental approach; empirical model

B. Empirical model; judgmental approach

C. Judgmental approach; financial model

D. Financial model; empirical model

Question CR-5 d.3

Bank Q176 • AnalystPrep Q.5979

A small community bank serving a rural area is updating its credit risk assessment process. It has a close relationship with its customers and a deep understanding of the local economy, but it lacks extensive historical data on loans and defaults. The portfolio mixes agricultural loans, small business loans and personal loans, often to customers whose financial situations are unique and not well documented statistically. Which type of approach would be most suitable for this bank?

A. Empirical models

B. Judgmental approaches

C. Financial models based on economic theories

D. Machine learning prediction models

Question CR-5 d.4

Bank Q172 • AnalystPrep Q.5974

The credit committee of a financial institution is discussing the limitations of judgmental approaches before extending their use. Which of the following statements accurately reflects a key limitation of judgmental approaches in credit risk assessment?

A. They are overly reliant on real-time market data, making them less suitable for long-term risk assessment.

B. They rely heavily on advanced statistical models, making them difficult to implement without technical expertise.

C. They face challenges in assessing the quality of their results and in examining scenarios that affect creditworthiness.

D. They require constant updates of empirical data, making them unsuitable for dynamic market conditions.

Question CR-5 d.5

Bank Q185 • AnalystPrep Q.1785 (rebuilt)

A bank has built an empirical default-prediction model using historical loan data from a long period of stable economic conditions. Before the model goes live, the risk manager is asked to flag its most important limitation. Which is the most appropriate?

A. Empirical models cannot be validated, because their structure and predictive power cannot be tested against actual data.

B. Because the model relies on historical data, it may not predict future outcomes well, especially in periods of high uncertainty and volatility.

C. Empirical models can be applied only to corporate borrowers, because they require market data.

D. Empirical models cannot incorporate new risk factors such as corporate governance or real-time market information.

Question CR-5 d.6

Bank Q182 • BT P2.T6.703.3 (rebuilt)

A credit risk trainer is contrasting the two families of financial models of default for a class of new analysts. Which of the following statements is correct?

A. Structural models assume default is an endogenous process linked to firm characteristics such as asset value and debt, as in the Merton model.

B. Reduced-form models assume default depends on the firm's asset value relative to its debt.

C. Structural models assume default is triggered by a random exogenous event such as a Poisson jump.

D. Both structural and reduced-form models are market models that can be applied to consumer as well as corporate borrowers.

Question CR-5 d.7

Bank Q181 • BT P2.T6.24.6.3 (rebuilt)

Mancus Lending Group is selecting an approach to predict borrower default. Three board members make statements:

- Moe: "A judgmental approach relies on expert assessment and can be applied to both consumer and corporate loans."
- Larry: "Empirical models are data-driven but, because they rely on market data, they can only be applied to corporate borrowers."
- Curly: "Financial models such as the Merton model rely on financial market data and therefore can only be applied to corporate borrowers."

Which statements are correct?

A. Moe only

B. Larry and Curly only

C. Moe and Curly only

D. Moe, Larry and Curly

Question CR-5 d.8 New • built from GARP Ch.5, 5.6.2 and 5.8; Ch.9, 9.1.2 • shape A10·S4

A credit risk consultant is explaining a lender's corporate scoring engine to its model risk committee. The engine applies Altman's original Z-score to listed manufacturers and reads each firm's credit group off the score. The committee chair asks how the method should be described. Which of the following statements is correct?

A. The Z-score is a structural financial model: its weights come from option pricing theory, and Z measures how many standard deviations the firm's assets are above the default point.

B. In a credit-scoring context, "LDA" refers to the loss distribution approach, which combines a frequency and a severity distribution of losses into the firm's score.

C. A higher Z-score signals a higher likelihood of default, because each of the five accounting ratios enters the score with a negative weight.

D. The Z-score is a linear discriminant function: five accounting ratios are combined with fixed weights estimated from samples of failed and non-failed firms, and cut-off values of Z assign each firm to a group.

Solutions • CR-5 d

Solution CR-5 d.1

Answer: D

Bank Q184 • AnalystPrep Q.5973

Step 1 – What a judgmental approach is

GARP calls these approaches qualitative approaches or expert systems: the assessment rests on the expert judgment of credit analysts.

⇒ The defining feature is who aggregates the information: an expert, not a fitted model or a market price.

Step 2 – The standard framework

5C analysis covers character, capacity, capital, collateral and conditions.

⇒ D names two of the five Cs.

Interpretation

Quantitative information (financial statements, a business plan) can still feed a judgmental assessment; what makes it judgmental is that an analyst, not a statistical model, weighs it.

Why the other options are wrong

A is incorrect. Real-time market data drive the financial (market) models, not expert assessment.

B is incorrect. Credit scoring models fitted to empirical data are the empirical approach.

C is incorrect. Systematic quantitative analysis of historical ratios and trends is the empirical approach's method.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.1 Judgmental Approaches

Solution CR-5 d.2

Answer: A

Bank Q183 • Schweser Module Quiz 23.2 Q1

Step 1 – The analyst

Reliance on the perspectives of internal experts is the judgmental (expert-system) approach.

Step 2 – The manager

Data driven and usable for consumers and corporates: the empirical approach. Financial models are data-based too, but on market data, and they are focused on corporate debt, so they cannot serve consumers.

⇒ Judgmental, then empirical.

Interpretation

The applicability clause does the work: "both consumer and corporate" rules out the financial models even though they also use data.

Why the other options are wrong

B is incorrect. Reverses the two descriptions.

C is incorrect. A financial model cannot assess consumer loans, so it does not fit the manager's requirement.

D is incorrect. A financial model rests on theory and market data, not on internal experts, so it does not fit the analyst.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.1 to 5.6.3; SchweserNotes 2026 Book 2, Reading 23, LO 23.d

Solution CR-5 d.3 — Answer: B

Bank Q176 • AnalystPrep Q.5979

Step 1 – What the data allow

Little loan and default history: an empirical model (machine learning included) cannot be estimated or validated reliably.

Step 2 – Who the borrowers are

Farmers, small businesses and individuals have no traded securities: the financial (market) models, focused on corporate debt, do not apply.

Step 3 – Where judgment adds value

GARP: judgmental approaches are still used where historical data are lacking and cases are unique, and experienced analysts can interpret unstructured information.
$\Rightarrow$ The bank's local knowledge is exactly the input a judgmental approach needs.

Interpretation

The mirror image is a regional lender with a long, rich record of applications, repayments and defaults: there the answer flips to empirical models.

Why the other options are wrong

A is incorrect. Empirical models need substantial historical default data, which the bank lacks.
C is incorrect. Financial models need market data and are focused on corporate debt.
D is incorrect. Machine learning is an empirical technique and needs even more data.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.1 and 5.6.2

Solution CR-5 d.4 — Answer: C

Bank Q172 • AnalystPrep Q.5974

Step 1 – GARP's list of difficulties

Judgmental systems are based on expertise rather than theory or data, so it is very difficult to (a) assess the quality of the results (validation), (b) update the process to changes in the environment, and (c) examine scenarios affecting one borrower or the portfolio.
$\Rightarrow$ C restates (a) and (c).

Interpretation

Transparency and consistency are the other weaknesses GARP adds, when implementation and monitoring protocols are unclear.

Why the other options are wrong

A is incorrect. Real-time market data are an input of the financial models (and, partly, empirical ones), not of expert judgment.
B is incorrect. Advanced statistical models characterise empirical models; judgmental approaches need no statistical machinery.
D is incorrect. Constant updates of empirical data describe the data needs of empirical models.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.1 Judgmental Approaches

Solution CR-5 d.5 — Answer: B

Bank Q185 • AnalystPrep Q.1785

Step 1 – What the model has learned from

Only a long stable period: it has never seen a stress episode.

Step 2 – GARP's main weakness of empirical models

Reliance on historical data: past data may not fully describe future risks, particularly in crises, and they are backward looking and static.

$\Rightarrow$ B names exactly this risk.

Interpretation

"Stable conditions" in the stem is the hint. Expert judgment and careful data processing can soften the problem but not remove it.

Why the other options are wrong

A is incorrect. Validation against actual data is one of the stated *advantages* of empirical models.

C is incorrect. Empirical models use historical loan data and cover consumer and corporate loans; market data and the corporate-only scope belong to the financial models.

D is incorrect. GARP lists corporate governance and real-time market information as new risk factors that empirical models now use.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.2 Data-Driven Empirical Models

Solution CR-5 d.6 — Answer: A

Bank Q182 • BT P2.T6.703.3

Step 1 – Structural models

Default is an endogenous process linked to the firm's structural characteristics, such as the value of its assets and debt; Merton is the example.

Step 2 – Reduced-form models

Default is a random event that may happen at any time, driven by a shift in an exogenous variable (usually a Poisson jump process); inputs are bond and credit-derivative prices.

Step 3 – Scope of both

Both are financial (market) models, focused on corporate debt.

$\Rightarrow$ Only A states a definition correctly.

Interpretation

The two families answer the same question from opposite ends: structural models explain *why* default happens; reduced-form models describe *when* it arrives and calibrate the rate to market prices.

Why the other options are wrong

B is incorrect. Swaps the definitions: asset value against debt is the structural mechanism.

C is incorrect. Swaps the definitions: the exogenous Poisson jump is the reduced-form assumption.

D is incorrect. Financial models rely on market data and are focused on corporate debt; consumers are covered by the judgmental and empirical schemes.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3 Financial Models

Solution CR-5 d.7 — Answer: C

Bank Q181 • BT P2.T6.24.6.3

Step 1 – Moe

Judgmental approaches use expert assessment of qualitative and quantitative factors and work for consumer and corporate loans. Correct.

Step 2 – Larry

Empirical models use historical loan data (accepted, rejected, performing, defaulted) from internal or external databases, not market data, and apply to consumer and corporate loans. Wrong.

Step 3 – Curly

Financial models are based on theory and market data and are focused on corporate debt. Correct.

$\Rightarrow$ Moe and Curly only.

Interpretation

Larry has attached the financial models' limitation to the empirical models: the classic applicability swap.

Why the other options are wrong

A is incorrect. Omits Curly, whose statement matches GARP's description of the financial models.

B is incorrect. Accepts Larry's swap and rejects Moe's correct statement.

D is incorrect. Accepts Larry's claim that empirical models are corporate-only.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.1 to 5.6.3

Solution CR-5 d.8 — Answer: D

New • built from GARP Ch.5, 5.6.2 and 5.8; Ch.9, 9.1.2

Step 1 – What kind of model the Z-score is

Altman (1968) used discriminant analysis on five accounting ratios, with weights estimated from 33 failed and 33 non-failed listed manufacturers. The weights come from the data, so the Z-score belongs to the empirical (data-driven) scheme; GARP traces the roots of empirical models to these late-1960s bankruptcy models.

$\Rightarrow$ A linear score with fixed, data-estimated weights: a linear discriminant function.

Step 2 – How the group is assigned

Cut-off values of Z (3.0, 2.7, 1.8) place the firm in "unlikely to default", "on alert", "good chance of default" or "financial embarrassment very high".

Step 3 – Direction

All five weights are positive (1.2, 1.4, 3.3, 0.6, 0.999), so a higher Z means a healthier firm.

Interpretation

This is shape A10·S4 without the arithmetic: the map's "calculate z score (LDA)" is Altman's discriminant function. The numerical versions (scoring a firm, reading its zone, reverse-solving an input) are Workbook A10.1 to A10.3 under CR-9 b.

Why the other options are wrong

A is incorrect. Sibling swap with Merton's distance to default: the Z-score has no option-pricing content and no default point.

B is incorrect. Acronym swap: the loss distribution approach is an operational-risk capital method; here LDA is linear discriminant analysis.

C is incorrect. Polarity flip: every weight is positive and a higher Z signals *lower* default risk.

LO: CR-5 d: Describe the judgmental approaches, empirical models, and financial models to predict default. CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.2 and 5.8; Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.2 eq. 9.1

CR-5 • Introduction to Credit Risk Modeling and Assessment • Vardeez: VERY IMPORTANT

CR-5 e

P1 CALC

Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model.

Workbook: A5 Merton Model, Distance to Default and Risk-Neutral PD • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: the Merton model is A5 (Map #2, 8/16) and came up in the August 2026 sitting; "finding distance to default and assign a rating based on the score" is one of the variants the map reports.
Vardeez: "LO 23.e & LO 23.f: Imp"; pointer: Merton model. Know equity as a call, $N(-d_2)$, the real-world swap of r for μ, the distance to default, and the limitations.

Reported exam shapes

- **A5·S1** Distance to default, then assign a rating band from the score.
- **A5·S6** Short-form DtD $\ln(V/F)/\sigma_V$ versus the full form with the drift.
- **A5·S7** Scope trap: "Merton cannot be applied beyond one year" is false; two- and three-year horizons are routine.

What the questions look like: half numerical, half conceptual. Numerical stems give the asset value, asset volatility, the debt due at T, r and sometimes μ; you compute d_2 and $N(-d_2)$, choosing the drift that matches "risk-neutral" or "real-world". Equity volatility, μ when a risk-neutral PD is asked, and long-term debt in a plain Merton stem are the usual red herrings. Conceptual stems test what $N(-d_2)$ means, which inputs are unobservable, the assumptions and the limitations.

Worked elsewhere: A5.1 (risk-neutral PD to a hazard rate for a CVA desk) at Question CR-9 m.9; A5.2 (debt value and the leverage multiplier) and A5.5 (solving for V_0 and σ_V) at Questions CR-9 l.9 and CR-9 l.10; A5.3 (real-world PD for a stress scenario) under CR-9 k.

1. Equity as a call on the firm's assets

Merton's model is the structural model of CR-5 d: default is triggered by the firm's own balance sheet. GARP (Doumpos) assumes a firm with a **simple debt structure**: a single liability with face value L maturing at time T. At T the shareholders repay only if the market value of the assets A_T exceeds L; otherwise the firm's net worth is negative and they walk away.

$$\boxed{E_T = \max(A_T - L,\, 0)} \qquad \boxed{D_T = L - \max(L - A_T,\, 0)}$$

Equity is a **long call** on the firm's assets struck at L: the shareholders are the buyers of the option and the creditors are the writers. The debt is a risk-free claim on L less a **short put**. With $L = 50$: assets of 60 give equity 10 and debt 50; assets of 40 give equity 0 and debt 40 (Figure 1).

Notation. GARP's Ch.5 writes A, L, σ_A; Hull's Ch.9 (CR-9) writes V_0, D, σ_V; Vardeez's payoff formulas use F. All three letters L, D, F are the same object: the amount owed at T, which is the strike of the option.

2. Valuing the equity and the two unknowns

Today's equity value is the Black-Scholes-Merton call price (no dividends; Doumpos eq. 5.1, Hull eq. 9.4):

$$E = A\,N(d_1) - L\,\mathrm{e}^{-rT}N(d_2), \qquad d_1 = \frac{\ln(A/L) + (r + \sigma_A^2/2)T}{\sigma_A\sqrt{T}}, \qquad d_2 = d_1 - \sigma_A\sqrt{T}$$

Because equity is a function of assets and time, its volatility satisfies (eq. 5.2) $\sigma_E\,E = N(d_1)\,\sigma_A\,A$.

- **Known:** the market value of equity E (market capitalisation); the equity volatility σ_E (from historical market data); T, the time to repayment; L, usually taken as the **short-term debt** on the balance sheet; r, the T-bill yield.
- **Unknown:** the current asset value A and the asset volatility σ_A. Equations 5.1 and 5.2 are solved together with a simple iterative Newton-type algorithm (Hull uses Excel Solver). The mechanics belong to CR-9 1.

Book check (Hull Example 9.4). $E_0 = 3$, $\sigma_E = 80\%$, $r = 5\%$, $T = 1$, $D = 10$ solve to $V_0 = 12.40$ and $\sigma_V = 21.23\%$; then $d_2 = 1.1408$ and $N(-d_2) = 12.7\%$. Debt is worth $V_0 - E_0 = 9.40$ against a risk-free value of $10\mathrm{e}^{-0.05} = 9.51$, an expected loss of about 1.2%.

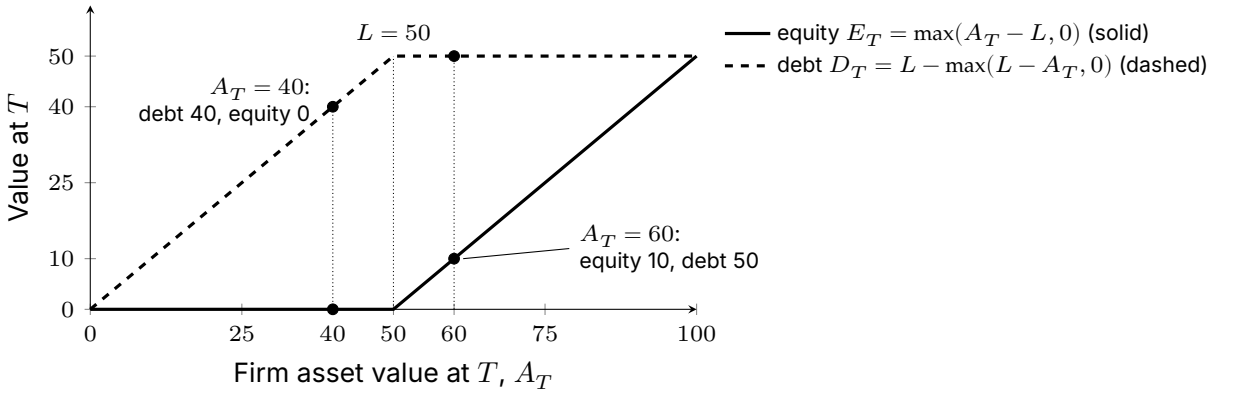


Figure 1. The asset value at T is split between the two claimants at a kink at the face value of the debt. Below L the creditors take everything and equity (solid) is worthless; above L the creditors are capped at L (dashed) and every further unit goes to equity. The two marked points are Vardeez's worked examples.

Correction to the source notes. SchweserNotes (LO 23.e) says the standard approach is "to solve for the market value of equity (E) and the standard deviation of equity σ_E using historical market data" and then define μ. E and σ_E are the *observed* inputs; GARP solves equations 5.1 and 5.2 for the asset value A and the asset volatility σ_A, then defines μ as the rate of change in A over past periods. Schweser's own list two paragraphs earlier names A and σ_A as the unknowns. Question CR-5 e.2 tests exactly this point.

3. Default probability and distance to default

- **Risk-neutral PD** $= N(-d_2) = 1 - N(d_2)$: the probability that the shareholders do *not* exercise their option to repay, derived with the assets growing at the risk-free rate r.
- **Real-world PD** (Doumpos eq. 5.3): replace r with the expected return on assets μ, estimated as the rate of change in A over past periods:

$$\boxed{\mathrm{PD} = N\left(-\frac{\ln(A/L) + (\mu - \sigma_A^2/2)\,T}{\sigma_A\sqrt{T}}\right)}$$

- **Which is larger.** Real-world asset growth is usually above r, so the risk-neutral PD is the higher one (Hull 9.8.2). Risk-neutral PDs are for valuation (pricing, CVA); real-world PDs for scenario analysis (CR-9 k).
- **Distance to default (DD)**: the number of standard deviations the firm's assets are away from the default point L at T; PD $= N(-\text{DD})$. As DD falls, default becomes more likely.

$$\boxed{\text{DD} = \frac{\ln A - \ln L + (\text{drift} - \sigma_A^2/2)\,T}{\sigma_A\sqrt{T}}}$$

Doumpos derives DD from eq. 5.3, so its drift is μ; Hull's definition is d_2, with drift r (Example 9.4: one-year DD $= 1.14$). Use the drift the stem specifies.

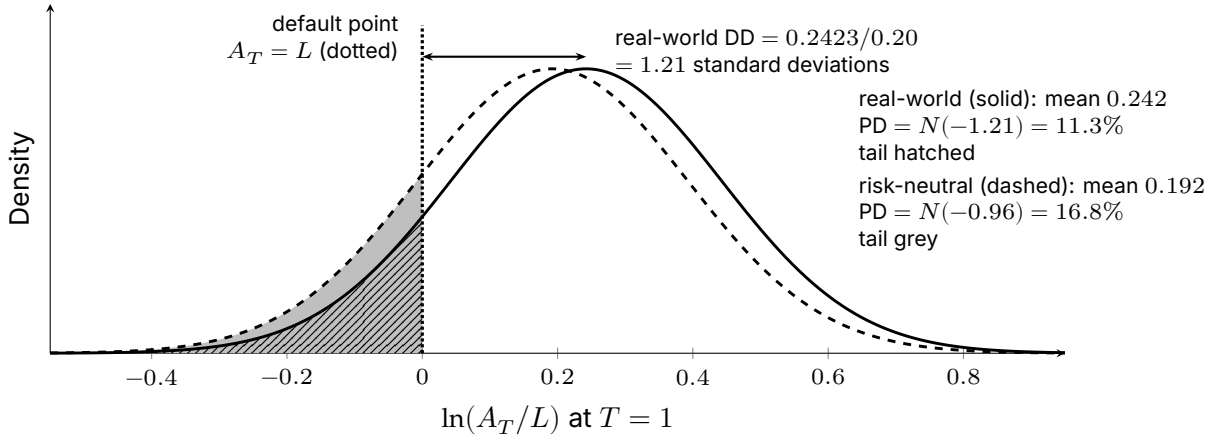


Figure 2. The firm of Question CR-5 e.4 ($A = 120$, $L = 100$, $\sigma_A = 20\%$, $\mu = 8\%$, $r = 3\%$). Default is the region left of the dotted line. Swapping μ for r shifts the whole distribution left by $\mu - r = 0.05$, so the risk-neutral tail (grey) is larger than the real-world tail (hatched).

Worked example. $A = 500$, $L = 400$ (short-term debt), $\sigma_A = 25\%$, $r = 4\%$, $\mu = 9\%$, $T = 1$; $\ln(500/400) = 0.2231$ and $\sigma_A^2/2 = 0.03125$.

$$\text{Risk-neutral: DD} = \frac{0.2231 + 0.04 - 0.03125}{0.25} = 0.9276, \quad \text{PD} = N(-0.9276) = 17.7\%$$

$$\text{Real-world: DD} = \frac{0.2231 + 0.09 - 0.03125}{0.25} = 1.1276, \quad \text{PD} = N(-1.1276) = 13.0\%$$

$$\text{Short form: DD} \approx \frac{0.2231}{0.25} = 0.8926, \quad \text{PD} = N(-0.8926) = 18.6\%$$

4. Short form, rating bands and horizons

Short form versus full form (shape S6)

Over one year with a small drift, DD $\approx \ln(V/F)/\sigma_V$. The full form adds $(\text{drift} - \sigma^2/2)T/(\sigma\sqrt{T})$, here $0.00875/0.25 = 0.035$ standard deviations with r. Use the short form only when the stem says "approximation" or gives no drift; when r or μ is given and the full form or d_2 is asked for, keep the drift.

From DD to a rating band (shape S1)

Banks bucket DD into internal grades: a higher DD earns a better band. This works because Merton's model and its extensions produce a good *ranking* of default probabilities (Hull 9.8.1), even if the

normal mapping to a PD is unrealistic. Moody's KMV keeps the DD but maps it to a probability through historical default frequencies instead of $N(\cdot)$ (CR-5 f).

Any horizon, not just one year (shape S7)

T is the time to repayment of the loan (Doumpos), and Hull defines DD for default "T years in the future". The drift scales with T and the volatility with $\sqrt{T}$; nothing restricts the model to one year. The worked example over longer horizons:

Horizon T (years)	1	2	3	5
Risk-neutral DD	0.93	0.68	0.58	0.48
Risk-neutral PD	17.7%	24.8%	28.2%	31.7%
Real-world PD	13.0%	16.8%	17.8%	17.8%

The spread of outcomes grows with $\sqrt{T}$, so the risk-neutral PD rises steadily; with $\mu = 9\%$ the faster drift almost offsets the wider spread after three years.

5. Limitations of the Merton model

GARP's Ch.5 gives the model's assumptions and the Moody's KMV fix rather than a list; Vardeez's gap-filled list (written from GARP) checks out against GARP 5.6.3.1 and Hull 9.8:

1. **A trivially simple capital structure.** One zero-coupon liability L maturing at one date T. In practice L is the short-term debt on the balance sheet: a choice, not an observation, and it ignores long-term debt (Moody's KMV adds half of it).
2. **Default only at T.** The firm cannot default before maturity however far its assets fall. Hull's extensions: default whenever assets breach a barrier, and debt payments at more than one date.
3. **Unobservable inputs and constant parameters.** A and σ_A must be backed out of E and σ_E by an iterative solve, and σ_A and r are held constant over the life (Black-Scholes assumptions).
4. **The normal mapping.** DD becomes a PD through the normal distribution. The Moody's KMV EDF replaces it with an empirical distribution of default frequencies.
5. **The real-world PD needs μ**, itself estimated from past changes in A, so it inherits that estimation error.
6. **Scope and evidence.** It needs traded equity, so it serves listed corporate borrowers (CR-5 d). The ranking it produces is good, but several studies find little information beyond empirical models or market capitalisation (GARP 5.8).

Formula card: CR-5 e

Payoffs at T	$E_T = \max(A_T - L, 0), \quad D_T = L - \max(L - A_T, 0)$
Equity today (5.1)	$E = A\,N(d_1) - L\mathrm{e}^{-rT}N(d_2)$
d_1 and d_2	$d_1 = \dfrac{\ln(A/L) + (r + \sigma_A^2/2)T}{\sigma_A\sqrt{T}}, \quad d_2 = d_1 - \sigma_A\sqrt{T}$
Equity volatility (5.2)	$\sigma_E E = N(d_1)\,\sigma_A A$ (solve 5.1 and 5.2 for A and σ_A)
Risk-neutral PD	$N(-d_2) = 1 - N(d_2)$ (drift r)
Real-world PD (5.3)	$N\left(-\dfrac{\ln(A/L) + (\mu - \sigma_A^2/2)T}{\sigma_A\sqrt{T}}\right)$ (drift μ)
Distance to default	$\mathrm{DD} = \dfrac{\ln A - \ln L + (\text{drift} - \sigma_A^2/2)T}{\sigma_A\sqrt{T}}, \quad \mathrm{PD} = N(-\mathrm{DD})$
Short form	$\mathrm{DD} \approx \ln(A/L)/\sigma_A \quad (T = 1, \text{drift} \approx 0)$
KMV default point (CR-5 f)	short-term debt + $0.5\times$ long-term debt; DD mapped to an EDF empirically
Merton PD to hazard (CR-9 e)	$\bar{\lambda} = -\ln(1 - N(-d_2))/T$

Exam traps

- **d_1 versus d_2, and the sign.** PD is $N(-d_2)$: not $N(-d_1)$ (too low) and not $N(d_2)$ (the probability of *no* default).
- **r versus μ.** Risk-neutral keeps r and is the higher PD; real-world uses μ. A stem that gives both is testing which one you pick.
- **σ_E versus σ_A.** Equity volatility only calibrates σ_A; it never enters d_2.
- **The $-\sigma_A^2/2$ term.** Dropping it, or using $+\sigma_A^2/2$ (which belongs to d_1), shifts DD by $\sigma_A\sqrt{T}/2$ or $\sigma_A\sqrt{T}$.
- **$\sqrt{T}$ errors.** Volatility scales with $\sqrt{T}$, the drift with T. Scaling σ_A by T understates DD.
- **Default point.** Plain Merton uses the single face value (short-term debt); KMV uses short-term plus half of long-term debt. Total debt understates DD.
- **One-year scope.** "Merton only works over one year" is a false absolute; use the T of the debt.
- **Short versus full form.** Use the form the stem asks for; the drift matters more as T grows.
- **Polarity.** Higher asset volatility or more debt lowers DD and raises the PD.

Practice questions: CR-5 e

Question CR-5 e.1

Bank Q188 • AnalystPrep Q.5984 (rebuilt)

In the Merton model, a borrower's equity is a call option on the firm's assets with a strike equal to the face value of debt L. A credit analyst computes $N(-d_2)$ using the risk-free rate r as the asset growth rate. This quantity represents:

A. the probability that the shareholders exercise the option and repay the debt.

B. the risk-neutral probability that the shareholders do not exercise the option, that is, that the firm defaults.

C. the real-world probability of default, since asset growth is set to μ.

D. the expected loss on the debt as a percentage of its face value.

Question CR-5 e.2

Bank Q187 • AnalystPrep Q.4364 (rebuilt)

A credit analyst is setting up Merton's model for a listed borrower and lists its inputs for the model validation team. Which of the following statements about the inputs of the Merton model is correct?

A. The real-world probability of default is obtained by substituting the risk-free rate r for the expected asset return μ.

B. The risk-neutral probability of default is $N(d_2)$.

C. The current asset value A and the asset volatility σ_A are unknowns that must be solved from the equity value and equity volatility equations.

D. The face value of debt L is usually taken as the firm's long-term debt.

Question CR-5 e.3

Bank Q189 • AnalystPrep Q.5986 (rebuilt)

A model validator is reviewing a bank's use of the Merton (Black-Scholes-Merton) model for corporate credit risk. Which of the following is an assumption of the model?

A. The firm's assets can jump suddenly in value, triggering default at a random time.

B. The firm has several debt issues with different maturities, and default can occur on any of them at any time.

C. The risk-free rate is a stochastic process that is correlated with asset returns.

D. The firm's asset returns have constant volatility over the life of the loan.

Question CR-5 e.4

Bank Q190 • AnalystPrep Q.5988 (rebuilt)

For a scenario analysis, a credit analyst models a firm with asset value $A = 120$ and a single zero-coupon debt with face value $L = 100$ maturing in $T = 1$ year. The asset volatility is constant at $\sigma_A = 20\%$, the expected return on assets is $\mu = 8\%$, and the risk-free rate is $r = 3\%$. Using the Merton model, which of the following is closest to the real-world probability of default?

A. 7.9%

B. 9.5%

C. 11.3%

D. 16.8%

Question CR-5 e.5

New • built from GARP Ch.5, 5.6.3.1 eq. 5.1 and Ch.9, 9.8.3 • shape A5·S7

A credit officer at a Korean bank is reviewing a corporate borrower whose only debt is a zero-coupon loan due in three years. The bank's Merton solver has already returned the asset value and asset volatility. The credit committee wants the risk-neutral probability that the borrower defaults at the loan's maturity. Data (KRW billion):

- Value of assets: 900; asset volatility: 22%; equity volatility: 55%
- Face value of debt due in 3 years: 700
- Risk-free rate: 3% per annum, continuously compounded; expected return on assets: 11%

Which of the following is closest to the risk-neutral probability of default?

A. 9.1%

B. 12.1%

C. 24.0%

D. 34.2%

Question CR-5 e.6

Workbook A5.4 • S1, S6 • Map #2

A credit analyst at a US asset manager is replicating the Moody's KMV approach for a borrower. Balance sheet and market data (USD million):

- Value of assets: 2,400; asset volatility: 18%; equity volatility: 40%
- Short-term debt: 900; long-term debt: 1,000
- Risk-free rate: 4% (continuous); horizon: 1 year

Using the Moody's KMV default point and the full-form distance to default defined as d_2, which of the following is closest to the distance to default?

A. 1.43

B. 3.13

C. 3.31

D. 5.58

Question CR-5 e.7

Bank Q191 • AnalystPrep Q.6009 (rebuilt)

A credit risk manager is asked to list the recognised limitations of the Merton model before it is used to estimate borrowers' probabilities of default. Which of the following is a recognised limitation?

A. The default point is defined only as the firm's short-term debt, which can understate the obligations of a firm with large long-term debt.

B. It requires a rating transition matrix, which is hard to obtain for many borrowers.

C. It relies on observed bond prices and credit derivative spreads, which are unavailable for most firms.

D. It assumes default is driven by an exogenous Poisson jump that cannot be linked to firm fundamentals.

Question CR-5 e.8

Workbook A5.6 • S1, S6, S7 • Map #2

A senior credit analyst at a Singapore bank is presenting the bank's internal scoring step to a group of newly hired analysts. The step converts a Merton distance to default into a rating band:

- Internal bands: DD of 4.0 or more: A; 3.0 to below 4.0: BBB; 2.0 to below 3.0: BB; below 2.0: B
- Borrower data (SGD million): value of assets 900; default point 600; asset volatility 20%; risk-free rate 4%; horizon 1 year

Which of the following statements should the analyst include in the presentation?

A. The short-form distance to default is about 2.03, placing the borrower in band BB; adding the drift term $(r - \sigma_V^2/2)T$ raises it by about 0.10 and leaves the band unchanged.

B. The short-form distance to default is about 2.03, placing the borrower in band BB; the same scoring step could not be used for its three-year notes, because Merton's model cannot be applied beyond one year.

C. The short-form distance to default is about 2.03, placing the borrower in band BB; a rise in asset volatility to 25% would increase the distance to default and support an upgrade toward BBB.

D. The short-form distance to default is about 2.03, placing the borrower in band BB; Moody's KMV would convert the same score into a default probability by reading $N(-\text{DD})$ off the standard normal distribution.

Solutions • CR-5 e

Solution CR-5 e.1 — Answer: B

Bank Q188 • AnalystPrep Q.5984

Step 1 – When the option is exercised

The equity payoff at T is $\max(A_T - L, 0)$: the shareholders repay (exercise) only if $A_T > L$. With assets growing at r, the probability of this is $N(d_2)$.

Step 2 – Default is the complement

$N(-d_2) = 1 - N(d_2)$ is the probability that the option expires unexercised, so the firm defaults. $\Rightarrow$ Built on r, it is the *risk-neutral* default probability.

Interpretation

The only difference between the two PDs in the Merton framework is the drift: r for risk-neutral, μ for real-world.

Why the other options are wrong

A is incorrect. That is $N(d_2)$, the complement of the PD.
C is incorrect. The real-world PD has the same form but requires μ in place of r.
D is incorrect. $N(-d_2)$ is a probability only; an expected loss also needs the loss severity. In Hull Example 9.4 the PD is 12.7% but the expected loss on the debt is about 1.2%.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1

Solution CR-5 e.2 — Answer: C

Bank Q187 • AnalystPrep Q.4364

Step 1 – Count equations and unknowns

Equation 5.1 (equity as a call) and equation 5.2 ($\sigma_E E = N(d_1)\sigma_A A$) contain two unknowns, A and σ_A.

Step 2 – What is observed

E (market capitalisation), σ_E (historical equity data), T, L (short-term debt) and r (T-bill yield). $\Rightarrow$ The two unknowns are solved jointly by an iterative Newton-type algorithm.

Interpretation

This is the point SchweserNotes garbles (see the correction box in Section 2): E and σ_E are what you observe, not what you solve for.

Why the other options are wrong

A is incorrect. Reversed: the real-world PD substitutes μ for r.
B is incorrect. $N(d_2)$ is the probability that the debt is repaid; the risk-neutral PD is $N(-d_2)$.
D is incorrect. In practice L is the short-term debt on the balance sheet.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1 eqs. 5.1 and 5.2

Solution CR-5 e.3 — Answer: D

Bank Q189 • AnalystPrep Q.5986

Step 1 – The Black-Scholes mechanics

Assets follow a continuous diffusion with constant volatility σ_A, and r is constant.

Step 2 – The capital structure

One zero-coupon liability L maturing at T; default can only happen at T.
$\Rightarrow$ Constant asset volatility is a core assumption, and also one of the model's limitations.

Interpretation

Jumps and random default times belong to reduced-form models; several maturities and default before T are extensions of Merton, not the basic model.

Why the other options are wrong

A is incorrect. Sudden jumps and random default times are the reduced-form assumption.
B is incorrect. The basic model has a single debt issue maturing at T.
C is incorrect. The risk-free rate is constant, not stochastic.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1; Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8

Solution CR-5 e.4 — Answer: C

Bank Q190 • AnalystPrep Q.5988

Step 1 – Choose the drift

Scenario analysis needs the real-world PD, so the drift is $\mu = 8\%$ (Doumpos eq. 5.3).

Step 2 – Real-world distance to default

$$\text{DD} = \frac{\ln(120/100) + (0.08 - 0.20^2/2)(1)}{0.20\sqrt{1}} = \frac{0.1823 + 0.06}{0.20} = 1.2116$$

$\Rightarrow$ Assets sit 1.21 standard deviations above the default point.

Step 3 – Real-world PD

$$\text{PD} = N(-1.2116) = 11.3\%$$

Interpretation

Figure 2 draws this firm: the risk-neutral PD with $r = 3\%$ would be 16.8%, higher because assets grow more slowly in the risk-neutral world.

Why the other options are wrong

A is incorrect. Used $d_1 = 1.2116 + 0.20 = 1.4116$: $N(-1.4116) = 7.9\%$.
B is incorrect. Dropped the $-\sigma_A^2/2$ term: $(0.1823+0.08)/0.20 = 1.3116$, $N(-1.3116) = 9.5\%$.
D is incorrect. Used r instead of μ: $(0.1823 + 0.03 - 0.02)/0.20 = 0.9616$, $N(-0.9616) = 16.8\%$, the risk-neutral PD.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1 eq. 5.3

Solution CR-5 e.5 — Answer: C

New • built from GARP Ch.5, 5.6.3.1 and Ch.9, 9.8.3

Step 1 – Scale drift and volatility to the debt's maturity

$$(r - \sigma_A^2/2)T = (0.03 - 0.0242) \times 3 = 0.0174, \qquad \sigma_A\sqrt{T} = 0.22\sqrt{3} = 0.3811$$

$\Rightarrow$ The horizon is three years because that is when the debt is due.

Step 2 – Distance to default (d_2)

$$d_2 = \frac{\ln(900/700) + 0.0174}{0.3811} = \frac{0.2513 + 0.0174}{0.3811} = 0.7052$$

$\Rightarrow$ Assets sit about 0.71 three-year standard deviations above the face value of the debt.

Step 3 – Risk-neutral PD

$$N(-0.7052) = 24.0\%$$

Interpretation

The model has no one-year restriction: T is the time to repayment. The 11% expected return is a red herring for a risk-neutral PD, and the 55% equity volatility only calibrated σ_A.

Why the other options are wrong

A is incorrect. Used $\mu = 11\%$, the real-world drift: $(0.2513 + (0.11 - 0.0242) \times 3)/0.3811 = 1.3350$, $N(-1.3350) = 9.1\%$.

B is incorrect. Scope trap: used a one-year horizon although the debt is due in three: $(0.2513 + 0.0058)/0.22 = 1.1687$, $N(-1.1687) = 12.1\%$.

D is incorrect. Scaled volatility by T instead of $\sqrt{T}$: $0.2687/(0.22 \times 3) = 0.4071$, $N(-0.4071) = 34.2\%$.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1; Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8 and 9.8.3

Solution CR-5 e.6 **Answer: B**

Workbook A5.4 • S1, S6

Step 1 – Default point

$$\text{DP} = \text{STD} + 0.5 \times \text{LTD} = 900 + 0.5(1{,}000) = 1{,}400$$

$\Rightarrow$ KMV treats only half of long-term debt as part of the trigger, which "better approximates the actual loan obligations of a firm".

Step 2 – Numerator

$$\ln(2{,}400/1{,}400) + (0.04 - 0.18^2/2)(1) = 0.5390 + 0.0238 = 0.5628$$

Step 3 – Distance to default

$$\text{DD} = 0.5628/(0.18\sqrt{1}) = 3.13$$

Interpretation

Assets sit 3.13 standard deviations above the default point. KMV would then map this DD to an EDF through its empirical database of historical default frequencies, rather than reading $N(-3.13) = 0.09\%$ off the normal distribution. The short form $\ln(2{,}400/1{,}400)/0.18 = 2.99$ is close, but the question asks for the full form, so the drift stays in.

Why the other options are wrong

A is incorrect. Used total debt (1,900) as the default point: $(\ln(2{,}400/1{,}900) + 0.0238)/0.18 = 1.43$.

C is incorrect. Used d_1 instead of d_2, adding $\sigma_V^2/2$: $(0.5390 + 0.04 + 0.0162)/0.18 = 3.31$.

D is incorrect. Used short-term debt alone (900) as the default point: $(\ln(2{,}400/900) + 0.0238)/0.18 = 5.58$.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Doumpos et al., Ch.1, 5.6.3.1; Hull, RMFI 6e, Ch.17, 9.8.3

Solution CR-5 e.7 — Answer: A

Bank Q191 • AnalystPrep Q.6009

Step 1 – Merton's default point

L is usually taken as the short-term debt reported on the balance sheet; long-term liabilities do not enter.

Step 2 – The consequence

For a firm with large long-term debt the trigger understates its obligations, so the DD is too high and the PD too low. Moody's KMV addresses this with short-term debt plus half of long-term debt.

$\Rightarrow$ A is the limitation GARP's own description implies.

Interpretation

The other three options are true facts about the *wrong* model: a transition matrix (CreditMetrics), bond and CDS prices and a Poisson jump (reduced form).

Why the other options are wrong

B is incorrect. The rating transition matrix is the CreditMetrics input.

C is incorrect. Merton uses equity market data; bond prices and credit-derivative spreads are the inputs of reduced-form models.

D is incorrect. The exogenous Poisson jump is the reduced-form assumption; Merton ties default to asset value against debt.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1; SchweserNotes 2026 Book 2, Reading 23, LO 23.e

Solution CR-5 e.8 — Answer: A

Workbook A5.6 • S1, S6, S7

Step 1 – Short-form distance to default

$$\text{DD} \approx \frac{\ln(V/F)}{\sigma_V} = \frac{\ln(900/600)}{0.20} = \frac{0.4055}{0.20} = 2.03 \;\Rightarrow\; \text{band BB}$$

Step 2 – Full form check

$$\text{DD} = \frac{0.4055 + (0.04 - 0.20^2/2)(1)}{0.20} = \frac{0.4055 + 0.0200}{0.20} = 2.13 \;\Rightarrow\; \text{still BB}$$

$\Rightarrow$ The drift contributes $0.02/0.20 = 0.10$ standard deviations: small over one year, which is why the short form is the accepted shortcut. Both forms agree on the band, which is what the rating step needs: Merton is judged on its ranking of default risk (Hull 9.8.1).

Interpretation

A is correct: the shared first clause is true, and the second clause is the only one consistent with the readings.

Why the other options are wrong

B is incorrect. Scope/absolute-claim error. Hull defines distance to default for default "T years in the future", Doumpos sets T as the time up to repayment of the loan, and Hull's own end-of-chapter problem uses two-year debt. The drift scales with T and volatility with $\sqrt{T}$; nothing limits Merton to one year.

C is incorrect. Polarity flip. Higher asset volatility lowers DD: $\ln(1.5)/0.25 = 1.62$, which moves the borrower down to band B. As DD declines, the company becomes more likely to default.

D is incorrect. Sibling swap. Mapping through the normal distribution is the plain Merton/-Doumpos eq. 5.3 approach; Moody's KMV uses historical data to derive an empirical distribution of default frequencies.

LO: CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. CR-9 m: Calculate distance to default and default probability using the Merton model. CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. • **Reference:** Hull, RMFI 6e, Ch.17, 9.8.1 and 9.8.3; Doumpos et al., Ch.1, 5.6.3.1

CR-5 • Introduction to Credit Risk Modeling and Assessment • Vardeez: VERY IMPORTANT

CR-5 g

P1 CALC

Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan.

Workbook: A2 RAROC, Adjusted RAROC and Hurdle Rate • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: RAROC is A2 (Map #3, 9/16) and came up in the August 2026 sitting ("RAROC - 1; Hurdle rate - 1"). In Credit Risk the exam uses Doumpos's loan-level form; the hurdle-rate and adjusted-RAROC variants belong to ORR-18.
Vardeez: "LO 23.g for RAROC is imp, but Raroc is more imp in Ops risk" (ORR-18).

Reported exam shapes

- **A2·S1** Compute after-tax RAROC, in plain form and with red-herring inputs.

What the questions look like: numerical. You get a spread (or a loan rate and a deposit rate), fees, expected loss (or PD, LGD and EAD), operating and transfer costs, a tax rate, and capital at risk: a percentage of the loan, a duration-based value change, or an unexpected loan loss. Many end with "is the loan profitable?" against the cost of capital. Red herrings: the loan rate when the spread is given, a risk-free rate (it belongs to the ORR-18 form), regulatory capital, the hurdle when only RAROC is asked.

Worked elsewhere: Workbook A2.1 to A2.9 use the ORR-18 capital-budgeting form (return on risk capital, hurdle rate, adjusted RAROC) and are worked in the ORR book, not here. Section 5 sets the two forms side by side.

1. What RAROC measures

RAROC assesses the performance of a bank, a product line or a loan by comparing the **financial outcomes** (income, profits) to the **economic capital** required to obtain them. For a loan, GARP (Doumpos 5.7) takes the outcome to be the revenue the loan generates net of expenses and expected losses, and the capital to be the **capital at risk**: the capital the bank would need to cover its exposure in case of default.

$$\boxed{\text{RAROC} = \frac{\text{Loan revenues}}{\text{Capital at risk}}}$$

Decision rule (GARP): the loan is profitable if its RAROC exceeds the bank's cost of capital. GARP's worked example phrases the rule as "if the bank's own interest rate is lower than 7.47%"; read that rate as the bank's cost of capital (its hurdle), never the loan's interest rate. Vardeez's XYZ example shows why the distinction matters: its RAROC of 9.61% is below the loan's 10.5% rate, so the two readings give opposite verdicts.

2. The numerator: loan revenues

$$\boxed{\text{Loan revenues} = L \times (s + f - l - c)(1 - x)}$$

- s: the spread between the loan rate and the bank's funding cost. GARP's list says "cost of capital", but its example says "cost of funds", and every exam stem uses the funding (deposit) rate: $s =$ loan rate $-$ deposit rate.
- f: fees and commissions attributable to the loan.

- l: expected loan losses. When the stem gives components, $l = \text{PD} \times \text{LGD}$ as a share of EAD (CR-4 c).
- c: operating costs. Other transfer costs and one-off transaction costs join c.
- x: the tax rate. The factor $(1 - x)$ multiplies the *whole* bracket.

Put every item on one basis (all as a percentage of the loan, or all in currency) and on one year.

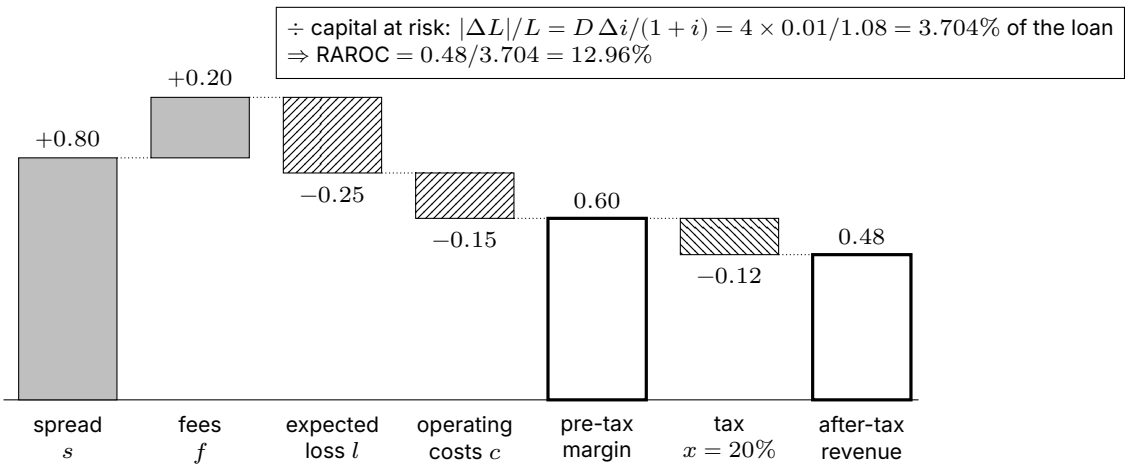


Figure 1. RAROC build-up for the loan of Question CR-5 g.3, in percent of the loan. Revenue items are grey bars, deductions are hatched, and the results are white bars with heavy outlines. The CR-5 form has no "return on capital" bar; the ORR-18 form adds one (Section 5).

3. Capital at risk, method 1: the duration (market-based) approach

Capital at risk is the fall in the loan's value over the period (usually one year) caused by a change in interest rates, with duration as the sensitivity:

$$\boxed{\Delta L \approx -L\,D\,\frac{\Delta i}{1+i}}$$

L is the loan's value, D its duration, i its current interest rate and Δi the expected rate change. Divide by $1 + i$, not by i, and use $|\Delta L|$ as the denominator.

- **GARP's example.** $L =$ EUR 500,000, $D = 3$, $i = 12\%$, $\Delta i = 2\%$: $\Delta L = -500{,}000 \times 3 \times 0.02/1.12 = -26{,}786$. Commission 0.1% plus spread 0.3% earn $500{,}000 \times 0.004 = 2{,}000$ (no losses, costs or tax given), so RAROC $= 2{,}000/26{,}786 = 7.47\%$.
- **Vardeez and Schweser (XYZ Bank).** $L =$ USD 1,000,000, $f = 0.1\%$, $s = 0.65\%$, $c = 0.2\%$, $l = 0$, $x = 11.1\%$: revenues $= 1{,}000{,}000 \times 0.0055 \times 0.889 = 4{,}889.50$. $D = 3.75$, $i = 10.5\%$, $\Delta i = 1.5\%$: $\Delta L = -1{,}000{,}000 \times 3.75 \times 0.015/1.105 = -50{,}904.98$. RAROC $= 4{,}889.50/50{,}904.98 = 9.61\%$.

4. Capital at risk, method 2: the unexpected loan loss

When default prediction models are available, the denominator can be the unexpected loan loss, built from historical default rates rather than market rates:

$$\boxed{\text{Unexpected loan loss} = \alpha \times \text{LGD} \times \text{EAD}}$$

α is the unexpected default rate, read from the distribution of historical default rates on similar loans. If default rates are normal, $\alpha = 2.6\sigma$ at 99.5% confidence (σ is the standard deviation of historical default rates; 2.6 rounds the one-tailed 2.576). Loan loss distributions are usually skewed, so banks often use 5 to 6 standard deviations.

Worked example. EAD USD 25 million, LGD 45%, spread 0.90%, fee 0.20%, expected loss 0.30%, operating costs 0.25%, tax 30%, $\sigma = 1.2\%$, cost of capital 12%.

- Revenue: $25\text{m} \times (0.90 + 0.20 - 0.30 - 0.25)\% \times 0.70 = 25\text{m} \times 0.385\% =$ USD 96,250.
- Normal, $\alpha = 2.6\sigma = 3.12\%$: capital $= 0.0312 \times 0.45 \times 25\text{m} =$ USD 351,000, RAROC $= 27.4\%$: accept.
- Skewed, $\alpha = 6\sigma = 7.2\%$: capital = USD 810,000, RAROC $= 11.9\%$: reject at 12%.

The choice of α alone flips the decision, which is why GARP's comment on skewness matters.

5. Reading the result, and the two RAROCs

Change (all else equal)	Enters	RAROC
Higher spread s or fees f	numerator, added	rises
Higher expected loss l, costs c or tax x	numerator, deducted	falls
Longer duration D or larger expected rate rise Δi	duration capital	falls
Higher α (more skew, higher confidence), higher LGD	UL capital	falls

	CR-5 g (Doumpos, loan)	ORR-18 (Crouhy, Galai and Mark)
Numerator	$(s + f - l - c)(1 - x) \times L$	(revenues − costs − EL − taxes + return on risk capital ± transfers)
Return on capital	**none**	$r_f \times$ economic capital, added
Denominator	Capital at risk: $\lvert\Delta L\rvert$ from duration, or $\alpha \times \text{LGD} \times \text{EAD}$	Economic capital: risk capital plus strategic risk capital
Benchmark	Bank's cost of capital	Hurdle rate (after-tax cost of equity); ARAROC against r_f

Keep the two apart: adding a return on capital to a CR-5 loan, or dropping it from an ORR-18 business unit, each moves the answer by one option.

Correction to the source notes. SchweserNotes (LO 23.g) describes the numerator as "the revenue (i.e., interest and principal repayments) coming from the loan". Principal repayment is the return of the amount lent, not income; the numerator is the after-tax spread and fees net of expected losses and operating costs, $L(s + f - l - c)(1 - x)$. Schweser's summary also states the goal as RAROC above "the bank's own interest rate"; GARP's rule, used here, compares RAROC with the bank's cost of capital.

Formula card: CR-5 g

RAROC	loan revenues ÷ capital at risk
Loan revenues	$L\,(s + f - l - c)(1 - x)$
Spread	s = loan rate − cost of funds (deposit rate); transfer costs go into c
Expected loss	l = PD × LGD (share of EAD), or as given
Capital at risk: duration	$\lvert\Delta L\rvert = L\,D\,\Delta i/(1 + i)$
Capital at risk: UL	$\alpha \times$ LGD $\times$ EAD; $\alpha = 2.6\sigma$ (normal, 99.5%) or 5σ to 6σ (skewed)
Capital as % of the loan	capital $= k \times L$, so RAROC $= (s + f - l - c)(1 - x)/k$
Decision	profitable if RAROC $>$ bank's cost of capital

Exam traps

- **Tax.** Multiply by $(1 - x)$, not by x, and apply it to the whole bracket. A pre-tax RAROC is always one of the options.
- **The $1 + i$ divisor.** Dropping it, or multiplying by it, misstates the duration capital.
- **Loan rate for spread.** When a deposit rate is given, s is the difference, not the loan rate.
- **Wrong denominator.** The loan amount, regulatory capital, or $\alpha\times$EAD without LGD; or σ in place of α.
- **Dropped items.** Fees, expected loss or a transfer cost left out. EL is deducted even though no capital is held for it.
- **ORR-18 contamination.** A risk-free rate in a CR-5 loan stem is a red herring: there is no return-on-capital term.
- **Wrong benchmark.** Compare RAROC with the bank's cost of capital, not the loan's interest rate.
- **Mixed units.** Operating costs in currency against fees in percent: convert before you add.

Practice questions: CR-5 g

Question CR-5 g.1

Bank Q213 • AnalystPrep Q.6012

A senior analyst stresses to a junior colleague that the denominator of the risk-adjusted return on capital (RAROC) ratio must be measured carefully. Which statement best describes the RAROC denominator for a loan?

A. The loan amount, which represents the revenue the loan can generate.

B. The capital at risk, that is, the capital the bank needs to cover the risk of default or loss in value on the loan.

C. The bank's cost of capital, which is the benchmark for judging the loan's return.

D. The interest rate spread between the loan rate and the bank's cost of funds.

Question CR-5 g.2

Bank Q209 • Schweser Module Quiz 23.2 Q4 (rebuilt)

ABC Bank has a loan with a value of USD 750,000. The commission is 0.15%, the spread between the loan's interest rate and the bank's cost of funds is 0.45%, operating costs are 0.3%, and the tax rate is 12%. The loan is expected to perform with no risk of default. It has a duration of 2.85 and an interest rate of 9.5%, and interest rates are expected to increase by 1.25%. Using the duration approach for capital at risk, which of the following is closest to the loan's RAROC?

A. 6.77%

B. 7.41%

C. 8.11%

D. 9.22%

Question CR-5 g.3

Bank Q224 • Mock 1 Q35 (rebuilt)

A credit analyst is evaluating a USD 2,000,000 loan with the following characteristics: spread between the loan rate and the bank's cost of funds 0.80%; fees attributable to the loan 0.20%; expected loan losses 0.25%; operating costs 0.15%; tax rate 20%. The loan has a duration of 4.0 years and an interest rate of 8%, and interest rates are expected to rise by 1.0 percentage point over the one-year horizon. Using the change in loan value from the duration approximation as capital at risk, which of the following is closest to the loan's RAROC?

A. 12.00%

B. 12.96%

C. 16.20%

D. 18.36%

Question CR-5 g.4

Bank Q218 • GARP PE1 Q73 • Mock 3 Q8 (rebuilt)

A regional commercial bank is considering a 1-year loan fully funded by deposits:

Loan amount	JPY 4.2 billion
Interest rate paid on deposits	0.4%
Interest rate received on loan	3.2%
Expected loss	2.0% of face value
Annual operating costs	0.5% of face value
Capital at risk (economic capital)	10.0% of the loan
Effective tax rate	38%
Other transfer costs	none

Using loan revenues $= (s + f - l - c)(1 - x)$, which of the following is closest to the after-tax RAROC?

A. 0.19%

B. 1.86%

C. 3.00%

D. 14.26%

Question CR-5 g.5

Bank Q219 • Mock 4 Q18 (rebuilt)

A regional commercial bank is considering a loan fully funded by deposits:

Loan amount	CNY 3.8 billion
Interest rate paid on deposits	0.6%
Interest rate received on loan	4.1%
Origination fee	0.5% of face value
Expected loss	3.0% of face value
Annual operating costs	0.3% of face value
Capital at risk (economic capital)	15.0% of the loan
Effective tax rate	38%

Which of the following is closest to the after-tax RAROC for this loan?

A. 0.43%

B. 0.83%

C. 2.89%

D. 4.67%

Question CR-5 g.6

Bank Q225 • Mock 1 Q35 (rebuilt)

A bank is evaluating a loan with exposure at default of USD 10,000,000 and loss given default of 40%. The spread over the bank's cost of funds is 0.60%, fees are 0.25%, expected loan losses are 0.20%, operating costs are 0.15% and the tax rate is 25%. The bank measures capital at risk as the unexpected loan loss, using an unexpected default rate $\alpha = 5\%$ derived from historical default rates on similar loans. Which of the following is closest to the loan's RAROC?

A. 7.50%

B. 18.75%

C. 25.00%

D. 26.25%

Question CR-5 g.7

Bank Q210 • AP Mock 2 Q38 (rebuilt)

A bank is reviewing a small-business lending division with a EUR 150 million loan portfolio (EAD equal to the portfolio value, LGD 60%). Spread income is 5.5% of the portfolio, fees are 0.5%, expected losses are PD $\times$ LGD $\times$ EAD with PD of 7%, operating costs are EUR 1.25 million, and the tax rate is 25%. Capital at risk is the unexpected loss $\alpha \times$ LGD $\times$ EAD, where $\alpha = 5\sigma$ and the standard deviation of historical default rates is $\sigma = 1.5\%$. The bank's cost of capital is 12%. What should management do?

A. Close the division; RAROC is 7.78%, below 12%.

B. Close the division; RAROC is 9.67%, below 12%.

C. Keep the division; RAROC is 16.11%, above 12%.

D. Keep the division; RAROC is 21.48%, above 12%.

Question CR-5 g.8

Bank Q216 • BT P2.T7.24.11.3 (rebuilt)

A bank originates USD 100 million of loans. Per unit of loan value, expected revenue (spread plus fees) is 7%, expected loss is 3%, operating costs are 2.5% and the tax rate is 30%. Capital at risk is 40% of the loans. At year-end, operating costs turned out to be 3.5%, with every other item as expected. The bank's cost of capital is 2%. Which of the following correctly gives the projected RAROC, the actual RAROC, and the conclusion?

A. Projected 1.05%, actual 0.35%; the loans failed the hurdle in both cases.

B. Projected 2.63%, actual 0.88%; the loans met the hurdle at origination but not in actuality.

C. Projected 3.75%, actual 1.25%; the loans met the hurdle at origination but not in actuality.

D. Projected 7.88%, actual 6.13%; the loans met the hurdle in both cases.

Question CR-5 g.9 New • built from GARP Ch.5, 5.7 • shape A2·S1 (red-herring inputs)

A credit analyst at a German bank is preparing a RAROC note on a proposed EUR 80 million one-year term loan to a mid-sized manufacturer. The bank measures capital at risk as the unexpected loan loss. The analyst has collected the following:

Spread over the bank's cost of funds	1.50%
Arrangement fee	0.30%
Probability of default; loss given default	1.0%; 40%
Exposure at default	equal to the loan amount
Operating costs	EUR 0.28 million per year
Standard deviation of historical default rates on similar loans	2.0%
Unexpected default rate used by the bank (losses are skewed)	$\alpha = 6\sigma$
Tax rate	25%
Regulatory capital requirement for the loan	EUR 6.4 million
Risk-free rate; bank's cost of capital	3.0%; 12%

Which of the following is closest to the loan's RAROC?

A. 6.56%

B. 9.84%

C. 16.41%

D. 18.66%

Solutions • CR-5 g

Solution CR-5 g.1 — Answer: B

Bank Q213 • AnalystPrep Q.6012

Step 1 – The ratio

RAROC = loan revenues ÷ capital at risk.

Step 2 – What sits where

Numerator: spread, fees, expected losses, operating costs and tax. Denominator: capital at risk, either the duration-based change in loan value $L\,D\,\Delta i/(1+i)$ or the unexpected loss $\alpha \times \text{LGD} \times \text{EAD}$.

$\Rightarrow$ The denominator is the capital the bank must hold against the loan's risk.

Interpretation

The bank's cost of capital is not in the ratio at all: it is the benchmark RAROC is compared with.

Why the other options are wrong

A is incorrect. The loan amount is exposure, not capital at risk; dividing by it gives a margin, not a return on capital.

C is incorrect. The cost of capital is the hurdle for the decision, not a component of the ratio.

D is incorrect. The spread is one component of the numerator.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7 Measuring the Financial Performance of Loans

Solution CR-5 g.2 — Answer: C

Bank Q209 • Schweser Module Quiz 23.2 Q4

Step 1 – Loan revenues

$$750{,}000 \times (0.0045 + 0.0015 - 0 - 0.003) \times (1 - 0.12) = 750{,}000 \times 0.003 \times 0.88 = \text{USD } 1{,}980$$

$\Rightarrow$ A performing loan with no risk of default: $l = 0$.

Step 2 – Capital at risk (duration approach)

$$|\Delta L| = 750{,}000 \times 2.85 \times \frac{0.0125}{1.095} = \text{USD } 24{,}400.68$$

Step 3 – RAROC

$$\text{RAROC} = 1{,}980/24{,}400.68 = 8.11\%$$

Interpretation

The same structure as Vardeez's XYZ example. Whether the loan is profitable depends on the bank's cost of capital, which the stem does not give.

Why the other options are wrong

A is incorrect. Multiplied by $1+i$ instead of dividing: $750{,}000 \times 2.85 \times 0.0125 \times 1.095 = 29{,}257$, so $1{,}980/29{,}257 = 6.77\%$.

B is incorrect. Dropped the $1 + i$ divisor: $1{,}980/26{,}718.75 = 7.41\%$.

D is incorrect. Forgot the tax: $2{,}250/24{,}400.68 = 9.22\%$.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7; SchweserNotes 2026 Book 2, Reading 23, LO 23.g

Solution CR-5 g.3

Answer: B

Bank Q224 • Mock 1 Q35

Step 1 – After-tax revenue rate

$$(0.80\% + 0.20\% - 0.25\% - 0.15\%)(1 - 0.20) = 0.60\% \times 0.80 = 0.48\%$$

$\Rightarrow$ Revenue $= 0.0048 \times 2{,}000{,}000 =$ USD 9,600.

Step 2 – Capital at risk

$$|\Delta L| = 2{,}000{,}000 \times 4.0 \times \frac{0.01}{1.08} = \text{USD } 74{,}074$$

Step 3 – RAROC

$$\text{RAROC} = 9{,}600/74{,}074 = 12.96\%$$

Interpretation

Figure 1 draws the build-up. In percent of the loan: 0.48% after-tax revenue on 3.704% capital at risk.

Why the other options are wrong

A is incorrect. Dropped the $1 + i$ divisor: $9{,}600/80{,}000 = 12.00\%$.
C is incorrect. Omitted the tax: $12{,}000/74{,}074 = 16.20\%$.
D is incorrect. Left out expected loan losses: $(1.00\% - 0.15\%) \times 0.80 \times 2{,}000{,}000 = 13{,}600$, and $13{,}600/74{,}074 = 18.36\%$.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7

Solution CR-5 g.4

Answer: B

Bank Q218 • GARP PE1 Q73

Step 1 – Spread over funding

$s = 3.2\% - 0.4\% = 2.8\%$; no fees.

Step 2 – After-tax margin

$$(2.8\% - 2.0\% - 0.5\%)(1 - 0.38) = 0.3\% \times 0.62 = 0.186\%$$

$\Rightarrow$ Revenue $= 0.186\% \times 4{,}200 =$ JPY 7.812 million.

Step 3 – RAROC on capital at risk

$$\text{RAROC} = 7.812/420 = 1.86\%$$

Interpretation

Expected loss and operating costs absorb most of the 2.8% spread; the thin after-tax remainder earns only 1.86% on the capital it ties up.

Why the other options are wrong

A is incorrect. Divided by the loan (4,200) instead of the capital at risk (420): $7.812/4{,}200 = 0.19\%$.
C is incorrect. Ignored tax: $0.3\%/10\% = 3.00\%$.
D is incorrect. Omitted expected loss: $(2.8\% - 0.5\%) \times 0.62/10\% = 14.26\%$.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7

Solution CR-5 g.5

Answer: C

Bank Q219 • Mock 4 Q18

Step 1 – Spread and fee

$s = 4.1\% - 0.6\% = 3.5\%$; $f = 0.5\%$.

Step 2 – After-tax margin

$$(3.5\% + 0.5\% - 3.0\% - 0.3\%)(1 - 0.38) = 0.7\% \times 0.62 = 0.434\%$$

$\Rightarrow$ Revenue $= 0.434\% \times 3{,}800 =$ CNY 16.492 million.

Step 3 – RAROC

$$\text{RAROC} = 16.492/570 = 2.89\%$$

Interpretation

With capital stated as a share of the loan, RAROC is simply the after-tax margin divided by that share: $0.434\%/15\%$.

Why the other options are wrong

A is incorrect. Divided by the full loan instead of capital at risk: 0.43%.
B is incorrect. Left out the 0.5% fee: $0.2\% \times 0.62/15\% = 0.83\%$.
D is incorrect. Ignored tax: $0.7\%/15\% = 4.67\%$.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7

Solution CR-5 g.6

Answer: B

Bank Q225 • Mock 1 Q35

Step 1 – Loan revenue

$$(0.60\% + 0.25\% - 0.20\% - 0.15\%)(1 - 0.25) = 0.50\% \times 0.75 = 0.375\%$$

$\Rightarrow$ Revenue $= 0.00375 \times 10{,}000{,}000 =$ USD 37,500.

Step 2 – Capital at risk (unexpected loan loss)

$$\alpha \times \text{LGD} \times \text{EAD} = 0.05 \times 0.40 \times 10{,}000{,}000 = \text{USD } 200{,}000$$

Step 3 – RAROC

$$\text{RAROC} = 37{,}500/200{,}000 = 18.75\%$$

Interpretation

Here α is given directly as an unexpected default rate (5%); when the stem gives σ instead, build α as a multiple of it (2.6, or 5 to 6).

Why the other options are wrong

A is incorrect. Left LGD out of capital at risk: $37{,}500/500{,}000 = 7.50\%$.
C is incorrect. Ignored tax: $50{,}000/200{,}000 = 25.00\%$.

D is incorrect. Omitted expected loan losses: $(0.85\% - 0.15\%) \times 0.75 \times 10{,}000{,}000 = 52{,}500$, so 26.25%.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7

Solution CR-5 g.7 — Answer: C

Bank Q210 • AP Mock 2 Q38

Step 1 – Revenue items (EUR million)

Spread $0.055 \times 150 = 8.25$; fees $0.005 \times 150 = 0.75$; expected loss $0.07 \times 0.60 \times 150 = 6.30$; operating costs 1.25.

Step 2 – After-tax revenue

$$(8.25 + 0.75 - 6.30 - 1.25)(1 - 0.25) = 1.45 \times 0.75 = 1.0875$$

Step 3 – Capital at risk

$\alpha = 5 \times 1.5\% = 7.5\%$, so capital $= 0.075 \times 0.60 \times 150 = 6.75$.

Step 4 – RAROC and decision

$$\text{RAROC} = 1.0875/6.75 = 16.11\% > 12\% \;\Rightarrow\; \text{keep the division}$$

Interpretation

A 7% PD makes expected loss the biggest deduction, yet the division still clears the hurdle. $\alpha = 5\sigma$ reflects GARP's point that skewed loss distributions call for 5 to 6 standard deviations rather than 2.6.

Why the other options are wrong

A is incorrect. Omitted the fee income: $(8.25 - 6.30 - 1.25) \times 0.75/6.75 = 7.78\%$, wrongly below the hurdle.

B is incorrect. Left LGD out of capital at risk $(0.075 \times 150 = 11.25)$: $1.0875/11.25 = 9.67\%$, wrongly below the hurdle.

D is incorrect. Ignored tax: $1.45/6.75 = 21.48\%$; the decision is right but the RAROC is wrong.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7

Solution CR-5 g.8 — Answer: B

Bank Q216 • BT P2.T7.24.11.3

Step 1 – Projected RAROC

$$(7\% - 3\% - 2.5\%)(1 - 0.30) = 1.05\% \;\Rightarrow\; 1.05/40 = 2.63\%$$

Step 2 – Actual RAROC

$$(7\% - 3\% - 3.5\%)(1 - 0.30) = 0.35\% \;\Rightarrow\; 0.35/40 = 0.88\%$$

Step 3 – Compare with the 2% cost of capital

$2.63\% > 2\%$ at origination; $0.88\% < 2\%$ in actuality.

$\Rightarrow$ A one-point rise in operating costs cuts RAROC by $1\% \times 0.70/40\% = 1.75$ points.

Interpretation

RAROC works ex ante for the lending decision and ex post for performance measurement; the same formula on realised costs reverses the verdict here.

Why the other options are wrong

A is incorrect. Divided by the loans (100) instead of capital at risk (40): 1.05% and 0.35%, both below 2%.
C is incorrect. Forgot the 30% tax: $1.5/40 = 3.75\%$ and $0.5/40 = 1.25\%$.
D is incorrect. Omitted expected loss: $4.5 \times 0.7/40 = 7.88\%$ and $3.5 \times 0.7/40 = 6.13\%$.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7

Solution CR-5 g.9 Answer: C

New • built from GARP Ch.5, 5.7

Step 1 – Revenue items (EUR million)

Spread $0.015 \times 80 = 1.20$; fee $0.003 \times 80 = 0.24$; expected loss PD $\times$ LGD $\times$ EAD $= 0.01 \times 0.40 \times 80 = 0.32$; operating costs 0.28.

Step 2 – After-tax revenue

$$(1.20 + 0.24 - 0.32 - 0.28)(1 - 0.25) = 0.84 \times 0.75 = 0.63$$

Step 3 – Capital at risk

$\alpha = 6 \times 2.0\% = 12\%$, so capital $= 0.12 \times 0.40 \times 80 = 3.84$.

Step 4 – RAROC

$$\text{RAROC} = 0.63/3.84 = 16.41\%$$

$\Rightarrow$ Above the 12% cost of capital, so the loan is profitable, but the question asked only for RAROC.

Interpretation

Three inputs are red herrings. Regulatory capital is not the bank's capital at risk; the risk-free rate belongs to the ORR-18 form, which adds a return on capital that the CR-5 loan form does not have; the cost of capital is the benchmark, not an input.

Why the other options are wrong

A is incorrect. Left LGD out of capital at risk: $0.12 \times 80 = 9.6$, so $0.63/9.6 = 6.56\%$.
B is incorrect. Used regulatory capital as the denominator: $0.63/6.4 = 9.84\%$.
D is incorrect. Added a return on capital as in ORR-18: $(0.84 + 0.03 \times 3.84) \times 0.75/3.84 = 18.66\%$.

LO: CR-5 g: Apply risk-adjusted return on capital (RAROC) to measure the performance of a loan. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.7

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 b

P1 CALC

Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality.

Workbook: A10 Altman Z-Score and Linear Discriminant Analysis • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: "Altman z score, direct plug value question" came up in the August 2026 sitting (A10), and Map #15 has it from 6 of 16 sources, once as "calculate z score (LDA)".

Vardeez: CR-9 is the most important chapter in Credit Risk. The notes frame the Z-score as the output of linear discriminant analysis and work GARP's Example 9.1.

Reported exam shapes

- **A10·S1** Direct plug: compute Z from a set of filings.
- **A10·S2** The five ratios and their weights recalled closed-book, none given in the stem.
- **A10·S3** Cut-off zones used to classify a firm.
- **A10·S4** "Calculate z score (LDA)": Z as Altman's linear discriminant function assigning a firm to a group.
- **A10·S5** The two LDAs: linear discriminant analysis for scoring versus the loss distribution approach for operational capital.

What the questions look like: almost all numerical. You get balance sheet and income statement extracts, usually with inputs that do not belong in the formula (book equity, net income, current assets on their own), and must compute Z, name the zone, or reverse-solve the input that moves a firm across a boundary. Expect the weights not to be given.

1. The methodology: discriminant analysis on accounting ratios

Edward Altman (1968) pioneered the use of accounting ratios to predict default. He used a statistical technique known as **discriminant analysis**: find the linear combination of a firm's ratios that best separates two groups, here companies that went on to fail and companies that did not. The result is a **scoring function**, a fixed-weight sum of the ratios, and **cut-off values** of the score assign each firm to a group. Vardeez's summary: the variables are chosen for their contribution to the likelihood of default from a large pool of accounting ratios and qualitative features, and the score sorts firms into solvent and insolvent before the insolvency happens.

- **Estimation sample.** 66 publicly traded manufacturing companies: 33 failed within one year and 33 did not fail within one year.
- **Out-of-sample accuracy.** The model "proved to be fairly accurate" on firms outside the estimation sample. Both error types were small.
- **Type I error:** predicted *not* to go bankrupt, but did. **Type II error:** predicted to go bankrupt, but did not. For a commercial bank's lending department a Type I error is liable to be the more costly: it lends to a firm that then fails, rather than turning away a good borrower.
- **Scope.** The weights in eq. 9.1 are for *publicly traded manufacturing* companies. Variations exist for manufacturers that are not publicly traded and for non-manufacturers, and the method has been revised and extended since 1968.

- **Where it fits.** Scores of this kind are the raw material of internal ratings for firms with no agency rating (CR-9 a). They rest on backward-looking accounting data, so they react more slowly than equity-based models such as Merton's (CR-9 l to n).

2. The five ratios and their weights

$$\boxed{Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5} \qquad \text{(GARP eq. 9.1)}$$

Ratio	Definition	Weight	What it picks up
X_1	Working capital / Total assets	1.2	Liquidity; working capital = CA − CL, can be negative
X_2	Retained earnings / Total assets	1.4	Profits accumulated over the firm's life
X_3	EBIT / Total assets	3.3	Operating earning power of the assets
X_4	Market value of equity / Book value of total liabilities	0.6	Market cushion over what is owed
X_5	Sales / Total assets	0.999	Asset turnover

- X_4 is the only ratio whose denominator is not total assets, and the only one with a *market* value in the numerator. Book equity is a classic red herring.
- X_3 uses EBIT, not net income. EBIT carries the largest weight, so it is the strongest single lever on Z.
- Every weight is positive: a higher ratio raises Z, and a higher Z means a lower likelihood of default. Negative working capital or negative retained earnings pull Z down.
- Memory aid, in order WC, RE, EBIT, ME, S: **1.2, 1.4, 3.3, 0.6, 1.0** (0.999 rounds to 1.0).

3. Reading the score: the cut-off zones

GARP's guidelines for the original model (Schweser's labels in brackets):

Z-score	GARP interpretation	Schweser's label
Above 3.0	Unlikely to default	No default is likely
2.7 to 3.0	"On alert"	Potential default
1.8 to 2.7	Good chance of default	Reasonable probability of default
Below 1.8	Probability of "a financial embarrassment" very high	High likelihood of default

Vardeez adds that Altman's single discriminant threshold was $Z = 2.675$, the point that best split the failed and non-failed firms in his sample. It is not in GARP's reading; answer zone questions from the four-zone table.

4. Applying it: one firm, then a sample of firms

GARP Example 9.1. Working capital 170,000; total assets 670,000; EBIT 60,000; sales 2,200,000; market value of equity 380,000; total liabilities 240,000; retained earnings 300,000.

	X_1	X_2	X_3	X_4	X_5	Z
Ratio	0.254	0.448	0.0896	1.583	3.284	
Weight × ratio	0.305	0.627	0.296	0.950	3.281	**5.46**

$Z = 5.46 > 3.0$: the company is not in danger of defaulting in the near future. Note that X_5 alone contributes 3.28 of the 5.46: for a firm with high sales relative to its assets, asset turnover dominates the total.

Classifying a sample. The LO asks you to sort several firms by credit quality. Four listed manufacturers (USD million):

Firm	WC	RE	EBIT	MVE	TL	Sales	TA	$1.2X_1$	$1.4X_2$	$3.3X_3$	$0.6X_4$	$0.999X_5$	Z
North	120	150	90	900	400	1,100	800	0.180	0.263	0.371	1.350	1.374	**3.54**
East	80	120	60	520	400	900	700	0.137	0.240	0.283	0.780	1.284	**2.72**
Mid	40	60	35	300	450	700	600	0.080	0.140	0.193	0.400	1.166	**1.98**
South	−30	−20	10	120	520	450	500	−0.072	−0.056	0.066	0.138	0.899	**0.98**

Ranked from strongest to weakest: North (unlikely to default), East (on alert), Mid (good chance of default), South (financial embarrassment very likely). South's negative working capital and retained earnings enter with a minus sign, and its thin market cushion ($X_4 = 0.23$) leaves almost nothing but sales to hold the score up.

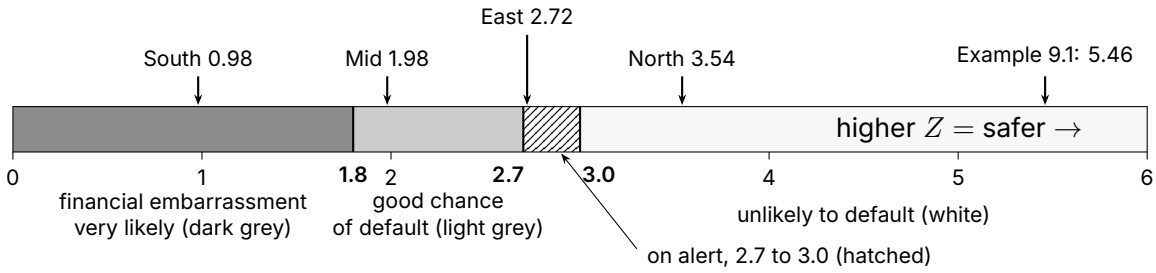


Figure 1. Altman's zones on the Z line: dark grey below 1.8, light grey from 1.8 to 2.7, hatched from 2.7 to 3.0, white above 3.0. The arrows mark the four firms classified above and GARP's Example 9.1. East sits just above the 2.7 boundary: a small input error moves it down a zone.

Reverse-solving one input. Required ratio = (target Z − contribution of the other four) ÷ its weight; then multiply by the ratio's denominator. For Mid to reach the 2.7 boundary through EBIT alone: X_3 must rise by $(2.7 - 1.978)/3.3 = 0.2188$, so EBIT must rise by $0.2188 \times 600 = 131$, from 35 to about 166. The large weight on EBIT is what makes this lever, and its reverse (Question CR-9 b.6), so sensitive.

5. The other "LDA" (shape S5)

In a credit-scoring stem, LDA is **linear discriminant analysis**: the method behind the Z-score, which assigns a firm to a group. In an operational risk stem, LDA is the **loss distribution approach**: a loss frequency distribution and a loss severity distribution are combined (convolution) into an annual loss distribution from which operational capital is read. The two share only the acronym. A recall of "calculate z score (LDA)" is the discriminant one.

Formula card: CR-9 b

Altman's Z (public manufacturers)	$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$
Ratios over total assets	X_1 working capital, X_2 retained earnings, X_3 EBIT, X_5 sales
The odd one out	X_4 = market value of equity / book value of total liabilities
Working capital	Current assets − current liabilities (can be negative)
Zones	> 3.0 unlikely to default; 2.7 to 3.0 on alert; 1.8 to 2.7 good chance of default; < 1.8 financial embarrassment very likely
Reverse-solve	Ratio needed $= (Z_{\text{target}} -$ other four terms$)/$weight
Errors	Type I: predicted to survive, failed (costlier for a lender). Type II: predicted to fail, survived

	Linear discriminant analysis	Loss distribution approach
Risk type	Credit risk: scoring and classifying borrowers	Operational risk: capital
Inputs	Accounting ratios of each firm	Loss frequency and loss severity data
Output	A score Z and a group (zone)	An annual loss distribution and its high quantile
Typical stem	"Calculate the Z-score", "classify the firm"	"Combine frequency and severity", "op risk capital"

Exam traps

- **Wrong input in X_4.** Book value of equity instead of market value, or total assets instead of total liabilities as the denominator.
- **Wrong input in X_3.** Net income instead of EBIT.
- **Working capital.** Current assets used on their own, or the minus sign dropped when working capital is negative.
- **Coefficient swap.** 1.4 belongs to retained earnings, 3.3 to EBIT; 0.6 to the equity ratio, 0.999 to sales. A swap moves Z by enough to change the zone.
- **A term left out.** Dropping $0.999X_5$ or $0.6X_4$ can cut Z by more than a full zone.
- **Zone polarity and boundaries.** High Z is safe. 1.8 to 2.7 is "good chance of default"; 2.7 to 3.0 is "on alert".
- **Reverse-solve slips.** Using the wrong weight for the ratio being solved, or targeting the wrong boundary (2.7 or 1.8 when the stem says the firm must stay above 3.0).
- **Stress percentages.** A 40% cut leaves 60% of the item, not 40%.
- **Acronym swap.** LDA in a scoring stem is discriminant analysis, not the operational risk loss distribution approach.

Practice questions: CR-9 b

Question CR-9 b.1

Bank Q317 • BT P2.T6.24.13.2 • GARP book Example 9.1 (rebuilt)

A publicly traded manufacturer reports: working capital 170,000; retained earnings 300,000; EBIT 60,000; sales 2,200,000; market value of equity 380,000; book value of total liabilities 240,000; total assets 670,000. Using Altman's original Z-score for public manufacturers, $Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$, where $X_1 =$ working capital/total assets, $X_2 =$ retained earnings/total assets, $X_3 =$ EBIT/total assets, $X_4 =$ market value of equity/book value of total liabilities and $X_5 =$ sales/total assets, which of the following gives the Z-score and the resulting credit classification?

A. Z of about 1.23; probability of financial embarrassment very high (below 1.8)

B. Z of about 2.18; good chance of default (1.8 to 2.7)

C. Z of about 4.51; unlikely to default (above 3.0)

D. Z of about 5.46; unlikely to default (above 3.0)

Question CR-9 b.2

Bank Q318 • Schweser Module Quiz 27.1 Q1 (rebuilt)

A credit analyst is classifying Nielsen Corp., a publicly traded manufacturer, with Altman's original Z-score. Nielsen reports: working capital 525,000; retained earnings 1,120,000; EBIT 480,000; market value of equity 4,215,000; book value of total liabilities 1,850,000; sales 1,760,000; total assets 5,100,000. The analyst's guidelines are: $Z > 3.0$ no default is likely; 2.7 to 3.0 potential default (on alert); 1.8 to 2.7 reasonable probability of default; $Z < 1.8$ high likelihood of default. What is the most appropriate classification of Nielsen?

A. No default is likely.

B. Potential default.

C. Reasonable probability of default.

D. High likelihood of default.

Question CR-9 b.3

Workbook A10.1 • S1, S2 • Aug-2026 • Map #15

A junior credit analyst at a Canadian commercial bank is preparing an internal credit memo on a listed auto-parts manufacturer. Extracts from its latest filings (CAD million):

Current assets	180	Total assets	600
Current liabilities	126	Total liabilities	390
Retained earnings	96	Book value of equity	210
EBIT	42	Market value of equity	330
Sales	690		

Which of the following is closest to the firm's Altman Z-score?

A. 2.03

B. 2.10

C. 2.22

D. 2.47

Question CR-9 b.4

Workbook A10.2 • S2, S3, S4, S5 • Map #15 • Map #74

A credit risk consultant is validating an Australian lender's scoring engine. The engine applies Altman's original linear discriminant function: the five accounting ratios are combined with Altman's fixed discriminant weights into a single score Z, and the cut-off values of Z assign each listed manufacturer to a credit-quality group. The consultant re-scores one listed packaging manufacturer (AUD million):

- Total assets 800; current assets 250; current liabilities 280
- Retained earnings 40; EBIT 40; net income 8
- Market value of equity 330; book value of equity 240; total liabilities 560; sales 1,040

Which of the following correctly states the discriminant score and the group to which the firm is assigned?

A. 1.71; probability of financial embarrassment very high

B. 1.75; probability of financial embarrassment very high

C. 1.84; on alert

D. 1.84; good chance of default

Question CR-9 b.5

Bank Q319 • BT P2.T6.704.3 (rebuilt)

A credit risk manager stress-tests a publicly traded manufacturer with Altman's original Z-score. The firm has total assets 1,000; working capital 200; retained earnings 300; EBIT 100; market value of equity 800; book value of total liabilities 500; sales 1,500. The stress scenario cuts sales by 40%, EBIT by 50% and the market value of equity by 25%; no other accounts change. Using the guidelines $Z > 3.0$ unlikely to default, 2.7 to 3.0 on alert, 1.8 to 2.7 good chance of default and $Z < 1.8$ financial embarrassment very likely, which of the following gives the stressed Z-score and classification?

A. Z of about 1.66; financial embarrassment very likely

B. Z of about 2.44; good chance of default

C. Z of about 2.68; good chance of default

D. Z of about 3.28; still unlikely to default

Question CR-9 b.6

Workbook A10.3 • S2, S3 • Map #15

The treasurer of a listed Indian textile manufacturer is preparing for a refinancing. The firm's current Altman Z-score places it in the zone in which a company is considered unlikely to default. Holding all other items fixed (INR crore):

- Total assets 400; working capital 60; retained earnings 48; current EBIT 170
- Market value of equity 210; total liabilities 280; sales 380

Which of the following is closest to the largest fall in EBIT the firm can absorb while remaining in that zone?

A. INR 18.1 crore

B. INR 42.7 crore

C. INR 54.5 crore

D. INR 163.6 crore

Solutions • CR-9 b

Solution CR-9 b.1

Answer: D

Bank Q317 • BT P2.T6.24.13.2 • GARP book Example 9.1

Step 1 – The five ratios

$$X_1 = \frac{170}{670} = 0.254, \quad X_2 = \frac{300}{670} = 0.448, \quad X_3 = \frac{60}{670} = 0.0896$$
$$X_4 = \frac{380}{240} = 1.583, \quad X_5 = \frac{2{,}200}{670} = 3.284$$

Step 2 – Weighted terms

$$1.2(0.254) = 0.305, \quad 1.4(0.448) = 0.627, \quad 3.3(0.0896) = 0.296$$
$$0.6(1.583) = 0.950, \quad 0.999(3.284) = 3.281$$

Step 3 – Sum and zone

$$Z = 0.305 + 0.627 + 0.296 + 0.950 + 3.281 = 5.46 > 3.0$$

$\Rightarrow$ Unlikely to default: the firm is not in danger in the near future.

Interpretation

This is GARP's own example. Sales turnover (3.28) and the market cushion over liabilities (0.95) carry the score; take either out and the number falls sharply even if the zone does not change.

Why the other options are wrong

A is incorrect. Kept only the first three terms and dropped both $0.6X_4$ and $0.999X_5$: $0.305 + 0.627 + 0.296 = 1.23$.

B is incorrect. Dropped the sales term $0.999X_5$: $5.46 - 3.28 = 2.18$, which lands in the wrong zone.

C is incorrect. Dropped the equity term $0.6X_4$: $5.46 - 0.95 = 4.51$. The zone happens to be right, but the score is wrong.

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.2 Altman's Z-Score, eq. 9.1, Example 9.1

Solution CR-9 b.2

Answer: C

Bank Q318 • Schweser Module Quiz 27.1 Q1

Step 1 – Recall the ratios and weights (not given)

$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$ with WC, RE, EBIT and sales over total assets and X_4 = market equity over book liabilities.

Step 2 – The ratios

$$X_1 = \frac{525}{5{,}100} = 0.1029, \quad X_2 = \frac{1{,}120}{5{,}100} = 0.2196, \quad X_3 = \frac{480}{5{,}100} = 0.0941$$
$$X_4 = \frac{4{,}215}{1{,}850} = 2.2784, \quad X_5 = \frac{1{,}760}{5{,}100} = 0.3451$$

Step 3 – Weighted sum

$$Z = 0.1235 + 0.3075 + 0.3106 + 1.3670 + 0.3448 = 2.4534$$

Step 4 – Classify

$1.8 < 2.45 < 2.7$: reasonable probability of default (GARP: "good chance of default").

Interpretation

Nielsen's large market cushion ($X_4 = 2.28$) is not enough: its sales are low relative to its asset base ($X_5 = 0.35$), so the score sits in the second-worst zone.

Why the other options are wrong

A is incorrect. "No default is likely" needs $Z > 3.0$. Dividing every ratio by total liabilities instead of total assets gives about 4.36 and this answer.

B is incorrect. "Potential default" is the 2.7 to 3.0 band; 2.45 is below it. This is the boundary mix-up.

D is incorrect. A score below 1.8 comes from inverting X_4 (liabilities over equity, $Z \approx 1.35$) or using market equity over total assets ($Z \approx 1.58$).

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.2, eq. 9.1; SchweserNotes 2026 Book 2, Reading 27, LO 27.b

Solution CR-9 b.3

Answer: C

Workbook A10.1 • S1, S2 • Aug-2026

Step 1 – Working capital

$$\text{WC} = \text{CA} - \text{CL} = 180 - 126 = 54$$

Step 2 – The five ratios

$$X_1 = \tfrac{54}{600} = 0.0900, \quad X_2 = \tfrac{96}{600} = 0.1600, \quad X_3 = \tfrac{42}{600} = 0.0700$$
$$X_4 = \tfrac{330}{390} = 0.8462, \quad X_5 = \tfrac{690}{600} = 1.1500$$

$\Rightarrow X_4$ uses the market value of equity (330) over total liabilities (390); the book equity of 210 is not an input.

Step 3 – Weighted sum

$$Z = 0.1080 + 0.2240 + 0.2310 + 0.5077 + 1.1489 = 2.2195$$

Interpretation

$Z = 2.22$ lies between 1.8 and 2.7: by Altman's original cut-offs there is a good chance of default. The memo should flag the name for closer review rather than standard approval.

Why the other options are wrong

A is incorrect. Used the book value of equity in X_4: $X_4 = 210/390 = 0.5385$, giving $Z = 2.03$.

B is incorrect. Swapped the weights of X_4 and X_5 (0.999 on the equity ratio, 0.6 on sales): $Z = 2.10$.

D is incorrect. Used current assets instead of working capital in X_1: $X_1 = 180/600 = 0.30$, giving $Z = 2.47$.

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.2 eq. 9.1, Example 9.1

Solution CR-9 b.4 Answer: D

Workbook A10.2 • S2, S3, S4, S5

Step 1 – Working capital

$$\text{WC} = 250 - 280 = -30$$

$\Rightarrow$ Negative working capital: short-term obligations exceed short-term assets, so X_1 enters with a minus sign.

Step 2 – The five ratios

$$X_1 = \tfrac{-30}{800} = -0.0375, \quad X_2 = \tfrac{40}{800} = 0.0500, \quad X_3 = \tfrac{40}{800} = 0.0500$$
$$X_4 = \tfrac{330}{560} = 0.5893, \quad X_5 = \tfrac{1{,}040}{800} = 1.3000$$

Step 3 – Discriminant score

$$Z = -0.0450 + 0.0700 + 0.1650 + 0.3536 + 1.2987 = 1.8423$$

Step 4 – Assign the group

$1.8 < 1.84 \leq 2.7$, so the firm falls in the zone where there is a good chance of default.

Interpretation

The engine is doing exactly what Altman's discriminant analysis does: collapse five ratios into one linear score and read the group off fixed cut-offs. The firm sits just above the 1.8 boundary, so any input error that lowers Z by more than 0.05 moves it into the worst group. "LDA" here is linear discriminant analysis, a scoring tool; it has nothing to do with the loss distribution approach used for operational risk capital.

Why the other options are wrong

A is incorrect. Used net income instead of EBIT in X_3: $X_3 = 8/800 = 0.01$, $Z = 1.71$, which wrongly drops the firm below 1.8.

B is incorrect. Used the book value of equity in X_4: $X_4 = 240/560 = 0.4286$, $Z = 1.75$.

C is incorrect. Correct score, wrong zone: "on alert" is the 2.7 to 3.0 band; 1.8 to 2.7 is "good chance of default".

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.2 eq. 9.1

Solution CR-9 b.5 Answer: B

Bank Q319 • BT P2.T6.704.3

Step 1 – Base score

$X_1 = 0.20$, $X_2 = 0.30$, $X_3 = 0.10$, $X_4 = 800/500 = 1.60$, $X_5 = 1.50$: $Z = 0.24 + 0.42 + 0.33 + 0.96 + 1.4985 = 3.45$.

$\Rightarrow$ Before the stress the firm is comfortably unlikely to default.

Step 2 – Apply the cuts

EBIT $100 \times 0.5 = 50$, market equity $800 \times 0.75 = 600$, sales $1{,}500 \times 0.6 = 900$: $X_3 = 0.05$, $X_4 = 1.20$, $X_5 = 0.90$.

Step 3 – Stressed score

$$Z = 0.24 + 0.42 + 0.165 + 0.72 + 0.8991 = 2.4441$$

$\Rightarrow 1.8 < 2.44 < 2.7$: good chance of default.

Interpretation

Three shocks together take about 1.0 off the score (-0.165 from EBIT, -0.24 from equity, -0.599 from sales) and move the firm two zones down. Working capital and retained earnings are unchanged, so their terms carry over.

Why the other options are wrong

A is incorrect. Treated each cut as the fraction *remaining* (50% of EBIT, 25% of equity, 40% of sales): $X_4 = 0.40$, $X_5 = 0.60$, $Z = 1.66$.
C is incorrect. Applied the EBIT and sales cuts but left market equity at 800, so X_4 stays 1.60: $Z = 2.68$.
D is incorrect. Shocked only EBIT: $3.45 - 0.165 = 3.28$.

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.2, eq. 9.1

Solution CR-9 b.6

Answer: A

Workbook A10.3 • S2, S3

Step 1 – Z contribution excluding EBIT

$$X_1 = \tfrac{60}{400} = 0.15,\quad X_2 = \tfrac{48}{400} = 0.12,\quad X_4 = \tfrac{210}{280} = 0.75,\quad X_5 = \tfrac{380}{400} = 0.95$$

$$Z_{\text{ex}X_3} = 0.180 + 0.168 + 0.450 + 0.949 = 1.7471$$

Step 2 – Current Z

$$X_3 = \tfrac{170}{400} = 0.425, \qquad Z = 1.7471 + 3.3(0.425) = 1.7471 + 1.4025 = 3.1496$$

$\Rightarrow$ The firm sits about 0.15 above the 3.0 boundary of the "unlikely to default" zone.

Step 3 – EBIT at the 3.0 boundary

$$3.3X_3 = 3.0 - 1.7471 = 1.2530 \;\Rightarrow\; X_3 = 0.3797 \;\Rightarrow\; \text{EBIT} = 0.3797 \times 400 = 151.9$$

Step 4 – Largest fall

$$170 - 151.9 = 18.1$$

Interpretation

EBIT can fall by only about INR 18 crore (11%) before the firm drops into the "on alert" band. EBIT carries the largest weight (3.3), so it is the most powerful single lever in both directions.

Why the other options are wrong

B is incorrect. Used 1.4 (the retained-earnings weight) on EBIT: $(3.1496 - 3.0)/1.4 \times 400 = 42.7$.
C is incorrect. Targeted 2.7, the lower edge of the "on alert" zone: $170 - 115.5 = 54.5$.
D is incorrect. Targeted the 1.8 cut-off: $170 - 6.4 = 163.6$.

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.2 eq. 9.1

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 d

P1 CALC

Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default.

Workbook: A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: "hazard rate and PD calculation" came up twice in the August 2026 sitting (A15), and Map #17 (5 of 16 sources) puts the marginal PD from a cumulative table on the closed-book list.
Vardeez: CR-9 is the most important chapter in Credit Risk; expect the marginal and conditional probability calculations.

Reported exam shapes

- **A15·S4** Marginal PD as the difference of successive cumulative PDs (shared with CR-9 e).

What the questions look like: almost all numerical. You get a cumulative default table (S&P or Moody's), a cohort of issuers, or a one-year migration matrix. The question turns on *which* probability is asked: default by year t, survival to year t, default during year t as seen today, or default during year t given survival. Matrix questions ask for a two-year PD.

Worked elsewhere: conditional versus unconditional PDs and the hazard rate are the core of CR-9 e (Sections 1 and 2); Workbook A15.1 (marginal PD from a constant hazard rate) is worked there.

1. Ratings and the cumulative default table

S&P ratings run from AAA (best quality) to D (in payment default). **Investment grade** is AAA to BBB; **non-investment grade** (junk) is BB to D. Rating agencies publish the default experience of issuers that started in each rating, as **cumulative** default rates: the share that has defaulted by the end of year t.

Rating	1	2	3	4	5	7	10	15
AAA	0.00	0.03	0.13	0.24	0.34	0.51	0.70	0.90
AA	0.02	0.06	0.11	0.21	0.30	0.49	0.70	0.99
A	0.05	0.13	0.22	0.33	0.46	0.76	1.20	1.84
BBB	0.16	0.43	0.75	1.14	1.54	2.27	3.24	4.54
BB	0.63	1.93	3.46	4.99	6.43	8.89	11.64	14.65
B	3.34	7.80	11.75	14.89	17.35	20.99	24.62	28.24
CCC/C	28.30	38.33	43.42	46.36	48.58	50.75	52.76	54.76

Average cumulative issuer-weighted default rates (%), 1981 to 2020, by horizon in years (GARP Table 9.1, S&P).

Write C_t for the cumulative PD by the end of year t. Every entry is cumulative; the other probabilities come from differences and complements within a row.

- **Probability of default by year** t (cumulative): read it off. A BBB issuer: 0.16% by year 1, 0.43% by year 2.
- **Survival to year** t: $1 - C_t$. BB to the end of year 2: $100 - 1.93 = 98.07\%$.
- **Marginal PD in year** t: the probability of default *during* year t, seen from today: $C_t - C_{t-1}$. It is the same number CR-9 e calls the unconditional PD.

Schweser calls this table a "rating migration (transition) matrix". It is not one: see the correction box in CR-9 e, Section 4. The true migration matrix is in Section 4 below.

2. Marginal PDs from the table

AA, year 3 (Vardeez and Schweser): $C_3 - C_2 = 0.11 - 0.06 = 0.05\%$. **B, years 1 to 5**: 3.34, $7.80 - 3.34 = 4.46$, $11.75 - 7.80 = 3.95$, $14.89 - 11.75 = 3.14$ and $17.35 - 14.89 = 2.46\%$.

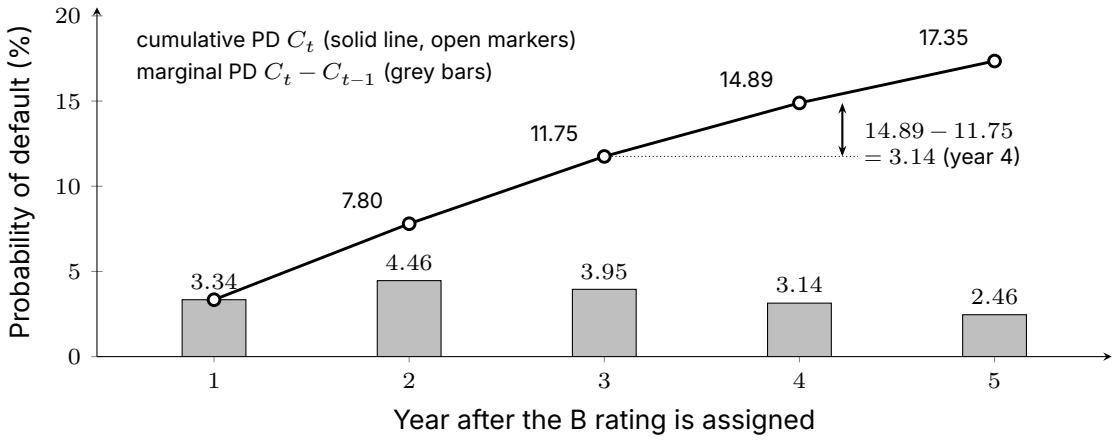


Figure 1. B-rated issuers from GARP Table 9.1. Each grey bar is the rise in the cumulative curve over one year (the arrow marks year 4). The marginal PD peaks in year 2 and then falls: an issuer that survives the critical early years has shown its health is better than the rating suggested.

- **The pattern by rating** (the subject of CR-9 c). Investment grade: marginal PDs rise in the first years (A: 0.05, 0.08, 0.09, 0.11%), because an issuer that starts creditworthy can only deteriorate. Lowest grades: they fall (CCC/C: 28.30, 10.03, 5.09, 2.94%), because the next year or two are critical.
- **Gaps in the horizons.** After year 5 the table jumps to 7, 10 and 15 years. $C_{10} - C_7$ for B is $24.62 - 20.99 = 3.63\%$: the probability of default during years 8 to 10 together, not in one year.
- **Conditional PD.** Divide the marginal PD by survival to the *start* of the year (CR-9 e, Section 1). B, year 4: $3.14/(100 - 11.75) = 3.14/88.25 = 3.56\%$.

3. Counting a cohort; a constant conditional rate

Cohort counts. Follow N_0 issuers with no replacement. With d_t defaults in year t and N_{t-1} names alive at the start of it:

$$\text{marginal PD} = \frac{d_t}{N_0} \qquad \text{conditional PD} = \frac{d_t}{N_{t-1}} \qquad \text{cumulative PD} = \frac{d_1 + \cdots + d_t}{N_0}$$

Example: 500 issuers with cumulative defaults of 12, 30 and 51 by the end of years 1 to 3. In year 3, $d_3 = 51 - 30 = 21$: marginal $21/500 = 4.20\%$; conditional $21/(500 - 30) = 21/470 = 4.47\%$; cumulative $51/500 = 10.2\%$.

A constant conditional rate. If a fixed share h of the names still performing defaults each year, survival to the end of year t is $(1 - h)^t$, so

$$\text{marginal PD in year } t = (1 - h)^{t-1}\, h \qquad \text{cumulative PD by year } t = 1 - (1 - h)^t$$

With $h = 4\%$: marginal PDs of 4.00%, 3.84% and 3.69% in years 1 to 3, and a cumulative PD of $1 - 0.96^3 = 11.53\%$ by year 3. The marginal PD falls even though the conditional rate is constant:

fewer names are left to default. (The continuous-time version, with a hazard rate and $e^{-\lambda t}$, is CR-9 e.)

4. The rating migration matrix

A **rating migration** (or **transition**) **matrix** gives, for each initial rating, the probabilities of being in each rating at the end of a period, usually one year. It is built from historical data. Each row sums to 100%, and D is absorbing: a defaulted issuer stays in default. GARP's Table 10.1 (S&P, 1981 to 2020; used in CR-10) shows, for example, that an A-rated company has a 92.61% chance of still being A after one year, 1.67% of being upgraded to AA, 5.23% of being downgraded to BBB, and a 0.05% chance of default.

- **Longer periods.** If rating changes in successive periods are independent, the two-year matrix is the one-year matrix multiplied by itself, and the five-year matrix is its fifth power. Shorter periods need roots (six months: the square root). Independence is not exactly true in practice: ratings show momentum.
- **Two-year PD from one row.** Default by year 2 happens either in year 1, or after migrating to rating j in year 1 and defaulting from j in year 2:

$$C_2 = p_{iD} + \sum_{j \neq D} p_{ij}\, p_{jD} \qquad \text{marginal year 2} = C_2 - C_1 \qquad \text{conditional year 2} = \frac{C_2 - C_1}{1 - C_1}$$

Worked example. A simplified matrix with three non-default grades:

Initial rating	To A	To B	To C	To D
A	88%	9%	2%	1%
B	6%	80%	10%	4%
C	1%	10%	74%	15%

For an issuer rated B today: $C_1 = 4\%$. The year-2 paths (Figure 2): via A $0.06 \times 0.01 = 0.06\%$; via B $0.80 \times 0.04 = 3.20\%$; via C $0.10 \times 0.15 = 1.50\%$. The marginal PD in year 2 is $0.06 + 3.20 + 1.50 = 4.76\%$, so $C_2 = 4 + 4.76 = 8.76\%$, and the conditional year-2 PD is $4.76/96 = 4.96\%$. The year-2 marginal PD exceeds the year-1 PD because a 10% chance of slipping to C, where the default rate is 15%, raises next year's risk. Doubling the one-year PD (8%) ignores migration.

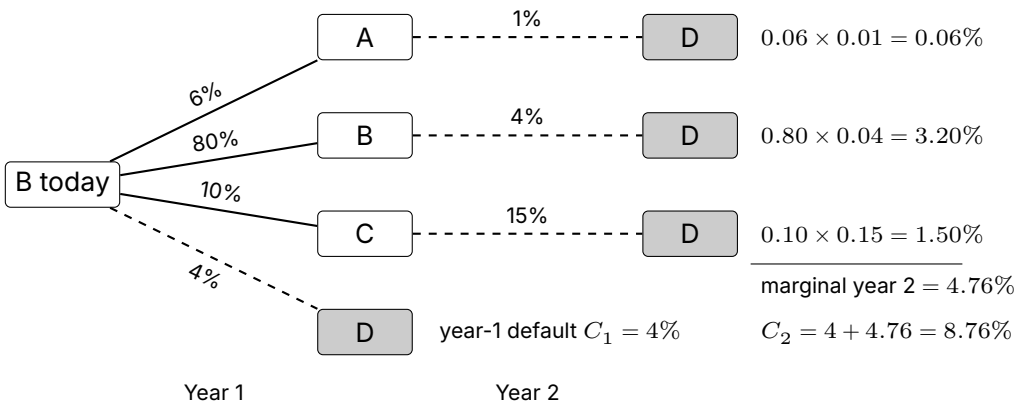


Figure 2. Two-step migration tree for an issuer rated B. Solid lines are rating moves, dashed lines are moves into default (grey boxes). Each year-2 default path multiplies a year-1 migration probability by that rating's one-year default probability; the paths add to the marginal year-2 PD.

Formula card: CR-9 d

Cumulative PD; survival	C_t read from the table; survival to $t = 1 - C_t$
Marginal PD, year t	$C_t - C_{t-1}$ (the unconditional PD)
Conditional PD, year t	$(C_t - C_{t-1})/(1 - C_{t-1})$ (survival to the *start* of the year)
Cohort of N_0 names	marginal d_t/N_0; conditional d_t/N_{t-1}; cumulative $\sum d/N_0$
Constant conditional rate h	marginal $(1-h)^{t-1}h$; cumulative $1-(1-h)^t$
Two-year PD, matrix	$C_2 = p_{iD} + \sum_{j \neq D} p_{ij}p_{jD}$; marginal year 2 $= C_2 - p_{iD}$
Other horizons	n years: M^n (independence); six months: $M^{1/2}$

	Cumulative PD	Survival	Marginal PD	Conditional PD
Question	Default by t?	No default by t?	Default in year t, seen today?	Default in year t, given survival to its start?
From a table	C_t	$1 - C_t$	$C_t - C_{t-1}$	$\dfrac{C_t - C_{t-1}}{1 - C_{t-1}}$
From a cohort	$\sum d/N_0$	N_t/N_0	d_t/N_0	d_t/N_{t-1}
From a matrix (year 2)	$p_{iD} + \sum_j p_{ij}p_{jD}$	$1 - C_2$	$C_2 - C_1$	$\dfrac{C_2 - C_1}{1 - C_1}$

Exam traps

- **Cumulative read as marginal.** The table entry for year 3 is the PD *by* year 3, not *in* year 3.
- **Wrong row, wrong column.** Moody's and S&P tables are large; the next row or the next horizon is always a distractor.
- **Survival versus default.** $1 - C_t$ is the probability of *no* default; it appears as an option next to the default probability.
- **Summing cumulative rates.** Adding C_1 and C_2 double counts year-1 defaults.
- **Wrong denominator.** Marginal PD uses the original cohort; conditional uses the survivors at the *start* of the year, not the end, and not the cumulative PD.
- **Off-by-one exponent.** With a constant conditional rate, the year-t marginal PD has $t-1$ years of survival, not t.
- **Doubling the one-year PD.** A two-year PD from a matrix needs the migration paths; $2 \times p_{iD}$ ignores them and counts defaulted names twice.
- **Missing a path.** Every non-default rating at the end of year 1 contributes, including upgrades.
- **Table 9.1 is not a migration matrix.** It shows cumulative default rates by initial rating, not one-year moves between ratings.
- **Horizon gaps.** $C_{10} - C_7$ is a three-year block, not a one-year marginal PD.

Practice questions: CR-9 d

Question CR-9 d.1

Bank Q325 • AnalystPrep Q.3052

A credit analyst uses Moody's average cumulative default rates (%), 1970 to 2012:

Term (years)	1	2	3	4	5	7	10	15	20
Aaa	0.000	0.013	0.013	0.037	0.106	0.247	0.503	0.935	1.104
Aa	0.022	0.069	0.139	0.256	0.383	0.621	0.922	1.756	3.135
A	0.063	0.203	0.414	0.625	0.870	1.441	2.480	4.255	6.841
Baa	0.177	0.495	0.894	1.369	1.877	2.927	4.740	8.628	12.483
Ba	1.112	3.083	5.424	7.934	10.189	14.117	19.708	29.172	36.321
B	4.051	9.608	15.216	20.134	24.613	32.747	41.947	52.217	58.084
Caa-C	16.448	27.867	36.908	44.128	50.366	58.302	69.483	79.178	81.248

What is the probability that a Ba-rated bond survives (does not default) until the end of year 2?

A. 94.576%

B. 95.805%

C. 96.917%

D. 98.888%

Question CR-9 d.2

Bank Q324 • AnalystPrep Q.3051

Using the Moody's table in Question CR-9 d.1, what is the probability that a bond rated Caa-C defaults during the third year (unconditional, as seen today)?

A. 9.041%

B. 36.908%

C. 45.949%

D. 52.124%

Question CR-9 d.3

Bank Q326 • AnalystPrep Q.3053

A risk analyst holds a bond rated Caa-C. Using the Moody's table in Question CR-9 d.1, the probability that the bond defaults during the fourth year, conditional on no earlier default, is closest to:

A. 11.44%

B. 19.56%

C. 63.09%

D. 69.94%

Question CR-9 d.4

Bank Q327 • AnalystPrep Q.5433 (rebuilt)

A risk manager tracks a cohort of 230 corporate issuers, none of which is replaced after default. The number of issuers not yet in default is:

End of year	0	1	2	3	4
Non-defaulted issuers	230	215	190	185	170

What is the estimated one-year marginal (unconditional) probability of default in year 4?

A. 6.52%

B. 8.11%

C. 10.87%

D. 26.09%

Question CR-9 d.5

Workbook A15.2 • S4, S6 • Map #17

A junior credit analyst at a UK pension fund is studying a rating agency's cumulative default table for B-rated issuers:

Year	1	2	3	4
Cumulative default rate (%)	3.10	7.40	11.20	14.60

Which of the following is closest to the probability that a B-rated issuer defaults during year 3, conditional on no default in years 1 and 2?

A. 3.73%

B. 3.80%

C. 4.10%

D. 4.28%

Question CR-9 d.6

Bank Q323 • AnalystPrep Q.1856 (rebuilt)

A bank follows a pool of 100 loans with no replacement of defaulted loans. Each year, 5% of the loans that are still performing at the start of that year default (a constant annual conditional default rate of 5%). What is the expected number of defaults during year 5?

A. 3.87

B. 4.07

C. 4.29

D. 5.00

Question CR-9 d.7

Bank Q328 • GARP PE2 Q78 • Mock 4 Q79 (rebuilt)

A credit risk analyst is given the following one-year credit migration matrix (rows are the current rating, columns the rating at the end of the year; D is default, which is absorbing):

Current rating	A	B	C	D
A	90%	8%	0%	2%
B	5%	76%	10%	9%
C	1%	14%	69%	16%

Assuming the matrix is constant over time, what is the probability that a bond currently rated B defaults within two years?

A. 15.94%

B. 17.44%

C. 17.54%

D. 18.00%

Question CR-9 d.8

Workbook A15.3 • S4, S6 • Map #17

A credit portfolio manager at a Swiss private bank (CHF book) uses a simplified one-year rating migration matrix with two non-default grades. Rating changes in successive years are assumed independent.

Initial rating	To X	To Y	Default
X	92%	6%	2%
Y	8%	82%	10%

Which of the following is closest to the unconditional (marginal) probability that an issuer rated X today defaults during year 2?

A. 2.44%

B. 2.49%

C. 4.00%

D. 4.44%

Solutions • CR-9 d

Solution CR-9 d.1

Answer: C

Bank Q325 • AnalystPrep Q.3052

Step 1 – Read the cumulative PD

Ba row, 2-year column: $C_2 = 3.083\%$.
$\Rightarrow$ 3.083% of Ba-rated bonds default by the end of year 2.

Step 2 – Survival is the complement

$$1 - C_2 = 100\% - 3.083\% = 96.917\%$$

Interpretation

Survival to t is one minus the cumulative PD at the *same* horizon. It is also the denominator of a year-3 conditional PD, which is why the exam tests it on its own.

Why the other options are wrong

A is incorrect. Used the 3-year cumulative PD: $100 - 5.424 = 94.576\%$.
B is incorrect. Subtracted the sum of the year-1 and year-2 cumulative PDs, $100 - (1.112 + 3.083) = 95.805\%$, double counting the year-1 defaults.
D is incorrect. Used the 1-year cumulative PD: $100 - 1.112 = 98.888\%$.

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2 Historical Default Probabilities, Table 9.1

Solution CR-9 d.2

Answer: A

Bank Q324 • AnalystPrep Q.3051

Step 1 – The two cumulative entries

Caa-C: $C_3 = 36.908\%$ and $C_2 = 27.867\%$.

Step 2 – Marginal PD in year 3

$$C_3 - C_2 = 36.908 - 27.867 = 9.041\%$$

$\Rightarrow$ Seen today, a Caa-C bond has a 9.04% chance of defaulting during year 3.

Interpretation

Every entry in the table is cumulative, so a one-year probability is always a difference within a row. Conditioning on survival to the end of year 2 would give $9.041/72.133 = 12.53\%$, but the stem asks for the unconditional value.

Why the other options are wrong

B is incorrect. Read the 3-year cumulative PD, 36.908%, as if it were the year-3 marginal PD.
C is incorrect. Added the marginal PD to the cumulative PD instead of subtracting the prior cumulative PD: $36.908 + 9.041 = 45.949\%$.
D is incorrect. Wrong row: added the B-rated 3-year cumulative rate to the Caa-C one, $15.216 + 36.908 = 52.124\%$.

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2, Table 9.1

Solution CR-9 d.3

Answer: A

Bank Q326 • AnalystPrep Q.3053

Step 1 – Marginal PD in year 4

$$C_4 - C_3 = 44.128 - 36.908 = 7.220\%$$

Step 2 – Survival to the start of year 4

$$1 - C_3 = 100 - 36.908 = 63.092\%$$

Step 3 – Conditional PD

$$\frac{7.220}{63.092} = 11.44\%$$

$\Rightarrow$ Given that the bond has survived three years, the chance of default in year 4 is 11.44%.

Interpretation

For a Caa-C bond the conditional PD (11.44%) is well above the marginal one (7.22%) because more than a third of the cohort has already gone. For investment grade the two are almost equal.

Why the other options are wrong

B is incorrect. Divided the marginal PD by the 3-year cumulative PD instead of the survival probability: $7.220/36.908 = 19.56\%$.

C is incorrect. Reported the survival probability to the end of year 3, 63.09%, which is not a default probability.

D is incorrect. Divided the 4-year cumulative PD by the 3-year survival probability: $44.128/63.092 = 69.94\%$.

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2 and 9.2.1, Table 9.1

Solution CR-9 d.4

Answer: A

Bank Q327 • AnalystPrep Q.5433

Step 1 – Defaults during year 4

$185 - 170 = 15$ issuers default in year 4.

Step 2 – Marginal PD against the original cohort

$$\frac{15}{230} = 6.52\%$$

$\Rightarrow$ Seen at time zero, 6.52% of the cohort defaults in year 4.

Interpretation

The marginal (unconditional) PD keeps the original 230 in the denominator. The conditional PD would use the 185 names alive at the start of year 4, $15/185 = 8.11\%$.

Why the other options are wrong

B is incorrect. Divided by the 185 survivors: $15/185 = 8.11\%$ is the conditional PD.

C is incorrect. Used the year-2 defaults: $(215 - 190)/230 = 25/230 = 10.87\%$.

D is incorrect. Used the cumulative defaults through year 4: $(230 - 170)/230 = 60/230 = 26.09\%$.

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of

default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2 and 9.2.1

Solution CR-9 d.5

Answer: C

Workbook A15.2 • S4, S6

Step 1 – Unconditional PD in year 3

$$C_3 - C_2 = 11.20 - 7.40 = 3.80\%$$

$\Rightarrow$ Seen from time zero, there is a 3.80% chance the issuer defaults during year 3.

Step 2 – Survival to the start of year 3

$$1 - C_2 = 100 - 7.40 = 92.60\%$$

Step 3 – Conditional PD

$$\frac{3.80}{92.60} = 4.10\%$$

Interpretation

Given the issuer has survived two years, the chance of defaulting in year 3 is 4.10%. Conditional PDs approximate the hazard rate over the year and are the right input when assessing a surviving name.

Why the other options are wrong

A is incorrect. Divided the three-year cumulative rate by three: $11.20/3 = 3.73\%$ (an average, not a year-3 probability).
B is incorrect. Stopped at the unconditional PD $C_3 - C_2 = 3.80\%$ (intermediate result).
D is incorrect. Divided by survival to the end of year 3: $3.80/88.80 = 4.28\%$.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2 and 9.2.1, Table 9.1

Solution CR-9 d.6

Answer: B

Bank Q323 • AnalystPrep Q.1856

Step 1 – Loans still performing at the start of year 5

Four years of survival at 95% a year: $100 \times 0.95^4 = 81.45$ loans.

Step 2 – Expected defaults in year 5

$$81.45 \times 0.05 = 4.07$$

$\Rightarrow$ The marginal PD of year 5, seen from today, is 4.07%; the conditional PD stays at 5%.

Interpretation

Because defaulted loans are not replaced, the same 5% conditional rate applies to a shrinking pool, so the expected count falls each year: 5.00, 4.75, 4.51, 4.29, 4.07. This is the discrete version of the constant hazard rate in CR-9 e.

Why the other options are wrong

A is incorrect. Used five years of survival before year 5: $100 \times 0.95^5 \times 0.05 = 3.87$ (off-by-one exponent).

C is incorrect. Gave the year-4 figure: $100 \times 0.95^3 \times 0.05 = 4.29$.
D is incorrect. Applied 5% to the original 100 loans, ignoring that earlier defaults have left the pool.

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2 and 9.2.1

Solution CR-9 d.7 — Answer: C

Bank Q328 • GARP PE2 Q78 • Mock 4 Q79

Step 1 – Default in year 1

$p_{BD} = 9\%$.

Step 2 – Survive year 1, then default in year 2

via A: $0.05 \times 0.02 = 0.10\%$ via B: $0.76 \times 0.09 = 6.84\%$ via C: $0.10 \times 0.16 = 1.60\%$

$\Rightarrow$ The year-2 marginal PD is $0.10 + 6.84 + 1.60 = 8.54\%$.

Step 3 – Two-year cumulative PD

$$C_2 = 9\% + 8.54\% = 17.54\%$$

Interpretation

This is row B of the squared matrix, column D. The year-2 marginal PD (8.54%) is a little below the year-1 PD only because 9% of the bonds are already gone; conditional on survival it is $8.54/91 = 9.38\%$, above 9%, because the 10% chance of slipping to C (16% default rate) outweighs the 5% chance of moving up to A (2%).

Why the other options are wrong

A is incorrect. Omitted the path through C: $9 + 0.10 + 6.84 = 15.94\%$.
B is incorrect. Omitted the path through A: $9 + 6.84 + 1.60 = 17.44\%$.
D is incorrect. Doubled the one-year PD, $2 \times 9\% = 18.00\%$, ignoring migration and counting defaulted bonds as able to default again.

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.1 Ratings Transition Matrices; Ch.17 (GARP CR Ch.9), 9.2

Solution CR-9 d.8 — Answer: A

Workbook A15.3 • S4, S6

Step 1 – Two-year cumulative PD (row X of the squared matrix)

$$C_2 = p_{XD} + p_{XX}p_{XD} + p_{XY}p_{YD} = 0.02 + 0.92(0.02) + 0.06(0.10) = 0.0444$$

$\Rightarrow$ 4.44% of X-rated issuers default within two years, including those that slip to Y first.

Step 2 – Marginal PD in year 2

$$C_2 - C_1 = 0.0444 - 0.0200 = 0.0244$$

Interpretation

2.44% is the probability, seen today, of default in year 2. It exceeds the year-1 PD of 2% because migration to Y (with a 10% default rate) raises year-2 risk: the investment-grade pattern of rising marginal PDs.

Why the other options are wrong

B is incorrect. Computed the conditional year-2 PD: $0.0244/(1-0.02) = 2.49\%$.
C is incorrect. Ignored migration and doubled the one-year PD: $2 \times 2\% = 4.00\%$.
D is incorrect. Reported the two-year cumulative PD $C_2 = 4.44\%$ (intermediate result).

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2; Ch.19 (GARP CR Ch.10), 10.1 Ratings Transition Matrices

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 e

P1 CALC

Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities.

Workbook: A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD • Tier A • August 2026 recall A5 Merton PD to hazard rate

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: "hazard rate and PD calculation" came up twice in the August 2026 sitting (A15), and the Merton PD converted into a hazard rate is an A5 shape.
Vardeez: CR-9 is the most important chapter in Credit Risk. Expect hazard rates and the marginal and conditional probability calculations. Skip "more precise hazard rates" (the bond-pricing method).

Reported exam shapes

- **A15·S1** From a constant hazard rate to cumulative and year-t default probabilities.
- **A15·S2** Hazard rate from a PD by the direct formula $\bar{\lambda} = -\ln(1-Q)/T$.
- **A15·S4** Marginal PD as the difference of successive cumulative PDs (shared with CR-9 d).
- **A15·S5** Sibling swap: $1 - e^{-\lambda t}$ is the cumulative PD, not the hazard rate.
- **A15·S6** Conditional versus unconditional default probability.
- **A5·S4** A Merton PD converted into an average hazard rate.

What the questions look like: almost all numerical. You get a constant hazard rate or a table of cumulative default rates, and the question turns on *which* of three probabilities is asked: by year t, during year t as seen today, or during year t given survival. Recovery rates, notionals and exposures often appear and are not needed for the probability.

Worked elsewhere: A15.2 (conditional PD from a table) at Question CR-9 d.5; A15.4 (historical versus spread-implied hazard) at Question CR-9 j.4; A15.5 and A15.6 (hazard from a CDS spread) at Questions CR-9 h.1 and CR-9 h.7; A5.1 (Merton PD to hazard, then a CVA input) under CR-9 m.

1. Three default probabilities from one table

Rating agencies publish **cumulative** default rates: the share of issuers starting in a rating that have defaulted by the end of year t. GARP's Table 9.1 (S&P, 1981 to 2020) is the standard example. Write C_t for the cumulative PD by the end of year t.

Initial rating	Year 1	Year 2	Year 3	Year 4	Year 5
BBB	0.16	0.43	0.75	1.14	1.54
BB	0.63	1.93	3.46	4.99	6.43
B	3.34	7.80	11.75	14.89	17.35
CCC/C	28.30	38.33	43.42	46.36	48.58

Cumulative issuer-weighted default rates (%), from GARP Table 9.1.

Three different questions can be asked of the same row:

- **Cumulative PD** (default *by* the end of year t): read it off, C_t.

- **Unconditional PD** (default *during* year t, as seen at time zero), also called the **marginal** PD: $C_t - C_{t-1}$. CCC/C, year 3: $43.42 - 38.33 = 5.09\%$. BBB, year 2: $0.43 - 0.16 = 0.27\%$.
- **Conditional PD** (default during year t, *given no earlier default*): divide the unconditional PD by the probability of surviving to the *start* of the year, $1 - C_{t-1}$. CCC/C, year 3: survival to the end of year 2 is $100 - 38.33 = 61.67\%$, so $5.09/61.67 = 8.25\%$.

The conditional PD is never smaller than the unconditional one, because its denominator is a survival probability below one. The two coincide only in year 1. The gap is tiny for investment grade (BBB year 2: 0.27% against 0.27%) and large for junk (CCC/C year 3: 5.09% against 8.25%).

2. The hazard rate

Shrink the conditional period from one year to an instant and you get the **hazard rate**, or **default intensity**. GARP's definition: $\lambda(t)\,\Delta t$ is the probability of default between t and $t + \Delta t$, *conditional on no default* between time zero and t.

Let $V(t)$ be the probability of surviving to t. The conditional default probability over Δt is the drop in survival divided by survival:

$$\frac{V(t) - V(t+\Delta t)}{V(t)} = \lambda(t)\,\Delta t \quad \xrightarrow{\Delta t \to 0} \quad \frac{\mathrm{d}V}{\mathrm{d}t} = -\lambda(t)\,V(t) \quad \Rightarrow \quad V(t) = \mathrm{e}^{-\int_0^t \lambda(\tau)\,\mathrm{d}\tau}$$

Define $Q(t) = 1 - V(t)$, the probability of default by t, and the **average hazard rate** $\bar{\lambda}(t)$ between time zero and t. Then (GARP eq. 9.2):

$$\boxed{Q(t) = 1 - \mathrm{e}^{-\bar{\lambda}(t)\,t}} \qquad \boxed{\bar{\lambda}(t) = -\frac{1}{t}\ln(1 - Q(t))}$$

These are the "probability functions for default time" in the LO wording. With a constant hazard rate λ:

- **Survival** (the default time t^* is later than t): $\Pr[t^* > t] = \mathrm{e}^{-\lambda t}$, falling towards zero, so every company defaults eventually.
- **Cumulative distribution** of default time: $F(t) = \Pr[t^* \le t] = 1 - \mathrm{e}^{-\lambda t}$.
- **Density** (the marginal default probability per unit time): $f(t) = F'(t) = \lambda \mathrm{e}^{-\lambda t}$.
- **Hazard** = density ÷ survival $= \lambda \mathrm{e}^{-\lambda t} / \mathrm{e}^{-\lambda t} = \lambda$. Not cumulative ÷ survival (Question CR-9 e.1).

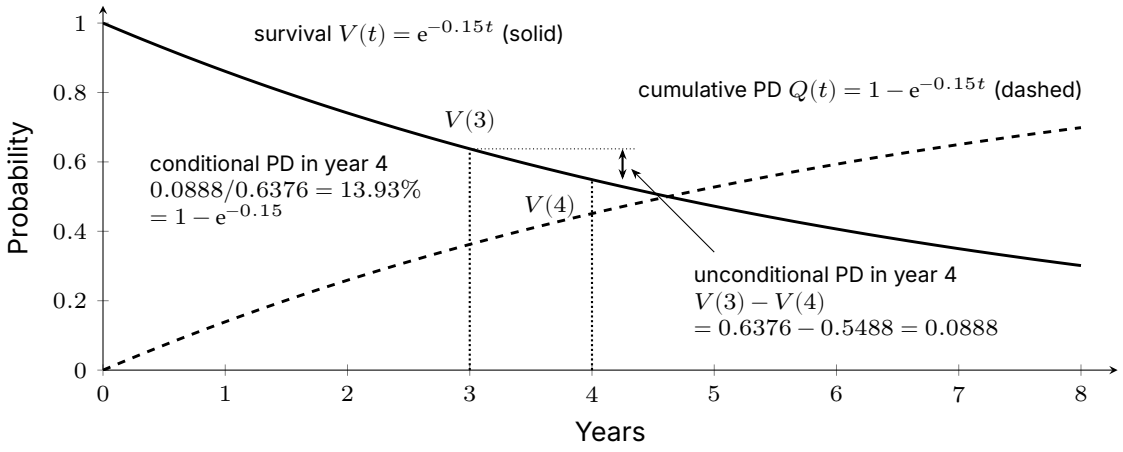


Figure 1. Survival (solid) and cumulative PD (dashed) under a constant 15% hazard rate, the rate in Question CR-9 e.9. The unconditional PD for year 4 is the drop in survival across the year (arrow). The conditional PD divides that drop by survival at the start of the year, $V(3)$. With a constant hazard rate it is the same 13.93% in every year.

3. Working with a constant hazard rate

GARP Example 9.2. $\lambda = 1.5\%$ a year.

End of year t	1	2	3	4	5
$Q(t) = 1 - \mathrm{e}^{-0.015t}$	0.0149	0.0296	0.0440	0.0582	0.0723

- Unconditional PD during year 4: $0.0582 - 0.0440 = 0.0142$.
- Conditional PD in year 4: $0.0142/(1 - 0.0440) = 0.0149$, which is $1 - \mathrm{e}^{-0.015}$ again.

The two shortcuts. With a constant λ:

$$\text{Unconditional PD in year } t = \mathrm{e}^{-\lambda(t-1)} - \mathrm{e}^{-\lambda t} \qquad \text{Conditional PD in any year} = 1 - \mathrm{e}^{-\lambda}$$

The second is the **memoryless** property: a firm that has survived faces the same one-year default probability every year. The unconditional PD falls year by year only because fewer firms are still alive to default.

Second example (Schweser and Vardeez, $\lambda = 2\%$): $Q(4) = 1 - \mathrm{e}^{-0.08} = 7.69\%$ and $Q(5) = 1 - \mathrm{e}^{-0.10} = 9.52\%$. The unconditional year-5 PD is $9.52 - 7.69 = 1.83\%$, and the conditional one is $1.83/(1 - 0.0769) = 1.98\% = 1 - \mathrm{e}^{-0.02}$.

Any horizon, not just whole years. $Q(T) = 1 - \mathrm{e}^{-\lambda T}$ works for $T = 0.5$ or $T = 0.25$. A 3% hazard rate gives a six-month PD of $1 - \mathrm{e}^{-0.015} = 1.49\%$, not 3% (Question CR-9 e.6). The same exponential describes the waiting time to the next event in a Poisson process with intensity λ.

4. When the hazard rate changes over time

Real cumulative tables do not follow one constant λ. Two quantities are then worth separating:

- The **average** hazard rate from zero to t: $\bar{\lambda}(t) = -\ln(1 - C_t)/t$. BBB over 5 years: $-\ln(1 - 0.0154)/5 = 0.31\%$. CCC/C over 5 years: $-\ln(1 - 0.4858)/5 = 13.30\%$.

- The hazard rate **during** year t: $\lambda_t = \ln(V(t-1)/V(t)) = -\ln(1-\text{conditional PD})$. CCC/C, year 3: $\ln(61.67/56.58) = 8.61\%$, slightly above the conditional PD of 8.25% because the hazard is the continuously compounded version of it (Question CR-9 e.5).

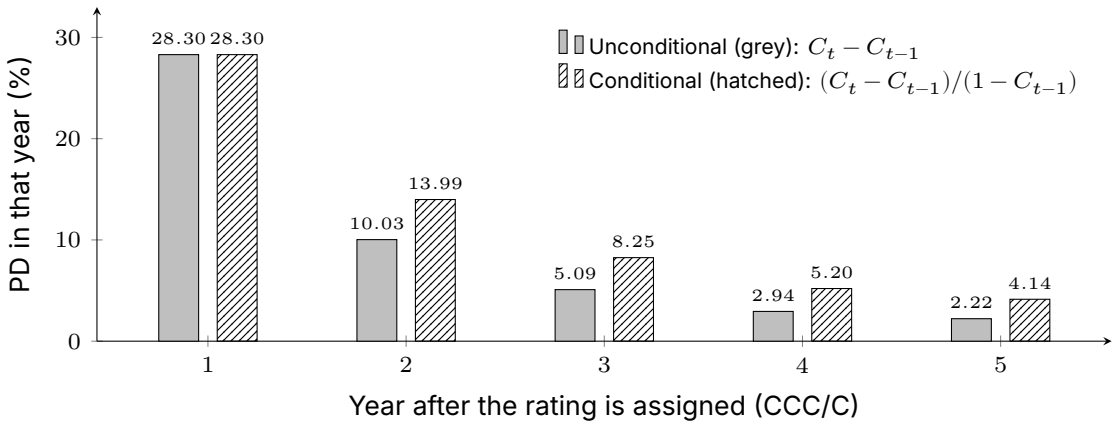


Figure 2. CCC/C issuers from GARP Table 9.1. Both bars fall over time, so the hazard rate is not constant. The first years are the critical ones; a firm that survives them is likely to be in better health. For BBB issuers the pattern runs the other way: the unconditional PD rises from 0.16% to 0.27%, 0.32%, 0.39% and 0.40% in years 1 to 5.

Correction to the source notes. SchweserNotes (LO 27.d) gives the B-rated marginal default probabilities for years 1 to 4 as 3.34%, 4.46%, 3.95% and 3.24%. From GARP Table 9.1 (B: 3.34, 7.80, 11.75, 14.89) the year-4 figure is $14.89 - 11.75 = 3.14\%$; the first three are right. Schweser also calls this table a "rating migration (transition) matrix". It is a table of cumulative default rates by initial rating; a migration matrix gives one-year probabilities of moving between ratings (CR-9 d, CR-10).

5. Converting between PDs, hazard rates and spreads

Questions often hand you one form and ask for another.

- **Cumulative PD to hazard rate** (shape S2): $\bar{\lambda} = -\ln(1-Q)/T$. Dividing Q by T is the linear shortcut and understates the hazard slightly.
- **Merton PD to hazard rate** (A5 shape S4): Merton's model gives the risk-neutral probability of default by T as $N(-d_2)$ (CR-9 m). Treat it as $Q(T)$ and apply the direct formula. Example: $N(-d_2) = 2.5\%$ over 2 years gives $\bar{\lambda} = -\ln(0.975)/2 = 1.27\%$. A CVA desk stores default risk this way (Question CR-9 e.7).
- **Spread to hazard rate** (CR-9 h, eq. 9.3): $\bar{\lambda} = s(T)/(1-R)$, with $1-R$ the loss given default. 240 bp with 40% recovery gives $0.024/0.6 = 4\%$. Reverse it to price a spread: $s = \bar{\lambda}(1-R)$ (Question CR-9 e.8).
- **Hazard to cumulative PD**: once you have $\bar{\lambda}$, $Q(T) = 1 - \mathrm{e}^{-\bar{\lambda}T}$ for the same T (Question CR-9 e.9).
- **Forward hazard between two maturities** (CR-9 h): $\bar{\lambda}_{T_1,T_2} = (T_2\bar{\lambda}(T_2) - T_1\bar{\lambda}(T_1))/(T_2 - T_1)$. GARP Example 9.3: 50, 60 and 100 bp at 3, 5 and 10 years with $R = 60\%$ give 1.25%, 1.5% and 2.5%, so 1.875% for years 3 to 5 and 3.5% for years 5 to 10.

Formula card: CR-9 e

Unconditional PD, year t	$C_t - C_{t-1}$ (the marginal PD)
Conditional PD, year t	$\dfrac{C_t - C_{t-1}}{1 - C_{t-1}}$ (divide by survival to the *start* of the year)
Survival and cumulative PD	$V(t) = \mathrm{e}^{-\bar{\lambda}(t)t}$, $Q(t) = 1 - \mathrm{e}^{-\bar{\lambda}(t)t}$
Average hazard from a PD	$\bar{\lambda}(t) = -\ln(1 - Q(t))/t$
Hazard during year t	$\lambda_t = \ln(V(t-1)/V(t)) = -\ln(1 - \text{conditional PD})$
Constant λ, unconditional	year t: $\mathrm{e}^{-\lambda(t-1)} - \mathrm{e}^{-\lambda t}$
Constant λ, conditional	any year: $1 - \mathrm{e}^{-\lambda}$ (memoryless)
Density of default time	$f(t) = \lambda \mathrm{e}^{-\lambda t}$ (a rate, not a one-year probability)
Merton PD	$Q(T) = N(-d_2)$ (risk-neutral), then $\bar{\lambda} = -\ln(1 - N(-d_2))/T$
Spread link (CR-9 h)	$\bar{\lambda} = s/(1 - R)$; $s = \bar{\lambda}(1 - R)$

	Cumulative PD	Unconditional PD	Conditional PD	Hazard rate
Question	Default by t?	Default in year t, seen today?	Default in year t, given survival?	Default in the next instant, given survival?
From a table	C_t	$C_t - C_{t-1}$	$\frac{C_t - C_{t-1}}{1 - C_{t-1}}$	$\ln \frac{V(t-1)}{V(t)}$
Constant λ	$1 - \mathrm{e}^{-\lambda t}$	$\mathrm{e}^{-\lambda(t-1)} - \mathrm{e}^{-\lambda t}$	$1 - \mathrm{e}^{-\lambda}$	λ
Over time	Rises to 1	Falls (constant λ)	Constant (constant λ)	Constant by assumption

Exam traps

- **Intermediate result.** The difference of cumulative PDs is the unconditional PD. Stopping there when the stem says "conditional on no earlier default", or dividing when it says "as seen today".
- **Wrong survival denominator.** Dividing by survival to the *end* of the year $(1 - C_t)$ or to the end of year 1 instead of the start of the year in question.
- **Cumulative PD read as the hazard rate (sibling swap).** $1 - \mathrm{e}^{-\lambda t}$ is the probability of default by t. Feeding a cumulative PD into $s = \bar{\lambda}(1 - R)$ overstates the spread.
- **Forgetting T.** $\bar{\lambda} = -\ln(1 - Q)/T$ needs the division by T; $Q(T) = 1 - \mathrm{e}^{-\bar{\lambda}T}$ needs the multiplication. Using a one-year or two-year horizon when three is asked is the usual slip.
- **Linear shortcuts.** Q/T for the hazard, λt or $1 - (1 - \lambda)^t$ for the cumulative PD. The exam's model is continuous.
- **Density as a probability.** $\lambda \mathrm{e}^{-\lambda t}$ at $t = 1$ is a rate, not the probability of default in year 2.
- **Memoryless forgotten.** With a constant hazard the conditional one-year PD is $1 - \mathrm{e}^{-\lambda}$ in every year; it does not grow with time.
- **Annual PD used for a shorter term.** A 3-month loan with a 4% hazard rate has a PD of $1 - \mathrm{e}^{-0.01}$, not 4%.

- **Red herrings.** Recovery rate, notional, exposure and extra maturities on a CDS curve change the loss, not the probability.

Practice questions: CR-9 e

Question CR-9 e.1

Bank Q330 • BT P2.T6.307.3 (rebuilt)

A credit analyst models a borrower's default time t^* with a constant hazard rate λ, so that the cumulative distribution of default time is $F(t) = \Pr[t^* \leq t] = 1 - \mathrm{e}^{-\lambda t}$ and the survival function is $1 - F(t) = \Pr[t^* > t] = \mathrm{e}^{-\lambda t}$. Each of the following statements is true EXCEPT:

A. As t grows large, $1 - F(t)$ converges to zero, so even a highly rated company with a low hazard rate will eventually default.

B. Given survival to time t, the conditional probability of default over a very short interval $(t,\, t + \mathrm{d}t)$ is $F(t)/\mathrm{e}^{-\lambda t}$.

C. The marginal default probability (density), the derivative of the cumulative distribution, is $F'(t) = \lambda \mathrm{e}^{-\lambda t}$.

D. The derivative of the survival function is $-F'(t) = -\lambda \mathrm{e}^{-\lambda t} < 0$, so the survival probability falls over time.

Question CR-9 e.2

Bank Q331 • AP Mock 3 Q38 (rebuilt)

A credit risk analyst at a European bank models a corporate borrower with a constant hazard rate of 9% per year. The borrower has survived its first year. Which of the following is closest to the probability that it defaults before the end of the second year?

A. 7.87%

B. 8.23%

C. 8.61%

D. 16.47%

Question CR-9 e.3

Workbook A15.1 • S1, S4, S5, S6 • Aug-2026 • Map #17

A quantitative analyst at an Indian bank models a corporate borrower with a constant hazard rate of 2.5% per annum. The borrower's expected recovery rate is 40% and the loan's notional is INR 500 crore. For the bank's expected-loss schedule, which of the following is closest to the probability, as seen today, that the borrower defaults during the fourth year?

A. 2.29%

B. 2.32%

C. 2.47%

D. 9.52%

Question CR-9 e.4

Bank Q335 • BT P2.T6.703.2 (rebuilt)

A credit analyst tracks a cohort of 1,000 issuers that all started in the same rating class:

Year	0	1	2	3	4	5
Names surviving at t	1,000	990	976	958	947	922
Cumulative defaults	0	10	24	42	53	78

Which of the following correctly gives the conditional and the unconditional default probabilities for year 3?

A. Conditional 1.844%; unconditional 1.800%

B. Conditional 1.800%; unconditional 1.844%

C. Conditional 1.879%; unconditional 1.800%

D. Conditional 4.200%; unconditional 1.800%

Question CR-9 e.5

GARP book • Ch.9 Practice Question 9.4 (rewritten as multiple choice)

A credit analyst is using the S&P cumulative default rates from GARP Table 9.1:

Cumulative default rate (%)	Year 1	Year 2	Year 3	Year 4
BBB	0.16	0.43	0.75	1.14
BB	0.63	1.93	3.46	4.99

Which of the following is closest to the average hazard rate for a BB-rated company during the third year?

A. 1.17%

B. 1.53%

C. 1.56%

D. 1.57%

Question CR-9 e.6

Bank Q336 • BT P2.T6.506.1 (rebuilt)

A bank holds four short-term loans. Each one-year probability of default is a hazard rate (default intensity), so the default probability over the remaining term T is $1 - e^{-\lambda T}$.

Loan	Remaining term	Exposure at default	One-year PD (hazard rate)	LGD
a	3 months	USD 100 million	4.00%	90%
b	6 months	USD 120 million	3.00%	60%
c	9 months	USD 150 million	2.00%	60%
d	12 months	USD 200 million	1.00%	50%

Which loan has the highest expected loss in dollar terms over its remaining term?

A. Loan a

B. Loan b

C. Loan c

D. Loan d

Question CR-9 e.7 New • built from GARP Ch.9, 9.2.1 eq. 9.2 and 9.8 (Merton) • shape A5·S4

A counterparty risk analyst at a Hong Kong bank runs Merton's model on a listed corporate counterparty over a 3-year horizon. Using the risk-free rate as the asset drift, the solver returns $d_2 = 1.60$. The firm's expected return on assets is 9% a year. The bank's CVA system stores each counterparty's default risk as a constant average hazard rate. Which of the following is closest to the hazard rate the analyst should enter?

A. 1.83%

B. 1.88%

C. 5.48%

D. 5.64%

Question CR-9 e.8 Workbook A15.7 • S2, S3, S5 • Map #17

A credit risk manager at a German insurer is checking whether a quoted five-year CDS spread on a counterparty is consistent with the firm's internal view. The internal model gives a five-year cumulative default probability of 8% and an expected recovery rate of 45%. Assume a constant hazard rate and the approximation in Hull eq. 9.3. Which of the following is closest to the five-year CDS spread consistent with these inputs?

A. 75 basis points

B. 92 basis points

C. 167 basis points

D. 440 basis points

Question CR-9 e.9 Bank Q337 • Mock 1 Q56 (rebuilt)

MRS Brokers has a three-year derivative exposure to a counterparty, with an expected exposure of AUD 18 million each year. Market CDS quotes for the counterparty, with the recovery rate assumed at each maturity, are:

Item	1-year	2-year	3-year
CDS spread (bps)	150	225	375
Recovery rate (%)	90	85	75

Using the approximation that the average hazard rate equals the CDS spread divided by one minus the recovery rate, and assuming a constant hazard rate over the first three years, which of the following is closest to the cumulative probability of default over three years?

A. 13.93%

B. 15.00%

C. 25.92%

D. 36.24%

Solutions • CR-9 e

Solution CR-9 e.1

Answer: B

Bank Q330 • BT P2.T6.307.3

Step 1 – Differentiate the cumulative distribution

$F(t) = 1 - \mathrm{e}^{-\lambda t}$ gives $F'(t) = \lambda \mathrm{e}^{-\lambda t}$ by the chain rule.
$\Rightarrow$ C is true: the density is the derivative of F.

Step 2 – Differentiate the survival function

$\frac{\mathrm{d}}{\mathrm{d}t}(1 - F(t)) = -\lambda \mathrm{e}^{-\lambda t} < 0$, and $\mathrm{e}^{-\lambda t} \to 0$ as $t \to \infty$ for any $\lambda > 0$.
$\Rightarrow$ D and A are true: survival falls, and towards zero.

Step 3 – The conditional default probability over $\mathrm{d}t$

$$\frac{F'(t)\,\mathrm{d}t}{1 - F(t)} = \frac{\lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t}{\mathrm{e}^{-\lambda t}} = \lambda\,\mathrm{d}t$$

$\Rightarrow$ This is the definition of the hazard rate: density divided by survival. B divides the *cumulative* probability by survival, which is not $\lambda\,\mathrm{d}t$.

Interpretation

The hazard rate is a conditional quantity: the default probability per unit of time for a firm still alive. Mixing up the density, the cumulative distribution and the survival function is exactly what the numerical questions punish.

Why the other options are wrong

A is incorrect. True: survival decays exponentially to zero, so any positive hazard means eventual default.
C is incorrect. True: the density is $F'(t) = \lambda \mathrm{e}^{-\lambda t}$.
D is incorrect. True: the survival function is decreasing, so its derivative is negative.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1 Hazard Rates, eq. 9.2

Solution CR-9 e.2

Answer: C

Bank Q331 • AP Mock 3 Q38

Step 1 – Survival to the end of years 1 and 2

$$V(1) = \mathrm{e}^{-0.09} = 0.91393, \qquad V(2) = \mathrm{e}^{-0.18} = 0.83527$$

Step 2 – Unconditional PD in year 2

$$V(1) - V(2) = 0.91393 - 0.83527 = 0.07866$$

$\Rightarrow$ Seen from today, a 7.87% chance of default during year 2.

Step 3 – Condition on survival to the end of year 1

$$\frac{0.07866}{0.91393} = 0.08607 = 1 - \mathrm{e}^{-0.09}$$

$\Rightarrow$ 8.61%: with a constant hazard the conditional one-year PD is the same every year.

Interpretation

The stem says the borrower "has survived its first year", so year-1 defaults are excluded and the answer is the conditional PD. The memoryless shortcut $1 - e^{-\lambda}$ gets there in one line.

Why the other options are wrong

A is incorrect. Stopped at the unconditional year-2 PD, $e^{-0.09} - e^{-0.18} = 7.87\%$, without conditioning on survival.

B is incorrect. Used the density at $t = 1$, $0.09\, e^{-0.09} = 8.23\%$, which is a rate, not a probability over a year.

D is incorrect. Reported the two-year cumulative PD, $1 - e^{-0.18} = 16.47\%$, which includes the year-1 defaults the condition rules out.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1, eq. 9.2, Example 9.2

Solution CR-9 e.3

Answer: A

Workbook A15.1 • S1, S4, S5, S6 • Aug-2026

Step 1 – Cumulative PDs from the hazard rate

$$Q(3) = 1 - e^{-0.025\times 3} = 0.07226, \qquad Q(4) = 1 - e^{-0.025\times 4} = 0.09516$$

$\Rightarrow$ There is a 7.23% chance of default by year 3 and a 9.52% chance by year 4.

Step 2 – Unconditional (marginal) PD in year 4

$$Q(4) - Q(3) = e^{-0.075} - e^{-0.100} = 0.09516 - 0.07226 = 0.02291$$

$\Rightarrow$ Seen from today, there is a 2.29% chance that default happens during year 4.

Step 3 – Check against the conditional PD

$$\frac{0.02291}{1 - 0.07226} = 0.02469 = 1 - e^{-0.025}$$

$\Rightarrow$ The conditional PD is larger because it assumes the borrower is still alive at the start of year 4; it is not what "as seen today" asks for.

Interpretation

An expected-loss schedule built today weights each year's loss by the unconditional PD, so year 4 gets 2.29%. Recovery and notional are red herrings for the probability; they enter only once the loss is computed.

Why the other options are wrong

B is incorrect. Used discrete compounding: $\lambda(1-\lambda)^3 = 0.025 \times 0.975^3 = 2.32\%$.

C is incorrect. Computed the conditional PD (sibling swap with the unconditional one): $0.02291/0.92774 = 2.47\%$.

D is incorrect. Reported the cumulative four-year PD $Q(4) = 1 - e^{-4\lambda} = 9.52\%$, the expression the formula card mislabels as a hazard rate.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1, eq. 9.2, Example 9.2

Solution CR-9 e.4 — Answer: A

Bank Q335 • BT P2.T6.703.2

Step 1 – Defaults during year 3

$42 - 24 = 18$ issuers default in year 3.

Step 2 – Unconditional PD

Divide by the original cohort: $18/1{,}000 = 1.800\%$.

$\Rightarrow$ Seen at time zero, 1.8% of the cohort defaults in year 3.

Step 3 – Conditional PD

Divide by the names alive at the *start* of year 3, $1{,}000 - 24 = 976$: $18/976 = 1.844\%$.

$\Rightarrow$ Given survival through year 2, the chance of defaulting in year 3 is 1.844%.

Interpretation

Cohort counts make the definitions concrete: the unconditional PD keeps the original 1,000 in the denominator, the conditional PD uses only the survivors. The conditional figure is the larger one after year 1.

Why the other options are wrong

B is incorrect. Swapped the labels: the conditional PD cannot be below the unconditional PD.

C is incorrect. Divided by the survivors at the *end* of year 3 (958): $18/958 = 1.879\%$.

D is incorrect. Used the cumulative defaults, $42/1{,}000 = 4.200\%$, as the conditional figure.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1

Solution CR-9 e.5 — Answer: D

GARP book • Ch.9 Practice Question 9.4

Step 1 – Survival to the start and end of year 3

$$V(2) = 1 - 0.0193 = 0.9807, \qquad V(3) = 1 - 0.0346 = 0.9654$$

Step 2 – Conditional PD in year 3

$$\frac{0.0346 - 0.0193}{0.9807} = \frac{0.0153}{0.9807} = 1.56\%$$

$\Rightarrow$ 1.56% is the probability of default during year 3 given survival to the start of it.

Step 3 – Convert to a hazard rate

$$\lambda_3 = \ln \frac{V(2)}{V(3)} = \ln \frac{0.9807}{0.9654} = -\ln(1 - 0.0156) = 1.57\%$$

$\Rightarrow$ The hazard rate is the continuously compounded version of the conditional PD, so it is slightly higher.

Interpretation

This is GARP's own answer: the conditional PD is 1.56% and the hazard rate during the year is 1.57%. The BBB row is not needed.

Why the other options are wrong

A is incorrect. Computed the average hazard from time zero to the end of year 3, $-\ln(0.9654)/3 = 1.17\%$, not the hazard during year 3.

B is incorrect. Stopped at the unconditional PD, $3.46 - 1.93 = 1.53\%$.

C is incorrect. Stopped at the one-year conditional PD, $1.53/98.07 = 1.56\%$, without converting it to a continuous rate.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), Table 9.1 and 9.2.1; Practice Question 9.4

Solution CR-9 e.6

Answer: C

Bank Q336 • BT P2.T6.506.1

Step 1 – PD over each remaining term

$$\text{a: } 1-\mathrm{e}^{-0.04\times0.25}=0.995\% \qquad \text{b: } 1-\mathrm{e}^{-0.03\times0.5}=1.489\%$$
$$\text{c: } 1-\mathrm{e}^{-0.02\times0.75}=1.489\% \qquad \text{d: } 1-\mathrm{e}^{-0.01\times1}=0.995\%$$

$\Rightarrow$ The hazard rate is scaled by each loan's own term before it becomes a probability.

Step 2 – Expected loss = EAD × PD × LGD

$$\text{a: } 0.896 \qquad \text{b: } 1.072 \qquad \text{c: } 1.340 \qquad \text{d: } 0.995 \quad \text{(USD million)}$$

$\Rightarrow$ Loan c has the highest expected loss, USD 1.34 million.

Interpretation

The short loan with the highest annual PD (a) looks worst on a one-year basis but carries the lowest expected loss here, because it is exposed for only three months. Loans b and c share the same term PD of 1.489%; c wins on exposure.

Why the other options are wrong

A is incorrect. Used the annual PD without scaling to the term: $100 \times 4\% \times 90\% = 3.6$, the largest on that basis.

B is incorrect. Same term PD as c but a smaller exposure: USD 1.07 million.

D is incorrect. Largest exposure, but a 0.995% PD and 50% LGD give only USD 0.995 million.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1, eq. 9.2

Solution CR-9 e.7

Answer: B

New • built from GARP Ch.9, 9.2.1 and 9.8

Step 1 – Merton PD over the horizon

$$Q(3)=N(-d_2)=N(-1.60)=0.0548$$

$\Rightarrow$ A 5.48% risk-neutral probability of default by year 3.

Step 2 – Average hazard rate by the direct formula

$$\bar{\lambda}=-\frac{1}{3}\ln(1-0.0548)=\frac{0.05636}{3}=0.01879$$

$\Rightarrow$ 1.88% a year; check: $1-\mathrm{e}^{-3\times0.01879}=5.48\%$.

Interpretation

A CVA system needs a risk-neutral default intensity, which is what $N(-d_2)$ with the risk-free drift delivers. The 9% expected asset return would give a real-world PD, appropriate for scenario analysis but not for pricing, so it is a red herring here (CR-9 k).

Why the other options are wrong

A is incorrect. Split the PD linearly: $0.0548/3=1.83\%$.

C is incorrect. Entered the 3-year cumulative PD $N(-d_2) = 5.48\%$ as if it were the annual hazard rate (sibling swap).
D is incorrect. Took $-\ln(1-0.0548) = 5.64\%$ and forgot to divide by $T = 3$.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1 eq. 9.2; 9.8 Using Equity Prices to Estimate Default Probabilities

Solution CR-9 e.8 — Answer: B

Workbook A15.7 • S2, S3, S5

Step 1 – Average hazard from the cumulative PD (direct formula)

$$\bar{\lambda} = -\frac{1}{5}\ln(1-0.08) = \frac{1}{5}(0.08338) = 0.01668$$

$\Rightarrow$ An 8% five-year cumulative PD corresponds to a 1.67% per year default intensity.

Step 2 – Loss given default

$$1 - R = 1 - 0.45 = 0.55$$

Step 3 – Spread

$$s = \bar{\lambda}(1-R) = 0.01668 \times 0.55 = 0.00917 = 91.7 \text{ bp}$$

Interpretation

If the market quotes materially more than about 92 bp, either the market's risk-neutral PD exceeds the internal (closer to real-world) estimate, which Hull shows is normal, or the market sees deterioration the internal model has not captured.

Why the other options are wrong

A is incorrect. Multiplied by the recovery rate instead of LGD: $0.01668 \times 0.45 = 75.0$ bp.
C is incorrect. Stopped at the hazard rate, omitting LGD: 166.8 bp.
D is incorrect. Sibling swap: treated the five-year cumulative PD $1 - \mathrm{e}^{-5\lambda} = 8\%$ as if it were the hazard rate, $0.08 \times 0.55 = 4.40\% = 440$ bp.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1 eq. 9.2; 9.6.1 eq. 9.3

Solution CR-9 e.9 — Answer: D

Bank Q337 • Mock 1 Q56

Step 1 – Hazard rate from the 3-year quote

$$\bar{\lambda} \approx \frac{s}{1-R} = \frac{0.0375}{1-0.75} = 0.15$$

$\Rightarrow$ 15% a year. The 1-year and 2-year quotes give 15% too ($0.015/0.10$ and $0.0225/0.15$), consistent with a constant hazard.

Step 2 – Cumulative PD over three years

$$Q(3) = 1 - \mathrm{e}^{-0.15\times 3} = 1 - \mathrm{e}^{-0.45} = 1 - 0.63763 = 36.24\%$$

Interpretation

Spreads that look very different across maturities can imply the same hazard rate once each is divided by its own loss given default. The expected exposure of AUD 18 million is needed for a CVA, not for the probability.

Why the other options are wrong

A is incorrect. Used a one-year horizon: $1 - \mathrm{e}^{-0.15} = 13.93\%$.

B is incorrect. Reported the hazard rate itself, 15.00%, as the three-year probability (sibling swap).

C is incorrect. Used a two-year horizon: $1 - \mathrm{e}^{-0.30} = 25.92\%$.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2.1 eq. 9.2; 9.6.1 eq. 9.3

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 h

P1 CALC

Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates.

Workbook: A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: "hazard rate from CDS spread" came up in the August 2026 sitting (A15, Map #17). The LO asks how CDS spreads are used to estimate hazard rates, and the answer is one formula, $\bar{\lambda} = s/(1-R)$, so the questions are calculations.
Vardeez: says skip CDS here (covered later, CR-13). The August recall makes the spread-to-hazard step examinable, so these notes keep to that step and leave contract mechanics to CR-9 g and valuation to CR-13.

Reported exam shapes

- **A15·S3** Hazard rate from a CDS spread, $\bar{\lambda} = s/(1-R)$.

What the questions look like: a spread in basis points and a recovery rate (sometimes a bond yield and a risk-free rate instead of a CDS quote), and a request for the hazard rate, a survival or default probability over T years, or the hazard rate between two maturities. Notionals, exposures and risk-free rates are frequent red herrings.

Worked elsewhere: A15.7 (the spread implied by a cumulative PD) at Question CR-9 e.8; D5.2 (one-period PDs from bond yields) under CR-9 j. CR-9 e, Section 5 sets out the full chain from PD to hazard rate to spread.

1. What a CDS spread is

A credit default swap insures against default by a **reference entity**; a default is a **credit event** (failure to pay, restructuring, bankruptcy). The protection buyer makes periodic payments to the seller, usually quarterly in arrears, until the contract matures or a credit event occurs. The **CDS spread** is the total paid per year as a percentage of the notional principal. On a credit event the seller pays the buyer the notional minus the value of the cheapest deliverable bond (cash settlement, now usual), and the buyer pays a final accrual.

GARP's example. A five-year CDS with notional USD 100 million at 90 bp: the buyer pays about $0.25 \times 0.0090 \times 100{,}000{,}000 =$ USD 225,000 each quarter. If a credit event is notified on May 20, 2027 and the auction values the cheapest deliverable bond at USD 35 per 100, the seller pays USD 65 million and the buyer owes the accrual from March 20 to May 20 (about USD 150,000). A market maker quoting bid 250, ask 260 will buy protection at 250 bp a year and sell it at 260 bp. Five-year contracts are the most popular.

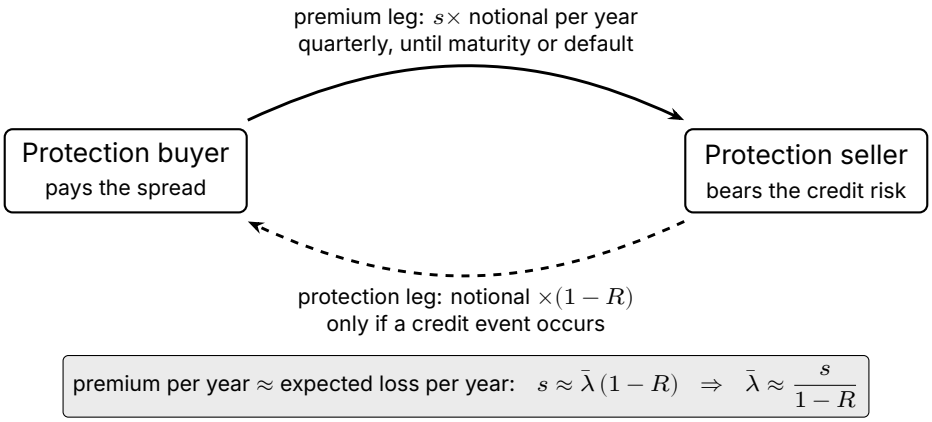


Figure 1. The two legs of a CDS. The premium leg (solid) runs from buyer to seller for as long as the reference entity survives; the protection leg (dashed) runs the other way and only on a credit event. Setting the yearly premium equal to the expected yearly loss gives the hazard rate.

2. From the spread to the hazard rate

A credit spread can be read as an **average loss rate**. If the five-year spread is 240 bp and the expected recovery rate is 40%, a holder of the credit risk must be expecting to lose about 2.4% a year from defaults. Each default costs $1-R = 60\%$ of the principal, so defaults must be arriving at $0.024/0.6 = 4\%$ a year: the average probability of default per year, conditional on no earlier default. In general (GARP eq. 9.3):

$$\boxed{\bar{\lambda}(T) = \frac{s(T)}{1-R}}$$

where $s(T)$ is the credit spread for maturity T (in theory continuously compounded), R the recovery rate and $\bar{\lambda}(T)$ the average hazard rate between time zero and T.

- **Which spreads.** The approximation works well for CDS spreads. It also works for bond yield spreads and asset swap spreads when the bond sells close to par. Otherwise GARP uses a more exact bond-pricing calculation (Section 9.6.2), which Vardeez says you can skip.
- **CDS spread versus bond yield spread.** Buying a five-year corporate bond yielding 7% and buying protection at 200 bp gives roughly a 5% risk-free bond, so the n-year CDS spread should be close to the corporate par yield minus the risk-free par yield. A bond yield spread can therefore stand in for s (Question CR-9 h.5); any gap is the CDS-bond basis (CR-9 i).
- **What the hazard rate is.** A risk-neutral default intensity, because it is backed out of a market price (CR-9 k). It feeds straight into the probability functions of CR-9 e: survival to T is $e^{-\bar{\lambda}T}$, and with 4% over five years the survival probability is $e^{-0.20} = 81.87\%$ and the cumulative PD 18.13%.

Recovery and the implied hazard. Hold the spread at 240 bp and vary R:

Recovery rate R	0%	20%	40%	60%	75%
Implied hazard rate $s/(1-R)$	2.40%	3.00%	4.00%	6.00%	9.60%

The same premium must cover the same expected loss. If each default costs less, defaults must be more frequent, so a higher recovery rate means a *higher* implied hazard rate (Question CR-9 h.3). With $R = 0$ the hazard rate equals the spread.

3. A term structure of hazard rates

Spreads quoted for several maturities give average hazard rates for each. Because survival probabilities multiply, $e^{-\bar{\lambda}(T_2)T_2} = e^{-\bar{\lambda}(T_1)T_1}\, e^{-\bar{\lambda}_{T_1,T_2}(T_2-T_1)}$, and the average hazard rate *between* the two maturities is

$$\bar{\lambda}_{T_1,T_2} = \frac{T_2\,\bar{\lambda}(T_2) - T_1\,\bar{\lambda}(T_1)}{T_2 - T_1}$$

GARP Example 9.3. CDS spreads of 50, 60 and 100 bp at 3, 5 and 10 years, $R = 60\%$: average hazard rates $0.005/0.4 = 1.25\%$, $0.006/0.4 = 1.5\%$ and $0.010/0.4 = 2.5\%$. Between years 3 and 5: $(5 \times 0.015 - 3 \times 0.0125)/2 = 1.875\%$. Between years 5 and 10: $(10 \times 0.025 - 5 \times 0.015)/5 = 3.5\%$. Vardeez's version (80, 90 and 110 bp, $R = 65\%$) gives 2.29%, 2.57% and 3.14%, and 3.00% for years 3 to 5 with unrounded inputs (2.99% from the rounded ones).

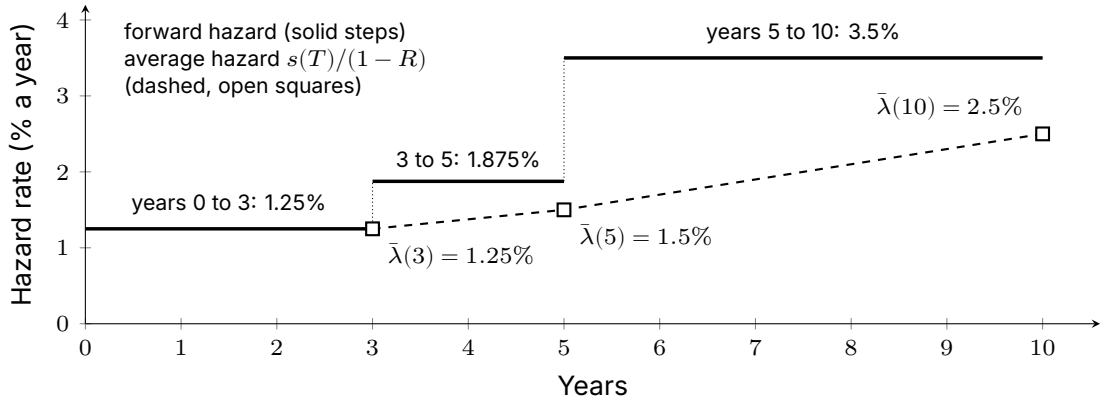


Figure 2. GARP Example 9.3. The average hazard rates (dashed) rise gently with maturity; the hazard rates between maturities (solid steps) rise much faster, because each average is a time-weighted blend of everything before it. An inverted spread curve works the other way: the forward hazard falls below the average (Question CR-9 h.7).

Formula card: CR-9 h

CDS spread	Payment per year as a % of notional; quarterly payment $= 0.25 \times s \times$ notional
Hazard from a spread (eq. 9.3)	$\bar{\lambda}(T) = s(T)/(1-R)$; reverse: $s = \bar{\lambda}(1-R)$
Units	1 bp $= 0.0001$; 180 bp $= 0.0180$
Survival and PD over T	survival $e^{-\bar{\lambda}T}$; cumulative PD $1 - e^{-\bar{\lambda}T}$
Hazard between maturities	$\bar{\lambda}_{T_1,T_2} = (T_2\bar{\lambda}(T_2) - T_1\bar{\lambda}(T_1))/(T_2 - T_1)$
CDS spread and bonds	$s \approx$ corporate par yield $-$ risk-free par yield (bond near par)

Exam traps

- **Basis-point conversion.** 180 bp is 0.0180, not 0.0018.
- **Recovery versus LGD.** Divide by $1 - R$, not by R; and divide, do not multiply.
- **Recovery direction.** For a given spread, higher recovery means a higher hazard rate, not a

lower one.

- **Hazard rate is not a probability over T.** A 4% hazard rate gives a five-year PD of $1 - e^{-0.2}$, not 4% and not 20%.
- **Forward hazard.** Weight each average by its maturity and divide by $T_2 - T_1$. A simple average of the two hazard rates, or their plain difference, is wrong.
- **Raw spreads bootstrapped.** Bootstrapping spreads without dividing by $1 - R$ gives a forward loss rate, not a hazard rate.
- **Stressed versus base inputs.** When a stem gives two sets of spread and recovery, use the set the question asks about.
- **Red herrings.** Notional, loan size, expected exposure and the risk-free rate do not enter $s/(1 - R)$.

Practice questions: CR-9 h

Question CR-9 h.1

Workbook A15.5 • S3 • Aug-2026 • Map #17

A treasury analyst at a Canadian bank hedging a loan to a mining company observes:

- Five-year CDS spread on the company: 180 basis points
- Expected recovery rate: 35%
- Loan notional: CAD 75 million; five-year risk-free rate: 3.2%

Which of the following is closest to the average hazard rate implied by the CDS spread over the five years?

A. 0.28%

B. 1.17%

C. 2.77%

D. 5.14%

Question CR-9 h.2

Bank Q350 • Mock 2 Q6 (rebuilt)

A credit analyst notes that the credit spread on a one-year B-rated corporate bond relative to the maturity-matched Treasury bill is 2.2%. The expected recovery rate in default is 75%. Using the approximation that the spread equals the hazard rate times the loss given default, what is the hazard rate (default intensity) of the bond?

A. 0.55%

B. 2.2%

C. 2.9%

D. 8.8%

Question CR-9 h.3

Bank Q352 • Mock 1 Q19

A risk manager values a credit default swap by computing default probabilities from the market CDS spread. Holding the market CDS spread constant, the risk-neutral probability of default (hazard rate) will be:

A. unrelated to the recovery rate.

B. proportional to the notional principal.

C. positively related to the recovery rate.

D. negatively related to the recovery rate.

Question CR-9 h.4

Bank Q347 • GARP PE2 Q62 • Mock 4 Q63

A bond issued by Company GRN trades at a spread of 250 bp over the risk-free rate with a recovery rate of 75%. A risk analyst applies a constant hazard rate model and assumes that under a stress scenario the spread widens to 480 bp and the recovery rate falls to 40%. Under the stress scenario, what is the probability that GRN does not default over the next 3 years?

A. 69.8%

B. 74.1%

C. 78.7%

D. 86.6%

Question CR-9 h.5

Bank Q348 • AP Mock 1 Q23 (rebuilt)

A 2-year corporate bond yields 7% and the comparable risk-free rate is 3%. The expected recovery rate is 40%. Assume the average hazard rate is constant and estimated by $\bar{\lambda} \approx s/(1-R)$, with survival to time t equal to $e^{-\bar{\lambda}t}$. What is the unconditional (marginal) probability that default occurs during the second year?

A. 0.0603

B. 0.0645

C. 0.0667

D. 0.1248

Question CR-9 h.6

Bank Q351 • BT P2.T6.308.1 (rebuilt)

The 3-year and 5-year CDS spreads for a company are 90 and 120 basis points, and the expected recovery rate is 40%. Using the approximation that the average hazard rate equals the spread divided by $(1-R)$, what is the estimated average hazard rate between year 3 and year 5?

A. 0.50%

B. 1.65%

C. 1.75%

D. 2.75%

Question CR-9 h.7 Workbook A15.6 • S3 • Map #17

A credit derivatives trader at a Hong Kong dealer (USD contracts) observes the following CDS spreads on a reference entity, with an expected recovery rate of 40%:

Maturity	3 years	5 years
CDS spread (bp)	400	350

Which of the following is closest to the average hazard rate between year 3 and year 5?

A. 1.83%

B. 2.75%

C. 4.58%

D. 5.83%

Solutions • CR-9 h

Solution CR-9 h.1

Answer: C

Workbook A15.5 • S3 • Aug-2026

Step 1 – Convert the spread

$$s = 180\text{ bp} = 180 \times 0.0001 = 0.0180$$

Step 2 – Loss given default

$$1 - R = 1 - 0.35 = 0.65$$

Step 3 – Average hazard rate

$$\bar{\lambda} = \frac{0.0180}{0.65} = 0.0277$$

$\Rightarrow$ The market prices a 2.77% per year (risk-neutral) default intensity, conditional on survival.

Interpretation

Protection buyers pay 1.80% a year because the market expects to lose 65% of notional on a default arriving at about 2.77% a year. The notional and risk-free rate are not needed.

Why the other options are wrong

A is incorrect. Converted 180 bp as 0.0018: $0.0018/0.65 = 0.28\%$.
B is incorrect. Multiplied by LGD instead of dividing: $0.018 \times 0.65 = 1.17\%$.
D is incorrect. Divided by the recovery rate instead of LGD: $0.018/0.35 = 5.14\%$.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1 eq. 9.3

Solution CR-9 h.2

Answer: D

Bank Q350 • Mock 2 Q6

Step 1 – The credit triangle

$s \approx \bar{\lambda}(1 - R)$, so $\bar{\lambda} = s/(1 - R)$.

Step 2 – Plug in

$$\bar{\lambda} = \frac{0.022}{1 - 0.75} = \frac{0.022}{0.25} = 0.088 = 8.8\%$$

$\Rightarrow$ Each default costs only 25% of the principal, so a 2.2% spread needs defaults at 8.8% a year to be fair.

Interpretation

With high recovery the hazard rate is a large multiple of the spread (four times here). The bond's B rating is consistent with a hazard rate of this size.

Why the other options are wrong

A is incorrect. Multiplied the spread by the LGD instead of dividing: $0.022 \times 0.25 = 0.55\%$.
B is incorrect. Took the spread itself, which is the hazard rate only if recovery is zero.
C is incorrect. Divided by the recovery rate instead of the LGD: $0.022/0.75 = 2.9\%$.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1 eq. 9.3

Solution CR-9 h.3 Answer: C

Bank Q352 • Mock 1 Q19

Step 1 – Write the relationship

$\bar{\lambda} \approx s/(1-R)$: the recovery rate sits in the denominator through the LGD.

Step 2 – Hold s fixed and raise R

A higher R shrinks $1-R$, so $\bar{\lambda}$ rises. With $s = 2\%$: $R = 40\%$ gives 3.33%; $R = 60\%$ gives 5.0%.
$\Rightarrow$ The hazard rate is positively related to the recovery rate.

Interpretation

A fixed spread must pay for the same expected loss per year. If less is lost on each default, the market must be expecting more defaults.

Why the other options are wrong

A is incorrect. The recovery rate enters through $1-R$, so it matters.
B is incorrect. The notional scales both the premium and the payoff and cancels out of the hazard rate.
D is incorrect. Intuitive but reversed: higher recovery lowers the loss per default, so more defaults are implied by the same spread.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1 eq. 9.3

Solution CR-9 h.4 Answer: C

Bank Q347 • GARP PE2 Q62 • Mock 4 Q63

Step 1 – Hazard rate under the stress inputs

$$\bar{\lambda} = \frac{0.0480}{1-0.40} = 0.08$$

Step 2 – Three-year survival

$$e^{-0.08\times 3} = e^{-0.24} = 0.7866 = 78.7\%$$

$\Rightarrow$ Under stress there is a 78.7% chance GRN survives three years (a 21.3% cumulative PD).

Interpretation

The stress works through both inputs: the spread almost doubles and the LGD rises from 25% to 60%. The base-case inputs (250 bp, 75% recovery) are a red herring for the stressed answer.

Why the other options are wrong

A is incorrect. Divided the spread by R instead of $1-R$: $\bar{\lambda} = 0.048/0.40 = 12\%$, survival $e^{-0.36} = 69.8\%$.
B is incorrect. Used the unstressed inputs: $\bar{\lambda} = 0.025/0.25 = 10\%$, survival $e^{-0.30} = 74.1\%$.
D is incorrect. Ignored recovery and used the stressed spread as the hazard rate: $e^{-0.144} = 86.6\%$.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1 eq. 9.3; 9.2.1 eq. 9.2

Solution CR-9 h.5

Answer: A

Bank Q348 • AP Mock 1 Q23

Step 1 – Spread and hazard rate

$$s = 7\% - 3\% = 4\%, \qquad \bar{\lambda} = \frac{0.04}{1 - 0.40} = 0.0667$$

$\Rightarrow$ The bond yield spread stands in for the CDS spread.

Step 2 – Survival to the end of years 1 and 2

$$\mathrm{e}^{-0.0667} = 0.9355, \qquad \mathrm{e}^{-0.1333} = 0.8752$$

Step 3 – Unconditional PD in year 2

$$0.9355 - 0.8752 = 0.0603$$

Interpretation

The unconditional PD for year 2 is the drop in survival across the year, both measured from today. Conditioning on survival to year 1 would give $1 - \mathrm{e}^{-0.0667} = 0.0645$.

Why the other options are wrong

B is incorrect. Computed the conditional PD given survival to year 1, $1 - \mathrm{e}^{-\bar{\lambda}} = 0.0645$.

C is incorrect. Reported the hazard rate itself, 0.0667, which is a rate, not the probability of default in year 2.

D is incorrect. Reported the two-year cumulative PD, $1 - 0.8752 = 0.1248$, which includes year 1.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.5.1, 9.6.1 eq. 9.3; 9.2.1 eq. 9.2

Solution CR-9 h.6

Answer: D

Bank Q351 • BT P2.T6.308.1

Step 1 – Average hazard rates

$$\bar{\lambda}(3) = \frac{0.0090}{0.60} = 1.50\%, \qquad \bar{\lambda}(5) = \frac{0.0120}{0.60} = 2.00\%$$

Step 2 – Cumulative hazards

$$5 \times 0.0200 = 0.1000, \qquad 3 \times 0.0150 = 0.0450$$

Step 3 – Hazard between years 3 and 5

$$\bar{\lambda}_{3,5} = \frac{0.1000 - 0.0450}{5 - 3} = 2.75\%$$

$\Rightarrow$ A name that survives to year 3 is expected to default at 2.75% a year over years 4 and 5.

Interpretation

An upward-sloping spread curve means a forward hazard above both averages: the five-year average of 2% blends three years at 1.5% with two years at 2.75%.

Why the other options are wrong

A is incorrect. Took the plain difference of the two averages, $2.00 - 1.50 = 0.50\%$, with no maturity weights.
B is incorrect. Bootstrapped the raw spreads without dividing by $1 - R$: $(5 \times 1.20\% - 3 \times 0.90\%)/2 = 1.65\%$.
C is incorrect. Took the simple average of 1.50% and 2.00%, ignoring the maturity weights.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1, Example 9.3

Solution CR-9 h.7

Answer: C

Workbook A15.6 • S3

Step 1 – Three-year average hazard

$$\bar{\lambda}(3) = \frac{0.0400}{1 - 0.40} = 0.0667$$

Step 2 – Five-year average hazard

$$\bar{\lambda}(5) = \frac{0.0350}{0.60} = 0.0583$$

Step 3 – Bootstrap the forward hazard

$$\bar{\lambda}_{3,5} = \frac{5(0.0583) - 3(0.0667)}{5 - 3} = \frac{0.2917 - 0.2000}{2} = 0.0458$$

$\Rightarrow$ An issuer that survives to year 3 is expected to default at about 4.58% a year over years 4 and 5.

Interpretation

An inverted CDS curve (shorter spread higher) means the market sees near-term distress: if the firm survives the next three years, its default intensity falls. This mirrors the book's observation that poorly rated issuers can have decreasing marginal PDs.

Why the other options are wrong

A is incorrect. Divided the bootstrapped difference by $T_2 = 5$ instead of $T_2 - T_1 = 2$: $0.0917/5 = 1.83\%$.
B is incorrect. Bootstrapped the raw spreads without dividing by LGD: $(5 \times 0.035 - 3 \times 0.040)/2 = 2.75\%$, a forward loss rate, not a hazard rate.
D is incorrect. Reported the five-year average hazard $\bar{\lambda}(5) = 5.83\%$ (intermediate result).

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1, Example 9.3

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 k

P1 CONCEPT

Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk.

Workbook: A5 Merton Model, Distance to Default and Risk-Neutral PD • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: Merton came up in the August 2026 sitting, and "Merton model risk-neutral prob" is one of the A5 variants on Map #2 (8 of 16 sources).
Vardeez: a three-row comparison ending in a decision rule: risk-neutral for valuation, real-world for scenario analysis.

Reported exam shapes

- **A5·S3** Risk-neutral PD $N(-d_2)$ versus the real-world PD with μ.

What the questions look like: conceptual. Which estimate is higher and why, where each comes from, and which one a given task needs. **One numerical question is kept outside the TYPE rule:** Workbook A5.3 computes a real-world Merton PD (Question CR-9 k.6); workbook questions always go in. The Merton mechanics are in CR-9 l and CR-9 m.

Worked elsewhere: A15.4 (historical versus spread-implied hazard rates) under CR-9 j.

1. Two kinds of default probability

- **Real-world** (physical) default probabilities are estimated from **historical data**: the default experience of issuers with a given rating (GARP Table 9.1, CR-9 d), or a model calibrated to actual default frequencies. They answer "how often do defaults actually happen?"
- **Risk-neutral** default probabilities are implied by **market prices**: credit spreads, CDS quotes and bond prices (the hazard rate $s/(1-R)$ of CR-9 h). They are the probabilities that make the market price equal to expected cash flows discounted at the risk-free rate.

Risk-neutral default probabilities are **higher** than real-world ones. Merton's model can produce either, depending on how it is calibrated.

2. Why risk-neutral PDs are higher

Merton's explanation (shape S3). In a risk-neutral world the expected growth rate of the firm's assets is the risk-free rate r. In the real world it is usually higher, $\mu = r+$ a risk premium. Assets that drift up more slowly are more likely to finish below the face value of the debt, so the probability of default is higher in the risk-neutral world. In the formula, only the drift changes:

$$\text{PD} = N\left(-\frac{\ln(V_0/D) + (\text{drift} - \sigma_V^2/2)T}{\sigma_V\sqrt{T}}\right) \qquad \text{drift} = r \text{ (risk-neutral)}, \quad \mu \text{ (real-world)}$$

Example: $V_0 = 100$, $D = 80$, $\sigma_V = 25\%$, $T = 1$, $r = 3\%$, $\mu = 10\%$. Risk-neutral: $N(-0.888) = 18.7\%$. Real-world: $N(-1.168) = 12.1\%$. $N(-d_2)$ from the option pricing model is in theory a risk-neutral PD; a real-world PD needs an estimate of μ (CR-9 m, Figure 1).

The market's explanation. Bond investors are paid more than the expected cost of defaults, so the spread implies more default than history shows. GARP's reasons, in order of weight: defaults are **correlated** through the economy, which creates systematic risk that cannot be diversified away and

earns a premium ("by far the most important reason"); bond returns are skewed and their idiosyncratic risk is hard to diversify; corporate bonds are **illiquid** (perhaps 25 bp in normal markets); and traders may allow for depression scenarios worse than the data. The comparison itself is CR-9 j.

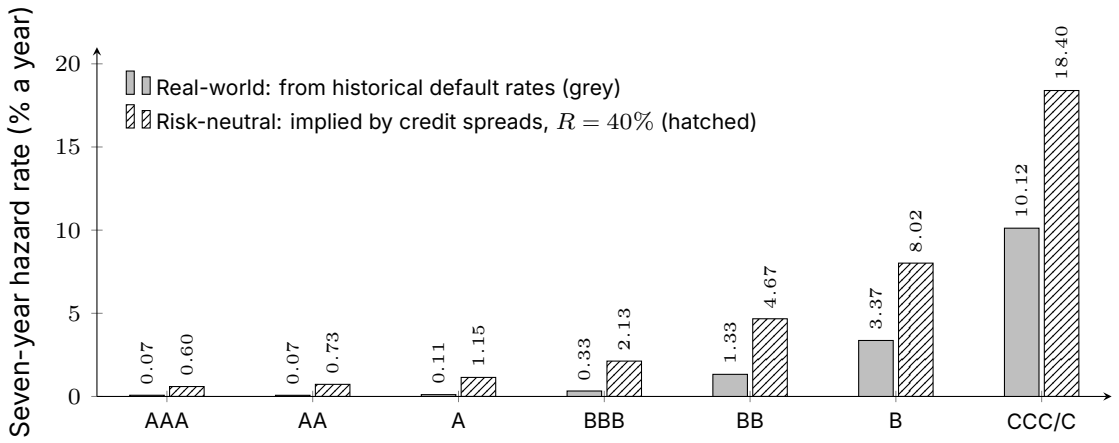


Figure 1. Average seven-year hazard rates from GARP Table 9.5 (default rates 1981 to 2020; spreads 1996 to 2007). The risk-neutral rate is higher for every rating. The absolute gap widens as quality falls (1.04 points for A, 8.28 for CCC/C); the ratio is largest for high grades (about 10 times for AA and A, under 2 times for CCC/C).

Correction to the source notes. GARP Table 9.5 (as reproduced in the course text) prints the CCC/C difference as 8.207. With the table's own figures, $18.395 - 10.118 = 8.277$; the other six differences check. Figure 1 uses 8.28.

3. Which one to use

- **Valuation: risk-neutral.** Pricing a bond, a CDS or a derivative with counterparty risk (CVA) uses risk-neutral default probabilities, the same way risk-neutral estimates of other variables are used: estimate the default losses with the spread-implied PDs and discount at the risk-free rate. The price must agree with the market.
- **Scenario analysis: real-world.** Estimating how much might actually be lost (stress tests, expected losses on a loan book, the scenarios behind credit VaR and economic capital) uses real-world probabilities from historical data.
- **Credit VaR uses both.** GARP's CR-10 reading: VaR is a scenario analysis measure, so the default probabilities that generate the scenarios to the horizon are real-world; the portfolio is then *valued* on the horizon date with risk-neutral estimates.
- **Merton either way.** Calibrated with r it gives a risk-neutral PD; with μ, a real-world one. Moody's KMV and Kamakura turn the Merton output into real-world PDs (CR-9 n).

Vardeez's decision rule: if the question is what something is *worth*, use risk-neutral; if it is what is likely to *happen*, use real-world.

	Real-world (physical)	Risk-neutral
Source	Historical default data, ratings tables	Market prices: credit spreads, CDS, bonds
Typical estimate	$\bar{\lambda} = -\ln(1-Q)/T$ from a cumulative default rate	$\bar{\lambda} = s/(1-R)$ from a spread
Merton drift	Expected asset return μ	Risk-free rate r
Size	Lower	Higher: includes compensation for bearing default risk
Use	Scenario analysis, stress tests, credit VaR scenarios	Valuation: bonds, CDS, CVA, horizon revaluation in credit VaR
Gap	Grows with credit risk in absolute terms; rises when spreads are high (crises)	

Exam traps

- **Labels swapped.** "Market-based so real-world" is wrong: anything implied by prices is risk-neutral; historical frequencies are real-world.
- **Ranking reversed.** Risk-neutral PDs are higher, not lower; the risk-neutral drift is lower.
- **Uses swapped.** Risk-neutral for valuation, real-world for scenario analysis, never the reverse.
- **Drift.** A real-world Merton PD needs μ; a risk-neutral one keeps r. A stem listing both is testing the choice.
- **Equity volatility in d_2.** σ_E only calibrates σ_V; it never goes into the PD.
- **Gap read as arbitrage or mispricing.** The excess of risk-neutral over real-world PDs is a risk premium, not a free lunch, and not a forecast of actual defaults.
- **Gap direction across ratings.** The absolute gap grows as quality falls; the ratio is what shrinks.

Practice questions: CR-9 k

Question CR-9 k.1

Bank Q361 • AnalystPrep Q.5989 (rebuilt)

An analyst estimates a firm's default probability with the Merton model and compares a risk-neutral calculation with a real-world calculation, holding the face value of debt, asset volatility and maturity constant. Which statement correctly describes the risk-neutral calculation?

A. The firm's assets are assumed to grow at the risk-free rate, so the default probability is higher than in the real-world calculation.

B. The firm's assets are assumed to grow at the risk-free rate plus a market risk premium, so the default probability is higher than in the real-world calculation.

C. The firm's assets are assumed to grow at the risk-free rate, so the default probability is lower than in the real-world calculation.

D. The firm's assets are assumed to grow at the risk-free rate plus a market risk premium, so the default probability is lower than in the real-world calculation.

Question CR-9 k.2 Bank Q362 • Schweser Module Quiz 27.3 Q1 (rebuilt)

A junior analyst is drafting a research report and addresses two questions: (1) Which type of default probabilities should be used in valuation? (2) What is the effect on the likelihood of a company defaulting when its distance to default rises? Which conclusion should the report support?

A. Real-world default probabilities for valuation; default becomes less likely as distance to default rises.

B. Risk-neutral default probabilities for valuation; default becomes less likely as distance to default rises.

C. Risk-neutral default probabilities for valuation; default becomes more likely as distance to default rises.

D. Real-world default probabilities for valuation; default becomes more likely as distance to default rises.

Question CR-9 k.3 Bank Q364 • BT P2.T6.327.2

While there are important exceptions, which is the best general advice in regard to the selection of real-world or risk-neutral probabilities in credit exposure models?

A. Exposure management (e.g., scenario generation) should focus on risk-neutral, market-implied parameters, but pricing should focus on real-world, historical parameters, especially when counterparty risk is actively managed.

B. Exposure management (e.g., scenario generation) should focus on real-world, historical parameters, but pricing should focus on risk-neutral, market-implied parameters, especially when counterparty risk is actively managed.

C. Both exposure management and pricing should focus on risk-neutral, market-implied parameters.

D. Both exposure management and pricing should focus on real-world, historical parameters.

Question CR-9 k.4 Bank Q363 • BT P2.T6.24.14.2 • BT notes P2.T6.24.15.2 (rebuilt: loss figures removed)

ACME Financial holds a USD 10 million exposure to Zigzag Corp. Zigzag's historical (real-world) default rate is 2% per year, while the risk-neutral default probability implied by its CDS spreads is 3% per year. ACME wants to (i) value a CDS on Zigzag and (ii) estimate the expected one-year credit loss on the exposure in a scenario analysis of its portfolio. Which approach is appropriate?

A. Use the 3% risk-neutral probability for both, because market-implied estimates are the most up to date.

B. Use the 2% real-world probability for the CDS valuation and the 3% risk-neutral probability for the scenario analysis.

C. Use the 3% risk-neutral probability for the CDS valuation and the 2% real-world probability for the scenario analysis.

D. Use the 2% real-world probability for both, because it reflects Zigzag's actual default experience.

Question CR-9 k.5 New • built from GARP Ch.9, 9.7, Table 9.5 and 9.7.2

A credit portfolio manager at a Dutch insurer compares two estimates for BB-rated issuers. The average seven-year hazard rate implied by BB credit spreads (with a 40% recovery rate) is 4.7% a year; the rate implied by the seven-year historical default experience of BB issuers is 1.3% a year. A colleague concludes that the market is mispricing BB bonds. Which of the following statements is correct?

A. The 4.7% is a real-world estimate because it comes from current market prices, while the 1.3% is a stale risk-neutral estimate that should be discarded.

B. The gap is an arbitrage: an investor can buy the bonds, buy CDS protection, and lock in a riskless profit equal to the difference in hazard rates.

C. The gap should be smaller for B and CCC/C issuers, because riskier bonds are priced closer to their actuarial cost of default.

D. The 4.7% is a risk-neutral estimate that includes compensation for bearing default risk, mainly systematic risk, so it should be used to value the bonds but not to forecast how many BB issuers will actually default.

Question CR-9 k.6 Workbook A5.3 • S3 • Map #2 • numerical, kept outside the TYPE rule

A senior risk analyst at a Japanese bank is preparing a one-year stress scenario and needs the real-world default probability of a corporate borrower. Merton's model has been calibrated as follows (JPY billion):

- Value of assets: 800; asset volatility: 20%; equity volatility: 45%
- Face value of debt due in 1 year: 600
- Risk-free rate: 4%; expected return on assets: 10% (both continuously compounded)

Which of the following is closest to the probability of default to use?

A. 2.1%

B. 3.3%

C. 6.2%

D. 26.2%

Solutions • CR-9 k

Solution CR-9 k.1 — Answer: A

Bank Q361 • AnalystPrep Q.5989

Step 1 – The drift in each world

Risk-neutral: assets are expected to grow at the risk-free rate (no premium for bearing risk). Real world: at the risk-free rate plus a market risk premium.

Step 2 – Effect on the probability of default

With the same face value of debt, a lower drift leaves the assets more likely to end below the debt at T.

$\Rightarrow$ The risk-neutral PD is higher than the real-world PD.

Interpretation

This is GARP's own explanation (9.8.2) of why risk-neutral default probabilities exceed real-world ones. In d_2 only the drift changes: r for risk-neutral, μ for real-world.

Why the other options are wrong

B is incorrect. Attaches the real-world drift to the risk-neutral calculation; that higher drift would lower the PD, not raise it.

C is incorrect. Right drift, wrong ranking: slower asset growth raises the PD.

D is incorrect. Describes the real-world calculation and labels it risk-neutral.

LO: CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.2 Real-World vs. Risk-Neutral Default Probabilities

Solution CR-9 k.2 — Answer: B

Bank Q362 • Schweser Module Quiz 27.3 Q1

Step 1 – Valuation needs risk-neutral probabilities

Valuation discounts at the risk-free rate with probabilities consistent with market prices: default rates backed out of credit spreads.

Step 2 – Distance to default and PD

DD is the number of standard deviations the assets sit above the default point; PD $= N(-\text{DD})$ falls as DD rises.

$\Rightarrow$ Risk-neutral for valuation; default becomes less likely as DD rises.

Interpretation

Real-world estimates from historical data belong to scenario analysis. Both parts of the question are standard pairings that GARP likes to split across two options.

Why the other options are wrong

A is incorrect. Real-world probabilities belong to scenario analysis, not valuation.

C is incorrect. Right measure, but a higher DD lowers the default probability.

D is incorrect. Wrong measure and wrong direction.

LO: CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7.2 and 9.8.3

Solution CR-9 k.3 Answer: B

Bank Q364 • BT P2.T6.327.2

Step 1 – Pricing

A price must be consistent with market prices, so pricing uses risk-neutral, market-implied parameters (for example, PDs from credit spreads).

Step 2 – Exposure management

Scenario generation for exposure management aims at realistic outcomes, so it uses real-world, historical parameters.

$\Rightarrow$ Only B assigns each measure to its proper use.

Interpretation

The split mirrors GARP's valuation versus scenario analysis rule. When counterparty risk is actively managed (hedged), the price of that risk must match the market, which strengthens the case for risk-neutral pricing.

Why the other options are wrong

A is incorrect. Swaps the two uses.

C is incorrect. Risk-neutral parameters overstate the likelihood of default for scenario purposes.

D is incorrect. Real-world parameters give prices inconsistent with the market.

LO: CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7.2 Valuation vs. Scenario Analysis

Solution CR-9 k.4 Answer: C

Bank Q363 • BT P2.T6.24.14.2

Step 1 – CDS valuation

Valuation uses the risk-neutral probability implied by the market, 3%, so the CDS value agrees with traded spreads.

Step 2 – Scenario analysis

An expected loss in a scenario analysis should reflect how often defaults actually happen: the real-world 2%.

$\Rightarrow$ 3% for the CDS, 2% for the scenario analysis.

Interpretation

The two numbers are not competing estimates of the same thing. The 3% contains the price of bearing default risk; the 2% is the actuarial frequency. (For reference, the scenario expected loss would be $10\text{m} \times 2\% \times 60\% = \text{USD } 120{,}000$ at a 40% recovery rate.)

Why the other options are wrong

A is incorrect. Applies the risk-neutral rate to scenario analysis too, overstating the likely loss.

B is incorrect. Reverses the two uses.

D is incorrect. A CDS valued with the historical 2% would be inconsistent with market spreads.

LO: CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7.2

Solution CR-9 k.5 — Answer: D

New • built from GARP Ch.9, 9.7, Table 9.5 and 9.7.2

Step 1 – Label the two estimates

The 4.7% is backed out of market spreads with $s/(1-R)$: risk-neutral. The 1.3% comes from historical default experience: real-world. (These are GARP's Table 9.5 BB figures, 4.671% and 1.330%.)

Step 2 – Explain the gap

Bond investors are compensated for more than the expected cost of default, above all for systematic risk because defaults are correlated through the economy, plus illiquidity and hard-to-diversify idiosyncratic risk.

$\Rightarrow$ The gap is a risk premium, not evidence of mispricing.

Step 3 – Decide the use

Value the bonds with the risk-neutral 4.7%; forecast actual defaults (scenario analysis) with the real-world 1.3%.

Interpretation

A risk-neutral PD is not a forecast. Reading the 4.7% as the expected share of BB issuers defaulting each year would overstate losses by a factor of about 3.5.

Why the other options are wrong

A is incorrect. Labels reversed: market-implied estimates are risk-neutral; historical frequencies are real-world.

B is incorrect. Buying the bond and buying protection hedges the default risk away and earns roughly the risk-free rate; the gap between hazard rates is a premium for bearing risk, not a riskless profit.

C is incorrect. Polarity flip: in Table 9.5 the gap widens as quality falls (B 4.65 points, CCC/C 8.28) because excess returns rise with credit risk.

LO: CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7 Comparison of Default Probability Estimates, Table 9.5; 9.7.1; 9.7.2

Solution CR-9 k.6 — Answer: B

Workbook A5.3 • S3

Step 1 – Choose the drift

Scenario analysis uses real-world estimates (Hull 9.7.2), so the drift is $\mu = 10\%$, not r (Doumpos eq. 5.3).

Step 2 – Real-world distance to default

$$\text{DD} = \frac{\ln(800/600) + (0.10 - 0.20^2/2)(1)}{0.20\sqrt{1}} = \frac{0.2877 + 0.0800}{0.20} = 1.8384$$

Step 3 – Real-world PD

$$\text{PD} = N(-1.8384) = 0.0330$$

$\Rightarrow$ A 3.3% chance, in the real world, that assets end the year below the JPY 600 billion owed.

Interpretation

The risk-neutral value with r would be 6.2%: higher, because assets grow only at r in the risk-neutral world (Hull 9.8.2). The 45% equity volatility is not an input here: it was used to calibrate

σ_V.

Why the other options are wrong

A is incorrect. Correct drift μ, but used $N(-d_1)$: $d_1 = 1.8384 + 0.20 = 2.0384$, $N(-2.0384) = 0.021$.

C is incorrect. Used $r = 4\%$ instead of μ: DD $= (0.2877 + 0.02)/0.20 = 1.5384$, PD $= 0.062$, a risk-neutral PD.

D is incorrect. Plugged equity volatility 45% in place of asset volatility: DD $= (0.2877 + 0.10 - 0.1013)/0.45 = 0.637$, PD $= 0.262$.

LO: CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. • **Reference:** Doumpos et al., Ch.1, 5.6.3.1 eq. 5.3; Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7.2 and 9.8.2

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 I

P1 CALC

Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model.

Workbook: A5 Merton Model, Distance to Default and Risk-Neutral PD • Tier A • August 2026 recall D5 Credit spread from debt value • Tier D

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: Merton came up in the August 2026 sitting, and Map #2 (8 of 16 sources) reports "Merton model, calculation of V_0 and sigma(V)" and "leverage ratio $\rightarrow$ ROE". D5 (Tier D, Map #42) adds the credit spread from the debt value.
Vardeez: Merton. The notes give the equity formula; the volatility link, the simultaneous solve and the worked example were filled in from GARP.

Reported exam shapes

- **A5·S2** Back out V_0 and σ_V from the simultaneous equations using the solver's converged values.
- **A5·S5** Leverage ratio to return on equity through the delta $N(d_1)$.
- **D5·S1** Credit spread from the market value of debt, with the card's inverted sign as the distractor.

What the questions look like: numerical. Given V_0, D, T, r and σ_V (often with $N(d_1)$ and $N(d_2)$ supplied, or a put price), value the equity and the debt; back out σ_V from $d_1 - d_2$ or from E_0 and σ_E; turn a debt value into a spread. Equity volatility and the expected asset return are frequent red herrings.

Worked elsewhere: D5.1 (spread from a Merton debt value) at Question CR-12 c.4; distance to default and PD in CR-9 m; Merton's limitations and Moody's KMV in CR-5 e.

1. Equity as a call on the firm's assets

Merton (1974): the firm has one zero-coupon bond with face value D due at T. Write V_0, V_T for the asset value today and at T, E_0, E_T for the equity, σ_V for the (constant) asset volatility and σ_E for the equity volatility.

- If $V_T < D$ it is rational to default: the creditors take the assets and the equity is worth zero. If $V_T > D$ the debt is repaid and the equity is worth $V_T - D$. So $E_T = \max(V_T - D,\, 0)$: **equity is a call on the assets** with strike D.
- The creditors receive $\min(V_T, D) = D - \max(D - V_T,\, 0)$: **a risk-free bond minus a put** on the assets. The shareholders hold the put's other side: limited liability lets them hand over the assets instead of paying D.

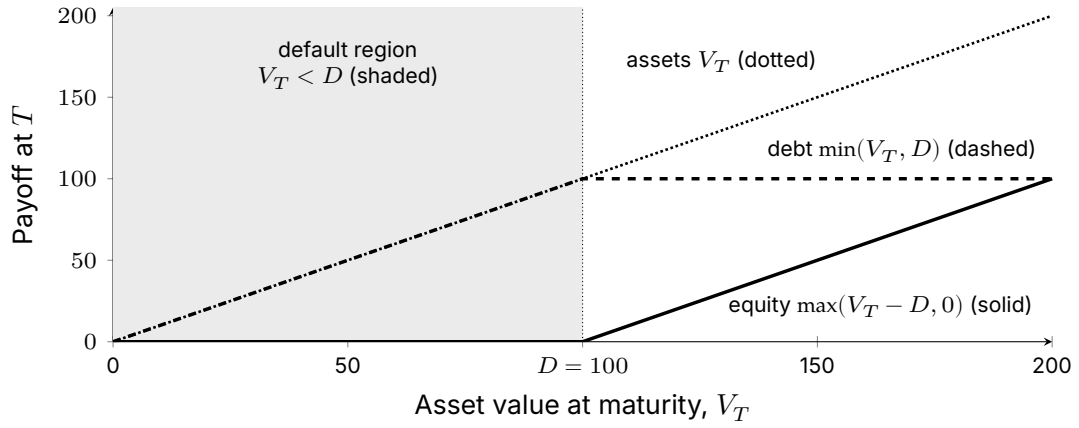


Figure 1. Payoffs at maturity with $D = 100$. Equity (solid) is a call struck at D; debt (dashed) is capped at D and falls one-for-one with the assets in the default region (shaded); the two add up to the assets (dotted).

2. Valuing the equity and the debt today

Black-Scholes-Merton with no dividends (GARP eq. 9.4):

$$E_0 = V_0 N(d_1) - D\mathrm{e}^{-rT} N(d_2), \qquad d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V\sqrt{T}}, \qquad d_2 = d_1 - \sigma_V\sqrt{T}$$

$$D_0 = V_0 - E_0 = D\mathrm{e}^{-rT} - \text{put} = D\mathrm{e}^{-rT} N(d_2) + V_0 N(-d_1)$$

The expected loss on the debt, as a share of its no-default value, is $(D\mathrm{e}^{-rT} - D_0)/D\mathrm{e}^{-rT}$.

Worked example. $V_0 = 150$, $D = 100$ due in $T = 2$ years, $r = 4\%$, $\sigma_V = 30\%$.

Step	Working	Result
d_1, d_2	$(\ln 1.5 + (0.04 + 0.045) \times 2)/(0.30\sqrt{2})$; $d_1 - 0.4243$	1.3564, 0.9321
$N(d_1), N(d_2)$	standard normal	0.9125, 0.8244
PV of the promised payment	$100\,\mathrm{e}^{-0.08}$	92.31
Equity E_0	$150(0.9125) - 92.31(0.8244)$	60.78
Debt D_0	$150 - 60.78$ (check: put $= 3.09$, $92.31 - 3.09$)	89.22
Expected loss on the debt	$(92.31 - 89.22)/92.31$	3.35%

What moves the values (assets V_0 held fixed unless it is the input that changes; from the example):

Change	Equity E_0	Debt D_0	Why
σ_V 30% → 40%	64.73 ↑	85.27 ↓	The call (and the creditors' short put) is worth more
T 2 → 3 years	66.09 ↑	83.91 ↓	More time value in the call; the promise is further off
r 4% → 6%	63.80 ↑	86.20 ↓	The PV of the strike falls
V_0 150 → 160	70.02 ↑	89.98 ↑	More cushion: the put is further out of the money
Face D 100 → 110	53.46 ↓	96.54 ↑	A larger promised payment

3. From the debt value to a credit spread (shape D5·S1)

The zero-coupon debt is worth $D_0 = De^{-yT}$, so its yield and credit spread are

$$y = -\frac{1}{T}\ln\frac{D_0}{D}, \qquad \boxed{\text{credit spread} = -\frac{1}{T}\ln\frac{D_0}{D} - r}$$

Worked example: $y = -\frac{1}{2}\ln(89.22/100) = 5.70\%$, so the spread is $5.70 - 4.00 = 1.70\%$, or 170 bp. Because $D_0 < D$, $\ln(D_0/D)$ is negative and the leading minus sign makes the yield positive. A formula card that writes $\frac{1}{T}\ln(D/F) - r_f$ without the minus sign returns $-5.70\% - 4\% = -9.70\%$: a negative spread on risky debt is the inverted sign at work. GARP's Problem 9.17 gives the same spread directly: $-\ln\big[N(d_2)+N(-d_1)/L\big]/T$ with $L = De^{-rT}/V_0$; here $L = 0.6154$ and the formula returns 1.70%.

4. The two volatilities and leverage (shape S5)

Equity is a function of the assets, so by Itô's lemma $\sigma_E E_0 = \frac{\partial E}{\partial V}\sigma_V V_0$, and the delta $\partial E/\partial V$ is $N(d_1)$ (GARP eq. 9.5):

$$\sigma_E E_0 = N(d_1)\,\sigma_V V_0 \qquad \Longrightarrow \qquad \sigma_E = \underbrace{N(d_1)\frac{V_0}{E_0}}_{\text{leverage multiplier}}\ \sigma_V$$

The same delta transmits returns: $\Delta E/E_0 \approx N(d_1)(V_0/E_0)(\Delta V/V_0)$. In the worked example the leverage ratio is $V_0/E_0 = 150/60.78 = 2.47$, the multiplier $0.9125 \times 2.47 = 2.25$, so $\sigma_E = 2.25 \times 30\% = 67.6\%$, and a 5% fall in asset value cuts the equity by about $2.25 \times 5\% = 11.3\%$. The raw leverage ratio overstates the effect: part of any asset loss falls on the creditors through their short put. Two shortcuts the exam uses:

- $d_1 - d_2 = \sigma_V\sqrt{T}$, so $\sigma_V = (d_1 - d_2)/\sqrt{T}$ when both d's are given.
- With $N(d_1)$ known, $\sigma_V = \sigma_E E_0/(N(d_1)V_0)$.

5. Calibrating the asset value and asset volatility (shape S2)

V_0 and σ_V cannot be observed. For a listed company E_0 (market capitalisation) and σ_E (from equity returns) can, so eq. 9.4 and eq. 9.5 are two equations in the two unknowns, solved numerically: Hull uses Excel's Solver to minimise $[F(x,y)]^2 + [G(x,y)]^2$; Doumpos uses a Newton-type iteration. Once the solver reports the converged $N(d_1)$ and $N(d_2)$, both unknowns follow directly:

$$V_0 = \frac{E_0 + De^{-rT}N(d_2)}{N(d_1)}, \qquad \sigma_V = \frac{\sigma_E E_0}{N(d_1)V_0}$$

GARP Example 9.4. $E_0 = 3$, $\sigma_E = 80\%$, $D = 10$ due in one year, $r = 5\%$. Solving gives $V_0 = 12.40$ and $\sigma_V = 21.23\%$, with $d_2 = 1.1408$ and a risk-neutral PD of $N(-d_2) = 12.7\%$ (CR-9 m). The

debt is worth $V_0 - E_0 = 9.40$ against a no-default value of $10e^{-0.05} = 9.51$: an expected loss of about $(9.51 - 9.40)/9.51 = 1.2\%$. An 80% equity volatility corresponds to only 21% asset volatility, because the equity is a levered claim on the assets.

Correction to the source notes. SchweserNotes (LO 27.l/m) prints $d_1 = 1.3547$ alongside $d_2 = 1.1408$ for Example 9.4. The two must differ by $\sigma_V\sqrt{T} = 0.2123$, so with $d_2 = 1.1408$ the correct value is $d_1 = 1.3531$, and $N(d_1) = 0.9120$ (Schweser's 0.9117 comes from the rounded V_0 and σ_V). d_2, the PD and the debt value are unaffected.

Formula card: CR-9 I

Payoffs at T	$E_T = \max(V_T - D, 0)$; debt $= \min(V_T, D) = D - \max(D - V_T, 0)$
Equity today	$E_0 = V_0 N(d_1) - De^{-rT} N(d_2)$; $d_2 = d_1 - \sigma_V\sqrt{T}$
Debt today	$D_0 = V_0 - E_0 = De^{-rT} - \text{put} = De^{-rT} N(d_2) + V_0 N(-d_1)$
Expected loss on debt	$(De^{-rT} - D_0)/De^{-rT}$
Credit spread	$-\frac{1}{T}\ln(D_0/D) - r$ (minus sign in front of the log)
Volatilities	$\sigma_E E_0 = N(d_1)\sigma_V V_0$; $\sigma_V = (d_1 - d_2)/\sqrt{T}$
Leverage to ROE	$\Delta E/E_0 \approx N(d_1)(V_0/E_0)\,\Delta V/V_0$
Back-solve (S2)	$V_0 = (E_0 + De^{-rT} N(d_2))/N(d_1)$; $\sigma_V = \sigma_E E_0/(N(d_1)V_0)$

Exam traps

- **Undiscounted strike.** E_0 uses De^{-rT}, not D; the slip overstates the debt value.
- **Equity reported as debt (or the reverse).** $D_0 = V_0 - E_0$; the call value is the equity.
- **Risk-free value as the debt value.** De^{-rT} is an upper bound; the creditors' short put must come off.
- **Put added, not subtracted.** Risky debt = risk-free debt − put.
- **$\sqrt{T}$ errors.** $\sigma_V = (d_1 - d_2)/\sqrt{T}$, not $/T$ and not $\times\sqrt{T}$; in d_1 the denominator is $\sigma_V\sqrt{T}$.
- **Delta dropped.** $\sigma_V = \sigma_E E_0/V_0$ forgets $N(d_1)$; the raw leverage ratio overstates the equity's response to asset moves.
- **$N(d_2)$ for $N(d_1)$.** The volatility link uses the delta $N(d_1)$.
- **Solver's starting values.** Seeds such as $V_0 = E_0 + De^{-rT}$ and $\sigma_V = \sigma_E E_0/V_0$ are iteration zero, not the answer.
- **Inverted sign in the spread.** $\frac{1}{T}\ln(D_0/D)$ is negative for risky debt; a negative spread signals the sign error. Also: the yield is not the spread until r comes off, and forgetting $1/T$ reports a T-year figure as annual.
- **Red herrings.** The expected asset return plays no part in valuation; equity volatility only calibrates σ_V.

Practice questions: CR-9 I

Question CR-9 I.1

Bank Q368 • AP Mock 1 Q3

Since it was founded ten years ago, Bright Technologies has paid no dividends and has been financed with 100% equity. Management recently issued a zero-coupon bond with a principal amount of USD 100 million maturing in exactly three years. If the value of the firm at maturity is USD 80 million, determine the values of the components of the firm's capital structure at the bond's maturity date.

A. Value of equity = USD 0; value of debt = USD 80 million.

B. Value of equity = USD 20 million; value of debt = USD 80 million.

C. Value of equity = USD 20 million; value of debt = USD 0.

D. Value of equity = USD 180 million; value of debt = USD 100 million.

Question CR-9 I.2

Bank Q370 • AP Mock 3 Q63 • AnalystPrep Q.2677

For a firm financed partly by debt and partly by equity, in the Merton model the value of the debt:

A. increases if the volatility of the firm increases.

B. increases if the face amount of debt falls.

C. decreases if its time to maturity increases.

D. increases if the interest rate increases.

Question CR-9 I.3

Bank Q365 • AnalystPrep Q.2878 (rebuilt)

A firm has only one liability, a zero-coupon bond with face value USD 206.74 million maturing in one year. A European put option on the firm's assets with exercise price equal to the bond's face value and the same maturity is currently priced at USD 19.37 million. The current price of a zero-coupon risk-free bond paying USD 1 in one year is USD 0.86. Using the Merton model, what is the market value of the firm's debt?

A. USD 158.43 million

B. USD 177.80 million

C. USD 187.37 million

D. USD 197.17 million

Question CR-9 I.4 Bank Q369 • AP Mock 2 Q41 (rebuilt: distractors replaced)

A firm has a current value of USD 200 million. Its only outstanding debt is a three-year zero-coupon bond with a face value of USD 180 million. The annual risk-free interest rate is 5% (continuously compounded) and the volatility of the firm's value is 10%. Using the Merton model, what is the value of the firm's equity?

A. USD 23.0 million

B. USD 45.1 million

C. USD 46.0 million

D. USD 50.8 million

Question CR-9 I.5 Bank Q367 • Mock 3 Q4 • GARP PE1 Q77 (rebuilt)

A fixed-income analyst applies the Merton model to a non-dividend-paying company whose assets have a current market value of CAD 85 million. The company's only debt is a zero-coupon bond with face value CAD 55 million maturing in 2 years. Additional information: $N(d_1) = 0.9495$; $N(d_2) = 0.8983$; annual risk-free rate 5% (continuously compounded); annual asset volatility 26%. What is the Merton-model market value of the company's debt?

A. CAD 36.003 million

B. CAD 48.997 million

C. CAD 49.766 million

D. CAD 53.699 million

Question CR-9 I.6 Bank Q390 • Mock 1 Q7 (rebuilt)

A credit analyst applies the Merton model to a firm whose only debt is a five-year zero-coupon bond.

Current value of firm assets (V_0)	USD 100 million
Principal due in five years (D)	USD 100 million
Risk-free rate (continuously compounded)	10% per annum
Asset volatility	20% per annum
$N(d_1)$	0.9101
$N(d_2)$	0.8145

Which of the following gives the current value of the firm's equity and the risk-neutral probability of default?

A. Equity of USD 41.61 million; default probability of 8.99%.

B. Equity of USD 41.61 million; default probability of 18.55%.

C. Equity of USD 58.39 million; default probability of 8.99%.

D. Equity of USD 58.39 million; default probability of 18.55%.

Question CR-9 I.7 Bank Q376 • AnalystPrep Q.3061 (rebuilt) • shape D5·S1

A credit analyst is checking a colleague's formula card, which gives the credit spread on zero-coupon debt as $\frac{1}{T}\ln(D_0/F) - r$, where D_0 is the market value of the debt and F its face value. A USD 2 million (face value) zero-coupon corporate bond has a remaining maturity of 7 years and, using the Merton model, a current value of USD 1.2 million. The risk-free rate is 5% (continuously compounded). Applying the standard definition, the credit spread is closest to:

A. -12.30%

B. 2.30%

C. 4.52%

D. 7.30%

Question CR-9 I.8 Bank Q389 • GARP PE1 Q22 • Mock 3 Q58 (rebuilt)

A portfolio manager applies the Merton model to a non-dividend-paying firm whose shares are held in the fund's portfolio. Preliminary analysis gives:

Value of equity	USD 45 million
Value of the firm's only debt, maturing in 5 years	USD 60 million
d_1	3.217790
d_2	3.038905

Assuming a constant volatility of firm value, what is the estimate of that volatility?

A. 3.6%

B. 8.0%

C. 17.9%

D. 40.0%

Question CR-9 I.9 Workbook A5.5 • S2 • Map #2

A treasury analyst at a UK bank is calibrating Merton's model for a listed borrower. Market and balance sheet data (GBP million):

- Market value of equity: 25; equity volatility: 80%
- Zero-coupon debt due in 1 year: 100
- Risk-free rate: 3% per annum, continuously compounded; expected return on assets: 8%

The analyst's Newton-type solver was seeded with $V_0 = E_0 + D\mathrm{e}^{-rT} = 122.04$ and $\sigma_V = \sigma_E E_0/V_0 = 16.39\%$. After convergence, it reports $N(d_1) = 0.9018$ and $N(d_2) = 0.8661$. Which of the following is closest to the converged value of the firm's assets and its asset volatility?

A. $V_0 =$ GBP 120.9 million; $\sigma_V = 18.3\%$

B. $V_0 =$ GBP 120.9 million; $\sigma_V = 16.5\%$

C. $V_0 =$ GBP 122.0 million; $\sigma_V = 16.4\%$

D. $V_0 =$ GBP 123.8 million; $\sigma_V = 17.9\%$

Question CR-9 I.10 Workbook A5.2 • S5, S7 • Map #2 • options in ascending order (workbook key C)

A fixed-income portfolio manager at a European insurer is valuing a block of a company's zero-coupon debt using Merton's model. The CRO also wants to know how leverage transmits asset shocks to the shareholders. Inputs:

- Value of assets: EUR 50 million; asset volatility: 30%
- Face value of debt due in 3 years: EUR 40 million
- Risk-free rate: 2% per annum, continuously compounded

Which of the following is closest to the market value of the company's debt today, and to the approximate instantaneous return on the company's equity if the value of its assets falls by 6%?

A. Debt EUR 33.6 million; return on equity -14.4%

B. Debt EUR 33.6 million; return on equity -18.3%

C. Debt EUR 35.0 million; return on equity -15.8%

D. Debt EUR 37.7 million; return on equity -14.4%

Solutions • CR-9 l

Solution CR-9 l.1

Answer: A

Bank Q368 • AP Mock 1 Q3

Step 1 – Equity payoff

$$E_T = \max(V_T - D,\, 0) = \max(80 - 100,\, 0) = 0$$

Step 2 – Debt payoff

$$\min(V_T, D) = \min(80, 100) = 80$$

$\Rightarrow$ Check: $0 + 80 = 80 = V_T$. The shareholders' call expires worthless and the bondholders take all the assets.

Interpretation

Limited liability floors the equity at zero; the creditors bear the USD 20 million shortfall. The ten-year history and the three-year maturity are scene-setting only.

Why the other options are wrong

B is incorrect. Gives equity the 20 million shortfall; the claims would then sum to 100, more than the firm is worth.

C is incorrect. Reverses the priority of claims: the debt is senior and the equity is $\max(V_T - D, 0)$.

D is incorrect. Adds the face value to the firm value and pays the debt in full, although the assets are insufficient.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8 Using Equity Prices to Estimate Default Probabilities

Solution CR-9 l.2

Answer: C

Bank Q370 • AP Mock 3 Q63

Step 1 – Write the debt as assets minus a call

$D_0 = V_0 - E_0$, with E_0 a call on the assets struck at the face value.

Step 2 – Effect of a longer maturity

For a non-dividend-paying underlying the call is worth more with more time to maturity, so with V_0 fixed the debt is worth less.

$\Rightarrow$ The debt value decreases as maturity increases (in Section 2: 89.22 at two years, 83.91 at three).

Interpretation

Every comparative static of the debt is the mirror image of the equity call: whatever raises the call (volatility, maturity, the interest rate) lowers the debt; a larger promised payment lowers the call and raises the debt.

Why the other options are wrong

A is incorrect. Higher volatility raises the call and lowers the debt: equity holders gain from risk.

B is incorrect. A smaller face amount is a smaller promised payment, so the debt is worth less, not more.

D is incorrect. A higher interest rate raises the call and lowers the debt.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8, eq. 9.4

Solution CR-9 l.3

Answer: A

Bank Q365 • AnalystPrep Q.2878

Step 1 – Risk-free value of the promised payment

$$0.86 \times 206.74 = 177.80$$

Step 2 – Subtract the creditors' short put

$$D_0 = 177.80 - 19.37 = 158.43$$

$\Rightarrow$ The debt is worth USD 158.43 million.

Interpretation

The bondholders own a risk-free bond and have written a put to the shareholders, who can hand over the assets instead of paying the face value. The put's value (USD 19.37 million) is the market value of the default risk.

Why the other options are wrong

B is incorrect. Reported the risk-free value of the debt and ignored the put.
C is incorrect. Subtracted the put from the undiscounted face value: $206.74 - 19.37 = 187.37$.
D is incorrect. Added the put instead of subtracting it: $177.80 + 19.37 = 197.17$.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8

Solution CR-9 l.4

Answer: C

Bank Q369 • AP Mock 2 Q41

Step 1 – d_1 and d_2

$$d_1 = \frac{\ln(200/180) + (0.05 + 0.10^2/2) \times 3}{0.10\sqrt{3}} = \frac{0.10536 + 0.165}{0.17321} = 1.5609$$

$$d_2 = d_1 - \sigma_V\sqrt{T} = 1.5609 - 0.1732 = 1.3877$$

Step 2 – $N(d_1)$, $N(d_2)$ and the PV of the strike

$$N(d_1) = 0.9407, \qquad N(d_2) = 0.9174, \qquad 180\,\mathrm{e}^{-0.15} = 154.93$$

Step 3 – Equity

$$E_0 = 200(0.9407) - 154.93(0.9174) = 188.14 - 142.13 = 46.0$$

Interpretation

With only 10% asset volatility the call is deep in the money and close to its lower bound $V_0 - D\mathrm{e}^{-rT} = 45.1$: almost all its value is intrinsic. The debt is worth $200 - 46.0 = 154.0$, just below its risk-free value of 154.93.

Why the other options are wrong

A is incorrect. Did not discount the strike: $200(0.9407) - 180(0.9174) = 23.0$.

B is incorrect. Reported the lower bound $V_0 - De^{-rT} = 200 - 154.93 = 45.1$, ignoring the option's time value.
D is incorrect. Scaled the volatility by T instead of $\sqrt{T}$: $d_1 = 0.9012$, $d_2 = 0.6012$, $E_0 = 50.8$.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8, eq. 9.4

Solution CR-9 l.5 Answer: B

Bank Q367 • Mock 3 Q4 • GARP PE1 Q77

Step 1 – PV of the promised payment

$$55\,e^{-0.05\times 2} = 55 \times 0.904837 = 49.766$$

Step 2 – Equity as a call

$$E_0 = 85(0.9495) - 49.766(0.8983) = 80.708 - 44.705 = 36.003$$

Step 3 – Debt

$$D_0 = V_0 - E_0 = 85 - 36.003 = 48.997$$

Interpretation

The debt is worth CAD 0.77 million less than its risk-free value of 49.766: the price of the creditors' short put. The asset volatility is already reflected in the $N(\cdot)$ values given.

Why the other options are wrong

A is incorrect. Reported the equity (the call value) instead of $V_0 - E_0$.
C is incorrect. Reported the risk-free PV of the debt, $55e^{-0.10}$, ignoring default risk.
D is incorrect. Did not discount the strike: $E_0 = 80.708 - 55(0.8983) = 31.30$, so $D_0 = 53.70$.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8, eq. 9.4

Solution CR-9 l.6 Answer: B

Bank Q390 • Mock 1 Q7

Step 1 – PV of the promised payment

$$100\,e^{-0.10\times 5} = 100\,e^{-0.5} = 60.653$$

Step 2 – Equity

$$E_0 = 100(0.9101) - 60.653(0.8145) = 91.01 - 49.40 = 41.61$$

Step 3 – Risk-neutral probability of default

$$N(-d_2) = 1 - N(d_2) = 1 - 0.8145 = 18.55\%$$

$\Rightarrow$ Default happens when the shareholders' call is not exercised.

Interpretation

$N(d_2)$ is the risk-neutral probability that the call is exercised ($V_T > D$); $N(d_1)$ is the delta. The debt is worth $100 - 41.61 = 58.39$.

Why the other options are wrong

A is incorrect. Used $1 - N(d_1) = 8.99\%$ as the default probability; $N(d_1)$ is the delta, not the exercise probability.

C is incorrect. Reported the debt value (58.39) as the equity, and used $1 - N(d_1)$.

D is incorrect. Reported the debt value as the equity; the PD is right.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8, eq. 9.4

Solution CR-9 l.7

Answer: B

Bank Q376 • AnalystPrep Q.3061

Step 1 – Yield on the debt

$$y = -\frac{1}{T}\ln\frac{D_0}{F} = -\frac{1}{7}\ln\frac{1.2}{2} = -\frac{1}{7}(-0.51083) = 0.07298$$

$\Rightarrow$ The debt yields 7.30%: the logarithm is negative because $D_0 < F$, and the minus sign in front makes the yield positive.

Step 2 – Credit spread

$$7.30\% - 5.00\% = 2.30\%$$

Interpretation

2.30% is the extra rate that, added to the risk-free rate, discounts the promised USD 2 million back to its Merton value. It is the input a spread-based PD or a CVA would use.

Why the other options are wrong

A is incorrect. Used the card's inverted sign: $\frac{1}{7}\ln(0.6) - 0.05 = -0.0730 - 0.05 = -12.30\%$. A negative spread on risky debt is the giveaway.

C is incorrect. Used a simple annual rate: $(2/1.2 - 1)/7 = 9.52\%$, minus 5%.

D is incorrect. Stopped at the yield, 7.30%, without subtracting the risk-free rate.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.5 and 9.8, Problem 9.17; Workbook D5 formula card

Solution CR-9 l.8

Answer: B

Bank Q389 • GARP PE1 Q22 • Mock 3 Q58

Step 1 – The gap between d_1 and d_2

$$d_1 - d_2 = \sigma_V\sqrt{T} = 3.217790 - 3.038905 = 0.178885$$

Step 2 – Divide by $\sqrt{T}$

$$\sigma_V = \frac{0.178885}{\sqrt{5}} = \frac{0.178885}{2.23607} = 0.0800 = 8.0\%$$

Interpretation

The equity and debt values are not needed: once both d's are given, the volatility of firm value follows from $d_2 = d_1 - \sigma_V\sqrt{T}$.

Why the other options are wrong

A is incorrect. Divided by $T = 5$ instead of $\sqrt{T}$: 3.6%.
C is incorrect. Took $d_1 - d_2 = 17.9\%$ as the volatility, without dividing by $\sqrt{T}$.
D is incorrect. Multiplied by $\sqrt{T}$ instead of dividing: 40.0%.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8, eq. 9.4

Solution CR-9 I.9

Answer: A

Workbook A5.5 • S2

Step 1 – PV of the promised payment

$$De^{-rT} = 100e^{-0.03} = 97.045$$

Step 2 – Asset value from the equity equation (Hull 9.4)

$$V_0 = \frac{E_0 + De^{-rT}N(d_2)}{N(d_1)} = \frac{25 + 97.045(0.8661)}{0.9018} = \frac{109.050}{0.9018} = 120.93$$

$\Rightarrow$ The market values the firm's assets at GBP 120.9 million, so the leverage ratio V_0/E_0 is about 4.8.

Step 3 – Asset volatility from the volatility equation (Hull 9.5)

$$\sigma_V = \frac{\sigma_E E_0}{N(d_1)V_0} = \frac{0.80(25)}{0.9018(120.93)} = \frac{20.0}{109.05} = 0.1834$$

$\Rightarrow$ An 80% equity volatility corresponds to only 18.3% volatility of the underlying assets: leverage magnifies asset risk into equity risk.

Interpretation

At the converged pair, $d_2 = 1.108$ and the one-year risk-neutral PD is $N(-1.108) \approx 13.4\%$. The 8% expected return is a red herring for calibration: it matters only if a real-world PD is later required (Doumpos eq. 5.3).

Why the other options are wrong

B is incorrect. Correct V_0, but dropped the delta in eq. 9.5: $\sigma_V = \sigma_E E_0/V_0 = 20/120.93 = 16.5\%$.
C is incorrect. Reported the solver's starting values (iteration zero) as the answer: $V_0 = 25 + 97.045 = 122.04$, $\sigma_V = 20/122.04 = 16.4\%$.
D is incorrect. Did not discount the debt in eq. 9.4: $V_0 = (25 + 100 \times 0.8661)/0.9018 = 123.76$, then $\sigma_V = 20/(0.9018 \times 123.76) = 17.9\%$.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8, eqs. 9.4 and 9.5, footnote 21; Doumpos et al., Ch.1, 5.6.3.1 eqs. 5.1 and 5.2

Solution CR-9 I.10

Answer: A

Workbook A5.2 • S5, S7

Step 1 – d_1 and d_2

$$d_1 = \frac{\ln(50/40) + (0.02 + 0.30^2/2)(3)}{0.30\sqrt{3}} = \frac{0.2231 + 0.1950}{0.5196} = 0.8047, \qquad d_2 = 0.2851$$

$$N(d_1) = 0.7895, \qquad N(d_2) = 0.6122$$

Step 2 – Equity value

$$E_0 = 50(0.7895) - 40\mathrm{e}^{-0.06}(0.6122) = 39.475 - 37.671(0.6122) = 16.41$$

$\Rightarrow$ The shareholders' call on the assets is worth EUR 16.41 million.

Step 3 – Debt value

$$D_0 = V_0 - E_0 = 50 - 16.41 = 33.59$$

$\Rightarrow$ Versus a risk-free value of $40\mathrm{e}^{-0.06} = 37.67$: an expected loss of $(37.67 - 33.59)/37.67 = 10.8\%$.

Step 4 – Leverage ratio and multiplier

$$\frac{V_0}{E_0} = \frac{50}{16.41} = 3.046, \qquad N(d_1)\frac{V_0}{E_0} = 0.7895 \times 3.046 = 2.405$$

$\Rightarrow$ Each 1% move in asset value moves equity by about 2.4%. The same multiplier gives $\sigma_E = 2.405 \times 30\% = 72.2\%$ (eq. 9.5).

Step 5 – Return on equity

$$\frac{\Delta E}{E_0} \approx N(d_1)\frac{V_0}{E_0}\frac{\Delta V}{V_0} = 2.405 \times (-6\%) = -14.4\%$$

Interpretation

Debt is worth EUR 33.6 million; creditors are effectively short a put on the assets. The leverage ratio of about 3 is damped by the delta $N(d_1)$ because part of any asset loss is absorbed by the creditors' put; a 6% asset loss becomes roughly a 14.4% equity loss. The three-year horizon is no obstacle: the model is defined for any T.

Why the other options are wrong

B is incorrect. Used the raw leverage ratio without the delta: $3.046 \times (-6\%) = -18.3\%$, treating equity as if it absorbed every asset loss.

C is incorrect. Did not discount the strike: $E_0 = 50(0.7895) - 40(0.6122) = 14.99$, so debt $= 35.01$ and the multiplier becomes $0.7895 \times 50/14.99$, giving -15.8%.

D is incorrect. Reported the risk-free PV of the promised payment, $40\mathrm{e}^{-0.06} = 37.67$, as the debt value, ignoring default risk.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8, eqs. 9.4 and 9.5, Example 9.4

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 m

P1 CALC

Calculate distance to default and default probability using the Merton model.

Workbook: A5 Merton Model, Distance to Default and Risk-Neutral PD • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: Merton came up in the August 2026 sitting, and Map #2 (8 of 16 sources) reports "finding distance to default and assign a rating based on the score", "Merton model risk-neutral prob" and "hazard rate and PD based on Merton model".
Vardeez: Merton distance to default.

Reported exam shapes

- **A5·S1** Distance to default, then assign a rating band from the score.
- **A5·S3** Risk-neutral PD $N(-d_2)$ versus the real-world PD with μ.
- **A5·S4** Merton PD converted into an average hazard rate.
- **A5·S6** Short-form DD $\ln(V/F)/\sigma_V$ versus the full form.

What the questions look like: numerical. V_0, D, σ_V, T, r and often μ; compute the DD and the PD, rank counterparties, map a DD to a band, or convert the PD to a hazard rate. The traps are the drift (r or μ), d_1 for d_2, $N(d_2)$ for $N(-d_2)$, the form of the DD, the default point and the horizon.

Worked elsewhere: A5.4 (KMV default point) and A5.6 (short-form DD to a band) at Questions CR-5 e.6 and CR-5 e.8; a Merton PD converted to a hazard rate is also Question CR-9 e.7.

1. Distance to default

GARP 9.8.3: the **distance to default** (DD) is "the number of standard deviations the asset price must change for default to be triggered T years in the future". One definition is d_2:

$$\boxed{\text{DD} = d_2 = \frac{\ln V_0 - \ln D + (r - \sigma_V^2/2)T}{\sigma_V\sqrt{T}}} \qquad \boxed{\text{PD} = N(-\text{DD}) = 1 - N(d_2)}$$

In $\ln V_T$ space the numerator is the expected log asset value at T minus the log of the debt, and the denominator is the standard deviation of $\ln V_T$. As DD declines the company becomes more likely to default. $N(-d_2)$ is the probability that the shareholders do not exercise their call. **Example 9.4** (CR-9 1): $V_0 = 12.40$, $\sigma_V = 21.23\%$, $D = 10$, $r = 5\%$, $T = 1$: DD $= 1.14$ standard deviations and PD $= N(-1.1408) = 12.7\%$. (With the rounded $V_0 = 12.40$ the formula gives 1.1426; GARP's 1.1408 uses the solver's unrounded 12.395.)

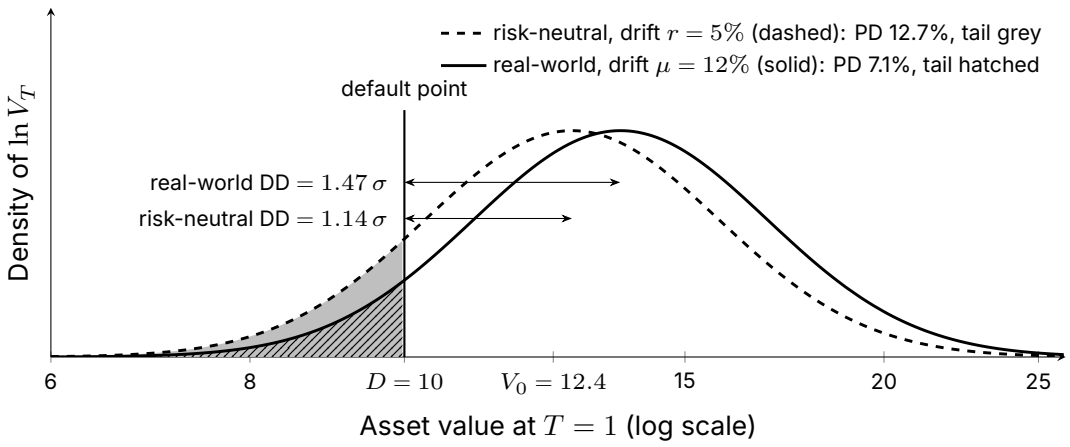


Figure 1. GARP Example 9.4 at the one-year horizon. Each arrow runs from the default point to the expected log asset value, measured in standard deviations of $\ln V_T$ ($\sigma_V\sqrt{T} = 0.2123$). The risk-neutral curve (dashed) drifts at r, the real-world curve (solid) at an assumed $\mu = 12\%$; the area left of D is the PD (grey under the dashed curve, hatched under the solid one).

2. Risk-neutral or real-world (shape S3)

Because it comes from an option pricing model, $N(-d_2)$ with drift r is a **risk-neutral** PD, the right input for valuation and CVA. For a **real-world** PD, used in scenario analysis, replace r by the expected return on assets μ (Doumpos eq. 5.3). In Example 9.4 with $\mu = 12\%$: DD $= (\ln(12.395/10) + 0.12 - 0.02254)/0.2123 = 1.47$ and PD $= 7.1\%$, against 12.7% risk-neutral. Since $\mu > r$, the real-world PD is the lower one (CR-9 k). A stem that lists both r and μ is testing the choice; the equity volatility, if listed, plays no part.

3. Short form versus full form (shape S6)

For a one-year horizon with a small drift, the drift term can be dropped:

$$\text{DD} \approx \frac{\ln(V/F)}{\sigma_V} \qquad \text{(short form)}$$

Example 9.4: $\ln(12.395/10)/0.2123 = 1.01$ against the full 1.14. Use the short form when the question says "simplified" or "drift ignored" (ranking questions often do); use the full d_2 when r or μ is given and asked for. For a one-year ranking the drift rarely changes the order, because $(r - \sigma_V^2/2)$ is small next to $\ln(V/F)$; over several years it can (Question CR-9 m.4). With $T \neq 1$ the drift is multiplied by T and the volatility by $\sqrt{T}$; nothing restricts Merton to one year.

4. From DD to a rating band, and to a hazard rate

- **Bands (shape S1).** Merton-type models rank default risk well (CR-9 n), so a bank can map DD ranges to internal grades: compute the DD in the form the bank specifies, then read the band. Moody's KMV instead uses a default point of short-term debt plus half of long-term debt and maps DD to an expected default frequency through historical default data rather than $N(\cdot)$ (CR-5 e).
- **Hazard rate (shape S4).** Treat $N(-d_2)$ as the cumulative PD $Q(T)$ and apply $\bar{\lambda} = -\ln(1 - Q(T))/T$ (CR-9 e, Section 5). Example 9.4: $-\ln(1 - 0.127) = 13.6\%$ a year. Dividing the PD by T understates the hazard; forgetting $1/T$ overstates it.

Formula card: CR-9 m

Distance to default	$\text{DD} = d_2 = \big(\ln(V_0/D) + (r - \sigma_V^2/2)T\big)/(\sigma_V\sqrt{T})$
Default probability	risk-neutral $N(-d_2)$ with r; real-world: replace r by μ
Short form	$\text{DD} \approx \ln(V/F)/\sigma_V$ (one year, drift ignored)
d_1	$d_1 = d_2 + \sigma_V\sqrt{T}$ ($N(-d_1)$ is not the PD)
Horizon	drift $\times T$, volatility $\times\sqrt{T}$; $T = 0.25$ for three months
KMV default point	short-term debt $+$ $0.5\times$ long-term debt; DD mapped to EDF empirically
Hazard from a Merton PD	$\bar{\lambda} = -\ln(1 - N(-d_2))/T$

Exam traps

- **Wrong drift.** r for a risk-neutral PD, μ for a real-world one; the other rate is the distractor.
- **$\pm\sigma^2/2$.** The DD subtracts $\sigma_V^2/2$; adding it gives d_1 and a PD that is too low.
- **$N(d_2)$ for $N(-d_2)$.** $N(d_2)$ is the probability of *no* default; it appears as the largest option.
- **Drift dropped or half-applied.** Subtracting rT alone, or omitting the drift when the full form is asked.
- **Horizon not scaled.** Annual drift and volatility used for a three-month or multi-year horizon.
- **Equity volatility.** σ_E in place of σ_V in the DD.
- **Default point.** Total debt, short-term debt only, or the KMV point: use what the stem specifies. Face value, not the market value of debt.
- **Ranking direction.** Lowest DD is most likely to default; "least to most likely" reverses the list.
- **Hazard conversion.** $-\ln(1 - Q)/T$, not Q/T.

Practice questions: CR-9 m

Question CR-9 m.1

Bank Q401 • GARP PE1 Q6 • Mock 3 Q74 (rebuilt)

A credit manager uses a simplified Merton model to rank three non-dividend-paying counterparties with the one-year approximation DD $\approx \ln(V_0/F)/\sigma_V$. Each company's only liability is a one-year zero-coupon bond.

Company	P	Q	R
Market value of assets (EUR million)	100	150	250
Face value of debt (EUR million)	60	100	160
Annual asset volatility	10.0%	7.0%	8.0%

Rank the counterparties from most likely to least likely to default.

A. P; R; Q

B. Q; P; R

C. Q; R; P

D. R; Q; P

Question CR-9 m.2

Bank Q411 • Mock 2 Q72 (rebuilt)

A credit analyst has the following data for Firm ABC:

Current market value of assets (V_0)	3,750
Face value of zero-coupon debt due in one year (D)	2,500
Annualised volatility of the firm's assets	20%
Risk-free rate	4% per annum

According to the Merton model, what is the one-year distance to default (DD $= d_2$)?

A. 2.03 standard deviations

B. 2.13 standard deviations

C. 2.23 standard deviations

D. 2.33 standard deviations

Question CR-9 m.3 New • built from GARP Ch.9, 9.8 and 9.8.3 • shapes A5·S1, S6

A credit analyst at a Korean bank converts each corporate borrower's Merton distance to default into an internal band. The bank's policy specifies the full-form distance to default, d_2, computed with the risk-free rate over the life of the borrower's debt:

- Bands: DD of 3.0 or more: A; 2.0 to below 3.0: BBB; 1.0 to below 2.0: BB; below 1.0: B
- Borrower (KRW billion): value of assets 500; zero-coupon debt of 300 due in 2 years; asset volatility 22%; equity volatility 50%; risk-free rate 4% (continuously compounded)

Which of the following gives the borrower's distance to default and band?

A. DD 0.48; band B

B. DD 1.74; band BB

C. DD 2.05; band BBB

D. DD 2.32; band BBB

Question CR-9 m.4 Bank Q397 • AP Mock 3 Q61 (rebuilt)

An analyst computes the Merton distance to default $\text{DD} = \left(\ln(A/L) + (\mu - \sigma^2/2)T\right)/(\sigma\sqrt{T})$ for three companies:

Company	Assets (A)	Face value of debt (L)	Asset volatility	Expected return (μ)	T
X	800,000	500,000	25%	7%	3 years
Y	1,000,000	600,000	20%	5%	4 years
Z	900,000	400,000	30%	6%	2 years

Based on distance to default, rank the companies from most likely to least likely to default.

A. X, Y, Z

B. Z, Y, X

C. Y, Z, X

D. Z, X, Y

Question CR-9 m.5 Bank Q410 • BT P2.T6.303.2 (rebuilt)

A firm's current asset value is USD 62.0 million with asset volatility of 38.0% per annum. Its debt consists of a single zero-coupon bond with face value USD 40.0 million maturing in one year. The risk-free rate is 2.0% per annum and the expected return on the firm's assets is 10.0%. Using the Merton model, what is the firm's risk-neutral probability of default?

A. 8.14%

B. 11.00%

C. 12.44%

D. 15.48%

Question CR-9 m.6

Bank Q404 • GARP PE2 Q6 • Mock 4 Q7 (rebuilt)

A credit analyst uses the Merton model to estimate the real-world probability of default of a non-dividend-paying company whose debt is a single zero-coupon bond.

Value of the company's assets	CAD 400 million
Face value of debt	CAD 300 million
Expected return on assets	15%
Asset volatility	25%
Risk-free rate	3%
Remaining maturity of debt	1 year

The probability of default is closest to:

A. 3.03%

B. 5.20%

C. 12.60%

D. 94.80%

Question CR-9 m.7

Bank Q396 • AP Mock 2 Q76 (rebuilt)

ICB Corp has assets of GBP 25 billion with 45% annual asset volatility. The expected return on assets is 8% and the risk-free rate is 3%. ICB has short-term debt of GBP 6 billion and long-term debt of GBP 4 billion; treat all GBP 10 billion as a single zero-coupon liability due in one year. Using the Merton model with the expected asset return, which is nearest to ICB's one-year probability of default?

A. 0.09%

B. 0.65%

C. 2.34%

D. 3.02%

Question CR-9 m.8

Bank Q399 • AnalystPrep Q.3054 (rebuilt)

The assets of Company X are worth USD 1,300,000. In three months it must repay USD 1,000,000 of debt. Asset volatility is 30% per annum and the expected return on assets is 15% per annum. Using the Merton model with the expected return as the drift, which is nearest to the probability of default in three months?

A. 2.28%

B. 2.72%

C. 4.01%

D. 11.04%

Question CR-9 m.9 Workbook A5.1 • S3, S4, S7 • Aug-2026 • Map #2

A risk analyst at a Singapore bank is estimating the risk-neutral default probability of a corporate borrower using Merton's model. The bank's CVA desk stores each counterparty's default risk as a constant average hazard rate, so the analyst must also convert the Merton output. The asset value and asset volatility have already been solved from the equity data:

- Value of assets V_0: SGD 150 million; asset volatility σ_V: 25%
- Zero-coupon debt due in 2 years: SGD 120 million
- Risk-free rate: 3% per annum, continuously compounded

Which of the following is closest to the two-year risk-neutral probability of default and the corresponding average hazard rate per year?

A. PD 16.4%; average hazard rate 9.0%

B. PD 26.6%; average hazard rate 15.5%

C. PD 26.6%; average hazard rate 13.3%

D. PD 33.0%; average hazard rate 20.0%

Solutions • CR-9 m

Solution CR-9 m.1

Answer: A

Bank Q401 • GARP PE1 Q6 • Mock 3 Q74

Step 1 – Short-form DD for each company

$$\text{P: } \frac{\ln(100/60)}{0.10} = \frac{0.5108}{0.10} = 5.11 \qquad \text{Q: } \frac{\ln(150/100)}{0.07} = \frac{0.4055}{0.07} = 5.79$$

$$\text{R: } \frac{\ln(250/160)}{0.08} = \frac{0.4463}{0.08} = 5.58$$

Step 2 – Rank by ascending DD

P (5.11), R (5.58), Q (5.79): the smallest cushion in standard deviations defaults first.

Interpretation

Q has the thinnest asset coverage (1.5 times, against 1.56 for R and 1.67 for P) but the lowest volatility, so it has the largest DD. Coverage alone is not the answer: the volatility in the denominator decides.

Why the other options are wrong

B is incorrect. Puts Q first, but Q has the highest DD because of its 7% volatility.
C is incorrect. Also ranks the lowest-volatility firm as the riskiest.
D is incorrect. The reverse order: least to most likely to default.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.3 Distance to Default

Solution CR-9 m.2

Answer: B

Bank Q411 • Mock 2 Q72

Step 1 – Log asset coverage

$$\ln(3{,}750/2{,}500) = \ln 1.5 = 0.4055$$

Step 2 – Drift term

$$(r - \sigma_V^2/2)T = 0.04 - 0.02 = 0.02$$

Step 3 – Distance to default

$$d_2 = \frac{0.4055 + 0.02}{0.20} = 2.127 \approx 2.13$$

$\Rightarrow$ Assets sit 2.13 standard deviations above the default point; the risk-neutral PD is $N(-2.13) = 1.67\%$.

Interpretation

The drift adds only 0.10 standard deviations over one year, which is why the short form (2.03) is close; but the question asks for d_2.

Why the other options are wrong

A is incorrect. Omitted the drift: $\ln 1.5/0.20 = 2.03$ (the short form).
C is incorrect. Added rT but forgot to subtract $\sigma^2/2$: $(0.4055 + 0.04)/0.20 = 2.23$.
D is incorrect. Computed d_1, with $+\sigma^2/2$: $(0.4055 + 0.06)/0.20 = 2.33$.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.3

Solution CR-9 m.3

Answer: B

New • built from GARP Ch.9, 9.8 and 9.8.3

Step 1 – Numerator over the two-year life

$$\ln(500/300)+(0.04-0.22^2/2)\times 2 = 0.5108+(0.04-0.0242)\times 2 = 0.5108+0.0316 = 0.5424$$

Step 2 – Two-year standard deviation

$$\sigma_V\sqrt{T} = 0.22\sqrt{2} = 0.3111$$

Step 3 – Distance to default and band

$$d_2 = \frac{0.5424}{0.3111} = 1.74 \;\Rightarrow\; \text{band BB}$$

$\Rightarrow$ The risk-neutral two-year PD is $N(-1.74) = 4.1\%$.

Interpretation

The two-year horizon matters: the one-year short form would put the borrower a full band higher. The 50% equity volatility is a red herring; it only calibrates σ_V.

Why the other options are wrong

A is incorrect. Used the equity volatility in place of asset volatility: $(0.5108 + (0.04 - 0.125) \times 2)/(0.50\sqrt{2}) = 0.48$, band B.

C is incorrect. Reported $d_1 = d_2 + \sigma_V\sqrt{T} = 1.74 + 0.31 = 2.05$, band BBB.

D is incorrect. Used the one-year short form $\ln(500/300)/0.22 = 2.32$, ignoring the drift and the two-year horizon: band BBB.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8 and 9.8.3; Workbook A5 (Map #2: "finding distance to default and assign a rating based on the score")

Solution CR-9 m.4

Answer: A

Bank Q397 • AP Mock 3 Q61

Step 1 – DD for X ($T = 3$)

$$\frac{\ln 1.6 + (0.07 - 0.03125) \times 3}{0.25\sqrt{3}} = \frac{0.4700 + 0.1163}{0.4330} = 1.354$$

Step 2 – DD for Y ($T = 4$)

$$\frac{\ln 1.6667 + (0.05 - 0.02) \times 4}{0.20 \times 2} = \frac{0.5108 + 0.1200}{0.4000} = 1.577$$

Step 3 – DD for Z ($T = 2$)

$$\frac{\ln 2.25 + (0.06 - 0.045) \times 2}{0.30\sqrt{2}} = \frac{0.8109 + 0.0300}{0.4243} = 1.982$$

Step 4 – Rank
Lowest DD first: X (1.354), Y (1.577), Z (1.982).

Interpretation
Each company has its own horizon, so the drift is multiplied by its own T and the volatility by its own $\sqrt{T}$. Z's high coverage (2.25 times) outweighs its high volatility.

Why the other options are wrong
B is incorrect. The reverse order, least to most likely.
C is incorrect. Puts Y first, but Y's DD (1.577) exceeds X's (1.354).
D is incorrect. Puts Z first, though Z has the largest DD.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.3; Doumpos et al., Ch.1, 5.6.3.1 eq. 5.3

Solution CR-9 m.5

Answer: D

Bank Q410 • BT P2.T6.303.2

Step 1 – Numerator with the risk-free drift

$$\ln(62/40) + (0.02 - 0.38^2/2) = 0.4383 + (0.02 - 0.0722) = 0.3861$$

Step 2 – Distance to default

$$d_2 = \frac{0.3861}{0.38} = 1.016$$

Step 3 – Risk-neutral PD

$$N(-1.016) = 15.48\%$$

Interpretation
The 10% expected return is a red herring here: a risk-neutral PD keeps r. With 38% volatility the drift term is negative even though $r > 0$, because $\sigma^2/2 = 7.22\%$.

Why the other options are wrong
A is incorrect. Used $N(-d_1)$ with $d_1 = 1.016 + 0.38 = 1.396$: 8.14%.
B is incorrect. Used the 10% expected return as the drift: $d = 1.226$, 11.00%, a real-world figure.
C is incorrect. Dropped the drift: $\ln(62/40)/0.38 = 1.153$, $N(-1.153) = 12.44\%$.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8 and 9.8.3

Solution CR-9 m.6

Answer: B

Bank Q404 • GARP PE2 Q6 • Mock 4 Q7

Step 1 – Numerator with the expected return

$$\ln(400/300) + (0.15 - 0.25^2/2) = 0.28768 + 0.11875 = 0.40643$$

Step 2 – Distance to default

$$d_2 = \frac{0.40643}{0.25} = 1.6257$$

Step 3 – Real-world PD

$$N(-1.6257) = 5.20\%$$

Interpretation

A real-world PD uses the 15% expected asset return; the 3% risk-free rate is the distractor, and gives the higher risk-neutral PD of 12.60%.

Why the other options are wrong

A is incorrect. Added the variance term instead of subtracting it: $d = 1.876$, 3.03%.
C is incorrect. Used the 3% risk-free rate as the drift: $d = 1.146$, 12.60% (risk-neutral).
D is incorrect. Reported $N(d_2) = 94.80\%$, the probability of no default.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.2 and 9.8.3; Doumpos et al., Ch.1, 5.6.3.1 eq. 5.3

Solution CR-9 m.7 — Answer: C

Bank Q396 • AP Mock 2 Q76

Step 1 – Default point and log coverage

The stem treats all GBP 10 billion as due in one year: $\ln(25/10) = 0.9163$.

Step 2 – Real-world drift

$$\mu - \sigma_V^2/2 = 0.08 - 0.10125 = -0.02125$$

Step 3 – DD and PD

$$\text{DD} = \frac{0.9163 - 0.02125}{0.45} = 1.989, \qquad \text{PD} = N(-1.989) = 2.34\%$$

Interpretation

With 45% volatility the variance correction outweighs the 8% expected return, so the drift term is negative. The split between short-term and long-term debt is a trap: the stem says to treat the whole GBP 10 billion as the barrier.

Why the other options are wrong

A is incorrect. Used short-term debt only (GBP 6 billion): DD 3.12, 0.09%.
B is incorrect. Used a KMV-style default point of $6 + 0.5 \times 4 = 8$: DD 2.48, 0.65%.
D is incorrect. Used the 3% risk-free rate instead of the 8% expected return: DD 1.88, 3.02%.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.3; Doumpos et al., Ch.1, 5.6.3.1

Solution CR-9 m.8 — Answer: B

Bank Q399 • AnalystPrep Q.3054

Step 1 – Scale to the horizon $T = 0.25$

$$\ln 1.3 = 0.2624, \qquad (0.15 - 0.045) \times 0.25 = 0.02625, \qquad \sigma_V\sqrt{T} = 0.30 \times 0.5 = 0.15$$

Step 2 – Distance to default

$$d_2 = \frac{0.2624 + 0.02625}{0.15} = 1.924$$

Step 3 – PD over three months

$$N(-1.924) = 2.72\%$$

Interpretation

Both the drift and the volatility must be scaled to three months: the drift by T, the volatility by $\sqrt{T}$. Over a short horizon the drift matters little; the volatility scaling matters a lot.

Why the other options are wrong

A is incorrect. Omitted the $-\sigma^2/2$ term: $d = 1.999$, 2.28%.
C is incorrect. Dropped the drift term entirely: $d = 0.2624/0.15 = 1.749$, 4.01%.
D is incorrect. Did not scale to three months (annual formula): $d = 1.225$, 11.04%.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.3

Solution CR-9 m.9 **Answer: B**

Workbook A5.1 • S3, S4, S7 • Aug-2026

Step 1 – d_1

$$d_1 = \frac{\ln(150/120) + (0.03 + 0.25^2/2)(2)}{0.25\sqrt{2}} = \frac{0.2231 + 0.1225}{0.3536} = 0.9776$$

Step 2 – d_2 (distance to default)

$$d_2 = d_1 - \sigma_V\sqrt{T} = 0.9776 - 0.3536 = 0.6241$$

$\Rightarrow$ Assets need to fall about 0.62 two-year standard deviations for default to be triggered.

Step 3 – Risk-neutral PD

$$Q(2) = N(-d_2) = N(-0.6241) = 0.2663$$

$\Rightarrow$ A 26.6% risk-neutral probability that $V_T < D$ at year 2, so the equity call option is not exercised.

Step 4 – Average hazard rate

$$\bar{\lambda} = -\frac{1}{T}\ln(1 - Q(T)) = -\frac{1}{2}\ln(1 - 0.2663) = \frac{0.3096}{2} = 0.1548$$

$\Rightarrow$ A constant default intensity of 15.5% per year reproduces a 26.6% two-year cumulative default probability.

Interpretation

Because the PD comes from an option pricing model with assets growing at r, both numbers are risk-neutral, which is the right type for a CVA desk (valuation). The two-year horizon is no obstacle: the drift is multiplied by T and the volatility by $\sqrt{T}$.

Why the other options are wrong

A is incorrect. Used $N(-d_1) = N(-0.9776) = 0.164$ instead of $N(-d_2)$; the hazard rate $-\ln(1 - 0.164)/2 = 0.090$ inherits the error.
C is incorrect. Correct PD, but divided the cumulative PD by T: $0.2663/2 = 0.133$. This ignores survival compounding in $Q = 1 - \mathrm{e}^{-\bar{\lambda}T}$ and understates the hazard rate.
D is incorrect. Scaled volatility by T instead of $\sqrt{T}$: $d_2 = (0.2231 - 0.0025)/0.50 = 0.4413$, PD $= N(-0.4413) = 0.330$, hazard $-\ln(0.670)/2 = 0.200$.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8 and 9.8.3; 9.2.1 eq. 9.2

CR-10 • Credit Value at Risk • Vardeez: VERY IMPORTANT

CR-10 d

P1 CONCEPT

Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach.

Workbook: A31 Basel IRB Capital • Tier A • August 2026 recall D20 Credit Model Comparison • Tier D • Map #41

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: "credit risk IRB" came up in the August 2026 sitting (A31). D20 (Map #41) adds Vasicek as the model under the IRB formula.
Vardeez: Vasicek's model: know the details. For CR-10 as a whole, do not get into formulas too much, but know each model.

Reported exam shapes

- **A31·S1** IRB capital for a single corporate loan under Advanced IRB, with the correlation formula and the maturity adjustment: the plain form.
- **D20·S3** Vasicek as the model under the Basel IRB formula. The planted errors: "Vasicek captures downgrades" and "the Gaussian copula gives strong tail dependence".

What the questions look like: mostly conceptual: which model sits under IRB, what ρ and X do, how ρ is estimated, what the capital formula deducts.

Numericals kept outside the TYPE rule (CONCEPT LO): Workbook A31.1 and A31.2 (Tier A, August 2026; also in the ORR book under ORR-21 f) and Bank Q451 (GARP PE1 Q19), a Vasicek capital calculation from GARP's own practice exam. They are Questions d.7 to d.9.

Worked elsewhere: A31.3 to A31.5 (Foundation inputs, 2017 revisions, output floor) test ORR-21 and ORR-23 and sit in the ORR book. The model comparison (D20) is CR-10 e.

1. Vasicek's model: the worst case default rate

Vasicek's Gaussian copula model gives a **high percentile of the default rate** of a loan portfolio. The worst case default rate, written WCDR(T, X), is the Xth percentile of the default rate over a period of length T (GARP eq. 10.1):

$$\text{WCDR}(T, X) = N\left(\frac{N^{-1}(\text{PD}) + \sqrt{\rho}\, N^{-1}(X)}{\sqrt{1-\rho}}\right) \qquad N^{-1}(0.999) = 3.090$$

PD is the probability of default over T, ρ the credit (copula) correlation, X the confidence level. **Where it comes from:** each borrower's credit index is $\sqrt{\rho}\, F + \sqrt{1-\rho}\, Z_i$, with F common to all borrowers. Given F, defaults are independent, so a large portfolio's default rate is $N((N^{-1}(\text{PD}) - \sqrt{\rho}F)/\sqrt{1-\rho})$. The 1-in-1,000 bad year is $F = -N^{-1}(0.999)$, which turns the minus into a plus (the conditional PD of CR-12 f is the same equation).

- $\rho = 0$ **gives** WCDR $=$ PD, whatever X is: with independent defaults a large portfolio's default rate is just the PD.
- ρ **appears twice:** $\sqrt{\rho}$ scales up the bad-year shock and $\sqrt{1-\rho}$ shrinks the denominator, so both push WCDR up. At PD $= 1\%$: $\rho = 0.10$ gives 7.75%, 0.20 gives 14.55%, 0.30 gives 22.44% (Figure 1).
- X **enters only through** $N^{-1}(X)$**:** it picks how bad a year the capital must survive. Basel fixes $X = 99.9\%$, $T = 1$ year.

- **A higher PD raises WCDR less than proportionally.** At $\rho = 0.20$, PD 1% gives 14.55% (14.6 times); PD 2% gives 22.63% (11.3 times).

Example (Vardeez; Carey's Table 21.4 prints 0.146). PD $= 1\%$, $\rho = 0.20$: $(-2.326 + 0.4472 \times 3.090)/\sqrt{0.8} = -0.944/0.894 = -1.056$, and $N(-1.056) = 14.55\%$.

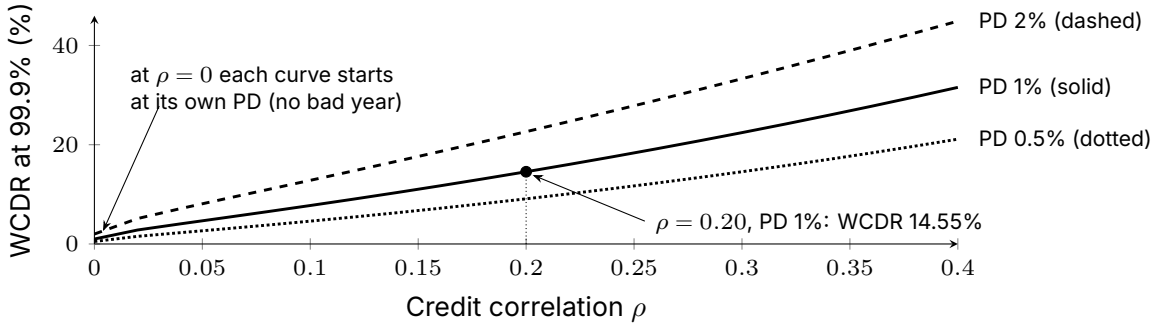


Figure 1. The 99.9% worst case default rate against the credit correlation for PDs of 2% (dashed), 1% (solid) and 0.5% (dotted), from eq. 10.1. A modest correlation of 0.20 turns a 1% PD into a 14.55% worst-year default rate, which is why the regulator, not the bank, sets ρ under IRB.

2. From the worst case default rate to a capital number

For one loan the Xth percentile of the loss is $\text{WCDR}(T, X) \times \text{EAD} \times \text{LGD}$. **Gordy (2003):** for a large portfolio of n small loans in a one-factor world, the Xth percentile of the portfolio loss is approximately (GARP eq. 10.2)

$$\sum_{i=1}^{n} \text{WCDR}_i(T, X) \times \text{EAD}_i \times \text{LGD}_i$$

Each loan's contribution depends only on its own PD, LGD and EAD (the asymptotic single risk factor model), so a regulator can compute capital loan by loan and add it up, whatever else the bank holds. The price is two assumptions: one systematic factor and an infinitely granular portfolio (real banks have sizable single-name exposures, hence the 2014 Large Exposure Framework). Hull: regulatory capital for the banking book is set from this sum with $T = 1$ year and $X = 99.9\%$, each loan's term sometimes multiplied by a maturity adjustment factor. The precise Basel form also deducts expected loss.

3. Basel II IRB: what the regulator fixes and what the bank supplies

	Foundation IRB	Advanced IRB
PD	Bank's own (floor 0.03%, bank and corporate)	Bank's own (same floor)
LGD, EAD, M	Set by the accord: LGD 45% senior, 75% subordinated; $M = 2.5$	Bank's own estimates
ρ, T, X	Basel formula by asset class; 1 year; 99.9%	Same: never the bank's choice

$$\boxed{K = \text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA}} \qquad \text{RWA} = 12.5 \times K$$

- **Why** $-$PD**.** Expected loss is covered by loan loss reserves and pricing, so capital covers *unexpected* loss only. Carey writes $\sum \text{EAD}_i \times \text{LGD}_i \times \text{DR}_{99.9,i} - \text{EL}$.
- **Correlation, set by regulators.** Bank, corporate and sovereign: $\rho = 0.12\,w + 0.24\,(1 - w)$, $w = (1 - \text{e}^{-50\,\text{PD}})/(1 - \text{e}^{-50})$, so ρ *falls* from 0.24 to 0.12 as PD rises (very risky borrowers default for their

own reasons; middle-risk ones default together in a downturn). Retail is lower: 0.15 residential mortgages, 0.04 qualifying revolving, 0.03 to 0.16 other retail.

- **Maturity adjustment** (bank, corporate and sovereign only): MA $= (1 + (M - 2.5)\,b)/(1 - 1.5\,b)$, $b = (0.11852 - 0.05478 \ln \text{PD})^2$. A loan lasting more than a year can deteriorate without defaulting; low-PD loans have the most room to fall, so they get the largest MA. MA $= 1$ at $M = 1$. **Retail has none.**
- **Downgrades.** Vasicek's model counts defaults only; GARP notes that the MA is where downgrades enter the regulatory implementation (D20 shape S3).

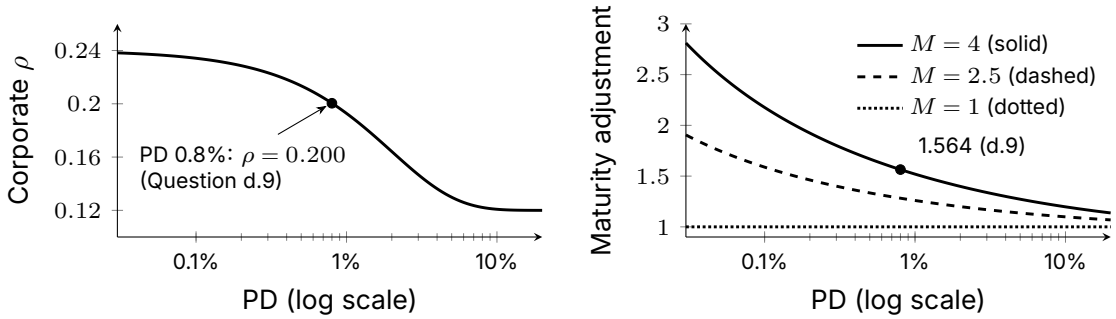


Figure 2. Left: the wholesale correlation falls from 0.24 to 0.12 as PD rises. Right: the maturity adjustment for 4-year (solid), 2.5-year (dashed) and 1-year (dotted, flat at 1) loans, largest for low-PD loans. The marked points are the inputs of Workbook A31.1 (Question CR-10 d.9).

4. Estimating the correlation, and the model's weak spot

Estimating ρ. Structural models show that ρ for two companies is roughly the correlation between the returns on their *assets*, approximately the correlation between the returns on their *equities*. Use the average equity-return correlation of the obligors; for private companies, that of similar publicly traded companies. A bank calibrates its own economic capital model this way; under IRB the Basel formula replaces it. **Weak spot:** the Gaussian copula "incorporates very little tail correlation", so years in which defaults cluster are understated; another copula (a Student t, for example) can remedy this.

> **Correction to the source notes.** (1) Vardeez gives "Capital requirement $= \sum \text{WCDR}_i \times \text{EAD}_i \times \text{LGD}_i$". That is GARP's eq. 10.2, the 99.9% *loss percentile*, which still contains expected loss. Basel II IRB capital deducts EL and, for wholesale exposures, applies the MA: $K = \text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA}$, as in Workbook A31. Hull's "set equal to" eq. 10.2 is shorthand. When a stem defines capital as the loss percentile (Question d.7), answer on that definition. (2) Vardeez says return on investment or return on equity can proxy for ρ; Schweser (LO 28.d) says ROA or ROE. GARP's proxy is the correlation between the companies' *equity returns* (stock returns). ROE and ROI are accounting ratios, not market returns (Question d.2).

	Loss percentile (Hull eq. 10.2)	IRB capital: wholesale	IRB capital: retail
Formula	$\sum \text{WCDR} \times \text{EAD} \times \text{LGD}$	$\text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA}$	$\text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD})$
Expected loss	Included	Deducted	Deducted
ρ	Estimated: equity-return correlation	Formula: 0.24 down to 0.12 as PD rises	Fixed: 0.15 mortgages, 0.04 QRRE; 0.03 to 0.16 other
Maturity	Not used	MA from M (F-IRB: $M = 2.5$)	None

Exam traps

- **Dropped −PD; capital versus RWA.** WCDR alone is the stress loss; IRB capital covers WCDR − PD unless the stem defines capital as the full percentile. RWA $= 12.5 \times K$: read which one is asked.
- **Maturity adjustment misapplied.** Omitted for a corporate loan; $M = 2.5$ used when an Advanced IRB stem gives M; applied to retail, which has none.
- **Correlation twins.** Weights swapped so that ρ rises with PD; the corporate formula, or 0.04, used for mortgages.
- **Polarity of ρ.** A higher ρ raises WCDR and capital: defaults cluster, so diversification helps less.
- **Wrong critical value.** $N^{-1}(0.999) = 3.090$, not 2.326 (99%) or 1.645 (95%). At $\rho = 0$, WCDR is simply the PD.
- **Sibling swaps from D20.** Vasicek counts defaults only (downgrades enter only through the MA), and the Gaussian copula has *little* tail dependence.

Practice questions: CR-10 d

Question CR-10 d.1

Bank Q440 • AnalystPrep Q.6024 (rebuilt)

A junior analyst on a bank's regulatory capital team is asked which model produces the high-percentile default rates that drive the Basel II internal-ratings-based (IRB) capital for the banking book. Which model should the analyst identify?

A. CreditMetrics, simulated with a rating transition matrix

B. Vasicek's one-factor Gaussian copula model

C. Credit Risk Plus, with a gamma-distributed default rate

D. Historical simulation of past portfolio losses

Question CR-10 d.2

Bank Q447 • Schweser Module Quiz 28.2 Q1 (rebuilt)

A credit analyst is calibrating Vasicek's model for the bank's economic capital model. The portfolio contains loans to two publicly traded companies, A and B. Company A's return on equity is 8% and Company B's is 6%. The average correlation between the two companies' equity (stock) returns over the past three years is 0.24; their asset returns cannot be observed directly. Which value of the credit correlation ρ should the analyst use?

A. 0.12

B. 0.24

C. 0.48

D. 0.76

Question CR-10 d.3

Bank Q444 • AnalystPrep Q.6543 (rebuilt)

Vasicek's model gives $\text{WCDR}(T, X) = N\left[(N^{-1}(\text{PD}) + \sqrt{\rho}\, N^{-1}(X))/\sqrt{1-\rho}\right]$. A bank has a large loan portfolio with a one-year PD of 2.0% and sets $X = 99.9\%$. Evaluating the formula gives a WCDR of 2.00% when $\rho = 0$ and 22.63% when $\rho = 0.20$. Which statement about the role of the credit correlation ρ is correct?

A. At $\rho = 0$ the worst case default rate equals the PD, and it rises as ρ increases because defaults cluster in the bad years.

B. The worst case default rate falls as ρ increases, because a higher correlation means greater diversification.

C. At $\rho = 0$ the worst case default rate is already far above the PD, because X is so high.

D. Once X is fixed at 99.9%, the worst case default rate no longer depends on ρ.

Question CR-10 d.4

Bank Q442 • AnalystPrep Q.6040 (rebuilt)

A risk manager presents Vasicek's model to the board with PD $= 2\%$, $\rho = 0.12$ and a confidence level $X = 99.9\%$. A board member asks what role X plays in the capital calculation. Which answer is correct?

A. It sets the expected loss that the capital must cover.

B. It is the minimum percentage of exposure that must be held as capital.

C. It sets the degree of diversification the loan portfolio must achieve.

D. It selects the percentile of the default rate distribution, and so the worst case default rate the bank must withstand.

Question CR-10 d.5

Bank Q445 • BT P2.T6.24.18.1 (rebuilt)

A bank uses Vasicek's Gaussian copula model for its loan portfolio. Which of the following statements about the model is INCORRECT?

A. The Xth percentile loss for a large portfolio of small loans is approximated by $\sum_i \text{WCDR}_i(T, X) \times \text{EAD}_i \times \text{LGD}_i$.

B. The credit correlation ρ can be estimated from the average correlation between the equity returns of the companies in the portfolio, or of similar publicly traded companies if they are private.

C. The model is used to set the interest rate charged on new loans by assessing the portfolio's expected profit.

D. For regulatory capital the sum is evaluated with T equal to one year and X equal to 99.9%.

Question CR-10 d.6 New • built from GARP Ch.10, 10.2 and 10.4 (footnote 7); Workbook A31 and D20 notes • shape D20·S3

The head of model validation at a bank is reviewing an internal memo on how Basel II IRB capital uses Vasicek's model. Which of the following statements in the memo is correct?

A. In the IRB implementation, the maturity adjustment is the channel through which downgrade risk enters, because Vasicek's model itself counts defaults only.

B. Vasicek's model captures losses from both downgrades and defaults, which is why regulators chose it for IRB capital.

C. Because it uses a Gaussian copula, Vasicek's model generates strong tail dependence between defaults in a crisis.

D. Under Advanced IRB each bank sets the correlation ρ from the equity-return correlations of its own obligors.

Question CR-10 d.7 Bank Q451 • GARP PE1 Q19 • Mock 3 Q61 (rebuilt) • numerical, kept outside the TYPE rule

A bank estimates credit capital for a large, granular portfolio of equally sized exposures using Vasicek's Gaussian copula model. Capital is defined as the one-year 99.9th percentile of the loss distribution; ignore any maturity adjustment. The portfolio data are:

Total exposure at default	EUR 200 million
One-year probability of default (PD)	2.0%
Credit correlation ρ	0.12
Loss given default (LGD)	60%

Useful values: $N^{-1}(0.02) = -2.054$, $N^{-1}(0.999) = 3.090$, $N^{-1}(0.95) = 1.645$. Which of the following is closest to the capital requirement?

A. EUR 6.82 million

B. EUR 15.27 million

C. EUR 17.67 million

D. EUR 29.46 million

Question CR-10 d.8 Workbook A31.2 • S2 • Aug-2026 • numerical, kept outside the TYPE rule

A model validation analyst at a UK building society is checking the IRB risk-weighted assets reported for its residential mortgage book. The data are:

- EAD: GBP 400 million
- Average one-year PD: 1.50%; average LGD: 25%
- Average remaining contractual maturity: 18 years
- Confidence level 99.9%; Basel II retail treatment applies

Which of the following is closest to the risk-weighted assets for this book?

A. GBP 163.2 million

B. GBP 182.0 million

C. GBP 191.9 million

D. GBP 575.5 million

Question CR-10 d.9 Workbook A31.1 • S1 • Aug-2026 • numerical, kept outside the TYPE rule

A credit risk analyst at a German universal bank using the Advanced IRB approach is computing the Basel II capital for a single corporate term loan. The data are:

- EAD: EUR 250 million
- One-year PD: 0.80%; LGD: 40%
- Remaining maturity: 4 years
- Correlation from the Basel corporate formula; confidence level 99.9%

Which of the following is closest to the required capital for this loan?

A. EUR 11.8 million

B. EUR 14.4 million

C. EUR 15.1 million

D. EUR 18.4 million

Solutions • CR-10 d

Solution CR-10 d.1 — Answer: B

Bank Q440 • AnalystPrep Q.6024

Step 1 – What IRB capital needs

A high percentile (99.9%) of the one-year default rate for each exposure, from its PD and a correlation, which can then be added loan by loan.

Step 2 – Match the model

Vasicek's one-factor Gaussian copula delivers exactly that: WCDR(T, X) from PD and ρ, summed over loans by Gordy's result.

$\Rightarrow$ The IRB formula is Vasicek's model with $T = 1$ year and $X = 99.9\%$.

Interpretation

The other three are real credit VaR tools, but banks use them for internal (economic capital) models, not for the regulatory formula.

Why the other options are wrong

A is incorrect. CreditMetrics simulates rating migrations with a transition matrix; it is an internal mark-to-market model, not the IRB formula.

C is incorrect. Credit Risk Plus is the actuarial Poisson and negative binomial model; it does not produce the IRB worst case default rate.

D is incorrect. Historical simulation is the main market risk VaR tool; a one-year credit loss history is far too short for a 99.9% percentile.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2 Vasicek's Model, eqs. 10.1 and 10.2

Solution CR-10 d.2 — Answer: B

Bank Q447 • Schweser Module Quiz 28.2 Q1

Step 1 – What ρ should measure

Structural models show that ρ for two companies is roughly the correlation between the returns on their assets.

Step 2 – The observable proxy

Asset returns are not observed, so GARP uses the correlation between the companies' equity returns as the approximation.

$\Rightarrow \rho \approx 0.24$.

Interpretation

The ROE figures are a red herring: they are accounting ratios for each firm, not a correlation of market returns. Schweser's original question gave only ROEs and an "average correlation between the two", which invites exactly this confusion (see the correction box). No scaling of the equity correlation is applied.

Why the other options are wrong

A is incorrect. Halved the correlation, $0.24/2 = 0.12$; there is no such adjustment.

C is incorrect. Doubled it, $2 \times 0.24 = 0.48$.

D is incorrect. Used the complement, $1 - 0.24 = 0.76$.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2

Solution CR-10 d.3 — Answer: A

Bank Q444 • AnalystPrep Q.6543

Step 1 – Set $\rho = 0$

The $\sqrt{\rho}\,N^{-1}(X)$ term vanishes and the denominator is 1: WCDR $= N(N^{-1}(0.02)) = 2.00\%$, the PD itself.

$\Rightarrow$ With independent defaults a large portfolio's default rate converges to the PD, so X has no effect.

Step 2 – Set $\rho = 0.20$

$$\frac{-2.0537 + 0.4472 \times 3.0902}{\sqrt{0.8}} = \frac{-0.6718}{0.8944} = -0.7510, \qquad N(-0.7510) = 22.63\%$$

$\Rightarrow$ The worst-year default rate is more than eleven times the PD once defaults share a common factor.

Interpretation

ρ raises WCDR through both the numerator (a bigger common shock) and the denominator (less idiosyncratic noise to average out). This is why capital is so sensitive to the correlation input.

Why the other options are wrong

B is incorrect. Polarity flip: a higher correlation means *less* diversification, and WCDR rises from 2.00% to 22.63%.

C is incorrect. At $\rho = 0$ the $N^{-1}(X)$ term is multiplied by zero, so WCDR is the PD of 2.00%.

D is incorrect. ρ appears in both the numerator and the denominator, so WCDR changes with ρ at any fixed X.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2, eq. 10.1

Solution CR-10 d.4 — Answer: D

Bank Q442 • AnalystPrep Q.6040

Step 1 – Where X enters

Only through $N^{-1}(X)$ in the numerator, which sets the size of the bad-year shock to the common factor.

Step 2 – What it therefore controls

X chooses the percentile of the default rate distribution: at 99.9% the capital covers the default rate exceeded only one year in 1,000.

$\Rightarrow$ X fixes how bad a year the bank must survive; PD, LGD and EAD set the size of the loss in that year.

Interpretation

Basel sets $X = 99.9\%$ for everyone. A bank can choose a different level for economic capital, for example to match a target credit rating.

Why the other options are wrong

A is incorrect. Expected loss depends on PD, LGD and EAD only; X concerns the tail beyond it.

B is incorrect. X is a probability level, not a capital ratio; capital also depends on PD, ρ, LGD and EAD.

C is incorrect. Diversification enters through ρ and through granularity, not through X.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2, eq. 10.1

Solution CR-10 d.5 — Answer: C

Bank Q445 • BT P2.T6.24.18.1

Step 1 – Test A, B and D against GARP 10.2

A is Gordy's result (eq. 10.2). B is GARP's method for ρ: asset-return correlation, approximated by equity-return correlation, with similar listed companies for private ones. D gives the regulatory parameters.

$\Rightarrow$ All three are true.

Step 2 – Test C

Vasicek's model produces a high percentile of the default rate and so a capital number. It says nothing about the expected profit of a loan or the rate to charge.

$\Rightarrow$ C is the incorrect statement.

Interpretation

The bank's source key treated B as the false statement. It is GARP's own wording, so the corrected key is C.

Why the other options are wrong

A is incorrect. True: the large-portfolio approximation of the Xth percentile loss.

B is incorrect. True: the equity-return correlation is GARP's practical proxy for ρ.

D is incorrect. True: regulatory capital uses one year and 99.9%, sometimes with a maturity adjustment.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2, eq. 10.2

Solution CR-10 d.6 — Answer: A

New • built from GARP Ch.10, 10.2 and 10.4 (footnote 7)

Step 1 – What Vasicek's model counts

GARP 10.4: "Vasicek's model and Credit Risk Plus estimate the probability distribution of losses arising from defaults. The impact of downgrades is not considered." Footnote 7 adds that the maturity adjustment does allow for downgrades in the regulatory implementation.

$\Rightarrow$ A is correct, and B is the sibling swap with CreditMetrics.

Step 2 – Tail dependence and who sets ρ

GARP 10.2: the model "incorporates very little tail correlation". Under IRB the correlation comes from the Basel formula for each asset class, not from the bank.

$\Rightarrow$ C and D are false.

Interpretation

This is the D20 audit format: three statements each borrow a true fact from the wrong place (CreditMetrics' downgrades, a fat-tailed copula, an economic capital calibration), and one states the regulatory link correctly.

Why the other options are wrong

B is incorrect. Sibling swap: downgrades plus defaults is CreditMetrics. Vasicek counts defaults only.

C is incorrect. Polarity flip: the Gaussian copula has *little* tail correlation; another copula is needed to add it.

D is incorrect. Scope error: equity correlations calibrate a bank's own economic capital model. Under Foundation and Advanced IRB alike, ρ is set by the Basel formula.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2 and 10.4 with

footnote 7; Carey, Capital Regulation Before the Global Financial Crisis (ORR-21), The IRB Approach

Solution CR-10 d.7 — Answer: C

Bank Q451 • GARP PE1 Q19 • Mock 3 Q61

Step 1 – Argument of $N(\cdot)$

$$\frac{N^{-1}(0.02)+\sqrt{0.12}\,N^{-1}(0.999)}{\sqrt{1-0.12}}=\frac{-2.054+0.3464\times 3.090}{0.9381}=\frac{-0.9836}{0.9381}=-1.0485$$

Step 2 – Worst case default rate

$$\text{WCDR}=N(-1.0485)=14.72\%\quad(14.73\%\text{ with unrounded inputs})$$

$\Rightarrow$ In the 1-in-1,000 year, about 14.7% of the portfolio defaults against 2% on average.

Step 3 – Capital as defined in the stem

$$200\times 0.60\times 0.1473=\text{EUR }17.67\text{ million}$$

Interpretation

The stem defines capital as the full 99.9th percentile of loss, Hull's eq. 10.2, so expected loss is not deducted. Under the Basel IRB formula proper, capital would be $200\times 0.60\times(0.1473-0.02)=15.27$, option B: read the definition in the stem before deciding. The $N^{-1}(0.95)$ value is a planted distractor input.

Why the other options are wrong

A is incorrect. Used the 95% quantile 1.645: WCDR $=N\big((-2.054+0.3464\times 1.645)/0.9381\big)=5.68\%$, so $120\times 0.0568=6.82$.

B is incorrect. Deducted the expected loss of $200\times 0.60\times 0.02=2.40$, although the stem defines capital as the full percentile: $17.67-2.40=15.27$.

D is incorrect. Omitted LGD: $200\times 0.1473=29.46$.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2, eqs. 10.1 and 10.2

Solution CR-10 d.8 — Answer: A

Workbook A31.2 • S2 • Aug-2026

Step 1 – Correlation

Residential mortgages use the fixed retail correlation $\rho=0.15$.

Step 2 – WCDR

$$z=\frac{N^{-1}(0.015)+\sqrt{0.15}\times 3.0902}{\sqrt{0.85}}=\frac{-2.1701+1.1968}{0.9220}=-1.0556$$

$$\text{WCDR}=N(-1.0556)=0.1456$$

$\Rightarrow$ In the 99.9th percentile year, 14.56% of the mortgage pool defaults.

Step 3 – Capital (no maturity adjustment for retail)

$$K=400\times 0.25\times(0.1456-0.0150)=\text{GBP }13.06\text{ million}$$

Step 4 – RWA

$$\text{RWA} = 12.5 \times 13.06 = \text{GBP } 163.2 \text{ million}$$

Interpretation

A risk weight of $163.2/400 = 40.8\%$. The 18-year maturity is a red herring: retail exposures carry no maturity adjustment.

Why the other options are wrong

B is incorrect. Dropped the $-$PD term: $12.5 \times 400 \times 0.25 \times 0.1456 = 182.0$.
C is incorrect. Applied the corporate correlation formula ($\rho = 0.1767$): WCDR $= 0.1685$, RWA $= 12.5 \times 100 \times (0.1685 - 0.015) = 191.9$.
D is incorrect. Applied a maturity adjustment with $M = 18$ ($b = 0.1215$, MA $= 3.526$), which retail does not have: $163.2 \times 3.526 = 575.5$.

LO: ORR-21 f: Calculate credit risk capital under Basel II utilizing the IRB approach. CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. • **Reference:** Carey, Capital Regulation Before the Global Financial Crisis, Retail Exposures Under IRB; Hull, RMFI 6e, Ch.19, 10.2

Solution CR-10 d.9

Answer: D

Workbook A31.1 • S1 • Aug-2026

Step 1 – Correlation weight and corporate correlation

$$w = \frac{1 - \mathrm{e}^{-50 \times 0.008}}{1 - \mathrm{e}^{-50}} = 0.3297, \qquad \rho = 0.12(0.3297) + 0.24(0.6703) = 0.2004$$

$\Rightarrow$ About 20% of each obligor's asset variance comes from the single systematic factor.

Step 2 – Argument of $N(\cdot)$

$$z = \frac{N^{-1}(0.008) + \sqrt{0.2004} \times 3.0902}{\sqrt{1 - 0.2004}} = \frac{-2.4089 + 1.3835}{0.8942} = -1.1468$$

Step 3 – WCDR

$$\text{WCDR} = N(-1.1468) = 0.1257$$

$\Rightarrow$ In the 1-in-1,000-year bad year, 12.57% of loans like this default, against 0.80% on average.

Step 4 – Maturity adjustment

$$b = \left[0.11852 - 0.05478 \ln(0.008)\right]^2 = 0.38302^2 = 0.1467$$

$$\text{MA} = \frac{1 + (4 - 2.5)(0.1467)}{1 - 1.5(0.1467)} = \frac{1.2201}{0.7799} = 1.5643$$

$\Rightarrow$ A 4-year loan can be downgraded without defaulting, so capital is scaled up by about 56%.

Step 5 – Capital

$$K = 250 \times 0.40 \times (0.1257 - 0.0080) \times 1.5643 = \text{EUR } 18.4 \text{ million}$$

Interpretation

RWA $= 12.5 \times 18.4 =$ EUR 230.2 million. Expected loss, $250 \times 0.40 \times 0.008 =$ EUR 0.8 million, is covered by reserves and pricing, not capital.

Why the other options are wrong

A is incorrect. Omitted the maturity adjustment: $250 \times 0.40 \times 0.1177 = 11.8$.

B is incorrect. Swapped the correlation weights so that ρ rises with PD: $\rho = 0.24(0.3297) + 0.12(0.6703) = 0.1596$, WCDR $= 0.1001$, capital $= 100 \times 0.0921 \times 1.5643 = 14.4$.

C is incorrect. Used the Foundation default $M = 2.5$ (MA $= 1.2821$) although the Advanced IRB stem gives 4 years: $100 \times 0.1177 \times 1.2821 = 15.1$.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. ORR-21 f: Calculate credit risk capital under Basel II utilizing the IRB approach. • **Reference:** Hull, RMFI 6e, Ch.19, 10.2 eq. 10.1; Carey, Capital Regulation Before the Global Financial Crisis, Bank, Corporate, and Sovereign Exposures Under IRB

CR-16 • Netting, Close-out and Related Aspects • Vardeez: VERY IMPORTANT

CR-16 d

P1 CONCEPT

Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall • Map #27

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: Map #27 (A21) singles out termination features and trade compression from this reading as "crucial", and CR-16 is a VERY IMPORTANT chapter for Vardeez. **Vardeez:** termination features and trade compression "seem to be crucial here".

Reported exam shapes

- **A21·S7** Termination features and trade compression: which statement about a break clause, a reset, a walkaway clause or a compression cycle is correct. Distractors use absolute claims ("every participant's notional falls") and sibling swaps (a rating-based break given the benefits of a mandatory one).

What the questions look like: all conceptual. A scenario states a need (cap the exposure but keep the hedge; exit if the counterparty weakens; cut gross notional) and asks for the feature, or a list of statements mixes up the properties of the four features. The net notional calculation is CR-16 e; termination events and their effects on the parties are CR-16 f.

Worked elsewhere: the other A21 questions are A21.1 (CR-17 h), A21.2 (CR-17 c), A21.3 and A21.5 (CR-20 j) and A21.4 (CR-17 f).

1. Four features, one purpose

A long-dated derivative may carry a small exposure today and a large, unmanageable one later. Gregory treats the contractual features below as "a more basic form of margining" (GARP Ch.17, Section 17.1; Schweser and Vardeez teach them here): resets act less often than margin, break clauses end the trade early, walkaway clauses mitigate nothing, and compression removes redundant trades. The table after Section 5 compares all four.

2. Reset agreements

A reset agreement periodically resets the parameters that determine a trade's value, so that neither party becomes strongly in the money; reset dates may coincide with payment dates or be triggered when the MtM breaches a set level. In a **resettable cross-currency swap** the MtM, driven mainly by the FX rate on the final exchange of notional, is cash-settled on each reset date and the FX rate is reset to spot, so the notional on one leg changes. Economically this terminates the swap at its MtM and executes a replacement at market rates: the exposure returns to zero on each reset date (GARP Figure 17.1). Resets are mandatory and linked to no external event, so they are easy to model, unlike break clauses. They are a weaker form of margining: they refix far less often than daily margin, and a refix is a *settlement*, whereas OTC margin is posted as *security*.

3. Break clauses and additional termination events

A **break clause** (Schweser: a liquidity put or early termination option) allows a trade to be terminated **mandatorily**, **optionally** or **contingent on a defined event**, for one or both parties.

- **ATE.** An Additional Termination Event in the ISDA schedule, usually at counterparty level. The most common is a rating downgrade (for example below investment grade); for unrated parties such as hedge funds, net asset value, market capitalisation or a key person departure.
- **Mutual puts.** Transaction-level breaks on set dates, mandatory or optional (GARP: a 15-year swap with a mutual put in year five and every two years thereafter; Figure 1). A mandatory break is sometimes used because a party cannot book a long-dated trade at all, with a "gentleman's agreement" to re-execute at the break date, much like a reset.
- **On exercise** the trade need not terminate: the party against which the clause is exercised may provide a replacement counterparty or a third-party guarantee, post margin, or terminate at the current replacement value. A termination is at the prevailing MtM set out in the documentation, typically priced by a panel of banks: a base valuation, so it largely avoids the xVA and close-out disputes that follow a default.

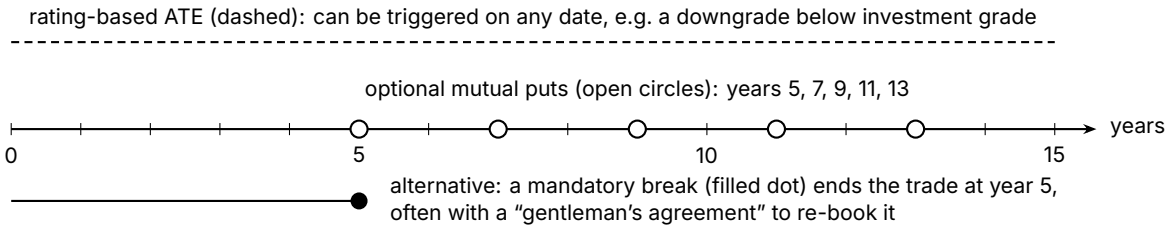


Figure 1. Where termination rights sit on a 15-year swap (GARP's mutual-put example). Optional dates give a right at fixed points; a rating trigger can fire at any time; a mandatory break is certain.

Benefits by type (GARP Table 17.1). Mandatory breaks deliver everything; the others much less.

Break type	Counterparty risk reduction	CVA reduction	LCR	Capital relief
Mandatory	Yes	Yes	Yes	Yes
Optional	Partly	No	Yes	No
Rating-based	Partly	No	No	No

A CVA desk credits only mandatory breaks (it controls neither the option nor the trigger); the LCR forces banks to pre-fund rating-based triggers; Basel rules out capital relief for rating triggers, and optional breaks earn it only if they are shown to be exercised rigorously.

Why break clauses disappoint (the disadvantages)

- **The banker's paradox.** A break helps only if exercised early, before credit quality falls a long way, but exercise damages the client relationship, so it tends to be used too late. Inside the bank it is a moral hazard: sales staff use the break to win the trade, then argue against exercising it. GARP adds that banks, now sensitive to xVA and capital, exercise optional breaks far more readily.
- **Discrete, not continuous** (Schweser): a break clause is a discrete form of collateralisation; continuous margin does the job better.
- **Slow ratings, systematic risk.** By the time of a downgrade the difficulties may be too acute for the clause to help (the monolines, CR-14 i); a systematic deterioration, as in the global financial crisis, is far harder to escape than an idiosyncratic one.

- **Cliff-edge effects.** Many counterparties with similar triggers can all act on a single-notch downgrade (AIG; see CR-16 f).
- **Hard to model and hedge.** Rating transitions cannot be implied from market data and there is no obvious hedge; only a mandatory break is simple (it happens with probability one).

4. Walkaway clauses

A walkaway (tear-up) clause lets an institution cancel its transactions if the counterparty defaults. It would use it only when it *owes* the defaulter: it walks away from its net liability while still claiming any positive MtM. It does not reduce credit exposure; it creates a gain that offsets potential losses. Common before the 1992 ISDA Master Agreement, it is not in standard documentation now and should be avoided: it costs the defaulter's estate, creates moral hazard (an incentive to contribute to the counterparty's default) and, Schweser adds, masks risk when embedded in the price. At Drexel Burnham Lambert (1990) counterparties chose not to walk away (small gains against legal and reputational costs). Without a clause a party can still stop paying and not close out, as TXU did against Enron Australia (2002) and some Lehman counterparties did.

5. Trade compression

Bilateral compression removes redundant offsetting trades between two parties: net exposure is unchanged, but a smaller gross position cuts operational costs and legal risk and avoids penalties from regulatory methods built on gross notional. **Multilateral (portfolio) compression**, the modern equivalent of exchange clearing rings, achieves multilateral netting without a CCP through the cooperation of many counterparties (TriOptima's TriReduce had eliminated over USD 1 quadrillion of notional by 2017). Trades are eliminated or replaced to reduce the *total* notional of all participants; strictly, compression reduces notional, not the number of contracts, though the two usually move together.

A typical cycle (GARP 16.2.4): (1) participants submit the relevant trades (and, Schweser adds, their risk tolerances); (2) trades are matched with each bilateral counterparty and may be checked against a trade-reporting warehouse; (3) an algorithm finds changes that generate multilateral netting benefits while keeping each participant neutral in value and risk, and participants review them; (4) the changes become binding: trades are unwound, new ones executed and others novated. It needs standard, fungible contracts: CDS trade with fixed coupons and quarterly maturity dates (20 March, June, September, December), so positions bucket cleanly; swaps traded at par carry different fixed rates, so "coupon blending" algorithms are needed. EMIR requires firms with more than 500 non-cleared contracts to consider compression at least twice a year.

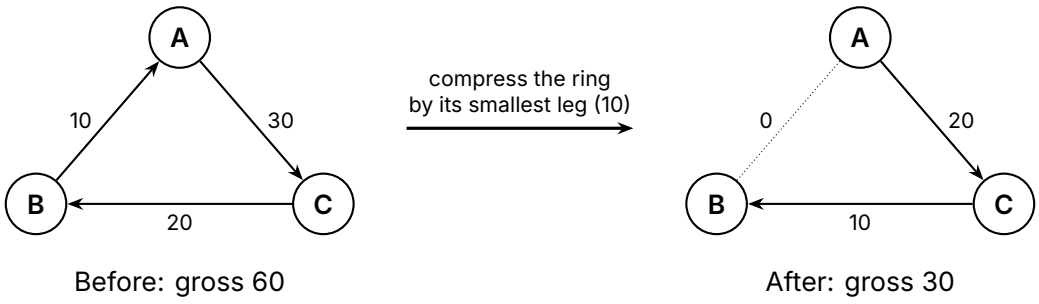


Figure 2. Multilateral compression of a ring of fungible obligations (GARP Figure 16.6; arrows show the direction of payment). Net positions stay at A -20, B $+10$, C $+10$, but gross notional halves and A is left with no exposure at all: the 10 it was owed by B is achieved by paying C 10 less. The trade between B and A (dotted) is torn up.

Advantages. Compression preserves each participant's market risk while reducing non-market risks: settlement and counterparty risk, operational costs (fewer cash flows and trades), legal risk (a simpler portfolio that relies less on netting being enforceable), regulatory capital (the leverage ratio and thresholds based on gross notional) and systemic risk (fewer contracts to replace after a default).

Disadvantages and limits. Diminishing marginal returns as the maximum netting is reached; counterparties must be interchangeable, so of comparable credit quality; an individual party's notional need not fall, and positions between two parties can grow or change sign, so participants set constraints (such as credit limits) that weaken the result; CDS settlement can differ depending on whether the buyer or the seller triggers it, so long and short protection do not net perfectly; and new trades may fall under rules (such as bilateral margin) from which legacy trades were exempt, while accounting, tax or legal reasons (hedge accounting for an end user) can make compression unattractive.

Corrections to the source notes. (1) Vardeez's "Gross versus net" box (CR-16 e) ends: "The point of the exercise is operational and capital efficiency, not risk reduction." GARP 16.2.4 calls portfolio compression "a risk-reduction exercise": it leaves *market* risk unchanged but reduces settlement and counterparty risk and may lower systemic risk. Question CR-16 d.6 turns on this. (2) The A21 revision notes say optional breaks give "partial counterparty risk reduction and capital relief (if rigorously exercised)". In GARP Table 17.1 an optional break's tick is in the LCR column; its capital relief is marked as not achieved, with the text saying only that it "might be possible" if the bank shows the breaks are rigorously exercised.

	Reset agreement	Break clause / ATE	Walkaway clause	Trade compression
Mechanics	MtM settled, terms reset to at the money; trade continues	Trade ended at replacement value (or replaced, guaranteed, margined)	Survivor cancels what it owes a defaulter	Offsetting trades torn up, net position kept
Trigger	Fixed dates or MtM breach	Fixed date, option, or event (downgrade)	Counterparty default	Voluntary cycle (EMIR: consider twice a year)
Exposure	Back to zero each reset	Ended before default	Not reduced; a gain to the survivor	Gross down; market risk unchanged
Weakness	Less frequent than margin	Banker's paradox, slow ratings, cliff edges	Moral hazard; cost to the estate	Diminishing returns; needs fungible trades

Exam traps

- **Compression changes the risk profile (polarity flip).** It cuts gross notional and trade count while keeping each participant's market risk; "only by changing" the risk profile is false.
- **Every participant gains (absolute claim).** The total notional falls; an individual party's notional, or a bilateral position, can rise.
- **Reset versus break (sibling swap).** A reset keeps the trade alive and removes accumulated MtM; a break ends the trade. "Keep the hedge" points to the reset.
- **Pre-agreed termination price.** A break settles at the *current* replacement value, not a value fixed at inception, and the right belongs to the exercising party, not the defaulter.
- **All breaks are equal.** Only mandatory breaks earn CVA reduction and capital relief; rating-based triggers earn neither and must be pre-funded under the LCR. The banker's paradox is exercising too *late*, not too early.

Practice questions: CR-16 d

Question CR-16 d.1

Bank Q761 • AnalystPrep Q.1888

Which of the following best describes the consequence of exercising a break clause in a contract?

A. The defaulting party will have authority to terminate the transaction at its current replacement value.

B. The exercising party will have authority to terminate the transaction at its pre-defined replacement value.

C. The exercising party will have authority to trade the new transaction at a pre-defined replacement value.

D. The exercising party will have the authority to terminate the transaction at its current replacement value.

Question CR-16 d.2

Bank Q764 • BT P2.T6.904.3 (rebuilt)

Doubledown Financial is entering a long-dated cross-currency swap under an ISDA Master Agreement. Its Board wants (1) to limit its credit exposure if the swap becomes deeply in the money, (2) the swap to stay in place for its full life as a hedge, and (3) no reliance on credit-rating triggers, because ratings lag credit deterioration and downgrades can cause sudden cliff-edge effects. Which of the following features is BEST suited to the Board's requirements?

A. A walkaway clause

B. A reset agreement

C. A trigger-based break clause linked to a rating downgrade

D. A mandatory break clause on a fixed future date

Question CR-16 d.3

Bank Q736 • AnalystPrep Q.1892 • re-filed from CR-16 b

Some OTC derivatives contain walkaway features, which allow an institution to terminate transactions if the counterparty defaults. Such clauses do not reduce credit exposure, but they allow the institution to:

A. cease payments and avoid settling the amounts it owes the defaulting counterparty.

B. terminate the contract and gain from the replacement cost estimated before the transaction.

C. cease payments but pay only the net amount to the counterparty.

D. cease payments and institute legal proceedings against the counterparty.

Question CR-16 d.4 Bank Q854 • AnalystPrep Q.6095 • re-filed from CR-18 c

Compression is a key mechanism utilized in central clearing. How does compression benefit financial institutions that hold multiple derivative contracts?

A. It allows them to expand their investment portfolios while maintaining net economic exposure.

B. It consolidates derivative contracts, reducing the number of contracts needed to maintain an equivalent position.

C. It increases market liquidity by expanding the available derivatives that can be traded.

D. It separates and categorizes derivatives based on their risk profiles for better risk management.

Question CR-16 d.5 Bank Q857 • AnalystPrep Q.6128 • re-filed from CR-18 c

A derivatives analyst notes growth in centrally cleared OTC derivatives, but certain characteristics of OTC derivatives make central clearing difficult. For derivatives that cannot be centrally cleared, which approach brings some of the functionality of central clearing into bilateral OTC markets?

A. Implementation of standardized terms for all OTC derivative contracts to facilitate CCP clearing.

B. Use of multilateral compression to reduce notional exposure and simplify the portfolio without central clearing.

C. Application of a central counterparty (CCP) for all OTC transactions to centralize trade and risk management.

D. Mandatory conversion of all long-dated OTC derivatives into short-term instruments for central clearing eligibility.

Question CR-16 d.6 Bank Q763 • BT P2.T6.408.3 (rebuilt)

Which of the following statements about termination features and trade compression is NOT correct?

A. A break clause is most useful if exercised early, before the counterparty's credit quality deteriorates substantially, but firms often avoid early exercise to preserve relationships with counterparties.

B. A walkaway clause allows a party to avoid its net liabilities to a defaulted counterparty while still claiming if its own MtM exposure is positive, and it is criticised for creating moral hazard.

C. Trade compression reduces the gross notional and the number of trades, but it can do so only by changing the institution's overall risk profile.

D. A reset agreement resets a trade that is heavily in the money back to at the money, for example by exchanging the MtM value on each reset date.

Question CR-16 d.7 Workbook A21.6 • S5, S6, S7 • Map #27

An internal auditor at an Australian bank is reviewing the unit that manages collateral and other counterparty risk mitigants, and is drafting findings for the audit committee. Which of the following statements is correct to include?

A. A rating-based break clause offers the same benefits as a mandatory break: counterparty risk reduction, CVA reduction, capital relief and LCR relief.

B. Liquidity risk from holding non-cash margin falls on the margin giver, which bears the bid-offer costs and price falls when the securities are sold after a default.

C. Because a compression cycle lowers the total notional of all participants, it guarantees that each participant ends with a lower notional to every counterparty.

D. Daily portfolio reconciliation reduces margin disputes, and a history of disputes is one reason Basel III requires a longer MPoR, raising capital and rewarding lower operational risk.

Solutions • CR-16 d

Solution CR-16 d.1 — Answer: D

Bank Q761 • AnalystPrep Q.1888

Step 1 – Who holds the right

A break clause gives the party that *exercises* it (one party or either, depending on the clause) the right to end the trade on the specified dates. It is a protection for the party worried about its counterparty's credit, not a right reserved for a defaulter.

⇒ A is out.

Step 2 – At what value

A termination is made at the prevailing MtM set out in the documentation, typically priced by a panel of banks: the replacement value *at the time of termination*, not a figure fixed in advance.

⇒ B and C are out; only D pairs the exercising party with the current replacement value.

Interpretation

Settling at the current replacement value lets the exercising party crystallise the MtM owed to it before the counterparty can default. Because this is a base valuation, it avoids most of the valuation disputes of a post-default close-out. The counterparty against which the clause is exercised may alternatively offer a replacement counterparty, a guarantee or margin.

Why the other options are wrong

A is incorrect. Gives the right to the defaulting party; the clause exists for the party concerned about the counterparty's credit quality.

B is incorrect. Uses a pre-defined value; the termination is at the replacement value prevailing when the clause is exercised.

C is incorrect. Describes entering a new trade at a fixed price; a break clause ends the existing trade and creates nothing new.

LO: CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.1.1 Break Clauses; SchweserNotes 2026 Book 2, Reading 34, LO 34.d

Solution CR-16 d.2 — Answer: B

Bank Q764 • BT P2.T6.904.3

Step 1 – Requirement 1: limit the exposure as the MtM grows

A reset cash-settles the MtM on each reset date and, for a cross-currency swap, resets the FX rate to spot, so the exposure returns to zero each time (GARP Figure 17.1). Break clauses also cap it; a walkaway clause does not.

Step 2 – Requirement 2: keep the hedge alive

After a reset the swap continues on refreshed terms. A mandatory break ends the swap on its break date, and an exercised trigger-based break ends it too.

⇒ Only the reset keeps the swap alive after it acts; any break, once it operates, ends the swap.

Step 3 – Requirement 3: no rating triggers

Resets are tied to payment dates or an MtM breach, not to ratings, so they carry no lagging-indicator or cliff-edge problem.

⇒ The reset agreement meets all three requirements.

Interpretation

GARP stresses that resets are mandatory and not linked to any external event, which makes them

much easier to assess and model than break clauses. Their cost is frequency: a quarterly refix is a weaker mitigant than daily margin.

Why the other options are wrong

A is incorrect. A walkaway clause only lets a party escape its net liability when the counterparty defaults; it does nothing about a growing in-the-money exposure, and it is not in standard ISDA documentation.

C is incorrect. Depends on ratings, which the Board rejects, and terminates rather than preserves the hedge when triggered.

D is incorrect. Terminates the swap on the fixed date whatever happens, ending the hedge, and does nothing to the exposure before that date.

LO: CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.1.1 and 17.1.2 Resettable Transactions

Solution CR-16 d.3 Answer: A

Bank Q736 • AnalystPrep Q.1892

Step 1 – When a walkaway clause is used

It allows an institution to cancel transactions if its counterparty defaults. The institution would only choose to do so when it is in debt to the defaulter, that is, when its net MtM is negative.

Step 2 – What it achieves

The institution ceases payments and is not obliged to settle the amount it owes. It does not reduce credit exposure; it produces a gain that offsets potential losses elsewhere.

$\Rightarrow$ A describes exactly this.

Interpretation

This is why walkaway clauses are discouraged: the defaulter's estate (and so its other creditors) bears the cost, and the survivor has an incentive to contribute to its counterparty's default. They were common before the 1992 ISDA Master Agreement and are not part of standard documentation now. Even without a clause, a survivor can achieve something similar by not closing out and stopping payments (Enron Australia against TXU).

Why the other options are wrong

B is incorrect. No gain comes from a replacement cost estimated before the trade; the gain is the liability the survivor does not pay.

C is incorrect. Paying only the net amount is ordinary close-out netting, the standard ISDA outcome, not a walkaway.

D is incorrect. Legal proceedings are not a feature of the clause; if anything, the administrator may sue the survivor (as Lehman's did).

LO: CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.3.3 Payment Under Close-Out; SchweserNotes 2026 Book 2, Reading 34, LO 34.d

Solution CR-16 d.4 Answer: B

Bank Q854 • AnalystPrep Q.6095

Step 1 – What compression does

It cancels redundant, offsetting contracts and replaces them with fewer ones. Gross notional and the number of contracts fall; each participant's net position is unchanged.

Step 2 – Why that helps

Fewer, smaller contracts mean lower operational costs, less settlement and counterparty risk, lower capital where rules use gross notional (the leverage ratio) and, at a CCP, lower margin multipliers and smaller default auctions.

⇒ B: the same position is held through fewer contracts.

Interpretation

Compression is a tidying exercise on positions that already exist: the economic position is the same before and after. That is why it reduces non-market risks but leaves market risk where it was.

Why the other options are wrong

A is incorrect. Compression shrinks the gross portfolio; it does not expand it.

C is incorrect. It reduces the number of outstanding contracts; it creates no new tradable derivatives or liquidity.

D is incorrect. Sorting contracts by risk profile is portfolio analytics; compression cancels contracts.

LO: CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.4 and 16.2.6; Ch.8 (GARP CR Ch.18), 18.2 Netting, Multilateral Offset and Compression

Solution CR-16 d.5 **Answer: B**

Bank Q857 • AnalystPrep Q.6128

Step 1 – The constraint

The contracts are to stay bilateral, so any route that moves them to a CCP (A, C, and D's conversion for clearing eligibility) does not answer the question.

Step 2 – What multilateral compression offers

GARP calls portfolio compression the modern equivalent of clearing rings: it "achieves multilateral netting benefits via the cooperation of multiple counterparties", with no membership organisation, by tearing up and replacing real transactions.

⇒ B brings part of the CCP's multilateral offset into the bilateral market.

Interpretation

The difference from a CCP is what is offset: compression needs actual cash flows or trades to be eliminated, whereas a CCP can offset risk between trades that are not identical, through novation. Compression also needs standard, fungible contracts and interchangeable counterparties of comparable credit quality.

Why the other options are wrong

A is incorrect. Standardising every contract is a route *to* central clearing, and non-standard trades cannot simply be standardised.

C is incorrect. Putting every trade through a CCP is exactly what is not possible for these contracts.

D is incorrect. Converting long-dated trades into short-dated ones would change their economics and is not a recognised technique.

LO: CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.3 Clearing Rings and 16.2.4 Portfolio Compression

Solution CR-16 d.6 Answer: C

Bank Q763 • BT P2.T6.408.3

Step 1 – Test A (break clause)

This is the banker's paradox: a break is useful only if exercised early, yet firms delay to protect the relationship. True.

Step 2 – Test B (walkaway)

A walkaway lets the survivor escape its net liability while claiming any positive MtM; it is criticised for moral hazard and the cost it imposes on the defaulter. True.

Step 3 – Test D (reset)

A reset brings a heavily in-the-money trade back to at the money by exchanging the MtM (a resettable cross-currency swap also resets the FX rate). True.

Step 4 – Test C (compression)

Compression cuts gross notional and trade count while keeping each participant neutral in value and risk. "Only by changing the risk profile" reverses this.
$\Rightarrow$ C is the false statement.

Interpretation

The polarity flip in C is the standard compression trap. The correct version is that compression keeps market risk unchanged and reduces non-market risks (see the correction box in Section 5).

Why the other options are wrong

A is incorrect. A correct statement of the banker's paradox.
B is incorrect. A correct description of a walkaway clause and its main criticism.
D is incorrect. A correct description of a reset agreement.

LO: CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.4, 16.3.3; Ch.7 (GARP CR Ch.17), 17.1.1, 17.1.2; SchweserNotes 2026 Book 2, Reading 34, LO 34.d

Solution CR-16 d.7 Answer: D

Workbook A21.6 • S5, S6, S7 • Map #27

Step 1 – Disputes, reconciliation and operational risk

Reconciliations "aim to minimise the chance of a dispute by agreeing on valuations and margin requirements across portfolios"; most large portfolios are reconciled daily. Basel III recognises the operational risk of collateralisation by requiring a larger MPoR where a netting set exceeds 5,000 trades or holds illiquid margin or exotics, and "a history of margin call disputes also requires a larger MPoR". Larger MPoRs mean higher capital and so an incentive to reduce operational risk.
$\Rightarrow$ D is correct.

Interpretation

The auditor should test reconciliation frequency and dispute logs alongside capital: an operational weakness in margining shows up directly as a longer MPoR and a higher capital charge.

Why the other options are wrong

A is incorrect. Sibling swap: Table 17.1 gives all four benefits only to mandatory breaks; a rating-based break gives at most some counterparty risk reduction, Basel explicitly rules out capital relief from rating triggers, and the LCR forces banks to pre-fund them.
B is incorrect. Role misassignment: liquidation risk sits with the non-defaulting party that holds the margin and must sell it for cash to rehedge; haircuts are set to cover it. The giver's risk is funding liquidity risk when it must post.
C is incorrect. Absolute claim: compression aims to reduce the total notional of all parties,

and “it does not necessarily follow that an individual party will experience a lower notional”; positions between pairs can even rise or change sign, which is why participants set constraints.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6, 16.2.4; Ch.7, 17.1.1 Table 17.1, 17.2.7, 17.5.3 and 17.5.4

CR-17 • Margin (Collateral) and Settlement • Vardeez: VERY IMPORTANT

CR-17 b

P1 CONCEPT

Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A21 (Map #27, reported by 4 of 16; the August 2026 recall named the two-way CSA). The map's reports include "calculate collateral required given the values of MtM exposure, threshold, etc." and an "overcollateralisation account balance".
Vardeez: CR-17 is VERY IMPORTANT: "know the terms here very well". The notes list the seven terms of this LO and draw the three zones in Figure 1; CR-17 g points back here.

Reported exam shapes

- **A21·S3** Overcollateralisation: initial margin and haircuts. (The securitisation overcollateralisation account is a different idea, drilled in A9 under CR-22 and CR-23.)

What the questions look like: definition and EXCEPT questions on the CSA terms: what each term does, which way it moves the collateral, and which document holds it. Distractors swap the threshold with the MTA or the initial margin, or apply a haircut the wrong way. The numerical version is CR-17 c; here only one statement in Question CR-17 b.7 (a haircut gross-up) needs arithmetic.

Worked elsewhere: A21.1 (rating-linked threshold after a downgrade, then the call) at Question CR-17 h.7; A21.2 (the MTA test that fails) at Question CR-17 c.8; A21.5 (initial margin and the margin period of risk) under CR-20 j.

1. The CSA and the Master Agreement

Margin (collateral) is a contractual right to demand assets against a credit exposure. In bilateral OTC derivatives it is *security*: the receiver becomes the economic owner only if the giver defaults. The **credit support annex (CSA)** is appended to the ISDA Master Agreement and makes margin posting contractual; it is the market-standard bilateral margin agreement. Its terms are fixed at signing and change only by mutual amendment (a CCP can change its terms unilaterally). The OTC term **independent amount** and the exchange term **initial margin** are largely interchangeable.

ISDA Master Agreement (CR-16)	Credit support annex (GARP 17.3.1)
Events of default and termination events; close-out and close-out netting; payment netting; the single net amount owed on default.	Valuation method and timing; the margin calculation; transfer timing; eligible margin and haircuts; substitution; dispute resolution; interest on cash; initial margin; rehypothecation; triggers (such as downgrades) that change the terms.

2. The seven terms

- **Threshold (K).** The exposure below which no margin is called. Above K only $V - K$ can be called, so the first K is never collateralised. A zero threshold means margin is always posted (above the MTA); an *infinite* threshold means a party never posts, which is how a one-way CSA is written (CR-17 h). Thresholds are increasingly zero (CR-17 g).
- **Initial margin (independent amount).** Extra margin posted irrespective of the exposure, against the delay in receiving margin and the close-out costs after a default. It now usually moves with the riskiness of the portfolio. It acts as a *negative threshold*, so the two are not used together.
- **Minimum transfer amount (MTA).** The smallest amount that can be transferred. The exposure must exceed K + MTA before a call can be made, but the MTA is *not* deducted from the amount called (Figure 1).
- **Rounding.** Calls and returns are rounded to a lot size to avoid awkward amounts; the effect is small. MTAs and rounding are generally zero where margin is cash only.
- **Haircut.** A discount on non-cash margin for the fall in its price between the last call and its liquidation after a default. A haircut h gives credit for $(1 - h)$ of market value, the *valuation percentage*.
- **Credit quality.** Thresholds, initial margin and sometimes MTAs can be linked to the counterparty's rating, tightening as it falls (GARP Table 17.4; CR-17 h).
- **Credit support amount.** The margin that may be requested at a given time, which is not the portfolio value. The margin already held is the **credit support balance**. Calculation: CR-17 c.

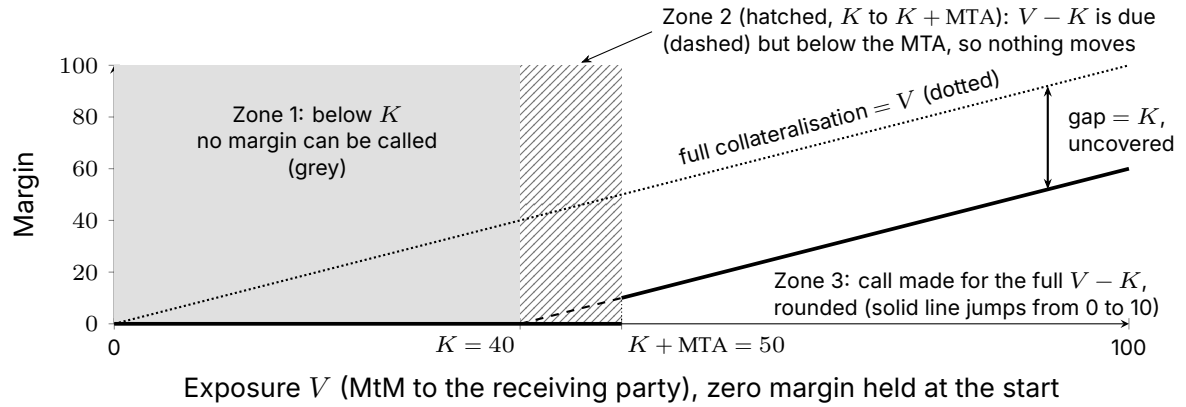


Figure 1. The three zones a CSA creates ($K = 40$, MTA $= 10$, nothing held yet). The trigger for a call is $K + \text{MTA}$, but the amount called is $V - K$, not $V - K - \text{MTA}$, and the threshold itself is never collateralised. Initial margin sits outside this picture: it is posted whatever the exposure.

3. Overcollateralisation: initial margin and haircuts (shape S3)

A threshold leaves the receiver *under*collateralised; two terms push the other way.

- **Initial margin** sits on top of variation margin. In GARP Table 17.8 a portfolio worth 350 is covered by 350 of variation margin plus 50 of initial margin: collateral 400 against an exposure of 350. It is computed and transferred separately, *not netted* against variation margin, and may be segregated; two-way initial margins are paid gross.
- **Haircuts** make the giver deliver more market value than the credit it receives: to deliver credit X, post $X/(1 - h)$. Schweser: a $100,000 call met with a sovereign bond at a 2% haircut needs

$100{,}000/0.98 = \$102{,}041$ of the bond. Cash takes no haircut, government bonds a few per cent (more at longer maturities), equities and gold much more; assets with large credit or liquidity risk are avoided because no practical haircut covers them.

GARP sizes a haircut as a worst-case price fall over the liquidation period τ, $h = \Phi^{-1}(\alpha)\,\sigma_C\sqrt{\tau}$ (eq. 17.1). A 10-year government bond at 99% over two days: $2.326 \times 1\% \times \sqrt{2/250} = 0.21\%$ of yield, times a duration of about 8.5, is about 2%. Overcollateralisation is not free for the giver: the excess is exposed if the *receiver* defaults, which is why initial margin is segregated and not reused (CR-17 i).

Exam traps

- **Threshold and MTA swapped.** The threshold is a level of exposure left uncollateralised; the MTA is the minimum size of a single transfer.
- **MTA treated as part of the threshold.** K + MTA decides *whether* a call is made; the amount called is still $V - K$ less the margin held.
- **Initial margin as a threshold (polarity flip),** or netted against variation margin. Initial margin adds collateral and is transferred separately; a threshold removes collateral.
- **Haircut applied the wrong way.** Credit is $(1-h)$ times market value; to deliver credit X post $X/(1-h)$, not $X(1+h)$ or $X(1-h)$.
- **CSA versus Master Agreement.** Events of default, termination events and close-out netting are in the Master Agreement, not the CSA.
- **Credit support amount read as the portfolio value** or as the margin already held (the credit support balance).

Practice questions: CR-17 b

Question CR-17 b.1 Bank Q781 • AnalystPrep Q.1900 (rebuilt)

A legal analyst at a bank is reviewing the credit support annex (CSA) to be appended to an existing ISDA Master Agreement with a dealer. Which of the following is NOT a term that is typically established in the CSA itself?

A. The eligible collateral and the haircuts applied to non-cash collateral

B. The threshold, independent amount (initial margin) and minimum transfer amount

C. The method and timing of valuations and of margin transfers, and the procedure for resolving disputes

D. The events of default and termination events that permit close-out netting of the trades

Question CR-17 b.2 Bank Q788 • BT P2.T6.409.1

Each of the following is true about a credit support annex (CSA) EXCEPT, which is not?

A. A CSA is appended within an ISDA Master Agreement and permits the counterparties to mitigate their credit risk by agreeing to various collateral posting

B. A CSA is critically defined by key parameters, including threshold, independent amount, and minimum transfer amount

C. In some trading relationships, CSAs are not used because one or both parties cannot commit to collateral posting

D. If a counterparty employs a credit value adjustment (CVA), then a CSA gives no benefit and is not necessary (a CSA would be redundant with CVA)

Question CR-17 b.3 Bank Q782 • AnalystPrep Q.1902 (rebuilt)

A CSA specifies a threshold of $10 million for a counterparty and no initial margin. Which statement about the threshold is correct?

A. It is the minimum size of a single collateral transfer, below which no transfer is made.

B. It is the level of mark-to-market exposure below which no margin can be called; if exposure is $25 million, only $15 million can be called.

C. It is an extra amount of collateral that must be posted regardless of the portfolio value, so a $25 million exposure requires $35 million.

D. It is the level of mark-to-market exposure above which posted collateral must be liquidated.

Question CR-17 b.4 Bank Q791 • BT P2.T6.330.3 (rebuilt)

Which of the following best describes the primary reason two counterparties agree to a non-zero (higher) threshold in a CSA?

A. To reduce the operational workload and liquidity cost of margining, accepting some uncollateralised exposure

B. To eliminate counterparty credit exposure entirely

C. To increase the amount of collateral held against the current exposure

D. To remove wrong-way risk from the collateral posted

Question CR-17 b.5 Bank Q783 • AnalystPrep Q.1903

Which of the following statements correctly describes the relationship between thresholds and initial margin (independent amounts) in a CSA?

A. They move in the same direction: both increase the collateral called.

B. They work in opposite directions: a threshold allows undercollateralisation, while initial margin creates overcollateralisation.

C. They are always equal for a given counterparty.

D. A threshold is always less than or equal to the initial margin.

Question CR-17 b.6 Bank Q786 • AnalystPrep Q.6146 (rebuilt)

A bank reviews the collateral terms of the credit support annexes (CSAs) appended to its ISDA Master Agreements. Which statement accurately reflects a feature of a CSA?

A. It mandates a fixed collateral amount for the duration of all derivative contracts.

B. It specifies the collateral terms, including events that trigger changes in collateral requirements.

C. It requires collateral to be posted exclusively in the form of government securities.

D. It requires the same threshold to apply to both parties regardless of their credit quality.

Question CR-17 b.7 Bank Q787 • BT P2.T6.325.1 • shape A21·S3

Counterparty FundCorp receives a collateral call of $100,000. Each of the following is true, except which is FALSE?

A. FundCorp can post $100,000 in cash

B. FundCorp can post about $108,700 in collateral with an 8.0% haircut

C. If FundCorp has already posted an Independent Amount (IA) of $60,000, then FundCorp can apply the IA and meet the margin call with only $40,000

D. Collateral with higher price volatility, ceteris paribus, will be assigned a higher haircut

Solutions • CR-17 b

Solution CR-17 b.1 — Answer: D

Bank Q781 • AnalystPrep Q.1900

Step 1 – What the CSA contains

The CSA is the annex in which the parties agree contractual margin-posting. GARP's list of its terms: valuation method and timing, the margin calculation, transfer mechanics and timing, eligible margin, substitutions, dispute resolution, remuneration, initial margin, haircuts, rehypothecation and triggers such as downgrades.
⇒ A, B and C are all CSA terms.

Step 2 – What stays in the Master Agreement

Events of default, termination events and the close-out netting they trigger are defined by the ISDA Master Agreement itself (CR-16). The CSA relies on them but does not define them.
⇒ D is the term that is not in the CSA.

Interpretation

Margin under a CSA is held against the single netted amount that the Master Agreement produces on close-out. The two documents work together: the Master Agreement says when and how trades are closed out and netted, the CSA says how much security is held against the result.

Why the other options are wrong

A is incorrect. Eligible collateral and haircuts are core CSA terms.
B is incorrect. Threshold, independent amount and MTA are the parameters that define the credit support amount.
C is incorrect. Valuation and transfer timing and dispute resolution are explicit CSA terms.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.1 The Credit Support Annex; Ch.6 (GARP CR Ch.16)

Solution CR-17 b.2 — Answer: D

Bank Q788 • BT P2.T6.409.1

Step 1 – Test the structural statements

A CSA is appended to the ISDA Master Agreement for contractual margin-posting (A), and it is defined by its parameters: threshold, independent amount, MTA (B). Some parties, such as corporates or triple-A sovereigns, cannot or will not post, so no CSA is used (C).
⇒ A, B and C are true.

Step 2 – CVA and the CSA

CVA is the price of the counterparty risk that remains. Margin reduces the exposure, so a CSA *reduces* the CVA; it does not duplicate it. GARP notes that non-posting entities prefer, or have no choice but, to pay xVA charges instead of posting margin.
⇒ D is false: CVA and a CSA are complements, not substitutes.

Interpretation

The choice is between paying for counterparty risk through CVA (and FVA, KVA) or mitigating it with margin. A CSA lowers the charge; it does not remove the need to measure what is left.

Why the other options are wrong

A is incorrect. True: the CSA sits within the ISDA Master Agreement and governs collateral

posting.
B is incorrect. True: these are the defining parameters.
C is incorrect. True: no-CSA relationships are typical with corporates.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.1 and 17.3.2

Solution CR-17 b.3 Answer: B

Bank Q782 • AnalystPrep Q.1902

Step 1 – Definition
The threshold is the amount of exposure below which no margin is required. Below it the portfolio is uncollateralised; above it only the incremental amount can be called.

Step 2 – Apply it
With exposure \$25 million and no initial margin: $\max(25 - 10,\ 0) = \$15$ million can be called.
$\Rightarrow$ The first \$10 million is never collateralised.

Interpretation
This is GARP's "undercollateralisation": the threshold trades risk reduction for less operational and liquidity cost.

Why the other options are wrong
A is incorrect. Describes the minimum transfer amount.
C is incorrect. Describes initial margin, which works in the opposite direction $(25 + 10 = 35)$.
D is incorrect. The threshold governs when margin is called; it has nothing to do with liquidating collateral.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4

Solution CR-17 b.4 Answer: A

Bank Q791 • BT P2.T6.330.3

Step 1 – What a higher threshold does
It raises the exposure that can build before any margin is called, so fewer and smaller calls are made.

Step 2 – Why parties accept that
GARP: thresholds exist because one or both parties gain in operational and liquidity costs, and that saving is worth the weaker collateralisation. A bank may be comfortable with uncollateralised exposure up to its credit limit for the counterparty.
$\Rightarrow$ A trade-off of workload and liquidity against risk.

Interpretation
The trend runs the other way now: funding and capital costs and the bilateral margin rules push thresholds to zero (Question CR-17 g.9).

Why the other options are wrong
B is incorrect. A higher threshold leaves *more* exposure uncollateralised.
C is incorrect. It reduces the collateral held; initial margin is what increases it.
D is incorrect. Wrong-way risk depends on the link between the collateral and the counterparty's

credit quality; the threshold does not address it.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4

Solution CR-17 b.5 Answer: B

Bank Q783 • AnalystPrep Q.1903

Step 1 – The receiving-only margin amount (GARP eq. 17.2)

$$\max(\text{value} - K_C,\, 0) + \text{IM}_C$$

$\Rightarrow$ A higher threshold K_C lowers the margin; a higher initial margin IM_C raises it.

Step 2 – Read the signs

The threshold leaves exposure uncollateralised (undercollateralisation); initial margin adds collateral beyond the exposure (overcollateralisation). GARP: initial margin "can be thought of (intuitively and mathematically) as a negative threshold".

Interpretation

Because they pull in opposite directions, GARP says they are not seen together: either a threshold with zero initial margin, or initial margin with a zero threshold.

Why the other options are wrong

A is incorrect. Sign error: the threshold subtracts from the margin called.
C is incorrect. No rule links their sizes; usually one of them is zero.
D is incorrect. No such ordering exists.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4; 17.3.6 eq. 17.2

Solution CR-17 b.6 Answer: B

Bank Q786 • AnalystPrep Q.6146

Step 1 – What a CSA must do

It must define all the parameters of the collateralisation and account for all possible scenarios, including triggers that change the margin conditions, such as rating downgrades that lead to enhanced requirements.
$\Rightarrow$ B describes a feature of every well-drafted CSA.

Step 2 – Check the others

Margin is recalculated as the portfolio is revalued (typically daily), so it is not fixed. Eligible margin is negotiated: cash is the most common. Thresholds can differ by party and be linked to credit quality.

Interpretation

Rating-linked terms are the "credit quality" item in the LO: they keep the workload light while the counterparty is strong and tighten collateral as it weakens (CR-17 h).

Why the other options are wrong

A is incorrect. The credit support amount changes with the portfolio value.
C is incorrect. Cash is the most common margin; eligible assets are agreed bilaterally.

D is incorrect. Thresholds are often asymmetric and rating-linked.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.1, 17.3.2 and 17.3.4 Table 17.4

Solution CR-17 b.7 Answer: C

Bank Q787 • BT P2.T6.325.1

Step 1 – Cash and haircut collateral

Cash in the eligible currency has no haircut, so \$100,000 of cash meets the call (A true). With an 8% haircut the valuation percentage is 92%: $100{,}000/0.92 = \$108{,}696 \approx \$108{,}700$ (B true).
$\Rightarrow$ Haircuts make the giver over-deliver: one source of overcollateralisation.

Step 2 – The independent amount

The independent amount (initial margin) is computed separately and is *not netted* against variation margin. FundCorp still owes the full \$100,000 call; the \$60,000 stays in place as the cushion.
$\Rightarrow$ C is false.

Step 3 – Haircut size

Haircuts reflect liquidation time, volatility, default risk, maturity and liquidity, so a more volatile asset gets a larger haircut (D true).

Interpretation

Both overcollateralisation devices of shape S3 appear here: the independent amount sits on top of variation margin, and the haircut makes the giver deliver more market value than the credit it receives.

Why the other options are wrong

A is incorrect. True: cash carries a 0% haircut.
B is incorrect. True: $100{,}000/(1-0.08) = 108{,}696$.
D is incorrect. True: haircuts rise with price volatility.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.5 Margin Types and Haircuts; 17.3.6

CR-17 • Margin (Collateral) and Settlement • Vardeez: VERY IMPORTANT

CR-17 c

P1 CALC

Calculate the credit support amount (margin) under various scenarios.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: the first item the map reports for A21 is "calculate collateral required given the values of MtM exposure, threshold, etc." (Map #27, 4 of 16), and the August 2026 recall named the two-way CSA.

Vardeez: the notes show two worked tables and say there is "no need to get into above tables in very depth". The formula below is GARP's eq. 17.3, which the rebuilt notes add; it reproduces both tables.

Reported exam shapes

- **A21·S1** Collateral required from the MtM value, threshold, MTA, rounding and margin already held.
- **A21·S2** The threshold-plus-MTA test that fails, so no transfer is made.

What the questions look like: a table of value, threshold(s), MTA, rounding and collateral held, sometimes with a haircut or a one-way CSA. The answer is a call, a return, or nothing. The wrong options are the raw amount before the MTA test, the amount with the MTA deducted, the amount before subtracting the balance, and the wrong rounding direction.

Worked elsewhere: A21.1 (rating-linked threshold after a downgrade, then the call) under CR-17 h.

1. Two quantities and one formula

Recap from CR-17 b: the **credit support amount** is the margin that may be requested now, not the portfolio value. The **credit support balance** C is the margin already held: from the calculating party's view, positive if it holds margin and *negative* if it has posted, with non-cash margin valued *after* haircuts. A party that only receives margin, with threshold K_C and initial margin IM_C for the counterparty, needs $\max(\text{value} - K_C,\, 0) + \text{IM}_C$ in total (GARP eq. 17.2). For variation margin between two parties (GARP eq. 17.3):

$$\boxed{\text{Credit support amount} = \max(\text{value} - K_C,\, 0) - \max(-\text{value} - K_P,\, 0) - C}$$

Here value is the current (base) value of the netted transactions to the party making the calculation, K_C is the counterparty's threshold and K_P the calculating party's own threshold. Only one of the two max terms can be non-zero: when the value is positive the counterparty's threshold applies, when it is negative the party's own threshold applies.

- **Positive** result: margin can be called (or its return requested). **Negative**: margin must be posted (or returned).
- Either way, a transfer is made only if the amount is at least the **MTA**; then the *full* amount moves, and **rounding** may be applied.
- **Initial margin** does not enter eq. 17.3. It is computed separately (the independent amount), is not netted against variation margin, and two-way initial margins are paid gross.

2. The procedure

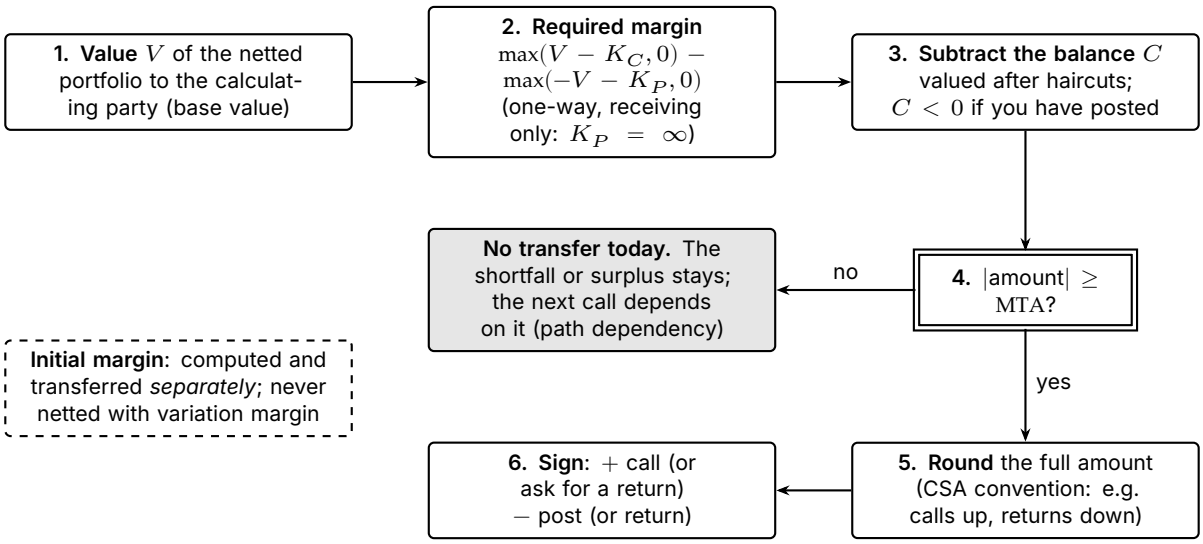


Figure 1. The margin call in six moves. The MTA only gates the transfer (step 4); it is never subtracted. The grey box is the S2 shape: every input is supplied and the answer is no transfer. The dashed box is outside the variation margin calculation.

3. Worked examples

Example 1: threshold, no initial margin (GARP Table 17.6)

Threshold 100 for the counterparty; no MTA or rounding.

	Time t_1	Time t_2
Value	350	325
Credit support balance C	0	250
Credit support amount	$\max(350-100,0)-0=250$	$\max(325-100,0)-250=-25$

At t_1 the portfolio is only partly collateralised: 100 is uncovered by design. At t_2 the portfolio is uncollateralised by $325-250=75$, yet the amount is -25: margin is *returned*, back up to the threshold (after the return, $325-225=100$ is uncovered). Vardeez and Schweser print the same table with values 250 and 225 (call 150, then -25); the logic is identical.

Example 2: the calculating party is out of the money (own threshold)

A bank's portfolio is worth −EUR 12.4 million to it. Each party's threshold is EUR 2.0 million, the bank has already posted EUR 8.5 million ($C=-8.5$), the MTA is EUR 0.5 million, and deliveries are rounded up to EUR 0.25 million.

$$\max(-12.4-2.0,\,0)-\max(12.4-2.0,\,0)-(-8.5)=0-10.4+8.5=-1.9$$

$|-1.9|\geq 0.5$, so the bank posts, rounded up to EUR 2.0 million. Its *own* threshold applies because it owes; deducting both thresholds is the classic error (Question CR-17 c.8).

Example 3: MTAs make margin path-dependent (GARP Table 17.7)

Threshold zero, MTA 50, no initial margin. The values at t_2 and t_3 are the same in all three scenarios; only t_1 differs.

		t_1	t_2	t_3
Scenario 1	Value	90	110	150
	Balance / amount	0 / **90**	90 / 0 ($20 < 50$)	90 / **60**
Scenario 2	Value	40	110	150
	Balance / amount	0 / 0 ($40 < 50$)	0 / **110**	110 / 0 ($40 < 50$)
Scenario 3	Value	130	110	150
	Balance / amount	0 / **130**	130 / 0 (-20)	130 / 0 ($20 < 50$)

At t_3 the same value of 150 produces amounts of 60, 0 and 0 and leaves 0, 40 and 20 uncollateralised, because the balance standing at t_3 (90, 110, 130) depends on what happened at t_1. This is why collateralised exposure is harder to model.

> **Correction to the source notes.** GARP's text under Table 17.7 says the credit support amount at t_3 "can either be smaller (scenario 1) or larger (scenario 3)". Its own table gives amounts at t_3 of 60 (scenario 1), 0 (scenario 2) and 0 (scenario 3), so the amount is *larger* in scenario 1. The sentence fits the credit support *balance* standing at t_3: 90 in scenario 1 is smaller, and 130 in scenario 3 larger, than the 110 of scenario 2. The point GARP makes (MTAs cause path dependency) is unaffected. (The scenario 1 balance of 90 at t_3 is also missing from the extracted table.)

Example 4: initial margin on top (GARP Table 17.8)

Threshold zero; initial margin recalculated over time.

	t_1	t_2
Value	350	375
Credit support balance (variation margin)	0	350
Credit support amount (variation margin)	350	25
Initial margin (independent amount)	50	45

At t_1 the receiver is overcollateralised by the 50 of initial margin. At t_2 the margin deficit of $375 - 350 = 25$ is within the initial margin. Two separate transfers follow: 25 of variation margin in, and 5 of initial margin back; they are not netted (Question CR-17 c.2). Vardeez and Schweser use 250 and 275 with initial margins of 40 and 35.

4. Haircuts in the calculation

Haircuts enter in two places.

- **Valuing the balance.** The valuation agent values the margin held and *applies the haircut*: EUR 10 million (market value) of a 5-year government bond at a 5% haircut counts as $C = 9.5$.
- **Delivering the call.** To deliver credit X with an asset whose haircut is h, post market value $X/(1-h)$. Meeting a 2.0 million call with the same bond takes $2.0/0.95 = 2.105$ million. Several assets in sequence: take the credit $(1-h)\times$ market value of each in turn and gross up only the last slice (Question CR-17 c.4).

The BCBS-IOSCO standardised schedule (GARP Table 17.12), used in Question CR-17 c.4: cash in the same currency 0%; high-quality government bonds 0.5%, 2% and 4% for 0 to 1, 1 to 5 and over 5 years; high-quality corporate and covered bonds 1%, 4% and 8%; equity and gold 15%; plus an 8% FX add-on where the currencies differ.

5. MTA and rounding conventions

- Test the MTA against the *credit support amount* (required margin less balance), not against the change in value or the required margin. If it fails, nothing moves, even when every input is supplied (shape S2); if it passes, the whole amount moves.
- Round only a transfer that is triggered. The direction is a CSA term: "to the nearest" unless the stem says otherwise; a common convention rounds calls up and returns down, which favours the party holding margin.
- With daily calls and instant delivery the uncollateralised amount stays below $K+$MTA; the margin period of risk adds market moves on top (CR-17 j).

Formula card: CR-17 c

Receiving only (eq. 17.2)	$\max(\text{value} - K_C, 0) + \text{IM}_C$
Two-way variation margin (eq. 17.3)	$\max(\text{value} - K_C, 0) - \max(-\text{value} - K_P, 0) - C$
Sign	$+$: call or request a return; $-$: post or return
MTA test	transfer only if $\lvert\text{amount}\rvert \geq$ MTA; then the full amount, rounded
Balance C	after haircuts; negative if the calculating party has posted
Haircut	credit $= (1-h)\times$ market value; to deliver credit X post $X/(1-h)$
Initial margin	separate from the credit support amount; not netted; two-way amounts paid gross
One-way CSA	non-posting party's threshold $= \infty$

Exam traps

- **MTA deducted.** The MTA decides whether a transfer happens; once passed, the full amount moves.
- **Balance ignored (intermediate result).** $\max(V - K, 0)$ is the total margin required, not the call. Subtract the margin held, with its sign.
- **Whose threshold.** When the party owes, its *own* threshold K_P applies; deducting both thresholds is wrong. After a downgrade use the new rating-linked threshold.
- **MTA tested on the wrong number.** Compare the MTA with required margin less balance, not with the exposure change or the required margin.
- **Rounding before the MTA test,** or in the wrong direction.
- **Haircut the wrong way.** Value the balance at $(1-h)$; gross deliveries up by $1/(1-h)$.
- **Initial margin netted** against variation margin, or treated as fixed at inception.
- **Red herrings.** Yesterday's exposure, the rounding amount and the IM figure when only variation margin is asked.

Practice questions: CR-17 c

Question CR-17 c.1

Bank Q801 • BT P2.T6.414.3 (rebuilt; options reordered)

Consider a two-way CSA with identical parameters for both parties: threshold \$1,000,000; independent amount zero; minimum transfer amount (MTA) \$100,000; rounding to a multiple of \$25,000 (collateral calls are rounded up, returns are rounded down); initial collateral held is zero. From the perspective of Party A, the portfolio mark-to-market is +\$1,430,000 on day 1 and +\$1,030,000 on day 2. Which is the correct sequence of collateral movements for Party A?

A. Day 1: \$430,000 called by Party A; day 2: \$420,000 returned by Party A

B. Day 1: \$450,000 called by Party A; day 2: no return, since the mark-to-market is still above the threshold

C. Day 1: \$450,000 called (received) by Party A; day 2: \$400,000 returned by Party A

D. Day 1: \$450,000 called by Party A; day 2: \$450,000 returned by Party A

Question CR-17 c.2

New • built from GARP Ch.17, 17.3.6 eq. 17.3 and Table 17.8

A collateral manager at a dealer receives margin from a hedge fund under a CSA with a zero threshold, no minimum transfer amount and no rounding. The hedge fund also posts initial margin (an independent amount) that the CSA recalculates daily from the risk of the portfolio. The positions are (USD million):

	Yesterday	Today
Portfolio value to the dealer	420	455
Variation margin held by the dealer	420	
Initial margin requirement	60	52
Initial margin held (segregated)	60	

Which of the following correctly describes today's margin transfers?

A. The hedge fund delivers a single net amount of USD 27 million.

B. The hedge fund delivers USD 35 million of variation margin, and the dealer separately returns USD 8 million of initial margin.

C. The hedge fund delivers USD 35 million of variation margin; the initial margin stays at USD 60 million because it is fixed at inception.

D. The hedge fund delivers USD 87 million: USD 35 million of variation margin plus a new USD 52 million of initial margin.

Question CR-17 c.3 Bank Q806 • bank new question • re-filed from CR-17 d (LO 35.d) (options reordered)

A collateral analyst acting as valuation agent for a bilateral OTC portfolio reports the following: net exposure after netting is USD 18 million; the market value of collateral previously posted is USD 8 million and carries a 10% haircut; the threshold is USD 5 million; no initial margin applies; and the minimum transfer amount is USD 0.5 million. Assuming the valuation agent applies the haircut to the posted collateral, what collateral amount should it call for?

A. USD 4.2 million

B. USD 5.0 million

C. USD 5.8 million

D. USD 10.8 million

Question CR-17 c.4 Bank Q794 • BT P2.T6.24.28.3 (rebuilt) • filed under LO 35.b

Eagle Heights Bank must post collateral with a post-haircut value of USD 45 million against an interest rate swap. Haircuts by asset class:

Asset class	Haircut	Asset class	Haircut
Cash in same currency	0%	High-quality corporate bond (<1 yr)	1%
High-quality government bond (<1 yr)	0.5%	High-quality corporate bond (1-5 yr)	4%
High-quality government bond (1-5 yr)	2%	High-quality corporate bond (>5 yr)	8%
High-quality government bond (>5 yr)	4%	Equities in a major index; gold	15%

The bank posts its assets in the following order, using as much of each as needed: (1) USD 10 million cash; (2) USD 15 million US Treasury bonds, remaining maturity 4 years; (3) USD 20 million high-quality corporate bonds, remaining maturity 4 years; (4) USD 5 million gold bullion; (5) USD 8 million S&P 500 stocks. What total market value of collateral must the bank post?

A. USD 43.90 million

B. USD 45.00 million

C. USD 46.29 million

D. USD 52.94 million

Question CR-17 c.5

Bank Q800 • BT P2.T6.410.3 (options reordered)

A financial institution (FI) enters a derivatives trade with a counterparty (CP) under a two-way CSA with identical parameters for both parties: daily valuation, a mutual threshold of \$100,000, and a mutual minimum transfer amount (MTA) of \$40,000. At inception T(0) the exposure is zero. FI then gains on the trade:

- T(1): FI's exposure to CP increases to \$115,000
- T(2): the trade gains a further \$35,000, so FI's exposure to CP is \$150,000

The collateral posted by CP attracts a haircut of 20.0%. Which of the following accurately describes the first collateral call?

A. On T(1), the counterparty posts collateral with market value of \$18,750

B. On T(2), the counterparty posts collateral with market value of \$50,000

C. On T(2), the counterparty posts collateral with market value of \$62,500

D. On T(1), the counterparty posts collateral with market value of \$143,750

Question CR-17 c.6

Bank Q802 • BT P2.T6.906.3 (rebuilt)

ToughFax International has a two-way credit support annex (CSA) with its counterparty, GreenBam Financial. Both parties post collateral under mutual parameters: threshold of \$1,000,000; minimum transfer amount (MTA) of \$100,000; and rounding of \$25,000. Before today's remargin determination, ToughFax held no collateral, but GreenBam held \$50,000 of collateral previously posted by ToughFax. Today the mark-to-market (MTM) value is +\$1,094,000 from ToughFax's perspective (and −\$1,094,000 from GreenBam's). What is the credit support amount?

A. Zero (neither call nor return)

B. ToughFax calls \$94,000

C. ToughFax calls \$150,000

D. GreenBam returns \$150,000

Question CR-17 c.7

Bank Q803 • GARP PE1 Q16 • Mock 3 Q64

An investment bank has a one-way credit support annex (CSA) on a bilateral transaction with a hedge fund counterparty. Under the CSA, the mark-to-market value of the transaction forms the basis of the hedge fund's collateral requirements, given below:

Item	Value (CNY)
Mark-to-market value of net exposure	25,000,000
Mark-to-market value of collateral posted	10,800,000
Threshold amount	14,000,000
Minimum transfer amount	2,500,000
Rounding amount	10,000

Assuming the net exposure increases to CNY 27,000,000 and the mark-to-market value of collateral posted has not changed, how much additional collateral will the hedge fund have to post?

A. CNY 0

B. CNY 1,990,000

C. CNY 2,000,000

D. CNY 2,500,000

Question CR-17 c.8

Workbook A21.2 • S1, S2, S8 • Aug-2026 • Map #27

A treasury operations analyst at a UK bank is processing today's margin under a two-way CSA with another bank. The data are:

- Portfolio value to the UK bank: −GBP 24.3 million
- Threshold for each party: GBP 5.0 million
- Margin already posted by the UK bank (credit support balance from its view): −GBP 18.9 million
- Minimum transfer amount: GBP 0.5 million; no rounding; independent amount: none

Which of the following is the correct action for the UK bank today?

A. No transfer

B. Post GBP 0.4 million

C. Post GBP 5.4 million

D. Request the return of GBP 4.6 million

Solutions • CR-17 c

Solution CR-17 c.1

Answer: C

Bank Q801 • BT P2.T6.414.3

Step 1 – Day 1

$$\max(1{,}430{,}000 - 1{,}000{,}000,\ 0) - 0 = 430{,}000 \ \geq \ 100{,}000$$

$\Rightarrow$ A call is valid; rounded *up* to a multiple of 25,000: \$450,000 called.

Step 2 – Day 2

$$\max(1{,}030{,}000 - 1{,}000{,}000,\ 0) - 450{,}000 = 30{,}000 - 450{,}000 = -420{,}000$$

$\Rightarrow |-420{,}000| \geq 100{,}000$, so a return is due; rounded *down*: \$400,000 returned.

Interpretation

Party A keeps \$50,000 against a requirement of \$30,000. Margin is returned whenever the balance exceeds the requirement by at least the MTA, even while the value is still above the threshold. Rounding calls up and returns down favours the party holding margin.

Why the other options are wrong

A is incorrect. Ignored the rounding: 430,000 and 420,000 are the unrounded amounts.
B is incorrect. Assumed margin comes back only when the value falls below the threshold; the surplus over the 30,000 requirement is returned.
D is incorrect. Returned everything, but 30,000 of exposure above the threshold still needs margin.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4; 17.3.6 eq. 17.3

Solution CR-17 c.2

Answer: B

New • built from GARP Ch.17, 17.3.6 and Table 17.8

Step 1 – Variation margin (eq. 17.3, threshold zero)

$$\max(455 - 0,\ 0) - 420 = 35$$

$\Rightarrow$ The hedge fund delivers USD 35 million of variation margin.

Step 2 – Initial margin

The requirement fell from 60 to 52 while 60 is held: USD 8 million is returned.

Step 3 – Keep them apart

Initial margin is computed separately (the independent amount), is not netted against variation margin, and may be segregated.
$\Rightarrow$ Two transfers in opposite directions: 35 in, 8 out.

Interpretation

After the transfers the dealer holds 455 of variation margin plus 52 of initial margin against an exposure of 455. Variation margin tracks the value; initial margin tracks the riskiness of the value, which fell. GARP Table 17.8 shows the same pattern (value 350 to 375, initial margin 50 to 45).

Why the other options are wrong

A is incorrect. Netted the two: $35 - 8 = 27$. Initial margin is not netted with variation margin.

C is incorrect. Treated initial margin as static. GARP: the term "may suggest a static quantity", but it is now recalculated over time.
D is incorrect. Added the new requirement in full, ignoring the 60 already held: $35 + 52 = 87$.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4; 17.3.6 eqs. 17.2 and 17.3, Table 17.8

Solution CR-17 c.3 — Answer: C

Bank Q806 • bank new question (LO 35.d)

Step 1 – Value the balance after the haircut

$$C = 8 \times (1 - 0.10) = 7.2$$

Step 2 – Required margin

$$\max(18 - 5,\ 0) = 13$$

Step 3 – Credit support amount and MTA test

$$13 - 7.2 = 5.8 \ \geq \ 0.5$$

$\Rightarrow$ Call USD 5.8 million. Equivalently: uncollateralised exposure $18 - 7.2 = 10.8$, less the threshold of 5.

Interpretation

These are the valuation agent's tasks in GARP's list: value the portfolio with netting, value the margin held and adjust it by the haircuts, compute the uncollateralised exposure, and compute the credit support amount.

Why the other options are wrong

A is incorrect. Grossed up the collateral instead of haircutting it: $18 - 8 \times 1.10 - 5 = 4.2$.
B is incorrect. Ignored the haircut: $18 - 8 - 5 = 5.0$.
D is incorrect. Stopped at the uncollateralised exposure, $18 - 7.2 = 10.8$, without deducting the threshold.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. CR-17 d: Describe the role of a valuation agent. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.7; 17.3.6 eq. 17.3

Solution CR-17 c.4 — Answer: C

Bank Q794 • BT P2.T6.24.28.3

Step 1 – Credit from each asset in the posting order

$$\text{cash } 10 \times 1.00 = 10.00; \quad \text{Treasuries } 15 \times 0.98 = 14.70; \quad \text{corporates } 20 \times 0.96 = 19.20$$

$\Rightarrow$ Cumulative credit 43.90, still 1.10 short.

Step 2 – Gross up the last slice (gold, 15%)

$$\frac{1.10}{0.85} = 1.294 \quad (\text{within the USD 5 million available})$$

Step 3 – Total market value posted

$$10 + 15 + 20 + 1.294 = \text{USD } 46.29 \text{ million}$$

Interpretation

The stocks are not needed. The haircut table is the BCBS-IOSCO standardised schedule (GARP Table 17.12); the bank over-delivers by USD 1.29 million of market value, an overcollateralisation the receiver holds as a cushion.

Why the other options are wrong

A is incorrect. 43.90 is the credit from the first three assets only, USD 1.10 million short of the requirement.

B is incorrect. Ignored the haircuts.

D is incorrect. Applied gold's haircut to the whole requirement: $45/0.85 = 52.94$.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.5; 17.4.4 Table 17.12

Solution CR-17 c.5

Answer: C

Bank Q800 • BT P2.T6.410.3

Step 1 – T(1)

$115{,}000 - 100{,}000 = 15{,}000 < 40{,}000$: no call.

Step 2 – T(2)

$150{,}000 - 100{,}000 = 50{,}000 \geq 40{,}000$: the first call, for credit of \$50,000.

$\Rightarrow$ The call is for the full excess over the threshold, not just the \$35,000 gain.

Step 3 – Gross up for the haircut

$$\frac{50{,}000}{1 - 0.20} = \$62{,}500 \text{ of market value}$$

Interpretation

No call can be made until exposure exceeds $K + \text{MTA} = \$140{,}000$. The MTA only gates the transfer; once passed, the whole 50,000 is called.

Why the other options are wrong

A is incorrect. Called at T(1) although 15,000 is below the MTA: $15{,}000/0.8 = 18{,}750$.

B is incorrect. Right timing and credit, but no haircut: \$50,000 of this collateral is credited at only \$40,000.

D is incorrect. Ignored the threshold and the MTA at T(1): $115{,}000/0.8 = 143{,}750$.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4, 17.3.5 and 17.3.6

Solution CR-17 c.6

Answer: C

Bank Q802 • BT P2.T6.906.3

Step 1 – The two threshold terms (ToughFax's view)

$$\max(1{,}094{,}000 - 1{,}000{,}000,\ 0) = 94{,}000; \qquad \max(-1{,}094{,}000 - 1{,}000{,}000,\ 0) = 0$$

Step 2 – Balance

ToughFax has *posted* \$50,000, so $C = -50{,}000$:

$$94{,}000 - 0 - (-50{,}000) = 144{,}000$$

Step 3 – MTA and rounding

$144{,}000 \geq 100{,}000$; rounded to a multiple of 25,000: \$150,000, called by ToughFax.

Interpretation

GreenBam must return ToughFax's \$50,000 and deliver \$94,000 more (plus rounding). A negative balance *adds* to the call.

Why the other options are wrong

A is incorrect. Compared only the 94,000 excess with the MTA, ignoring the 50,000 ToughFax had posted.

B is incorrect. Right direction, but ignored the balance and the rounding.

D is incorrect. Wrong flow: GreenBam holds only \$50,000 of ToughFax's collateral; the \$150,000 is a delivery to ToughFax, made up of that return and new collateral.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.6 eq. 17.3

Solution CR-17 c.7 **Answer: A**

Bank Q803 • GARP PE1 Q16

Step 1 – Required margin at the new exposure

$$\max(27{,}000{,}000 - 14{,}000{,}000,\ 0) = 13{,}000{,}000$$

Step 2 – Less the collateral held

$$13{,}000{,}000 - 10{,}800{,}000 = 2{,}200{,}000$$

Step 3 – MTA test

$2{,}200{,}000 < 2{,}500{,}000$: no call can be made, so rounding never comes into it.
$\Rightarrow$ Additional collateral: CNY 0.

Interpretation

Under the one-way CSA only the hedge fund's side matters. The opening exposure of CNY 25 million is a red herring (the shortfall then was only 0.2 million). The 2.2 million will be called once the shortfall reaches the MTA.

Why the other options are wrong

B is incorrect. Took the CNY 2 million rise in exposure as the call and deducted the CNY 10,000 rounding amount: 1,990,000.

C is incorrect. Used the CNY 2 million rise in exposure, ignoring the threshold, the balance and the MTA.

D is incorrect. Called the MTA itself; the MTA is a minimum size, not an amount to post.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4; 17.3.6 eq. 17.3

Solution CR-17 c.8 **Answer: A**

Workbook A21.2 • S1, S2, S8 • Aug-2026

Step 1 – Counterparty term

$$\max(\text{value} - K_C,\, 0) = \max(-24.3 - 5,\, 0) = 0$$

$\Rightarrow$ The portfolio is negative to the UK bank, so it cannot call margin.

Step 2 – Own term

$$\max(-\text{value} - K_P,\, 0) = \max(24.3 - 5,\, 0) = 19.3$$

$\Rightarrow$ The UK bank's own GBP 5.0 million threshold is the uncollateralised amount it is allowed to owe.

Step 3 – Credit support amount

$$0 - 19.3 - (-18.9) = -\text{GBP } 0.4 \text{ million}$$

$\Rightarrow$ A negative amount would mean the UK bank posts.

Step 4 – MTA test

$|-0.4| = 0.4 < 0.5$: the amount is below the minimum transfer amount, so no margin moves today.

Interpretation

The counterparty holds GBP 18.9 million against a liability of GBP 24.3 million: GBP 5.4 million uncollateralised, of which 5.0 is the agreed threshold and 0.4 the MTA gap. Full data are supplied, yet the threshold-plus-MTA test fails, which is the red-herring pattern the map flags. The 0.4 will be called once the shortfall reaches the MTA, which is why MTAs make collateralised exposure path dependent.

Why the other options are wrong

B is incorrect. Ignored the MTA test and posted the raw credit support amount of 0.4.
C is incorrect. Ignored the UK bank's own threshold: $24.3 - 18.9 = 5.4$.
D is incorrect. Deducted both parties' thresholds: required $24.3 - 5 - 5 = 14.3$ against 18.9 posted, so it wrongly asks for 4.6 back. Only the out-of-the-money party's own threshold applies.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. CR-17 g: Explain the features of a collateralization agreement. CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4; 17.3.6 eq. 17.3 and Table 17.7

CR-17 • Margin (Collateral) and Settlement • Vardeez: VERY IMPORTANT

CR-17 f

P1 CONCEPT

Explain the process for the reconciliation of collateral disputes.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A21, Map #27), and Vardeez singles it out inside a VERY IMPORTANT reading.
Vardeez: "LO 35.f: this is really imp abt disputes in collateral."

Reported exam shapes

- **A21·S5** Disputes and reconciliation.

What the questions look like: process-order questions (what comes first, what is not a step), the "split the difference" rule for small differences, the sources of disputes, and prevention by reconciliation. Distractors put market-maker quotes first, post the full disputed amount, split any difference whatever its size, or name two valuation agents as the cure.

Worked elsewhere: A21.6 (daily reconciliation, disputes and the Basel III margin period of risk) under CR-16 d.

1. Who calculates: the valuation agent (recap)

The valuation agent (CR-17 d) is normally the party calling for margin; it values the netted portfolio and the margin held (after haircuts) and computes the credit support amount (CR-17 c). A large dealer facing a smaller counterparty usually insists on the role. Where both parties are valuation agents, each calls when it has an exposure and "the potential for margin disputes is significant"; a third-party valuation agent is a common remedy. Centrally cleared trades avoid disputes because the CCP is the valuation agent.

2. Where disputes come from

GARP: a dispute over a margin call is *common*. Its sources:

- **Trade population**: the two parties do not hold the same list of trades in the netting set.
- **Trade valuation methodology**: different models, especially for exotics and illiquid credit derivatives.
- **Application of CSA rules**: thresholds, eligible collateral (Vardeez and Schweser also name netting rules).
- **Market data and market close time**: the same model gives different values on different curves or snapshot times.
- **Valuation of previously posted collateral**: haircuts and prices of the margin already held.

3. The process

Vardeez and Schweser give three steps.

1. The disputing party **notifies** its counterparty (or the third-party valuation agent) of its intent to dispute the exposure or the collateral calculation, by the **end of the business day following the collateral call**.
2. **All undisputed amounts are transferred**, and the **reason for the dispute is identified**.
3. If the dispute is not resolved, the parties **request MtM quotes from several market makers** (usually four) for the disputed trades.

A small difference may simply be **split**; a larger one cannot, and the disputed part stays **uncollateralised** until it is resolved. Neither party posts the disputed amount "to be safe". The 2026 GARP text does not repeat the three steps: it lists the sources of disputes and stresses reconciliation. The steps come from Schweser and Vardeez, and bank questions test them.

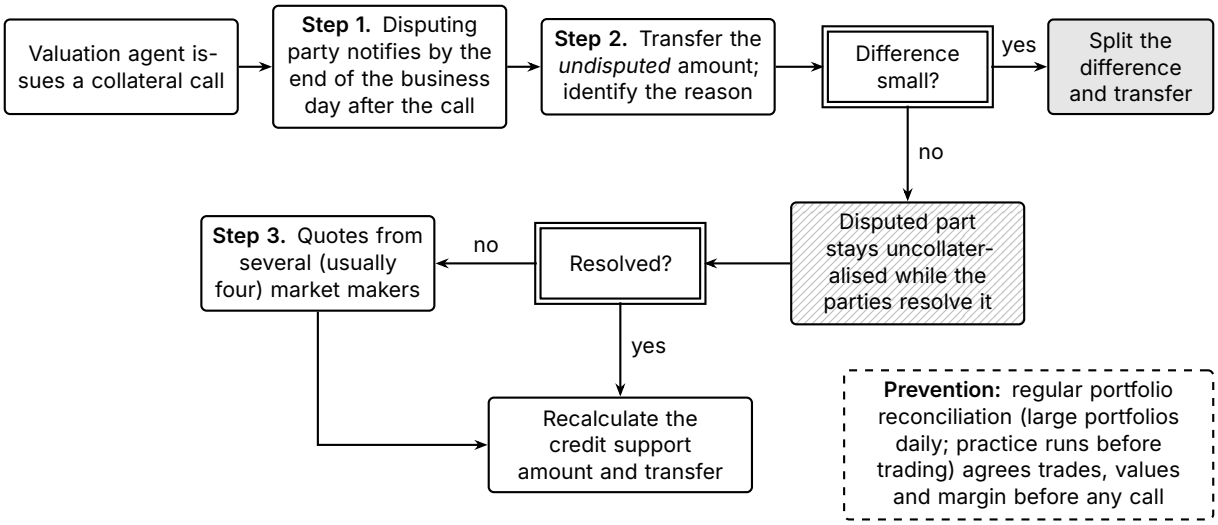


Figure 1. Reconciling a collateral dispute. Market-maker quotes are the last step, used only when the parties cannot resolve the difference themselves. The hatched box is the cost of a dispute: exposure left uncollateralised meanwhile. The dashed box is how disputes are avoided in the first place.

4. Preventing disputes: reconciliation

- **Reconcile regularly.** Reconciliations agree valuations and margin requirements across portfolios *before* calls are made. ISDA (2015): most portfolios, especially large ones, are reconciled daily. Schweser adds practice (dummy) reconciliations before trading and periodic (weekly or monthly) ones during trading.
- **Regulation.** Dodd-Frank and EMIR require more rigorous and frequent portfolio reconciliations. Under Basel III a history of margin call disputes requires a longer margin period of risk, so more capital (CR-17 j). The bilateral margin rules require robust dispute resolution procedures, because risk-sensitive initial margin models invite disputes. Third parties offer valuation, reconciliation and dispute management services.
- **Trade coverage** (Schweser). Collateralising as many trades as possible reduces risk, but one trade that cannot be valued can hold up every call. Hard-to-value products (exotics, illiquid credit derivatives) may be left out, or left uncollateralised if the counterparty cannot value them. Since undisputed amounts must be transferred immediately, it still pays to collateralise most products.

	What goes wrong	What fixes it
Trade population	Different trade lists in the netting set	Daily portfolio reconciliation
Valuation method	Model differences on complex trades	Exclude hard-to-value trades; market-maker quotes (step 3)
CSA rules	Threshold, eligibility or netting applied differently	Agree terms upfront; reconcile
Market data, close time	Different curves or snapshot times	Identify the cause (step 2); align data
Posted collateral	Different prices or haircuts on margin held	Reconcile collateral values
Two valuation agents	Each calls on its own numbers	Third-party valuation agent

Exam traps

- **Wrong order.** Market-maker quotes are step 3, for unresolved disputes only. Step 1 is notice by the end of the next business day; step 2 is transfer the undisputed amount and find the cause.
- **Posting the disputed amount.** Only the *undisputed* amount moves; the disputed part stays uncollateralised.
- **Splitting any difference.** Splitting is for small differences; large ones need the cause found.
- **Two valuation agents as a cure.** Two agents *create* disputes; the remedy is a third-party agent or regular reconciliation, which is the best tool because it acts before any call.
- **Absolute claims.** Reconciliation reduces disputes; nothing removes them. Cleared trades avoid them only because the CCP is the valuation agent.

Practice questions: CR-17 f

Question CR-17 f.1

Bank Q813 • BT P2.T6.409.3

Each of the following is a step in the process for the reconciliation of a collateral dispute EXCEPT:

A. The disputing party notifies its counterparty (or the third-party valuation agent) that it disputes the exposure or collateral calculation.

B. Both parties post collateral for the full amount of any disputed exposure, even if this means mutual overcollateralization, so that neither counterparty is partially exposed during the dispute.

C. The disputing party identifies the reason for the dispute and transfers the undisputed amount, and the parties attempt to resolve the dispute within a resolution time.

D. If the parties fail to resolve the dispute within the resolution time, they obtain MtM quotations from several market makers for the disputed components.

Question CR-17 f.2

Bank Q811 • AnalystPrep Q.5504

Global Finance Bank (GFB) has a collateral dispute with a major OTC derivatives counterparty over the valuation of complex credit derivatives covered by their collateral agreement. As risk manager, which of the following is the standard first sequence of steps in reconciling the dispute?

A. Immediately request quotes from several market makers for the MtM value of the disputed trades and temporarily exclude the problematic assets from collateral calls.

B. Split the difference on the disputed amount regardless of its size to ensure a quick resolution and preserve the relationship.

C. Notify the counterparty of the intent to dispute by the end of the day following the collateral call, transfer all undisputed amounts, and identify the reason for the dispute.

D. Permanently exclude all complex credit derivatives from the collateral agreement to prevent future disputes.

Question CR-17 f.3

Bank Q810 • AnalystPrep Q.1907

Collateral management still relies heavily on manual processes, which often leads to valuation disputes over previously posted collateral. Which of the following correctly describes how such a dispute should be addressed?

A. If the disputed amount is higher than an acceptable level stated in the collateral agreement, the counterparties may "split the difference."

B. If the disputed amount is under an acceptable level stated in the collateral agreement, the counterparties may "split the difference"; otherwise, a case must be filed against the defaulting party.

C. If the disputed amount is under an acceptable level stated in the collateral agreement, the counterparties may "split the difference"; otherwise, the cause of the difference must be found.

D. If the disputed amount is under an acceptable level stated in the collateral agreement, the parties should ignore it; otherwise, the cause of the difference must be found.

Question CR-17 f.4

Bank Q812 • AnalystPrep Q.5505 (options reordered)

During reconciliation of its OTC derivatives portfolio, GFB finds valuation discrepancies causing a dispute with a key counterparty. The dispute stems from different market data and market closing times used for valuation. Which action best aligns with standard practice for resolving such a dispute?

A. Review both parties' valuation methodologies, market data sources and the market closing times used for the disputed trades to identify the cause.

B. Insist that the counterparty accept GFB's valuation because its market data and processes are accurate.

C. Immediately split the difference, regardless of the reason for the discrepancy.

D. Automatically accept the counterparty's valuation to preserve the relationship and adjust GFB's processes accordingly.

Question CR-17 f.5 Bank Q805 • bank new question • re-filed from CR-17 d (LO 35.d) (options reordered)

Two dealer banks of broadly similar size have just signed a bilateral credit support annex. A collateral manager at one of them notes that the agreement names both banks as valuation agent. Which of the following is the most likely consequence of this arrangement, and the usual remedy?

A. Only the bank with the larger positive exposure may call for collateral, so disputes are eliminated and no remedy is needed.

B. Each bank calculates exposure but neither may call for collateral, so the central counterparty must be introduced as the remedy.

C. Collateral calls are made only on the net of the two banks' calculations, so disputes cannot arise and no third party is required.

D. Each bank calls for collateral when it has positive exposure, which can lead to disputes and delays; appointing a third-party valuation agent is a common remedy.

Question CR-17 f.6 Bank Q814 • BT P2.T6.905.3 (rebuilt)

A large bank trades OTC derivatives with a smaller counterparty under a collateral agreement. Given the decentralized, non-transparent OTC market, significant disagreements about collateral requirements can occur. Which of the following is the most effective way to reduce the chance of collateral disputes?

A. Name both parties as valuation agents so each can call for collateral on its own valuation.

B. Reconcile portfolios and valuations regularly, for example daily, and consider practice reconciliations before trading.

C. Wait until a dispute arises and then split the difference regardless of its size.

D. Name the smaller counterparty as sole valuation agent, since it has the most to lose from a miscalculation.

Question CR-17 f.7 Workbook A21.4 • S4, S5, S8 • Aug-2026 • Map #27

A collateral manager at a Canadian bank is recommending CSA terms for three new relationships: a triple-A supranational development bank, an unrated hedge fund, and a dealer bank. Which of the following recommendations is consistent with the reading?

A. Accept the supranational's one-way CSA in its favour, since a one-way CSA adds risk only for the party that receives the margin.

B. Sign a two-way CSA with zero thresholds with the dealer bank, the interbank standard, which aims to benefit both parties on counterparty risk.

C. Link the hedge fund's threshold to its net asset value so that the threshold rises as the net asset value falls, easing its workload under stress.

D. Let both parties act as valuation agent on every CSA, since two valuation agents remove the potential for disputes over margin calls.

Solutions • CR-17 f

Solution CR-17 f.1 — Answer: B

Bank Q813 • BT P2.T6.409.3

Step 1 – The three steps

(1) Notify the dispute by the end of the day after the call; (2) transfer the undisputed amount and identify the reason; (3) if unresolved, obtain MtM quotes from several market makers.
⇒ A is step 1, C is step 2, D is step 3.

Step 2 – The odd one out

Only the *undisputed* amount moves. The disputed part stays uncollateralised until the cause is found and corrected; nobody posts the full disputed amount.

Interpretation

The cost of a dispute is exactly this temporary uncollateralised exposure, which is why prevention through reconciliation matters so much.

Why the other options are wrong

A is incorrect. Step 1 of the process.
C is incorrect. Step 2 of the process.
D is incorrect. Step 3, for disputes not resolved within the resolution time.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. • **Reference:** SchweserNotes 2026 Book 2, Reading 35, LO 35.f; Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.7

Solution CR-17 f.2 — Answer: C

Bank Q811 • AnalystPrep Q.5504

Step 1 – Notice

The disputing party notifies its counterparty of its intent to dispute by the end of the day following the collateral call.

Step 2 – Undisputed amounts and the cause

All undisputed amounts are transferred and the reason for the dispute is identified. Market-maker quotes come only if the dispute remains unresolved.
⇒ Option C is steps 1 and 2.

Interpretation

Complex credit derivatives are exactly the trades where valuation methodology differs, so step 3 (several market-maker quotes) may well follow, but not first.

Why the other options are wrong

A is incorrect. Market-maker quotes are the last step; excluding assets from calls is not part of the dispute process.
B is incorrect. Splitting is acceptable only when the difference is small.
D is incorrect. Excluding trades is a design choice when the agreement is negotiated, not a step in resolving a live dispute.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. • **Reference:** SchweserNotes 2026 Book 2, Reading 35, LO 35.f

Solution CR-17 f.3 — Answer: C

Bank Q810 • AnalystPrep Q.1907

Step 1 – Small differences

If the disputed amount is below the tolerance in the agreement, the parties may simply split the difference.

Step 2 – Larger differences

The undisputed amount is transferred, the disputed part stays uncollateralised, and the cause must be found (then, if unresolved, market-maker quotes).

$\Rightarrow$ Only C pairs "split" with small amounts and "find the cause" with large ones.

Interpretation

Disputes over previously posted collateral are one of GARP's five sources, and manual processes make them more frequent; electronic messaging and daily reconciliation are the industry's response.

Why the other options are wrong

A is incorrect. Reverses the logic: splitting is for amounts below the tolerance.
B is incorrect. A disputing party is not in default; litigation is not a step in reconciliation.
D is incorrect. Small differences are split, not ignored.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. • **Reference:** SchweserNotes 2026 Book 2, Reading 35, LO 35.f; Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.7

Solution CR-17 f.4 — Answer: A

Bank Q812 • AnalystPrep Q.5505

Step 1 – Recognise the source

Market data and market close time are one of GARP's listed sources of disputes.

Step 2 – Apply step 2 of the process

With the undisputed amount transferred, the parties identify the reason: compare methodologies, data sources and snapshot times.

$\Rightarrow$ A is the standard practice.

Interpretation

A data or timing difference can be diagnosed and fixed for good, which also prevents the next dispute; splitting it would leave the cause in place.

Why the other options are wrong

B is incorrect. Unilateral insistence resolves nothing; either side's data could be the cause.
C is incorrect. Splitting applies only to small differences, and not whatever the cause.
D is incorrect. Blind acceptance does not identify why the values differ.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.7; SchweserNotes 2026 Book 2, Reading 35, LO 35.f

Solution CR-17 f.5 — Answer: D

Bank Q805 • bank new question (LO 35.d)

Step 1 – Two valuation agents

Where both parties are valuation agent, each calls for margin when it has an exposure, on its own calculations. GARP: "the potential for margin disputes is significant".

Step 2 – The remedy

A third-party valuation agent can run the process: calls, substitutions, dispute resolution and daily valuation reports.
⇒ D.

Interpretation

Dual valuation agents are common between similar-sized dealers precisely because neither will accept the other's numbers. The price is more disputes and delays; regular reconciliation or a third party contains it.

Why the other options are wrong

A is incorrect. Nothing restricts calls to the larger exposure, and disputes are not eliminated.
B is incorrect. A valuation agent does call for collateral; a CCP is not the remedy for a bilateral agreement.
C is incorrect. Netting the two calculations is not how the arrangement works; two independent calculations are the source of disputes.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. CR-17 d: Describe the role of a valuation agent. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.7; SchweserNotes 2026 Book 2, Reading 35, LO 35.d

Solution CR-17 f.6 Answer: B

Bank Q814 • BT P2.T6.905.3

Step 1 – Prevention, not cure

Reconciliation agrees trades, valuations and margin requirements before a call is made. Most large portfolios are reconciled daily; Schweser adds practice reconciliations before trading.

Step 2 – Check the alternatives

Two valuation agents create disputes; splitting is reactive and only for small amounts; a large bank normally insists on being valuation agent itself.
⇒ B.

Interpretation

Regulators agree: Dodd-Frank and EMIR require more rigorous and frequent reconciliations, and under Basel III a history of disputes lengthens the margin period of risk and raises capital.

Why the other options are wrong

A is incorrect. Two valuation agents each call on their own numbers, which leads to disputes and delays.
C is incorrect. Reactive, and valid only for small differences.
D is incorrect. Not a dispute-prevention tool; a one-sided agent does not align valuations.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.7; SchweserNotes 2026 Book 2, Reading 35, LO 35.f

Solution CR-17 f.7 Answer: B

Workbook A21.4 • S4, S5, S8 • Aug-2026

Step 1 – The dealer relationship

Gregory: "Two-way CSAs with zero thresholds are standard in the interbank market and aim to be beneficial (from a counterparty risk point of view, at least) to both parties." Both parties post against a negative value, so both exposures are mitigated.
⇒ The plain two-way CSA from the August recall: each party's own threshold (here zero) governs when it posts.

Step 2 – The other two relationships

The supranational will want to receive but not post (an infinite threshold for itself): a one-way CSA that is extra risk for the bank as margin giver, worse than no CSA, and costly in funding and capital. For the hedge fund, an unrated party, terms can be linked to other metrics such as net asset value, but they must tighten as quality falls (lower threshold, then initial margin).

Interpretation

The bank should price the supranational's one-way CSA through xVA rather than treat it as risk-free, and keep the hedge fund's trigger mechanics aware of cliff-edge and liquidity effects.

Why the other options are wrong

A is incorrect. Role misassignment: a one-way CSA "represents additional risk for the margin giver", putting it in a worse situation than no CSA. The receiver (the supranational) is the party that benefits.

C is incorrect. Polarity flip: credit-quality links minimise workload when quality is high and tighten terms (lower threshold, initial margin) as it deteriorates; a threshold that rises as net asset value falls loosens collateralisation exactly when risk is highest.

D is incorrect. Absolute claim: where both counterparties are valuation agents, each calling for margin when it has an exposure, "the potential for margin disputes is significant". Reconciliation reduces disputes; nothing removes them.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. CR-17 d: Describe the role of a valuation agent. CR-17 g: Explain the features of a collateralization agreement. CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.7, 17.3.2 and 17.3.4 Table 17.4

CR-17 • Margin (Collateral) and Settlement • Vardeez: VERY IMPORTANT

CR-17 g

P1 CONCEPT

Explain the features of a collateralization agreement.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A21, Map #27), and Vardeez's strongest pointer in the reading. **Vardeez:** "LO 35.g: know the terms here very well." The terms are defined under CR-17 b; this LO is about how they are set and what each choice costs.

Reported exam shapes

- **A21·S4** Features and terms of a collateral agreement.

What the questions look like: which setting increases protection and which reduces workload; tiering by credit rating; margin call frequency; gap risk and initial margin; haircuts; interest, coupons and dividends on margin; the trend in thresholds. Distractors flip a direction (raise the threshold to protect more) or make an absolute claim (posting should be continuous).

Worked elsewhere: A21.2 (the MTA test) at Question CR-17 c.8; A21.4 (choosing CSA terms for three counterparties) under CR-17 f.

1. The balance every agreement strikes

A collateral agreement is negotiated before trading (and often revised before trading increases), and must define every parameter and cover every scenario. Each choice trades the **workload of calling and returning margin** (and its liquidity cost) against the **risk-mitigation benefit**; funding costs, once ignored, now matter too. The terms are defined in CR-17 b; the table adds the direction of each.

Feature	More protection	Less workload and liquidity cost
Threshold	Lower (zero)	Higher
Minimum transfer amount	Lower	Higher
Initial margin	Higher	Lower (zero)
Margin call frequency	More frequent (daily, intraday)	Less frequent (weekly)
Rounding	Smaller	Larger (effect is small)
Haircuts	Larger for riskier collateral	Smaller; wider eligibility

Initial margin is the one parameter where more means more protection; the others are frictions that let exposure build before margin arrives (Vardeez).

2. Margin call frequency

The contractual period at which margin may be called and returned. Daily calls have become standard in most OTC derivatives; intraday calls are common for repos and for CCP-cleared derivatives. A longer frequency cuts the operational workload and gives time for valuations, which some smaller institutions need, and can work in less volatile markets. The call frequency is *not* the margin period of risk: calls can be daily while the MPoR is assumed to be 10 days (CR-17 j).

3. How the parameters are set

- **Linked to credit quality** (GARP Table 17.4, CR-17 h). Thresholds fall to zero and initial margin rises as the rating falls; MTAs are higher for higher ratings (Schweser).
- **Thresholds increasingly zero.** Funding (FVA) and capital (KVA) costs, central clearing and the bilateral margin rules (full variation margin, zero threshold, MTA no more than EUR 500,000) push thresholds to zero; they can be non-zero only for exempt parties.
- **Initial margin against gap risk.** Initial margin keeps the stronger party overcollateralised even if the value gaps before the defaulting party's margin can be called (Schweser: it converts counterparty risk into gap risk). It is sized to the margin period of risk.

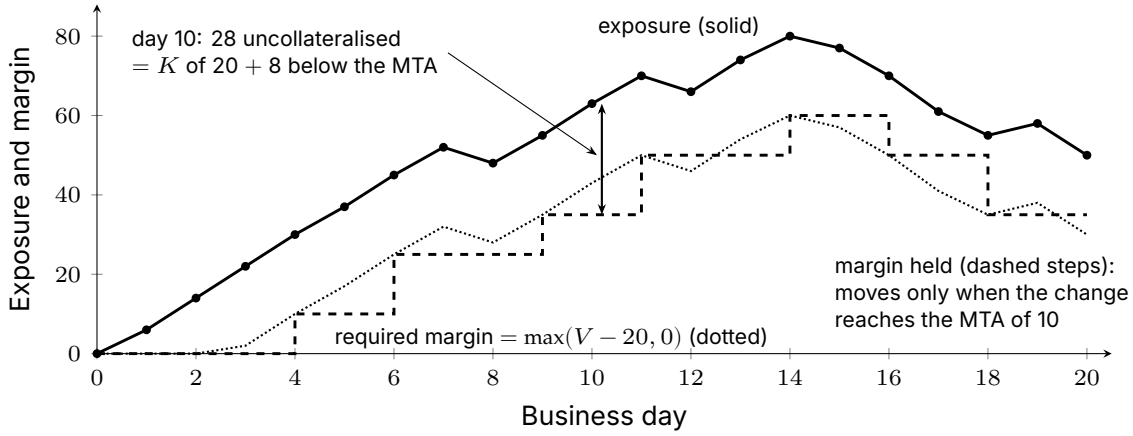


Figure 1. Daily calls with a threshold of 20 and an MTA of 10 (no delivery delay). The gap between exposure and margin is the threshold plus up to one MTA, so it stays below 30. Raising either parameter widens the gap and cuts the number of transfers (7 calls and returns here). Delivery delays and a default add the margin period of risk on top (CR-17 j).

4. Eligible margin and haircuts

Cash (mostly USD and EUR) is 74.9% of margin received against non-cleared OTC derivatives, government securities 14.8% and other assets 10.3% (agency and supranational bonds, covered and corporate bonds, letters of credit, equities). In the order of rising haircuts: cash (none), high-quality government bonds, AAA corporate bonds, structured products, then equities and commodities. The drivers of a haircut are the time to liquidate the margin, the volatility of the underlying market variable, the asset's default risk, maturity and liquidity, and any link between its value and the counterparty or the exposure (wrong-way risk), the hardest to write down. Haircuts are fixed in a bilateral agreement; a CCP can change them as markets change. If an asset held falls below the rating the agreement requires, it must normally be replaced.

5. Interest, coupons and substitution

- **Interest on cash** is paid at the overnight indexed (OIS) rate (Fed Funds in the US, EONIA in Europe in GARP's text). Some receivers subtract a spread to discourage cash; others pay above OIS to attract it in stressed, illiquid markets.
- **Coupons and dividends** on securities posted are passed back to the giver: the receiver becomes the economic owner only if the giver defaults. The exception: where passing on the cash flow would trigger an immediate margin call, the receiver may keep the minimum needed to stay collateralised.

- **Substitution.** The giver may ask for securities back by posting other eligible margin of equivalent value after haircuts (CR-17 i).

Exam traps

- **Direction flips.** A *higher* threshold or MTA *reduces* protection; higher initial margin and more frequent calls increase it.
- **Continuous posting as best practice.** Margin is posted periodically (daily at least), not continuously, to keep costs down.
- **Call frequency read as the MPoR** (sibling swap). Daily calls do not mean a one-day MPoR.
- **Wrong protection for gap risk.** Gap risk is covered by initial margin; a haircut covers the price risk of the collateral asset, and higher thresholds or MTAs make gap risk worse.
- **Rating tiers reversed.** Weaker credits get *lower* thresholds and MTAs and *more* initial margin. The trend in thresholds is to zero, not infinity (an infinite threshold is a one-way CSA).
- **The receiver keeps coupons.** It passes them on to the giver (the economic owner) unless the giver defaults.

Practice questions: CR-17 g

Question CR-17 g.1

Bank Q816 • AnalystPrep Q.1899

A financial institution trades OTC derivatives with a counterparty under a bilateral collateral agreement. From the institution's perspective, which statement about how collateral is exchanged is correct?

A. If its MtM is positive it posts collateral, and if its MtM is negative it calls for collateral.

B. If its MtM is positive it calls for collateral from the counterparty, and if its MtM is negative the counterparty calls for collateral from it.

C. If its MtM is negative it calls for collateral, and if its MtM is positive the counterparty must post collateral to it only when the counterparty has a lower rating.

D. Collateral is a constant requirement whenever the institution has a lower credit rating than the counterparty.

Question CR-17 g.2

Bank Q821 • BT Mock Q72 • BT P2.T6.905.1 (rebuilt)

Two counterparties have a set of OTC derivative transactions and a bilateral collateral agreement. Each of the following is true EXCEPT which is false?

A. Following best practice, collateral posting should be continuous for both parties.

B. Variation margin is collateral that tracks the change in the MtM of the position over time.

C. In a bilateral agreement, a party with positive MtM calls for collateral and a party with negative MtM must post.

D. The threshold and the minimum transfer amount are additive: exposure must exceed their sum before a call can be made.

Question CR-17 g.3 Bank Q817 • AnalystPrep Q.1909 (rebuilt)

Although derivative markets increasingly use daily margining, some smaller institutions may prefer a longer margin call frequency because:

A. a longer frequency gives them more time to meet operational and funding requirements.

B. a longer frequency removes the counterparty's uncollateralized exposure between calls.

C. daily margin calls are appropriate only for markets that are highly volatile.

D. daily margin calls are prohibited under the ISDA Master Agreement for smaller institutions.

Question CR-17 g.4 Bank Q815 • BT P2.T6.325.2

In regard to collateralization, each of the following is true except which is FALSE?

A. The valuation agent is the party calling for delivery or return of collateral and thus must handle all calculations, i.e., calculation of credit exposure, market value of posted collateral, uncollateralized exposure, and the delivery or return amount

B. A counterparty who posts a dividend-paying asset as collateral (i.e., the "giver" of collateral) forfeits all dividends as the receiver immediately becomes the economic owner of the collateral

C. The threshold represents an amount of uncollateralized exposure: if an exposure is above the threshold, only the incremental exposure will be collateralized

D. An independent amount is effectively a negative threshold and is typically held as a cushion against "gap risk", i.e., the risk that a transaction's market value may gap substantially and quickly

Question CR-17 g.5 Bank Q818 • AnalystPrep Q.5500

Global Finance Bank is revising its collateralization agreement for its OTC derivatives portfolio, aiming to balance operational workload against risk mitigation. Which set of features best matches the typical features of such an agreement?

A. A high threshold, a low minimum transfer amount, haircuts from the latest VaR, and interest on cash collateral at a fixed rate below the market rate.

B. A zero threshold for all counterparties regardless of rating, a high minimum transfer amount, and uniform haircuts for all collateral including cash and government bonds.

C. Thresholds and initial margins linked to counterparty credit rating, tiered haircuts reflecting collateral riskiness, and interest on cash collateral at the overnight market rate.

D. No initial margin, a high threshold linked to market volatility instead of credit quality, and daily margin calls with no minimum transfer amount.

Question CR-17 g.6 Bank Q820 • AnalystPrep Q.5503 (rebuilt)

A bank is setting the minimum transfer amount (MTA) in its collateral agreements. Which approach is most consistent with standard practice?

A. Tier the MTA by credit rating, with higher MTAs for lower-rated counterparties.

B. Use one high MTA for all counterparties to minimise the number of transfers, whatever their credit quality.

C. Tier the MTA by credit rating, with higher MTAs for higher-rated counterparties and lower MTAs for weaker ones.

D. Eliminate the MTA so that any change in exposure is collateralized immediately.

Question CR-17 g.7 Bank Q822 • BT P2.T6.410.2

In detailing the terms of a collateralization agreement, Barbara wants to mitigate her firm's exposure to a counterparty default as much as possible, accepting a heavier operational workload. Each of these adjustments will tend to increase mitigation EXCEPT which one?

A. Increase the independent amount

B. Increase the threshold

C. Decrease the minimum transfer amount

D. Increase the margin call frequency (e.g., from weekly to daily)

Question CR-17 g.8 Bank Q823 • BT P2.T6.906.2

An institution is re-positioning a large portfolio into significant positions in the over-the-counter (OTC) derivatives market. Its primary concern with respect to the portfolio is gap risk: the risk that the portfolio value may gap substantially in a short period of time. Which of the following is the most appropriate protection against gap risk?

A. Higher haircut

B. Higher thresholds

C. Higher initial margins

D. Higher minimum transfer amounts (MTA)

Question CR-17 g.9 Bank Q824 • BT P2.T6.906.1

Collateral agreements are increasingly about funding (FVA) and capital (KVA) costs in addition to pure counterparty risk (CVA) issues. Given the explicit recognition of these costs and the shift toward central clearing and bilateral margin requirements, what is the general impact on thresholds?

A. More zero thresholds

B. More infinite thresholds

C. No impact on thresholds

D. Thresholds often set equal to initial margin

Solutions • CR-17 g

Solution CR-17 g.1

Answer: B

Bank Q816 • AnalystPrep Q.1899

Step 1 – Who is exposed

Positive MtM means the counterparty owes the institution value, so the institution bears the credit risk and calls for margin.

Step 2 – The reverse case

Negative MtM means the counterparty is exposed, so it calls and the institution posts.
$\Rightarrow$ Margin flows from the party with negative MtM to the party with positive MtM.

Interpretation

In a two-way agreement the direction of flow follows the sign of the MtM, not relative ratings; ratings change the parameters (thresholds, initial margin), not the direction.

Why the other options are wrong

A is incorrect. Reverses the direction: the party in the money does not post.
C is incorrect. Reverses the negative case and adds a rating condition that does not exist.
D is incorrect. Confuses rating-linked terms with a constant requirement; posting still depends on MtM.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.2 Rationale

Solution CR-17 g.2

Answer: A

Bank Q821 • BT Mock Q72

Step 1 – Frequency of posting

To keep operational costs under control, posting is not continuous and occurs in blocks under pre-agreed rules; daily calls are the market norm.
$\Rightarrow$ A is false.

Step 2 – The true statements

Variation margin tracks MtM changes (B). Flows follow the sign of MtM (C). Threshold and MTA are additive for the trigger (D).

Interpretation

"Continuous" is an absolute qualifier: the best practice is frequent (daily) calls, balancing the workload against the risk left between calls.

Why the other options are wrong

B is incorrect. True: variation margin follows MtM.
C is incorrect. True: calls run both ways according to the sign of MtM.
D is incorrect. True: a call requires exposure above threshold plus MTA.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.2, 17.3.3 and 17.3.4

Solution CR-17 g.3 — Answer: A

Bank Q817 • AnalystPrep Q.1909

Step 1 – What daily margining demands

Daily valuation, calls and funding of transfers. GARP: some smaller institutions "may struggle with the operational and funding requirements" of daily calls from larger counterparties.

Step 2 – The trade-off

A longer frequency reduces that burden, at the cost of more uncollateralised exposure between calls.

$\Rightarrow$ A.

Interpretation

Longer frequencies are practical for less volatile markets, but daily calls have become standard in most OTC derivatives markets.

Why the other options are wrong

B is incorrect. Backwards: a longer frequency *increases* uncollateralised exposure between calls.

C is incorrect. Reverses the logic: daily calls are the general norm, and longer frequencies suit less volatile markets.

D is incorrect. Invented: nothing in ISDA documentation prohibits daily calls.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.3 Margin Call Frequency

Solution CR-17 g.4 — Answer: B

Bank Q815 • BT P2.T6.325.2

Step 1 – Economic ownership of margin

In OTC derivatives the receiver becomes the permanent economic owner of margin only if the giver defaults. It must pass on coupons, dividends and other cash flows to the giver.

$\Rightarrow$ B is false: the giver keeps the dividends.

Step 2 – The true statements

The valuation agent's tasks (A), the threshold as uncollateralised exposure (C) and the independent amount as a negative threshold against gap risk (D) are all correct.

Interpretation

The one exception GARP gives: if passing on a cash flow would trigger an immediate margin call, the holder may keep the minimum part of it needed to remain collateralised.

Why the other options are wrong

A is incorrect. True: the standard list of the valuation agent's duties.

C is incorrect. True: only the excess over the threshold is collateralised.

D is incorrect. True: initial margin works opposite to the threshold and cushions gap risk.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.2, 17.2.4, 17.2.7 and 17.3.4

Solution CR-17 g.5 — Answer: C

Bank Q818 • AnalystPrep Q.5500

Step 1 – Rating-linked terms

Thresholds and initial margins are usually tiered by rating: lower ratings get lower thresholds and more initial margin.

Step 2 – Haircuts and interest

Haircuts rise with collateral riskiness (cash none, then government bonds, corporates, structured products, equities). Cash is remunerated at the overnight rate.

⇒ Only C is consistent with all three conventions.

Interpretation

Each element of C reflects the balance between workload and protection: generous terms where credit is strong, tighter terms and larger haircuts where the risk is.

Why the other options are wrong

A is incorrect. A high threshold leaves large uncollateralised exposure, a low MTA adds workload, and a fixed below-market rate is not standard.

B is incorrect. Uniform haircuts ignore collateral risk (cash carries none), and rating-blind terms ignore credit quality.

D is incorrect. Ignores the rating link, and a daily regime with no MTA maximises workload.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.4, 17.3.4 and 17.3.5; SchweserNotes 2026 Book 2, Reading 35, LO 35.g

Solution CR-17 g.6 Answer: C

Bank Q820 • AnalystPrep Q.5503

Step 1 – What the MTA trades off

A larger MTA means fewer small transfers but more uncollateralised exposure, which is tolerable for strong credits.

Step 2 – Rating tiers

Schweser: MTAs are typically tied to credit ratings, with higher ratings corresponding to higher amounts. Weaker names get lower MTAs, tightening collateralisation where the risk is.

⇒ C.

Interpretation

The same logic drives the rating-linked threshold schedule in GARP Table 17.4 (CR-17 h): loose for AAA, tight below A+.

Why the other options are wrong

A is incorrect. Reverses the tiering and loosens terms for weaker credits.

B is incorrect. Ignores credit quality and leaves weak counterparties undercollateralised.

D is incorrect. No MTA maximises protection but also the workload of tiny transfers.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** SchweserNotes 2026 Book 2, Reading 35, LO 35.g; Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4

Solution CR-17 g.7 Answer: B

Bank Q822 • BT P2.T6.410.2

Step 1 – What each change does

More independent amount adds overcollateralisation; a lower MTA lets smaller amounts be called; daily calls shorten the time exposure can build unprotected.

⇒ A, C and D all increase mitigation.

Step 2 – The threshold

The threshold is the uncollateralised exposure: raising it *reduces* mitigation.

Interpretation

Barbara's goal (maximum mitigation, workload accepted) points to a zero threshold, a small

MTA, daily calls and initial margin: the interbank standard plus the bilateral margin rules.

Why the other options are wrong

A is incorrect. More initial margin improves mitigation.
C is incorrect. A smaller MTA collateralises smaller exposures.
D is incorrect. Daily calls reduce unprotected exposure between calls.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.3 and 17.3.4

Solution CR-17 g.8 Answer: C

Bank Q823 • BT P2.T6.906.2

Step 1 – What gap risk is

The risk that the portfolio value jumps between the last margin call and the close-out after a default, before more margin can be called.

Step 2 – What covers it

Initial margin is posted upfront, independent of variation margin, and leaves the receiver over-collateralised; it is sized to absorb a large move over the period without margin.
⇒ C.

Interpretation

The parameters split cleanly: haircuts cover the price risk of the *collateral asset*; initial margin covers moves in the *exposure*; thresholds and MTAs only add uncollateralised exposure.

Why the other options are wrong

A is incorrect. A haircut covers volatility and illiquidity of the posted asset, not a jump in the portfolio's value: plausible, but second best.
B is incorrect. A higher threshold increases uncollateralised exposure and worsens gap risk.
D is incorrect. A higher MTA delays transfers and adds residual exposure.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.3 and 17.3.4; SchweserNotes 2026 Book 2, Reading 35, LO 35.g

Solution CR-17 g.9 Answer: A

Bank Q824 • BT P2.T6.906.1

Step 1 – The pressures

Funding and capital costs make uncollateralised exposure expensive; central clearing and the bilateral margin rules require full variation margin with a zero threshold.

Step 2 – GARP's conclusion

"Thresholds are increasingly zero and can only be non-zero for parties that are exempt from the bilateral margin requirements."
⇒ A.

Interpretation

Two-way, zero-threshold CSAs are already the interbank standard; the trend extends them.

Why the other options are wrong

B is incorrect. An infinite threshold means a party never posts (a one-way CSA), which funding and capital costs have made more problematic, not more common.
C is incorrect. Regulation and clearing change threshold terms directly.
D is incorrect. Thresholds and initial margin work in opposite directions; where initial margin is present the threshold is typically zero.

LO: CR-17 g: Explain the features of a collateralization agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.2, 17.3.4 and 17.4.1

CR-17 • Margin (Collateral) and Settlement • Vardeez: VERY IMPORTANT

CR-17 h

P1 CONCEPT

Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: the two-way CSA came up in the August 2026 sitting (A21, Map #27).
Vardeez: a one-way versus two-way table and two lines on credit-quality links (less workload when credit is strong, more protection through collateral on downgrade).

Reported exam shapes

- **A21·S8** Two-way versus one-way CSA and rating-linked terms, as seen in August.

What the questions look like: which relationship gets which CSA; who benefits from a one-way CSA and who bears its risk; rating-tiered thresholds and initial margin; downgrade triggers and their dangers. **Kept outside the TYPE rule:** Workbook A21.1 is numerical (a downgrade changes the threshold, then the call is computed). It stays here as a workbook question; the calculation itself is taught under CR-17 c.

Worked elsewhere: A21.2 (the MTA test under a two-way CSA) at Question CR-17 c.8; A21.3 (margin period of risk and collateralised CVA) at Question CR-20 j.6; A21.4 (CSA terms for a supranational, a hedge fund and a dealer) under CR-17 f.

1. Three arrangements

CSA parameters are negotiated, so there are endless variants; GARP sorts them into three descriptive classes.

- **No CSA.** One party cannot or will not commit to posting, typically a corporate whose treasury could not manage the liquidity demands. It pays xVA charges (CVA, FVA, KVA) instead. Non-financials use CSAs far less than banks (ISDA 2015: 28.6% against 95.5%).
- **Two-way CSA.** Both parties post margin against a negative value. Typical between two financial counterparties of similar credit quality; two-way CSAs with zero thresholds are the interbank standard and aim to benefit both parties. Thresholds and initial margin can still differ with each party's risk.
- **One-way CSA.** Only one party can receive margin, at once or after an event such as a downgrade. Contractually it is a two-way CSA with an *infinite threshold* for the non-posting party. It "represents additional risk for the margin giver", which is worse off than with no CSA: it posts against its liabilities but receives nothing against its assets. Examples: a triple-A sovereign or supranational trading with a bank; banks demanding one-way CSAs in their favour from some hedge funds. Funding and capital costs have made one-way CSAs problematic for the giving bank.

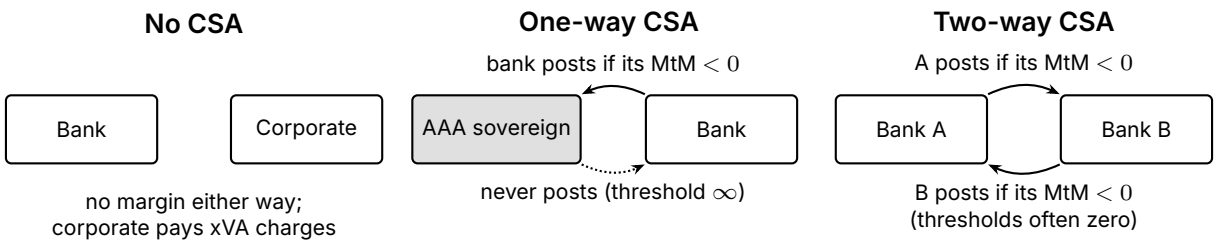


Figure 1. The three arrangements. Solid arrows are margin that can flow; the dotted arrow is margin that never flows. In the one-way CSA the grey party (the receiver) benefits and the bank (the giver) carries extra risk.

2. Linking parameters to credit quality

Thresholds, initial margin and sometimes MTAs can be tied to a rating (or, for unrated parties, to net asset value, market capitalisation or credit spreads). The aim: little operational workload while the counterparty is strong, tighter collateralisation as it weakens.

Rating	Initial margin	Threshold	Rating	Initial margin	Threshold
AAA/Aaa	0%	∞	A+/A1	0%	\$0
AA+/Aa1	0%	\$100m	A/A2	1%	\$0
AA/Aa2	0%	\$50m	A−/A3	1%	\$0
AA−/Aa3	0%	\$25m	BBB+/Baa1	2%	\$0

GARP Table 17.4: rating-linked parameters (initial margin as a percentage of the portfolio's total notional), for one or both parties.

A downgrade activates the tighter row at once: more margin must be posted, or under a break clause the trades may be terminated (CR-16). Question CR-17 h.7 (Workbook A21.1) computes a call after a downgrade from AA to AA−.

The dangers of rating triggers. AIG posted margin as a function of its rating; its downgrade in September 2008 required about \$20bn of extra margin and a Federal Reserve facility of up to \$85bn. GARP's two lessons:

- **Ineffective as a mitigant**: ratings react slowly, so the downgrade arrives when the difficulties are already acute (monoline insurers traded under one-way CSAs with such triggers).
- **Liquidity problems and cliff-edge effects**: margin is demanded exactly when the counterparty is weakest, and many similar triggers fire together on a single-notch downgrade (Schweser: a "death spiral").

Regulation recognises this. Basel capital rules give *no* capital benefit for rating triggers and assume a counterparty defaults without first being downgraded. The liquidity coverage ratio makes a bank *pre-fund* its own contingent outflows: under Table 17.4 an A+ bank must fund 2% initial margin for a three-notch downgrade to BBB+.

	No CSA	One-way CSA	Two-way CSA
Who posts	Neither	One party only (now or on a trigger)	Whichever party has negative MtM
Contract form	No annex	Infinite threshold for the non-poster	Thresholds for both, often zero
Who benefits	Neither (xVA charged)	The receiver; the giver is worse off than with no CSA	Both
Typical pair	Bank and corporate	AAA sovereign or supranational and bank; bank and some hedge funds	Two financial institutions of similar credit quality

Exam traps

- **Role misassignment.** A one-way CSA benefits the *receiver* and adds risk for the *giver*, who is worse off than with no CSA.
- **One-way means zero threshold.** The non-posting party's threshold is *infinite*; zero thresholds are the two-way interbank standard.
- **Two-way means both always post.** Only the party with negative MtM posts; the direction can reverse as values move.
- **Polarity flip on credit links.** Thresholds fall and initial margin rises as quality falls, never the reverse; after a downgrade use the new row (A21.1).
- **Rating triggers as reliable protection.** Ratings lag, triggers create cliff-edge liquidity strain, Basel gives no capital credit, and the LCR forces pre-funding.

Practice questions: CR-17 h

Question CR-17 h.1 Bank Q827 • AnalystPrep Q.1894 (rebuilt; options reordered)

A collateral (margin) agreement between two counterparties is described as two-way. Which of the following best describes what this implies?

A. Both parties must post collateral at all times, regardless of the sign of the mark-to-market value.

B. Only the party with the weaker credit rating ever posts collateral, and the other party never does.

C. Whichever party has a negative mark-to-market value on the portfolio posts collateral to the other, so the direction of posting can change as exposure changes.

D. Either party may unilaterally change the collateral terms at any time if its counterparty's credit quality deteriorates.

Question CR-17 h.2 Bank Q830 • Schweser Module Quiz 35.2 Q2

When dealing with a hedge fund, a bank would most likely negotiate a:

A. one-way agreement in the bank's favor given the bank's stronger credit rating.

B. one-way agreement in the bank's favor agreeing to post collateral to the hedge fund.

C. two-way agreement given the relatively small difference in credit quality between the two entities.

D. two-way agreement where both parties agree to post collateral.

Question CR-17 h.3 Bank Q829 • BT P2.T6.905.2

Two counterparties are negotiating the terms of a new Master Agreement in reference to a series of derivative transactions. They are also negotiating the parameters of a credit support annex (CSA) which might be appended to the Master Agreement. There are three approaches: two-way CSA, one-way CSA, or no CSA. About the possible CSA, which of the following statements is TRUE?

A. As CSAs are rare, the most likely approach is "No CSA"

B. If they agree to a one-way CSA, then by definition, the CSA will specify a threshold of zero

C. If they agree to a two-way CSA, then either party can unilaterally edit the CSA subsequent to signing the CSA

D. If they agree to a one-way CSA, the non-collateral posting party is likely to pay charges for the counterparty risk (CVA), funding (FVA), collateral terms (ColVA), and/or capital requirements (KVA) they impose on the other party

Question CR-17 h.4 Bank Q828 • AnalystPrep Q.5502

Global Finance Bank (GFB) is reassessing the threshold levels in its collateralization agreements for OTC derivatives transactions. The reassessment comes amid changing market conditions and a diverse range of counterparties with varying credit qualities. Which of the following approaches to setting threshold levels in the collateralization agreement is most appropriate for GFB to balance risk management with operational efficiency?

A. GFB sets a universal threshold level for all counterparties, irrespective of their credit ratings, to standardize the collateral process and reduce complexity.

B. The bank decides to adopt a tiered threshold approach, where threshold levels are linked to the credit ratings of counterparties, with lower-rated counterparties having lower thresholds.

C. GFB eliminates threshold levels altogether, requiring all counterparties to fully collateralize their exposures from the first dollar, maximizing risk mitigation.

D. The bank sets extremely high threshold levels for all transactions to minimize the frequency of collateral calls, thereby reducing operational workload.

Question CR-17 h.5 Bank Q826 • AnalystPrep Q.6050

In the context of a derivative contract, a financial risk manager is considering a downgrade trigger as part of the credit risk mitigation strategy. This trigger requires a counterparty to post additional collateral if their credit rating falls below a predetermined level. Assuming that the counterparty's credit rating is downgraded, triggering the clause, and the counterparty has outstanding transactions, which of the following would be a likely course of action?

A. The counterparty would be granted a grace period to improve their credit rating before any action is taken.

B. The outstanding transactions would be netted to reflect a single net claim without any change in the collateral requirements.

C. The counterparty would be required to post additional collateral or potentially terminate outstanding transactions at market values.

D. Netting will be suspended, and the full gross value of all outstanding transactions becomes due immediately.

Question CR-17 h.6 New • built from GARP Ch.17, 17.3.4 Table 17.4 and "The dangers of credit rating triggers"

The treasurer of an A+ rated bank is reviewing the bank's own CSAs, many of which use GARP's rating-linked schedule: at A+ the bank's threshold is zero with no initial margin; at A and A− it must also post initial margin of 1% of notional; at BBB+, 2%. Which of the following statements about these rating-linked terms is correct?

A. Basel capital rules give the bank's counterparties capital relief for the triggers, since a downgrade would bring in margin before any default.

B. The liquidity coverage ratio requires the bank to pre-fund the contingent outflow, for example 2% initial margin for a three-notch downgrade to BBB+.

C. The triggers are effective because rating agencies typically downgrade well before a firm's financial difficulties become acute.

D. Because each counterparty's trigger operates separately, a single-notch downgrade can have only a small and gradual effect on the bank's liquidity.

Question CR-17 h.7 Workbook A21.1 • S1, S8 • Aug-2026 • Map #27 • numerical, kept as a workbook question

A collateral manager at a US bank is computing today's margin call under a two-way CSA with a European insurer. Thresholds are rating-linked (AA: USD 50 million; AA−: USD 25 million; A+: zero). The data are:

- Portfolio value to the bank: +USD 58.6 million
- Insurer's rating: downgraded overnight from AA to AA−
- Credit support balance already held by the bank: USD 8.0 million (cash)
- Minimum transfer amount: USD 0.5 million; rounding: amounts rounded up to the nearest USD 0.25 million

Which of the following is closest to the margin the bank can call?

A. USD 0.75 million

B. USD 25.25 million

C. USD 25.50 million

D. USD 25.75 million

Solutions • CR-17 h

Solution CR-17 h.1 — Answer: C

Bank Q827 • AnalystPrep Q.1894

Step 1 – What two-way means

Both parties are contractually obliged to post, but only when they are out of the money (negative MtM), so that the other party is exposed.

Step 2 – Direction can reverse

If the portfolio value changes sign, the flows reverse: the previous receiver becomes the poster.
$\Rightarrow$ C.

Interpretation

Two-way does not mean symmetric: thresholds and initial margin can differ between the two parties according to their risk. What makes it two-way is that either side can end up posting.

Why the other options are wrong

A is incorrect. A party that is in the money does not post.
B is incorrect. Describes a one-way agreement.
D is incorrect. Terms change only through pre-agreed provisions (such as rating triggers) or mutual amendment.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.2 and 17.3.2

Solution CR-17 h.2 — Answer: A

Bank Q830 • Schweser Module Quiz 35.2 Q2

Step 1 – When one-way CSAs are used

When the two parties differ significantly in size, credit quality or risk.

Step 2 – Apply it

A bank is typically of much higher credit quality than a hedge fund, so it can demand that only the hedge fund posts. GARP: "banks themselves have typically been able to demand one-way CSAs in their favour when transacting with some hedge funds."
$\Rightarrow$ A.

Interpretation

"In the bank's favour" means the bank receives. The same bank will be on the losing side of a one-way CSA when it faces a triple-A sovereign.

Why the other options are wrong

B is incorrect. Self-contradictory: an agreement in the bank's favour has the hedge fund posting.
C is incorrect. The credit gap between a bank and a hedge fund is large, not small.
D is incorrect. Two-way agreements suit similar counterparties, such as two banks.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** SchweserNotes 2026 Book 2, Reading 35, LO 35.h; Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.2

Solution CR-17 h.3 — Answer: D

Bank Q829 • BT P2.T6.905.2

Step 1 – The one-way CSA's cost

The non-posting party leaves the giver with uncollateralised exposure plus the funding of the margin it posts, and the capital that goes with it.
⇒ It is likely to be charged for the xVA it imposes: CVA, FVA, ColVA and KVA.

Step 2 – Check the others

CSAs are common (banks almost always, financial institutions often). A one-way CSA gives the non-poster an infinite threshold, not zero. A signed CSA changes only by mutual agreement.

Interpretation

GARP: non-margin-posting entities "generally prefer (or have no choice) to pay xVA charges" rather than post margin.

Why the other options are wrong

A is incorrect. CSAs are not rare: 90% or more of dealers, banks and hedge funds trade under them.
B is incorrect. A one-way CSA specifies an infinite threshold for the non-posting party.
C is incorrect. A signed agreement cannot be changed unilaterally.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.1 and 17.3.2

Solution CR-17 h.4 — Answer: B

Bank Q828 • AnalystPrep Q.5502

Step 1 – What the threshold represents

The amount of exposure left uncollateralised, so it should reflect the counterparty's credit risk.

Step 2 – Tier it

Linking thresholds to rating (infinite for AAA, falling to zero from A+ down in GARP Table 17.4) keeps the workload light for strong names and calls margin early from weak ones.
⇒ B.

Interpretation

This is the "credit quality" half of the LO: the same schedule then needs care, because a downgrade tightens terms abruptly.

Why the other options are wrong

A is incorrect. One threshold for all ignores credit differences.
C is incorrect. Zero thresholds everywhere maximise protection but ignore the operational cost: no balance.
D is incorrect. Very high thresholds for all leave large uncollateralised exposure to weak names.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4 Table 17.4

Solution CR-17 h.5 — Answer: C

Bank Q826 • AnalystPrep Q.6050

Step 1 – What the trigger does

Once the rating falls below the agreed level the tighter terms apply at once (a lower threshold, initial margin), so more collateral is due.

Step 2 – The alternative

Where the contract has a rating-based break clause (an Additional Termination Event), the trades may be terminated at their market value.

$\Rightarrow$ C.

Interpretation

The protection works only if the downgrade comes early enough. GARP's warning: ratings lag, and a trigger can push a weakened counterparty into a liquidity crisis (AIG).

Why the other options are wrong

A is incorrect. A grace period defeats the purpose of the trigger.

B is incorrect. The clause has been triggered, so the collateral terms must change.

D is incorrect. Triggers do not suspend netting; any close-out is at the net value.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.1.1 and 17.3.4

Solution CR-17 h.6 Answer: B

New • built from GARP Ch.17, 17.3.4

Step 1 – Regulation recognises the asymmetry

Basel capital rules give no capital benefit for rating triggers and assume a counterparty defaults without first being downgraded. The liquidity coverage ratio, by contrast, requires a bank to pre-fund outflows such as ratings-contingent initial margin.

$\Rightarrow$ B: an A+ bank funds 2% initial margin for a three-notch downgrade (A+ to A, A−, BBB+).

Step 2 – The two problems with triggers

Ratings react too slowly to be a reliable mitigant, and many similar triggers firing on one downgrade create liquidity problems and cliff-edge effects.

Interpretation

The bank pays twice for rating triggers: it gets no capital relief when it receives them, and it must hold liquidity against them when it gives them.

Why the other options are wrong

A is incorrect. Basel explicitly rules out capital relief from rating triggers (BCBS 2011b).

C is incorrect. Ratings are "somewhat unreactive"; by the time of the downgrade the difficulties may be too acute (the monolines).

D is incorrect. Cliff-edge effects: a single-notch downgrade can trigger many counterparties at once, as with AIG.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4 Table 17.4 and box "The dangers of credit rating triggers"; 17.1.1

Solution CR-17 h.7 Answer: D

Workbook A21.1 • S1, S8 • Aug-2026

Step 1 – Applicable threshold

After the downgrade to AA−, K_C = USD 25 million (the AA threshold of USD 50 million no longer applies).

$\Rightarrow$ The rating link tightens collateral terms as the insurer's credit quality falls.

Step 2 – Required margin

$$\max(\text{value} - K_C,\, 0) = \max(58.6 - 25,\, 0) = \text{USD } 33.6 \text{ million}$$

Step 3 – Credit support amount before MTA and rounding

$$33.6 - \max(-58.6 - K_P,\, 0) - 8.0 = 33.6 - 0 - 8.0 = \text{USD } 25.6 \text{ million}$$

$\Rightarrow$ The bank's own threshold term is zero because the portfolio has positive value to the bank.

Step 4 – MTA test and rounding

$25.6 > 0.5$, so a call is valid and the full amount is called. Rounded up to the nearest 0.25: USD 25.75 million.

Interpretation

The downgrade alone triggers a USD 25.75 million call on the insurer: the cliff-edge liquidity effect Gregory warns about with rating triggers (the AIG pattern).

Why the other options are wrong

A is incorrect. Used the pre-downgrade AA threshold: $58.6 - 50 - 8.0 = 0.6$, which passes the MTA and rounds up to 0.75.

B is incorrect. Deducted the MTA from the amount: $25.6 - 0.5 = 25.1$, rounded up to 25.25. The MTA only gates the transfer.

C is incorrect. Rounded down instead of up as the CSA specifies: 25.50.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. CR-17 c: Calculate the credit support amount (margin) under various scenarios. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.4 Table 17.4; 17.3.6 eq. 17.3

CR-17 • Margin (Collateral) and Settlement • Vardeez: VERY IMPORTANT

CR-17 j

P1 CONCEPT

Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A21, Map #27), flagged by Vardeez inside a VERY IMPORTANT reading. The map calls this LO "a role-misassignment favourite".
Vardeez: "LO 35.j: this too can be imp." The notes are a table of six risks; the one below adds who bears each.

Reported exam shapes

- **A21·S6** Market, operational and liquidity risk through collateralisation, with role misassignment.

What the questions look like: match a source (MTA and MPoR, missed calls, a large sale of collateral, a margin call in a downturn) to the risk it creates and the party that bears it. Distractors move a source to the wrong risk type or the risk to the wrong party.

Worked elsewhere: A21.3 and A21.5 (margin period of risk and collateralised CVA) at Questions CR-20 j.4 and CR-20 j.6; A21.6 (liquidity risk sits with the margin holder) at Question CR-16 d.7; A20.2 (wrong-way collateral) under CR-20 m.

1. Margin converts risk; it does not remove it

GARP: margin, like netting, "does not reduce risk overall but merely redistributes it". It reduces counterparty risk but creates the LO's three (market, operational, liquidity including funding liquidity), plus legal risk and the default, FX and wrong-way risk of the collateral. Each is smaller than the risk removed, but none is zero (Vardeez).

2. Market risk and the margin period of risk

Thresholds and MTAs delay calls, and even with daily calls and zero thresholds margin arrives late. The **margin period of risk (MPoR)** is "the effective time between a counterparty ceasing to post margin and the time when all the underlying transactions have been successfully closed out and replaced (or otherwise hedged)". Any rise in exposure over it is uncollateralised, so the market risk sits with the margin *receiver*, which has the exposure. The MPoR drives initial margin; its effect on CVA is CR-20 j.

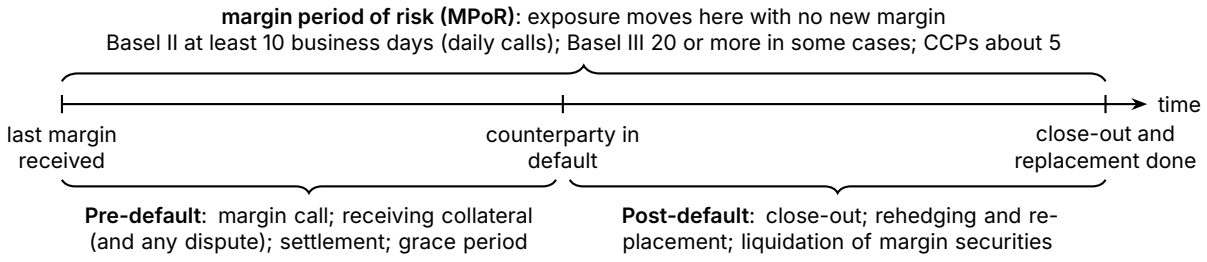


Figure 1. The margin period of risk (GARP Figures 17.17 and 17.18). The call-to-receipt delay is only the start of it; models use one simple "catch-all" period.

Correction to the source notes. SchweserNotes (LO 35.j) defines the margin period of risk as "the effective time between a collateral call and the receipt of the collateral". GARP's definition runs from the counterparty ceasing to post margin until all its transactions have been closed out and replaced or hedged (Figure 1). Use GARP's definition.

3. Operational and legal risk

Collateral management is time-consuming and intensely dynamic: missed calls, failed deliveries, computer and human error, wrong valuations, fraud. It bites hardest for the largest banks, with thousands of non-standard agreements. Controls (Schweser): enforceable legal agreements, robust automated IT, timely and accurate valuation, current data on CSA parameters, and close watch on failed deliveries. Basel III lengthens the MPoR for netting sets with more than 5,000 trades, illiquid margin or exotic trades, or a history of margin call disputes. Legal risk: netting or the margin held may be challenged, and segregation may fail (MF Global).

4. Liquidity (liquidation) risk

The margin *holder* bears it when it must sell margin securities after a default to raise cash for rehedging: bid-offer costs, price moves over the liquidation period, and price impact if the amount sold is large relative to the volume traded (selling slowly leaves it exposed to volatility instead). Haircuts buffer ordinary price falls but not a large sale's impact or the issuer's default; concentration limits of 5% to 10% of an issue are common.

5. Funding liquidity risk

The risk of being unable to raise the required funding in future. Margin creates it: contractual payments over short timescales, in amounts that depend on future values. GARP: margin "converts counterparty risk into funding liquidity risk", cheaply in liquid markets and dangerously when liquidity dries up. It falls on the party that must *post*.

- **End users** are directional (a corporate pays fixed to hedge floating-rate debt), so a market move creates large cash calls while the offsetting benefit arrives slowly. Ashanti lost $450m on gold hedges in 1999 and faced about $280m of cash calls it could not meet despite owning the gold.
- **Banks** hedge uncollateralised client trades with collateralised trades and must fund the margin on the hedge when the client trade moves in their favour (the FVA problem). They must also pre-fund rating-contingent margin under the LCR (CR-17 h).

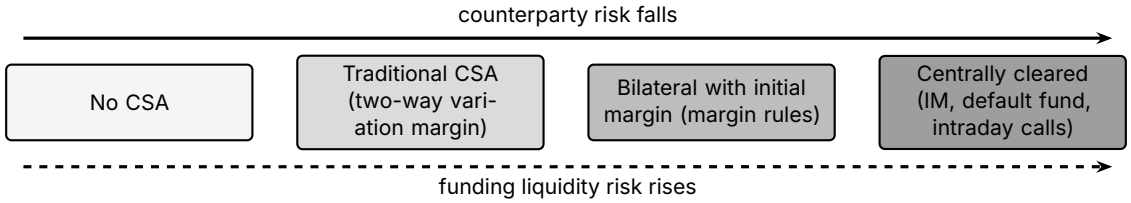


Figure 2. The trade margin makes (GARP Figure 17.19). Moving right (darker grey fills) reduces counterparty risk (solid arrow) at the price of more funding liquidity risk (dashed arrow).

Risk	How it arises through collateralisation	Who bears it
Market risk	Exposure moves after the last margin received: thresholds, MTAs and the MPoR	The margin receiver (it has the exposure)
Operational risk	Missed calls, failed deliveries, errors, fraud, wrong valuations	Both; worst for large banks
Liquidation risk	Selling margin securities after a default: bid-offer, price falls, price impact	The margin holder
Funding liquidity risk	Calls met in eligible assets at short notice, in amounts that depend on future values	The margin giver
Legal risk	Netting or margin challenged; failed segregation	Receiver (enforceability); giver (return of excess)
Default, FX, wrong-way	Collateral issuer defaults; margin in another currency; value falls with the counterparty	The margin receiver

Exam traps

- **Role misassignment.** Liquidation risk sits with the margin *holder*; funding liquidity risk with the party that must *post*; MPoR market risk with the party that has the exposure.
- **Category misassignment.** Thresholds, MTAs and the MPoR create *market* risk; missed calls and failed deliveries *operational* risk; a large sale's price impact *liquidity* risk; meeting calls *funding liquidity* risk.
- **"Collateral reduces all risks."** It reduces counterparty risk only; the other risks rise or remain.
- **Haircut as a cure-all.** A haircut buffers price falls but not a large sale's price impact or the issuer's default.
- **MPoR read as the call frequency** or as the call-to-receipt delay only. It runs to close-out and replacement.

Practice questions: CR-17 j

Question CR-17 j.1

Bank Q837 • BT P2.T6.900.2

Acme Financial Incorporated has an outstanding derivatives contract with its counterparty. Currently neither posts collateral, but they agree to modify the contract to include a bilateral collateral arrangement. Which risk has Acme reduced?

A. Acme reduced counterparty risk

B. Acme reduced operational risk

C. Acme reduced market/liquidity risk

D. All of the above (reduction in counterparty, operational, and market/liquidity risk)

Question CR-17 j.2 Bank Q836 • AnalystPrep Q.6158

How does the process of mitigating counterparty risk through collateralization in derivatives transactions potentially affect other types of risk?

A. It reduces operational risk by simplifying trade entry and settlement processes.

B. It increases funding liquidity risk due to the need for timely payments, often within tight timescales.

C. It eliminates market risk by ensuring an objective valuation of the underlying transactions.

D. It diminishes legal risk by standardizing contractual terms across different jurisdictions.

Question CR-17 j.3 Bank Q840 • AnalystPrep Q.5499 (rebuilt; options reordered)

Global Finance Bank (GFB) is a party to a collateralized OTC derivatives agreement that includes a threshold, a minimum transfer amount and a delay between a collateral call and receipt of the collateral. Which statement best describes how market risk arises through the collateralization?

A. Posting collateral eliminates all market risk because exposure is reset to zero at every valuation date.

B. Market risk arises only when the collateral is cash held in a different currency from the trade.

C. Market moves between the last collateral posting and the receipt of collateral after default can leave exposure only partly covered, so some residual risk remains.

D. Market risk is created only by operational failures such as missed collateral calls.

Question CR-17 j.4 Bank Q838 • AnalystPrep Q.5497 (options reordered)

Global Finance Bank (GFB) is reviewing the operational risks associated with the collateralization of its OTC derivatives portfolio. During a period of market volatility, GFB has encountered several challenges in its collateral management process. Which scenario accurately demonstrates how operational risk can arise in the context of collateralization?

A. Operational risk is mitigated through the use of diversified collateral, reducing the bank's exposure to any single asset's volatility.

B. Operational risk is unaffected by collateralization practices, as it is primarily concerned with market fluctuations and credit risk management.

C. Operational risk increases as GFB engages in more OTC derivatives transactions, irrespective of the collateralization process.

D. Operational risk emerges due to inaccurate valuation of collateral, leading to inadequate collateral postings and potential disputes with counterparties.

Question CR-17 j.5 Bank Q844 • Schweser Module Quiz 35.2 Q1 (rebuilt; options reordered)

Collateral agreements could create multiple risks, including liquidity and liquidation risks. Which statement is most accurate?

A. Liquidity risk is eliminated entirely once an appropriate haircut is applied to the collateral.

B. Liquidation risk occurs when the amount of a security sold is large relative to its typical trading volume, which may drive down its price.

C. Liquidation risk arises only if the issuer of the collateral security defaults, not from the size of the position sold.

D. Liquidity risk occurs when there are potential pitfalls in the handling of collateral, including human error.

Question CR-17 j.6 Bank Q842 • BT P2.T6.24.25.3 (rebuilt)

Global Manufacturing Corp. pays fixed and receives floating on a $100 million, 5-year interest rate swap with Acme Bank to hedge floating-rate debt. The swap is collateralized with daily exchange of variation margin. Interest rates fall sharply. Which risk does the collateral arrangement most directly create for Global?

A. Funding liquidity risk, because the swap has a negative MtM and Global must post liquid variation margin immediately, while the offsetting benefit (lower interest on its debt) arrives only gradually.

B. Increased credit risk to Acme, because Global's swap now has a large positive MtM to be collected from Acme.

C. No new risk, because the swap hedges the debt and daily margining removes all risks of the position.

D. Reduced operational risk, because daily margin automatically eliminates valuation disputes and failed deliveries.

Question CR-17 j.7 Bank Q841 • AnalystPrep Q.6141 (options reordered)

A bank's hedging strategy involves a mix of transactions with varying collateral requirements. The bank notices an imbalance in its collateral flows, which could affect its liquidity. Which is the most likely cause of the imbalance?

A. Inconsistent valuation methods for collateral across different transactions.

B. Over-collateralization of certain transactions leading to liquidity constraints.

C. Fluctuating market values affect the quality and quantity of posted collateral.

D. Discrepancy between uncollateralized client transactions and bilaterally collateralized hedges.

Question CR-17 j.8 Bank Q843 • BT P2.T6.325.3 (rebuilt; options reordered)

Collateral management mitigates counterparty risk but converts it into other forms of risk. Which of the following correctly matches a source of risk to the risk type created by collateralization?

A. Missed collateral calls, failed deliveries and human error create market risk.

B. Price impact and transaction costs when liquidating a large collateral position create operational risk.

C. Minimum transfer amounts, thresholds and the margin period of risk leave exposure that can move before collateral is received, creating market risk.

D. Difficulty in meeting frequent, large margin calls creates operational risk.

Question CR-17 j.9 New • built from GARP Ch.17, 17.5.2, 17.5.3 and 17.6.2 • shape A21·S6 (role misassignment)

A collateral risk officer at a dealer bank is training new staff on a two-way CSA with a corporate client. The corporate is out of the money on a large swap and meets the bank's daily margin calls by posting government bonds. Which of the following assigns a risk to the correct party?

A. Liquidation risk falls on the corporate as margin giver, which bears the bid-offer cost and price falls if the bonds are sold after a default.

B. Funding liquidity risk falls mainly on the corporate, which must source eligible collateral within short timescales whenever the swap moves further against it.

C. Market risk over the margin period of risk falls on the corporate, since the bank as margin receiver is fully protected until close-out.

D. Operational risk falls only on the bank as valuation agent, so the corporate, which does not calculate the calls, carries none.

Solutions • CR-17 j

Solution CR-17 j.1 — Answer: A

Bank Q837 • BT P2.T6.900.2

Step 1 – What falls

Bilateral margin reduces each party's exposure to the other's default: counterparty risk falls.

Step 2 – What rises or remains

Margin calls, valuation, delivery and reconciliation add operational risk; residual market risk remains over the MPoR and within thresholds and MTAs; liquidity and funding liquidity risk rise because Acme must post liquid collateral.

$\Rightarrow$ Only counterparty risk is reduced.

Interpretation

GARP: margin "gives rise to new risks, such as market, operational, and liquidity risk". The trade is worth making, but it is a conversion, not a free reduction.

Why the other options are wrong

B is incorrect. Operational risk rises: collateral handling adds processes and chances for error.

C is incorrect. Market risk persists and liquidity risk increases.

D is incorrect. Combines the right answer with two risks that actually increase.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.2 and 17.5

Solution CR-17 j.2 — Answer: B

Bank Q836 • AnalystPrep Q.6158

Step 1 – The conversion

Margin agreements require contractual payments over short timescales, in amounts that depend on future values.

Step 2 – The resulting risk

A party without the liquidity or operational capacity to meet frequent calls faces funding liquidity risk, which rises when markets are illiquid and funding costs high.

$\Rightarrow$ B.

Interpretation

"The problem with margin is that it converts counterparty risk into funding liquidity risk" (GARP 17.6.2): beneficial in normal markets, potentially severe in stressed ones.

Why the other options are wrong

A is incorrect. Collateral management adds missed calls, failed deliveries and errors: operational risk increases.

C is incorrect. Market risk remains over the MPoR and within thresholds and MTAs.

D is incorrect. Collateral adds legal risk (enforceability, segregation); it does not standardise terms across jurisdictions.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5 and 17.6.2

Solution CR-17 j.3 Answer: C

Bank Q840 • AnalystPrep Q.5499

Step 1 – Contractual delays

Thresholds and MTAs let exposure build before a call can be made.

Step 2 – The inherent delay

Even when called, margin arrives after a delay, and after a default the MPoR runs until the trades are closed out and replaced. Exposure that rises meanwhile is uncollateralised.

$\Rightarrow$ C: a residual, hard-to-hedge market risk remains.

Interpretation

Schweser: this market risk is small compared with an uncollateralised position, but it is hard to hedge and to quantify. Initial margin exists to cover it.

Why the other options are wrong

A is incorrect. Ignores the delay, threshold and MTA: exposure is not reset to zero continuously.

B is incorrect. A currency mismatch creates FX risk, one source among several, not the market risk the LO describes.

D is incorrect. Missed calls are operational risk; they can add to exposure but do not define market risk.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5.2 Market Risk and Margin Period of Risk

Solution CR-17 j.4 Answer: D

Bank Q838 • AnalystPrep Q.5497

Step 1 – Where operational risk comes from

The handling of collateral: valuation errors, missed calls, failed deliveries, computer or human error, fraud.

Step 2 – Match the scenario

Inaccurate valuation produces wrong call amounts and disputes: a direct operational failure of the collateral process.

$\Rightarrow$ D.

Interpretation

Basel III ties this to capital: illiquid margin, exotic trades that are hard to value and a history of disputes all require a longer MPoR.

Why the other options are wrong

A is incorrect. Diversifying collateral addresses market and concentration risk, not process failures.

B is incorrect. Collateralisation adds operational risk; operational risk is not about market fluctuations.

C is incorrect. Ignores the collateral process, which is what the question asks about.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5.4 Legal and Operational Risks

Solution CR-17 j.5 — Answer: B

Bank Q844 • Schweser Module Quiz 35.2 Q1

Step 1 – Costs of liquidation

Liquidating margin securities after a default costs the bid-ask spread and selling costs.

Step 2 – Size

Selling an amount large relative to normal trading volume moves the price and can cause a loss well beyond the haircut; selling slowly instead leaves the holder exposed to market volatility.
$\Rightarrow$ B.

Interpretation

Haircuts and concentration limits (5% to 10%) are the defences. Schweser's own answer key labels this question LO 35.i; its content is LO 35.j.

Why the other options are wrong

A is incorrect. A haircut is a buffer; it does not remove price-impact losses beyond it.
C is incorrect. Issuer default is a separate (default) risk; liquidation risk is driven by size and market depth.
D is incorrect. Handling errors are operational risk.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5.3; Schweser-Notes 2026 Book 2, Reading 35, LO 35.j

Solution CR-17 j.6 — Answer: A

Bank Q842 • BT P2.T6.24.25.3

Step 1 – Sign of the swap

Global pays fixed and receives floating; when rates fall the swap's MtM turns negative for Global.

Step 2 – Margin and timing

Under the CSA Global must deliver variation margin daily in eligible assets, while the saving on its floating-rate debt accrues only as coupons reset.
$\Rightarrow$ A: cash is needed now, the gain comes later.

Interpretation

This is GARP's end-user case: directional hedges create large margin calls in volatile markets, which is why many end users do not sign CSAs, and why those that do must plan for worst-case margin in their funding.

Why the other options are wrong

B is incorrect. Sign error: Global owes margin; Acme has the exposure.
C is incorrect. Collateral transforms risk; it does not remove it.
D is incorrect. Daily margining increases the operational burden.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.6.2 Margin and Funding Liquidity Risk

Solution CR-17 j.7 — Answer: D

Bank Q841 • AnalystPrep Q.6141

Step 1 – The bank's structure

Banks run mainly flat books: a client trade is often uncollateralised and is hedged with a trade under a CSA.

Step 2 – The asymmetry

When the client trade moves in the bank's favour, the bank posts margin on the hedge but receives none from the client.

⇒ D: a structural, one-sided margin flow, a funding liquidity cost.

Interpretation

GARP uses this to explain FVA, and why banks want more clients to sign CSAs to balance the flows.

Why the other options are wrong

A is incorrect. Valuation inconsistencies cause disputes (operational risk), not a systematic imbalance.

B is incorrect. Overcollateralisation is a possible side effect, not the structural cause.

C is incorrect. Market moves affect both sides; they do not create a one-sided imbalance.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.2.2 and 17.6.2

Solution CR-17 j.8 Answer: C

Bank Q843 • BT P2.T6.325.3

Step 1 – Sort the sources

Thresholds, MTAs and the MPoR let exposure move before margin arrives: market risk. Missed calls, failed deliveries and errors: operational risk. Price impact of a large sale: liquidity (liquidation) risk. Meeting calls: funding liquidity risk.

Step 2 – Match

Only C pairs a source with its own risk type.

Interpretation

This is the map's "role misassignment" shape in its category form: each distractor moves a true source to a neighbouring risk type.

Why the other options are wrong

A is incorrect. Process failures are operational, not market, risk.

B is incorrect. Liquidation costs are liquidity risk.

D is incorrect. Inability to fund calls is funding liquidity risk.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5.2 to 17.5.4 and 17.6.2

Solution CR-17 j.9 Answer: B

New • built from GARP Ch.17, 17.5.2, 17.5.3 and 17.6.2

Step 1 – The giver's risk

The corporate must post eligible collateral within short timescales whenever the swap moves against it. GARP: end users face funding liquidity risk when posting margin, and may default while solvent if they cannot meet a call in eligible securities.

⇒ B.

Step 2 – The receiver's risks

The bank, holding the bonds, would have to sell them after a default (liquidation risk) and carries the market risk over the MPoR, since its exposure can rise before close-out.

Interpretation

Each risk has a home: liquidation and MPoR market risk with the receiver, funding liquidity risk with the giver, operational risk with both.

Why the other options are wrong

A is incorrect. Role misassignment: liquidation risk sits with the holder of the margin, the bank, which must sell it to rehedge.

C is incorrect. Role misassignment: the MPoR is the receiver's uncollateralised period; the bank is not protected during it.

D is incorrect. Absolute claim: missed calls and failed deliveries hit the giver as well; operational risk falls on both parties.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5.2, 17.5.3, 17.5.4 and 17.6.2

CR-19 • Future Value and Exposure • Vardeez: VERY IMPORTANT

CR-19 h

P1 CONCEPT

Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: the margin period of risk came up in the August 2026 sitting (A21 recall: "two-way CSA; margin period of risk"), and A21 shape S9 ties the MPoR to collateralised exposure and CVA.
Vardeez: the MPoR and the exposure that builds over it ($\text{EE} = 0.4\,\sigma_E\sqrt{T_M}$, $\text{PFE} = k\,\sigma_E\sqrt{T_M}$), and the direction table: a lower MPoR, threshold and minimum transfer amount are better, a higher initial margin is better.

Reported exam shapes

- **A21·S9** Margin period of risk: its components and its effect on collateralised exposure and CVA, as seen in August.

What the questions look like: mostly "which is correct" statements on the direction of each CSA term, the steps inside the MPoR and why collateral never removes exposure completely. Two GARP-tagged practice questions are numerical (Questions CR-19 h.8 and h.9: EE and PFE over the MPoR); they are kept outside the CONCEPT rule and flagged here. Portfolio values, notionals and confidence levels often appear and are not needed.

Worked elsewhere: A21.3 (collateralised CVA with a 20-day MPoR by the square-root-of-time rule) and A21.5 (MPoR components; MPoR versus call frequency) at Questions CR-20 j.6 and CR-20 j.4; A21.1 and A21.2 (margin calls with thresholds and the MTA test) at Questions CR-17 h.7 and CR-17 c.8. The exposure metrics themselves are defined in full under CR-19 a.

1. Exposure once margin is held

Credit exposure is the amount lost if the counterparty defaults: the positive part of the netted value (CR-19 a). Margin is simply subtracted from the value (GARP eqs. 19.5 and 19.6):

$$\text{Positive exposure} = \max(\text{value} - \text{margin},\ 0) \qquad \text{Negative exposure} = \min(\text{value} - \text{margin},\ 0)$$

Margin *received* is positive and reduces positive exposure. Margin *posted* is negative, so posted margin that is not returned when the counterparty defaults *creates* positive exposure. Expected exposure (EE, the average positive exposure at a date) and potential future exposure (PFE, a high quantile of it) are computed on this collateralised value.

GARP Table 19.3 (two-way margin, no initial margin, five equally likely scenarios):

	S1	S2	S3	S4	S5	EPE	ENE
Value, no margin	25	15	5	−5	−15	9.0	−4.0
Margin	20	12	3	−3	−16		
Value with margin	5	3	2	−2	1	2.2	−0.4

In S1 to S3 margin held cuts the exposure sharply but not to zero (a rapid rise in value, thresholds and MTAs). In S4 the bank posts 3 against a value of −5 without creating exposure. In S5 it has posted 16 against −15: the extra 1 (margin not yet returned) is a positive exposure. EPE falls from 9 to 2.2 and ENE from −4 to −0.4: two-way margin reduces both.

2. Why collateral never removes exposure

GARP (19.3.3, Figure 19.22) names three imperfections, plus a fourth feature that makes them hard to model:

- **Granularity.** Thresholds, minimum transfer amounts and rounding mean the full amount cannot always be called. It cuts both ways: an MTA can also leave the holder briefly *over*collateralised when a return is too small to be made.
- **Delay.** Requesting, disputing, settling and finally liquidating margin takes time: the **margin period of risk** (Section 3).
- **Variability of the margin.** Non-cash margin (or cash in another currency) can lose value just when it is needed.
- **Path dependency.** The amount called today depends on what was called (or posted) in the past, which matters most under two-way agreements and is driven mainly by the MTA (CR-17 c).

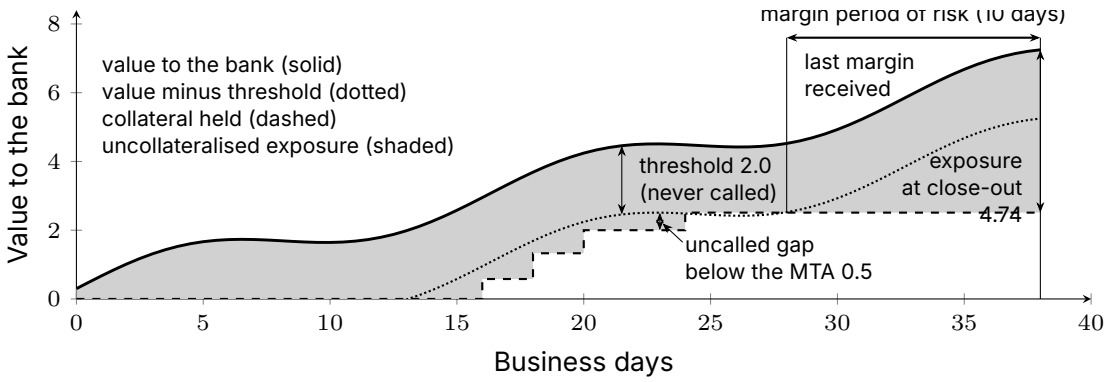


Figure 1. One stylised path under a CSA with a threshold of 2.0 and an MTA of 0.5. Below the threshold nothing is called (days 0 to 13). Above it, calls are made only once the shortfall reaches the MTA, so collateral moves in steps and lags the dotted level; on days 24 to 27 a return would be below the MTA, so the bank briefly holds slightly more than the CSA requires. After the last margin on day 28 nothing more arrives, and the exposure at close-out on day 38 is 4.74: the threshold plus the move over the MPoR.

Each term is summarised in the table in Section 5. Two points matter most: the MTA only *gates* a transfer (below it nothing moves, above it the full amount moves), and initial margin is a *negative threshold*, the only term where more is better. More frequent calls and lower thresholds and MTAs reduce exposure "at the expense of increasing operational and liquidity costs" (GARP 17.5.2).

3. The margin period of risk (shape S9)

GARP (Ch.17, 17.5.2) defines the MPoR as "the effective time between a counterparty ceasing to post margin and the time when all the underlying transactions have been successfully closed out and replaced (or otherwise hedged)". Any increase in exposure over it, including close-out costs, is uncollateralised. It is **counterparty-risk specific** (it assumes default) and is not used for funding (CR-19 g).

Pre-default	**Valuation and margin call** (compute the value and the margin held, decide whether a valid call can be made, including the contractual period between calls); **receiving collateral** (from the request to the release of cash or securities, including any dispute); **settlement** (cash intraday, government bonds about one day, corporate bonds about three days); **grace (cure) period** before the counterparty is deemed in default.
Post-default	**Macro-hedging** of key sensitivities; **close-out** of the transactions into one value; **rehedging and replacement** (one-to-one or by portfolio); **liquidation** of margin securities.

- **Not the call frequency.** Margin can be called daily while the MPoR is 10 days: the call frequency is only one component.
- **Values.** Basel II: at least 10 business days for OTC derivatives with daily calls (add days if calls are less frequent). Basel III: 20 days or more in certain cases (netting sets of over 5,000 trades, illiquid margin or exotic trades, a history of margin disputes). CCPs: about 5 days. The ISDA Resolution Stay Protocol may lengthen it.
- **A catch-all parameter.** Models hold 100% of the risk for the whole period, while in reality risk falls during close-out; but post-default volatility is likely to be higher. GARP: 20% more volatility is equivalent to an MPoR about 44% longer ($1.2^2 - 1$). The MPoR is the primary driver of the need for initial margin.

Correction to the source notes. Vardeez (and SchweserNotes) define the MPoR as "the time from a collateral call to the actual delivery of collateral". The 2026 GARP text is wider: from the counterparty *ceasing to post* margin to the close-out and *replacement or rehedging* of the trades, with the post-default steps (macro-hedging, close-out, rehedging, liquidation of margin) inside it. Schweser's own step list ends with "liquidation/close-out and re-hedge", which only fits GARP's definition. Use GARP's.

Schweser Module Quiz 37.3 Q1 keys "posting collateral" as the step that is *not* in the MPoR. GARP's "receiving collateral" step is exactly the delay until the counterparty releases (posts) the collateral, so that key does not hold. The bank rebuilt the question with an unambiguous exception (Question CR-19 h.7).

4. How much exposure the MPoR leaves

For a strongly collateralised position (zero threshold, no initial margin) the exposure at default is the change in value over the MPoR. If that change is normal with mean zero and annual volatility σ_E, over T_M years (Vardeez; Gregory's earlier editions):

$$\text{EE} = \frac{\sigma_E \sqrt{T_M}}{\sqrt{2\pi}} \approx 0.4\,\sigma_E \sqrt{T_M} \qquad \text{PFE}_\alpha = z_\alpha\,\sigma_E \sqrt{T_M}$$

The 0.4 is $1/\sqrt{2\pi} = 0.3989$, the expected positive part of a standard normal; it is GARP's normal formula $\text{EPE} = \mu\Phi(\mu/\sigma) + \sigma\phi(\mu/\sigma)$ (19.1.5) at $\mu = 0$. EE and PFE are the same distribution read at two points.

Example. $\sigma_E = 12\%$, MPoR of 20 business days, 250 days a year: $\sqrt{20/250} = 0.2828$, so the MPoR volatility is 3.394%. EE $= 0.3989 \times 3.394\% = 1.35\%$ and the 99% PFE $= 2.326 \times 3.394\% = 7.89\%$.

- **Square root, not linear.** Halving the MPoR to 10 days cuts both by a factor of $\sqrt{2}$, to 0.96% and 5.58%, not by half.
- **A threshold adds on top.** With a threshold H the exposure can reach about H plus the MPoR move (Figure 1; Question CR-19 h.9).
- **Initial margin takes off the bottom.** With initial margin I the exposure is $\max(\Delta V - I, 0)$: in the example, 3% of initial margin cuts EE from 1.35% to 0.35%.
- **Limitations** (Vardeez): the formula assumes strong collateralisation, ignores the volatility of non-cash collateral, and ignores liquidity and liquidation risk.

5. Three degrees of collateralisation, and the effect on CVA

GARP's Figure 19.23 compares the EPE of a swap under three broad cases:

- **Partially collateralised:** a threshold roughly caps the exposure.
- **Strongly collateralised:** zero threshold, no initial margin; exposure falls a long way, but the MPoR leaves a "reasonably material" residual.
- **Overcollateralised:** initial margin reduces it further, potentially to zero.

Effect on CVA (worked under CR-20 j). Margin changes only the EPE in the CVA formula; the default probability and LGD are untouched. With variation margin only, the best case is a reduction of about $\sqrt{\text{maturity}/\text{MPoR}}$: $\sqrt{5 \times 250/10} \approx 11$ times for a five-year portfolio (GARP 17.5.2). "In reality, the result is worse than this estimate", so only initial margin makes counterparty risk negligible. High thresholds take CVA back to the uncollateralised value, and so does a longer MPoR; high initial margin takes it towards zero.

	What it does	A larger value	Cost of a smaller value
Threshold	Exposure below it is never called; roughly caps the collateral benefit	More exposure	More calls: operational and liquidity cost
Minimum transfer amount	Gates each transfer; not deducted once passed; makes margin path dependent	More exposure	More, smaller transfers
Rounding	Rounds each call to an increment	Small effect either way	Small
Initial margin	Collateral posted regardless of value; a negative threshold	**Less** exposure, possibly zero	Funding cost for the poster (MVA)
MPoR	Time with no margin received, to close-out and replacement	More exposure, $\propto \sqrt{\text{MPoR}}$	Set by operations, law and Basel floors
Call frequency	One component of the MPoR	Less frequent: more exposure	Operational cost

Exam traps

- **MPoR read as the call frequency (sibling swap).** Daily calls do not mean a one-day MPoR; Basel's floor with daily calls is 10 business days.
- **MPoR cut short.** Defining it as "call to delivery" leaves out the grace period and the post-

default close-out, rehedging and liquidation.

- **Initial margin and threshold swapped (polarity flip).** Initial margin is the only term where more is better; a threshold is a negative initial margin.
- **MTA deducted or confused with the threshold.** The MTA only decides whether a transfer happens; once passed the full amount moves.
- **Linear time scaling.** Exposure over the MPoR grows with $\sqrt{T_M}$; doubling the MPoR raises it by about 41%, not 100%. Use the day count in the stem (250 or 252), not 365.
- **EE and PFE swapped.** $0.4\,\sigma\sqrt{T_M}$ is the EE; $z_\alpha\sigma\sqrt{T_M}$ is the PFE. A confidence level in an EE question is a red herring.
- **Margin only reduces exposure.** Posted margin not returned at default is itself an exposure (Table 19.3, scenario 5).
- **Collateral lowers the default probability.** It changes the EPE in CVA, never the PD or LGD.
- **Zero threshold means zero exposure.** The MPoR still leaves a material residual; only initial margin pushes towards zero.

Practice questions: CR-19 h

Question CR-19 h.1

Bank Q977 • GARP PE1 Q78 • Mock 3 Q3

A financial institution has a two-way credit support annex (CSA) with a counterparty covering a portfolio valued at JPY 400 million. The margining terms of the collateralized portfolio include a threshold of JPY 180 million, a minimum transfer amount of JPY 30 million, and a margin period of risk of 10 days. Which of the following is correct?

A. A lower threshold implies that a larger portion of the exposure is protected by collateral.

B. A shorter margin period of risk implies that a smaller portion of the exposure is protected by collateral.

C. A lower independent amount (initial margin) implies that a larger portion of the exposure is protected by collateral.

D. The protection from collateral specified in the CSA is uniform throughout the life of the exposure profile.

Question CR-19 h.2

Bank Q981 • BT P2.T6.330.2 (rebuilt)

A counterparty risk analyst is reviewing the CSA terms on a collateralized netting set whose mark-to-market value to the bank is positive and increasing. Which of the following changes to the terms would NOT tend to increase the risk in the bank's expected exposure (EE) profile?

A. A longer remargin period

B. A higher threshold

C. A larger independent amount (initial margin) posted by the counterparty

D. A larger minimum transfer amount

Question CR-19 h.3

Bank Q936 • New question written for this bank • re-filed from CR-19 c

A counterparty risk analyst at a bank reviews a netting set with a hedge fund. The credit support annex provides for cash collateral only, with no threshold, a minimum transfer amount of USD 1.0 million, and daily margin calls. Since the last call, the mark-to-market exposure to the hedge fund has risen by USD 0.6 million, and the hedge fund has met every previous call in full and on time. Which of the following best explains why the bank is still not fully collateralized at the end of the day?

A. Variation in collateral value: the cash collateral held by the bank has lost value since it was posted.

B. Path dependency: the call amount is reduced because collateral was received late in earlier periods.

C. The granularity effect: the minimum transfer amount prevents the bank from calling for all of the collateral it is actually owed.

D. The margin period of risk: the hedge fund has a grace period in which it may delay delivery of an already-valid call.

Question CR-19 h.4

Bank Q980 • BT P2.T6.330.1 (rebuilt)

A collateral analyst reviews one simulated path for a collateralized interest rate swap from Counterparty A's point of view. MtM is the value to A (positive means B owes A). Collateral is positive if held by A and negative if posted by A. Exposure is $\max(\text{MtM} - \text{Collateral}, 0)$. All figures are percent of notional.

Year	MtM	Collateral	Exposure
1.0	1.10	0.00	1.10
1.5	1.60	0.60	1.00
2.0	2.00	1.00	1.00
3.0	0.50	0.00	0.50
4.0	−1.80	−0.80	0.00

Each party has a threshold and there is a single minimum transfer amount. Each of the following is consistent with the table EXCEPT:

A. Counterparty A's threshold is 1.0%.

B. The minimum transfer amount is zero, and collateral is delivered with no delay.

C. Counterparty B's threshold is 1.0%.

D. Collateral is delivered with no delay in this path.

Question CR-19 h.5 New • built from GARP Ch.19, 19.3.3, eq. 19.5, Table 19.3 and Figure 19.23

A counterparty risk analyst at a Dutch bank is explaining to the credit committee how a new two-way CSA with zero thresholds, daily variation margin and no initial margin will change the bank's exposure to a hedge fund. Which of the following statements is correct?

A. Because margin received is subtracted from the portfolio value, the CSA can only reduce the bank's positive exposure; margin the bank posts can never create exposure to the hedge fund.

B. Exposure will remain, because margin arrives with a delay over the margin period of risk, and margin the bank has posted but not had returned when the hedge fund defaults is itself a positive exposure.

C. A threshold acts like initial margin, so agreeing a threshold of EUR 10 million instead of zero would move the profile from strongly collateralised towards overcollateralised.

D. With zero thresholds and daily calls the exposure profile is close to zero at every horizon, so adding initial margin would bring no further reduction.

Question CR-19 h.6 Bank Q975 • BT P2.T6.329.3

A model validator is reviewing the documentation of a bank's model for the credit exposure of collateralized positions. Each of the following statements in the documentation is true EXCEPT:

A. If a position is strongly collateralized, the primary concerns are the length of the remargin period and the volatilities of the exposure and the collateral.

B. For risk assessment, the remargin period will equal the legal margin call frequency, that is, typically one day.

C. Imperfect collateralization at a given date can arise from the terms of the collateral agreement (threshold, minimum transfer amount, rounding), which prevent calling the full amount.

D. Exposure may increase between margin calls, and that portion cannot be collateralized.

Question CR-19 h.7 Bank Q983 • Schweser Module Quiz 37.3 Q1 (rebuilt, key corrected)

The margin period of risk (MPoR) is the time from a collateral call to the point at which the position has been closed out and re-hedged following the default of the party that must post collateral. A risk manager is estimating the number of days in the MPoR for a bilateral portfolio. Which of the following is NOT one of the time steps that enter into the calculation?

A. Valuation of the exposure and collateral, and the margin call

B. Grace period allowed to the counterparty before it is deemed to have failed to pay

C. Settlement of the collateral that is delivered

D. Annual review and renegotiation of the collateral threshold in the credit support annex

Question CR-19 h.8 Bank Q982 • Mock 1 Q13 (rebuilt, key corrected) • GARP-tagged numerical

A risk analyst models the change in value of a collateralized position over a 10-day remargining period as a zero-mean normal distribution. The annual volatility of the position is 10%, there are 252 trading days per year, and the bank reports potential future exposure at the 99% confidence level. Which of the following is nearest to the expected exposure (EE) over the remargining period, as a percentage of the position?

A. 0.66%

B. 0.80%

C. 1.99%

D. 4.63%

Question CR-19 h.9 Bank Q970 • AP Mock 2 Q43 (rebuilt) • GARP-tagged numerical

A coffee trader has purchased 10,000 bags of coffee for delivery in one year at USD 500 per bag, so the contract is worth zero today. The annual volatility of the coffee price return is 25% (normally distributed, mean zero). A margin call is made whenever the credit exposure exceeds a threshold of USD 500,000, and collateral is received within a two-day remargining period. Assume 252 trading days per year and that the margin agreement is fully enforced. Which of the following is closest to the 95% maximum credit exposure over the one-year life of the contract?

A. USD 183,185

B. USD 516,319

C. USD 683,185

D. USD 2,056,250

Question CR-19 h.10 New • built from GARP Ch.17, 17.5.2 and Ch.20, 20.5.1 • shape A21·S9

The xVA desk of a Japanese bank is moving a five-year interest rate swap portfolio with a corporate counterparty from no CSA to a two-way CSA with a zero threshold, daily variation margin and a 10-business-day margin period of risk, with no initial margin. There are 250 business days per year. Which of the following statements best describes the expected effect?

A. CVA falls because the corporate's default probability is lower once it posts margin; the EPE profile is unchanged.

B. The best-case fall in exposure is about the ratio of maturity to MPoR, roughly 125 times, because margin covers all but 10 of the 1,250 business days.

C. The best-case fall in exposure is about the square root of maturity over MPoR, roughly 11 times; in practice it is smaller, and a longer MPoR (for example 20 days for a large or disputed netting set) moves CVA back towards the uncollateralised value.

D. With a zero threshold and daily calls, collateralised CVA is negligible, so the desk can set it to zero without modelling the MPoR.

Solutions • CR-19 h

Solution CR-19 h.1 — Answer: A

Bank Q977 • GARP PE1 Q78

Step 1 – What the threshold does

The threshold is the exposure below which no collateral is called, so it is uncollateralised by construction. Lowering it means collateral is called on a larger share of the exposure.
⇒ A is correct: a lower threshold protects more of the exposure.

Step 2 – Test the other terms

A shorter MPoR gives the value less time to move before collateral arrives, so more (not less) of the exposure is protected. A lower initial margin is a smaller cushion, so less is protected. Protection is not uniform: the threshold and MTA bind differently as the exposure changes, and margin is path dependent.
⇒ B and C reverse the direction; D ignores that collateral is path dependent.

Interpretation

No calculation is needed. The JPY 400 million value, the JPY 180 million threshold and the JPY 30 million MTA describe the CSA; they would give a required margin of $\max(400 - 180, 0) = 220$ if a call were asked, which it is not.

Why the other options are wrong

B is incorrect. Polarity flip: a shorter MPoR means collateral arrives sooner, so a *larger* portion is protected.

C is incorrect. Polarity flip: initial margin is extra collateral, so a lower amount protects *less* of the exposure.

D is incorrect. Absolute claim: protection varies over the profile because thresholds, MTAs and the MPoR bite differently as the exposure moves, and the amount called depends on past calls.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.3; Ch.7 (GARP CR Ch.17), 17.5.2

Solution CR-19 h.2 — Answer: C

Bank Q981 • BT P2.T6.330.2

Step 1 – Terms that create uncollateralised exposure

A longer remargin period lets the value rise for longer before collateral arrives. A higher threshold leaves a larger amount never called. A larger MTA leaves a larger shortfall uncalled.
⇒ A, B and D each increase the EE profile.

Step 2 – The odd one out

Initial margin posted by the counterparty is collateral held regardless of the value, a negative threshold. It offsets exposure, so a larger amount reduces EE.
⇒ C is the exception.

Interpretation

This is Vardeez's direction table in one question: three frictions that let exposure build, and initial margin, the only term where more is better.

Why the other options are wrong

A is incorrect. Increases risk: a longer MPoR means a larger uncollateralised move, growing with $\sqrt{T_M}$.

B is incorrect. Increases risk: the threshold is exposure that is never collateralised.
D is incorrect. Increases risk: shortfalls below the MTA are not called, so a larger MTA leaves a larger gap.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.3, Figure 19.23

Solution CR-19 h.3 — Answer: C

Bank Q936 • New question written for this bank

Step 1 – Rule out the other frictions

The collateral is cash, so its value cannot vary. The hedge fund has always paid in full and on time, so there is no delay and no history to make the calls path dependent.

Step 2 – Apply the MTA test

The shortfall of USD 0.6 million is below the USD 1.0 million MTA, so no valid call can be made today.
$\Rightarrow$ The uncollateralised USD 0.6 million is a granularity effect: the terms prevent the bank from calling for all the collateral it is owed.

Interpretation

The shortfall will be called once it reaches USD 1.0 million, and then in full. Until then the bank carries it unsecured, which is why a larger MTA raises exposure.

Why the other options are wrong

A is incorrect. Cash does not change in value; variation in collateral value applies to non-cash margin (or cash in another currency).
B is incorrect. There is no history of late or partial delivery, so path dependency has not reduced any call.
D is incorrect. The MPoR is the delay between a call and the close-out; here no call has been made at all.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.3, Figure 19.22

Solution CR-19 h.4 — Answer: B

Bank Q980 • BT P2.T6.330.1

Step 1 – Read B's threshold from the collateral A holds

At years 1.5 and 2.0 collateral held equals MtM minus 1.0 (0.60 and 1.00).
$\Rightarrow$ B's threshold is 1.0% (C is consistent), and collateral equals the required amount at those dates, so there is no visible delay (D is consistent).

Step 2 – Read A's threshold from the collateral A posts

At year 4.0 MtM is -1.80 and A has posted 0.80, the amount beyond 1.0.
$\Rightarrow$ A's threshold is 1.0% (A is consistent).

Step 3 – Explain year 1.0

MtM of 1.10 exceeds B's threshold by 0.10, yet no collateral is held. Collateral matches the requirement exactly at every other date, so the missing 0.10 is not a delay: it is a call below the minimum transfer amount.
$\Rightarrow$ The MTA must exceed 0.10 (and be at most 0.60, since a 0.60 call was made at year 1.5). B is

inconsistent.

Interpretation

The table shows both frictions in one path: the threshold leaves 1.00 permanently uncollateralised, and the MTA adds a small extra gap at year 1.0. At year 4.0 A's posted collateral covers part of its liability and creates no exposure.

Why the other options are wrong

A is incorrect. Consistent: A posts 0.80 when MtM is -1.80, the excess over a 1.0% threshold.
C is incorrect. Consistent: collateral held at years 1.5 and 2.0 equals MtM minus 1.0.
D is incorrect. Consistent: collateral equals MtM less the threshold at every date where a call was possible, so no lag is evident.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.3, eq. 19.5; Ch.7 (GARP CR Ch.17), 17.3

Solution CR-19 h.5 Answer: B

New • built from GARP Ch.19, 19.3.3

Step 1 – Margin enters exposure with its sign

Positive exposure $= \max(\text{value} - \text{margin}, 0)$ (eq. 19.5). Received margin is positive and reduces it; posted margin is negative, so posted margin that exceeds what the bank owes adds to it. In GARP's Table 19.3, scenario 5, the bank has posted 16 against a value of -15 and has a positive exposure of 1.
$\Rightarrow$ Two-way margin can create exposure as well as reduce it.

Step 2 – What is left with zero thresholds

GARP's three imperfections remain: granularity (MTA, rounding), the delay over the MPoR, and variation in the value of non-cash margin. A strongly collateralised profile is much lower than an uncollateralised one but still material; only initial margin reduces it further, potentially to zero.
$\Rightarrow$ B is correct.

Interpretation

The committee should expect a residual exposure driven mainly by the MPoR, and should know that its own posted margin is at risk if it is not returned. Both are reasons to model collateralised exposure rather than assume it away.

Why the other options are wrong

A is incorrect. Ignores posted margin: margin posted but not returned is an unsecured claim on the defaulter (Table 19.3, scenario 5).
C is incorrect. Polarity flip: a threshold is a *negative* initial margin; raising it moves the profile towards partially collateralised, not overcollateralised.
D is incorrect. Absolute claim: the MPoR leaves a material exposure under strong collateralisation, and initial margin reduces it further (the overcollateralised case).

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.3, eq. 19.5, Table 19.3, Figures 19.22 and 19.23

Solution CR-19 h.6 Answer: B

Bank Q975 • BT P2.T6.329.3

Step 1 – Strongly collateralised positions

With zero threshold and no initial margin, the residual risk comes from the move over the remargin period and from the volatility of the exposure and of the collateral.
⇒ A is true.

Step 2 – The remargin period

The MPoR runs from the last successful margin posting to close-out and replacement. It contains valuation and call, receipt (with disputes), settlement, a grace period and the post-default steps, so it is much longer than the contractual call frequency (Basel's floor with daily calls is 10 business days).
⇒ B is false, and is the answer.

Step 3 – The other statements

Thresholds, MTAs and rounding prevent calling the full amount (C true), and exposure that rises between calls cannot be collateralised (D true).

Interpretation

Equating the MPoR with a one-day call cycle would understate collateralised exposure by a factor of about $\sqrt{10} \approx 3.2$ against a 10-day MPoR.

Why the other options are wrong

A is incorrect. True: under strong collateralisation the MPoR and the volatilities are the main drivers.
C is incorrect. True: these are GARP's granularity effects.
D is incorrect. True: this is the delay effect.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.3; Ch.7 (GARP CR Ch.17), 17.5.2

Solution CR-19 h.7 Answer: D

Bank Q983 • Schweser Module Quiz 37.3 Q1

Step 1 – List the components of the MPoR

Pre-default: valuation and margin call; receiving collateral; settlement; grace period. Post-default: macro-hedging, close-out, rehedging and replacement, liquidation of margin securities.
⇒ A is the first step, C the third, B the fourth.

Step 2 – Find the item outside the chain

An annual review of the CSA threshold is a periodic contractual negotiation. It happens outside the window from the last margin posting to close-out, so it adds no days.
⇒ D is the exception.

Interpretation

Every component of the MPoR is a delay between the counterparty's last successful posting and the moment the surviving party is hedged again. A contractual review changes future terms; it does not delay today's margin. (Schweser's original key, "posting collateral", is not an exception: see the correction box in Section 3.)

Why the other options are wrong

A is incorrect. The first step: computing the value and the margin held and making a valid call.
B is incorrect. The grace (cure) period before a failure to pay becomes a default is a pre-default component.

C is incorrect. Settlement: cash may settle intraday, government bonds in about one day and corporate bonds in about three.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5.2; SchweserNotes 2026 Book 2, Reading 37, LO 37.h

Solution CR-19 h.8

Answer: B

Bank Q982 • Mock 1 Q13

Step 1 – Volatility over the remargining period

$$\sigma_M = 10\% \times \sqrt{10/252} = 10\% \times 0.19920 = 1.992\%$$

Step 2 – Expected positive part of a zero-mean normal

$$\text{EE} = \frac{\sigma_M}{\sqrt{2\pi}} = 1.992\% \times 0.3989 = 0.7947\%$$

$\Rightarrow$ Nearest option: 0.80%. Vardeez's $0.4\,\sigma\sqrt{T_M}$ gives $0.4 \times 1.992\% = 0.797\%$, which rounds to 0.80% directly.

Interpretation

EE is an average and needs no confidence level; the 99% figure in the stem is only needed for PFE. The answer is about 0.4 times the 10-day volatility.

Why the other options are wrong

A is incorrect. Used 365 days: $10\% \times \sqrt{10/365} \times 0.3989 = 0.66\%$.
C is incorrect. Reported the 10-day volatility itself, 1.99%, without the factor $1/\sqrt{2\pi}$.
D is incorrect. Computed the 99% PFE instead: $2.326 \times 1.992\% = 4.63\%$.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5 and 19.3.3; Vardeez CR-19 h (exposure over the MPoR)

Solution CR-19 h.9

Answer: C

Bank Q970 • AP Mock 2 Q43

Step 1 – Contract size

$$N = 10{,}000 \times 500 = \text{USD } 5{,}000{,}000$$

Step 2 – Volatility over the two-day remargining period

$$\sigma_{2d} = 0.25 \times \sqrt{2/252} = 0.022272$$

Step 3 – 95% move over the remargining period

$$1.645 \times 0.022272 \times 5{,}000{,}000 = \text{USD } 183{,}185$$

Step 4 – Add the threshold

$$500{,}000 + 183{,}185 = \text{USD } 683{,}185$$

⇒ With a fully enforced agreement the exposure can exceed the threshold only by the value change during the remargining period.

Interpretation

The one-year life and annual volatility matter only through the two-day window: the threshold is a permanent uncollateralised amount and the remargining period adds the move on top (Figure 1).

Why the other options are wrong

A is incorrect. Omitted the threshold and reported only the two-day move, 183,185.
B is incorrect. Scaled volatility by $2/252$ instead of $\sqrt{2/252}$: $500{,}000 + 1.645 \times 0.25 \times (2/252) \times 5{,}000{,}000 = 516{,}319$.
D is incorrect. Ignored the margin agreement and used the one-year 95% move: $1.645 \times 0.25 \times 5{,}000{,}000 = 2{,}056{,}250$.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.3; Vardeez CR-19 h (PFE over the MPoR)

Solution CR-19 h.10 Answer: C

New • built from GARP Ch.17, 17.5.2 and Ch.20, 20.5.1

Step 1 – Where margin enters CVA

$\text{CVA} = -\text{LGD} \sum \text{EPE}(t_i)\,\text{PD}(t_{i-1}, t_i)$. Margin "only changes EPE", so the same formula is used with a collateralised EPE; the PD and LGD do not change.
⇒ A is wrong on the mechanism.

Step 2 – Size of the best-case reduction

With variation margin only, exposure is limited to the move over the MPoR instead of up to five years, and moves scale with the square root of time:

$$\sqrt{\frac{5 \times 250}{10}} = \sqrt{125} = 11.2$$

⇒ About 11 times at best (GARP 17.5.2); "in reality, the result is worse than this estimate".

Step 3 – Sensitivity to the MPoR

A 20-day MPoR gives only $\sqrt{62.5} = 7.9$ times. As the MPoR lengthens, collateral arrives too late to matter and CVA moves back towards the uncollateralised value; only initial margin takes it towards zero.
⇒ C is correct.

Interpretation

Ten days has become standard even for accounting CVA, and market participants now model variation margin even with zero thresholds. A collateralised CVA of roughly one tenth (or more) of the uncollateralised figure is the realistic expectation (A21.3, worked as Question CR-20 j.6).

Why the other options are wrong

A is incorrect. Role misassignment: collateral works on exposure, never on the counterparty's default probability.
B is incorrect. Linear scaling: $1{,}250/10 = 125$ treats exposure as proportional to time instead of its square root.
D is incorrect. Absolute claim: the MPoR leaves a material residual even with zero thresholds and daily calls, which is why it is modelled.

LO: CR-19 h: Explain the impact of collateralization on exposure and assess the risk associated with the remargining period, threshold, and minimum transfer amount. CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.5.2; Ch.17 (GARP CR Ch.20), 20.5.1 and 20.5.3

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 c

P1 CALC

Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization.

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A6, Map #6, on 8 of 16 recall sheets). The August 2026 recall reported a BCVA spread numerical, and the consolidated sheet lists "UCVA as a spread: calculation (formula)" twice, next to "2 numericals on BCVA, CVA".

Vardeez: CR-20 cannot be skipped: CVA, BCVA and DVA are a GARP favourite, so know the formulas very well. For this LO that means UCVA $= -\text{LGD}\sum \text{EPE}(t_i)\,\text{PD}(t_{i-1}, t_i)$ and the running spread $-X^{\text{CDS}}\times$ average EPE.

Reported exam shapes

- **A6·S1** UCVA from an EPE profile and a PD term structure: LGD $\times$ EPE $\times$ PD, summed over the intervals.
- **A6·S2** CVA as a running spread: $-$average EPE $\times$ spread, and upfront CVA divided by the risky annuity.

What the questions look like: numerical. You get a discounted EPE profile with cumulative or marginal PDs (or a hazard rate or CDS quotes to turn into PDs), or a single average EPE and a credit spread. Typical red herrings: the institution's own spread, PFE or EE at a single date, an LGD next to the spread formula, the notional when a percentage is asked.

Sign convention: Gregory (GARP Ch.20) reports CVA as a negative number, a reduction in value. Questions that quote CVA as a positive charge say so in the stem.

Worked elsewhere: A6.3 (UCVA and BCVA as running spreads from exposure profiles) at Question CR-20 g.4; A6.4 (incremental CVA turned into a running spread) under CR-20 i.

1. The CVA formula and its three inputs

CR-20 b defines CVA as the difference between the value of a portfolio with and without counterparty default. Gregory builds it from three questions: does the counterparty default (PD), what is the exposure if it does (EPE), and how much of that exposure is lost (LGD $= 1 - R$). Assume the three are independent, which means no wrong-way risk (CR-20 k), so a single average LGD is enough. The time integral of GARP eq. 20.2 is then approximated by a sum over m intervals (eq. 20.3):

$$\text{UCVA} = -\text{LGD} \sum_{i=1}^{m} \text{EPE}(t_i) \times \text{PD}(t_{i-1}, t_i)$$

"Unilateral" means the institution doing the calculation assumes it cannot default itself; CR-20 e to g drop that assumption.

Input	What it is	How the exam disguises it
LGD	$1 - R$: the share of the exposure lost; one average value	A recovery rate is given: use $1 - R$, not R
EPE(t_i)	Discounted expected positive exposure at t_i; risk-free discounting is done inside the EPE	Undiscounted EE given: multiply by a discount factor $DF(t_i)$ as well
PD(t_{i-1}, t_i)	Marginal (unconditional) PD for the interval, as seen today	Cumulative PDs given: take differences. Hazard rate given: $\text{e}^{-\lambda t_{i-1}} - \text{e}^{-\lambda t_i}$ (CR-9 e)

Each term is an EPE multiplied by the probability of defaulting in that interval, so CVA is LGD times a PD-weighted average of the EPE profile: market risk (EPE) times credit risk (PD). Default enters only through the PD, so no default events need to be simulated. BT and Schweser write the same formula with the discount factor pulled out, CVA $= -\text{LGD} \sum DF(t_i)\,\text{EE}(t_i)\,\text{PD}(t_{i-1}, t_i)$.

One-period special case. With one exposure figure and one horizon, the sum collapses to CVA $\approx -\text{LGD} \times \text{PD}(0, T) \times \text{EPE}$, where PD$(0, T)$ is the *cumulative* PD over the horizon. For a bought European option it is exact: the exposure can only be positive and its discounted expectation at every date is today's option value V (GARP 20.2.4 makes the same point for deep in-the-money trades, where EPE is close to the expected future value). Question CR-20 c.2 uses this.

2. Worked example: from a CDS spread to UCVA (shape S1)

A 4-year swap has discounted EPE of 1.5%, 2.4%, 2.2% and 1.2% of notional at the ends of years 1 to 4. The counterparty's CDS spread is 150 bp with 40% recovery; interest rates are close to zero.

1. Hazard rate (CR-9 h): $\lambda = s/(1 - R) = 0.0150/0.60 = 2.5\%$.
2. Survival $V(t) = \text{e}^{-0.025t}$: 0.97531, 0.95123, 0.92774, 0.90484. Marginal PDs $V(t_{i-1}) - V(t_i)$: 2.469%, 2.408%, 2.349%, 2.291%.
3. EPE (%) $\times$ PD (%) gives basis points of notional: 3.704, 5.779, 5.167, 2.749; total 17.40 bp.
4. UCVA $= -0.60 \times 17.40 = -10.44$ bp of notional, paid upfront. On USD 100 million that is $-$USD 104,400.

Using the cumulative PDs (2.469%, 4.877%, 7.226%, 9.516%) instead would give -25.63 bp, more than double: the intermediate-result trap of Question CR-20 c.3.

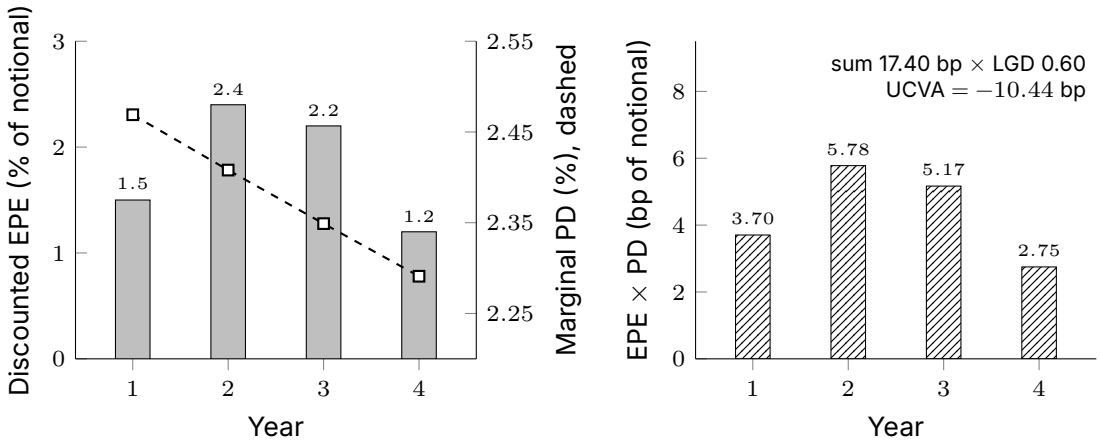


Figure 1. The worked example. Left: the EPE profile (grey bars) and the counterparty's marginal PD (dashed line, right-hand scale, which starts at 2.2% to make the slope visible). Right: the product for each year (hatched). The PD falls slowly because fewer firms survive to default later; the CVA is concentrated in years 2 and 3, where the exposure peaks.

3. CVA as a running spread (shape S2)

For pricing, an upfront CVA is often charged as a per-annum spread, for example added to the fixed rate of a swap. There are two routes.

Route 1: divide by the risky annuity

$$\text{CVA}_{\text{running}} = \frac{\text{CVA}_{\text{upfront}}}{\text{notional} \times \text{risky annuity}}$$

The risky annuity (risky duration) is the value of 1 per year paid until maturity or default; with rates near zero it is $\sum V(t_i)\,\Delta t_i$. In the example: $0.97531 + 0.95123 + 0.92774 + 0.90484 = 3.759$, so the running CVA is $-10.44/3.759 = -2.78$ bp a year. When CVA is given as a percentage of notional, the notional cancels: 0.90% upfront over a risky duration of 3.5 is 25.7 bp a year (Question CR-20 c.7). Dividing ignores the "CVA of the CVA"; the recursive correction is covered at CR-20 i.

Route 2: the average-EPE approximation

Assume the EPE is constant over time (GARP eq. 20.4):

$$\boxed{\text{UCVA} \approx -\text{average EPE} \times \text{spread}}$$

In the example: average EPE $= (1.5 + 2.4 + 2.2 + 1.2)/4 = 1.825\%$, so UCVA $\approx -0.01825 \times 150 = -2.74$ bp a year, close to the -2.78 of Route 1.

- **Units.** The answer is in the units of the spread: bp per year when EPE is a fraction of notional. Use the spread for the maturity of the instrument.
- **No LGD input.** The spread already contains the LGD ($s \approx \lambda \times \text{LGD}$), so when the LGD used for the loss equals the LGD used to imply the PD, the two cancel to first order. Check in the example: LGD $\times$ PD for year 1 is $0.60 \times 2.469\% = 1.481\%$, against spread $\times$ survival to the start of the year $= 1.500\%$. Multiplying $-$EPE $\times\ s$ by an LGD counts it twice.
- **Which EPE.** The simple average for equal intervals (a time-weighted average otherwise), strictly the undiscounted EPE; with low rates and short maturities the difference is small. Not the EE at maturity, not the PFE, not the peak PFE (Question CR-20 c.5).

- **When it works.** Best when the EPE (or the default probability) profile is roughly flat. GARP keeps it for intuition rather than actual calculation: it separates the credit component (the spread) from the market risk component (the average EPE).
- **Examples.** Schweser: EPE 6%, spread 375 bp gives -22.5 bp a year. A spread built from yields is the counterparty yield minus the risk-free rate, $5.8\% - 2.6\% = 3.2\%$, not the yield itself (Question CR-20 c.6).

Running CVA for GARP's 10-year swap (Table 20.1)	bp per year
Divide the upfront CVA by the risky annuity	-1.92
Exact (recursive: includes the CVA of the CVA)	-1.96
Average-EPE approximation	-2.01

The three routes agree to within 0.1 bp for this small charge; the recursive effect grows for risky counterparties and long-dated trades.

Formula card: CR-20 c

UCVA, discrete (eq. 20.3)	$\text{UCVA} = -\text{LGD} \sum_i \text{EPE}(t_i)\, \text{PD}(t_{i-1}, t_i)$
With explicit discounting	$-\text{LGD} \sum_i DF(t_i)\, \text{EE}(t_i)\, \text{PD}(t_{i-1}, t_i)$
Marginal PD	$Q(t_i) - Q(t_{i-1})$; constant hazard: $\mathrm{e}^{-\lambda t_{i-1}} - \mathrm{e}^{-\lambda t_i}$
Hazard from a spread (CR-9 h)	$\lambda = s/(1-R)$
One period	$\text{CVA} \approx -\text{LGD} \times \text{PD}(0,T) \times \text{EPE}$ (bought option: EPE $= V$)
Running spread (eq. 20.4)	$\text{UCVA} \approx -$average EPE $\times$ spread (no LGD; units of the spread)
Upfront to running	$\text{CVA}_{\text{upfront}}/(\text{notional} \times \text{risky annuity})$
Risky annuity, zero rates	$\sum_i V(t_i)\, \Delta t_i, \quad V(t) = \mathrm{e}^{-\lambda t}$
Units	EPE (%) $\times$ PD (%) $=$ bp of notional; $\quad 1\% = 100$ bp

Exam traps

- **Cumulative PD instead of marginal (intermediate result).** Each EPE is weighted by the PD for its own interval. Cumulative PDs more than double the CVA in the worked example.
- **Conditional PD.** The formula's PD is unconditional, as seen today; dividing by survival overstates CVA.
- **Recovery for LGD, or no LGD** in the profile formula. Multiply by $1 - R$.
- **LGD in the spread formula.** $-\text{EPE} \times s$ already contains the LGD; multiplying again double counts it.
- **Wrong exposure measure.** Average EPE, not EE at one date, PFE or maximum PFE.
- **Hazard rate used as a multi-year PD.** Over T years the cumulative PD is $1 - \mathrm{e}^{-\lambda T}$, not λ.
- **Yield for spread.** Subtract the risk-free rate first.
- **Annuity the wrong way round.** Divide the upfront CVA by the risky annuity; multiplying, or quoting the upfront figure as a running spread, is wrong.
- **Units.** 29.25 bp is 0.2925%, not 29.25%.

- **Own spread.** The institution's own spread is a red herring for UCVA; it enters DVA (CR-20 e).

Practice questions: CR-20 c

Question CR-20 c.1

Bank Q1002 • BT P2.T6.331.1 (rebuilt)

A bank computes the simplified unilateral CVA, with no wrong-way risk, as

$$\text{CVA} = -\underbrace{\text{LGD}}_{(1)} \sum_{j=1}^{m} \underbrace{DF(t_j)}_{(2)} \underbrace{\text{EE}(t_j)}_{(3)} \underbrace{\text{PD}(t_{j-1}, t_j)}_{(4)}$$

so that CVA is a negative number (a cost). Each of the following statements about the four components is accurate EXCEPT:

A. The first component is the loss given default, one minus the expected recovery rate.

B. The second component is the risk-free discount factor for date t_j.

C. The third component is the expected exposure at the relevant future dates.

D. The fourth component is the conditional default probability: the probability of default in the interval given survival to t_{j-1}.

Question CR-20 c.2

Bank Q999 • BT Mock Q27 • BT P2.T6.915.1 (rebuilt)

Strongcore Financial has bought a 3-year European at-the-money call option from a risky counterparty. The data are $S = K =$ USD 25.0 million, volatility 20.0%, risk-free rate 3.0% (continuously compounded), $T = 3.0$ years, $N(d_1) = 0.66750$ and $N(d_2) = 0.53450$. The counterparty has a constant hazard rate of 6.0% and may default at any time during the three years; recovery on default is 30.0%. Using the single-period approximation $\text{CVA} \approx \text{LGD} \times \text{PD}(0,T) \times V$, where V is the current value of the option (treated as the discounted expected exposure) and $\text{PD}(0,T)$ is the cumulative default probability, which of the following is closest to the CVA charge, quoted as a positive amount?

A. USD 188,000

B. USD 221,000

C. USD 516,000

D. USD 737,000

Question CR-20 c.3

Workbook A6.1 • S1 • Map #6

An xVA analyst at a US dealer is computing the unilateral CVA on an uncollateralised 5-year interest rate swap with a corporate counterparty, notional USD 200 million. The counterparty's recovery rate is 40%; the dealer's own credit spread is 90 bps. The simulated profile is:

Year	1	2	3	4	5
Discounted EPE (% of notional)	2.0	3.2	3.6	3.0	1.6
Counterparty cumulative PD (%)	1.5	3.2	5.0	6.9	8.9

Assuming no wrong-way risk, which of the following is closest to the UCVA (Gregory's convention: CVA is negative)?

A. −USD 0.794 million

B. −USD 0.476 million

C. −USD 0.286 million

D. −USD 0.191 million

Question CR-20 c.4

Bank Q1001 • BT P2.T6.914.3 (rebuilt)

The CRO of Global Financial wants to price the counterparty risk of a 3-year derivative exposure to a major counterparty. There is no netting, collateral or wrong-way risk. The counterparty's default follows a constant hazard rate process, with the hazard rate implied by each year's data as $\lambda = s/(1-R)$. Losses in each year use that year's recovery rate. Forecast data:

	Year 1	Year 2	Year 3
Expected exposure (USD million)	32.0	32.0	32.0
CDS spread (bps)	100	200	300
Recovery rate (%)	90	80	70

The risk-free rate is 4.0% (continuously compounded, flat), and exposures are discounted from the end of each year. Which of the following is closest to the CVA for this position, quoted as a positive charge?

A. USD 1.46 million

B. USD 1.54 million

C. USD 1.60 million

D. USD 2.30 million

Question CR-20 c.5

Bank Q994 • BT P2.T6.331.3 (rebuilt)

A trader needs a very quick estimate of the CVA on an interest rate swap and has no time for complex calculations. The exposure management group reports, all as a percentage of notional:

- Expected exposure (EE) at maturity: 100 bps
- Potential future exposure (PFE) at maturity: 300 bps
- Average expected positive exposure (EPE) over the life of the swap: 250 bps
- Maximum PFE: 600 bps

The counterparty's credit spread is 200 bps per annum. Which of the following is closest to the CVA as a running spread, quoted as a positive charge in bps per annum?

A. 2.0 bps

B. 5.0 bps

C. 6.0 bps

D. 12.0 bps

Question CR-20 c.6

Bank Q1004 • GARP PE2 Q48

A risk analyst at a regional bank is estimating the unilateral credit valuation adjustment (UCVA) for an interest rate swap position with a counterparty bank. The analyst gathers the following information:

1-year risk-neutral default probability of the counterparty	6.5%
Annual continuously compounded risk-free rate	2.6%
Annual yield on the counterparty's credit instrument underlying the swap	5.8%
Loss given default of the swap position	75.0%
Average expected positive exposure (EPE), constant over time	5.0%

What is the best estimate of the position's UCVA as a spread (Gregory's convention: negative)?

A. -33 bps

B. -29 bps

C. -16 bps

D. -12 bps

Question CR-20 c.7

Bank Q1042 • BT P2.T6.418.2 (rebuilt) • re-filed from CR-20 i

An interest rate swap that references a USD 100.0 million notional has a risky duration of 3.50 years. The upfront CVA is a cost of 0.90% of the notional. Which of the following is closest to the approximate CVA as a running spread, quoted as a positive charge?

A. 2.6 bps

B. 25.7 bps

C. 90.0 bps

D. 315.0 bps

Solutions • CR-20 c

Solution CR-20 c.1 — Answer: D

Bank Q1002 • BT P2.T6.331.1

Step 1 – Components (1) to (3)

(1) is LGD $= 1-R$; (2) is the risk-free discount factor to t_j; (3) is the expected exposure profile at each date. With an explicit discount factor the EE is undiscounted; Gregory folds the discounting into a discounted EPE instead.

$\Rightarrow$ A, B and C are accurate.

Step 2 – Which default probability

PD(t_{j-1}, t_j) is the *marginal* (unconditional) probability that default happens in the interval, seen from today: the cumulative PD at t_j less the cumulative PD at t_{j-1}.

$\Rightarrow$ The conditional PD divides that by survival to t_{j-1}. It is larger and would overstate CVA, because survival to the interval is already inside the unconditional PD. D is the inaccurate statement.

Interpretation

Each term is the expected loss from a default in one interval, so the weights must be probabilities of defaulting *in* that interval as seen today. Over the life of the trade they add up to the cumulative PD; conditional PDs do not.

Why the other options are wrong

A is incorrect. Accurate: component (1) is LGD $= 1 - R$.
B is incorrect. Accurate: component (2) is the risk-free discount factor.
C is incorrect. Accurate: component (3) is the expected exposure profile.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2, eq. 20.3

Solution CR-20 c.2 — Answer: C

Bank Q999 • BT Mock Q27 • BT P2.T6.915.1

Step 1 – Value of the call (Black-Scholes-Merton)

$$V = S\,N(d_1) - K\mathrm{e}^{-rT}N(d_2) = 25(0.66750) - 25\,\mathrm{e}^{-0.09}(0.53450)$$
$$= 16.6875 - 12.2124 = \text{USD } 4.4751 \text{ million}$$

$\Rightarrow$ The volatility is not needed, since $N(d_1)$ and $N(d_2)$ are given.

Step 2 – Cumulative default probability over three years

$$\text{PD}(0,3) = 1 - \mathrm{e}^{-0.06\times 3} = 1 - \mathrm{e}^{-0.18} = 16.473\%$$

Step 3 – CVA

$$\text{CVA} \approx \text{LGD} \times \text{PD}(0,T) \times V = 0.70 \times 0.16473 \times 4.4751 = \text{USD } 0.516 \text{ million}$$

$\Rightarrow$ A charge of about USD 516,000 ($-$USD 516,000 in Gregory's convention).

Interpretation

A bought European option is a one-way exposure: its value to the buyer is never negative, and its discounted expected value at every future date is today's price. That is why the option value

can stand in for the whole EPE profile, and the single-period formula is exact here. The hazard rate must be turned into a three-year probability before it is used.

Why the other options are wrong

A is incorrect. Used the annual hazard rate as the 3-year PD: $4.4751 \times 0.06 \times 0.70 = 0.188$.
B is incorrect. Used the recovery rate in place of LGD: $4.4751 \times 0.16473 \times 0.30 = 0.221$.
D is incorrect. Left out the LGD: $4.4751 \times 0.16473 = 0.737$.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2 eq. 20.3 and 20.2.4; Hull (GARP CR Ch.9), 9.2.1 eq. 9.2

Solution CR-20 c.3 — Answer: C

Workbook A6.1 • S1 • Map #6

Step 1 – Marginal PDs from cumulative PDs

$\text{PD}(t_{i-1}, t_i) = \text{PD}(0, t_i) - \text{PD}(0, t_{i-1})$: 1.5, 1.7, 1.8, 1.9, 2.0 (%).
$\Rightarrow$ The chance of the counterparty defaulting within each year, rising slowly from 1.5% to 2.0%.

Step 2 – EPE $\times$ marginal PD per year (% $\times$ %)

$2.0(1.5) = 3.00$, $3.2(1.7) = 5.44$, $3.6(1.8) = 6.48$, $3.0(1.9) = 5.70$, $1.6(2.0) = 3.20$; sum $= 23.82$.
$\Rightarrow$ Sum $= 23.82 \times 10^{-4} = 0.002382$ of notional: the probability-weighted exposure before applying LGD.

Step 3 – LGD

$\text{LGD} = 1 - R = 1 - 0.40 = 0.60$
$\Rightarrow$ 60% of the exposure is lost if the counterparty defaults.

Step 4 – UCVA

$$\text{UCVA} = -0.60 \times 0.002382 \times 200 = -\text{USD } 0.286 \text{ million}$$

Interpretation

The swap is worth USD 0.286 million less to the dealer than its risk-free value (about 14.3 bps of notional upfront); this is the charge the xVA desk levies on the trader. The dealer's own spread is a red herring for UCVA (it would enter DVA), and the CVA is negative in Gregory's convention.

Why the other options are wrong

A is incorrect. Used cumulative instead of marginal PDs: $\sum \text{EPE} \times \text{PD}_{\text{cum}} = 66.18 \times 10^{-4}$, so $-0.60 \times 0.006618 \times 200 = -0.794$.
B is incorrect. Dropped the LGD term, treating the whole exposure as lost: $-0.002382 \times 200 = -0.476$.
D is incorrect. Used the recovery rate (40%) in place of LGD: $-0.40 \times 0.002382 \times 200 = -0.191$.

LO: CR-20 b: Describe credit value adjustment (CVA). CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2, eq. 20.3

Solution CR-20 c.4 — Answer: A

Bank Q1001 • BT P2.T6.914.3

Step 1 – Hazard rate implied by each year

$\lambda = s/(1 - R)$: $0.01/0.10 = 0.02/0.20 = 0.03/0.30 = 10\%$.
$\Rightarrow$ The hazard rate is constant at 10%: the wider spreads are explained by the higher LGDs.

Step 2 – Survival and marginal PDs
$V(t) = \mathrm{e}^{-0.1t}$: 0.90484, 0.81873, 0.74082, so the marginal PDs are 9.516%, 8.611% and 7.791%.

Step 3 – Discount factors
$\mathrm{e}^{-0.04t}$: 0.96079, 0.92312, 0.88692.

Step 4 – Sum $\mathrm{LGD}_t \times DF_t \times \mathrm{EE} \times \mathrm{PD}_t$

$$\text{Year 1: } 0.10 \times 0.96079 \times 32 \times 0.09516 = 0.2926$$
$$\text{Year 2: } 0.20 \times 0.92312 \times 32 \times 0.08611 = 0.5087$$
$$\text{Year 3: } 0.30 \times 0.88692 \times 32 \times 0.07791 = 0.6634$$

$\Rightarrow$ Total $= 1.4647$, a charge of about USD 1.46 million.

Interpretation
The exposure is flat and the marginal PD falls each year, yet the later years cost more because their LGD is higher. The calculation uses the marginal PD, discounting and the right LGD for each interval: three places to slip.

Why the other options are wrong
B is incorrect. Used one average LGD of 20% for every year: 1.54.
C is incorrect. Did not discount the exposures: 1.60.
D is incorrect. Applied the year-3 LGD of 30% to every year: 2.30.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2 eq. 20.3; Hull (GARP CR Ch.9), 9.2.1 and 9.6.1

Solution CR-20 c.5 **Answer: B**

Bank Q994 • BT P2.T6.331.3

Step 1 – Choose the exposure measure
The running-spread approximation uses the *average EPE* over the life of the trade: 250 bps $= 2.5\%$ of notional.
$\Rightarrow$ EE at maturity, PFE at maturity and the maximum PFE are distractors.

Step 2 – Apply eq. 20.4

$$\mathrm{CVA} \approx -\text{average EPE} \times \text{spread} = -0.025 \times 200 = -5.0 \text{ bps per annum}$$

$\Rightarrow$ A running charge of 5.0 bps a year.

Interpretation
The approximation treats the exposure as constant at its average, so a single-date figure (EE at maturity) understates it and a percentile (PFE) overstates it. PFE is a limit-setting measure; CVA prices the expected loss.

Why the other options are wrong
A is incorrect. Used the EE at maturity: $0.01 \times 200 = 2.0$.
C is incorrect. Used the PFE at maturity: $0.03 \times 200 = 6.0$.
D is incorrect. Used the maximum PFE: $0.06 \times 200 = 12.0$.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.3 eq. 20.4

Solution CR-20 c.6

Answer: C

Bank Q1004 • GARP PE2 Q48

Step 1 – Credit spread

Spread = yield on the counterparty's instrument − risk-free rate $= 5.8\% - 2.6\% = 3.2\% = 320$ bps.

Step 2 – UCVA as a spread

$$\text{UCVA} \approx -\text{average EPE} \times \text{spread} = -0.05 \times 320 = -16 \text{ bps}$$

$\Rightarrow$ A running charge of 16 bps a year.

Interpretation

The spread already reflects the market's default probability and LGD, so the 6.5% PD and the 75% LGD are red herrings. They are not even consistent with the spread ($320/0.75$ implies a hazard rate of about 4.3%), a reminder that the formula uses the spread and nothing else.

Why the other options are wrong

A is incorrect. Used the default probability as if it were a spread: $0.05 \times 6.5\% = 32.5 \approx 33$ bps.
B is incorrect. Used the full yield: $0.05 \times 5.8\% = 29$ bps.
D is incorrect. Multiplied by the LGD again: $0.05 \times 3.2\% \times 0.75 = 12$ bps.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.3 eq. 20.4

Solution CR-20 c.7

Answer: B

Bank Q1042 • BT P2.T6.418.2

Step 1 – Upfront CVA in currency

$0.90\% \times 100 =$ USD 0.90 million.

Step 2 – Divide by notional $\times$ risky duration

$$\frac{0.90}{100 \times 3.50} = 0.002571 = 25.7 \text{ bps per annum}$$

$\Rightarrow$ Equivalently $0.90\%/3.50 = 0.2571\%$: the notional cancels.

Interpretation

Dividing by the risky annuity spreads the one-off charge over the expected life of the payments, allowing for the chance that the counterparty stops paying. It is a first-order answer: adding the spread changes the trade value and so the CVA itself (CR-20 i).

Why the other options are wrong

A is incorrect. Basis-point slip: read 0.2571% as 0.0257%, giving 2.6 bps.
C is incorrect. Quoted the upfront CVA (0.90% = 90 bps) as if it were a running spread.
D is incorrect. Multiplied by the duration: $0.90\% \times 3.5 = 3.15\% = 315$ bps.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.3; SchweserNotes 2026 Book 2, Reading 38, LO 38.i

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 e

P1 CONCEPT

Describe debt value adjustment (DVA) and bilateral CVA (BCVA).

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A6, Map #6). The closed-book list on the consolidated recall sheet is "BCVA = CVA + DVA with survival adjustments and the sign convention CVA negative, DVA positive", and the August 2026 sitting had a BCVA numerical.

Vardeez: DVA represents counterparty risk from the point of view of a party's own default; taking both defaults together gives bilateral CVA. Know why DVA comes out positive when both formulas start with a minus sign.

Reported exam shape

- **A6·S3** BCVA = CVA + DVA with the sign convention CVA negative, DVA positive.

What the questions look like: mostly conceptual: what DVA is and what drives it, why it is positive, what BCVA implies for price symmetry, and how spread moves change CVA and DVA. Every stem states the sign convention it uses.

Kept outside the TYPE rule: Question CR-20 e.6 (Workbook A6.2, a BCVA from two exposure profiles) is a calculation under a describe LO. It stays because Tier workbook questions always go in.

Worked elsewhere: A6.5 (contingent BCVA and the change from UCVA) at Question CR-20 f.6; DVA, BCVA and BCVA as a spread are calculated under CR-20 g.

1. Why DVA: the institution can default too

Unilateral CVA (CR-20 c) rests on one assumption: the party making the calculation cannot default. That fits the accounting "going concern" idea, but accounting standards allow, and in effect require, an own-credit adjustment to liabilities. FAS 157 (2006) told US banks to reflect their own credit standing in the fair value of liabilities, and IFRS 13 (from January 2013) defines the fair value of a liability to include non-performance risk, including the entity's own credit risk.

DVA does for the liability side what CVA does for the asset side:

- **CVA:** the counterparty defaults while it owes you (positive exposure, EPE). You recover only a fraction: a *loss*.
- **DVA:** you default while you owe the counterparty (negative exposure, ENE). You pay only the recovery fraction of what you owe: a *gain*, the mirror image of the loss your counterparty suffers.

Recognising both is **bilateral CVA (BCVA)**. Banks first reported it in the US after FAS 157; most other large banks moved to BCVA with market-implied parameters after IFRS 13.

2. The bilateral formula and the sign of DVA

With C the counterparty and P the party making the calculation (GARP eqs. 20.7a, 20.8a and 20.8b):

$$\boxed{\text{BCVA} = \text{CVA} + \text{DVA}}$$

$$\text{CVA} = -\text{LGD}_C \sum_i \text{EPE}(t_i) \times \text{PD}_C(t_{i-1}, t_i) \times \left[1 - \text{PD}_P(0, t_{i-1})\right]$$

$$\text{DVA} = -\text{LGD}_P \sum_i \text{ENE}(t_i) \times \text{PD}_P(t_{i-1}, t_i) \times \left[1 - \text{PD}_C(0, t_{i-1})\right]$$

- **Mirror image.** DVA swaps every input: ENE for EPE, the party's *own* PD and *own* LGD for the counterparty's.
- **Why DVA is positive.** ENE is negative by construction, so $-\text{LGD}_P \times (\text{negative}) \times \text{PD}_P > 0$. CVA is a cost, DVA a benefit that opposes it.
- **The brackets.** Each term is weighted by the *other* party's survival to the start of the interval (the first-to-default effect). Drop them and you have UCVA and UDVA; the difference is CR-20 f.

Vardeez's example (no survival terms): EPE $= 100$, ENE $= -80$, both LGDs 60%, $\text{PD}_C = 2\%$, $\text{PD}_P = 1.5\%$:

$$\text{CVA} = -0.6(100)(0.02) = -1.20 \quad \text{DVA} = -0.6(-80)(0.015) = +0.72 \quad \text{BCVA} = -0.48$$

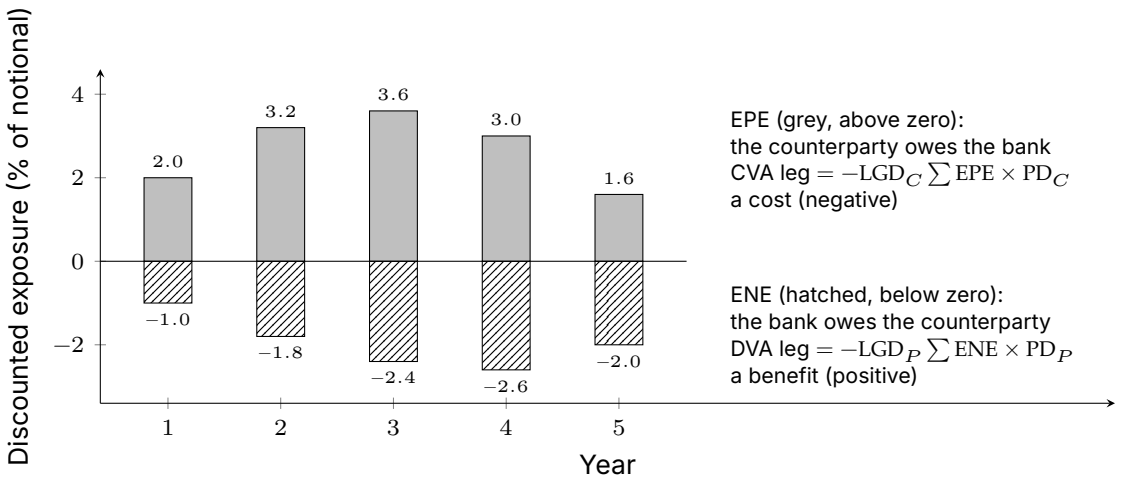


Figure 1. The two legs of BCVA for the swap of Question CR-20 e.6. The positive exposure above the axis (grey) is weighted by the counterparty's default probabilities; the negative exposure below it (hatched) by the bank's own. One trade, two defaults, two adjustments of opposite sign.

3. Price symmetry: "my CVA is your DVA"

Two parties with base valuations $V_A = -V_B$ agree on value. CVA alone breaks the agreement, because each side marks the other's default as a cost. Add DVA (GARP eqs. 20.6a and 20.6b): party A reports $V_A + \text{CVA}_A + \text{DVA}_A$ and party B reports $V_B + \text{DVA}_B + \text{CVA}_B$. With $\text{CVA}_A = -\text{DVA}_B$ and $\text{DVA}_A = -\text{CVA}_B$ the two valuations agree again. The implications (Schweser lists them; each is a standard exam statement):

- **BCVA can be positive** when DVA outweighs CVA: the risky value is then *above* the risk-free value. Unilateral CVA can only be negative.

- **Zero sum.** If both parties agree on the inputs, $\text{BCVA}_A = -\text{BCVA}_B$, so the sum of all BCVAs in the market is zero. This holds more in theory than in practice.
- **Netting can hurt** when the DVA term dominates (the institution is the riskier party). Without netting, a defaulting institution would collect in full on trades that are in its favour while paying only recovery on the others; netting removes that option.
- **Weaker pays stronger.** If average EPE $=$ $-$average ENE, BCVA is about $-\text{EPE} \times (\text{Spread}_C - \text{Spread}_P)$: the party with the wider spread pays (CR-20 g).

Historically banks charged end users CVA and ignored their own default, which was then remote. After 2007 to 2009 bank spreads became material, raising the question of how two banks could trade; DVA resolves it in theory.

What moves each leg. DVA rises when the party's *own* spread widens (a higher own PD) or its negative exposure grows. CVA moves with the *counterparty's* spread and the positive exposure. If both spreads tighten with exposures unchanged, every CVA and DVA in the relationship shrinks in magnitude (Question CR-20 e.5).

4. DVA is a double-edged sword

DVA resolves the pricing asymmetry but creates its own problems. A bank books a gain when its own credit quality falls: in 2011 the press reported that UK bank profits could be inflated by as much as £4bn this way. Gregory lists the ways DVA might be monetised and finds them weak:

- **Defaulting:** like trying to monetise your own life insurance.
- **Unwinds and novations:** the counterparty pays its CVA gain away. Monolines did this with banks; MBIA paid Morgan Stanley USD 1.1bn against an exposure of several billion. The replacement trades would normally carry CVA that wipes out the benefit.
- **Close-out:** claims in a default process may include it (Lehman), though courts have not always agreed.
- **Hedging:** hedging DVA means going long your own credit, by buying back your own bonds (which needs funding) or selling CDS protection on yourself (not possible). Some banks sold protection on correlated names instead, including on Lehman before its default.

How it is treated. Basel derecognises DVA from the CVA capital charge, so a riskier bank cannot hold less capital because of its DVA. The FASB puts DVA gains and losses in other comprehensive income, and equity analysts usually strip DVA out. Shareholders receive nothing in default, while bondholders may see DVA as real through a higher recovery. In pricing, many banks pass on some but not all of the DVA benefit, and market practice increasingly treats it as a funding benefit (FVA, CR-15).

	CVA	DVA
Whose default	The counterparty's	The institution's own
Exposure used	EPE (asset side)	ENE (liability side, negative)
PD and LGD	Counterparty's	Institution's own
Sign (Gregory)	Negative: a cost	Positive: a benefit
Grows when	The counterparty's spread widens	The institution's own spread widens
Natural hedge	Buy CDS protection on the counterparty, or short its bonds	Sell protection on yourself (impossible); buy back own bonds (needs funding)
Accounting	Required (IFRS 13, market-implied)	Required as own credit (FAS 157, IFRS 13); FASB: other comprehensive income
Basel capital	Inside the CVA charge	Derecognised

Exam traps

- **ENE sign.** ENE is negative, so DVA is positive. Entering ENE as a positive number in $-\text{LGD} \times \text{ENE} \times \text{PD}$ turns the benefit into a cost.
- **Wrong party's inputs.** DVA uses the institution's own PD and LGD with ENE; using the counterparty's PD in the DVA leg is the planted error in Question CR-20 e.6.
- **Direction of own spread.** A wider own spread *raises* DVA: a gain from your own deterioration.
- **DVA is not wrong-way risk.** It is the own-default term; the formulas assume independent defaults.
- **Zero-sum statement.** With agreed parameters the BCVAs sum to zero, not to the sum of the exposures.
- **Leg for total.** Reporting the DVA leg alone, or subtracting DVA instead of adding it, when BCVA is asked.
- **Two conventions.** Hull (CR-12) writes $f_{nd} - \text{CVA} + \text{DVA}$ with both as positive amounts; Gregory writes $V + \text{CVA} + \text{DVA}$ with CVA negative. Same economics (CR-20 g).

Practice questions: CR-20 e

Question CR-20 e.1

Bank Q1009 • AnalystPrep Q.1930

Which of the following best describes the concept of bilateral exposure in a contract between an institution and its counterparty? (A loss to the institution reduces its value; a gain increases it.)

A. It is the exposure the institution owes to the counterparty.

B. When the counterparty defaults, the institution is paid only a recovery fraction of its exposure, a loss for the institution; when the institution defaults, the counterparty receives only a recovery fraction of the negative exposure, a gain for the institution.

C. It is the worst exposure the institution could have at a certain time in the future, measured at a specified level of confidence.

D. It is the amount expected to be lost if the counterparty defaults while the mark-to-market is positive.

Question CR-20 e.2

Bank Q1013 • BT P2.T6.916.3 (rebuilt)

A valuation analyst uses Gregory's convention: CVA is negative (a cost), DVA is positive (a benefit) and BCVA $=$ CVA $+$ DVA. Each of the following statements about the debt value adjustment (DVA) is true EXCEPT:

A. Bilateral CVA is the sum of the credit value adjustment (CVA) and DVA.

B. Compared with a risky value that includes only the unilateral CVA term, adding DVA increases the value of the risky position.

C. DVA formalises a term that brings specific wrong-way risk (dependence between exposure and counterparty default) into the CVA framework.

D. DVA is the mirror image of CVA: it is based on the expected negative exposure, the institution's own default probability and its own loss given default.

Question CR-20 e.3

Bank Q1015 • BT notes P2.T5.507.3 (rebuilt)

A bank uses Gregory's convention: CVA is negative, DVA is positive and BCVA $=$ CVA $+$ DVA. Which of the following statements about CVA, DVA and BCVA is FALSE?

A. CVA is the cost to a party arising from the possibility that its counterparty defaults.

B. DVA is the benefit to a party arising from the possibility that it defaults itself and, with both parties using the same inputs, it is equal and opposite to the counterparty's CVA on the same position.

C. Because BCVA is CVA plus DVA, with a positive DVA the BCVA is less negative than the unilateral CVA.

D. All else equal, a widening of a party's own credit spread reduces its DVA.

Question CR-20 e.4

Bank Q1010 • BT P2.T6.333.2 (rebuilt)

An institution reports bilateral CVA as BCVA $=$ CVA $+$ DVA, with CVA negative and DVA positive. Each of the following statements about BCVA is true EXCEPT:

A. The second BCVA term is the debt value adjustment (DVA); it corresponds to the institution defaulting first, when it books a gain if its mark-to-market is negative.

B. Unlike a unilateral CVA, which can only reduce the value of a position, BCVA can increase it when the DVA term outweighs the CVA term, so the risky value can exceed the risk-free value.

C. Netting does not always benefit the institution under BCVA; it can be a disadvantage when the DVA term dominates, which normally means the institution is riskier than its counterparty.

D. If all market participants agree on the BCVA approach and parameters, the total counterparty risk in the market (the sum of all BCVAs) equals the sum of all expected exposures.

Question CR-20 e.5

Bank Q1014 • AP Mock 1 Q4 (rebuilt)

Northern Star Bank (NSB) and Horizon Financial Group (HFG) entered into a 4-year interest rate swap in which NSB pays a fixed rate of 4.5% and receives 6-month SOFR plus a spread. Since inception, both entities' credit ratings have improved: NSB's credit spread fell from 90 bps to 40 bps and HFG's from 130 bps to 100 bps. The SOFR curve is unchanged, so the exposure profiles (EPE and ENE) of an identical 4-year swap initiated today are the same as at inception. Each party reports CVA as a negative and DVA as a positive adjustment; "magnitude" means absolute size. Which of the following statements is most likely correct for the new swap?

A. Because NSB's spread fell by more than HFG's, NSB's DVA will decrease and HFG's DVA will increase.

B. Because NSB's spread fell by more than HFG's, NSB's CVA will decrease and HFG's CVA will increase.

C. Because both spreads fell, the CVA and DVA recorded by each party will be lower in magnitude.

D. Because both spreads fell, the CVA and DVA recorded by each party will be higher in magnitude.

Question CR-20 e.6 Workbook A6.2 • S3 • Map #6

A valuation control officer at a French bank is computing BCVA for an uncollateralised 5-year swap with a BBB corporate, notional EUR 150 million. Both LGDs are 60%. Survival adjustments are ignored (the bank reports UCVA + UDVA). The profiles are:

Year	1	2	3	4	5
Discounted EPE (% of notional)	2.0	3.2	3.6	3.0	1.6
Discounted ENE (% of notional)	−1.0	−1.8	−2.4	−2.6	−2.0
Counterparty marginal PD (%)	1.5	1.7	1.8	1.9	2.0
Bank's own marginal PD (%)	0.6	0.7	0.8	0.8	0.9

Using the convention CVA negative, DVA positive and BCVA = CVA + DVA, which of the following is closest to the BCVA?

A. −EUR 0.145 million

B. −EUR 0.097 million

C. −EUR 0.054 million

D. +EUR 0.069 million

Solutions • CR-20 e

Solution CR-20 e.1 — Answer: B

Bank Q1009 • AnalystPrep Q.1930

Step 1 – Two defaults, two directions

If the counterparty defaults while the institution's mark-to-market is positive, the institution recovers only a fraction of what it is owed: a loss (the CVA side). If the institution defaults while its mark-to-market is negative, it pays only a fraction of what it owes: a gain to the institution (the DVA side).

$\Rightarrow$ Bilateral exposure means both of these exist at once in the same contract.

Step 2 – Match the options

Only B describes both the loss on positive exposure and the gain on negative exposure.

Interpretation

Bilateral exposure is what turns unilateral CVA into BCVA. A swap can be an asset or a liability over its life, so it carries both an EPE (priced by CVA) and an ENE (priced by DVA).

Why the other options are wrong

A is incorrect. Describes only the liability side (negative exposure) and none of the recovery mechanics.

C is incorrect. This is potential future exposure (PFE), a percentile measure, not bilateral exposure.

D is incorrect. A one-sided expected loss on positive exposure: the unilateral CVA view.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.1 and 20.3.3; SchweserNotes 2026 Book 2, Reading 38, LO 38.e

Solution CR-20 e.2 — Answer: C

Bank Q1013 • BT P2.T6.916.3

Step 1 – Statements A and B

BCVA $=$ CVA $+$ DVA is GARP eq. 20.7a, so A is true. DVA is positive, so adding it to a value that carried only the (negative) unilateral CVA raises the risky value: B is true.

Step 2 – Statement D

GARP: the DVA term is "the mirror image of CVA based on the ENE, the party's own default probability and the LGD". D is true.

Step 3 – Statement C

Wrong-way risk is a dependence between exposure and the counterparty's credit quality (CR-20 k). DVA is the term for the institution's *own* default on its negative exposure, and the BCVA formulas assume independent defaults.

$\Rightarrow$ C confuses two separate ideas and is the false statement.

Interpretation

DVA answers "what if I default first?"; wrong-way risk answers "is my exposure largest when my counterparty is weakest?". Gregory treats them in different sections (20.3 and 20.6).

Why the other options are wrong

A is incorrect. True: BCVA is defined as CVA plus DVA.

B is incorrect. True: DVA is a benefit, so the risky value rises relative to a CVA-only valuation.

D is incorrect. True: this is the mirror-image structure of the DVA formula.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). • **Reference:** Gregory, The xVA

Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3, eqs. 20.7a to 20.7c; 20.6.1

Solution CR-20 e.3 Answer: D

Bank Q1015 • BT notes P2.T5.507.3

Step 1 – What drives DVA

$\text{DVA} = -\text{LGD}_P \sum \text{ENE} \times \text{PD}_P$. A wider own spread means a higher own default probability PD_P, so DVA (positive) gets *larger*.
$\Rightarrow$ D says the opposite and is false.

Step 2 – Check the other statements

A is the definition of CVA. B is "my CVA is your DVA": $\text{DVA}_A = -\text{CVA}_B$ when both use the same inputs. C: the bilateral CVA leg is no more negative than UCVA and DVA is positive, so BCVA lies above UCVA.

Interpretation

This is the counter-intuitive feature that makes DVA controversial: a bank's reported value rises when its own credit quality falls, which is why Basel derecognises DVA from capital.

Why the other options are wrong

A is incorrect. True: CVA is the cost of the counterparty's possible default.
B is incorrect. True: with agreed inputs one party's DVA is equal and opposite to the other's CVA.
C is incorrect. True: adding a positive DVA makes the net adjustment less negative than UCVA.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.2 eqs. 20.6a, 20.6b; 20.3.3; 20.3.5

Solution CR-20 e.4 Answer: D

Bank Q1010 • BT P2.T6.333.2

Step 1 – Statements A to C

A describes DVA: a gain on the institution's own default when its mark-to-market is negative. B: DVA can outweigh CVA, so BCVA can be positive and the risky value can exceed the risk-free value; UCVA cannot. C: when the DVA term dominates, netting removes the option of paying only recovery on the trades it owes, so it can be a disadvantage.
$\Rightarrow$ All three are true.

Step 2 – Statement D

With agreed parameters the formula is symmetric: if one party's BCVA is $+X$, the other's is $-X$. The sum of all BCVAs is therefore zero, not the sum of the expected exposures.
$\Rightarrow$ D is false.

Interpretation

The zero-sum property is the theoretical attraction of BCVA (price symmetry). Gregory and Schweser both note that it holds in theory more than in practice, since parties do not agree on models and inputs.

Why the other options are wrong

A is incorrect. True: the standard description of DVA.
B is incorrect. True: DVA can make BCVA positive.
C is incorrect. True: netting can hurt when the institution is the riskier party.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.2; SchweserNotes 2026 Book 2, Reading 38, LO 38.e

Solution CR-20 e.5 — Answer: C

Bank Q1014 • AP Mock 1 Q4

Step 1 – Which spread drives which leg

Each party's CVA is driven by the *other* party's spread (CVA $\approx -\text{EPE} \times s_{\text{other}}$); its DVA by its *own* spread (DVA $\approx -\text{ENE} \times s_{\text{own}}$).

Step 2 – Apply the spread moves

NSB's CVA depends on HFG's spread (130 to 100 bps, lower); NSB's DVA on its own (90 to 40 bps, lower). HFG's CVA depends on NSB's spread (lower); HFG's DVA on its own (lower).
$\Rightarrow$ With exposures unchanged, all four adjustments shrink in magnitude.

Interpretation

The relative size of the two spread changes is irrelevant to the direction of each leg: each responds to one spread only. Only the *net* BCVA depends on both: for NSB the CVA leg shrinks by about 30 bps $\times$ EPE and the DVA leg by about 50 bps $\times$ |ENE|, so its BCVA can move either way.

Why the other options are wrong

A is incorrect. HFG's DVA depends only on HFG's own spread, which fell, so it cannot rise; comparing the two spread changes is beside the point.
B is incorrect. HFG's CVA is driven by NSB's spread, which fell, so it decreases.
D is incorrect. Lower spreads mean lower default probabilities, so the adjustments shrink rather than grow.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3, eqs. 20.7a to 20.9

Solution CR-20 e.6 — Answer: A

Workbook A6.2 • S3 • Map #6

Step 1 – CVA leg (counterparty PD with EPE)

$\sum \text{EPE} \times \text{PD}_C = 3.00 + 5.44 + 6.48 + 5.70 + 3.20 = 23.82\ (\times 10^{-4})$

$$\text{CVA} = -0.60 \times 0.002382 \times 150 = -\text{EUR } 0.2144 \text{ million}$$

$\Rightarrow$ The expected loss from the counterparty defaulting while the bank is owed money.

Step 2 – DVA leg (own PD with ENE)

$\sum \text{ENE} \times \text{PD}_P = -0.60 - 1.26 - 1.92 - 2.08 - 1.80 = -7.66\ (\times 10^{-4})$

$$\text{DVA} = -0.60 \times (-0.000766) \times 150 = +\text{EUR } 0.0689 \text{ million}$$

$\Rightarrow$ The "gain" from the bank's own default while it owes the counterparty money; positive because ENE is negative.

Step 3 – BCVA

$$\text{BCVA} = \text{CVA} + \text{DVA} = -0.2144 + 0.0689 = -\text{EUR } 0.145 \text{ million}$$

Interpretation

Recognising DVA cuts the net adjustment by about a third. Basel derecognises the EUR 0.069 million DVA from capital, and many banks pass only part of it into pricing. Subtracting DVA instead of adding it would give -0.283: the sign error the convention is designed to prevent.

Why the other options are wrong

B is incorrect. Used the recovery rate (40%) in place of LGD in both legs: $-0.40 \times 0.002382 \times 150 + 0.40 \times 0.000766 \times 150 = -0.1429 + 0.0460 = -0.097$.

C is incorrect. Used the counterparty's PD in the DVA leg: $\sum \text{ENE} \times \text{PD}_C = -17.82 \times 10^{-4}$, DVA $= +0.160$, BCVA $= -0.054$.
D is incorrect. Reported the DVA leg alone (an intermediate result): $+0.069$.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3, eqs. 20.7a to 20.8b

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 f

P1 CONCEPT

Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA.

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A6, Map #6). The consolidated recall sheet flags an intermediate-result trap: "BCVA planted as a distractor when the change from UCVA is asked". **Vardeez:** the distinction is entirely about whose default is admitted into the calculation; a unilateral measure switches one of the two terms off.

Reported exam shape

- **A6·S6** Change from UCVA to BCVA, with the BCVA itself planted as a distractor.

What the questions look like: conceptual: which defaults each measure admits, what the survival factor is and does, when BCVA can be positive, and how UDVA and the DVA inside BCVA react to the counterparty's credit. Every stem states the sign convention (Gregory: CVA negative, DVA positive).

Kept outside the TYPE rule: Question CR-20 f.6 (Workbook A6.5, the change from UCVA to a contingent BCVA) is a calculation under an explain LO. It stays because Tier workbook questions always go in.

Worked elsewhere: A6.2 (a non-contingent BCVA, UCVA + UDVA) at Question CR-20 e.6; GARP PE1 Q37 (the one-year change from UCVA to BCVA) as Question CR-20 g.7.

1. Whose default is admitted

The formulas are those of CR-20 e. What separates the measures is which defaults they allow and whether each default is weighted by the other party's survival.

	Defaults admitted	Survival weight	Sign
UCVA	The counterparty's only: the party assumes it cannot default	None	≤ 0 (a cost)
UDVA	The party's own only: the counterparty is treated as default-free	None	≥ 0 (a benefit)
CVA in BCVA	The counterparty's, if the party has not defaulted first	Party's survival to t_{i-1}	≤ 0; smaller in size than UCVA
DVA in BCVA	The party's own, if the counterparty has not defaulted first	Counterparty's survival to t_{i-1}	≥ 0; smaller in size than UDVA
BCVA	Both: whichever defaults first	Both legs	Either sign

- **UCVA versus BCVA:** BCVA adds the DVA term *and* shrinks the CVA term by the party's own survival probability.
- **UDVA versus BCVA:** BCVA adds the CVA term *and* shrinks the DVA term by the counterparty's survival probability.

2. What the survival terms change

First to default. When either party defaults the contracts cease, so a second default never matters. In the bilateral CVA term a counterparty default between t_{i-1} and t_i counts only if the party is still alive at t_{i-1}, hence the factor $1 - \text{PD}_P(0, t_{i-1})$; the DVA term mirrors it with $1 - \text{PD}_C(0, t_{i-1})$. In the integral form the risky discount factor $D_{r+\lambda_C+\lambda_P}$ carries both survivals.

Each leg shrinks, so BCVA is not UCVA + UDVA. GARP Table 20.3 (upfront bp, 10-year swaps; the party's own spread is below the counterparty's; both LGDs 60%):

	Pay fixed	Receive fixed	Pay fixed (ITM)	Receive fixed (OTM)
UCVA	-29.9	-17.2	-43.7	-9.9
UDVA	8.1	14.4	4.8	20.7
UCVA + UDVA	-21.8	-2.7	-39.0	10.8
CVA	-28.9	-16.6	-42.4	-9.6
DVA	7.4	13.2	4.3	18.9
BCVA	-21.5	-3.5	-38.1	9.4

Every contingent value is smaller in size than its unilateral twin (GARP rounds each entry, so some sums differ from the printed totals by 0.1). The net effect on BCVA has no fixed direction, because both legs shrink and they have opposite signs (pay fixed: -21.5 against -21.8; receive fixed: -3.5 against -2.7).

Cross-dependence. The survival terms make each leg depend on the *other* party's credit:

- The bilateral CVA depends on the party's *own* credit: a riskier party shrinks its own CVA. Gregory calls this sensitivity "unnatural", and regulation may prohibit it.
- The bilateral DVA depends on the *counterparty's* credit: a weaker counterparty shrinks your DVA, because it is more likely to default first (Question CR-20 f.4).
- UCVA depends only on the counterparty's credit and UDVA only on your own.

Sign. UCVA can only be a cost and UDVA only a benefit. BCVA can take either sign: the OTM receive-fixed swap above has a positive BCVA of 9.4 bp.

3. Moving from UCVA to BCVA (shape S6)

$$\boxed{\Delta = \text{BCVA} - \text{UCVA} = \text{DVA} + (\text{CVA} - \text{UCVA})}$$

the DVA benefit plus a (small) survival reduction of the CVA leg. **Example.** A 2-year swap, both LGDs 60%:

	Year 1	Year 2
Discounted EPE / ENE (% of notional)	4.0 / -2.0	3.0 / -3.0
Counterparty / own marginal PD (%)	5.0 / 2.0	6.0 / 3.0
Own / counterparty survival to the start of the year	1.00 / 1.00	0.98 / 0.95

- UCVA $= -0.6\,[4(5) + 3(6)] = -22.80$ bp; UDVA $= -0.6\,[-2(2) - 3(3)] = +7.80$ bp; UCVA + UDVA $= -15.00$ bp.
- CVA $= -0.6\,[4(5)(1) + 3(6)(0.98)] = -22.58$ bp; DVA $= -0.6\,[-2(2)(1) - 3(3)(0.95)] = +7.53$ bp.
- BCVA $= -22.58 + 7.53 = -15.05$ bp, and $\Delta = -15.05 - (-22.80) = +7.75$ bp $= 7.53 + 0.22$.

Four candidate answers sit side by side: the change $(+7.75)$, the DVA leg $(+7.53)$, the BCVA level (-15.05) and the non-contingent sum (-15.00). That is exactly how GARP and the workbook build the distractors.

One-period questions. The bracket uses survival to the *start* of each interval, so with a single one-year interval eq. 20.8a's bracket equals 1. GARP's practice exam (Question CR-20 g.7) instead multiplies each leg by the other party's one-year survival probability. Follow the stem.

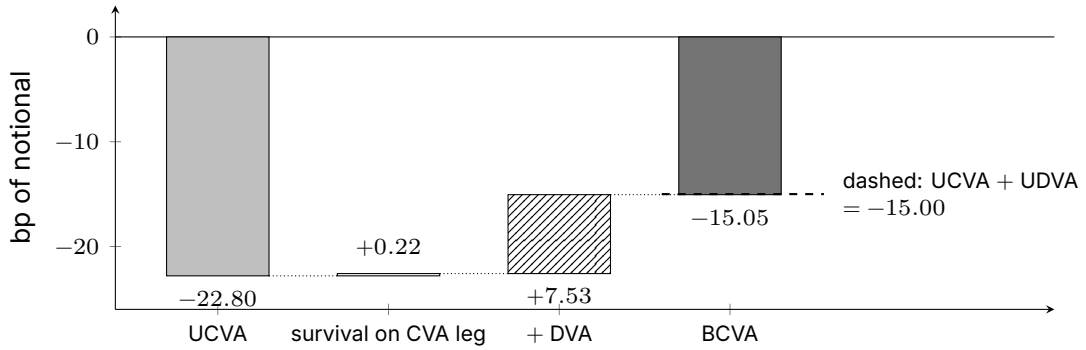


Figure 1. From UCVA to contingent BCVA in the example. The survival term trims the CVA leg (white bar, $+0.22$), the DVA leg adds the benefit (hatched, $+7.53$), and BCVA ends at -15.05 (dark grey). The change from UCVA is $+7.75$; the dashed line is the non-contingent UCVA + UDVA.

4. Close-out, default dependency and market practice

GARP 20.3.4 lists three simplifications in the bilateral formulas:

- **Survival adjustment.** First-to-default logic justifies it, but it makes CVA depend on the party's own credit quality (above).
- **Default dependency.** The formulas assume the two parties default independently; a material dependency would change both terms.
- **Close-out.** EPE and ENE assume trades settle at base values on default, while the 2002 ISDA close-out amount "may take into account the creditworthiness of the Determining Party", which suggests CVA and DVA can be part of the close-out value. Brigo and Morini (2010) show that for a one-sided exposure with "risky close-out" the survival term cancels; Gregory and German (2013) find no simple two-sided result and conclude that the formulas *without* survival probabilities are probably the best simple approximation.

Practice is split. In a 2012 Ernst & Young survey, 6 of 19 banks made CVA and DVA contingent (survival-adjusted), 7 non-contingent, and the rest did not report DVA. Anecdotal evidence suggests the mix persists.

Correction to the source notes. Vardeez's comparison table says "BCVA is UCVA plus the DVA term". That holds only for the non-contingent version (UCVA + UDVA). In the contingent BCVA that the same table describes ("each weighted by the other's survival"), the CVA leg also shrinks: BCVA $=$ UCVA $+$ (CVA $-$ UCVA) $+$ DVA. In Question CR-20 f.6 the change of $+0.575$ is 0.539 of DVA plus 0.036 of survival effect; in GARP Table 20.3 the pay-fixed CVA leg moves from -29.9 to -28.9.

Exam traps

- **Level for change.** When the move from UCVA to BCVA is asked, the answer is BCVA − UCVA; the BCVA itself is the planted distractor.
- **DVA alone.** The change also includes the survival reduction of the CVA leg.
- **Survival dropped.** Then the change equals UDVA and the "BCVA" is the non-contingent UCVA + UDVA.
- **Survival in the wrong leg.** The CVA leg carries the party's *own* survival; the DVA leg the *counterparty's*.
- **Start versus end of interval.** Gregory's bracket uses survival to t_{i-1}; a question that specifies another convention overrides it.
- **Sign range.** BCVA can be positive; UCVA never can.
- **UDVA's inputs.** UDVA ignores the counterparty's credit entirely; only the bilateral DVA responds to it.

Practice questions: CR-20 f

Question CR-20 f.1

Bank Q1016 • BT P2.T6.24.34.3 (rebuilt)

A bank values a 5-year, USD 50 million interest rate swap with a corporate client. The risk team compares the unilateral CVA (UCVA), unilateral DVA (UDVA) and bilateral CVA (BCVA) of the trade, with CVA negative and DVA positive. Which of the following statements correctly distinguishes them?

A. UCVA reflects the default of both parties; UDVA reflects only the client's default; BCVA is exactly UCVA plus UDVA.

B. UCVA reflects only the client's default and UDVA only the bank's own default; BCVA is exactly UCVA plus UDVA because the two defaults are treated independently.

C. UCVA reflects only the client's default and UDVA only the bank's own default; BCVA reflects both, with each default conditioned on the other party not having defaulted first, so its CVA and DVA components are smaller in magnitude than UCVA and UDVA.

D. UCVA reflects only the bank's own default and UDVA only the client's default; BCVA reflects both defaults with first-to-default survival adjustments.

Question CR-20 f.2 Bank Q1012 • BT P2.T6.333.1 (rebuilt) • re-filed from CR-20 e

The first term of Gregory's bilateral CVA (BCVA) formula, from the viewpoint of an institution I facing a counterparty C, is

$$-\text{LGD}_C \sum_j \text{EE}(t_j)\, S_I(t_{j-1})\, \text{PD}_C(t_{j-1}, t_j),$$

a negative number (a cost). It generalises the unilateral CVA by adding the multiplicative factor $S_I(t_{j-1})$. What does $S_I(t_{j-1})$ represent?

A. The probability that the institution has survived (not defaulted) up to time t_{j-1}.

B. The probability that the counterparty has survived up to time t_{j-1}.

C. The probability that the institution has defaulted before time t_{j-1}.

D. The recovery rate on the institution's collateral at time t_{j-1}.

Question CR-20 f.3 Bank Q1018 • New question written for this bank (GARP practice-exam style)

A bank's own credit spread is much wider than that of a highly rated corporate client. The bank's valuation team compares the unilateral CVA (UCVA) and the bilateral CVA (BCVA) of a swap portfolio it holds with the client, with CVA reported as a negative and DVA as a positive adjustment. Which of the following statements is correct?

A. BCVA is always more negative than UCVA, because it adds the cost of the bank's own default to the cost of the client's default.

B. UCVA can turn positive when the bank's own credit risk is large, whereas BCVA is always a cost to the bank.

C. UCVA and BCVA will be equal, because the bank's own default cannot affect the value of a trade in which the client is the only party at risk.

D. BCVA can be positive because the debt value adjustment (DVA) may exceed the CVA, whereas UCVA can only be a cost to the bank.

Question CR-20 f.4 New • built from GARP Ch.20, 20.3.3 eqs. 20.8a and 20.8b

An xVA analyst at a Singapore bank reports both the unilateral DVA (UDVA) and a contingent, survival-adjusted BCVA for an uncollateralised swap with a corporate client, using Gregory's convention (CVA negative, DVA positive). The client is downgraded and its CDS spread widens sharply; the bank's own spread and both exposure profiles are unchanged. Which of the following best describes the effect on the bank's UDVA and on the DVA term inside its BCVA?

A. UDVA is unchanged; the DVA term inside BCVA decreases, because each own-default term is weighted by the client's probability of surviving to the start of the interval.

B. UDVA is unchanged; the DVA term inside BCVA increases, because the client's higher default probability makes the bank's liabilities to it worth less.

C. UDVA and the DVA term inside BCVA both increase by the same amount, because both are driven by the expected negative exposure.

D. UDVA decreases, while the DVA term inside BCVA is unchanged, because survival adjustments apply only to the CVA term.

Question CR-20 f.5

New • built from GARP Ch.20, 20.3.4

A CVA desk is deciding whether to report contingent (survival-adjusted) or non-contingent CVA and DVA, with CVA negative and DVA positive. Which of the following statements about including survival probabilities is correct?

A. Including survival probabilities makes the CVA term sensitive to the institution's own credit quality, which may be seen as unnatural and which regulation may prohibit.

B. Including survival probabilities leaves BCVA exactly equal to UCVA plus UDVA, because the reductions in the two legs offset each other.

C. The survival-adjusted formulas also capture default dependency between the two parties, so no separate correlation assumption is needed.

D. Market practice has converged on contingent CVA and DVA: surveys show that almost all banks apply survival adjustments.

Question CR-20 f.6

Workbook A6.5 • S3, S4, S6 • Map #6

The head of the CVA desk at a Canadian bank is moving the bank's reported counterparty risk adjustment on an uncollateralised 3-year swap (notional CAD 250 million) from UCVA to a contingent BCVA that includes survival adjustments. The inputs are:

Year	1	2	3
Discounted EPE (% of notional)	3.0	4.0	3.0
Discounted ENE (% of notional)	−2.0	−3.5	−4.5
Counterparty marginal PD (%)	6.0	7.0	8.0
Bank's own marginal PD (%)	3.0	3.5	4.0

The counterparty's recovery rate is 40% and the bank's is 35%. Using Gregory's convention (CVA negative, DVA positive, BCVA $=$ CVA $+$ DVA), which of the following is closest to the change in the reported adjustment when the bank moves from UCVA to BCVA?

A. −CAD 0.475 million

B. +CAD 0.539 million

C. +CAD 0.575 million

D. +CAD 0.589 million

Solutions • CR-20 f

Solution CR-20 f.1 — Answer: C

Bank Q1016 • BT P2.T6.24.34.3

Step 1 – The unilateral measures

UCVA charges only for the client's default (the bank treats itself as default-free); UDVA credits only the bank's own default (the client treated as default-free).
$\Rightarrow$ A and D get these definitions wrong.

Step 2 – What BCVA adds

BCVA $=$ CVA $+$ DVA, where each term is multiplied by the probability that the *other* party has survived to the start of the interval. Both components are therefore smaller in size than UCVA and UDVA.
$\Rightarrow$ BCVA is not a plain sum: GARP Table 20.3 shows UCVA $+$ UDVA $= -21.8$ bp against BCVA $= -21.5$ bp for the pay-fixed swap.

Interpretation

The two unilateral measures each switch one default off. BCVA switches both on and lets only the first default count.

Why the other options are wrong

A is incorrect. UCVA admits only the counterparty's default, and BCVA is not the plain sum.
B is incorrect. The definitions are right, but adding them ignores the first-to-default conditioning; the sum is the non-contingent version.
D is incorrect. Reverses the roles of UCVA and UDVA.

LO: CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3, eqs. 20.8a, 20.8b and Table 20.3

Solution CR-20 f.2 — Answer: A

Bank Q1012 • BT P2.T6.333.1

Step 1 – Read the notation

S is a survival probability and the subscript I is the institution. The term is the loss from a counterparty default in $(t_{j-1}, t_j]$, captured by PD_C.

Step 2 – Why the factor is there

That loss happens only if the institution has not already defaulted, because the contracts end at the first default. The term is weighted by the institution's survival to t_{j-1}, GARP's $1 - \text{PD}_P(0, t_{j-1})$.
$\Rightarrow$ The CVA term of BCVA is smaller than the UCVA term by this factor.

Interpretation

This one factor is the whole difference between the CVA term of BCVA and UCVA. It also makes the bilateral CVA depend on the institution's own credit quality, the feature Gregory calls unnatural.

Why the other options are wrong

B is incorrect. The counterparty's survival appears in the DVA term, not here; its default is already in PD_C.
C is incorrect. That is $1 - S_I$; using it would reverse the first-to-default logic.
D is incorrect. S is a survival probability, not a recovery rate; recovery enters through LGD_C.

LO: CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3 eq. 20.8a; 20.3.4

Solution CR-20 f.3 — Answer: D

Bank Q1018 • New question written for this bank

Step 1 – UCVA

UCVA admits only the client's default, so it is always a cost (negative).

Step 2 – BCVA

BCVA adds DVA, a benefit from the bank's own default on its negative exposure. With a bank much riskier than its client, DVA can exceed the (shrunken) CVA, so BCVA can be positive: the risky value is then above the risk-free value.

Interpretation

A wide own spread is exactly the case where UCVA and BCVA diverge most, and where the weaker party (here the bank) would in theory pay the stronger one.

Why the other options are wrong

A is incorrect. DVA is a benefit, not a further cost, so BCVA is less negative than UCVA.
B is incorrect. Reversed: UCVA can only be a cost, while BCVA can be positive.
C is incorrect. In a bilateral setting the bank's own default creates DVA, so the two measures differ.

LO: CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.2 and 20.3.3

Solution CR-20 f.4 — Answer: A

New • built from GARP Ch.20, 20.3.3

Step 1 – UDVA

$\text{UDVA} = -\text{LGD}_P \sum \text{ENE}(t_i)\, \text{PD}_P(t_{i-1}, t_i)$ uses only the bank's own LGD and PD and its ENE. None of these changed.
$\Rightarrow$ UDVA is unchanged.

Step 2 – DVA inside BCVA

Each term also carries the client's survival to the start of the interval, $1 - \text{PD}_C(0, t_{i-1})$. A wider client spread raises the client's cumulative PD and lowers that survival.
$\Rightarrow$ The DVA term shrinks: the client is now more likely to default first, before the bank's own default could produce a gain.

Interpretation

This is the UDVA versus BCVA distinction in one move. The same downgrade typically makes the CVA leg more negative too, so the bank's BCVA falls on both counts, while its UDVA does not move.

Why the other options are wrong

B is incorrect. Polarity flip: the client's credit enters the DVA term only through its survival probability, which falls.
C is incorrect. UDVA has no counterparty input at all, so it cannot change with the client's spread.
D is incorrect. Survival adjustments apply to both legs (eqs. 20.8a and 20.8b), and UDVA does not move.

LO: CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA)

and BCVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3, eqs. 20.8a and 20.8b

Solution CR-20 f.5 — Answer: A

New • built from GARP Ch.20, 20.3.4

Step 1 – What the survival term does

It conditions each default on the other party not having defaulted first. Gregory: this "leads to CVA having sensitivity to a party's own credit quality, which may be seen as unnatural. Regulation may also prohibit doing this."
⇒ A restates the text.

Step 2 – Test the others against 20.3.4 and Table 20.3

Both legs shrink but have opposite signs, so BCVA differs from UCVA + UDVA (-21.5 against -21.8). The formulas assume independent defaults. The 2012 survey split 6 contingent, 7 non-contingent of 19.

Interpretation

Survival adjustment, default dependency and close-out are interconnected simplifications; Gregory and German (2013) conclude that the formulas without survival terms are probably the best simple approximation.

Why the other options are wrong

B is incorrect. Absolute claim: the two reductions do not offset exactly.
C is incorrect. True fact about the wrong model: survival terms are not a dependency model; the formulas assume independence.
D is incorrect. Practice is mixed, not converged.

LO: CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.4

Solution CR-20 f.6 — Answer: C

Workbook A6.5 • S3, S4, S6 • Map #6

Step 1 – UCVA (no survival term)

$$\begin{aligned}\text{UCVA} &= -0.60 \times [3.0(6.0) + 4.0(7.0) + 3.0(8.0)] \times 10^{-4} \times 250 \\ &= -0.60 \times 0.0070 \times 250 = -1.0500\end{aligned}$$

⇒ The currently reported charge: CAD 1.050 million, assuming the bank cannot default.

Step 2 – Survival probabilities to the start of each year

Bank: $1 - \text{PD}_P(0, t_{i-1}) = 1.000,\ 0.970,\ 0.935$. Counterparty: $1 - \text{PD}_C(0, t_{i-1}) = 1.000,\ 0.940,\ 0.870$.
⇒ A default only counts if the other party has not defaulted first.

Step 3 – Contingent CVA leg (eq. 20.8a)

$$\begin{aligned}\text{CVA} &= -0.60 \times [18(1.000) + 28(0.970) + 24(0.935)] \times 10^{-4} \times 250 \\ &= -0.60 \times 0.00676 \times 250 = -1.0140\end{aligned}$$

Step 4 – Contingent DVA leg (eq. 20.8b), own LGD $= 1 - 0.35 = 0.65$

$$\begin{aligned}\text{DVA} &= -0.65 \times [-6.0(1.000) - 12.25(0.940) - 18.0(0.870)] \times 10^{-4} \times 250 \\ &= -0.65 \times (-0.0033175) \times 250 = +0.5391\end{aligned}$$

⇒ The benefit from the bank's own default while it owes the counterparty money: positive because ENE is negative.

Step 5 – BCVA and the change

$\text{BCVA} = \text{CVA} + \text{DVA} = -1.0140 + 0.5391 = -0.4749$

$$\Delta = \text{BCVA} - \text{UCVA} = -0.4749 - (-1.0500) = +\text{CAD } 0.575 \text{ million}$$

⇒ The reported adjustment becomes CAD 0.575 million less negative: the DVA of 0.539 plus the survival reduction of the CVA leg, 0.036.

Interpretation

Moving to BCVA books a CAD 0.575 million gain, most of it DVA. Basel would derecognise the DVA from the CVA capital charge, and the survival term makes the bank's CVA depend on its own credit quality, which is why practice is split between contingent and non-contingent BCVA.

Why the other options are wrong

A is incorrect. Reported the BCVA itself (-0.475) as the change: the intermediate result planted when the move from UCVA is asked.

B is incorrect. Reported the DVA leg alone $(+0.539)$, forgetting that the survival term also shrinks the CVA leg from -1.050 to -1.014.

D is incorrect. Dropped the survival adjustments, so the change equals the UDVA: $-0.65 \times (-0.003625) \times 250 = +0.589$; the non-contingent BCVA would be -0.461.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3 eqs. 20.8a, 20.8b; 20.3.4

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 g

P1 CALC

Calculate DVA, BCVA, and BCVA as a spread.

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A6, Map #6). The August 2026 recall reported a BCVA spread numerical; the consolidated sheet adds "2 numericals on BCVA, CVA" and the closed-book list "BCVA = CVA + DVA with survival adjustments and the sign convention CVA negative, DVA positive".

Vardeez: the spread form extends the CVA spread to both sides. Each party's own credit spread multiplies the other side's exposure: the counterparty's spread goes with your positive exposure, your own spread with your negative exposure. Swapping them is the easiest error in the formula.

Reported exam shapes

- **A6·S3** BCVA = CVA + DVA with the sign convention CVA negative, DVA positive.
- **A6·S4** BCVA with survival (first-to-default) adjustments.
- **A6·S5** BCVA as a spread, the plain form seen in August.

What the questions look like: numerical. Either average EPE, average ENE and two spreads (the spread form), or one-period or profile inputs with PDs, recoveries and survival adjustments. Distractors are built from the CVA leg alone, the DVA leg alone, DVA treated as a cost, swapped spreads, and the level reported when the change is asked.

Worked elsewhere: A6.2 (a non-contingent BCVA from profiles) at Question CR-20 e.6; A6.5 (contingent BCVA and the change from UCVA) under CR-20 f.

1. Computing BCVA from profiles (shapes S3, S4)

The formulas are set out at CR-20 e and the survival logic at CR-20 f. The recipe:

1. Marginal PDs for *both* parties: differences of cumulative PDs, or $e^{-\lambda t_{i-1}} - e^{-\lambda t_i}$ with $\lambda = s/\text{LGD}$.
2. Survival of each party to the *start* of each interval (only for a contingent BCVA).
3. CVA leg: EPE $\times$ PD_C $\times$ the party's own survival, summed, times $-\text{LGD}_C$ (negative).
4. DVA leg: ENE $\times$ PD_P $\times$ the counterparty's survival, summed, times $-\text{LGD}_P$ (positive, since ENE < 0).
5. BCVA = CVA + DVA. If the change from UCVA is asked, subtract UCVA.

Example. A 3-year swap; counterparty spread 180 bp, own spread 90 bp, both LGDs 60%, rates near zero. The hazard rates are $\lambda_C = 0.018/0.6 = 3\%$ and $\lambda_P = 0.009/0.6 = 1.5\%$.

Year	EPE (%)	ENE (%)	PD_C (%)	PD_P (%)	Own surv. to t_{i-1}	Cpty surv. to t_{i-1}
1	3.0	−1.5	2.9554	1.4888	1.00000	1.00000
2	3.5	−2.5	2.8681	1.4666	0.98511	0.97045
3	2.5	−3.0	2.7833	1.4448	0.97045	0.94176

- $\sum \text{EPE} \times \text{PD}_C = 8.866 + 10.038 + 6.958 = 25.863$ bp; with the party's own survival: $8.866 + 9.889 + 6.753 = 25.508$ bp.

- $\sum \text{ENE} \times \text{PD}_P = -2.233 - 3.667 - 4.334 = -10.234$ bp; with the counterparty's survival: $-2.233 - 3.558 - 4.082 = -9.873$ bp.

$\text{CVA} = -0.6 \times 25.508 = -15.30$ bp $\text{DVA} = -0.6 \times (-9.873) = +5.92$ bp $\text{BCVA} = -9.38$ bp

Without the survival terms: UCVA $= -15.52$ bp and UDVA $= +6.14$ bp, which sum to -9.38 bp as well. The two survival reductions (0.21 bp and 0.22 bp) almost cancel here because they have opposite signs; the change from UCVA is $-9.38 - (-15.52) = +6.14$ bp.

2. BCVA as a running spread (shape S5)

Extend the average-EPE approximation of CR-20 c to both sides (GARP eq. 20.9):

$$\boxed{\text{BCVA} \approx -\text{average EPE} \times \text{Spread}_C - \text{average ENE} \times \text{Spread}_P}$$

- **Structure.** The counterparty's spread multiplies your positive exposure (its default hurts you); your own spread multiplies your negative exposure (your default helps you). Average ENE is negative, so the second term is positive.
- **No LGD and no survival term.** As for UCVA, the spreads already contain the LGDs.
- **Example above.** Average EPE $= 3.0\%$, average ENE $= -2.333\%$: $-0.03 \times 180 + 0.02333 \times 90 = -5.40 + 2.10 = -3.30$ bp a year. The upfront BCVA divided by the risky annuity $(0.97045 + 0.94176 + 0.91393 = 2.826)$ gives $-9.38/2.826 = -3.32$ bp a year.
- **Schweser's example.** EPE 5%, ENE magnitude 3%, spreads 300 bp (counterparty) and 200 bp (own): $-15 + 6 = -9$ bp, what the institution may charge the counterparty.
- **Weaker pays stronger.** If average EPE $=$ −average ENE, $\text{BCVA} = -\text{EPE} \times (\text{Spread}_C - \text{Spread}_P)$. A party charges its counterparty for the difference in their spreads, and pays it if the difference is negative.

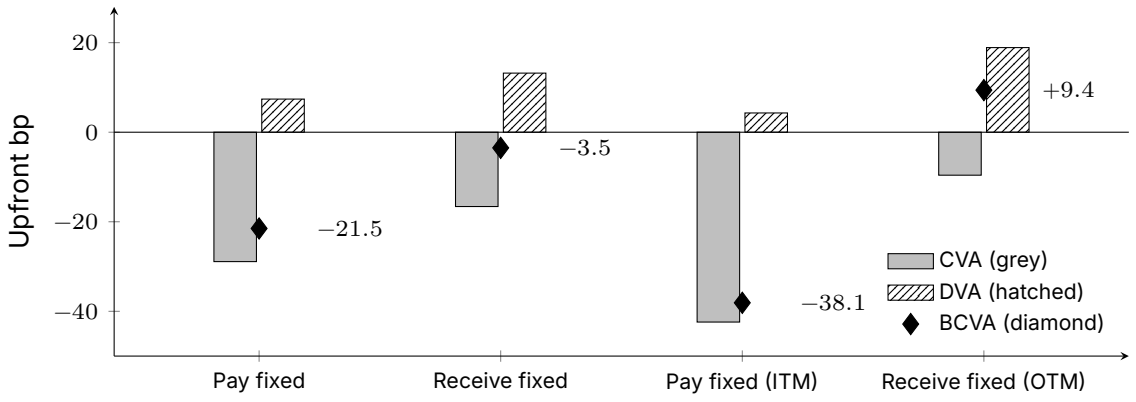


Figure 1. CVA and DVA as the two legs of BCVA for GARP's 10-year swaps (Table 20.3; the party's own spread is below the counterparty's, both LGDs 60%). The pay-fixed swap has mostly positive exposure, so CVA dominates. The receive-fixed swap's large ENE brings BCVA close to zero, and the out-of-the-money receive-fixed swap has a positive BCVA. Values as printed; GARP rounds each entry, so CVA + DVA can differ from BCVA by 0.1.

3. Reading BCVA numbers

- **Asymmetry.** Pay-fixed and receive-fixed swaps have skewed exposure profiles, so their BCVAs differ sharply (-21.5 against -3.5 bp) even with the same counterparty.
- **Positive BCVA.** Off-market, the out-of-the-money receive-fixed swap has BCVA $= +9.4$ bp: to step into it the party would "pay through mid". Gregory questions whether market participants would really price such benefits, or charge so little on the receive-fixed swap.
- **Survival effects are small** when PDs are small: the contingent values in Table 20.3 differ from the unilateral ones by at most 1.8 bp.

4. Sign conventions and sensitivities

	Gregory (GARP Ch.20; the workbook)	Hull (GARP CR-12); BT
CVA	Negative number	Positive amount (a cost)
DVA	Positive number	Positive amount (a benefit)
Value with default risk	$V + \text{CVA} + \text{DVA} = V + \text{BCVA}$	$f_{nd} - \text{CVA} + \text{DVA}$
Risk-free 20, cost 3, benefit 1	$20 + (-3) + 1 = 18$	$20 - 3 + 1 = 18$

- **Own spread widens:** DVA rises, BCVA rises (less negative). In the spread form the change is $|\text{ENE}| \times \Delta s_P$: with ENE of 5% and a 60 bp widening, $+3$ bp (Question CR-20 g.5).
- **Counterparty spread widens:** CVA becomes more negative, so BCVA falls.
- **Both tighten, exposures unchanged:** both legs shrink; the net effect depends on the exposures (Question CR-20 e.5).

Correction to the source notes. Vardeez (CR-20 e) calls Gregory's convention "the same sign convention used at CR-12 b, where the credit-adjusted value is $f_{nd} - \text{CVA} + \text{DVA}$". The economics are the same, but the arithmetic is not: Hull quotes CVA as a *positive* amount and subtracts it, while Gregory's CVA is already negative and is added ($V + \text{CVA} + \text{DVA}$). Putting Gregory's negative CVA into Hull's formula would add the charge to the value. Use whichever form the stem states.

Formula card: CR-20 g

BCVA (eq. 20.7a)	$\text{BCVA} = \text{CVA} + \text{DVA}$
CVA leg (eq. 20.8a)	$-\text{LGD}_C \sum_i \text{EPE}(t_i)\,\text{PD}_C(t_{i-1}, t_i)\,[1 - \text{PD}_P(0, t_{i-1})]$
DVA leg (eq. 20.8b)	$-\text{LGD}_P \sum_i \text{ENE}(t_i)\,\text{PD}_P(t_{i-1}, t_i)\,[1 - \text{PD}_C(0, t_{i-1})] \quad (> 0)$
Non-contingent	Drop the brackets: UCVA + UDVA
Change from UCVA	$\text{BCVA} - \text{UCVA} = \text{DVA} + (\text{CVA} - \text{UCVA})$
One period, GARP PE style	$\text{CVA} = -\text{LGD}_C\,\text{PD}_C(1 - \text{PD}_P)\,\text{EPE}$; $\text{DVA} = \text{LGD}_P\,\text{PD}_P(1 - \text{PD}_C)\,\|\text{ENE}\|$
BCVA spread (eq. 20.9)	$-\text{avg EPE} \times \text{Spread}_C - \text{avg ENE} \times \text{Spread}_P$
Symmetric exposure	$-\text{EPE} \times (\text{Spread}_C - \text{Spread}_P)$
Hazard from a spread	$\lambda = s/\text{LGD}$

Exam traps

- **Swapped spreads.** The counterparty's spread goes with EPE, your own with ENE.
- **DVA as a cost.** Subtracting the DVA term (or entering ENE as positive in $-\text{ENE} \times s_P$) adds two costs together.
- **Leg for total.** The CVA leg alone or the DVA leg alone is always among the options.
- **LGD in the spread form.** The spreads already contain it.
- **Level for change.** When the move from UCVA is asked, subtract UCVA from BCVA.
- **Survival in the wrong leg.** The CVA leg uses your survival; the DVA leg the counterparty's.
- **Own downgrade.** It *raises* BCVA through DVA; it does not make BCVA worse.
- **Red herrings.** PFE, LGDs next to the spread formula, the notional.

Practice questions: CR-20 g

Question CR-20 g.1

Bank Q1022 • BT P2.T6.901.3

Herman, Ferry, and Quitzon LLC is a market maker with a position in an over-the-counter (OTC) derivatives contract. For this position it has estimated the following values, with both adjustments quoted as positive amounts:

- The risk-free value of the position is +USD 32.0 million
- The credit valuation adjustment (CVA) is USD 5.0 million
- The debt valuation adjustment (DVA) is USD 8.0 million

Based only on this information, which of the following is the best estimate of the risky value of the position from Herman's perspective?

A. USD 27.0 million

B. USD 29.0 million

C. USD 35.0 million

D. It cannot be estimated without the counterparty's default probability and loss given default.

Question CR-20 g.2

Bank Q1028 • Mock 1 Q26 (rebuilt)

A firm has entered into a portfolio of derivative transactions with a single counterparty. The expected positive exposure is 7% of notional and the expected negative exposure has a magnitude of 5% of notional. The counterparty's credit spread is 330 bps and the firm's own credit spread is 250 bps. Using BCVA $= -\text{EPE} \times s_C - \text{ENE} \times s_P$, with ENE entered as a negative number (so CVA is negative and DVA positive), which of the following is correct regarding the firm's BCVA?

A. The CVA component will be positive for the firm.

B. The BCVA will be positive for the firm.

C. The BCVA will be negative for the firm.

D. The DVA component will exceed the CVA component in magnitude.

Question CR-20 g.3

Bank Q1023 • AP Mock 3 Q66 (rebuilt)

A trading firm needs a quick estimate of a swap's bilateral credit value adjustment. The expected positive exposure for a trade of this type is 13.6% and the expected negative exposure is -9.0%. The counterparty's credit spread is 267 bps per annum and the firm's own credit spread is 191 bps per annum. Using BCVA $= -\text{EPE} \times s_C - \text{ENE} \times s_P$ (CVA negative, DVA positive), which of the following is nearest to the BCVA?

A. -53.502 bps

B. -36.312 bps

C. -19.122 bps

D. -1.946 bps

Question CR-20 g.4

Workbook A6.3 • S2, S5 • Aug-2026 • Map #6

A risk consultant reviewing pricing at an Indian bank wants quick estimates, as running spreads, of the counterparty risk adjustments on a 5-year swap with a weaker counterparty. The data are:

- Undiscounted EPE profile (% of notional), years 1 to 5: 1.8, 2.6, 3.0, 2.8, 1.8
- Undiscounted ENE profile (% of notional), years 1 to 5: -1.2, -2.0, -2.6, -2.4, -1.8
- Counterparty 5-year credit spread: 180 bps; bank's own 5-year credit spread: 120 bps
- LGD for both parties: 60%

Using Gregory's average-exposure approximations (CVA negative, DVA positive), which of the following is closest to the UCVA and the BCVA, each in bps per annum?

A. UCVA -4.32 bps; BCVA -1.92 bps

B. UCVA -4.32 bps; BCVA $+2.40$ bps

C. UCVA -2.88 bps; BCVA $+0.72$ bps

D. UCVA -2.59 bps; BCVA -1.15 bps

Question CR-20 g.5

Bank Q1019 • AnalystPrep Q.3084 (rebuilt)

Bank X has a swap with Bank Y. From Bank X's perspective the expected positive exposure (EPE) is 8% and the magnitude of the expected negative exposure (ENE) is 5%. Bank Y's credit spread is 200 bps and Bank X's own credit spread is initially 120 bps. Bank X is then downgraded and its own credit spread widens to 180 bps; Bank Y's spread and both exposures are unchanged. Using BCVA $= -\text{EPE} \times s_Y - \text{ENE} \times s_X$ with ENE entered as a negative number (CVA negative, DVA positive), by how much does Bank X's BCVA, as a running spread, change?

A. -7 bps

B. -3 bps (BCVA becomes more negative)

C. $+3$ bps (BCVA becomes less negative)

D. $+4.8$ bps

Question CR-20 g.6

Bank Q1020 • Mock 4 Q49 (rebuilt)

A senior risk manager at Bank Gamma is showing new risk analysts how to calculate bilateral CVA (BCVA). Bank Gamma and Bank Phi are each other's only counterparty. The discounted expected positive exposure of Gamma to Phi is CNY 60 million and the discounted expected negative exposure (magnitude) is CNY 45 million. Use the convention BCVA $=$ CVA $+$ DVA, with CVA negative (a cost) and DVA positive, and apply the first-to-default adjustment (each leg is multiplied by the one-year survival probability of the other party), assuming independent defaults.

Parameter	Bank Gamma	Bank Phi
Annual probability of default	2.5%	1.8%
Recovery rate	82%	92%

Which of the following is closest to the BCVA from Bank Gamma's perspective?

A. $-$CNY 283,095

B. $-$CNY 84,240

C. $+$CNY 114,615

D. $+$CNY 198,855

Question CR-20 g.7

Bank Q1027 • GARP PE1 Q37 • Mock 3 Q43 (rebuilt)

The manager of the CVA desk of a financial institution is reviewing a 1-year bilateral CVA (BCVA) on derivative positions with a counterparty, and suspects that the analyst computed the unilateral CVA (UCVA) instead. The discounted expected positive exposure of the institution to the counterparty is AUD 48 million for the coming year, equal to the discounted expected positive exposure of the counterparty to the institution. Use the convention BCVA $=$ CVA $+$ DVA, with CVA negative (a cost) and DVA positive, and apply the first-to-default adjustment (each leg is multiplied by the one-year survival probability of the other party), assuming independent defaults.

Parameter	Financial institution	Counterparty
Annual probability of default	3.3%	1.5%
Expected recovery rate	75%	90%

From the institution's perspective, which of the following is closest to the size of the change in the charge if the analyst derives the BCVA instead of the UCVA?

A. AUD 248,400

B. AUD 320,400

C. AUD 390,100

D. AUD 392,400

Solutions • CR-20 g

Solution CR-20 g.1

Answer: C

Bank Q1022 • BT P2.T6.901.3

Step 1 – Write the risky value with positive amounts

$$\text{Risky value} = \text{risk-free value} - \text{CVA} + \text{DVA} = 32.0 - 5.0 + 8.0 = \text{USD } 35.0 \text{ million}$$

$\Rightarrow$ In Gregory's signed form: BCVA $= -5.0 + 8.0 = +3.0$, and $32.0 + 3.0 = 35.0$.

Interpretation

DVA exceeds CVA, so the risky value is *above* the risk-free value: the position's own-default benefit outweighs the cost of the counterparty's default. The PD and LGD are already inside the two adjustments.

Why the other options are wrong

A is incorrect. Deducted the CVA and ignored the DVA: $32 - 5 = 27$.
B is incorrect. Reversed the roles, treating DVA as a cost and CVA as a gain: $32 - 8 + 5 = 29$.
D is incorrect. CVA and DVA are given in currency, so no further inputs are needed.

LO: CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.2 eqs. 20.6a, 20.6b; 20.3.3

Solution CR-20 g.2

Answer: C

Bank Q1028 • Mock 1 Q26

Step 1 – CVA term

$-0.07 \times 330 = -23.1$ bps.

Step 2 – DVA term

$-(-0.05) \times 250 = +12.5$ bps.

Step 3 – BCVA

$-23.1 + 12.5 = -10.6$ bps: negative.
$\Rightarrow$ The counterparty is the weaker credit and the positive exposure is larger, so the cost outweighs the benefit.

Interpretation

A quick sign check before any arithmetic: the CVA term pairs the larger exposure with the wider spread, so BCVA must be a net cost.

Why the other options are wrong

A is incorrect. The CVA component is never positive; it is the cost of the counterparty's default.
B is incorrect. BCVA is positive only if DVA exceeds CVA; here $12.5 < 23.1$.
D is incorrect. The DVA of 12.5 bps is smaller than the CVA of 23.1 bps in magnitude.

LO: CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3 eq. 20.9

Solution CR-20 g.3

Answer: C

Bank Q1023 • AP Mock 3 Q66

Step 1 – CVA term (counterparty spread with EPE)

$-0.136 \times 267 = -36.312$ bps.

Step 2 – DVA term (own spread with ENE)
$-(-0.090) \times 191 = +17.190$ bps.

Step 3 – BCVA
$-36.312 + 17.190 = -19.122$ bps.

Interpretation
The firm charges about 19 bps a year net: the counterparty's weaker credit and the larger positive exposure dominate.

Why the other options are wrong
A is incorrect. Treated the DVA term as a further cost: $-36.312 - 17.190 = -53.502$.
B is incorrect. Reported the CVA leg alone: -36.312.
D is incorrect. Swapped the spreads: $-0.136 \times 191 + 0.090 \times 267 = -25.976 + 24.030 = -1.946$.

LO: CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3 eq. 20.9

Solution CR-20 g.4 **Answer: A**

Workbook A6.3 • S2, S5 • Aug-2026

Step 1 – Average EPE and average ENE

$$\overline{\text{EPE}} = \frac{1.8+2.6+3.0+2.8+1.8}{5} = 2.4\%, \quad \overline{\text{ENE}} = \frac{-1.2-2.0-2.6-2.4-1.8}{5} = -2.0\%$$

$\Rightarrow$ On average the bank is owed 2.4% of notional and owes 2.0%.

Step 2 – UCVA as a spread (eq. 20.4)
$\text{UCVA} \approx -\overline{\text{EPE}} \times \text{Spread}_C = -0.024 \times 180 = -4.32$ bps p.a.
$\Rightarrow$ Ignoring its own default, the bank would charge about 4.3 bps per annum for counterparty risk.

Step 3 – DVA leg (eq. 20.9)
$-\overline{\text{ENE}} \times \text{Spread}_P = -(-0.020) \times 120 = +2.40$ bps p.a.

Step 4 – BCVA as a spread
$\text{BCVA} = -4.32 + 2.40 = -1.92$ bps p.a.

Interpretation
Because the counterparty is weaker (180 against 120 bps) it pays the bank about 1.9 bps per annum to trade; with equal spreads and symmetric exposure the bilateral charge would be close to zero. The LGD is a red herring: the approximation has no LGD input because the spread already embeds it.

Why the other options are wrong
B is incorrect. Correct UCVA, but reported the DVA leg alone as the BCVA (an intermediate result): $+2.40$.
C is incorrect. Swapped the parties' spreads, applying the bank's own spread to the EPE and the counterparty's spread to the ENE: $-0.024 \times 120 = -2.88$; $-2.88 + 0.020 \times 180 = +0.72$.
D is incorrect. Multiplied by LGD as well, double counting it: $0.60 \times (-4.32) = -2.59$ and $0.60 \times (-1.92) = -1.15$.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.3 eq. 20.4; 20.3.3 eq. 20.9

Solution CR-20 g.5 — Answer: C

Bank Q1019 • AnalystPrep Q.3084

Step 1 – BCVA before the downgrade

$-0.08 \times 200 - (-0.05) \times 120 = -16 + 6 = -10$ bps.

Step 2 – BCVA after the downgrade

$-0.08 \times 200 - (-0.05) \times 180 = -16 + 9 = -7$ bps.

Step 3 – Change

$-7 - (-10) = +3$ bps; equivalently, only the DVA term moves: $0.05 \times (180 - 120) = +3$ bps.
$\Rightarrow$ Bank X's own deterioration raises its DVA benefit, so its BCVA becomes less negative.

Interpretation

This is the DVA paradox in a spread question: a downgrade improves the reported adjustment. It is one reason Basel derecognises DVA from capital.

Why the other options are wrong

A is incorrect. Reported the new BCVA level (-7 bps) instead of the change.
B is incorrect. Sign error: assumed that one's own downgrade must worsen BCVA, but the DVA leg is a gain that grows with the own spread.
D is incorrect. Applied the EPE (8%) instead of the ENE (5%) to the spread change: $0.08 \times 60 = 4.8$.

LO: CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3 eq. 20.9; 20.3.5

Solution CR-20 g.6 — Answer: C

Bank Q1020 • Mock 4 Q49

Step 1 – CVA leg (Phi defaults, Gamma survives)

$$\text{CVA} = -60{,}000{,}000 \times (1 - 0.92) \times 0.018 \times (1 - 0.025) = -84{,}240$$

Step 2 – DVA leg (Gamma defaults, Phi survives)

$$\text{DVA} = +45{,}000{,}000 \times (1 - 0.82) \times 0.025 \times (1 - 0.018) = +198{,}855$$

Step 3 – BCVA

$-84{,}240 + 198{,}855 = +\text{CNY } 114{,}615$, a net benefit to Gamma.

Interpretation

Gamma is the weaker credit (higher PD and lower recovery) with a large ENE, so its DVA outweighs its CVA. Note the convention: with one annual interval, Gregory's eq. 20.8a would use survival to the *start* of the year (a factor of 1); this question, like GARP's practice exam, applies the one-year survival of the other party. Follow the stem.

Why the other options are wrong

A is incorrect. Added the two legs as costs: $-84{,}240 - 198{,}855 = -283{,}095$.
B is incorrect. Reported the CVA leg alone and ignored the DVA.
D is incorrect. Reported the DVA leg alone and ignored the CVA.

LO: CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3 eqs. 20.7a, 20.8a, 20.8b

Solution CR-20 g.7 **Answer: D**

Bank Q1027 • GARP PE1 Q37

Step 1 – UCVA (no survival term)

$\text{UCVA} = -48{,}000{,}000 \times 0.10 \times 0.015 = -72{,}000.$

Step 2 – CVA leg of BCVA

$-72{,}000 \times (1 - 0.033) = -69{,}624.$

Step 3 – DVA leg of BCVA

$48{,}000{,}000 \times 0.25 \times 0.033 \times (1 - 0.015) = +390{,}060.$

Step 4 – BCVA and the change

$\text{BCVA} = -69{,}624 + 390{,}060 = +320{,}436;$ change $= 320{,}436 - (-72{,}000) = 392{,}436 \approx$ AUD 392,400.

$\Rightarrow$ The charge falls by about AUD 392,400 and becomes a net benefit.

Interpretation

The institution is the riskier party (3.3% PD and 25% LGD against 1.5% and 10%), so DVA dwarfs CVA. The change is the DVA plus the survival reduction of the CVA leg $(390{,}060 + 2{,}376)$, not the BCVA level. This is the shape A6.5 drills with a profile (CR-20 f).

Why the other options are wrong

A is incorrect. Subtracted the UCVA cost instead of adding it back: $320{,}436 - 72{,}000 = 248{,}436$.

B is incorrect. Reported the BCVA level (320,436) instead of the change: the planted intermediate result.

C is incorrect. Reported the DVA leg alone (390,060), ignoring the survival effect on the CVA leg.

LO: CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.3.3 eqs. 20.8a, 20.8b; 20.3.4

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 i

P1 CALC

Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread.

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A6, Map #6): CVA as a running spread is the reported specific ("UCVA as a spread: calculation (formula)", listed twice), and the workbook drills it on an incremental CVA (A6.4).

Vardeez: incremental CVA is the portfolio CVA after the change less the portfolio CVA before it, the standalone formula with the incremental EPE; marginal CVA uses the marginal EPE to attribute an existing book; to quote CVA as a running spread, divide by the risky duration and the notional ($-90{,}000/(3.75 \times 100\text{m}) = -2.40$ bp).

Reported exam shape

- **A6·S2** CVA as a running spread: −average EPE × spread, and upfront CVA divided by the risky annuity.

What the questions look like: a mix. Concept items on incremental versus marginal CVA (which to use when, order dependence, sign, additivity) and numericals: a second trade's incremental CVA from the totals, an incremental CVA from netting-set CVAs, then a running spread. Every stem states whether CVA is quoted as a negative number or as a positive cost.

Worked elsewhere: a plain upfront-to-running conversion (Bank Q1042) is Question CR-20 c.7; netting itself is CR-20 h.

1. Why CVA has to be allocated

Netting and margin reduce CVA, but only at the level of the netting set (CR-20 h). Netting can only help (GARP eq. 20.10):

$$\text{CVA}_{NS} \geq \sum_{i=1}^{N} \text{CVA}_i^{SA} \qquad \text{(CVA negative: the netted total is less negative)}$$

The benefit is large for balanced, offsetting portfolios and small for directional ones. Pricing a trade, or valuing it for accounting, needs a transaction-level number, so the netting benefit has to be allocated. GARP gives two ways: incremental and marginal CVA.

2. Incremental CVA

The CVA charge for a change to the netting set (eqs. 20.11 and 20.12):

$$\Delta\text{CVA}_{NS\to NS'} = \text{CVA}_{NS'} - \text{CVA}_{NS} = -\text{LGD}\sum_{i=1}^{m} \text{EPE}_{NS\to NS'}(t_i) \times \text{PD}(t_{i-1}, t_i)$$

the standalone formula with the incremental EPE in place of the standalone EPE. It assumes the change does not alter the counterparty's PD or LGD.

- **Charged at execution.** It is the trade's contribution to total CVA at the time it is done, so an xVA desk charging it exactly offsets the jump in its total CVA. That makes it the number to charge salespeople, traders, businesses and clients.

- **Order-dependent, but fixed afterwards.** It depends on which trades came first, and later trades do not change it.
- **Not only new trades.** Unwinds, restructurings and changes to contractual terms (for example margin agreements) are priced the same way.
- **Bounds and sign.** Incremental CVA is never more negative than the trade's standalone CVA. If the trade offsets the book, incremental EPE is negative and incremental CVA is *positive*: a benefit (an unwind typically does this).
- **Size matters.** As the new trade grows relative to the book, the netting benefit is lost and incremental CVA tends to the standalone value (GARP Figure 20.12: from a lower limit of -6.2 bp a year for a small offsetting trade towards -17.9 bp a year standalone).
- **Better terms for existing clients.** A bank with trades already on can quote an offsetting trade more cheaply (Cooper and Mello, 1991); clients in jurisdictions with doubtful netting enforceability may deal only with banks that treat netting as enforceable.

GARP Table 20.4 (bp upfront)	Standalone	Directional	Neutral/offsetting
CVA	-26.3	-25.1	-13.7
DVA	6.0	5.3	-2.5
BCVA	-20.3	-19.9	-16.3

Incremental values for a 7-year payer swap; counterparty spread 150 bp, own spread 100 bp, both LGDs 60% (entries rounded in the source, so CVA + DVA can differ from BCVA by 0.1). Against a directional book the reduction is small; against an offsetting book it is large. The incremental DVA even changes sign (offsetting), while CVA falls without changing sign (neutral). BCVA barely moves because the two effects cancel; with equal spreads BCVA would have no incremental effect at all.

3. Marginal CVA

Put the *marginal* EPE into the same formula. Marginal CVAs are trade-level contributions that **sum to the total CVA** and do not depend on the order of execution, so the allocation is "fair".

- **Not for pricing a new trade:** every new trade changes the marginal CVA of the existing trades, which would mean revaluing them.
- **Used for:** trades executed at the same time (one deal, or a policy of computing CVA periodically), allocating CVA across a book at a given date (accounting), and finding trades worth restructuring, novating or unwinding.

Identity worth knowing. Taken in execution order, incremental CVAs also sum to the total. The first trade's incremental CVA is its standalone CVA, so with two trades the second gets total − first (Question CR-20 i.6).

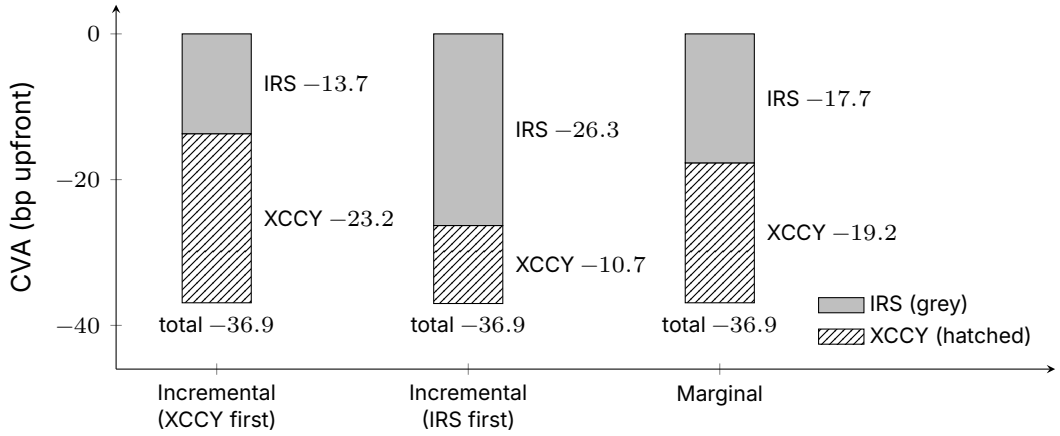


Figure 1. One netted total, three allocations (GARP Table 20.5: an interest rate swap and a cross-currency swap, counterparty spread 150 bp, LGD 60%). Whichever trade comes first takes its full standalone CVA and the second gets only the remainder; marginal CVA splits the same total without regard to order. The IRS-first column adds to -37.0 because of rounding in the source.

	Incremental CVA	Marginal CVA
Question answered	What does this change do to the netting-set CVA?	How is the existing CVA split across trades?
Exposure used	Incremental EPE (with minus without)	Marginal EPE
Order of execution	Matters (first trade gets its standalone CVA)	Irrelevant
After later trades	Unchanged	Changes
Sums to total CVA	Only when taken in execution order	Always
Typical use	Pricing a new trade, unwind, restructuring or margin change; charging desks and clients	Simultaneous trades; accounting allocation; picking trades to restructure

4. Converting CVA into a running spread

To build CVA into a swap rate, divide the upfront CVA by the risky duration (risky annuity) times the notional. Vardeez's and Schweser's example: a 5-year payer swap, notional 100 million, risky duration 3.75, standalone CVA $-90{,}000$:

$$\frac{-90{,}000}{3.75 \times 100{,}000{,}000} = -0.000240 = -2.40 \text{ bp per year}$$

The CVA of the CVA. Charging the spread changes the trade's value, which changes its CVA. The exact charge is the adjusted rate C' at which the trade's value including CVA is zero (Schweser: $V_{C'} = \text{CVA}_{C'}$), found recursively; C' then becomes the hurdle rate. GARP Table 20.1: dividing by the risky annuity gives -1.92 bp, the exact recursive answer -1.96 bp, and the average-EPE shortcut -2.01 bp. A desk that charges the first-order figure therefore always loses a little when the trade is booked. The effect is small for small charges but significant for risky counterparties and long-dated trades (Vrins and Gregory, 2011), and can be bounded by using both risky and risk-free annuities.

Incremental CVA as a running charge. A desk prices a new trade, notional 200 million, risky annuity 6.2. Netting-set CVA is -3.10 million before and -3.45 million after; the trade's standalone CVA is -0.90 million. Incremental CVA $= -0.35$ million, so the charge is $-0.35/(200 \times 6.2) = -2.82$ bp a year, against $-0.90/1{,}240 = -7.26$ bp standalone.

Formula card: CR-20 i

Netting benefit (eq. 20.10)		$\text{CVA}_{NS} \geq \sum_i \text{CVA}_i^{SA}$ (less negative)
Incremental CVA (eq. 20.11)		$\text{CVA}_{NS'} - \text{CVA}_{NS}$
Incremental CVA (eq. 20.12)		$-\text{LGD} \sum_i \text{EPE}_{NS \to NS'}(t_i)\, \text{PD}(t_{i-1}, t_i)$
Marginal CVA		Same formula with marginal EPE; $\sum_i \text{CVA}_i^{\text{marg}} = \text{CVA}_{NS}$
Sequential allocation		first trade: standalone CVA; second trade: total − first
Upfront to running		$\text{CVA}_{\text{upfront}}/(\text{notional} \times \text{risky annuity})$ (first order)
Exact running spread		solve $V_{C'} = \text{CVA}_{C'}$ recursively ("CVA of the CVA")
Shortcut (CR-20 c)		−average EPE × spread

Exam traps

- **Incremental, standalone, total or benefit.** The price of a new trade under netting is the incremental CVA: not its standalone CVA, not the post-trade netting-set CVA, and not the netting benefit (standalone minus incremental).
- **Sibling swap.** "Which trades drive the CVA?" is marginal CVA; "what should this new trade be charged?" is incremental CVA.
- **Order.** Incremental CVA depends on execution order; marginal CVA does not.
- **No floor at zero.** An offsetting trade can have a positive incremental CVA (a benefit).
- **No fixed ranking.** Marginal CVA is not always below (or above) incremental CVA.
- **Mixing allocations.** The second trade's incremental CVA is total minus the first trade's incremental CVA, not minus its marginal CVA.

- **Annuity direction and units.** Divide by the risky annuity times notional; 0.000637 is 6.37 bp.
- **LGD again.** When CVA figures are given, the LGD is already inside them.

Practice questions: CR-20 i

Question CR-20 i.1

Bank Q1033 • AP Mock 1 Q8 (rebuilt)

Alliance Derivatives Group (ADG) has an existing netting set with a counterparty and is deciding which CVA measure to use in different situations; CVA is reported as a negative number. In which situation would it be most appropriate for ADG to use incremental CVA?

A. Splitting the netting-set CVA at a given date into trade-level contributions that sum to the total, for accounting purposes.

B. Pricing several trades executed simultaneously with the counterparty as part of the same deal.

C. Charging the desk that has just executed a new trade with the change in netting-set CVA caused by that trade.

D. Analysing each trade's CVA contribution in a way that does not depend on the order in which trades were executed.

Question CR-20 i.2

Bank Q1034 • BT P2.T6.418.3

Your colleague Peter is evaluating whether to use marginal or incremental CVA (whatever the sign convention, the statements concern how the measures behave). He makes two statements:

I. Incremental CVA depends on the order in which trades are executed but does not change due to subsequent trades. It makes the most sense when CVA needs to be charged to individual traders and businesses.

II. Marginal CVA does change when new trades are executed. While it probably should not be used to price new transactions, it is good for pricing simultaneous transactions and for decomposing a total CVA into its trade-level contributions.

Which of the above statements is (are) true?

A. Neither

B. I only

C. Both

D. II only

Question CR-20 i.3

Bank Q1039 • BT P2.T6.332.1 (rebuilt)

Incremental CVA of a new trade is the CVA of the netting set including the new trade minus the CVA of the existing netting set, with CVA expressed here as a positive cost. Each of the following statements about incremental CVA is true EXCEPT:

A. It depends on the order in which trades are executed.

B. It is less than or equal to the standalone CVA of the new trade.

C. It must be greater than or equal to zero; it cannot be negative.

D. It is useful for measuring the impact of adding a trade to, or removing a trade from, the existing netting set.

Question CR-20 i.4

Bank Q1040 • BT P2.T6.332.2

With CVA expressed as a positive cost, each of the following statements about marginal CVA is true EXCEPT:

A. By definition, marginal CVA must be less than or equal to incremental CVA.

B. Unlike incremental CVA, marginal CVA is additive: the sum of the marginal CVAs is the total CVA.

C. Marginal CVA is more relevant than incremental CVA for apportioning CVA fairly across existing trades or assessing the CVA of more than one new trade.

D. Whereas it is difficult to price simultaneous trades with incremental CVA, marginal CVA is the appropriate way to calculate the trade-level CVA contributions of several trades at the same time.

Question CR-20 i.5

Bank Q1041 • BT P2.T6.418.1 (rebuilt)

The table shows standalone CVA, expressed as a positive cost in USD, for four transactions (three interest rate swaps and one cross-currency swap). Some incremental and marginal CVAs are shown as variables. The trades partly offset each other, so any trade executed after another gains a strictly positive netting benefit. Two execution orders are shown: 1-2-3-4 and 4-1-2-3.

		Incremental		Marginal	
Trade	Standalone	1-2-3-4	4-1-2-3	1-2-3-4	4-1-2-3
(1) Payer IRS GBP 5Y	92,593	S	T	V	W
(2) Payer IRS GBP 6Y	124,816				
(3) Payer IRS EUR 5Y	76,006				
(4) CCS GBPUSD 5Y	99,558				
Total	392,973	X	X	Y	Y

Each of the following is true EXCEPT:

A. $S = 92{,}593$: the incremental CVA of the trade executed first equals its standalone CVA.

B. $T < S$: trade 1 has a lower incremental CVA if executed second than if executed first.

C. $W < V$: trade 1 has a lower marginal CVA if executed second than if executed first.

D. $X < 392{,}973$ and $Y < 392{,}973$: total CVA is lower than the sum of the standalone CVAs.

Question CR-20 i.6

Bank Q1038 • BT Mock Q7 (rebuilt)

A netting set contains two derivatives with a counterparty, with CVA reported as a negative number: an interest rate swap (IRS) with a marginal CVA of -1.450 and a cross-currency swap (CCS) with a marginal CVA of -1.550. The CCS was executed first and its incremental CVA is -2.100. Which of the following is the incremental CVA of the IRS?

A. -5.100

B. -1.450

C. -0.900

D. -0.550

Question CR-20 i.7

Workbook A6.4 • S2 • Map #6

A trader at a German bank is pricing a new 5-year EUR swap, notional EUR 150 million, with a counterparty under an existing ISDA netting agreement. The xVA desk reports (Gregory's convention, CVA negative):

- CVA of the existing netting set: −EUR 1.84 million
- CVA of the netting set including the new swap: −EUR 2.27 million
- Standalone CVA of the new swap: −EUR 0.96 million
- 5-year risky annuity: 4.5; counterparty LGD: 60%

Which of the following is closest to the CVA charge for the new swap, quoted as a running cost in bps per annum on its notional?

A. 6.37 bps p.a.

B. 7.85 bps p.a.

C. 14.22 bps p.a.

D. 33.63 bps p.a.

Solutions • CR-20 i

Solution CR-20 i.1 — Answer: C

Bank Q1033 • AP Mock 1 Q8

Step 1 – What incremental CVA measures

The change in netting-set CVA caused by one change (a new trade, unwind, restructuring or new margin terms), fixed at the time of execution.
⇒ That is exactly the amount to charge the desk that executed the trade.

Step 2 – Classify the other uses

Trade-level contributions that sum to the total at a given date, simultaneous trades and order-independent analysis are all marginal CVA uses.

Interpretation

GARP: incremental CVA "makes sense when CVA needs to be charged to individual salespeople, traders, businesses, and, ultimately, clients".

Why the other options are wrong

A is incorrect. An accounting allocation that sums to the total is marginal CVA.
B is incorrect. Simultaneous trades are priced with marginal CVA; incremental CVA would depend on an arbitrary ordering.
D is incorrect. Order independence is the defining feature of marginal CVA.

LO: CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1 and 20.4.2

Solution CR-20 i.2 — Answer: C

Bank Q1034 • BT P2.T6.418.3

Step 1 – Statement I

GARP: incremental CVA "depends on the order in which transactions are executed, but does not change due to subsequent transactions" and suits charging traders and businesses. True.

Step 2 – Statement II

Marginal CVA of existing trades is revised whenever a new trade arrives, so it is unsuitable for pricing new trades; it suits simultaneous trades and decomposing total CVA. True.

Interpretation

The two statements are the two halves of GARP's comparison: stability for charging (incremental) and additivity for attribution (marginal).

Why the other options are wrong

A is incorrect. Both statements match GARP.
B is incorrect. Statement II is also true: marginal CVA does change with new trades.
D is incorrect. Statement I is also true: incremental CVA is fixed once the trade is done.

LO: CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1 and 20.4.2

Solution CR-20 i.3 — Answer: C

Bank Q1039 • BT P2.T6.332.1

Step 1 – The true statements

Incremental CVA is measured against the trades already in the set, so it depends on order (A).

Netting can only reduce exposure, so it is at most the standalone CVA (B). It prices additions and removals (D).

Step 2 – The sign

If the new trade offsets the existing book strongly enough, netting-set EPE falls, incremental EPE is negative, and the incremental CVA is a benefit: below zero as a positive-cost number.
$\Rightarrow$ C is false.

Interpretation

GARP: "Incremental EPE can be negative, due to beneficial netting effects, which will lead to CVA being positive" in its own convention, a benefit rather than a cost. Unwinding a trade is the typical example.

Why the other options are wrong

A is incorrect. True: execution order changes each trade's increment.
B is incorrect. True: netting never makes the increment exceed the standalone CVA.
D is incorrect. True: this is the purpose of incremental CVA.

LO: CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1

Solution CR-20 i.4 Answer: A

Bank Q1040 • BT P2.T6.332.2

Step 1 – The true statements

Marginal CVAs sum to the total (B), are order-independent and so "fairer" (C), and are the way to allocate CVA to several trades done at once (D).

Step 2 – Is there a ranking?

No. In GARP Table 20.5 the IRS has a marginal CVA of -17.7 against incremental CVAs of -13.7 (XCCY first) and -26.3 (IRS first): the marginal figure lies between them.
$\Rightarrow$ A is false.

Interpretation

Marginal CVA is an order-free split of the total; incremental CVA is an order-dependent one. Neither bounds the other by definition.

Why the other options are wrong

B is incorrect. True: additivity defines marginal CVA.
C is incorrect. True: order independence makes it the fair allocation.
D is incorrect. True: simultaneous trades are a stated use of marginal CVA.

LO: CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.2, Table 20.5

Solution CR-20 i.5 Answer: C

Bank Q1041 • BT P2.T6.418.1

Step 1 – Incremental columns

In order 1-2-3-4, trade 1 has nothing to net against, so $S = 92{,}593$ (A true). In order 4-1-2-3 it nets against trade 4, so $T < S$ (B true).

Step 2 – Marginal columns

Marginal CVA does not depend on the order of execution, so $W = V$.
$\Rightarrow$ C is false.

Step 3 – Totals
Both allocations sum to the same netted total, which is below the 392,973 sum of standalone CVAs because the trades partly offset (D true).

Interpretation
Only the incremental columns change with the order; the total and the marginal split are properties of the book as it stands.

Why the other options are wrong
A is incorrect. True: the first trade's increment is its standalone CVA.
B is incorrect. True: executing second lets trade 1 benefit from netting with trade 4.
D is incorrect. True: netting makes the total smaller than the standalone sum.

LO: CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1 and 20.4.2

Solution CR-20 i.6 — Answer: C

Bank Q1038 • BT Mock Q7

Step 1 – Total CVA from the marginal CVAs
$-1.450 + (-1.550) = -3.000.$

Step 2 – Sequential increments also sum to the total
$-2.100 + x = -3.000 \Rightarrow x = -0.900.$
$\Rightarrow$ The IRS, executed second, takes only the remainder because it nets against the CCS.

Interpretation
The CCS was first, so its incremental CVA (-2.100) is its standalone CVA. Both allocations split the same -3.000; they just split it differently.

Why the other options are wrong
A is incorrect. Added the CCS increment to the total instead of subtracting it: $-3.000 - 2.100 = -5.100.$
B is incorrect. Reported the IRS marginal CVA, ignoring that incremental CVA depends on the order.
D is incorrect. Mixed the two allocations: CCS incremental minus CCS marginal, $-2.100 + 1.550 = -0.550.$

LO: CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1 and 20.4.2, Table 20.5

Solution CR-20 i.7 — Answer: A

Workbook A6.4 • S2 • Map #6

Step 1 – Incremental CVA (upfront)
$\text{CVA}_{\text{incr}} = \text{CVA}_{NS'} - \text{CVA}_{NS} = -2.27 - (-1.84) = -\text{EUR } 0.43 \text{ million}$
$\Rightarrow$ The new swap partly offsets existing trades, so it adds only EUR 0.43 million of CVA, less than its standalone EUR 0.96 million, as netting guarantees.

Step 2 – Annuity value of notional
$150 \times 4.5 = 675$ (EUR million-years)

Step 3 – Running spread

$$\frac{-0.43}{675} = -0.000637 = -6.37 \text{ bps p.a.}$$

$\Rightarrow$ In Gregory's convention the CVA is -6.37 bps; the counterparty pays a running cost of 6.37 bps per annum.

Interpretation

The trader can quote about 6.4 bps per annum instead of 14.2 bps standalone, which is why an existing netting relationship lets a bank offer better terms (Cooper and Mello). Dividing by the annuity ignores the CVA of the CVA, so the exact recursive charge is slightly larger. The LGD is a red herring: it is already inside the CVA figures, and applying it again would give 3.82 bps.

Why the other options are wrong

B is incorrect. Charged the netting benefit (standalone minus incremental) instead of the incremental CVA: $(0.96 - 0.43)/675 = 7.85$ bps.

C is incorrect. Used the standalone CVA, ignoring the netting benefit: $0.96/675 = 14.22$ bps.

D is incorrect. Charged the entire post-trade netting set CVA to the new swap: $2.27/675 = 33.63$ bps.

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.3; 20.4.1 eqs. 20.10 to 20.12

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 j

P1 CONCEPT

Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins.

Workbook: A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A21, Map #27, 4 of 16 recall sheets). The August 2026 recall reported "two-way CSA; margin period of risk", and A21 drills the margin period of risk in collateralised CVA.

Vardeez: collateral reduces CVA by lowering expected exposure and does not affect the default probability; minimum transfer amounts and thresholds increase CVA; higher initial margins (negative thresholds) decrease it; as the margin period of risk grows, CVA moves towards the uncollateralised CVA.

Reported exam shape

- **A21·S9** Margin period of risk: its components and its effect on collateralised exposure and CVA, as seen in August.

What the questions look like: conceptual: which CVA input collateral changes, the direction of each margin term, what the MPoR is (and is not), and how to read collateralised CVA results. Every stem states the convention (CVA negative unless quoted as a cost).

Kept outside the TYPE rule: Question CR-20 j.6 (Workbook A21.3, collateralised CVA by the square-root-of-time rule) is a calculation under an explain LO; approved TYPE note: it stays.

Background elsewhere: the CSA terms and the credit support amount (CR-17 b, c), collateralised exposure and the remargin period (CR-19 h), the risks created by margin (CR-17 j).

1. Margin works through EPE only

GARP 20.5.1: the impact of margin on CVA "follows directly from the assessment of the impact of margin on exposure". As with netting, the standard formula of CR-20 c is unchanged; only the EPE is replaced by a collateralised EPE:

$$\mathrm{CVA}_{\mathrm{coll}} = -\mathrm{LGD} \sum_i \mathrm{EPE}_{\mathrm{coll}}(t_i) \times \mathrm{PD}(t_{i-1}, t_i)$$

- **What does not change:** the counterparty's default probability (its credit quality does not depend on the CSA), the LGD on the unsecured claim (the collateral effect is already in the exposure) and the discount factors.
- **Why exposure remains.** Variation margin lags the portfolio value. Between the last margin actually received and the close-out of the trades, the exposure can grow with nothing posted against it. That window is the **margin period of risk (MPoR)**: pre-default it covers valuation and the margin call, receiving collateral (including disputes), settlement and the grace (cure) period; post-default it covers macro-hedging, close-out, rehedging and replacement, and liquidation of the margin securities (CR-17 j).
- **10 business days** is the regulatory minimum with daily calls (more if calls are less frequent; 20 days or more for netting sets of over 5,000 trades, illiquid collateral or exotics, or a history of disputes). Although it is a capital rule, it has become standard in other CVA calculations, including accounting. Daily calls do *not* mean a one-day MPoR.

- **Still modelled when thresholds are zero.** Strong collateralisation might suggest a negligible CVA, but it is now common to model variation margin explicitly. A bank's CVA to banks and other financial institutions can be material even though most of them post margin (GARP Figure 9.4).

2. The margin period of risk and the size of the reduction

With variation margin only, the collateralised exposure is the value change over one MPoR, while the uncollateralised exposure builds up over the whole life of the trade. Under the square-root-of-time rule the best-case reduction is about

$$\text{reduction factor} \approx \sqrt{\frac{\text{maturity}}{\text{MPoR}}} \qquad \text{(same units: 250 business days a year)}$$

GARP's CR-17 example: $\sqrt{5 \times 250/10} = 11$ times for a five-year portfolio (A21.3 uses the same rule). Three points follow.

- **Longer MPoR, less benefit.** The collateralised exposure grows with $\sqrt{\text{MPoR}}$, so collateralised CVA moves towards the uncollateralised CVA as the MPoR lengthens, but not in proportion to it: 20 days instead of 10 makes it about $\sqrt{2} = 1.41$ times larger (Schweser).
- **Profile matters.** For its 10-year swap GARP quotes about 8.4 times, not $\sqrt{2{,}500/10} = 15.8$. A swap's exposure amortises: if the uncollateralised EPE follows $\sqrt{t}\,(T-t)/T$ and the collateralised EPE $\sqrt{\text{MPoR}}\,(T-t)/T$, the averages are $\frac{4}{15}\sqrt{T}$ and $\frac{1}{2}\sqrt{\text{MPoR}}$, a ratio of $\frac{8}{15}\sqrt{2{,}500/10} = 8.4$ (Figure 1).
- **Reality is worse.** Gregory: the reduction is "not as strong as might be thought", and the only way to make counterparty risk negligible is initial margin.

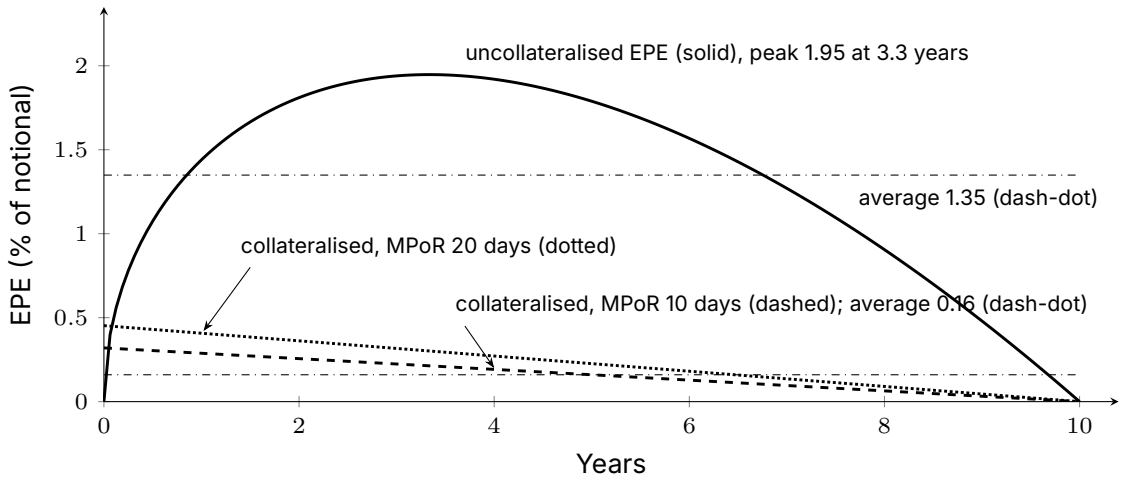


Figure 1. Illustrative 10-year swap profile. Variation margin cuts the exposure to the value change over one MPoR. The average EPE falls from 1.35% to 0.16% of notional, a factor of 8.4, GARP's figure for its 10-year swap. Doubling the MPoR to 20 days lifts the collateralised profile by $\sqrt{2}$ (dotted), back towards the uncollateralised one.

3. Reading GARP's collateralised results

Upfront bp (GARP Table 20.6, MPoR 10 days)	Pay fixed		Receive fixed	
	Uncoll.	Coll.	Uncoll.	Coll.
CVA	-28.9	-3.0	-16.6	-3.2
DVA	7.4	1.5	13.2	1.3
BCVA	-21.5	-1.5	-3.5	-1.9

- **Rounding.** GARP rounds each entry, so the receive-fixed uncollateralised CVA + DVA is -3.4 against the printed -3.5.
- **Symmetry.** Collateralised values are much more symmetric between pay and receive (the MPoR model is symmetric). Uncollateralised, the pay-fixed CVA is almost twice the receive-fixed one.
- **Skew.** The pay-fixed CVA falls 9.6 times, better than 8.4, because its EPE is positively skewed; the receive-fixed CVA falls only 5.2 times because of the negative skew. The receiver's DVA falls 10.2 times (it mirrors the payer's EPE).
- **Small BCVA.** Collateralised BCVA is small, which is some comfort: it is hard to charge collateralised counterparties for CVA (impossible to charge interbank counterparties for hedging), so there is less incentive to price collateralised CVA.
- **Collateralised DVA is even more controversial.** It assumes that, about 10 days before its own default, the party stops posting margin and benefits; even then the gain goes to bondholders, whose recovery the unreturned margin enhances.

4. Thresholds, minimum transfer amounts and initial margin

- **Threshold and MTA.** Exposure up to the threshold is never collateralised, and calls below the MTA are not made, so both raise the collateralised EPE and make CVA more negative. As the threshold grows, CVA tends to the uncollateralised value.
- **Initial margin.** Initial margin received reduces exposure, and so CVA and DVA, towards zero. A threshold is a negative initial margin and vice versa, so GARP Figure 20.14 plots both on one axis: CVA of a 10-year receiver swap with a 10-day MPoR falls towards zero as the initial margin rises and tends to the uncollateralised value as the threshold rises (one-way margin in the party's favour).
- **Modelling choices.** Dynamic initial margin is complex to model, and the CVA left over is small. Parties may argue that CVA is zero where initial margin coverage is high, or ignore the initial margin benefit altogether, which overstates CVA but avoids the modelling.
- **Initial margin posted.** Margin you post and that is segregated does not appear in your CVA; it is a funding cost, the margin value adjustment (MVA).

Term	Effect on CVA	Mechanism and limit
Collateral generally	Smaller (less negative)	Lowers EPE only; PD and LGD unchanged
Longer MPoR	Larger, towards uncollateralised	Collateralised EPE $\propto \sqrt{\text{MPoR}}$
Higher threshold	Larger	Uncollateralised slice grows; large threshold $\rightarrow$ uncollateralised CVA
Higher MTA	Larger	Small calls not made; exposure drifts before a transfer
More initial margin received	Smaller	Overcollateralisation; large initial margin $\rightarrow$ CVA near zero
Initial margin posted (segregated)	None	Not in CVA; creates MVA

Correction to the source notes. Vardeez's table says the margin period of risk "defines the timeframe over which CVA is measured", and Schweser calls it "the number of calendar days over which the CVA is measured". The CVA is still summed over the whole life of the trade. The MPoR is the period over which the collateralised exposure can grow uncollateralised, from the last margin received to close-out, and GARP states it in business days (10 business days, 250 a year). Each date's collateralised EPE depends on it, which is why a longer MPoR pushes CVA towards the uncollateralised value.

Exam traps

- **Collateral and PD.** Margin changes the exposure, never the counterparty's default probability.
- **Polarity of the terms.** Thresholds and MTAs increase CVA; initial margin received decreases it; a longer MPoR increases it.
- **MPoR is not the call frequency.** Daily calls, 10-day MPoR; and the MPoR is not used for funding costs.
- **Linear scaling.** Collateralised CVA grows with $\sqrt{\text{MPoR}}$, and the reduction factor is $\sqrt{\text{maturity}/\text{MPoR}}$, not the ratio itself.
- **Units.** Maturity and MPoR in the same units: business days, 250 a year.
- **Variation margin is not enough.** Only initial margin takes CVA close to zero.
- **Posted versus received.** Segregated initial margin you post does not lower your CVA (it creates MVA).

- **Same factor for every trade.** Skew in the exposure profile makes the reduction differ between pay and receive swaps.

Practice questions: CR-20 j

Question CR-20 j.1

Bank Q1046 • BT P2.T6.332.3 (rebuilt)

A bank computes unilateral CVA with no wrong-way risk as $\text{CVA} = -\text{LGD}\sum_i DF(t_i)\,\text{EE}(t_i)\,\text{PD}(t_{i-1}, t_i)$, a negative number. The four components are (1) the loss given default, LGD, (2) the discount factor, DF, (3) the expected exposure, EE, and (4) the marginal default probability, PD. When the bank begins to collateralise the trades with this counterparty, which component(s) are changed by the collateral?

A. (1) only

B. (3) only

C. (4) only

D. All except (2)

Question CR-20 j.2

Bank Q1047 • Schweser Module Quiz 38.1 Q4

When incorporating netting and collateralisation into the CVA calculation (CVA here quoted as a positive cost, the "CVA price"), which of the following statements is incorrect?

I. Netting increases the CVA price because it reduces exposure when trades are settled.

II. Collateralisation does not change the CVA because it only changes the counterparty's expected exposure.

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Question CR-20 j.3

New • built from GARP Ch.20, 20.5.3 (Initial Margin)

A collateral manager is explaining to the CVA desk how the terms of a new CSA will affect the CVA on a 10-year receiver swap with a 10-business-day MPoR. CVA is reported as a negative number. Which of the following statements is correct?

A. Raising the counterparty's threshold moves CVA towards zero, because fewer margin calls means less operational exposure.

B. Initial margin acts as a negative threshold: as initial margin received grows, CVA tends to zero, and as the threshold grows, CVA tends to the uncollateralised value.

C. Initial margin that the bank posts to the counterparty and that is held segregated reduces the bank's CVA, because it lowers the counterparty's exposure to the bank.

D. A higher minimum transfer amount reduces CVA, because collateral is transferred in larger and therefore more protective amounts.

Question CR-20 j.4 Workbook A21.5 • S3, S9 • Aug-2026

An xVA quantitative analyst at a UK bank is documenting the assumptions of the collateralised CVA model for portfolios under two-way, zero-threshold CSAs with daily variation margin calls (CVA reported as a negative number). Which of the following statements should the documentation include?

A. Because margin is called daily, the model's MPoR should be one business day, matching the contractual period between margin calls.

B. A grace (cure) period after a missed call is a pre-default component of the MPoR, while rehedging and liquidating margin securities are post-default.

C. Because the MPoR measures the delay in receiving margin, the same MPoR should be used to assess the funding cost of the collateralised portfolio.

D. Because the MPoR is the primary driver of the need for initial margin, a longer MPoR reduces the initial margin the bank should require.

Question CR-20 j.5 New • built from GARP Ch.20, 20.5.1 and Ch.17, 17.5.2

A bank currently models its collateralised CVA with a counterparty using a 10-business-day MPoR, two-way zero-threshold CSA and daily variation margin. Because the netting set now exceeds 5,000 trades, it moves to a 20-business-day MPoR. CVA is reported as a negative number. Which of the following best describes the effect on the collateralised CVA?

A. It becomes more negative and moves towards the uncollateralised CVA; the collateralised exposure grows with the square root of the MPoR, so the CVA rises by less than in proportion.

B. It is unchanged, because margin is still called daily and the MPoR matters only for the funding of the portfolio.

C. It doubles in magnitude, because the uncollateralised exposure accumulates over twice as many days.

D. It becomes less negative, because the longer period gives the bank more time to liquidate the collateral at good prices.

Question CR-20 j.6

Workbook A21.3 • S9 • Aug-2026

A CVA desk analyst at a Swiss bank is estimating how far a new two-way, zero-threshold CSA with daily variation margin calls will reduce the CVA on an 8-year portfolio with a hedge fund. The data are:

- Uncollateralised CVA: −EUR 4.2 million
- The netting set contains over 5,000 trades, so the bank applies a 20-business-day MPoR
- 250 business days per year; no initial margin

Using the square-root-of-time approximation in the reading (CVA negative), which of the following is closest to the best-case collateralised CVA?

A. −EUR 0.420 million

B. −EUR 0.348 million

C. −EUR 0.297 million

D. −EUR 0.042 million

Question CR-20 j.7

New • built from GARP Ch.20, 20.5.2 Table 20.6

A risk manager reviews GARP's upfront CVA and DVA (bp, Gregory's convention: CVA negative, DVA positive) for 10-year interest rate swaps, uncollateralised and collateralised with a 10-business-day MPoR. GARP notes that the simple square-root-of-time estimate of the CVA reduction for this maturity is about 8.4 times.

	Pay fixed		Receive fixed	
	Uncoll.	Coll.	Uncoll.	Coll.
CVA	−28.9	−3.0	−16.6	−3.2
DVA	7.4	1.5	13.2	1.3

Which of the following statements is correct?

A. Collateral reduced the CVA of both swaps by about the same factor, as the square-root-of-time estimate predicts.

B. The pay-fixed swap's CVA fell by more than the estimate because its EPE is positively skewed, the receive-fixed swap's by less, and the collateralised CVAs are much more symmetric between the two swaps.

C. Because the collateralised BCVA is small, collateralised interbank counterparties can easily be charged for the remaining CVA.

D. Collateralised DVA is less controversial than uncollateralised DVA, because daily margin removes any benefit from the party's own default.

Solutions • CR-20 j

Solution CR-20 j.1 — Answer: B

Bank Q1046 • BT P2.T6.332.3

Step 1 – What collateral does

Collateral is an asset the bank can keep if the counterparty defaults, so it reduces the amount at risk at each date: the expected exposure (3).

Step 2 – What it does not touch

The counterparty's PD (4) depends on its credit quality, not on the CSA; the discount factor (2) depends on interest rates; the LGD (1) is the loss rate on the unsecured claim, and the collateral effect is already captured in the lower EE.
$\Rightarrow$ Only (3) changes.

Interpretation

GARP: margin "only changes EPE, and hence the same formula may be used with EPE based on assumptions of collateralisation". Counting the collateral in the LGD as well would double count it.

Why the other options are wrong

A is incorrect. Treats collateral as a lower loss rate; in the standard formula it enters through the exposure.
C is incorrect. Confuses collateral with credit protection; the counterparty's default probability is unchanged.
D is incorrect. Only EE changes; PD in particular is independent of the collateral.

LO: CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.5.1

Solution CR-20 j.2 — Answer: C

Bank Q1047 • Schweser Module Quiz 38.1 Q4

Step 1 – Statement I

Netting reduces exposure, so it *reduces* the CVA price; it never increases it. I is incorrect.

Step 2 – Statement II

CVA is driven directly by expected exposure, so lowering the exposure lowers the CVA. Collateral changes only the exposure, not the PD, but that is precisely how it reduces CVA. II is incorrect.
$\Rightarrow$ Both statements are incorrect.

Interpretation

Netting and margin both work through the exposure term of the same formula. Neither touches the counterparty's default probability, and both reduce CVA.

Why the other options are wrong

A is incorrect. Spots I but accepts II; CVA depends directly on the exposure.
B is incorrect. Spots II but accepts I; netting lowers exposure and so lowers CVA.
D is incorrect. Accepts both incorrect statements.

LO: CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1 and 20.5.1; SchweserNotes 2026 Book 2, Reading 38, LO 38.h and 38.j

Solution CR-20 j.3 Answer: B

New • built from GARP Ch.20, 20.5.3

Step 1 – Threshold and initial margin on one axis

GARP: "a threshold can be seen to be a negative initial margin and vice versa. For high thresholds, CVA tends to the uncollateralised value, whilst for high initial margin it tends to zero."
$\Rightarrow$ B restates Figure 20.14.

Step 2 – Posted initial margin and the MTA

Segregated initial margin the bank *posts* "would not show up in any of the above calculations"; it is an MVA cost. A higher MTA leaves exposure uncollateralised until the call is large enough, so it raises CVA.

Interpretation

Every term that leaves part of the exposure uncovered (threshold, MTA, longer MPoR) makes CVA more negative; only margin *received* beyond the portfolio value (initial margin) pushes it towards zero.

Why the other options are wrong

A is incorrect. Polarity flip: a higher threshold leaves more exposure uncollateralised, so CVA moves towards the uncollateralised value.
C is incorrect. Posted and received swapped: initial margin the bank posts protects the counterparty and creates MVA for the bank; it does not reduce the bank's CVA.
D is incorrect. Polarity flip: a higher MTA delays transfers and increases CVA.

LO: CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.5.3, Figure 20.14; SchweserNotes 2026 Book 2, Reading 38, LO 38.j

Solution CR-20 j.4 Answer: B

Workbook A21.5 • S3, S9 • Aug-2026

Step 1 – Components

Gregory splits the MPoR into pre-default (valuation and margin call; receiving collateral, including any dispute; settlement; grace or cure period) and post-default (macro-hedging; close-out of transactions; rehedging and replacement; liquidation of margin securities).
$\Rightarrow$ The grace period runs before the counterparty is legally in default; liquidation of the margin it posted can only happen after.

Step 2 – Effect on collateralised exposure and CVA

Market risk over the MPoR is uncollateralised. With variation margin only, the best-case reduction is about $\sqrt{\text{maturity}/\text{MPoR}}$; reality is worse. Margin changes only EPE in the CVA formula: in Table 20.6 the 10-year pay-fixed swap's CVA falls from -28.9 to -3.0 bps with a 10-day MPoR. Initial margin (overcollateralisation) pushes CVA towards zero; a threshold acts as a negative initial margin.

Interpretation

The model should use a 10-business-day MPoR at least (more for large or disputed netting sets) and size initial margin to the MPoR, since the choice of initial margin is closely tied to the assumed MPoR.

Why the other options are wrong

A is incorrect. Sibling swap: the margin call frequency is only one component of the MPoR; "it could, therefore, be that the margin call frequency is daily, but the MPoR is assumed to be 10 days".

C is incorrect. Scope error: the MPoR is defined assuming the counterparty's default and is "not relevant when assessing funding costs"; funding uses the normal, much shorter posting delay.
D is incorrect. Polarity flip: the MPoR is indeed the primary driver of the need for initial margin, but a longer MPoR leaves more uncollateralised market risk and so raises the initial margin needed.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7, 17.3.3 and 17.5.2; Ch.17 (GARP CR Ch.20), 20.5.2 and 20.5.3

Solution CR-20 j.5 Answer: A

New • built from GARP Ch.20, 20.5.1 and Ch.17, 17.5.2

Step 1 – Direction

The MPoR is the window over which the exposure can grow uncollateralised. Lengthening it raises the collateralised EPE at every date, so the CVA becomes more negative and moves towards the uncollateralised CVA (Vardeez, Schweser).

Step 2 – Size

Under the square-root-of-time rule, the value change over the MPoR scales with $\sqrt{\text{MPoR}}$: doubling 10 days to 20 makes the collateralised exposure, and so the CVA, about $\sqrt{2} = 1.41$ times larger, not twice as large.

Interpretation

This is why Basel III imposes 20 days or more on large or troublesome netting sets: the longer delay to close-out leaves more market risk uncollateralised. The daily call frequency is unchanged and irrelevant to the comparison.

Why the other options are wrong

B is incorrect. Sibling swap: the call frequency is not the MPoR, and the MPoR is a counterparty risk concept that is *not* used for funding.
C is incorrect. Linear scaling: the exposure grows with the square root of time, so doubling the MPoR multiplies it by about 1.41.
D is incorrect. Polarity flip: a longer MPoR adds uncollateralised market risk (and liquidation risk on the margin held); it does not reduce CVA.

LO: CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7, 17.5.2; Ch.17 (GARP CR Ch.20), 20.5.1; SchweserNotes 2026 Book 2, Reading 38, LO 38.j

Solution CR-20 j.6 Answer: A

Workbook A21.3 • S9 • Aug-2026

Step 1 – Maturity in business days

$8 \times 250 = 2{,}000$ business days

Step 2 – Reduction factor

$$\sqrt{\frac{\text{maturity}}{\text{MPoR}}} = \sqrt{\frac{2{,}000}{20}} = \sqrt{100} = 10$$

$\Rightarrow$ With only variation margin, exposure is limited to the market move over the 20-day MPoR instead of up to 8 years, so risk falls by about 10 times at best.

Step 3 – Collateralised CVA

$$\text{CVA}_{\text{coll}} \approx \frac{-4.2}{10} = -\text{EUR } 0.420 \text{ million}$$

Interpretation

Even a zero-threshold daily CSA leaves about 10% of the CVA, and Gregory notes reality is worse than this best case; only initial margin can push CVA towards zero. The longer MPoR for large netting sets cuts the benefit compared with the 10-day standard. Margin changes only the EPE in the CVA formula, so the PD and LGD are untouched.

Why the other options are wrong

B is incorrect. Used 365 calendar days with a business-day MPoR: $4.2/\sqrt{2{,}920/20} = 0.348$.
C is incorrect. Used the standard 10-day MPoR instead of the 20 days applied to netting sets over 5,000 trades: $4.2/\sqrt{200} = 0.297$.
D is incorrect. Scaled linearly instead of by the square root: $4.2 \times 20/2{,}000 = 0.042$.

LO: CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7, 17.5.2 and 17.5.4; Ch.17 (GARP CR Ch.20), 20.5.1 and 20.5.2

Solution CR-20 j.7 — Answer: B

New • built from GARP Ch.20, 20.5.2 Table 20.6

Step 1 – Reduction factors

Pay fixed: $28.9/3.0 = 9.6$ times. Receive fixed: $16.6/3.2 = 5.2$ times. The estimate is 8.4.
$\Rightarrow$ The payer's CVA falls by more than the estimate, the receiver's by less: GARP attributes this to the positive skew of the payer's EPE and the negative skew of the receiver's.

Step 2 – Symmetry

Uncollateralised, the two CVAs differ by a factor of almost two (-28.9 against -16.6); collateralised they are close (-3.0 against -3.2), because the MPoR model is symmetric.

Interpretation

Margin compresses the exposure to the move over one MPoR, which looks much the same for a payer and a receiver; what remains of the original asymmetry is the skew. BCVA ends up small (-1.5 and -1.9 bp).

Why the other options are wrong

A is incorrect. The factors are 9.6 and 5.2, not the same; the estimate ignores the skew of each profile.
C is incorrect. GARP says the opposite: it is "clearly not possible" to charge interbank counterparties for hedging, and the small BCVA is only some comfort because such costs are hard to charge.
D is incorrect. GARP calls collateralised DVA "potentially even more controversial": it relies on the party ceasing to post margin about 10 days before its own default.

LO: CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.5.1 and 20.5.2, Table 20.6

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 k

P1 CONCEPT

Describe wrong-way risk and contrast it with right-way risk.

Workbook: A20 Wrong-Way and Right-Way Risk • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: wrong-way risk came up in the August 2026 sitting as two theory questions ("RWR & WWR (2 theory questions): confusing, study...", Map #26).
Vardeez: CVA, BCVA and DVA are a GARP favourite and the chapter cannot be skipped. For this LO, know wrong-way risk as a single correlation between your exposure and the counterparty's default probability, and that WWR and RWR push CVA in opposite directions.

Reported exam shapes

- **A20·S1** Classify trades as wrong-way or right-way, the plain form seen in August.
- **A20·S2** The test: is the exposure linked to the counterparty's credit quality, and in which direction?

What the questions look like: all conceptual. Definitions with a polarity flip (WWR raises CVA, RWR lowers it), "how many statements are correct" items, the challenges of quantifying WWR, and a four-trade classification in which each distractor misreads one trade. There are no calculations.

Shared background: this block carries the wrong-way risk background for CR-20 k to p. **Worked elsewhere:** A20.2 (collateral, CCPs and models) at Question CR-20 m.6; further classification drills under CR-20 l. Bank Q1059 and Q1062 (definitions, not classification) are re-filed here from CR-20 l.

1. What wrong-way risk is

The CVA formula of CR-20 c multiplies three ingredients that it treats as unrelated: the loss given default, the discounted expected positive exposure at each future date, and the counterparty's probability of default in each period (GARP eq. 20.3):

$$\text{UCVA} = -\text{LGD}\sum_{i=1}^{m}\text{EPE}(t_i)\times\text{PD}(t_{i-1},t_i)$$

GARP calls this "the commonly-made simplification of no WWR, which assumes that EPE, PD, and LGD are not related". Wrong-way and right-way risk are what happens when the exposure and the default probability are *not* independent.

- **Wrong-way risk (WWR):** "an unfavourable dependence between exposure (EPE) and counterparty credit quality: the exposure is high when the counterparty is more likely to default, and vice versa." It **increases CVA**.
- **Right-way risk (RWR):** the dependence is favourable. The exposure is low when the counterparty is more likely to default and high when it is strong. "Right-way situations will reduce CVA."
- **Why derivatives.** Counterparty risk is roughly exposure times default probability at a given loss rate. A loan's exposure is roughly fixed; an OTC derivative's exposure moves with the market, so it can move with the counterparty's credit quality.
- **One correlation.** Positive correlation between exposure and default probability is WWR, negative correlation is RWR, and zero is the standard CVA of CR-20 c.

- **Schweser's framing.** Any dependence that produces an overall *increase* in counterparty risk is WWR; any that produces an overall *decrease* is RWR. Schweser adds that WWR reduces DVA and RWR increases it; GARP says only that the dependence "would have a clear impact on CVA and DVA".

The two-question test (shape S2)

1. **What makes my exposure to this counterparty rise?** I must be *owed* money for there to be exposure. A party that has sold an option, or sold CDS protection, has no growing claim on the buyer.
2. **Does the same move or event weaken the counterparty?** If yes, the trade is wrong-way. If the same move *strengthens* the counterparty, it is right-way. If there is no link, the independence assumption of standard CVA is reasonable.

Vardeez's shorthand: in a wrong-way trade, the move that is good for you is the move that is bad for them. State the roles (who is owed, who pays) before classifying.

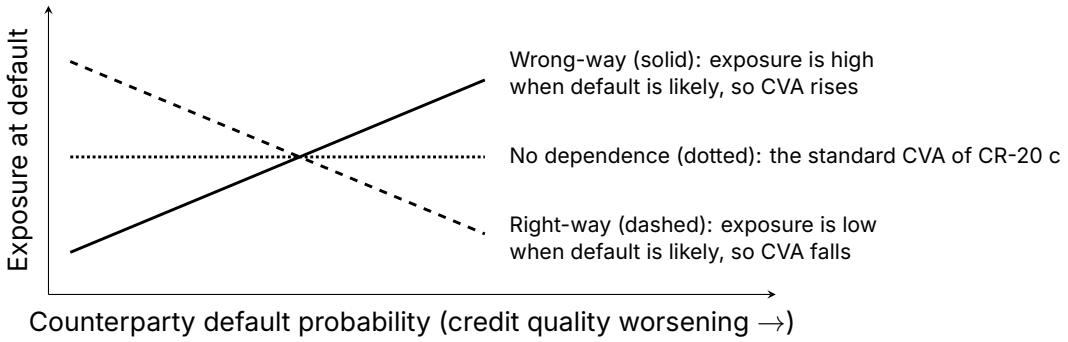


Figure 1. Exposure conditional on default, plotted against the counterparty's default probability (schematic). Standard CVA assumes the flat dotted line. Wrong-way risk tilts the line up, right-way risk tilts it down.

2. Why it matters

WWR is "difficult to identify, model, and hedge due to the often subtle macro-economic and structural effects that cause it". It may often be reasonable to ignore it, but "its manifestation can be potentially dramatic": dealers lost heavily in the Asian crisis of 1997/1998 (sovereign and corporate defaults together with collapsing local currencies), and from 2007 banks lost on protection bought from monoline insurers. It is often natural and unavoidable: a mortgage lender faces falling property prices and rising defaults in the same recession.

- **Empirical evidence.** Duffee (1998) found corporate defaults cluster when interest rates are falling (a recession brings rate cuts and defaults together). Banks' uncollateralised receiver swaps moved in the money as the sovereign or corporate counterparty weakened, a "cross-gamma" between spreads and rates that appears even without defaults.

3. General versus specific wrong-way risk

Regulators separate WWR driven by macro-economic relationships (general) from WWR driven by causal links between the exposure or the margin and the counterparty's default (specific). GARP's Table 20.7:

General WWR	Specific WWR
Based on macro-economic behaviour	Based on structural relationships often not captured by real-world experience
Relationships may be detectable using historical data	Hard to detect except by knowledge of the market, the counterparty and the economic rationale of the transaction
Potentially can be incorporated into pricing models	Difficult to model; naive correlation assumptions are dangerous; address qualitatively, e.g. by stress testing
Should be priced and managed correctly	Should, in general, be avoided as it may be extreme

GARP's illustration of why specific WWR escapes historical data: in 2010 the European sovereign debt crisis brought weaker sovereigns and a weaker euro together, a link that history had not shown because neither the sovereigns nor the currency had suffered strong adverse credit effects before. A bank buying CDS protection on its own sovereign from a domestic bank is the textbook specific case (Question CR-20 k.6).

4. Bringing WWR into CVA

Conditional exposure. The most direct fix is to replace the unconditional EPE in eq. 20.3 by the EPE *conditional on the counterparty defaulting at that date*:

$$\text{CVA}_{\text{WWR}} = -\text{LGD} \sum_{i=1}^{m} \text{EPE}\left(t_i \,|\, \tau = t_i\right) \times \text{PD}(t_{i-1}, t_i)$$

With WWR the conditional EPE exceeds the unconditional one; with RWR it is smaller. LGD, the discount factors and the PDs keep their usual roles. A qualitative version is the regulatory alpha factor: a conservative alpha plus stressed data in the exposure estimate partially capitalise general WWR.

One abstract correlation. In GARP's simple model a single correlation links EPE and default. With $+50\%$ correlation WWR roughly doubles the EPE; with -50% RWR reduces it by at least half (Figure 2).

Credit quality. GARP's Figure 20.16 shows the conditional exposure rising as the counterparty's credit quality *improves*: for a strong name, default is a less probable event and therefore a bigger shock when it happens. The general conclusion: **WWR increases as the credit quality of the counterparty increases**. Schweser makes the same point.

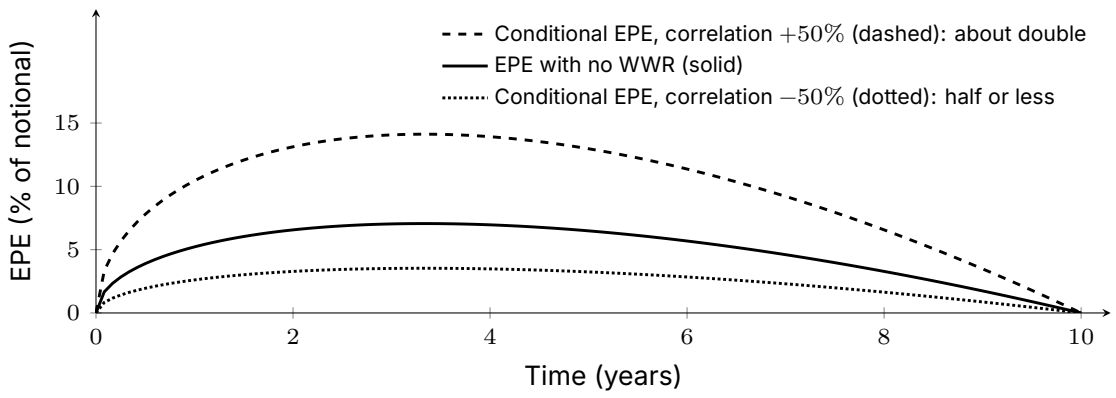


Figure 2. Stylised EPE profile of a 10-year swap and the effect of wrong-way and right-way dependence, after GARP Figure 20.15: positive correlation between exposure and default probability raises the conditional EPE, negative correlation lowers it.

Why modelling is hard. GARP lists three problems:

- **Uninformative historical data.** WWR may be subtle and not revealed by any empirical data, such as a historical time series of correlations (Question CR-20 k.4).
- **Misspecification.** The dependence may be causal (cause and effect) rather than a correlation. Zero correlation does not prove independence: if X is normal and $Y = X^2$, X and Y are uncorrelated but far from independent.
- **Direction.** Low rates usually come with recessions and wide spreads, but an adverse credit environment with high rates is not impossible.

The four modelling approaches (hazard rate, structural, parametric, jump) are covered under CR-20 p.

> **Correction to the source notes.** Vardeez's CR-20 k table describes right-way risk as "exposure decreases as the counterparty's credit quality improves". That pattern (exposure low when the counterparty is strong, so high when it is weak) is wrong-way. GARP defines right-way risk as the favourable dependence: the exposure is *low when the counterparty is more likely to default*, so it decreases as credit quality worsens (equivalently, increases as credit quality improves). The CVA row (WWR increases CVA, RWR reduces it) is correct.

	Wrong-way risk	Right-way risk
Dependence	Unfavourable: exposure high when default is likely	Favourable: exposure low when default is likely
Correlation, exposure and PD	Positive	Negative
EPE conditional on default	Conditional EPE higher	Conditional EPE lower
CVA	Increases	Decreases
DVA (Schweser)	Decreases	Increases
Typical source	Put bought from a correlated writer; receiving fixed into a recession; buying CDS from a linked seller	Hedges: a commodity producer hedging output; equity calls; selling CDS protection on a linked name
Management	General: price and manage. Specific: in general avoid	Welcome: reduces CVA

Exam traps

- **Polarity flip on CVA.** WWR raises CVA and RWR lowers it, relative to the independence case. Statements that reverse this are false.
- **Polarity flip on credit quality.** "WWR is greatest for weak, low-rated counterparties" is false: WWR increases with credit quality.
- **Hedging versus speculation.** Right-way risk arises from hedging (the producer owes money when it is profitable). Speculation does not automatically produce it.
- **Role misassignment.** An option writer or a CDS protection seller has no growing claim on the buyer, so it cannot suffer WWR from that trade.
- **Value rise is not WWR by itself.** A trade gaining value only raises exposure. It is WWR only if the same event weakens the counterparty.

- **Zero correlation read as independence.** $Y = X^2$ has zero correlation with X. Historical correlations can miss WWR entirely.
- **Specific WWR priced with a correlation.** GARP says specific WWR should be addressed qualitatively (stress testing) and in general avoided.

Practice questions: CR-20 k

Question CR-20 k.1

Bank Q1049 • AnalystPrep Q.2688

The risk that the exposure to a counterparty is adversely correlated with the credit quality of that counterparty is known as:

A. Concentration risk

B. Settlement risk

C. Wrong-way risk

D. Credit migration risk

Question CR-20 k.2

Bank Q1059 • Schweser Module Quiz 38.3 Q1 (rebuilt)

How many of the following statements regarding wrong-way risk (WWR) and right-way risk (RWR) are correct?

1. Co-movement in exposure and counterparty default probability that produces a decline in overall counterparty risk is an example of WWR.
2. Co-movement in exposure and counterparty default probability that produces an increase in overall counterparty risk is an example of RWR.
3. Co-movement in exposure and counterparty default probability that produces neither a decline nor an increase in overall counterparty risk is an example of WWR.
4. A rise in exposure that coincides with an improvement in the counterparty's credit quality is an example of WWR.

A. None

B. Two

C. Three

D. All

Question CR-20 k.3 Bank Q1062 • Schweser Module Quiz 38.3 Q4 (rebuilt as single best answer)

A junior analyst is preparing a briefing note on counterparty risk for the credit committee. Which of the following statements should the note include?

A. Speculation in normally functioning derivatives markets automatically produces right-way risk.

B. Relative to the case of no dependence, right-way risk increases CVA and wrong-way risk reduces it.

C. Loan exposure fluctuates with market conditions, whereas OTC derivative exposure is normally a fixed amount, so wrong-way risk is mainly a lending problem.

D. Counterparty risk is roughly exposure times default probability at a given loss rate, and unlike a loan, an OTC derivative's exposure moves with market conditions.

Question CR-20 k.4 Bank Q1050 • AnalystPrep Q.1984 (options reordered)

One of the challenges experienced when attempting to quantify wrong-way risk has a lot to do with:

A. Difficult and time-consuming calculations

B. Uninformative historical data

C. The need for high-level computer software that can sometimes be expensive

D. A lack of well-developed models

Question CR-20 k.5 Bank Q1048 • AP Mock 3 Q2 (rebuilt)

An investor holds a USD 2 million bond issued by Issuer A and buys credit protection on Issuer A through a CDS from Issuer B (the protection seller). The investor will be paid only if Issuer A defaults and Issuer B is able to honour its obligation. If the correlation between the default of Issuer A and the default of Issuer B increases, what is the impact on the value of this CDS protection to the investor (the price the investor should be willing to pay)?

A. It decreases, because a default of A becomes more likely to coincide with a default of B (wrong-way risk), lowering the expected recovery of the protection payment.

B. It increases, because a higher chance of joint default means the protection is more likely to be needed.

C. It is unchanged, because the CDS value depends only on the default probability of the reference entity A.

D. It may rise or fall depending on the economic conditions of the country, so the direction cannot be determined.

Question CR-20 k.6 Workbook A20.1 • S1, S2 • Aug-2026 • Map #26

A counterparty risk officer at a Singapore bank is classifying four proposed uncollateralised trades before a credit committee. The bank's rule is to tag a trade wrong-way if its exposure to the counterparty is expected to rise when the counterparty's credit quality falls, and right-way if the reverse holds.

- Trade 1: the bank receives fixed from a domestic manufacturer on a 7-year interest rate swap, agreed while the economy is strong.
- Trade 2: a copper producer that is hedging its future output pays the floating copper price to the bank and receives a fixed price on a commodity swap.
- Trade 3: in an FX forward with an emerging-market sovereign, the bank pays the sovereign's local currency and receives US dollars.
- Trade 4: the bank buys CDS protection on its domestic sovereign from a domestic bank.

Which of the following classifications should the officer record?

A. Trade 1 is right-way, because a recession that weakens the manufacturer would also push rates up and move the swap against the bank.

B. Trade 2 is right-way whatever the producer's motive, because a producer's commodity swap always pays the bank when the producer is profitable.

C. Trade 3 is wrong-way, because a weakening of the local currency would raise the bank's exposure as the sovereign's credit quality deteriorates.

D. Trade 4 is right-way, because buying protection from a seller linked to the reference entity is the right-way side of a CDS.

Solutions • CR-20 k

Solution CR-20 k.1 — Answer: C

Bank Q1049 • AnalystPrep Q.2688

Step 1 – Read the definition in the stem

The stem describes a dependence between two things: the *size of the exposure* and the *credit quality* of the counterparty, with the link running the adverse way.

⇒ Exposure rising as credit quality falls is GARP's definition of wrong-way risk.

Step 2 – Check the effect

An unfavourable dependence makes the conditional EPE larger than the unconditional EPE, so CVA rises above the standard (independent) figure.

Interpretation

Every WWR question rests on this definition. The opposite, favourable dependence is right-way risk, which reduces CVA.

Why the other options are wrong

A is incorrect. Concentration risk is a large exposure to one name, sector or region. It says nothing about how the size of the exposure moves with the counterparty's credit quality.

B is incorrect. Settlement risk is the risk of delivering one leg of a transaction without receiving the other.

D is incorrect. Credit migration risk is the risk of a change in rating or credit quality, with no link to the size of the exposure.

LO: CR-20 k: Describe wrong-way risk and contrast it with right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1 Overview

Solution CR-20 k.2 — Answer: A

Bank Q1059 • Schweser Module Quiz 38.3 Q1

Step 1 – Fix the definitions

A dependence that raises overall counterparty risk (CVA up) is WWR. A dependence that lowers it (CVA down) is RWR. No net effect is the independence case, neither WWR nor RWR.

Step 2 – Test statements 1 to 3

Statement 1 calls a decline in overall risk WWR: false, it is RWR. Statement 2 calls an increase RWR: false, it is WWR. Statement 3 calls no net change WWR: false, it is neither.

⇒ The first two simply swap the labels; the third invents a label for the neutral case.

Step 3 – Test statement 4

Exposure rising while the counterparty becomes stronger means the exposure is high when default is *less* likely: a favourable dependence, so RWR, not WWR. False.

⇒ None of the four statements is correct.

Interpretation

Schweser's original fourth statement (a decline in exposure with a rise in default probability "is right-way risk") was arguable: by GARP's dependence definition that pattern is the favourable one, while Schweser's key says only that it "may or may not" lower overall risk. It has been replaced by an unambiguous version; the key is unchanged.

Why the other options are wrong

B is incorrect. Two: accepts the swapped labels in statements 1 and 2.

C is incorrect. Three: accepts the swapped labels and the invented label for the neutral case.

D is incorrect. All: reads every statement with the definitions reversed.

LO: CR-20 k: Describe wrong-way risk and contrast it with right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1; SchweserNotes 2026 Book 2, Reading 38, LO 38.k

Solution CR-20 k.3 Answer: D

Bank Q1062 • Schweser Module Quiz 38.3 Q4

Step 1 – What counterparty risk is made of

Counterparty risk is roughly the product of exposure and the counterparty's default probability at a given loss rate. For a loan the exposure is close to a fixed amount; for an OTC derivative it moves with market conditions.
⇒ D is correct: it is because derivative exposure moves with the market that it can move *with* the counterparty's credit quality.

Step 2 – Check the direction of the CVA effect

WWR increases CVA and RWR reduces it, relative to no dependence. B reverses both.

Step 3 – Hedging versus speculation

GARP's commodity example: the right-way character "arises due to hedging (as opposed to speculation)". A hedger owes money when its business is doing well. A speculator's payments have no such link to its health.

Interpretation

The three false options each reverse one basic fact. The original Schweser item asked how many of four statements were correct (answer: none); it has been rebuilt as a single best answer with the corrected facts merged into option D.

Why the other options are wrong

A is incorrect. Scope error: hedging, not speculation, produces right-way risk.
B is incorrect. Polarity flip: right-way risk lowers CVA and wrong-way risk raises it.
C is incorrect. Reversed: it is derivative exposure that fluctuates and loan exposure that is roughly fixed. WWR is central to derivatives for exactly that reason.

LO: CR-20 k: Describe wrong-way risk and contrast it with right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1; SchweserNotes 2026 Book 2, Reading 38, LO 38.k

Solution CR-20 k.4 Answer: B

Bank Q1050 • AnalystPrep Q.1984

Step 1 – GARP's list of problems

Modelling the dependence between default probability and exposure faces three problems: uninformative historical data, misspecification of the relationship, and an unclear direction.

Step 2 – Why history fails

WWR "may be subtle and not revealed via any empirical data, such as a historical time series analysis of correlations". Specific WWR in particular is driven by structural links that may never have appeared in the data (the 2010 euro example).
⇒ The data problem is the stated challenge.

Interpretation

Even a measured correlation of zero is not reassuring: if the link is causal or non-linear ($Y = X^2$), correlation can miss it. That is why GARP wants specific WWR handled qualitatively, for example by stress testing.

Why the other options are wrong

A is incorrect. Computation can be heavy, but GARP does not list it as the fundamental problem; the intensity approach is described as relatively tractable.
C is incorrect. Software cost is a practical matter, not one of the conceptual problems GARP names.
D is incorrect. Models exist (hazard rate, structural, parametric, jump; CR-20 p). The difficulty is calibrating them when the data say little.

LO: CR-20 k: Describe wrong-way risk and contrast it with right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.2 Quantification of WWR in CVA

Solution CR-20 k.5 Answer: A

Bank Q1048 • AP Mock 3 Q2

Step 1 – Where the investor's exposure to B comes from

The CDS pays when reference entity A defaults, so the investor's exposure to seller B is large exactly in the states where A has defaulted or its spread has widened.

Step 2 – Apply the test

Higher default correlation between A and B raises the chance that B defaults when A does. The exposure to B is then highest when B is most likely to fail: wrong-way risk.
$\Rightarrow$ The CVA on the protection rises, so its counterparty-risk-adjusted value (risk-free value less CVA) falls.

Step 3 – The limit

At perfect correlation the protection pays only the recovery on B's obligation. GARP's illustration (reference entity 250 bp, seller 500 bp, both LGDs 60%): the fair premium falls to about 200 bp at 60% correlation and to just above $250 \times 40\% = 100$ bp at 100%.

Interpretation

Buying protection is the wrong-way side when reference entity and seller are linked; selling protection in the same situation is right-way.

Why the other options are wrong

B is incorrect. Confuses the need for protection with its reliability. A joint default is the state in which the protection fails to pay.
C is incorrect. Ignores counterparty risk on B. The dependence between A and B changes the expected payment even though A's own default probability is unchanged.
D is incorrect. The direction is determinate: positive dependence between the exposure and the seller's default is WWR and lowers the value of protection.

LO: CR-20 k: Describe wrong-way risk and contrast it with right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1; 20.6.5 Credit Derivatives

Solution CR-20 k.6 Answer: C

Workbook A20.1 • S1, S2 • Aug-2026

Step 1 – Apply the test to Trade 3

What raises the bank's exposure? The bank pays local currency and receives dollars, so a fall in the local currency makes the forward more valuable to the bank. Does the same event weaken the counterparty? Yes: a currency collapse and a sovereign default go together, either way round (Gregory: "paying its local currency in an FX forward or cross-currency swap"). Unfavourable dependence, so wrong-way.
$\Rightarrow$ Levy and Levin put residual currency values at sovereign default at 17% (AAA) to 62% (CCC):

the exposure jumps exactly when the counterparty fails.

Step 2 – Check the others with the same two questions

Trade 1: rates fall in a recession, so the receive-fixed swap gains as the manufacturer weakens: wrong-way. Trade 2: the producer owes the bank only when copper is high and it is profitable: right-way, because it is a hedge. Trade 4: the bank's exposure is the protection's value, which rises as the sovereign weakens, and the domestic bank weakens with its sovereign: extreme wrong-way.

Interpretation

The committee should price or limit Trades 1 and 3 (general WWR) and decline Trade 4, a case of specific WWR that Gregory says should in general be avoided; Trade 2 reduces CVA.

Why the other options are wrong

A is incorrect. Polarity flip: Gregory says rates are likely to be cut in a recession, so the receive-fixed swap moves in the bank's favour as the manufacturer weakens. That is wrong-way, not right-way.

B is incorrect. Scope or condition error (absolute claim): Gregory ties the right-way character to hedging ("as opposed to speculation"). Dropping the condition with "whatever the motive" turns a true statement false.

D is incorrect. Role misassignment: Gregory gives the right-way side to the protection seller. The bank is the buyer: its exposure rises as the reference sovereign weakens, which drags the domestic bank (the seller) down with it, so it is extreme wrong-way.

LO: CR-20 k: Describe wrong-way risk and contrast it with right-way risk. CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1 and Table 20.7

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 I

P1 CONCEPT

Identify examples of wrong-way risk and examples of right-way risk.

Workbook: A20 Wrong-Way and Right-Way Risk • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A20): the August 2026 wrong-way risk questions were classification items, and Map #26 frames the skill as "identify if there is any correlation".
Vardeez: a table of six classic transactions (OTC put, CDS, FX transaction, currency swap, interest rate swap, commodity forward) with one rule: in a wrong-way trade, the move that is good for you is the move that is bad for them. Right-way examples are the same trades with the link reversed.

Reported exam shapes

- **A20·S1** Classify trades as wrong-way or right-way, the plain form seen in August.

What the questions look like: pick the wrong-way (or right-way) trade from four; "I only / II only / both / neither" items on guarantors and monolines; statements about puts, calls, currencies and swaps with one direction reversed; and scenarios in which the same CDS is bought from two different sellers. All conceptual.

Worked elsewhere: A20.1 (the four-trade classification) under CR-20 k. The definitions and the two-question test are in CR-20 k, Section 1.

1. One test for every product

Every example below is the two-question test of CR-20 k applied to a different product: (1) what makes my exposure to this counterparty rise, and (2) does the same move weaken the counterparty? Write down who is owed money before answering. A party that has sold an option or sold CDS protection has no growing claim on its counterparty, so the question of wrong-way risk does not arise for it on that trade.

2. The classic examples

- **Equity options.** Buying a put on a stock or index whose fortunes are highly correlated with the counterparty's is "an obvious case of WWR", for example a put on one bank's stock bought from another bank. "Correspondingly, equity call options should be right-way products." Schweser adds that out-of-the-money puts carry more WWR than in-the-money puts: an OTM put gains value only in a large fall, the very state in which the writer is likely to be in trouble.
- **FX forwards and cross-currency swaps.** Consider any FX contract in terms of a weakening currency and a simultaneous deterioration of the counterparty. Paying a sovereign its local currency (or hedging that trade with a bank in the same region) is wrong-way. The link runs either way: a weak currency signals a weak economy, and a sovereign or large-firm default can itself trigger the devaluation. A cross-currency swap is like a loan collateralised by the opposite currency. Levy and Levin find average residual currency values at sovereign default of 17% (AAA) to 62% (CCC).
- **Interest rate swaps.** A corporate paying fixed while the economy is strong is wrong-way for the bank receiving fixed, because rates are likely to be cut in a recession. During a recovery, rates may rise, so paying fixed may be right-way. Schweser's case: euro receiver swaps against Italian financial institutions gained as rates were cut in the sovereign crisis while those institutions weakened.

- **Commodity swaps.** A producer (for example a mining company) hedging its output owes money only when the price is high and its business is profitable: right-way, "due to hedging (as opposed to speculation)". Schweser's mirror: an airline long oil forwards from a dealer with concentrated positions faces wrong-way risk, because a price spike brings a flood of claims on that dealer.
- **Credit default swaps.** When buying protection, the exposure grows as the reference entity's spread widens. If the reference entity and the seller are strongly linked there is extreme WWR; "a bank selling protection on its own sovereign would be an obvious problem". With the same strong link, selling protection should be right-way. Monolines in 2007 to 2009 (AMBAC, MBIA) were the real-world case: their protection was concentrated in mortgage-related risk, so the claims and their own distress rose together.
- **Guarantees (Schweser).** A guarantor that shares ownership or an industry with the borrower is likely to fail when the borrower does: wrong-way. An unrelated guarantor's promise holds.
- **Currency swaps after Lehman (Schweser).** Japanese banks had pledged yen to obtain dollars from US banks. The yen *appreciated*, giving them large gains (exposure) on US banks whose default probabilities were rising: wrong-way.

How strong is the CDS effect? GARP prices protection bought from a risky seller (reference entity spread 250 bp, seller spread 500 bp, both LGDs 60%). At 60% correlation the fair premium is only about 200 bp: the CVA of 50 bp is one-fifth of the risk-free premium. At 100% correlation it is just above 100 bp, the recovery value alone ($250 \times 40\%$). Selling protection carries a much smaller CVA that falls as correlation rises (right-way).

3. Same trade, different counterparty

WWR is a property of the *pair*: the trade and the counterparty. The same CDS on a Singapore manufacturer is wrong-way if bought from a bank whose loan book is exposed to the same currency shock, and close to independent if bought from a diversified foreign bank (Question CR-20 l.10). A put on a bank index is wrong-way if bought from a bank and much less so if bought from an unrelated counterparty. A rising mark-to-market is not WWR by itself; the counterparty's credit quality must be linked to the same move.

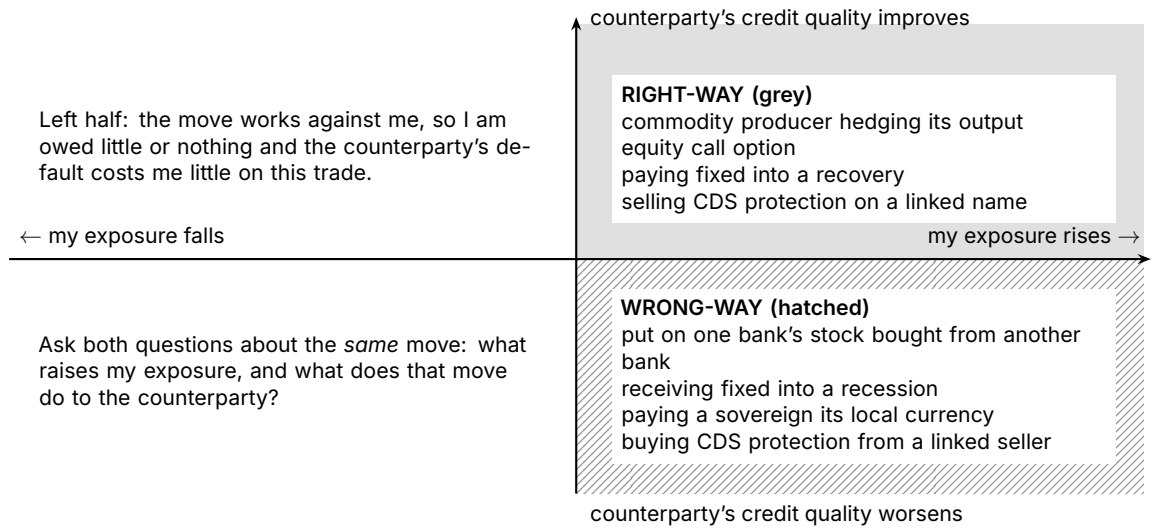


Figure 3. Classifying a trade: place the market move that raises your exposure on the horizontal axis and its effect on the counterparty on the vertical axis. Wrong-way trades sit in the hatched quadrant, right-way trades in the grey one.

Correction to the source notes. Vardeez's CR-20 l table gives the interest rate swap example as "You receive fixed and pay floating. If rates rise (good for you), their creditworthiness might also decline." A fixed-rate receiver gains when rates *fall*, not rise. GARP's example is: receiving fixed from a corporate is wrong-way because "interest rates would be likely to be cut in a recession", so the swap gains just as the corporate weakens; paying fixed during a recovery may be right-way. Schweser states the same direction.

	Wrong-way example	Right-way mirror
Equity	Buy a put on a bank (or bank index) from a bank; OTM worse than ITM	Buy an equity call
FX	Pay a sovereign its local currency; hedge it with a bank in the same region	No adverse link between the currency move and the counterparty
Rates	Receive fixed from a corporate into a recession	Pay fixed into a recovery
Commodity	Buy oil forwards from a dealer with concentrated positions	Swap with a producer that is hedging its output
Credit	Buy CDS protection from a seller linked to the reference entity; monolines 2007 to 2009	Sell protection where the buyer is linked to the reference entity
Guarantees	Guarantor shares ownership or industry with the borrower	Unrelated guarantor

Exam traps

- **Role misassignment (CDS).** Buying protection from a linked seller is wrong-way; selling it is right-way. A seller's own link to the reference entity is a problem for whoever buys from it.
- **Option writer.** Having sold a call or put, you owe the payoff; you have no exposure to the buyer once the premium is paid.
- **Swap direction.** A receiver gains when rates fall (recession, wrong-way with a weakening corporate); a payer gains when rates rise.
- **Currency direction.** The bank that pays the weak currency gains when it weakens. Japanese banks gained because the yen *appreciated*.
- **Hedging versus speculation.** The producer's right-way character comes from hedging; a speculator does not inherit it.
- **Moneyness.** OTM puts carry more WWR than ITM puts, not less.
- **Diversification illusion.** A put on a financial-sector index bought from a bank is still wrong-way: the index falls with the bank.

Practice questions: CR-20 I

Question CR-20 I.1

Bank Q1051 • AnalystPrep Q.1982 (options reordered)

Cartogenes Investments is contemplating the purchase of a bespoke over-the-counter put option on GigaBank's stock from Alamine Bank, another financial institution. This decision could be unsuitable because of wrong-way risk. What is the most accurate reason?

A. The put option will only be valuable if the stock of GigaBank goes up, in which case the counterparty's credit quality will likely be deteriorating.

B. The put option will only be valuable if the stock of GigaBank goes down, in which case the counterparty's credit quality will likely be improving.

C. Such a purchase makes the buyer prone to specific risks affecting banks.

D. The performance of GigaBank and Alamine Bank is likely to be correlated, and an increase in the value of the put will most likely coincide with a deterioration in the credit quality of Alamine Bank.

Question CR-20 I.2

Bank Q1060 • Schweser Module Quiz 38.3 Q2

Which of the following events would likely lead to an increase in wrong-way risk (WWR)?

I. The borrower and the guarantor are business partners (they share ownership or operate in the same industry).

II. A monoline insurer has sold protection concentrated in a single business or industry.

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Question CR-20 I.3

Bank Q1061 • Schweser Module Quiz 38.3 Q3

Which of the following statements regarding wrong-way risk (WWR) and right-way risk (RWR) is correct?

A. A long put option is subject to WWR if both risk exposure and counterparty default probability decrease.

B. A long call option experiences RWR if the interaction between risk exposure and counterparty default probability produces an overall decline in counterparty risk.

C. A declining local currency can decrease the position gain in a foreign currency transaction while increasing the risk exposure of the counterparty.

D. The 2007 to 2009 credit crisis provides an example of WWR from the perspective of a long who had sold credit default swaps (CDSs) as protection against bond issuers' default.

Question CR-20 I.4 Bank Q1063 • Schweser Module Quiz 38.3 Q5

Which of the following statements is correct?

I. After the default of Lehman Brothers, depreciation of the yen against the dollar gave Japanese banks a substantial gain on the currency swap positions they had used to obtain dollar funding.

II. A fixed-rate receiver in an interest rate swap gains value to the extent that the swap rate increases.

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Question CR-20 I.5 Bank Q1065 • Mock 2 Q61

An investor has recently purchased several over-the-counter (OTC) puts on the S&P 500 exchange-traded fund (ETF). He uses options to hedge risk and to overweight or underweight asset classes in an efficient and risk-conscious fashion. Which of the following statements best describes the use of long OTC puts in the context of wrong-way risk (WWR) and right-way risk (RWR)?

A. Out-of-the-money put options have less WWR than in-the-money put options.

B. Out-of-the-money put options have more WWR than in-the-money put options.

C. In-the-money put options have neither more nor less WWR or RWR than out-of-the-money put options.

D. A long put option is subject to WWR if both risk exposure and counterparty default probability decrease.

Question CR-20 I.6 Bank Q1056 • AnalystPrep Q.5482 (rebuilt)

Apex Portfolio Management wants to buy OTC put protection on its equity holdings from a dealer bank and wants to limit wrong-way risk (WWR). Which choice best limits WWR?

A. Buy a put on the stock of a company in an unrelated industry, whose fortunes have little link to the dealer bank's credit quality.

B. Buy a put on the stock of another large bank from the dealer bank.

C. Buy a put on a financial-sector index from the dealer bank, since the index diversifies across many banks.

D. Buy a put on the dealer bank's own stock, struck in the money rather than out of the money.

Question CR-20 I.7 Bank Q1058 • BT P2.T6.419.1 (rebuilt)

From the perspective of Acme Financial Institution, which of the following positions is most likely to exhibit right-way risk rather than wrong-way risk?

A. Acme buys an out-of-the-money put option on a bank's stock from another bank.

B. Acme is the floating-rate payer (fixed-rate receiver) in a vanilla interest rate swap with a bank, and Acme's economist has shown that low rates signal a recession.

C. Acme is long an oil commodity swap with an oil producer that is hedging its output: Acme pays a fixed oil price and receives the spot price of oil.

D. Acme buys CDS protection where the credit quality of the reference entity is strongly positively related to that of the protection seller.

Question CR-20 I.8 Bank Q1053 • AnalystPrep Q.5477 (rebuilt, options reordered)

Oracle Financial Group has a large book of OTC derivatives. Its risk committee wants to prioritise contracts for wrong-way risk (WWR) review. Which of the following contracts should be prioritised?

A. Selling CDS protection to a counterparty on Oracle's own sovereign, because the strong link between Oracle and the sovereign creates extreme WWR for the seller.

B. Buying CDS protection on mortgage-backed securities from a monoline insurer whose guarantees are concentrated in the mortgage market.

C. Derivatives with a commodity producer that is hedging its output price, because the producer owes money mainly when prices are high.

D. Equity call options bought from a bank on that same bank's stock.

Question CR-20 I.9 Bank Q1057 • BT Mock Q2 • BT Mock Q32 (options reordered)

Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among the following four positions, which is the best example of wrong-way risk?

A. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty's default probability.

B. Zapex has written (sold) a call option to a counterparty in the same industry.

C. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery.

D. Zapex has written (sold) a credit default swap to a counterparty whose credit quality is positively correlated with the reference entity.

Question CR-20 I.10 Bank Q1064 • GARP PE2 Q58 (rebuilt, options reordered)

VLT Bank (VLTB), a Singapore bank, has bought a 5-year credit default swap (CDS) on the EUR-denominated bonds of ConSol Corp, a Singapore manufacturer that buys locally produced inputs in SGD and earns most of its revenue in EUR. The SGD then appreciates sharply against all major currencies, and ConSol's credit spread widens substantially. VLTB is comparing two possible protection sellers:

- Seller X: a German bank with a diversified global book and no meaningful exposure to Singapore manufacturers.
- Seller Y: a Singapore bank whose loan book is concentrated in Singapore exporters that earn revenue in foreign currencies.

Which statement is most accurate?

A. Both CDS trades show WWR, because any purchased protection gains value when the reference spread widens.

B. The CDS with Seller X shows WWR, because a weaker euro damages a German bank, whereas the CDS with Seller Y does not.

C. The CDS with Seller Y shows WWR, whereas the CDS with Seller X does not.

D. Neither trade shows WWR, because CDS exposure depends only on ConSol and is independent of the seller's credit quality.

Solutions • CR-20 l

Solution CR-20 l.1 — Answer: D

Bank Q1051 • AnalystPrep Q.1982

Step 1 – What raises Cartogenes' exposure?

A put pays off when the underlying falls. Cartogenes' claim on Alamine is largest when GigaBank's stock has dropped.

Step 2 – What does that move do to the counterparty?

Bank stocks and bank credit quality share the same macro drivers. A fall in GigaBank's stock is likely to come with a deterioration in Alamine Bank's credit quality.

$\Rightarrow$ Exposure up as the counterparty weakens: wrong-way risk. This is GARP's own example, "buying a put on one bank's stock from another bank".

Interpretation

Cartogenes could reduce the problem by buying the put from a counterparty outside the banking sector, or by requiring margin, although margin helps only if the exposure builds gradually (CR-20 m).

Why the other options are wrong

A is incorrect. Reverses the payoff: a put gains when the stock falls, not when it rises.

B is incorrect. Gets the payoff right but pairs it with *improving* counterparty credit quality, which would be right-way risk.

C is incorrect. Too vague: it names a sector concentration rather than the link between the exposure and the counterparty's credit quality that defines WWR.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1 Overview

Solution CR-20 l.2 — Answer: C

Bank Q1060 • Schweser Module Quiz 38.3 Q2

Step 1 – Event I: a linked guarantor

The guarantee is called when the borrower defaults. If the guarantor shares ownership or an industry with the borrower, the shock that sinks the borrower is likely to weaken the guarantor at the same time.

$\Rightarrow$ The guarantee is least reliable exactly when it is needed: WWR rises.

Step 2 – Event II: a concentrated monoline

A monoline that has sold protection concentrated in one industry faces a flood of claims when that industry suffers, and its own default probability rises with the claims.

$\Rightarrow$ The protection buyers' exposure and the insurer's default probability rise together: WWR rises.

Interpretation

Both are the credit-protection version of the same pattern. The monoline case is the 2007 to 2009 experience with AMBAC and MBIA; an unrelated guarantor or a diversified insurer would be the right-way (or neutral) alternative.

Why the other options are wrong

A is incorrect. Misses the monoline case: concentration of the protection seller in the insured risk is a textbook WWR source.

B is incorrect. Misses the guarantor case: a guarantor tied to the borrower tends to fail with it.

D is incorrect. Treats both protection providers as independent of the risk they cover, which is the no-WWR case and does not apply here.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** SchweserNotes 2026 Book 2, Reading 38, LO 38.l; Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1

Solution CR-20 l.3 Answer: B

Bank Q1061 • Schweser Module Quiz 38.3 Q3

Step 1 – Classify by the net effect on counterparty risk

An interaction between exposure and default probability that lowers overall counterparty risk is RWR; one that raises it is WWR.

⇒ B applies this directly to a long call: an overall decline in counterparty risk is RWR. Correct.

Step 2 – Check the directions in A and C

A: for a long put, WWR needs exposure and default probability to *rise* together (the underlying falls, the put gains, the writer weakens), not to fall. C: when the local currency declines, the party receiving hard currency *gains*, so its exposure rises; C says the gain decreases.

Step 3 – Check the roles in D

In 2007 to 2009 the WWR was borne by those who *bought* CDS protection from monolines. A seller of protection owes payments; it is not the victim of the seller's own distress.

Interpretation

Each false option reverses one element: the direction of the move (A), the sign of the FX gain (C) or the role (D). Name the move, the gain and the role before choosing.

Why the other options are wrong

A is incorrect. Direction reversed: WWR on a long put needs exposure and default probability to rise together.

C is incorrect. Sign reversed: a declining local currency increases the gain of the party receiving hard currency.

D is incorrect. Role reversed: the buyers of protection, not the sellers, suffered WWR in the credit crisis.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** SchweserNotes 2026 Book 2, Reading 38, LO 38.l

Solution CR-20 l.4 Answer: D

Bank Q1063 • Schweser Module Quiz 38.3 Q5

Step 1 – Statement I: the direction of the yen

Japanese banks had pledged yen to receive dollars. After Lehman the yen *appreciated*, so the pledged yen was worth more dollars and the US banks owed more: a gain, and rising exposure to weakening US counterparties (WWR). Depreciation would have produced a loss.

⇒ Statement I is false.

Step 2 – Statement II: the receiver swap

A fixed-rate receiver locks in a coupon. When the swap rate falls the fixed leg is worth more, so the receiver gains when rates *decline*.

⇒ Statement II is false.

Interpretation

Both facts sit behind classic WWR examples: yen swaps after Lehman, and receiver swaps against counterparties that weaken in a recession (when rates are cut). The statements each flip one

direction.

Why the other options are wrong

A is incorrect. Accepts the wrong direction of the yen move.
B is incorrect. Confuses receiver and payer: a fixed-rate payer gains when the swap rate rises.
C is incorrect. Accepts both direction errors.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** SchweserNotes 2026 Book 2, Reading 38, LO 38.l; Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1

Solution CR-20 l.5 — Answer: B

Bank Q1065 • Mock 2 Q61

Step 1 – When does a long put have exposure?

A put pays only if the underlying falls. The macro stress that drives the index down is also the state in which the writer's default probability rises.

Step 2 – Compare moneyness

An out-of-the-money put has little exposure today and becomes valuable only in a large fall, exactly the stressed state in which the writer is likely to fail. An in-the-money put already has exposure in normal states, so a smaller share of its exposure depends on the stress scenario.
$\Rightarrow$ OTM puts have more WWR than ITM puts (Schweser LO 38.l).

Interpretation

The more a payoff is concentrated in crisis states, the stronger the link between exposure and the writer's default. That is also why deep OTM protection bought from a financial counterparty is the hardest to rely on.

Why the other options are wrong

A is incorrect. Reverses the relationship.
C is incorrect. Moneyness does matter: the dependence between exposure and default differs between ITM and OTM puts.
D is incorrect. WWR requires exposure and default probability to rise together; both falling is not the WWR pattern.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** SchweserNotes 2026 Book 2, Reading 38, LO 38.l; Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1

Solution CR-20 l.6 — Answer: A

Bank Q1056 • AnalystPrep Q.5482

Step 1 – Where the WWR in a put comes from

The put pays off when the underlying falls. WWR arises when the underlying's fortunes are highly correlated with the writer's credit quality, so the writer is weak when the put is valuable.

Step 2 – Rank the choices by that link

The dealer's own stock gives the strongest (specific) link; another large bank and a financial-sector index share the same macro shocks as the dealer. A company in an unrelated industry has little link.
$\Rightarrow$ Choosing an unrelated underlying weakens the dependence between exposure and default probability.

Interpretation

Apex could also buy bank-sector protection from a non-bank counterparty. The point is the pair: underlying and writer should not fall together.

Why the other options are wrong

B is incorrect. Another bank shares the same shocks: this is GARP's standard WWR example.
C is incorrect. Diversifying across banks does not remove the common financial-sector factor; the index falls when the dealer weakens.
D is incorrect. The dealer's own stock gives the strongest specific link; moving the strike in the money only modestly changes it.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1

Solution CR-20 I.7 Answer: C

Bank Q1058 • BT P2.T6.419.1

Step 1 – Option C: who owes whom, and when?

Acme receives spot and pays fixed, so it gains when oil rises. The producer is hedging its output, so it owes money mainly when oil is high, which is when its business is most profitable and default least likely.
⇒ Exposure high when the counterparty is strong: right-way risk, arising from the producer's hedging.

Step 2 – Apply the test to the others

A: the put gains as bank stocks fall, when the writing bank weakens: WWR. B: the receiver gains when rates fall, and the economist links low rates to recession and weaker counterparties: WWR. D: protection gains as the reference entity weakens, when the linked seller weakens too: extreme WWR.

Interpretation

The right-way case is the only one in which the counterparty is paying out of strength. If the producer were speculating rather than hedging, the conclusion would not follow.

Why the other options are wrong

A is incorrect. A put on a bank's stock bought from another bank is the textbook WWR example.
B is incorrect. A receiver swap gains in a downturn, when low rates coincide with higher counterparty default risk.
D is incorrect. Strong linkage between reference entity and seller makes the protection least reliable when it is most valuable.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1

Solution CR-20 I.8 Answer: B

Bank Q1053 • AnalystPrep Q.5477

Step 1 – Option B

If mortgage securities fall, the purchased protection gains value (Oracle's exposure rises), and the monoline, concentrated in mortgage risk, faces a flood of claims (its default probability rises).
⇒ Exposure and the counterparty's default probability rise together: the classic 2007 to 2009 WWR. Prioritise it.

Step 2 – Option A: whose problem is it?

Oracle's link to its own sovereign makes Oracle a wrong-way counterparty for anyone who *buys* protection on that sovereign from it. As the seller, Oracle owes the payout; it has no growing claim on the buyer, so this is not a WWR exposure of Oracle's.

Step 3 – Options C and D

A hedging commodity producer owes money when it is profitable (right-way). A call on the writer's own stock gains when the stock rises, when the writer is stronger (right-way).

Interpretation

A WWR review should start where Oracle is owed money by a counterparty that weakens with the source of the gain: bought protection and bought puts from correlated writers.

Why the other options are wrong

A is incorrect. Role misassignment: the seller's link to the reference entity is the buyer's WWR, not the seller's.

C is incorrect. Hedging by a commodity producer creates right-way risk.

D is incorrect. Call payoffs rise with the stock, when the writing bank's credit quality is likely to be improving: right-way.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1; SchweserNotes 2026 Book 2, Reading 38, LO 38.l

Solution CR-20 I.9

Answer: A

Bank Q1057 • BT Mock Q2

Step 1 – Wrong-way risk needs a claim on the counterparty

Zapex must be owed money, and that claim must grow as the counterparty's default probability rises.

Step 2 – Option A

The forward's mark-to-market gain (Zapex's exposure) is positively correlated with the counterparty's default probability.

$\Rightarrow$ This states the definition of WWR directly.

Step 3 – The written positions

Having written the call, Zapex owes the payoff and has no claim on the buyer beyond the premium already received. Having written the CDS, Zapex owes the protection payment; the buyer's link to the reference entity does not give Zapex a growing claim on the buyer.

Interpretation

Two of the four positions are traps for the same reason: the seller of an option or of protection is the party that owes. The swap in a recovery is the right-way case from GARP.

Why the other options are wrong

B is incorrect. The option writer has an obligation, not an asset, so there is no meaningful credit exposure to the buyer.

C is incorrect. In a recovery rates rise: the fixed payer gains while counterparty credit quality tends to improve, a right-way situation.

D is incorrect. Looks wrong-way because of the positive correlation, but Zapex is the protection seller and owes the payment.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1

Solution CR-20 I.10

Answer: C

Bank Q1064 • GARP PE2 Q58

Step 1 – What raises VLTB's exposure?

The SGD appreciation hurts ConSol (its EUR revenue buys fewer SGD while its costs are in SGD),

so its spread widens and VLTB's purchased CDS gains value. VLTB's exposure to whichever bank sold the protection rises.

Step 2 – Does the same shock weaken the seller?

Seller Y lends to Singapore exporters with foreign-currency revenue, the same weakness as ConSol: the SGD shock raises Y's default probability too. Seller X is diversified with no stated link to the shock.

$\Rightarrow$ With Y, exposure and default probability rise together: WWR. With X, the exposure rises but the seller's credit quality is not tied to it.

Interpretation

The CDS is identical in both cases; only the counterparty changes. Wrong-way risk is a property of the trade-counterparty pair, which is why counterparty selection is a mitigant.

Why the other options are wrong

A is incorrect. Treats a rising CDS value as WWR by itself. WWR also needs a link to the seller's credit quality; otherwise every hedge would be wrong-way.

B is incorrect. Assumes a weaker euro damages the diversified German bank, which the stem does not support, and misses Seller Y's concentration.

D is incorrect. Assumes independence; it fails for Seller Y, whose creditworthiness depends on the same exporter stress that raises the CDS value.

LO: CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1; 20.6.5 Credit Derivatives

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 m

P1 CONCEPT

Discuss the impact of collateral on wrong-way risk.

Workbook: A20 Wrong-Way and Right-Way Risk • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A20). Map #26 names the examinable sub-items behind the August wrong-way risk questions: wrong-way collateral, the effect on CCPs, and the modelling approaches.

Vardeez: collateral reduces exposure but its effectiveness depends on timing: a gradual exposure increase can be collateralised, a sudden jump (a devaluation) cannot. Collateral is weakest exactly where wrong-way risk is worst.

Reported exam shapes

- **A20·S3** Wrong-way collateral (worked here in A20.2 and drilled in CR-20 n).

What the questions look like: "how effective is margin against WWR?" with absolute-claim distractors ("in all cases", "eliminates"); which model's view of margin matches an observation; why margin helps least against the most systemic counterparties; the special case of credit derivatives. All conceptual.

Worked here: A20.2, homed in this LO, also covers S4 (CCPs) and S5 (models); CR-20 n, o and p point back to it. Bank Q1077 (the jump view of collateral) is re-filed here from CR-20 p.

1. Collateral works on exposure, and timing decides

Margin reduces the exposure; it does nothing to the counterparty's default probability (CR-20 j, CR-19 h). Because WWR can make the exposure rise sharply, "the impact of margin on WWR is very important to consider. However, this is very hard to characterise because it is very timing dependent" (GARP 20.6.6).

- **Gradual increase before default.** Each rise in exposure can be called and received, so margin mitigates the WWR. Schweser: in this case the benefit of collateral grows with the WWR, because additional collateral is relatively easy to request and receive.
- **Jump at default.** "A jump in exposure deems margin useless." A currency devaluation that comes with a sovereign default, or the credit event on a CDS, moves the exposure in one step. Margin always lags the exposure by the margin period of risk, and a jump inside that period is uncollateralised.

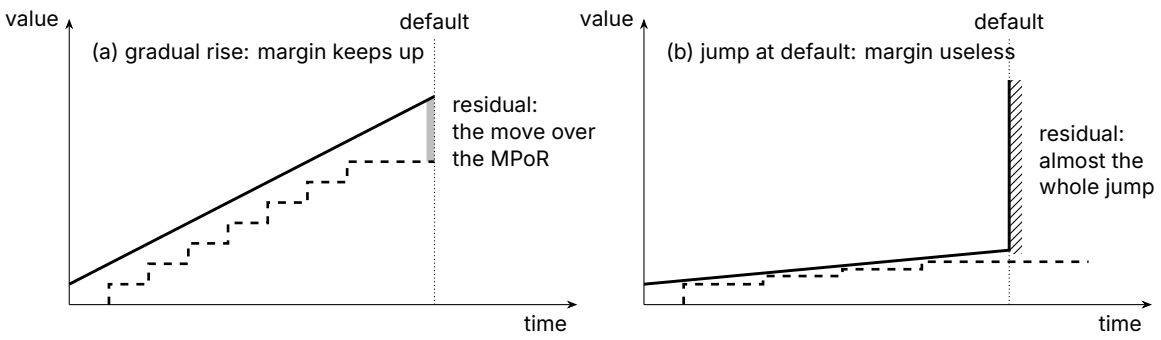


Figure 4. Exposure (solid) and margin held (dashed, stepping up with each call after a lag), schematic. Left: a gradual rise is collateralised except for the move over the margin period of risk (grey). Right: a jump at the default time cannot be called in time (hatched).

2. What the models say

- **Jump approaches** (CR-20 p) "tend to show collateralisation as being near useless in mitigating WWR". Chung and Gregory (2019) show margining has very little impact on an FX portfolio under the jump approach.
- **Continuous approaches** (intensity and structural) "suggest that margin is an effective mitigant against WWR".
- **Verdict:** "the truth is somewhere in between, but is likely very dependent on the type of transaction and counterparty."

3. Systemic counterparties: where margin is used, WWR is worst

Pykhtin and Sokol (2013) argue that measuring the benefit of margin under WWR must allow for **jumps and a period of higher volatility during the MPoR**, and that WWR should be **higher for the default of more systemic parties**, such as banks. Their overall result: *WWR has a negative impact on the benefit of collateralisation*. GARP draws the uncomfortable conclusion: the counterparties that actively post margin (banks) are highly systemic and subject to extreme WWR, while non-systemic counterparties (corporates) often do not post margin at all. Margin is weakest against WWR precisely in the relationships where it is used most.

4. Credit derivatives

For CDSs the WWR is unavoidable (protection on one party is bought from another) and may be specific (protection on a sovereign bought from a bank in that country). "Due to the highly contagious and systemic nature of CDS risks, the impact of margin may be hard to assess and indeed may be quite limited." The credit event and the seller's distress tend to arrive together, as a jump.

5. When the margin itself is wrong-way

WWR can also sit in the link between the *value of the margin* and the exposure it secures, or between the margin and the poster's own default: a payer swap secured by government bonds, a cross-currency swap secured by cash in the currency being paid, an entity posting its own bonds, a bank posting its own sovereign's bonds. The examples are classified under CR-20 n. Two effects matter here:

- **Some benefit survives.** "Wrong-way margin does still provide some benefit as a mitigant as long

as it retains some value when the counterparty defaults" (GARP 19.4.3). It is weaker, not worthless.

- **Credit versus funding.** To mitigate counterparty risk the margin must have no adverse correlation with the poster's credit quality; to provide a funding benefit it must be reusable (not segregated, and transferred by title transfer or rehypothecable). An entity's own bonds are "a very weak mitigant against exposure and CVA, although it may mitigate FVA".

CCPs rely on margin as their primary protection, which makes them particularly prone to WWR (CR-20 o).

	Gradual exposure increase	Jump at default
Exposure path	Builds up before default	Gaps up at the default time
Can margin keep up?	Yes, call by call	No: the jump falls inside the MPoR
Residual at default	The move over the MPoR	Almost the whole jump
Typical case	General WWR in rates	Devaluation at sovereign default; CDS credit event
Models showing it	Intensity, structural: margin effective	Jump: margin near useless
Margin versus WWR	Effective mitigant	Very limited mitigant

Exam traps

- **Absolute claims.** "Margin eliminates WWR" or "is effective in all cases" is false: a jump at default defeats it.
- **Speed reversed.** A rapid rise in exposure weakens margin's role; it does not enhance it, and effectiveness is not "proportional to the rate of increase".
- **Exposure versus PD.** Margin reduces exposure; it does not lower the counterparty's default probability.
- **Model polarity.** Jump approaches show margin as near useless; continuous approaches (intensity, structural) show it as effective.
- **Systemic paradox reversed.** WWR is higher, not lower, for systemic margin posters such as banks.
- **Wrong-way margin.** It is weaker, not worthless; own bonds may still help FVA although they barely help CVA.

Practice questions: CR-20 m

Question CR-20 m.1

Bank Q1066 • AnalystPrep Q.1985

A risk manager at Prestige Global Bank is analysing derivative agreements for the potential impact of wrong-way risk (WWR). One agreement is an interest rate swap with SovereignX, a small country with an emerging economy, in which the bank receives a fixed rate and pays a floating rate. The collateral backing the swap is bonds issued by SovereignX itself. News breaks of increasing political instability in SovereignX, which might lead to drastic economic decisions. Which of the following best describes the impact of collateral on WWR, particularly in the context of potential rapid changes in SovereignX's economic conditions?

A. Collateral will effectively counterbalance any WWR, regardless of the speed of the exposure increase.

B. The potential for a sudden increase in exposure means collateral might not provide significant mitigation against WWR.

C. Collateral's effectiveness in reducing WWR is directly proportional to the rate of exposure increase.

D. A rapid increase in exposure will enhance the role of collateral in mitigating WWR.

Question CR-20 m.2

Bank Q1077 • AnalystPrep Q.5483 (rebuilt, options reordered; re-filed from CR-20 p)

Hudson Capital has observed that collateral becomes much less effective during market stress, particularly on its FX portfolio, where exposure tends to gap upward when a counterparty defaults. Which WWR modelling approach would best reflect this observation, and why?

A. Continuous approaches such as the intensity and structural approaches, because they show margin to be an effective mitigant against WWR.

B. Parametric approaches, because they give a clear-cut measure of collateral effectiveness without needing to consider the timing of exposure changes.

C. Hudson's current approach is adequate, because WWR is not materially affected by how quickly exposure increases relative to margin calls.

D. Jump approaches, because they tend to show collateralisation as nearly useless when exposure jumps at default.

Question CR-20 m.3 New • built from GARP Ch.20, 20.6.6 (Pykhtin and Sokol)

The head of xVA at a European bank argues that, because the bank's large dealer counterparties all post daily variation margin and initial margin under their CSAs, the bank can largely disregard wrong-way risk on those relationships and should concentrate its WWR analysis on uncollateralised corporate clients. Which response is most consistent with GARP's discussion of collateral and WWR?

A. The argument is flawed: WWR reduces the benefit of collateralisation, that benefit must allow for jumps and higher volatility during the margin period of risk, and systemic dealers that post margin are the counterparties most exposed to extreme WWR.

B. The argument is sound: daily variation margin means any rise in exposure before default is collateralised, so WWR can only arise with counterparties that do not post margin.

C. The argument is flawed, because WWR is lower for systemic counterparties such as dealers, so the bank should concentrate instead on its smaller and weaker counterparties.

D. The argument is sound, because initial margin is sized to a high confidence level and therefore already covers any jump in exposure at a dealer's default.

Question CR-20 m.4 New • built from GARP Ch.20, 20.6.5 and 20.6.6

A bank buys five-year CDS protection on its domestic sovereign from a domestic bank. The trade is covered by a CSA with daily variation margin and a zero threshold. The trader argues that the margin makes the trade's wrong-way risk negligible. Which assessment is most accurate?

A. The trader is right: with a zero threshold and daily margin, the bank is fully collateralised at all times, so the seller's default cannot cause a loss.

B. The trader is right, provided the protection seller posts high-quality government bonds of the same sovereign as margin.

C. The trader is wrong: WWR in credit derivatives is unavoidable and here specific, the credit event and the seller's distress are likely to arrive together as a jump, so margin may help very little.

D. The trader is wrong, because the trade is right-way: the bank's exposure rises as the sovereign weakens, which strengthens the domestic bank selling protection.

Question CR-20 m.5 New • built from GARP Ch.20, 20.6.6; Ch.19, 19.4.3; Ch.17, 17.3.5

An Italian bank posts Italian government bonds as variation margin to a US dealer under a CSA that allows the dealer to rehypothecate margin. The dealer's xVA desk is assessing what this margin achieves. Which statement is most accurate?

A. It provides no counterparty risk benefit at all, because any margin linked to the poster's credit quality is worthless when the poster defaults.

B. It is a weaker mitigant than cash or US Treasuries because its value is likely to fall as the bank weakens, but it still reduces exposure to the extent it retains value at default, and being reusable it can still give a funding benefit.

C. It is as effective as cash, because high-quality sovereign bonds have good liquidity, low default risk and low price volatility.

D. It mitigates counterparty risk fully but gives no funding benefit, because wrong-way margin cannot be rehypothecated.

Question CR-20 m.6 Workbook A20.2 • S3, S4, S5 • Map #26

A risk consultant is reviewing how a European derivatives clearing house and its largest clearing members measure and mitigate wrong-way risk, and is drafting recommendations. Which of the following statements is correct to include in the report?

A. Variation and initial margin are effective mitigants of wrong-way risk in all cases, because margin received as the exposure rises offsets the higher exposure before default.

B. The hazard rate (intensity) approach suits specific wrong-way risk, because correlating credit spreads with exposure drivers produces a strong dependence that matches quanto CDS data.

C. Setting initial margin and default fund contributions mainly from the market risk of each portfolio may ignore wrong-way risk, which argues for more from higher-quality members.

D. A payer interest rate swap collateralised by a high-quality government bond is right-way collateral, because a rise in rates raises both the swap value and the bond value.

Solutions • CR-20 m

Solution CR-20 m.1 — Answer: B

Bank Q1066 • AnalystPrep Q.1985

Step 1 – When can collateral help?

Collateral reduces exposure only if it can be called and received before the default. A gradual rise in exposure can be collateralised call by call.

Step 2 – What the stem describes

Political instability threatens a sudden move (a devaluation, a default, drastic policy). A jump in exposure leaves no time to collect more collateral.

$\Rightarrow$ GARP: "a jump in exposure deems margin useless". The benefit is limited.

Step 3 – The collateral itself

The margin is SovereignX's own bonds, which would fall in value in the same crisis: wrong-way collateral (CR-20 n). The protection weakens just as the exposure rises.

Interpretation

The bank receives fixed from a sovereign that may cut rates in a crisis, so the swap itself carries WWR; the collateral does little against a jump and is itself wrong-way. Cash or third-party high-quality securities, or a smaller position, would be the remedies.

Why the other options are wrong

A is incorrect. Absolute claim: it ignores the speed of the exposure increase and the wrong-way nature of the sovereign-bond collateral.

C is incorrect. Reverses the relationship: the faster the exposure rises, the less collateral can be collected in time.

D is incorrect. Backwards: a rapid increase reduces the role of collateral, because it cannot be called in time.

LO: CR-20 m: Discuss the impact of collateral on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.6 Collateralisation and WWR

Solution CR-20 m.2 — Answer: D

Bank Q1077 • AnalystPrep Q.5483

Step 1 – The observation

On the FX book, collateral fails in stress because the exposure gaps up at the counterparty's default. Margin helps only if exposure rises gradually enough to be called.

Step 2 – Which model reproduces it?

Jump approaches assume the FX rate jumps at the default time; they "tend to show collateralisation as being near useless in mitigating WWR". Chung and Gregory (2019) find margining has very little impact on an FX portfolio under the jump approach.

$\Rightarrow$ The jump approach matches what Hudson sees.

Interpretation

Continuous models smooth the exposure path, so collateral appears to keep pace with it. GARP's view is that the truth lies between the two and depends on the transaction and counterparty, but for an FX book exposed to devaluation at default the jump view is the relevant one.

Why the other options are wrong

A is incorrect. Continuous approaches show margin as effective, which is exactly why they fail to reproduce Hudson's experience.

B is incorrect. The parametric approach links default probability to portfolio value through a functional form calibrated to history; it has no mechanism for the timing of a jump and gives no "clear-cut" collateral measure.
C is incorrect. Timing is the key driver: GARP calls the effect of margin on WWR "very timing dependent".

LO: CR-20 m: Discuss the impact of collateral on wrong-way risk. CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.4 Jump Approaches; 20.6.6

Solution CR-20 m.3 Answer: A

New • built from GARP Ch.20, 20.6.6

Step 1 – What Pykhtin and Sokol add

The benefit of margin in a WWR situation must allow for jumps and a period of higher volatility during the MPoR, and WWR should be higher for the default of more systemic parties, such as banks. Overall, WWR has a negative impact on the benefit of collateralisation.

Step 2 – Apply it to the dealer relationships

The large dealers post margin, but they are the systemic counterparties whose default comes with jumps and extreme volatility. Their margin does not remove their WWR; it is least effective against it.
⇒ The head of xVA's argument fails. GARP notes the irony: the counterparties that use margin are the ones subject to extreme WWR, while corporates often do not post margin anyway.

Interpretation

Margin should be treated as a partial mitigant whose benefit is reduced by WWR, with stressed MPoR assumptions for systemic names. Uncollateralised corporates still matter, but they are not the only place WWR lives.

Why the other options are wrong

B is incorrect. Gradual rises are collateralised, but a jump at default inside the MPoR is not; WWR survives daily margin.
C is incorrect. Polarity flip: WWR is higher for systemic parties, and it increases with credit quality (CR-20 k).
D is incorrect. Initial margin is set to a high but finite confidence level from market-risk models; it does not automatically cover the jump that a systemic default brings.

LO: CR-20 m: Discuss the impact of collateral on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.6 Collateralisation and WWR

Solution CR-20 m.4 Answer: C

New • built from GARP Ch.20, 20.6.5 and 20.6.6

Step 1 – Classify the trade

The bank is owed the protection value, which rises as its sovereign weakens. A domestic bank selling protection on its own sovereign weakens with it. GARP: "a bank selling protection on its own sovereign would be an obvious problem"; for credit derivatives the WWR is unavoidable and may be specific.

Step 2 – What can margin do?

The sovereign's distress and the domestic bank's distress are likely to come together, as a jump. GARP: "due to the highly contagious and systemic nature of CDS risks, the impact of margin

may be hard to assess and indeed may be quite limited."
⇒ Daily margin and a zero threshold do not make the WWR negligible.

Interpretation

This is specific WWR, which GARP says should in general be avoided. If the bank needs the protection, it should buy it from a seller whose credit quality is not tied to the sovereign.

Why the other options are wrong

A is incorrect. "Fully collateralised at all times" ignores the margin period of risk and the jump at the credit event.
B is incorrect. Bonds of the same sovereign would be wrong-way collateral: they fall exactly when the protection is needed, making matters worse.
D is incorrect. Role misassignment: buying protection from a linked seller is the wrong-way side; the seller weakens, not strengthens, as the sovereign weakens.

LO: CR-20 m: Discuss the impact of collateral on wrong-way risk. CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1, 20.6.5 Credit Derivatives and 20.6.6

Solution CR-20 m.5 Answer: B

New • built from GARP Ch.20, 20.6.6; Ch.19, 19.4.3; Ch.17, 17.3.5

Step 1 – Is the margin wrong-way?

A bank posting bonds of its own sovereign is one of GARP's "clearly problematic" examples: the bank's credit quality and its sovereign's are linked, so the bonds are likely to lose value as the bank weakens. GARP also notes that a CSA may not prevent this, because high-quality sovereign bonds are normally eligible.

Step 2 – Counterparty risk benefit

"Wrong-way margin does still provide some benefit as a mitigant as long as it retains some value when the counterparty defaults." The bonds are a weaker mitigant than cash or US Treasuries, but not worthless.

Step 3 – Funding benefit

For margin to help funding it must be reusable. The CSA allows rehypothecation, so the bonds can still give a funding benefit; the wrong-way link affects the credit benefit, not reusability.

Interpretation

The desk should apply a larger haircut or prefer other eligible assets, and it should not count the bonds at full value in the CVA. Credit benefit and funding benefit are two separate tests.

Why the other options are wrong

A is incorrect. Absolute claim: wrong-way margin keeps some value at default, so it still mitigates some exposure.
C is incorrect. Liquidity and low volatility do not remove the link between the bank's default and its sovereign's bonds.
D is incorrect. Reverses both tests: the credit benefit is reduced, while reusability (and so the funding benefit) depends on rehypothecation rights, not on the wrong-way link.

LO: CR-20 m: Discuss the impact of collateral on wrong-way risk. CR-20 n: Identify examples of wrong-way collateral. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.6; Ch.11 (GARP CR Ch.19), 19.4.3; Ch.7 (GARP CR Ch.17), 17.3.5 Haircuts

Solution CR-20 m.6 Answer: C

Workbook A20.2 • S3, S4, S5

Step 1 – The CCP statement

Gregory: CCPs require a minimum credit quality for membership but then charge initial margins and default fund contributions "driven primarily by the market risk of their portfolio"; in doing this "CCPs are in danger of implicitly ignoring WWR". Because WWR increases with credit quality, this "perversely" suggests greater initial margin and default fund from better-credit-quality, more systemically important members.
$\Rightarrow$ The surviving members' default fund sits in a senior, systemic second-loss position of a CDO-like waterfall, so under-margined WWR lands on them.

Interpretation

The consultant should recommend that the clearing house quantify WWR in initial margin and default fund setting (hard for CDSs), and review eligible margin for adverse selection, since members will post the riskiest assets relative to the haircut.

Why the other options are wrong

A is incorrect. Absolute claim: margin helps only if exposure rises gradually; a jump at default makes it useless. Jump approaches show collateral as near useless, and Pykhtin and Sokol find WWR reduces the benefit of collateralisation. "In all cases" is false.
B is incorrect. Sibling swap: the intensity approach generates only weak dependence and may underestimate WWR; Ehlers and Schönbucher show it cannot match quanto CDS data even at $\pm 100\%$ correlation. The jump approach is the one suited to specific WWR and supported by quanto data.
D is incorrect. Polarity flip: a rate rise raises the payer swap's value but lowers the bond's value, so the margin shrinks as the exposure grows: general wrong-way collateral (minor, given low rate volatility). The receiver swap is the right-way case.

LO: CR-20 m: Discuss the impact of collateral on wrong-way risk. CR-20 n: Identify examples of wrong-way collateral. CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.3 to 20.6.7

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 n

P1 CONCEPT

Identify examples of wrong-way collateral.

Workbook: A20 Wrong-Way and Right-Way Risk • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A20): wrong-way collateral is one of the examinable sub-items Map #26 lists behind the August wrong-way risk questions.

Vardeez: a payer interest rate swap collateralised by government bonds. When rates rise the exposure grows while the bonds posted as collateral fall: wrong-way risk located inside the collateral rather than in the counterparty.

Reported exam shapes

- **A20·S3** Wrong-way collateral.

What the questions look like: pick the wrong-way arrangement, with its right-way mirror (the receiver swap) as the trap; which currency of cash margin is wrong-way; a counterparty posting its own bonds, or bonds tied to a CDS reference entity; general versus specific. All conceptual.

Worked elsewhere: A20.2 (option D is the payer-swap polarity flip) under CR-20 m. The effect of wrong-way margin on CVA and FVA is in CR-20 m, Section 5.

1. What makes collateral wrong-way

Collateral is wrong-way when its value is adversely linked to what it is meant to protect against. GARP's haircut checklist names both links: "any relationship between the value of the margin and either the default of the counterparty or the underlying exposure" (17.3.5).

- **General (link to the exposure).** The market move that raises the exposure also lowers the margin's value. Macro-driven and measurable.
- **Specific (link to the poster's default).** "A more direct relationship between the type of margin and the counterparty credit quality": the margin loses value because the poster is failing.

The test mirrors CR-20 k: what raises my exposure, and what does that same move, or the poster's distress, do to the margin I hold?

2. The examples

General wrong-way margin

- **Payer swap secured by a high-quality government bond.** A rate rise increases the swap's value while the bond's value declines. For a receiver swap "the situation is reversed and there would be a beneficial right-way margin position". Given the relatively low volatility of rates, "this is not generally a major problem" (Figure 5).
- **Cross-currency swap secured by cash in one of its currencies.** "If margin is held in the currency being paid, then an FX move will simultaneously increase the exposure and reduce the value of the collateral." GARP calls this "a more significant example", because FX moves are larger and can jump.

Specific wrong-way margin

- **An entity posting its own bonds:** "obviously a very weak mitigant against exposure and CVA, although it may mitigate FVA". A sovereign posting its own bonds is the same case (17.3.5).
- **A bank posting bonds of its own sovereign:** "another clearly problematic example". High-quality sovereign bonds are normally eligible, so a CSA may not prevent it.
- **Even cash:** receiving euros from European sovereigns or European banks; "even a European bank posting cash in euros will give rise to a potentially problematic linkage and wrong-way risk" (17.5.3).
- **Collateral tied to a CDS reference entity:** a protection seller posting bonds of the reference entity's subsidiary. At the credit event the exposure jumps and the collateral collapses together (Question CR-20 n.4).

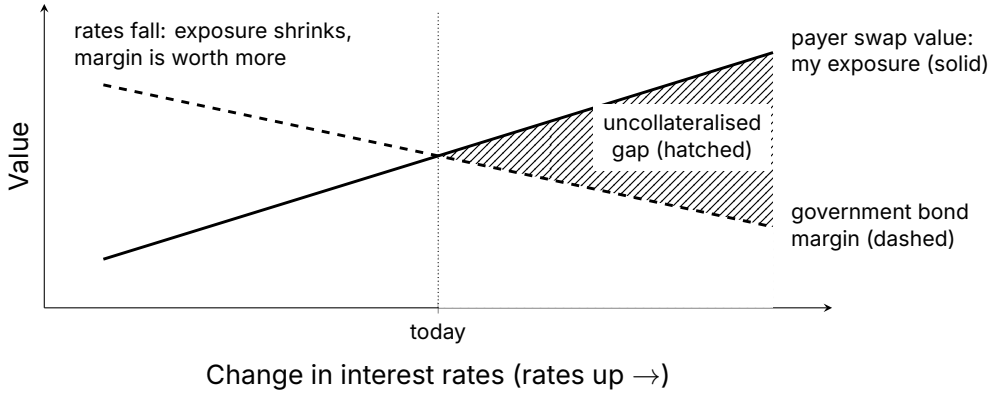


Figure 5. A payer swap secured by a government bond (schematic). A rate rise lifts the exposure and lowers the margin at the same time, opening an uncollateralised gap: general wrong-way margin. Swap the roles (a receiver swap) and the bond gains as the exposure grows: right-way margin.

3. What the rules and haircuts do

- **Bilateral margin rules** (17.4.4): margin "must not be 'wrong-way', meaning correlated to the default of the counterparty (e.g. a counterparty posting its own bonds or equity)". Satisfactory margin: cash; high-quality government and central bank securities; high-quality corporate or covered bonds; equity in major stock indices; gold.
- **Haircuts** reflect volatility, liquidity, default risk, maturity and any wrong-way relationship. The wrong-way factor is "often the hardest to implement in a prescriptive fashion". Assets with significant credit or liquidity risk are generally avoided because haircuts cannot practically be large enough.
- **CCPs** face adverse selection in the margin they accept and can impose higher haircuts on assets carrying WWR (CR-20 o).

Arrangement	What moves together	Type	Severity
Payer swap, government bond margin	Rates up: exposure up, bond down	General	Minor (low rate volatility)
Receiver swap, government bond margin	Rates down: exposure up, bond up	Right-way	Beneficial
Cross-currency swap, cash in the currency paid	FX move: exposure up, margin down	General	More significant
Counterparty's own bonds or equity	Poster weakens: margin falls	Specific	Very weak for CVA; banned under bilateral rules
Bank posts its own sovereign's bonds	Bank and sovereign weaken together	Specific	Clearly problematic
Euro cash from a European bank	Euro and bank stress together	Linkage	Potentially problematic
Bonds tied to a CDS reference entity	Credit event: exposure jumps, margin collapses	Specific	Severe

Exam traps

- **Polarity flip.** A government bond securing a *payer* swap is wrong-way; securing a *receiver* swap it is right-way.
- **Quality is not independence.** High credit quality does not stop a bond falling when rates rise, or a bank's own sovereign's bonds falling with the bank.
- **Daily valuation.** Marking own bonds to market every day does not remove their link to the poster's credit quality.
- **Which currency?** Cash in the currency the collateral-taker *pays* on the swap is the wrong-way choice.
- **Cash is not automatically safe.** Euro cash from a European bank or sovereign carries a linkage.
- **General versus specific.** Market-driven links (rates, FX) are general; links to the poster's own credit (own bonds, own sovereign) are specific.

Practice questions: CR-20 n

Question CR-20 n.1

Bank Q1070 • New question written for this bank

A dealer enters a 10-year pay-fixed interest rate swap with a hedge fund, and the hedge fund posts high-quality government bonds as collateral against the dealer's exposure. Which of the following statements about this collateral arrangement is correct?

A. It is right-way collateral, because government bonds are high quality and so retain their value in a rate shock.

B. It would be wrong-way collateral if the dealer were instead receiving fixed, because a fall in rates would then raise the exposure and reduce the bonds' value.

C. It is wrong-way collateral, because a rise in interest rates increases the swap's value to the dealer while reducing the value of the government bonds held.

D. It is neutral, because the collateral and the swap are both rate-driven and therefore always change in value by the same amount.

Question CR-20 n.2

New • built from GARP Ch.20, 20.6.6 • shape A20·S3

A US bank has a five-year cross-currency swap with a Brazilian corporate in which the bank pays Brazilian real (BRL) and receives US dollars. The CSA lets the corporate post cash in BRL or in USD, or short-dated US Treasury bills. Measuring all values in USD, a risk analyst is asked which of these forms of margin is wrong-way collateral for the bank. Which answer is correct?

A. USD cash, because a weaker real raises the corporate's cost of buying the dollars it must post.

B. BRL cash, because a fall in the real raises the bank's exposure on the swap and lowers the USD value of the margin at the same time.

C. US Treasury bills, because rising US rates lower their value just as the swap's value rises.

D. None of them, because margin in either currency of the swap offsets the FX risk of the exposure.

Question CR-20 n.3

Bank Q1071 • New question written for this bank (options reordered)

A risk manager reviews a derivative with a corporate counterparty that posts its own senior bonds as collateral. Which of the following best describes the wrong-way risk associated with this collateral?

A. The value of the bonds is likely to fall when the counterparty's credit quality deteriorates, so the collateral weakens exactly as the default probability rises.

B. There is none, because the counterparty's own bonds are liquid and are valued daily at market prices.

C. The risk is severe because the bonds lose all value as soon as the derivative's exposure increases, regardless of the counterparty's credit quality.

D. The risk arises only if the derivative exposure itself rises in an interest rate shock, since the bonds are unrelated to the counterparty's own default.

Question CR-20 n.4 Bank Q1069 • BT P2.T6.24.33.1 (rebuilt)

Moe's Bank buys credit protection through a credit default swap (CDS) with a notional of USD 700 million on the bonds of Sully's Corporation. The protection seller, Larry's Securities, posts as collateral USD 300 million of bonds issued by DEF Inc., a wholly owned subsidiary of Sully's Corporation. If Sully's Corporation defaults, which of the following outcomes is most likely?

A. The DEF Inc. bonds will rise in value as investors move into related debt, so the collateral will exceed the CDS exposure.

B. The DEF Inc. bonds will hold their value, so the collateral will continue to cover the CDS exposure.

C. The DEF Inc. bonds will fall moderately, but the collateral will still cover most of the CDS exposure.

D. The DEF Inc. bonds will fall sharply just as the CDS exposure jumps, so the collateral will cover only a small part of the exposure.

Question CR-20 n.5 New • built from GARP Ch.17, 17.3.5, 17.4.4 and 17.5.3

A collateral manager is drafting the eligible-margin schedule for a new bilateral CSA with a European bank. Which of the following statements is correct?

A. Under the bilateral margin rules, the European bank may post its own bonds as margin, provided a haircut reflecting their price volatility is applied.

B. Cash margin cannot create wrong-way risk, so euro cash posted by the European bank is free of any adverse linkage.

C. Even cash can be wrong-way: euro cash from a European bank may be linked to the bank's own credit quality, and such links are the hardest haircut factor to set prescriptively.

D. High-quality sovereign bonds are free of wrong-way risk by definition, so accepting them prevents the European bank from posting margin linked to its own credit quality.

Question CR-20 n.6 New • built from GARP Ch.20, 20.6.6 • shape A20·S3

A model validator is classifying four margin arrangements reviewed this quarter, using GARP's distinction between general and specific wrong-way risk. Which pairing of arrangement and classification is correct?

A. A dealer holding high-quality government bonds as margin against a receiver interest rate swap: general wrong-way margin.

B. A bank holding, as margin from another bank, bonds issued by that bank's own sovereign: specific wrong-way margin.

C. A dealer holding high-quality government bonds as margin against a payer interest rate swap: specific wrong-way margin.

D. A dealer holding a corporate counterparty's own senior bonds as margin: general wrong-way margin driven by interest rates.

Solutions • CR-20 n

Solution CR-20 n.1 — Answer: C

Bank Q1070 • New question written for this bank

Step 1 – What raises the dealer's exposure?

The dealer pays fixed and receives floating, so the swap gains value when rates rise.

Step 2 – What does that move do to the margin?

Government bond prices fall when rates rise. The margin loses value exactly as the exposure grows.

$\Rightarrow$ Exposure up, collateral down: general wrong-way collateral (Figure 5).

Interpretation

GARP adds that this is not generally a major problem, because rate volatility is relatively low. The credit quality of the bond is irrelevant; the issue is that both values are driven by rates in opposite directions for the dealer.

Why the other options are wrong

A is incorrect. High quality protects against default of the issuer, not against the price fall caused by a rate rise.

B is incorrect. Reversed: for a receive-fixed swap a fall in rates raises both the exposure and the bond price, a right-way margin position.

D is incorrect. The two values move in opposite directions for the dealer, which is why the protection erodes.

LO: CR-20 n: Identify examples of wrong-way collateral. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.6 Collateralisation and WWR

Solution CR-20 n.2 — Answer: B

New • built from GARP Ch.20, 20.6.6

Step 1 – What raises the bank's exposure?

The bank pays BRL and receives USD. A fall in the real makes the BRL it pays cheaper in dollar terms, so the swap gains value for the bank.

Step 2 – What does that move do to each form of margin?

BRL cash loses USD value as the real falls. USD cash has no FX link for a USD-measuring bank. Short-dated Treasury bills have little price sensitivity and no link to the real.

$\Rightarrow$ BRL cash is the currency the bank *pays* on the swap: GARP's "margin held in the currency being paid", where an FX move raises the exposure and cuts the collateral at the same time.

Interpretation

GARP calls this a more significant example of general wrong-way margin than the payer swap secured by bonds, because FX moves are larger and, for emerging-market currencies, can jump. The bank should restrict eligibility to USD or apply a heavy FX haircut to BRL.

Why the other options are wrong

A is incorrect. The corporate's cost of obtaining dollars is a funding issue for the poster; USD cash held by the bank does not lose value when the real falls.

C is incorrect. Applies the payer-swap example to the wrong trade: short-dated bills barely move with rates, and US rates are not the driver of this swap's exposure.

D is incorrect. Margin in the currency being paid moves against the exposure; it does not offset it.

LO: CR-20 n: Identify examples of wrong-way collateral. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.6 Collateralisation and WWR

Solution CR-20 n.3 — Answer: A

Bank Q1071 • New question written for this bank

Step 1 – The link

Bonds issued by the counterparty are priced off its own credit quality. As it deteriorates, the bonds fall, and at default they are worth only their recovery value.

$\Rightarrow$ The collateral weakens exactly as the default probability rises: specific wrong-way collateral.

Step 2 – How much does it still help?

GARP: "obviously a very weak mitigant against exposure and CVA, although it may mitigate FVA". It is banned under the bilateral margin rules (own bonds or equity).

Interpretation

The mechanism is the poster's credit quality, not market rates. Daily valuation only records the fall; it does not prevent it.

Why the other options are wrong

B is incorrect. Liquidity and daily valuation do not remove the link between the bonds and the issuer's credit quality.

C is incorrect. Overstated and mis-specified: the bonds fall because the poster's credit deteriorates, not merely because the derivative's exposure rises, and they keep some recovery value.

D is incorrect. Wrong mechanism: the link is to the poster's own default, and the bonds are directly related to it.

LO: CR-20 n: Identify examples of wrong-way collateral. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.6; Ch.7 (GARP CR Ch.17), 17.4.4

Solution CR-20 n.4 — Answer: D

Bank Q1069 • BT P2.T6.24.33.1

Step 1 – The exposure

Moe's Bank is owed the protection payment if Sully's defaults. The CDS exposure jumps from near its current value to roughly notional times $(1 - \text{recovery})$ at the credit event.

Step 2 – The collateral

DEF Inc. is a wholly owned subsidiary of the reference entity, so its credit quality is tied to Sully's. A parent default signals distress across the group, and DEF's bonds are likely to fall sharply (or default) at the same moment.

$\Rightarrow$ Collateral value is lowest exactly when exposure is highest: specific wrong-way collateral, and because the exposure jumps, extra margin cannot be called in time.

Interpretation

The notional and collateral amounts only set the scale; the answer turns on the link between the collateral and the reference entity. Protection buyers should require margin unrelated to both the seller and the reference entity.

Why the other options are wrong

A is incorrect. Reverses the direction of the dependence: a group member's bonds fall, not rise, when the parent defaults.

B is incorrect. Treats the collateral as independent of the reference entity, ignoring the parent-subsidiary credit link.

C is incorrect. Recognises a fall but understates it; with a strong link and a jump in exposure, the collateral covers only a small part.

LO: CR-20 n: Identify examples of wrong-way collateral. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.5 and 20.6.6

Solution CR-20 n.5 Answer: C

New • built from GARP Ch.17, 17.3.5, 17.4.4 and 17.5.3

Step 1 – Can cash be wrong-way?

Yes. GARP: "adverse correlations can also be present with cash margin: an example would be receiving euros from European sovereigns or European banks", and "even a European bank posting cash in euros will give rise to a potentially problematic linkage and wrong-way risk".

Step 2 – How haircuts treat it

The haircut checklist includes any relationship between the margin's value and the counterparty's default or the exposure, but this factor is "often the hardest to implement in a prescriptive fashion".

$\Rightarrow$ C is correct.

Interpretation

A severe euro crisis would hit the bank and the euro together. The manager might prefer cash in a currency not linked to the bank's credit, or accept the linkage knowingly.

Why the other options are wrong

A is incorrect. The bilateral margin rules say margin must not be wrong-way, with a counterparty posting its own bonds or equity as the example; a haircut does not make it eligible.

B is incorrect. Absolute claim: GARP gives euro cash from European banks or sovereigns as a wrong-way example.

D is incorrect. High-quality sovereign bonds are normally eligible, which is exactly why a CSA "may not prevent" a bank posting bonds of its own sovereign, a specific wrong-way case.

LO: CR-20 n: Identify examples of wrong-way collateral. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.3.5 Haircuts; 17.4.4 Eligible Assets and Haircuts; 17.5.3 Liquidity, FX, and Wrong-way Risks

Solution CR-20 n.6 Answer: B

New • built from GARP Ch.20, 20.6.6

Step 1 – General versus specific

General wrong-way margin comes from a market link between the margin's value and the exposure (rates, FX). Specific wrong-way margin comes from "a more direct relationship between the type of margin and the counterparty credit quality".

Step 2 – Classify each arrangement

Own-sovereign bonds posted by a bank: the bank's credit and its sovereign's are directly linked; GARP lists it with own bonds as specific ("another clearly problematic example"). Payer swap with government bonds: general. Receiver swap with government bonds: right-way. A corporate's own bonds: specific.

$\Rightarrow$ Only B pairs the arrangement with its correct class.

Interpretation

Specific links call for qualitative treatment (eligibility limits, large haircuts); general links can be modelled and priced.

Why the other options are wrong

A is incorrect. Polarity flip: with a receiver swap a rate fall raises both the exposure and the bond's value, a right-way position.
C is incorrect. Sibling swap: the payer-swap case is general (market-driven), not specific.
D is incorrect. Sibling swap: own bonds are linked to the poster's credit quality, the specific case, not an interest rate effect.

LO: CR-20 n: Identify examples of wrong-way collateral. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.1 Table 20.7; 20.6.6

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 o

P1 CONCEPT

Discuss the impact of wrong-way risk on central counterparties (CCPs).

Workbook: A20 Wrong-Way and Right-Way Risk • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A20): "the effect on CCPs" is one of the examinable sub-items Map #26 lists behind the August wrong-way risk questions.

Vardeez: CCPs rely on collateral and default funds from members to cover losses, and WWR makes them vulnerable in two ways: higher-quality members may bring higher WWR (so margin and default fund contributions need adjusting), and risky collateral posted by members adds WWR (so haircuts need adjusting).

Reported exam shapes

- **A20·S4** The effect of WWR on CCPs.

What the questions look like: why CCPs are prone to WWR; what higher initial margin does and does not achieve; the "perverse" conclusion that better-quality members should post more; haircuts on members' own bonds; clearing CDSs; the CCP waterfall seen as a CDO. All conceptual.

Worked elsewhere: A20.2 (option C is this LO's key point) under CR-20 m. CR-20 q (P2) covers the implications of central clearing on WWR; the loss waterfall itself is CR-18 f (P3). Bank Q863 is re-filed here from CR-18 e (a P4 LO not in this book).

1. Why CCPs are prone to wrong-way risk

"Given their reliance on collateralisation as their primary protection via variation and initial margin, CCPs may be particularly prone to WWR, especially those that clear products such as CDSs" (GARP 20.6.7). A CCP aims to contain the losses from a member's default within that member's own resources, the **defaulter-pays** principle. WWR threatens exactly that: if the defaulter's initial margin and default fund contribution prove insufficient, "the CCP would impose losses on its members and may be in danger of becoming insolvent itself".

Members can lose well short of CCP insolvency. Initial margin is taken to a high coverage, so the expected loss to a CCP looks small, but extreme market moves and the difficulty of auctioning a defaulter's portfolio can create large losses. In the Nasdaq Nordic commodity default of September 2018, clearing members lost money through their default fund contributions although the CCP itself was not insolvent (GARP 20.5.4).

2. Credit quality and exposure are kept apart

CCPs "tend to disassociate credit quality and exposure". A member must meet a minimum credit standard (set by the CCP, not by external ratings), but its initial margin and default fund contribution are then driven primarily by the *market risk* of its portfolio. "In doing this, CCPs are in danger of implicitly ignoring WWR." Some CCPs base margins partly on credit quality, but not as a large effect. For CDSs, quantifying the WWR component of initial margin and default fund contributions is "far from an easy task".

The perverse conclusion. WWR increases with credit quality (CR-20 k), and Pykhtin and Sokol (2012) argue that a large dealer represents more WWR than a smaller or weaker counterparty. Hence CCPs should arguably require **greater** initial margin and default fund contributions from **better-credit-quality and more systemically important** members. Such members are less likely to default, but the impact if they do is likely to be more severe.

3. The CCP waterfall behaves like a CDO

Murphy, Pirrong and Gregory compare the CCP loss waterfall with a collateralised debt obligation (Figure 6). The defaulter-pays initial margin and default fund, together with CCP equity, are the **first loss**. Surviving members, through their default fund contributions and other loss-allocation exposures, hold a **second-loss** position. That position should correspond to an unlikely event (otherwise initial margin coverage would be too thin), so it is "rather senior in CDO terms", and senior tranches are known to be heavily concentrated in systemic risk. WWR losses that get past the first loss will therefore arrive in systemic scenarios, when the surviving members are themselves under stress.

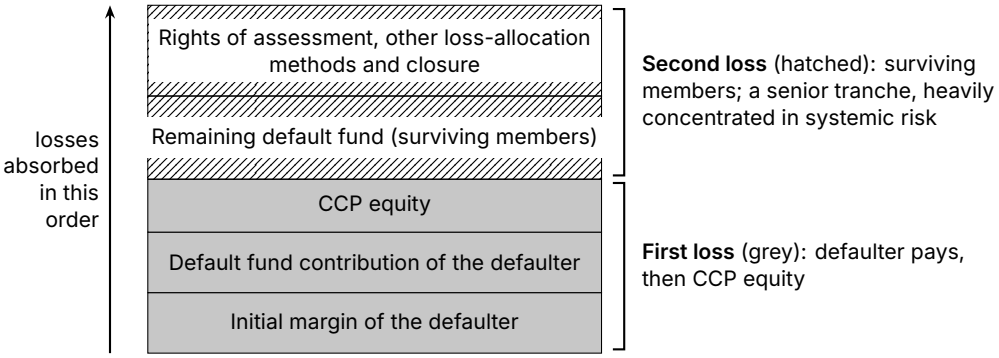


Figure 6. The CCP loss waterfall seen as a CDO, after GARP Figure 20.24. Wrong-way losses that exhaust the grey first-loss layers reach the hatched layers held by surviving members, and do so mainly in systemic scenarios.

4. Wrong-way risk in the margin a CCP receives

CCPs are under pressure to accept a wide range of eligible securities as initial margin. Riskier and less liquid assets put more weight on haircut calculations, which add risk if underestimated, and they invite **adverse selection**: members "will naturally choose to post margin that has the greatest risk (relative to its haircut) and may also present the greatest WWR to a CCP", for example a European bank posting European sovereign debt. Members posting their own or affiliates' bonds is the extreme case (CR-20 n). Schweser's remedy: higher haircuts on the specific assets that carry WWR.

5. Clearing CDSs

GARP's central clearing chapter lists "no WWR" among the features a clearable product should ideally have: when a member's default is adversely linked to the value of its contracts, "the value of the contract will potentially decline rapidly at the time of the counterparty default", which complicates default management. CDSs catalysed the clearing mandate, "and yet they are arguably the most difficult to clear centrally due to WWR". A member that sells protection on names linked to itself (its parent, its sovereign) is the clearest case.

	How WWR hurts the CCP	Response in GARP and Schweser
Reliance on margin	Defaulter-pays resources may be exhausted by a jump at default	Quantify WWR in initial margin and default fund (hard for CDSs)
Market-risk-based margin	Credit quality and exposure kept apart, so WWR is implicitly ignored	More initial margin and default fund from better-quality, systemic members
Margin received	Adverse selection: riskiest assets relative to haircut, possibly the most wrong-way	Higher haircuts on assets carrying WWR; tighter eligibility
Loss mutualisation	Losses past the first loss hit surviving members in systemic states	Members should quantify their exposure (CVA) to the CCP
CDS clearing	Contract value can collapse at a linked member's default	Review margin and default fund adequacy for such positions

Exam traps

- **"Central clearing removes WWR."** It does not: the CCP inherits it through margin and the default fund.
- **"Higher initial margin eliminates WWR."** It lowers exposure towards zero, but coverage is finite; extreme moves and auction difficulties can still exceed it.
- **Polarity flip.** WWR rises with credit quality, so the argument is for *more* margin from stronger, systemic members, not less.
- **External ratings.** Membership standards are set by the CCP itself, and margins are driven by market risk, not by ratings.
- **Haircuts reversed.** Own or affiliate bonds call for higher haircuts, not lower ones.
- **Netting.** Multilateral netting reduces exposure but does not break the link between a member's default and the value of its positions.
- **Waterfall order.** The defaulter's margin and default fund are used first; surviving members' default fund contributions come after CCP equity.
- **Senior means safe?** The members' second-loss layer is senior, and for that reason heavily concentrated in systemic risk.

Practice questions: CR-20 o

Question CR-20 o.1

Bank Q1072 • AnalystPrep Q.5475 (rebuilt, options reordered)

A central counterparty (CCP) raises the initial margin it collects from clearing members. A risk manager at a clearing member notes that the Nasdaq default showed clearing members can lose money through default fund contributions even when the CCP itself is not insolvent. Which of the following best describes the effect of the higher initial margin on the wrong-way risk (WWR) faced by the CCP and its members?

A. It eliminates WWR entirely, because continuous full collateralisation removes any possibility of loss to the CCP or its clearing members.

B. It increases the CCP's exposure, because margin posted by members has an opportunity cost and therefore raises member credit risk.

C. It reduces exposure and the CVA to the defaulter, but does not eliminate WWR, because extreme market moves and difficulty auctioning a defaulted portfolio can still produce losses exceeding the margin.

D. It has no effect on exposure, because initial margin only matters for uncollateralised counterparties and clearing members are already fully collateralised.

Question CR-20 o.2

Bank Q1073 • BT P2.T6.24.33.3 (rebuilt, options reordered)

A central counterparty (CCP) clearing interest rate swaps allows clearing members to post corporate bonds as initial margin. Several members have begun posting bonds issued by themselves or their affiliates, which are illiquid. Which of the following is the most appropriate way for the CCP to address the wrong-way risk (WWR) this creates?

A. Apply lower haircuts to the bonds, since a larger accepted collateral value reduces the exposure to the member.

B. Impose higher haircuts on such assets, since their value is likely to fall exactly when the posting member is more likely to default.

C. Take no action, since corporate bond prices are driven by interest rates and are unrelated to the member's credit quality.

D. Reduce the members' default fund contributions, since the bonds already cover the losses that would arise from their default.

Question CR-20 o.3 Bank Q1074 • New question written for this bank

A risk manager at a central counterparty (CCP) reviews how initial margins and default fund contributions are set. Margins are driven primarily by the market risk of each member's portfolio, and members must meet a minimum credit standard to be admitted. A colleague argues that wrong-way risk (WWR) is therefore adequately captured. Which of the following is the most appropriate response?

A. The colleague is incorrect, because WWR tends to increase with credit quality, so the CCP implicitly ignores it and should arguably require higher margin and default fund contributions from stronger, more systemic members.

B. The colleague is correct, because margin set from portfolio market risk already captures the jump in exposure at the default of a systemic member.

C. The colleague is incorrect, because WWR falls as the credit quality of a member rises, so the CCP should require less margin from stronger members.

D. The colleague is correct, because the minimum credit standard ensures that members' defaults are independent of their exposures.

Question CR-20 o.4 Bank Q863 • AnalystPrep Q.5484 (rebuilt, options reordered; re-filed from CR-18 e)

A central counterparty (CCP) clears single-name credit default swaps. A clearing member sells protection on a reference entity that is closely tied to the member's own financial health (for example, its parent bank). Which of the following best describes why this creates wrong-way risk (WWR) for the CCP, and how the CCP should respond?

A. Central clearing removes WWR because the CCP is the buyer to every seller, so no change to margin or default fund methodology is needed.

B. WWR arises mainly because the CCP's interest income on margin falls when credit spreads widen, so the CCP should hold less margin.

C. WWR is a purely bilateral issue that is eliminated by multilateral netting of the member's positions, so the CCP only needs to monitor the member's credit rating.

D. The member's default makes the value of its cleared position fall sharply at the time of default, so losses may exceed initial margin sized for normal conditions; the CCP should review the adequacy of initial margin and default fund contributions for such positions.

Question CR-20 o.5 New • built from GARP Ch.20, 20.6.7 • shape A20·S4

A clearing member's risk committee asks how its default fund contribution to a CCP should be viewed in the light of wrong-way risk. A consultant compares the CCP's loss waterfall with a collateralised debt obligation (CDO). Which statement best reflects that comparison?

A. The member's default fund contribution is an equity (first-loss) tranche, hit whenever any member defaults, so its risk is mainly idiosyncratic.

B. The member's default fund contribution and other loss-allocation exposures form a rather senior second-loss position, behind the defaulter's resources and CCP equity, so losses reach it mainly in systemic scenarios.

C. The member's default fund contribution ranks ahead of the defaulter's initial margin, so it absorbs the first losses on any member default.

D. The member's exposure through the default fund is diversified across many members, so like a senior CDO tranche it carries little systemic risk.

Question CR-20 o.6 New • built from GARP Ch.20, 20.6.7 • shape A20·S4

A regulator's review of a CCP that clears interest rate swaps and single-name CDSs lists ways in which wrong-way risk could leave the CCP's resources short after a member default. Each of the following is consistent with GARP's discussion EXCEPT:

A. Initial margins and default fund contributions are driven mainly by the market risk of each portfolio, so the CCP may implicitly ignore wrong-way risk.

B. Members may choose to post the eligible margin with the greatest risk relative to its haircut, which may also carry the greatest wrong-way risk to the CCP.

C. Multilateral netting across members breaks the link between a member's default and the value of its cleared portfolio, so wrong-way risk is confined to bilateral trades.

D. The default of a large, highly rated, systemically important member is less likely but would be a bigger shock, so the wrong-way risk it brings is greater.

Solutions • CR-20 o

Solution CR-20 o.1 — Answer: C

Bank Q1072 • AnalystPrep Q.5475

Step 1 – What initial margin does

Initial margin is posted up front against potential future exposure, so it reduces the CCP's exposure, and the CVA to a defaulter, towards zero.

Step 2 – Why towards zero is not zero

Margins are set to a high but finite coverage. Large, correlated market moves at the time of a default, and the difficulty of auctioning a defaulted portfolio, can exceed them. GARP cites the Nasdaq case for exactly this.

⇒ Higher initial margin mitigates CCP WWR but does not eliminate it. Losses beyond the defaulter's resources fall on the default fund and so on surviving members.

Interpretation

That is why GARP suggests clearing members quantify their CVA to a CCP, and why default-fund capital requirements are relatively high.

Why the other options are wrong

A is incorrect. Absolute claim: "eliminates" ignores the finite coverage of margin and the Nasdaq-type losses to default funds.

B is incorrect. Reverses the effect: the opportunity cost of margin is a cost to the poster, not extra exposure for the CCP; margin reduces exposure.

D is incorrect. Margin reduces exposure for any counterparty, and members are not fully collateralised against extreme moves, as default-fund losses show.

LO: CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.5.3 Initial Margin; 20.5.4 CVA to CCPs; 20.6.7 Central Clearing and WWR

Solution CR-20 o.2 — Answer: B

Bank Q1073 • BT P2.T6.24.33.3

Step 1 – Why the collateral matters to a CCP

The CCP's first protection against a member's default is that member's margin and default fund contribution, so the value of that margin *at the default* is what counts.

Step 2 – The wrong-way link

Bonds issued by the member or its affiliates depend on the member's credit quality. If the member defaults they are likely to lose value and become even less liquid: wrong-way collateral (CR-20 n).

⇒ Schweser's mitigant: impose higher haircuts on the specific assets accepted as collateral that carry WWR.

Interpretation

Higher haircuts protect the defaulter-pays layer of the waterfall. The CCP could go further and exclude own or affiliate bonds, in line with the bilateral margin rules that ban a counterparty posting its own bonds.

Why the other options are wrong

A is incorrect. Lower haircuts credit the bonds with more value and widen the shortfall when they fall at default.

C is incorrect. Ignores that the issuer's credit quality drives these bonds' value; that link is the

source of the WWR.
D is incorrect. Cutting default fund contributions weakens the waterfall further, when the bonds' value is unreliable at default.

LO: CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7; SchweserNotes 2026 Book 2, Reading 38, LO 38.o

Solution CR-20 o.3 Answer: A

Bank Q1074 • New question written for this bank

Step 1 – How the CCP sets requirements

CCPs disassociate credit quality and exposure: a minimum credit standard for entry, then margin and default fund contributions driven by portfolio market risk. GARP: "CCPs are in danger of implicitly ignoring WWR."

Step 2 – Which members bring the most WWR?

WWR increases with credit quality, and a large dealer represents more WWR than a smaller or weaker counterparty (Pykhtin and Sokol). The default of a strong, systemic member is a bigger shock.
$\Rightarrow$ GARP's "perverse" conclusion: higher initial margin and default fund contributions from better-credit-quality, more systemically important members.

Interpretation

Stronger members are less likely to default, but the impact if they do is likely to be more severe. A minimum credit standard addresses the first point, not the second.

Why the other options are wrong

B is incorrect. Market-risk margin does not build in the jump and stressed volatility around a systemic member's default.
C is incorrect. Polarity flip: WWR increases with credit quality, so stronger members warrant more, not less, margin.
D is incorrect. A minimum credit standard does not make a member's default independent of its exposure.

LO: CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7 Central Clearing and WWR

Solution CR-20 o.4 Answer: D

Bank Q863 • AnalystPrep Q.5484

Step 1 – Where the WWR is

WWR for a CCP arises when a member's default is adversely linked to the value of its cleared contracts. A member selling protection on its own parent is the clearest case: at the member's default its position is a large loss, just when it can no longer pay variation margin.

Step 2 – Why margin may fall short

Initial margin and default fund contributions sized on normal market moves will understate the loss over the margin period of risk when the contract's value collapses at the default.
$\Rightarrow$ The CCP should reassess margin and default fund adequacy for such positions (or limit them). GARP calls CDSs "arguably the most difficult to clear centrally due to WWR".

Interpretation

The CCP does not escape WWR by standing in the middle; it inherits it, and the defaulter-pays layer is where it first bites.

Why the other options are wrong

A is incorrect. Central clearing relocates WWR to the CCP; it does not remove it.

B is incorrect. Confuses WWR with investment-income risk; holding less margin would make the shortfall worse.

C is incorrect. Netting reduces exposure across positions but does not remove the adverse link between a member's default and the value of its positions; ratings alone do not size margin.

LO: CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7; Ch.8 (GARP CR Ch.18), 18.2 Mechanics of Central Clearing

Solution CR-20 o.5 Answer: B

New • built from GARP Ch.20, 20.6.7

Step 1 – Map the waterfall to tranches

The first loss is covered by the defaulter-pays initial margin and default fund, together with CCP equity. Surviving members, through their default fund contributions and other loss-allocation exposures, have a second-loss position.

Step 2 – How senior is the second loss?

It should correspond to a relatively unlikely event, otherwise initial margin coverage would be too thin. In CDO terms it is "rather senior", and senior tranches are heavily concentrated in systemic risk.

$\Rightarrow$ Losses reach the member's contribution mainly in systemic scenarios, exactly when WWR has made the defaulter's resources insufficient and other members are under stress.

Interpretation

A clearing member should treat its default fund contribution as a systemic, senior-tranche exposure and quantify it (its CVA to the CCP), rather than as a diversified, idiosyncratic one.

Why the other options are wrong

A is incorrect. The equity (first-loss) layer is the defaulter's own resources plus CCP equity, not the surviving members' default fund.

C is incorrect. Reverses the waterfall order: the defaulter's margin and default fund are used first.

D is incorrect. Seniority does not mean low systemic risk: senior tranches are known to be heavily concentrated in it.

LO: CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7, Figure 20.24

Solution CR-20 o.6 Answer: C

New • built from GARP Ch.20, 20.6.7

Step 1 – Check each channel against GARP

A: initial margins and default fund contributions driven by market risk mean CCPs are "in danger of implicitly ignoring WWR". Consistent. B: members "will naturally choose to post margin that has the greatest risk (relative to its haircut) and may also present the greatest WWR". Consistent. D: WWR increases with credit quality; a better-quality member is less likely to default but its default is a bigger shock. Consistent.

Step 2 – The exception

Multilateral netting reduces the exposure the CCP faces, but a member whose default is linked to the value of its positions still leaves the CCP with that linked loss. GARP says CCPs are par-

ticularly prone to WWR, especially for CDSs.
⇒ C is the statement GARP does not support.

Interpretation

The three true statements are the CCP's three WWR channels: market-risk-based margining, adverse selection in the margin it receives, and the size of systemic members' defaults.

Why the other options are wrong

A is incorrect. Consistent with GARP: CCPs disassociate credit quality and exposure.
B is incorrect. Consistent with GARP: adverse selection in eligible margin, for example a European bank posting European sovereign debt.
D is incorrect. Consistent with GARP: WWR increases with credit quality, which is why it argues for more margin from systemic members.

LO: CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7 Central Clearing and WWR

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 p

P1 CONCEPT

Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches.

Workbook: A20 Wrong-Way and Right-Way Risk • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A20): Map #26 lists "the modelling approaches (hazard-rate, structural, parametric, jump)" among the examinable sub-items behind the August wrong-way risk questions.

Vardeez: a four-row table of methods. The two ends are the examinable pair: the hazard rate approach is the easiest to implement and the most likely to understate WWR; the jump approach is the best suited to it, because the wrong-way events that matter are discontinuous.

Reported exam shapes

- **A20·S5** Modelling approaches: hazard rate (intensity), structural, parametric, jump.

What the questions look like: match a description to its method, with true facts about a sibling method as distractors; choose the method that fits an observation (a weak CVA response, abrupt exposure, quanto CDS prices); and say how WWR enters the CVA formula. All conceptual.

Worked elsewhere: A20.2 (option B is the intensity-versus-jump sibling swap) and Bank Q1077 (the jump view of collateral) under CR-20 m. Conditional EPE and the problems of modelling WWR are in CR-20 k, Section 4.

1. What every model has to deliver

WWR enters the CVA formula through the exposure conditional on default (CR-20 k):

$$\text{CVA}_{\text{WWR}} = -\text{LGD} \sum_{i=1}^{m} \text{EPE}\left(t_i \,|\, \tau = t_i\right) \times \text{PD}(t_{i-1}, t_i)$$

The models differ in how they generate the link between default and exposure. Three are continuous (intensity, structural, parametric) and aimed mainly at general WWR; jump approaches let a market variable move discontinuously at the default time and suit specific WWR. All face the problems of CR-20 k: uninformative history, misspecified dependence and unclear direction.

2. Hazard rate (intensity) approach

- **How it works.** Give the counterparty's credit spread a stochastic process and correlate it with the processes that drive the exposure. Defaults are generated by the spread ("intensity") process, and the conditional EPE is computed in the usual way, but only on paths where a default has occurred (typically paths with wide spreads).
- **Strengths.** "The most obvious and easy to implement." Relatively tractable: spread paths are generated first and exposure paths are simulated only where a default is observed. The correlations can be observed directly from historical time series of credit spreads and market variables.
- **Payer swap illustration.** Positive correlation between spreads and rates puts rates higher in default scenarios and raises the positive exposure (WWR); negative correlation increases the negative exposure (right-way). Empirically, rates and default rates are generally negatively related.

- **Weakness.** It generates only **weak dependence** between exposure and default, even with correlations close to $\pm 100\%$, so "it may underestimate the true WWR effect".

3. Structural approach

- **How it works.** "An even simpler and more tractable approach to general WWR": specify the dependence directly between the default time and the exposure distribution. Both are mapped separately onto a bivariate distribution (for example a Gaussian copula). Positive dependence couples an early default with a high exposure (WWR); negative dependence with a low one (right-way).
- **Strength.** No exposures are recalculated: the original unconditional values are sampled directly, so WWR is added on top of the existing exposure engine.
- **Effect.** Qualitatively like the intensity model, but the effect is **much stronger** at the same correlation (GARP Figures 20.19 and 20.20; Figure 7).
- **Weaknesses.** The correlation parameter is **opaque and difficult to calibrate**, and the unconditional exposure distribution may not contain all the information needed to define WWR. Richer versions (Iscoe et al.; De Prisco and Rosen) correlate default with the exposure drivers directly, which makes estimation more achievable.

4. Parametric approach

- **How it works.** Hull and White (2011) link the default probability to the exposure through a simple functional relationship.
- **Calibration.** Either an intuitive what-if scenario, or history: compute the portfolio's value on past dates and compare it with the counterparty's credit spread on those dates. High portfolio values together with larger-than-average spreads indicate WWR; a higher dependency parameter means a higher CVA.
- **Condition.** The current portfolio must be similar in nature to the one used in the historical calibration, and the data must show a meaningful relationship.

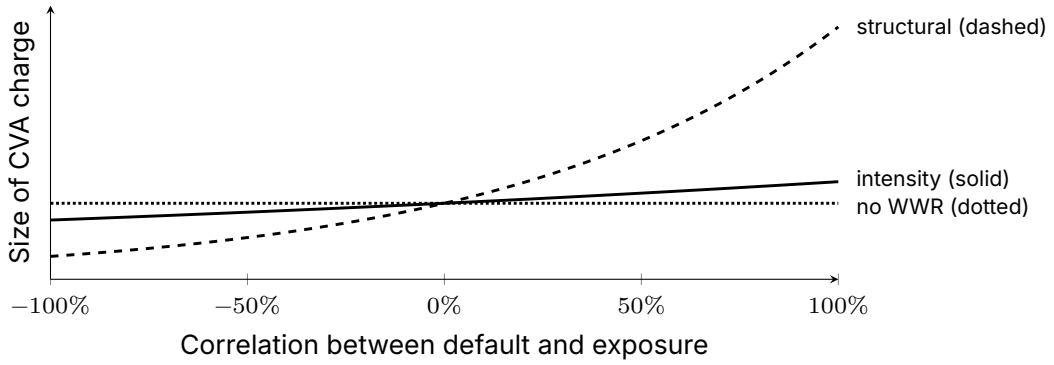


Figure 7. Schematic of GARP Figure 20.20: both continuous models raise the CVA charge as correlation rises, but the effect is much stronger in the structural model. The intensity model's weak response is why it may underestimate WWR.

5. Jump approaches

Jump approaches "may be more realistic, especially in cases of specific WWR". Levy and Levin (1999) assume the relevant FX rate jumps at the counterparty's default time. Schweser expresses the jump through a residual value (RV) factor: the FX rate depreciates by $(1 - \text{RV})$.

- **Size of the jump.** The implied jump is larger for better-rated sovereigns (Levy and Levin: 83% for AAA, 27% for BBB), probably because their default causes, or is caused by, a more severe shock. Implied euro jumps in the sovereign crisis: Greece 9%, Italy 17%, Spain 20%, Germany 25%. Chung and Gregory (2019), Japanese CDS data: sovereign 38.4%, financial institutions 13.5%, non-financial corporates 8% (Figure 8). WWR increases with credit quality and systemic importance.
- **Market evidence: the quanto basis.** CDSs on the same sovereign quoted in two currencies trigger on the same credit events and differ only in the currency of payment. Euro protection on Italy was about 30% cheaper than dollar protection at all maturities (GARP Table 20.8, April 2011): the market prices a euro fall at an Italian default.
- **Intensity cannot match it.** Ehlers and Schönbucher (2006) show an intensity-type model, with correlation between the FX rate and the spread, cannot reproduce the CDS data even with $\pm 100\%$ correlation. The relationship looks causal: the FX rate jumps around the default.

Maturity (years)	1	2	3	4	5	7	10
USD CDS on Italy (bp)	50	73	96	118	131	137	146
EUR CDS on Italy (bp)	35	57	63	78	91	97	103
EUR below USD (%)	30	22	34	34	31	29	29

GARP Table 20.8 (mid-market, April 2011); last row computed as $1 - \text{EUR/USD}$.

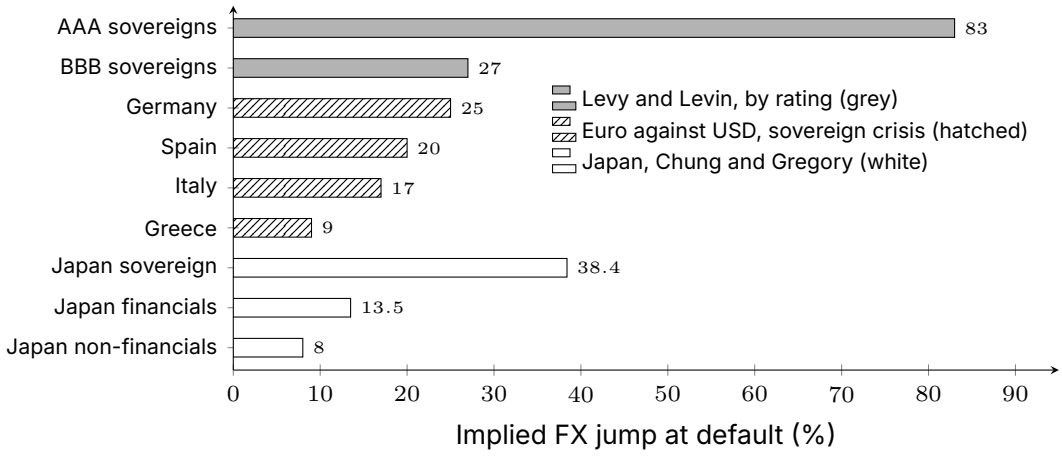


Figure 8. Implied jumps in the exchange rate at the counterparty's default, from GARP 20.6.4. Better-rated and more systemic names carry bigger jumps, which is why WWR rises with credit quality.

Collateral under each model. Jump approaches show margin as near useless against WWR; the continuous intensity and structural approaches show it as effective (CR-20 m, Section 2).

	How it links default and exposure	Strength of effect	Calibration and drawback
Hazard rate (intensity)	Stochastic spread correlated with exposure drivers; conditional EPE on default paths	Weak, even near $\pm 100\%$; may underestimate	Correlations from historical spread data; easiest to implement
Structural	Default time and exposure mapped onto a bivariate distribution (e.g. Gaussian copula)	Much stronger than intensity	Uses pre-computed exposures; correlation opaque and hard to calibrate
Parametric	Default probability a simple function of portfolio value (Hull and White)	Set by the dependency parameter	What-if or historical calibration; needs a similar portfolio and a meaningful history
Jump	Market variable (FX rate) jumps at default by $(1 - \text{RV})$	Strong; suits specific WWR	Implied from quanto CDS and default history; jumps larger for better credits

Exam traps

- **Intensity overstates?** No: it generates weak dependence and may *underestimate* WWR.
- **Structural needs re-simulation?** No: it samples the pre-computed unconditional exposures. It is the intensity approach that simulates exposure on default paths.
- **Sibling swaps.** Bivariate mapping is structural; a historical portfolio-value-against-spread link is parametric (Hull and White); a discrete FX move at default is jump.
- **Jump size reversed.** Implied jumps are larger for highly rated, systemic sovereigns and firms, not smaller.
- **Quanto evidence.** It supports jump models; intensity models cannot match it even at $\pm 100\%$ correlation.
- **Formula fixes that are not WWR.** WWR replaces EPE by EPE conditional on default; it does not change the discount factor or remove the recovery term.

Practice questions: CR-20 p

Question CR-20 p.1

Bank Q1075 • BT P2.T6.419.2 (rebuilt)

The standard CVA expression with no wrong-way risk is

$$\text{CVA} = (1 - \text{Rec}) \sum_i \text{DF}(t_i)\, \text{EE}(t_i)\, \text{PD}(t_{i-1}, t_i),$$

where EE is the unconditional expected exposure. When wrong-way risk is incorporated, which adjustment to this expression is appropriate?

A. Remove the $(1 - \text{Rec})$ term to reflect the most conservative recovery assumption.

B. Increase the discount factor $\text{DF}(t)$ to reflect the correlation.

C. Replace $\text{EE}(t)$ with the expected exposure conditional on default of the counterparty.

D. No adjustment is possible; the expression cannot be used when wrong-way risk is present.

Question CR-20 p.2 New • built from GARP Ch.20, 20.6.3 • shape A20·S5

A bank's model validation team tests the wrong-way risk produced by its CVA engine for a portfolio of payer interest rate swaps. The engine simulates the counterparty's credit spread as a stochastic process correlated with interest rates and computes exposure only on paths where a default occurs. Even with the correlation set to $+90\%$, the CVA rises only modestly above the no-WWR figure. Which conclusion is most appropriate?

A. The result is typical of the intensity (hazard rate) approach, which generates only weak dependence between exposure and default even near its correlation limits, so it may underestimate WWR.

B. The result shows that the payer swaps carry no material WWR, since the approach captures the full dependence once the correlation approaches 100%.

C. The engine must in fact be using a structural approach, which is known to give a weaker effect than the intensity approach at the same correlation.

D. The result is caused by the jump in interest rates the approach assumes at the default time, which offsets the correlation effect.

Question CR-20 p.3 New • built from GARP Ch.20, 20.6.3 • shape A20·S5

A counterparty risk manager wants to add general wrong-way risk to the bank's CVA without re-running its exposure simulation, by linking the counterparty's default time directly to the pre-computed exposure distribution through a bivariate (for example, Gaussian copula) distribution. Which of the following is the main drawback of this approach?

A. It requires exposures to be re-simulated on every path where a default occurs, which makes it computationally expensive.

B. It produces only a weak dependence between exposure and default, so its CVA barely changes with correlation.

C. It can only be calibrated by revaluing the portfolio at past dates against the counterparty's historical credit spread.

D. Its correlation parameter is opaque and hard to calibrate, and the unconditional exposure distribution may not contain all the information needed to define WWR.

Question CR-20 p.4 New • built from GARP Ch.20, 20.6.3 • shape A20·S5

Following Hull and White (2011), a bank links each counterparty's default probability to the value of its portfolio through a simple functional relationship. For one hedge fund, the team calibrates the dependency parameter by computing the value of its portfolio with the fund on dates over the past five years and comparing those values with the fund's credit spread on the same dates. Which of the following is the most important condition for this calibration to be reliable?

A. The fund's credit spread must be quoted in a liquid quanto CDS market, so that a jump in the FX rate at default can be implied.

B. The current portfolio must be similar in nature to the one used in the historical calibration, and the history must show a meaningful relationship between portfolio value and spread.

C. The correlation between the fund's spread and interest rates must be close to $+100\%$, otherwise the approach generates only weak dependence.

D. The default time and the exposure distribution must first be mapped onto a bivariate Gaussian distribution before the parameter can be estimated.

Question CR-20 p.5 Bank Q1078 • BT P2.T6.24.33.2 (rebuilt, options reordered)

A bank holds a cross-currency swap with a sovereign counterparty and models wrong-way risk with the jump approach. Which of the following statements best describes this approach?

A. It simulates a stochastic process for credit spreads and computes conditional exposure on paths where default occurs; it is the easiest method but tends to overstate WWR.

B. It maps the default-time and exposure distributions onto a bivariate distribution, so that an early default coupled with a high exposure signals WWR.

C. It assumes the relevant FX rate jumps by a factor related to a residual value at the counterparty's default time; the jump is typically larger for highly rated, systemically important sovereigns.

D. It infers WWR from the historical relationship between portfolio value and credit spreads, and is reliable only when history reflects current scenarios.

Question CR-20 p.6 Bank Q1076 • AnalystPrep Q.5476

LuxeBank is reassessing its wrong-way risk (WWR) models and is debating whether to enhance its hazard rate, structural, parametric or jump approach. Given the findings of Chung and Gregory (2019), which enhancement should LuxeBank prioritise if its goal is to better capture scenarios in which exposure can increase significantly and abruptly?

A. Hazard rate approaches, as they offer transparent correlation parameters that are easy to calibrate, making them ideal for precise WWR estimation.

B. Structural approaches, because they provide a strong correlation effect and suit situations where gradual changes in exposure are expected.

C. Parametric approaches, due to their simplicity and use of historical data for calibration, suitably addressing moderate changes in exposure over time.

D. Jump approaches, since they account for abrupt changes in exposure, which is critical for capturing the risk of sudden market shifts.

Solutions • CR-20 p

Solution CR-20 p.1 — Answer: C

Bank Q1075 • BT P2.T6.419.2

Step 1 – What the standard formula assumes

It multiplies unconditional EE by the period PD, which is valid only if exposure and default are independent.

Step 2 – The WWR fix

GARP: incorporating WWR "is probably most obviously achieved by simply representing the exposure conditional upon the default of the counterparty", replacing EPE in eq. 20.3 by EPE at t_i conditional on t_i being the default time.

$\Rightarrow$ With WWR the conditional EE exceeds the unconditional EE and CVA rises; with right-way risk it is lower.

Interpretation

The structure of the formula survives; only the exposure term changes. The four models of this LO are different ways of producing that conditional exposure.

Why the other options are wrong

A is incorrect. Recovery is not the dependence being modelled; dropping $(1 - \text{Rec})$ would set LGD to 100% for every trade without addressing the exposure-default link.

B is incorrect. The discount factor is a risk-free term with no role in the exposure-default dependence.

D is incorrect. The expression remains usable once conditional exposure is used.

LO: CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.2 Quantification of WWR in CVA, eq. 20.3

Solution CR-20 p.2 — Answer: A

New • built from GARP Ch.20, 20.6.3

Step 1 – Identify the method

A stochastic credit spread correlated with the exposure drivers, with exposure computed only on default paths, is the intensity (hazard rate) approach.

Step 2 – Explain the result

GARP: the approach "can be seen to generate only weak dependence between exposure and default, given that the correlations used are close to their maximum and minimum values. Hence ... it may underestimate the true WWR effect."

$\Rightarrow$ The modest response at $+90\%$ is a property of the model, not evidence that the WWR is small.

Interpretation

The validator should benchmark against a structural model (much stronger effect at the same correlation) or a jump model for any specific WWR. For payer swaps, positive spread-rate correlation is the WWR direction; history suggests the rate-default relation is generally negative.

Why the other options are wrong

B is incorrect. The intensity approach does not capture the full dependence even near $\pm 100\%$; a weak response is not proof of weak WWR.

C is incorrect. Reversed: the structural approach gives a much stronger effect than the intensity approach, and it samples pre-computed exposures rather than simulating spreads.

D is incorrect. The intensity approach has no jump at default; jumps belong to the jump approaches.

LO: CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.3 Wrong-Way Risk Models, Figures 20.17 to 20.20

Solution CR-20 p.3 Answer: D

New • built from GARP Ch.20, 20.6.3

Step 1 – Identify the method

Mapping the default time and the exposure distribution onto a bivariate distribution, without re-running exposures, is the structural approach.

Step 2 – Its trade-off

The advantage is that pre-computed exposure distributions are used and WWR is added on top of the existing methodology. GARP: "this is also a disadvantage since it may not be appropriate to assume that all the relevant information to define WWR is contained within the unconditional exposure distribution", and "the big drawback ... is that the correlation parameter ... is opaque and, therefore, difficult to calibrate".
$\Rightarrow$ D states both drawbacks.

Interpretation

The structural model gives a much stronger WWR effect than the intensity model at the same correlation, so the hard-to-calibrate correlation matters a great deal for the answer.

Why the other options are wrong

A is incorrect. Reversed: the structural approach needs no re-simulation; simulating exposures on default paths is the intensity approach.
B is incorrect. Weak dependence is the intensity approach's weakness; the structural effect is much stronger.
C is incorrect. Calibration against historical portfolio values and spreads describes the parametric approach.

LO: CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.3, Figure 20.18

Solution CR-20 p.4 Answer: B

New • built from GARP Ch.20, 20.6.3

Step 1 – Identify the method

A simple functional link between default probability and portfolio value, calibrated from past portfolio values against the counterparty's spread, is Hull and White's parametric approach.

Step 2 – When is the calibration reliable?

GARP: "This approach obviously requires that the current portfolio of trades with the counterparty is similar in nature to that used in the historical calibration, in addition to the historical data showing a meaningful relationship."
$\Rightarrow$ B is the condition.

Interpretation

If the hedge fund has changed strategy, the historical link between its portfolio value and its spread says little about today's WWR; a what-if calibration is the alternative GARP mentions.

Why the other options are wrong

A is incorrect. Quanto CDS prices are evidence for jump approaches, not a requirement of the parametric approach.

C is incorrect. Weak dependence near extreme correlations is a property of the intensity approach.

D is incorrect. Bivariate mapping is the structural approach.

LO: CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.3 Wrong-Way Risk Models

Solution CR-20 p.5

Answer: C

Bank Q1078 • BT P2.T6.24.33.2

Step 1 – The jump approach

Levy and Levin's approach assumes the relevant FX rate jumps at the counterparty's default time. Schweser expresses the jump through a residual value factor: the FX rate depreciates by $(1-\text{RV})$.

Step 2 – Size of the jump

Implied jumps are larger for better-rated and more systemic names: 83% for AAA against 27% for BBB sovereigns (Levy and Levin); 38.4% for the Japanese sovereign against 13.5% for financials and 8% for non-financials (Chung and Gregory).

$\Rightarrow$ C describes the jump approach.

Interpretation

For a cross-currency swap with a sovereign this is the natural model: the exposure gaps up at default, which no continuous model reproduces and which margin cannot keep up with.

Why the other options are wrong

A is incorrect. Describes the intensity approach, and gets its bias wrong: it tends to understate, not overstate, WWR.

B is incorrect. Describes the structural approach.

D is incorrect. Describes the parametric approach.

LO: CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.4 Jump Approaches; SchweserNotes 2026 Book 2, Reading 38, LO 38.p

Solution CR-20 p.6

Answer: D

Bank Q1076 • AnalystPrep Q.5476

Step 1 – The goal

Exposure that rises significantly and abruptly is the specific-WWR case, such as an FX rate jumping at a sovereign or bank default.

Step 2 – Which method builds that in?

Only jump approaches put a discrete move in the exposure driver at the default time. Chung and Gregory (2019) find material implied jumps in Japanese CDS data (sovereign 38.4%, financials 13.5%, non-financial corporates 8%), and GARP says jump approaches may be more realistic for specific WWR.

$\Rightarrow$ Enhance the jump approach.

Interpretation

The three continuous approaches have their uses for general WWR, but none produces a discrete

gap at default; the quanto CDS evidence shows intensity models cannot match the data even at $\pm 100\%$ correlation.

Why the other options are wrong

A is incorrect. The intensity approach's correlations can be estimated from historical spread data, but it generates only weak dependence, may underestimate WWR and has no jump at default.

B is incorrect. The structural effect is strong, but the approach is continuous and its correlation is opaque and hard to calibrate; it does not model an abrupt gap.

C is incorrect. The parametric approach relies on a historical value-spread relationship and has no mechanism for a sudden jump.

LO: CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.3 and 20.6.4 Jump Approaches

CR-22 • Structured Credit Risk • Vardeez: IMPORTANT

CR-22 c

P1 CONCEPT

Describe a waterfall structure in a securitization.

Workbook: A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A9 (Map #9, 7/16) reports "securitization waterfall structure (GARP Mocks)" and "calculate shares to senior and equity tranche". One recall says a tranche question came "exactly the same from the mocks".
Vardeez: an IMPORTANT reading, but do it at the very end because it consumes time. His one possible calculation for the reading sits in CR-22 e and f.

Reported exam shape

- **A9·S4** Shares paid to the senior and equity tranches in the final-year waterfall.

What the questions look like: mostly concept questions on payment priority: who is paid first, where a shortfall lands, where excess spread and recoveries go. The one numerical is the workbook's final-year waterfall (Question CR-22 c.6, Workbook A9.5, shape S4). It is a calculation under a describe LO, kept because workbook questions always go in.

Worked elsewhere: A9.4 (interim waterfall with fees, the equity flow and the account balance) under CR-22 e.

1. The reading's three-tranche CLO

Malz builds the whole chapter on one example, and CR-22 e, f, h, i and l all reuse it. The collateral is 100 identical leveraged loans of \$1 million par, priced at par, paying a fixed spread of 350 bp over Libor:

Assets	Liabilities	Par	Coupon
100 loans, \$100 million	Senior debt	\$85 million	Libor + 50 bp
Libor + 350 bp	Mezzanine (junior) debt	\$10 million	Libor + 500 bp
	Equity note	\$5 million	none: residual

- **Assumptions.** Swap curve ("Libor") flat at 5%; loans and liabilities mature in five years; all coupons and loan interest are annual, at year end; no upfront, management or trustee fees.
- **Attachment points.** Mezzanine 5% (the equity below it), senior 15% (equity plus mezzanine). There is initially no overcollateralisation: the liabilities equal the assets.
- **Spreads.** The weighted average spread on the debt is $(10 \times 500 + 85 \times 50)/95 = 97.4$ bp. The junior bond has a much wider spread than the senior and much less credit enhancement.
- **Annual cash flows with no defaults.** Collateral $0.085 \times 100 = \$8.500$m; mezzanine $0.10 \times 10 = \$1.000$m; senior $0.055 \times 85 = \$4.675$m. The bond coupons due are $B = \$5.675$m and the **excess spread** is $8.500 - 5.675 = \$2.825$m.

The design rests on the senior tranche: it must be a large share of the capital structure with a low coupon, or the pool cannot be financed and the securitisation is not viable.

2. The waterfall: cash runs top down, losses run bottom up

GARP's definition: the waterfall is *the rules about how the cash flows from the collateral are distributed to the various securities in the capital structure*. The capital structure is paid in sequence, "top down": the senior debt receives all of its promised payments before any lower tranche receives anything. Fees and other costs typically take priority over coupons. At an interim year end the order is:

1. fees and expenses;
2. the senior coupon;
3. the mezzanine coupon;
4. the overcollateralisation test: excess spread is diverted to the overcollateralisation account, up to a cap K;
5. the equity note: whatever is left.

Where a shortfall lands. If loan interest cannot cover the coupons, the senior is still paid in full first and the shortfall falls on the mezzanine. Nothing is shared pro rata, and the equity gets nothing (Question CR-22 c.3). In Malz's structure the overcollateralisation account then tops up the bond interest (CR-22 f).

Losses run the other way. Credit losses write the tranches down from the bottom: equity first, then the mezzanine, and the senior only once both are exhausted. The subordination below a tranche is its *credit enhancement*. A senior bond with 20% enhancement facing pool losses of 21% is written down by about $1/80 = 1.25\%$, because it is 80% of the balance sheet. Because a securitisation rarely defaults on a hard maturity date, GARP calls a default event *material impairment*: uncured missed interest for more than a few months, or pool performance so poor that payments are likely to stop.

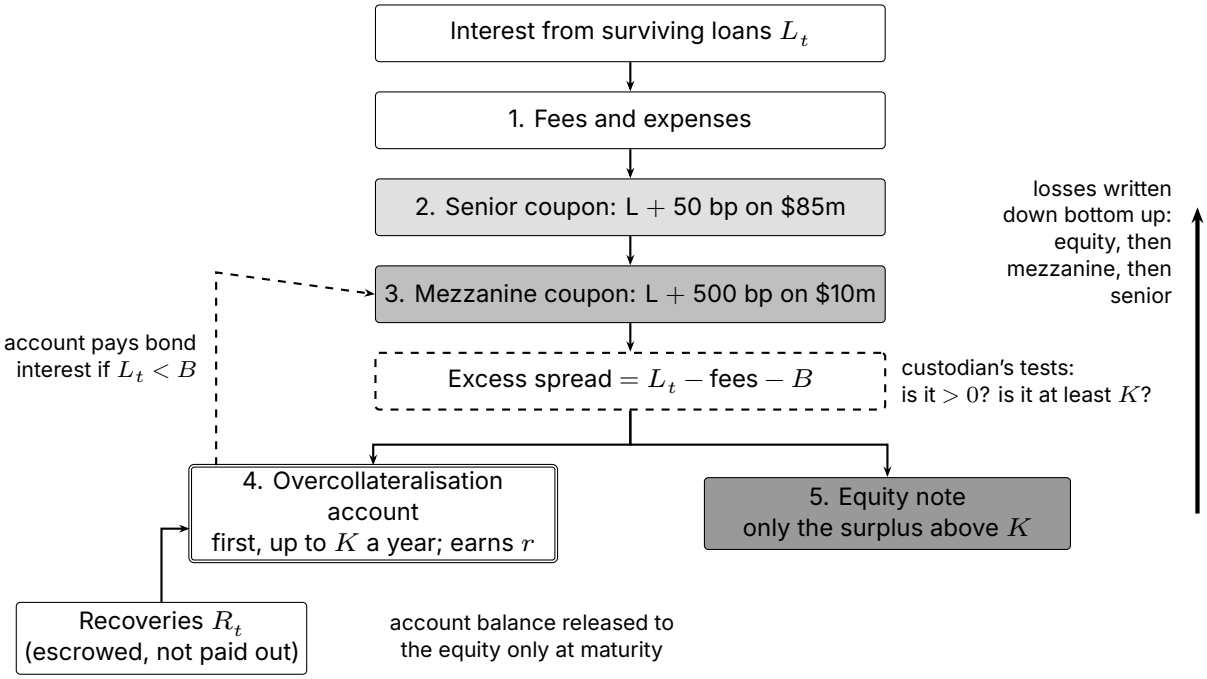


Figure 1. The interim waterfall of the reading's CLO. Cash falls from the pool through fees and the two coupons (grey shading darkens down the stack); the excess spread fills the overcollateralisation account (double border) before anything reaches the equity (dark box). Recoveries go straight into the account, and the account (dashed arrow) pays bond interest when loan interest falls short. Losses travel the other way (thick arrow).

3. Overcollateralisation and the trigger

- **Hard versus soft.** Hard overcollateralisation is collateral par above bond par, in place at issue (an equity note already makes the debt less than 100% of the deal). Soft enhancement is excess spread retained over time: it is not present at initiation.
- **Triggers.** Overcollateralisation triggers state when excess spread is diverted into reserves. The aim is to build credit enhancement early in the deal's life, when collateral cash flows are strongest. Some deals use the diverted interest to pay down the senior debt; Malz's example puts it in a reserve.
- **Malz's mechanism.** Up to $K = \$1.75$m of excess spread a year goes to the overcollateralisation account, which earns the 5% money market rate. Only excess spread above K is paid to the equity; if excess spread is below K, all of it goes to the account and the equity gets nothing. The account pays bond interest whenever loan interest falls short, and any balance is released to the equity only at maturity. Malz calls the name "a bit of a misnomer": it is soft, not hard, enhancement.
- **Recoveries are escrowed.** The 40% recovered on a defaulted loan is paid into the account. If recoveries flowed through the waterfall, the equity would "perversely benefit from defaults". In real deals recoveries go to the senior bonds and then the mezzanine; time tranching sends early recoveries to short-dated bonds.

> **Correction to the source notes.** Vardeez's waterfall figure and caption for CR-22 c (taken from the Schweser LO 40.c summary) read: "If the cash flows to equity holders exceed the overcollateralization trigger, the excess is diverted to a trust account." That caps the *equity* and sends the surplus to the account. The GARP reading's mechanism runs the other way: the account is filled first, up to K, and the equity receives only the surplus above K. Vardeez's own CR-22 e and f notes, and GARP PE1 Q72 (Question CR-22 f.5), follow GARP. Schweser's LO 40.c example and Module Quiz 40.1 Q2 (Question CR-22 e.2) describe a different trigger design. Follow the rule the stem states; if it says only that excess spread "flowing into the overcollateralization account is limited to K", use GARP's order.

4. The final-year waterfall (shape S4)

At maturity nothing is diverted and there is no overcollateralisation test. Four sources make up the terminal available funds F: interest from the surviving loans, their redemption at par, recoveries on loans that default in the final year, and the account balance grown for one more year. F then meets two shortfall tests in sequence:

1. **Senior.** Due: par plus the final coupon, $85 \times 1.055 = \$89.675$m. If $F \geq 89.675$ the senior is money good; otherwise it is written down by $89.675 - F$.
2. **Mezzanine.** Due: $10 \times 1.10 = \$11.0$m, paid from $F - 89.675$. Shortfall $\max[11.0 - (F - 89.675), 0]$.
3. **Equity.** Only the residual, $\max(F - 100.675, 0)$.

Annual default rate (GARP Table 22.2)	2.0%	7.5%	10.0%
Terminal available funds F (\$m)	109.990	93.702	86.356
Senior shortfall (\$m)	0	0	3.319
Mezzanine shortfall (\$m)	0	6.973	11.000
Equity terminal cash flow (\$m)	9.315	0	0

The split is **sequential, not pro rata**: dividing F in proportion to par is pass-through logic (Question CR-22 c.6, option D). How F is built year by year is CR-22 e.

5. Two other uses of the word

- **Sequential-pay CMO.** Schweser calls the basic CMO structure a "sequential pay (or waterfall)" structure: Tranche 1 receives all principal until it is retired, then Tranche 2. That is *time* tranching for prepayment risk, not credit priority (CR-22 a).
- **Choudhry's cash waterfall (CR-23 b).** Fees, Class A interest, Class A coverage tests, Class B interest, Class B tests, and so on, with the equity taking the residual. If a coverage test fails, the notes are paid down starting with the senior class.

	Interim year end	Final year	Credit losses
Direction	Top down	Top down	Bottom up
What is paid	Fees, senior coupon, mezzanine coupon, then the account (up to K), then equity	Senior par + coupon, then mezzanine par + coupon, then equity residual	Writedowns hit equity, then mezzanine, then senior
A shortfall hits	Mezzanine coupon first; the account tops up	Mezzanine first; senior only if $F < 89.675$	The thinnest junior tranche still standing
The account	Filled from excess spread and recoveries	Released into F after one more year of interest	Soft enhancement for both bonds

Exam traps

- **Pro rata instead of sequential.** A credit waterfall pays the senior in full before the mezzanine receives anything; splitting cash in proportion to par or coupons is pass-through logic.
- **Trigger direction.** In GARP's structure the account is filled first (up to K) and the equity gets only the surplus. Reading K as a cap on the equity reverses who is paid when excess spread is small.
- **Fees after coupons.** Fees and expenses come off the top, before the senior coupon.
- **Recoveries to the equity.** Recoveries are escrowed in the account; letting them reach the equity rewards defaults.
- **Final coupon forgotten.** At maturity each bond is owed par *plus* its last coupon (89.675 and 11.0 in the example), not par alone.
- **Hard versus soft enhancement.** Diverted excess spread builds over time (soft); collateral par above bond par is there at issue (hard).
- **The wrong waterfall.** A sequential-pay CMO "waterfall" allocates principal for prepayment risk, not credit losses.

Practice questions: CR-22 c

Question CR-22 c.1

Bank Q1134 • AnalystPrep Q.2886

A newly hired analyst on a structured credit desk is asked to define the term "waterfall" for the desk's glossary. Which of the following best describes a waterfall structure in a securitization?

A. The excess spread when there is no default in the collateral pool

B. A financial intermediary that aggregates underlying loans and designs the securitization structure

C. The rules on how cash flows from the collateral are distributed to the various securities in the capital structure

D. The process of creating a securitized credit product

Question CR-22 c.2

New • built from GARP Ch.22, 22.1 Waterfall and 22.2 Tracking the Interim Cash Flows

A trustee's analyst is documenting how the collateral interest collected at an interim year end is applied in a three-tranche CLO with an overcollateralization account of the type described in the reading. The deal pays trustee and management fees. Which sequence correctly describes the order of application?

A. Senior coupon; mezzanine coupon; fees; equity residual; overcollateralization account

B. Fees; senior coupon; mezzanine coupon; overcollateralization account up to its annual cap; equity residual

C. Fees; senior and mezzanine coupons pro rata to their par amounts; equity residual; overcollateralization account

D. Fees; overcollateralization account up to its annual cap; senior coupon; mezzanine coupon; equity residual

Question CR-22 c.3

Bank Q1136 • New question written for this bank

A securitization has a waterfall that pays the senior coupon first, then the mezzanine coupon, and then the equity tranche, with no reserves yet in the excess trust account. In a given period the senior coupon due is $15.4 million and the mezzanine coupon due is $9.0 million. Because of a higher-than-assumed default rate, the loan interest collected is only $23.8 million. Which of the following correctly describes the distribution of this period's cash flows?

A. The senior and mezzanine tranches share the $0.6 million shortfall pro rata to their coupons, so each is paid slightly less than its promised coupon.

B. The senior and mezzanine tranches are paid in full, with the $0.6 million shortfall funded by the equity holders.

C. The senior tranche receives $14.8 million and the mezzanine receives $9.0 million, because the most senior claim absorbs the first shortfall.

D. The senior tranche receives $15.4 million, the mezzanine receives $8.4 million, and neither the equity tranche nor the trust account receives anything.

Question CR-22 c.4 Bank Q1152 • AnalystPrep Q.5522 (rebuilt) • re-filed from CR-22 f

A pension fund is considering the equity note of a CLO whose offering circular follows the structure in the reading: at each interim year end, after fees and the bond coupons, the excess spread $L_t - B$ is diverted to the overcollateralization account up to a maximum annual amount K. The fund's analyst asks where the excess spread goes in a year in which it exceeds K, and in a year in which it is positive but below K. Which of the following is correct?

A. Above K: the surplus is also deposited in the account. Below K: all of it goes to the account.

B. Above K: the surplus is paid to the equity. Below K: all of it goes to the account and the equity receives nothing.

C. Above K: the surplus is shared pro rata among the three tranches. Below K: all of it goes to the account.

D. Above K: the surplus is paid to the equity. Below K: all of it goes to the equity, because the cap is not reached.

Question CR-22 c.5 New • built from GARP Ch.22, 22.1 Waterfall and 22.2 Tracking the Interim Cash Flows

A risk committee is reviewing the overcollateralization mechanism of the CLO in the reading, in which part of the excess spread and all recoveries on defaulted loans are paid into an overcollateralization account that earns the money market rate. Which of the following statements about this mechanism is correct?

A. Recoveries are paid through the waterfall together with loan interest, so in a year with defaults the equity can receive more cash than in a year without defaults.

B. The diverted excess spread is hard credit enhancement: like the equity note, it is fully in place when the deal is issued.

C. The account is built up early in the deal's life, when collateral cash flows are strongest; it pays bond interest if loan interest falls short, and any balance is released to the equity only at maturity.

D. The account protects the senior bond only: it may not be used to pay the mezzanine coupon, which must be met from loan interest alone.

Question CR-22 c.6 Workbook A9.5 • S4 • Map #9

A risk consultant is presenting the final-year waterfall of a four-year CLO to a group of newly hired analysts at a Frankfurt bank. The deal has:

- Collateral: 120 identical floating-rate loans of EUR 1 million each, bought at par, paying the reference rate + 380 bps; surviving loans pay their final interest and redeem at par at the end of year 4
- Liabilities: senior notes EUR 96 million at reference + 70 bps; mezzanine notes EUR 15 million at reference + 600 bps; equity EUR 9 million
- The reference rate is flat at 4%; no fees
- Defaults: 13 loans in years 1 to 3 and 5 loans in year 4; 40% of defaulted par is recovered at year end
- The overcollateralisation account held EUR 7.2 million at the end of year 3 and earns 4%

Which of the following correctly states the amounts paid at the end of year 4 to the senior notes and to the equity tranche?

A. Senior EUR 100.512 million; equity EUR 0.432 million

B. Senior EUR 100.512 million; equity EUR 2.432 million

C. Senior EUR 96.000 million; equity EUR 6.944 million

D. Senior EUR 95.555 million; equity EUR 8.958 million

Solutions • CR-22 c

Solution CR-22 c.1

Answer: C

Bank Q1134 • AnalystPrep Q.2886

Step 1 – Recall the definition

GARP: the waterfall refers to the rules about how the cash flows from the collateral are distributed to the various securities in the capital structure.
⇒ It is a set of payment-priority rules, not a cash amount, a participant or a process.

Step 2 – How the rules work

Payments run top down: fees, the senior coupon, the mezzanine coupon, and the residual to the equity (with excess spread diverted to the overcollateralisation account first, up to its cap).

Interpretation

Everything else in this reading, from interim cash flows to the final-year shortfall tests, is the waterfall applied to a particular set of defaults.

Why the other options are wrong

A is incorrect. Excess spread is an *output* of the waterfall (collateral interest left after the coupons), not the waterfall itself.
B is incorrect. That is the underwriter or arranger, who aggregates the loans and designs and markets the structure.
D is incorrect. That is securitisation as a whole; the waterfall is one feature of the structure it creates.

LO: CR-22 c: Describe a waterfall structure in a securitization. • **Reference:** Malz, Financial Risk Management, Ch.9 (GARP CR Ch.22), 22.1 Waterfall

Solution CR-22 c.2

Answer: B

New • built from GARP Ch.22, 22.1 and 22.2

Step 1 – Fees come first

GARP: in addition to the coupons promised to the bonds, there are fees and other costs, "which typically take priority over coupons".
⇒ Any sequence that pays coupons before fees (A) is out.

Step 2 – Then the bonds, top down

The senior coupon is paid in full before the mezzanine receives anything; the two are not paid pro rata.
⇒ C is out.

Step 3 – Excess spread: account first, equity last

Only what is left after the coupons is excess spread. Up to the cap K goes to the overcollateralisation account; the equity receives the remainder.
⇒ D is out: the account is funded from excess spread, after the coupons, never ahead of them.

Interpretation

The order fees, senior, mezzanine, account, equity is the interim waterfall of Figure 1. It is also why the equity flow is the first casualty when defaults erode loan interest.

Why the other options are wrong

A is incorrect. Puts fees after the coupons and the equity ahead of the account.
C is incorrect. Pays the bonds pro rata and puts the equity ahead of the account: two reversals of priority.

D is incorrect. Diverts to the account before the coupons are paid, which would starve the bonds the account is meant to protect.

LO: CR-22 c: Describe a waterfall structure in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.1 Waterfall; 22.2 Tracking the Interim Cash Flows

Solution CR-22 c.3 Answer: D

Bank Q1136 • New question written for this bank

Step 1 – Pay the senior first

The senior is owed \$15.4 million and is paid in full from the \$23.8 million collected, leaving $23.8 - 15.4 = \$8.4$ million.

Step 2 – The mezzanine takes what is left

The mezzanine is owed \$9.0 million but only \$8.4 million remains, so it bears the \$0.6 million shortfall.

$\Rightarrow$ Nothing is left for the equity or the trust account.

Interpretation

A waterfall is sequential: a shortfall reaches the most junior claim still owed. In Malz's structure an overcollateralisation account with a balance would then top up the mezzanine coupon; here the account is empty.

Why the other options are wrong

A is incorrect. Pari passu sharing (senior about \$15.02 million, mezzanine about \$8.78 million) is not how a waterfall works.

B is incorrect. The equity is a residual claimant with no obligation to fund shortfalls; it simply receives nothing.

C is incorrect. Reverses the priority by making the senior absorb the shortfall.

LO: CR-22 c: Describe a waterfall structure in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.1 Waterfall

Solution CR-22 c.4 Answer: B

Bank Q1152 • AnalystPrep Q.5522

Step 1 – Read the cap as GARP defines it

GARP: "If excess spread is less than \$1,750,000, that smaller amount is diverted to the overcollateralization account. If excess spread is greater than \$1,750,000, the amount that exceeds \$1,750,000 is paid out to the equity."

$\Rightarrow K$ is the most the account can take in a year, and the account has first call on excess spread.

Step 2 – Excess spread above K

The custodian's second test, "is $L_t - B \geq K$?", passes, so $OC_t = K$ and the equity is paid $L_t - B - K$.

$\Rightarrow$ The surplus above the cap is the equity's residual cash flow.

Step 3 – Excess spread below K

The test fails, so $OC_t = L_t - B$ and the equity flow $L_t - B - OC_t$ is zero.

$\Rightarrow$ All of the excess spread goes to the account; the equity receives nothing that year.

Interpretation

GARP's 2% scenario is the first case in every year ($2.485 - 1.750 = \$0.735$m to the equity in year 2). Its 7.5% scenario is the second case in years 2 to 4: excess spread of \$1.55m, \$1.04m and \$0.53m all goes to the account and the equity sees cash again only at maturity, if the bonds are

repaid in full.

Why the other options are wrong

A is incorrect. Ignores the cap: if all excess spread were retained there would be no need for a limit K, and the equity would receive nothing until maturity in every scenario.
C is incorrect. Excess spread is not allocated across tranches: the bonds receive only their contractual coupons, and nothing in the waterfall is split pro rata.
D is incorrect. Reads K as a cap on the equity, the Schweser LO 40.c design (see the correction box in Section 3). Under the circular's wording the account is filled first.

LO: CR-22 c: Describe a waterfall structure in a securitization. CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.2 Tracking the Interim Cash Flows

Solution CR-22 c.5 Answer: C

New • built from GARP Ch.22, 22.1 and 22.2

Step 1 – When the account fills

Soft overcollateralisation builds credit enhancement "at the beginning of the life of the product, when collateral cash flows are strongest". In Malz's example it accumulates funds "while cumulative defaults are low".

Step 2 – What the account is for, and when it pays out

GARP: the funds "will be used to pay interest on the bonds if there is not enough interest flowing from the loans"; any remaining funds "will be released to the equity tranche only at maturity".
$\Rightarrow$ C states both points correctly.

Interpretation

Malz admits the mechanism "is not a robust mechanism for protecting the senior bonds", but unless defaults come very early it has money in it when loan interest starts to fall short.

Why the other options are wrong

A is incorrect. Recoveries are paid *into the account*, precisely so that the equity does not "perversely benefit from defaults".
B is incorrect. Diverted excess spread is soft enhancement: it accrues over time and is not present at initiation. Hard overcollateralisation is collateral par above bond par at issue.
D is incorrect. The account pays interest on "the bonds", both tranches; it is not reserved for the senior coupon.

LO: CR-22 c: Describe a waterfall structure in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.1 Waterfall; 22.2 Tracking the Interim Cash Flows

Solution CR-22 c.6 Answer: B

Workbook A9.5 • S4 • Map #9

Step 1 – Overcollateralisation account at maturity

$$7.2 \times 1.04 = \text{EUR } 7.488 \text{ million}$$

$\Rightarrow$ The account earns one more year of interest and is now released into the terminal waterfall.

Step 2 – Terminal available funds

Survivors $= 120 - 18 = 102$: interest $102 \times 7.8\% = 7.956$, redemption 102.0, recovery $0.4 \times 5 = 2.0$.

$$F = 7.488 + 7.956 + 102.0 + 2.0 = \text{EUR } 119.444 \text{ million}$$

Step 3 – Senior shortfall test

$$\text{Senior due} = 96 \times (1 + 0.047) = 100.512 \leq F$$

$\Rightarrow$ Paid in full; $119.444 - 100.512 = 18.932$ is left.

Step 4 – Mezzanine test, then the equity residual

$$\text{Mezzanine due} = 15 \times 1.10 = 16.5 \leq 18.932, \quad \text{Equity} = 18.932 - 16.5 = 2.432$$

$\Rightarrow$ Both bonds are money good; the equity's terminal claim is only what is left.

Interpretation

With 18 cumulative defaults (15% of the pool) the equity collects only EUR 2.432 million at maturity on a EUR 9 million tranche: it absorbed the first loss, exactly as the capital structure intends. A few more defaults would push the shortfall into the mezzanine before the senior is touched.

Why the other options are wrong

A is incorrect. Dropped term: left out the final-year recoveries of EUR 2.0 million, so $F = 117.444$ and the equity gets 0.432.

C is incorrect. Paid the senior only its principal of 96.000 and forgot its final coupon, which inflates the equity residual to $119.444 - 96 - 16.5 = 6.944$.

D is incorrect. Sibling swap with a pass-through: split F pro rata to par, $119.444 \times 96/120 = 95.555$ and $119.444 \times 9/120 = 8.958$, ignoring the sequential priority of the waterfall.

LO: CR-22 c: Describe a waterfall structure in a securitization. CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure. CR-23 b: Explain the terms over-collateralization, first-loss piece, equity piece, and cash waterfall within the securitization process. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), Tracking the Final-Year Cash Flows, Table 22.2; Choudhry, Ch.12 (GARP CR Ch.23), Figure 23.4

CR-22 • Structured Credit Risk • Vardeez: IMPORTANT

CR-22 e

P1 CALC

Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure.

Workbook: A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A9 (Map #9, 7/16) reports "securitization: calculating cash flow into equity tranche" and "calculate shares to senior and equity tranche".
Vardeez: "There can be one calculation question come from this reading." Leave the reading for the very end. His original notes carry no worked example here; the rebuilt notes work two iterations at a 5% default rate, and everything below follows GARP's Tables 22.1 and 22.2.

Reported exam shapes

- **A9·S2** Cash flow into the equity tranche after the interim waterfall, with defaults and fees.
- **A9·S4** Shares paid to the senior and equity tranches in the final-year waterfall.

What the questions look like: a deal sheet (loans, spreads, tranche sizes, the cap K, the money market rate, recovery, defaults) and one year's flows to find: loan interest, excess spread, the split between the overcollateralisation account and the equity, the account balance, or the final-year payments. Typical red herrings are the equity's par, an extra year of defaults and the original pool size where the survivors are needed.

Worked elsewhere: A9.5 (final-year shares to the senior and the equity) under CR-22 c.

1. Notation and the one-year algorithm

The structure is the reading's CLO of CR-22 c, Section 1: 100 loans of \$1m at Libor + 350 bp (8.5%), senior \$85m at 5.5%, mezzanine \$10m at 10%, equity \$5m, Libor 5%. GARP's notation:

N	number of loans in the initial pool ($N = 100$)
d_t	number of defaults during year t
L_t	loan interest at the end of year t: $(0.050 + 0.035)(N - \sum_{\tau \le t} d_\tau) \times \$1\text{m}$
B	coupon interest due to the senior and junior bonds, constant: \$5,675,000
K	maximum diverted from excess spread to the account each year: \$1,750,000
OC_t	amount actually diverted in year t (negative when the account pays out)
R_t	recovery paid into the account: $0.4\, d_t \times \$1\text{m}$
r	money market rate earned by the account: 5%

A loan that defaults during year t pays no interest for that year: there is no partial interest. One iteration of the interim waterfall is five moves:

1. **Survivors.** $S_t = N - \sum_{\tau \le t} d_\tau$: *cumulative* defaults, not this year's alone.
2. **Loan interest.** $L_t = (\text{reference} + \text{spread}) \times S_t \times \text{par}$.
3. **Excess spread.** $X_t = L_t - \text{fees} - B$; fees come off the top.
4. **Custodian's tests.** $X_t \ge K$: $OC_t = K$ and the equity gets $X_t - K$. $0 \le X_t < K$: $OC_t = X_t$ and the equity gets nothing. $X_t < 0$: the account pays the shortfall $B - L_t$ and the equity gets nothing.

5. **Account.** $A_t = A_{t-1}(1+r) + OC_t + R_t$: last year's balance grows at r, then this year's diversion and recoveries go in.

2. Two iterations at a 7.5% default rate

GARP counts defaults in whole loans: 7.5% of 100 is 7.5, entered as 8; 7.5% of the 92 survivors is 6.9, entered as 7. Because defaulted loans are not replaced, the annual number of defaults falls as the pool shrinks.

Year 1. $d_1 = 8$, $S_1 = 92$.

$$L_1 = 92 \times \$85{,}000 = \$7{,}820{,}000, \qquad X_1 = 7{,}820{,}000 - 5{,}675{,}000 = \$2{,}145{,}000 \geq K$$

So $OC_1 = \$1{,}750{,}000$ and the equity receives $2{,}145{,}000 - 1{,}750{,}000 = \$395{,}000$. Recovery $R_1 = 0.4 \times 8 \times \$1\text{m} = \$3{,}200{,}000$, so the account ends the year at $1{,}750{,}000 + 3{,}200{,}000 = \$4{,}950{,}000$.

Year 2. $d_2 = 7$, $S_2 = 85$.

$$L_2 = 85 \times \$85{,}000 = \$7{,}225{,}000, \qquad X_2 = \$1{,}550{,}000 < K$$

All of it is diverted ($OC_2 = \$1{,}550{,}000$) and the equity receives nothing. $R_2 = \$2{,}800{,}000$, and

$$A_2 = 4{,}950{,}000 \times 1.05 + 1{,}550{,}000 + 2{,}800{,}000 = \$9{,}547{,}500$$

The full interim table for GARP's three scenarios ($ million):

Year	d_t	S_t	L_t	$X_t = L_t - B$	OC_t	R_t	Equity	Account A_t
Default rate 2.0%								
1	2	98	8.330	2.655	1.750	0.800	0.905	2.550
2	2	96	8.160	2.485	1.750	0.800	0.735	5.228
3	2	94	7.990	2.315	1.750	0.800	0.565	8.039
4	2	92	7.820	2.145	1.750	0.800	0.395	10.991
Default rate 7.5%								
1	8	92	7.820	2.145	1.750	3.200	0.395	4.950
2	7	85	7.225	1.550	1.550	2.800	0	9.548
3	6	79	6.715	1.040	1.040	2.400	0	13.465
4	6	73	6.205	0.530	0.530	2.400	0	17.068
Default rate 10.0%								
1	10	90	7.650	1.975	1.750	4.000	0.225	5.750
2	9	81	6.885	1.210	1.210	3.600	0	10.848
3	8	73	6.205	0.530	0.530	3.200	0	15.120
4	7	66	5.610	−0.065	−0.065	2.800	0	18.611

Interim cash flows, recomputed from GARP Table 22.1. In year 4 at 10% the loan interest falls short of B and the account pays the $0.065m shortfall.

> **Correction to the source notes.** GARP's text announces scenarios with default rates of "1.5, 5.25, and 9.0 percent annually", but Tables 22.1 and 22.2, and every number in the discussion, use 2.0, 7.5 and 10.0 percent. The tables are right; the sentence is a leftover.

Vardeez's rebuilt notes run the same two iterations at 5% with expected (fractional) defaults, 5 and then 4.75 loans: the equity receives $650,000 and $246,250, and the account ends years 1 and 2 at $3,750,000 and $7,587,500. Same mechanics; GARP's tables round to whole loans.

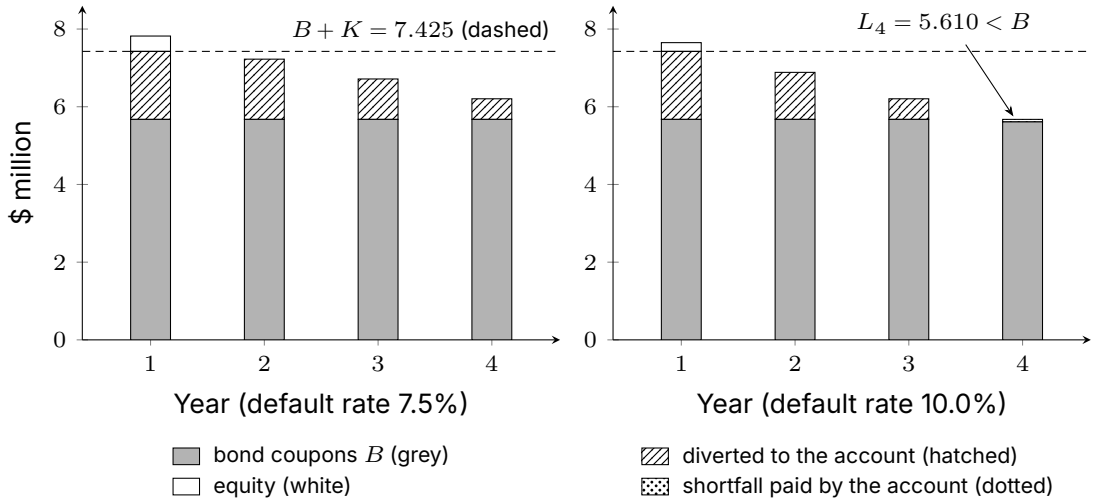


Figure 2. Where each year's loan interest goes. The grey part of every bar is the $5.675m of bond coupons; the hatched slice is diverted to the overcollateralisation account; the white slice reaches the equity only when loan interest clears $B + K$ (dashed line). At 7.5% the equity is paid in year 1 only; at 10% the excess spread turns negative in year 4 and the account covers the gap (dotted slice).

3. Fees and the equity flow (shape S2)

Fees and expenses come off the top, before the senior coupon, so they reduce the excess spread one for one. While the cap binds, every dollar of fees comes out of the equity's pocket; once the excess spread falls below K, fees reduce the amount diverted to the account instead.

Example. Add trustee and management fees of $0.5m a year to the 2% scenario. Year 1: $X_1 = 8.330 - 0.500 - 5.675 = \$2.155\text{m} \geq K$, so the account still receives $1.750m and the equity receives $0.405m instead of $0.905m. The account is unchanged at $2.550m.

4. The final year (shape S4)

At maturity there is no diversion and no test: everything is pooled and paid down the waterfall (CR-22 c, Section 4).

$$F = \underbrace{A_{T-1}(1+r)}_{\text{account}} + \underbrace{S_T \times (r+s) \times \text{par}}_{\text{final interest}} + \underbrace{S_T \times \text{par}}_{\text{redemption}} + \underbrace{0.4\, d_T \times \text{par}}_{\text{final recoveries}}$$

The senior is owed $89,675,000, the mezzanine $11,000,000, so the bonds together $100,675,000. Worked at 7.5% ($d_5 = 5$, $S_5 = 68$):

$$\begin{aligned} F &= 17{,}068{,}119 \times 1.05 + 68 \times 85{,}000 + 68{,}000{,}000 + 2{,}000{,}000 \\ &= 17{,}921{,}525 + 5{,}780{,}000 + 68{,}000{,}000 + 2{,}000{,}000 = \$93{,}701{,}525 \end{aligned}$$

The senior takes $89,675,000 in full, leaving $4,026,525 for a mezzanine owed $11,000,000: a mezzanine writedown of $6,973,475 and nothing for the equity.

Default rate	2.0%	7.5%	10.0%
Year-5 defaults; cumulative; survivors	2; 10; 90	5; 32; 68	7; 41; 59
Final-year loan interest	7,650,000	5,780,000	5,015,000
Loan redemption proceeds	90,000,000	68,000,000	59,000,000
Final-year recovery	800,000	2,000,000	2,800,000
Account balance at T	11,540,360	17,921,525	19,541,412
Terminal available funds F	109,990,360	93,701,525	86,356,412
Equity terminal cash flow	9,315,360	0	0
Mezzanine shortfall	0	6,973,475	11,000,000
Senior shortfall	0	0	3,318,588

Terminal cash flows ($), recomputed from GARP Table 22.2. Redemption of the surviving loans is the main driver of F.

5. Evaluating the result: the equity's IRR

The equity is owed nothing, so its credit risk is measured by its internal rate of return on the $5m outlay:

$$0 = -5{,}000{,}000 + \sum_{t=1}^{4} \frac{\max(L_t - B - OC_t, 0)}{(1+x)^t} + \frac{\max(F - 100{,}675{,}000, 0)}{(1+x)^5}$$

- **2%:** flows 0.905, 0.735, 0.565, 0.395 and 9.315 ($m) give $x = 23.0\%$. The deal is "designed" for this default rate.
- **7.5% and 10%:** the equity receives one early payment out of excess spread ($0.395m and $0.225m) and nothing after, so $-5 + 0.395/(1+x) = 0$ gives $x = -92.1\%$, and -95.5% at 10%. The owner is "out" nearly the whole $5m.

Three evaluation points GARP makes:

- **The bonds cannot break before maturity here.** Recoveries are held back and the account covers the coupons, so even at 10% the bonds are paid interest in full in years 1 to 4. Real trust agreements allow an early unwind in an extreme scenario.
- **Timing of defaults.** Collateral cash flows are largest early in the deal's life. Accelerating defaults raise the risk to the bonds; other things equal, the equity benefits if defaults come later.
- **Breakpoints.** Excess spread turns negative once survivors fall below $5.675/0.085 = 66.76$ loans (66 survivors in year 4 at 10%). The equity stops receiving interim cash much earlier, as soon as $X_t < K$, that is, below $7.425/0.085 = 87.4$ survivors.

Formula card: CR-22 e

Survivors and interest	$S_t = N - \sum_{\tau \le t} d_\tau$, $L_t = (\text{ref} + s)\, S_t \times \text{par}$ (no interest in the year of default)
Bond coupons	$B = \text{senior par} \times (\text{ref} + s_{\text{sen}}) + \text{mezz par} \times (\text{ref} + s_{\text{mez}})$
Excess spread	$X_t = L_t - \text{fees} - B$
Diversion and equity	$OC_t = \max[\min(X_t, K), 0]$, $\text{equity}_t = \max(X_t - OC_t, 0)$
Negative excess spread	account pays $B - L_t$, up to $A_{t-1}(1 + r) + R_t$; equity gets nothing
Recoveries and account	$R_t = RR \times d_t \times \text{par}$, $A_t = A_{t-1}(1 + r) + OC_t + R_t$
Terminal funds	$F = A_{T-1}(1 + r) + S_T(1 + \text{ref} + s)\,\text{par} + RR\, d_T\,\text{par}$
Final shares	senior $\min(F, \text{sen due})$; mezz $\min[\max(F - \text{sen due}, 0), \text{mezz due}]$; equity $\max(F - \text{total due}, 0)$
Equity IRR	$0 = -E_0 + \sum_t \text{equity}_t (1 + x)^{-t}$, interim and terminal flows together

Exam traps

- **This year's defaults only.** Interest comes from loans that have survived *every* year so far; subtracting only this year's defaults overstates L_t.
- **Reference rate dropped.** Coupons are reference rate plus spread; using the spread alone understates the coupon (Question CR-22 e.2, option B).
- **Fees after coupons, or forgotten.** Fees come off the top and hit the equity first while the cap binds.
- **Trigger direction.** GARP fills the account first, up to K; some stems (Schweser's) cap the equity instead. Read the stated rule.
- **Recoveries to the equity.** They go into the account; routing them through the waterfall rewards defaults (Question CR-22 e.3, option D).
- **Stale balance.** Last year's account balance must be grown at the money market rate before adding this year's flows.
- **Final-year omissions.** Forgetting the final coupon owed to a bond, the final-year recoveries, or the last year of interest on the account; paying F pro rata.
- **IRR without timing.** Averaging or summing the equity's inflows ignores when they arrive (Question CR-22 e.6).

Practice questions: CR-22 e

Question CR-22 e.1

Bank Q1146 • AnalystPrep Q.1855

A junior analyst is checking the annual interest amounts in a securitization. The swap curve is flat at 5%. The collateral, mezzanine and senior tranches have the following (principal, spread over the swap rate): collateral ($100m, 350 bps); mezzanine ($10m, 500 bps); senior ($85m, 50 bps). How do the annual interest amounts compare?

A. Mezzanine > Collateral > Senior

B. Collateral > Mezzanine > Senior

C. Collateral > Senior > Mezzanine

D. Senior > Collateral > Mezzanine

Question CR-22 e.2

Bank Q1145 • Schweser Module Quiz 40.1 Q2 (rebuilt)

A structurer is modelling a securitization pool of 100 identical loans of $500,000 each. Each loan pays the market reference rate plus a 300 bp spread, reset annually; the market reference rate is 4%. The senior, mezzanine and equity tranches are 75%, 20% and 5% of the pool. The senior and mezzanine tranches pay the reference rate plus spreads of 2% and 6%, respectively. The equity receives the residual, but under this deal's trigger any residual cash flow above $1,000,000 is diverted to the excess trust account. The default rate is 2% and defaulted loans pay no interest. What are the first-period cash flows to the mezzanine tranche and to the excess trust account?

A. Mezzanine $180,000; trust account $0

B. Mezzanine $600,000; trust account $0

C. Mezzanine $1,000,000; trust account $0

D. Mezzanine $1,000,000; trust account $180,000

Question CR-22 e.3 Workbook A9.4 • S2, S3 • Map #9

A structured credit analyst at a Shanghai insurer, which holds the equity tranche of a five-year CLO, is checking the trustee's report for the end of year 2. The deal terms are:

- Collateral: 80 identical floating-rate loans of CNY 25 million each, bought at par, paying the reference rate + 400 bps
- Liabilities: senior notes CNY 1,600 million at reference + 60 bps; mezzanine notes CNY 280 million at reference + 450 bps; equity CNY 120 million
- The reference rate is flat at 3%; all payments are annual, at year end
- Trustee and management fees of CNY 4 million a year are paid ahead of all note coupons
- Up to CNY 20 million of excess spread a year is diverted to the overcollateralisation account, which earns 3%; any remainder goes to the equity
- A loan that defaults pays no interest for that year; 40% of the defaulted par is recovered at year end and paid into the overcollateralisation account
- Defaults: 3 loans in year 1 and 5 loans in year 2; the account held CNY 50 million at the end of year 1

Which of the following correctly states the overcollateralisation account balance and the cash flow to the equity tranche at the end of year 2?

A. Account CNY 121.5 million; equity CNY 23.40 million

B. Account CNY 120.0 million; equity CNY 23.40 million

C. Account CNY 121.5 million; equity CNY 28.65 million

D. Account CNY 71.5 million; equity CNY 73.40 million

Question CR-22 e.4 Bank Q1148 • BT P2.T6.313.1 (rebuilt)

An investor is reviewing the three-tier securitization of the reading: 100 identical leveraged loans of \$1.0m par paying a fixed 8.5% (Libor 5.0% + 350 bps); loans and liabilities mature in 5 years; senior bonds \$85.0m pay Libor + 50 bps; mezzanine bonds \$10.0m pay Libor + 500 bps; equity contributes \$5.0m. The default rate is 2.0% per year (2 loans), recovery is 40%, and the money market rate is 5.0%. Results (\$ million except defaults):

t	d_t	Survivors	L_t	Excess spread	OC_t	R_t	$OC_t + R_t$	Equity flow	OC account
0		100						(5.000)	
1	2	98	8.330	2.655	1.750	0.800	2.550	0.905	2.550
2	2	96	8.160	2.485	1.750	0.800	2.550	0.735	5.228
3	2	94	7.990	2.315	1.750	0.800	2.550	0.565	8.039
4	2	92	7.820	2.145	1.750	0.800	2.550	0.395	10.991
5	2	90	7.650			0.800		9.315	11.540

Total terminal available funds are 109.990; the amount owed to the bond tranches is 100.675; the equity terminal cash flow is 9.315. Each of the following statements about this structure is true EXCEPT:

A. The bondholders are paid interest of \$5.675 million in each of the five years, plus principal in year 5.

B. The amount diverted to the overcollateralization account from excess spread is capped at \$1.750 million per year.

C. The recovery assumption is 40%.

D. Recovered funds are paid out to the equity holders each year as they are received.

Question CR-22 e.5

Bank Q1149 • BT P2.T6.313.2 (rebuilt)

Consider the same structure as the reading's example (100 loans of \$1.0m paying 8.5%; senior \$85.0m at Libor + 50 bps; mezzanine \$10.0m at Libor + 500 bps; equity \$5.0m; Libor 5.0%; recovery 40%; overcollateralization cap \$1.750m), but defaults are much higher: 10, 9, 8, 7 and 7 loans in years 1 to 5. Results (\$ million except defaults):

t	d_t	Survivors	L_t	Excess spread	OC_t	R_t	$OC_t + R_t$	Equity flow	OC account
0		100						(5.000)	
1	10	90	7.650	1.975	1.750	4.000	5.750	0.225	5.750
2	9	81	6.885	1.210	1.210	3.600	4.810	0	10.848
3	8	73	6.205	0.530	0.530	3.200	3.730	0	15.120
4	7	66	5.610	(0.065)	(0.065)	2.800	2.735	0	18.611
5	7	59	5.015			2.800		0	19.541

Total terminal available funds are 86.356; the amount owed to the bond tranches is 100.675. Which statement is TRUE?

A. There is never a year in which either the mezzanine or the senior bonds are paid their full interest.

B. Both bondholder classes receive all of their interest in the first four years, but neither is repaid its full obligation in year 5.

C. The mezzanine bondholders suffer interest and principal shortfalls, but the senior bondholders receive all their interest and principal.

D. Both bondholder classes receive all interest and get their full principal back.

Question CR-22 e.6

Bank Q1150 • BT P2.T6.313.3 (rebuilt)

An equity investor runs the reading's structure (senior \$85.0m, mezzanine \$10.0m, equity contribution \$5.0m, collateral coupon 8.5%, Libor 5.0%, recovery 40%) at an annual default rate of 3.0% (3 loans per year). Results (\$ million except defaults):

t	d_t	Survivors	L_t	Excess spread	OC_t	R_t	$OC_t + R_t$	Equity flow	OC account
0		100						(5.000)	
1	3	97	8.245	2.570	1.750	1.200	2.950	0.820	2.950
2	3	94	7.990	2.315	1.750	1.200	2.950	0.565	6.048
3	3	91	7.735	2.060	1.750	1.200	2.950	0.310	9.300
4	3	88	7.480	1.805	1.750	1.200	2.950	0.055	12.715
5	3	85	7.225			1.200		6.101	13.351

Total terminal available funds are 106.776; the amount owed to the bond tranches is 100.675; the equity terminal cash flow is 6.101. Which is nearest to the equity holders' internal rate of return?

A. 4.06%

B. 11.44%

C. 31.40%

D. 57.02%

Question CR-22 e.7 New • built from GARP Ch.22, Tracking the Final-Year Cash Flows, Table 22.2 • shape A9·S4

A CLO analyst at a Toronto pension fund is projecting the final-year waterfall of a five-year CLO with the structure of the reading: 100 loans of USD 1 million paying Libor + 350 bps; senior notes USD 85 million at Libor + 50 bps; mezzanine notes USD 10 million at Libor + 500 bps; equity USD 5 million; Libor flat at 5%. Recoveries are 40% and are paid into the overcollateralization account, which earns 5% and held USD 16.385 million at the end of year 4. By the end of year 4, 24 loans have defaulted, and 6 more default in year 5. Surviving loans pay their final interest and redeem at par at the end of year 5. Which of the following is closest to the mezzanine bond's credit loss at maturity?

A. USD 4.12 million

B. USD 5.12 million

C. USD 5.94 million

D. USD 7.52 million

Solutions • CR-22 e

Solution CR-22 e.1 — Answer: C

Bank Q1146 • AnalystPrep Q.1855

Step 1 – Coupon rates

Each rate is the swap rate plus the spread: collateral $5\% + 3.5\% = 8.5\%$, mezzanine $5\% + 5\% = 10\%$, senior $5\% + 0.5\% = 5.5\%$.

Step 2 – Dollar interest ($ million)

$$\text{Collateral } 0.085 \times 100 = 8.500, \quad \text{Senior } 0.055 \times 85 = 4.675, \quad \text{Mezzanine } 0.10 \times 10 = 1.000$$

$\Rightarrow$ Collateral $>$ Senior $>$ Mezzanine.

Interpretation

These are the reading's numbers. The bonds together cost \$5.675m a year, \$2.825m less than the collateral pays: that gap is the excess spread with no defaults. The large senior tranche pays more dollars than the small mezzanine despite its far lower spread.

Why the other options are wrong

A is incorrect. Ranked by spread $(500 > 350 > 50)$ and ignored the principal amounts.
B is incorrect. Put the mezzanine second because of its high rate; on \$10m it pays only \$1.0m, below the senior's \$4.675m.
D is incorrect. Put the senior above the collateral; the collateral has both a higher rate and a larger principal.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure.
• **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.1 Waterfall (the annual cash flows with no defaults)

Solution CR-22 e.2 — Answer: C

Bank Q1145 • Schweser Module Quiz 40.1 Q2

Step 1 – Collateral cash flow

Pool $= 100 \times \$500{,}000 = \$50\text{m}$; loan rate $4\% + 3\% = 7\%$. Two per cent of the loans default and pay nothing:

$$50{,}000{,}000 \times 7\% \times (1 - 0.02) = \$3{,}430{,}000$$

Step 2 – Bond coupons, top down

$$\text{Senior: } 37.5\text{m} \times (4\% + 2\%) = \$2{,}250{,}000, \qquad \text{Mezzanine: } 10\text{m} \times (4\% + 6\%) = \$1{,}000{,}000$$

$\Rightarrow$ Both are covered: the mezzanine receives its full \$1,000,000.

Step 3 – Residual and the deal's trigger

$$3{,}430{,}000 - 2{,}250{,}000 - 1{,}000{,}000 = \$180{,}000 < \$1{,}000{,}000$$

$\Rightarrow$ Under this deal's rule only residual cash above \$1,000,000 is diverted, so all \$180,000 goes to the equity and the trust account receives nothing.

Interpretation

This is Schweser's equity-cap trigger. Under GARP's mechanism, with the account taking the first \$1,000,000 of excess spread, the same \$180,000 would go to the account and the equity would

receive nothing (see the correction box in CR-22 c). Read which way the stem's trigger runs before splitting the residual.

Why the other options are wrong

A is incorrect. Confused the mezzanine coupon with the residual (equity) flow of \$180,000.
B is incorrect. Dropped the 4% reference rate from the mezzanine coupon: $10\text{m} \times 6\% = \$600{,}000$.
D is incorrect. Sent the \$180,000 residual to the trust account; under the stated rule only residual above \$1,000,000 is diverted.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure. CR-22 c: Describe a waterfall structure in a securitization. • **Reference:** SchweserNotes 2026 Book 2, Reading 40, LO 40.c, Module Quiz 40.1 Q2; Malz, Ch.9 (GARP CR Ch.22), 22.1 Waterfall

Solution CR-22 e.3

Answer: A

Workbook A9.4 • S2, S3 • Map #9

Step 1 – Surviving loans and loan interest

$$N - \sum d = 80 - (3 + 5) = 72, \qquad L_2 = 72 \times 25 \times (0.03 + 0.04) = \text{CNY } 126.0 \text{ million}$$

$\Rightarrow$ Only loans that have not defaulted by the end of year 2 pay interest; the eight defaulted loans pay nothing.

Step 2 – Excess spread and the custodian's test

$$B = 1{,}600 \times 3.6\% + 280 \times 7.5\% = 57.6 + 21.0 = 78.6$$

$$L_2 - \text{fees} - B = 126.0 - 4.0 - 78.6 = 43.4 > 0, \qquad 43.4 \geq K = 20 \;\Rightarrow\; OC_2 = 20.0$$

$\Rightarrow$ Excess spread is positive and exceeds the cap, so exactly CNY 20 million is diverted.

Step 3 – Cash flow to the equity

$$\text{Equity}_2 = 43.4 - 20.0 = \text{CNY } 23.40 \text{ million}$$

Step 4 – Overcollateralisation account at the end of year 2

$$R_2 = 0.40 \times 5 \times 25 = 50.0, \qquad 50.0 \times 1.03 + 20.0 + 50.0 = \text{CNY } 121.5 \text{ million}$$

$\Rightarrow$ Last year's balance earns 3%, then this year's diversion and recoveries are added.

Interpretation

The equity flow falls from CNY 32.15 million in year 1 to 23.40 million as cumulative defaults erode loan interest. The account, now CNY 121.5 million, is soft enhancement: it pays bond interest if loan interest ever falls short and reaches the equity only at maturity. (Dropping the fee would give equity 27.40; ignoring the cap would divert all 43.4 and leave the equity nothing.)

Why the other options are wrong

B is incorrect. Dropped term: did not grow the year-1 balance at 3%, $50.0 + 20.0 + 50.0 = 120.0$. The equity flow is unaffected.
C is incorrect. Wrong input: counted only year-2 defaults, so 75 loans pay interest, $L_2 = 131.25$, excess spread 48.65, equity $48.65 - 20 = 28.65$. The cap still binds, so the account is unchanged.
D is incorrect. Recoveries routed through the waterfall to the equity: equity $23.4 + 50.0 = 73.40$, account $51.5 + 20.0 = 71.5$. This is exactly the perverse benefit from defaults that escrowing recoveries prevents.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure. CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. CR-22 g: Explain the tests on the excess spread that a custodian must go through at the end of each year to determine the cash flow to the overcollateralization account and to the equity noteholders. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), Tracking the Interim Cash Flows; Table 22.1

Solution CR-22 e.4 Answer: D

Bank Q1148 • BT P2.T6.313.1

Step 1 – Bond coupons

$B = 0.055 \times 85 + 0.10 \times 10 = 4.675 + 1.000 = 5.675$ every year, with principal at year 5.
$\Rightarrow$ A is true.

Step 2 – The cap

Year 1: excess spread $8.330 - 5.675 = 2.655 > K$, so $OC_1 = 1.750$ and the equity gets $2.655 - 1.750 = 0.905$. The diversion is 1.750 in every year shown.
$\Rightarrow$ B is true.

Step 3 – Recovery rate

$R_t/d_t = 0.800/2 = 0.4$ per \$1m loan.
$\Rightarrow$ C is true.

Step 4 – Where the recoveries go

The account grows by $OC_t + R_t = 2.550$ a year: year 2 is $2.550 \times 1.05 + 2.550 = 5.228$. The equity receives only $L_t - B - OC_t$.
$\Rightarrow$ Recoveries are deposited in the account, not paid to the equity: D is false.

Interpretation

The terminal check closes the loop: $7.650 + 90 + 0.8 + 11.540 = 109.990$, less 100.675 owed, leaves the equity 9.315. Escrowing the recoveries stops the equity from benefiting from defaults.

Why the other options are wrong

A is incorrect. True: coupons of $4.675 + 1.000 = 5.675$ each year, principal at maturity.
B is incorrect. True: $OC_t = 1.750$ whenever excess spread exceeds 1.750, which it does every year at 2%.
C is incorrect. True: $0.8/2 = 40\%$ of each defaulted \$1m loan.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), Tracking the Interim Cash Flows, Table 22.1; Table 22.2

Solution CR-22 e.5 Answer: B

Bank Q1149 • BT P2.T6.313.2

Step 1 – Interim years

$B = 5.675$. Years 1 to 3: $L_t = 7.650, 6.885, 6.205$, all above 5.675, so the coupons are paid from loan interest.

Step 2 – Year 4: the account covers the gap

$L_4 = 5.610 < 5.675$: shortfall 0.065, paid from the account, $15.120 \times 1.05 + 2.800 - 0.065 = 18.611$.
$\Rightarrow$ Both bonds receive full interest in years 1 to 4.

Step 3 – Terminal funds

$$F = 18.611 \times 1.05 + 59 \times 0.085 + 59 + 0.4 \times 7 = 19.541 + 5.015 + 59 + 2.800 = 86.356$$

Step 4 – Sequential shortfall tests

Senior due $89.675 > 86.356$: the senior takes everything and is short $89.675 - 86.356 = 3.319$. The mezzanine (due 11.000) gets nothing; the equity gets nothing.
$\Rightarrow$ Neither bond class is repaid in full: B.

Interpretation

The account built from early excess spread and recoveries is enough to keep coupons current, but not to save principal: 41 cumulative defaults leave terminal funds \$14.319m short of the \$100.675m owed, deep enough to reach the senior.

Why the other options are wrong

A is incorrect. Coupons are paid in full in years 1 to 4: from loan interest in years 1 to 3, with the account covering the \$0.065m gap in year 4.
C is incorrect. The mezzanine does lose everything, but the senior is also short about \$3.319m.
D is incorrect. Terminal funds of 86.356 against 100.675 owed must produce a shortfall.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure.
• **Reference:** Malz, Ch.9 (GARP CR Ch.22), Table 22.1 and Table 22.2 (default rate 10 percent)

Solution CR-22 e.6 Answer: B

Bank Q1150 • BT P2.T6.313.3

Step 1 – The equity's dated cash flows

Outlay -5.000 at $t = 0$; interim flows 0.820, 0.565, 0.310, 0.055; terminal flow $106.776 - 100.675 = 6.101$ at $t = 5$.

Step 2 – Solve for the IRR

$$-5 + \frac{0.820}{1+x} + \frac{0.565}{(1+x)^2} + \frac{0.310}{(1+x)^3} + \frac{0.055}{(1+x)^4} + \frac{6.101}{(1+x)^5} = 0 \;\Rightarrow\; x = 11.44\%$$

$\Rightarrow$ At 11.44% the discounted inflows sum to the \$5.0m outlay.

Interpretation

At 2% defaults the same structure returns 23.0% to the equity; one more default a year halves the IRR. With a 25% hurdle rate (the one Malz uses later to value the equity) the tranche would be worth well below par.

Why the other options are wrong

A is incorrect. Ignored the interim flows and annualised only the terminal flow: $(6.101/5)^{1/5} - 1 = 4.06\%$.
C is incorrect. Average annual inflow over the outlay: $(7.851/5)/5 = 31.40\%$, ignoring timing and compounding.
D is incorrect. Total profit over the outlay, $(7.851 - 5)/5 = 57.02\%$, neither annualised nor discounted.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure.
• **Reference:** Malz, Ch.9 (GARP CR Ch.22), Tracking the Final-Year Cash Flows (equity IRR)

Solution CR-22 e.7 Answer: B

New • built from GARP Ch.22, Table 22.2

Step 1 – Account at maturity

$$16.385 \times 1.05 = \text{USD } 17.204 \text{ million}$$

$\Rightarrow$ The year-4 balance earns one more year of interest before it is released.

Step 2 – Terminal available funds

Survivors $= 100 - 24 - 6 = 70$: final interest $70 \times 0.085 = 5.950$, redemption 70.000, final-year recoveries $0.4 \times 6 = 2.400$.

$$F = 17.204 + 5.950 + 70.000 + 2.400 = \text{USD } 95.554 \text{ million}$$

Step 3 – Senior test

Senior due $85 \times 1.055 = 89.675 \leq F$: paid in full, leaving $95.554 - 89.675 = 5.879$.

Step 4 – Mezzanine test

Mezzanine due $10 \times 1.10 = 11.000$; it receives 5.879.

$$\text{Credit loss} = 11.000 - 5.879 = \text{USD } 5.12 \text{ million}$$

$\Rightarrow$ The equity receives nothing.

Interpretation

With 30 cumulative defaults the mezzanine recovers only USD 5.879 million of the USD 11.0 million it is owed (it loses about 47%), while the senior is untouched: the thin junior tranche absorbs the hit once the equity is gone.

Why the other options are wrong

A is incorrect. Measured the loss against the mezzanine's par of 10.000 only, forgetting the final coupon it is owed: $10.000 - 5.879 = 4.12$.

C is incorrect. Did not grow the account for year 5: $F = 94.735$, so the mezzanine receives 5.060 and loses 5.94.

D is incorrect. Left out the final-year recoveries: $F = 93.154$, so the mezzanine receives 3.479 and loses 7.52.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure.

• **Reference:** Malz, Ch.9 (GARP CR Ch.22), Tracking the Final-Year Cash Flows, Table 22.2

CR-22 • Structured Credit Risk • Vardeez: IMPORTANT

CR-22 f

P1 CALC

Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year.

Workbook: A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A9 (Map #9, 7/16) reports "overcollateralisation account balance". Two GARP practice-exam questions test the account directly (PE1 Q72 and PE2 Q45, Questions CR-22 f.5 and f.7).

Vardeez: the one possible calculation from this reading sits here and in CR-22 e. His original notes give the mechanics in prose only; the rebuilt notes write the account recursion from GARP.

Reported exam shape

- **A9·S3** Overcollateralisation account balance at a year end.

What the questions look like: the excess spread (after fees, swap payments or defaults), the amount diverted against the cap, and the account balance after one or two years, often with a year in which excess spread turns negative and the account pays the shortfall. Concept questions ask where excess spread above and below the cap goes.

Worked elsewhere: A9.4 (account balance and equity flow at the end of year 2) under CR-22 e.

1. Excess spread

Excess spread is the collateral interest left after everything the trust must pay out of it: fees and expenses, any swap payments, and the coupons on the bond tranches. GARP: "the interest from the collateral that is not paid out to the liabilities or as fees and expenses". It can be quoted in dollars or as a rate on the collateral: collateral earning 8% net of fees against payments to the notes of 7% leaves 1% a year.

Why there is any. In the reading's CLO (no defaults) the excess spread of \$2.825m has three sources:

$$\underbrace{85 \times (8.5\% - 5.5\%)}_{\text{senior: } +2.550} + \underbrace{10 \times (8.5\% - 10\%)}_{\text{mezzanine: } -0.150} + \underbrace{5 \times 8.5\%}_{\text{equity's slice: } +0.425} = \$2.825\text{m}$$

The cheap senior debt creates most of it, the expensive mezzanine eats a little, and the collateral funded by the equity earns the full loan rate, reference rate included, without paying any coupon (Question CR-22 f.2).

Soft credit enhancement. Excess spread is zero at origination and has to be earned year by year, so it is *soft* enhancement, in contrast to the *hard* enhancement of overcollateralisation present at issue. It falls over time as defaults erode loan interest: in the 2% scenario from \$2.655m in year 1 to \$2.145m in year 4.

2. Where excess spread goes: the custodian's two tests

In Malz's structure up to $K = \$1.75$m of excess spread is diverted each year to the overcollateralisation account; anything above K goes to the equity; the account earns the 5% money market rate, pays bond interest when loan interest falls short, and is released to the equity only at maturity (CR-22 c, Section 3). At each year end the custodian runs a two-step decision tree:

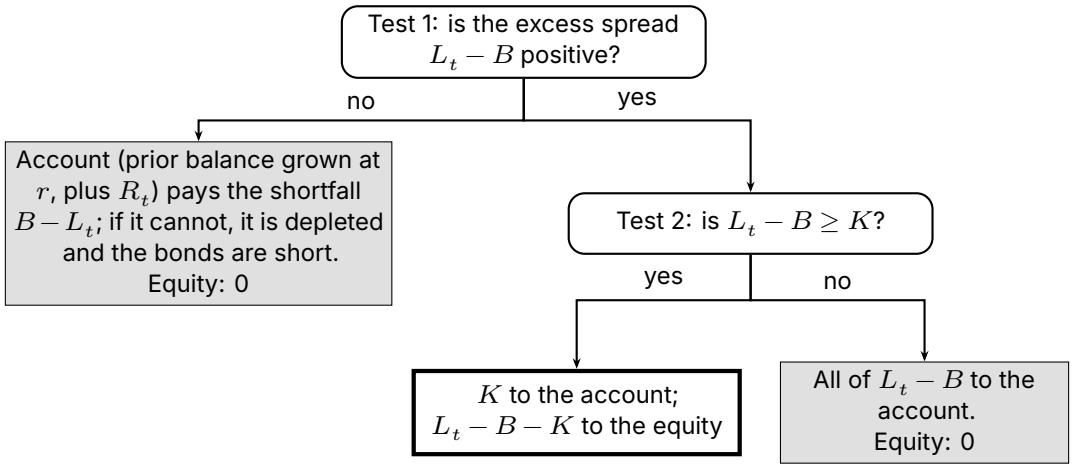


Figure 3. The custodian's year-end tests and their three outcomes. Only one branch (the white box with a bold frame) pays the equity anything before maturity; the two grey boxes leave it with nothing. Recoveries R_t go into the account on every branch.

Two consequences worth holding:

- The account has **first call** on excess spread up to K. When excess spread is positive but below K the equity is paid nothing (GARP's 7.5% scenario from year 2).
- When excess spread is negative, nothing is diverted (OC_t is negative: a payment out), and the account's funds plus this year's recoveries are tested against the shortfall: $\sum_{\tau<t}(1+r)^{t-\tau}(OC_\tau + R_\tau) + R_t \geq B - L_t$?

3. The value of the account at each year end (shape S3)

The account grows by three things: interest on last year's balance, this year's diversion from excess spread, and this year's recoveries:

$$\boxed{A_t = A_{t-1}(1+r) + OC_t + R_t} \qquad \Longleftrightarrow \qquad A_t = \sum_{\tau=1}^{t}(1+r)^{t-\tau}\,(OC_\tau + R_\tau)$$

with $OC_t = \max[\min(L_t - B, K), 0]$ when excess spread is positive and $OC_t = L_t - B < 0$ when the account pays a shortfall.

GARP's 2% scenario. Two defaults a year, so $R_t = \$0.8$m, and excess spread above K every year, so $OC_t = \$1.75$m: \$2.55m goes in each year.

$$\begin{aligned} A_1 &= 1.750 + 0.800 = \$2.550\text{m} \\ A_2 &= 2.550 \times 1.05 + 2.550 = \$5.2275\text{m} \\ A_3 &= 5.2275 \times 1.05 + 2.550 = \$8.0389\text{m} \\ A_4 &= 8.0389 \times 1.05 + 2.550 = \$10.9908\text{m} \\ A_4(1.05) &= \$11.5404\text{m, released into the final-year waterfall} \end{aligned}$$

GARP's 10% scenario, year 4. Excess spread is $5.610 - 5.675 = -\$0.065$m, so nothing is diverted and the account pays the shortfall while receiving recoveries of \$2.8m:

$$A_4 = 15.1199 \times 1.05 + 2.800 - 0.065 = \$18.6109\text{m}$$

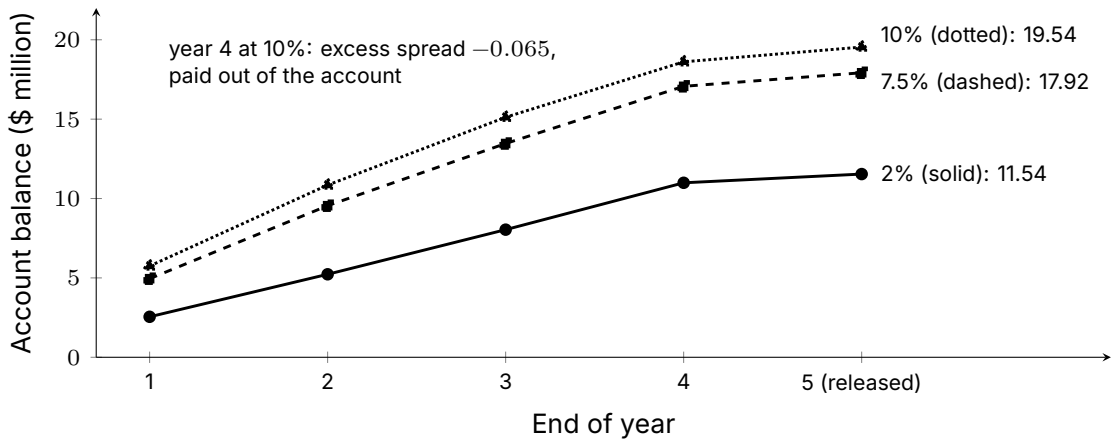


Figure 4. The overcollateralisation account in GARP's three scenarios. The worst scenario has the largest account, because recoveries are escrowed in it: a big balance here is a symptom of defaults, not of health. The year-5 point is the year-4 balance grown at 5%, released into the final-year waterfall.

4. When excess spread turns negative

Excess spread is negative once loan interest falls below the coupons, $S_t\,(r+s)\,\text{par} < B$, so the breakpoint in surviving loans is

$$S^* = \frac{B}{(r+s)\times \text{par}} = \frac{5{,}675{,}000}{0.085\times 1{,}000{,}000} = 66.76 \text{ loans}$$

With 67 survivors excess spread is $+\$0.02$m; with 66 (year 4 at 10%) it is $-\$0.065$m. Below that point the account is drawn down rather than built up, and the equity has already received nothing for some time, since it needs $S_t > (B+K)/(0.085\times \$1\text{m}) = 87.4$ survivors to be paid at all.

Formula card: CR-22 f

Excess spread	$X_t = L_t - \text{fees} - \text{swap payments} - B$ (as a rate: collateral yield less fees less weighted coupon)
Diversion	$OC_t = \max[\min(X_t, K), 0]$ if $X_t \geq 0$; $OC_t = X_t$ (a payment out) if $X_t < 0$
Equity, interim	$\max(X_t - OC_t, 0)$: positive only if $X_t > K$
Recoveries	$R_t = RR \times d_t \times$ par, paid into the account
Account recursion	$A_t = A_{t-1}(1+r) + OC_t + R_t$
Closed form	$A_t = \sum_{\tau \leq t}(1+r)^{t-\tau}(OC_\tau + R_\tau)$
Shortfall test	$A_{t-1}(1+r) + R_t \geq B - L_t$? If not, the account is depleted and the bonds are short
Released at maturity	$A_{T-1}(1+r)$ joins the terminal funds F
Breakpoint	excess spread < 0 when survivors $< B/[(r+s)\,\text{par}]$

Exam traps

- **Cap read as a floor.** K is the most the account can take. Excess spread below K goes entirely to the account; it is not withheld until K is exceeded.
- **Amount above K.** It goes to the equity, not into the account and not to the bonds.
- **No interest on last year's balance.** The opening balance earns the money market rate before this year's flows are added (Question CR-22 f.7, option A).
- **Shortfall forgotten or added.** When $L_t < B$ the shortfall comes *out* of the account; ignoring it, or adding it, misstates the balance.
- **Recoveries left out.** They go into the account even in a year with negative excess spread, and they are what makes the account large in bad years.
- **Fees and swap payments.** Both come out before excess spread; leaving either in overstates it (Question CR-22 f.3).
- **The equity's reference-rate income.** Excess spread is not just the difference in spreads: the collateral funded by the equity earns the reference rate too (Question CR-22 f.2, option A).

Practice questions: CR-22 f

Question CR-22 f.1

Bank Q1153 • AnalystPrep Q.5525 (rebuilt)

A credit analyst is explaining a three-tiered securitization with an overcollateralization (OC) mechanism to a client. Which of the following best describes the treatment of positive excess spread in such a structure?

A. All excess spread is paid to the equity tranche each year.

B. Excess spread is paid to the senior bonds to retire principal early.

C. Excess spread is shared equally among the senior, mezzanine and equity tranches.

D. Up to a maximum annual amount K is diverted to the OC account to absorb future shortfalls, and any remainder goes to the equity.

Question CR-22 f.2

Bank Q1156 • BT P2.T6.312.2 (rebuilt)

A CLO holds 100 identical floating-rate loans of \$1.0 million each, priced at par, paying Libor + 300 bps. The capital structure is: senior debt \$75.0 million paying Libor + 100 bps; junior (mezzanine) debt \$15.0 million paying Libor + 500 bps; and an equity tranche. Coupons are annual, there are no fees and no defaults, and the Libor swap curve is flat at 2.0%. What is the annual excess spread, that is, collateral interest minus tranche coupon payments?

A. \$1.50 million

B. \$1.70 million

C. \$2.75 million

D. \$3.95 million

Question CR-22 f.3 Bank Q1154 • BT L2.T6.106.4 (rebuilt)

A securitization trust holds collateral earning a weighted average interest rate of 8.30%. The tranches pay a weighted average coupon of Libor + 23 bps, with Libor at 5.32%. The trust also pays servicing fees of 51 bps and a payment to its swap counterparty of 13 bps (all as annual percentages of the collateral balance). What is the initial excess spread?

A. 2.11%

B. 2.24%

C. 2.62%

D. 2.75%

Question CR-22 f.4 Bank Q1151 • AnalystPrep Q.4398 (rebuilt; bank key corrected)

A collateralized loan obligation holds 100 identical leveraged loans with par value \$1,000,000 each, priced at par. The loans pay a fixed spread of 4% over the one-year swap rate (r). The senior, mezzanine and equity tranches are 85%, 10% and 5% of the pool. The senior tranche pays $r + 200$ bps and the mezzanine tranche pays $r + 500$ bps. Each year, up to $K = \$2{,}000{,}000$ of excess spread is diverted to the overcollateralization (OC) account, and any excess spread above K is paid to the equity tranche. The swap curve is flat at 5%, there are no fees and no defaults, and all payments occur at year end. What are the first-year cash flow to the mezzanine tranche and the amount diverted to the OC account?

A. Mezzanine \$1,000,000; OC account \$50,000

B. Mezzanine \$1,000,000; OC account \$2,000,000

C. Mezzanine \$2,050,000; OC account \$0

D. Mezzanine \$5,950,000; OC account \$2,000,000

Question CR-22 f.5 Bank Q1157 • GARP PE1 Q72 • Mock 3 Q9

A manager of collateralized loan obligations (CLOs) is reviewing a CLO with a pool of 50 identical loans, each priced at its par value of GBP 1 million. The loans are floating-rate obligations that pay a fixed spread of 150 bps over LIBOR. Coupons and interest payments on the liabilities are made annually at year end:

Liabilities	Amount (GBP)	Coupon
Senior debt	37,500,000	LIBOR + 45 bps
Mezzanine debt	10,000,000	LIBOR + 300 bps
Equity	2,500,000	none

The CLO initially has no overcollateralization, and the annual excess spread flowing into the overcollateralization account is limited to GBP 250,000. Suppose LIBOR remains flat at 4% in the first year, there are no defaults, and there are no management or transaction fees. What amounts are posted to the overcollateralization account and to the equity tranche after the first year?

A. OC account GBP 0 and equity GBP 0

B. OC account GBP 0 and equity GBP 381,250

C. OC account GBP 250,000 and equity GBP 131,250

D. OC account GBP 381,250 and equity GBP 0

Question CR-22 f.6

Bank Q1155 • BT P2.T6.24.19.1 (rebuilt)

A CLO holds 100 identical loans of $1,000,000 each, priced at par. The senior tranche is 85% of the pool and the junior tranche 10%. The market reference rate (MRR) is 3.5%; the loans pay 9.5% in total; the senior tranche pays MRR + 3.5% and the junior tranche pays MRR + 6.0%, so total bond coupons due each year are $6.90 million. Loans that default in a year pay no interest that year; interest is paid at year end by surviving loans only. The numbers of defaults are 8 in year 1, 7 in year 2 and 7 in year 3 (all interim years). What is the excess spread at the end of year 3?

A. $0.415 million

B. $0.51 million

C. $1.175 million

D. $1.46 million

Question CR-22 f.7

Bank Q1158 • GARP PE2 Q45 (rebuilt; bank key corrected)

A manager of a structured credit desk illustrates cash flows in a securitization using the following information (all liabilities and the collateral have a maturity of 5 years):

Initial number of loans in the collateral pool	100
Principal amount of each loan	EUR 1,000,000
Total coupon interest paid annually on all junior and senior bonds	EUR 5,500,000
Maximum annual amount flowing from excess spread into the OC account	EUR 1,500,000
Swap rate for all maturities	3.5%
Recovery rate on defaulted loans	45%

The loans pay a fixed spread of 2.2% over the swap curve. Defaulted loans pay no interest in the year of default. There were no defaults in year 1, and recoveries are deposited in the overcollateralization (OC) account, which earns the swap rate. What is the value of the OC account at the end of year 2 if there are 8 defaults in year 2?

A. EUR 3,544,000

B. EUR 3,551,000

C. EUR 3,807,000

D. EUR 4,063,000

Solutions • CR-22 f

Solution CR-22 f.1

Answer: D

Bank Q1153 • AnalystPrep Q.5525

Step 1 – Test 1: is excess spread positive?

If loan interest covers the bond coupons, there is positive excess spread to allocate.

Step 2 – Test 2: is it at least K?

If $L_t - B \geq K$, K goes to the OC account and $L_t - B - K$ to the equity; if $L_t - B < K$, all of it goes to the account and the equity gets nothing.
$\Rightarrow$ Up to K is diverted first; the remainder goes to the equity.

Step 3 – What the account is for

It pays bond interest in later years when excess spread turns negative, and is released to the equity at maturity.

Interpretation

Excess spread is soft credit enhancement: the structure first uses it to build a reserve that protects the bondholders, and only then lets the residual claimant have it.

Why the other options are wrong

A is incorrect. Ignores the diversion to the account, which has first priority up to K.
B is incorrect. In this structure interim excess spread funds the account; it does not amortise senior principal.
C is incorrect. The tranches receive only their contractual coupons; excess spread is never shared equally.

LO: CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.2 Tracking the Interim Cash Flows

Solution CR-22 f.2

Answer: B

Bank Q1156 • BT P2.T6.312.2

Step 1 – Collateral interest

$100 \times (2\% + 3\%) = \$5.000$ million.

Step 2 – Tranche coupons

Senior $75 \times (2\% + 1\%) = \$2.250$m; junior $15 \times (2\% + 5\%) = \$1.050$m.

Step 3 – Excess spread

$$5.000 - 2.250 - 1.050 = \$1.700 \text{ million}$$

Interpretation

Decomposed: the senior earns a 2% spread pickup ($75 \times 2\% = 1.50$), the junior costs 2% ($15 \times 2\% = -0.30$), and the \$10m funded by the equity earns the full 5% (0.50): $1.50 - 0.30 + 0.50 = 1.70$. The equity's slice earns the Libor component without paying a coupon.

Why the other options are wrong

A is incorrect. Compared spreads only, $3.0 - 0.75 - 0.75 = 1.50$, missing the 2% Libor earned on the \$10m not funded by debt (0.20).
C is incorrect. Subtracted only the senior coupon: $5.000 - 2.250 = 2.75$.
D is incorrect. Subtracted only the junior coupon: $5.000 - 1.050 = 3.95$.

LO: CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the over-collateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.1 Waterfall

Solution CR-22 f.3 — Answer: A

Bank Q1154 • BT L2.T6.106.4

Step 1 – Weighted average tranche coupon

$5.32\% + 0.23\% = 5.55\%$.

Step 2 – Deduct every payment out of collateral interest

$$8.30\% - 5.55\% - 0.51\% - 0.13\% = 2.11\%$$

$\Rightarrow$ The initial excess spread is 2.11% a year of the collateral balance.

Interpretation

Excess spread is what is left after *all* payments the trust makes: coupons, servicing fees and swap payments. It is the cushion that soft enhancement is built from, so overstating it overstates the protection.

Why the other options are wrong

B is incorrect. Stopped after the servicing fee, omitting the 13 bp swap payment: $2.75\% - 0.51\% = 2.24\%$.

C is incorrect. Deducted the swap payment but not the 51 bp servicing fee: $2.75\% - 0.13\% = 2.62\%$.

D is incorrect. Compared collateral yield with the coupon only: $8.30\% - 5.55\% = 2.75\%$.

LO: CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the over-collateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.1 Capital Structure and Credit Losses; SchweserNotes 2026 Book 2, Reading 40, LO 40.b

Solution CR-22 f.4 — Answer: B

Bank Q1151 • AnalystPrep Q.4398

Step 1 – Collateral interest

$100 \times \$1\text{m} \times (5\% + 4\%) = \$9{,}000{,}000$.

Step 2 – Coupons

Senior $85\text{m} \times 7\% = \$5{,}950{,}000$; mezzanine $10\text{m} \times 10\% = \$1{,}000{,}000$; $B = \$6{,}950{,}000$.

$\Rightarrow$ The mezzanine receives its \$1,000,000 coupon in full.

Step 3 – Excess spread against the cap

$$L_1 - B = 9{,}000{,}000 - 6{,}950{,}000 = \$2{,}050{,}000 \geq K = \$2{,}000{,}000$$

$\Rightarrow$ The account receives $K = \$2{,}000{,}000$ and the equity the remaining \$50,000.

Interpretation

The bank's corrected key matches GARP's rule that the account takes the first K. With 5% of loans defaulting the excess spread would fall to \$1.6m, all of it diverted.

Why the other options are wrong

A is incorrect. Reversed the rule: gave the cap to the equity and only the \$50,000 above it to the account.

C is incorrect. Paid the whole excess spread to the mezzanine; bonds receive only their contractual coupons.

D is incorrect. Reported the senior coupon ($5,950,000) as the mezzanine flow.

LO: CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.2 Tracking the Interim Cash Flows

Solution CR-22 f.5 — Answer: C

Bank Q1157 • GARP PE1 Q72

Step 1 – Collateral interest

$50\text{m} \times (4\% + 1.5\%) = \text{GBP } 2{,}750{,}000.$

Step 2 – Coupons

Senior $37.5\text{m} \times 4.45\% = 1{,}668{,}750$; mezzanine $10\text{m} \times 7.0\% = 700{,}000$; $B = 2{,}368{,}750$.

Step 3 – Excess spread and the two tests

$$2{,}750{,}000 - 2{,}368{,}750 = 381{,}250 > 0, \qquad 381{,}250 \geq K = 250{,}000$$

$\Rightarrow$ GBP 250,000 goes to the account and $381{,}250 - 250{,}000 = \text{GBP } 131{,}250$ to the equity.

Interpretation

GARP's own wording, "the annual excess spread flowing into the overcollateralization account is limited to GBP 250,000", is the Malz rule: the account takes up to the limit, the equity gets the rest.

Why the other options are wrong

A is incorrect. Treated the excess spread as zero.
B is incorrect. Skipped the overcollateralisation test and paid all the excess spread to the equity.
D is incorrect. Ignored the GBP 250,000 limit and diverted the whole excess spread.

LO: CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.2 Tracking the Interim Cash Flows; GARP 2026 Practice Exam Part II, Q72

Solution CR-22 f.6 — Answer: B

Bank Q1155 • BT P2.T6.24.19.1

Step 1 – Survivors at the end of year 3

Cumulative defaults $8 + 7 + 7 = 22$, so 78 loans pay interest.

Step 2 – Loan interest

$L_3 = 78 \times \$1\text{m} \times 9.5\% = \$7.41\text{m}.$

Step 3 – Excess spread

$B = 85 \times 7.0\% + 10 \times 9.5\% = 5.95 + 0.95 = \$6.90\text{m}$, so $L_3 - B = 7.41 - 6.90 = \$0.51\text{m}.$
$\Rightarrow$ Positive but small: below any realistic cap, so all of it would go to the account.

Interpretation

Excess spread shrinks every year as the pool erodes: $2.60m with no defaults, $1.84m in year 1, $1.175m in year 2 and $0.51m in year 3. Six more defaults would turn it negative: 72 survivors pay $6.84m, below B.

Why the other options are wrong

A is incorrect. Assumed 8 defaults in year 3 (77 survivors): $7.315 - 6.90 = 0.415$.
C is incorrect. Used the year-2 survivors (85 loans): $8.075 - 6.90 = 1.175$.
D is incorrect. Subtracted only the senior coupon: $7.41 - 5.95 = 1.46$.

LO: CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.2 Tracking the Interim Cash Flows

Solution CR-22 f.7 — Answer: B

Bank Q1158 • GARP PE2 Q45

Step 1 – Year 1: build-up

Loan rate $3.5\%+2.2\% = 5.7\%$: $L_1 = 100\times 5.7\%\times 1\text{m} = 5{,}700{,}000$. Excess spread $5{,}700{,}000 - 5{,}500{,}000 = 200{,}000 < K$, so all of it is diverted: $A_1 = 200{,}000$.

Step 2 – Year 2: excess spread turns negative

$L_2 = 92 \times 5.7\% \times 1\text{m} = 5{,}244{,}000$, so $L_2 - B = -256{,}000$: a shortfall paid from the account.

Step 3 – Recoveries

$R_2 = 8 \times 0.45 \times 1{,}000{,}000 = 3{,}600{,}000$, deposited in the account.

Step 4 – Account at the end of year 2

$$A_2 = 200{,}000 \times 1.035 + 3{,}600{,}000 - 256{,}000 = 207{,}000 + 3{,}344{,}000 = \text{EUR } 3{,}551{,}000$$

$\Rightarrow$ The opening balance earns the swap rate, recoveries go in, the coupon shortfall comes out.

Interpretation

The account covers the year-2 shortfall easily because the recoveries arrive in the same year as the defaults that cause it. This is the bank's corrected key: the full recursion $A_t = A_{t-1}(1+r) + OC_t + R_t$.

Why the other options are wrong

A is incorrect. Omitted a year of interest on the opening balance: $200{,}000 + 3{,}600{,}000 - 256{,}000 = 3{,}544{,}000$.

C is incorrect. Ignored the shortfall: $207{,}000 + 3{,}600{,}000 = 3{,}807{,}000$.

D is incorrect. Added the negative excess spread instead of paying it out: $207{,}000 + 3{,}600{,}000 + 256{,}000 = 4{,}063{,}000$.

LO: CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.2 Tracking the Interim Cash Flows; GARP 2026 Practice Exam Part II, Q45

CR-23 • An Introduction to Securitisation • Vardeez: IMPORTANT

CR-23 g

P1 CALC

Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures.

Workbook: A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A9): the delinquency ratio came up as a "straightforward numerical" in the August 2026 sitting, and the recall map lists DSCR and WAL among the formulas to know closed-book.

Vardeez: "Some calculations pls check them out": DSCR and the ratios. His chapter note calls this LO "39.g"; the text matches CR-23 g (bank and Schweser reading 41).

Reported exam shapes

- **A9·S1** Delinquency ratio, the plain numerical seen in August: only receivables more than 90 days overdue count.
- **A9·S7** Closed-book formulas: DSCR and WAL here, CPR under CR-23 h. The stem gives the data, never the formula.

What the questions look like: almost all numerical. You get a receivables ageing table, a pool of loans with balances, rates and terms, a CMBS income line, or an amortisation schedule with pool factors. Each stem carries inputs you do not need: write-offs in a delinquency question, remaining terms in a WAC question, the pool's value in a DSCR question.

Worked elsewhere: A9.6 (CPR and PSA) and A9.7 (prepayment theory) at Questions CR-23 h.5 and CR-23 h.8; A9.4 (interim waterfall and overcollateralisation account) at Question CR-22 e.3; A9.5 (final-year waterfall) under CR-22 c.

1. Which measure for which pool

Choudhry's primer on performance metrics (GARP 23.5) ties each measure to the collateral it was built for. ABS performance depends on consumer credit, so deals carry **trigger mechanisms** that accelerate amortisation and **reserve accounts** that cover interest shortfalls; the measures below are the language of those triggers and of the monthly investor report. A CDO holds 100 to 200 loans; an ABS pool often has thousands of obligors. The map below, condensed from GARP Table 23.2 (Summary of Performance Measures), is shared by CR-23 g and CR-23 h.

Measure	Calculation	Typical asset class
Delinquency ratio	overdue more than 90 days ÷ total receivables	credit cards
Default ratio	written off in the period ÷ receivables at period end	credit cards
MPR	principal and interest collected ÷ pool balance	all non-amortising classes
DSCR	net operating income ÷ debt payments	commercial mortgages
WAC, WAM, WAL	weighted coupon, term; $\sum(a/365)\,\mathrm{PF}(s)$	mortgages
SMM, CPR, PSA	prepayment measures (CR-23 h)	mortgages, home equity, student loans
Loss curve, ABS speed	cumulative loss; prepayments ÷ pool balance	auto and truck loans

2. Credit card ABS: delinquency, default and MPR

Credit card loans have **no predetermined term**, and a single obligor's balance is often repaid within six months. The structure therefore has to control the repayment speed, through scheduled amortisation or a revolving period in which principal collections buy new receivables (CR-23 c). Three ratios monitor the pool:

$$\text{Delinquency ratio} = \frac{\text{receivables overdue more than 90 days}}{\text{total credit card receivables}}$$

$$\text{Default ratio} = \frac{\text{receivables written off during the period}}{\text{total credit card receivables at the end of the period}}$$

$$\text{MPR} = \frac{\text{principal and interest collected in the period}}{\text{outstanding pool balance}}$$

- **Delinquency** is the *early* indication of portfolio quality: a receivable is overdue before it is written off. **Default** measures the realised loss. Together they assess the credit loss on the pool, are normally tied to early amortisation triggers, and are reported through the life of the deal.
- **MPR** is the proportion of the pool's principal and interest repaid in the period. The rating agencies require every non-amortising ABS to set a **minimum** MPR as an early amortisation trigger: an MPR *below* the floor ends the revolving period. GARP's footnote: MPR is *not* a prepayment measure, because credit cards are non-amortising assets.
- Vardeez's shortcut: the three share one denominator, total receivables. The numerators differ: receivables that are late, receivables that are gone, and cash that has arrived.

Example (Schweser). A pool of USD 57.80 million: current 49.90, over 30 days past due 5.75, over 60 days 1.27, over 90 days 0.88 (the buckets are mutually exclusive and sum to the pool); written off 1.10; principal and interest collected 1.56.

$$\text{Delinquency} = \frac{0.88}{57.80} = 1.52\% \qquad \text{Default} = \frac{1.10}{57.80} = 1.90\% \qquad \text{MPR} = \frac{1.56}{57.80} = 2.70\%$$

Counting every past-due bucket as delinquent would give $(5.75 + 1.27 + 0.88)/57.80 = 13.67\%$, the scope error behind most wrong options.

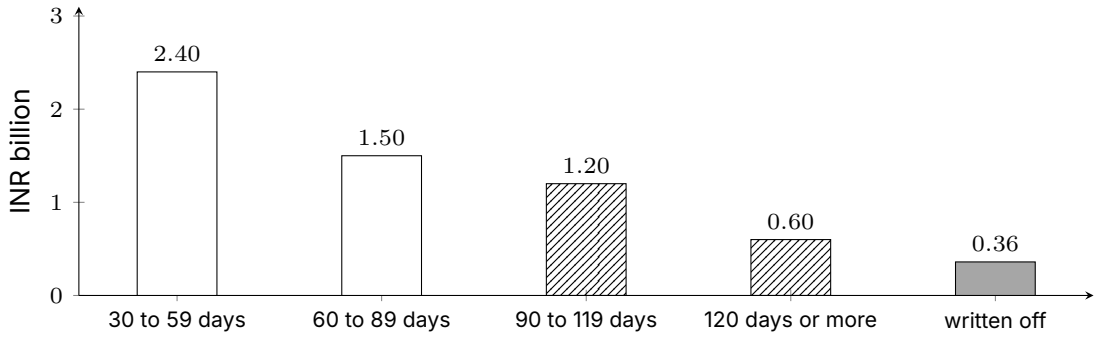


Figure 1. Past-due buckets of an INR 48.0 billion credit card pool (the numbers of Question CR-23 g.2). Only the hatched buckets, overdue more than 90 days, enter the delinquency ratio: $1.8/48.0 = 3.75\%$. The open bars are past due but not delinquent. Write-offs (grey) feed the default ratio, $0.36/48.0 = 0.75\%$.

3. Commercial MBS: the debt service coverage ratio

The notable difference between RMBS and CMBS is that a CMBS loan is **non-recourse**: it is fully secured by the underlying property, so the lender can look only to the property's cash flow. That makes the DSCR "crucial to evaluating credit risk":

$$\text{DSCR} = \frac{\text{net operating income}}{\text{debt payments}} = \frac{\text{income} - \text{operating expenses}}{\text{interest} + \text{principal}}$$

- **Below 1.0** the property does not generate enough cash to cover the required debt payments. The higher the DSCR, the more comfort for noteholders.
- **Calibration** (GARP 23.4): about 2.5 to 3.0 for a residential mortgage deal; for a volatile asset such as airline ticket receivables, unlikely to be lower than 4.0. The arranger's cash flow model sizes the notes that can be issued while keeping the minimum DSCR.
- **Example** (Schweser): NOI USD 89,572,500 against debt payments of USD 87,958,000 gives DSCR $=$ 1.02. Debt service is covered, but thinly for a mortgage deal. The inverted ratio, 0.98, would wrongly read as a shortfall.
- The denominator is interest *and* principal (Vardeez); NOI is after operating expenses.

4. Mortgage pools: WAC and WAM

$$\text{WAC} = \frac{\sum_i B_i\, c_i}{\sum_i B_i} \qquad \text{WAM} = \frac{\sum_i B_i\, m_i}{\sum_i B_i}$$

with B_i the outstanding balance, c_i the mortgage rate and m_i the remaining term of loan group i.

- **WAC** is the pool's coupon weighted by loan balance. It drifts as loans repay, but at any time, compared with the **net coupon payable to investors**, it indicates the pool's ability to pay. The gap pays fees and builds excess spread.
- **WAM** is the balance-weighted remaining term of the loans, expressed in **months**. Longer securities are more volatile. On the *stated maturity date* the WAM avoids a subjective call on whether loans run to term and recognises each security's potential liquidity risk; on the *reset date* it shows the shortening effect of prepayments.

Example (rates from Schweser's WAC example, remaining terms added):

Loan group	Balance (USD m)	Rate	Term (months)	$B \times c$	$B \times m$
1	6.0	7.8%	300	0.468	1,800
2	10.0	6.0%	240	0.600	2,400
3	4.0	5.0%	120	0.200	480
Total	20.0			1.268	4,680

WAC $= 1.268/20 = 6.34\%$ and WAM $= 4{,}680/20 = 234$ months. Against notes paying 5.50%, the pool has 0.84 percentage points of coupon headroom. The simple averages, 6.27% and 220 months, ignore the balances and are the standard distractors.

5. Weighted average life

Amortising (pass-through) structures are priced and traded on expected maturity and **WAL**, "the time-weighted period during which principal is outstanding" (GARP 23.3). It lets an investor compare an MBS with other investments of similar maturity, and it matters for MBS precisely because principal comes back throughout the life of the deal. GARP's formula:

$$\boxed{\text{WAL} = \sum_{s} \frac{a_s}{365}\,\text{PF}(s)} \qquad \text{PF}(s) = \frac{\text{notional outstanding during period } s}{\text{original notional}}$$

with a_s the actual days in period s. Each period contributes its length in years times the fraction of principal still outstanding.

GARP Table 23.1. A note issue of USD 89,529,500 amortises over quarterly payment dates from 21 November 2003 to 24 July 2008:

Period	Actual days a	Outstanding (USD)	PF	PF $\times\, a/365$
0	66	89,529,500	1.0000	0.18082
1	91	84,470,588	0.9435	0.23523
2	91	79,529,412	0.8883	0.22147
⋮				⋮
18	91	16,470,588	0.1840	0.04587
Sum of the 19 contributions				**2.58 years**

The last payment falls about 4.68 years after the start, but the WAL is only 2.58 years, because most of the principal is returned well before then.

The same number, counted the other way. Weighting each repayment date t_k (in years) by the share of principal returned on it gives the identical result:

$$\text{WAL} = \sum_{k} t_k \,\frac{\text{principal repaid at } t_k}{\text{original principal}}$$

Both formulas measure the area under the pool factor curve (Figure 2), one in vertical strips and one in horizontal strips. Use whichever the data support: a pool factor column calls for the first, a principal schedule for the second (Question CR-23 g.8). Two consequences:

- **A WAL is tied to a prepayment assumption.** Faster prepayment cuts the pool factor sooner and shortens the WAL (CR-23 h).

- **WAL is not duration.** It weights by principal only and does not discount. A ten-year level-payment loan at 6% has a WAL of 5.98 years but a Macaulay duration of 5.02 years.

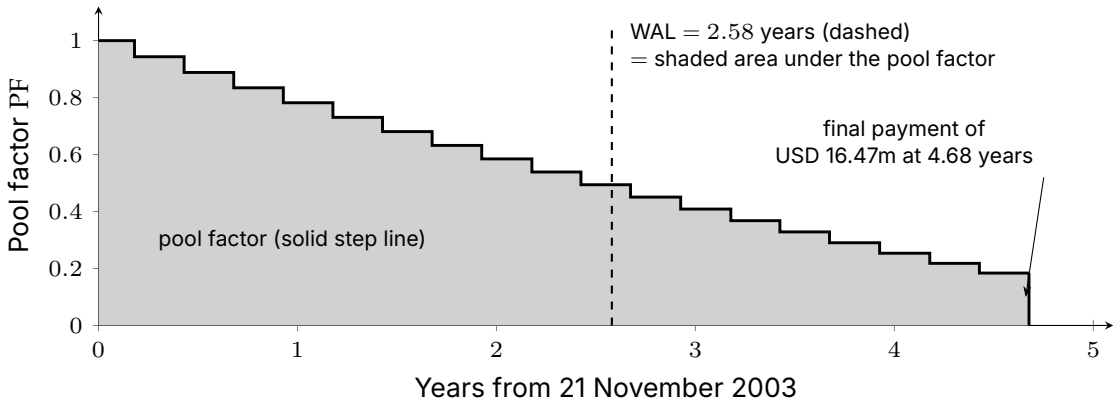


Figure 2. GARP Table 23.1 drawn as a pool factor curve. Each step is one quarterly period: its width is $a/365$ and its height the pool factor, so the shaded area is $\sum(a/365)\,\mathrm{PF} = 2.58$ years. Squeezing the same area into a full-height rectangle gives the WAL: a single bullet repayment at 2.58 years would have the same time-weighted principal.

Formula card: CR-23 g

Delinquency ratio	receivables overdue more than 90 days ÷ total receivables (credit cards)
Default ratio	written off in the period ÷ total receivables at period end (credit cards)
MPR	principal and interest collections ÷ outstanding pool balance; a *minimum* is the early amortisation trigger
DSCR	NOI ÷ debt payments = (income − operating expenses) ÷ (interest + principal); below 1.0 is a shortfall (CMBS)
WAC	$\sum B_i c_i / \sum B_i$; compare with the net investor coupon
WAM	$\sum B_i m_i / \sum B_i$, in months; stated maturity or reset date
WAL	$\sum_s (a_s/365)\,\mathrm{PF}(s) = \sum_k t_k \times$ (principal at t_k)/original principal

	Asset class	Healthy reading	What it feeds
Delinquency	Credit cards	Low; it leads write-offs	Early amortisation trigger
Default	Credit cards	Low	Early amortisation trigger
MPR	Non-amortising pools	High; must stay above the floor	Early amortisation trigger
DSCR	CMBS	Well above 1.0 (2.5 to 3.0 residential, 4.0 or more for volatile assets)	Sizing of the notes
WAC	Mortgages	Above the net investor coupon	Excess spread, ability to pay
WAM	Mortgages	Longer means more volatile	Liquidity and prepayment view
WAL	Amortising MBS and ABS	Shortens with faster prepayment	Pricing and maturity comparison

Exam traps

- **Bucket scope.** Delinquent means overdue *more than 90 days*: every bucket from 90 days on, and nothing younger. Counting all past-due buckets, or only the 90 to 119 day bucket, is the August trap.
- **Sibling swap.** Write-offs give the default ratio, not the delinquency ratio; collections give the MPR.
- **Wrong denominator.** Total receivables is the whole pool (current plus every past-due bucket when the buckets are mutually exclusive), not current receivables alone and not delinquent receivables.
- **MPR is not a prepayment measure,** and it includes interest. New receivables bought in the revolving period are not netted off.
- **Trigger polarity.** The MPR floor is breached when the MPR falls *below* it; delinquency and default triggers are breached when the ratios rise *above* theirs.
- **DSCR inverted or mis-scaled.** Debt payments over NOI, or NOI over the pool's value, are not the DSCR. DSCR is a CMBS credit measure, not a prepayment measure.
- **Simple averages.** WAC and WAM weight by balance. Weighting rates by term, or terms by rate, is wrong too.
- **WAL slips.** Dropping the first (stub) period, summing the period lengths or the pool factors alone, or weighting by total payments (duration) or by interest.
- **Red herrings.** Pool value in a DSCR question, remaining terms in a WAC question, write-offs in a delinquency question, an overcollateralisation balance in any of them.

Practice questions: CR-23 g

Question CR-23 g.1

Bank Q1224 • AP Mock 3 Q12 (rebuilt)

An ABS based on credit card receivables has the following composition:

Category	Amount (USD)
Current receivables	25,200,000
Receivables over 30 days past due	8,250,000
Receivables over 60 days past due	2,750,000
Receivables over 90 days past due	1,300,000

Assuming the buckets are mutually exclusive, which of the following is closest to the delinquency ratio of the ABS?

A. 3.47%

B. 3.59%

C. 5.16%

D. 7.33%

Question CR-23 g.2

Workbook A9.1 • S1 • Aug-2026 • Map #9

A junior surveillance analyst at an Indian asset manager is preparing the monthly investor report on a credit card master trust. The month-end ageing of receivables is:

Status	INR billion
Total credit card receivables (month end)	48.0
30 to 59 days past due	2.4
60 to 89 days past due	1.5
90 to 119 days past due	1.2
120 days or more past due	0.6
Receivables written off during the month	0.36

Which of the following is closest to the delinquency ratio as defined for credit card ABS performance analysis?

A. 0.75%

B. 2.50%

C. 3.75%

D. 11.88%

Question CR-23 g.3

Bank Q1234 • Mock 2 Q60 (rebuilt)

TRL Security is an asset-backed security supported by a pool of credit card receivables. The receivables pool is the sum of the four outstanding categories below (written-off receivables are excluded from the pool). Information on the receivables and the month's collections is:

Receivable category	Amount (USD)	Collections in the month	Amount (USD)
Current receivables	35,200,000	Principal payments	350,000
Over 30 days past due	4,600,000	Interest payments	860,000
Over 60 days past due	2,450,000		
Over 90 days past due	905,000		
Written off in the month	560,000		

Which of the following statements about TRL's key ratios is most accurate?

A. The monthly payment rate is 0.81%.

B. The default ratio is 1.30%.

C. The monthly payment rate is 3.44%.

D. The delinquency ratio is 18.43%.

Question CR-23 g.4

Workbook A9.2 • self-generated

A rating agency analyst in Sydney is testing a credit card ABS against its monthly payment rate trigger. The trust documents set a minimum MPR of 15.0% as an early amortisation trigger: if the month's MPR falls below it, the revolving period ends. During the month, the trust's data are:

- Outstanding pool balance: AUD 1,500 million
- Principal collections: AUD 210 million
- Finance charge (interest) collections: AUD 27 million
- Receivables written off: AUD 6 million
- New receivables purchased by the trust (revolving period): AUD 180 million

Which of the following correctly states the MPR and whether the early amortisation trigger is hit?

A. MPR 3.8%; trigger hit

B. MPR 14.0%; trigger hit

C. MPR 15.8%; trigger not hit

D. MPR 15.8%; trigger hit

Question CR-23 g.5

Workbook A9.3 • self-generated

A portfolio manager at a Canadian pension fund is reviewing an RMBS whose collateral pool consists of three loan groups. The notes pay investors a net coupon of 4.10%.

Loan group	Balance (CAD million)	Mortgage rate	Remaining term (months)
1	60	4.0%	300
2	90	5.2%	240
3	150	6.0%	120

Which of the following is closest to the pool's weighted average coupon (WAC)?

A. 1.26%

B. 4.80%

C. 5.07%

D. 5.36%

Question CR-23 g.6

Bank Q1229 • AnalystPrep Q.3091 (rebuilt)

A servicer's MBS portfolio comprises four pools of commercial mortgages:

Pool	Balance (USD million)	Maturity (days)
Superstore	26	82
Office space	67	169
Cinema	22	253
Restaurant	17	336

Which of the following is closest to the weighted average maturity (WAM) of the pools?

A. 165 days

B. 169 days

C. 187 days

D. 210 days

Question CR-23 g.7

Bank Q1232 • GARP PE2 Q32 • Mock 4 Q34

A financial analyst is calculating the weighted average life (WAL) of the loans in an MBS pool. The initial notional value outstanding of the MBS pool is USD 50 million. The payment schedule and the assumed pool factor for the MBS are provided in the table below:

Payment	Date	Actual days	Pool factor	Principal paid (USD million)
0	November 21, 2023	66	1.00	
1	January 26, 2024	181	0.76	12
2	July 26, 2024	184	0.54	11
3	January 26, 2025	183	0.38	8
4	July 25, 2025	182	0.26	6
5	January 24, 2026	183	0.18	4
6	July 25, 2026			9

Assuming coupon payments are made semi-annually and there are 365 days in a year, what is the WAL of the MBS pool?

A. 1.06 years

B. 1.24 years

C. 2.12 years

D. 2.68 years

Question CR-23 g.8

Bank Q1231 • BT P2.T6.612.1 (rebuilt)

A USD 1,000,000 mortgage loan fully amortises over ten years with level annual payments of USD 135,868 (rate 6.0%). Annual principal repayments are:

Year t	1	2	3	4	5
Principal repaid (USD)	75,868	80,420	85,245	90,360	95,782
Year t	6	7	8	9	10
Principal repaid (USD)	101,528	107,620	114,077	120,922	128,177

Which of the following is closest to the weighted average life (WAL) of the loan, in years?

A. 4.17

B. 5.02

C. 5.50

D. 5.98

Question CR-23 g.9

Bank Q1233 • GARP PE2 Q44 • Mock 4 Q45

A fixed-income portfolio manager is conducting a performance analysis on an agency-MBS and the underlying pool of commercial mortgages. The manager obtains the following information:

- The constant prepayment rate (CPR) of the commercial mortgage loan pool: 14.38%
- Debt payments per year on the commercial mortgage loans: USD 730,000
- Net operating income per year of the commercial mortgage property: USD 1,825,000

The manager uses the following metrics as benchmarks for the commercial mortgage sector analysis:

- Debt service coverage ratio (DSCR) of a benchmark obligor: 2.5
- Single monthly mortality (SMM) rate of a benchmark security: 0.85%

Which of the following statements is correct?

A. The prepayments on the commercial mortgage measured by the CPR show that the reduction in yield on the MBS will likely be lower than that on a benchmark security.

B. The prepayments on the commercial mortgage measured by the CPR show that the reduction in yield on the MBS will likely be higher than that on a benchmark security.

C. The estimated DSCR of the commercial mortgage implies that the borrower's ability to repay the mortgage loan is stronger than that of a comparable benchmark obligor.

D. The estimated DSCR of the commercial mortgage implies that the borrower's ability to repay the mortgage loan is weaker than that of a comparable benchmark obligor.

Solutions • CR-23 g

Solution CR-23 g.1

Answer: A

Bank Q1224 • AP Mock 3 Q12

Step 1 – Total receivables

The buckets are mutually exclusive, so the pool is their sum:

$$25.20 + 8.25 + 2.75 + 1.30 = \text{USD } 37.50 \text{ million}$$

$\Rightarrow$ The denominator is the whole pool, current and past due together.

Step 2 – Delinquency ratio

$$\frac{1.30}{37.50} = 3.47\%$$

$\Rightarrow$ 3.47% of the pool is overdue more than 90 days.

Interpretation

"Mutually exclusive" tells you that "over 30 days" means 30 to 59 days, so only the last bucket is delinquent and every bucket belongs in the total. Had the buckets been cumulative ("over 30 days" including the older ones), the 90-day figure would sit inside the others and the pool would not be their sum.

Why the other options are wrong

B is incorrect. Left the delinquent bucket out of the denominator: $1.30/36.20 = 3.59\%$.
C is incorrect. Divided by current receivables only: $1.30/25.20 = 5.16\%$.
D is incorrect. Used the 60-day bucket as the numerator: $2.75/37.50 = 7.33\%$.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. • **Reference:** Choudhry, Structured Credit Products 2e, Ch.12 (GARP CR Ch.23), 23.5 Performance Analysis for Credit Card ABS; Table 23.2

Solution CR-23 g.2

Answer: C

Workbook A9.1 • S1 • Aug-2026

Step 1 – Delinquent receivables (overdue more than 90 days)

$$1.2 + 0.6 = \text{INR } 1.8 \text{ billion}$$

$\Rightarrow$ Only the 90-plus buckets meet the book's definition of delinquent.

Step 2 – Delinquency ratio

$$\text{Delinquency ratio} = \frac{1.8}{48.0} = 3.75\%$$

$\Rightarrow$ 3.75% of the pool is seriously past due.

Interpretation

Because delinquencies precede write-offs, a rising delinquency ratio is the early warning that the default ratio and the early amortisation triggers may be hit in coming months (Figure 1).

Why the other options are wrong

A is incorrect. Sibling swap: used write-offs, which gives the default ratio $0.36/48.0 = 0.75\%$.

B is incorrect. Scope error: counted only the 90 to 119 day bucket, $1.2/48.0 = 2.50\%$, omitting receivables 120 days or more past due.
D is incorrect. Scope error: counted every past-due bucket, $(2.4 + 1.5 + 1.2 + 0.6)/48.0 = 11.88\%$.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. • **Reference:** Choudhry, Structured Credit Products 2e, Ch.12 (GARP CR Ch.23), Performance Analysis for Credit Card ABS

Solution CR-23 g.3

Answer: B

Bank Q1234 • Mock 2 Q60

Step 1 – The receivables pool

$$35{,}200{,}000 + 4{,}600{,}000 + 2{,}450{,}000 + 905{,}000 = \text{USD } 43{,}155{,}000$$

$\Rightarrow$ Written-off receivables have left the pool, so they are not in the denominator.

Step 2 – The three ratios

$$\text{Default} = \frac{560{,}000}{43{,}155{,}000} = 1.30\% \qquad \text{MPR} = \frac{350{,}000 + 860{,}000}{43{,}155{,}000} = 2.80\%$$

$$\text{Delinquency} = \frac{905{,}000}{43{,}155{,}000} = 2.10\%$$

$\Rightarrow$ Only the default ratio statement matches.

Interpretation

All three ratios share the pool as denominator; the numerators are write-offs, collections and the 90-plus bucket. (The mock frames TRL as an auto loan ABS; these ratios are GARP's credit card measures, and MPR applies to non-amortising pools, so the stem is rebuilt on a credit card pool. The numbers are unchanged.)

Why the other options are wrong

A is incorrect. Left interest out of the MPR: $350{,}000/43{,}155{,}000 = 0.81\%$. MPR counts principal and interest.
C is incorrect. Divided collections by current receivables only: $1{,}210{,}000/35{,}200{,}000 = 3.44\%$.
D is incorrect. Counted every past-due bucket as delinquent: $7{,}955{,}000/43{,}155{,}000 = 18.43\%$.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. • **Reference:** Choudhry, Structured Credit Products 2e, Ch.12 (GARP CR Ch.23), 23.5 Performance Analysis for Credit Card ABS; Table 23.2

Solution CR-23 g.4

Answer: C

Workbook A9.2 • self-generated

Step 1 – Collections (principal and interest)

$$210 + 27 = \text{AUD } 237 \text{ million}$$

$\Rightarrow$ MPR measures the proportion of the principal and interest on the pool repaid in the period.

Step 2 – MPR

$$\text{MPR} = \frac{\text{collections}}{\text{outstanding pool balance}} = \frac{237}{1{,}500} = 15.8\%$$

$\Rightarrow$ Cardholders repay 15.8% of the pool in the month.

Step 3 – Trigger test

$15.8\% \geq 15.0\%$ (minimum) $\Rightarrow$ trigger not hit.

$\Rightarrow$ The revolving period continues: principal collections keep buying new receivables.

Interpretation

A falling MPR signals slower repayment; the rating agencies require every non-amortising ABS to set a minimum MPR as an early amortisation trigger. Write-offs and new purchases do not enter the ratio (deducting write-offs would give 15.4%, still above the floor).

Why the other options are wrong

A is incorrect. Netted off the new receivables bought in the revolving period, $(237 - 180)/1{,}500 = 3.8\%$, which then falls below the floor.

B is incorrect. Dropped interest collections, $210/1{,}500 = 14.0\%$, which falls below the floor.

D is incorrect. Right MPR, polarity flip on the trigger: treated the 15.0% minimum as a ceiling, so an MPR above it was read as a breach.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. CR-23 f: Explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Performance Analysis for Credit Card ABS; Table 23.2

Solution CR-23 g.5

Answer: D

Workbook A9.3 • self-generated

Step 1 – Balance times rate for each group

$$60 \times 4.0\% = 2.40, \qquad 90 \times 5.2\% = 4.68, \qquad 150 \times 6.0\% = 9.00 \quad \text{(CAD million)}$$

Step 2 – Total balance

$$60 + 90 + 150 = \text{CAD } 300 \text{ million}$$

Step 3 – WAC

$$\text{WAC} = \frac{2.40 + 4.68 + 9.00}{300} = \frac{16.08}{300} = 5.36\%$$

$\Rightarrow$ The pool earns 5.36% on average, weighted by loan balance.

Interpretation

Against the 4.10% net investor coupon, the pool has 1.26 percentage points of coupon headroom, an indication of its ability to pay. WAC drifts as loans repay (here the high-rate group 3 has the shortest term). The remaining terms are irrelevant to WAC; they would give a WAM of 192 months.

Why the other options are wrong

A is incorrect. Intermediate result: reported WAC minus the net coupon, $5.36\% - 4.10\% = 1.26\%$.

B is incorrect. Wrong weights: weighted the rates by remaining term, $(300 \times 4.0 + 240 \times 5.2 + 120 \times 6.0)/660 = 4.80\%$.

C is incorrect. Wrong weights: simple average of the three rates, $(4.0 + 5.2 + 6.0)/3 = 5.07\%$.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Performance Analysis for MBS

Solution CR-23 g.6

Answer: C

Bank Q1229 • AnalystPrep Q.3091

Step 1 – Total balance

$$26 + 67 + 22 + 17 = \text{USD } 132 \text{ million}$$

Step 2 – Balance times maturity

$$26(82) + 67(169) + 22(253) + 17(336) = 2{,}132 + 11{,}323 + 5{,}566 + 5{,}712 = 24{,}733$$

Step 3 – WAM

$$\text{WAM} = \frac{24{,}733}{132} = 187.4 \text{ days}$$

$\Rightarrow$ The large office-space pool pulls the WAM towards its own 169 days, the long restaurant pool pulls it up.

Interpretation

WAM is a balance-weighted average, so a pool's influence depends on its size, not on its maturity. GARP quotes WAM in months for mortgage pools; the arithmetic is the same in days.

Why the other options are wrong

A is incorrect. Dropped the restaurant pool from both sums: $19{,}021/115 = 165.4$ days.
B is incorrect. Took the maturity of the largest pool, 169 days, instead of weighting all four.
D is incorrect. Simple average of the maturities: $(82 + 169 + 253 + 336)/4 = 210$ days.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Performance Analysis for MBS; SchweserNotes 2026 Book 2, Reading 41, LO 41.g

Solution CR-23 g.7

Answer: B

Bank Q1232 • GARP PE2 Q32

Step 1 – Length of each period in years

$$\frac{66}{365} = 0.1808,\ \frac{181}{365} = 0.4959,\ \frac{184}{365} = 0.5041,\ \frac{183}{365} = 0.5014,\ \frac{182}{365} = 0.4986,\ \frac{183}{365} = 0.5014$$

Step 2 – Weight each period by its pool factor

$$0.1808(1.00) + 0.4959(0.76) + 0.5041(0.54) + 0.5014(0.38) + 0.4986(0.26) + 0.5014(0.18)$$
$$= 0.1808 + 0.3769 + 0.2722 + 0.1905 + 0.1296 + 0.0903 = 1.2403$$

$\Rightarrow$ WAL ≈ 1.24 years.

Step 3 – Cross-check with the principal schedule

Repayment times are the cumulative period lengths: 0.1808, 0.6767, 1.1808, 1.6822, 2.1808 and 2.6822 years.

$$\frac{12(0.1808)+11(0.6767)+8(1.1808)+6(1.6822)+4(2.1808)+9(2.6822)}{50}=\frac{62.02}{50}=1.24$$

$\Rightarrow$ Both forms measure the area under the pool factor curve.

Interpretation

The pool's last payment is 2.68 years out, but half the principal is back within about a year, so the WAL is only 1.24 years. Use the day counts given: a few do not match the calendar dates exactly (26 January to 25 July 2025 is 180 days, not 183), which moves the WAL by less than 0.001 years.

Why the other options are wrong

A is incorrect. Dropped the 66-day stub period at a pool factor of 1.00: $1.2403-0.1808=1.06$.
C is incorrect. Summed the pool factors of rows 1 to 5, $0.76+0.54+0.38+0.26+0.18=2.12$, with no weighting by period length.
D is incorrect. Summed the period lengths, $979/365=2.68$, with no pool factor weighting: that is the final maturity.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Performance Analysis for MBS, Table 23.1

Solution CR-23 g.8

Answer: D

Bank Q1231 • BT P2.T6.612.1

Step 1 – Weight each repayment date by its share of principal

$$\text{WAL}=\sum_t t\,\frac{P_t}{1{,}000{,}000}=\frac{1(75{,}868)+2(80{,}420)+\cdots+10(128{,}177)}{1{,}000{,}000}=\frac{5{,}977{,}985}{1{,}000{,}000}=5.98$$

$\Rightarrow$ The principal repayments sum to USD 1,000,000, so the weights are fractions of the loan.

Step 2 – Same answer from pool factors

With annual periods, WAL $=\sum$ PF $\times$ 1 year: $1.000+0.924+0.844+0.758+0.668+0.572+0.471+0.363+0.249+0.128=5.98$.
$\Rightarrow$ GARP's $\sum(a/365)$ PF and the principal-weighted form agree.

Interpretation

Principal repayments of a level-payment loan grow over time, so the WAL sits above the 5.5-year midpoint. Duration (5.02) weights the whole payment by its present value, so it is shorter than the WAL.

Why the other options are wrong

A is incorrect. Weighted the dates by interest paid instead of principal: $\sum t\,I_t/\sum I_t=4.17$.
B is incorrect. Computed the Macaulay duration of the level payments at 6%: 5.02. WAL does not discount and counts principal only.
C is incorrect. Weighted the ten dates equally: $(1+2+\cdots+10)/10=5.50$, ignoring that principal repayments grow.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Performance Analysis for MBS, Table 23.1

Solution CR-23 g.9

Answer: B

Bank Q1233 • GARP PE2 Q44

Step 1 – DSCR against the benchmark

$$\text{DSCR} = \frac{1{,}825{,}000}{730{,}000} = 2.50$$

$\Rightarrow$ Exactly equal to the benchmark's 2.5: the borrower is neither stronger nor weaker. C and D both fail.

Step 2 – Put the benchmark prepayment rate on the same footing

$$\text{CPR}_{\text{bench}} = 1 - (1 - 0.0085)^{12} = 9.74\%$$

$\Rightarrow$ The pool's CPR of 14.38% is well above 9.74%: the pool prepays faster.

Step 3 – Prepayment and yield

GARP: prepayments, for a given price, reduce the yield on the MBS. Faster prepayment therefore means a larger reduction in yield than on the benchmark.

Interpretation

Both comparisons need a closed-book formula: NOI over debt payments for the DSCR, and the monthly-to-annual conversion for the SMM. Converting either way gives the same verdict: the benchmark's CPR is 9.74% against the pool's 14.38%, or the pool's SMM is 1.29% against the benchmark's 0.85%. Comparing a monthly SMM with an annual CPR directly is the unit error to avoid. The other trap is the DSCR that sits exactly on the benchmark.

Why the other options are wrong

A is incorrect. Reverses the comparison: a smaller yield reduction would need a CPR below the benchmark's 9.74%.

C is incorrect. The DSCR is 2.50, equal to the benchmark, not above it.

D is incorrect. The DSCR is 2.50, equal to the benchmark, not below it.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Performance Analysis for MBS; Example 23.2 Forecasting Prepayment Levels

CR-23 • An Introduction to Securitisation • Vardeez: IMPORTANT

CR-23 h

P1 CALC

Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate.

Workbook: A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A9): the recall map reports "CPR, SMM, PSA" and "prepayment rate (theory)", and lists CPR $= 1-(1-\text{SMM})^{12}$ among the formulas to know closed-book. **Vardeez:** SMM and CPR. His notes add the PSA multiple and "three places to lose a mark": annual versus monthly, the ramp cut-off at month 30, and what the multiple scales.

Reported exam shapes

- **A9·S5** CPR, SMM and PSA conversion, with the deal's overcollateralisation as the twist: the SMM is measured on the pool, not on the notes.
- **A9·S6** Prepayment rate theory and forecasting methodology: what PSA assumes, how prepayment moves yield and WAL.
- **A9·S7** Closed-book CPR (with DSCR and WAL, worked under CR-23 g).

What the questions look like: conversions between SMM, CPR and PSA at a stated pool age, often with a scheduled principal figure, a note balance or a reserve account that tests the denominator; then statement questions on the PSA convention and on what faster prepayment does to yield and WAL. The asset-class map for all CR-23 measures is in CR-23 g, Section 1.

1. Why prepayments have to be forecast

Mortgage borrowers can repay early, and in a pass-through every early repayment flows straight to investors. GARP: "Forecasting prepayments is crucial to computing the cash flows of MBS. Though the underlying payment remains unchanged, prepayments, for a given price, reduce the yield on the MBS." Faster prepayment also returns principal sooner, so it shortens the WAL (CR-23 g); this is **prepayment risk**. (Choudhry's yield statement is general. Strictly, it is an MBS bought above par whose yield falls when principal comes back early at par; follow GARP's line in the exam.)

- **Mortgages, home equity and student loans:** SMM, CPR and PSA (GARP Table 23.2). These are the two "commonly used" methods: the CPR approach and the PSA method.
- **Auto loans:** a car is an essential purchase and the loan is short (3 to 5 years), so "no real prepayment culture exists" and prepayment speed is extremely stable. Prepayments are tracked with the **absolute prepayment speed** (ABS speed), actual period prepayments as a proportion of the whole pool balance, which indicates the expected maturity of the notes and the value of their implicit call option.
- **Credit cards:** MPR, which is *not* a prepayment measure (CR-23 g).

2. SMM and CPR

The **single monthly mortality** (SMM) is the single-month proportional prepayment. GARP: "A SMM of 0.65% means that approximately 0.65% of the remaining mortgage balance at the beginning of the month, less the scheduled principal payment, will prepay that month."

$$\text{SMM} = \frac{\text{prepayment in the month}}{\text{balance at the start of the month} - \text{scheduled principal}}$$

The **constant** (or conditional) **prepayment rate** (CPR) is the same speed annualised. GARP: "The CPR is based on the characteristics of the pool and the current expected economic environment."

$$\text{CPR} = 1 - (1 - \text{SMM})^{12} \qquad \text{SMM} = 1 - (1 - \text{CPR})^{1/12}$$

- **Why a power, not a multiple.** Each month the SMM applies only to the balance that survived the previous months, so the balance left after a year is $(1 - \text{SMM})^{12}$ of the start. GARP's SMM of 0.65% gives CPR $= 1 - 0.9935^{12} = 7.53\%$, not $12 \times 0.65\% = 7.80\%$. Schweser: an SMM of 1.5% gives $1 - 0.985^{12} = 16.59\%$.
- **Prepayment in currency.** A pool of USD 400 million at the start of the month with USD 0.8 million of scheduled principal, prepaying at 6% CPR: SMM $= 1 - 0.94^{1/12} = 0.5143\%$, so the month's prepayment is $0.005143 \times 399.2 =$ USD 2.053 million. Dividing the CPR by 12 (0.5%) gives USD 1.996 million.
- **Which balance.** The SMM is a property of the *collateral pool*. In an overcollateralised deal the notes are smaller than the pool, so dividing by the note balance overstates the speed (Question CR-23 h.5).

3. The PSA benchmark

The PSA (now part of SIFMA) benchmark projects prepayments that **rise as a pool seasons**. A pool is at 100% PSA if its CPR starts at 0 and rises by 0.2% each month until it reaches 6% in month 30, then stays at 6%. Other scenarios are multiples of that path, reflecting the assumption that mortgages prepay more slowly during their first 30 months.

$$\text{CPR}_{100\%\,\text{PSA}}(m) = 0.2\% \times \min(m, 30) \qquad \text{PSA speed} = \frac{\text{CPR}}{0.2\% \times \min(m, 30)} \times 100$$

with m the number of months since origination. GARP prints PSA $= [\text{CPR}/(0.2\,m)] \times 100$; that is the ramp form, valid up to month 30. From month 30 on, the denominator is the 6% plateau.

Pool age (months)	5	10	20	30	60
CPR at 100% PSA	1.0%	2.0%	4.0%	6.0%	6.0%
CPR at 150% PSA	1.5%	3.0%	6.0%	9.0%	9.0%
CPR at 200% PSA	2.0%	4.0%	8.0%	12.0%	12.0%
SMM at 100% PSA	0.0837%	0.1682%	0.3396%	0.5143%	0.5143%
SMM at 200% PSA	0.1682%	0.3396%	0.6924%	1.0596%	1.0596%

Three worked moves

- **PSA to SMM on the ramp** (Vardeez). A pool 5 months old at 150 PSA: benchmark $5 \times 0.2\% = 1\%$, scaled $1.5 \times 1\% = 1.5\%$ CPR, SMM $= 1 - 0.985^{1/12} = 0.1259\%$. Check: $[1.5/(0.2 \times 5)] \times 100 = 150$.

- **PSA to SMM on the plateau.** A pool 60 months old at 200 PSA: $2 \times 6\% = 12\%$ CPR and SMM $= 1 - 0.88^{1/12} = 1.0596\%$. Month 40 and month 300 give the same answer.
- **CPR to PSA speed.** A 20-month-old pool prepaying at 6% CPR: benchmark $0.2\% \times 20 = 4\%$, so $6/4 \times 100 = 150\%$ PSA. Dividing by the 6% plateau instead would report 100% PSA.

The multiple scales the **CPR**, not the month and not the SMM. At month 30, 200 PSA has an SMM of 1.0596%, but 100 PSA has 0.5143%: 48.5% of it, not half, because the SMM is a concave function of the CPR.

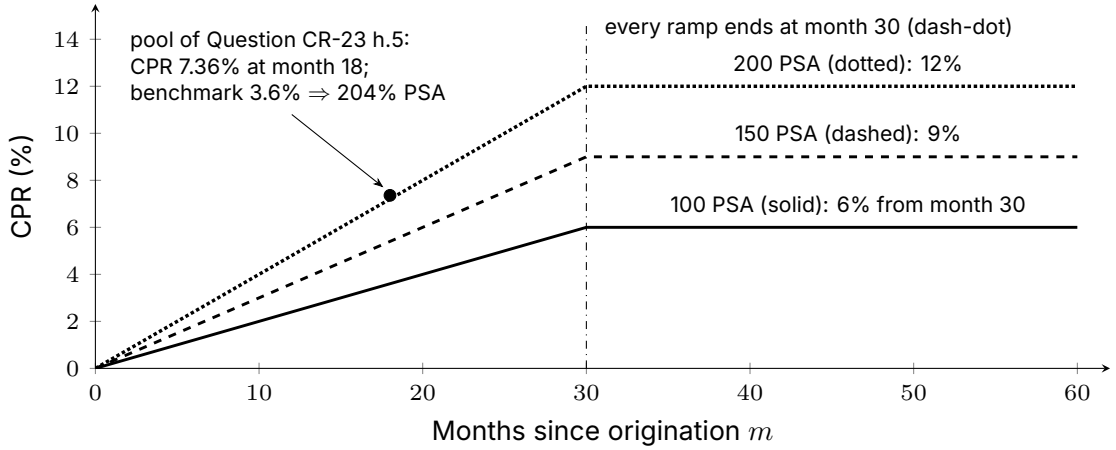


Figure 3. The PSA benchmark at three speeds. Every multiple has the same shape: a straight climb over the first 30 months and a flat line after. The multiple scales the height, not the length, of the ramp: 200 PSA does not reach its plateau sooner, it reaches a higher one. The marked pool prepays a little faster than 200 PSA at month 18.

4. The forecasting methods side by side

- **CPR approach.** One annual rate for the pool, set from the pool's characteristics and the expected economic environment, converted to a monthly SMM and applied to each month's balance net of scheduled principal.
- **PSA method.** A seasoning path rather than a single rate: the CPR is low for a new pool and rises to a plateau. A scenario is stated as a multiple of the benchmark (50, 100, 150 PSA), which gives an implied prepayment speed for any pool age. It applies to mortgages, home equity and student loans.
- **One assumption, three outputs.** The chosen speed drives the projected cash flows, the yield at a given price and the WAL. Amortising structures are priced on expected maturity and WAL, and "a WAL approach incorporates various pre-payment assumptions" (GARP 23.3); a change in speed changes all three together (Question CR-23 h.8).

Correction to the source notes. SchweserNotes (LO 41.h) says "the PSA typically assumes that prepayments will increase as a pool approaches maturity" and describes its PSA graph as the CPR "as it approaches maturity". GARP's benchmark runs on pool *age*: the CPR rises as the pool **seasons**, from 0 at origination to 6% in month 30, and then stays flat for the rest of the pool's life. m is months since origination, not months to maturity.

Formula card: CR-23 h

SMM	prepayment ÷ (start-of-month pool balance − scheduled principal)
SMM to CPR	$\text{CPR} = 1 - (1 - \text{SMM})^{12}$
CPR to SMM	$\text{SMM} = 1 - (1 - \text{CPR})^{1/12}$ (a twelfth root, not a division by 12)
Prepayment amount	SMM × (start balance − scheduled principal)
100% PSA	$\text{CPR} = 0.2\% \times \min(m, 30)$, m = months since origination
x% PSA	$\text{CPR} = \frac{x}{100} \times 0.2\% \times \min(m, 30)$
PSA speed	$\text{PSA} = \dfrac{\text{CPR}}{0.2\% \times \min(m, 30)} \times 100$
Auto loans	ABS speed = prepayments ÷ whole pool balance

	SMM	CPR	PSA speed	ABS speed
Period	One month	One year	A path over pool age	One period
What it is	Share of the net balance that prepays this month	SMM annualised by compounding	Multiple of the 100% PSA seasoning ramp	Prepayments over the whole pool balance
Changes with age	Yes, under PSA	Yes, under PSA	No: one number describes the whole path	No ramp
Used for	Mortgages, home equity, student loans	Same	Same	Auto and truck loans

Exam traps

- **Simple annualising.** $12 \times$ SMM for the CPR, CPR/12 for the SMM, or $(1 + \text{SMM})^{12} - 1$. The errors are small enough to look right and sit next to the answer.
- **Ramp or plateau.** Before month 30 the benchmark is $0.2\% \times m$; from month 30 it is 6%. Dividing an 18-month pool's CPR by 6%, or using $0.2\% \times 60 = 12\%$ for a 60-month pool at 100 PSA, are the two threshold errors.
- **Forgotten or doubled multiple.** Using the 100 PSA CPR when 140 PSA is stated, or applying the multiple twice.
- **The multiple scales the CPR,** not the month (150 PSA at month 5 is 1.5%, not the month-7.5 value) and not the SMM.
- **Wrong denominator.** The SMM is measured on the start-of-month *pool* balance less scheduled principal: not the full balance, not the note balance of an overcollateralised deal. Reserve accounts are irrelevant.
- **Units.** Comparing a monthly SMM with an annual CPR without converting one of them.
- **Polarity.** Faster prepayment, for a given price, reduces the yield (GARP) and shortens the WAL.

- **Age, not maturity.** The PSA ramp counts months since origination.
- **Asset-class swap.** Auto loans use ABS speed; credit cards use MPR, which is not a prepayment measure; DSCR is a CMBS credit measure.

Practice questions: CR-23 h

Question CR-23 h.1

Bank Q1235 • AnalystPrep Q.2703 (rebuilt)

A mortgage-backed security has a single monthly mortality (SMM) rate of 0.45%. Which of the following is closest to the constant prepayment rate (CPR)?

A. 0.45%

B. 5.27%

C. 5.40%

D. 5.54%

Question CR-23 h.2

Bank Q1236 • AnalystPrep Q.3048 (rebuilt)

An investor's USD 100,000 share of a seasoned mortgage pool has a scheduled principal payment of USD 28.61 this month. The pool's conditional prepayment rate (CPR) is 6%. Which of the following is closest to the single monthly mortality (SMM) rate and the estimated prepayment this month?

A. SMM 0.005000; prepayment USD 499.86

B. SMM 0.005000; prepayment USD 500.00

C. SMM 0.005143; prepayment USD 514.15

D. SMM 0.005143; prepayment USD 514.30

Question CR-23 h.3

Bank Q1244 • Mock 4 Q33 (rebuilt)

An analyst is modelling prepayments on a pool of 30-year residential mortgages. The pool is currently in month 20 of its life and prepayments follow 150% PSA. Under the standard PSA benchmark, the CPR rises linearly from 0% to 6% over the first 30 months and stays at 6% thereafter. Which of the following is closest to the single monthly mortality (SMM) rate in month 20?

A. 0.34%

B. 0.50%

C. 0.51%

D. 0.78%

Question CR-23 h.4

Bank Q1238 • AnalystPrep Q.3050

A risk analyst computes single monthly mortality (SMM) rates using the PSA benchmark. For month 22 at 140% PSA she obtains an SMM of 0.37%. For month 200 at 90% PSA she obtains an SMM of 0.46%. Which of the following statements is correct?

A. She is correct for both calculations.

B. She is correct for month 22, but incorrect for month 200.

C. She is incorrect for both calculations.

D. She is incorrect for month 22, but correct for month 200.

Question CR-23 h.5

Workbook A9.6 • S5, S7 • Map #9

A treasury analyst at a US life insurer is reviewing the monthly trustee report on a non-agency RMBS, 18 months after the pool's origination. The report shows:

- Collateral pool balance at the start of the month: USD 820.0 million
- Notes outstanding at the start of the month: USD 779.0 million (the deal is overcollateralised)
- Overcollateralisation reserve account: USD 6.5 million
- Scheduled principal received in the month: USD 1.4 million
- Unscheduled prepayments received in the month: USD 5.2 million

Which of the following is closest to the pool's CPR and its speed expressed against the PSA benchmark?

A. CPR 0.64%; 17.6% PSA

B. CPR 7.36%; 123% PSA

C. CPR 7.36%; 204% PSA

D. CPR 7.74%; 215% PSA

Question CR-23 h.6

Bank Q1239 • BT P2.T5.107.1 (rebuilt)

A mortgage pool has a USD 800 million balance and a weighted average coupon (WAC) of 7.2%. A sequential-pay CMO splits it into four tranches, each paying a 6.0% annual coupon:

Tranche	A	B	C	D
Par (USD million)	300	200	175	125

All principal, scheduled and prepaid, goes to Tranche A until it is retired. In month 1 the pool's total scheduled payment (interest plus scheduled principal) is USD 5.430 million, and prepayments run at 200% PSA. Which of the following is closest to the total cash flow to Tranche A in month 1?

A. USD 2.13 million

B. USD 2.26 million

C. USD 2.40 million

D. USD 5.33 million

Question CR-23 h.7

Bank Q1242 • BT P2.T5.104.3

With respect to prepayment measures, each of the following statements is true EXCEPT:

A. PSA assumes lower prepayment rates for young mortgage pools.

B. 100% PSA assumes a CPR of 0.2% for the first month, increasing by 0.2% per month for the next 29 months, when it reaches 6% per year, and remains at 6% per year thereafter.

C. At month 30, the remaining principal of a mortgage pool will be lower under a 150% PSA assumption than under 100% PSA.

D. The single monthly mortality rate (SMM) under 100% PSA is exactly 50% of the SMM under 200% PSA.

Question CR-23 h.8

Workbook A9.7 • S6, S7 • Map #9

An internal auditor at a Singapore life insurer is reviewing the investment team's methodology note on how prepayments are forecast for the insurer's MBS holdings. Which of the following statements in the note is correct?

A. The WAL of a pass-through note is quoted under a stated prepayment assumption, and a faster assumed PSA speed shortens it.

B. For a given price, faster prepayments increase the yield on the MBS, because principal is returned to investors earlier.

C. For commercial MBS, prepayment speed is tracked with the DSCR, the ratio of net operating income to debt payments.

D. A pool at 100% PSA prepays at a constant 6% CPR from origination, and faster speeds are multiples of that rate.

Solutions • CR-23 h

Solution CR-23 h.1

Answer: B

Bank Q1235 • AnalystPrep Q.2703

Step 1 – Monthly survival

$$(1 - 0.0045)^{12} = 0.9955^{12} = 0.94732$$

$\Rightarrow$ 94.73% of the balance survives a year of prepayments at this monthly speed.

Step 2 – CPR

$$\text{CPR} = 1 - 0.94732 = 5.27\%$$

Interpretation

Compounding the survival rate gives slightly less than twelve times the SMM, because each month's prepayment applies to a balance already reduced by earlier prepayments.

Why the other options are wrong

A is incorrect. Reported the monthly SMM itself as the annual rate.
C is incorrect. Simple annualising: $12 \times 0.45\% = 5.40\%$.
D is incorrect. Compounded the wrong way: $(1 + 0.0045)^{12} - 1 = 5.54\%$.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Structured Credit Products 2e, Ch.12 (GARP CR Ch.23), Example 23.2 Forecasting Prepayment Levels

Solution CR-23 h.2

Answer: C

Bank Q1236 • AnalystPrep Q.3048

Step 1 – De-compound the CPR

$$\text{SMM} = 1 - (1 - 0.06)^{1/12} = 0.005143$$

Step 2 – Balance exposed to prepayment

$$100{,}000 - 28.61 = 99{,}971.39$$

$\Rightarrow$ The SMM applies to the start-of-month balance less the scheduled principal.

Step 3 – Prepayment

$$0.005143 \times 99{,}971.39 = \text{USD } 514.15$$

Interpretation

Both refinements are small, which is why the four options are close: the twelfth root raises the SMM from 0.5000% to 0.5143%, and netting the scheduled principal trims the prepayment by about 15 cents.

Why the other options are wrong

A is incorrect. Used $\text{CPR}/12 = 0.005$ as the SMM: $0.005 \times 99{,}971.39 = 499.86$.
B is incorrect. Used $\text{CPR}/12$ and also the full USD 100,000: 500.00.
D is incorrect. Right SMM, but applied it to the full USD 100,000 without netting the scheduled principal: 514.30.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Example 23.2 Forecasting Prepayment Levels

Solution CR-23 h.3

Answer: C

Bank Q1244 • Mock 4 Q33

Step 1 – Benchmark CPR at month 20

$$0.2\% \times 20 = 4.0\%$$

$\Rightarrow$ Month 20 is still on the ramp, so the 6% plateau does not apply.

Step 2 – Scale to 150% PSA

$$1.5 \times 4.0\% = 6.0\%$$

Step 3 – Convert to a monthly rate

$$\text{SMM} = 1 - (1 - 0.06)^{1/12} = 0.5143\% \approx 0.51\%$$

Interpretation

At month 20, 150 PSA happens to give the same 6% CPR as the 100 PSA plateau. The match is a coincidence of the numbers, not a rule.

Why the other options are wrong

A is incorrect. Forgot the 150% multiplier: 4% CPR gives $1 - 0.96^{1/12} = 0.34\%$.
B is incorrect. Divided the 6% CPR by 12: 0.50%.
D is incorrect. Applied 150% to the 6% plateau although the pool is only 20 months old: 9% CPR gives $1 - 0.91^{1/12} = 0.78\%$.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Example 23.2 Forecasting Prepayment Levels

Solution CR-23 h.4

Answer: D

Bank Q1238 • AnalystPrep Q.3050

Step 1 – Month 22, 140% PSA (on the ramp)

Benchmark $0.2\% \times 22 = 4.4\%$; at 140%: $1.4 \times 4.4\% = 6.16\%$ CPR.

$$\text{SMM}_{22} = 1 - (1 - 0.0616)^{1/12} = 0.528\%$$

$\Rightarrow$ Not 0.37%. Her 0.37% is the SMM for 100% PSA (4.4% CPR): she dropped the multiplier.

Step 2 – Month 200, 90% PSA (on the plateau)

Benchmark 6%; at 90%: $0.9 \times 6\% = 5.4\%$ CPR.

$$\text{SMM}_{200} = 1 - (1 - 0.054)^{1/12} = 0.462\% \approx 0.46\%$$

$\Rightarrow$ Correct.

Interpretation

The two months test the two halves of the benchmark: the ramp before month 30 and the plateau after it. Each needs the multiple applied to the right benchmark CPR before the twelfth root is taken.

Why the other options are wrong

A is incorrect. The month-22 figure is wrong: 0.53%, not 0.37%.
B is incorrect. Reverses the verdicts: month 22 is wrong and month 200 is right.
C is incorrect. The month-200 figure is right: 90% of the 6% plateau gives 5.4% CPR and a 0.46% SMM.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Example 23.2 Forecasting Prepayment Levels

Solution CR-23 h.5 Answer: C

Workbook A9.6 • S5, S7

Step 1 – SMM

$$\text{SMM} = \frac{\text{prepayment}}{\text{pool balance} - \text{scheduled principal}} = \frac{5.2}{820.0 - 1.4} = \frac{5.2}{818.6} = 0.6352\%$$

$\Rightarrow$ 0.635% of the pool's start-of-month balance, net of scheduled principal, prepaid this month.

Step 2 – CPR

$$\text{CPR} = 1 - (1 - 0.006352)^{12} = 7.36\%$$

$\Rightarrow$ At this pace roughly 7.36% of the pool would prepay over a year.

Step 3 – 100% PSA benchmark at month 18

$m = 18 \leq 30 \;\Rightarrow\; \text{CPR}_{100\%\,\text{PSA}} = 0.2\% \times 18 = 3.6\%$
$\Rightarrow$ The pool is still on the seasoning ramp, so the 6% plateau does not apply yet.

Step 4 – PSA speed

$$\text{PSA} = \frac{7.36\%}{3.6\%} \times 100 = 204\%\ \text{PSA}$$

Interpretation

The pool is prepaying about twice as fast as the benchmark (Figure 3; 204.5% on the unrounded CPR). For a given price this reduces the yield on the MBS and shortens the WAL, so the insurer should re-run its cash flow projections at the faster speed. The reserve account balance is irrelevant to the prepayment measures.

Why the other options are wrong

A is incorrect. Unit error: used the monthly SMM as if it were the annual CPR, $0.6352/3.6 \times 100 = 17.6\%$ PSA.
B is incorrect. Threshold error: compared with the 6% plateau although month 18 is still on the ramp, $7.36/6 \times 100 = 123\%$ PSA.
D is incorrect. Wrong input twin (the overcollateralisation twist): divided by the note balance instead of the pool balance, SMM $= 5.2/(779.0 - 1.4) = 0.6687\%$, CPR 7.74%, 215% PSA.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Example 23.2 Forecasting Prepayment Levels; Table 23.2

Solution CR-23 h.6 — Answer: C

Bank Q1239 • BT P2.T5.107.1

Step 1 – Tranche A interest

$$\frac{0.06}{12} \times 300 = \text{USD } 1.500 \text{ million}$$

Step 2 – Scheduled principal from the pool

Pool interest $= \frac{0.072}{12} \times 800 = 4.800$, so scheduled principal $= 5.430 - 4.800 = \text{USD } 0.630$ million.

Step 3 – Prepayment at 200% PSA in month 1

CPR $= 2 \times 0.2\% \times 1 = 0.4\%$; SMM $= 1 - (1 - 0.004)^{1/12} = 0.03339\%$.

$$\text{Prepayment} = 0.0003339 \times (800 - 0.630) = \text{USD } 0.267 \text{ million}$$

$\Rightarrow$ A brand-new pool prepays very little, even at twice the benchmark speed.

Step 4 – Cash flow to Tranche A

$$1.500 + 0.630 + 0.267 = \text{USD } 2.397 \text{ million} \approx 2.40$$

Interpretation

Under sequential pay, Tranche A takes its own coupon plus *all* the pool's principal, scheduled and prepaid; Tranches B to D receive interest only. The pool's 7.2% WAC is used only to split the scheduled payment into interest and principal.

Why the other options are wrong

A is incorrect. Left out prepayments: $1.500 + 0.630 = 2.13$.
B is incorrect. Used 100% PSA (0.2% CPR, SMM 0.01668%): prepayment 0.133, total 2.26.
D is incorrect. Treated the 0.4% CPR as a monthly rate on the full pool: $0.004 \times 800 = 3.2$, total 5.33.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Example 23.2 Forecasting Prepayment Levels; CMO sequential pay: Malz, Ch.9 (GARP CR Ch.22), 22.1

Solution CR-23 h.7 — Answer: D

Bank Q1242 • BT P2.T5.104.3

Step 1 – The benchmark statements

A and B restate the 100% PSA convention: CPR 0.2% in month 1, up 0.2% a month to 6% at month 30, flat after. Young pools prepay slowly.
$\Rightarrow$ Both true.

Step 2 – Remaining principal at month 30

150% PSA has a higher CPR in every month up to month 30, so more principal has prepaid and the balance is lower.
$\Rightarrow$ C is true.

Step 3 – SMM under 100% and 200% PSA after month 30

$$1 - 0.94^{1/12} = 0.5143\% \qquad 1 - 0.88^{1/12} = 1.0596\% \qquad \frac{0.5143}{1.0596} = 48.5\%$$

$\Rightarrow$ Not exactly 50%: D is false.

Interpretation

The PSA multiple is applied to the CPR, and the SMM is a non-linear (concave) function of the CPR, so doubling the speed slightly more than doubles the SMM.

Why the other options are wrong

A is incorrect. True: the seasoning ramp gives young pools low prepayment rates.
B is incorrect. True: this is the definition of 100% PSA.
C is incorrect. True: a faster speed means more cumulative prepayment and a lower balance by month 30.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Example 23.2 Forecasting Prepayment Levels

Solution CR-23 h.8 Answer: A

Workbook A9.7 • S6, S7

Step 1 – Why A is correct

Amortising (pass-through) structures are priced and traded on expected maturity and WAL, and "a WAL approach incorporates various pre-payment assumptions"; Malz adds that "a WAL is associated with a particular prepayment assumption". With WAL $= \sum(a/365)\,\mathrm{PF}(s)$, faster prepayment cuts the pool factor sooner, so the WAL shortens.

Interpretation

Prepayment forecasts (SMM, CPR, PSA) drive the projected cash flows, the yield and the WAL together; an audit should check that all three are run off the same prepayment assumption.

Why the other options are wrong

B is incorrect. Polarity flip: Choudhry states that prepayments, for a given price, reduce the yield on the MBS.
C is incorrect. Sibling swap: DSCR (net operating income / debt payments) is the CMBS credit measure of the borrower's ability to repay. Prepayment is measured by SMM, CPR and PSA for mortgages (Table 23.2).
D is incorrect. Scope error: dropped the seasoning ramp. 100% PSA starts at 0, rises 0.2% a month and reaches 6% only at month 30, because mortgages prepay slower in their first 30 months.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. CR-23 f: Explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to. CR-22 k: Describe risk factors that impact structured products. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), Amortising Structures, Example 23.2, Table 23.2; Malz, Ch.9 (GARP CR Ch.22), Capital Structure and Credit Losses in a Securitization

CREDIT RISK • PART

P2

Second priority

Tier B topics, or LOs Vardeez flags inside a Very Important chapter. 37 LOs from 11 readings.

CR-4 • Capital Structure in Banks • Vardeez: VERY IMPORTANT

CR-4 c

P2 CALC

Define and calculate expected loss (EL).

Workbook: A1 cross-reference only: A1.1 (worked as Question CR-4 d.3) uses EL in its interpretation

WHAT IS TESTED HERE

Why P2. No workbook section homes this LO, but Vardeez flags it inside a Very Important reading (map focus: VARDEEZ FOCUS). EL is also the first link in the chain EL, UL, UL contribution, economic capital that the P1 blocks CR-4 d to g build on.
Vardeez: "Important reading: unexpected loss calculations as well as risk contributions can show up"; EL is the input they start from. No separate pointer for this LO.

Reported exam shapes: none listed in the map.

What the questions look like: $\text{EL} = \text{EA} \times \text{PD} \times \text{LR}$ for one loan or a small portfolio, with the usual planted inputs: a recovery rate where the loss rate belongs, a credit line's full commitment where the drawn amount belongs, and a joint default probability or a default correlation that has no effect on EL. Concept questions test that EL is a priced, reserved cost and not a risk.

Background: the full chain from EL to economic capital is in CR-4 d (P1), Section 1.

1. What expected loss is

A bank expects to lose, on average, a certain amount of money over a set period (usually one year) when it extends credit. That actuarial-type average is the **expected loss**:

- **A long-run average.** Losses fluctuate from year to year, but there is an average level over time that can be determined statistically. GARP stresses that EL "is not the level of losses predicted for the following year based on the economic cycle, but rather the long-run average loss level across a range of typical economic conditions".
- **A cost, not a risk.** EL can be viewed as payments to an insurance pool, is calculated from the bottom up (transaction by transaction), and is the foreseeable cost of doing business. It is covered by loan loss reserves ("standard risk costs") and reimbursed through adequate loan pricing. "By definition, EL does not itself constitute risk."
- **Not the whole story.** EL describes the average behaviour of a risky asset but not the variation around it. That variation, from unexpected default or unexpected credit migration, is the unexpected loss (CR-4 d), and it is what economic capital covers (CR-4 g).

2. Deriving the formula

Default is a Bernoulli event: with probability PD_H the borrower defaults before the horizon H and the bank recovers only $\mathrm{EA}_H(1-\mathrm{LR}_H)$; otherwise it receives the full promised EA_H. EL is the promised amount minus the amount the bank can expect to receive (GARP eq. 4.1):

$$\begin{aligned}\mathrm{EL}_H &= \mathrm{EA}_H - \mathrm{E}(\mathrm{EA}_H) = \mathrm{EA}_H - \left[(1-\mathrm{PD}_H)\,\mathrm{EA}_H + \mathrm{PD}_H\,\mathrm{EA}_H(1-\mathrm{LR}_H)\right] \\ &= \boxed{\mathrm{PD}_H \times \mathrm{EA}_H \times \mathrm{LR}_H}\end{aligned}$$

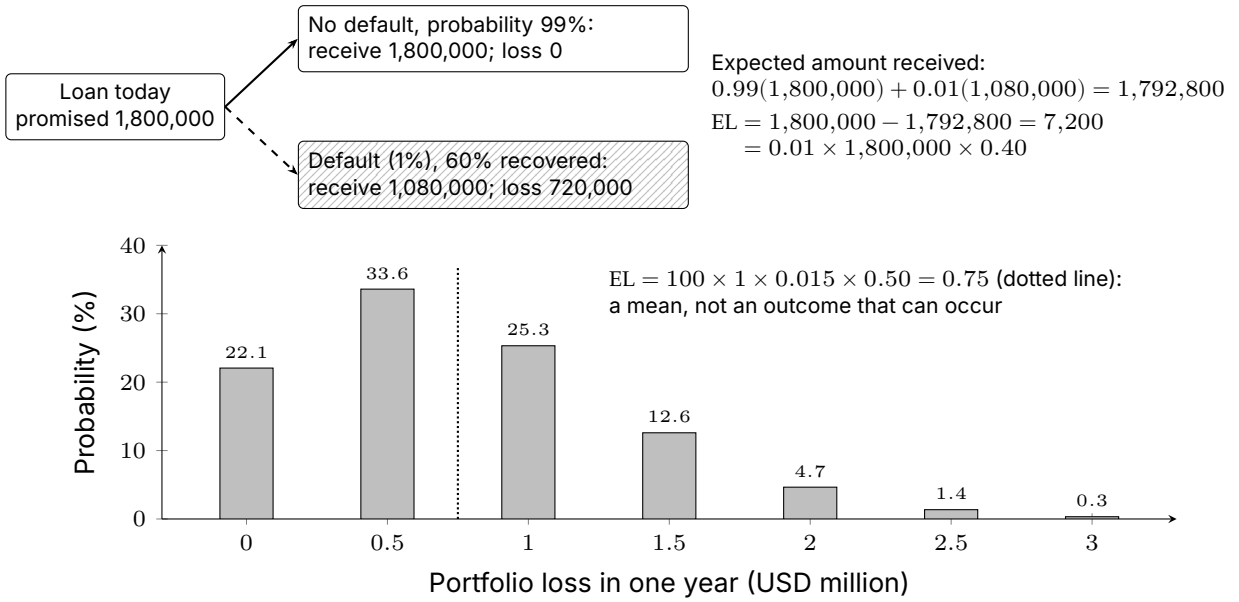


Figure 1. Top: GARP's derivation of EL (Figure 4.1) with the XYZ Bank loan. The solid branch is the no-default outcome, the dashed and hatched one the default outcome; EL is the promised amount minus the probability-weighted amount received. Bottom: 100 independent loans of USD 1 million, PD 1.5%, LR 50%: each default costs USD 0.5 million. EL is the mean of the loss distribution (USD 0.75 million), yet no single year can lose exactly that, and the distribution is skewed to the right.

3. The three components

- **Probability of default (PD)**, also expected default frequency (EDF): the probability that the borrower defaults before the end of the horizon (typically one year) or before maturity. It is borrower-specific, linked to the rating and estimated independently of collateral. PDs change over time: the second-year PD of an initially low-PD credit is typically higher than the first-year PD, which migration (transition) matrices capture as cumulative and marginal PDs.
- **Exposure amount (EA)**, also exposure at default (EAD): the expected exposure at the time of default, including all outstanding payments and interest. It can differ greatly from the amount at origination: amortisation schedules and credit lines (limit against utilisation) move it, and for derivatives such as swaps it is hard to quantify and may need Monte Carlo simulation. In questions it is the outstanding (drawn) balance, plus any stated drawdown of the unused line, not the full commitment.
- **Loss rate (LR)**, also severity, loss given default (LGD) or loss in the event of default (LIED): actual losses at default, including the costs of collecting and selling collateral, divided by EA. It is largely a function of collateral (unsecured loans lose far more than secured ones), and $\mathrm{LR} = 1 - \mathrm{RR}$.

For an individual loan PD is independent of EA and LR (default is binary), and EA and LR are usually treated as independent, so the expectation of the product is the product of the three. Country (transfer) risk has the same three components, with an adjustment for overlap because the bank can only lose its money once.

4. Calculations

One loan. XYZ Bank: commitment \$2,000,000, outstanding \$1,800,000, PD 1%, LR 40%:

$$\text{EL} = 1{,}800{,}000 \times 0.01 \times 0.40 = \$7{,}200$$

The commitment plays no part. Dividing by EA gives the **expected loss rate** $\text{PD} \times \text{LR} = 0.40\%$ of exposure.

A portfolio: EL simply adds (GARP eq. 4.4), whatever the correlation:

$$\text{EL}_P = \sum_i \text{EA}_i \times \text{PD}_i \times \text{LR}_i$$

Bigger Bank: asset A $8{,}250{,}000 \times 0.005 \times 0.50 = \$20{,}625$ plus asset B \$7,200 gives $\text{EL}_P = \$27{,}825$ at any correlation (CR-4 f). Joint default probabilities and default correlations change the *spread* of portfolio losses (UL), never their mean.

Recovery rates in the stem. Convert first: a bond with 70% recovery has $\text{LR} = 30\%$. A GBP 1,000,000 bond with a 4% PD then has $\text{EL} = 1{,}000{,}000 \times 0.04 \times 0.30 =$ GBP 12,000, not GBP 28,000.

Directions. EL rises with each of EA, PD and LR. A higher recovery rate lowers LR and so lowers EL; a lower PD lowers EL too, so the two changes reinforce each other (Question CR-4 c.4).

Vardeez's description of unexpected loss under this LO ("the difference between the actual loss and the expected loss", "EL + UL is the total potential loss") is corrected in the CR-4 d block: UL is the standard deviation of losses around EL.

Formula card: CR-4 c

Expected loss	$\text{EL} = \text{EA} \times \text{PD} \times \text{LR}$ (same as EAD × PD × LGD)
Derivation	$\text{EL} = \text{EA} - \left[(1 - \text{PD})\text{EA} + \text{PD} \cdot \text{EA}(1 - \text{LR})\right]$
Loss rate	$\text{LR} = 1 - \text{RR}$
Expected loss rate	$\text{EL}/\text{EA} = \text{PD} \times \text{LR}$
Credit line	$\text{EA} = \text{drawn} + \text{drawdown fraction} \times \text{undrawn}$ (not the full commitment)
Portfolio	$\text{EL}_P = \sum_i \text{EL}_i$ (correlation and joint default probabilities do not enter)

	PD	EA	LR
Meaning	Probability of default before the horizon	Expected exposure at default, with interest	Share of EA lost at default, net of recoveries and costs
Driven by	The borrower and its rating	The facility: drawn amount, amortisation, line usage	The facility: collateral and seniority
Other names	EDF	EAD	Severity, LGD, LIED; $1 - \text{RR}$
Typical trap	PD for the wrong horizon	The commitment instead of the drawn balance	The recovery rate used as LR

Exam traps

- **Recovery rate as the loss rate.** A 70% recovery is a 30% loss rate.
- **Commitment as exposure.** Use the outstanding balance plus any stated drawdown, not the facility limit.
- **Dropped factor.** EA $\times$ PD (no LR) or EA $\times$ LR (no PD); adding PD and LR instead of multiplying.
- **Correlation in EL.** Joint default probabilities and default correlations are red herrings for EL; they matter for UL.
- **One asset only.** Portfolio EL is the sum; the single-asset ELs are always among the options.
- **EL as risk.** EL is a priced, reserved cost; the risk is the variation around it.
- **EL as next year's forecast.** It is the long-run average across typical conditions, not a cyclical prediction.

Practice questions: CR-4 c

Question CR-4 c.1

Bank Q95 • AnalystPrep Q.3667 (rebuilt)

American International Bank has extended a loan to a corporate client. The credit analyst's note gives an exposure amount of USD 100 million, a loss rate of 10% and a probability of default of 20%. What is the expected loss on the loan?

A. USD 2 million

B. USD 10 million

C. USD 20 million

D. USD 30 million

Question CR-4 c.2

Bank Q101 • BT P2.T6.505.2

A credit risk trainer asks a class of new analysts about expected credit loss. Which of the following statements about expected loss is true?

A. Expected loss itself constitutes risk.

B. Expected loss represents credit loss levels that do not fluctuate from year to year.

C. Expected loss is calculated from the bottom up (transaction by transaction) and reimbursed through adequate loan pricing.

D. Expected loss is the risk that arises from the variation in loss levels, and it therefore justifies the rationale for economic capital.

Question CR-4 c.3

Bank Q102 • BT P2.T6.505.3

Expected loss (EL) has three components: probability of default (PD), exposure amount (EA) and loss rate (LR). Each of the following is true EXCEPT:

A. The exposure amount is the standard deviation of credit losses estimated at the end of the horizon, excluding outstanding interest payments.

B. The probability of default is the probability that a borrower will default before the end of a predetermined period of time (the estimation horizon typically chosen is one year).

C. The loss rate is the ratio of actual losses incurred at the time of default (including all costs associated with the collection and sale of collateral) to the exposure amount.

D. Expected loss equals the product of the probability of default up to time H (the horizon), the exposure amount at time H and the loss rate experienced at time H: $\text{EL}_H = \text{PD}_H \times \text{EA}_H \times \text{LR}_H$.

Question CR-4 c.4

Bank Q103 • Schweser Module Quiz 22.1 Q3 (rebuilt)

A bank's credit committee is told that, for one of its corporate loans, the expected recovery rate (RR) has increased and the probability of default (PD) has decreased. What is the effect on the loan's expected loss (EL), all else equal?

A. EL will increase, because a higher RR raises the loss rate.

B. EL will be indeterminate, because the two changes have offsetting effects.

C. EL will be unchanged, because EL depends only on the exposure amount.

D. EL will decrease.

Question CR-4 c.5

Bank Q104 • Schweser Module Quiz 22.1 Q4 (rebuilt)

Big Bank has a USD 20,000,000 credit facility with Upstart Corp., of which USD 18,000,000 is currently outstanding. Big Bank estimates a one-year probability of default of 2% and, because the collateral is unique to its industry with limited resale opportunities, an 80% loss rate. Which of the following is closest to the expected loss for Big Bank?

A. USD 72,000

B. USD 288,000

C. USD 320,000

D. USD 360,000

Question CR-4 c.6

Bank Q96 • Mock 1 Q17 (AP) (rebuilt)

A portfolio consists of two bonds, A and B. The probability that A and B default jointly is 1.45%, and their default correlation is 25%.

Bond	Value	Default probability	Recovery rate
A	GBP 1,000,000	4%	70%
B	GBP 800,000	6%	60%

Which of the following is the expected credit loss of the portfolio?

A. GBP 12,000

B. GBP 19,200

C. GBP 31,200

D. GBP 56,800

Solutions • CR-4 c

Solution CR-4 c.1 — Answer: A

Bank Q95 • AnalystPrep Q.3667

Step 1 – Multiply the three components

$$\text{EL} = \text{EA} \times \text{PD} \times \text{LR} = 100 \times 0.20 \times 0.10 = \text{USD } 2 \text{ million}$$

$\Rightarrow$ On average the bank expects to lose 2% of the exposure on this loan each year.

Interpretation

EL needs all three factors: the chance of default, the amount exposed and the share lost if default happens. Each wrong option drops or misuses one of them.

Why the other options are wrong

B is incorrect. Dropped PD: $100 \times 0.10 = 10$, the loss given default, not the expected loss.
C is incorrect. Dropped LR: $100 \times 0.20 = 20$, as if the whole exposure were lost on default.
D is incorrect. Added PD and LR instead of multiplying: $(0.20 + 0.10) \times 100 = 30$.

LO: CR-4 c: Define and calculate expected loss (EL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Expected Losses, eq. 4.1

Solution CR-4 c.2 — Answer: C

Bank Q101 • BT P2.T6.505.2

Step 1 – How EL is built

GARP: EL is "typically calculated from the bottom up, that is, transaction by transaction", "must be treated as the foreseeable cost of doing business" and "needs to be reflected in differentiated risk costs and reimbursed through adequate loan pricing".
$\Rightarrow$ C is true.

Step 2 – Test the others

Losses do fluctuate from year to year; EL is their long-run average. Risk is the variation around EL, which is unexpected loss.

Interpretation

The split is the core of the reading: EL is a cost (reserves and pricing), UL is the risk (economic capital).

Why the other options are wrong

A is incorrect. "By definition, EL does not itself constitute risk."
B is incorrect. Absolute qualifier: actual credit losses fluctuate with the cycle; EL is the anticipated average level over time.
D is incorrect. Describes unexpected loss, which is what justifies economic capital.

LO: CR-4 c: Define and calculate expected loss (EL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Expected Losses

Solution CR-4 c.3 — Answer: A

Bank Q102 • BT P2.T6.505.3

Step 1 – Check each definition against GARP

PD (B), LR (C) and the EL formula (D) are GARP's own wording.

Step 2 – Find the false one

EA is the expected amount of the bank's exposure at the time of default, "including all outstanding payments (including interest)". A standard deviation of credit losses is unexpected loss.
$\Rightarrow$ A is the exception on two counts: wrong concept (UL, not EA) and wrong scope (interest is included, not excluded).

Interpretation

EXCEPT questions on this LO usually swap a definition from the UL side into the EL side.

Why the other options are wrong

B is incorrect. True: the definition of PD, with a one-year horizon typical.
C is incorrect. True: LR includes the costs of collection and sale of collateral.
D is incorrect. True: GARP eq. 4.1.

LO: CR-4 c: Define and calculate expected loss (EL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Expected Losses, eq. 4.1

Solution CR-4 c.4 — Answer: D

Bank Q103 • Schweser Module Quiz 22.1 Q3

Step 1 – Recovery and loss rate

$\text{LR} = 1 - \text{RR}$, so a higher recovery rate lowers the loss rate.

Step 2 – Combine the two changes

$\text{EL} = \text{EA} \times \text{PD} \times \text{LR}$: both PD and LR fall, so EL falls.
$\Rightarrow$ The effects reinforce each other; EL decreases.

Interpretation

Both changes reduce a multiplicative factor of EL, so there is no ambiguity. A question becomes indeterminate only when one factor rises and another falls.

Why the other options are wrong

A is incorrect. Reversed link: a higher RR lowers LR.
B is incorrect. The effects are not offsetting; both reduce EL.
C is incorrect. EL depends on PD and LR as well as EA.

LO: CR-4 c: Define and calculate expected loss (EL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Expected Losses; SchweserNotes 2026 Book 2, Reading 22

Solution CR-4 c.5 — Answer: B

Bank Q104 • Schweser Module Quiz 22.1 Q4

Step 1 – Exposure

The exposure is the USD 18,000,000 outstanding, not the USD 20,000,000 facility.

Step 2 – Expected loss

$$\text{EL} = 18{,}000{,}000 \times 0.02 \times 0.80 = \text{USD } 288{,}000$$

Interpretation

The high loss rate reflects collateral with limited resale value: LR is largely a function of collateral. If the stem had given a drawdown on default for the unused USD 2 million, that part would be added to EA.

Why the other options are wrong

A is incorrect. Used 20% (the implied recovery rate) as the loss rate: $18{,}000{,}000 \times 0.02 \times 0.20 = 72{,}000$.

C is incorrect. Used the full facility: $20{,}000{,}000 \times 0.02 \times 0.80 = 320{,}000$.
D is incorrect. Omitted the loss rate: $18{,}000{,}000 \times 0.02 = 360{,}000$.

LO: CR-4 c: Define and calculate expected loss (EL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), Expected Losses, eq. 4.1; SchweserNotes 2026 Book 2, Reading 22

Solution CR-4 c.6 — Answer: C

Bank Q96 • Mock 1 Q17 (AP)

Step 1 – Loss rates

$\text{LR}_A = 1 - 0.70 = 30\%, \quad \text{LR}_B = 1 - 0.60 = 40\%.$

Step 2 – Expected loss of each bond

$$\text{EL}_A = 1{,}000{,}000 \times 0.04 \times 0.30 = 12{,}000 \qquad \text{EL}_B = 800{,}000 \times 0.06 \times 0.40 = 19{,}200$$

Step 3 – Portfolio EL

$$\text{EL}_P = 12{,}000 + 19{,}200 = \text{GBP } 31{,}200$$

Interpretation

EL is additive and independent of default correlation, so the 1.45% joint default probability and the 25% correlation are red herrings here. They would matter for the portfolio's unexpected loss (CR-4 f).

Why the other options are wrong

A is incorrect. Counted only bond A: 12,000.
B is incorrect. Counted only bond B: 19,200.
D is incorrect. Used the recovery rates as loss rates: $1{,}000{,}000 \times 0.04 \times 0.70 + 800{,}000 \times 0.06 \times 0.60 = 28{,}000 + 28{,}800 = 56{,}800$.

LO: CR-4 c: Define and calculate expected loss (EL). • **Reference:** Schroeck (2002), Ch.5 (GARP CR Ch.4), eqs. 4.1 and 4.4

CR-5 • Introduction to Credit Risk Modeling and Assessment • Vardeez: VERY IMPORTANT

CR-5 f

P2 CONCEPT

Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model.

Workbook: D20 Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a VERY IMPORTANT chapter ("LO 23.e & LO 23.f: Imp"). The workbook treatment is Tier D (D20, Map #41, 2/16, notes only), and the Merton-versus-KMV contrast also runs through Tier A section A5.

Reported exam shapes

- **D20·S1** CreditRisk+ as the actuarial, default-mode model.
- **D20·S4** Moody's KMV as the equity-market based implementation of Merton.
- **D20·S5** Four-way comparison on loss recognition and correlation modelling.

What the questions look like: conceptual. Four statements, three of them a true fact attached to the wrong model, or "which model meets these requirements". They turn on inputs, default point, the DD-to-PD mapping, default mode versus mark-to-market, the Poisson assumption and the stated criticisms.

Worked elsewhere: A5.4 (KMV default point and full-form DD) and A5.6 (DD bands, KMV mapping) under CR-5 e. The CreditRisk+ and CreditMetrics mechanics (Poisson and negative binomial, Monte Carlo rating migration, Gaussian copula) are CR-10 e.

1. Four models, one question each

CR-5 d split the financial models into structural and reduced form; CR-5 e built Merton's model. GARP 5.6.3 adds three named models, each answering a different question:

- **Merton**: what is this firm's PD, read from option pricing? Default point L = short-term debt; distance to default mapped to a PD through the **standard normal** distribution (CR-5 e).
- **Moody's KMV Expected Default Frequency (EDF)**: the best-known implementation of Merton. Asset value and volatility are backed out of **equity market data**. Two differences from Merton: DD is mapped to a PD through an **empirical distribution of historical default frequencies**, and the default point is **short-term debt plus half of long-term debt**, closer to the firm's actual obligations.
- **CreditMetrics** (JPMorgan, 1997): what is this loan worth next period, and how volatile is that value? Inputs: the issuer's rating, a **rating transition matrix**, recovery rates and bond-market yield spreads. A **mark-to-market** model: upgrades, downgrades and defaults all change the loan's value, over a horizon of usually one year (up to 10). It is "not a rating tool, nor a risk pricing model". Criticisms: **homogeneous** credit grades, and recovery values at **fixed interest rates**, fragile over long horizons.
- **CreditRisk+** (Credit Suisse Financial Products, 1997): how many defaults will the portfolio suffer? **Actuarial** approximations from insurance; it does **not** use the capital structure. Losses arise **only on default**; defaults are rare and independent, which suits the **Poisson** distribution (mean equals variance). No ratings, so **minimal data**.

Correlation (Hull, CR-10 e). CreditRisk+ defaults are independent *given* the default rate; an uncertain (gamma-distributed) rate turns the Poisson into a negative binomial, creating correlation and a fat tail (Hull Table 10.4: 4 expected defaults, 99.9th percentile 11 with a known rate, 98 with a standard deviation of 10). CreditMetrics correlates rating changes with a Gaussian copula.

	Merton	Moody's KMV	CreditMetrics	CreditRisk+
Family	Structural, option pricing	Structural, Merton implementation	Rating migration, mark-to-market	Actuarial, default mode
Main inputs	Equity value and volatility, debt, r	Equity market data, historical default database	Rating, transition matrix, recovery rates, yield spreads	Default rates; no capital structure
Default point	Short-term debt	Short-term $+\frac{1}{2}$ long-term debt	None (ratings)	None (default counts)
PD from	$N(-\text{DD})$, normal	Empirical EDF from DD	Transition probabilities	Poisson default frequency
Losses booked	Default at T	Default (EDF at the horizon)	Defaults *and* rating changes	Defaults only
Correlation	Single firm	Single firm in CR-5	Gaussian copula on rating changes	Uncertain default rate (negative binomial)
Data needs	Listed equity	Listed equity plus history	Heavy: ratings, matrix, spreads	Minimal
Stated weak spot	Normal mapping; short-term debt only	Corporate borrowers with traded equity only	Homogeneous grades; fixed interest rates	Ignores downgrades

Comparison of the four models (GARP Ch.5, 5.6.3; correlation rows from Hull, GARP Ch.10). The rows "Losses booked" and "Correlation" are the four-way comparison of shape S5.

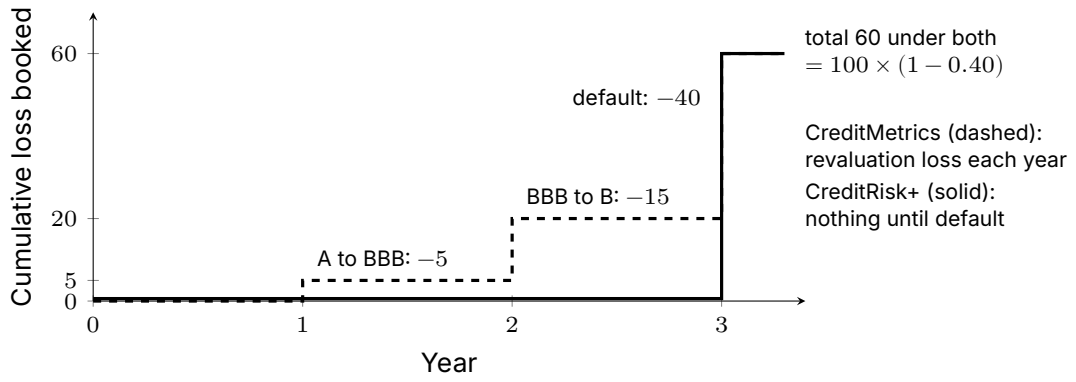


Figure 1. Same losses, different timing (Hull, GARP Ch.10, 10.4). A loan worth 100 is downgraded A to BBB in year 1, BBB to B in year 2, and defaults in year 3 with 40% recovery. CreditMetrics (dashed) books a revaluation loss at each step; CreditRisk+ (solid) books the whole loss at default. On the same assumptions the totals agree. The step sizes are illustrative.

Exam traps

- **Capital structure swap.** KMV (and Merton) use the firm's capital structure and equity data; CreditRisk+ explicitly does not.
- **Default mode versus mark-to-market.** CreditRisk+ books losses only on default; CreditMetrics books downgrades too (upgrades as gains). Statements that swap them are the commonest distractor.

- **KMV's two differences.** Empirical DD-to-EDF mapping and short-term plus half of long-term debt. "KMV uses the normal distribution", "total debt" or "book value of equity" are all wrong.
- **Ownership swaps.** Ratings and a transition matrix are CreditMetrics only; the Poisson default count is CreditRisk+ only, not KMV's mapping.
- **CreditMetrics criticisms.** Homogeneous grades and fixed interest rates; "too complex" is not one of GARP's.
- **Default-rate uncertainty.** It does change the tail: "uncertainty leaves the 99.9% loss unchanged because the mean is unchanged" is false.

Practice questions: CR-5 f

Question CR-5 f.1

Bank Q194 • AnalystPrep Q.5992 (rebuilt)

A credit analyst applies the Moody's KMV model to a listed corporation that has just issued new equity and restructured its capital. The analyst needs to back out the unobservable asset value and asset volatility to compute the distance to default. Which pair of inputs, observable in the market, is used for this purpose?

A. The book value of equity and the volatility of accounting earnings.

B. The borrower's credit rating and the rating transition matrix.

C. The market value of equity (market capitalization) and the volatility of equity.

D. The recovery rate and the credit spread for each rating grade.

Question CR-5 f.2

Bank Q192 • AP Mock 3 Q62

An analyst has already computed the distance to default for a firm and now uses the Moody's KMV (EDF) model to convert it into a probability of default. Which of the following best describes how KMV maps distance to default to a probability of default?

A. By applying the standard normal distribution to the distance to default.

B. By using a Poisson distribution based on the firm's past default history.

C. By deriving an empirical distribution of default frequencies from historical data.

D. By assigning a constant default probability regardless of the distance to default.

Question CR-5 f.3 Bank Q200 • Schweser Module Quiz 23.2 Q2 (rebuilt)

A credit risk trainer explains that the Merton model differs from the Moody's KMV Expected Default Frequency (EDF) approach in two key areas. Which of the following statements describes one of those differences?

A. The Merton model uses the market value of the borrower's equity, while Moody's KMV uses its book value.

B. The Merton model is based on an option-pricing framework, while Moody's KMV is based on a rating transition matrix.

C. The Merton model uses historical data to derive an empirical distribution of default frequencies, while Moody's KMV maps distance to default using the standard normal distribution.

D. Moody's KMV defines the default point as short-term debt plus half of long-term debt, while the Merton model uses short-term debt only.

Question CR-5 f.4 Bank Q196 • AnalystPrep Q.6006

An analyst at an asset manager is considering CreditRisk+ for a diverse corporate loan portfolio. Which feature of CreditRisk+ is most distinctive?

A. It monitors fluctuations in the equity market prices of the borrowers.

B. It treats default frequency as a continuous random variable described by a Poisson distribution.

C. It evaluates the historical performance and dividend yields of the borrowers.

D. It assesses the impact of changing global economic conditions on the portfolio.

Question CR-5 f.5 Bank Q195 • AnalystPrep Q.6003

A risk analyst is considering CreditMetrics for a client's long-term fixed-income portfolio with various maturities. Which of the following limitations of CreditMetrics should concern the analyst most in this context?

A. The complexity of the model, which requires extensive computational resources.

B. The exclusion of equity instruments from the model's credit risk assessment.

C. The use of fixed interest rates when calculating recovery rates.

D. The model's inability to incorporate economic forecasting into its analysis.

Question CR-5 f.6 Bank Q203 • AnalystPrep Q.6000 (rebuilt)

A bank holding both debt securities and loans is choosing between two portfolio credit models. Which of the following statements correctly distinguishes the CreditMetrics model from the CreditRisk+ model?

A. CreditMetrics is a mark-to-market model using rating transitions, whereas CreditRisk+ considers only default events and does not use credit ratings.

B. CreditRisk+ uses a rating transition matrix to revalue loans, whereas CreditMetrics considers only bankruptcy.

C. CreditMetrics assumes defaults follow a Poisson distribution, whereas CreditRisk+ revalues loans using bond-market spreads.

D. CreditRisk+ requires extensive data on the borrower's capital structure, whereas CreditMetrics requires only default counts.

Question CR-5 f.7 Bank Q199 • GARP PE2 Q71 • Mock 4 Q72

A risk manager at a bank is conducting an internal training program for newly hired credit risk analysts. The manager gives an overview of credit risk models, compares the bank's internal credit rating models to the rating agencies' models, and explains the assumptions, structure, and performance measures of these models as applied to counterparty default prediction. Assuming a stable market environment, which of the following statements would the manager be correct to make?

A. The Credit Risk Plus (CreditRisk+) model relies on the use of a company's capital structure and credit ratings to predict default of the company.

B. Unlike the Merton model that produces default predictions based on asset and liability values, the agencies' rating models produce default predictions from a substantially larger set of business factors.

C. Unlike the agencies' rating models, the KMV model is only used to compute the risk of pure debt contracts and is not applicable to portfolios that include derivatives contracts.

D. A good rating model is expected to provide accurate borrower default predictions without the need to be backtested.

Question CR-5 f.8 New • built from GARP Ch.5, 5.6.3 and Ch.10, 10.3 to 10.5 • shape D20·S5

The head of credit portfolio management at a European bank is summarising for the risk committee how four credit risk models recognise losses and treat dependence between defaults. Which of the following statements is correct?

A. CreditMetrics is a default-mode model: it books a loss only when a counterparty defaults, and default correlation enters through uncertainty about the overall default rate.

B. Moody's KMV revalues each loan at the horizon using rating-specific credit spreads, so it books losses from downgrades as well as from defaults.

C. CreditRisk+ uses each borrower's capital structure and asset volatility, so the correlation between defaults comes from the correlation between asset returns.

D. CreditRisk+ books losses only on default, and default correlation enters through uncertainty about the default rate; the greater that uncertainty, the fatter the tail of the loss distribution.

Solutions • CR-5 f

Solution CR-5 f.1

Answer: C

Bank Q194 • AnalystPrep Q.5992

Step 1 – Equity is a call on the assets

KMV, like Merton, links the observable equity to the unobservable assets through two equations: the equity value (a call on the assets) and the equity volatility ($\sigma_E E = N(d_1)\sigma_A A$).

Step 2 – What is observable

Market capitalisation and equity volatility, both from the stock market.
$\Rightarrow$ Two observables pin down the two unknowns, A and σ_A.

Interpretation

A capital restructuring changes both inputs at once, which is why a market-based model reacts immediately while an agency rating may not.

Why the other options are wrong

A is incorrect. Book values and accounting volatility are not the inputs of a market-based structural model.
B is incorrect. Ratings and a transition matrix are CreditMetrics inputs.
D is incorrect. Recovery rates and grade-specific spreads are CreditMetrics revaluation inputs.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1

Solution CR-5 f.2

Answer: C

Bank Q192 • AP Mock 3 Q62

Step 1 – Merton's mapping

Merton reads the PD off the normal distribution: $\text{PD} = N(-\text{DD})$.

Step 2 – KMV's replacement

Rather than resorting to the normal distribution, the EDF model uses historical data to derive an empirical distribution of default frequencies: firms with a given DD are grouped and their observed default frequency becomes the EDF.
$\Rightarrow$ An empirical, history-based mapping.

Interpretation

This is one of KMV's two differences from Merton; the other is the default point (short-term debt plus half of long-term debt).

Why the other options are wrong

A is incorrect. That is Merton's mapping, which KMV deliberately replaces.
B is incorrect. A Poisson distribution is the CreditRisk+ choice for default counts.
D is incorrect. A constant PD would make the distance to default irrelevant.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1

Solution CR-5 f.3 — Answer: D

Bank Q200 • Schweser Module Quiz 23.2 Q2

Step 1 – KMV's first difference

The default point is short-term debt plus half of long-term debt; Merton's L is the short-term debt.

Step 2 – KMV's second difference

KMV maps DD to a PD through an empirical distribution of historical default frequencies; Merton uses the standard normal.

$\Rightarrow$ D states the first difference correctly; C states the second one backwards.

Interpretation

Both models are option-pricing models on market equity data; they share the skeleton and differ only in the trigger and the mapping.

Why the other options are wrong

A is incorrect. Both use the market value of equity; book value plays no part.

B is incorrect. KMV is also an option-pricing (structural) model; the transition matrix belongs to CreditMetrics.

C is incorrect. Reversed: KMV uses the empirical distribution and Merton the normal.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.1; SchweserNotes 2026 Book 2, Reading 23, LO 23.f

Solution CR-5 f.4 — Answer: B

Bank Q196 • AnalystPrep Q.6006

Step 1 – What CreditRisk+ ignores

It does not use the capital structure or equity prices, and it does not calculate ratings.

Step 2 – What it models

Default frequency as a continuous random variable; defaults small-probability and independent, so the Poisson distribution (mean equals variance) fits.

$\Rightarrow$ B is the defining feature.

Interpretation

Because it needs only default rates, its advantage is the minimum data required.

Why the other options are wrong

A is incorrect. Equity-price monitoring belongs to market-based structural models such as KMV.

C is incorrect. Dividend yields and historical company performance are not CreditRisk+ inputs.

D is incorrect. The model is not built around macroeconomic scenarios.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.3

Solution CR-5 f.5 — Answer: C

Bank Q195 • AnalystPrep Q.6003

Step 1 – GARP's two criticisms of CreditMetrics

Credit grades are assumed homogeneous, and recovery values are calculated using fixed interest rates.

Step 2 – Which one bites for long maturities

Interest rates over long periods can be quite volatile, so a fixed-rate assumption becomes less defensible the longer the horizon.

⇒ For a long-term portfolio the fixed interest rate assumption matters most.

Interpretation

CreditMetrics usually runs at a one-year horizon but can stretch to ten years; the longer the horizon, the more the fixed-rate assumption matters.

Why the other options are wrong

A is incorrect. GARP describes CreditMetrics as promoting transparency; computational complexity is not a stated criticism.

B is incorrect. Equity exclusion is not a cited weakness; the model covers a wide range of credit products.

D is incorrect. Economic forecasting is not one of the criticisms GARP lists.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.2

Solution CR-5 f.6

Answer: A

Bank Q203 • AnalystPrep Q.6000

Step 1 – CreditMetrics

Inputs: rating, transition matrix, recovery rates, yield spreads. Output: a distribution of loan value changes, including upgrades and downgrades. A mark-to-market model.

Step 2 – CreditRisk+

Considers only default events, uses a Poisson distribution, ignores capital structure and ratings, needs little data.

⇒ Only A attaches each feature to the right model.

Interpretation

On the same assumptions the two should predict the same long-run losses; they differ in *when* losses are booked (Figure 1).

Why the other options are wrong

B is incorrect. Roles reversed: the transition matrix is CreditMetrics, the default-only focus CreditRisk+.

C is incorrect. Roles reversed: Poisson defaults are CreditRisk+, spread-based revaluation CreditMetrics.

D is incorrect. CreditRisk+ ignores capital structure and needs little data; CreditMetrics needs ratings, transitions and spreads.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.2 and 5.6.3.3

Solution CR-5 f.7

Answer: B

Bank Q199 • GARP PE2 Q71

Step 1 – What Merton uses

Only the market value of the assets, their volatility and the face value of the debt: asset and liability values.

Step 2 – What rating models use

Agency (and internal) rating models weigh a much broader set of business factors (industry, competitive position, country and others) alongside the financials.
⇒ B is a correct contrast.

Interpretation

The other statements fail on the model facts of this LO and on basic model governance: every rating model must be validated.

Why the other options are wrong

A is incorrect. CreditRisk+ does not use the capital structure and does not use ratings; it models default events only. The statement describes KMV.
C is incorrect. Nothing in the readings limits KMV to pure debt contracts; the claim is invented.
D is incorrect. Any rating model's accuracy must be confirmed by backtesting against realised defaults.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3

Solution CR-5 f.8 **Answer: D**

New • built from GARP Ch.5, 5.6.3 and Ch.10, 10.3 to 10.5

Step 1 – Loss recognition

CreditRisk+ (like Vasicek) estimates losses from defaults only; CreditMetrics counts defaults and downgrades, with upgrades as negative losses. Moody's KMV produces an EDF for a single firm; it does not revalue loans on rating spreads.

Step 2 – Dependence between defaults

In CreditRisk+, defaults are independent given the default rate; uncertainty about the rate (gamma-distributed, giving a negative binomial) creates the correlation. With 4 expected defaults, Hull's 99.9th percentile rises from 11 defaults with no uncertainty to 98 when the rate's standard deviation is 10. CreditMetrics instead correlates rating changes with a Gaussian copula calibrated to equity correlations.
⇒ D is the only statement that pairs the right loss recognition with the right correlation mechanism.

Interpretation

This is the four-way grid of shape S5: default mode versus mark-to-market on one axis, the source of correlation on the other (the comparison table in Section 1).

Why the other options are wrong

A is incorrect. Sibling swap: it gives CreditMetrics the default-mode recognition and the correlation mechanism of CreditRisk+.
B is incorrect. Sibling swap: rating-spread revaluation is CreditMetrics; KMV maps DD to an EDF.
C is incorrect. Sibling swap with the structural models: CreditRisk+ explicitly does not use the capital structure.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.6.3.2 and 5.6.3.3; Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.3 to 10.5, Table 10.4

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 a

P2 CONCEPT

Compare agencies' ratings to internal credit rating systems.

Workbook: none homed here • revision notes in A10 Altman Z-Score and Linear Discriminant Analysis

WHAT IS TESTED HERE

Why P2. Vardeez flags this LO (map: VARDEEZ FOCUS) inside CR-9, the chapter he calls the most important in Credit Risk. No workbook question is homed here; the A10 revision notes summarise it.

Vardeez: a two-column comparison. Agencies (Moody's, S&P, Fitch) rate large issuers from public information, aim for stability, take a long-term view and avoid reversals. Internal systems cover the small and mid-sized firms that agencies do not rate, are built on financial-statement ratios and a cash flow statement, and are used to estimate default probabilities directly.

Reported exam shapes: none listed in the map.

What the questions look like: "which statement correctly compares" items with no numbers. Wrong options flip the stability property (ratings "change with every piece of news"), make profit rather than cash the source of repayment, or give agencies coverage of small private firms. Equity-based estimates from Moody's KMV and Kamakura are the usual third source and the usual sibling-swap distractor.

Background elsewhere: default rates by rating (GARP Table 9.1) in CR-9 c and CR-9 e; the Z-score in CR-9 b; the quality of equity-based PDs in CR-9 n; point-in-time versus through-the-cycle systems in CR-6 b.

1. Agency ratings: built for stability

Rating agencies such as Moody's, S&P and Fitch publish ratings that describe the creditworthiness of corporate bonds. Strictly, a rating belongs to a bond issue, not to the company; because a company's bonds usually share one rating, it is normally quoted as the company's rating. One might expect ratings to move each time news reaches the market. In fact they change relatively infrequently, and on purpose:

- **Stability is an objective.** Agencies want to avoid *ratings reversals*, a downgrade followed by an upgrade a few weeks later. A rating changes only when there is reason to believe that a *long-term* change in creditworthiness has taken place.
- **The reason is the users.** Bond traders are major users of ratings and are often bound by rules on them (many bond funds may hold only investment-grade bonds). Frequent changes would force large amounts of trading, with high transaction costs, just to stay within the rules.
- **Rate through the cycle.** If a downturn raises the default probability over the next six months but makes very little difference over the next three to five years, the agency would not usually change the rating (Figure 1).
- **Coverage.** Agency ratings are usually available only for companies that have issued publicly traded debt.

An agency rating becomes a default probability through the history of its grade. The cumulative default rates of GARP Table 9.1 (S&P, 1981 to 2020) are averages over good and bad years: real-world PDs for a typical year, not for a recession. Default rates for all rated companies ranged from 0.15% in 1981 to 4.19% in 2009 (GARP Table 9.6), while the ratings themselves moved little.

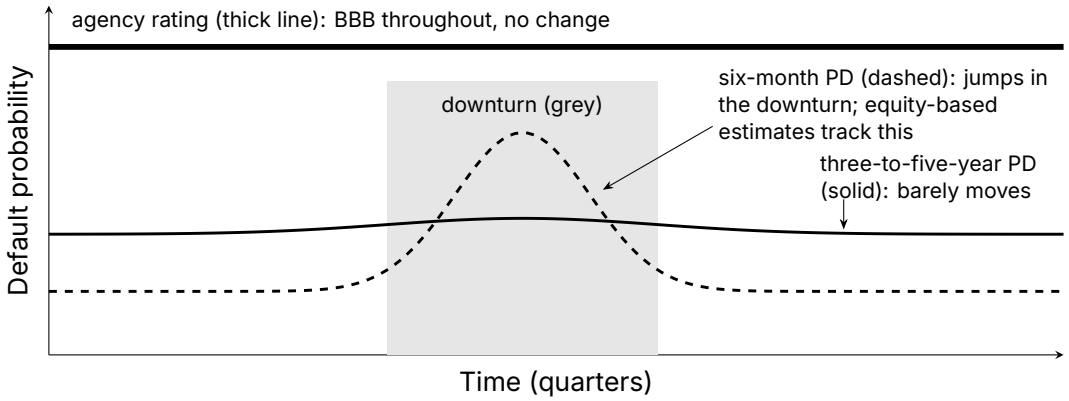


Figure 1. Rating through the cycle (schematic, no data). A downturn lifts the six-month default probability but leaves the three-to-five-year probability almost unchanged, so the agency keeps the rating. Equity-based estimates (Moody's KMV, Kamakura) have no stability objective and follow the short-horizon move.

2. Internal ratings: built to estimate default

Most banks rate the creditworthiness of their corporate and retail clients. GARP calls this a necessity: agency ratings exist mainly for issuers of public debt, so many small and medium-sized companies have none.

- **Inputs.** Profitability ratios (return on assets; Schweser adds return on equity), liquidity or balance-sheet ratios (the current ratio, current assets over current liabilities; Schweser uses the quick ratio) and leverage (debt to equity). Altman's Z-score (CR-9 b) is the classic ratio-based score.
- **Cash, not profit, repays a loan.** Banks take the financial information the company provides and convert it into a cash flow statement, to judge how easily the company can service its debt.
- **Output.** The lender's own grade and default probability for the borrower, used directly in the lending decision. Built from financial statements, it is backward looking and only as current as the latest accounts (CR-5 d).

3. The third source: equity-based estimates

Moody's KMV and Kamakura estimate a company's default probability from its equity price and other variables, using Merton-type models (CR-9 l to n). These estimates do not have stability as an objective and respond more quickly to market information than credit ratings. They need a traded share price, so they cover listed companies only.

	Agency ratings	Internal ratings	Equity-based (KMV, Kamakura)
Producer	Moody's, S&P, Fitch	The lending bank	Vendors, from Merton-type models
Coverage	Issuers of publicly traded debt	Any client, including unrated SMEs and retail borrowers	Companies with traded equity
Information	The issuer's public information	Profitability, liquidity and leverage ratios; a derived cash flow statement	Equity price and other market variables
Objective	Stability: rate through the cycle, avoid reversals	The borrower's PD and capacity to repay	A timely PD; stability is not an objective
Response to news	Slow: only on a long-term change	As current as the latest accounts	Fast
PD from it	The grade's historical default rate (Table 9.1), a long-run average	A direct estimate for the borrower	Merton output turned into a real-world PD (CR-9 n)

Exam traps

- **Stability flipped.** Agency ratings do not move with each piece of news; frequent change is exactly what agencies avoid.
- **Stable is not frozen.** A persistent, long-term change in creditworthiness does lead to a rating change. There is no rule of fixed annual reviews.
- **Coverage.** Agencies rate issuers of publicly traded debt. Most SMEs are unrated, which is why internal systems exist.
- **Profit versus cash.** Internal analysis converts statements into a cash flow statement because cash, not accounting profit, repays the loan.
- **Sibling swap.** The equity-price-based, fast-moving estimates are Moody's KMV and Kamakura, not agency ratings.
- **No share price, no equity-based PD.** A privately held SME cannot be run through an equity-based model; it needs an internal rating.
- **Long-run averages in a recession.** A grade's historical default rate averages good and bad years; in a downturn the realised rate of that grade is higher.

Practice questions: CR-9 a

Question CR-9 a.1

Bank Q314 • BT P2.T6.24.13.1 (rebuilt)

A bank lends to many small and mid-sized firms that have no publicly traded debt. Which of the following statements correctly compares agency credit ratings with the bank's internal credit rating system?

A. Agency ratings are designed to be stable and "rate through the cycle", whereas an internal system for small and mid-sized firms can be built on profitability, liquidity and leverage ratios and a cash flow statement derived from the borrower's financial statements.

B. Agency ratings change frequently so that they reflect each piece of new market information, whereas internal ratings are updated only at annual reviews.

C. Agencies routinely rate small and mid-sized private firms, so a bank has little need for an internal system for these borrowers.

D. Internal rating systems rely mainly on accounting profits, because profits rather than cash are what repay a loan, whereas agency ratings focus on cash flow.

Question CR-9 a.2

Bank Q316 • New question written for this bank

A regional bank is building an internal credit rating system for privately held manufacturing borrowers that are too small to be rated by any rating agency. Which of the following approaches is most consistent with how such systems are typically built?

A. Score each borrower on profitability, liquidity and solvency measures such as return on equity, the quick ratio and debt-to-equity, and adjust the borrower's financial statements to derive a cash flow statement.

B. Adopt the rating of the largest rated firm in the same industry, since agencies rate whole industries rather than individual issuers.

C. Use the equity-market-based default probability from a vendor model, because small private firms have market-traded shares.

D. Rely on accounting net income alone, because a loan is repaid out of reported profits rather than out of cash.

Question CR-9 a.3

Bank Q315 • New question written for this bank

A portfolio manager at a bond fund is told that her fund may hold only investment-grade bonds. She observes that a rating agency did not change an issuer's rating even though a sharp economic downturn raised the issuer's probability of default over the next six months, with little effect on its default probability over the next three to five years. Which of the following best explains the agency's behavior?

A. Agencies measure only the probability of default over the next six months, which the downturn left unchanged.

B. Agencies base ratings on the issuer's equity price, which does not respond to short-term economic conditions.

C. Agencies update ratings only at fixed annual reviews and ignore information released between reviews.

D. Agencies try to rate through the cycle, and stable ratings spare rule-bound bond investors from heavy trading and transaction costs caused by frequent rating changes.

Question CR-9 a.4

New • built from GARP Ch.9, 9.1

A credit analyst at a Canadian pension fund follows a BBB-rated industrial issuer. Over the past year two things happened. First, a two-month supply disruption raised the issuer's default probability over the next six months, with almost no effect on its three-to-five-year default probability. Later, the issuer lost its largest customer under a long-term contract, permanently reducing its cash flows. Based on how rating agencies set ratings, which response by the agency is most likely?

A. A downgrade after the disruption and an upgrade once it ends, so that the rating always reflects current information.

B. No change after the disruption; a downgrade after the loss of the customer, because that is a long-term change in creditworthiness.

C. No change after either event, because agencies change ratings only at fixed annual reviews.

D. A downgrade after the disruption and no change after the loss of the customer, because agencies weight short-term default risk most heavily.

Question CR-9 a.5

New • built from GARP Ch.9, 9.1 and 9.1.1

The CRO of a regional bank asks which sources of default risk can be used for three borrowers. Borrower X is a large listed company with publicly traded bonds. Borrower Y is a listed mid-sized company with no bonds outstanding; it borrows only from banks. Borrower Z is a privately held family manufacturer. Which statement is correct?

A. Agency ratings are available for all three, because agencies rate every company that borrows from banks.

B. An equity-based estimate is the natural tool for Z, because such models need no market prices.

C. An agency rating is usually available only for X. Y and Z need the bank's internal rating; an equity-based estimate (Moody's KMV or Kamakura) is also possible for Y, whose shares trade, but not for Z.

D. The bank should use its internal rating for X rather than the agency rating, because agency ratings are too stable to be informative for large companies.

Question CR-9 a.6

New • built from GARP Ch.9, 9.1 and 9.7.1 (Table 9.6)

A risk manager at a Singapore bank sets the default probability of each agency-rated borrower in a recession stress test equal to the long-run average one-year default rate of its rating in GARP Table 9.1 (S&P, 1981 to 2020). Annual default rates for all rated companies have ranged from 0.15% (1981) to 4.19% (2009). Which statement best assesses this approach?

A. It is appropriate: because ratings are stable, each rating grade has the same default rate in every year.

B. It overstates stress PDs: in a recession agencies quickly downgrade weak borrowers, so each grade keeps only its safer members.

C. It is appropriate: the 1981 to 2020 average includes 2009, the worst year, so it already reflects recession conditions.

D. It understates stress PDs: agencies rate through the cycle, so ratings move little in a recession while each grade's realised default rate rises; the long-run average blends good and bad years.

Solutions • CR-9 a

Solution CR-9 a.1

Answer: A

Bank Q314 • BT P2.T6.24.13.1

Step 1 – What the agency is trying to do

Agencies aim for stability: they rate through the cycle and change a rating only when they believe a long-term change in creditworthiness has occurred.
⇒ The first half of A is right, and B reverses it.

Step 2 – What the bank must build for unrated borrowers

Agency ratings exist mainly for issuers of publicly traded debt. For the rest, banks build internal systems on profitability, liquidity and leverage ratios, and convert the financial statements into a cash flow statement.
⇒ The second half of A is right as well.

Interpretation

The two systems serve different users: an agency serves investors who must not be forced to trade on noise; an internal system serves a lender that needs a default estimate for a borrower nobody else rates.

Why the other options are wrong

B is incorrect. Polarity flip: agencies deliberately avoid frequent changes and reversals, so ratings change relatively infrequently.
C is incorrect. Coverage error: small and mid-sized firms without public debt are generally unrated, which is exactly why banks need internal systems.
D is incorrect. Reversed logic: banks recognise that cash rather than profit repays a loan, which is why they build a cash flow statement.

LO: CR-9 a: Compare agencies' ratings to internal credit rating systems. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1 Credit Ratings; 9.1.1 Internal Credit Ratings

Solution CR-9 a.2

Answer: A

Bank Q316 • New question written for this bank

Step 1 – The inputs

Internal systems score profitability (return on assets or equity), liquidity (current or quick ratio) and leverage or solvency (debt to equity).

Step 2 – The cash flow adjustment

A loan is repaid with cash, so the bank adjusts the company-prepared statements into a cash flow statement to judge repayment capacity.
⇒ A describes both steps.

Interpretation

This is the same family of accounting-ratio scoring as Altman's Z-score (CR-9 b): the bank builds the measure itself because nobody else covers these borrowers.

Why the other options are wrong

B is incorrect. Agencies rate issues and issuers, not industries; a peer's rating says nothing reliable about this borrower.
C is incorrect. Privately held firms have no traded shares, so an equity-based vendor estimate cannot be produced.
D is incorrect. Cash, not reported profit, repays a loan; that is the point of the cash flow state-

ment.

LO: CR-9 a: Compare agencies' ratings to internal credit rating systems. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1.1 Internal Credit Ratings; SchweserNotes 2026 Book 2, Reading 27, LO 27.a

Solution CR-9 a.3 — Answer: D

Bank Q315 • New question written for this bank

Step 1 – Read the horizon in the stem

The downturn raised the six-month default probability but barely moved the three-to-five-year probability.

⇒ Nothing long-term has changed, so a through-the-cycle agency leaves the rating alone.

Step 2 – Why agencies behave this way

Bond traders are major users of ratings and are often bound by rating rules, such as this fund's investment-grade limit. Frequent changes would force heavy trading and transaction costs.

Interpretation

The fund manager is the user the stability objective protects: a temporary downgrade below BBB would have forced her to sell, and an upgrade weeks later would have let her buy back, both at a cost.

Why the other options are wrong

A is incorrect. Agencies take a long-term view, and the stem says the six-month default probability did rise.

B is incorrect. Agency ratings are not equity-price based; that describes Moody's KMV and Kamakura, whose estimates respond quickly.

C is incorrect. Stability is an objective, not a calendar: ratings do change when a long-term change is believed to have occurred.

LO: CR-9 a: Compare agencies' ratings to internal credit rating systems. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1 Credit Ratings

Solution CR-9 a.4 — Answer: B

New • built from GARP Ch.9, 9.1

Step 1 – The supply disruption

It raises the six-month default probability but not the three-to-five-year one. Agencies rate through the cycle, so they would not usually act.

⇒ No change after the first event.

Step 2 – The lost customer

A permanent cut in cash flows is a long-term change in creditworthiness, the trigger for a rating change.

⇒ A downgrade after the second event.

Interpretation

Stability means avoiding reversals, not ignoring information. A downgrade followed by an upgrade a few weeks later is exactly the reversal agencies try to avoid.

Why the other options are wrong

A is incorrect. Describes the reversal agencies avoid; a rating is not meant to track every short-term development.

C is incorrect. There is no fixed-review rule; a long-term change does lead to a rating change.

D is incorrect. Reverses the horizon: agencies look through short-term risk and act on long-term

change.

LO: CR-9 a: Compare agencies' ratings to internal credit rating systems. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1 Credit Ratings

Solution CR-9 a.5 — Answer: C

New • built from GARP Ch.9, 9.1 and 9.1.1

Step 1 – Agency coverage

Agency ratings are usually available only for companies that have issued publicly traded debt.
⇒ X is likely rated; Y (bank loans only) and Z (private) are likely not.

Step 2 – Equity-based estimates

Moody's KMV and Kamakura work from the equity price and other variables. Y's shares trade, so an estimate can be produced; Z has no share price.

Step 3 – Internal ratings

The bank can rate any client from its financial statements, so Y and Z both get an internal rating.

Interpretation

The data each method needs decides where it can be used: public debt for agency ratings, a share price for equity-based estimates, financial statements for internal ratings.

Why the other options are wrong

A is incorrect. Coverage error: many companies, especially small and private ones, have no agency rating.
B is incorrect. Reversed: equity-based models need a market price, which a private firm does not have.
D is incorrect. Stability does not make agency ratings uninformative; for a rated issuer they are a primary source, and the bank can add its own view.

LO: CR-9 a: Compare agencies' ratings to internal credit rating systems. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1 and 9.1.1

Solution CR-9 a.6 — Answer: D

New • built from GARP Ch.9, 9.1 and 9.7.1

Step 1 – What the Table 9.1 figure is

It is the average default experience of each grade over 1981 to 2020, good and bad years together.

Step 2 – How a grade's default rate behaves in a recession

Agencies rate through the cycle, so borrowers keep their ratings while their near-term default risk rises. Default rates vary markedly from year to year (0.15% to 4.19% for all rated companies), so the default rate of a given grade is higher in bad years.
⇒ The long-run average understates the stress default rate for each grade.

Interpretation

This is the price of rating stability for a risk manager: the rating is a long-horizon, through-the-cycle signal, so stress PDs must be adjusted upwards (or taken from a point-in-time system, CR-6 b).

Why the other options are wrong

A is incorrect. Stable ratings do not imply stable default rates; the economy moves the default rate of every grade.
B is incorrect. Through-the-cycle agencies do not downgrade quickly on cyclical news; that is the opposite of the stability objective.

C is incorrect. One bad year among forty barely moves the average; the average is a typical-year figure, not a recession figure.

LO: CR-9 a: Compare agencies' ratings to internal credit rating systems. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1 Credit Ratings; 9.7.1 Reason for the Difference, Table 9.6

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 c

P2 CONCEPT

Describe the relationship between borrower rating and probability of default.

Workbook: none homed here • revision notes in A10 (rating and PD) and A15 (marginal PDs by rating)

WHAT IS TESTED HERE

Why P2. Vardeez flags this LO (map: VARDEEZ FOCUS) in CR-9, his most important Credit Risk chapter. No workbook question is homed here; A10 and A15 carry it in their revision notes. **Vardeez:** three points. Higher ratings mean lower default probabilities; for investment grade the PD in a year rises over the first years; for non-investment grade it can fall once the issuer survives the early critical period.

Reported exam shapes: none listed in the map.

What the questions look like: statements about GARP Table 9.1 and the two time profiles, with the explanation behind each. The numbers only support a statement; computing marginal and conditional PDs is tested under CR-9 d and CR-9 e. The usual trap is reading a cumulative rate as the rate for a single year.

1. Lower rating, higher default probability

On the S&P scale, AAA, AA, A and BBB are investment grade; BB and below are non-investment grade (junk). GARP Table 9.1 gives the cumulative share of issuers that defaulted by each horizon, by initial rating.

Rating	1	2	3	4	5	7	10	15
AAA	0.00	0.03	0.13	0.24	0.34	0.51	0.70	0.90
AA	0.02	0.06	0.11	0.21	0.30	0.49	0.70	0.99
A	0.05	0.13	0.22	0.33	0.46	0.76	1.20	1.84
BBB	0.16	0.43	0.75	1.14	1.54	2.27	3.24	4.54
BB	0.63	1.93	3.46	4.99	6.43	8.89	11.64	14.65
B	3.34	7.80	11.75	14.89	17.35	20.99	24.62	28.24
CCC/C	28.30	38.33	43.42	46.36	48.58	50.75	52.76	54.76

Average cumulative issuer-weighted default rates (%) by horizon in years, S&P 1981 to 2020 (GARP Table 9.1).

- **The inverse relationship is steep.** Below A, each step down multiplies the one-year rate: BBB 0.16% to BB 0.63% (about 4 times), to B 3.34% (about 5 times), to CCC/C 28.30% (about 8.5 times).
- **It holds across the scale, not strictly at the top.** AAA's cumulative rate is above AA's at 3, 4, 5 and 7 years and equal at 10 years (0.70%). Defaults in the top two grades are rare, so their ordering is not strict.
- **Horizon matters as much as rating.** Cumulative rates rise with the horizon, so a 15-year BBB exposure (4.54%) carries more default risk over its life than a one-year B exposure (3.34%). The inverse relationship compares ratings at the same horizon.

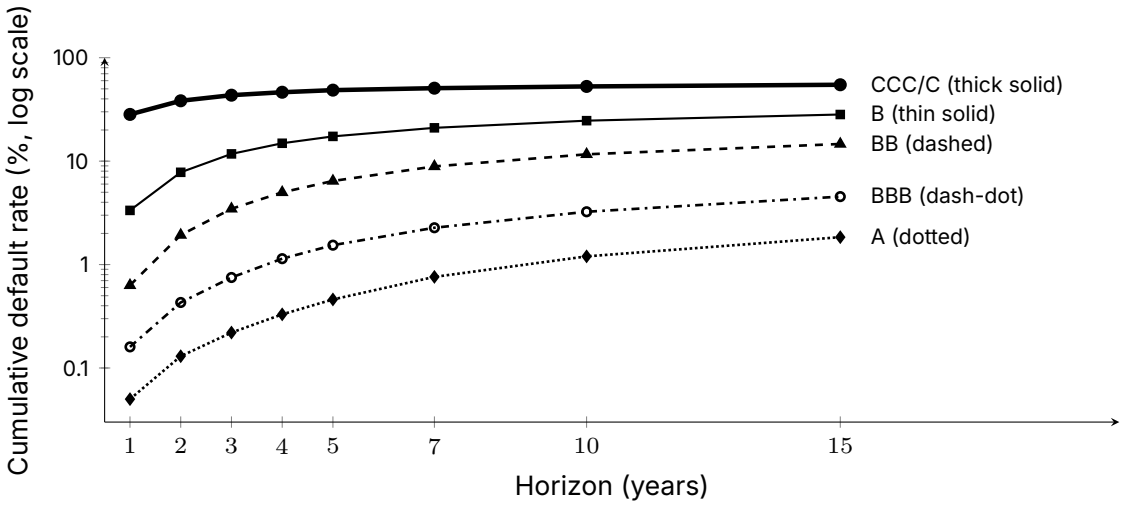


Figure 1. Cumulative default rates by initial rating (GARP Table 9.1) on a log scale; AAA and AA are left out because their one-year rates are 0.00% and 0.02%. The curves never cross: at every horizon a lower grade has defaulted more. The CCC/C curve flattens early because most of its defaults come in year 1 (28.30 of 54.76 points); the investment-grade curves keep climbing.

2. How the default probability evolves after the rating

The marginal (unconditional) PD for year t is $C_t - C_{t-1}$, the difference of successive cumulative rates (CR-9 d). Its profile over time is the second half of the relationship.

Marginal PD (%)	Year 1	Year 2	Year 3	Year 4	Year 5	Profile
A	0.05	0.08	0.09	0.11	0.13	rising
BBB	0.16	0.27	0.32	0.39	0.40	rising
BB	0.63	1.30	1.53	1.53	1.44	rising, then flat
B	3.34	4.46	3.95	3.14	2.46	peak in year 2
CCC/C	28.30	10.03	5.09	2.94	2.22	falling

- **Investment grade: rising.** The issuer starts out creditworthy, and the more time elapses, the greater the possibility that its financial health declines (GARP's A example: 0.05%, 0.08%, 0.09%, 0.11%).
- **Poorly rated: falling.** For the lowest grades the next year or two may be critical. An issuer that survives that period is likely to have improved its financial health, so the PD in later years is lower (CCC/C: 28.3%, 10.03%, 5.09%, 2.94%).
- **In between: up, then down.** BB and B rise at first and then level off or fall.
- **It is not just a shrinking pool.** With a constant hazard rate the marginal PD would fall a little anyway, because fewer issuers are left to default, while the conditional PD would stay constant (CR-9 e). For CCC/C the conditional PD itself falls (28.30%, 13.99%, 8.25%, 5.20%, 4.14%), so survivors really are healthier.

Correction to the source notes. Vardeez's third point says non-investment-grade bonds "show an initial increase in default risk" that later falls. That fits BB and B, but not CCC/C, whose marginal PD falls from year 1 (28.30%, 10.03%, 5.09%, 2.94%). GARP's wording is that for bonds with a poor rating the PD "can be a decreasing function of time"; Schweser says the early increase holds for investment grade and *some*

non-investment grade. (Schweser's B-rated year-4 figure, printed 3.24%, is 3.14%; that correction is at CR-9 e.)

	Investment grade (AAA to BBB)	Low grades (B, CCC/C)
One-year PD	Below 0.2%	3.34% (B) to 28.30% (CCC/C)
Marginal PD over time	Rises in the first years	Falls (CCC/C) or peaks early (B)
Reason	Starts healthy; time gives room to deteriorate	The first year or two are critical; survivors are healthier
Conditional PD	Rises, close to the marginal PD	Falls too: the decline is not just a shrinking pool
Typical example	A: 0.05, 0.08, 0.09, 0.11%	CCC/C: 28.30, 10.03, 5.09, 2.94%

Exam traps

- **Cumulative read as marginal.** 38.33% is CCC/C's default rate *by* year 2; the rate *in* year 2 is 10.03%.
- **Cumulative rates never fall.** Only marginal and conditional PDs can decline with time.
- **Profiles swapped.** Rising belongs to investment grade (deterioration), falling to the lowest grades (survival through the critical period).
- **Absolute qualifiers.** "Strictly lower at every horizon" fails between AAA and AA; "every junk grade falls from year 1" fails for BB and B.
- **Rating without horizon.** A lower rating means a higher PD at the same horizon only; a long BBB exposure can be riskier than a short B one.
- **Shrinking-pool argument.** A falling unconditional PD alone proves little; the falling conditional PD is the evidence that survivors improve.
- **Linear scaling.** A 15-year cumulative PD is not 15 times the one-year PD.

Practice questions: CR-9 c

Question CR-9 c.1 GARP book • Ch.9 Practice Question 9.1 (extended with the 9.2 time profile; rewritten as multiple choice)

A trainee analyst at a German bank is asked which ratings on the S&P scale are investment grade, and how the probability of default in each individual year (the marginal PD) typically evolves for those ratings in the first few years after the rating is assigned. Which answer is correct?

A. AAA to A; the marginal PD falls with time because a creditworthy issuer becomes safer as it ages.

B. AAA to BBB; the marginal PD tends to rise with time because the issuer starts creditworthy and the longer the period, the greater the chance that its financial health declines.

C. AAA to BBB; the marginal PD falls with time because the next year or two are critical and the survivors are healthier.

D. AAA to BB; the marginal PD is constant because rating agencies rate through the cycle.

Question CR-9 c.2

Bank Q320 • New question written for this bank

A credit analyst uses the cumulative default rate table below (percent, by years since the rating was assigned) from a rating agency.

Rating	Year 1	Year 2	Year 3	Year 4
BBB	0.16	0.43	0.75	1.14
B	3.34	7.80	11.75	14.89
CCC/C	28.30	38.33	43.42	46.36

Which of the following statements about the probability of default in each individual year (the marginal probability) is correct?

A. For B, the probability of default in each year falls steadily from year 1 through year 4.

B. For BBB, the probability of default in each year rises from year 1 through year 4.

C. For CCC/C, the probability of default in year 2 is 38.33%, which is higher than in year 1.

D. For BBB, the probability of default in year 3 is 0.75%.

Question CR-9 c.3

Bank Q322 • New question written for this bank

A risk manager observes that for an A-rated issuer the probability of default during each of years 1 to 4 is 0.05%, 0.08%, 0.09% and 0.11%, whereas for an issuer initially rated CCC/C the corresponding probabilities are 28.30%, 10.03%, 5.09% and 2.94%. Which of the following best explains the CCC/C pattern?

A. A CCC/C issuer is initially considered creditworthy, so the longer time passes the greater the chance its financial health declines.

B. Rating agencies lower the rating of CCC/C issuers each year, which mechanically reduces the default probability of those that remain.

C. Cumulative default rates for CCC/C issuers decline over time, so fewer issuers default in later years.

D. A CCC/C issuer faces a critical period in the first year or two; if it survives, its financial health is likely to have improved.

Question CR-9 c.4

New • built from GARP Ch.9, 9.2 and 9.2.1

A credit portfolio manager at a Hong Kong bank argues that the falling year-by-year default probabilities of CCC/C issuers in GARP Table 9.1 say nothing about the issuers' health, because a shrinking pool of survivors mechanically produces fewer defaults each year. An analyst has computed the following for CCC/C issuers:

Year	1	2	3	4	5
Unconditional (marginal) PD (%)	28.30	10.03	5.09	2.94	2.22
Conditional PD, given no earlier default (%)	28.30	13.99	8.25	5.20	4.14

Which response is correct?

A. The manager is right: the conditional PDs decline only because they are computed from the same cumulative table.

B. The manager is right: with a constant hazard rate the conditional PD would fall in exactly this way.

C. The manager is wrong: the conditional PD, which already removes the effect of the shrinking pool, also falls, so a CCC/C issuer that survives the first year or two is genuinely less likely to default.

D. The manager is wrong: if the survivors' health were improving, the unconditional PDs would rise over time.

Question CR-9 c.5

New • built from GARP Ch.9, 9.2 (Table 9.1)

A model validator at an Indian bank reviews a sentence in the bank's credit policy: "In the agency data, a higher rating always means a strictly lower cumulative default rate, at every horizon." She checks an excerpt of GARP Table 9.1 (cumulative default rates, %):

Rating	5 years	7 years	10 years	15 years
AAA	0.34	0.51	0.70	0.90
AA	0.30	0.49	0.70	0.99
A	0.46	0.76	1.20	1.84

Which conclusion is correct?

A. The sentence fails: AAA's rate is above AA's at 5 and 7 years and equal at 10 years. The inverse relationship holds across the scale as a whole but not strictly between the top two grades.

B. The sentence holds: AAA has the lowest cumulative default rate at every horizon shown.

C. The sentence fails because A-rated issuers default less than AA-rated issuers at long horizons.

D. The sentence fails because AAA cumulative default rates fall at longer horizons.

Question CR-9 c.6 New • built from GARP Ch.9, 9.2 (Table 9.1)

A relationship manager at a UK bank compares two loans: a 15-year amortisation-free loan to a BBB-rated borrower and a one-year loan to a B-rated borrower. From GARP Table 9.1, the BBB cumulative default rate at 15 years is 4.54% and the B cumulative default rate at one year is 3.34%. Which statement is correct?

A. The B loan has the higher probability of default, because a lower rating means a higher probability of default whatever the horizon.

B. The two probabilities cannot be compared, because one figure is cumulative and the other is marginal.

C. The BBB loan's probability of default over its life is about $15 \times 0.16\% = 2.40\%$, below the B loan's 3.34%.

D. The BBB loan has the higher probability of default over its life (4.54% against 3.34%): the inverse relationship between rating and PD compares borrowers at the same horizon, and cumulative PDs rise with the horizon.

Solutions • CR-9 c

Solution CR-9 c.1 — Answer: B

GARP book • Ch.9 Practice Question 9.1 (extended)

Step 1 – Investment-grade ratings

On the S&P scale AAA, AA, A and BBB are investment grade; BB and below are non-investment grade.

$\Rightarrow$ A (stops at A) and D (includes BB) fail on the scale.

Step 2 – The time profile

For investment-grade bonds the PD in a year tends to increase with time in the first few years: the issuer starts creditworthy, and time gives its financial health room to decline.

Interpretation

GARP's own example is the A grade: 0.05%, 0.08%, 0.09% and 0.11% in years 1 to 4. The "critical period" story in C is the explanation for the lowest grades, the opposite profile.

Why the other options are wrong

A is incorrect. Leaves out BBB and gives the profile the wrong direction.

C is incorrect. Right scale, but the falling profile and its explanation belong to poorly rated issuers.

D is incorrect. BB is non-investment grade, and rating through the cycle concerns how agencies change ratings, not the PD profile after a rating is assigned.

LO: CR-9 c: Describe the relationship between borrower rating and probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2 Historical Default Probabilities, Table 9.1; Practice Question 9.1

Solution CR-9 c.2 — Answer: B

Bank Q320 • New question written for this bank

Step 1 – Turn each row into marginal PDs

Marginal PD in year $t = C_t - C_{t-1}$. BBB: 0.16, 0.27, 0.32, 0.39. B: 3.34, 4.46, 3.95, 3.14. CCC/C: 28.30, 10.03, 5.09, 2.94.

Step 2 – Read the profiles

BBB rises every year; B rises in year 2 and then falls; CCC/C falls from year 1.

$\Rightarrow$ Only option B, the BBB statement, is correct.

Interpretation

The table shows both halves of the LO: lower grades default more, and the time profile turns from rising (investment grade) to falling (lowest grade).

Why the other options are wrong

A is incorrect. B rises from 3.34% to 4.46% in year 2 before declining, so it does not fall steadily.

C is incorrect. 38.33% is the cumulative rate by year 2; the year-2 marginal PD is $38.33 - 28.30 = 10.03\%$, lower than year 1.

D is incorrect. 0.75% is cumulative through year 3; the year-3 marginal PD is $0.75 - 0.43 = 0.32\%$.

LO: CR-9 c: Describe the relationship between borrower rating and probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2, Table 9.1

Solution CR-9 c.3 — Answer: D

Bank Q322 • New question written for this bank

Step 1 – Identify the profile

CCC/C: 28.30% in year 1, then 10.03%, 5.09% and 2.94%: a falling profile.

Step 2 – GARP's explanation

For a poorly rated bond the next year or two may be critical; if the issuer survives, its financial health is likely to have improved.
$\Rightarrow$ D.

Interpretation

The A-rated pattern rises for the opposite reason: a healthy issuer has more room to deteriorate as time passes. The question pairs the two so that the explanations can be swapped.

Why the other options are wrong

A is incorrect. This is the explanation for the rising investment-grade pattern.
B is incorrect. The pattern is the default experience of issuers that *started* CCC/C; later downgrades are not what produces it.
C is incorrect. Cumulative default rates never decline; only the marginal rates do.

LO: CR-9 c: Describe the relationship between borrower rating and probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2

Solution CR-9 c.4 — Answer: C

New • built from GARP Ch.9, 9.2 and 9.2.1

Step 1 – What the conditional PD removes

The conditional PD divides the year's defaults by the issuers still alive at the start of the year. A shrinking pool cannot, by itself, make it fall.

Step 2 – What a shrinking pool alone would produce

With a constant hazard rate the conditional PD is the same every year $(1 - e^{-\lambda})$ while the unconditional PD falls slowly (CR-9 e).

Step 3 – Compare with the data

The CCC/C conditional PD falls from 28.30% to 13.99%, 8.25%, 5.20% and 4.14%.
$\Rightarrow$ The default intensity of survivors really declines: the manager is wrong.

Interpretation

This is the evidence behind GARP's "critical period" explanation: an issuer that survives the first year or two is in better health.

Why the other options are wrong

A is incorrect. Being computed from the same table does not force a decline; the decline in the conditional PD is a feature of the data.
B is incorrect. Under a constant hazard the conditional PD is constant, not falling (the memoryless property).
D is incorrect. Unconditional PDs can fall even when survivors are unchanged, so a rise is not required; the conditional PD is the right test.

LO: CR-9 c: Describe the relationship between borrower rating and probability of default. CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2 and 9.2.1

Solution CR-9 c.5 Answer: A

New • built from GARP Ch.9, 9.2 (Table 9.1)

Step 1 – Compare AAA with AA horizon by horizon

5 years: 0.34 against 0.30 (AAA higher). 7 years: 0.51 against 0.49 (AAA higher). 10 years: 0.70 against 0.70 (equal). 15 years: 0.90 against 0.99 (AAA lower).
$\Rightarrow$ "Strictly lower at every horizon" fails.

Step 2 – Check the rest of the scale

A is above AA at every horizon shown, and the same holds down the full table.

Interpretation

The inverse relationship is a description of the whole scale; between the top two grades, where defaults are very rare, small reversals appear. An "always" or "strictly" option is the absolute-qualifier trap.

Why the other options are wrong

B is incorrect. AAA is not the lowest at 5 and 7 years.
C is incorrect. A is above AA at every horizon shown.
D is incorrect. Cumulative rates cannot fall; AAA rises from 0.34% to 0.90%.

LO: CR-9 c: Describe the relationship between borrower rating and probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2, Table 9.1

Solution CR-9 c.6 Answer: D

New • built from GARP Ch.9, 9.2 (Table 9.1)

Step 1 – Read both figures as lifetime PDs

Both are cumulative rates: 4.54% is the BBB probability of default within 15 years; 3.34% is the B probability within one year.

Step 2 – Compare

$4.54\% > 3.34\%$, so the long BBB loan has the higher probability of default over its life.

Interpretation

The rating ranks borrowers at a fixed horizon. Over different horizons the cumulative curve does the work: a 15-year exposure gives a BBB borrower time to deteriorate, exactly the rising investment-grade profile.

Why the other options are wrong

A is incorrect. Absolute qualifier: "whatever the horizon" ignores that cumulative PDs rise with time.
B is incorrect. Both figures come from the cumulative table, so they are directly comparable.
C is incorrect. Linear scaling of the one-year rate; the actual 15-year cumulative rate is 4.54%, because the BBB marginal PD rises over time.

LO: CR-9 c: Describe the relationship between borrower rating and probability of default. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.2, Table 9.1

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 f

P2 CONCEPT

Describe recovery rates and their dependencies on default rates.

Workbook: none homed here • revision notes in A15 (Table 9.2 and the default and recovery link)

WHAT IS TESTED HERE

Why P2. Vardeez flags this LO (map: VARDEEZ FOCUS) in his most important Credit Risk chapter. No workbook question is homed here; the A15 revision notes summarise it.
Vardeez: the definition (price about one month after default as a share of principal); senior bonds recover more than junior ones (first lien 54.6%, subordinated 32.1%); recovery rates fall when default rates rise, shown by mortgages in 2007 to 2008.

Reported exam shapes: none listed in the map.

What the questions look like: the direction of the default and recovery link, its mechanism and its consequence ("a bad year is doubly bad"); the definition; the seniority ordering in GARP Table 9.2. Numerical bank items (an expected-loss rate with a stressed recovery, a liquidation waterfall) were set aside under the TYPE rule; the first is rebuilt here as a concept question.

1. What a recovery rate is

When a company goes bankrupt, its creditors file claims (in the United States a bondholder claims face value plus accrued interest). Sometimes there is a reorganisation in which creditors accept partial payment; otherwise the liquidator sells the assets and meets the claims as far as possible, with some claims ranking ahead of others.

Definition. The recovery rate of a bond is normally the *price at which it trades about one month after default, as a percentage of its principal amount*. Loss given default is $1 - R$. For CDS contracts an auction sets the recovery: after Lehman's default it was only about 8 cents per dollar, so protection buyers received about 92% of notional (CR-9 g).

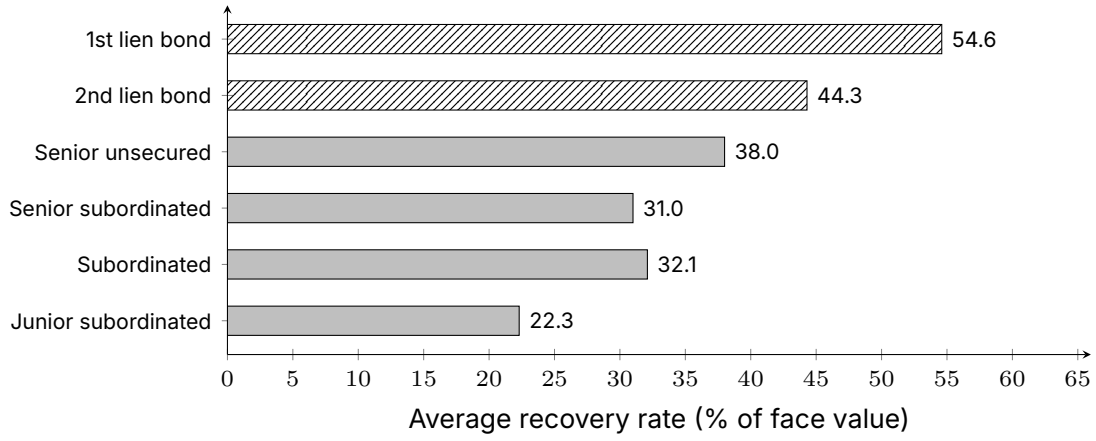


Figure 1. Recovery rates on corporate bonds as a percentage of face value, Moody's, 1983 to 2021, issuer weighted (GARP Table 9.2). Hatched: bonds with a claim on collateral (a first lien has the first claim, a second lien the second). Grey: unsecured bonds, ranked senior unsecured, senior subordinated, subordinated, junior subordinated. Recovery broadly falls with seniority, but not strictly: subordinated (32.1%) is slightly above senior subordinated (31.0%).

2. Recoveries fall when defaults rise

- **Mortgages, 2007 to 2008.** Rising mortgage default rates led to foreclosures, more houses offered for sale and falling house prices. Recovery rates fell with them.
- **Corporate bonds.** In a year when few bonds default, economic conditions are usually good and the average recovery on the bonds that do default may be as high as 60%. In a high-default year conditions are usually poor and the average recovery may be as low as 30%. Moody's finds the average recovery in a year and the non-investment-grade default rate highly (negatively) correlated.
- **Doubly bad.** A bad year for defaults is usually a year of low recoveries as well, so lenders take more defaults *and* a larger loss on each one.

Why it matters for loss estimates. With an expected loss rate of PD $\times$ $(1 - R)$, a recession scenario that raises the PD from 2% to 5% but keeps R at 40% moves the loss rate from 1.2% to 3.0% (2.5 times). Lowering R to 30%, as the evidence suggests, gives 3.5% (2.9 times). A portfolio model that holds recovery constant understates losses in exactly the years that drive the tail (Figure 2); random-recovery extensions address this (CR-13 h).

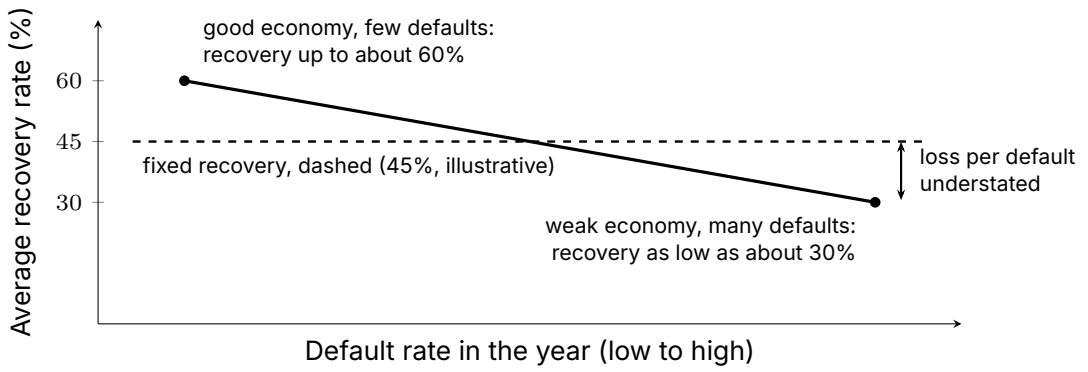


Figure 2. The negative dependence of recovery on default rates (schematic; the end points are GARP's "as high as 60%" and "as low as 30%"). A constant recovery assumption (dashed) is too high in exactly the high-default years.

	Good year	Bad year
Economy	Strong	Weak
Default rate	Low	High
Collateral and house prices	Firm	Falling (forced sales, foreclosures)
Average recovery	As high as about 60%	As low as about 30%
Loss given default $1 - R$	Low	High
Lender's loss	Few defaults, small loss on each	Many defaults, large loss on each: "doubly bad"

Exam traps

- **Direction.** Recovery and default rates move in *opposite* directions; options that make them independent or positively related are wrong.
- **Mechanism.** More defaults bring forced sales and falling collateral values; they do not im-

prove recoveries through "workout experience".

- **Definition.** Price about one month after default, as a share of principal (face value), not of the pre-default price and not the final workout amount.
- **Recovery versus LGD.** Losses use $1 - R$; multiplying by R is the wrong-input twin.
- **Seniority "always".** Secured beats unsecured on average, but the averages are not strictly ordered (subordinated 32.1% against senior subordinated 31.0%).
- **Constant recovery in a stress test.** Keeping the normal-times recovery understates stressed losses.

Practice questions: CR-9 f

Question CR-9 f.1

GARP book • Ch.9 Practice Question 9.12 (rewritten as multiple choice)

A junior analyst at a Dubai bank is building a loss-given-default database for defaulted corporate bonds. Following the usual convention, how should the recovery rate on a defaulted bond be measured?

A. As the amount finally paid to bondholders at the end of the bankruptcy process, as a percentage of the bond's market price just before default.

B. As the price at which the bond trades about one month after default, as a percentage of its principal amount.

C. As the bond's price immediately before default, as a percentage of its principal amount.

D. As one minus the loss given default, with the loss measured against the bond's market value one month before default.

Question CR-9 f.2

New • built from GARP Ch.9, 9.3 (Table 9.2)

A credit analyst at a Luxembourg fund reviews Moody's average recovery rates on corporate bonds (issuer weighted, 1983 to 2021): first lien 54.6%; second lien 44.3%; senior unsecured 38.0%; senior subordinated 31.0%; subordinated 32.1%; junior subordinated 22.3%. Which statement is most accurate?

A. Recovery falls strictly with each step down in seniority, so subordinated bonds always recover less than senior subordinated bonds.

B. Second lien bonds recover more on average than first lien bonds, because they also have a claim on collateral.

C. Bonds with a claim on collateral recover more on average than unsecured bonds, and recovery broadly falls with seniority, although subordinated bonds recovered slightly more on average than senior subordinated bonds.

D. Seniority makes little difference: first lien and junior subordinated bonds differ by less than 10 percentage points.

Question CR-9 f.3 Bank Q341 • BT P2.T6.24.14.2 (rebuilt)

A credit risk analyst is refining loss estimates for a corporate bond portfolio during an economic downturn in which default rates rise sharply. Which of the following best describes how average recovery rates behave in this environment?

A. Recovery rates rise, because the larger number of defaults gives lenders more experience to improve their workout strategies.

B. Recovery rates fall, because defaults rise with a weak economy that depresses the prices of defaulted bonds and collateral.

C. Recovery rates are unaffected, because they are determined only by the issuer's original credit rating.

D. Recovery rates are unaffected, because defaulted securities are highly liquid in a downturn and are always resold at face value.

Question CR-9 f.4 New • built from GARP Ch.9, 9.3.1

A risk manager at a US mortgage lender is explaining to the board why recovery rates on defaulted mortgages fell during the 2007 to 2008 crisis. Which sequence is most consistent with the reading?

A. Rising default rates led to more foreclosures, more houses offered for sale and falling house prices, which reduced recovery rates.

B. Rising default rates led lenders to hold foreclosed houses off the market, which kept house prices and recovery rates stable.

C. Falling house prices reduced foreclosures, so fewer houses were sold and recovery rates fell for lack of buyers.

D. Rising default rates gave lenders more workout experience, which raised recovery rates, but higher interest rates more than offset this.

Question CR-9 f.5 Bank Q343 • New question written for this bank (rebuilt as a concept question)

A credit risk manager estimates the annual expected loss rate on a corporate loan book as the default probability multiplied by (1 minus the recovery rate). In normal conditions the default probability is 2% and the recovery rate is 40%. For a recession stress scenario she raises the default probability to 5%. Which recovery assumption and conclusion are most consistent with the historical relationship between default rates and recovery rates?

A. Keep the recovery rate at 40%, because recovery depends only on seniority; the loss rate rises in proportion to the default probability.

B. Raise the recovery rate, because more defaults let lenders sell collateral more efficiently; the loss rate rises by less than the default probability.

C. Lower the recovery rate, but also cut the stressed default probability back toward 2%, because high-default years are followed by high-recovery years.

D. Lower the recovery rate, because recoveries fall when defaults rise; the loss rate then rises by more than the increase in the default probability alone implies.

Question CR-9 f.6 New • built from GARP Ch.9, 9.3.1

A bank's credit portfolio model simulates correlated defaults but assigns every defaulted loan a fixed 45% recovery rate, whatever the number of defaults in the simulated year. Which statement best describes the effect on the simulated loss distribution?

A. Losses in high-default years are overstated, because recoveries rise when many firms default and collateral markets deepen.

B. Expected loss is overstated in every year, because a fixed recovery rate is always too low.

C. Losses in high-default years are understated, so the tail of the loss distribution, and any credit VaR read from it, is too thin: in practice those years also bring lower recoveries.

D. The tail is unaffected, because once seniority is fixed, recovery rates are unrelated to default rates.

Solutions • CR-9 f

Solution CR-9 f.1

Answer: B

GARP book • Ch.9 Practice Question 9.12

Step 1 – Timing

The recovery rate is read from the market price of the defaulted bond about one month after default, not at the end of the bankruptcy process and not before default.

Step 2 – Base

The price is expressed as a percentage of the bond's principal (face value).
$\Rightarrow$ B.

Interpretation

A market price soon after default gives a timely, observable number for every defaulted bond; that is why databases and the CDS auction work this way. Loss given default is then $1 - R$.

Why the other options are wrong

A is incorrect. Wrong timing (end of the process) and wrong base (pre-default market price).
C is incorrect. A pre-default price measures the market's view before the event, not what is recovered.
D is incorrect. Circular, and measured against the wrong base: LGD is defined from R, with both relative to principal.

LO: CR-9 f: Describe recovery rates and their dependencies on default rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.3 Recovery Rates; Practice Question 9.12

Solution CR-9 f.2

Answer: C

New • built from GARP Ch.9, 9.3 (Table 9.2)

Step 1 – Secured against unsecured

First lien 54.6% and second lien 44.3% are both above the best unsecured class, senior unsecured at 38.0%.

Step 2 – Down the unsecured ranking

Senior unsecured 38.0%, senior subordinated 31.0%, subordinated 32.1%, junior subordinated 22.3%: broadly falling, with one small reversal.
$\Rightarrow$ C.

Interpretation

Seniority is the main driver of recovery, but the averages are historical, issuer-weighted figures, so they are not perfectly ordered.

Why the other options are wrong

A is incorrect. Absolute qualifier: subordinated (32.1%) is above senior subordinated (31.0%).
B is incorrect. The first lien has the *first* claim on collateral and the higher recovery.
D is incorrect. The range is $54.6 - 22.3 = 32.3$ percentage points.

LO: CR-9 f: Describe recovery rates and their dependencies on default rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.3, Table 9.2

Solution CR-9 f.3 — Answer: B

Bank Q341 • BT P2.T6.24.14.2

Step 1 – Direction

Recovery rates are negatively related to default rates.

Step 2 – Mechanism

The weak economy that raises defaults also depresses the prices of defaulted bonds and of collateral.

$\Rightarrow$ Recoveries fall in the downturn: B.

Interpretation

Lenders take a double hit in a downturn: more defaults, and a larger loss on each.

Why the other options are wrong

A is incorrect. Reverses the sign; the market effect dominates any learning in workouts.
C is incorrect. Ignores the economic cycle; average recoveries move with default rates.
D is incorrect. Defaulted bonds are illiquid and trade well below face value in a downturn.

LO: CR-9 f: Describe recovery rates and their dependencies on default rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.3.1 The Dependence of Recovery Rates on Default Rates

Solution CR-9 f.4 — Answer: A

New • built from GARP Ch.9, 9.3.1

Step 1 – Start from the defaults

Higher mortgage default rates lead to foreclosures.

Step 2 – Supply and price

Foreclosed houses are offered for sale; the extra supply pushes house prices down.

Step 3 – Recovery

The lender recovers what the collateral fetches, so recovery rates fall.

$\Rightarrow$ A.

Interpretation

The same loop runs through corporate credit: the economy that causes defaults also lowers what defaulted assets are worth.

Why the other options are wrong

B is incorrect. Contradicts the reading: foreclosed houses came to market and prices fell.
C is incorrect. Reverses the causal chain; defaults drove foreclosures, which drove prices down.
D is incorrect. Gets the direction wrong; recoveries fell because collateral values fell.

LO: CR-9 f: Describe recovery rates and their dependencies on default rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.3.1

Solution CR-9 f.5 — Answer: D

Bank Q343 • New question written for this bank

Step 1 – Which way should recovery move

Recovery rates fall when default rates rise, so the recession scenario needs a recovery below the normal 40%.

Step 2 – What that does to the loss rate

At a constant 40%: $2\% \times 0.60 = 1.2\%$ rises to $5\% \times 0.60 = 3.0\%$, 2.5 times. With recovery lowered to 30% (the bank's original figure): $5\% \times 0.70 = 3.5\%$, 2.9 times.

⇒ The loss rate rises by more than the default probability alone implies.

Interpretation

High defaults and low recoveries arrive together, so a stress test that moves only the PD understates the loss. The bank's original numerical version keys 3.5%.

Why the other options are wrong

A is incorrect. Holding recovery at 40% ignores the default and recovery link and understates the stressed loss (3.0% rather than 3.5%).

B is incorrect. Reverses the sign: more defaults depress collateral values.

C is incorrect. The scenario is defined by the higher default probability; there is no basis for offsetting it.

LO: CR-9 f: Describe recovery rates and their dependencies on default rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.3.1

Solution CR-9 f.6 Answer: C

New • built from GARP Ch.9, 9.3.1

Step 1 – Where the fixed recovery is wrong

In high-default years actual recoveries are low (as low as about 30%), so a fixed 45% is too high exactly then.

Step 2 – Effect on the distribution

Simulated losses in the worst years are too small, so the right tail is too thin and a credit VaR read from it is too low.

⇒ C.

Interpretation

A fixed recovery near the long-run average can leave expected loss roughly right while getting the tail wrong. Models with random recovery tied to the default environment address this.

Why the other options are wrong

A is incorrect. Reverses the sign of the default and recovery link.

B is incorrect. A fixed rate near the average is too low in good years and too high in bad years; "overstated in every year" is false.

D is incorrect. Seniority does not remove the cyclical link; the average recovery in a year moves with the default rate.

LO: CR-9 f: Describe recovery rates and their dependencies on default rates. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.3.1

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 j

P2 CALC

Compare default probabilities calculated from historical data with those calculated from credit yield spreads.

Workbook: D5 Credit Spread and PD from Spreads • Tier D (D5.2) A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD • Tier A (A15.4)

WHAT IS TESTED HERE

Why P2. Tier D workbook topic (D5, Map #42) with Vardeez's focus, in his most important Credit Risk chapter. Comparing the two estimates means computing both, so the LO is CALC (approved judgment call).

Vardeez: three findings. Hazard rates from credit spreads are significantly greater than those from historical data; the difference shrinks as credit quality improves and grows as it worsens; a good (bad) economy lowers (raises) default probabilities.

Reported exam shapes

- **D5·S3** One-period PD $= \frac{1}{\text{LGD}}\left[1-\frac{1+r_f}{1+y}\right]$, where recovery decides the ranking (Question CR-9 j.7).
- **A15·S2** Historical hazard rate by the direct formula $\bar{\lambda} = -\ln(1-Q)/T$, against the spread-implied $s/(1-R)$ (Questions CR-9 j.2 to j.4).

What the questions look like: a cumulative default rate, a credit spread and a recovery rate; compute the two hazard rates, their difference, and say what the difference means. Ratings, notionals and the other estimate are the usual red herrings.

Worked elsewhere: A15.7 (spread from a cumulative PD) at Question CR-9 e.8; A15.5 and A15.6 (hazard from a CDS spread) at Questions CR-9 h.1 and CR-9 h.7; D5.1 (spread from the market value of debt) under CR-12 c. Which estimate to use for what: CR-9 k.

1. Two estimates of the same default intensity

- **Historical (real-world).** From the cumulative default rate $Q(T)$ of the rating class (CR-9 e, Section 2):

$$\bar{\lambda}_{\text{hist}} = -\frac{1}{T}\ln(1-Q(T))$$

- **Spread-implied (risk-neutral).** The credit spread is roughly an average annual loss rate, so dividing by the loss given default gives the default intensity (GARP eq. 9.3, CR-9 h):

$$\bar{\lambda}_{\text{spread}} \approx \frac{s(T)}{1-R}$$

- **Where the spread comes from.** A CDS spread, a bond yield spread (corporate yield minus the yield on a similar risk-free bond) or an asset swap spread. For zero-coupon debt worth D_t with face F: $y = -\ln(D_t/F)/T$ and $s = y - r$ (Question CR-9 j.1; CR-12 c). The approximation works well for CDS spreads and for bonds trading near par.

Worked example (GARP Tables 9.4 and 9.5, BB). Seven-year cumulative default rate 8.89%, seven-year spread 280.28 bp, $R = 40\%$:

$$\bar{\lambda}_{\text{hist}} = -\tfrac{1}{7}\ln(1-0.0889) = 1.330\%, \qquad \bar{\lambda}_{\text{spread}} = \frac{0.028028}{0.60} = 4.671\%$$

The market prices in 3.341 percentage points a year of default intensity more than history shows, about 3.5 times as much.

2. One period, exactly (shape S3)

Price a one-year zero-coupon bond with face 1 two ways: at its yield y, and as the risk-neutral expected payoff (1 with no default, R on default) discounted at the risk-free rate:

$$\frac{(1-\text{PD})+\text{PD}\cdot R}{1+r_f}=\frac{1}{1+y}\quad\Rightarrow\quad\boxed{\text{PD}=\frac{1}{\text{LGD}}\left[1-\frac{1+r_f}{1+y}\right]=\frac{y-r_f}{(1+y)\,\text{LGD}}}$$

The bracket $1-(1+r_f)/(1+y)$ is the expected loss rate PD $\times$ LGD; dividing by LGD $=1-R$ turns it into a PD. This exact one-period form comes from the workbook, not GARP's text; it is the discrete version of $s/(1-R)$.

- **Example.** $y=5.5\%$, $r_f=3.0\%$, $R=40\%$: expected loss rate $1-1.03/1.055=2.370\%$, so PD $=2.370/0.60=3.95\%$. The approximation $(y-r_f)/(1-R)=4.17\%$ is slightly higher because it drops the $1/(1+y)$.
- **Recovery decides the ranking.** Two issuers both yielding 5.0%: the expected loss rate is $1-1.03/1.05=1.905\%$ for each. With $R=50\%$ the PD is 3.81%; with $R=25\%$ it is 2.54%. At the same spread, the higher-recovery issuer must default more often to justify it, so ranking by spread is not ranking by PD.

3. The evidence: GARP Tables 9.4 and 9.5

Historical seven-year default rates (S&P, 1981 to 2020) against average seven-year bond spreads (Merrill Lynch, December 1996 to June 2007, risk-free rate taken as the seven-year swap rate minus 10 bp), with $R=40\%$:

Rating	$Q(7)$ (%)	Spread (bp)	$\bar{\lambda}_{\text{hist}}$ (%)	$\bar{\lambda}_{\text{spread}}$ (%)	Difference	Ratio
AAA	0.51	35.74	0.073	0.596	0.523	8.2
AA	0.49	43.67	0.070	0.728	0.658	10.4
A	0.76	68.68	0.109	1.145	1.036	10.5
BBB	2.27	127.53	0.328	2.126	1.798	6.5
BB	8.89	280.28	1.330	4.671	3.341	3.5
B	20.99	481.04	3.366	8.017	4.651	2.4
CCC/C	50.75	1,103.70	10.118	18.395	8.277	1.8

- Spread-implied hazard rates are far above historical ones for every grade.
- The *difference* grows as credit quality declines, from about 0.5 to 8.3 percentage points. The *ratio* does the opposite, from about 10 times for the top grades to under 2 times for CCC/C.
- The spreads stop before the 2008 crisis; with the crisis included, the gap would be much more extreme. The excess return is not constant: it rises when spreads are high.

> **Correction to the source notes.** GARP Table 9.5 prints the CCC/C difference as 8.207. From the table's own columns, $18.395-10.118=8.277$ percentage points; the other six differences check.

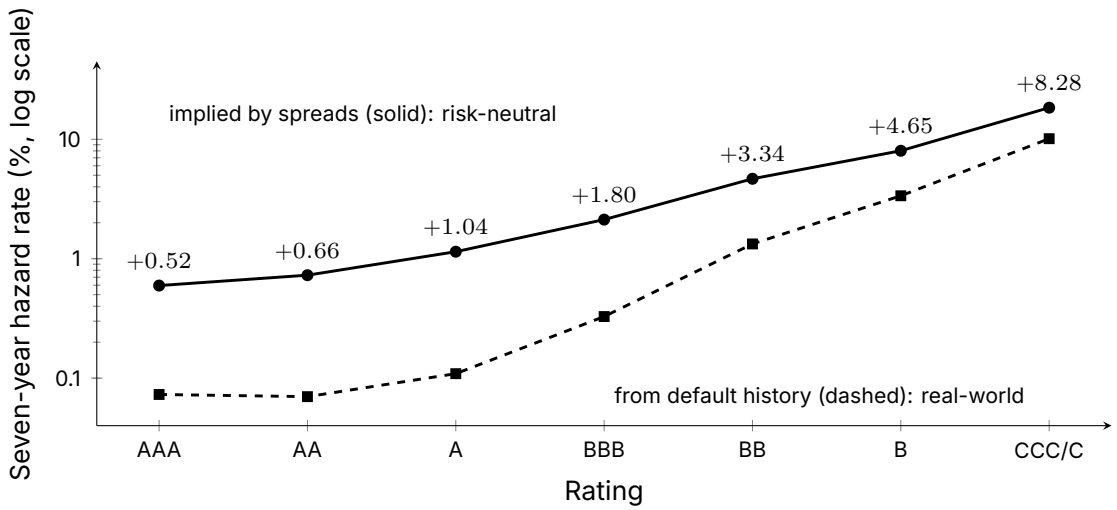


Figure 1. Seven-year hazard rates by rating (GARP Table 9.5). Labels give the difference in percentage points, which widens as quality falls; on the log scale the vertical gap is the ratio, which narrows from about 10 to under 2.

4. Why the gap, and what to use

Investors are compensated for more than the actuarial cost of default, and the excess grows with the credit risk taken. GARP's reasons, in order of weight:

- **Liquidity.** Corporate bonds are relatively illiquid. In normal markets this explains perhaps 25 bp of the excess: a small part.
- **Subjective probabilities.** Traders may allow for depression scenarios worse than anything in the data; hard to see this explaining much.
- **Systematic risk: by far the most important.** Bonds do not default independently. Default rates vary markedly from year to year (all rated companies: 0.15% in 1981, 4.19% in 2009), driven by the economy and by contagion, and this risk cannot be diversified away. As quality falls, debt behaves more like equity and the excess return rises.
- **Hard-to-diversify idiosyncratic risk.** Bond returns are highly skewed (say a 99.75% chance of $+7\%$ and a 0.25% chance of -60% in a year); diversifying it away takes tens of thousands of bonds, so investors may be paid for it too.

Use. Spread-implied (risk-neutral) estimates for valuation, for example pricing a bond or adjusting a derivative's price for default; historical (real-world) estimates for scenario analysis, such as credit VaR. The full treatment is at CR-9 k.

Formula card: CR-9 j

Historical hazard rate	$\bar{\lambda}_{\text{hist}} = -\ln(1-Q(T))/T \quad$ (not Q/T)
Spread-implied hazard rate	$\bar{\lambda}_{\text{spread}} \approx s(T)/(1-R) \quad$ (divide by LGD, not by R)
Spread from a zero-coupon price	$y = -\ln(D_t/F)/T, \quad s = y - r$
One-period PD, exact (S3)	$\text{PD} = \dfrac{1}{1-R}\left[1 - \dfrac{1+r_f}{1+y}\right] = \dfrac{y-r_f}{(1+y)(1-R)}$
One-period PD, approximate	$\text{PD} \approx (y-r_f)/(1-R)$
Excess default intensity	$\bar{\lambda}_{\text{spread}} - \bar{\lambda}_{\text{hist}}$: rises as quality falls; the ratio falls
Which to use (CR-9 k)	Risk-neutral (spreads) for valuation; real-world (history) for scenario analysis

	From historical data	From credit spreads
Data	Cumulative default rates by rating (Table 9.1)	CDS, bond yield or asset swap spreads, plus a recovery assumption
Measure	Real-world (physical)	Risk-neutral
Formula	$-\ln(1-Q)/T$	$s/(1-R)$; one period: $[1-(1+r_f)/(1+y)]/(1-R)$
Level	Lower	Higher for every grade
What it contains	Default experience only	Expected default loss plus premiums for systematic risk, idiosyncratic risk and illiquidity
Use	Scenario analysis, credit VaR	Valuation, pricing the default risk in derivatives

Exam traps

- **Cumulative read as annual.** A seven-year cumulative 2.80% is not a 2.80% hazard rate; convert with $-\ln(1-Q)/T$.
- **Recovery versus LGD.** The spread is divided by $1-R$. Not dividing at all, or dividing by R, are the two standard distractors.
- **Linear shortcut.** Q/T understates the hazard rate, badly for low grades (CCC/C: 7.25% against 10.12%).
- **Difference versus ratio.** "The gap widens as quality falls" is about percentage points; the ratio narrows.
- **Approximate versus exact (S3).** If the stem says "one-period" or gives annual compounding and a single default date, use $[1-(1+r_f)/(1+y)]/(1-R)$, not $(y-r_f)/(1-R)$.
- **Spread is not PD.** The spread prices PD $\times$ LGD; ranking issuers by spread ignores recovery and can reverse the PD ranking.
- **Over- versus under-compensation.** A spread-implied rate above the historical one means investors earn *more* than the expected cost of default.

- **Liquidity as the main reason.** It explains about 25 bp in normal markets; systematic risk is the main reason.

Practice questions: CR-9 j

Question CR-9 j.1

Bank Q356 • AnalystPrep Q.1805 (rebuilt)

A firm's only debt is a zero-coupon bond with face value \$100 million and five years remaining to maturity. The bond currently has a market value of \$88 million. The risk-free rate implied by the zero-coupon bond price is 1% (continuously compounded). What is the credit spread on the bond, using continuous compounding?

A. 1.56%

B. 1.73%

C. 2.56%

D. 11.78%

Question CR-9 j.2

GARP book • Ch.9 Practice Question 9.13 (rewritten as multiple choice)

A credit analyst is checking GARP Table 9.5 against Table 9.4. For B-rated issuers, Table 9.4 gives a seven-year cumulative default probability of 20.99% and an average seven-year credit spread of 481.04 basis points; the recovery rate is assumed to be 40%. Which of the following is closest to the average seven-year hazard rate calculated from the historical default probability?

A. 3.00%

B. 3.37%

C. 8.02%

D. 23.56%

Question CR-9 j.3

Bank Q360 • BT P2.T6.24.14.2 (rebuilt)

A risk analyst compares two estimates of the average 7-year default intensity for a BB-rated issuer. Historical data show a 7-year cumulative default probability of 4.0%. The issuer's 7-year average credit spread is 300 basis points, and the assumed recovery rate is 40%. Which statement correctly gives the two average hazard rates and the proper interpretation of their comparison?

A. Historical hazard rate about 0.58% per year versus spread-implied about 5.00% per year; the spread-implied rate is much higher, so investors are compensated for more than the expected cost of default (systematic risk, liquidity).

B. Historical hazard rate about 0.58% per year versus spread-implied about 5.00% per year; the spread-implied rate is much higher, which shows investors are under-compensated for default risk.

C. Historical hazard rate about 0.58% per year versus spread-implied about 3.00% per year; the spread-implied rate is higher, so investors are compensated for more than the expected cost of default.

D. Historical hazard rate about 0.58% per year versus spread-implied about 7.50% per year; the spread-implied rate is higher, so investors are compensated for more than the expected cost of default.

Question CR-9 j.4

Workbook A15.4 • S2 • Map #17

A risk consultant at an Australian fund manager is explaining the gap between market-implied and historical default estimates to the investment committee for a BBB-type issuer class:

- Seven-year cumulative historical default probability: 2.80%
- Average seven-year credit spread: 150 basis points
- Expected recovery rate: 40%

Which of the following is closest to the excess of the spread-implied (risk-neutral) average hazard rate over the historical (real-world) average hazard rate, per annum?

A. -0.30%

B. 1.09%

C. 2.09%

D. 3.34%

Question CR-9 j.5

New • built from GARP Ch.9, 9.7, Tables 9.4 and 9.5

An investment committee member at a Canadian pension fund looks at an excerpt of GARP Table 9.5 (average seven-year hazard rates, %):

Rating	From default probabilities	Implied by credit spreads
A	0.109	1.145
BBB	0.328	2.126
B	3.366	8.017
CCC/C	10.118	18.395

Which statement is correct?

A. The spread-implied hazard rate is roughly ten times the historical one for every rating, so the excess is a fixed multiple of default risk.

B. For CCC/C bonds the historical hazard rate exceeds the spread-implied one, so investors in the lowest grade are under-compensated.

C. The excess falls as credit quality declines, because low-quality bonds are more liquid.

D. The excess rises from about 1.0 percentage point for A to about 8.3 points for CCC/C, while the ratio of the two hazard rates falls from about 10.5 to about 1.8.

Question CR-9 j.6

GARP book • Ch.9 Practice Question 9.7 (rewritten as multiple choice)

A board member at a Swiss insurer asks why default probabilities implied by credit spreads are so much higher than those calculated from historical default experience. Which explanation does the reading identify as by far the most important?

A. Corporate bonds are relatively illiquid, and the liquidity premium explains most of the gap.

B. Bond traders allow for depression scenarios much worse than any in the historical data.

C. Bonds do not default independently of each other, so bondholders bear systematic risk that cannot be diversified away and earn an excess return for it.

D. Rating agencies understate historical default rates because they rate through the cycle.

Question CR-9 j.7 Workbook D5.2 • S2, S3 • Map #42

A risk analyst at an Australian bank is comparing three issuers using their one-year zero-coupon bonds. The analyst assumes risk-neutral pricing with a single default date at maturity. The one-year risk-free rate is 3.0%; all rates use annual compounding.

Issuer	One-year yield	Expected recovery rate	Bond seniority
P	6.0%	60%	Senior secured
Q	7.0%	20%	Subordinated
R	6.5%	40%	Senior unsecured

Which of the following correctly identifies the issuer with the highest implied one-year probability of default and that probability?

A. Issuer P; 7.08%

B. Issuer P; 7.50%

C. Issuer Q; 3.74%

D. Issuer Q; 18.69%

Solutions • CR-9 j

Solution CR-9 j.1

Answer: A

Bank Q356 • AnalystPrep Q.1805

Step 1 – Continuously compounded yield on the zero

$$y = -\frac{1}{T}\ln\frac{D}{F} = -\frac{1}{5}\ln(0.88) = \frac{0.12783}{5} = 2.557\%$$

Step 2 – Credit spread

$$s = y - r = 2.557\% - 1.00\% = 1.56\%$$

$\Rightarrow$ Investors demand 1.56% a year over the risk-free rate; divided by $1-R$ this becomes a spread-implied hazard rate.

Interpretation

This is the input side of the spread-implied PD. The logarithm is negative because the bond trades below face, and the leading minus sign makes the yield positive.

Why the other options are wrong

B is incorrect. Used a simple yield, $(100/88-1)/5 = 2.73\%$, then subtracted 1%: 1.73%.
C is incorrect. Stopped at the yield, 2.56%, without subtracting the risk-free rate.
D is incorrect. Did not annualise: $-\ln(0.88) - 1\% = 11.78\%$, a five-year log loss treated as a rate.

LO: CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.5 Credit Spreads; Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.1

Solution CR-9 j.2

Answer: B

GARP book • Ch.9 Practice Question 9.13

Step 1 – Direct formula

$$\bar{\lambda}_{\text{hist}} = -\frac{1}{7}\ln(1-0.2099) = \frac{-\ln(0.7901)}{7} = \frac{0.23560}{7} = 3.366\%$$

$\Rightarrow$ This matches the B entry in the second column of GARP Table 9.5.

Step 2 – Check: back to the cumulative rate

$$1 - \mathrm{e}^{-7\times 0.03366} = 1 - \mathrm{e}^{-0.2356} = 20.99\%$$

Interpretation

The spread is a red herring for this part: it gives the spread-implied 8.02% in the third column. For B the historical rate is about 42% of the spread-implied rate.

Why the other options are wrong

A is incorrect. Linear shortcut: $20.99/7 = 3.00\%$, which understates the hazard rate.
C is incorrect. Answered the other column: $0.048104/0.60 = 8.02\%$ is the spread-implied hazard rate.
D is incorrect. Forgot to divide by $T = 7$: $-\ln(0.7901) = 23.56\%$.

LO: CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7, Tables 9.4 and 9.5; Practice Question 9.13

Solution CR-9 j.3 Answer: A

Bank Q360 • BT P2.T6.24.14.2

Step 1 – Historical hazard rate

$$\bar{\lambda}_{\text{hist}} = -\frac{1}{7}\ln(1-0.04) = \frac{0.04082}{7} = 0.583\%$$

Step 2 – Spread-implied hazard rate

$$\bar{\lambda}_{\text{spread}} = \frac{0.0300}{1-0.40} = 5.00\%$$

$\Rightarrow$ The spread-implied rate is about 8.6 times the historical one.

Step 3 – Interpretation

Spreads carry more than the expected default loss: premiums for systematic (non-diversifiable) risk, hard-to-diversify idiosyncratic risk and illiquidity.

Interpretation

A spread-implied rate above the historical one means investors are *over*-compensated relative to the actuarial cost of default; that excess is the reward for the risks listed in Step 3.

Why the other options are wrong

B is incorrect. Right numbers, reversed conclusion.
C is incorrect. Used the spread itself as the hazard rate (3.00%), without dividing by $1-R$.
D is incorrect. Divided by the recovery rate: $0.03/0.40 = 7.50\%$.

LO: CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1 eq. 9.3; 9.7 and 9.7.1

Solution CR-9 j.4 Answer: C

Workbook A15.4 • S2

Step 1 – Real-world average hazard rate (direct formula)

$$\bar{\lambda}_{\text{hist}} = -\frac{1}{7}\ln(1-0.028) = 0.00406 = 0.406\%$$

$\Rightarrow$ Historical data imply defaults arrive at about 0.41% per year.

Step 2 – Risk-neutral average hazard rate

$$\bar{\lambda}_{\text{spread}} = \frac{s}{1-R} = \frac{0.0150}{1-0.40} = 0.0250 = 2.50\%$$

Step 3 – Difference

$$2.50\% - 0.406\% = 2.09\%$$

Interpretation

Bond investors are compensated for about 2.09% per year more default intensity than history justifies. Hull attributes most of this to systematic (non-diversifiable) default risk, with a small liquidity component. For valuation use the 2.50%; for scenario analysis use the 0.41%.

Why the other options are wrong

A is incorrect. Treated the seven-year cumulative 2.80% as an annual hazard rate: $2.50 - 2.80 = -0.30\%$.
B is incorrect. Did not divide the spread by LGD: $1.50 - 0.41 = 1.09\%$.

D is incorrect. Divided the spread by the recovery rate instead of $1 - R$: $1.50/0.40 - 0.41 = 3.34\%$.

LO: CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. • **Reference:** Hull, RMFI 6e, Ch.17, 9.7, Tables 9.4 and 9.5, 9.7.1 and 9.7.2

Solution CR-9 j.5 — Answer: D

New • built from GARP Ch.9, 9.7, Tables 9.4 and 9.5

Step 1 – Differences

A: $1.145 - 0.109 = 1.036$. BBB: $2.126 - 0.328 = 1.798$. B: $8.017 - 3.366 = 4.651$. CCC/C: $18.395 - 10.118 = 8.277$ percentage points.
$\Rightarrow$ The excess rises as quality falls.

Step 2 – Ratios

A: $1.145/0.109 = 10.5$. BBB: 6.5. B: 2.4. CCC/C: $18.395/10.118 = 1.8$.
$\Rightarrow$ The ratio falls as quality falls.

Interpretation

GARP's statement that the difference "tends to increase as credit quality declines" is about the absolute excess. The excess return grows with the credit risk taken, even though for low grades a larger share of the spread is genuine expected default loss. (GARP prints the CCC/C difference as 8.207; see the correction box.)

Why the other options are wrong

A is incorrect. The ratio ranges from about 10.5 to 1.8; it is not a fixed multiple.
B is incorrect. The spread-implied rate is above the historical one for every grade, CCC/C included (18.395% against 10.118%).
C is incorrect. The excess rises, and low-quality bonds are not more liquid; illiquidity adds to the excess.

LO: CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7, Tables 9.4 and 9.5

Solution CR-9 j.6 — Answer: C

GARP book • Ch.9 Practice Question 9.7

Step 1 – Rule out the minor reasons

Illiquidity explains perhaps 25 bp of the excess in normal markets. Subjective depression scenarios are hard to see explaining a large part.

Step 2 – The main reason

Bonds do not default independently: default rates swing from year to year (0.15% in 1981 to 4.19% in 2009), so default risk is partly systematic and cannot be diversified away. Bondholders earn an excess return for bearing it, more so as quality falls and debt becomes more equity-like.
$\Rightarrow$ C.

Interpretation

Two other contributions complete the picture: idiosyncratic bond risk is highly skewed and hard to diversify, and the excess return rises when spreads are high, as in 2008.

Why the other options are wrong

A is incorrect. True that bonds are illiquid, but this is a small part of the gap.

B is incorrect. A possible factor that GARP considers unlikely to explain much.
D is incorrect. Not a reason GARP gives; historical default rates are measured outcomes of rated issuers.

LO: CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.7.1 Reason for the Difference; Practice Question 9.7

Solution CR-9 j.7 Answer: A

Workbook D5.2 • S2, S3

Step 1 – Expected loss rate implied by each yield

$$\text{PD} \cdot \text{LGD} = 1 - \frac{1 + r_f}{1 + y}$$

$$P: \ 1 - \frac{1.03}{1.060} = 0.02830, \qquad Q: \ 1 - \frac{1.03}{1.070} = 0.03738, \qquad R: \ 1 - \frac{1.03}{1.065} = 0.03286$$

$\Rightarrow$ Q has the widest spread and the largest expected loss rate: the market expects to lose 3.74% of Q's face value.

Step 2 – Divide by LGD $= 1 - R$

$$P: \frac{0.02830}{0.40} = 7.08\%, \qquad Q: \frac{0.03738}{0.80} = 4.67\%, \qquad R: \frac{0.03286}{0.60} = 5.48\%$$

$\Rightarrow$ P loses only 40% when it defaults, so its smaller expected loss rate needs a larger default probability to explain it.

Step 3 – Rank

P (7.08%) > R (5.48%) > Q (4.67%).

Interpretation

The ranking by spread (Q, R, P) reverses once recovery is taken into account: the spread prices expected loss, which is PD times LGD. These are risk-neutral PDs, suited to valuation and CVA; Hull's evidence (Table 9.5) is that they sit well above real-world PDs from default history, so a stress scenario would use historical rates instead. The formula is the one-period version of the idea that the hazard rate is roughly $s/(1-R)$ (outside-source in this exact discrete form).

Why the other options are wrong

B is incorrect. Dropped the $1/(1+y)$ term and treated the raw spread as the expected loss rate: $(0.060 - 0.030)/0.40 = 7.50\%$. That is the approximation $s/(1-R)$, not the exact one-period PD the stem asks for.
C is incorrect. Dropped the division by LGD and ranked on the expected loss rate: Q's $1 - 1.03/1.07 = 3.74\%$ is PD times LGD, not PD.
D is incorrect. Wrong-input twin: divided by the recovery rate instead of LGD, $0.03738/0.20 = 18.69\%$ for Q (with P at 7.08% becoming 4.72%).

LO: CR-12 c: Calculate the probability of default using credit spreads. CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. • **Reference:** Hull, OFOD 11e, Ch.24, 12.1; Hull, RMFI 6e, Ch.17, 9.6.1 eq. 9.3, 9.7, Table 9.5

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 n

P2 CONCEPT

Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model.

Workbook: A5 Merton Model, Distance to Default and Risk-Neutral PD • Tier A (revision notes; A5.6 worked as Question CR-5 e.8)

WHAT IS TESTED HERE

Why P2. The Master Map index lists this LO as P2 but has no detail entry for it; the wording here is GARP's and Schweser's. The Tier A section A5 tags it, and Vardeez expects Merton questions in this chapter. Assessing model PDs is a matter of empirical evidence, so the LO is CONCEPT (approved judgment call).

Vardeez: Merton needs liquid, publicly traded equity, maps distance to default through the normal distribution and puts only short-term debt in the default point; Moody's KMV uses an empirical distribution of default frequencies and short-term plus half of long-term debt. The rebuilt notes add GARP's assessment: good ranking, with KMV and Kamakura converting Merton output to real-world PDs.

Reported exam shapes: none listed in the map.

What the questions look like: "which statement best assesses" items. Wrong options claim accurate levels instead of good rankings, reverse the direction of the conversion (risk-neutral to real-world), give KMV a normal-distribution mapping, or credit market-based PDs with rating-style stability.

Worked elsewhere: Workbook A5.6 (distance to default to a rating band, with KMV's empirical mapping as a distractor) under CR-5 e; Merton calculations under CR-9 l and CR-9 m; real-world versus risk-neutral under CR-9 k.

1. What the evidence says

How well do the default probabilities from Merton's model and its extensions match actual default experience? GARP's answer is precise: **they produce a good ranking of default probabilities**, risk-neutral or real-world. That is weaker than a claim of accurate levels.

- Because the ranking is good, a **monotonic transformation** can be estimated that converts Merton's output into a good estimate of either the real-world or the risk-neutral default probability (Figure 1).
- **Moody's KMV and Kamakura** sell exactly this service: they transform the Merton default probability into a *real-world* default probability. Other organisations use Merton's model to estimate credit spreads, which are tied to risk-neutral probabilities.
- $N(-d_2)$ is in theory a **risk-neutral** probability, because it comes from an option pricing model. Using it to produce real-world PDs rests on one assumption: **the rankings of risk-neutral PDs, real-world PDs and Merton PDs are all the same.**
- Schweser's summary: the three models give "reliable and reasonably accurate rankings"; KMV and Kamakura restate the risk-neutral Merton output as a real-world probability.

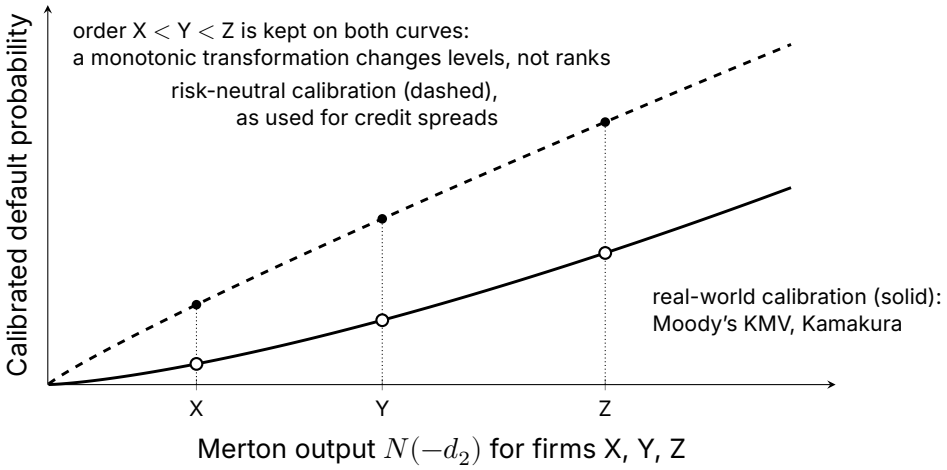


Figure 1. Why a good ranking is enough (schematic, no data). Each curve maps the Merton output into a probability. Filled dots: risk-neutral values; open dots: real-world values. Both maps are increasing, so the firms keep their order; the real-world map sits lower because real-world PDs are below risk-neutral ones (CR-9 k).

2. The three models side by side

	Merton	Moody's KMV (EDF)	Kamakura
Inputs	Equity value and volatility, debt due at T, r, T	Merton-type distance to default from equity data	Merton default probability
Default point	Debt due at T (in Doumpos, usually short-term debt)	Short-term debt plus half of long-term debt	–
PD mapping	Normal distribution: $N(-d_2)$	Empirical distribution of historical default frequencies	Calibrated transformation
Output	Risk-neutral in theory (real-world if μ replaces r)	Real-world	Real-world
Evidence	Good ranking	Good ranking	Good ranking
Needs	Traded equity	Traded equity	Traded equity

3. Strengths and limits of market-based PDs

- **Timeliness.** Credit ratings are revised infrequently, so equity prices can give more up-to-date information. Estimates from Moody's KMV and Kamakura do not have stability as an objective and respond more quickly to market information than ratings (CR-9 a).
- **Coverage.** All three need a traded share price: listed companies only.
- **Simplifications in the basic model** (CR-5 e): one zero-coupon bond, default only at T, constant asset volatility, and V_0, σ_V not observable (solved from E_0 and σ_E, CR-9 l). Extensions let default occur whenever assets fall below a barrier and allow debt payments at several dates.
- **Levels need calibration.** Ranking ability is what the evidence supports; the raw $N(-d_2)$ is not a real-world frequency, which is why the KMV empirical mapping and the Kamakura transformation exist.

(CreditMetrics, which Vardeez's crash notes list here, is a rating and transition-matrix model, not a Merton-based PD; see CR-5 f.)

Exam traps

- **Ranking versus level.** The evidence supports a good *ranking*; "accurate levels without calibration" is an overstatement.
- **Direction of the conversion.** Merton's $N(-d_2)$ is risk-neutral in theory; KMV and Kamakura convert it to real-world, not the reverse.
- **Different numbers, same ranking.** A Merton PD above a KMV or Kamakura PD for the same firm is expected (different measures), not proof of error.
- **KMV mapping.** Empirical default frequencies, not $N(-\text{DD})$ from the normal table (A5.6 option D).
- **Default point.** KMV uses short-term debt plus half of long-term debt; total debt understates the distance to default.
- **The calibration assumption.** Rankings of risk-neutral, real-world and Merton PDs are assumed identical; levels are not.
- **Stability swapped.** Market-based PDs are the fast-moving ones; agency ratings are the stable ones.
- **No share price.** None of the three can be run on a private firm.

Practice questions: CR-9 n

Question CR-9 n.1

Bank Q412 • BT answers P2.T6.24.16.2 (rebuilt)

A credit analyst is preparing a board report comparing the Merton model, the Moody's KMV model and the Kamakura model as sources of default probabilities. Which statement best assesses the quality of these models' default probabilities?

A. All three produce reliable and reasonably accurate rankings of default probabilities; the Merton output is in theory a risk-neutral probability, and KMV and Kamakura restate it as a real-world probability.

B. The Merton model produces real-world default probabilities directly, while KMV and Kamakura convert them into risk-neutral probabilities.

C. Only the KMV and Kamakura models rank default probabilities reliably; the Merton model's rankings are poor because it ignores market data.

D. None of the three models ranks default probabilities well enough to be useful, so ratings from agencies should be used instead.

Question CR-9 n.2

Bank Q414 • New question written for this bank (rebuilt)

A risk manager at a bank wants Merton-type default probabilities for three tasks on its listed counterparties: (1) ranking them for credit limits; (2) estimating the frequency of actual defaults for economic capital; (3) estimating credit spreads to price the default risk in derivatives. She is considering the Merton model, Moody's KMV and Kamakura. Which plan is most consistent with the evidence on these models' default probabilities?

A. Use the raw Merton $N(-d_2)$ for all three tasks, because it is a real-world probability that the evidence shows is accurate in level as well as in ranking.

B. Use any of the three for task 1; for task 2 the KMV or Kamakura output, which turns the Merton probability into a real-world one; for task 3 Merton output calibrated to credit spreads, which are linked to risk-neutral probabilities.

C. Use KMV or Kamakura for task 3, because they convert the Merton output into a risk-neutral probability, and the raw Merton output for task 2, because it is already a real-world probability.

D. Use none of the three for task 1, because a risk-neutral probability cannot rank counterparties by default risk, and take the ranking from agency ratings instead.

Question CR-9 n.3

Bank Q413 • BT answers P2.T6.24.16.3 (rebuilt)

A quantitative analyst reviews five-year default probability estimates for a technology issuer from three models using the same financial statements: Merton 2.5%, Moody's KMV 1.8%, Kamakura 2.1%. Which interpretation is most consistent with how these models are assessed?

A. The Merton figure is in theory a risk-neutral probability, whereas the KMV and Kamakura figures are restated as real-world probabilities, so a somewhat higher Merton figure does not by itself signal that any model is wrong.

B. KMV's estimate is the lowest, so it is the most accurate because it uses market data.

C. Because the three estimates differ, all three models are unreliable and should be discarded.

D. Kamakura's estimate lies between the other two, so it must be the correct probability.

Question CR-9 n.4

New • built from GARP Ch.9, 9.8.1

A model risk analyst at a Swiss bank reviews a vendor service that converts each firm's Merton default probability $N(-d_2)$ into a real-world default probability, using a monotonic transformation estimated from historical default experience. Which assumption underlies this calibration?

A. The levels of risk-neutral and real-world default probabilities are equal, so no adjustment for risk premiums is needed.

B. The rankings of risk-neutral default probabilities, real-world default probabilities and Merton default probabilities are all the same.

C. Merton default probabilities are already real-world probabilities, so the transformation only corrects estimation error.

D. Real-world default probabilities exceed risk-neutral ones, so the transformation must raise every Merton probability.

Question CR-9 n.5 New • built from GARP Ch.5, 5.6.3.1 and Ch.9, 9.8.1

A credit analyst at a Korean bank compares a plain Merton implementation with Moody's KMV Expected Default Frequency (EDF) model for listed borrowers. Which feature of the KMV model is designed to make its default probabilities more realistic in level than the plain Merton $N(-d_2)$?

A. KMV reads the probability off the standard normal distribution, $N(-\text{DD})$, but uses total debt as the default point.

B. KMV replaces the equity-based distance to default with the borrower's agency rating and a rating transition matrix.

C. KMV uses the book value of equity rather than its market value, which makes its probabilities more stable.

D. KMV maps the distance to default to a probability through an empirical distribution of historical default frequencies, and sets the default point at short-term debt plus half of long-term debt.

Question CR-9 n.6 New • built from GARP Ch.9, 9.1 and 9.8

The head of credit at an Australian bank proposes using default probabilities from Moody's KMV or Kamakura, rather than agency ratings, as an early-warning signal for the bank's corporate borrowers, some listed and some privately held. Which assessment is most accurate?

A. Not sensible: like agency ratings, these estimates aim for stability and avoid reversals, so they react slowly.

B. Sensible for every borrower, including the private firms, because the models need only accounting data.

C. Sensible for the listed borrowers: the estimates are based on equity prices, do not target stability and respond more quickly to market information than ratings; they cannot be produced for borrowers without traded equity.

D. Not sensible: the estimates are risk-neutral, so they cannot rank borrowers by credit quality.

Solutions • CR-9 n

Solution CR-9 n.1 — Answer: A

Bank Q412 • BT answers P2.T6.24.16.2

Step 1 – The evidence

Merton's model and its extensions produce a good ranking of default probabilities; Schweser calls the rankings of all three "reliable and reasonably accurate".

Step 2 – The measures

$N(-d_2)$ comes from an option pricing model, so it is in theory risk-neutral. KMV and Kamakura transform it into a real-world default probability.

$\Rightarrow$ Only A gets both the quality claim and the direction right.

Interpretation

The models share one engine (the Merton ranking); KMV and Kamakura add the calibration from that ranking to real-world default frequencies.

Why the other options are wrong

B is incorrect. Reverses the direction of the conversion.

C is incorrect. The KMV and Kamakura outputs are built on the Merton ranking, and Merton uses market data (the equity price).

D is incorrect. Overstates the weakness; the evidence supports the rankings.

LO: CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.1 Performance of the Model; SchweserNotes 2026 Book 2, LO 27.n

Solution CR-9 n.2 — Answer: B

Bank Q414 • New question written for this bank (rebuilt)

Step 1 – Task 1 needs only a ranking

The evidence is that Merton's model and its extensions produce a good ranking of default probabilities, so any of the three can order the counterparties. Being risk-neutral affects the level of $N(-d_2)$, not its ranking.

Step 2 – Task 2 needs a real-world level

Actual default frequency is a real-world quantity. $N(-d_2)$ is in theory risk-neutral; KMV and Kamakura transform it into a real-world default probability.

Step 3 – Task 3 needs a risk-neutral level

Credit spreads are closely linked to risk-neutral default probabilities. Other organisations calibrate the Merton output to spreads, using the same monotonic-transformation idea.

$\Rightarrow$ One ranking engine, two calibrations: B.

Interpretation

Ranking and level are separate questions. The evidence supports the ranking; each use then needs the level in its own measure, real-world for capital and risk-neutral for pricing (CR-9 k).

Why the other options are wrong

A is incorrect. $N(-d_2)$ is risk-neutral in theory, and the evidence supports its ranking, not its raw level.

C is incorrect. Reverses the direction: KMV and Kamakura turn the Merton output into a *real-world* probability.

D is incorrect. A risk-neutral probability still ranks counterparties; the evidence supports the

Merton ranking.

LO: CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.1; 9.7.2 Valuation vs. Scenario Analysis

Solution CR-9 n.3 — Answer: A

Bank Q413 • BT answers P2.T6.24.16.3

Step 1 – What each number measures

Merton's 2.5% is in theory a risk-neutral probability; the KMV 1.8% and Kamakura 2.1% are real-world restatements.

Step 2 – Is the gap a problem

Risk-neutral probabilities are higher than real-world ones (CR-9 k), so a higher Merton figure is expected.

$\Rightarrow$ The models are judged on ranking, which the evidence supports; A.

Interpretation

Comparing levels across measures is comparing different quantities. Model quality here means ranking quality, tested across many firms, not the size of one firm's number.

Why the other options are wrong

B is incorrect. A lower number is not evidence of accuracy.

C is incorrect. Different levels follow from different measures; the rankings remain reliable.

D is incorrect. Lying between two estimates says nothing about correctness.

LO: CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.1 and 9.8.2

Solution CR-9 n.4 — Answer: B

New • built from GARP Ch.9, 9.8.1

Step 1 – What a monotonic transformation preserves

It changes levels but keeps the order of the firms.

Step 2 – The assumption it needs

Mapping a risk-neutral Merton output into a real-world PD works only if both measures order firms the same way as the Merton output. GARP states this directly: the rankings of risk-neutral, real-world and Merton PDs are assumed to be the same.

$\Rightarrow$ B.

Interpretation

The service sells a change of level on top of a ranking that is assumed common to all three measures; that is why ranking quality is the thing to test.

Why the other options are wrong

A is incorrect. If the levels were equal there would be nothing to calibrate; risk-neutral PDs are higher.

C is incorrect. $N(-d_2)$ is risk-neutral in theory, being calculated from an option pricing model.

D is incorrect. Reversed: real-world PDs are below risk-neutral ones, so the real-world map sits lower.

LO: CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.1 and 9.8.2

Solution CR-9 n.5 — Answer: D

New • built from GARP Ch.5, 5.6.3.1 and Ch.9, 9.8.1

Step 1 – The mapping

Plain Merton reads the default probability off the normal distribution, $N(-d_2)$. KMV uses historical data to build an empirical distribution of default frequencies and maps the distance to default into it.

Step 2 – The default point

KMV sets it at short-term debt plus half of long-term debt, which better approximates the firm's actual loan obligations.

$\Rightarrow$ Both features aim at realistic real-world levels: D.

Interpretation

This is the evidence of Section 1 in practice: the Merton distance to default ranks well, and KMV calibrates its level to observed defaults.

Why the other options are wrong

A is incorrect. Describes plain Merton's normal mapping, with the wrong default point.

B is incorrect. A true description of the wrong model: rating transition matrices belong to CreditMetrics.

C is incorrect. KMV uses market values, and stability is not one of its objectives.

LO: CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. • **Reference:** Doumpos et al., Ch.1 (GARP CR Ch.5), 5.6.3.1 and 5.6.3.2; Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.8.1

Solution CR-9 n.6 — Answer: C

New • built from GARP Ch.9, 9.1 and 9.8

Step 1 – Responsiveness

Credit ratings are revised infrequently and aim for stability. KMV and Kamakura estimates come from equity prices and other variables, have no stability objective and respond more quickly to market information.

$\Rightarrow$ As an early-warning signal they suit listed borrowers.

Step 2 – Coverage

They need a traded share price, so they cannot be produced for the privately held borrowers.

Interpretation

The two sources complement each other: ratings give a stable through-the-cycle view, market-based PDs a timely one, and private borrowers still need an internal rating (CR-9 a).

Why the other options are wrong

A is incorrect. Attributes the agencies' stability objective to the market-based models.

B is incorrect. The models need equity prices, not just accounting data.

D is incorrect. Risk-neutral probabilities still rank correctly, and KMV and Kamakura report real-world probabilities anyway.

LO: CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.1 Credit Ratings; 9.8 Using Equity Prices to Estimate Default Probabilities; 9.8.1

CR-10 • Credit Value at Risk • Vardeez: VERY IMPORTANT

CR-10 b

P2 CALC

Define and calculate Credit VaR.

Workbook: B1 Credit VaR and Granularity • Tier B • Map #10 (6/16)

WHAT IS TESTED HERE

Why P2. Tier B workbook topic (B1, Map #10, 6 of 16 sources): reported as "CVaR calculation, direct formula application" and "CVaR (1) FC".
Vardeez: two factors drive credit VaR, credit correlation and rating transitions; the worked calculation sits at CR-11 c.

Reported exam shapes

- **B1·S1** Credit VaR by direct formula: worst-case loss at the confidence level minus expected loss.
- **B1·S2** The formula-card form, and the dropped-EL trap.

What the questions look like: numerical. The loss distribution comes in one of three forms: a single obligor, a binomial table of default counts, or a worst case default rate from Vasicek's model. You also get EAD, LGD (or a recovery rate) and the PD. The question turns on whether the stem's credit VaR is the loss quantile itself or the quantile minus expected loss.

Worked elsewhere: B1.2 (granularity) under CR-11 h. Vasicek's WCDR is built in CR-10 d (P1 part).

1. What credit VaR is

GARP defines credit VaR the way market VaR is defined: **the credit loss over a certain time period that will not be exceeded with a certain confidence level**. Some models count only losses from defaults (Vasicek, Credit Risk Plus); others also count downgrades and credit spread changes (CreditMetrics; CR-10 e and f).

- **Horizon and level.** Usually one year for instruments not held for trading (market VaR: often one day). Basel uses one year and 99.9%; economic capital may use another level.
- **Tools.** Historical simulation is the main market VaR tool; credit VaR usually needs a more elaborate model.
- **Probabilities.** Real-world PDs generate the scenarios; risk-neutral estimates value the portfolio on the horizon date.
- **The two drivers** (Vardeez). *Credit correlation:* defaults are not independent. In a downturn most companies are hit, defaults rise and correlation rises, so the risk of a credit portfolio rises. Correlation changes the tail of the loss distribution, not the expected loss. *Rating transitions:* migration matrices bring in downgrade risk (CR-10 c).

2. Two definitions in the curriculum: read the stem

	Loss quantile (Hull, CR-10)	Quantile minus EL (Malz, CR-11)
Formula	$\text{CVaR}_X = q_X(L)$	$\text{CVaR}_X = q_X(L) - \text{EL}$
Also called	Worst-case loss; "credit VaR" in Hull	Unexpected loss; economic capital
With Vasicek	EAD $\times$ LGD $\times$ WCDR	EAD $\times$ LGD $\times$ (WCDR $-$ PD)
Can be negative?	No	Yes, when $q_X(L) < \text{EL}$
Stem wording	"loss that will not be exceeded"	"in excess of", "minus" expected loss; "unexpected loss"

The bank's questions state their definition, and you answer on it. The reported exam shape (B1·S1) is the Malz form, and **forgetting to subtract EL** is the classic error. When a stem is silent, B1's rule: "Credit VaR" with EL or PD supplied means quantile minus EL; a stem that reproduces Hull's $L(1-R)$ WCDR wants the loss quantile.

3. Finding the loss quantile: three shapes

(a) One obligor, or perfect correlation. The loss is all or nothing: EAD $\times$ LGD with probability PD, zero otherwise. So $q_X = \text{EAD} \times \text{LGD}$ if $\text{PD} > 1 - X$, and $q_X = 0$ if not. Malz's USD 1 billion single credit, $\pi = 2\%$, zero recovery, EL $= 20$ million: at 99% ($2\% > 1\%$) the quantile is 1,000 and Credit VaR is **980**; at 95% ($2\% < 5\%$) the quantile is 0 and Credit VaR is -20 million, a negative number.

(b) Independent defaults, n equal loans. The number of defaults is binomial(n, π), and the quantile is the *smallest* k with $P(N \le k) \ge X$. The same book as 50 loans of USD 20 million: $P(N \le 2) = 0.9216 < 0.95 \le P(N \le 3) = 0.9822$, so the 95th percentile is 3 defaults, a loss of 60 and Credit VaR $60 - 20 = 40$. With 1,000 loans: 28 defaults, Credit VaR 8. EL stays 20 throughout; only the tail shrinks (granularity, CR-11 h).

(c) A large, granular, correlated book (Vasicek). $q_X = \sum \text{EAD}_i \times \text{LGD}_i \times \text{WCDR}_i$. GARP Problem 10.10: USD 10 million of small exposures, PD 1%, recovery 40%, $\rho = 0.2$, 99.5%:

$$\text{WCDR} = N\left(\frac{-2.326 + 0.4472 \times 2.576}{0.8944}\right) = N(-1.313) = 9.46\%$$

$$q_{99.5} = 10 \times 0.6 \times 0.0946 = 0.568$$

That USD 0.568 million is Hull's credit VaR; minus EL of $10 \times 0.6 \times 0.01 = 0.06$ it is 0.508 (Malz).

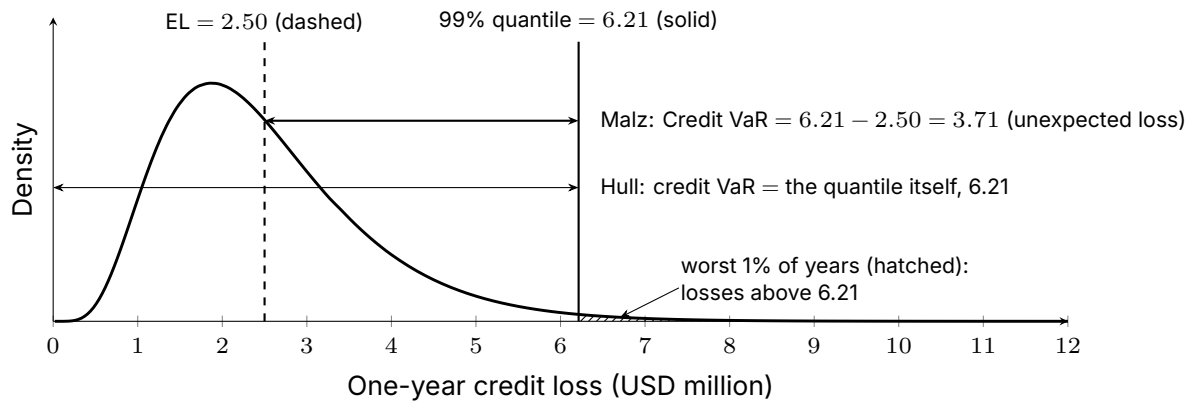


Figure 1. One-year loss distribution of a large, granular USD 100 million book from Vasicek's model (PD 5%, LGD 50%, $\rho = 0.05$). The distribution is skewed to the right. The 99% quantile (solid) is the worst-case loss; subtracting the expected loss (dashed) gives the Malz Credit VaR, the unexpected loss.

Formula card: CR-10 b

Credit VaR, Hull	$\text{CVaR}_X = q_X(L)$, the loss quantile
Credit VaR, Malz	$\text{CVaR}_X = q_X(L) - \text{EL}$, the unexpected loss
Expected loss	$\text{EL} = \sum_i \text{EAD}_i \times \text{LGD}_i \times \text{PD}_i$, $\text{LGD} = 1 - R$
One obligor	$q_X = \text{EAD} \times \text{LGD}$ if $\text{PD} > 1 - X$, otherwise 0
n equal independent loans	$q_X = k^* \times \text{loan} \times \text{LGD}$, $k^* =$ smallest k with $P(N \le k) \ge X$
Large granular book	$q_X = \sum_i \text{EAD}_i \times \text{LGD}_i \times \text{WCDR}_i(T, X)$ (Vasicek)
Critical values	$N^{-1}(0.95) = 1.645$, $N^{-1}(0.99) = 2.326$, $N^{-1}(0.995) = 2.576$, $N^{-1}(0.999) = 3.090$

Exam traps

- **Dropped EL.** Reporting the worst-case loss when the stem defines Credit VaR net of expected loss (B1 shape S2). The reverse slip, subtracting EL when the stem asks for the loss quantile, is just as costly; adding EL is never right.
- **Quantile threshold.** The 99% quantile is the smallest count whose cumulative probability is *at least* 99%, not the last row below it.
- **Recovery for LGD.** Using R in place of $1 - R$ in the quantile, in EL, or in both.
- **All-or-nothing books.** Ten positions in one obligor are one credit: compare the PD with $1-X$, and do not treat them as independent.
- **EL is not a tail measure.** Correlation and granularity change the quantile; they leave EL unchanged.
- **Negative Credit VaR is possible** under the Malz definition, when the PD is below $1 - X$ for a single credit.

Practice questions: CR-10 b

Question CR-10 b.1

Bank Q420 • AnalystPrep Q.6023 (rebuilt)

In a risk management workshop, an instructor is asked to define credit VaR as the GARP reading (Hull) does. Which definition is most accurate?

A. The average credit loss expected over the horizon, which does not depend on any confidence level.

B. The loss from adverse market price movements that will not be exceeded over a one-day horizon with a certain confidence level.

C. The total exposure that would be lost if every counterparty in the portfolio defaulted over the horizon.

D. The credit loss over a certain time period that will not be exceeded with a certain confidence level.

Question CR-10 b.2

Bank Q423 • Schweser Module Quiz 28.1 Q2 (rebuilt)

A credit portfolio manager at a regional bank expects economic growth to slow sharply next year. Which of the following trends should she expect in credit correlation and in defaults across the bank's borrowers?

A. An increase in credit correlation and an increase in defaults.

B. A decrease in credit correlation and a decrease in defaults.

C. An increase in credit correlation and a decrease in defaults.

D. A decrease in credit correlation and an increase in defaults.

Question CR-10 b.3

Bank Q424 • AP Mock 3 Q3 (rebuilt)

A credit risk analyst is reviewing a USD 1 million portfolio split into 10 equal positions of USD 100,000. All ten positions are obligations of the same obligor, which has a one-year default probability of 5%; the recovery rate is zero. Credit VaR at 99% is defined as the 99th percentile loss minus the expected loss. Which of the following is closest to the one-year 99% credit VaR?

A. USD 50,000

B. USD 250,000

C. USD 950,000

D. USD 1,000,000

Question CR-10 b.4

Bank Q421 • BT P2.T6.24.17.2 (rebuilt)

A bank holds a USD 100 million corporate loan portfolio with a one-year probability of default of 1% and a loss given default of 50%. The bank's Vasicek model estimates that, with 99% confidence, the one-year default rate will not exceed 4.0%. Credit VaR at 99% is defined as the 99th percentile credit loss in excess of the expected loss. Which of the following is closest to the credit VaR?

A. USD 500,000

B. USD 1,500,000

C. USD 2,000,000

D. USD 3,000,000

Question CR-10 b.5

Bank Q428 • Mock 2 Q56 (rebuilt)

Neo Bank holds a homogeneous loan portfolio with a total exposure at default of USD 350 million. The expected annual default rate is 2.5% and the recovery rate on defaulted loans is 60%. At the 99.9% confidence level, the worst case annual default rate (WCDR) is estimated at 10.25%. Using Hull's definition of credit VaR as the credit loss that will not be exceeded at the confidence level, which of the following is closest to the bank's one-year 99.9% credit VaR?

A. USD 3.50 million

B. USD 10.85 million

C. USD 14.35 million

D. USD 21.53 million

Question CR-10 b.6

Bank Q419 • BT P2.T6.305.1 (rebuilt)

A firm's only debt is a one-year zero-coupon bond with a face value of USD 800 million. A Merton-type model gives these results for the bond one year from now:

Expected loss (EL)	USD 8.40 million
Probability of default	10.6%
0.001 quantile of the bond's value in one year	USD 503.00 million

Credit VaR at 99.9% is defined as the 99.9th percentile loss in excess of the expected loss (the unexpected loss). Which of the following is closest to the credit VaR?

A. USD 79.25 million

B. USD 288.60 million

C. USD 297.00 million

D. USD 305.40 million

Question CR-10 b.7

Bank Q422 • GARP PE2 Q37 • Mock 4 Q38 (rebuilt)

A credit analyst at a bank estimates credit VaR (CVaR) for three loans, with CVaR defined as the 95th percentile of the unrecovered credit loss minus the expected loss.

Loan	Maturity (years)	Exposure (SGD)	Loss given default	S&P rating
S	2	55,000,000	0.8	BBB
T	3	36,000,000	0.9	BB−
U	4	50,000,000	0.7	A

Probability of default by loan maturity	2 years	3 years	4 years
Investment grade	1.5%	2.5%	3.5%
Non-investment grade	5.0%	12.0%	18.0%

Assuming the 95th percentile of the unrecovered credit loss is the same for all three loans, which comparison of the 95% CVaR is correct?

A. CVaR of Loan S $>$ CVaR of Loan U $>$ CVaR of Loan T.

B. CVaR of Loan T $>$ CVaR of Loan U $>$ CVaR of Loan S.

C. CVaR of Loan T $>$ CVaR of Loan S $>$ CVaR of Loan U.

D. CVaR of Loan U $>$ CVaR of Loan S $>$ CVaR of Loan T.

Question CR-10 b.8 Workbook B1.1 • S1, S2 • Map #10

A credit portfolio manager at a Canadian bank is computing the one-year Credit VaR of a CAD 400 million book that the bank's economic capital model treats in Malz's portfolio credit risk framework. Data:

- 40 loans of CAD 10 million each, to 40 different obligors
- One-year probability of default of each loan: 3%; recovery rate: 40%
- Defaults are assumed independent; the cumulative probability of the number of defaults is:

Number of defaults	0	1	2	3	4	5	6
Cumulative probability	0.2957	0.6615	0.8822	0.9686	0.9933	0.9988	0.9998

Which of the following is closest to the 99% Credit VaR of the book?

A. CAD 10.8 million

B. CAD 11.2 million

C. CAD 16.8 million

D. CAD 24.0 million

Solutions • CR-10 b

Solution CR-10 b.1

Answer: D

Bank Q420 • AnalystPrep Q.6023

Step 1 – The two ingredients of any VaR

A time period and a confidence level, applied to a loss distribution.

Step 2 – Apply them to credit losses

Hull: credit VaR is "the credit risk loss over a certain time period that will not be exceeded with a certain confidence level".

⇒ Only D has a credit loss, a period and a confidence level.

Interpretation

Hull's definition is the loss quantile. Malz (CR-11) subtracts the expected loss from it; when a stem uses that version it says so.

Why the other options are wrong

A is incorrect. Describes expected loss, a mean with no confidence level.

B is incorrect. Describes market risk VaR: market price moves over one day.

C is incorrect. Describes the maximum possible loss (every name defaults), a 100% confidence level rather than a chosen one.

LO: CR-10 b: Define and calculate Credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), introduction and Summary

Solution CR-10 b.2

Answer: A

Bank Q423 • Schweser Module Quiz 28.1 Q2

Step 1 – Defaults in a downturn

A slowing economy hurts most companies at once, so default rates rise.

Step 2 – Correlation in a downturn

Because the same economic factor pushes many firms towards default together, defaults cluster: credit correlation rises in stressed conditions.

⇒ Both rise, and the risk of the bank's credit portfolio rises with them.

Interpretation

GARP: the relationship between default rates and economic factors "is a major reason for credit correlation", and "if credit correlation increases (as it tends to do in stressed economic conditions), the risk for a financial institution with a portfolio of credit exposures increases".

Why the other options are wrong

B is incorrect. Describes a strong economy, where companies are favourably affected.

C is incorrect. Defaults do not fall in a downturn; the same weak economy drives both effects up.

D is incorrect. Right on defaults, wrong on correlation: the common factor makes defaults more, not less, correlated.

LO: CR-10 b: Define and calculate Credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), introduction

Solution CR-10 b.3 — Answer: C

Bank Q424 • AP Mock 3 Q3

Step 1 – The loss distribution

All ten positions are claims on one obligor, so they default together (perfect correlation): the loss is USD 1,000,000 with probability 5% and zero with probability 95%.

Step 2 – The 99th percentile

$P(\text{loss} = 0) = 95\% < 99\%$, so the 99th percentile loss is the full USD 1,000,000.
$\Rightarrow$ The 5% default probability is larger than the 1% tail, so the tail is the default.

Step 3 – Expected loss and Credit VaR

$$\text{EL} = 0.05 \times 1{,}000{,}000 = 50{,}000, \qquad \text{CVaR}_{99} = 1{,}000{,}000 - 50{,}000 = \text{USD } 950{,}000$$

Interpretation

Splitting an exposure into pieces does nothing when the pieces share one obligor: the portfolio behaves as if $n = 1$. Diversification needs different, imperfectly correlated obligors.

Why the other options are wrong

A is incorrect. Reported only the expected loss, USD 50,000.
B is incorrect. Treated the ten positions as independent credits: binomial$(10,\ 0.05)$ gives $P(N \le 2) = 0.9885 < 0.99 \le P(N \le 3) = 0.9990$, so 3 defaults, $300{,}000 - 50{,}000 = 250{,}000$. They are one obligor.
D is incorrect. Forgot to subtract the expected loss: the 99th percentile loss itself, USD 1,000,000.

LO: CR-10 b: Define and calculate Credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), introduction; Malz, Ch.8, 8.2 (CR-11 c)

Solution CR-10 b.4 — Answer: B

Bank Q421 • BT P2.T6.24.17.2

Step 1 – 99th percentile loss

$$q_{99} = 100 \times 4.0\% \times 50\% = \text{USD } 2.0 \text{ million}$$

$\Rightarrow$ In the 1-in-100 year, 4% of the book defaults and half of it is lost.

Step 2 – Expected loss

$$\text{EL} = 100 \times 1\% \times 50\% = \text{USD } 0.5 \text{ million}$$

Step 3 – Credit VaR as defined

$$\text{CVaR}_{99} = 2.0 - 0.5 = \text{USD } 1.5 \text{ million}$$

Interpretation

This is the Vasicek form of the Malz definition: EAD $\times$ LGD $\times$ (WCDR $-$ PD), the same (WCDR $-$ PD) structure as Basel IRB capital (CR-10 d).

Why the other options are wrong

A is incorrect. Reported only the expected loss, USD 0.5 million.
C is incorrect. Forgot to subtract the expected loss: USD 2.0 million.
D is incorrect. Omitted the LGD: $100 \times (4\% - 1\%) = 3.0$, as if every defaulted dollar were lost.

LO: CR-10 b: Define and calculate Credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2; Malz, Ch.8, 8.2

Solution CR-10 b.5 — Answer: C

Bank Q428 • Mock 2 Q56

Step 1 – Loss given default

$\text{LGD} = 1 - 0.60 = 0.40.$

Step 2 – Loss at the worst case default rate

$$\text{CVaR}_{99.9} = \text{WCDR} \times \text{EAD} \times \text{LGD} = 0.1025 \times 350 \times 0.40 = \text{USD } 14.35 \text{ million}$$

$\Rightarrow$ With 99.9% confidence the one-year credit loss will not exceed USD 14.35 million.

Interpretation

The stem uses Hull's definition, the loss quantile, so the expected loss of $0.025 \times 350 \times 0.40 = 3.50$ is not deducted. Had it asked for the unexpected loss, the answer would be $14.35 - 3.50 = 10.85$.

Why the other options are wrong

A is incorrect. Used the expected default rate: $0.025 \times 350 \times 0.40 = 3.50$, the expected loss.
B is incorrect. Deducted the expected loss although the stem asks for the loss quantile: $14.35 - 3.50 = 10.85$.
D is incorrect. Used the recovery rate as the LGD: $0.1025 \times 350 \times 0.60 = 21.53$.

LO: CR-10 b: Define and calculate Credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2, eq. 10.2

Solution CR-10 b.6 — Answer: B

Bank Q419 • BT P2.T6.305.1

Step 1 – 99.9th percentile loss

The 0.001 quantile of the bond's value is the worst value in 1,000: $800 - 503 = \text{USD } 297.00$ million below face.

Step 2 – Subtract the expected loss

$$\text{CVaR}_{99.9} = 297.00 - 8.40 = \text{USD } 288.60 \text{ million}$$

$\Rightarrow$ Equivalently, the expected value $800 - 8.40 = 791.60$ minus the worst-case value 503.00.

Interpretation

Credit VaR measured from the mean of the value distribution, not from face value, is the unexpected loss. The 10.6% PD is not needed; it only tempts the conditional-loss calculation in option A.

Why the other options are wrong

A is incorrect. Computed the expected loss given default, $8.40/0.106 = 79.25$, a conditional average, not a percentile.
C is incorrect. Forgot to subtract the expected loss: 297.00, the percentile loss from face.
D is incorrect. Added the expected loss instead of subtracting it: $297.00 + 8.40 = 305.40$.

LO: CR-10 b: Define and calculate Credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10); Malz, Ch.8, 8.2

Solution CR-10 b.7 — Answer: A

Bank Q422 • GARP PE2 Q37 • Mock 4 Q38

Step 1 – PD for each loan

S is BBB (investment grade), 2 years: 1.5%. T is BB− (non-investment grade), 3 years: 12.0%. U is A (investment grade), 4 years: 3.5%.

Step 2 – Expected loss = exposure × LGD × PD

$$\text{EL}_S = 55 \times 0.8 \times 0.015 = 0.660 \qquad \text{EL}_T = 36 \times 0.9 \times 0.12 = 3.888$$

$$\text{EL}_U = 50 \times 0.7 \times 0.035 = 1.225$$

$\Rightarrow$ In SGD million: T has the largest EL, then U, then S.

Step 3 – Rank CVaR = common percentile loss − EL

With the same percentile loss for all three, the smallest EL gives the largest CVaR: S > U > T.

Interpretation

CVaR moves inversely with EL when the percentile loss is held fixed. The ranking is the same if the PDs are first scaled by maturity (EL 1.32, 11.66 and 4.90), so the reading of the PD table does not matter here.

Why the other options are wrong

B is incorrect. Ranked CVaR in the order of EL (T > U > S), forgetting that EL is subtracted.
C is incorrect. Placed S below T and U below S; T has the largest EL, so its CVaR is the lowest.
D is incorrect. Placed U above S, which needs $\text{EL}_U < \text{EL}_S$; in fact $1.225 > 0.660$.

LO: CR-10 b: Define and calculate Credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10); GARP Practice Exam Part II (PE2 Q37)

Solution CR-10 b.8 — Answer: C

Workbook B1.1 • S1, S2 • Map #10

Step 1 – 99th percentile of the number of defaults

The quantile is the smallest count whose cumulative probability is at least 0.99: $P(\leq 3) = 0.9686 < 0.99$, $P(\leq 4) = 0.9933 \geq 0.99$, so the 99th percentile is 4 defaults.
$\Rightarrow$ In 99% of years the book suffers 4 or fewer defaults.

Step 2 – Worst-case loss at 99%

$$\text{Loss}_{99\%} = 4 \times 10 \times (1 - 0.40) = 4 \times 6.0 = 24.0$$

$\Rightarrow$ Each default costs CAD 6.0 million (LGD 60%), so the 99% loss quantile is CAD 24.0 million.

Step 3 – Expected loss

$$\text{EL} = 400 \times 0.03 \times 0.60 = 7.2$$

$\Rightarrow$ On average the book loses CAD 7.2 million a year, which is priced into the loans and reserved for.

Step 4 – Credit VaR

$$\text{Credit VaR}_{99\%} = \text{Loss}_{99\%} - \text{EL} = 24.0 - 7.2 = 16.8$$

Interpretation

CAD 16.8 million is the unexpected loss at 99%: the economic capital buffer beyond the provisioned EL. Making the book more granular (more, smaller independent loans) would shrink this number while leaving the CAD 7.2 million EL unchanged.

Why the other options are wrong

A is incorrect. Took the last row below 99% as the quantile: $3 \times 6.0 - 7.2 = 10.8$.

B is incorrect. Used the 40% recovery rate as the LGD in both terms: $4 \times 10 \times 0.40 - 400 \times 0.03 \times 0.40 = 16.0 - 4.8 = 11.2$.

D is incorrect. Forgot to subtract the expected loss and reported the 99% loss quantile, 24.0.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. CR-11 h: Assess the effect of granularity on Credit VaR. CR-10 b: Define and calculate Credit VaR. • **Reference:** Malz, Ch.8, 8.2 Granularity and Portfolio Credit Value-at-Risk, Table 11.1; Hull, RMFI 6e, Ch.19

CR-10 • Credit Value at Risk • Vardeez: VERY IMPORTANT

CR-10 e

P2 CONCEPT

Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation.

Workbook: D20 Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV • Tier D • Map #41 • notes only

WHAT IS TESTED HERE

Why P2. Vardeez flags it in a VERY IMPORTANT chapter: "we can expect different credit risk models comparison to show up on exam". D20 (Map #41) reports a four-way comparison built for the four-statement audit format.
Vardeez: comparison of credit risk models: know each model, not formulas.

Reported exam shapes

- **D20·S1** CreditRisk+ as the actuarial, default-mode model.
- **D20·S2** CreditMetrics as the ratings-migration, mark-to-market model.
- **D20·S3** Vasicek as the model under the Basel IRB formula.
- **D20·S5** Four-way comparison on loss recognition and correlation modelling.

What the questions look like: all conceptual: "which model" identifications, and audits in which three statements attach a true fact to the wrong model.
Worked elsewhere: Vasicek and IRB in CR-10 d (P1 part); Moody's KMV (D20 S4) in CR-5 f and CR-9 n; CreditMetrics credit VaR in CR-12 g.

1. One job, two questions that separate the models

Each model produces a probability distribution of credit losses, and credit VaR is read off it (CR-10 b). They differ on **(a) which losses are recognised** and **(b) how default correlation is modelled**. That is GARP's Practice Question 10.1, and the table is the answer.

	Approach	Losses recognised	Default correlation
Vasicek	One-factor Gaussian copula of time to default; WCDR from PD and ρ; basis of Basel IRB	Defaults only (downgrades enter IRB only through the maturity adjustment)	A single copula ρ, roughly the equity-return correlation
CreditRisk+	Actuarial (insurance) mathematics: Poisson, or negative binomial; no capital structure	Defaults only: nothing is booked until default	Uncertainty in the default rate that all obligors share
CreditMetrics	Rating transition matrix and Monte Carlo revaluation; mark-to-market	Defaults and downgrades; upgrades count as gains	Gaussian copula on rating changes, with the equity-return correlation
Moody's KMV (CR-5 f)	Merton's structural model on equity-market data; empirical distance to default to EDF	Default (the EDF at the horizon)	Not part of this comparison

2. Vasicek's model (recap of CR-10 d)

A one-factor Gaussian copula model of the time to default. It gives WCDR(T, X), a high percentile of the portfolio default rate, from the PD and one correlation parameter ρ (roughly the equity-return correlation of the obligors). It counts defaults only, carries very little tail correlation, and is the model under the Basel IRB formula.

3. Credit Risk Plus: the actuarial model

Proposed by Credit Suisse Financial Products in 1997, using "analytic approximations that are well established in the insurance industry".

- **Independent defaults.** With n loans of default probability q the number of defaults is binomial, approximately Poisson with mean qn when q is small and n large: $P(m) = \mathrm{e}^{-qn}(qn)^m/m!$ (still approximately true if the small PDs differ and q is their average).
- **Where the correlation comes from.** Default rates vary greatly from year to year. If qn is gamma-distributed with mean μ and standard deviation σ, the Poisson becomes a **negative binomial**, which returns to the Poisson as $\sigma \to 0$. Correlation arises because all companies share one default rate, high or low; there is no pairwise parameter.
- **Effect on the tail** (GARP Table 10.4, $\mu = 4$): the 99.9th percentile of the number of defaults is 11, 13, 17, 39 and 98 for $\sigma = 0$, 1, 2, 5 and 10. Without default correlation the loss distribution is fairly symmetrical; with it, positively skewed (Figure 1).
- **Also:** a Monte Carlo version samples an overall default rate, links each category to it, then samples defaults and LGDs. Doumpos (CR-5 f): unlike KMV it does not use the capital structure, and it needs minimal data.

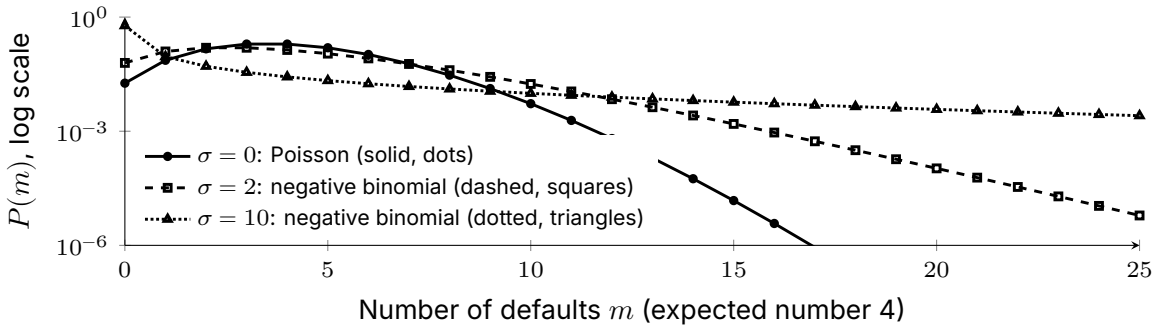


Figure 1. Credit Risk Plus with an expected 4 defaults (recomputed from eqs. 10.3 and 10.4; values match GARP Table 10.4). As the uncertainty in the default rate rises, the probability of many defaults rises by orders of magnitude: at $m = 15$ it is 0.00002 with no uncertainty, 0.0016 at $\sigma = 2$ and 0.0058 at $\sigma = 10$. The 99.9th percentiles are 11, 17 and 98 defaults.

4. CreditMetrics: the ratings-migration model

Proposed by JPMorgan in 1997 and built on a rating transition matrix (internal or agency ratings), so it captures **downgrades as well as defaults**: in Doumpos's words, "a mark-to-market model rather than a default mode model".

- **Simulation.** Each trial samples every counterparty's year-end rating. No default: revalue its transactions at one year with the spread term structure of the new rating. Default: loss = exposure $\times(1 - R)$, with R sampled from an empirical distribution. Upgrades give negative losses.

- **Correlation.** A Gaussian copula builds the joint distribution of rating changes, with the copula correlation set to the equity-return correlation. Each normal draw maps to a rating through N^{-1} of cumulative transition probabilities: an A-rated firm (AAA 0.03%, AA 1.67%, A 92.61%, ..., D 0.05%) goes to AAA if $x_A < N^{-1}(0.0003) = -3.4316$, to AA if $x_A < -2.1201$, stays A up to 1.5813, and defaults if $x_A > N^{-1}(0.9995) = 3.2905$.

Same losses, different timing. On the same assumptions CreditMetrics and Credit Risk Plus should predict the same long-run loss distribution. A loan downgraded from A to BBB in year 1, to B in year 2, and defaulting in year 3 shows no loss until year 3 under Credit Risk Plus; under CreditMetrics it shows revaluation losses in years 1, 2 and 3, which in theory add up to the same total.

Correction to the source notes. GARP (10.4.1) prints $N^{-1}(0.0003) = -3.3416$ and the AAA threshold as $x_A < -3.3416$; Schweser copies it. The digits are transposed: $N^{-1}(0.0003) = -3.4316$ (and $N(-3.3416) = 0.00042$, not 0.0003). The other A-rated thresholds (-2.1201, 1.5813, 2.6045, 2.8943, 3.1947, 3.2905) are correct.

Exam traps

- **CreditRisk+ and the capital structure.** Using the firm's capital structure and asset volatility is KMV and Merton. CreditRisk+ is an actuarial default-count model (D20 distractor 1).
- **CreditMetrics as a default-mode model.** It is mark-to-market: downgrades create losses and upgrades gains (distractor 2).
- **Vasicek and downgrades.** Defaults only; the regulatory maturity adjustment is the one channel for downgrades (distractor 3).
- **Default-rate uncertainty "changes nothing".** With the same expected number of defaults it fattens the tail: 11 defaults at the 99.9th percentile become 98 (distractor 5). As σ falls the count moves *towards* the Poisson.
- **Tail dependence.** The Gaussian copula has *little* tail correlation (distractor 6). Monte Carlo is a technique, not a fourth model.

Practice questions: CR-10 e

Question CR-10 e.1

Bank Q452 • AP Mock 2 Q9 (rebuilt)

A portfolio manager is comparing models for the probability distribution of losses from defaults and for credit correlation in a debt portfolio. How does Vasicek's model treat default losses and default correlation?

A. Losses arise only from defaults, and default events are correlated through a multiple-factor Gaussian copula.

B. Losses arise only from defaults, and default correlation is captured by a one-factor Gaussian copula with a single credit (asset) correlation parameter.

C. Defaults follow a Poisson distribution, and losses are quantified without explicitly modelling default correlation.

D. Defaults are treated as independent events, so the loss distribution is estimated separately for each borrower.

Question CR-10 e.2 Bank Q455 • AnalystPrep Q.6007 (rebuilt)

A credit risk analyst is extending a Credit Risk Plus model. Instead of fixing the default rate, she assumes that the expected number of defaults is itself gamma-distributed with mean μ and standard deviation σ. Which distribution then describes the number of defaults?

A. Poisson distribution

B. Gamma distribution

C. Negative binomial distribution

D. Normal distribution

Question CR-10 e.3 Bank Q460 • BT P2.T6.24.18.2 (rebuilt)

A bank's risk team is comparing Vasicek's model, Credit Risk Plus and CreditMetrics for estimating the one-year distribution of credit losses. Which of the following statements is most accurate?

A. In Credit Risk Plus, greater uncertainty (volatility) in the default rate raises the probability of a large number of defaults and gives a positively skewed loss distribution.

B. Vasicek's model and Credit Risk Plus both capture losses from rating downgrades as well as from defaults.

C. In Credit Risk Plus, as the volatility of the default rate decreases, the distribution of the number of defaults moves further away from the Poisson distribution.

D. CreditMetrics models the rating changes of different counterparties as independent, so no copula is needed.

Question CR-10 e.4 Bank Q453 • AnalystPrep Q.5997 (rebuilt)

A bank uses CreditMetrics to assess how the credit quality of its bond portfolio could change over the next year, including downgrades as well as defaults. Which input is most central to this model?

A. The volatility of the gamma-distributed expected number of defaults.

B. A rating transition matrix (internal or from a rating agency), with the term structure of credit spreads for each rating category.

C. The worst case default rate at the 99.9% level, computed from each bond's PD and a single correlation parameter.

D. The Poisson mean number of defaults per year for each rating bucket.

Question CR-10 e.5 Bank Q454 • AnalystPrep Q.6547 • AnalystPrep Q.5999

Which feature distinguishes CreditMetrics from both Vasicek's model and Credit Risk Plus?

A. It ignores credit rating migrations and focuses solely on default events.

B. It assumes fixed recovery rates for all obligors in the portfolio.

C. It uses a Poisson distribution to model default frequency.

D. It incorporates mark-to-market revaluations due to credit downgrades.

Question CR-10 e.6 Bank Q459 • AnalystPrep Q.6544 • AnalystPrep Q.6548

In CreditMetrics, how is the correlation between the rating transitions of two companies typically modelled?

A. By assuming independence between all rating transitions.

B. By linking each company's default probability directly to macroeconomic factors.

C. By a Gaussian copula with the correlation set equal to the correlation between the companies' equity returns.

D. Through sector-specific Poisson distributions.

Question CR-10 e.7 New • built from GARP Ch.10, 10.2 to 10.4 and Ch.5, 5.6.3 • Workbook D20 audit format • shapes D20·S1, S3

A model validation team audits four statements in a credit-model comparison memo. Which statement is correct?

A. Credit Risk Plus estimates each obligor's default probability from its capital structure and asset volatility.

B. Vasicek's model captures losses from both downgrades and defaults, which is why regulators use it for IRB capital.

C. Because it uses a Gaussian copula, Vasicek's model generates strong tail dependence between defaults.

D. In Credit Risk Plus, making the default rate uncertain while keeping the expected number of defaults unchanged raises the 99.9th percentile of the number of defaults.

Question CR-10 e.8 GARP book • Ch.10 Practice Question 10.1 (rewritten as multiple choice) • shape D20·S5

A bank holds a loan that is downgraded from A to BBB in year 1 and from BBB to B in year 2, and defaults in year 3. A risk analyst compares how three credit VaR models would treat this loan, on (a) when a credit loss is recognised and (b) how default correlation is modelled. Which of the following statements is correct?

A. Credit Risk Plus recognises no loss until the default in year 3, and it generates default correlation through uncertainty in the default rate shared by all obligors.

B. CreditMetrics recognises no loss until the default in year 3, and it generates default correlation by making the expected number of defaults gamma-distributed.

C. Vasicek's model recognises revaluation losses in years 1 and 2 through a transition matrix, and it correlates defaults through a gamma-distributed default rate.

D. CreditMetrics recognises revaluation losses in years 1, 2 and 3, but it treats the rating changes of different obligors as independent.

Solutions • CR-10 e

Solution CR-10 e.1 — Answer: B

Bank Q452 • AP Mock 2 Q9

Step 1 – Which losses

Vasicek's model estimates the distribution of losses from defaults; downgrades are not considered.

Step 2 – Which correlation structure

A one-factor Gaussian copula: one common factor plus an idiosyncratic factor per borrower, so all correlation is summarised by one parameter ρ (roughly the asset, or equity, return correlation).
$\Rightarrow$ Only B has both features.

Interpretation

The single factor is what makes the model tractable enough for regulation: each loan's worst case default rate depends only on its own PD and ρ (CR-10 d).

Why the other options are wrong

A is incorrect. Right on defaults only, but the copula is one-factor, not multi-factor.
C is incorrect. Poisson default counts describe Credit Risk Plus, which models correlation through default-rate uncertainty.
D is incorrect. Independence is the opposite of the model's premise: it exists to capture credit correlation.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2 and Summary

Solution CR-10 e.2 — Answer: C

Bank Q455 • AnalystPrep Q.6007

Step 1 – Fixed default rate

Many loans with a small, known PD give a Poisson number of defaults with mean qn.

Step 2 – Uncertain default rate

If qn is gamma-distributed (mean μ, standard deviation σ), mixing the Poisson over it gives a negative binomial: $P(m) = p^m(1-p)^\alpha\,\Gamma(m+\alpha)/(\Gamma(m+1)\Gamma(\alpha))$, with $\alpha = \mu^2/\sigma^2$ and $p = \sigma^2/(\mu+\sigma^2)$.
$\Rightarrow$ The count is negative binomial; it returns to the Poisson as $\sigma \to 0$.

Interpretation

The gamma describes the uncertain *rate*; the negative binomial describes the resulting *count*. Keeping the two apart is the whole question.

Why the other options are wrong

A is incorrect. The Poisson applies only when the default rate is fixed ($\sigma = 0$).
B is incorrect. The gamma is the distribution of the expected number of defaults, not of the number of defaults.
D is incorrect. A symmetric normal cannot produce the positively skewed, fat-tailed count that default-rate uncertainty creates.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.3, eqs. 10.3 and 10.4

Solution CR-10 e.3 — Answer: A

Bank Q460 • BT P2.T6.24.18.2

Step 1 – Credit Risk Plus and default-rate uncertainty

A larger σ means all obligors share a default rate that may turn out high: defaults cluster, large default counts become more likely, and the loss distribution becomes positively skewed (Table 10.4: the 99.9th percentile rises from 11 to 98 defaults).
$\Rightarrow$ A is correct.

Step 2 – Check the others

Only CreditMetrics captures downgrades (B false). Lower σ moves the negative binomial *towards* the Poisson (C false). CreditMetrics correlates rating changes with a Gaussian copula (D false).

Interpretation

In Credit Risk Plus, default correlation *is* the uncertainty in the common default rate: with no uncertainty there is no correlation and the distribution is fairly symmetrical.

Why the other options are wrong

B is incorrect. Scope error: Vasicek and Credit Risk Plus count defaults only.
C is incorrect. Polarity flip: as $\sigma \to 0$ the negative binomial converges to the Poisson.
D is incorrect. CreditMetrics explicitly correlates rating transitions with a Gaussian copula, using equity-return correlations.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.3 and 10.4, Table 10.4, Figure 10.1

Solution CR-10 e.4 — Answer: B

Bank Q453 • AnalystPrep Q.5997

Step 1 – What CreditMetrics must model

Year-end rating outcomes for every counterparty, including downgrades short of default.

Step 2 – The inputs that do this

A rating transition matrix gives the migration probabilities; the credit spread term structure for each rating category revalues the non-defaulted positions at the horizon.
$\Rightarrow$ B names both.

Interpretation

The other three options are the inputs of the two default-only models, which is why they cannot capture a downgrade.

Why the other options are wrong

A is incorrect. The volatility of the gamma-distributed default count is a Credit Risk Plus input.
C is incorrect. WCDR from PD and a single ρ is Vasicek's model, which ignores downgrades.
D is incorrect. A Poisson mean number of defaults is again Credit Risk Plus.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.4

Solution CR-10 e.5 Answer: D

Bank Q454 • AnalystPrep Q.6547 • AnalystPrep Q.5999

Step 1 – What the other two count

GARP: "Vasicek's model and Credit Risk Plus estimate the probability distribution of losses arising from defaults. The impact of downgrades is not considered."

Step 2 – What CreditMetrics adds

A downgrade lowers the revalued position at the horizon and creates a loss without a default; an upgrade creates a gain.
$\Rightarrow$ The distinguishing feature is mark-to-market revaluation for rating changes.

Interpretation

This is the "mark-to-market versus default mode" distinction in Doumpos (CR-5 f).

Why the other options are wrong

A is incorrect. Reversed: ignoring migrations describes Vasicek and Credit Risk Plus.
B is incorrect. CreditMetrics samples recovery rates from an empirical distribution; fixed recovery is not its distinguishing feature.
C is incorrect. A Poisson default frequency is Credit Risk Plus.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.4; Doumpos et al., GARP CR Ch.5, 5.6.3

Solution CR-10 e.6 Answer: C

Bank Q459 • AnalystPrep Q.6544 • AnalystPrep Q.6548

Step 1 – Joint rating changes

A Gaussian copula builds the joint distribution of rating changes: two correlated standard normals x_A and x_B are drawn on each trial.

Step 2 – The correlation and the mapping

The copula correlation is set to the correlation between the two companies' equity returns (0.2 in GARP's example). Each draw is mapped to a new rating through thresholds from the transition matrix; an A-rated firm defaults if $x_A > N^{-1}(0.9995) = 3.2905$.
$\Rightarrow$ C describes the model.

Interpretation

Equity correlation stands in for asset correlation here exactly as it does for Vasicek's ρ.

Why the other options are wrong

A is incorrect. Rating changes are explicitly not assumed independent.
B is incorrect. Macroeconomic factors are not modelled directly; they enter only through the correlation structure.
D is incorrect. Poisson counts belong to Credit Risk Plus.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.4.1 The Correlation Model, Figure 10.2

Solution CR-10 e.7 — Answer: D

New • built from GARP Ch.10, 10.2 to 10.4 and Ch.5, 5.6.3

Step 1 – Test each statement against its model

A: Doumpos says Credit Risk Plus, "in contrast to the KMV model, ...does not use the capital structure". B: Vasicek counts defaults only. C: Vasicek "incorporates very little tail correlation".
$\Rightarrow$ A, B and C are false.

Step 2 – Test D

Holding $\mu = 4$ fixed, raising σ from 0 to 10 moves the 99.9th percentile of the number of defaults from 11 to 98 (GARP Table 10.4).
$\Rightarrow$ D is correct: default-rate uncertainty is how Credit Risk Plus creates correlation and fattens the tail.

Interpretation

Each false statement borrows a true feature from another model: capital structure from KMV, downgrades from CreditMetrics, tail dependence from a fat-tailed copula. That is the D20 audit pattern.

Why the other options are wrong

A is incorrect. Sibling swap with Moody's KMV: Credit Risk Plus is an actuarial default-count model.
B is incorrect. Sibling swap with CreditMetrics: Vasicek counts defaults only (downgrades enter IRB only through the maturity adjustment).
C is incorrect. Polarity flip: the Gaussian copula has little tail dependence; another copula is needed to add it.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2 to 10.4, Table 10.4; Doumpos et al., GARP CR Ch.5, 5.6.3.3

Solution CR-10 e.8 — Answer: A

GARP book • Ch.10 Practice Question 10.1

Step 1 – (a) When is the loss recognised?

Credit Risk Plus and Vasicek: only on default, so nothing until year 3. CreditMetrics: a revaluation loss at each downgrade, in years 1, 2 and 3.

Step 2 – (b) How is default correlation modelled?

Vasicek: one-factor Gaussian copula with ρ. Credit Risk Plus: uncertainty in the default rate common to all obligors. CreditMetrics: Gaussian copula on rating changes with equity-return correlations.
$\Rightarrow$ A is correct on both counts.

Interpretation

GARP's answer to 10.1 in one line: the models differ in the *timing* of losses (CreditMetrics and Credit Risk Plus should agree on the long-run total) and in the *mechanism* that correlates defaults.

Why the other options are wrong

B is incorrect. Gives CreditMetrics both Credit Risk Plus features: it books downgrade losses as they happen and uses a copula, not a gamma-distributed default rate.
C is incorrect. Vasicek has no transition matrix and no gamma-distributed rate: defaults only, one-factor copula.
D is incorrect. Right on CreditMetrics' loss timing, wrong on correlation: its rating changes are correlated through a Gaussian copula.

LO: CR-10 e: Explain the Vasicek's model, Credit Risk Plus (CreditRisk+) model, and the CreditMetrics ways of estimating the probability distribution of losses arising from defaults as well as modeling the default correlation. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.2 to 10.4; Practice Question 10.1

CR-10 • Credit Value at Risk • Vardeez: VERY IMPORTANT

CR-10 f

P2 CONCEPT

Define credit spread risk and assess its impact on calculating credit VaR.

Workbook: none homed here B1 and D5 revision notes cover credit spread risk

WHAT IS TESTED HERE

Why P2. Vardeez flags it in a VERY IMPORTANT chapter: "Credit Spread Risk can be imp too." **Vardeez:** credit spread risk.

Reported exam shapes: none listed in the map.

What the questions look like: conceptual. Why spread risk matters for trading-book credit VaR, the two weaknesses of historical simulation, the CreditMetrics-style alternative and how it brings in correlation, Hull's Example 10.1, and the constant level of risk assumption.

Worked elsewhere: Workbook D5 (credit spreads and PD from spreads): D5.1 at Question CR-12 c.4, D5.2 under CR-9 j. Credit VaR definitions: CR-10 b.

1. What credit spread risk is, and why it changes credit VaR

GARP: "The values of most credit-sensitive products in the trading book depend critically on credit spreads." **Credit spread risk** is the risk that spread changes, with or without a change of rating, change those values. A default-only model (Vasicek, Credit Risk Plus) cannot see it, yet over a 10-day trading horizon it usually dominates: defaults inside 10 days are rare, spreads move every day. VaR or ES for these products must model spread changes; regulators add jump-to-default risk separately.

2. Two ways to bring spread risk into VaR

	Historical simulation	CreditMetrics-style simulation
Method	500 days of spread changes per company; scenario i applies day i's changes to all; one-day 99% VaR $\times\sqrt{10}$	10-day transition matrix plus a spread distribution per rating; each trial samples stay, migrate or default, then a spread, and revalues
Defaults	None: survivors never defaulted in the sample. Fixes: default above a spread level, or a separate jump-to-default charge (regulators)	Included through the matrix's default column
Data	Spreads not updated daily for all names: poor history	Rating histories and spread distributions
Correlation	Built in (same-day changes)	Gaussian copula on ratings; spreads across categories (nearly) perfectly correlated

3. Hull's Example 10.1: where spread risk shows up

A two-year zero-coupon bond, principal USD 1,000, rated BB; risk-free rate 3% and spread 200 bp, so the price is $1{,}000/1.05^2 = 907.03$. Over the next month: 0.1% default (worth 400), 0.8% downgrade to B, 98.5% stay BB, 0.6% upgrade to BBB, with three equally likely spreads per rating. At the month end the bond has 1.917 years left.

Rating	Spread (bp)	Probability	Loss (USD)	Tail probability
Default		0.100%	507.03	0.100%
B	500 / 450 / 400	0.267% each	44.17 / 36.47 / 28.65	0.367 / 0.633 / 0.900%
BB	240 / 200 / 160	32.833% each	2.92 / −3.70 / −10.38	33.733% at 240 bp
BBB	120 / 100 / 80	0.200% each	−17.14 / −20.55 / −23.98	

GARP Table 10.5, recomputed: value $= 1{,}000/(1.03+s)^{1.917}$ (B at 500 bp: 862.85), loss $= 907.03 -$ value; tail probability = probability of this loss or worse.

- **Reading the VaR.** Above 99.9% confidence the VaR is 507.03 (default); between 99.633% and 99.9% it is 44.17; at 99% it is a modest 2.92.
- **The impact.** At 99% the 0.9% of default and B outcomes do not reach the 1% tail, so the VaR comes from the bond *staying BB* while its spread widens to 240 bp: pure spread risk. With a single spread per rating (BBB 100, BB 200, B 450 bp) the 99% outcome would be BB at 200 bp, a loss of −3.70: a gain. Spread risk inside the rating moves the 99% VaR from −3.70 to +2.92.

4. The constant level of risk assumption

Banks sometimes assume a portfolio is rebalanced periodically back to its initial risk. For BBB bonds: **buy and hold** for a year (outcomes from the BBB row of Table 10.1), or **rebalance monthly**, selling any bond no longer BBB and buying a BBB bond (the one-month BBB row of Table 10.3, twelve times).

- **Default:** buy and hold 0.0017; rebalanced, one default $12 \times 0.0001 \times 0.9999^{11} = 0.0012$.
- **One downgrade to BB:** buy and hold 0.0378; rebalanced $12 \times 0.0035 \times 0.9965^{11} = 0.0404$ (more than one downgrade: about 0.001).
- **Result:** rebalancing makes defaults and big downgrades less likely and small rating changes more likely. VaR and ES are generally **smaller** under the constant level of risk strategy.

> **Correction to the source notes.** GARP (10.5.1) prints the rebalanced probability of one downgrade to BB as $12 \times 0.0035 \times 0.9965^{11} = 0.04404$. The product is $0.042 \times 0.9622 = 0.0404$. Workbook B1 has 0.040; Workbook D5 and the explanation of Bank Q463 repeat 0.0440. The conclusion (above the buy-and-hold 0.0378) is unchanged.

Exam traps

- **"Spread risk is market risk, not credit VaR."** Credit VaR models may count downgrades and spread changes; for the trading book they must.
- **Historical simulation's blind spot** is default (survivors only), not correlation, which is built in. In the simulation route, correlation comes from a copula and highly correlated spreads, never from independence.
- **Default is the worst outcome, not always the VaR.** If its probability is below $1 - X$, the VaR sits further up (Example 10.1 at 99%).

- **Constant level of risk:** fewer big losses, more small ones; VaR generally smaller. Upgrades carry the lowest spreads; no change is the most likely outcome.

Practice questions: CR-10 f

Question CR-10 f.1

Bank Q466 • Schweser Module Quiz 28.2 Q3 (rebuilt)

A bond analyst is building a CreditMetrics-style VaR for a bond currently rated A. For the coming period she assigns each possible rating outcome a probability and a set of credit spreads. She will most likely assign the:

A. biggest spreads to outcomes where the bond's rating increases.

B. lowest probability to outcomes where the bond's rating increases.

C. lowest spreads to outcomes where the bond keeps its rating.

D. highest probability to outcomes where the bond keeps its rating.

Question CR-10 f.2

New • built from GARP Ch.10, introduction and 10.5

A bank's trading desk holds a portfolio of investment-grade corporate bonds. Its current credit VaR model, built for the banking book, counts only losses from default; over a 10-day horizon the probability that any of the issuers defaults is well below 1%. The CRO asks how measuring credit spread risk would change the desk's 10-day 99% VaR. Which response is correct?

A. The VaR would be unchanged, because spread changes are a market risk and credit VaR covers default losses only by definition.

B. The VaR would rise, because bond values fall when spreads widen or ratings fall, so losses appear at the 99th percentile even though no issuer defaults.

C. The VaR would fall, because spread changes diversify the default losses already captured by the model.

D. The VaR would rise only if spread changes in different rating categories were independent; with highly correlated spreads the effects cancel.

Question CR-10 f.3

Bank Q462 • AnalystPrep Q.6545 (rebuilt)

A risk manager calculates a one-day 99% VaR for trading-book bonds whose values are sensitive to credit spreads. She uses historical simulation of the last 500 days of daily credit spread changes for each issuer the bank is exposed to. Which of the following is a key limitation of this approach?

A. Because the issuers are alive today, none defaulted in the sample period, so the calculation implicitly assumes no probability of default and may understate tail risk.

B. It assumes credit spread changes follow a normal distribution, underestimating extreme movements in bond prices.

C. It fails to capture correlations between the spread changes of different issuers, overstating portfolio losses.

D. It requires historical recovery rates from past defaults, which skew the VaR.

Question CR-10 f.4

Bank Q465 • BT P2.T6.24.18.3 (rebuilt)

Three analysts discuss how credit spread risk is handled in credit VaR.

- Damian: "A historical simulation of credit spread changes is hampered because spreads are not updated daily for all companies, so the data may be of poor quality."
- Tanyth: "Because a company operating today has not defaulted in the past, a historical simulation implicitly assumes no default; a CreditMetrics-type approach using a rating transition matrix brings in downgrades and defaults."
- James: "To introduce credit correlation, the spread changes for different rating categories should be assumed uncorrelated, so that diversification is preserved."

Which analyst's statement is incorrect?

A. Damian

B. Tanyth

C. James

D. None of the above

Question CR-10 f.5

New • built from GARP Ch.10, 10.5, Example 10.1 and Table 10.5

A risk manager holds a two-year zero-coupon bond with a principal of USD 1,000, rated BB and priced at USD 907.03. Her one-month outcomes follow Hull's Example 10.1:

Outcome	Spread (bp)	Probability	Loss (USD)
Default		0.100%	507.03
Downgrade to B	500 / 450 / 400	0.267% each	44.17 / 36.47 / 28.65
Stay BB	240 / 200 / 160	32.833% each	2.92 / −3.70 / −10.38
Upgrade to BBB	120 / 100 / 80	0.200% each	−17.14 / −20.55 / −23.98

Which statement correctly assesses the role of credit spread risk in the bond's one-month 99% VaR?

A. The 99% VaR is the default loss of USD 507.03, because default is the worst outcome.

B. Spread changes within a rating category matter only below 99% confidence; at 99% and above, only migrations and default drive the VaR.

C. The 99% VaR is USD 2.92, a loss caused by the spread widening to 240 bp while the bond stays BB; with a single spread per rating, the 99% VaR would be a gain.

D. Because 98.5% of outcomes leave the rating unchanged, credit spread risk cannot affect the VaR at any confidence level above 98.5%.

Question CR-10 f.6 Bank Q463 • AnalystPrep Q.6546 (rebuilt)

A risk manager compares two strategies for a portfolio of BBB-rated bonds. Under buy-and-hold the bonds are held for one year without rebalancing. Under a constant level of risk strategy the portfolio is rebalanced monthly: any bond no longer rated BBB is sold and replaced with a BBB-rated bond. Relative to buy-and-hold, which statement best describes the effect of the constant level of risk strategy?

A. It increases the probability of large losses from defaults and reduces the probability of losses from small rating changes.

B. It has no effect on the probability of losses, because the transition matrix is the same.

C. It increases both the probability of large losses from defaults and the probability of losses from small rating changes.

D. It reduces the probability of large losses from defaults and increases the probability of losses from small rating changes.

Solutions • CR-10 f

Solution CR-10 f.1 — Answer: D

Bank Q466 • Schweser Module Quiz 28.2 Q3

Step 1 – Probabilities

In a transition matrix the diagonal dominates: an A-rated company keeps its rating over a year with probability 92.61% (and over a month with 99.35%).
⇒ The no-change outcome gets the highest probability.

Step 2 – Spreads

Spreads rank with credit quality: upgrades carry the lowest spreads, an unchanged rating intermediate spreads, downgrades the highest.

Interpretation

For an A-rated name, upgrades (to AA or AAA) total 1.70% over a year, far more than the 0.05% chance of default; the smallest single entries in the row are CCC/C (0.02%) and AAA (0.03%). So "lowest probability to upgrades" fails, and so does Schweser's own remark that default has the lowest probability.

Why the other options are wrong

A is incorrect. Reversed: the biggest spreads go with downgrades.
B is incorrect. Upgrades as a group (1.70%) are much more likely than default or the deep downgrades.
C is incorrect. The lowest spreads go with upgrades; an unchanged rating keeps an intermediate spread.

LO: CR-10 f: Define credit spread risk and assess its impact on calculating credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), Tables 10.1 and 10.3; 10.5

Solution CR-10 f.2 — Answer: B

New • built from GARP Ch.10, introduction and 10.5

Step 1 – What the current model sees

With default probability well below 1% over 10 days, the 99th percentile of a default-only loss is zero: the model reports essentially no VaR.

Step 2 – What spread risk adds

Bond values fall whenever spreads widen or ratings slip, and that happens on many days with no default. Those losses populate the 99th percentile.
⇒ Measuring spread risk raises the 10-day 99% VaR.

Interpretation

GARP: some credit VaR models consider only losses from defaults; "others consider losses from downgrades or credit spread changes as well as from defaults". The values of trading-book credit products "depend critically on credit spreads". For a trading book, a default-only model badly understates risk.

Why the other options are wrong

A is incorrect. GARP's credit VaR explicitly can include downgrade and spread losses; leaving them out does not make them disappear.
C is incorrect. Spread losses add to default losses; they do not offset them.
D is incorrect. Correlation shapes how spread losses combine; GARP introduces correlation by making spread changes highly correlated across categories, which raises VaR rather than can-

celling it.

LO: CR-10 f: Define credit spread risk and assess its impact on calculating credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), introduction and 10.5

Solution CR-10 f.3 — Answer: A

Bank Q462 • AnalystPrep Q.6545

Step 1 – How the scenarios are built

Each of the 500 scenarios applies one past day's spread changes to today's portfolio.

Step 2 – What is missing

Every issuer in today's portfolio survived the sample period, so no scenario contains a default.
⇒ The method implicitly assumes a zero probability of default and understates the tail.

Interpretation

GARP names two problems: this survivorship gap and the poor quality of spread data that is not updated daily. The fixes are to assume default when the spread passes a level, or to add jump-to-default risk separately, as regulators do.

Why the other options are wrong

B is incorrect. Historical simulation is non-parametric; it assumes no distribution.
C is incorrect. Applying the same day's changes to every issuer preserves their historical correlation.
D is incorrect. With no default scenarios, no recovery rates are used: the problem is the absence of defaults, not their recovery.

LO: CR-10 f: Define credit spread risk and assess its impact on calculating credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.5 with footnote 9

Solution CR-10 f.4 — Answer: C

Bank Q465 • BT P2.T6.24.18.3

Step 1 – Damian and Tanyth

Both state GARP's points: spreads are not updated daily for all companies (data quality), and survivors imply no default, which the transition-matrix approach fixes by sampling migrations and defaults.
⇒ Both statements are correct.

Step 2 – James

GARP introduces credit correlation in two ways: a Gaussian copula on rating changes, or spread changes for different rating categories assumed perfectly (or nearly perfectly) correlated, so that when A spreads rise the others do too.
⇒ Assuming uncorrelated spread changes is the opposite, and understates risk.

Interpretation

Diversification is exactly what a stressed credit market takes away: spreads across ratings move together, which is why the model forces them to.

Why the other options are wrong

A is incorrect. Damian's statement is one of GARP's two stated weaknesses of historical simulation.
B is incorrect. Tanyth's statement is the other weakness plus the standard remedy.
D is incorrect. James's statement is incorrect, so "none" cannot be the answer.

LO: CR-10 f: Define credit spread risk and assess its impact on calculating credit VaR. • **Reference:** Hull, RMFI 6e,

Ch.19 (GARP CR Ch.10), 10.5

Solution CR-10 f.5 — Answer: C

New • built from GARP Ch.10, 10.5, Example 10.1

Step 1 – Walk up the tail

Default 0.100%; adding the three B outcomes gives $0.100 + 3 \times 0.267 = 0.900\%$, still below 1%. The next outcome, BB at 240 bp, takes the tail to 33.733%.
$\Rightarrow$ The 99% VaR is the BB, 240 bp loss: USD 2.92.

Step 2 – Remove spread risk within ratings

With one spread per rating (B 450, BB 200, BBB 100 bp), default and B still carry 0.9%, so the 99% outcome is BB at 200 bp: a loss of -3.70.
$\Rightarrow$ Spread risk alone turns a gain of 3.70 into a loss of 2.92 at 99%.

Interpretation

GARP: above 99.9% the VaR is 507.03; between 99.633% and 99.9% it is 44.17 (B at 500 bp, itself a spread outcome); at 99% it is "a modest \$2.92". Spread dispersion inside a rating matters at every confidence level below 99.9%.

Why the other options are wrong

A is incorrect. Default has only a 0.1% probability, inside the 1% tail, so it sets the VaR only above 99.9%.
B is incorrect. Reversed: at 99% and between 99.633% and 99.9% the VaR comes from a spread level (240 bp in BB, 500 bp in B).
D is incorrect. The 98.5% of unchanged-rating outcomes are spread across three spread levels; the widest of them sets the 99% VaR.

LO: CR-10 f: Define credit spread risk and assess its impact on calculating credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.5, Example 10.1, Table 10.5

Solution CR-10 f.6 — Answer: D

Bank Q463 • AnalystPrep Q.6546

Step 1 – Defaults

Buy and hold keeps a bond that slides from BBB towards default: one-year PD 0.0017. Rebalancing sells it before it can get there, so a default needs a BBB bond to default within a month: $12 \times 0.0001 \times 0.9999^{11} = 0.0012$.
$\Rightarrow$ Large losses from defaults become less likely.

Step 2 – Small downgrades

Rebalancing gives twelve chances a year to suffer a one-notch downgrade: $12 \times 0.0035 \times 0.9965^{11} = 0.0404$, against 0.0378 under buy and hold.
$\Rightarrow$ Losses from small rating changes become more likely.

Interpretation

Fewer big losses and more small ones: VaR and ES are generally smaller under the constant level of risk strategy (GARP prints 0.04404 for the downgrade figure; see the correction box).

Why the other options are wrong

A is incorrect. Reversed: it is buy and hold that leaves bonds exposed to large downgrades and default.
B is incorrect. The per-period matrix is the same, but compounding it with rebalancing gives different outcome probabilities (0.0012 against 0.0017 for default).

C is incorrect. Right on small rating changes, wrong on defaults: their probability falls.

LO: CR-10 f: Define credit spread risk and assess its impact on calculating credit VaR. • **Reference:** Hull, RMFI 6e, Ch.19 (GARP CR Ch.10), 10.5.1 Constant Level of Risk Assumptions

CR-11 • Portfolio Credit Risk • Vardeez: VERY IMPORTANT

CR-11 a

P2 CALC

Define and calculate default correlation for credit portfolios.

Workbook: B3 Default Correlation, Joint PD and Copulas • Tier B • Map #16 (6/16)

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: "default correlation of 2 firms" and "default correlation between credits" are reported in B3 (Map #16, 6 of 16 sources), and Vardeez names this LO (his LO 29.a) as one of the three Very Important LOs of a VERY IMPORTANT chapter.
Vardeez: LOs a, c and h carry this reading; read the crash-course notes for the rest, and "do not get stuck on any formulas or notations".

Reported exam shapes

- **B3·S1** Default correlation of two firms from π_1, π_2 and the joint default probability π_{12}.
- **B3·S3** Default correlation between credits with unequal default probabilities: each credit keeps its own Bernoulli standard deviation $\sqrt{\pi_i(1-\pi_i)}$.

What the questions look like: almost all numerical. You get two default probabilities and one more number: the joint default probability (find ρ_{12}), the default correlation (find π_{12}), or another outcome such as "neither defaults" (back out π_{12} first). Equity or asset correlations in the stem are red herrings, and the dropped-term trap (B3·S5) sits among the options.

Worked elsewhere: B3.2 (joint default and "at least one defaults") under CR-11 f.

1. Two credits, four outcomes

Malz's set-up: two firms with default probabilities π_1 and π_2 over the same horizon T, and a **joint default probability** π_{12}, the probability that both default over T. Treat each firm's default as a Bernoulli variable x_i (1 for default, 0 for survival). The product $x_1 x_2$ equals 1 only when both default, and the four outcomes are distinct with probabilities that add to one:

Outcome	x_1	x_2	x_1x_2	Probability
No default	0	0	0	$1-\pi_1-\pi_2+\pi_{12}$
Firm 1 only defaults	1	0	0	$\pi_1-\pi_{12}$
Firm 2 only defaults	0	1	0	$\pi_2-\pi_{12}$
Both firms default	1	1	1	π_{12}

The four outcomes for two credits (GARP Ch.11, 11.1). "Firm 1 only" excludes the joint default, so π_{12} is taken out of π_1.

- **At least one defaults:** $\pi_1+\pi_2-\pi_{12}$, which is one minus "no default". **Exactly one defaults:** $\pi_1+\pi_2-2\pi_{12}$.
- π_{12} is "a genuinely new parameter, a primitive": it cannot be computed from π_1 and π_2 unless you posit independence, $\pi_{12}=\pi_1\pi_2$.
- Every cell must be non-negative, so $\pi_{12} \le \min(\pi_1, \pi_2)$. A joint PD equal to the smaller PD is the strongest dependence possible; with unequal PDs that still gives a default correlation below 1.

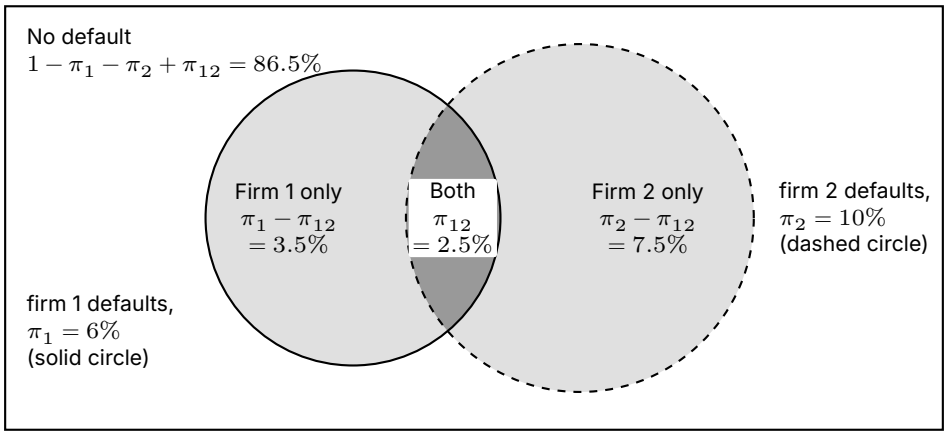


Figure 1. The four outcomes as regions (not to scale), with the numbers of Question CR-11 a.6: $\pi_1 = 6\%$, $\pi_2 = 10\%$ and 86.5% for no default give $\pi_{12} = 2.5\%$ in the dark grey overlap. Under independence the overlap would be only $0.06 \times 0.10 = 0.6\%$.

2. Default correlation

The moments of the two Bernoulli variables are

$$E[x_i] = \pi_i, \qquad \text{Var}[x_i] = \pi_i(1-\pi_i), \qquad E[x_1x_2] = \pi_{12}, \qquad \text{Cov}[x_1, x_2] = \pi_{12} - \pi_1\pi_2.$$

The default correlation is the covariance over the product of the two standard deviations (GARP eq. 11.1). Turned round, it gives the joint default probability:

$$\boxed{\rho_{12} = \frac{\pi_{12} - \pi_1\pi_2}{\sqrt{\pi_1(1-\pi_1)}\,\sqrt{\pi_2(1-\pi_2)}}} \quad \Longleftrightarrow \quad \boxed{\pi_{12} = \pi_1\pi_2 + \rho_{12}\sqrt{\pi_1(1-\pi_1)}\,\sqrt{\pi_2(1-\pi_2)}}$$

Read the second form as "the independence value plus the covariance". Most models, including the single-factor model, specify the correlation and derive the joint probability.

- **Independence:** $\pi_{12} = \pi_1\pi_2$, so the numerator and the correlation are zero. The larger the excess of π_{12} over $\pi_1\pi_2$, the higher the correlation; the relation is linear.
- **Equal PDs** ($\pi_1 = \pi_2 = \pi$): the denominator becomes $\pi(1-\pi)$ and $\rho = (\pi_{12} - \pi^2)/(\pi(1-\pi))$, the form used with the single-factor model (CR-11 f).
- **Unequal PDs (shape S3):** each credit keeps its own $\sqrt{\pi_i(1-\pi_i)}$. Dropping the $(1-\pi_i)$ factors, as in $\sqrt{\pi_1\pi_2}$, is harmless for tiny PDs and costly for high-yield ones.

Example (Schweser). A BBB+ and a BBB credit with $\pi_1 = 0.002$, $\pi_2 = 0.003$ and $\pi_{12} = 0.00015$:

$$\rho_{12} = \frac{0.00015 - 0.002 \times 0.003}{\sqrt{0.002(0.998)}\,\sqrt{0.003(0.997)}} = \frac{0.000144}{0.044677 \times 0.054690} = \frac{0.000144}{0.0024434} = 0.059$$

GARP Example 11.1 (the other direction). A BBB+ and a BBB− credit with $\pi_1 = 0.0025$ and $\pi_2 = 0.0125$. Independent: $\pi_{12} = 0.0025 \times 0.0125 = 0.000031$, under a third of a basis point. With a default correlation of 5%:

$$\pi_{12} = 0.000031 + 0.05 \times \sqrt{0.0025(0.9975)}\,\sqrt{0.0125(0.9875)} = 0.000031 + 0.05 \times 0.0055482$$

$$\pi_{12} = 0.0000313 + 0.0002774 = 0.000309$$

That is nearly ten times the independent figure and, at 3 basis points, "no longer negligible".

3. The order of magnitude

- **Hard to estimate.** Defaults are rare, so default histories say little about co-movement. Most studies find one-year default correlations of about 0.05, but estimates vary widely by period, industry and domicile, and are often negative.
- **Small, yet important.** PDs are small (around 1% for all but CCC and below), so an "optically" small correlation of 0.05 can multiply the joint default probability, as Example 11.1 shows. The regression yardstick, where 0.05 is negligible, does not apply.
- **More than two credits (CR-11 b).** With n credits there are 2^n outcomes but only $n+1+n(n-1)/2$ conditions (10 credits: 1,024 against 56), so pairwise correlations cannot fix the whole distribution. In practice all pairwise correlations are set to one non-negative parameter, which leads to the single-factor model of CR-11 d and f.

4. Three ways the inputs arrive

Given, besides π_1 and π_2	First step ($\sigma_i = \sqrt{\pi_i(1-\pi_i)}$)
Joint PD π_{12}	Straight into eq. 11.1 for ρ_{12}
Default correlation ρ_{12}	$\pi_{12} = \pi_1\pi_2 + \rho_{12}\sigma_1\sigma_2$, then any outcome probability
P(no default)	$\pi_{12} = P(\text{none}) - 1 + \pi_1 + \pi_2$
P(at least one defaults)	$\pi_{12} = \pi_1 + \pi_2 - P(\text{at least one})$
P(exactly one defaults)	$\pi_{12} = (\pi_1 + \pi_2 - P(\text{exactly one}))/2$
P(firm 1 only)	$\pi_{12} = \pi_1 - P(\text{firm 1 only})$

A conditional probability such as π_{12}/π_1 (the chance that firm 2 defaults given that firm 1 does) is not a correlation: under independence it equals π_2, not zero (Question CR-11 a.5).

Formula card: CR-11 a

Bernoulli moments	$E[x_i] = \pi_i, \quad \text{Var}[x_i] = \pi_i(1-\pi_i), \quad \text{Cov} = \pi_{12} - \pi_1\pi_2$
Default correlation	$\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)}}$
Joint default probability	$\pi_{12} = \pi_1\pi_2 + \rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)}$
Equal PDs	$\rho = \dfrac{\pi_{12} - \pi^2}{\pi(1-\pi)}, \quad \pi_{12} = \pi^2 + \rho\,\pi(1-\pi)$
Outcomes	none $1 - \pi_1 - \pi_2 + \pi_{12}$; at least one $\pi_1 + \pi_2 - \pi_{12}$; exactly one $\pi_1 + \pi_2 - 2\pi_{12}$
Bounds	$\pi_{12} \le \min(\pi_1, \pi_2)$; independence: $\pi_{12} = \pi_1\pi_2$, $\rho_{12} = 0$
Events and conditions	2^n outcomes against $n + 1 + n(n-1)/2$ conditions

Exam traps

- **Dropped term (S5).** Leaving $\pi_1\pi_2$ out of the numerator overstates ρ_{12}; leaving $+\pi_1\pi_2$ out when rebuilding π_{12} understates it.
- **Wrong denominator.** The standard deviations are $\sqrt{\pi_i(1-\pi_i)}$, not $\sqrt{\pi_i}$, and not the variances $\pi_i(1-\pi_i)$ (which give "correlations" far above 1).

- **Intermediate result.** Stopping at the covariance $\pi_{12} - \pi_1\pi_2$.
- **Conditional probability read as a correlation.** π_{12}/π_1 is a probability of default given the other's default.
- **Wrong outcome.** "At least one" is $\pi_1 + \pi_2 - \pi_{12}$; "exactly one" subtracts π_{12} twice; "both" is π_{12}.
- **Sibling swap.** An equity or asset return correlation is not the default correlation; for the same pair it is far larger (CR-11 f).
- **Joint PD above the smaller PD.** Impossible; an option like that is the "perfect dependence" distractor.

Practice questions: CR-11 a

Question CR-11 a.1

Bank Q486 • Schweser Module Quiz 29.1 Q1

Which of the following best defines the default correlation ρ_{12} for a two-firm credit portfolio, where π_1 and π_2 are the firms' default probabilities and π_{12} is their joint default probability?

A. $\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\sqrt{\pi_1(1-\pi_1)\,\pi_2(1-\pi_2)}}$

B. $\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\pi_1(1-\pi_1)\,\pi_2(1-\pi_2)}$

C. $\rho_{12} = \dfrac{\pi_{12}}{\sqrt{\pi_1(1-\pi_1)\,\pi_2(1-\pi_2)}}$

D. $\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\pi_1\pi_2}$

Question CR-11 a.2

Bank Q484 • Mock 1 Q2 (rebuilt: option B)

Two prime counterparties each have a one-year marginal default probability of 5.0%. Default of each is modelled as a Bernoulli event, and the default correlation between the two is 0. Which of the following is closest to the joint probability of default of the two counterparties?

A. 0.25%

B. 4.75%

C. 5.00%

D. 9.75%

Question CR-11 a.3

Bank Q485 • Mock 1 Q3

Two prime counterparties each have a one-year marginal default probability of 5.0%. Default of each is modelled as a Bernoulli event, and the default correlation between the two is 0.5. Which of the following is closest to the joint probability of default of the two counterparties?

A. 2.13%

B. 2.38%

C. 2.63%

D. 11.15%

Question CR-11 a.4

Bank Q474 • BT P2.T5.22.11.2 (rebuilt)

A credit analyst at a commercial bank reviews two loans, to Company X and Company Y, with one-year default probabilities of 10.0% and 4.0%. The bank's portfolio model puts the probability that both default within the year at 3.930%. Which of the following is nearest to the implied default correlation between the two companies?

A. 0.00

B. 0.56

C. 0.60

D. 0.67

Question CR-11 a.5

Bank Q475 • AnalystPrep Q.1832 (rebuilt)

Two credits rated BBB+ and BBB− have default probabilities of 0.3% and 0.4%, respectively, over the same horizon. If the probability that both default is 0.2%, what is their default correlation?

A. 57.59%

B. 57.74%

C. 57.94%

D. 66.67%

Question CR-11 a.6

Workbook B3.1 • S1, S3, S5 • Map #16

A credit risk analyst at a Brazilian bank is assessing concentration in two high-yield borrowers in the same region. The bank's one-year default model gives:

- Probability of default: borrower 1, 6%; borrower 2, 10%
- Probability that neither borrower defaults over the year: 86.5%
- Correlation of the two borrowers' equity returns, used by the desk as an asset-correlation proxy: 0.55

Which of the following is closest to the one-year default correlation between the two borrowers?

A. 0.019

B. 0.245

C. 0.267

D. 0.351

Solutions • CR-11 a

Solution CR-11 a.1

Answer: A

Bank Q486 • Schweser Module Quiz 29.1 Q1

Step 1 – Moments of the default indicators

Each default is Bernoulli: $E[x_i] = \pi_i$, $\text{Var}[x_i] = \pi_i(1-\pi_i)$, and $E[x_1 x_2] = \pi_{12}$ because the product is 1 only in the joint default.

Step 2 – Covariance

$\text{Cov}[x_1, x_2] = E[x_1 x_2] - E[x_1]E[x_2] = \pi_{12} - \pi_1\pi_2$.

$\Rightarrow$ The numerator is the excess of the joint PD over its independence value.

Step 3 – Correlation

Divide by the product of the standard deviations, the square root of the product of the variances: option A.

Interpretation

Every numerical question in this LO is this formula read forwards (find ρ_{12}) or backwards (find π_{12}). The three wrong options are the three classic slips.

Why the other options are wrong

B is incorrect. Divides by the product of the variances, not the standard deviations; with small PDs this produces numbers far above 1.

C is incorrect. Omits $\pi_1\pi_2$, using the joint PD instead of the covariance (the dropped-term trap).

D is incorrect. Scales by the product of the PDs, which is the independence value of π_{12}, not a product of standard deviations.

LO: CR-11 a: Define and calculate default correlation for credit portfolios. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation, eq. 11.1; SchweserNotes 2026 Book 2, LO 29.a

Solution CR-11 a.2

Answer: A

Bank Q484 • Mock 1 Q2

Step 1 – Zero correlation means zero covariance

$\rho_{12} = 0 \Rightarrow \pi_{12} - \pi_1\pi_2 = 0$: the defaults are independent.

Step 2 – Joint PD

$$\pi_{12} = \pi_1\pi_2 = 0.05 \times 0.05 = 0.0025 = 0.25\%$$

$\Rightarrow$ A one-in-400 chance that both counterparties default in the same year.

Interpretation

This is the baseline every positive correlation is measured against: the covariance term $\rho_{12}\sigma_1\sigma_2$ is added on top of 0.25% (Question CR-11 a.3).

Why the other options are wrong

B is incorrect. Took the Bernoulli variance $\pi(1-\pi) = 0.05 \times 0.95 = 4.75\%$ as if it were a probability.

C is incorrect. Used the marginal PD of 5%: the joint PD reaches the smaller marginal PD only under the strongest possible dependence.

D is incorrect. Computed the probability that at least one defaults, $0.05 + 0.05 - 0.0025 = 9.75\%$.

LO: CR-11 a: Define and calculate default correlation for credit portfolios. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation

Solution CR-11 a.3 — Answer: C

Bank Q485 • Mock 1 Q3

Step 1 – Product of the standard deviations

With equal PDs, $\sigma_1\sigma_2 = \pi(1-\pi) = 0.05 \times 0.95 = 0.0475$.

Step 2 – Covariance term

$\rho_{12}\,\sigma_1\sigma_2 = 0.5 \times 0.0475 = 0.02375$.

Step 3 – Add the independence value

$$\pi_{12} = \pi_1\pi_2 + \rho_{12}\sigma_1\sigma_2 = 0.0025 + 0.02375 = 0.02625 = 2.63\%$$

$\Rightarrow$ Over ten times the 0.25% of the independent case.

Interpretation

A default correlation of 0.5 is ten times the typical 0.05, and it pushes the joint PD to more than half of each marginal PD: when one counterparty defaults, the other defaults with probability $2.625/5 = 52.5\%$.

Why the other options are wrong

A is incorrect. Subtracted the independence term instead of adding it: $0.02375 - 0.0025 = 2.13\%$.

B is incorrect. Dropped $\pi_1\pi_2$ and reported the covariance term alone, 2.38% (shape S5).

D is incorrect. Multiplied ρ by a single standard deviation, $0.5 \times \sqrt{0.0475} + 0.0025 = 0.5 \times 0.21794 + 0.0025 = 11.15\%$, which exceeds the 5% marginal PD and is impossible.

LO: CR-11 a: Define and calculate default correlation for credit portfolios. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation, eq. 11.1

Solution CR-11 a.4 — Answer: C

Bank Q474 • BT P2.T5.22.11.2

Step 1 – Covariance

$$\pi_{12} - \pi_X\pi_Y = 0.0393 - 0.10 \times 0.04 = 0.0393 - 0.0040 = 0.0353$$

Step 2 – Standard deviations (unequal PDs)

$$\sigma_X = \sqrt{0.10 \times 0.90} = 0.30000, \qquad \sigma_Y = \sqrt{0.04 \times 0.96} = 0.19596, \qquad \sigma_X\sigma_Y = 0.058788$$

Step 3 – Default correlation

$$\rho_{XY} = \frac{0.0353}{0.058788} = 0.6005$$

$\Rightarrow$ A correlation of 0.60: joint default is almost ten times as likely as under independence (3.93% against 0.40%).

Interpretation

With PDs of 10% and 4% the $(1-\pi_i)$ factors matter: they move the answer from 0.56 to 0.60. The maximum feasible joint PD here is 4% (the smaller PD), and 3.93% is close to it.

Why the other options are wrong

A is incorrect. Zero would require $\pi_{12} = \pi_X\pi_Y = 0.40\%$.

B is incorrect. Dropped the $(1-\pi_i)$ factors: $0.0353/\sqrt{0.10 \times 0.04} = 0.0353/0.063246 = 0.56$.

D is incorrect. Did not subtract $\pi_X\pi_Y$: $0.0393/0.058788 = 0.67$ (shape S5).

LO: CR-11 a: Define and calculate default correlation for credit portfolios. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation, eq. 11.1

Solution CR-11 a.5

Answer: A

Bank Q475 • AnalystPrep Q.1832

Step 1 – Covariance

$$\pi_{12} - \pi_1\pi_2 = 0.002 - 0.003 \times 0.004 = 0.002 - 0.000012 = 0.001988$$

Step 2 – Standard deviations

$$\sigma_1 = \sqrt{0.003 \times 0.997} = 0.054690, \qquad \sigma_2 = \sqrt{0.004 \times 0.996} = 0.063119$$

$$\sigma_1\sigma_2 = 0.054690 \times 0.063119 = 0.0034520$$

Step 3 – Default correlation

$$\rho_{12} = \frac{0.001988}{0.0034520} = 0.5759 = 57.59\%$$

Interpretation

Three options sit within 0.35 percentage points, so the question rewards using the exact formula. With PDs this small the independence term and the $(1 - \pi_i)$ factors barely matter; the real trap is D, a conditional probability.

Why the other options are wrong

B is incorrect. Dropped both the $\pi_1\pi_2$ term and the $(1 - \pi_i)$ factors: $0.002/\sqrt{0.003 \times 0.004} = 0.002/0.0034641 = 57.74\%$.

C is incorrect. Dropped the $\pi_1\pi_2$ term only: $0.002/0.0034520 = 57.94\%$ (shape S5).

D is incorrect. Computed $\pi_{12}/\pi_1 = 0.002/0.003 = 66.67\%$, the probability that the second credit defaults given that the first does. It is a conditional probability, not a correlation.

LO: CR-11 a: Define and calculate default correlation for credit portfolios. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation, eq. 11.1

Solution CR-11 a.6

Answer: C

Workbook B3.1 • S1, S3, S5

Step 1 – Joint default probability from the no-default outcome

$$P(\text{neither}) = 1 - \pi_1 - \pi_2 + \pi_{12} \quad \Rightarrow \quad \pi_{12} = 0.865 - 1 + 0.06 + 0.10 = 0.025$$

$\Rightarrow$ A 2.5% chance that both borrowers default in the same year, against $0.06 \times 0.10 = 0.6\%$ under independence.

Step 2 – Covariance of the default indicators

$$\pi_{12} - \pi_1\pi_2 = 0.025 - 0.006 = 0.019$$

Step 3 – Standard deviations

$$\sqrt{\pi_1(1-\pi_1)}\,\sqrt{\pi_2(1-\pi_2)} = \sqrt{0.06(0.94)}\,\sqrt{0.10(0.90)} = 0.23749 \times 0.30000 = 0.071246$$

Step 4 – Default correlation

$$\rho_{12} = \frac{0.019}{0.071246} = 0.267$$

$\Rightarrow$ A default correlation of 0.27 is very high by Malz's yardstick (studies find about 0.05): joint default is over four times as likely as under independence.

Interpretation

The 0.55 equity correlation is a red herring: it proxies the asset return correlation, a sibling concept that is much larger than the default correlation for the same pair. For limits on joint exposure the bank needs the default correlation, 0.267.

Why the other options are wrong

A is incorrect. Stopped at the covariance, $\pi_{12} - \pi_1\pi_2 = 0.019$, without dividing by the standard deviations (intermediate result).

B is incorrect. Used $\sqrt{\pi_1\pi_2} = 0.07746$ as the denominator, dropping the $(1 - \pi_i)$ factors: $0.019/0.07746 = 0.245$.

D is incorrect. Omitted the $\pi_1\pi_2$ subtraction in the numerator: $0.025/0.071246 = 0.351$ (the formula-card dropped-term trap).

LO: CR-11 a: Define and calculate default correlation for credit portfolios. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation, eq. 11.1, Example 11.1

CR-11 • Portfolio Credit Risk • Vardeez: VERY IMPORTANT

CR-11 c

P2 CONCEPT

Assess the impact of correlation on a credit portfolio and its Credit VaR.

Workbook: B1 Credit VaR and Granularity • Tier B • Map #10 (6/16)

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: B1 (Map #10, 6 of 16 sources) reports Credit VaR as the worst-case loss minus EL, with the correlation and granularity effects "separately examinable"; Vardeez names this LO (his LO 29.c) as Very Important.
Vardeez: the Credit VaR of a perfectly correlated portfolio and of an independent one, worked side by side: same expected loss, very different tails.

Reported exam shape

- **B1·S4** Granularity effect 2: Credit VaR does not fall to zero when defaults are correlated.

What the questions look like: reasoning, not arithmetic (approved TYPE: assess the direction of the correlation effect). Which way Credit VaR moves when correlation rises, why the expected loss does not move, how a portfolio behaves at correlation 1, and what is left in a granular book when defaults are correlated. The Credit VaR arithmetic in Section 2 is there to show the direction; the calculation LO is CR-10 b.

Worked elsewhere: B1.1 (Credit VaR by formula) at Question CR-10 b.8; B1.2 (the granularity memo) under CR-11 h.

1. Credit VaR and what correlation moves

Malz: portfolio Credit VaR "is a quantile of the credit loss, minus the expected loss of the portfolio":

$$\text{Credit VaR}_\alpha = q_\alpha(\text{credit loss}) - \text{EL}, \qquad \text{EL} = \sum_i \text{EAD}_i \times \text{LGD}_i \times \pi_i$$

- EL depends only on each credit's exposure, LGD and PD, so **correlation cannot change it**. "Default correlation has a tremendous impact on portfolio risk. But it affects the volatility and extreme quantiles of loss rather than the expected loss."
- Because EL is always subtracted, Credit VaR is **negative** when the quantile falls in the no-loss region.

2. The two extremes on one portfolio

Vardeez's version of Malz's example: a USD 1,000,000 portfolio, each credit with $\pi = 2\%$ and zero recovery, so EL = USD 20,000 whatever the correlation.

- **Default correlation 1.** "The portfolio behaves as if it consisted of just one credit": all or nothing, however many positions it holds. The loss is USD 1,000,000 with probability 2% and zero otherwise. The α-quantile is the whole portfolio if $\pi > 1 - \alpha$ and zero if $\pi < 1 - \alpha$. At 99%: 980,000. At 95%: $0 - 20{,}000 = -20{,}000$.
- **Default correlation 0**, 50 credits of USD 20,000. The number of defaults is binomial$(50,\ 2\%)$: $P(\leq 2) = 92.16\%$, $P(\leq 3) = 98.22\%$, $P(\leq 4) = 99.68\%$. The 95th percentile is 3 defaults (Credit VaR $60{,}000 - 20{,}000 = 40{,}000$) and the 99th is 4 ($80{,}000 - 20{,}000 = 60{,}000$).

What the pair shows. At 99%, moving from independence to perfect correlation lifts Credit VaR from 60,000 to 980,000. At 95% it *lowers* it from $+40{,}000$ to $-20{,}000$, because the 2% default probability sits inside the 5% tail. Correlation does not move EL; it moves the tail, and Credit VaR can go either way depending on where the confidence level falls relative to the default probability. At the confidence levels used for economic capital (99% and above, with $\pi > 1 - \alpha$) more correlation means more Credit VaR.

3. Between the extremes: the single-factor model

The single-factor model (CR-11 d) sets default correlation through each credit's beta to a common market factor. Malz: "a higher correlation leads to a higher probability of either very few or very many defaults, and a lower probability of intermediate outcomes." Figure 1 puts numbers on it for the same 50-credit portfolio, with an asset correlation of 0.25 (default correlation 0.049, Malz's "typical" 0.05):

- The chance of no default rises from 36.4% to 57.5%; the chance of five or more defaults rises from 0.3% to 4.9%. The mean stays at one default (EL 20,000).
- The 99th percentile moves from 4 to 9 defaults: Credit VaR rises from 60,000 to 160,000.
- **Granular book (shape S4).** With many small credits, idiosyncratic risk diversifies away and the loss rate converges to the conditional default probability $p(m)$: "portfolio returns are driven by the market factor". Credit VaR therefore stays positive however many names are added, and it is larger the higher the beta. Only at $\beta \to 0$ does the loss converge to EL and Credit VaR to zero; at $\beta \to 1$ the loss rate is 0 or 100%. (Granularity itself is CR-11 h.)

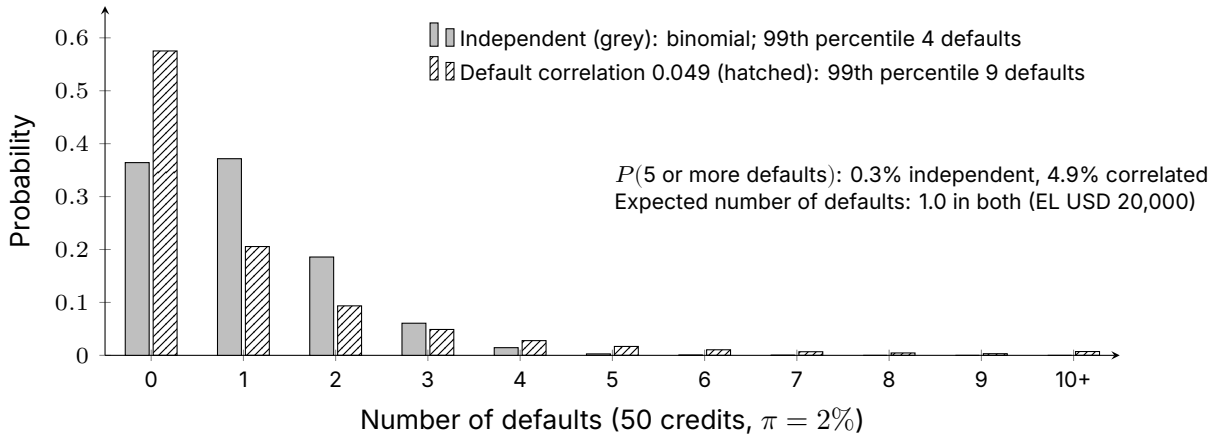


Figure 1. The same portfolio with independent defaults (grey bars) and with a single-factor asset correlation of 0.25 (hatched bars). Correlation moves probability to both ends: more years with no default, and a long right tail. The mean, and so the EL, does not change.

USD 1m, $\pi = 2\%$	Independent (50 credits)	Single factor, asset corr. 0.25	Default correlation 1
Defaults	Binomial$(50,\ 2\%)$	Binomial given m, mixed over m	0 or 50
Expected loss	20,000	20,000	20,000
P**(no default)**	36.4%	57.5%	98.0%
95% Credit VaR	3 defaults: 40,000	4 defaults: 60,000	quantile 0: $-20{,}000$
99% Credit VaR	4 defaults: 60,000	9 defaults: 160,000	total loss: 980,000
Diversification	Significant	Partial: the systematic part remains	None

Exam traps

- **Sibling swap: EL.** Correlation changes the dispersion and the tail quantiles of loss, never the expected loss.
- **Absolute claim: "correlation always raises Credit VaR".** Not at a low confidence level: with $\pi = 2\%$ the 95% Credit VaR at correlation 1 is negative.
- **Polarity at correlation 1.** The quantile is the whole portfolio when $\pi > 1 - \alpha$, zero when $\pi < 1 - \alpha$. Compare π with the tail, not with the confidence level.
- **Dropped EL or dropped sign.** Credit VaR is the quantile minus EL, so a zero quantile gives $-$EL, not zero.
- **Granular means zero Credit VaR.** Only with independent defaults; with correlation the loss rate follows the market factor.
- **Limits of the beta.** $\beta \to 1$: all or nothing; $\beta \to 0$: loss rate equal to π. Swapping the two is a polarity flip.

Practice questions: CR-11 c

Question CR-11 c.1 Bank Q492 • AnalystPrep Q.1857 (rebuilt)

In credit portfolio analysis, which of the following captures the systematic, non-diversifiable component of portfolio credit risk, that is, the tendency for defaults to cluster?

A. Default correlation

B. Credit VaR

C. Expected loss

D. Granularity

Question CR-11 c.2 Bank Q496 • AnalystPrep Q.4366

A firm owns a portfolio of credits that exhibit a default correlation of 1. In this case:

A. The number of defaults is binomially distributed because there is no correlation among the credits.

B. The number of defaults is Poisson distributed with a mean of 1 per unit time.

C. The portfolio behaves as if it consisted of just one credit.

D. Significant credit diversification may be achieved.

Question CR-11 c.3 Bank Q487 • BT P2.T6.309.2 (re-filed from CR-11 b)

According to Malz, each of the following is an important implication of default correlation in models of portfolio credit risk EXCEPT:

A. Default correlation is hard to measure or estimate from historical default data.

B. Default correlation has a tremendous impact on the credit portfolio's expected loss.

C. Default correlations are small in magnitude, so an "optically" small correlation can still have a large impact.

D. With many credits, all pairwise correlations are often set equal to a single parameter, which must be non-negative.

Question CR-11 c.4 Bank Q502 • AnalystPrep Q.5541 (rebuilt)

Portfolio A and Portfolio B each hold 100 equally sized loans with the same default probability and the same loss given default. Defaults in Portfolio A are uncorrelated (default correlation 0); defaults in Portfolio B are highly correlated. Which of the following statements is correct?

A. Both have the same expected loss, but Portfolio B has a more volatile loss distribution with a fatter right tail.

B. Portfolio B has a higher expected loss because correlated loans default more often.

C. Portfolio A has the higher volatility of loss because it has more independent sources of default.

D. Both have the same expected loss and the same loss volatility, since default correlation affects neither.

Question CR-11 c.5 Bank Q503 • AnalystPrep Q.5543 (rebuilt as a direction question)

A risk manager runs Malz's two extreme cases on a USD 1,000,000 portfolio of 50 equal credits, each with a 2% default probability and zero recovery. With a default correlation of 0 the number of defaults is binomial; its 95th percentile is 3 and its 99th percentile is 4. She then sets the default correlation to 1, leaving each credit's default probability unchanged. Which of the following statements is correct?

A. Credit VaR rises at both confidence levels, because a higher correlation always fattens the loss tail.

B. Credit VaR is unchanged at both confidence levels, because the expected loss is the same in the two cases.

C. Credit VaR rises at the 99% level but falls at the 95% level, where it becomes negative.

D. Credit VaR falls at the 99% level but rises at the 95% level, because the portfolio now behaves as a single credit.

Question CR-11 c.6 New • built from GARP Ch.11, 11.3 • shape B1·S4

A portfolio manager at a Dutch bank runs a very large, homogeneous loan book: thousands of equal loans to different borrowers, each with a 2% one-year default probability and 100% loss given default. The bank models the book with Malz's single-factor model and asks how the book's 99% Credit VaR depends on the loans' common beta β to the market factor. Which of the following statements is correct?

A. A higher β raises the book's expected loss rate above 2%, which is why Credit VaR rises with β.

B. A higher β makes both very few and very many defaults more likely, so the 99% Credit VaR rises; for any $\beta > 0$ it stays positive however many loans are added.

C. As β approaches 1, the loss rate converges to 2% in every state of the economy, so Credit VaR falls towards zero.

D. Once the book is this granular the market factor no longer moves the loss rate, so Credit VaR is close to zero whatever the value of β.

Solutions • CR-11 c

Solution CR-11 c.1 — Answer: A

Bank Q492 • AnalystPrep Q.1857

Step 1 – What makes defaults cluster

Defaults cluster when credits respond to common drivers such as the state of the economy. The measure of that co-movement is the default correlation.

$\Rightarrow$ It is the systematic part of credit risk: adding more credits with the same exposure to the common driver does not remove it.

Step 2 – Check the other quantities

Credit VaR is a loss quantile minus EL, an outcome that *depends* on correlation. EL does not depend on correlation at all. Granularity describes how finely the portfolio is divided.

Interpretation

At correlation 1 there is no diversification and the portfolio behaves as one credit; at correlation 0 diversification can be significant. Correlation is the input, Credit VaR the result.

Why the other options are wrong

B is incorrect. Credit VaR is a risk measure that rises or falls with correlation; it is not the clustering itself.

C is incorrect. Expected loss is $\sum \text{EAD} \times \text{LGD} \times \pi$ and is unaffected by correlation, so it cannot capture clustering.

D is incorrect. Granularity concerns idiosyncratic concentration (how many and how large the positions are), which diversification can remove.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2 Granularity and Portfolio Credit Value-at-Risk

Solution CR-11 c.2 — Answer: C

Bank Q496 • AnalystPrep Q.4366

Step 1 – What correlation 1 means for defaults

With a default correlation of 1, the credits default together or not at all.

$\Rightarrow$ There are only two outcomes, total loss with probability π or no loss, exactly as for a single credit.

Step 2 – Consequence for diversification

Malz: "If default correlation in a portfolio of credits is equal to 1, then the portfolio behaves as if it consisted of just one credit. No credit diversification is achieved."

Interpretation

Splitting the portfolio into more positions changes nothing at correlation 1. This is also why several positions with the same obligor count as one credit (CR-11 h).

Why the other options are wrong

A is incorrect. The binomial distribution of defaults belongs to the opposite extreme, correlation 0.

B is incorrect. Nothing in perfect correlation implies a Poisson distribution; the distribution is two-point.

D is incorrect. Significant diversification is possible at correlation 0; at correlation 1 there is none.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. • **Reference:** Malz, Ch.8 (GARP

CR Ch.11), 11.2 Granularity and Portfolio Credit Value-at-Risk

Solution CR-11 c.3 Answer: B

Bank Q487 • BT P2.T6.309.2

Step 1 – What correlation does to the loss distribution

EL is the sum of exposure $\times$ LGD $\times$ PD over the credits, with no correlation term. Malz: correlation "affects the volatility and extreme quantiles of loss rather than the expected loss".
$\Rightarrow$ B is false, so it is the exception.

Step 2 – Check the true statements

Defaults are rare, so correlation is hard to estimate from default data (A). PDs are small, so default correlations are small in magnitude yet matter (C). With many credits a single common parameter is used, and it must be non-negative to keep the correlation matrix positive-definite (D).

Interpretation

The "tremendous impact" in Malz's sentence is on portfolio *risk*, the tail, not on the mean. Questions on this LO keep testing that distinction.

Why the other options are wrong

A is incorrect. True: rare defaults make default correlation hard to measure; estimates vary widely by period and industry.
C is incorrect. True: Example 11.1 shows a 5% correlation multiplying the joint PD almost tenfold.
D is incorrect. True: a single non-negative parameter avoids a correlation matrix that is not positive-definite.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. CR-11 b: Identify drawbacks in using the correlation-based credit portfolio framework. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 The Order of Magnitude of Default Correlation; 11.2

Solution CR-11 c.4 Answer: A

Bank Q502 • AnalystPrep Q.5541

Step 1 – Expected loss

Same number of loans, same size, same PD and same LGD: EL $= \sum \text{EAD} \times \text{LGD} \times \pi$ is identical for A and B.

Step 2 – Shape of the loss distribution

In B defaults cluster: years with very few defaults and years with very many become more likely, intermediate outcomes less likely.
$\Rightarrow$ B's loss is more volatile and has a fatter right tail, so its Credit VaR at high confidence is larger.

Interpretation

This is Figure 1 in words: the same mean with more weight at both ends. A bank that judged the two books by EL alone would miss the difference in capital they need.

Why the other options are wrong

B is incorrect. Clustering is not a higher default frequency: each loan's PD is unchanged, so EL is unchanged.
C is incorrect. Polarity flip: independence diversifies and lowers the volatility of loss.
D is incorrect. Right on EL, wrong on volatility: correlation raises the variance of the portfolio loss.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2; 11.3 Credit VaR Using the Single-Factor Model

Solution CR-11 c.5 — Answer: C

Bank Q503 • AnalystPrep Q.5543 (rebuilt)

Step 1 – Independent case

EL $= 1{,}000{,}000 \times 2\% = 20{,}000$. Each credit is worth 20,000, so the 95th percentile loss is $3 \times 20{,}000 = 60{,}000$ (Credit VaR 40,000) and the 99th is $4 \times 20{,}000 = 80{,}000$ (Credit VaR 60,000).

Step 2 – Correlation 1: the portfolio is one credit

All or nothing: a loss of 1,000,000 with probability 2%, otherwise zero. EL is still 20,000.
$\Rightarrow$ At 99% the 1% tail lies inside the 2% default probability: quantile 1,000,000, Credit VaR 980,000.
At 95% the 5% tail is wider than 2%: quantile 0, Credit VaR $-20{,}000$.

Step 3 – Direction

99%: up, from 60,000 to 980,000. 95%: down, from 40,000 to $-20{,}000$.

Interpretation

Correlation leaves EL alone and reshapes the tail; whether Credit VaR rises depends on where the confidence level falls relative to the default probability. Vardeez's two examples make exactly this point.

Why the other options are wrong

A is incorrect. Absolute claim: the tail is fatter at the extreme, but at 95% the quantile drops to zero, so Credit VaR falls.
B is incorrect. Sibling swap: an unchanged EL does not mean an unchanged quantile.
D is incorrect. Polarity flip: the directions are the other way round.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2 Granularity and Portfolio Credit Value-at-Risk, Table 11.1

Solution CR-11 c.6 — Answer: B

New • built from GARP Ch.11, 11.3

Step 1 – What the granular book's loss rate follows

With thousands of small loans the idiosyncratic shocks average out, and the loss rate converges to the conditional default probability $p(m) = \Phi\big((k - \beta m)/\sqrt{1-\beta^2}\big)$, with $k = \Phi^{-1}(2\%)$.
$\Rightarrow$ The market factor still moves the loss rate whenever $\beta > 0$, so the loss has a distribution and Credit VaR is positive.

Step 2 – How beta changes that distribution

A higher β makes $p(m)$ more sensitive to m: defaults "mount rapidly as business conditions deteriorate". For this book the 99% loss rate is $\Phi\big((k + \beta \times 2.326)/\sqrt{1-\beta^2}\big)$: 2% at $\beta = 0$, 8.24% at $\beta^2 = 0.10$, 15.19% at $\beta^2 = 0.25$.
$\Rightarrow$ Credit VaR as a share of the book: 0, 6.24% and 13.19%. EL stays at 2% throughout.

Interpretation

This is the second granularity effect seen from the correlation side: adding names removes idiosyncratic risk only, and the beta to the common factor sets the Credit VaR that remains.

Why the other options are wrong

A is incorrect. Sibling swap: the expected loss rate is 2% for every β; correlation reshapes the distribution around it.

C is incorrect. Polarity flip: the loss rate converges to π when $\beta \to 0$. When $\beta \to 1$ it is either 0 or 100%.
D is incorrect. Absolute claim: granularity removes the idiosyncratic part only; the loss rate still moves with m.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. CR-11 h: Assess the effect of granularity on Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Credit VaR Using the Single-Factor Model, eq. 11.2

CR-11 • Portfolio Credit Risk • Vardeez: VERY IMPORTANT

CR-11 d

P2 CALC

Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation.

Workbook: B3 Default Correlation, Joint PD and Copulas • Tier B • Map #16 (6/16)

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: B3 (Map #16) flags the conditional default probability as "separately examinable", in a chapter Vardeez rates VERY IMPORTANT.

Vardeez: skip the single-factor model here, because it is tested from later readings (the one-factor Gaussian copula, CR-12 f). The exception is the **conditional default probability**: it "can be imp", and he posts a practice question on it.

Reported exam shape

- **B3·S6** Conditional default probability given the common factor, $p(m) = \Phi((k - \beta m)/\sqrt{1-\beta^2})$.

What the questions look like: the plug-in conditional PD from π (or k), β and a value of the market factor; the conditional mean and standard deviation; reading β, k and the asset correlation off the model; and the loss-distribution step (the market factor at which a loss level is reached, and its probability). Concept items test conditional independence and the $\beta \to 0$, $\beta \to 1$ limits.

Worked elsewhere: B3.3 (one-factor Gaussian copula documentation, conditional independence) at Question CR-12 f.7; B1.2 (granularity memo) under CR-11 h.

1. The single-factor model

Each firm's (standardised) asset return is driven by one market factor m and its own shock ε_i:

$$\boxed{a_i = \beta_i\, m + \sqrt{1-\beta_i^2}\,\varepsilon_i} \qquad m \sim N(0,1), \quad \varepsilon_i \sim N(0,1), \quad \mathrm{Cov}[m,\varepsilon_i] = 0, \quad \mathrm{Cov}[\varepsilon_i,\varepsilon_j] = 0$$

- $E[a_i] = 0$ and $\mathrm{Var}[a_i] = \beta_i^2 + 1 - \beta_i^2 = 1$: each a_i is standard normal. The weight $\sqrt{1-\beta_i^2}$ on the shock is chosen to keep the variance at 1.
- β_i is the correlation of a_i with the factor (a correlation, unlike a CAPM beta); β_i^2 is the systematic share of the variance.
- The asset return correlation of any pair is $\beta_i\beta_j$ (CR-11 e). Example 11.3: a cyclical firm with $\beta_1 = 0.5$ and a defensive one with $\beta_2 = 0.1$ have asset correlation 0.05. With a common β it is β^2.
- Firm i defaults if $a_i \le k_i$, the distance to the default asset value in standard deviations. Since a_i is standard normal, $\pi_i = \Phi(k_i)$ and $k_i = \Phi^{-1}(\pi_i)$: -2.326 for 1% (GARP rounds to -2.33), -2.054 for 2%, -1.645 for 5%. The threshold depends on π only, not on β.

2. Conditioning on the market factor

Fix the factor at a value $\bar{m}$. The only random term left is the idiosyncratic shock, so

$$a_i \,|\, \bar{m} \;\sim\; N\!\left(\beta_i\bar{m},\ \sqrt{1-\beta_i^2}\right), \qquad \text{default} \iff \varepsilon_i \le \frac{k_i - \beta_i\bar{m}}{\sqrt{1-\beta_i^2}}$$

The trigger is linear and downward sloping in $\bar{m}$: in a weak economy a smaller idiosyncratic shock is enough. The conditional cumulative default probability is

$$\boxed{p(\bar{m}) = \Phi\left(\frac{k_i - \beta_i \bar{m}}{\sqrt{1-\beta_i^2}}\right)}$$

The numerator is the new distance to default given the factor; the denominator is the standard deviation left once the factor is known. Fixing $\bar{m}$ does three things: it shifts the mean of the asset return to $\beta_i \bar{m}$, it cuts the variance from 1 to $1-\beta_i^2$, and it makes different firms' defaults independent (**conditional independence**), because only the independent ε_i are left. Conditional independence is what makes the model a handy tool for portfolios.

GARP Example 11.4 ($\beta = 0.4$, $\pi = 1\%$, $k = -2.326$, conditional standard deviation $\sqrt{1-0.16} = 0.9165$):

- Modest downturn, $\bar{m} = -1$: mean -0.4; $p = \Phi((-2.326+0.4)/0.9165) = \Phi(-2.102) = 1.78\%$. A 1.93 standard deviation shock now kills the firm, against 2.33 in a stable economy.
- Severe downturn, $\bar{m} = -2.33$: mean -0.932; $p = \Phi((-2.326+0.932)/0.9165) = \Phi(-1.521) = 6.4\%$.

Vardeez's worked example ($\beta = 0.5$, $k = -2.33$, $\bar{m} = -0.5$): mean $0.5 \times (-0.5) = -0.25$, standard deviation $\sqrt{0.75} = 0.8660$, and

$$p(-0.5) = \Phi\left(\frac{-2.33 - (-0.25)}{0.8660}\right) = \Phi(-2.4018) = 0.82\%$$

Why below 1% when $\bar{m}$ is negative? Conditioning does two things that pull in opposite directions. The mean shift $\beta\bar{m}$ raises the PD when $\bar{m} < 0$. The narrower distribution lowers the PD of a threshold far out in the tail. The two balance at

$$m^* = \frac{k\left(1-\sqrt{1-\beta^2}\right)}{\beta} \qquad (-0.62 \text{ for } \beta = 0.5;\ -0.49 \text{ for } \beta = 0.4;\ -1.17 \text{ for } \beta = 0.8)$$

Only below m^* is the conditional PD above π. At $\bar{m} = -0.5 > m^* = -0.62$ it is still below 1%; at $\bar{m} = 0$ it is $\Phi(-2.33/0.866) = 0.36\%$.

> **Correction to the source notes.** GARP (11.3, "specifying a realization does three things"), SchweserNotes (LO 29.d) and the B3 revision notes say the conditional PD differs from the unconditional PD "unless either $\bar{m} = 0$ or $\beta_i = 0$". The $\bar{m} = 0$ case is wrong: with $\beta > 0$ the conditional standard deviation is below 1, so $p(0) = \Phi(k/\sqrt{1-\beta^2})$, which is below π whenever $\pi < 50\%$. For GARP's own Example 11.4 inputs ($\beta = 0.4$, $\pi = 1\%$), $p(0) = \Phi(-2.538) = 0.56\%$, not 1%. The conditional PD equals π at $m^* = k(1-\sqrt{1-\beta^2})/\beta$ (or for any $\bar{m}$ when $\beta = 0$). This, not a "mildly good state", is why Vardeez's $\bar{m} = -0.5$ example gives 0.82%.

3. The impact of correlation

Figure 1 plots $p(\bar{m})$ for three firms with the same 1% unconditional PD and different betas. A low-beta firm's curve is nearly flat: the economy barely matters to it. A high-beta firm is almost riskless in good states and above 50% by $\bar{m} = -3$. That fanning out *is* the correlation: high-beta firms default together in bad states.

- **Beta to 1** ($\beta \to 1$): $a_i = m$ for every firm, so all default when $m \le k$ and none otherwise. The loss rate of the portfolio is 0 or 100%.

- **Beta to 0** ($\beta \to 0$): $p(\bar{m}) = \pi$ in every state; defaults are independent and a granular portfolio loses π "almost certainly".
- In between, "when the correlation is high, then, for any probability of default, defaults mount rapidly as business conditions deteriorate. When the correlation is low, it takes an extremely bad economic scenario to push the probability of default high."

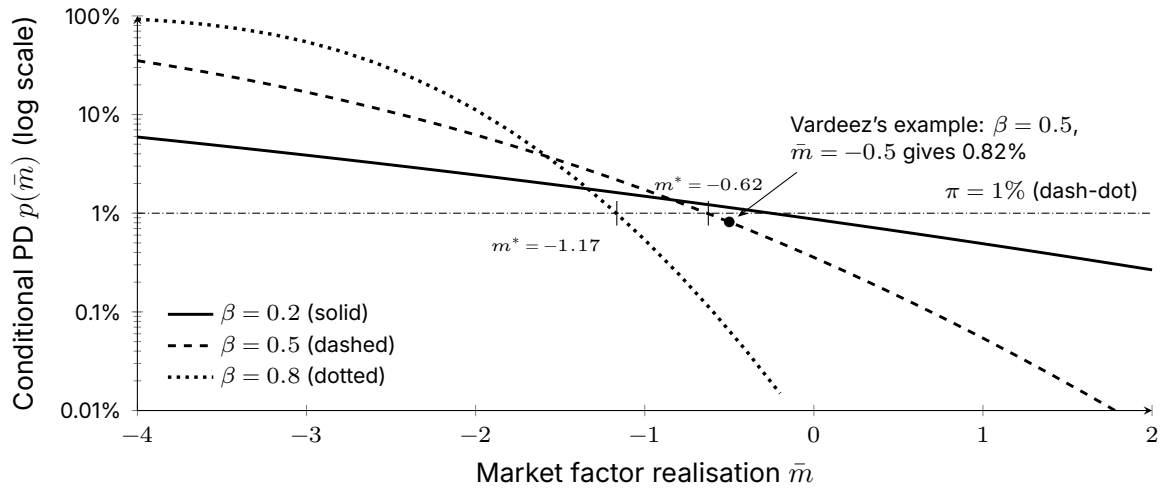


Figure 1. Conditional PD against the market factor for $k = -2.33$ ($\pi = 1\%$) and $\beta = 0.2$ (solid), 0.5 (dashed) and 0.8 (dotted). Each curve crosses the 1% line (dash-dot) at its break-even m^*, marked for $\beta = 0.5$ and 0.8; at $\bar{m} = 0$ every curve is below 1%. The higher the beta, the steeper the curve: that is how correlation enters.

4. From one firm to the portfolio loss distribution

Take a large homogeneous portfolio ($\beta_i = \beta$, $k_i = k$, loss given default 1 per name, each name a tiny share). Given $\bar{m}$ the defaults are independent, so by the law of large numbers the fraction that defaults converges to the conditional PD (GARP eq. 11.2): "if we know the realization of the market factor return, we know the level of losses realized."

$$x(\bar{m}) = p(\bar{m}) \qquad \Longrightarrow \qquad m^*(x) = \frac{k - \sqrt{1-\beta^2}\,\Phi^{-1}(x)}{\beta}, \qquad P(\text{loss} \geq x) = \Phi(m^*(x))$$

The loss rate falls as m rises, so a loss of x or more happens exactly when the factor is at or below $m^*(x)$. Repeat for each loss level x between 0 and 1 to trace the whole distribution.

- **GARP Example 11.6** ($\pi = 1\%$, $\beta = 0.5$, $x = 1\%$): $m^* = (-2.326 + 0.866 \times 2.326)/0.5 = -0.6233$, and $P(\text{loss} \geq 1\%) = \Phi(-0.6233) = 26.65\%$. Because $x = \pi$ here, m^* is the break-even of Section 2.
- **Schweser's example** ($\beta = 0.25$, $k = -2.33$): $m^* = (-2.33 + 0.9682 \times 2.33)/0.25 = -0.296$.
- The α-quantile of the loss rate sits at $\bar{m} = \Phi^{-1}(1-\alpha)$: $\Phi\big((k + \beta\,\Phi^{-1}(\alpha))/\sqrt{1-\beta^2}\big)$, Vasicek's worst-case default rate (CR-10, CR-12 g).

> **Correction to the source notes.** SchweserNotes (LO 29.d) calls the loss level $x = 0.01$ "the 99% confidence level" and asks for the market value used "to compute the probability of reaching a default threshold at the 99% confidence level". The 0.01 is a loss rate (1% of the portfolio), not a tail probability: in GARP Example 11.6 the probability of losing 1% or more is $\Phi(-0.6233) = 26.65\%$, not 1%. Schweser's step 4 ("repeated for each individual credit") should read "for each loss level", as in GARP.

Formula card: CR-11 d

Asset return	$a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i; \quad E[a_i] = 0, \mathrm{Var}[a_i] = 1$
Correlations	$\mathrm{corr}(a_i, m) = \beta_i; \quad \mathrm{corr}(a_i, a_j) = \beta_i\beta_j; \quad$ systematic share β_i^2
Default threshold	default if $a_i \le k_i; \quad \pi_i = \Phi(k_i), \quad k_i = \Phi^{-1}(\pi_i)$
Conditional distribution	$a_i \,\vert\, \bar{m} \sim N(\beta_i \bar{m}, \sqrt{1-\beta_i^2})$
Conditional PD	$p(\bar{m}) = \Phi\left(\dfrac{k_i - \beta_i \bar{m}}{\sqrt{1-\beta_i^2}}\right)$
Break-even factor	$p(m^*) = \pi$ at $m^* = k(1-\sqrt{1-\beta^2})/\beta$
Granular loss rate	$x(\bar{m}) = p(\bar{m}); \quad m^*(x) = (k - \sqrt{1-\beta^2}\,\Phi^{-1}(x))/\beta; \quad P(\text{loss} \ge x) = \Phi(m^*(x))$
Thresholds	$\Phi^{-1}(1\%) = -2.326, \quad \Phi^{-1}(2\%) = -2.054, \quad \Phi^{-1}(5\%) = -1.645$

Exam traps

- **Standard deviation, not variance.** Divide by $\sqrt{1-\beta^2}$, not by $1-\beta^2$; and not by nothing, which drops the conditioning.
- **Sign of the shift.** The numerator is $k - \beta\bar{m}$. In a downturn ($\bar{m} < 0$) it moves towards zero and the PD rises; $k + \beta\bar{m}$ treats the downturn as a boom.
- β **versus** β^2**.** β multiplies m; β^2 is the variance share and the pair correlation when betas are equal. A 70% systematic share means $\beta = \sqrt{0.70}$.
- **"Negative** m **means a higher PD."** Only below m^*; a mild negative $\bar{m}$ can still give a PD below π.
- **Loss level read as a probability.** In $m^*(x)$ the x is a loss rate; the probability of that loss or worse is $\Phi(m^*)$.
- **Forgetting to divide by** β in $m^*(x)$, or swapping β and $\sqrt{1-\beta^2}$.
- **Conditioning on the wrong variable.** The model fixes the market factor m, not ε; conditioning reduces the standard deviation.

Practice questions: CR-11 d

Question CR-11 d.1

Bank Q514 • Schweser Module Quiz 29.2 Q1 (rebuilt)

In the single-factor model $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$, where m and ε_i are independent standard normal variables, a specific value $\bar{m}$ of the market factor is realised. What are the mean and standard deviation of the conditional distribution of firm i's asset return?

A. Mean 0 and standard deviation 1, the same as the unconditional distribution.

B. Mean $\beta_i \bar{m}$ and standard deviation $\sqrt{1-\beta_i^2}$.

C. Mean $\bar{m}$ and standard deviation β_i.

D. Mean $\beta_i \bar{m}$ and standard deviation $1-\beta_i^2$.

Question CR-11 d.2

Bank Q512 • BT P2.T6.304.3 (rebuilt)

Each of the following is an assumption or implication of the single-factor credit model $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$ EXCEPT:

A. m and ε_i are independent standard normal variables.

B. Because $\text{Cov}[m, \varepsilon_i] = 0$, the asset return a_i is also standard normal.

C. The correlation between the asset returns of firms i and j is $\beta_i \beta_j$.

D. Conditioning is done by fixing the idiosyncratic shock ε_i, which increases the standard deviation of the default distribution.

Question CR-11 d.3

Bank Q516 • BT P2.T6.310.1 (re-filed from CR-11 e)

A credit analyst models three credits with the single-factor model, in which each firm's standardised asset return is $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$ with $m, \varepsilon_i \sim N(0,1)$, $\text{Cov}[m, \varepsilon_i] = 0$ and $\text{Cov}[\varepsilon_i, \varepsilon_j] = 0$ for $i \neq j$. The betas are $\beta_1 = 0.85$, $\beta_2 = 0.87$ and $\beta_3 = 0.56$. What is the implied asset return correlation between credits 1 and 2?

A. 0.260

B. 0.476

C. 0.740

D. 0.860

Question CR-11 d.4

Bank Q510 • BT P2.T6.304.1 (rebuilt)

In a single-factor model, $a = \beta m + \sqrt{1-\beta^2}\,\varepsilon$ with $m, \varepsilon \sim N(0,1)$ independent, and the firm defaults if $a < k$. A firm's systematic risk accounts for 70% of its total return variance (idiosyncratic risk 30%), and its unconditional default probability is 2.0%. What are the implied β and k?

A. $\beta = 0.55$, $k = -2.05$

B. $\beta = 0.70$, $k = -2.05$

C. $\beta = 0.84$, $k = -2.05$

D. $\beta = 0.84$, $k = +2.05$

Question CR-11 d.5

Bank Q513 • GARP PE2 Q69 • Mock 4 Q70

An analyst is assessing the credit VaR of a homogeneous portfolio using the single-factor model, in which each firm's asset return is $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\varepsilon_i$. The portfolio has 1,000 firms, each 0.1% of the portfolio; it is fully granular, so idiosyncratic shocks diversify away, and loss given default is 100% for every firm. Which of the following observations about the model and its parameters is correct?

A. When $\beta = 1$, the loss rate of the portfolio is either 0% or 100%.

B. For a given probability of default, the distance to default of a firm is low when its correlation to the market factor is low.

C. When the correlation to the market factor is zero, the portfolio loss rate is typically higher than the firm's probability of default.

D. The asset return correlation to the market factor is measured by the square root of β.

Question CR-11 d.6

Bank Q508 • BT P2.T6.310.2

In the single-factor model, a firm's conditional cumulative default probability is $p(m) = \Phi\left[(k - \beta m)/\sqrt{1-\beta^2}\right]$. A firm has $\beta = 0.60$ and $k = -1.645$, so its unconditional default probability is 5.0%. In an economic downturn the market factor shifts to $m = -1.41$. What is the conditional default probability?

A. 7.75%

B. 10.59%

C. 15.90%

D. 21.21%

Question CR-11 d.7

New • built from GARP Ch.11, 11.3 and Vardeez CR-11 e example • shape B3·S6

A credit analyst at a Malaysian bank stress-tests a manufacturing borrower with the single-factor model. The borrower's one-year unconditional default probability is 2% and its beta to the market factor is 0.60. The bank's mild-slowdown scenario sets the market factor at $m = -0.5$. The loan has a loss given default of 45%. Which of the following is closest to the borrower's default probability conditional on the scenario?

A. 0.16%

B. 1.42%

C. 2.00%

D. 3.97%

Question CR-11 d.8 Bank Q509 • BT P2.T6.311.1 (rebuilt)

In a single-factor Credit VaR model of a large, granular portfolio, a portfolio loss level x corresponds to the market factor value $m = [k - \sqrt{1-\beta^2}\,\Phi^{-1}(x)]/\beta$. The default probability is 1.0%, so $k = \Phi^{-1}(0.01) = -2.326$, and $\beta = 0.80$. Which of the following is nearest to the probability that the portfolio loss is 0.01 or worse, that is, the probability that the market factor ends at or below the value of m associated with a loss of 0.01?

A. 1.0%

B. 12.2%

C. 17.6%

D. 21.9%

Solutions • CR-11 d

Solution CR-11 d.1

Answer: B

Bank Q514 • Schweser Module Quiz 29.2 Q1

Step 1 – Fix the factor

With $m = \bar{m}$ the only random term left is ε_i: $a_i \mid \bar{m} = \beta_i \bar{m} + \sqrt{1-\beta_i^2}\,\varepsilon_i$.

Step 2 – Mean and standard deviation

$E[a_i \mid \bar{m}] = \beta_i \bar{m}$ because $E[\varepsilon_i] = 0$; the standard deviation is $\sqrt{1-\beta_i^2} \times 1 = \sqrt{1-\beta_i^2}$.
$\Rightarrow$ The conditional distribution is normal, shifted to $\beta_i \bar{m}$ and narrower than the unconditional standard normal.

Interpretation

Conditioning removes the systematic variance β_i^2 and leaves the idiosyncratic variance $1-\beta_i^2$. These two numbers are the numerator shift and the denominator of the conditional PD formula.

Why the other options are wrong

A is incorrect. This is the unconditional distribution; it ignores the information in $\bar{m}$.
C is incorrect. Mixes up the roles: the mean is $\bar{m}$ scaled by β_i, and the dispersion comes from the idiosyncratic loading $\sqrt{1-\beta_i^2}$, not from β_i.
D is incorrect. Right mean, but $1-\beta_i^2$ is the conditional variance, not the standard deviation (0.36 rather than 0.6 at $\beta = 0.8$).

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Conditional Default Distributions; SchweserNotes 2026 Book 2, LO 29.d

Solution CR-11 d.2

Answer: D

Bank Q512 • BT P2.T6.304.3

Step 1 – Check the assumptions

A: m and ε_i are standard normal with zero covariance, true. B: $\text{Var}[a_i] = \beta_i^2 + (1-\beta_i^2) = 1$ with mean 0, so a_i is standard normal, true. C: only the factor is shared, so $\text{corr}(a_i, a_j) = \beta_i \beta_j$, true.

Step 2 – Check the conditioning statement

The model conditions on the *market* factor m. Fixing it shifts the mean to $\beta_i \bar{m}$ and *reduces* the standard deviation from 1 to $\sqrt{1-\beta_i^2}$.
$\Rightarrow$ D is wrong twice: wrong variable, wrong direction. It is the exception.

Interpretation

Conditioning on the common factor is what delivers conditional independence; fixing an idiosyncratic shock would tell you nothing about the other firms.

Why the other options are wrong

A is incorrect. True: the basic assumption of the model.
B is incorrect. True: the two variances add to one.
C is incorrect. True: the pairwise asset return correlation is the product of the betas.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Conditional Default Distributions

Solution CR-11 d.3 — Answer: C

Bank Q516 • BT P2.T6.310.1

Step 1 – Unit variances

$\text{Var}[a_i] = \beta_i^2 + (1 - \beta_i^2) = 1$, so covariance and correlation coincide.

Step 2 – Only the factor is shared

$\text{Cov}[a_1, a_2] = \beta_1\beta_2\,\text{Var}[m] = \beta_1\beta_2$, because the ε's are independent of each other and of m.

Step 3 – Correlation

$$\rho_{12} = 0.85 \times 0.87 = 0.7395 \approx 0.740$$

Interpretation

Credit 3's beta of 0.56 is a red herring for this pair. Note that 0.74 is an *asset* return correlation; the default correlation of the same pair is far smaller (CR-11 f).

Why the other options are wrong

A is incorrect. Multiplied the idiosyncratic loadings, $\sqrt{1-0.85^2}\sqrt{1-0.87^2} = 0.5268 \times 0.4931 = 0.260$; those shocks are uncorrelated and contribute nothing.

B is incorrect. Used the wrong pair: $\beta_1\beta_3 = 0.85 \times 0.56 = 0.476$.

D is incorrect. Averaged the betas, $(0.85 + 0.87)/2 = 0.860$ (almost the same as the geometric mean $\sqrt{0.7395} = 0.860$); the correlation is the product.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. CR-11 e: Define beta and calculate the asset return correlation of any pair of firms using the single factor model. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3, Example 11.3

Solution CR-11 d.4 — Answer: C

Bank Q510 • BT P2.T6.304.1

Step 1 – Beta from the variance share

The systematic variance is $\beta^2 = 0.70$, so $\beta = \sqrt{0.70} = 0.837 \approx 0.84$.

$\Rightarrow$ Beta is the square root of the systematic share, not the share itself.

Step 2 – Threshold from the PD

$\pi = \Phi(k) = 0.02 \Rightarrow k = \Phi^{-1}(0.02) = -2.054 \approx -2.05$.

$\Rightarrow$ Negative, because a 2% PD lies in the left tail.

Interpretation

The threshold depends only on the PD; beta only decides how strongly the economy moves the firm towards it.

Why the other options are wrong

A is incorrect. Took the square root of the idiosyncratic share: $\sqrt{0.30} = 0.55$.

B is incorrect. Treated the 70% variance share as β itself rather than β^2.

D is incorrect. Right beta, wrong sign on the threshold: $+2.05$ would mean a 98% PD.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Conditional Default Distributions

Solution CR-11 d.5 — Answer: A

Bank Q513 • GARP PE2 Q69

Step 1 – The case $\beta = 1$

The idiosyncratic term vanishes, $a_i = m$ for every firm, so every firm defaults when $m \leq k$ and none does otherwise.

$\Rightarrow$ With 100% LGD the portfolio loses 100% or 0%: A is correct.

Step 2 – The other statements

$k = \Phi^{-1}(\pi)$ is set by the PD alone, so correlation does not change the distance to default (B false). At $\beta = 0$ the firms are independent and the granular loss rate converges to the PD, not above it (C false). β itself is the correlation with the factor (D false).

Interpretation

These are Malz's two limits: $\beta \to 1$ gives all-or-nothing outcomes, $\beta \to 0$ gives a loss rate equal to π "almost certainly". In between, higher beta means more weight on very few and very many defaults.

Why the other options are wrong

B is incorrect. The distance to default $k = \Phi^{-1}(\pi)$ does not depend on the correlation; correlation changes how the factor moves the firm towards it.

C is incorrect. With zero correlation diversification makes the loss rate equal to the PD.

D is incorrect. The correlation with the factor is β; $\beta_i\beta_j$ is the pair correlation, and $\sqrt{\beta}$ plays no role.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Credit VaR Using the Single-Factor Model

Solution CR-11 d.6 — Answer: C

Bank Q508 • BT P2.T6.310.2

Step 1 – Conditional standard deviation

$\sqrt{1-\beta^2} = \sqrt{1-0.36} = 0.80$.

Step 2 – New distance to default

$k - \beta m = -1.645 - 0.60 \times (-1.41) = -1.645 + 0.846 = -0.799$.

$\Rightarrow$ The downturn has pulled the firm's asset return 0.846 standard deviations towards the threshold.

Step 3 – Standardise and read off

$$p(-1.41) = \Phi\left(\frac{-0.799}{0.80}\right) = \Phi(-0.9988) = 15.90\%$$

$\Rightarrow$ More than three times the 5.0% unconditional PD.

Interpretation

At $m = -1.41$ the factor is far below the break-even $m^* = -1.645(1-0.8)/0.6 = -0.55$, so the mean shift dominates the narrower distribution and the PD rises sharply.

Why the other options are wrong

A is incorrect. Used $\beta^2 m$ instead of βm: $\Phi((-1.645 + 0.36 \times 1.41)/0.80) = \Phi(-1.4218) = 7.75\%$.

B is incorrect. Divided by the variance $1 - \beta^2 = 0.64$: $\Phi(-0.799/0.64) = \Phi(-1.2484) = 10.59\%$.

D is incorrect. Left out the $\sqrt{1-\beta^2}$ scaling: $\Phi(-0.799) = 21.21\%$.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Conditional Default Distributions, Example 11.4

Solution CR-11 d.7

Answer: B

New • built from GARP Ch.11, 11.3

Step 1 – Threshold and conditional standard deviation

$k = \Phi^{-1}(0.02) = -2.054; \quad \sqrt{1 - 0.60^2} = 0.80.$

Step 2 – New distance to default

$k - \beta m = -2.054 - 0.60 \times (-0.5) = -2.054 + 0.30 = -1.754.$

Step 3 – Conditional PD

$$p(-0.5) = \Phi\left(\frac{-1.754}{0.80}\right) = \Phi(-2.192) = 1.42\%$$

$\Rightarrow$ Below the 2% unconditional PD, even though the scenario is a slowdown.

Interpretation

The break-even factor is $m^* = k(1 - 0.80)/0.60 = -0.68$. The scenario $m = -0.5$ is milder than that, so the narrower conditional distribution outweighs the mean shift. This is the same effect as in Vardeez's example (0.82% against 1%). The 45% LGD is a red herring: it matters for the loss, not the probability.

Why the other options are wrong

A is incorrect. Sign error, treating the slowdown as a boom: $\Phi((-2.054 - 0.30)/0.80) = \Phi(-2.942) = 0.16\%$.

C is incorrect. Assumed the conditional PD equals the unconditional PD; that happens only at $m^* = -0.68$ (or with $\beta = 0$).

D is incorrect. Dropped the $\sqrt{1 - \beta^2}$ scaling: $\Phi(-1.754) = 3.97\%$, which wrongly confirms the intuition that a negative m must raise the PD.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Conditional Default Distributions

Solution CR-11 d.8

Answer: B

Bank Q509 • BT P2.T6.311.1

Step 1 – Inputs

$\sqrt{1 - \beta^2} = \sqrt{1 - 0.64} = 0.60; \quad \Phi^{-1}(0.01) = -2.326.$

Step 2 – Market factor for a 1% loss

$$m^* = \frac{-2.326 - 0.60 \times (-2.326)}{0.80} = \frac{-0.9304}{0.80} = -1.163$$

Step 3 – Probability

$$P(\text{loss} \geq 0.01) = \Phi(-1.163) = 12.2\%$$

$\Rightarrow$ The book loses 1% or more in about one year in eight.

Interpretation

The loss rate equals the PD of 1% only on average. With $\beta = 0.8$ the distribution is very skewed: most years lose less than 1%, a few lose much more. With $\beta = 0.5$ the same probability is 26.65% (GARP Example 11.6).

Why the other options are wrong

A is incorrect. Confused the loss level (1% of the portfolio) with the probability of reaching it.
C is incorrect. Forgot to divide by β: $\Phi(-0.9304) = 17.6\%$.
D is incorrect. Swapped β and $\sqrt{1-\beta^2}$: $m = (-2.326 + 0.80 \times 2.326)/0.60 = -0.775$, $\Phi(-0.775) = 21.9\%$.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Credit VaR Using the Single-Factor Model, eq. 11.2, Example 11.6

CR-11 • Portfolio Credit Risk • Vardeez: VERY IMPORTANT

CR-11 f

P2 CALC

Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model.

Workbook: B3 Default Correlation, Joint PD and Copulas • Tier B • Map #16 (6/16)

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: B3 (Map #16, 6 of 16 sources) reports "joint prob of default (direct calculation)" and the dropped $\pi_1\pi_2$ trap, in a chapter Vardeez rates VERY IMPORTANT. **Vardeez:** crash-course notes only. His notes skip this LO; the rebuilt notes fill it from GARP, and so do these.

Reported exam shapes

- **B3·S2** Joint probability of default, direct calculation from the PDs and the default correlation.
- **B3·S5** Dropped-term trap: omitting $\pi_1 \times \pi_2$.

What the questions look like: numerical. A joint PD from two PDs and a default correlation (often equal PDs, because the single-factor model gives every firm the same parameters), sometimes followed by "at least one defaults" or the expected loss of the joint-default scenario; a default correlation from a joint PD; and the single-factor link between the asset correlation β^2 and the default correlation, given as a model output or a table.

Worked elsewhere: B3.1 (default correlation from the no-default probability) under CR-11 a.

1. Joint default from a default correlation (recap of CR-11 a)

The two-credit formulas of CR-11 a carry all the arithmetic (Section 2 of CR-11 a):

$$\pi_{12} = \pi_1\pi_2 + \rho_{12}\sqrt{\pi_1(1-\pi_1)}\,\sqrt{\pi_2(1-\pi_2)}, \qquad \text{equal PDs: } \pi_{12} = \pi^2 + \rho\,\pi(1-\pi)$$

Example. Two credits with $\pi = 4\%$ each and $\rho = 0.10$: $\pi^2 = 0.0016$, $\rho\,\pi(1-\pi) = 0.10 \times 0.04 \times 0.96 = 0.00384$, so $\pi_{12} = 0.00544 = 0.544\%$, 3.4 times the independent 0.16%. Dropping the π^2 term (shape S5) gives 0.384% and understates the joint PD by almost 30%.

Then the other outcomes follow from the four-cell table:

Outcome ($\pi = 4\%$ each)	Independent	$\rho = 0.10$
Both default, π_{12}	0.160%	0.544%
At least one, $\pi_1 + \pi_2 - \pi_{12}$	7.840%	7.456%
Exactly one, $\pi_1 + \pi_2 - 2\pi_{12}$	7.680%	6.912%
Neither, $1 - \pi_1 - \pi_2 + \pi_{12}$	92.160%	92.544%

Positive correlation moves probability into "both" and "neither" and out of "exactly one". So a limit breached when *at least one* borrower defaults is breached slightly *less* often when defaults are correlated (Question CR-11 f.4).

2. Joint default in the single-factor model

Give every firm the same parameters: $\beta_i = \beta$, $k_i = k$, $\pi_i = \pi = \Phi(k)$. In the single-factor model (CR-11 d) the asset returns of any pair are bivariate standard normal with correlation $\beta_i\beta_j$, here β^2. Both firms default when both returns are at or below the threshold:

$$\boxed{\pi_{12} = \Phi_2(k, k; \beta^2) = P\left[-\infty \le a_1 \le k,\ -\infty \le a_2 \le k\right]} \qquad \boxed{\rho = \frac{\Phi_2(k, k; \beta^2) - \pi^2}{\pi(1-\pi)}}$$

The second box is the default correlation of CR-11 a with identical firms substituted in. For a pair with different parameters, $\pi_{ij} = \Phi_2(k_i, k_j; \beta_i\beta_j)$ goes into eq. 11.1. Φ_2 needs a bivariate normal routine, so exam questions hand you its value (as a model output or in a table) and ask for the default correlation, or go the other way through a table.

GARP Example 11.5 ($\pi = 1\%$, $k = -2.326$):

- $\beta = 0.50$, so asset correlation $\beta^2 = 0.25$: $\Phi_2 = 0.0438\%$ (4.38 bp) and $\rho = (0.0004375 - 0.0001)/0.0099 = 0.034$.
- What β gives a "typical" default correlation of 0.05? A numerical search (there is no closed form) gives $\beta = 0.561$, asset correlation $\beta^2 = 0.315$, and a joint PD of $0.0001 + 0.05 \times 0.0099 = 0.000595$, about 6 bp.

Two sanity checks. At $\beta = 0$ the returns are independent, $\Phi_2(k, k; 0) = \pi^2$, and the default correlation is zero. And asset correlation is not default correlation: an asset correlation of 0.315 produces a default correlation of only 0.05 (Figure 1). For the same asset correlation, a higher PD gives a higher default correlation.

Using a table (Question CR-11 f.5). To turn a default correlation into an asset correlation, first rebuild the joint PD, $\pi_{12} = \pi^2 + \rho\,\pi(1-\pi)$, then find that joint PD in the table of $\Phi_2(k, k; \beta^2)$ values.

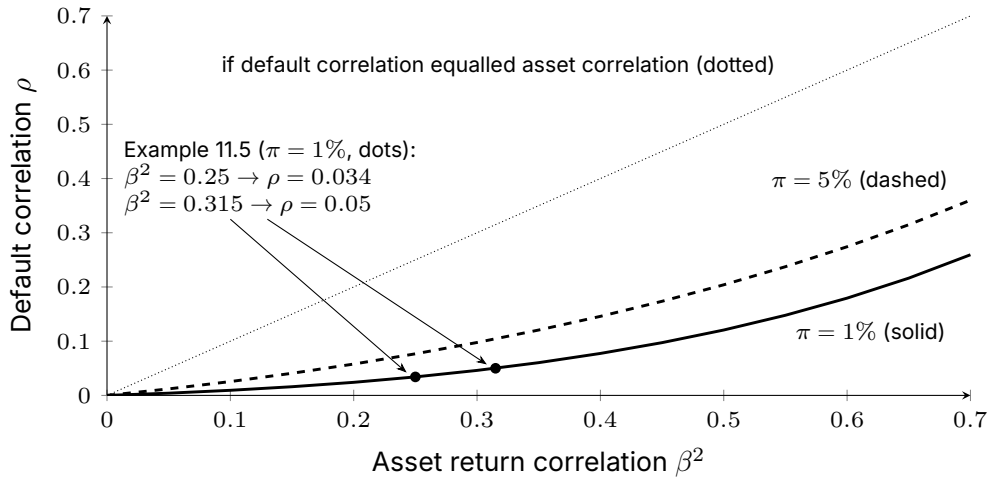


Figure 1. Default correlation implied by the single-factor model, $\rho = (\Phi_2(k, k; \beta^2) - \pi^2)/(\pi(1-\pi))$, for $\pi = 1\%$ (solid) and $\pi = 5\%$ (dashed). Both curves lie far below the 45-degree line (dotted): default correlation is a small fraction of asset correlation, and smaller still for rarer defaults.

Correction to the source notes. GARP Example 11.5 (copied in the B3 revision notes) gives the joint default probability at $\beta = 0.50$ and $\pi = 1\%$ as "4.3 basis points". The bivariate normal gives 4.38 bp, which rounds to 4.4. GARP's default correlation of 0.034 is consistent with 4.38 bp; with 4.3 bp it would be 0.033.

Formula card: CR-11 f

Joint PD from ρ	$\pi_{12} = \pi_1\pi_2 + \rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)}$; equal PDs $\pi^2 + \rho\,\pi(1-\pi)$
ρ from joint PD, equal PDs	$\rho = (\pi_{12} - \pi^2)/(\pi(1-\pi))$
Single factor, pair asset correlation	$\beta_i\beta_j$; homogeneous β^2
Single factor, joint PD	$\pi_{12} = \Phi_2(k, k; \beta^2)$, $k = \Phi^{-1}(\pi)$; general $\Phi_2(k_i, k_j; \beta_i\beta_j)$
Single factor, default correlation	$\rho = (\Phi_2(k, k; \beta^2) - \pi^2)/(\pi(1-\pi))$
Other outcomes	at least one $\pi_1 + \pi_2 - \pi_{12}$; exactly one $\pi_1 + \pi_2 - 2\pi_{12}$; neither $1 - \pi_1 - \pi_2 + \pi_{12}$
Expected loss of joint default	$\pi_{12} \times$ (loss if both default)

Exam traps

- **Dropped term (S5).** $\pi_{12} = \rho\,\pi(1-\pi)$ alone omits the independence part π^2 (or $\pi_1\pi_2$).
- **Variance read as a probability.** $\pi(1-\pi)$ is the Bernoulli variance; it still has to be multiplied by ρ.
- **Asset correlation read as default correlation** (sibling swap), or β reported where the asset correlation β^2 is asked.
- **Wrong outcome.** "At least one" keeps the joint default in: $\pi_1 + \pi_2 - \pi_{12}$, not $-2\pi_{12}$.
- **Joint PD equal to a single PD.** That is the strongest dependence possible, not what a correlation of 0.45 implies.
- **Closed form for β.** There is none: a target default correlation is turned into β numerically or from a table.

Practice questions: CR-11 f

Question CR-11 f.1

Bank Q520 • bank's own question (GARP practice-exam style)

A risk analyst at a bank estimates that two counterparties each have a one-year default probability of 10%, and that the probability that both default within the year is 2%. Which of the following is the default correlation between the two counterparties?

A. 0.010

B. 0.100

C. 0.111

D. 0.200

Question CR-11 f.2

Bank Q519 • GARP PE1 Q41 • Mock 3 Q39

The fixed-income trading desk of an investment bank uses several methodologies to estimate joint default probabilities of counterparties. The manager in charge of the desk asks an analyst to calculate the joint default probability of two firms (credits) over a one-year period using the single-factor model. The portfolio consists of two firms that have the same beta to the market factor. The probability of default of each firm is 2.85%. If the analyst applies a default correlation between this pair of firms of 0.45, what is the joint default probability of the firms using the single-factor model?

A. 1.25%

B. 1.33%

C. 2.77%

D. 2.85%

Question CR-11 f.3

Bank Q469 • Schweser Module Quiz 8.3 Q1 (re-filed from CR-11 a)

A creditor makes a USD 4 million loan to Company X and a USD 4 million loan to Company Y. Each company has a 7% default probability, and the default correlation between them is 0.6. Assuming a loss given default of 100%, the expected loss from the scenario in which both loans default (a loss of USD 8 million) is closest to:

A. USD 39,200

B. USD 312,480

C. USD 351,680

D. USD 560,000

Question CR-11 f.4

Workbook B3.2 • S2, S3, S5 • Map #16

A treasury analyst at a UK bank is sizing a stress scenario for two corporate exposures. The bank's limit framework is breached if at least one of the two borrowers defaults within a year, and the stress report also requires the probability that both default. Inputs:

- One-year probability of default: borrower 1, 1.5%; borrower 2, 4.0%
- Default correlation between the two borrowers: 0.08

Which of the following is closest to the joint default probability and the probability that the limit is breached?

A. Joint default 0.06%; limit breached 5.44%

B. Joint default 0.19%; limit breached 5.31%

C. Joint default 0.25%; limit breached 5.00%

D. Joint default 0.25%; limit breached 5.25%

Question CR-11 f.5

Bank Q518 • BT P2.T6.310.3 (rebuilt: options)

Two credits each have a one-year default probability of $\pi = 2.0\%$. Under the single-factor model their joint default probability is $\pi_{12} = \Phi_2(k, k; \rho_a)$, the bivariate standard normal probability at the default threshold $k = \Phi^{-1}(2\%)$, where ρ_a is the asset return correlation. The default correlation is $\rho = (\pi_{12} - \pi^2)/(\pi(1-\pi))$. A risk analyst has the following joint default probabilities, solved numerically:

Asset return correlation ρ_a	0.00	0.05	0.10	0.15	0.20	0.25
Joint default probability	0.040%	0.053%	0.069%	0.088%	0.110%	0.136%

Given a default correlation of 4.90%, which of the following is the implied asset return correlation?

A. 0.05

B. 0.15

C. 0.25

D. 0.50

Question CR-11 f.6

New • built from GARP Ch.11, 11.3 Example 11.5

A credit portfolio analyst at a Swiss bank models two industrial borrowers with the single-factor model. Each borrower has a one-year default probability of 3% and a beta of 0.50 to the market factor; each exposure is CHF 25 million with an expected recovery of 40%. The bank's bivariate normal routine returns a joint default probability of 0.265%. Which of the following is closest to the asset return correlation and the default correlation between the two borrowers?

A. Asset return correlation 0.25; default correlation 0.060

B. Asset return correlation 0.25; default correlation 0.091

C. Asset return correlation 0.25; default correlation 0.250

D. Asset return correlation 0.50; default correlation 0.060

Solutions • CR-11 f

Solution CR-11 f.1

Answer: C

Bank Q520 • bank's own question

Step 1 – Covariance

$\pi_{12} - \pi^2 = 0.02 - 0.10 \times 0.10 = 0.01.$

$\Rightarrow$ Joint default is twice as likely as under independence (2% against 1%).

Step 2 – Variance of each default indicator

With equal PDs the product of the standard deviations is $\pi(1-\pi) = 0.10 \times 0.90 = 0.09.$

Step 3 – Default correlation

$$\rho = \frac{0.01}{0.09} = 0.111$$

Interpretation

This is the homogeneous form the single-factor model uses. A correlation of 0.11 is about twice Malz's typical 0.05, consistent with joint default being twice the independent figure.

Why the other options are wrong

A is incorrect. Stopped at the covariance, 0.010, without scaling by the variance.
B is incorrect. Divided by $\pi = 0.10$ instead of $\pi(1-\pi) = 0.09$.
D is incorrect. Computed $\pi_{12}/\pi = 0.02/0.10 = 0.200$, the probability that one defaults given that the other does.

LO: CR-11 f: Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 eq. 11.1; 11.3 Asset and Default Correlation

Solution CR-11 f.2

Answer: B

Bank Q519 • GARP PE1 Q41

Step 1 – Same beta means same parameters

Same beta and same PD: use the homogeneous form $\pi_{12} = \pi^2 + \rho\,\pi(1-\pi)$.

Step 2 – Covariance term

$\pi(1-\pi) = 0.0285 \times 0.9715 = 0.027688; \quad \rho\,\pi(1-\pi) = 0.45 \times 0.027688 = 0.012459.$

Step 3 – Add the independence term

$$\pi_{12} = 0.0285^2 + 0.012459 = 0.000812 + 0.012459 = 0.013272 \approx 1.33\%$$

$\Rightarrow$ Almost half of each firm's 2.85% PD: given one default, the other follows with probability $1.33/2.85 = 47\%$.

Interpretation

"Using the single-factor model" here only fixes the common parameters; the arithmetic is the CR-11 a formula with equal PDs. The default correlation already contains the model's information.

Why the other options are wrong

A is incorrect. Dropped the independence term $\pi^2 = 0.081\%$: $0.012459 = 1.25\%$ (shape S5).
C is incorrect. Reported the Bernoulli variance $\pi(1-\pi) = 2.77\%$ without multiplying by ρ.
D is incorrect. Reported the single-name PD, 2.85%, which is the joint PD only under the strongest possible dependence.

LO: CR-11 f: Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 eq. 11.1; 11.3 Asset and Default Correlation

Solution CR-11 f.3

Answer: C

Bank Q469 • Schweser Module Quiz 8.3 Q1

Step 1 – Joint default probability

$$\pi_{12} = \pi^2 + \rho\,\pi(1-\pi) = 0.07^2 + 0.6 \times 0.07 \times 0.93 = 0.0049 + 0.03906 = 0.04396$$

Step 2 – Loss in the joint-default scenario

Both loans lost in full: $4 + 4 =$ USD 8 million.

Step 3 – Expected loss of that scenario

$$0.04396 \times 8{,}000{,}000 = \text{USD } 351{,}680$$

Interpretation

The joint-default scenario carries a 4.4% probability, nine times the 0.49% of independent defaults. Correlation matters most for exactly this worst-case cell.

Why the other options are wrong

A is incorrect. Used the independent joint PD: $0.0049 \times 8{,}000{,}000 = 39{,}200$.
B is incorrect. Kept only the covariance term and omitted π^2: $0.03906 \times 8{,}000{,}000 = 312{,}480$ (shape S5).
D is incorrect. Used a single company's PD: $0.07 \times 8{,}000{,}000 = 560{,}000$, the joint-default EL only under the strongest dependence.

LO: CR-11 f: Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model. CR-11 a: Define and calculate default correlation for credit portfolios. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation

Solution CR-11 f.4

Answer: D

Workbook B3.2 • S2, S3, S5

Step 1 – Product of the standard deviations

$$\sqrt{\pi_1(1-\pi_1)}\,\sqrt{\pi_2(1-\pi_2)} = \sqrt{0.015(0.985)}\,\sqrt{0.04(0.96)} = 0.12155 \times 0.19596 = 0.023819$$

Step 2 – Joint default probability

$$\pi_{12} = \rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} + \pi_1\pi_2 = 0.08(0.023819) + 0.015(0.04)$$

$$\pi_{12} = 0.001906 + 0.000600 = 0.002506$$

$\Rightarrow$ A 0.25% chance that both borrowers default, about four times the 0.06% independence figure.

Step 3 – Probability that at least one defaults

$$P(\text{at least one}) = \pi_1 + \pi_2 - \pi_{12} = 0.015 + 0.040 - 0.0025 = 0.0525$$

$\Rightarrow$ The limit is breached with probability 5.25%. Positive correlation lowers this number relative to independence, because defaults cluster in the same states.

Interpretation

Correlation pushes probability into the "both" and "neither" cells and out of the "one only" cells: the joint figure rises from 0.06% to 0.25%, and the breach probability falls from 5.44% to 5.25%.

Why the other options are wrong

A is incorrect. Ignored the default correlation: $\pi_{12} = \pi_1\pi_2 = 0.06\%$, breach $0.055 - 0.0006 = 5.44\%$.

B is incorrect. Dropped the $+\pi_1\pi_2$ term: $\pi_{12} = 0.19\%$, breach $0.055 - 0.0019 = 5.31\%$.

C is incorrect. Correct joint PD, but computed the probability that exactly one defaults, $\pi_1 + \pi_2 - 2\pi_{12} = 5.00\%$, which misses the joint-default state that also breaches the limit (sibling swap).

LO: CR-11 a: Define and calculate default correlation for credit portfolios. CR-11 f: Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.1 Defining Default Correlation

Solution CR-11 f.5

Answer: C

Bank Q518 • BT P2.T6.310.3 (rebuilt)

Step 1 – Rebuild the joint default probability

$$\pi_{12} = \rho\,\pi(1-\pi) + \pi^2 = 0.049 \times 0.02 \times 0.98 + 0.02^2 = 0.000960 + 0.000400 = 0.136\%$$

Step 2 – Find it in the table

A joint default probability of 0.136% sits in the column for an asset return correlation of 0.25.
$\Rightarrow$ An asset correlation of 0.25 produces a default correlation of only 4.9%.

Interpretation

Asset correlation is about five times the default correlation here, because default is a rare, discrete event. This is the gap of Figure 1, read off a table instead of a chart.

Why the other options are wrong

A is incorrect. Treated the default correlation as the asset correlation (4.9% $\approx$ 0.05), a sibling swap; the 0.05 column gives a default correlation of only 0.7%.

B is incorrect. Dropped the π^2 term: $\pi_{12} = 0.096\%$, whose nearest entry is 0.088% in the 0.15 column (shape S5).

D is incorrect. Found the right column but reported the factor loading $\beta = \sqrt{0.25} = 0.50$ instead of the asset correlation $\beta^2 = 0.25$.

LO: CR-11 f: Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Asset and Default Correlation, Example 11.5

Solution CR-11 f.6

Answer: A

New • built from GARP Ch.11, 11.3 Example 11.5

Step 1 – Asset return correlation

Same beta for both: $\beta_1\beta_2 = 0.50 \times 0.50 = 0.25$.

Step 2 – Covariance of the default indicators

$\pi_{12} - \pi^2 = 0.00265 - 0.03^2 = 0.00265 - 0.00090 = 0.00175$.

Step 3 – Default correlation

$$\rho = \frac{0.00175}{0.03 \times 0.97} = \frac{0.00175}{0.0291} = 0.060$$

$\Rightarrow$ An asset correlation of 0.25 translates into a default correlation of about 0.06.

Interpretation

At $\pi = 1\%$ the same beta gave 0.034 (Example 11.5): for a given asset correlation, the default correlation rises with the PD. Exposures and recovery are red herrings; they matter for losses, not for the correlation.

Why the other options are wrong

B is incorrect. Dropped the π^2 term: $0.00265/0.0291 = 0.091$ (shape S5).
C is incorrect. Assumed the default correlation equals the asset correlation (sibling swap).
D is incorrect. Reported $\beta = 0.50$ as the asset correlation; the pair correlation is β^2.

LO: CR-11 f: Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.3 Asset and Default Correlation, Example 11.5

CR-11 • Portfolio Credit Risk • Vardeez: VERY IMPORTANT

CR-11 h

P2 CONCEPT

Assess the effect of granularity on Credit VaR.

Workbook: B1 Credit VaR and Granularity • Tier B • Map #10 (6/16)

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: B1 (Map #10, 6 of 16 sources) says "two granularity effects are separately examinable", and Vardeez names this LO (his LO 29.h) as Very Important.
Vardeez: as a credit portfolio becomes more granular, Credit VaR decreases; when the default probability is low, Credit VaR is not affected as much.

Reported exam shapes

- **B1·S3** Granularity effect 1: Credit VaR falls as the portfolio becomes more granular.
- **B1·S4** Granularity effect 2: it does not fall to zero when defaults are correlated.

What the questions look like: direction statements (more and smaller loans lower Credit VaR, EL unchanged), the limits (zero only with independent defaults), and the special cases (positions of one obligor, a low default probability). **Kept outside the TYPE rule:** Question CR-11 h.4 (Mock 2 Q41) is a GARP-practice numerical on Malz's 1,000-name portfolio; it is flagged here and in the report.

Worked elsewhere: B1.1 (Credit VaR by formula) under CR-10 b.

1. What granularity means

A portfolio becomes more **granular** when the same total exposure is split into more, smaller positions, so that each credit is a smaller fraction of the portfolio. Malz's experiment keeps everything else fixed: a USD 1 billion book, the same PD π and zero recovery for every position, and one condition that is easy to miss: "each position is an obligation of a different obligor; if two positions were debts of the same obligor, they would be equivalent to one large position." Splitting a loan to one borrower into many loans adds no granularity.

2. Effect 1: Credit VaR falls as the book becomes more granular

With independent defaults the number of defaults is binomial(n, π). Credit VaR (USD million, quantile of loss minus EL) from GARP Table 11.1:

		$n = 1$		$n = 50$		$n = 1{,}000$	
PD	EL	95%	99%	95%	99%	95%	99%
$\pi = 0.5\%$	5	−5	−5	15	35	4	6
$\pi = 2\%$	20	−20	980	40	60	8	11
$\pi = 5\%$	50	−50	950	50	90	12	17

Credit VaR of an uncorrelated USD 1 billion portfolio, zero recovery (GARP Table 11.1). For $n = 1{,}000$ and $\pi = 2\%$ the 95th and 99th percentiles are 28 and 31 defaults.

- **EL does not move.** It is $\pi\times$ USD 1 billion in every column: granularity changes the dispersion of losses, not their mean.
- **Credit VaR falls** from $n = 50$ to $n = 1{,}000$ in every cell: it "decreases as the credit portfolio

becomes more granular for a given default probability".

- **Low PD, slower fall.** "The convergence is more drastic with a high default probability. But that has an important converse: It is harder to reduce VaR by making the portfolio more granular, if the default probability is low." At $n = 1{,}000$ and 99%, Credit VaR is still 1.2 times EL for $\pi = 0.5\%$ (6 against 5) but only 0.34 times EL for $\pi = 5\%$ (17 against 50).
- **The limit.** With a very large number of small *independent* positions the loss equals EL "almost certainly", so the volatility of credit loss and the Credit VaR go to zero.
- **The $n = 1$ column.** One credit is all or nothing, so Credit VaR is $-$EL whenever $\pi < 1 - \alpha$ (95% column) and $V-$EL otherwise. Malz calls it "well-defined ... but not terribly informative". Read the granularity effect from $n = 50$ upwards: going from a negative single-credit figure to $n = 50$ is not a rise in risk.

Correction to the source notes. In GARP Table 11.1, for $n = 1$ at the 95% level, the "Number of defaults" row shows 1 for $\pi = 2\%$ and $\pi = 5\%$. It should be 0: with one credit and $\pi \leq 5\%$ the 95th percentile is no default, as the table's own "Proportion of defaults" (0.000) and Credit VaR rows (-20 and -50 million) show.

3. Effect 2: not to zero when defaults are correlated

With correlated defaults the granular limit is different. In the single-factor model (CR-11 d), once the market factor is known the defaults are independent, so the loss rate of a very granular book converges to the conditional default probability $p(m)$, not to π. Idiosyncratic risk diversifies away; systematic risk does not, and "portfolio returns are driven by the market factor". Credit VaR therefore falls with n towards a floor set by π and β, not towards zero (Figure 1). For $\pi = 2\%$ and an asset correlation of 0.25, the 99% floor is the Vasicek loss rate $\Phi\big((\Phi^{-1}(0.02) + 0.5 \times 2.326)/\sqrt{0.75}\big) = 15.19\%$ minus the 2% EL: 13.19% of the book.

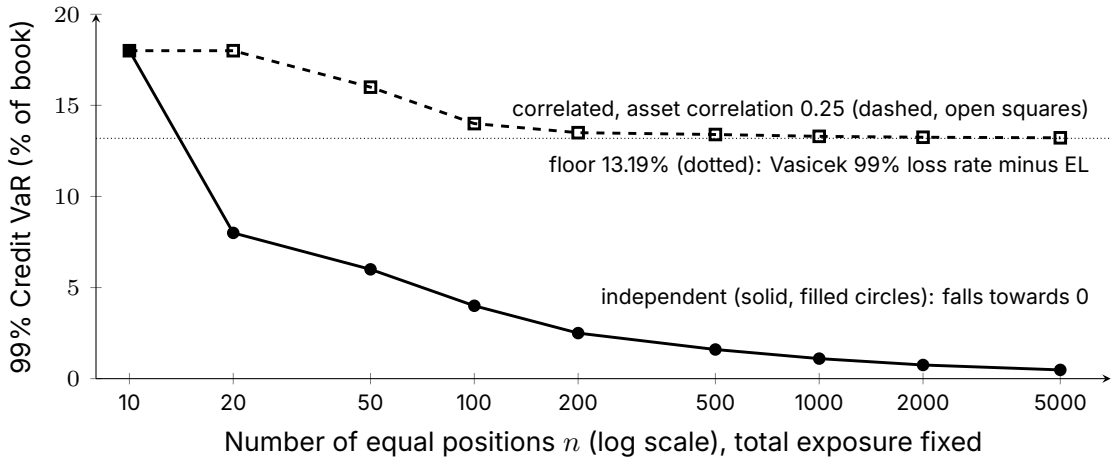


Figure 1. 99% Credit VaR of a book with $\pi = 2\%$ and zero recovery as the same exposure is split into more names. Independent defaults (solid): 6% at $n = 50$ and 1.1% at $n = 1{,}000$, as in Table 11.1, heading to zero. Single-factor model with asset correlation 0.25, a default correlation of 0.049 (dashed): the fall stops at the systematic floor. The jagged steps come from whole numbers of defaults.

	Concentrated ($n = 1$)	Granular, independent	Granular, correlated
Diversifies	Nothing	All default risk	Idiosyncratic risk only
Loss rate	0 or 100%	Converges to π	Converges to $p(m)$: moves with the economy
Expected loss	πV	πV	πV
99% Credit VaR, $\pi = 2\%$	98% of V	Towards 0 (1.1% at $n = 1{,}000$)	Towards 13.2% of V (asset corr. 0.25)
Low PD	All-or-nothing still	Slower convergence	Floor set by π and β

Exam traps

- **Sibling swap: EL.** Granularity changes the dispersion and tail of losses, not the expected loss.
- **Polarity flip: low PD.** Granularity helps most when the PD is high; with a low PD it is harder to reduce Credit VaR.
- **Absolute claim: zero Credit VaR.** True only for independent defaults; with correlation the common factor remains.
- **Same obligor.** Several positions with one obligor default together: no granularity is added.
- **Starting from $n = 1$.** The single-credit Credit VaR can be negative; compare granular books with each other.
- **Quantile threshold.** The 99% quantile is the smallest number of defaults with cumulative probability of at least 99%; then subtract EL.

Practice questions: CR-11 h

Question CR-11 h.1 Bank Q524 • BT P2.T6.24.12.3

Bastrop Bank specializes in making commercial loans to large real estate developments. Management is considering reducing its exposure to any one development and making smaller average loans to more small businesses. How would this likely affect the granularity of the portfolio and its credit VaR?

A. Granularity would decrease as the bank focuses on fewer loans with higher exposure.

B. Granularity would increase as the bank holds a larger number of smaller loans, which tends to lower credit VaR.

C. Granularity would not be affected by the loan size or number of borrowers.

D. The impact cannot be determined without knowing the current credit VaR.

Question CR-11 h.2 Bank Q526 • AnalystPrep Q.5529

Global Finance Bank is assessing a collateralized loan obligation (CLO) whose underlying pool consists of a large number of small-to-medium-sized corporate loans. How does the higher loan granularity of this pool affect its risk profile compared with a pool of lower granularity?

A. It increases the systemic risk due to a higher number of loans.

B. It reduces the diversification benefit, thus increasing the probability of tail events.

C. It enhances diversification, thereby reducing the probability of tail events and concentration risk.

D. It leads to higher operational costs due to the complexity of managing a larger number of loans.

Question CR-11 h.3 New • built from GARP Ch.11, 11.2

A credit portfolio manager at an Indian NBFC holds a single INR 500 crore loan to an infrastructure developer, with a one-year default probability of 3% and zero expected recovery. To make the book "granular", she refinances it as 50 loans of INR 10 crore each, all owed by the same developer and ranking equally. Which of the following statements about the effect on the book's Credit VaR is correct?

A. The book is now granular: with 50 positions the number of defaults is binomial, so the 99% Credit VaR falls.

B. Nothing has changed in substance: the 50 loans default together, so the book still behaves as a single credit and its Credit VaR is unchanged.

C. Credit VaR falls because the expected loss of each loan is now only 2% of the book's expected loss.

D. Credit VaR falls almost to zero, because no single position is more than 2% of the book.

Question CR-11 h.4 Bank Q505 • Mock 2 Q41 (re-filed from CR-11 c; GARP-practice numerical)

Bank XYZ has a credit portfolio of 1,000 positions with a total value of USD 1 million. Each position has a default probability of 0.5%, a recovery rate of zero, and default correlation zero. The 99th percentile of the number of defaults is 11. At a 99% confidence level, the credit VaR is approximately:

A. USD 1,000

B. USD 5,000

C. USD 6,000

D. USD 11,000

Question CR-11 h.5 Bank Q525 • BT P2.T6.309.3 (rebuilt: option D)

A portfolio has a total value of USD 1.0 billion divided into $n = 20$ equal positions of USD 50 million, each to a different obligor. Each position has a default probability of 1.0% and a recovery rate of zero, and default correlation is zero. Credit VaR is defined at the 95% confidence level as the 95th percentile of credit loss less the expected loss. Varying granularity (n) or default correlation one at a time, holding all else constant, each of the following statements is true EXCEPT:

A. The initial assumptions imply a 95% credit VaR of USD 40 million.

B. Increasing granularity from $n = 20$ to $n = 100$ positions decreases the 95% credit VaR.

C. As n approaches infinity with independent small positions, the 95% credit VaR tends towards USD 990 million.

D. Raising the default correlation to 1 would make the 95% credit VaR negative.

Question CR-11 h.6 Workbook B1.2 • S3, S4 • Map #10

The CRO of a Japanese regional bank wants to cut the Credit VaR of its SME loan book. She proposes replacing 20 loans of JPY 5 billion each with 2,000 loans of JPY 50 million each to similar borrowers. The default probability, the LGD and the pairwise default correlation (0.05, estimated with a single-factor model) would all be unchanged. A risk analyst is drafting the memo to the risk committee. Which of the following statements should the analyst include?

A. The move lowers Credit VaR, because spreading the book across more borrowers lowers the portfolio's expected loss, and the expected loss is subtracted from the loss quantile.

B. The move lowers Credit VaR, and the proportional reduction from granularity is largest when the default probability of the loans is low.

C. The move lowers Credit VaR, but with a positive default correlation it cannot bring Credit VaR to zero, because losses on the granular book still move with the common factor.

D. The move lowers Credit VaR and, with enough borrowers, drives it to zero whatever the default correlation, because the loss rate converges to the default probability.

Solutions • CR-11 h

Solution CR-11 h.1 — Answer: B

Bank Q524 • BT P2.T6.24.12.3

Step 1 – Direction of granularity

Granularity rises when each credit becomes a smaller share of the total, through more and smaller loans. Moving from a few large developments to many small businesses does exactly that.

Step 2 – Effect on Credit VaR

With more, smaller positions to different obligors, idiosyncratic defaults diversify and the loss quantile moves towards EL.

$\Rightarrow$ Credit VaR tends to fall; EL is unchanged if the PDs and LGDs are similar.

Interpretation

The direction is a general result. How far Credit VaR falls depends on the default probability and on how correlated the small businesses are (Section 3).

Why the other options are wrong

A is incorrect. Reverses the move: fewer, larger loans would lower granularity, but the bank is going the other way.

C is incorrect. Granularity is defined by the number and size of positions.

D is incorrect. The direction does not depend on the current level of Credit VaR.

LO: CR-11 h: Assess the effect of granularity on Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2 Granularity and Portfolio Credit Value-at-Risk

Solution CR-11 h.2 — Answer: C

Bank Q526 • AnalystPrep Q.5529

Step 1 – What higher granularity does

Each loan is a smaller share of the pool, so any one default moves the pool's loss very little.

Step 2 – Effect on the tail

Idiosyncratic defaults average out, losses cluster closer to the expected loss, and extreme losses from a few names become less likely: less concentration risk, lower Credit VaR.

Interpretation

Malz's structured-credit chapter makes the same point for securitisations: a more granular collateral pool can carry a larger senior tranche. What granularity cannot remove is the common-factor risk of a recession that hits all the borrowers.

Why the other options are wrong

A is incorrect. More loans do not add systematic risk; granularity removes idiosyncratic risk.

B is incorrect. Polarity flip: granularity increases diversification and lowers the probability of tail events.

D is incorrect. Operating costs may rise, but that is not the credit-risk effect the question asks about.

LO: CR-11 h: Assess the effect of granularity on Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2 Granularity and Portfolio Credit Value-at-Risk

Solution CR-11 h.3 — Answer: B

New • built from GARP Ch.11, 11.2

Step 1 – Who can default

All 50 loans are owed by one developer and rank equally: if the developer defaults, all 50 default; if not, none does.

$\Rightarrow$ Malz: debts of the same obligor "would be equivalent to one large position". The default correlation among the loans is 1.

Step 2 – Credit VaR before and after

The loss is INR 500 crore with probability 3% and zero otherwise, in both cases. EL $= 500 \times 3\% = 15$. At 99% the 1% tail lies inside the 3% default probability, so Credit VaR $= 500 - 15 =$ INR 485 crore, before and after.

Interpretation

Granularity means more *obligors*, not more loan contracts. To cut Credit VaR the manager has to lend to different borrowers.

Why the other options are wrong

A is incorrect. A binomial count needs independent defaults; here the loans default together.

C is incorrect. Sibling swap: the book's EL is unchanged at INR 15 crore, and EL does not drive the change in Credit VaR anyway.

D is incorrect. Position size alone does not diversify; the 50 positions share one default event.

LO: CR-11 h: Assess the effect of granularity on Credit VaR. CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2 Granularity and Portfolio Credit Value-at-Risk

Solution CR-11 h.4 — Answer: C

Bank Q505 • Mock 2 Q41

Step 1 – Position size and expected loss

Each position is $1{,}000{,}000/1{,}000 =$ USD 1,000. EL $= 1{,}000{,}000 \times 0.5\% =$ USD 5,000, five defaults' worth.

Step 2 – 99th percentile loss

11 defaults $\times$ USD 1,000 $=$ USD 11,000.

Step 3 – Credit VaR

$$11{,}000 - 5{,}000 = \text{USD } 6{,}000$$

$\Rightarrow$ 0.6% of the book, 1.2 times EL.

Interpretation

This is Malz's $n = 1{,}000$, $\pi = 0.5\%$ cell scaled down from USD 1 billion. Even with 1,000 independent names the Credit VaR is larger than EL, because with a low PD "it is harder to reduce VaR by making the portfolio more granular".

Why the other options are wrong

A is incorrect. Reported the value of a single position.

B is incorrect. Reported the expected loss instead of the quantile minus EL.

D is incorrect. Reported the 99th percentile loss without subtracting EL.

LO: CR-11 h: Assess the effect of granularity on Credit VaR. CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2, Table 11.1

Solution CR-11 h.5 Answer: C

Bank Q525 • BT P2.T6.309.3 (rebuilt)

Step 1 – Check A ($n = 20$)

EL $= 1\% \times 1{,}000 = 10$. $P(0) = 0.99^{20} = 81.79\%$ and $P(\leq 1) = 98.31\% \geq 95\%$, so the 95th percentile is one default, a loss of 50. Credit VaR $= 50 - 10 = 40$: true.

Step 2 – Check B ($n = 100$)

$P(\leq 2) = 92.06\%$, $P(\leq 3) = 98.16\%$, so three defaults of USD 10 million: $30 - 10 = 20 < 40$: true.

Step 3 – Check D (correlation 1)

The book becomes one credit with a 1% PD, below the 5% tail, so the 95th percentile loss is zero and Credit VaR is $0 - 10 = -10$: true.

Step 4 – Check C

With independent small positions the loss converges to EL, so Credit VaR tends to zero, not to USD 990 million.

$\Rightarrow$ C is false: it is the exception.

Interpretation

USD 990 million is the all-or-nothing figure V − EL of a single credit, the opposite end of the granularity scale.

Why the other options are wrong

A is incorrect. True: one default at the 95th percentile, $50 - 10 = 40$.

B is incorrect. True: three defaults of USD 10 million give $30 - 10 = 20$.

D is incorrect. True: at correlation 1 the 95% quantile is zero, so Credit VaR is −USD 10 million.

LO: CR-11 h: Assess the effect of granularity on Credit VaR. CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2 Granularity and Portfolio Credit Value-at-Risk

Solution CR-11 h.6 Answer: C

Workbook B1.2 • S3, S4

Step 1 – What granularity changes

With the total exposure fixed, more and smaller positions reduce the variance of portfolio losses, so the loss quantile moves towards the expected loss and Credit VaR falls (Malz Table 11.1: at $\pi = 2\%$ and 99%, Credit VaR is USD 980 million for $n = 1$, 60 million for $n = 50$, 11 million for $n = 1{,}000$).

Step 2 – What it cannot change

In the single-factor model the loss rate of a granular book converges to the conditional default probability $p(m) = \Phi((k - \beta m)/\sqrt{1 - \beta^2})$, not to π. Idiosyncratic risk is diversified away; the market factor m still moves the loss rate, so Credit VaR stays positive.

$\Rightarrow$ The CRO's plan removes concentration risk but leaves the systematic part of Credit VaR in place.

Interpretation

C is correct. The memo should present the reduction as a cut in idiosyncratic risk only; the residual Credit VaR is set by the default probability and the correlation to the common factor.

Why the other options are wrong

A is incorrect. Sibling swap. Granularity changes the dispersion and extreme quantiles of loss, not the expected loss, which stays $\pi\times$ portfolio value for every n.

B is incorrect. Polarity flip. Malz: the convergence "is more drastic with a high default probability" and "it is harder to reduce VaR by making the portfolio more granular, if the default probability is low".

D is incorrect. Absolute claim / scope error. Convergence of the loss to EL and Credit VaR to zero holds for a very large number of independent positions; with correlation 0.05 the common factor remains.

LO: CR-11 h: Assess the effect of granularity on Credit VaR. CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. • **Reference:** Malz, Ch.8 (GARP CR Ch.11), 11.2 and 11.3

CR-12 • Credit Risk • Vardeez: IMPORTANT

CR-12 f

P2 CALC

Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model.

Workbook: B3 Default Correlation, Joint PD and Copulas • Tier B • Map #16 (6/16) D15 Structured Credit and Implied Correlation • Tier D

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: B3 (Map #16, 6 of 16 sources, Telegram-confirmed) reports "Gaussian copula, implied correlation" and flags the conditional default probability given the common factor as separately examinable. Vardeez marks the one-factor model as important for the exam in a chapter he rates IMPORTANT.

Vardeez: "LO 30.f, in this we have One Factor Model ... Imp it is for the exam."

Reported exam shapes

- **B3·S4** Gaussian copula and implied correlation: what the copula assumes, and how its correlation is calibrated.
- **D15·S4** The same shape reported for structured credit: the copula is the engine that tranche prices (and implied correlations) run on.
- **B3·S6** The conditional default probability given the common factor (also carried by Workbook B3.3).

What the questions look like: statements about what the copula does (which variables are normal, what the correlation is, what conditioning on the factor buys), and short calculations: a default threshold $N^{-1}(Q)$, a simulated sample mapped to a default year, and the conditional PD $Q(T \mid F)$.

Worked elsewhere: B3.1 (default correlation of two firms) at Question CR-11 a.6; B3.2 (joint default probability) at Question CR-11 f.4; D15.1 (implied correlation from tranche prices) at Question CR-22 l.6; D14.1 (compound versus base correlation) under CR-9 g. The credit VaR built on this model is CR-12 g; its use in CDO valuation is CR-13 f.

1. Why a copula: times to default are not normal

To measure portfolio credit risk we need the *joint* distribution of the times to default $t_1, t_2, \ldots$ of several companies. If each t_i were normal we could assume a multivariate normal. It is not: a company's time to default "is not even approximately normal". The Gaussian copula solves this in two moves (GARP Ch.12, 12.2):

1. **Map each time to default into a standard normal variable**, percentile to percentile, using Q_i, the cumulative probability distribution of t_i:

$$x_i = N^{-1}[Q_i(t_i)]$$

   The 5th percentile of t_i becomes $x_i = -1.645$, the 10th percentile becomes -1.282, and so on.

2. **Assume the x_i are multivariate normal.** The correlation between x_i and x_j is the **copula correlation**. It is often proxied by the correlation between the two companies' equity returns.

The transformed variables are normal by construction, while the times to default keep whatever marginal distributions they had. That is the point of the model: the correlation structure is estimated

separately from the marginal (unconditional) default distributions, and the joint distribution is then fully defined by the marginals together with a single correlation parameter. The model works with **real-world** PDs (left tail of Q_i from rating agency data) or **risk-neutral** PDs (left tail from bond prices); it assumes every company defaults eventually, extends easily to many companies, and can be shown to be similar to Merton's structural model (CR-12 e).

2. Simulating default times (GARP Example 12.4)

Ten companies, pairwise copula correlation 0.2, each with cumulative PDs of 1%, 3%, 6%, 10% and 15% by the end of years 1 to 5. Sample ten correlated standard normals; each sample is read against the thresholds $N^{-1}(Q(t))$:

End of year t	1	2	3	4	5
Cumulative PD $Q(t)$	1%	3%	6%	10%	15%
Threshold $N^{-1}[Q(t)]$	-2.326	-1.881	-1.555	-1.282	-1.036

A sample below -2.326 means default in year 1; between -2.326 and -1.881, default in year 2; and so on. A sample above -1.036 means no default within five years. Correlated samples produce clustered default years; repeating the simulation many times builds the joint loss distribution.

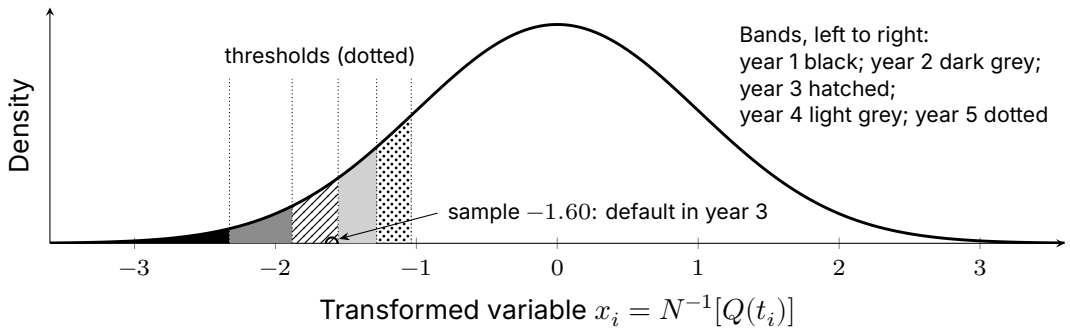


Figure 1. GARP Example 12.4: the thresholds $N^{-1}(1\%)$ to $N^{-1}(15\%)$ cut the standard normal into default-year bands. The circled sample, $x_i = -1.60$, lies between -1.881 and -1.555, so the company defaults in year 3 (Question CR-12 f.3). The unshaded area ($x_i > -1.036$, 85%) is no default within five years. The bands are equal in probability to the marginal PDs (1%, 2%, 3%, 4%, 5%), whatever the correlation.

3. The one-factor Gaussian copula

Specifying a separate correlation for every pair is impractical, so a common factor drives all the x_i (GARP eq. 12.7):

$$x_i = a_i F + \sqrt{1 - a_i^2}\, Z_i$$

F is the common factor and Z_i the company's own factor; F and all the Z_i are independent standard normals, and $-1 < a_i < 1$. Then each x_i is standard normal and $\text{corr}(x_i, x_j) = a_i a_j$. GARP notes that a_i is sometimes approximated by the correlation of company i's equity returns with a well-diversified market index.

Conditional PD. Default happens by T when $x_i < N^{-1}[Q_i(T)]$, that is when $Z_i < (N^{-1}[Q_i(T)] - a_i F)/\sqrt{1 - a_i^2}$. Given F, only Z_i is random, so (eq. 12.8):

$$\boxed{Q_i(T \mid F) = N\left(\frac{N^{-1}[Q_i(T)] - a_i F}{\sqrt{1 - a_i^2}}\right)}$$

In the homogeneous case every company has the same PD $Q(T)$ and the same pairwise correlation ρ, so $a_i = \sqrt{\rho}$ (eq. 12.9):

$$Q(T \mid F) = N\left(\frac{N^{-1}[Q(T)] - \sqrt{\rho}\,F}{\sqrt{1-\rho}}\right)$$

- **Low F is a bad state.** F enters with a minus sign, so the lower F, the higher the conditional PD. In a severe downturn a company with a larger loading a_i suffers more (a cyclical firm); precisely, the conditional PD rises with a_i only when $F < a_i N^{-1}[Q_i(T)]$ (see the correction box below).
- **Conditional independence.** Once F is fixed, the Z_i are independent, so defaults are independent. All dependence enters through the loadings on F. This is what makes the binomial calculations of CR-13 f possible.
- **Averaging over F gives back $Q(T)$.** The unconditional PD is the conditional PD weighted by the normal density of F.

Worked example (GARP Practice Question 13.21(a)). Five-year PD $Q = 3\%$, copula correlation $\rho = 0.2$: $N^{-1}(0.03) = -1.8808$, $\sqrt{0.2} = 0.4472$, $\sqrt{0.8} = 0.8944$. For $F = -2$:

$$Q(5 \mid F = -2) = N\left(\frac{-1.8808 + 0.4472 \times 2}{0.8944}\right) = N(-1.1028) = 13.51\%$$

Factor value F	-2	-1	0	1	2
$Q(5 \mid F)$, $\rho = 0.2$	13.51%	5.45%	1.77%	0.46%	0.10%

At the median factor value ($F = 0$) the conditional PD, 1.77%, is *below* the 3% unconditional PD: the average is pulled up by the rare bad states.

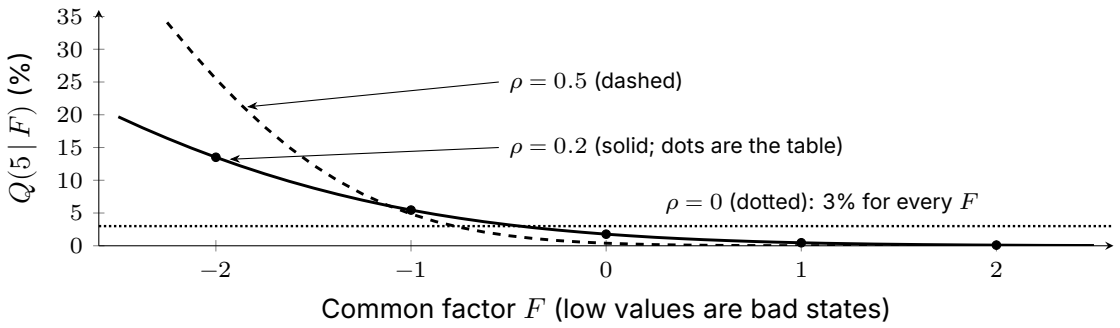


Figure 2. Conditional five-year PD against the common factor, unconditional PD 3%. With zero correlation the factor is irrelevant (dotted line). The higher the correlation, the more the conditional PD fans out: very high in bad states, near zero in good ones, with the same 3% average.

Correction to the source notes. Workbook B3.3 (Step 2 and the reason given for option B) says that, with $m < 0$, a higher loading always raises the conditional PD because it "raises the numerator and shrinks the denominator". Shrinking the denominator of a *negative* argument pushes it further down, so the two effects pull in opposite directions. Differentiating $(k - \beta m)/\sqrt{1-\beta^2}$ shows the conditional PD rises with β only when $m < \beta k$. With $\pi = 1\%$ ($k = -2.326$) and $m = -1$: $\beta = 0.4$ gives 1.78% but $\beta = 0.6$ gives 1.55%. At $m = -2$: 4.79% against 7.96%, as the workbook intends. Option B of B3.3 is still false (it fails in severe downturns) and the key D stands. Figure 2 shows the same thing: the $\rho = 0.5$ curve lies above the $\rho = 0.2$ curve only for F below about -1.1.

4. From the conditional PD to a worst case

For a large portfolio of similar loans, conditional independence means the *fraction* defaulting by T, given F, is close to $Q(T \mid F)$. We are $X\%$ certain that F will exceed $N^{-1}(1-X) = -N^{-1}(X)$. Substituting $F = -N^{-1}(X)$ flips the sign of the middle term and gives Vasicek's worst case default rate (eq. 12.10):

$$V(X,T) = N\left(\frac{N^{-1}[Q(T)] + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right)$$

Vardeez flags the sign: the conditional PD has $-\sqrt{\rho}F$, the worst case has $+\sqrt{\rho}N^{-1}(X)$. Both are right; F enters negatively and the percentile X positively. Credit VaR from $V(X,T)$ is CR-12 g; the regulatory use is CR-10 d.

Formula card: CR-12 f

Percentile mapping	$x_i = N^{-1}[Q_i(t_i)]$; default by t when $x_i < N^{-1}[Q_i(t)]$
One-factor structure	$x_i = a_i F + \sqrt{1-a_i^2}\,Z_i$, $\text{corr}(x_i, x_j) = a_i a_j$
Conditional PD	$Q_i(T \mid F) = N\left(\dfrac{N^{-1}[Q_i(T)] - a_i F}{\sqrt{1-a_i^2}}\right)$
Homogeneous case	$a_i = \sqrt{\rho}$: $Q(T \mid F) = N\left(\dfrac{N^{-1}[Q(T)] - \sqrt{\rho}F}{\sqrt{1-\rho}}\right)$
Worst case at X ($F = -N^{-1}(X)$)	$V(X,T) = N\left(\dfrac{N^{-1}[Q(T)] + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right)$
Inverse normal values	$N^{-1}(1\%) = -2.326$, $N^{-1}(5\%) = -1.645$, $N^{-1}(10\%) = -1.282$, $N^{-1}(20\%) = -0.842$

Exam traps

- **Wrong variables normal.** The copula makes the transformed x_i multivariate normal, not the times to default, the default probabilities or the recovery rates.
- **Copula correlation read as default correlation.** A copula correlation of 0.2 implies a far lower correlation between the default indicators: about 0.04 for a 3% PD and 0.095 for a 15% PD.
- **Sign and polarity of F.** The conditional PD has $-a_i F$: a low factor raises it. Adding $a_i F$ turns a recession into a boom. A higher loading raises the conditional PD only in a bad enough state ($F < a_i N^{-1}[Q_i(T)]$), not for every $F < 0$.
- **Loading versus correlation.** The loading is $a_i = \sqrt{\rho}$, not ρ; the denominator is $\sqrt{1-\rho}$, not $1-\rho$.
- **Wrong tail for the threshold.** A 10% PD gives the threshold $N^{-1}(0.10) = -1.282$, not $+1.282$.
- **Wrong band in a simulation.** A sample of -1.60 is above -1.881 (so it survives year 2) and below -1.555 (so it defaults in year 3).

Practice questions: CR-12 f

Question CR-12 f.1

Bank Q551 • Schweser Module Quiz 30.3 Q2

The Gaussian copula model transforms which of the following factors into normal variables?

A. Recovery rates.

B. Times to default.

C. Credit spread risk.

D. Default probabilities.

Question CR-12 f.2

New • built from GARP Ch.12, 12.2 (Gaussian copula, Example 12.4)

A portfolio credit analyst at a Dutch bank simulates five-year default times for ten companies with a Gaussian copula, following GARP Example 12.4. Each company has a 15% probability of defaulting within five years, and the pairwise copula correlation is set at 0.2, using the correlation of the companies' equity returns as a proxy. Which of the following statements about the 0.2 parameter is correct?

A. It is the correlation between the default indicators of any two companies over the five years, so it can be used directly in the default correlation formula.

B. It is the correlation between the two companies' times to default, because the copula is applied directly to the default times.

C. It is the correlation between the transformed standard normal variables $x_i = N^{-1}[Q_i(t_i)]$; the default correlation it implies over five years is much lower, below 0.1.

D. Raising it from 0.2 to 0.4 would raise each company's five-year default probability above 15%, because defaults would cluster.

Question CR-12 f.3

Bank Q549 • BT P2.T6.24.21.2 (rebuilt)

A risk analyst simulates defaults over 5 years for 10 companies using a Gaussian copula with pairwise copula correlation 0.2. Each company's cumulative default probability at 1, 2, 3, 4 and 5 years is 1%, 3%, 6%, 10% and 15%. For one company the simulated standard normal sample is $x_i = -1.60$. When does this company default?

A. In year 2

B. In year 3

C. In year 4

D. No default within 5 years

Question CR-12 f.4 Bank Q547 • AnalystPrep Q.6054 (rebuilt)

Under the one-factor Gaussian copula model, $x_i = a_i F + \sqrt{1 - a_i^2}\, Z_i$, where F and Z_i are independent standard normal variables. A company has $a_i = 0.7$. In a given scenario the common factor F is at the 80th percentile of the standard normal distribution and the idiosyncratic factor Z_i is at the 60th percentile. The company's probability of default by time T is 10%. Which of the following is closest to the company's value of x_i and the default threshold for x_i?

A. $x_i = 0.72$; default threshold -1.28

B. $x_i = 0.77$; default threshold -1.28

C. $x_i = 0.77$; default threshold $+1.28$

D. $x_i = 0.99$; default threshold -1.28

Question CR-12 f.5 Bank Q548 • AnalystPrep Q.6214 (rebuilt)

A risk analyst uses the one-factor Gaussian copula model with a common copula correlation $\rho = 0.25$ for all companies. Each company has a 10% cumulative probability of default over 5 years. Conditional on the common factor F being at its 1st percentile ($F = -2.326$), which of the following is closest to the probability of default over 5 years for a given company? ($N^{-1}(0.10) = -1.2816$.)

A. 0.2%

B. 20.9%

C. 43.7%

D. 44.6%

Question CR-12 f.6 New • built from GARP Ch.12, 12.2 (eq. 12.8 and footnote 4)

A credit portfolio manager at a Singapore bank uses a one-factor Gaussian copula with company-specific loadings a_i, each set equal to the correlation of the company's equity returns with a well-diversified market index. Two borrowers each have a one-year probability of default of 2%:

Borrower	Sector	Loading a_i	One-year PD
K	Copper miner	0.6	2%
L	Water utility	0.2	2%

In the bank's recession scenario the common factor takes the value $F = -1.5$. Which of the following is closest to the two borrowers' one-year default probabilities conditional on this scenario?

A. K: 3.57%; L: 3.39%

B. K: 3.67%; L: 7.46%

C. K: 7.46%; L: 3.67%

D. K: 7.92%; L: 6.10%

Question CR-12 f.7 Workbook B3.3 • S4, S6 • Map #16 • Map #60

A quantitative analyst at a Canadian bank is reviewing the documentation of the bank's portfolio credit model. The model joins each obligor's default-time distribution with a one-factor Gaussian copula, and its correlation parameter is calibrated so that modelled values of standard index tranches match their market prices. Which of the following statements in the documentation is correct?

A. The model assumes that the obligors' times to default are jointly normally distributed, so each obligor's marginal default-time distribution must itself be normal.

B. Conditional on a negative realisation of the common factor, an obligor with a higher factor loading has a lower conditional default probability than an otherwise identical obligor.

C. The correlation calibrated to the tranche prices is a real-world parameter, estimated from the historical default experience of the obligors in the index.

D. Once the common factor is realised, the obligors' defaults are independent, so all default dependence in the model enters through the obligors' loadings on that factor.

Solutions • CR-12 f

Solution CR-12 f.1

Answer: B

Bank Q551 • Schweser Module Quiz 30.3 Q2

Step 1 – What is not normal

Times to default are far from normal (heavily skewed, bounded below by zero), so a multivariate normal cannot be assumed for them directly.

Step 2 – What the copula does

Each time to default is mapped percentile to percentile into $x_i = N^{-1}[Q_i(t_i)]$, a standard normal variable, and the x_i are assumed multivariate normal.

$\Rightarrow$ The variables transformed are the times to default.

Interpretation

The default probabilities are inputs to the transformation (through Q_i), not the variables transformed. Keeping the marginals intact while imposing a normal dependence structure is the whole purpose of the copula.

Why the other options are wrong

A is incorrect. Recovery rates are not transformed; they enter only when losses are computed (for example in credit VaR, CR-12 g).

C is incorrect. Credit spread risk is not a variable in the copula transformation.

D is incorrect. Default probabilities define the mapping $x_i = N^{-1}[Q_i(t_i)]$; they are not themselves turned into normal variables.

LO: CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2 The Gaussian Copula Model for Time to Default

Solution CR-12 f.2

Answer: C

New • built from GARP Ch.12, 12.2

Step 1 – Where the correlation sits

The copula maps each t_i into $x_i = N^{-1}[Q_i(t_i)]$ and assumes the x_i are multivariate normal. "The default correlation between t_i and t_j is measured as the correlation between x_i and x_j. This is referred to as the copula correlation."

$\Rightarrow$ 0.2 is the correlation of the transformed normals, not of the default times or the default indicators.

Step 2 – What it implies for default correlation

With 15% five-year PDs the default threshold is $k = N^{-1}(0.15) = -1.036$. The joint PD is the bivariate normal probability $M_2(k, k; 0.2) = 3.46\%$ (against $0.15^2 = 2.25\%$ under independence), so the default correlation is

$$\frac{0.0346 - 0.0225}{0.15 \times 0.85} = 0.095$$

$\Rightarrow$ A copula correlation of 0.2 implies a default correlation of under 0.1.

Interpretation

Asset, equity and copula correlations are all much larger than the default correlation they generate (the B3 trap "asset correlation is not default correlation"). Treating 0.2 as a default correlation would grossly overstate joint defaults.

Why the other options are wrong

A is incorrect. Sibling swap: 0.2 is the copula correlation; the default correlation it implies is about 0.095.

B is incorrect. Scope error: the copula is applied to the transformed variables x_i, precisely because the times to default are not normal.

D is incorrect. The copula leaves the marginals untouched: each company keeps its 15% five-year PD whatever the correlation. Higher correlation changes how defaults cluster, not how likely each one is.

LO: CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2, footnote 3 and Example 12.4

Solution CR-12 f.3 — Answer: B

Bank Q549 • BT P2.T6.24.21.2

Step 1 – Thresholds from the cumulative PDs

$N^{-1}(0.01) = -2.33$, $N^{-1}(0.03) = -1.88$, $N^{-1}(0.06) = -1.55$, $N^{-1}(0.10) = -1.28$, $N^{-1}(0.15) = -1.04$.

Step 2 – Place the sample

$-1.88 < -1.60 < -1.55$: the sample is above the year-2 threshold and below the year-3 threshold.

$\Rightarrow$ The company survives the first two years and defaults in year 3.

Interpretation

Each band between successive thresholds has a probability equal to that year's unconditional PD (here 3% for year 3). The correlation of 0.2 affects which samples are drawn together, not how a given sample is read (Figure 1).

Why the other options are wrong

A is incorrect. Year 2 needs a sample between -2.33 and -1.88; -1.60 is above -1.88.

C is incorrect. Year 4 needs a sample between -1.55 and -1.28; -1.60 is below -1.55, so default comes earlier.

D is incorrect. No default needs a sample above $N^{-1}(0.15) = -1.04$.

LO: CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2, Example 12.4

Solution CR-12 f.4 — Answer: B

Bank Q547 • AnalystPrep Q.6054

Step 1 – Convert the percentiles into factor values

$$F = N^{-1}(0.80) = 0.8416, \qquad Z_i = N^{-1}(0.60) = 0.2533$$

Step 2 – Combine with the loadings

$$x_i = 0.7 \times 0.8416 + \sqrt{1 - 0.49} \times 0.2533 = 0.5891 + 0.7141 \times 0.2533 = 0.5891 + 0.1809 = 0.770$$

Step 3 – Default threshold

Default by T occurs if $x_i < N^{-1}(0.10) = -1.28$.

$\Rightarrow 0.77 > -1.28$: no default in this scenario.

Interpretation

Both draws are above their medians, so x_i is comfortably positive. The threshold depends only on the marginal PD; the loading decides how much of x_i comes from the common factor.

Why the other options are wrong

A is incorrect. Weighted Z_i by $1 - a_i^2 = 0.51$ instead of $\sqrt{0.51} = 0.7141$: $0.5891 + 0.51 \times 0.2533 = 0.72$.

C is incorrect. Right x_i, wrong tail: $+1.28$ is $N^{-1}(0.90)$. A 10% PD needs the 10th percentile, -1.28.

D is incorrect. Used the percentiles themselves as the draws: $0.7 \times 0.80 + 0.7141 \times 0.60 = 0.99$.

LO: CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2 A Factor-Based Correlation Structure, eq. 12.7

Solution CR-12 f.5

Answer: D

Bank Q548 • AnalystPrep Q.6214

Step 1 – Loading and denominator

$a = \sqrt{0.25} = 0.5$; $\sqrt{1-\rho} = \sqrt{0.75} = 0.8660$.

Step 2 – Argument of N

$$\frac{N^{-1}(0.10) - aF}{\sqrt{1-\rho}} = \frac{-1.2816 - 0.5 \times (-2.326)}{0.8660} = \frac{-0.1186}{0.8660} = -0.1369$$

Step 3 – Conditional PD

$$Q(5 \mid F = -2.326) = N(-0.1369) = 44.6\%$$

$\Rightarrow$ In the 1-in-100 bad state, the five-year PD rises from 10% to about 45%.

Interpretation

A factor at its 1st percentile is the 99% worst case: this conditional PD is the 99% worst case default rate $V(0.99, 5)$ of CR-12 g. A correlation of only 0.25 more than quadruples the PD in that state.

Why the other options are wrong

A is incorrect. Sign error: added aF instead of subtracting it, as if F were at its 99th percentile: $N(-2.823) = 0.2\%$.

B is incorrect. Used $\rho = 0.25$ instead of $a = 0.5$ as the loading: $N((-1.2816 + 0.25 \times 2.326)/0.8660) = N(-0.808) = 20.9\%$.

C is incorrect. Divided by $1 - \rho = 0.75$ instead of $\sqrt{1-\rho}$: $N(-0.158) = 43.7\%$.

LO: CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2, eq. 12.9

Solution CR-12 f.6

Answer: C

New • built from GARP Ch.12, 12.2

Step 1 – Common threshold

$N^{-1}(0.02) = -2.0537$ for both borrowers.

Step 2 – Borrower K ($a = 0.6$)

$$Q_K(1\,|\,F) = N\left(\frac{-2.0537 - 0.6 \times (-1.5)}{\sqrt{1-0.36}}\right) = N\left(\frac{-1.1537}{0.8}\right) = N(-1.4422) = 7.46\%$$

Step 3 – Borrower L ($a = 0.2$)

$$Q_L(1\,|\,F) = N\left(\frac{-2.0537 - 0.2 \times (-1.5)}{\sqrt{1-0.04}}\right) = N\left(\frac{-1.7537}{0.9798}\right) = N(-1.7899) = 3.67\%$$

$\Rightarrow$ The cyclical miner's PD almost quadruples in the recession; the utility's less than doubles.

Interpretation

With heterogeneous loadings (eq. 12.8) the same 2% unconditional PD hides very different exposure to the common factor. The copula correlation between the two borrowers is $a_K a_L = 0.12$.

Why the other options are wrong

A is incorrect. Divided by $1 - a_i^2$ instead of its square root: K $N(-1.1537/0.64) = 3.57\%$, L $N(-1.7537/0.96) = 3.39\%$.

B is incorrect. Polarity flip: swapped the results, as if a higher loading protected a borrower in a downturn.

D is incorrect. Treated each a_i as a copula correlation ρ and used $\sqrt{a_i}$ as the loading: K $N(-0.8918/0.6325) = 7.92\%$, L $N(-1.3829/0.8944) = 6.10\%$.

LO: CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2, eq. 12.8 and footnote 4

Solution CR-12 f.7 **Answer: D**

Workbook B3.3 • S4, S6

Step 1 – Conditional independence

In the single-factor model $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\epsilon_i$ (Hull: $x_i = a_i F + \sqrt{1-a_i^2}\,Z_i$) the idiosyncratic shocks are independent of one another and of the factor. Fixing m leaves only ϵ_i random, so defaults are conditionally independent and the pairwise correlation $\beta_i\beta_j$ comes entirely from the loadings.

Step 2 – Conditional default probability check

$p(m) = \Phi((k_i - \beta_i m)/\sqrt{1-\beta_i^2})$. With $\pi = 1\%$ ($k = -2.33$) and $m = -1$: $\beta = 0.4$ gives 1.78% (Malz Example 11.4). A higher β raises the numerator but also shrinks the denominator of a negative argument, so the conditional PD rises with β only when $m < \beta k$: at $m = -1$, $\beta = 0.6$ gives 1.55%; at $m = -2$, $\beta = 0.4$ gives 4.79% and $\beta = 0.6$ gives 7.96% (corrected; see the correction box in the notes).

Interpretation

D is correct. Conditional independence is what makes the factor model tractable and underlies the Vasicek large-portfolio result used for credit VaR (B1, CR-12 g) and IRB capital (A31).

Why the other options are wrong

A is incorrect. Sibling swap (t_i for x_i). Hull: a company's time to default "is not even approximately normal"; the copula maps each t_i percentile to percentile into $x_i = N^{-1}[Q_i(t_i)]$ and assumes only the x_i are multivariate normal, so the marginals can take any shape.

B is incorrect. Polarity flip. In a severe downturn ($m < \beta k$) a higher loading pushes the conditional mean $\beta_i m$ further down and the conditional PD up: cyclical firms suffer more. B claims the opposite for every negative m, which fails at $m = -2$ (7.96% for $\beta = 0.6$ against

4.79% for $\beta = 0.4$).

C is incorrect. Sibling swap. Implied correlation is inferred from market prices using risk-neutral default probabilities from single-name CDS spreads (Malz 22.4); it is a risk-neutral, market rather than credit risk, parameter, not a historical estimate.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. CR-22 l: Define implied correlation and describe how it can be measured.

• **Reference:** Malz, Ch.8, 8.3 Conditional Default Distributions; Hull, OFOD 11e, Ch.24, 24.2; Meissner, Ch.5, The Gaussian Copula; Malz, Ch.9, 22.4 Implied Correlation

CR-16 • Netting, Close-out and Related Aspects • Vardeez: VERY IMPORTANT

CR-16 e

P2 CALC

Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression.

Workbook: none homed here • background in the A21 revision notes (Table 16.1; see the correction box)

WHAT IS TESTED HERE

Why P2. Vardeez flags trade compression in a VERY IMPORTANT chapter ("termination features and trade compression seem to be crucial here"). No workbook question is homed here and the map lists no shapes.

Vardeez: trade compression, with the net notional and the holder of the net contract.

What the questions look like: short calculations. A table of CDS (or swap) positions gives reference entity, maturity, long or short, notional, coupon and counterparty; you net the longs against the shorts, state the direction and name the counterparty that holds the net contract. Harder versions add a position on a different reference entity or maturity that cannot be compressed, or a ring of obligations among three to five parties where the net position of every party must be preserved. Coupons and counterparty names are usually red herrings. Two of the eight questions are GARP-tagged (GARP PE2 Q18 and Mock 4 Q19).

1. What compression keeps and what it removes

Compression tears up redundant offsetting trades and replaces them with fewer ones (CR-16 d, Section 5). Every calculation rests on one invariant: **each participant's net position is unchanged**; only the gross notional and the number of trades fall. Three things are asked:

- **Net notional:** total long minus total short on positions that are fungible (same reference entity, same maturity, standard coupon).
- **Direction:** long if the long side is larger, short if the short side is.
- **Holder of the net contract:** a counterparty on the side of the net position. GARP: the net contract "may naturally be with Counterparty A as a reduction of the initial transaction". When several counterparties sit on that side, questions state a convention, usually the one with the largest position in the net direction.

2. One reference entity, one maturity

GARP Table 16.1. A party has three CDS contracts on the same index with identical maturities and coupons, each with a different counterparty.

Reference	Notional	Long/short	Maturity	Coupon	Counterparty
ABC index	150	Long	20/12/2023	100	A
ABC index	75	Short	20/12/2023	100	B
ABC index	50	Short	20/12/2023	100	C
Net contract	*25*	*Long*	*20/12/2023*	*100*	*A*

$$\text{Net notional} = 150 - (75 + 50) = 25 \text{ long} \qquad \text{Gross} = 150 + 75 + 50 = 275 \;\rightarrow\; 25$$

Gross notional falls by 250 (90.9%) and three trades become one. Counterparty A is the only long counterparty, so the net contract is a reduction of the original trade with A, from 150 to 25. The other two trades are torn up.

Vardeez's (Schweser's) version. XYZ index, 200 long with X, 100 short with Y, 75 short with Z: net $200 - 175 = 25$ long, held by X; gross falls from 375 to 25, a reduction of 350 or 93.3%.

Coupons. Today's CDS trade with standard coupons and an upfront payment, so all positions in a bucket carry the same coupon and the net contract simply keeps it. Schweser describes the net coupon as the weighted average of the original coupons. If the coupons did differ, only weights equal to the *signed* notionals (long $+$, short $-$) preserve the premium cash flows: long 9 at 180 bp and short 7 at 150 bp leave $9 \times 180 - 7 \times 150 = 570$ notional-bp, which is 285 bp on the net 2. Swaps are harder still because each carries its own par rate; that is why GARP mentions "coupon blending" algorithms.

3. What can be compressed together

Only fungible positions net. Bucket by **reference entity** (single name or index) and **maturity**: CDS now mature on standard dates (20 March, June, September, December), which is exactly what makes bucketing possible.

- A position on a **different reference entity** stays as it is, even if it faces the same counterparty (Question CR-16 e.5).
- A position with a **different maturity** stays as it is, even on the same index (Question CR-16 e.4).
- The **counterparty** does not matter for netting: replacing several counterparties with one is the whole point. It only matters for who holds the net contract.
- Between **two parties only** (bilateral compression), all fungible trades collapse into one contract held long by the party with the larger gross long (Question CR-16 e.2).

4. Many parties: rings and the minimum notional

Write "$X \to Y$: n" for an obligation of n from X to Y (for CDS, X sells protection to Y). A party's **net position** is what it receives less what it pays, and the nets of all parties sum to zero. Two results do all the work:

- **Ring rule.** If obligations form a cycle, subtract the smallest leg from *every* leg. Each party's net is unchanged (it pays and receives the same amount less), no leg reverses, the smallest leg disappears and the gross falls by (number of legs) $\times$ (smallest leg).
- **Minimum notional.** The gross can never fall below the sum of the positive nets (equal to the sum of the negative nets). Reaching that floor usually means reversing a position or creating one between parties that had none; a ban on reversals, or a constraint such as two parties refusing exposure to each other, leaves the gross above it.

Worked example. $A \to B$: 40, $B \to C$: 70, $C \to A$: 55 (gross 165).

- Nets: $A = 55 - 40 = +15,\ B = 40 - 70 = -30,\ C = 70 - 55 = +15$; they sum to zero.
- Ring rule (smallest leg 40): $A \to B$: 0, $B \to C$: 30, $C \to A$: 15. Gross $165 - 3 \times 40 = 45$. Check: $A = +15$, $B = -30$, $C = 30 - 15 = +15$.
- Minimum: $15 + 15 = 30$, reached by $B \to A$: 15 and $B \to C$: 15, but $B \to A$ reverses the original $A \to B$ position.

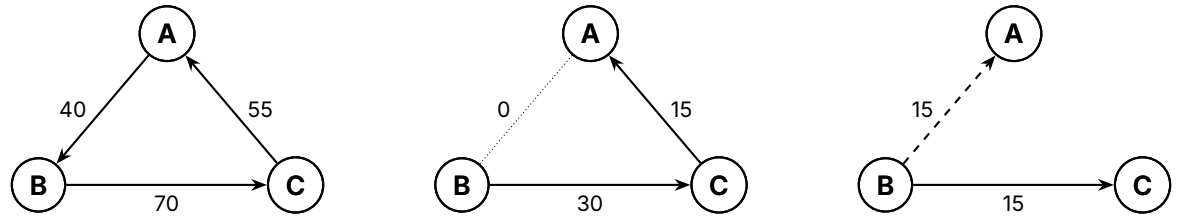


Figure 1. The worked example. Nets stay at A $+15$, B -30, C $+15$ throughout. The ring rule (centre) tears up the smallest leg (dotted) without reversing anything. The minimum (right) needs the reversed position from B to A (dashed), so it is allowed only if the stem permits reversals.

GARP's five-party market (Figures 16.7 to 16.9). Counting each transaction twice (both points of view), the total notional is 1,250. Counterparties 2, 3 and 4 form a ring with legs of 60, 70 and 85; reducing it by 60 cuts the total to $1{,}250 - 2 \times 3 \times 60 = 890$. Continuing gives several possible end points: one with a total of 130 that reverses the exposure between counterparties 4 and 5, and one of 110 that creates a trade between 1 and 3 where none existed. If 1 and 3 refuse exposure to each other, the better solution is ruled out. GARP also notes the objective itself is a choice: total notional, squared notional (penalises large positions) or the number of positions (penalises interconnectedness).

Correction to the source notes. The A21 revision notes describe Table 16.1 as "long 150 and 25 with A, short 75 with B and 50 with C, so a net long of 50 (175—125) held with Counterparty A." The 25 row is the *result* of the compression, printed below the three original contracts, not a fourth trade. The net is $150 - 75 - 50 = 25$ long with Counterparty A, as in GARP's caption ("the net long position resulting from transactions with three counterparties is reduced to a single identical long position with one of the counterparties").

Formula card: CR-16 e

Net notional (one bucket)	$\sum N_{\text{long}} - \sum N_{\text{short}}$; positive: net long, negative: net short
Bucket	same reference entity, same maturity, same (standard) coupon
Holder of the net contract	a counterparty on the net side; by convention the largest position in the net direction
Gross reduction	$\sum \lvert N_i \rvert - \lvert N_{\text{net}} \rvert$, as a share: divide by $\sum \lvert N_i \rvert$
Net position of party X	received $-$ paid (protection bought $-$ protection sold); unchanged by compression
Ring rule	subtract the smallest leg from every leg; gross falls by $k \times$ min for a k-leg ring
Minimum gross	$\sum$ positive nets; reaching it may need reversed or new positions
Counted twice	GARP's totals count each trade from both sides: halve them for a one-sided total
Net coupon (if coupons differ)	$\left(\sum N_{\text{long}} c - \sum N_{\text{short}} c\right) / N_{\text{net}}$ preserves the premium cash flows

Exam traps

- **Gross instead of net.** Reporting total long (or total long plus short) as the compressed notional.
- **Netting across buckets.** Offsetting a position on a different reference entity or maturity. Compress within the bucket and carry the rest unchanged.
- **Wrong direction.** Subtracting in the wrong order and calling the result long.
- **Wrong holder.** Right amount and direction, but a counterparty with the smaller same-direction position, or one on the opposite side.
- **Partial netting.** Offsetting only one of several opposite trades.
- **Ring arithmetic.** Subtracting the smallest leg once from the gross instead of from every leg; or subtracting the largest leg, which reverses positions.
- **Stopping early, or going too far.** Without a reversal ban the minimum is the sum of the positive nets; with one, the ring rule result is the answer.
- **Counted once or twice.** Check whether the stem's total counts each trade from both points of view.

Practice questions: CR-16 e

Question CR-16 e.1

Bank Q765 • AnalystPrep Q.5439

An analyst has the following information on a bank's derivatives trades. All trades are on the same reference credit and have the same maturity.

Counterparty	Reference credit	Bank's position	Notional ($)
4G Inc.	Blue Seal Int.	Long protection	300,000
Lights Advisors	Blue Seal Int.	Short protection	800,000
Air London	Blue Seal Int.	Short protection	250,000
Texas Electric Plc.	Blue Seal Int.	Long protection	900,000

What is the bank's net contract notional value after trade compression?

A. $150,000

B. $900,000

C. $950,000

D. $1,200,000

Question CR-16 e.2

Bank Q769 • BT P2.T6.24.27.1 (rebuilt)

Alpha Financial and Beta Investments have five 5-year interest rate swaps with each other, all with identical fixed rates, payment dates and maturity, so they are fully fungible. Notionals in USD millions:

Contract	Party long	Counterparty (short)	Notional
1	Alpha Financial	Beta Investments	100
2	Beta Investments	Alpha Financial	50
3	Alpha Financial	Beta Investments	100
4	Beta Investments	Alpha Financial	100
5	Alpha Financial	Beta Investments	50

After trade compression, what is the net notional amount and which party holds the net contract position?

A. Zero; the positions fully offset and no net contract remains.

B. USD 50m; Alpha Financial holds the net long position.

C. USD 100m; Alpha Financial holds the net long position.

D. USD 100m; Beta Investments holds the net long position.

Question CR-16 e.3

Bank Q766 • AnalystPrep Q.5511 (rebuilt)

Global Finance Bank (GFB) is compressing credit default swaps on the same reference entity and maturity (notionals in millions):

Reference entity	Notional	Position	Counterparty
XYZ Index	300	Long	Counterparty A
XYZ Index	100	Short	Counterparty B
XYZ Index	150	Short	Counterparty C
XYZ Index	50	Long	Counterparty D

Assume the net contract is booked with the counterparty that already holds the largest position in the net direction. After compression, what is GFB's net notional exposure and which counterparty holds the net contract position?

A. 0, with no counterparty holding a net position.

B. 100 short, held by Counterparty C.

C. 100 long, held by Counterparty D.

D. 100 long, held by Counterparty A.

Question CR-16 e.4

Bank Q767 • AnalystPrep Q.5512 (rebuilt)

A bank holds the following CDS positions on the same index (notionals in millions). Only trades with the same reference entity and maturity can be compressed together. The net contract is booked with the counterparty that already holds the largest position in the net direction.

Reference entity	Maturity	Notional	Position	Counterparty
DEF Index	20 Dec 2030	500	Long	Counterparty I
DEF Index	20 Dec 2030	400	Short	Counterparty J
DEF Index	20 Dec 2030	300	Short	Counterparty K
DEF Index	20 Jun 2031	200	Long	Counterparty L

What is the net notional of the compressible (December 2030) positions and which counterparty holds the net contract?

A. 0, because all long and short positions offset.

B. 200 long, held by Counterparty I.

C. 200 short, held by Counterparty J.

D. 200 short, held by Counterparty K.

Question CR-16 e.5

Bank Q771 • GARP PE2 Q18 (rebuilt)

An analyst applies trade compression to single-name CDS contracts, all with the same maturity and with the bank's counterparties shown:

Counterparty	Reference credit	Notional (EUR m)	Bank's position	Coupon (bps)
Digital Corporation	NUMU Inc.	9	Long protection	180
Digital Corporation	WKL Company	6	Long protection	150
WKL Company	NUMU Inc.	7	Short protection	150

What is the bank's total notional across all contracts after compression?

A. EUR 1 million.

B. EUR 2 million.

C. EUR 8 million.

D. EUR 22 million.

Question CR-16 e.6

Bank Q768 • BT P2.T6.904.2 • BT Mock Q67 (rebuilt)

Three counterparties A, B and C hold fungible positions in a ring. Notional sizes and directions: A $\rightarrow$ 50 $\rightarrow$ B, B $\rightarrow$ 75 $\rightarrow$ C, C $\rightarrow$ 60 $\rightarrow$ A. Counting each position from both parties' points of view (twice), the current total notional is 370. Trade compression must leave every party's net position unchanged, may not reverse the direction of any position, and should minimise total notional. Which is the efficient solution?

A. B $\rightarrow$ 10 $\rightarrow$ A and B $\rightarrow$ 15 $\rightarrow$ C (new total notional $= 50$).

B. B $\rightarrow$ 25 $\rightarrow$ C and C $\rightarrow$ 10 $\rightarrow$ A (new total notional $= 70$).

C. A $\rightarrow$ 25 $\rightarrow$ B, B $\rightarrow$ 50 $\rightarrow$ C and C $\rightarrow$ 35 $\rightarrow$ A (new total notional $= 220$).

D. B $\rightarrow$ 75 $\rightarrow$ C and C $\rightarrow$ 60 $\rightarrow$ A (new total notional $= 270$).

Question CR-16 e.7

Bank Q770 • BT P2.T6.24.27.3 (rebuilt)

Five counterparties hold fungible contracts. "X $\rightarrow$ Y: n" means X owes Y a notional of n. Positions: 1 $\rightarrow$ 2: 95; 1 $\rightarrow$ 5: 20; 2 $\rightarrow$ 3: 60; 3 $\rightarrow$ 4: 70; 4 $\rightarrow$ 2: 85; 4 $\rightarrow$ 5: 105; 5 $\rightarrow$ 2: 95; 5 $\rightarrow$ 4: 75. Compression reduces the ring 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 2 by its smallest leg and leaves every party's net position unchanged. After compression, what is the gross notional (each position counted once) and Party 4's net position?

A. Gross 425; Party 4 is a net payer of 45.

B. Gross 425; Party 4 is a net receiver of 85.

C. Gross 425; Party 4 is a net payer of 105.

D. Gross 545; Party 4 is a net payer of 45.

Question CR-16 e.8

Bank Q772 • Mock 4 Q19 (rebuilt)

Three dealers hold credit default swap positions referencing the same entity, all with identical maturities and coupons:

Protection buyer	Protection seller	Notional
Dealer A	Dealer B	EUR 50 million
Dealer B	Dealer C	EUR 30 million
Dealer C	Dealer A	EUR 40 million

The dealers submit these positions to a multilateral compression cycle that reduces gross notional as far as possible while leaving each dealer's net protection position unchanged. After compression, what is the total remaining gross notional, and which dealer is the net protection seller?

A. EUR 10 million; Dealer B is the net protection seller.

B. EUR 20 million; Dealer B is the net protection seller.

C. EUR 20 million; Dealer A is the net protection seller.

D. EUR 30 million; Dealer B is the net protection seller.

Solutions • CR-16 e

Solution CR-16 e.1

Answer: A

Bank Q765 • AnalystPrep Q.5439

Step 1 – Check the bucket

Same reference credit (Blue Seal Int.) and the same maturity: all four trades compress into one.

Step 2 – Total long and total short protection

$$\text{Long} = 300{,}000 + 900{,}000 = 1{,}200{,}000, \qquad \text{Short} = 800{,}000 + 250{,}000 = 1{,}050{,}000$$

Step 3 – Net

$$1{,}200{,}000 - 1{,}050{,}000 = 150{,}000 \text{ long protection}$$

$\Rightarrow$ Four trades become one \$150,000 long protection contract; gross notional falls from \$2,250,000 to \$150,000.

Interpretation

The bank's net credit view on Blue Seal is unchanged: it was net long \$150,000 of protection before compression and is afterwards. The counterparty names play no part in the netting.

Why the other options are wrong

B is incorrect. Reported the largest single trade (Texas Electric, \$900,000) instead of netting.
C is incorrect. Offset only one short: $1{,}200{,}000 - 250{,}000 = 950{,}000$, ignoring Lights Advisors.
D is incorrect. Reported gross long protection, \$1,200,000, ignoring both shorts.

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.4 Portfolio Compression, Table 16.1

Solution CR-16 e.2

Answer: C

Bank Q769 • BT P2.T6.24.27.1

Step 1 – Gross long of each party

$$\text{Alpha long} = 100 + 100 + 50 = 250, \qquad \text{Beta long} = 50 + 100 = 150 \ \text{(USD m)}$$

Step 2 – Net

$$250 - 150 = 100 \text{ in Alpha's favour}$$

$\Rightarrow$ All five swaps compress into one USD 100m swap with Alpha long.

Interpretation

This is bilateral compression: with only two parties there is no choice of holder, the net simply sits with the side whose gross long is larger. Gross notional falls from 400 to 100 and five trades become one, with no change in either party's market risk.

Why the other options are wrong

A is incorrect. Assumed that having trades in both directions means full cancellation; the gross longs are 250 and 150, not equal.
B is incorrect. Netted only contracts 1 and 2 $(100 - 50)$ and ignored contracts 3 to 5.

D is incorrect. Right amount, wrong direction: Beta's gross long (150) is the smaller one.

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.1 Payment Netting (bilateral compression) and 16.2.4

Solution CR-16 e.3 Answer: D

Bank Q766 • AnalystPrep Q.5511

Step 1 – Long and short totals

$$\text{Long} = 300 + 50 = 350, \qquad \text{Short} = 100 + 150 = 250$$

Step 2 – Net and direction

$$350 - 250 = 100 \text{ long}$$

Step 3 – Holder

The long counterparties are A (300) and D (50). By the stated convention the net 100 long sits with A, the largest long position.
$\Rightarrow$ One 100 long contract with Counterparty A replaces four trades; gross falls from 600 to 100.

Interpretation

Booking the net with A is GARP's "natural" outcome: it is simply a reduction of A's existing trade from 300 to 100, while the trades with B, C and D are torn up.

Why the other options are wrong

A is incorrect. Assumed the positions offset completely; longs (350) and shorts (250) do not match.
B is incorrect. Reversed the direction: short is the smaller side.
C is incorrect. Right amount and direction, but D holds only 50 long; A holds the largest long.

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.4, Table 16.1

Solution CR-16 e.4 Answer: C

Bank Q767 • AnalystPrep Q.5512

Step 1 – Bucket by maturity

Only the three 20 December 2030 trades are compressible; the 20 June 2031 trade with L (200 long) has a different maturity and is left untouched.

Step 2 – Net the December 2030 bucket

$$\text{Long} = 500, \qquad \text{Short} = 400 + 300 = 700, \qquad 500 - 700 = -200 \;\Rightarrow\; 200 \text{ short}$$

Step 3 – Holder

The short counterparties are J (400) and K (300); the net 200 short sits with J.
$\Rightarrow$ After compression the bank holds 200 short with J (December 2030) and, unchanged, 200 long with L (June 2031).

Interpretation

Standard CDS maturity dates exist so that positions can be bucketed by reference entity and

maturity. Across the two buckets the bank's positions look flat, but a December 2030 short and a June 2031 long are not fungible: the bank is still exposed to the shape of the credit curve.

Why the other options are wrong

A is incorrect. Netted the June 2031 long into the bucket $(500 + 200 - 700 = 0)$, although a different maturity cannot be compressed.
B is incorrect. Right amount, wrong sign: the short side (700) is larger.
D is incorrect. Right amount and direction, but K's short (300) is smaller than J's (400).

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.4 (standard coupons and maturity dates)

Solution CR-16 e.5 Answer: C

Bank Q771 • GARP PE2 Q18

Step 1 – Bucket by reference credit

Only trades on the same reference credit net. Counterparty and coupon do not decide the bucket; WKL Company appearing as a counterparty is irrelevant to the WKL Company *reference* position.

Step 2 – NUMU Inc. bucket

$$9 \text{ long} - 7 \text{ short} = 2 \text{ long protection}$$

Step 3 – WKL Company bucket

One trade only (6 long): nothing to offset, so it stays at 6.

Step 4 – Total after compression

$$2 + 6 = \text{EUR } 8 \text{ million} \quad (\text{down from } 9 + 6 + 7 = 22)$$

Interpretation

The coupons differ within the NUMU bucket (180 and 150 bp); in practice positions on a standard coupon are compressed, or the difference is settled upfront. The question asks only for notional. The trap is to treat the counterparty column as the bucket.

Why the other options are wrong

A is incorrect. Netted the WKL long (6) against the NUMU short (7), across different reference credits: $|6 - 7| = 1$.
B is incorrect. Counted only the compressed NUMU position (2) and dropped the WKL 6.
D is incorrect. Gross notional before compression, $9 + 6 + 7 = 22$.

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.4, Table 16.1

Solution CR-16 e.6 Answer: B

Bank Q768 • BT P2.T6.904.2

Step 1 – Net positions (receipts less payments)

$$A = 60 - 50 = +10, \qquad B = 50 - 75 = -25, \qquad C = 75 - 60 = +15$$

Step 2 – Ring rule: subtract the smallest leg (50) from every leg

$$A \to B = 0, \qquad B \to C = 25, \qquad C \to A = 10$$

Check: $A = +10$, $B = -25$, $C = 25 - 10 = +15$, all unchanged, and no leg has reversed.

Step 3 – New total (counted twice)

$$2 \times (25 + 10) = 70 \quad (\text{from } 2 \times 185 = 370)$$

$\Rightarrow$ The efficient solution is B $\to$ 25 $\to$ C and C $\to$ 10 $\to$ A.

Interpretation

The absolute minimum, the sum of the positive nets ($10 + 15 = 25$, or 50 counted twice), needs B to pay A, which reverses the original A $\to$ B position. The no-reversal constraint makes the ring rule the answer; contrast Question CR-16 e.8, where reversals are allowed.

Why the other options are wrong

A is incorrect. Subtracted 60 from every leg, which turns A $\to$ B into B $\to$ A: a reversal the stem forbids (total $2 \times 25 = 50$).

C is incorrect. Subtracted only 25 from each leg: nets preserved and no reversal, but not the maximum reduction ($2 \times 110 = 220$).

D is incorrect. Deleted the A $\to$ B leg without reducing the others, which changes the nets (A becomes $+60$); total $2 \times 135 = 270$.

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.5 Compression Algorithm, Figures 16.7 to 16.9

Solution CR-16 e.7 **Answer: A**

Bank Q770 • BT P2.T6.24.27.3

Step 1 – Gross before

$$95 + 20 + 60 + 70 + 85 + 105 + 95 + 75 = 605$$

Step 2 – Ring rule on 2 $\to$ 3 $\to$ 4 $\to$ 2 (legs 60, 70, 85)

Subtract 60 from each leg: $2 \to 3$: 0, $3 \to 4$: 10, $4 \to 2$: 25.

$$\text{Gross after} = 605 - 3 \times 60 = 425$$

Step 3 – Party 4's net position

Before: receives $70 + 75 = 145$, pays $85 + 105 = 190$, net payer of 45.
After: receives $10 + 75 = 85$, pays $25 + 105 = 130$, net payer of 45.
$\Rightarrow$ Gross 425, and Party 4 is still a net payer of 45.

Interpretation

This is the same ring and the same first step as GARP's Figure 16.8 (legs 60, 70 and 85, reduced by 60). Compression is not finished: the opposite positions between 4 and 5 (105 and 75) could be netted bilaterally next, as GARP's further solutions show. Party 4's net is the check that the compression is valid.

Why the other options are wrong

B is incorrect. Looked only at Party 4's receipts after compression ($10 + 75 = 85$) and ignored its payments.

C is incorrect. Mixed the receipts after compression (85) with the payments before it ($85 + 105 = 190$), giving a net payer of 105.

D is incorrect. Subtracted 60 once from the gross $(605 - 60 = 545)$ instead of from each of the three legs.

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.5 Compression Algorithm, Figures 16.7 and 16.8

Solution CR-16 e.8 Answer: B

Bank Q772 • Mock 4 Q19

Step 1 – Net protection positions (bought less sold)

$$A = 50 - 40 = +10, \qquad B = 30 - 50 = -20, \qquad C = 40 - 30 = +10$$

$\Rightarrow$ In EUR million, the nets sum to zero and B is the only net protection seller.

Step 2 – Gross before and the ring rule

Gross before $= 50 + 30 + 40 = 120$. The positions form a ring (B sells to A, A sells to C, C sells to B); removing the smallest leg (30) leaves 20 and 10, a gross of 30.

Step 3 – Minimum gross

The stem allows any restructuring that keeps the nets, so the floor is the sum of the net protection sold, 20: B sells 10 to A and 10 to C, replacing all three trades.

$\Rightarrow$ Remaining gross EUR 20 million, with Dealer B the net protection seller; a reduction of 100.

Interpretation

Reaching 20 requires B to sell protection to C, reversing the original C-to-B trade. That is allowed here; in Question CR-16 e.6 a reversal ban stops at the ring-rule answer. GARP's Figure 16.9 makes the same point: the lower-notional solution needs positions to change direction or appear where none existed.

Why the other options are wrong

A is incorrect. Counted only one net buyer's position (A's or C's 10) and ignored the other, so the nets are not preserved.

C is incorrect. Right amount, wrong direction: A $(+10)$ and C $(+10)$ are net buyers; B (-20) is the seller.

D is incorrect. Stopped after the ring rule (30); without a reversal ban the residual legs can be netted further.

LO: CR-16 e: Provide examples of trade compression of derivative positions, calculate net notional exposure amount, and identify the party holding the net contract position in a trade compression. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.2.4 and 16.2.5, Figures 16.7 to 16.9

CR-16 • Netting, Close-out and Related Aspects • Vardeez: VERY IMPORTANT

CR-16 f

P2 CONCEPT

Identify and describe termination events and discuss their potential effects on parties to a transaction.

Workbook: none homed here • termination features are tested in A21 Question A21.6 (worked as Question CR-16 d.7)

WHAT IS TESTED HERE

Why P2. Vardeez flags termination events in a VERY IMPORTANT chapter; the map lists no shapes and no workbook question is homed here.
Vardeez: termination events and their effects on the parties. The mechanics of break clauses, resets and walkaways are set out under CR-16 d; this LO recaps them and adds the events themselves and what happens to each party when one occurs.

What the questions look like: conceptual. Which feature lets a bank exit a long-dated trade before default; what a mandatory break means; whether a downgrade is an event of default; what close-out does after a default; which statement about ATEs or walkaways is false; and why break clauses tend to be exercised too late.

1. Two families of event

The ISDA Master Agreement specifies, among its risk-mitigating terms, "events of default and termination". Keep the two apart:

- **Events of default**, listed in the Master Agreement:

failure to pay or deliver	default under a specified transaction
breach of agreement	cross-default (default on another obligation)
credit support default (collateral terms)	bankruptcy
misrepresentation	merger without assumption

  The most common are failure to pay (subject to a threshold amount) and bankruptcy. A default leads to the termination of transactions before their maturity and the start of the close-out process.

- **Termination events** that the parties negotiate: **additional termination events (ATEs)**, usually set at counterparty level in the ISDA schedule. The most common is a rating downgrade (for example below investment grade); for unrated counterparties, a fall in net asset value or market capitalisation, or a key person leaving. At transaction level, **mutual puts** are mandatory or optional breaks on set dates. ATEs are not part of the standard documentation; they are the result of negotiation. They let a party act *before* the counterparty reaches bankruptcy.

A rating downgrade is therefore not an event of default. It can trigger an ATE, or tighter collateral terms under the CSA (CR-17 h).

2. Effects when an event of default occurs

- **Close-out.** The surviving party may terminate all transactions with the defaulter and stop contractual payments.
- **Close-out netting.** The values are offset into one net amount: a claim on the estate if positive, a payment to the estate if negative. The survivor can then replace and re-hedge at once, freezing its exposure.

- **Not acceleration.** An acceleration clause makes future payments immediately due; close-out terminates and nets. Courts may impose a short stay on both.
- **Valuation.** The close-out amount is often disputed (1992 ISDA: market quotation or loss; 2002 ISDA: a single close-out amount), so the default route is slower and more contentious than a break.
- **Walkaway exceptions.** Under a walkaway clause, or by not closing out and ceasing payments, a survivor that owes the defaulter may avoid paying; this is not standard ISDA practice and is criticised (CR-16 d).

3. Effects when a termination event is triggered

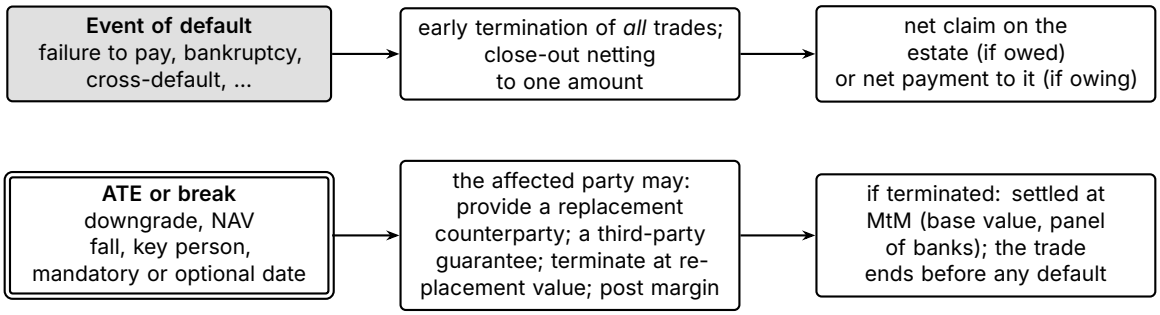


Figure 1. The two routes. An event of default (grey box, top) ends every trade and leads to a single netted close-out amount. A termination event (double-framed box, bottom) gives the affected party choices, and if the trade is terminated it is settled at a base MtM, before any default.

On the party exercising the clause. It exits a long-dated trade with a weakening counterparty before default, at the current replacement value, and avoids most close-out valuation disputes. It loses the trade, so it must replace the hedge or market exposure, and it may damage the client relationship.

On the counterparty against which it is exercised. It must find a replacement counterparty or guarantor, post margin, or pay out the MtM if it is out of the money, all at the moment its credit quality has just weakened. A termination can therefore strain its liquidity precisely when it is least able to cope.

On the market and both parties.

- **Cliff-edge effects.** Many counterparties hold similar clauses, so a single-notch downgrade can make them all act at once. AIG needed an extra USD 20bn of margin after its downgrade, and the Federal Reserve provided a facility of up to USD 85bn; downgrades sent monolines into collateral calls they could not meet (CR-14 i).
- **Systematic deterioration.** An idiosyncratic downgrade can be escaped; a system-wide decline, as among large financial institutions in the crisis, cannot.
- **Ratings react slowly.** By the time of the downgrade, the clause may be too late to help.
- **Inside the bank.** The banker's paradox (exercised too late to protect the relationship); the LCR forces pre-funding of rating-based triggers; Basel gives no capital relief for them; and they are hard to model or hedge (CR-16 d, Table 17.1).
- **Symmetry.** Under a bilateral clause either side may invoke it, so each party is exposed to the other's option.

	Event of default	ATE (credit trigger)	Mutual put (break date)
Examples	Failure to pay, bankruptcy, cross-default	Downgrade below investment grade; NAV, market cap, key person	Mandatory or optional break on fixed dates
Source	Standard ISDA Master Agreement	Negotiated, ISDA schedule	Negotiated, trade confirmation
Timing	After the counterparty fails	Before default, when credit weakens	On the dates, whatever the credit
Effect	Close-out of all trades, one net amount	Replace, guarantee, terminate at MtM or margin	Trade ends (mandatory) or may end (optional)
Main problem	Valuation disputes, legal delay	Slow ratings, cliff edges, LCR, no capital relief	Banker's paradox for optional breaks

Exam traps

- **Downgrade as an event of default.** A downgrade is not on the ISDA list; it may trigger an ATE or CSA terms.
- **ATEs act after default.** They are designed to act before the counterparty reaches bankruptcy.
- **ATEs and walkaways are standard ISDA.** ATEs are negotiated; walkaway features are not in standard documentation and should be avoided.
- **Mandatory versus optional.** A mandatory break ends the trade on the date whatever either party's credit; an optional break gives a choice.
- **Close-out versus acceleration.** Close-out terminates and nets; acceleration only brings payments forward.
- **Exercise always terminates.** The affected party may instead offer a replacement counterparty, a guarantee or margin.
- **Banker's paradox direction.** Breaks are exercised too late, not too early.

Practice questions: CR-16 f

Question CR-16 f.1

Bank Q773 • AnalystPrep Q.1887 (rebuilt)

A bank holds a long-dated derivative with a counterparty. Current exposure is small, but it could become large and unmanageable if the counterparty's credit quality declines over the life of the trade. The bank wants the contractual right to end the trade at specified future dates at its replacement value, so that it can exit if the counterparty weakens. Which feature best provides this?

A. A reset agreement that moves the trade back to at-the-money

B. A break clause (additional termination event)

C. A walkaway clause

D. Payment netting of the trade's cash flows

Question CR-16 f.2 Bank Q762 • AnalystPrep Q.1889 • moved from CR-16 d

A mandatory break clause implies that:

A. the transaction will certainly terminate at the date of the break clause regardless of the financial situations of both parties.

B. the transaction will not terminate at the date of the break clause regardless of the situations of both parties.

C. both parties have the option to terminate the trade transaction depending on the creditworthiness of each party.

D. the transaction will certainly terminate at the date of the break clause but give the defaulting party an option to accept it or reject it.

Question CR-16 f.3 Bank Q632 • AnalystPrep Q.6148 • re-filed from CR-14 e

Which of the following scenarios is considered a default event as outlined in the ISDA Master Agreement?

A. A transient delay in the settlement process due to technical issues.

B. The counterparty's credit rating downgrade below a specified threshold.

C. The breach of a covenant related to a specified transaction within the agreement.

D. A change in market interest rates impacting the valuation of the derivatives.

Question CR-16 f.4 Bank Q728 • AnalystPrep Q.5509 • re-filed from CR-16 b

One of Global Finance Bank's derivatives counterparties defaults. Which statement best explains the role of close-out netting under the ISDA Master Agreement?

A. It reallocates the defaulted trades to other counterparties to maintain market stability.

B. It accelerates all future payments due from the defaulting counterparty to give immediate liquidity.

C. It allows the bank to terminate all outstanding transactions with the defaulter and net them into a single claim or obligation.

D. It automatically transfers all of the bank's positions with the defaulter to a central clearinghouse.

Question CR-16 f.5

Bank Q775 • BT P2.T6.323.3

According to Gregory, each of the following is true about termination and walkaway features in credit contracts, EXCEPT which is false?

A. Termination events give an institution the right (option) to terminate a trade prior to their counterparty's creditworthiness deteriorating to the point of bankruptcy.

B. A break clause (a.k.a. liquidity put or early termination option) is an agreement to terminate (break) a transaction at pre-specified dates in the future at market rates.

C. Break clauses are often linked to additional termination events (ATEs), which are not in the standardized ISDA documentation and therefore are a result of negotiations.

D. Walkaway features are parts of the standardized ISDA documentation and "should be utilized in almost all transactions" as they mitigate counterparty risk.

Question CR-16 f.6

Bank Q774 • AnalystPrep Q.1890 (rebuilt)

Which of the following best describes a potential danger (practical weakness) of additional termination events and other break clauses?

A. They can only be exercised after the counterparty has already defaulted, so they give no protection.

B. Parties tend to avoid exercising them early to preserve the relationship, so they may not be used until the counterparty's credit quality has already declined substantially.

C. They are always mandatory, so the trade is terminated whether or not the counterparty has deteriorated.

D. They reset the trade to at-the-money, which transfers collateral to the weaker counterparty.

Solutions • CR-16 f

Solution CR-16 f.1 — Answer: B

Bank Q773 • AnalystPrep Q.1887

Step 1 – Read the need

Exit a long-dated trade, on specified future dates, at replacement value, if the counterparty's credit quality declines: a right to terminate before default.

Step 2 – Match the feature

That is a break clause (liquidity put or early termination option), often written as an additional termination event linked to the counterparty's credit quality.

⇒ B.

Interpretation

The value of the clause is timing: it lets the bank act before default, when the exposure can still be crystallised at a base MtM. Its weakness is the same timing: if the bank delays to protect the relationship, the clause loses its value (Question CR-16 f.6).

Why the other options are wrong

A is incorrect. A reset removes accumulated MtM but keeps the trade alive; the bank stays exposed to the counterparty.

C is incorrect. A walkaway clause operates only after default and lets the survivor avoid what it owes; it gives no exit before deterioration.

D is incorrect. Payment netting combines same-day cash flows and reduces settlement risk; it does not end the trade.

LO: CR-16 f: Identify and describe termination events and discuss their potential effects on parties to a transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.1.1 Break Clauses; SchweserNotes 2026 Book 2, Reading 34, LO 34.f

Solution CR-16 f.2 — Answer: A

Bank Q762 • AnalystPrep Q.1889

Step 1 – The three kinds of break

Mandatory: the trade terminates on the break date. Optional: one or both parties may choose to terminate on the date. Trigger-based: an event such as a downgrade must occur first.

Step 2 – Apply to the stem

A mandatory break involves no discretion and no trigger: the trade ends on the date, whatever the parties' credit quality.

⇒ A.

Interpretation

Because it is certain, a mandatory break is the only kind that a CVA desk, the LCR and Basel capital rules give full credit to (GARP Table 17.1): it can be modelled as happening with probability one. Its effect on the parties is that the hedge ends too, which is why a "gentleman's agreement" to re-book is common.

Why the other options are wrong

B is incorrect. States the opposite: the trade does terminate on the date.

C is incorrect. Describes an optional (or credit-linked) break, where a party chooses whether to terminate.

D is incorrect. Adds an accept-or-reject right for a defaulting party; a mandatory clause gives

no one a choice.

LO: CR-16 f: Identify and describe termination events and discuss their potential effects on parties to a transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.1.1 Break Clauses, Table 17.1; SchweserNotes 2026 Book 2, Reading 34, LO 34.d and 34.f

Solution CR-16 f.3 Answer: C

Bank Q632 • AnalystPrep Q.6148

Step 1 – The ISDA events of default

Failure to pay or deliver; breach of agreement; credit support default; misrepresentation; default under a specified transaction; cross-default; bankruptcy; merger without assumption.

Step 2 – Match the scenarios

A covenant breach on a specified transaction is a breach of agreement or a default under a specified transaction, both on the list.

$\Rightarrow$ C.

Interpretation

The distractor to watch is the downgrade. It is not an event of default; it is the most common *additional termination event*, or a trigger for tighter CSA terms. The distinction matters for the effect: a default terminates everything and starts close-out; an ATE gives the affected party options before any default.

Why the other options are wrong

A is incorrect. A brief operational delay is not a default; failure to pay counts only if it persists beyond the grace period and threshold.

B is incorrect. A downgrade is not a listed event of default; it may trigger an ATE or collateral terms.

D is incorrect. A change in interest rates is market risk that changes the exposure, not a failure by the counterparty.

LO: CR-16 f: Identify and describe termination events and discuss their potential effects on parties to a transaction. CR-14 e: Define the International Swaps and Derivatives Association (ISDA) Master Agreement, the risk-mitigating features it provides, and the default events it covers. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), ISDA Documentation; Ch.7 (GARP CR Ch.17), 17.1.1

Solution CR-16 f.4 Answer: C

Bank Q728 • AnalystPrep Q.5509

Step 1 – What a default does

An event of default lets the non-defaulting party terminate all transactions under the Master Agreement before maturity and start the close-out process.

Step 2 – What netting adds

The terminated values are offset into one amount: a claim on the estate if positive, an obligation to it if negative.

$\Rightarrow$ C.

Interpretation

The effect on the survivor is that its exposure is the net, not the sum of the positive values, and is frozen at once, so it can replace and re-hedge immediately. The effect on the defaulter's estate is that it receives only the net if it is owed money, which is why close-out netting shifts value from other creditors to derivative counterparties.

Why the other options are wrong

A is incorrect. Close-out is a bilateral termination; it does not reallocate trades to others.
B is incorrect. Describes an acceleration clause, which makes future payments due immediately; close-out terminates and nets instead.
D is incorrect. Transfer to a clearinghouse is central clearing, not a consequence of close-out.

LO: CR-16 f: Identify and describe termination events and discuss their potential effects on parties to a transaction. CR-16 b: Summarize netting and close-out procedures (including multilateral netting), explain their advantages and disadvantages, and describe how they fit into the framework of the ISDA master agreement. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.3.2 Close-Out Netting; SchweserNotes 2026 Book 2, Reading 34, LO 34.b

Solution CR-16 f.5 — Answer: D

Bank Q775 • BT P2.T6.323.3

Step 1 – Test A

Termination events let an institution act before its counterparty reaches bankruptcy. True.

Step 2 – Test B

A break clause terminates a transaction on pre-specified future dates at the prevailing market (replacement) value. True.

Step 3 – Test C

ATEs are negotiated, not part of standard ISDA documentation. True.

Step 4 – Test D

Walkaway features have been uncommon since the 1992 ISDA Master Agreement, are not in standard documentation, do not reduce credit exposure, and should be avoided because of the cost to the defaulter and the moral hazard they create.
$\Rightarrow$ D is false.

Interpretation

Statement D reverses every property of a walkaway clause. The effect of a walkaway on the parties is a transfer from the defaulter's estate to the survivor, not a reduction of risk.

Why the other options are wrong

A is incorrect. True: the point of a termination event is to act before default.
B is incorrect. True: the break settles at market value on the specified dates.
C is incorrect. True: ATEs come from negotiation, typically in the ISDA schedule.

LO: CR-16 f: Identify and describe termination events and discuss their potential effects on parties to a transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.3.3; Ch.7 (GARP CR Ch.17), 17.1.1; SchweserNotes 2026 Book 2, Reading 34, LO 34.d and 34.f

Solution CR-16 f.6 — Answer: B

Bank Q774 • AnalystPrep Q.1890

Step 1 – Where a break clause gets its value

It protects only if exercised before the counterparty's credit quality has fallen a long way.

Step 2 – What happens in practice

Exercising early damages the client relationship, so firms delay (the banker's paradox); sales staff may even argue against exercising a break they used to win the trade.
$\Rightarrow$ The clause tends to be used too late, when it helps least: B.

Interpretation

GARP lists further dangers that bite at the same moment: ratings react slowly, a systematic deterioration cannot be escaped, and many similar clauses can fire together (cliff-edge effects, as with AIG). GARP also notes that banks have recently become much more willing to exercise optional breaks.

Why the other options are wrong

A is incorrect. Break clauses act before default, when credit quality declines; they are not default-only.

C is incorrect. Breaks can be mandatory, optional or trigger-based; they are not always mandatory.

D is incorrect. Resetting to at-the-money is a reset agreement, a different feature, and it transfers no collateral to the weaker party.

LO: CR-16 f: Identify and describe termination events and discuss their potential effects on parties to a transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.7 (GARP CR Ch.17), 17.1.1 Break Clauses; SchweserNotes 2026 Book 2, Reading 34, LO 34.f

CR-19 • Future Value and Exposure • Vardeez: VERY IMPORTANT

CR-19 d

P2 CONCEPT

Identify typical credit exposure profiles for various derivative contracts and combination profiles.

Workbook: C8 Counterparty Exposure Profiles: EE, PFE, EPE • Tier C

WHAT IS TESTED HERE

Why P2. Vardeez flags the security exposure profiles in a VERY IMPORTANT chapter ("Various payoff profiles are important ... Pls do this"); C8 is a Tier C workbook section.
Vardeez: payoff / security exposure profiles: bonds and loans, swaps, FX products, long options and credit default swaps.

Reported exam shapes

- **C8·S3** Sold option red herring: zero exposure whatever Black-Scholes data the stem supplies.

What the questions look like: name the instrument from a described profile, pick the position that has (or keeps) exposure, and say how drift, correlation or the confidence level reshapes a profile. Question CR-19 d.4 (GARP Mock 3) supplies a full set of Black-Scholes inputs that are not needed.

Worked elsewhere: C8.1 (a sold put adds nothing to EPE inside a two-trade EPE and EFV calculation) under CR-19 a. Payment frequencies and exercise dates are CR-19 e.

1. Reading an exposure profile

An exposure profile plots the EPE (or the PFE, which has the same shape but sits higher) against time, as a percentage of notional, for a trade that starts at par unless stated (GARP 19.2). Exposure is the positive part of the value, max(value, 0) (CR-19 h, Section 1; the metrics are defined in CR-19 a). Two questions decide the shape: *how fast does uncertainty about the value grow*, and *how much of the contract is still outstanding*.

Loans and bonds are almost deterministic: exposure is roughly the notional. A fixed-rate bond can rise above it if rates fall; a floating-rate loan can fall below it through prepayment.

2. The two building blocks

- **One final exchange** (forward, FX forward; a vanilla option with an upfront premium). Uncertainty about the final cash flow grows with time, so with i.i.d. returns

$$\text{Exposure} \propto \sqrt{t} \qquad (t \le T) \qquad \text{(GARP eq. 19.3)}$$

  The profile rises monotonically to maturity. The maturity does not change the shape, except that exposure is zero after it.

- **Periodic cash flows** (interest rate swap). Payments "reverse the effect of future uncertainty": growing uncertainty (diffusion) is offset by the roll-off of remaining payments (amortisation):

$$\text{Exposure} \propto (T-t)\sqrt{t} \qquad \text{(GARP eq. 19.4)}$$

  Setting the derivative to zero, $\frac{1}{2}(T-t)t^{-1/2} - t^{1/2} = 0$, gives the peak at $t = T/3$: 3.33 years for a 10-year swap. A longer swap has much more risk (longer life, more payments).

3. What shifts a profile up or down

- **Curve shape (drift).** With an upward-sloping curve the projected floating payments are below the fixed rate early and above it late. A **receiver** swap expects net inflows first and outflows later: its EFV is negative, its EPE lower and its ENE more negative. The **payer** swap is the reverse, with the higher EPE.
- **Moneyness.** An in-the-money swap has EPE close to its EFV and a small ENE, more predictable and less sensitive to volatility, like an in-the-money option.

4. Combination profiles: the cross-currency swap

A cross-currency swap is an interest rate swap plus an FX forward (the final exchange of notionals), so its profile combines the peaked and the rising shapes. High FX volatility, long maturities and the final exchange make the **FX component dominate**: the profile keeps rising to maturity and does not fall back. Three further points:

- The **payer of the high-rate currency** has large exposure: its high payments are expected to be repaid by a gain on the final exchange. The payer of the low-rate currency has a negative drift and much lower exposure.
- The correlation between the interest rates and the FX rate matters; a low correlation, common in practice, increases the cross-currency exposure.
- Longer cross-currency swaps carry slightly more risk, from the extra interest payments.

5. Options and credit derivatives

- **Bought option, premium paid.** The option is an asset from day one and its exposure rises with time until exercise. A long call's upside is unbounded; a long put's is capped at the strike.
- **Sold option, premium received (shape S3).** The seller is owed nothing: the value to the seller is never positive, so exposure is **zero on every path**. Spot, strike, volatility and a Black-Scholes price describe the option's value, which is the *buyer's* exposure.
- **Long CDS protection.** The EPE looks like a swap's (spread widening, then roll-off). The 99% PFE jumps to $1 - R$ (60% of notional at 40% recovery) once the reference entity's default probability exceeds 1%, because the 99th percentile then falls in the default scenarios. The jump depends on the confidence level; the EPE, an average like expected shortfall, shows none, and an index CDS shows little. A protection seller's exposure is only the premiums still owed if spreads tighten. Wrong-way risk makes credit derivatives especially risky (CR-20 k).

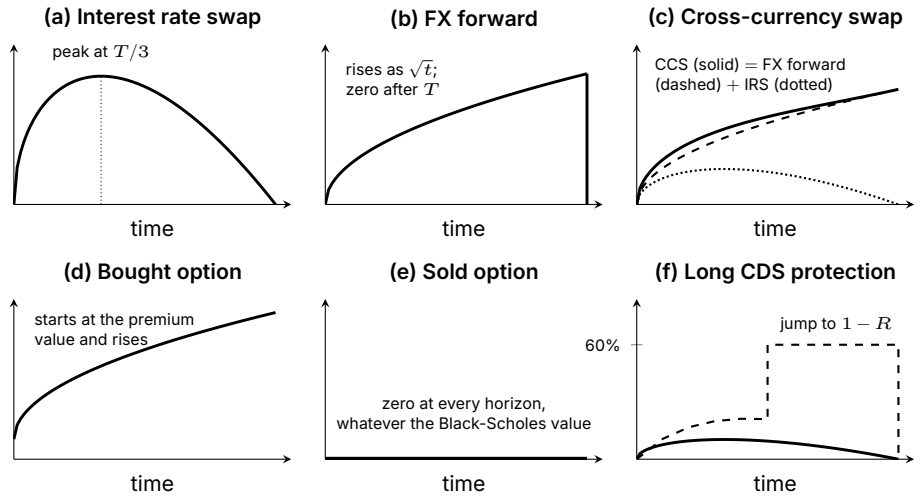


Figure 1. Stylised profiles (EPE unless stated). (a) Swap: $(T - t)\sqrt{t}$, peaking at a third of maturity. (b) FX forward: $\sqrt{t}$. (c) Cross-currency swap built from the two: the FX forward (dashed) dominates, the swap component (dotted) adds a little, and the total (solid) keeps rising. (d) Bought option: exposure from day one. (e) Sold option: none. (f) Long protection with 40% recovery: the EPE (solid) is swap-like, the 99% PFE (dashed) jumps to 60% once the reference entity's default probability exceeds 1%.

	Shape	What drives it	Maximum
Loan, bond, repo	Flat near notional	Principal owed; rates (bond up), prepayment (loan down)	Throughout
Forward, FX forward	Rising, $\propto \sqrt{t}$	One final exchange; uncertainty grows	At maturity
Interest rate swap	Peaked, $\propto (T - t)\sqrt{t}$	Diffusion against roll-off of payments	$T/3$
Cross-currency swap	Rising, large	FX volatility and final notional; IRS adds a little	At maturity
Bought option	Rising	Option value can only be positive	At exercise
Sold option	Zero	Seller is never owed anything	None
Long CDS protection	EPE swap-like; PFE jumps	Spread widening; default pays $1 - R$	At default, $1 - R$

Exam traps

- **Sold option red herring (S3).** A Black-Scholes value in the stem is the buyer's exposure; the seller's exposure is zero.
- **Swap and forward swapped (sibling swap).** Peaked is the swap (roll-off); monotonic is the forward, FX product or option (single final exchange).
- **Peak in the wrong place.** The swap peaks at $T/3$, not at half the maturity.
- **Payer and receiver reversed.** On an upward-sloping curve the payer swap has the higher EPE; the receiver has the negative EFV.
- **Cross-currency side reversed.** Paying the high-rate currency gives the large exposure; paying the low-rate currency gives a negative drift.
- **CCS read as starting in the money.** Exposure that stays high at maturity comes from the final notional exchange, not from an initial value.
- **CDS jump blamed on the wrong input.** The PFE jump comes from default being a discrete event and the confidence level, not from recovery-driven EE or a correlation parameter; recovery only sets its height.
- **Notional as value.** A par swap starts with zero value and zero exposure; the notional is never exchanged.

Practice questions: CR-19 d

Question CR-19 d.1

Bank Q939 • AnalystPrep Q.2901 (rebuilt)

Wiki Bank has entered into a two-year FX forward with a counterparty, GSO, agreeing to sell GBP against JPY at the market forward rate. Which of the following best describes the bank's potential future exposure (PFE) profile to GSO over the life of the contract?

A. Bell-shaped, peaking at about one year and falling toward zero at maturity because payments roll off.

B. Monotonically increasing and largest near maturity, driven mainly by FX volatility and the single final exchange of notional.

C. Flat at roughly 100% of notional throughout, as for a loan or repo.

D. Highest at inception and declining thereafter as the forward's time value decays.

Question CR-19 d.2

Bank Q949 • BT P2.T6.413.1 (rebuilt)

A counterparty risk analyst is reviewing the expected exposure profile of a 10-year interest rate swap, which is hump-shaped. Let T be the 10-year maturity and t the horizon between today and maturity. Each of the following is true EXCEPT:

A. Because the swap's maturity is ten years, the exposure peaks at about $\frac{1}{3} \times 10 = 3.33$ years.

B. With a typical upward-sloping yield curve, the receiver swap (counterparty who pays floating) has greater exposure than the payer swap (counterparty who pays fixed).

C. The profile initially increases because of the $\sqrt{t}$ term, called the diffusion effect.

D. The profile later decreases because of the $(T - t)$ term, called the amortization effect.

Question CR-19 d.3 Bank Q943 • BT P2.T6.407.1 (rebuilt)

A bank trades with a counterparty on an underlying whose price can rise without limit. Which of the following positions offers the bank the greatest potential future credit exposure to the counterparty?

A. Long call

B. Long put

C. Short forward

D. Short credit default swap (selling protection)

Question CR-19 d.4 Bank Q937 • Mock 3 Q35 (rebuilt) • shape C8·S3

A derivative trading firm sells (writes) a European-style call option on stock JKJ with 9 months to expiration, a strike of EUR 45, an underlying price of EUR 67 and implied annual volatility of 27%. The annual risk-free rate is 2.5%. The premium has been received in full at inception. What is the firm's counterparty credit exposure from this transaction?

A. EUR 0

B. EUR 22.00

C. EUR 22.84

D. EUR 23.02

Question CR-19 d.5 Bank Q945 • BT P2.T6.910.2 (rebuilt)

A counterparty risk manager is reviewing the drivers of exposure on a cross-currency swap. Which of the following tends to DECREASE both the expected exposure (EE) and the potential future exposure (PFE)?

A. A longer maturity

B. Higher foreign exchange (FX) volatility

C. Paying in the currency with the lower interest rate (receiving the higher-rate currency)

D. Paying in the currency with the higher interest rate (receiving the lower-rate currency)

Question CR-19 d.6

Bank Q950 • BT P2.T6.413.2 (rebuilt)

A bank reviews the 99% potential future exposure (PFE) profile of a 10-year cross-currency swap with a final exchange of notional. The PFE rises steadily through time, is largest close to the 10-year maturity, and does not decline toward zero as maturity approaches. The risk manager decomposes the profile into an interest rate swap component and an FX forward component. Which of the following statements about this profile is LEAST plausible?

A. The profile can be represented as a combination of an interest rate swap profile, an FX forward profile, and an assumption about the correlation between the interest rates and the FX rate.

B. The FX forward component contributes more to the total exposure than the interest rate swap component.

C. Because the PFE does not fall back toward zero at maturity, the swap must have had a significantly positive value at inception.

D. An increase in the assumed FX volatility would increase the exposure.

Question CR-19 d.7

Bank Q946 • BT P2.T6.911.1 (rebuilt)

A long-protection single-name CDS has a five-year life. Its EE profile looks like that of an ordinary swap (rising early as the spread widens, then amortizing). Its 99% PFE profile, however, shows an abrupt jump to 60% of notional beginning around year 3, assuming a 40% recovery rate. Which statement is TRUE?

A. The key parameter that determines the shape of the EE profile is the recovery rate.

B. The EE profile is incorrectly drawn; it should also show an abrupt jump like the PFE.

C. From year 3 on, the default probability of the reference entity exceeds one minus the PFE confidence level (PD $> 1\%$), so the 99th percentile scenario includes default.

D. To force the PFE curve to overlap the EE curve, the analyst should keep lowering the interest rate parameter until the PFE falls to the EE.

Question CR-19 d.8 Bank Q944 • BT P2.T6.910.1 (rebuilt)

A credit analyst compares the credit exposure profiles of six positions, each with five years to maturity:

I. Fixed-rate payer in a vanilla interest rate swap (IRS)

II. Cross-currency swap (CCS) with a final exchange of notional

III. Long call option, premium paid up front

IV. Short (written) call option, premium received up front

V. Long CDS (protection buyer)

VI. Short CDS (protection seller)

Just before the five-year maturity, for which position(s) is the expected exposure (EE) likely to remain clearly above zero?

A. II (CCS) and III (long call) only

B. IV (short call) and VI (short CDS) only

C. I (fixed-rate payer IRS), II (CCS), IV (short call) and V (long CDS)

D. All six positions

Solutions • CR-19 d

Solution CR-19 d.1

Answer: B

Bank Q939 • AnalystPrep Q.2901

Step 1 – Identify the building block

An FX forward has one exchange, at maturity, and no intermediate payments to roll off.

Step 2 – Shape

Uncertainty about the exchange rate grows roughly with $\sqrt{t}$, and FX volatility is high. The forward is struck at market, so exposure starts at zero and rises to its maximum at or just before maturity.

$\Rightarrow$ A monotonically increasing profile, the typical FX shape.

Interpretation

The currencies and the two-year term do not matter to the shape; the single final exchange does. Adding periodic payments (an interest rate swap) is what creates a peak.

Why the other options are wrong

A is incorrect. Describes an interest rate swap, whose periodic payments roll off; the forward has none.

C is incorrect. Describes loans, bonds and repos, where principal is owed; a market-struck forward starts with zero exposure.

D is incorrect. A market-struck forward has zero value at inception, so exposure starts at zero, not at its maximum.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.1, eq. 19.3

Solution CR-19 d.2

Answer: B

Bank Q949 • BT P2.T6.413.1

Step 1 – The swap profile

Exposure $\propto (T-t)\sqrt{t}$: the $\sqrt{t}$ term (diffusion) raises it early, the $(T-t)$ term (amortisation) lowers it later, and the maximum is at $t = T/3 = 3.33$ years.

$\Rightarrow$ A, C and D are true.

Step 2 – Payer versus receiver

On an upward-sloping curve the receiver of fixed (payer of floating) expects positive net cash flows early and negative ones later, so its EFV is negative and its EPE lower. The payer swap has the greater exposure.

$\Rightarrow$ B reverses the relationship and is the false statement.

Interpretation

The curve-shape effect is a drift: it moves the whole distribution of future value, which is why the payer and receiver swaps have different EPEs even though their volatilities are the same.

Why the other options are wrong

A is incorrect. True: the maximum of $(T-t)\sqrt{t}$ is at $T/3$.

C is incorrect. True: the diffusion effect dominates early.

D is incorrect. True: the amortisation effect dominates late.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.2 eq. 19.4 and 19.2.3, Figure 19.11

Solution CR-19 d.3 — Answer: A

Bank Q943 • BT P2.T6.407.1

Step 1 – Exposure needs positive value

Credit exposure arises only when the position is worth something to the bank, so compare the largest value each position can reach.

Step 2 – Bound each position

Long call: worth $S - K$ at least, unbounded as S rises. Long put: at most K (the price cannot fall below zero). Short forward: gains only if the price falls, at most the forward price. Short CDS: the seller is owed only the premiums still due.

$\Rightarrow$ Only the long call has unbounded potential exposure.

Interpretation

This is why bought options rise through time in Figure 1(d): their value can only be positive and, for a call, has no ceiling.

Why the other options are wrong

B is incorrect. A put's value is capped at the strike.

C is incorrect. A short forward's gain is capped at the forward price.

D is incorrect. A protection seller pays on default; its asset is the premium stream, small relative to notional.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.4 and 19.2.1

Solution CR-19 d.4 — Answer: A

Bank Q937 • Mock 3 Q35

Step 1 – Value to the writer

The firm has received the premium, so the buyer owes it nothing further. The firm's value is $V = -c \leq 0$ on every path.

$\Rightarrow$ Credit exposure $= \max(V, 0) = 0$.

Step 2 – What the inputs actually measure

With $S = 67$, $K = 45$, $T = 0.75$, $r = 2.5\%$, $\sigma = 27\%$: $d_1 = 1.899$, $d_2 = 1.666$ and the Black-Scholes value is $c = 23.02$. That is the buyer's asset and the writer's liability.

$\Rightarrow$ The inputs price the option; they do not create exposure for the writer.

Interpretation

This is shape S3: GARP supplies a full set of Black-Scholes data to tempt a calculation. Only the option buyer faces counterparty risk (Figure 1(e)).

Why the other options are wrong

B is incorrect. EUR 22.00 is the intrinsic value $S - K$, the buyer's view.

C is incorrect. EUR 22.84 is $S - Ke^{-rT} = 67 - 45e^{-0.01875}$, the lower bound of the call value, again the buyer's exposure.

D is incorrect. EUR 23.02 is the Black-Scholes value: the buyer's exposure, not the writer's.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.1 eq. 19.1 and 19.2.1; Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.1

Solution CR-19 d.5 Answer: C

Bank Q945 • BT P2.T6.910.2

Step 1 – The profile is driven by FX

A cross-currency swap is an IRS plus an FX forward; exposure comes mainly from FX volatility and the final exchange of notional.

Step 2 – The drift from the interest rate differential

The payer of the high-rate currency pays large coupons that are expected to be repaid by a gain on the final exchange: a positive drift and large exposure. The payer of the low-rate currency is increasingly likely to have a negative value: a negative drift and much lower exposure.
$\Rightarrow$ C lowers both EE and PFE.

Interpretation

The same contract can carry very different exposure to the two counterparties. A longer maturity and higher FX volatility only widen the distribution.

Why the other options are wrong

A is incorrect. A longer maturity adds payments and time for FX to move, so exposure rises.
B is incorrect. Higher FX volatility widens the distribution of the final exchange and raises exposure.
D is incorrect. The opposite side: paying the high-rate currency gives the positive drift and the larger exposure.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.3, Figure 19.12, and 19.2.5

Solution CR-19 d.6 Answer: C

Bank Q950 • BT P2.T6.413.2

Step 1 – The decomposition

A cross-currency swap combines an IRS profile, an FX forward profile and the correlation between rates and FX (A plausible). The FX component dominates because of FX volatility and the final exchange (B plausible), and more FX volatility raises the exposure (D plausible).

Step 2 – Why the profile stays high at maturity

The final exchange of notional is still outstanding and FX uncertainty keeps growing with time. A par swap with zero initial value has exactly this shape.
$\Rightarrow$ C infers an initial value that the shape does not require; it is the least plausible.

Interpretation

Moneyness shifts a profile up (an in-the-money swap has EPE close to EFV), but a non-vanishing profile at maturity is the signature of a final exchange, not of a starting value.

Why the other options are wrong

A is incorrect. Plausible: GARP builds the CCS profile from the IRS and FX forward profiles, with correlation as a driver.
B is incorrect. Plausible: the FX forward contributes most of the exposure.
D is incorrect. Plausible: higher FX volatility widens the exposure distribution.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.4 and 19.2.5, Figures 19.14 and 19.15

Solution CR-19 d.7 — Answer: C

Bank Q946 • BT P2.T6.911.1

Step 1 – Why the PFE jumps

The PFE at 99% is the 99th percentile of exposure. On a credit event the protection buyer is owed $1 - R = 60\%$ of notional. Once the reference entity's default probability exceeds $1 - 99\% = 1\%$, the 99th percentile lies in the default scenarios and the PFE jumps to 60%.
$\Rightarrow$ C is true.

Step 2 – Why the EE does not

The EE is an average over scenarios, so the rare default is diluted and the profile stays swap-like, driven by spread moves.
$\Rightarrow$ A and B are false; recovery sets the height of the PFE jump, not the EE shape.

Interpretation

GARP calls this a confusing effect: the PFE may or may not reflect a credit event depending on the confidence level. An average measure (EPE, or expected shortfall) avoids it.

Why the other options are wrong

A is incorrect. Recovery sets the 60% height of the jump; the EE follows spread movements.
B is incorrect. An expectation averages the rare default away; it should not jump.
D is incorrect. Interest rates do not cause the jump; only lowering the confidence level (or using an average measure) would narrow the gap.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.7, Figure 19.18

Solution CR-19 d.8 — Answer: A

Bank Q944 • BT P2.T6.910.1

Step 1 – Profiles that roll off

The IRS (I) and both CDS positions (V, VI), whose EE is swap-like, amortise as payments are made, so their EE converges to about zero at maturity.

Step 2 – Profiles with a large final payoff

The CCS (II) still has the final exchange of notional, driven by high FX volatility. The long call (III) has an expected payoff at expiry equal to its expected intrinsic value, which is positive.

Step 3 – The written call

The writer received the premium and is owed nothing: zero exposure throughout (IV).
$\Rightarrow$ Only II and III.

Interpretation

Exposure stays high at the end only when a single large uncertain payoff is still to come: a final notional exchange or an option's intrinsic value.

Why the other options are wrong

B is incorrect. Reverses the logic: the written call is owed nothing, and the protection seller's swap-like exposure amortises to zero.
C is incorrect. Gives terminal exposure to the IRS and long CDS, treats the written call as exposed and misses the long call.
D is incorrect. Ignores roll-off and the zero exposure of the option writer.

LO: CR-19 d: Identify typical credit exposure profiles for various derivative contracts and combination profiles. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.1, 19.2.2, 19.2.5 and 19.2.7

CR-19 • Future Value and Exposure • Vardeez: VERY IMPORTANT

CR-19 e

P2 CONCEPT

Explain how payment frequencies and exercise dates affect the exposure profile of various securities.

Workbook: none homed here • Vardeez focus: security exposure profiles

WHAT IS TESTED HERE

Why P2. Vardeez flags the security exposure profiles in a VERY IMPORTANT chapter; this objective is the second half of them. There is no workbook section and no reported shape.
Vardeez: unequal versus equal payment frequencies, and the swaption versus forward swap ranking that "flips at the exercise date".

What the questions look like: directional statements: which frequency mismatch raises exposure, which of a swaption and a forward swap has more exposure before and after exercise, and what cash versus physical settlement does. The bank has three questions; Question CR-19 e.1 is re-filed from CR-19 d (Schweser keys it to this objective) and two are new.

1. Payment frequency

Periodic payments are what turn the rising $\sqrt{t}$ profile into the peaked swap profile (CR-19 d, Section 2): each settled cash flow removes part of the remaining contract, so the exposure drops at each payment date (a sawtooth) and rolls off to zero by maturity.

Unequal frequencies (GARP 19.2.2, Figures 19.8 and 19.9). In a basis swap, or a swap where one leg pays more often:

- **Receive more often than you pay: less exposure.** Between your receipt and your payment you have already collected part of the period's cash while still owing the full payment on the other leg, so the contract's value to you is lower.
- **Pay more often than you receive: more exposure.** You have paid but are still waiting for the counterparty's leg, which is owed to you if it defaults.

Figure 1 shows the effect for a five-year swap with semiannual fixed payments: during the second quarter of each half-year the curves split, the receive-quarterly leg below and the pay-quarterly leg above the equal-frequency profile.

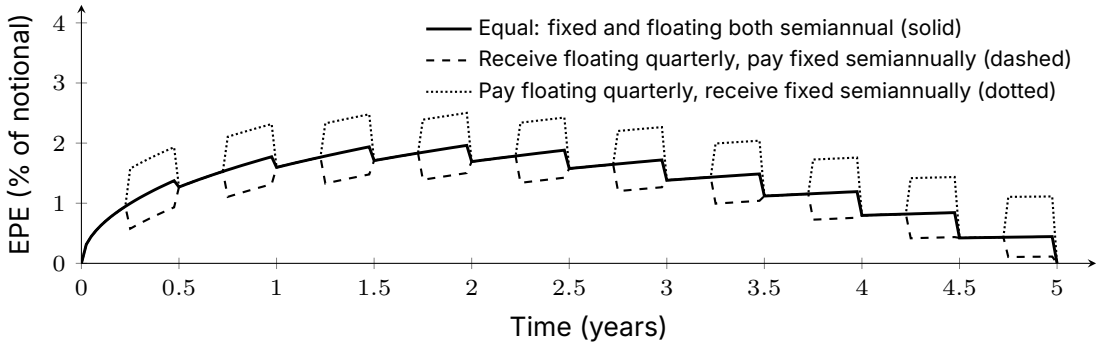


Figure 1. Stylised EPE of a five-year swap (1% rate volatility, 4% fixed rate). Every semiannual payment knocks the profile down (the sawtooth). In the second quarter of each half-year the quarterly leg has paid out a quarter's coupon: receiving it early lowers the exposure (dashed), paying it early raises it (dotted). Averaged over the life: 1.33% equal, 1.11% receive-quarterly, 1.61% pay-quarterly.

2. Exercise dates

Physically settled European swaption versus forward-starting swap (GARP 19.2.6, Figures 19.16 and 19.17). Same underlying swap; the option holder decides at the exercise date.

- **Before exercise** the swaption *must always* have the greater exposure: an option to enter a contract cannot be worth less than the obligation to enter it, and it is never negative.
- **After exercise** the ranking reverses: there are scenarios where the forward swap later has a positive value but the swaption was not exercised (out of the money at the exercise date) and no longer exists. Where the swaption was exercised, the two are the same swap.
- **Cash-settled swaption:** identical exposure until the exercise date, zero afterwards, since settlement ends the contract.

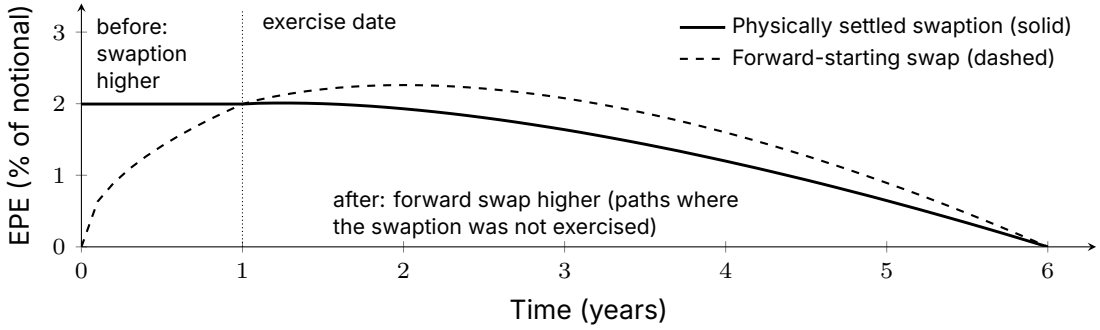


Figure 2. Stylised EPE of a one-year payer swaption into a five-year swap and of the equivalent forward swap (1% rate volatility, simulated). The swaption's EPE is its expected value, flat at 1.99% before exercise; the forward swap rises from zero to meet it at the exercise date. At year 3 the forward swap has 2.08% against the swaption's 1.64%. A cash-settled swaption would follow the solid line to year 1 and then drop to zero.

Modelling points.

- On the **base value**, the valuation paths are the same for cash- and physically settled swaptions, and exposure follows directly from the simulated future values.
- On the **actual value**, the exercise decision should include future xVA, so value, exposure and exercise become linked: a recursive problem. With several exercise dates, all combinations of decisions should be considered.

- Short-horizon VaR can ignore exercise decisions and cash flows; exposure, over long horizons, cannot ("ageing" and path dependency, CR-19 b).

Correction to the source notes. Vardeez's exercise-date table gives the reason for the swaption's higher exposure before exercise as "the potential for the option not to be exercised if it is not profitable". That is the reason for the *after*-exercise ranking. Before exercise the swaption is higher because "the option to enter into a contract cannot be worth less than the equivalent obligation" (GARP footnote 16), and GARP says it *must always* be higher, not "generally".

	Effect on the exposure profile
Each payment date	Exposure drops as a cash flow is settled (sawtooth); roll-off to zero by maturity
Receive more often than pay	Lower exposure than the equal-frequency swap
Pay more often than receive	Higher exposure than the equal-frequency swap
Swaption, before exercise	Always at least the forward swap's exposure
Physical swaption, after exercise	Below the forward swap: zero on paths not exercised
Cash-settled swaption, after exercise	Zero: the contract has ended

Exam traps

- **Frequency direction reversed (polarity flip).** Receiving more often lowers exposure; paying more often raises it.
- **"More frequent payments always help."** Only frequent *receipts* help; the effect depends on which leg is more frequent.
- **Ranking not flipped at the exercise date.** Swaption above before, forward swap above after.
- **Premium as the reason.** The swaption is higher before exercise because an option cannot be worth less than the obligation, not because a premium was paid; after exercise the premium does not keep it higher.
- **Settlement types swapped.** Cash settlement ends exposure at exercise; physical settlement leaves a live swap.
- **Exercise ignored.** Exposure models must carry exercise decisions forward; only short-horizon VaR can neglect them.

Practice questions: CR-19 e

Question CR-19 e.1

Bank Q951 • Schweser Module Quiz 37.2 Q2 • re-filed from CR-19 d

A risk analyst is matching exposure profiles to instruments. Which of the following security types will most likely result in a peaked shape for the exposure profile represented by potential future exposure (PFE)?

A. Long option position

B. Foreign exchange product

C. 10-year loan with a floating rate payment

D. Swap

Question CR-19 e.2

Bank Q952 • New question written for this bank

A risk analyst compares two five-year interest rate swaps with the same notional, fixed rate and term. In Swap 1, fixed and floating payments are both exchanged semiannually. In Swap 2, the bank pays fixed semiannually but receives floating quarterly. The analyst then considers Swap 3, in which the bank pays floating quarterly but receives fixed only semiannually. Relative to the equal-frequency profile of Swap 1, which of the following is correct about the bank's potential future exposure (PFE)?

A. Swap 2 shows higher exposure and Swap 3 shows lower exposure.

B. Swap 2 and Swap 3 both show the same exposure as Swap 1, because payment frequency affects only cash flow timing and not exposure.

C. Swap 2 and Swap 3 both show lower exposure, because more frequent payments in either direction reduce uncertainty.

D. Swap 2 shows lower exposure and Swap 3 shows higher exposure.

Question CR-19 e.3

Bank Q953 • New question written for this bank

A bank buys from a corporate client a European swaption that is swap-settled at an exercise date one year from now, and also enters into a forward swap on an identical underlying swap with another client. Which of the following correctly compares the potential future exposure of the swaption with that of the forward swap?

A. Before the exercise date the swaption has greater exposure, and after the exercise date the forward swap has greater exposure.

B. The swaption has greater exposure both before and after the exercise date, because the bank has paid a premium.

C. The two have identical exposure at all times, because the swaption is exercised into the same underlying swap.

D. Before the exercise date the forward swap has greater exposure, and after the exercise date the swaption has greater exposure.

Question CR-19 e.4

New • built from GARP Ch.19, 19.2.6, Figure 19.17

A counterparty risk analyst at a Canadian bank simulates the exposure of a physically settled one-year European payer swaption on a five-year swap, bought from Dealer X, alongside an otherwise identical forward-starting payer swap traded with Dealer Y. On one simulated path the underlying swap is worth -0.8% of notional to the bank on the exercise date, and then rises to $+1.5\%$ of notional by year 3 as rates rise. Which of the following correctly describes the bank's exposure at year 3 on this path?

A. Both contracts have the same exposure, because after the exercise date both reference the same underlying swap.

B. Only the forward swap has exposure: the swaption was not exercised when the swap was out of the money, so it no longer exists on this path.

C. Only the swaption has exposure: the forward-starting swap is cancelled when it is out of the money on the exercise date.

D. Both have exposure, but the swaption's is smaller because its negative value at the exercise date is carried forward as a loss.

Question CR-19 e.5

Bank Q954 • New question written for this bank

A credit analyst reviews the exposure of two European swaptions with a one-year exercise date on the same underlying interest rate swap. Swaption X is cash-settled and Swaption Y is physically settled (swap-settled). Which of the following statements about their exposure profiles is correct?

A. Swaption X has greater exposure than Swaption Y before the exercise date, because the cash settlement amount is certain.

B. The two have identical exposure until the exercise date; afterwards Swaption X has zero exposure while Swaption Y retains exposure on the underlying swap.

C. After the exercise date both swaptions have zero exposure, because the exercise decision has been made.

D. Swaption Y has zero exposure after the exercise date, while Swaption X retains exposure until the underlying swap matures.

Question CR-19 e.6

New • built from GARP Ch.19, 19.1.4 and 19.2.6

A model validator at an Australian bank is reviewing how the bank's exposure engine treats products with exercise decisions. Which of the following statements is correct?

A. Exercise decisions can be ignored when simulating exposure, as they are in 10-day VaR models, because they occur only on discrete dates.

B. If the exercise decision is based on the base value of the underlying swap, the simulated valuation paths are the same for cash- and physically settled swaptions, and exposure can be computed directly from the simulated future values.

C. If the exercise decision is based on the actual value including xVA, exposure becomes independent of the exercise boundary, which removes any recursion from the calculation.

D. For a Bermudan swaption with several exercise dates, only the first exercise decision needs to be modelled, because later decisions are implied by it.

Solutions • CR-19 e

Solution CR-19 e.1

Answer: D

Bank Q951 • Schweser Module Quiz 37.2 Q2

Step 1 – Two opposing forces in a swap

Uncertainty about the swap's value grows with the horizon (roughly $\sqrt{t}$), while periodic payments shrink the remaining contract (roughly $T - t$).

Step 2 – Their product

$(T - t)\sqrt{t}$ rises, peaks part-way through the life (at $T/3$) and falls to zero at maturity.
$\Rightarrow$ A swap gives the peaked profile; the payment frequency is what creates the peak.

Interpretation

Schweser's answer key files this question under the payment-frequency objective: without periodic payments the profile would keep rising like a forward's.

Why the other options are wrong

A is incorrect. A long option has no periodic cash flows to reverse the growth in uncertainty; its exposure keeps rising until exercise.
B is incorrect. An FX product has one large payment at maturity, so the profile rises monotonically.
C is incorrect. A loan's exposure is roughly its notional throughout; floating payments do not create a peak.

LO: CR-19 e: Explain how payment frequencies and exercise dates affect the exposure profile of various securities. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.2, eq. 19.4; SchweserNotes 2026 Book 2, Reading 37, Module Quiz 37.2

Solution CR-19 e.2

Answer: D

Bank Q952 • New question written for this bank

Step 1 – Receiving more often (Swap 2)

Amounts owed to the bank are settled sooner, so less value accumulates unpaid while the bank still owes its semiannual payment.
$\Rightarrow$ Exposure is below the equal-payment profile.

Step 2 – Paying more often (Swap 3)

The bank pays out sooner while its receipts lag, so the counterparty owes it more in the meantime.
$\Rightarrow$ Exposure is above the equal-payment profile.

Interpretation

The effect reverses with the direction of the mismatch (Figure 1). GARP: payments made more frequently than received carry more risk than the equivalent equal-payment swap, and vice versa.

Why the other options are wrong

A is incorrect. Reverses the direction: frequent receipts lower exposure.
B is incorrect. Frequency changes the timing of what is owed, and so the exposure profile.
C is incorrect. Treats frequency as always helpful; paying more often than receiving raises exposure.

LO: CR-19 e: Explain how payment frequencies and exercise dates affect the exposure profile of various securities. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.2, Figures 19.8 and 19.9

Solution CR-19 e.3 — Answer: A

Bank Q953 • New question written for this bank

Step 1 – Before exercise

The swaption can never be worth less than zero, and an option to enter a contract cannot be worth less than the obligation to enter it. The forward swap can be negative.
⇒ The swaption has at least the forward swap's exposure.

Step 2 – After exercise

The forward swap exists on every path, including those where its value later turns positive although the swaption was not exercised at the exercise date.
⇒ The forward swap has the greater exposure: the ranking reverses.

Interpretation

Figure 2 shows the flip: the two meet at the exercise date and then separate the other way.

Why the other options are wrong

B is incorrect. The premium does not keep the swaption ahead after exercise; it no longer exists on the paths where it was not exercised.
C is incorrect. The two are identical only on paths where the option is exercised.
D is incorrect. Reverses both periods; it ignores that an option cannot be worth less than the obligation.

LO: CR-19 e: Explain how payment frequencies and exercise dates affect the exposure profile of various securities. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.6, Figures 19.16 and 19.17

Solution CR-19 e.4 — Answer: B

New • built from GARP Ch.19, 19.2.6

Step 1 – The exercise decision on this path

At the exercise date the underlying swap is worth -0.8% to the bank, so a rational holder does not exercise. The swaption expires worthless.
⇒ From the exercise date on, the swaption does not exist on this path.

Step 2 – The forward swap

The forward swap was an obligation; it starts regardless and is worth $+1.5\%$ at year 3.
⇒ Exposure at year 3: forward swap 1.5%, swaption zero.

Interpretation

This is GARP's scenario A in Figure 19.17: a path that gives exposure on the forward swap but not on the swaption. On paths like GARP's scenario B (exercised), the two exposures are identical.

Why the other options are wrong

A is incorrect. After exercise the two are the same swap only on paths where the swaption was exercised.
C is incorrect. A forward-starting swap is not cancelled when it is out of the money; that is the option's privilege.
D is incorrect. An unexercised option carries no value forward; it simply ceases to exist.

LO: CR-19 e: Explain how payment frequencies and exercise dates affect the exposure profile of various securities. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.6, Figure 19.17

Solution CR-19 e.5 — Answer: B

Bank Q954 • New question written for this bank

Step 1 – Before the exercise date

Both are options on the same underlying swap, so their exposure profiles are identical.

Step 2 – After the exercise date

Cash settlement ends the contract with a single payment, so exposure drops to zero. Physical settlement leaves the underlying swap in place when exercised, with the usual swap profile until its maturity.

$\Rightarrow$ B is correct.

Interpretation

GARP footnote 15: "The cash-settled swaption has an identical exposure until the exercise date and then zero exposure thereafter."

Why the other options are wrong

A is incorrect. Settlement type does not change pre-exercise exposure.

C is incorrect. Physical settlement leaves a live swap, so exposure persists.

D is incorrect. Reverses the two settlement types.

LO: CR-19 e: Explain how payment frequencies and exercise dates affect the exposure profile of various securities. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.2.6, footnote 15

Solution CR-19 e.6 — Answer: B

New • built from GARP Ch.19, 19.1.4 and 19.2.6

Step 1 – Base value

With the exercise decision based on the base value of the underlying swap, the valuation paths are the same for cash- and physically settled swaps, and exposure can be calculated directly from the future values of the swaption.

$\Rightarrow$ B is correct.

Step 2 – The other statements

Over long horizons, exercise decisions, cash flows and margin postings must be carried forward ("ageing"); only short-horizon VaR can neglect them. Using actual value links the future value, exposure and exercise decision, which creates the recursive problem. With several exercise dates, all combinations of decisions should be considered.

$\Rightarrow$ A, C and D are false.

Interpretation

Base values keep the calculation tractable; the price is that exercise ignores the xVA that a real holder would weigh.

Why the other options are wrong

A is incorrect. Scope error: VaR's 10-day horizon lets it ignore exercise; exposure, defined over many horizons, cannot.

C is incorrect. Polarity flip: actual value makes exposure and exercise depend on each other, creating the recursion rather than removing it.

D is incorrect. GARP footnote 17: where multiple exercise decisions are taken, all possible combinations should be considered.

LO: CR-19 e: Explain how payment frequencies and exercise dates affect the exposure profile of various securities. •
Reference: Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.4 and 19.2.6, footnote 17

CR-19 • Future Value and Exposure • Vardeez: VERY IMPORTANT

CR-19 f

P2 CALC

Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values.

Workbook: C8 Counterparty Exposure Profiles: EE, PFE, EPE • Tier C

WHAT IS TESTED HERE

Why P2. Vardeez flags the netting factor in a VERY IMPORTANT chapter ("Netting Factor Formula and impact is important. Pls go to Crash Course"); C8 is a Tier C workbook section. CALC is the approved judgment call for the netting factor (C8 shape S2).
Vardeez: netting factor formula and impact; positive correlation reduces the netting benefit, negative correlation gives the most.

Reported exam shapes

- **C8·S2** The netting row: aggregation, EFV additive but EPE not, the netting factor.

What the questions look like: the netting factor from n and $\bar{\rho}$, from a scenario table, or across alternative portfolios; statements on additivity, correlation and moneyness. Watch the denominator: the netting factor is net over gross, and the netting benefit is one minus it.

Worked elsewhere: C8.1 (two trades with no enforceable netting, so EPEs are added standalone) under CR-19 a.

1. Aggregation: why netting reduces exposure

Close-out netting lets the values of all trades in a netting set offset, so exposure is taken on the *sum* (GARP 19.3):

$$\text{Netted exposure} = \max\Big(\sum_i v_i,\, 0\Big) \;\le\; \sum_i \max(v_i, 0) = \text{exposure without netting}$$

Three consequences (GARP 19.3.1):

- **EFV is additive**, because it is an average of values. **EPE and ENE are not**: the aggregated EPE (or ENE, in absolute terms) can never exceed the sum of the parts.
- **The benefit arises only in scenarios where values have opposite signs.**
- **It needs a legally enforceable netting agreement.** Without one, each trade is its own netting set and standalone exposures add (C8.1).

GARP Tables 19.1 and 19.2 (five equally likely scenarios, one date):

	Table 19.1: positive correlation			Table 19.2: negative correlation		
	Trade 1	Trade 2	Aggregated	Trade 1	Trade 2	Aggregated
Scenarios	50, 30, 10, −10, −30	15, 5, −5, −15, −25	65, 35, 5, −25, −55	50, 30, 10, −10, −30	−25, −15, −5, 5, 15	25, 15, 5, −5, −15
EFV	10	−5	5	10	−5	5
EPE	18	4	21	18	4	9
ENE	−8	−9	−16	−8	−9	−4

EFV adds $(10 - 5 = 5)$ in both tables. With positive correlation only scenario 3 has opposite signs, so the aggregated EPE (21) is barely below the sum (22): a netting factor of $21/22 = 95.5\%$. With

negative correlation every scenario offsets: EPE 9, a netting factor of $9/22 = 40.9\%$. (In Table 19.1 the *values* are perfectly correlated, but their positive parts are not, which is where the small benefit comes from.)

2. The netting factor

For n exposures with equal volatility and average correlation $\bar{\rho}$, jointly normal with zero mean, EE is proportional to the standard deviation. Netted, the standard deviation is $\sigma\sqrt{n + n(n-1)\bar{\rho}}$; unnetted, the EEs add to n times one. The ratio is the **netting factor** (Vardeez; Gregory's appendix, which the GARP excerpt cites but does not reproduce):

$$\text{Netting factor} = \frac{\text{net exposure}}{\text{gross exposure}} = \frac{\sqrt{n + n(n-1)\bar{\rho}}}{n} \qquad \text{netting benefit} = 1 - \text{netting factor}$$

Lower is better. Note $n(n-1)$, the number of *ordered* pairs: each covariance appears twice in the variance of a sum.

$\bar{\rho}$	Netting factor	Reading
1	100%	no benefit, whatever n
0	$1/\sqrt{n}$: 70.7% for 2, 50% for 4	benefit grows with n
$-1/(n-1)$	0%	the lowest attainable average correlation (-1 only for $n = 2$)
large n	$\to \sqrt{\bar{\rho}}$	beyond about 20 trades the correlation decides

Example. Ten trades, $\bar{\rho} = 0.25$: $\sqrt{10 + 90 \times 0.25}/10 = \sqrt{32.5}/10 = 57.0\%$, a netting benefit of 43%. At $\bar{\rho} = 0.5$ the factor rises to $\sqrt{55}/10 = 74.2\%$. GARP's summary: the exposure reduction increases with the size of the portfolio and as the correlation between transactions falls.

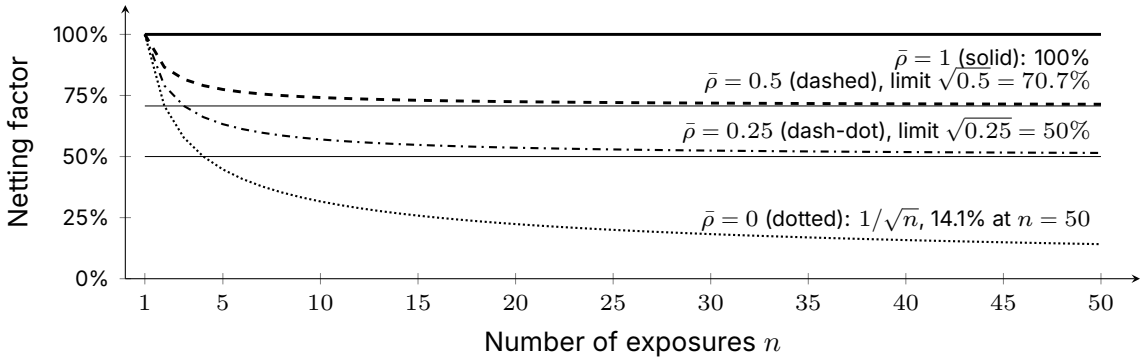


Figure 1. Netting factor against the number of exposures. At $\bar{\rho} = 1$ adding trades buys nothing. At $\bar{\rho} = 0$ the factor keeps falling as $1/\sqrt{n}$. In between, each curve flattens quickly towards $\sqrt{\bar{\rho}}$ (thin horizontal lines), so in a large netting set the correlation, not the size, governs the benefit.

3. Two trades with known distributions

When the stem gives each trade's mean and standard deviation, use the normal exposure formula (GARP 19.1.5, CR-19 a) twice:

$$\text{EE} = \mu\,\Phi(\mu/\sigma) + \sigma\,\phi(\mu/\sigma), \qquad \mu_P = \mu_1 + \mu_2, \quad \sigma_P = \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}$$

Without netting, add the two standalone EEs (correlation is irrelevant). With netting, compute one EE on (μ_P, σ_P).

Example. Two zero-mean trades with $\sigma_1 = 6$ and $\sigma_2 = 8$. Standalone EEs $0.3989 \times 6 = 2.39$ and $0.3989 \times 8 = 3.19$, gross 5.59. At $\rho = -0.5$: $\sigma_P = \sqrt{36 + 64 - 48} = 7.21$, netted EE 2.88, netting factor 51.5%. At $\rho = +0.5$: $\sigma_P = \sqrt{148} = 12.17$, netted EE 4.85, netting factor 86.9%. Higher correlation raises the netted EE towards the gross figure.

4. Moneyness also aggregates

Offsets are not driven only by structural correlation (GARP 19.3.2). An **out-of-the-money** netting set is unlikely to turn positive unless values move a long way, so it absorbs most of a new trade's positive values and damps its incremental exposure, even when the new trade is in the same direction. An **in-the-money** set can also benefit: a new trade's negative values offset part of the set's positive exposure. Even directional portfolios can therefore show large aggregation effects, which drives incremental CVA (CR-20 i).

Formula card: CR-19 f

Netted exposure	$\max(\sum_i v_i, 0) \le \sum_i \max(v_i, 0)$; EFV adds, EPE and ENE do not
Scenario netting factor	$\dfrac{\text{EE with netting}}{\text{EE without netting}}$, each averaged over all scenarios
Netting factor (equal σ)	$\sqrt{n + n(n-1)\bar{\rho}}\,/\,n$; benefit $= 1-$ factor
Special cases	$\bar{\rho} = 1$: 100%; $\bar{\rho} = 0$: $1/\sqrt{n}$; $\bar{\rho} = -1/(n-1)$: 0%; large n: $\sqrt{\bar{\rho}}$
Normal EE	$\mu\Phi(\mu/\sigma) + \sigma\phi(\mu/\sigma)$; zero mean: $0.3989\,\sigma$
Netted two-trade σ	$\sigma_P = \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}$, $\mu_P = \mu_1 + \mu_2$

Exam traps

- **Factor and benefit swapped.** 72% netting factor means a 28% benefit; GARP-style distractors offer both.
- **Distinct pairs.** Using $n(n-1)/2$ instead of $n(n-1)$ understates the factor.
- **Square root forgotten.** $(1 + (n-1)\bar{\rho})/n$ is the squared factor (the variance ratio), not the factor.
- **Correlation of -1 for many trades.** Only two exposures can have $\bar{\rho} = -1$; for n trades the floor is $-1/(n-1)$.
- **EPE treated as additive (sibling swap).** Only EFV adds; aggregated EPE is at most the sum.
- **Netting assumed without an agreement.** No enforceable netting, no offset: add standalone EPEs.
- **Mixed denominators.** Divide both the netted and the gross exposure by the number of scenarios, or neither; a factor above 100% means the ratio is inverted.
- **Correlation applied to un-netted EE.** Without netting each trade stands alone, so correlation has no effect.

Practice questions: CR-19 f

Question CR-19 f.1

Bank Q964 • Schweser Module Quiz 37.2 Q3

A counterparty risk analyst is explaining close-out netting to a new colleague. Which of the following statements best describes the benefit of netting risk exposures? The benefits of netting are realized when:

A. marked-to-market (MtM) values have high structural correlations for two trades.

B. marked-to-market (MtM) values have opposite signs for two trades.

C. expected exposure (EE) values are minimal.

D. expected future exposure (EFE) values have zero correlation.

Question CR-19 f.2

Workbook C8.2 • S1, S2 • Map #37

A senior counterparty risk analyst at a Singapore bank is reviewing a glossary of exposure metrics drafted by a junior colleague for the bank's xVA desk. Which of the following statements is correct to keep in the glossary?

A. Expected future value and expected positive exposure are both additive across transactions, so the EPE of a netting set equals the sum of the EPEs of its trades.

B. Average EPE is the time-weighted average of the EPE profile, often called the loan equivalent, while maximum PFE is the peak of the PFE profile over the life of the portfolio.

C. Expected negative exposure is the exposure seen from the counterparty's point of view, so it drives the credit value adjustment the bank charges that counterparty.

D. Doubling the volatility of future value leaves the expected future value unchanged but lowers expected positive exposure, because more scenarios end below zero.

Question CR-19 f.3

New • built from GARP Ch.19, 19.3.1 and 19.3.2

A counterparty risk manager at a Swiss bank is assessing the incremental exposure of a new pay-fixed interest rate swap with a corporate client. The existing netting set with the client consists of pay-fixed swaps that are deep out of the money to the bank after a fall in rates, so the new trade's value is highly positively correlated with the portfolio. Which of the following statements is correct?

A. Because the new trade is highly positively correlated with the existing swaps, aggregation gives no benefit and the incremental EPE equals the new trade's standalone EPE.

B. Although the values are highly correlated, the netting set is unlikely to have positive value unless rates rise a long way, so the incremental EPE will be well below the new trade's standalone EPE.

C. Because EFV is additive and the portfolio's EFV is negative, the netting set's EPE equals the new trade's standalone EPE plus the portfolio's EFV.

D. An out-of-the-money portfolio raises the incremental exposure of any new trade, because the bank already owes the client money.

Question CR-19 f.4

Bank Q960 • BT P2.T6.328.1 (rebuilt)

A risk analyst evaluates two trades with the same counterparty under five equally likely scenarios (a variation on Gregory's aggregation example). Exposure is the positive part of value, and the netting factor is EE with netting divided by EE without netting.

Scenario	1	2	3	4	5
Trade 1 MtM	25	15	10	−5	−15
Trade 2 MtM	15	−10	−5	5	−3

Which of the following is closest to the netting factor?

A. 14.29%

B. 28.57%

C. 71.43%

D. 140.00%

Question CR-19 f.5

Bank Q962 • Mock 2 Q28 (rebuilt)

Counterparty A has five exposures to counterparty B covered by a single netting agreement. The exposures have equal volatility and an average correlation of 40%. Which of the following is closest to the netting factor?

A. 28%

B. 40%

C. 52%

D. 72%

Question CR-19 f.6

Bank Q955 • AP Mock 3 Q67 (rebuilt)

A trading desk holds all of its equity positions under one netting agreement. Exposures are jointly normal with identical volatilities and a common pairwise correlation, so the netting factor is $\sqrt{n + n(n-1)\rho}\,/\,n$. The desk currently has 9 positions with an average correlation of 0.35. The chief trader proposes four alternative combinations:

Combination	Number of positions	Average correlation
K	4	0.25
Y	7	−0.08
E	10	−0.11
W	5	0.55

Which combination would increase the desk's expected netting benefit the most relative to the current position?

A. Combination K

B. Combination Y

C. Combination E

D. Combination W

Question CR-19 f.7 Bank Q961 • BT P2.T6.328.2 (rebuilt)

A counterparty risk analyst models two trades with jointly normal future MtM values. Trade 1 has mean 6% and standard deviation 10%; Trade 2 has mean -10% and standard deviation 30%; the correlation is 10%. Their standalone expected exposures (EE) are 7.69% and 7.63%. Which of the following statements about the two-trade portfolio is FALSE?

A. EE without netting is about 15.31%.

B. EE with netting is about 11.09%.

C. If the correlation increases, the EE with netting increases.

D. If the correlation increases, the EE without netting increases.

Solutions • CR-19 f

Solution CR-19 f.1 — Answer: B

Bank Q964 • Schweser Module Quiz 37.2 Q3

Step 1 – Where netting acts

Close-out netting offsets positive and negative values within a netting set: $\max(\sum v_i, 0) \leq \sum \max(v_i, 0)$, with equality whenever all values share a sign.
$\Rightarrow$ A benefit arises only in scenarios where the trades' MtMs have opposite signs.

Interpretation

Correlation matters only because it changes how often opposite signs occur; the sign condition is the mechanism.

Why the other options are wrong

A is incorrect. High positive correlation produces same-signed values and little or no benefit.
C is incorrect. A low EE is a result of netting, not the condition that creates the benefit.
D is incorrect. Zero correlation gives only a partial benefit; the offset itself requires opposite-signed values.

LO: CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.1; SchweserNotes 2026 Book 2, Reading 37, Module Quiz 37.2

Solution CR-19 f.2 — Answer: B

Workbook C8.2 • S1, S2

Step 1 – Check each statement against the metric definitions

Gregory (19.1.5): average EPE is "the average exposure across all time horizons", "the weighted average (the weights being the time intervals) of the EPE profile", and "this single average EPE number is often called a 'loan equivalent' ". Maximum PFE is the highest point of the PFE profile (Figure 19.4; Hull's maximum peak exposure).
$\Rightarrow$ B is correct. The trap inside it is naming: average EPE is "sometimes known simply as EPE", which differs from the single-horizon EPE.

Interpretation

The six-way family splits on two axes: what is averaged (all values: EFV; positive values with negatives set to zero: EPE; negative values: ENE; a quantile: PFE) and how time is handled (one horizon, a time-weighted average, or a maximum). A glossary should state both axes for every metric and flag that Basel's effective EE and effective EPE are non-decreasing versions defined for capital (outside-source).

Why the other options are wrong

A is incorrect. Sibling swap: additivity belongs to EFV only. EPE and ENE are not additive; aggregated EPE can never exceed the sum, since $\max(\sum v_i, 0) \leq \sum \max(v_i, 0)$ (Table 19.1: 18 and 4 versus 21). This sub-additivity is the whole netting benefit.
C is incorrect. Role misassignment: ENE "represents the positive exposure from the counterparty's point of view", which is the bank's own default risk and drives DVA. CVA is driven by EPE, the bank's exposure to the counterparty.
D is incorrect. Polarity flip: higher volatility raises EPE. With mean 2.0, EPE rises from 2.17 to 2.79 when the standard deviation doubles from 2.0 to 4.0, because exposure is option-like; only EFV is unchanged.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. CR-19 b: Compare the characterization of credit exposure to VaR methods and describe additional considerations used in the determination of credit exposure. CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11, 19.1.4, 19.1.5, 19.3.1

Solution CR-19 f.3

Answer: B

New • built from GARP Ch.19, 19.3.1 and 19.3.2

Step 1 – Two sources of offset

Aggregation reduces exposure in scenarios where values have opposite signs. That can come from structural correlation, or from relative moneyness: a portfolio that is deep out of the money stays negative in most scenarios.

Step 2 – Apply it to the new trade

The new pay-fixed swap gains when rates rise, and so does the portfolio, but the portfolio starts deeply negative. Unless rates rise a long way the netting set stays negative, so most of the new trade's positive values are absorbed.

$\Rightarrow$ The incremental EPE is well below the standalone EPE.

Interpretation

GARP: the OTM portfolio "damps the positive exposure contribution of a new transaction"; even directional portfolios can have significant aggregation effects. This is why incremental CVA can be small for a trade that looks risky on its own (CR-20 i).

Why the other options are wrong

A is incorrect. Ignores moneyness: positive correlation of values does not mean the positive parts move together when the portfolio is deeply negative.

C is incorrect. Borrows EFV's additivity for EPE; EPE is not additive and cannot be found by adding an EFV.

D is incorrect. Polarity flip: an OTM portfolio damps a new trade's exposure; it does not raise it.

LO: CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.1 and 19.3.2, Figures 19.20 and 19.21

Solution CR-19 f.4

Answer: C

Bank Q960 • BT P2.T6.328.1

Step 1 – EE without netting

$$\max(T_1, 0) + \max(T_2, 0) = 40,\ 15,\ 10,\ 5,\ 0; \qquad \text{EE} = 70/5 = 14.0$$

Step 2 – EE with netting

Netted values $T_1 + T_2 = 40,\ 5,\ 5,\ 0,\ -18$, so

$$\max(T_1 + T_2, 0) = 40,\ 5,\ 5,\ 0,\ 0; \qquad \text{EE} = 50/5 = 10.0$$

Step 3 – Netting factor

$$\frac{10.0}{14.0} = 71.43\%$$

$\Rightarrow$ Netting removes only 20 of 70 units of gross exposure.

Interpretation

Only scenarios 2, 3 and 4 have opposite signs, and in scenario 1 both trades are positive, so the benefit is a modest 28.57%.

Why the other options are wrong

A is incorrect. Divided the netted EE (10, already averaged) by the gross sum (70, not averaged): 14.29%.

B is incorrect. Reported the netting benefit, $1 - 71.43\% = 28.57\%$, not the factor.

D is incorrect. Inverted the ratio, $14/10 = 140\%$; netting cannot raise exposure.

LO: CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.1, Tables 19.1 and 19.2

Solution CR-19 f.5

Answer: D

Bank Q962 • Mock 2 Q28

Step 1 – Inside the square root

$$n + n(n-1)\bar{\rho} = 5 + 5 \times 4 \times 0.40 = 13$$

Step 2 – Netting factor

$$\frac{\sqrt{13}}{5} = \frac{3.6056}{5} = 72.1\%$$

$\Rightarrow$ Moderate positive correlation leaves most of the exposure un-netted.

Interpretation

The netting benefit is only about 28%. With zero correlation the five exposures would give $1/\sqrt{5} = 44.7\%$.

Why the other options are wrong

A is incorrect. Reported the netting benefit, $1 - 72\% = 28\%$.

B is incorrect. Repeated the correlation input.

C is incorrect. Forgot the square root: $13/25 = 52\%$.

LO: CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.1; Vardeez CR-19 f (netting factor)

Solution CR-19 f.6

Answer: C

Bank Q955 • AP Mock 3 Q67

Step 1 – Current netting factor

$$\frac{\sqrt{9 + 72 \times 0.35}}{9} = \frac{\sqrt{34.2}}{9} = 65.0\%$$

Step 2 – The four alternatives

$$\text{K: } \frac{\sqrt{4 + 12 \times 0.25}}{4} = 66.1\% \qquad \text{Y: } \frac{\sqrt{7 + 42 \times (-0.08)}}{7} = 27.3\%$$

$$\text{E: } \frac{\sqrt{10 + 90 \times (-0.11)}}{10} = 3.2\% \qquad \text{W: } \frac{\sqrt{5 + 20 \times 0.55}}{5} = 80.0\%$$

$\Rightarrow$ E has the lowest factor, so the largest benefit. Each correlation is attainable: E's -0.11 is just above its floor of $-1/9 = -0.111$.

Interpretation

More positions and lower correlation both cut the factor; E combines the most positions with the most negative correlation and is almost perfectly hedged.

Why the other options are wrong

A is incorrect. K has fewer positions and still positive correlation: 66.1%, slightly worse than today.

B is incorrect. Y improves the factor to 27.3%, but less than E.

D is incorrect. W has fewer positions and higher correlation: 80.0%, worse than today.

LO: CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.3.1; Vardeez CR-19 f (netting factor)

Solution CR-19 f.7

Answer: D

Bank Q961 • BT P2.T6.328.2

Step 1 – EE without netting

$$7.69\% + 7.63\% = 15.31\%$$

$\Rightarrow$ A is true; each trade stands alone, so correlation plays no part.

Step 2 – EE with netting

$$\mu_P = 6 - 10 = -4\%, \quad \sigma_P = \sqrt{10^2 + 30^2 + 2(0.1)(10)(30)} = \sqrt{1{,}060} = 32.56\%$$

$$\text{EE} = -4\,\Phi(-0.1229) + 32.56\,\phi(-0.1229) = -4(0.4511) + 32.56(0.3959) = 11.09\%$$

$\Rightarrow$ B is true.

Step 3 – Effect of correlation

Higher correlation raises σ_P, and the normal EE rises with σ, so the netted EE increases (C true). The un-netted EE is a sum of standalone figures and does not depend on correlation.

$\Rightarrow$ D is false.

Interpretation

Netting cuts EE from 15.31% to 11.09%, a netting factor of 72.4%. As correlation rises the netted figure climbs towards the gross one.

Why the other options are wrong

A is incorrect. True: $7.69 + 7.63 = 15.31$ (unrounded 15.31%).

B is incorrect. True: the netted EE is 11.09%.

C is incorrect. True: more correlation, less offsetting, higher netted EE.

LO: CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5 and 19.3.1

CR-19 • Future Value and Exposure • Vardeez: VERY IMPORTANT

CR-19 g

P2 CONCEPT

Describe the differences between funding exposure and credit exposure.

Workbook: none homed here • Vardeez focus: "just read"

WHAT IS TESTED HERE

Why P2. Vardeez flags it in a VERY IMPORTANT chapter ("LO 36.g: Just read this as well"; the text matches CR-19 g, GARP global number 37). No workbook section and no reported shape. **Vardeez:** the five-row table (value, MPoR, aggregation, wrong-way risk, segregation) and the mapping from positive and negative exposure to funding cost and benefit.

What the questions look like: "which statement is correct" on the parallel between the two exposures and on where they differ. The bank has one conceptual question; Question CR-19 g.6 is re-filed from CR-19 i and four are new. The bank's three funding calculations (Q965 to Q967) are calculations under a CONCEPT objective and not GARP-tagged, so they are left out; the formulas behind them belong to CR-19 i.

1. The parallel

Uncollateralised exposures that create counterparty risk also have to be *funded*, which is what FVA prices (GARP 19.4.1). Loosely (eq. 19.7):

$$\text{funding} = \text{value} - \text{margin}$$

- A **positive value** is an asset that must be funded (a funding cost). It is also the positive exposure lost if the counterparty defaults: the same amount is at risk in default and must be funded while the counterparty survives.
- A **negative value** is a funding benefit. It is also the negative exposure associated with the bank's own default (DVA); the two are thought to overlap.
- **Margin held** against a positive value reduces both the counterparty risk and the funding cost. **Margin posted** against a negative value reduces the counterparty's exposure to the bank and the bank's funding benefit.
- Funding is considered at **portfolio level**, so costs and benefits partly cancel. Not every asset creates a funding cost: liquid, low-risk bonds can be repoed, and the repo cash largely offsets the cost of buying them. Derivatives cannot be repoed.

2. The five differences

	Credit exposure (CVA, DVA)	Funding exposure (FVA)
Definition of value	Arises only in default: depends on close-out assumptions, subjective, agreed with the defaulter's administrators, open to legal challenge	Arises in non-default scenarios: an objective measure made by the bank itself (the recursion between value and xVA still exists)
Margin period of risk	Defined assuming the counterparty's default; the residual over the MPoR drives collateralised CVA	A normal, much shorter margin posting delay applies, so a collateralised trade's FVA may be zero while its CVA is not
Aggregation	Close-out netting applies in default, so exposure is measured per netting set	Values are additive and margin received can be reused across counterparties, so funding applies at the overall portfolio level
Wrong-way risk	Central: links default to the value of the portfolio or margin	Less important
Segregation	Segregated margin received can still be used in default, so it reduces exposure	Segregation prevents reuse, so segregated margin gives no funding benefit

Despite the differences, CVA, DVA and FVA are usually computed with shared methodologies.

GARP's examples (19.4.4) show the two measures diverging. Value 20, variation margin held 18, segregated initial margin received 3: credit exposure $\max(20-18-3,0)=0$, but funding $20-18=2$, because segregated initial margin gives no funding benefit. Value 20, variation margin held 15, initial margin of 3 exchanged both ways and segregated: credit exposure $\max(20-15-3,0)=2$ and funding $20-15+3=8$.

Which margin helps which (Table 19.4; worked in CR-19 i): non-segregated cash helps both; rehypothecable securities help both if haircuts are sufficient; segregated cash or securities reduce counterparty risk only; the counterparty's own bonds, rehypothecable, give a funding benefit but questionable risk reduction (wrong-way risk inside the margin).

Correction to the source notes.

- Vardeez's mapping reads "Counterparty defaults ⇒ positive exposure ⇒ funding costs". A positive exposure is *lost* if the counterparty defaults; the funding cost arises while the counterparty does *not* default (GARP 19.4.2: "the amount that has to be funded when the counterparty does not").
- SchweserNotes: "Posting margin against a positive exposure will decrease both funding costs and counterparty risk." A party does not post margin against its own positive exposure; it is margin *held* (received) against a positive exposure that reduces both.
- Vardeez's segregation row ("rules might be less restrictive" for funding, "more restrictive" for credit) is not GARP's distinction. GARP: segregation prevents the reuse of margin, so it affects the two differently: segregated margin still reduces credit exposure but gives no funding benefit.
- GARP's second example says the funding of 8 is "partially uncollateralised value (2) and partly the initial margin that is posted (5)". Those parts do not add to 8. The split is $20-15=5$ of value not covered by variation margin plus the 3 of initial margin posted (equivalently, 2 of credit exposure, plus 3 of segregated initial margin received that cannot be reused, plus 3 posted).

Exam traps

- **Funding cost tied to default (scope error).** Funding arises in survival; credit exposure arises in default.
- **Negative value as a funding cost (polarity flip).** A negative value is a funding benefit.
- **One MPoR for both.** The MPoR assumes default and applies to credit exposure only.

- **Aggregation levels swapped.** Credit: netting set. Funding: whole portfolio, with margin reuse.
- **Wrong-way risk assigned to funding.** It is mainly a credit-exposure concept.
- **Segregation's effect reversed.** Segregated margin protects in default but kills the funding benefit; it does not help funding.
- **Posted versus held margin.** Margin held against a positive value cuts both counterparty risk and funding cost; margin posted against a negative value cuts the funding benefit.

Practice questions: CR-19 g

Question CR-19 g.1

New • built from GARP Ch.19, 19.4.1 and 19.4.2

A treasury analyst at a Korean bank is explaining to the ALCO how funding exposure relates to counterparty credit exposure on the bank's uncollateralised derivatives. Which of the following statements is correct?

A. A positive portfolio value is at risk if the counterparty defaults and is also an asset that must be funded while the counterparty survives; margin received against it reduces both the counterparty risk and the funding cost.

B. A positive portfolio value creates a funding cost only if the counterparty defaults, since funding needs arise from the close-out of the transactions.

C. A negative portfolio value creates a funding cost, because the bank must eventually pay the amount it owes to the counterparty.

D. Margin posted by the bank against a negative value increases the bank's funding benefit and reduces the counterparty's exposure to the bank's default.

Question CR-19 g.2

New • built from GARP Ch.19, 19.4.1 and 19.4.2

The head of funding at a UK bank compares three holdings: a portfolio of liquid gilts; an uncollateralised portfolio of interest rate swaps with corporate clients that has a large positive value; and an offsetting set of uncollateralised swaps with other corporates that has a large negative value. Which statement is most consistent with the reading?

A. Every asset with a positive value has a funding cost, so the gilts and the positive swaps must be funded in full, and the negative swaps are irrelevant to funding.

B. The gilts can be repoed, so the repo cash largely offsets their funding cost; the swaps cannot be repoed, but because funding is assessed at portfolio level the negative values provide a benefit that offsets part of the cost of the positive values.

C. Funding is assessed counterparty by counterparty, like credit exposure, so the negative swaps provide no offset to the positive swaps.

D. The negative swaps create a funding cost equal to their value, which adds to the cost of funding the positive swaps.

Question CR-19 g.3

Bank Q968 • BT P2.T6.24.32.1 (rebuilt)

A risk manager reviewing a book of collateralized and uncollateralized swaps must distinguish funding exposure from credit exposure. Which statement correctly describes how they differ?

A. Credit exposure arises only in default scenarios, is measured at the netting set level and depends on subjective close-out assumptions; funding exposure arises in non-default scenarios, is objective and is considered at the overall portfolio level.

B. Funding exposure depends on close-out assumptions and a default-contingent margin period of risk, whereas credit exposure is an objective value measured at the overall portfolio level.

C. Both use the same margin period of risk, so the funding value adjustment is zero whenever the credit value adjustment is zero and vice versa.

D. Wrong-way risk is a key consideration for funding exposure but not for credit exposure, while segregation affects only credit exposure.

Question CR-19 g.4

New • built from GARP Ch.19, 19.4.2

A bank has two uncollateralised netting sets. With Counterparty X the portfolio is worth +USD 40 million to the bank; with Counterparty Y it is worth −USD 40 million. There is no netting agreement covering the two sets. Which statement best describes the bank's credit and funding positions?

A. Credit exposure and funding requirement are both zero, because the two values offset.

B. Credit exposure is USD 40 million to X, while the funding position across the two sets is roughly flat, because funding is assessed at portfolio level, where values are additive.

C. Credit exposure is USD 40 million to X and the funding requirement is USD 40 million, because funding, like credit exposure, is assessed netting set by netting set.

D. Credit exposure is USD 80 million, the sum of the absolute values, and the funding requirement is zero.

Question CR-19 g.5

New • built from GARP Ch.19, 19.4.2 and Ch.17, 17.5.2

A bank trades interest rate swaps with another dealer under a two-way CSA: zero thresholds, daily cash variation margin that can be reused, and no initial margin. The xVA desk reports a material CVA on the netting set but an FVA close to zero. Which explanation is consistent with the reading?

A. The FVA should equal the CVA, because funding and credit exposure use the same margin period of risk; the FVA calculation is therefore in error.

B. CVA is measured over a margin period of risk that assumes the dealer has defaulted and stopped posting, leaving residual exposure; funding uses the normal, much shorter posting delay, and the reusable cash margin offsets the value to be funded.

C. FVA is close to zero because wrong-way risk between the dealer's default and the swap values is high, which reduces the funding cost.

D. FVA is close to zero because the cash margin must be segregated, so it provides no funding benefit.

Question CR-19 g.6 Bank Q986 • BT P2.T6.912.3 • re-filed from CR-19 i

In a certain credit portfolio, Acme Bank holds four large over-the-counter (OTC) derivative positions, each with a different counterparty. The mark-to-market value of each position is significantly positive (above the threshold), so Acme Bank holds collateral for each position. The collateral types are:

I. Cash that does not need to be segregated

II. Securities that can be rehypothecated

III. Cash or securities that must be segregated (cannot be rehypothecated)

IV. The counterparty's own bonds, which can be rehypothecated

Which of the following statements about counterparty risk reduction and funding benefits is TRUE?

A. All four provide counterparty risk reduction and funding benefits.

B. I and II provide both counterparty risk reduction and funding benefits; III provides only counterparty risk reduction; and IV provides only a funding benefit.

C. I provides both counterparty risk reduction and funding benefits; II and III provide only a funding benefit; and IV provides only counterparty risk reduction.

D. None except I provides either counterparty risk reduction or a funding benefit.

Solutions • CR-19 g

Solution CR-19 g.1

Answer: A

New • built from GARP Ch.19, 19.4.1 and 19.4.2

Step 1 – The positive value

A positive value is at risk when the counterparty defaults and "is also the amount that has to be funded when the counterparty does not".
⇒ A positive exposure carries both a credit and a funding dimension.

Step 2 – Margin held against it

"Any margin held against a positive exposure reduces both counterparty risk and the associated funding cost", as long as the margin is reusable.
⇒ A is correct.

Interpretation

The parallel is why CVA and FVA share methodologies. The differences (value, MPoR, aggregation, wrong-way risk, segregation) are what stop them being the same number.

Why the other options are wrong

B is incorrect. Scope error: funding arises in non-default scenarios; default is when the exposure is lost.
C is incorrect. Polarity flip: a negative value is a funding benefit.
D is incorrect. Half right: posted margin does reduce the counterparty's exposure to the bank, but it reduces the bank's funding benefit rather than increasing it.

LO: CR-19 g: Describe the differences between funding exposure and credit exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.4.1 and 19.4.2, eq. 19.7

Solution CR-19 g.2

Answer: B

New • built from GARP Ch.19, 19.4.1 and 19.4.2

Step 1 – Not every asset costs funding

Liquid, low-risk bonds such as gilts can be repoed; the cash from the repo largely offsets the cost of buying them. Derivatives cannot be repoed, so their positive values are funding costs.

Step 2 – Funding is a portfolio quantity

Costs and benefits partly cancel because they only need to be considered at portfolio level; values are additive across counterparties.
⇒ The negative swaps' funding benefit offsets part of the positive swaps' funding cost.

Interpretation

For credit exposure the same two swap books could not offset unless they sat in one netting set with one counterparty.

Why the other options are wrong

A is incorrect. Ignores repo funding of liquid bonds and the portfolio-level offset.
C is incorrect. Sibling swap: netting-set aggregation belongs to credit exposure; funding applies at portfolio level.
D is incorrect. Polarity flip: a negative value is a funding benefit.

LO: CR-19 g: Describe the differences between funding exposure and credit exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.4.1 and 19.4.2

Solution CR-19 g.3 — Answer: A

Bank Q968 • BT P2.T6.24.32.1

Step 1 – Credit exposure

Defined in default scenarios: value rests on subjective close-out assumptions, netting applies at netting-set level, the MPoR assumes the counterparty's default and wrong-way risk matters.

Step 2 – Funding exposure

Arises in non-default scenarios: value is objective, the whole portfolio may be considered (margin can be reused across counterparties), the funding delay is not default-contingent and wrong-way risk is of little importance.

$\Rightarrow$ Only A states both sides correctly.

Interpretation

Segregation affects both measures, in opposite ways: it still helps in default (credit) but blocks reuse (funding).

Why the other options are wrong

B is incorrect. Swaps the roles: close-out subjectivity and the default-contingent MPoR belong to credit exposure, and the portfolio level to funding.

C is incorrect. The funding delay is not default-contingent, so a collateralised derivative can have zero FVA while its CVA is not zero.

D is incorrect. Reverses wrong-way risk (a credit concept) and wrongly says segregation does not affect funding; it blocks reuse.

LO: CR-19 g: Describe the differences between funding exposure and credit exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.4.2

Solution CR-19 g.4 — Answer: B

New • built from GARP Ch.19, 19.4.2

Step 1 – Credit exposure is per netting set

Close-out netting applies only within a netting agreement. X's set is worth $+40$ and Y's -40, with no agreement linking them.

$\Rightarrow$ Exposure to X is USD 40 million; Y's negative value cannot offset it in X's default.

Step 2 – Funding is portfolio-wide

Values for different transactions are additive for funding: the $+40$ asset needs funding and the -40 liability provides it.

$\Rightarrow$ The funding position is roughly flat.

Interpretation

The same pair of trades produces a large CVA (on X) but little funding cost: aggregation is one of the five differences, and here it dominates.

Why the other options are wrong

A is incorrect. Applies portfolio-level offset to credit exposure, which needs a netting agreement.

C is incorrect. Applies netting-set aggregation to funding.

D is incorrect. Adds absolute values: Y's set is a liability for the bank, not an exposure.

LO: CR-19 g: Describe the differences between funding exposure and credit exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.4.2

Solution CR-19 g.5 Answer: B

New • built from GARP Ch.19, 19.4.2

Step 1 – Why CVA remains

CVA uses the MPoR, defined assuming the counterparty has defaulted and stopped posting margin; the market move over that period (at least 10 days) is uncollateralised.

Step 2 – Why FVA vanishes

For funding, a normal margin posting frequency applies (likely much shorter), and reusable cash variation margin offsets the value that would otherwise need funding (funding = value − margin).
⇒ "This is one reason why the FVA of a collateralised derivative may be considered to be zero, even though the equivalent CVA is not."

Interpretation

The MPoR is a counterparty-risk-specific concept; GARP stresses it "is not relevant when assessing funding costs" (17.5.2).

Why the other options are wrong

A is incorrect. The two do not use the same MPoR; a non-zero CVA alongside a zero FVA is expected here.
C is incorrect. Wrong-way risk is mainly a credit-exposure concept and does not explain a low FVA.
D is incorrect. Variation margin is not segregated; and if it were, it would give no funding benefit and FVA would be higher, not lower.

LO: CR-19 g: Describe the differences between funding exposure and credit exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.4.2; Ch.7 (GARP CR Ch.17), 17.5.2

Solution CR-19 g.6 Answer: B

Bank Q986 • BT P2.T6.912.3

Step 1 – The two tests

Margin reduces counterparty risk if it keeps its value and can be used in default. It gives a funding benefit only if it can be reused in non-default scenarios (not segregated, rehypothecable).

Step 2 – Classify each type

I, non-segregated cash: usable in default and reusable, both benefits. II, rehypothecable securities: both, provided haircuts are sufficient. III, segregated margin: usable in default, not reusable, risk reduction only. IV, the counterparty's own bonds: likely in default exactly when needed (wrong-way risk), so questionable risk reduction, but rehypothecable, so a funding benefit.
⇒ B.

Interpretation

Segregation and wrong-way margin fail in opposite directions: segregation protects against default and kills the funding benefit; own bonds fund and fail as protection.

Why the other options are wrong

A is incorrect. Ignores segregation (III gives no funding benefit) and wrong-way risk (IV gives little risk reduction).
C is incorrect. Reverses the logic for II, III and IV.
D is incorrect. Too restrictive: II gives both benefits, and III and IV each give one.

LO: CR-19 g: Describe the differences between funding exposure and credit exposure. CR-19 i: Assess the impact of collateral on counterparty risk and funding, with and without segregation or rehypothecation. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.4.2 and 19.4.3, Table 19.4

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 a

P2 CONCEPT

Explain the motivation for and the challenges of pricing counterparty risk.

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • revision notes only, no question homed here

WHAT IS TESTED HERE

Why P2. Vardeez flags the LO in a VERY IMPORTANT chapter ("CVA/BCVA and DVA is a favorite topic of GARP ... you cannot afford to skip this chapter"). No Tier workbook question is homed here.

Vardeez: pricing lets an institution set aside reserves for counterparty default and see the risk profile of its portfolio; the challenges listed are exotic products and path dependency.

Reported exam shapes: none listed in the map. The two GARP themes to know cold are the bond-versus-swap contrast (Figure 1) and market-implied versus real-world parameters.

What the questions look like: all conceptual, "which response is most accurate". Distractors confuse a swap with a bond (notional at risk), flip the ordering of market-implied and real-world default probabilities, or claim that netting or collateral removes the need for an adjustment.

Worked elsewhere: the CVA formula under CR-20 b; the calculation under CR-20 c (Workbook A6.1).

1. Why price counterparty risk

The price of counterparty risk depends on the credit **exposure** to the counterparty and its **default probability**. It is charged on *all* outstanding positions with the counterparty, on top of the risk-free price of each instrument, and the charges build **reserves** against default losses. It must reflect the mitigants in place: netting (CR-20 h) and collateral (CR-20 j). CVA became a key topic because of volatile credit spreads and the associated accounting (IFRS 13) and capital (Basel III) requirements, and it matters to other financial institutions and corporates with significant OTC hedges, not only to banks. Historically it was a "credit charge" or "reserve" computed with historical parameters; accounting now treats it as an *exit price* computed with market-implied parameters where practical.

2. Challenge one: exposure on a two-way contract

A **bond** pays one way: allow for default when discounting the cash flows and add any recovery. A **swap** has payments in *both* directions, so a cash flow owed to you is only partly at risk, because opposing current and future flows offset it. The swap's risk is smaller (and there is no principal), but the fraction at risk depends on the yield-curve shape, forward rates and volatilities, so exposure has to be *simulated* (CR-19), not read off the contract.

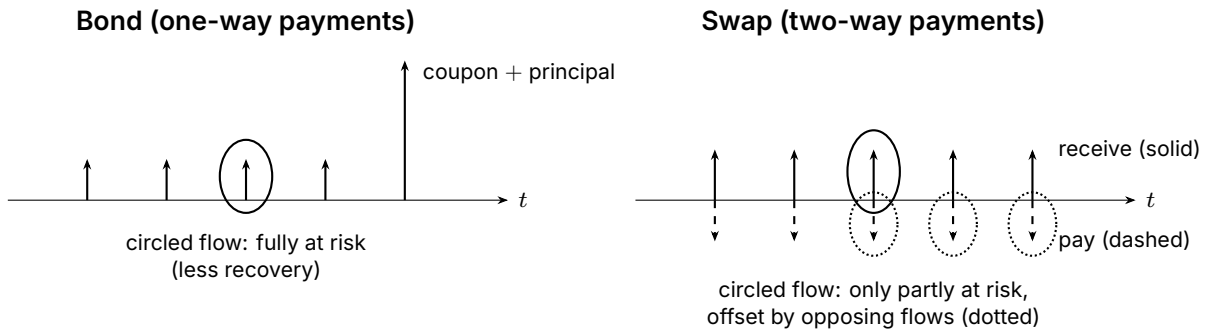


Figure 1. After GARP Figure 20.1. The bond's circled cash flow is lost (less recovery) if the issuer defaults. In the swap the same flow is partly cancelled by flows in the opposite direction, so only a fraction is at risk, and that fraction depends on rates, forwards and volatilities.

3. Challenges two and three: parameters and computation

- **Exposure inputs.** Risk-neutral simulation parallels option pricing, but correlations are not observed in the market and long-dated volatilities need extrapolation.
- **Default probabilities (the controversial part).** Most counterparties have no liquid single-name CDS, so *proxy credit spread curves* are needed. GARP gives three reasons to question market-implied PDs: they are significantly higher than real-world PDs; default cannot generally be hedged; and banks mostly *warehouse* credit risk, so they face only real-world default risk. These arguments are "somewhat academic": accounting standards generally require credit spreads. Historical PDs survive only in rare cases, such as a small regional bank whose exit price would be with a local competitor using the same approach.
- **Exotics and path dependency.** Valuing an exotic inside an exposure simulation may itself need Monte Carlo, so approximations are used (a swaption treated as a forward swap, a Bermudan as a European). Collateral and some exotics make the exposure at a date depend on the whole path, and the fast "direct" CVA simulation is less obviously applied to them (CR-20 b).

	CVA as an actuarial reserve	CVA as an exit price
Parameters	Historical volatilities and PDs	Market-implied where practical; proxy spreads otherwise
Default probabilities	Real-world: lower	Market-implied: significantly higher
Who uses it	Rare: small regional banks	Most banks (IFRS 13)

Exam traps

- **Swap treated as a bond.** A swap exchanges no notional; each flow is only partly at risk.
- **Polarity flip on PDs.** Market-implied default probabilities are *higher* than real-world ones.

- **Warehousing misread.** Warehousing means real-world default risk: an argument *against* market-implied PDs, which accounting nonetheless requires.
- **Absolute claims about mitigants.** Netting and collateral reduce CVA; they do not remove the need to price it.
- **"Only banks".** Corporates with significant OTC hedges face CVA too.

Practice questions: CR-20 a

Question CR-20 a.1

Bank Q989 • New question written for this bank

The finance director of a regional bank asks the counterparty risk manager why the bank charges a separate adjustment for counterparty risk on a newly executed 7-year interest rate swap with a corporate client, rather than simply relying on the swap's quoted price. Which of the following is the most accurate response?

A. The swap price already embeds the client's default risk, so the adjustment is only a presentation item that does not generate any reserve.

B. The adjustment should be computed on the swap in isolation, ignoring netting and collateral, so that the reserve is as conservative as possible.

C. The adjustment is a price of counterparty risk on all outstanding positions with the client, charged in addition to the price of the swap itself, and it builds reserves to absorb potential default losses.

D. The adjustment is needed mainly because the swap's full notional is exchanged at maturity and is therefore wholly at risk in default.

Question CR-20 a.2

New • built from GARP Ch.20, 20.1

The treasurer of a European manufacturing group that hedges its currency and interest rate risk with uncollateralised OTC derivatives asks the group's risk manager whether credit value adjustment (CVA) matters for a non-bank. Which of the following responses is most accurate?

A. CVA matters only to banks, because it was introduced by Basel III as a regulatory capital charge and has no role in a corporate's accounts.

B. CVA matters to any institution with significant OTC derivatives, including corporates that hedge economic risks; it became a key topic because of volatile credit spreads and the related accounting and capital requirements.

C. CVA matters only for trades outside a netting agreement, because netting and collateral remove counterparty risk entirely and leave nothing to price.

D. CVA needs to be computed only after a counterparty has been downgraded; until then the risk-free value of the derivative is its fair value.

Question CR-20 a.3

Bank Q987 • BT answers P2.T6.331.2 (rebuilt)

A bank's risk manager explains why quantifying counterparty risk on an interest rate swap is significantly more difficult than pricing the credit risk of a bond issued by the same counterparty. Which statement best captures the reason?

A. A swap has cash flows in both directions, so a cash flow owed to the bank is only partly at risk because opposing flows offset it, and the exposure must be modelled rather than read off the contract.

B. A swap exchanges its notional at maturity, so the amount at risk is always the full notional, which is larger than the bond's principal.

C. A bond has no recovery value in default, whereas a swap always has a positive recovery rate that must be estimated.

D. A swap counterparty's default probability cannot be estimated from market data, whereas a bond issuer's can.

Question CR-20 a.4

Bank Q988 • BT answers P2.T6.320.2

The price of a financial instrument can generally be defined in one of two ways:

I. Actuarial price: the price represents an expected value of future cash flows, incorporating some adjustment for the risk that is being taken (the risk premium).

II. Risk-neutral price: the price is the cost of an associated hedging strategy.

Into which camp does the credit value adjustment (CVA) fall?

A. I only

B. II only

C. Both, because the actuarial price must equal the risk-neutral price

D. Both, although the prices are likely to differ, perhaps dramatically

Question CR-20 a.5

New • built from GARP Ch.20, 20.1

The head of the CVA desk at a global dealer reviews the default probability inputs for about 1,500 corporate counterparties, fewer than 50 of which have liquid single-name credit default swaps. Which of the following statements about this part of the CVA calculation is correct?

A. For counterparties without liquid CDS, accounting standards require the desk to switch to historical default probabilities, so no proxy curves are needed.

B. Market-implied default probabilities are generally lower than real-world estimates, so CVA based on credit spreads understates the actuarial expected loss.

C. Most counterparties need proxy credit spread curves; the implied default probabilities are well above real-world ones and usually cannot be hedged, yet accounting standards generally still require credit spreads.

D. Because the dealer warehouses its credit risk rather than hedging it, it is exposed only to risk-neutral default risk, which is the main argument for market-implied default probabilities.

Question CR-20 a.6 New • built from GARP Ch.20, 20.2.2 and SchweserNotes LO 38.a

A bank's CVA project team is extending its exposure engine from vanilla interest rate swaps to Bermudan swaptions and to collateralised netting sets. Which of the following best describes the main challenge and a typical practical response?

A. Exotic and path-dependent trades change only the counterparty's default probability, so the team should recalibrate the credit curve and leave the exposure model unchanged.

B. Collateralised netting sets can be dropped from the CVA calculation because margin removes the exposure, so only the Bermudan swaptions need to be simulated.

C. Path dependency matters only for trades with a single payment at maturity, so neither the swaptions nor the collateralised netting sets raise new exposure-modelling issues.

D. Valuing exotics inside an exposure simulation may itself need Monte Carlo, so approximations are common (for example, a Bermudan treated as a European), and path-dependent exposure needs the whole path to each date.

Solutions • CR-20 a

Solution CR-20 a.1 — Answer: C

Bank Q989 • New question written for this bank

Step 1 – What is being priced

The adjustment prices the client's credit risk. It depends on the exposure to the client and the client's default probability, and it covers *all* outstanding positions with that client, not one trade.
⇒ It is a counterparty-level quantity.

Step 2 – How it relates to the swap's price

The quoted swap price is a risk-free price. The counterparty risk charge is levied in addition to it, and charging it builds reserves against default losses.

Step 3 – What it must reflect

Risk mitigants such as netting and collateral reduce the exposure, so the charge must recognise them.
⇒ C states all three features.

Interpretation

In practice the xVA desk charges the trader, and ultimately the client, for the change in this adjustment. Computing it on the swap alone would overcharge a client whose existing trades offset the new one (CR-20 h).

Why the other options are wrong

A is incorrect. The quoted price is the risk-free price; the adjustment is a real charge that funds reserves, not a presentation item.
B is incorrect. Ignoring netting and collateral overstates the charge; pricing must account for mitigants.
D is incorrect. An interest rate swap exchanges no notional. The amount at risk is the (netted, collateral-adjusted) replacement value, far below notional.

LO: CR-20 a: Explain the motivation for and the challenges of pricing counterparty risk. • **Reference:** SchweserNotes 2026 Book 2, Reading 38, LO 38.a; Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.1

Solution CR-20 a.2 — Answer: B

New • built from GARP Ch.20, 20.1

Step 1 – Why CVA became a key topic

GARP names three drivers: the volatility of credit spreads, the associated accounting requirements (IFRS 13) and the capital requirements (Basel III).

Step 2 – Who it applies to

CVA calculations are a major concern for banks, but GARP states that they are also relevant for other financial institutions and corporations with significant numbers of OTC derivatives used to hedge economic risks.
⇒ The manufacturing group's uncollateralised hedges create exactly this kind of counterparty exposure.

Interpretation

The group's banks charge it CVA in the price of each hedge, and its own fair values reflect the credit risk on both sides. The answer turns on recognising CVA as a valuation concept, not only a regulatory one.

Why the other options are wrong

A is incorrect. CVA is a fair-value (accounting) adjustment as well as a capital topic, and it applies beyond banks.
C is incorrect. Absolute claim: netting and collateral reduce CVA and must be reflected, but they do not remove counterparty risk.
D is incorrect. Default can happen at any time and spreads move continuously, so CVA is part of fair value from the trade date, not something triggered by a downgrade.

LO: CR-20 a: Explain the motivation for and the challenges of pricing counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.1 Overview

Solution CR-20 a.3 Answer: A

Bank Q987 • BT answers P2.T6.331.2

Step 1 – The bond

Payments are one way: the investor is owed every cash flow. Credit risk is priced by allowing for default when discounting and adding any recovery. Each cash flow is fully at risk, less recovery.

Step 2 – The swap

Payments flow in both directions, so the net value can be positive or negative and moves with market variables. A cash flow owed to the bank may be offset by current and future opposing payments.
$\Rightarrow$ Only part of each cash flow is at risk, and how much depends on the curve shape, forward rates and volatilities.

Step 3 – Consequence

The exposure is $\max(V, 0)$ at the default date, which has to be simulated (Figure 1).

Interpretation

The swap's risk is actually smaller than the bond's (partial offset, no principal), but it is much harder to quantify. That bilateral nature is the core challenge in GARP's account.

Why the other options are wrong

B is incorrect. Interest rate swaps exchange no notional; this confuses a swap with a bond's principal repayment.
C is incorrect. Both instruments have a loss given default; recovery is not what makes the swap harder.
D is incorrect. The default probability is estimated in the same way (CDS or proxy spreads) for both; the difficulty is the exposure.

LO: CR-20 a: Explain the motivation for and the challenges of pricing counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.1 CVA Compared to Traditional Credit Pricing, Figure 20.1

Solution CR-20 a.4 Answer: D

Bank Q988 • BT answers P2.T6.320.2

Step 1 – The actuarial camp

Historically CVA was a "reserve" or "provision" computed with historical (real-world) parameters: an expected-loss price.

Step 2 – The risk-neutral camp

Accounting now defines CVA as an exit price computed with market-implied parameters, which prices it relative to hedging instruments.

Step 3 – Do the two agree?

No. Market-implied default probabilities are significantly higher than real-world ones, and de-

fault usually cannot be hedged with liquid single-name CDS.
⇒ CVA can be seen in both camps, but the two numbers can be far apart.

Interpretation

Most banks report the risk-neutral version; the actuarial view survives only in rare cases such as small regional banks with minor derivatives books.

Why the other options are wrong

A is incorrect. Ignores the exit-price, market-implied view now required for pricing and accounting.
B is incorrect. Ignores the actuarial reserve view that banks historically used and a few still use.
C is incorrect. The two prices need not coincide, because real-world and risk-neutral default probabilities differ.

LO: CR-20 a: Explain the motivation for and the challenges of pricing counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.1 Overview

Solution CR-20 a.5 — Answer: C

New • built from GARP Ch.20, 20.1

Step 1 – The data problem

Obtaining market-implied default probabilities in most cases requires proxy credit spread curves, because most counterparties have no liquid single-name CDS.

Step 2 – The economic objections

GARP lists three: market-implied PDs are significantly higher than real-world PDs; default cannot in general be hedged; and banks warehouse credit risk, so they are exposed only to real-world default risk.

Step 3 – What practice does

The objections are "somewhat academic": accounting standards generally require credit spreads when reporting CVA.
⇒ C combines the proxy problem, the objections and the accounting requirement correctly.

Interpretation

The desk therefore spends its effort on the proxy mapping (rating, sector, region), not on choosing between real-world and risk-neutral PDs.

Why the other options are wrong

A is incorrect. The standard response to missing CDS is a proxy curve; historical PDs are used only in rare cases, not by rule.
B is incorrect. Polarity flip: market-implied PDs are higher than real-world PDs, so spread-based CVA is the larger figure.
D is incorrect. Warehousing means the bank bears *real-world* default risk; GARP uses it as an argument against market-implied PDs.

LO: CR-20 a: Explain the motivation for and the challenges of pricing counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.1 Overview

Solution CR-20 a.6 — Answer: D

New • built from GARP Ch.20, 20.2.2 and SchweserNotes LO 38.a

Step 1 – Exotic products

An exposure simulation needs the trade's value on every path at many dates. If the trade itself needs Monte Carlo to value (nested simulation), approximations are used: a swaption treated as

a forward swap, a Bermudan payoff as a European one.

Step 2 – Path dependency

For collateralised netting sets the exposure at a date depends on the margin already called, so the whole path to that date matters. Probabilities of path-dependent events are approximated.

Step 3 – Effect on method

GARP notes that the faster direct CVA approach, which values the portfolio only at a simulated default time, is less obviously applied to path-dependent portfolios such as collateralised trades and some exotics.

$\Rightarrow$ D names both challenges and the usual response.

Interpretation

These are exposure-side problems: the credit curve and the LGD are the same whatever the product.

Why the other options are wrong

A is incorrect. Exotics and collateral change the exposure profile, not the counterparty's default probability.

B is incorrect. Absolute claim: margin reduces CVA but does not remove it (the margin period of risk leaves residual exposure, CR-20 j).

C is incorrect. Path dependency means the payoff or exposure depends on the history of the underlying; Bermudan exercise and margin both do.

LO: CR-20 a: Explain the motivation for and the challenges of pricing counterparty risk. • **Reference:** SchweserNotes 2026 Book 2, Reading 38 (Applying CVA to exotic products and path dependency); Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 b

P2 CONCEPT

Describe credit value adjustment (CVA).

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • revision notes only; A6.1 tags this LO

WHAT IS TESTED HERE

Why P2. Vardeez flags the LO in a VERY IMPORTANT chapter ("know the formulas very well"). Workbook A6.1 tags it, but is homed under CR-20 c.

Vardeez: CVA is the difference between the risk-free value and the true value of a transaction or portfolio once counterparty risk is taken into account.

Reported exam shapes: none listed in the map. The A6 notes stress the three drivers (default, exposure at default, loss), the marginal PD in each interval, and the direct versus path-wise formulas.

What the questions look like: conceptual: what CVA is, which inputs the formula combines, what assumption makes it a simple sum, and why the direct simulation is faster. Distractors swap in PFE for EPE, the conditional or cumulative PD for the marginal PD, or the DVA inputs.

Worked elsewhere: UCVA from an EPE profile and cumulative PDs, Workbook A6.1 (at Question CR-20 c.3).

1. What CVA is

CVA was introduced as an adjustment to the risk-free value of a derivative for potential default; Gregory defines it as the **difference between the value with and without counterparty default**. In GARP's convention it is *negative*: $V_{\text{risky}} = V_{\text{risk-free}} + \text{CVA}$. Schweser quotes it as a positive cost and subtracts it, so read the stem. **Unilateral** CVA (UCVA) assumes the party making the calculation cannot default; its own default brings in DVA and bilateral CVA (CR-20 e to g).

2. Three questions and one formula

CVA is driven by three terms (GARP 20.2.2): *does* the counterparty default (PD); *if so, what is the exposure* then (EPE); and *how much* of it is lost (LGD $= 1 - R$). As an expected value (eq. 20.1), $\text{UCVA}(t) = -\text{E}\big[\mathbb{1}(\tau \le T)\, V(t,\tau)^{+}\,\text{LGD}\big]$, with τ the default time and V^{+} the discounted exposure at default. The three terms may be dependent. Assuming they are **independent** (no wrong-way risk, CR-20 k to p) means a single average LGD can be used and the expectation splits into the discrete sum used throughout this reading (eq. 20.3):

$$\text{UCVA} = -\text{LGD} \sum_{i=1}^{m} \text{EPE}(t_i) \times \text{PD}(t_{i-1}, t_i)$$

- **EPE** is the *discounted* expected positive exposure: an average, not a high quantile. PFE is a limit measure, not a CVA input.
- **PD**$(t_{i-1}, t_i) = Q(t_i) - Q(t_{i-1})$ is the *marginal* (unconditional) PD for the interval: not the conditional PD given survival, and not the cumulative PD to t_i.
- CVA is a **PD-weighted sum of the EPE profile** times LGD: market risk (EPE) times credit risk (PD). Under independence, default enters only through PD and need not be simulated.
- **Deep in the money.** EPE is then close to the expected future value (EFV), and CVA converges to a

simpler, bond-like discounting calculation (GARP Figure 20.7; the bond end of Figure 1 in CR-20 a).

3. Direct versus path-wise simulation

The **path-wise** approach evaluates eq. 20.3 by valuing the portfolio on every path at every grid date. The **direct** approach simulates a default time τ consistent with the credit curve, values the portfolio only at τ, multiplies by an average LGD and averages.

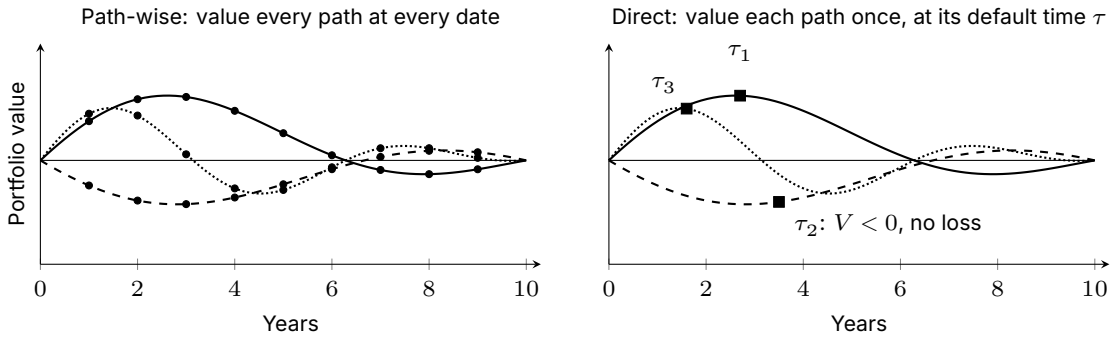


Figure 1. Three simulated value paths (solid, dashed, dotted). Left: the path-wise method values every path at every grid date (dots); successive values on a path are highly correlated. Right: the direct method draws one default time per path and values the portfolio once (squares); a path that is negative at its default time contributes zero exposure.

GARP's 10-year receiver swap (40,000 valuations each: 1,000 paths $\times$ 40 quarterly dates against 40,000 default times): the direct estimate has a standard deviation **six times smaller**, so it needs about $6 \times 6 = 36$ times fewer valuations for the same accuracy, because path-wise valuations at successive dates are highly autocorrelated. Two caveats: Amdahl's law caps the overall gain (90% of time in pricing sped up 25 times gives only $1/(0.1+0.9/25) = 7.3$ times), and the direct method is less obviously applied to path-dependent (collateralised, exotic) portfolios. The path-wise formula is the intuitive one; the direct one is the fast one.

	Path-wise formula	Direct calculation
What is simulated	Portfolio value at every grid date on every path	A default time per path, then the value at that time only
Strength	Intuitive: a sum over the EPE profile	Converges faster: 6 times smaller standard deviation
Weakness	Slow: successive valuations highly correlated	Less obvious for path-dependent portfolios

Exam traps

- **PFE for EPE.** CVA prices expected loss, so it uses expected positive exposure, not a high-percentile PFE.
- **Conditional or cumulative PD.** The formula uses the marginal PD for each interval as seen today.
- **DVA inputs.** ENE and the bank's own PD belong to DVA, not to the counterparty CVA.
- **Independence forgotten.** The product form, and the single average LGD, rely on no wrong-

way risk.

- **"Intuitive means fast".** The path-wise formula is the intuitive one but the slower one.
- **Dropped drivers.** CVA is proportional to LGD and depends on the whole exposure profile, not on PD alone.

Practice questions: CR-20 b

Question CR-20 b.1

New • built from GARP Ch.20, 20.1 and 20.2

A junior valuation analyst at a Swiss bank drafts a definition of credit value adjustment (CVA) for the bank's pricing manual. Which of the following drafts is most accurate?

A. CVA is the 99th percentile loss from counterparty default over one year, held as capital against the derivatives portfolio.

B. CVA is the difference between the value of a derivatives portfolio with and without counterparty default; in Gregory's convention it is negative, so the risky value is the risk-free value plus CVA.

C. CVA is the value to the bank of its own possible default on derivative liabilities, computed from its expected negative exposure and its own default probability.

D. CVA is the sum of the standalone adjustments of each trade with the counterparty, so it is unaffected by netting and collateral agreements.

Question CR-20 b.2

Bank Q990 • AnalystPrep Q.6208 (rebuilt, key corrected in the bank)

A bank is computing the unilateral credit value adjustment (CVA) on an uncollateralised portfolio of swaps with one counterparty, assuming no wrong-way risk. Which of the following correctly lists the inputs that the standard discrete CVA formula, $\text{CVA} = -\text{LGD}\sum_{i=1}^{m}\text{EPE}(t_i)\,\text{PD}(t_{i-1},t_i)$, combines?

A. Loss given default, discounted expected positive exposure at each future date, and the marginal default probability in each interval.

B. Loss given default, the 97.5th-percentile potential future exposure at each future date, and the marginal default probability in each interval.

C. Loss given default, the current mark-to-market value of the portfolio only, and the cumulative default probability to maturity.

D. Loss given default, discounted expected negative exposure at each future date, and the marginal default probability of the bank itself.

Question CR-20 b.3 New • built from GARP Ch.20, 20.1 and 20.2.2

A model validator at a Singapore bank reviews the CVA engine, which evaluates CVA $= -\text{LGD}\sum_{i=1}^{m}\text{EPE}(t_i)\,\text{PD}(t_{i-1},t_i)$ with a single average LGD for each counterparty. Which of the following statements about the assumption that justifies this form is most accurate?

A. It assumes the counterparty's hazard rate is constant, so it cannot be used when the credit curve is upward sloping or inverted.

B. It assumes exposure and default are positively dependent, so each EPE is measured conditional on the counterparty defaulting at that date.

C. It assumes the LGD is zero on collateralised trades, so only uncollateralised netting sets need to enter the sum.

D. It assumes default time, exposure at default and LGD are independent (no wrong-way risk), which lets the expectation split into EPE times marginal PD with one average LGD.

Question CR-20 b.4 Bank Q993 • BT Mock Q42 • BT answers P2.T6.914.1 (rebuilt, key corrected in the bank)

Which of the following statements about credit value adjustment (CVA) is correct?

A. CVA is generally computed with market-implied (risk-neutral) parameters where practical, which supports the exit-price concept required by accounting standards.

B. Accounting standards require CVA to be computed with historical default probabilities and volatilities, so that it acts as an actuarial reserve.

C. CVA depends only on the counterparty's probability of default and is independent of the exposure profile.

D. CVA is proportional to the counterparty's exposure and likelihood of default, but not to the loss given default.

Question CR-20 b.5 Bank Q991 • AnalystPrep Q.5474 (rebuilt)

Eastpointe Bank has entered into a 10-year receiver interest rate swap with Counterparty A (it receives fixed and pays floating). The risk team is computing the CVA for this uncollateralised swap by Monte Carlo, choosing between a direct CVA calculation and a path-wise CVA calculation. The bank wants to minimise computational time without sacrificing accuracy. Which approach is more effective?

A. A path-wise CVA formula, because it is more intuitive as a sum over the EPE profile and therefore must be cheaper to compute.

B. A path-wise CVA formula, because valuations at successive simulation points have low correlation, so it converges quickly.

C. A direct CVA formula, because assuming independence of default time, exposure at default and LGD lets one simulate the exposure at the default time directly with a single average LGD, giving a lower variance for the same number of valuations.

D. A direct CVA formula, because it requires multiple valuations of the portfolio at every time interval of each simulation, which improves accuracy.

Question CR-20 b.6 New • built from GARP Ch.20, 20.2.4

An xVA quant at a Japanese bank notes that, for one counterparty, the bank's only position is a swap that is deeply in the money for the bank. Which of the following statements about the CVA of this position is most accurate?

A. Because the swap is deeply in the money, CVA should be based on its potential future exposure at a high percentile rather than on its EPE.

B. Because the exposure is almost certain to stay positive, CVA no longer depends on the loss given default and equals the expected future value times the cumulative default probability.

C. A deeply in-the-money swap requires the path-wise CVA formula, because the direct approach cannot be applied when the exposure is always positive.

D. When a position is very in the money, its EPE can be approximated by its expected future value, so the CVA calculation converges to a simpler, bond-like discounting calculation.

Solutions • CR-20 b

Solution CR-20 b.1

Answer: B

New • built from GARP Ch.20, 20.1 and 20.2

Step 1 – The definition

Gregory: rather than "risk-free value", CVA can be defined as the difference between the value with and without counterparty default.

Step 2 – The sign

In GARP's convention CVA is negative, a reduction in value, so $V_{\text{risky}} = V_{\text{risk-free}} + \text{CVA}$.
$\Rightarrow$ B gives both the definition and the convention.

Interpretation

Schweser quotes the same quantity as a positive cost and subtracts it; either way the risky value is below the risk-free value. The definition is unilateral: the bank's own default is excluded.

Why the other options are wrong

A is incorrect. A 99th percentile loss is a capital or risk measure; CVA is an expected value (a price).
C is incorrect. This describes DVA, which uses the expected negative exposure and the bank's own default probability (CR-20 e).
D is incorrect. Netting and collateral reduce the netting-set CVA, so CVA is not the sum of standalone adjustments (CR-20 h, j).

LO: CR-20 b: Describe credit value adjustment (CVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.1 Overview; 20.2.2

Solution CR-20 b.2

Answer: A

Bank Q990 • AnalystPrep Q.6208

Step 1 – The three drivers

$\text{CVA} = -\text{LGD} \sum \text{EPE}(t_i)\, \text{PD}(t_{i-1}, t_i)$ combines the loss given default, the exposure profile and the default probability.

Step 2 – The right version of each

The exposure is the discounted *expected* positive exposure (risk-free discounting is usually inside the EPE); the default probability is the marginal PD for each interval.
$\Rightarrow$ A matches the formula term by term.

Interpretation

CVA is the expected loss from counterparty default, so every input is an expectation or a probability; no input is a tail quantile.

Why the other options are wrong

B is incorrect. PFE is a high-percentile exposure used for limits. Using it overstates CVA.
C is incorrect. A single current MtM ignores how exposure evolves, and a cumulative PD is not matched to each exposure date.
D is incorrect. Expected negative exposure and the bank's own default probability drive DVA, not the counterparty CVA.

LO: CR-20 b: Describe credit value adjustment (CVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2, eq. 20.3

Solution CR-20 b.3 — Answer: D

New • built from GARP Ch.20, 20.1 and 20.2.2

Step 1 – The general definition

Eq. 20.1 takes one expectation of the product $\mathbb{1}(\tau \leq T) \times V(t,\tau)^+ \times \text{LGD}$. GARP notes that the three components may be dependent.

Step 2 – What independence buys

If default time, exposure at default and LGD are independent, the expectation of the product is the product of expectations: the LGD becomes a single average, the exposure becomes the unconditional EPE at each date, and default enters only through the marginal PD of each interval.
$\Rightarrow$ That is eq. 20.3. GARP states the assumption at the start of the chapter: credit exposure and default probability are independent, which neglects wrong-way risk.

Interpretation

The validator's real question is whether the counterparty has wrong-way risk. If it does, EPE must be replaced by EPE conditional on default (CR-20 p), and the simple form understates CVA.

Why the other options are wrong

A is incorrect. The marginal PDs $Q(t_i) - Q(t_{i-1})$ can come from any credit curve shape; a constant hazard is a convenience, not an assumption of the formula.
B is incorrect. Describes the wrong-way risk extension (conditional EPE), which is exactly what the simple form leaves out.
C is incorrect. Collateral reduces EPE (CR-20 j); it does not set the LGD to zero or remove netting sets from CVA.

LO: CR-20 b: Describe credit value adjustment (CVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.1 Overview; 20.2.2, eqs. 20.1 and 20.3

Solution CR-20 b.4 — Answer: A

Bank Q993 • BT Mock Q42 • BT answers P2.T6.914.1

Step 1 – The parameters

CVA is computed with market-implied parameters where practical: this defines the price relative to hedging instruments and supports the exit-price concept that accounting requires.

Step 2 – The drivers

$\text{CVA} = -\text{LGD} \sum \text{EPE} \times \text{PD}$ depends on all three of LGD, the exposure profile and the default probability.
$\Rightarrow$ Only A is correct.

Interpretation

Historical parameters belong to the older reserve view; they survive today only in rare cases (CR-20 a).

Why the other options are wrong

B is incorrect. Reverses the history: historical parameters were the old approach; accounting now points to market-implied exit prices.
C is incorrect. CVA is the product of exposure, default probability and LGD, so it depends heavily on the exposure profile.
D is incorrect. CVA is directly proportional to LGD as well.

LO: CR-20 b: Describe credit value adjustment (CVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.1 Overview; 20.2.2

Solution CR-20 b.5 — Answer: C

Bank Q991 • AnalystPrep Q.5474

Step 1 – What the path-wise method needs

Each simulation values the portfolio at all m grid dates. Successive values on a path are highly correlated, so the extra valuations add little information and convergence is slow.

Step 2 – What the direct method needs

Assuming default time, exposure at default and LGD are independent (no wrong-way risk), it simulates a default time and values the portfolio only then, with a single average LGD.

Step 3 – The gain

In GARP's 10-year receiver swap the direct estimate had a standard deviation six times smaller for the same 40,000 valuations, about $6 \times 6 = 36$ times fewer valuations for equal accuracy.
$\Rightarrow$ C is the efficient choice and gives the right reason.

Interpretation

The realised speed-up is smaller if valuation is not the only bottleneck (Amdahl's law), and the direct method is less natural for collateralised or other path-dependent portfolios.

Why the other options are wrong

A is incorrect. Intuitiveness is a real merit of the path-wise formula, but it is the slower method.
B is incorrect. Wrong premise: successive path-wise valuations are highly (not lowly) correlated, which is why it converges slowly.
D is incorrect. Multiple valuations at every interval describe the path-wise method and are its computational burden.

LO: CR-20 b: Describe credit value adjustment (CVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2 Direct and Path-Wise CVA Formulas, Figures 20.4 to 20.6

Solution CR-20 b.6 — Answer: D

New • built from GARP Ch.20, 20.2.4

Step 1 – EPE of a deep in-the-money position

EPE averages $\max(V, 0)$. If V is almost never negative, $\max(V, 0) \approx V$, so EPE is close to the expected future value (EFV).

Step 2 – What that does to CVA

GARP: in this case one can use the CVA formula with the EFV, or a discounting approach (which has to allow for the LGD). The complex swap-like calculation converges to a simpler bond-like case.
$\Rightarrow$ D describes GARP's special case.

Interpretation

This is the link back to CR-20 a: a swap that is deep in the money behaves like a one-way claim, so its cash flows are almost fully at risk, as for a bond.

Why the other options are wrong

A is incorrect. CVA always uses the expected exposure; PFE is a limit measure whatever the moneyness.
B is incorrect. The EFV part is right, but LGD still scales the loss; dropping it treats the whole exposure as lost.
C is incorrect. Nothing prevents the direct method here; positive exposure makes the calculation simpler, not harder.

LO: CR-20 b: Describe credit value adjustment (CVA). • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP

CR Ch.20), 20.2.4 Special Cases, Figure 20.7

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 d

P2 CALC

Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA.

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • revision notes only, no question homed here

WHAT IS TESTED HERE

Why P2. Vardeez flags the LO in a VERY IMPORTANT chapter ("know the formulas very well"). TYPE CALC is an approved judgment call: spread and recovery sensitivity of CVA is tested with numbers as well as direction.

Vardeez: the shape of the credit curve has a material impact on CVA (upward -24.6, flat -20.0, inverted -15.7); CVA generally increases with the credit spread and converges to the exposure in default; a higher recovery rate raises the implied default probability but reduces CVA.

Reported exam shapes: none listed in the map. The one GARP-tagged bank item (AP Mock 1 Q25, Question CR-20 d.2) asks whose spread drives a unilateral CVA when both parties widen.

What the questions look like: direction questions (spread up, curve shape, recovery up) and short calculations: the change in a running CVA, CVA under two curve shapes, and CVA after a recovery revision with equal or unequal LGDs. Distractors scale CVA by the LGD ratio, imply the PD with the wrong LGD, treat the same cumulative PD as the same CVA, or put an LGD into the spread approximation.

Worked elsewhere: the basic UCVA and running-spread calculations, Workbook A6.1 and A6.3 (Questions CR-20 c.3 and CR-20 g.4).

1. The CVA formula with the spread inside

Substituting the spread-implied default probability (hazard $\lambda \approx s/\text{LGD}$, CR-9 h) into eq. 20.3 gives GARP eq. 20.5:

$$\text{UCVA} = -\text{LGD}_{\text{actual}} \sum_{i=1}^{m} \text{EPE}(t_i) \left[\exp\left(-\frac{s_{i-1} t_{i-1}}{\text{LGD}_{\text{mkt}}} \right) - \exp\left(-\frac{s_i \, t_i}{\text{LGD}_{\text{mkt}}} \right) \right]$$

The bracket is the marginal PD for each interval. $\text{LGD}_{\text{actual}}$ is the loss the bank expects on its claim; LGD_{mkt} calibrates the PD and should match the seniority behind the spread (usually senior unsecured CDS). With a flat spread and a constant EPE the sum collapses to $\text{UCVA} = -\text{LGD}_{\text{actual}} \times \text{EPE} \times \left(1 - \text{e}^{-sT/\text{LGD}_{\text{mkt}}}\right)$, and the running-spread approximation (eq. 20.4) is UCVA $\approx -$average EPE $\times\, s$: linear in the spread, with no LGD at all.

2. The level and shape of the credit curve

Level. CVA generally increases in size as the counterparty's spread widens, because the implied PDs rise. A unilateral CVA uses only the *counterparty's* spread; the bank's own spread enters through DVA (CR-20 e to g). The effect is not linear, because PDs cannot exceed 100%. Constant EPE, five years, LGD $= 60\%$, CVA as a percentage of EPE:

Spread (bps)	100	200	500	1,000
Exact: $0.6\left(1 - \text{e}^{-5s/0.6}\right)$	4.80	9.21	20.45	33.92
Linear: $s \times 5$	5.00	10.00	25.00	50.00

In default, CVA converges to the exposure of the trade today, which may be zero: for a par swap the CVA turns and shrinks as the spread keeps widening (Figure 1, left), a large gamma for big moves and the root of jump-to-default risk.

Shape. GARP Table 20.2 prices a 10-year receive-fixed swap off three curves with the same 5-year spread of 150 bps (LGD 60%): upward sloping -24.6, flat -20.0, inverted -15.7 bps upfront. The cumulative PDs agree at five years, but the marginal PDs are evenly spaced (flat), back-loaded (upward) or front-loaded (inverted). The upward curve gives the largest CVA mainly through its larger extrapolated 10-year spread. **The same cumulative PD does not mean the same CVA**: the timing of the marginal PDs is weighted by the EPE profile (Question CR-20 d.5).

> **Correction to the source notes.** SchweserNotes LO 38.d says CVA "will be lower for an upward-sloping curve" and "higher for a downward-sloping curve". GARP Table 20.2 shows the reverse in size: the upward-sloping curve gives the *largest* CVA (-24.6 against -15.7 bps), so option B of Schweser Module Quiz 38.1 Q3 is true as well as the keyed C. Schweser's "in default the CVA will fall to zero" holds only for a trade worth zero today: GARP says CVA converges to the exposure of the transaction.

3. The recovery rate: two LGDs

Equal LGDs (the claim ranks with the senior unsecured debt behind the CDS: the usual case, and required for regulatory CVA unless seniority differs). A higher LGD lowers the implied PD but raises the loss. Because $e^{-x} \approx 1 - x$, the two cancel to first order; GARP: moving the LGD from 60% to 50% changes CVA by less than 2%. Constant EPE, five years, spread 200 bps:

Recovery (LGD)	40% (60%)	50% (50%)	60% (40%)
Implied hazard s/LGD	3.33%	4.00%	5.00%
CVA, % of EPE	9.21	9.06	8.85
Change from the 40% case		-1.6%	-3.9%

Unequal LGDs. GARP's cases for a different $\text{LGD}_{\text{actual}}$: seniority in the exposure waterfall, or a structurally different LGD such as project finance (recoveries about 75% against about 40% for corporates; S&P: 77%); credit enhancements such as margin; and work-out experience. LGD_{mkt} still sets the PD, so CVA moves *in proportion* to $\text{LGD}_{\text{actual}}$: 40% against a 60% market LGD gives $9.21 \times 40/60 = 6.14\%$. Caveat: a proxy spread mapped from a rating must reflect PD only (S&P ratings are PD-based; Moody's may include recovery), or the better recovery is counted twice.

> **Correction to the source notes.** Vardeez says the LGD "falls faster" than the implied hazard rises, so "the LGD effect dominates". With one LGD for both, the two move together to first order: $\text{LGD} \times \lambda = s$ (at 200 bps, $0.60 \times 3.33\% = 0.40 \times 5.00\% = 2\%$). The direction (higher recovery, slightly lower CVA) is right, but the effect is second order: -3.9% in the table, not the third that the LGD alone suggests.

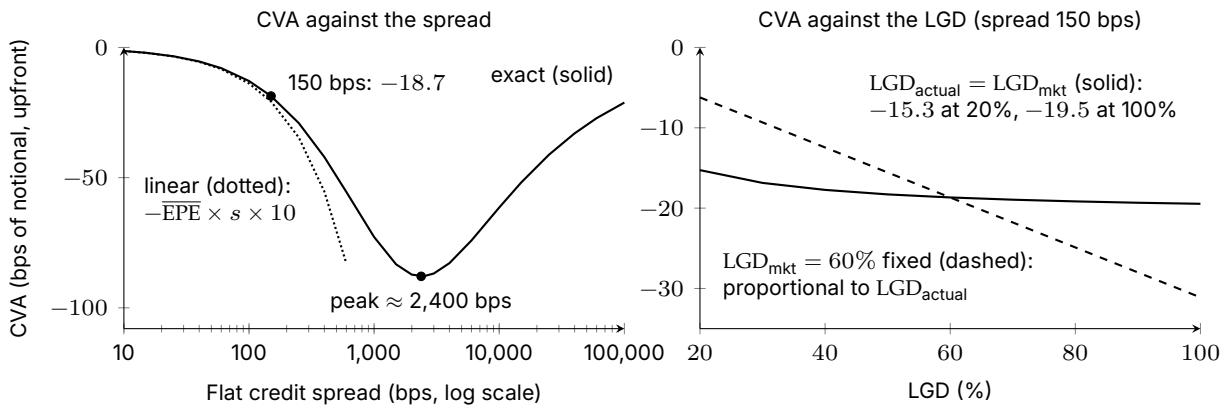


Figure 1. Illustrative 10-year par receiver swap, EPE peaking at about 2% of notional, LGD 60%. Left: CVA grows with the spread, less than linearly, peaks and then shrinks back towards the zero exposure of a par trade (after GARP Figure 20.8). Right: with equal LGDs the curve is nearly flat (first-order cancellation); with the market LGD fixed, CVA is a straight line through the origin in $\text{LGD}_{\text{actual}}$ (after GARP Figures 20.9 and 20.10).

Formula card: CR-20 d

UCVA from spreads (eq. 20.5)	$-\text{LGD}_{\text{actual}} \sum \text{EPE}(t_i)\left[\text{e}^{-s_{i-1}t_{i-1}/\text{LGD}_{\text{mkt}}} - \text{e}^{-s_i t_i/\text{LGD}_{\text{mkt}}}\right]$; flat spread: $\lambda = s/\text{LGD}_{\text{mkt}}$
Running approximation (eq. 20.4)	UCVA $\approx -$average EPE $\times\, s$ (no LGD; linear in s)
Equal LGDs	$\text{LGD} \times \lambda = s$: cancel to first order
Unequal LGDs	CVA proportional to $\text{LGD}_{\text{actual}}$, PD fixed by LGD_{mkt}

Change in assumption	Effect on CVA (size)	Why
Counterparty spread up	Larger	Higher implied PDs
Upward vs inverted curve	Upward larger (GARP)	Back-loaded PDs, higher long-dated spread
Recovery up, equal LGDs	Slightly smaller	First-order cancellation
Recovery up on the claim only	Proportionally smaller	PD unchanged, loss smaller
Own spread up (unilateral)	No change	Own spread enters only DVA

Exam traps

- **Loss channel only.** Scaling CVA by the LGD ratio ignores the fall in the implied PD.
- **PD channel only.** Re-implying the PD with the new LGD but keeping the old LGD on the loss.
- **Wrong LGD in the PD.** LGD_{mkt} calibrates the PD; $\text{LGD}_{\text{actual}}$ measures the loss.
- **Recovery for LGD.** Multiplying by R instead of $1 - R$.
- **LGD in the spread approximation.** $-\overline{\text{EPE}} \times s$ already contains the LGD; do not multiply again.
- **Same cumulative PD, same CVA; wrong party.** Timing matters; a unilateral CVA uses the

counterparty's spread only.

Practice questions: CR-20 d

Question CR-20 d.1

Bank Q1007 • AnalystPrep Q.5488

Quantum Financial Consulting is reassessing the risk factors that affect its credit value adjustment (CVA) calculations across its derivatives portfolio, to refine the sensitivity of CVA to market conditions. How should Quantum approach the impact of varying credit spread and recovery rate assumptions on its CVA calculations?

A. Quantum should anticipate that CVA will likely decrease with an increasing credit spread and an increasing recovery rate, resulting from the offsetting effects.

B. Quantum should focus primarily on changes in recovery rate assumptions, as the credit spread has minimal impact on CVA.

C. Quantum should expect that an increase in the credit spread will generally lead to an increase in CVA, reflecting heightened credit risk and potential losses.

D. Quantum should presume that CVA does not change meaningfully with fluctuations in the credit spread and recovery rate under normal market conditions.

Question CR-20 d.2

Bank Q1005 • AP Mock 1 Q25 (rebuilt)

Bank of Baroda often enters into swap contracts with Bank of Panamba, a major swap dealer. Panamba charges Baroda a unilateral CVA that reflects Panamba's expected loss from a Baroda default on swaps where Baroda owes Panamba. Recently Panamba was downgraded from A to A−, and its credit spread rose from 38 bps to 128 bps. Baroda was downgraded from BB+ to BB, and its credit spread rose from 118 bps to 228 bps. Assuming exposures and recovery rates are unchanged, which of the following is the most likely response with respect to the CVA charge?

A. Panamba requests an increase in the CVA charge, because Baroda's wider credit spread implies a higher default probability.

B. Baroda requests a reduction in the CVA charge, because Panamba's own spread has widened.

C. Panamba requests a reduction in the CVA charge, because the spread widening of both banks offsets in the CVA calculation.

D. No change, because both banks were downgraded by only one notch and the rating moves do not alter CVA.

Question CR-20 d.3

New • built from GARP Ch.20, 20.2.3 and 20.2.6

A swaps desk at a Canadian bank quotes CVA as a running spread using Gregory's average-EPE approximation, and reports CVA as a negative number. For a 5-year swap with a corporate client the desk has the following data:

	Last month	Today
Average EPE (% of notional)	3.5	3.5
Client 5-year CDS spread (bps)	140	220
Recovery rate assumed for the client (%)	40	20

Which of the following is closest to the change in the running CVA since last month?

A. -7.70 bps per annum

B. -3.22 bps per annum

C. -2.80 bps per annum

D. -1.46 bps per annum

Question CR-20 d.4

Bank Q1008 • BT answers P2.T6.915.2 (rebuilt, options revised)

Gregory's CVA approximation for a plain vanilla interest rate swap is

$$\text{UCVA} \approx -\text{LGD}_{\text{actual}} \sum_i \text{EPE}(t_i) \left[\exp\left(-\frac{s_{i-1} t_{i-1}}{\text{LGD}_{\text{mkt}}} \right) - \exp\left(-\frac{s_i t_i}{\text{LGD}_{\text{mkt}}} \right) \right],$$

where s_i is the credit spread to t_i. A risk analyst uses it to explain the CVA sensitivities of a par swap to the trading desk. Which of the following statements is TRUE?

A. CVA is linear in the credit spread, so doubling the spread doubles the CVA at any spread level.

B. For a given cumulative default probability to maturity, CVA does not depend on the shape of the credit curve.

C. If the settled (market) LGD is assumed equal to the actual LGD, raising this common LGD from 50% to 60% makes CVA about 20% more negative, in proportion to the LGD.

D. CVA generally becomes more negative as the spread widens, but as the counterparty approaches default it converges to the current exposure, which is zero for a par swap, so beyond some point CVA shrinks as the spread widens further.

Question CR-20 d.5

New • built from GARP Ch.20, 20.2.5

An xVA analyst at a US bank compares the CVA on identical 4-year cross-currency swaps, notional USD 500 million, with two counterparties, U and V. The swap's discounted EPE rises over its life because of the final exchange of notionals. Both counterparties have an LGD of 60% and the same 4-year cumulative default probability of 10%, but U's credit curve is upward sloping and V's is inverted. CVA is reported as a negative number and there is no wrong-way risk.

Year	1	2	3	4
Discounted EPE (% of notional)	2.0	4.0	6.0	8.0
Marginal PD, counterparty U (%)	1.0	2.0	3.0	4.0
Marginal PD, counterparty V (%)	4.0	3.0	2.0	1.0

Which of the following is closest to the CVA with each counterparty?

A. U: −USD 3.00 million; V: −USD 2.00 million

B. U: −USD 1.80 million; V: −USD 1.20 million

C. U: −USD 1.50 million; V: −USD 1.50 million

D. U: −USD 1.20 million; V: −USD 1.80 million

Question CR-20 d.6

Bank Q1006 • AnalystPrep Q.1969 (rebuilt)

A bank calibrates a counterparty's default probability from its CDS spread and then computes CVA. It assumes that the actual LGD on the derivatives claim equals the market LGD used to imply the default probability from the spread (same seniority), and it raises both from 50% to 60%, holding the spread constant. Which statement best describes the effect on CVA?

A. CVA rises sharply, because a higher LGD directly scales up the loss given default.

B. CVA changes only modestly, because a higher LGD lowers the market-implied default probability while raising the loss given default, and the two effects largely offset.

C. CVA is exactly unchanged, because the LGD terms cancel in the CVA formula to all orders.

D. CVA rises sharply, because a higher LGD raises both the implied default probability and the loss given default.

Question CR-20 d.7

New • built from GARP Ch.20, 20.2.6

A valuation officer at a German bank recomputes the unilateral CVA on a 5-year uncollateralised swap, notional EUR 400 million, after the credit team revises the counterparty's expected recovery rate from 40% to 25%. The derivative claim ranks with the senior unsecured debt referenced by the counterparty's CDS, so the bank uses the same LGD to imply default probabilities and to measure the loss. The CDS curve is flat at 300 bps and does not move. Default probabilities are implied with a constant hazard rate $\lambda = s/\text{LGD}$, and CVA is reported as a negative number.

Year	1	2	3	4	5
Discounted EPE (% of notional)	1.5	2.4	2.8	2.4	1.4

Which of the following is closest to the CVA after the revision?

A. −EUR 1.39 million

B. −EUR 1.14 million

C. −EUR 0.91 million

D. −EUR 0.38 million

Question CR-20 d.8

New • built from GARP Ch.20, 20.2.6

An xVA analyst at a UK bank is pricing a 4-year interest rate swap with a project-finance vehicle. The vehicle has no CDS, so the bank uses a proxy curve flat at 240 bps for senior unsecured risk, calibrated with the market-standard 40% recovery. The swap ranks with the project's senior secured debt, and the credit team expects a 75% recovery on it, in line with project-finance recovery data. The proxy curve was mapped from a default-probability-based rating, so the higher recovery is not already in the spread. Default probabilities are implied with a constant hazard rate, and CVA is reported as a negative number.

Year	1	2	3	4
Discounted EPE (GBP million)	10	14	12	6

Which of the following is closest to the CVA?

A. −GBP 1.17 million

B. −GBP 0.94 million

C. −GBP 0.85 million

D. −GBP 0.39 million

Solutions • CR-20 d

Solution CR-20 d.1

Answer: C

Bank Q1007 • AnalystPrep Q.5488

Step 1 – The spread

A wider spread means a higher market-implied default probability, so CVA generally increases. The relationship is not linear and reverses only for a counterparty very close to default.

Step 2 – The recovery rate

When one LGD both implies the PD and measures the loss, a recovery change moves the two in opposite directions and the effects largely cancel, so the net effect is modest.

$\Rightarrow$ Only C states the spread effect correctly.

Interpretation

The spread is the first-order driver of CVA; the recovery assumption is a second-order one unless the actual and market LGDs differ.

Why the other options are wrong

A is incorrect. Gets the sign of the spread effect wrong, and the recovery effect is small rather than an offset to the spread.

B is incorrect. Treats the spread as unimportant when it is the main driver; it is the recovery effect that is modest.

D is incorrect. CVA is clearly sensitive to the spread.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.5 Credit Spread Effects; 20.2.6 Loss Given Default

Solution CR-20 d.2

Answer: A

Bank Q1005 • AP Mock 1 Q25

Step 1 – Who charges what

Panamba charges a *unilateral* CVA: Panamba's expected positive exposure to Baroda times Baroda's marginal default probabilities times LGD.

Step 2 – Which spread matters

Baroda's spread rose from 118 to 228 bps, a widening of 110 bps, so Baroda's implied default probability rises.

Step 3 – Panamba's own spread

Panamba's widening from 38 to 128 bps is not an input to a unilateral CVA.

$\Rightarrow$ With exposures and recoveries unchanged, the CVA charge on Baroda rises and Panamba will ask for more.

Interpretation

Under bilateral pricing, Panamba's own wider spread would raise its DVA on the swaps where it owes Baroda and offset part of the increase (CR-20 e and g). The one-notch rating moves are a distraction: CVA is driven by spreads, not rating labels.

Why the other options are wrong

B is incorrect. Uses the wrong party's credit: the dealer's own spread does not enter the unilateral CVA, and Baroda's deterioration pushes the charge up.

C is incorrect. Netting the two spread moves is bilateral reasoning, and even then Baroda's move raises the charge.

D is incorrect. A 110 bps widening on a BB name is a large rise in implied default probability;

CVA moves with it.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2 eq. 20.3; 20.2.5

Solution CR-20 d.3 — Answer: C

New • built from GARP Ch.20, 20.2.3 and 20.2.6

Step 1 – The approximation

GARP eq. 20.4: UCVA $\approx -$average EPE $\times$ spread, in the units of the spread.

Step 2 – Last month and today

$$-0.035 \times 140 = -4.90 \text{ bps}, \qquad -0.035 \times 220 = -7.70 \text{ bps}$$

Step 3 – The change

$$-7.70 - (-4.90) = -2.80 \text{ bps per annum}$$

$\Rightarrow$ The running CVA becomes 2.80 bps more negative.

Step 4 – The recovery revision

Eq. 20.4 has no LGD input: with the same LGD implying the PD and measuring the loss, the LGD cancels to first order, and the spread already embeds it.

$\Rightarrow$ The revision from 40% to 20% does not change the approximate answer.

Interpretation

Under the approximation CVA is linear in the spread, which is fine for moderate moves. The recovery revision is a red herring at first order; the exact eq. 20.5 would show only a small second-order effect.

Why the other options are wrong

A is incorrect. Reported today's level, -7.70 bps, instead of the change.

B is incorrect. Multiplied each spread charge by its LGD, double counting it: $0.80 \times 220 \times 0.035 - 0.60 \times 140 \times 0.035 = 6.16 - 2.94 = 3.22$.

D is incorrect. Used hazard rates $s/(1-R)$ in place of the spreads: $0.035 \times (275.00 - 233.33) = 1.46$.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.3 eq. 20.4; 20.2.6

Solution CR-20 d.4 — Answer: D

Bank Q1008 • BT answers P2.T6.915.2 (options revised)

Step 1 – The PD term

The bracket is a difference of exponentials in $s/\text{LGD}_{\text{mkt}}$ and can never exceed 1, so CVA is concave in the spread, not linear. A is false.

Step 2 – Very large spreads

As default becomes imminent, almost all the default probability falls in the first interval, where a par swap has almost no exposure. CVA converges to the exposure today, which is zero, so it eventually shrinks (Figure 1, left). D is true.

Step 3 – Curve shape

Curves with the same cumulative PD distribute the marginal PDs differently over time, and the EPE profile weights them differently (GARP Table 20.2). B is false.

Step 4 – Equal LGDs

The LGDs cancel to first order: moving from 60% to 50% changes CVA by less than 2% in GARP's example, not 20%. C is false.

⇒ Only D is true: CVA rises with the spread, but not without limit.

Interpretation

Each false statement is a common shortcut: linearity, curve shape ignored, LGD read as a pure loss multiplier. The equal-LGD cancellation itself is tested in Question CR-20 d.6; here the point is that a wider spread does not always mean a larger CVA.

Why the other options are wrong

A is incorrect. PDs are bounded by 100%, so doubling a large spread less than doubles CVA.

B is incorrect. Timing matters: upward-sloping -24.6 against inverted -15.7 bps in GARP's 10-year swap.

C is incorrect. Counts only the loss channel; the lower implied PD offsets most of it.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.5, Table 20.2 and Figure 20.8; 20.2.6 eq. 20.5

Solution CR-20 d.5

Answer: B

New • built from GARP Ch.20, 20.2.5

Step 1 – EPE × marginal PD, counterparty U (% × %)

$$2.0(1.0) + 4.0(2.0) + 6.0(3.0) + 8.0(4.0) = 2 + 8 + 18 + 32 = 60 \ (\times 10^{-4})$$

Step 2 – The same for counterparty V

$$2.0(4.0) + 4.0(3.0) + 6.0(2.0) + 8.0(1.0) = 8 + 12 + 12 + 8 = 40 \ (\times 10^{-4})$$

Step 3 – CVA

$$\text{U: } -0.60 \times 0.0060 \times 500 = -1.80, \qquad \text{V: } -0.60 \times 0.0040 \times 500 = -1.20 \text{ (USD million)}$$

⇒ The same 10% cumulative PD, but U's defaults are back-loaded into the years when the exposure is highest, so U's CVA is 50% larger.

Interpretation

This matches the direction of GARP's finding that the upward-sloping curve gives the largest CVA (there for a 10-year swap, mostly through the extrapolated long spread). For a hump-shaped or amortising profile the gap would shrink or could reverse, so compute it rather than assume it. A flat curve (2.5% a year) gives −USD 1.50 million.

Why the other options are wrong

A is incorrect. Dropped the LGD: $0.0060 \times 500 = 3.00$ and $0.0040 \times 500 = 2.00$.

C is incorrect. Used average EPE × cumulative PD × LGD $= 0.05 \times 0.10 \times 0.60 \times 500 = 1.50$ for both, ignoring the timing of defaults; this is right only for evenly spread PDs.

D is incorrect. Swapped the two counterparties: the front-loaded curve meets the small early exposures.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.2 eq. 20.3; 20.2.5 Credit Spread Effects

Solution CR-20 d.6 — Answer: B

Bank Q1006 • AnalystPrep Q.1969

Step 1 – Two channels

The implied PD is about $1-\mathrm{e}^{-st/\mathrm{LGD}_{\mathrm{mkt}}}$, so raising $\mathrm{LGD}_{\mathrm{mkt}}$ lowers it. $\mathrm{LGD}_{\mathrm{actual}}$ multiplies the loss, so raising it raises CVA.

Step 2 – Net effect

With $\mathrm{LGD}_{\mathrm{actual}} = \mathrm{LGD}_{\mathrm{mkt}}$, LGD $\times$ hazard $\approx s$, so the two cancel to first order ($\mathrm{e}^{-x} \approx 1-x$).

Step 3 – Size

Only a second-order effect remains: GARP finds that 60% to 50% changes CVA by less than 2%; in the five-year example of Section 3, by 1.6%.

$\Rightarrow$ Modest, not zero: B.

Interpretation

This is why the running approximation $-\overline{\mathrm{EPE}} \times s$ needs no LGD. The answer changes completely if the claim's actual LGD differs from the market LGD (Question CR-20 d.8).

Why the other options are wrong

A is incorrect. Counts only the loss channel and forgets that the implied PD falls.

C is incorrect. The cancellation holds only to first order, so CVA still moves slightly.

D is incorrect. Gets the sign of the PD effect wrong: a higher LGD lowers the implied PD.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.6 Loss Given Default, eq. 20.5 and Figure 20.9

Solution CR-20 d.7 — Answer: B

New • built from GARP Ch.20, 20.2.6

Step 1 – Hazard rates

Before: $\lambda = 0.03/0.60 = 5.00\%$. After: $\lambda = 0.03/0.75 = 4.00\%$.

$\Rightarrow$ The higher LGD lowers the implied hazard rate.

Step 2 – Marginal PDs after the revision

$\mathrm{e}^{-0.04(i-1)} - \mathrm{e}^{-0.04i}$ for years 1 to 5: 3.921, 3.767, 3.620, 3.478, 3.341 (%).

Step 3 – EPE $\times$ PD

$$1.5(3.921) + 2.4(3.767) + 2.8(3.620) + 2.4(3.478) + 1.4(3.341) = 38.08 \ (\times 10^{-4})$$

Step 4 – CVA after the revision

$$\mathrm{CVA} = -0.75 \times 0.003808 \times 400 = -\text{EUR } 1.142 \text{ million}$$

Step 5 – Compare with before

At $\lambda = 5\%$ the PDs are 4.877, 4.639, 4.413, 4.198, 3.993 (%), the sum is 46.47, and CVA $= -0.60 \times 0.004647 \times 400 = -1.115$.

$\Rightarrow$ A 25% rise in LGD moves CVA by only 2.4%.

Interpretation

The lower implied PD offsets almost all of the higher loss. The bank's reserve barely changes, which is the cancellation GARP describes; the recovery assumption matters much more when the claim's seniority differs from the CDS (Question CR-20 d.8).

Why the other options are wrong

A is incorrect. Loss channel only: scaled the old CVA by $0.75/0.60$ with the old PDs, $-1.115 \times 1.25 = -1.39$.

C is incorrect. PD channel only: new PDs but the old 60% LGD on the loss, $-0.60 \times 0.003808 \times 400 = -0.91$.
D is incorrect. Used the new recovery rate in place of LGD: $-0.25 \times 0.003808 \times 400 = -0.38$.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.6 Loss Given Default, eq. 20.5

Solution CR-20 d.8 — Answer: D

New • built from GARP Ch.20, 20.2.6

Step 1 – PD from the proxy curve with the market LGD

$\lambda = 0.024/0.60 = 4.00\%$; marginal PDs for years 1 to 4: 3.921, 3.767, 3.620, 3.478 (%).
$\Rightarrow$ The default probability comes from the senior unsecured curve and its own 60% LGD.

Step 2 – EPE $\times$ PD

$$10(0.03921) + 14(0.03767) + 12(0.03620) + 6(0.03478)$$
$$= 0.3921 + 0.5274 + 0.4344 + 0.2087 = 1.5625 \text{ (GBP million)}$$

Step 3 – Loss with the actual LGD

$$\text{CVA} = -(1 - 0.75) \times 1.5625 = -\text{GBP } 0.391 \text{ million}$$

$\Rightarrow$ About GBP 0.39 million, roughly 40% of the charge with a 60% LGD.

Interpretation

Seniority in the exposure waterfall is the easiest case GARP gives for an $\text{LGD}_{\text{actual}}$ that differs from LGD_{mkt}; the PD must still come from the market LGD. Because the proxy was mapped from a PD-based rating, the better recovery is not double counted.

Why the other options are wrong

A is incorrect. Multiplied by the recovery rate instead of LGD: $0.75 \times 1.5625 = 1.17$.
B is incorrect. Ignored the seniority and used the 60% market LGD for the loss: $0.60 \times 1.5625 = 0.94$.
C is incorrect. Re-implied the PD with the actual LGD: $\lambda = 0.024/0.25 = 9.60\%$, PDs 9.154, 8.316, 7.555, 6.863 (%), sum 3.398, and $0.25 \times 3.398 = 0.85$.

LO: CR-20 d: Evaluate the impact of changes in the credit spread and recovery rate assumptions on CVA. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.2.6 Loss Given Default, eq. 20.5 and Figure 20.10

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 h

P2 CONCEPT

Explain how netting can be incorporated into the CVA calculation.

Workbook: A6 CVA, DVA, BCVA and CVA as a Spread • Tier A • revision notes only; A6.4 tags this LO

WHAT IS TESTED HERE

Why P2. Vardeez flags the LO in a VERY IMPORTANT chapter. TYPE CONCEPT is an approved judgment call: explain how netting enters CVA; the incremental CVA calculation is homed in CR-20 i.

Vardeez: netting reduces the CVA price because it reduces exposure; CVA is linear in EPE, so whatever lowers EPE lowers CVA.

Reported exam shapes: none listed in the map. Workbook A6.4 tags this LO (incremental CVA quoted as a running spread) and is worked under CR-20 i.

What the questions look like: conceptual: at what level CVA is computed, how a new trade under netting is charged, when the netting benefit is large or small, when incremental CVA becomes a benefit, and what netting does to BCVA. The numbers belong to CR-20 i.

1. Netting works through the exposure only

Under a legally enforceable netting agreement the values of all trades in the **netting set** are offset at the counterparty's default (close-out netting, CR-16). The exposure is $\max(\sum_k V_k, 0) \le \sum_k \max(V_k, 0)$ on every path, so the netting-set EPE never exceeds the sum of standalone EPEs. PD and LGD are untouched. Since CVA is a PD-weighted sum of EPE (CR-20 b), GARP eq. 20.10 follows, with CVA negative:

$$\boxed{\text{CVA}_{NS} \;\ge\; \sum_{i=1}^{N} \text{CVA}_i^{SA}} \qquad \text{(less negative: netting can reduce CVA, never increase it)}$$

In GARP's five-scenario example (Table 19.1), trade 1 is worth 50, 30, 10, -10, -30 and trade 2 is worth 15, 5, -5, -15, -25. The expected future values add up ($10 - 5 = 5$); the EPEs and ENEs do not ($18 + 4 = 22$ against 21 netted; $-8 - 9 = -17$ against -16). Netting helps only where the trades have opposite signs (here only the third scenario), so for positively correlated values the benefit is small. A counterparty may have several netting sets (separate master agreements; trades in jurisdictions where netting is unenforceable each stand alone). EPE is aggregated within each set, usually by Monte Carlo, and the counterparty CVA is the sum over its netting sets.

2. How large is the netting benefit?

- **Portfolio composition.** Strong for balanced, offsetting portfolios; weak for directional ones. Most clients hedge directionally, so their netting effects are modest.
- **Relative size.** As a new trade grows relative to the book the benefit is lost and its incremental CVA tends to its standalone value (GARP Figure 20.12: from -6.2 bps a year for a small trade towards the standalone -17.9 bps).
- **Bilateral view.** Netting shrinks EPE *and* ENE, so it lowers CVA and DVA. GARP Table 20.4 prices a seven-year pay-fixed swap (counterparty 150 bps, own 100 bps, LGD 60%, bps upfront): standalone CVA -26.3, DVA 6.0, BCVA -20.3; incremental in a directional book -25.1, 5.3, -19.9; in an offsetting book -13.7, -2.5, -16.3.

The offsetting book roughly halves the CVA and turns the incremental DVA negative, yet BCVA moves only 4 bps. With equal spreads BCVA would not move at all: without survival terms it depends on EPE + ENE = EFV, which is additive.

3. Charging a trade that sits in a netting set

The benefit exists only at netting-set level, so it must be allocated. The **incremental CVA** of a new trade is the netting-set CVA after the trade minus the CVA before it (eq. 20.11), the standalone formula with incremental EPE (eq. 20.12). It is never more negative than the standalone CVA, it can be *positive* (a benefit) when the trade offsets the book, and it depends on the order of execution. **Marginal CVA** splits a netting-set CVA into contributions that sum to the total. A bank with existing trades can therefore offer better terms on an offsetting trade (Cooper and Mello, 1991). The calculations are CR-20 i; Figure 1 shows the idea.

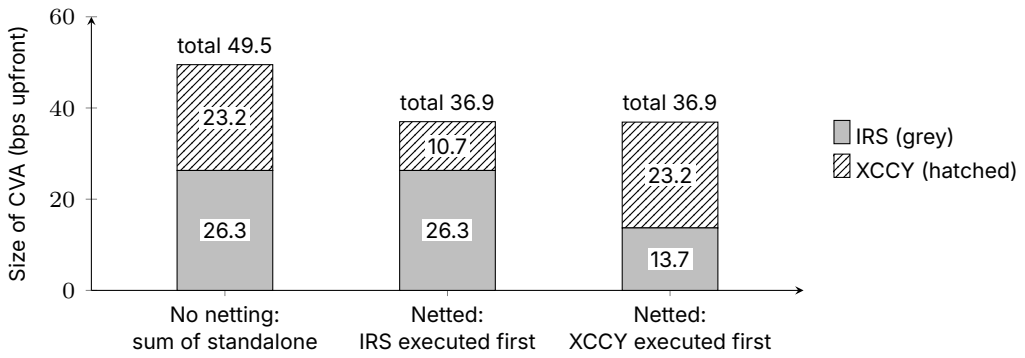


Figure 1. GARP Table 20.5 (counterparty curve 150 bps, LGD 60%), CVA shown as positive sizes. Without netting the two standalone CVAs add to 49.5 bps. Netted, the total is 36.9 bps whichever trade comes first (GARP's rounded components show 37.0 for the IRS-first order), but the incremental split depends on the order: the trade executed second gets the netting benefit of 12.6 bps.

	Standalone CVA	Netting-set CVA	Incremental CVA	Marginal CVA
Exposure used	Trade EPE alone	EPE of the summed values	Change in netting-set EPE	Marginal EPE
Adds up?	No: overstates	It is the total	In execution order	Yes, at a point in time
Used for	Quick appraisal	Reporting the counterparty	Pricing a new trade, unwinds	Attribution, simultaneous trades

Exam traps

- **Summing standalone CVAs.** EPE is not additive within a netting set; the sum overstates CVA.
- **Netting on the wrong input.** Netting changes the exposure only, never the PD or the LGD.
- **Netting across agreements.** Only trades under the same enforceable agreement net.
- **A floor at zero, or bigger is better.** Incremental CVA can be a benefit when the new trade offsets the book, and the benefit fades as the trade grows.
- **BCVA forgotten.** Netting cuts DVA as well as CVA; with equal spreads BCVA has no netting effect.

Practice questions: CR-20 h

Question CR-20 h.1

Bank Q1029 • AnalystPrep Q.1884 (rebuilt)

A bank has several OTC derivative trades with one counterparty, all covered by a single legally enforceable netting agreement. Which of the following statements about incorporating netting into the bank's CVA calculation is most accurate?

A. CVA must be calculated at the netting-set level from the netted exposure profile, so the netting-set CVA is not the sum of the trades' standalone CVAs and is never larger in size than that sum.

B. CVA can be calculated trade by trade and summed, because expected exposure is additive across the trades in a netting set.

C. Netting increases CVA, because offsetting positions concentrate the remaining exposure in fewer trades.

D. Netting changes CVA only through the counterparty's default probability, since exposure at default is unaffected by offsetting positions.

Question CR-20 h.2

New • built from GARP Ch.20, 20.4.1 and Ch.19, 19.3.1

A bank has two corporate counterparties with the same credit spread, each with ten uncollateralised interest rate swaps under an enforceable netting agreement. Counterparty X hedges its fixed-rate debt and pays fixed on all ten swaps. Counterparty Y is a hedge fund whose book mixes pay-fixed and receive-fixed swaps of similar size and maturity. Which of the following statements about the netting benefit in each counterparty's CVA is most accurate?

A. The netting benefit is the same for both, because it depends only on the number of trades in the netting set.

B. The netting benefit is larger for X, because trades in the same direction reinforce each other's exposure and netting removes this concentration.

C. The netting benefit is large for Y and small for X: X's netting-set CVA is close to the sum of its standalone CVAs, while Y's offsetting trades make its netting-set EPE far lower than the sum of standalone EPEs.

D. The netting benefit is larger for Y, because netting lowers Y's default probability, which depends on its offsetting positions.

Question CR-20 h.3 New • built from GARP Ch.20, 20.4.1 and Ch.19, 19.3

A bank faces one corporate counterparty through three groups of trades: (1) interest rate swaps under an ISDA Master Agreement; (2) FX forwards under a separate master agreement, with no cross-agreement netting; and (3) a few trades booked through a branch in a jurisdiction where the bank's lawyers consider close-out netting unenforceable. How should the bank incorporate netting into the CVA for this counterparty?

A. Net all the trades with the counterparty into a single exposure, because CVA depends on the counterparty's one default probability.

B. Compute standalone CVAs for every trade and scale the total by an average netting factor, because netting acts mainly through the counterparty's default probability.

C. Net groups (1) and (2) together because they face the same legal entity, and exclude group (3) from the CVA because its exposure cannot be netted.

D. Compute EPE separately for each enforceable netting set, netting trade values before taking the positive part; treat each trade in group (3) as its own netting set; then add the CVAs, using the same PD and LGD throughout.

Question CR-20 h.4 Bank Q1032 • AnalystPrep Q.5486 (rebuilt) • re-filed from CR-20 i

MetroQuant Advisors has a portfolio of OTC transactions with a counterparty under a netting agreement. The portfolio contains both directional and offsetting positions. A new trade is being priced with this counterparty. How should MetroQuant incorporate netting into its CVA calculation for the new trade?

A. Compute the new trade's standalone CVA and add it to the existing netting-set CVA, since CVA is additive across trades.

B. Compute the CVA of the netting set with and without the new trade, and charge the difference (incremental CVA), which reflects the netting benefit.

C. Compute the CVA of each trade using only its positive exposure, then sum these, so that netting is ignored conservatively.

D. Compute the CVA of the existing netting set only, since the new trade's effect is captured by the counterparty's credit spread.

Question CR-20 h.5

Bank Q1031 • New question written for this bank

A bank has an existing receive-fixed swap with a client under an enforceable netting agreement and is pricing a new pay-fixed swap that largely offsets it. Which of the following statements regarding the incremental CVA of the new trade is correct?

A. It can never be a benefit, because adding any trade to a netting set must raise the netting set's expected positive exposure.

B. It can be a benefit rather than a cost, because the new trade can reduce the netting set's expected positive exposure, which allows the bank to offer more favourable terms than to a client with no existing trades.

C. It equals the new trade's standalone CVA, because the netting agreement affects only the counterparty's default probability and not the exposure.

D. Its netting benefit grows as the new trade's notional is increased, because greater size produces more offsetting cash flows.

Question CR-20 h.6

New • built from GARP Ch.20, 20.4.1 and Ch.19, 19.3.1

An xVA desk prices BCVA without survival adjustments, applying each party's own default probability to its leg. The counterparty and the bank have the same flat credit spread and the same LGD, and therefore the same marginal default probabilities. A trader proposes a new swap that strongly offsets the existing netting set and argues that it should earn a large BCVA benefit from netting. Which of the following is the best response?

A. The trader is right: the offsetting swap lowers the netting set's EPE, so the CVA leg shrinks while the DVA leg is unaffected, giving a large BCVA benefit.

B. The trader is wrong: netting raises both EPE and ENE in size, so the new swap makes BCVA more negative.

C. The trader is wrong: with equal spreads and LGDs, BCVA depends on EPE plus ENE, which is the expected future value; that is additive across trades, so the CVA and DVA effects of netting offset and there is no incremental netting benefit.

D. The trader is right: netting lowers the DVA leg but leaves the CVA leg unchanged, so BCVA becomes less negative.

Solutions • CR-20 h

Solution CR-20 h.1 — Answer: A

Bank Q1029 • AnalystPrep Q.1884

Step 1 – Exposure under netting

At default the trade values are offset, so the exposure is $\max(\sum_k V_k, 0)$ rather than $\sum_k \max(V_k, 0)$.

Step 2 – The inequality

$\max(\sum_k V_k, 0) \leq \sum_k \max(V_k, 0)$ on every path, so the netted expected exposure is at most the sum of standalone exposures.

Step 3 – Consequence for CVA

CVA is a PD-weighted sum of EE with the same PD and LGD, so it must be computed on the netting-set profile, is non-additive, and is never larger in size than the sum of standalone CVAs.
$\Rightarrow$ A restates GARP eq. 20.10.

Interpretation

The netting benefit can be significant for balanced portfolios, which is why CVA systems aggregate exposures by netting set before applying the credit inputs.

Why the other options are wrong

B is incorrect. EE is not additive within a netting set; summing trade-level CVAs ignores the offset and overstates CVA.

C is incorrect. Netting can only reduce exposure, so it cannot raise CVA; this confuses concentration with offsetting.

D is incorrect. Netting acts on the exposure; default probability and recovery are unchanged.

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1 eq. 20.10; Ch.11 (GARP CR Ch.19), 19.3.1

Solution CR-20 h.2 — Answer: C

New • built from GARP Ch.20, 20.4.1 and Ch.19, 19.3.1

Step 1 – When netting helps

Aggregation reduces exposure only in scenarios where trade values have opposite signs (GARP Table 19.1).

Step 2 – Counterparty X

All ten swaps are pay-fixed for X, so their values to the bank move together: they are rarely of opposite sign, and the netting-set EPE is close to the sum of standalone EPEs.

Step 3 – Counterparty Y

Pay-fixed and receive-fixed swaps of similar size offset, so the netting-set EPE is far below the sum.
$\Rightarrow$ With the same spread, the whole difference is in the exposure: a large benefit for Y, a small one for X.

Interpretation

GARP: netting benefits are a portfolio effect, strong for balanced and offsetting portfolios and weaker for directional ones. Corporate hedgers like X trade directionally, so netting does little for their CVA.

Why the other options are wrong

A is incorrect. The number of trades does not set the benefit; the correlation (sign pattern) of

their values does.
B is incorrect. Polarity flip: same-direction trades offset least; netting has nothing to cancel.
D is incorrect. Netting leaves the default probability unchanged.

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1; Ch.11 (GARP CR Ch.19), 19.3.1, Table 19.1

Solution CR-20 h.3 — Answer: D

New • built from GARP Ch.20, 20.4.1 and Ch.19, 19.3

Step 1 – What a netting set is
The trades under one legally enforceable netting agreement. GARP notes there may be more than one netting agreement with a given counterparty.

Step 2 – Within each netting set
Add the trade values on each simulated path, then take the positive part, giving one EPE profile per set.

Step 3 – Where netting is not enforceable
Each trade in group (3) is its own netting set, so its gross positive exposure counts.

Step 4 – Counterparty CVA
Add the netting-set CVAs, using the counterparty's PD and the LGD (same seniority) throughout.
$\Rightarrow$ D follows the legal boundaries of netting.

Interpretation
GARP notes that a bank may sometimes judge netting enforceable for pricing while accepting the legal risk, but where its lawyers consider it unenforceable the trades are treated gross. Clients in such jurisdictions may trade only with banks that do recognise netting.

Why the other options are wrong
A is incorrect. Netting is a legal right within an agreement, not a consequence of having one default probability.
B is incorrect. Netting acts on the exposure, path by path; a scalar netting factor on standalone CVAs is not how GARP incorporates it.
C is incorrect. Groups (1) and (2) sit under separate agreements with no cross-netting, and group (3) still carries CVA on its gross exposure.

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1; Ch.11 (GARP CR Ch.19), 19.3 and footnote 20; Ch.6 (GARP CR Ch.16), netting enforceability

Solution CR-20 h.4 — Answer: B

Bank Q1032 • AnalystPrep Q.5486

Step 1 – Why the standalone figure is wrong
Under netting a new trade changes the exposure of the whole netting set, not just its own.

Step 2 – The incremental calculation
Compute CVA_{NS} before and $\text{CVA}_{NS'}$ after adding the trade and charge the difference (eq. 20.11), equivalently the CVA formula with the incremental EPE (eq. 20.12).

Step 3 – Its properties
It is never larger in size than the standalone CVA and can be a benefit when the trade offsets existing exposure.

⇒ B is the method GARP describes.

Interpretation

The numbers are worked under CR-20 i (Workbook A6.4 converts an incremental CVA into a running spread).

Why the other options are wrong

A is incorrect. CVA is not additive under netting; adding the standalone CVA overstates the charge.
C is incorrect. Summing positive-exposure CVAs ignores netting altogether.
D is incorrect. The credit spread carries the default risk, not the change in netting-set exposure caused by the trade.

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1 eqs. 20.11 and 20.12

Solution CR-20 h.5 — Answer: B

Bank Q1031 • New question written for this bank

Step 1 – The measure

Incremental CVA is the change in netting-set CVA caused by the new trade, from the netting-set exposure with and without it.

Step 2 – An offsetting trade

The pay-fixed swap gains when the existing receive-fixed swap loses, so it can lower the netting set's EPE: incremental EPE is negative and incremental CVA is a benefit.

Step 3 – The commercial result

GARP: an institution with existing trades under netting will be likely to offer more favourable conditions (Cooper and Mello, 1991).
⇒ B.

Interpretation

Unwinding a trade is GARP's example of a change that would be expected to produce a benefit of this kind.

Why the other options are wrong

A is incorrect. False premise: incremental EPE can be negative because of beneficial netting.
C is incorrect. Netting changes exposure, not the default probability; incremental CVA is generally smaller than standalone.
D is incorrect. As the new trade's size increases the netting benefit is lost and incremental CVA moves towards the standalone value (GARP Figure 20.12).

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1, Figure 20.12

Solution CR-20 h.6 — Answer: C

New • built from GARP Ch.20, 20.4.1 and Ch.19, 19.3.1

Step 1 – BCVA without survival terms

$\text{BCVA} = -\text{LGD}\sum \text{EPE}(t_i)\,\text{PD}_C(t_{i-1},t_i) - \text{LGD}\sum \text{ENE}(t_i)\,\text{PD}_B(t_{i-1},t_i).$

Step 2 – Equal spreads and LGDs

With $\text{PD}_C = \text{PD}_B$, $\text{BCVA} = -\text{LGD}\sum[\text{EPE}+\text{ENE}](t_i)\,\text{PD}(t_{i-1},t_i)$, and $\text{EPE}+\text{ENE} = \text{E}[\max(V,0)] + \text{E}[\min(V,0)] = \text{EFV}$.

Step 3 – Additivity

The EFV is additive across trades (GARP Table 19.1: $10 + (-5) = 5$), so the netting-set BCVA equals the sum of standalone BCVAs.

$\Rightarrow$ Netting cuts CVA and DVA by the same amount: no incremental netting benefit.

Interpretation

GARP: "for equal credit spread, BCVA is always the same and has no incremental effect." Even with 150 against 100 bps (Table 20.4), the offsetting book halved the CVA but moved BCVA only from -20.3 to -16.3 bps.

Why the other options are wrong

A is incorrect. Ignores the DVA leg: the offset shrinks ENE in size too, so DVA falls.

B is incorrect. Polarity flip: netting reduces both EPE and ENE in size.

D is incorrect. Netting reduces both legs, not only DVA.

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.4.1, Table 20.4; Ch.11 (GARP CR Ch.19), 19.3.1, Table 19.1

CR-20 • CVA • Vardeez: VERY IMPORTANT

CR-20 q

P2 CONCEPT

Explain the implications of central clearing on wrong-way risk.

Workbook: A20 Wrong-Way and Right-Way Risk • Tier A • revision notes only; A20.2 touches this LO

WHAT IS TESTED HERE

Why P2. Vardeez flags the LO in a VERY IMPORTANT chapter. No Tier workbook question is homed here; the A20 revision notes cover GARP 20.6.7 (central clearing and WWR).

Vardeez: central clearing does not remove wrong-way risk, it relocates it: the CCP inherits the WWR of its members' positions and of the collateral they post, so it must scale margin and default fund contributions to member credit quality and set haircuts or eligibility by the WWR in the collateral.

Reported exam shapes: none listed in the map. A20·S4 (the effect of WWR on CCPs) belongs to CR-20 o, which draws on the same GARP section and carries the shared background, including the CCP waterfall drawn as a CDO; this block recaps it and adds what clearing itself changes.

What the questions look like: conceptual: why a CCP is prone to WWR, why margin does not settle the matter, which members should post more, why CDSs are hard to clear, how eligible collateral creates adverse selection, and why a matched book does not make WWR disappear.

Worked elsewhere: Workbook A20.2 (at Question CR-20 m.6) includes the CCP margin statement.

1. What central clearing changes

Through **novation** the CCP becomes the buyer to every seller and the seller to every buyer, so the original parties no longer face each other's default (GARP Ch.18). Positions are netted multilaterally and secured by variation margin, initial margin and default fund contributions. The CCP runs a **matched book** with no net market risk, but it has **conditional market risk**: when a member defaults its book is no longer matched, and it must hedge and auction the defaulter's positions, which is exactly when WWR bites. Clearing therefore does not remove WWR. It moves the WWR of members' trades to the CCP and, beyond the defaulter's own resources and the CCP's equity, **mutualises** it across the surviving members.

Correction to the source notes. SchweserNotes (LO 38.o and 38.q) says that in a default "a CCP must replace non-performing contracts. This process is formally called novation". In GARP (Ch.18, 18.2) novation is the legal process at clearing by which the CCP is placed between buyer and seller. Replacing a defaulter's positions is default management: hedging and auctioning them to return to a matched book.

2. Where the wrong-way risk goes (recap of CR-20 o)

- **Reliance on margin.** Margin mitigates WWR only if exposure builds gradually; a jump at default makes it useless (20.6.6). Pykhtin and Sokol: the benefit of margin under WWR must allow for jumps and higher volatility during the margin period of risk, and WWR is greatest for systemic parties such as banks, which are the ones that post margin (corporates often post none).
- **Defaulter pays may not be enough.** If the defaulter's resources are insufficient, losses fall on the other members and the CCP may become insolvent.
- **Credit quality and exposure kept apart.** Members must meet a minimum credit quality set by the CCP, then pay initial margin and default fund contributions driven mainly by market risk, so

WWR is implicitly ignored. Since WWR *increases* with credit quality (a large dealer is more WWR than a smaller or weaker one), GARP's "perverse" conclusion is that better-quality, more systemic members should contribute *more*.

- **A senior, systemic second loss.** The defaulter's margin and default fund plus CCP equity are the first loss; survivors' default fund contributions are a second loss that is rather senior in CDO terms, and senior tranches are concentrated in systemic risk (GARP Figure 20.24, drawn under CR-20 o).
- **Products and collateral.** Ideally cleared products carry no WWR (Ch.18); CDSs catalysed the clearing mandate yet are arguably the hardest to clear, and their WWR is hard to put into margin and default funds. Wider collateral eligibility invites **adverse selection**: members post the assets with the greatest risk relative to their haircut, which may also carry the greatest WWR (a European bank posting European sovereign debt).

	Bilateral (uncleared)	Centrally cleared
Who faces a default	The member's bilateral counterparties	The CCP, then surviving members
Credit quality enters	Through the counterparty's spread in CVA	Membership threshold only; margin driven by market risk
Treatment of WWR	Can be priced trade by trade (conditional EPE)	Implicitly ignored unless built into margin, default fund and haircuts
Tail losses go to	The exposed counterparty	Mutualised: a senior, systemic second loss

Exam traps

- **"Clearing removes WWR".** It relocates WWR to the CCP and mutualises the tail.
- **Matched book means no risk.** The book is matched only while every member performs.
- **Novation misread.** Novation is the substitution of the CCP at clearing, not the replacement of a defaulter's trades.
- **Polarity flip on credit quality.** Stronger, more systemic members bring *more* WWR, so they should post more.
- **Margin as a complete cure.** Jumps and stressed volatility over the margin period of risk can exhaust it.
- **Eligibility as diversification.** A wider list of eligible assets invites adverse selection.

Practice questions: CR-20 q

Question CR-20 q.1

Bank Q1079 • New question written for this bank

A clearing member's risk analyst explains why a central counterparty (CCP) may be particularly prone to wrong-way risk (WWR). Which of the following is the best explanation?

A. Novation to the CCP removes the original counterparty entirely, so any CDS exposure that remains is a residual risk that is not related to the defaulting member.

B. The CCP relies mainly on the defaulting member's margin and default fund contribution, and in a WWR scenario these defaulter-pays resources may prove insufficient, which would push losses onto surviving members and could threaten the CCP's own solvency.

C. Multilateral netting by the CCP eliminates the exposure on CDS positions, so WWR arises only from the CCP's equity capital.

D. The CCP sets initial margin primarily from members' credit ratings, which makes margins unstable when ratings change.

Question CR-20 q.2

New • built from GARP Ch.20, 20.6.6 and 20.6.7

At the risk committee of a CCP, a member argues that because every cleared position carries daily variation margin and initial margin, wrong-way risk is not a material concern for the CCP. Which response is most consistent with GARP?

A. Disagree: margin mitigates WWR only if exposure builds gradually before a default; jumps and higher volatility during the margin period of risk can exhaust initial margin, and the members that post margin are mostly systemic banks whose defaults bring the most extreme WWR.

B. Agree: margin removes WWR at a CCP, because variation margin is received continuously as the exposure to a member rises.

C. Agree: WWR matters only for uncollateralised counterparties such as corporates, and corporates are not clearing members.

D. Disagree: margin increases WWR, because collateral posted by a member always loses its value when that member defaults.

Question CR-20 q.3 New • built from GARP Ch.18, 18.2 and Ch.20, 20.6.7

A regulator notes that credit default swaps (CDSs) were a main catalyst of the clearing mandate after the 2007 to 2009 crisis. A CCP risk officer replies that single-name CDSs are among the hardest products to clear. Which reasoning best supports the risk officer?

A. CDSs are illiquid by construction, so a CCP cannot compute variation margin on them at all.

B. Once cleared, CDSs carry no WWR, because multilateral netting offsets protection bought and sold across members, so the difficulty is purely operational.

C. CDS exposures fall when credit spreads widen, so the CCP faces right-way risk that makes initial margins too high.

D. Ideally cleared products carry no WWR, but a member's default can be adversely linked to the value of the CDSs it has traded (for example, protection sold on correlated financials), so values can move sharply just as it defaults, complicating default management and the WWR part of margin and default funds.

Question CR-20 q.4 Bank Q1080 • New question written for this bank

A CCP sets each clearing member's initial margin and default fund contribution primarily from the market risk of that member's portfolio. A member of the CCP's risk committee warns that, because these requirements do not incorporate wrong-way risk (WWR), the CCP's loss waterfall may be insufficient if a member defaults. Which of the following changes is most consistent with addressing this concern?

A. Use the default fund contributions of non-defaulting members before the defaulting member's margin, so that the defaulter's collateral is preserved for later losses.

B. Reduce initial margin for the highest-credit-quality members, because they are less likely to default and their own resources will therefore always cover their losses.

C. Require higher initial margin and default fund contributions from members with higher credit quality, because WWR tends to increase with increasing credit quality.

D. Set initial margins solely from members' external credit ratings, because ratings fully capture the WWR in each member's portfolio.

Question CR-20 q.5 Bank Q1081 • New question written for this bank

A CCP is under pressure from clearing members to widen the range of securities it accepts as initial margin. A risk manager warns that this would increase the CCP's wrong-way risk (WWR). Which of the following is the most likely reason?

A. A larger list of eligible assets means that haircuts no longer matter, because every accepted asset is then assumed to be risk-free.

B. Accepting more securities lowers the CCP's liquidity risk, which makes margin calls less frequent and so increases the CCP's exposure to members.

C. Wider eligibility reduces the diversity of collateral, so any WWR is concentrated in a single asset class.

D. Members would tend to post the assets with the greatest risk relative to their haircut, which may also present the greatest WWR to the CCP, and underestimated haircuts would add to the risk.

Question CR-20 q.6 New • built from GARP Ch.18, 18.2 and Ch.20, 20.6.7

The treasurer of a regional bank that is a clearing member argues that, because the CCP stands between all buyers and sellers and runs a matched book, moving the bank's credit derivatives into clearing has eliminated the wrong-way risk it used to face bilaterally. Which response is most accurate?

A. Agree: after novation the CCP has no net market risk in any circumstances, so a member's default cannot produce wrong-way losses for anyone.

B. Agree: the wrong-way risk now sits entirely in the defaulting member's own initial margin, which by design covers every possible close-out loss.

C. Disagree: the book is matched only while all members perform; a default leaves it unmatched just when WWR bites, and losses beyond the defaulter's resources and CCP equity fall on surviving members.

D. Disagree: clearing increases wrong-way risk mainly because the CCP must post initial margin to its members, which it loses when a member defaults.

Solutions • CR-20 q

Solution CR-20 q.1 — Answer: B

Bank Q1079 • New question written for this bank

Step 1 – The CCP's protection

A CCP relies on posted margin and default fund contributions, and aims to keep a member's default within that member's own resources (defaulter pays).

Step 2 – What WWR does

When the exposure to a member rises as its credit quality falls, its resources may not cover the losses at default.

Step 3 – Where losses go

The CCP then uses its own equity and the default fund contributions of the surviving members, and may itself become insolvent.

$\Rightarrow$ B.

Interpretation

This is the sense in which clearing relocates WWR: from the member's bilateral counterparties to the CCP and its mutualised resources.

Why the other options are wrong

A is incorrect. Novation puts the CCP in the original counterparty's place; the CCP still faces the member's default.

C is incorrect. Netting reduces but does not eliminate exposure, and the WWR sits in the defaulter's resources, not only in the CCP's equity.

D is incorrect. CCP margins are driven mainly by portfolio market risk, not by ratings; that is why WWR can be ignored.

LO: CR-20 q: Explain the implications of central clearing on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7 Central Clearing and WWR

Solution CR-20 q.2 — Answer: A

New • built from GARP Ch.20, 20.6.6 and 20.6.7

Step 1 – Margin and timing

GARP 20.6.6: if exposure increases gradually before default, margin can be received; a jump in exposure makes margin useless.

Step 2 – Stressed close-out

Pykhtin and Sokol: the benefit of margin under WWR must allow for jumps and higher volatility during the margin period of risk, so initial margin set at normal coverage can be exhausted.

Step 3 – Who posts margin

The parties that actively use margin, mainly banks, are highly systemic and face the most extreme WWR; non-systemic corporates often post no margin anyway.

$\Rightarrow$ A: at a CCP, margin is weakest exactly where WWR is worst.

Interpretation

CCP membership is dominated by large banks, so the WWR that matters most to a CCP is the default of a systemic member, which is when markets gap.

Why the other options are wrong

B is incorrect. Absolute claim: margin arrives with a lag and cannot keep up with a jump at default.

C is incorrect. The systemic, margin-posting members are the ones subject to extreme WWR.
D is incorrect. Absolute claim: only specific collateral, such as a member's own bonds, is linked to its default; margin in general still reduces exposure.

LO: CR-20 q: Explain the implications of central clearing on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.6 Collateralisation and WWR; 20.6.7

Solution CR-20 q.3 — Answer: D

New • built from GARP Ch.18, 18.2 and Ch.20, 20.6.7

Step 1 – The clearability criterion

GARP Ch.18 lists WWR among the features of a clearable product: ideally it would have none, because if a member's default is adversely linked to a contract's value, that value may fall rapidly at the default, which harms default management.

Step 2 – Why CDSs fail it

A member that has sold protection on names correlated with itself (other financials, its sovereign) owes much more exactly when it defaults.

Step 3 – The CCP's task

GARP 20.6.7: CCPs clearing products such as CDSs are particularly prone to WWR, and quantifying the WWR component of initial margin and default funds is far from easy.
$\Rightarrow$ D.

Interpretation

Hence GARP's "subtle point": the product that catalysed the clearing mandate is arguably the most difficult to clear centrally.

Why the other options are wrong

A is incorrect. Liquidity is a clearability criterion, but CDSs are cleared with variation margin; illiquidity is not the reason given.
B is incorrect. Netting offsets market risk across members, but the CCP still faces each member's default; WWR remains.
C is incorrect. Polarity flip: bought protection gains when spreads widen, so the exposure to protection sellers rises in a credit shock.

LO: CR-20 q: Explain the implications of central clearing on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8 (GARP CR Ch.18), 18.2 Mechanics of Central Clearing; Ch.17 (GARP CR Ch.20), 20.6.5 and 20.6.7

Solution CR-20 q.4 — Answer: C

Bank Q1080 • New question written for this bank

Step 1 – The waterfall

The CCP first uses the defaulter's margin and default fund contribution, then its own equity and the surviving members' default funds.

Step 2 – The gap

If initial margin and default fund contributions are driven only by market risk, WWR is implicitly ignored and the defaulter-pays layers may be too thin.

Step 3 – Which members

WWR tends to increase with credit quality (a large dealer is more WWR than a smaller one), so GARP argues that CCPs should demand more from better-credit-quality, more systemically important members.
$\Rightarrow$ C.

Interpretation

GARP calls this "perverse": better-quality members are less likely to default, but their default would be a far bigger shock. Workbook A20.2 (worked as Question CR-20 m.6) tests the same statement.

Why the other options are wrong

A is incorrect. Reverses the waterfall: defaulter-pays resources come first.

B is incorrect. Confuses default probability with the severity of WWR; cutting their margin worsens the problem.

D is incorrect. CCPs use their own membership criteria, and ratings do not capture the jump in exposure at a systemic default.

LO: CR-20 q: Explain the implications of central clearing on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7 and footnote 40

Solution CR-20 q.5 — Answer: D

Bank Q1081 • New question written for this bank

Step 1 – The pressure

CCPs are pushed to accept a wide range of eligible securities; riskier, less liquid assets create extra risk and put weight on haircut calculations.

Step 2 – Adverse selection

Members will naturally post the assets with the greatest risk relative to their haircut, and these may carry the greatest WWR (a European bank posting European sovereign debt).

Step 3 – Model risk

Haircuts that are underestimated add to the risk.

$\Rightarrow$ D.

Interpretation

The response is higher haircuts on assets whose value is linked to the poster's credit quality, or a shorter eligibility list.

Why the other options are wrong

A is incorrect. Haircuts matter more, not less, when riskier assets are accepted.

B is incorrect. Accepting riskier, less liquid assets raises collateral risk; it does not improve liquidity.

C is incorrect. Wider eligibility widens the range of assets; the problem is adverse selection, not concentration.

LO: CR-20 q: Explain the implications of central clearing on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.17 (GARP CR Ch.20), 20.6.7

Solution CR-20 q.6 — Answer: C

New • built from GARP Ch.18, 18.2 and Ch.20, 20.6.7

Step 1 – The matched book

Through novation the CCP faces every buyer and every seller, so with all members performing its positions offset and it has no net market risk.

Step 2 – Conditional market risk

GARP Ch.18: the CCP has conditional market risk, because a member default leaves it with an unmatched book that it must hedge and auction. With WWR the defaulter's positions move sharply against the CCP at exactly that moment.

Step 3 – Who bears the loss

Defaulter pays first (its initial margin and default fund contribution), then CCP equity, then the surviving members' default fund contributions: a senior, mutualised second loss concentrated in systemic scenarios.

$\Rightarrow$ C: the bank has swapped bilateral WWR for a share of the CCP's tail.

Interpretation

Clearing changes the form of the risk, not its existence. The surviving members' exposure is rare but systemic, which is why GARP likens it to a senior CDO tranche (CR-20 o).

Why the other options are wrong

A is incorrect. Absolute claim: the book is matched only while every member performs; default is the case that matters.

B is incorrect. Defaulter-pays resources may prove insufficient under WWR; initial margin is set to a confidence level, not to cover every loss.

D is incorrect. Members post initial margin to the CCP, not the reverse; the risk comes from the defaulter's positions and the mutualised default fund.

LO: CR-20 q: Explain the implications of central clearing on wrong-way risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8 (GARP CR Ch.18), 18.2 Mechanics of Central Clearing; Ch.17 (GARP CR Ch.20), 20.6.7, Figure 20.24

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 a

P2 CONCEPT

Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure.

Workbook: none homed here • revision notes in C9 Stress-Testing Practices and Scenario Design • Tier C • Vol 2 p.74

WHAT IS TESTED HERE

Why P2. Vardeez rates CR-21 a "full important chapter" ("Stress Testing has always been favorite topic of Garp"), and the map flags all eight LOs of the reading VARDEEZ FOCUS. The workbook's C9 section (Tier C, Map #38, 3/16) carries revision notes for every LO of the reading; its question C9.1 is worked under CR-21 h.

Vardeez: the four measures in one table, ordered by what they average over. Current exposure is a point in time, peak exposure a percentile of the future distribution, expected exposure its mean at a date, and EPE that mean averaged across dates. The fuller metric family (expected MtM, PFE, ENE, effective EPE, maximum PFE) is CR-19 a.

Reported exam shapes: none listed in the map for this LO.

What the questions look like: definitions and sibling swaps. Match "mean at a date", "high percentile at a date", "time-weighted average" and "larger of zero and today's value" to the right name, read a peak-exposure number, and know which measure plays the loan-equivalent role. This is a CONCEPT LO, so there are no calculations here; the arithmetic versions (EPE from an EE profile, PFE from a distribution) are CR-19 a material.

Shared background: Section 1 maps the whole reading. The later CR-21 LOs point back to it.

1. The reading at a glance

Lynch's starting point: since the late 1990s the *statistical* measures of counterparty credit risk (CCR) have progressed much faster than CCR *stress testing*. The chapter therefore builds stress tests on top of the statistical measures, in four steps of increasing quality, and ends with the pitfalls.

Stress test	Exposure used	Credit quality?	Output	LO
Stressed current exposure (21.3)	Current exposure after repricing under the scenario	No	Top-counterparty tables, no aggregate	c
Expected-loss stress, credit view (21.4)	EAD for loans; $\alpha \times$EPE for derivatives	Yes: physical PD	$EL^S - EL$, adds to the loan book	c, d
CVA stress, market view (21.5)	Discounted risk-neutral EE at each date	Yes: risk-neutral PD from spreads	$CVA^S - CVA$	e, f
Bilateral CVA stress (21.5)	EE and negative EE (the DVA side)	Both parties	$BCVA^S - BCVA$, adds to trading stresses	g

The pitfalls of Section 21.6 (current exposure where expected exposure belongs, delta shortcuts, no integration with loan or trading stresses) are CR-21 h. The two ways of viewing CCR, as a credit risk and as a market risk, are CR-21 b.

2. Four exposure measures

The chapter uses the Basel Committee's (2005) definitions. All four are measured for a **netting set**, so trades with the counterparty under one netting agreement are valued together.

- **Current exposure** (also called **replacement cost**): the larger of zero and the market value of the netting set, the amount that would be lost if the counterparty defaulted *now*, assuming no recovery. It is a single value today with no distribution. A negative market value gives zero, not a positive number. In GARP's Table 21.1 collateral is netted off as well: MtM 35, collateral 20, current exposure 15.
- **Peak exposure**: a high percentile (typically 95% or 99%) of the distribution of exposures at a particular *future date* before the longest maturity in the netting set. It is produced for many future dates, so it forms a profile; CR-19 calls it potential future exposure (PFE). It is the natural basis for limits.
- **Expected exposure** (EE): the *mean* of the exposure distribution at a particular future date, again produced date by date. Negative values are floored at zero before averaging, so EE is never below the expected MtM.
- **Expected positive exposure** (EPE): the weighted average *over time* of the expected exposures, each weighted by the proportion of the whole interval it represents. For minimum capital the average runs over the first year, or over the life of the longest contract if everything matures within the year. EPE is one number, and it is the exposure that lets derivatives sit beside loans in a portfolio credit model.

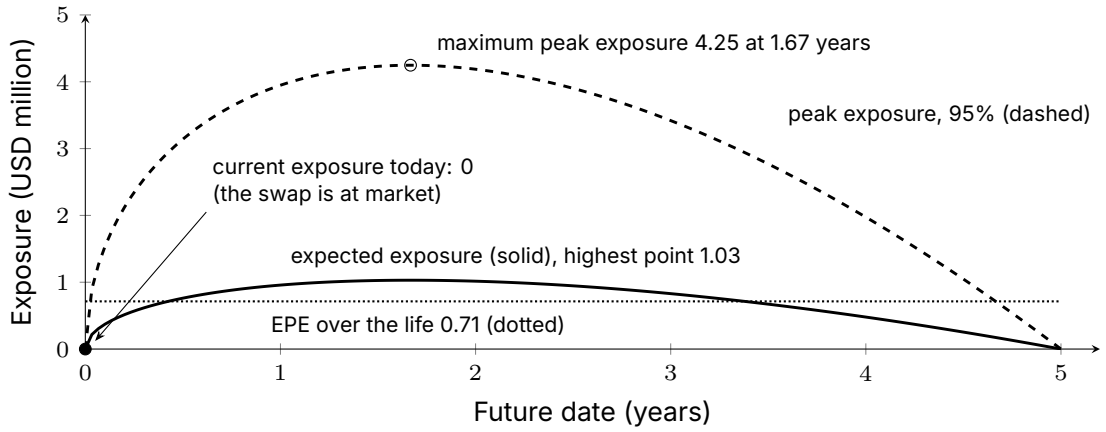


Figure 1. Exposure profile of an at-market 5-year swap (illustrative: MtM normal with zero mean and standard deviation $3\sqrt{t}\,(5-t)/5$). Peak exposure (dashed) and EE (solid) are curves over future dates; EPE (dotted) collapses the EE curve into one number; current exposure (dot) is a single value today and here it is zero, although the trade clearly carries exposure.

Where each measure turns up in this reading: **current exposure** is what the simplest stress tests shock (CR-21 c); **EE and EPE** are what belong in a portfolio credit model and in the expected-loss stress (CR-21 c, d); **discounted risk-neutral EE** drives CVA (CR-21 e, f); **peak exposure** sets limits. One more definition the chapter needs early: **wrong-way risk** arises when the counterparty's credit quality is correlated with the exposure, so that exposure grows when default is most likely. It cannot arise for a loan, whose exposure is fixed.

Correction to the source notes. Vardeez's table defines peak exposure as a percentile "over a future timeframe" and expected exposure as the "average potential exposure over a future timeframe". In GARP

(the BCBS 2005 definitions) both are statistics of the exposure distribution *at a particular future date*, produced for many dates; averaging *over time* is what defines EPE. Vardeez's own commentary under the table states it correctly ("expected exposure is its mean at a date; EPE averages that mean across dates"), and that is the version used here.

	Current exposure	Peak exposure	Expected exposure	EPE
Statistic	$\max(0, \text{MtM})$, no recovery	High percentile (95%, 99%)	Mean, negatives floored at 0	Time-weighted mean of EE
When	Today	Each future date	Each future date	Over an interval
Shape	One value	Profile	Profile	One value
Typical use	Replacement cost; simple stress tables	Limits (PFE limits)	Input to EPE and CVA	Loan equivalent ($\alpha \times$EPE); capital

Exam traps

- **Mean versus percentile.** Expected exposure is the mean at a date; peak exposure is the 95th or 99th percentile at a date.
- **EE versus EPE.** EE averages across scenarios at one date; EPE then averages the EE profile across time. "Averaged over time" always means EPE.
- **Negative MtM.** Current exposure is floored at zero; the absolute value of a negative MtM is not an exposure. But a negative MtM today does not make the *future* exposure zero.
- **Maximum EE is not peak exposure.** The highest point of the EE curve is still a mean; the peak exposure profile lies above it.
- **Equal weights.** EPE weights each EE by the length of its interval; a simple average is right only for equally spaced dates.
- **Current exposure as "the exposure".** An at-market swap has zero current exposure and a positive EE and peak exposure profile (Figure 1). This is the root of the CR-21 h pitfalls.

Practice questions: CR-21 a

Question CR-21 a.1

Bank Q1083 • Schweser Module Quiz 39.1 Q1

Which of the following exposure measures reflects the average (mean) of the distribution of exposures at a specific future date prior to the maturity of the longest-maturity transaction within a netting set?

A. Peak exposure.

B. Current exposure.

C. Expected exposure.

D. Expected positive exposure.

Question CR-21 a.2

Bank Q1085 • BT answers P2.T6.707.1 (rebuilt)

Which of the following statements about counterparty credit exposure measures is correct?

A. Current exposure is the mean of the distribution of exposures at a given future date.

B. Peak exposure is a high percentile (typically 95% or 99%) of the exposure distribution at a future date, and is usually generated for many future dates.

C. Expected positive exposure is the maximum of the expected exposures across all future dates.

D. Expected exposure is the larger of zero and the current market value of the netting set.

Question CR-21 a.3

Bank Q923 • BT notes P2.T6.407.2 • re-filed from CR-19 a

A credit officer is training new analysts on counterparty exposure. Each of the following statements about credit exposure, current exposure and potential exposure is necessarily true, EXCEPT:

A. Credit exposure is conditional on counterparty default.

B. If the mark-to-market (MtM) value of a contract is negative, current exposure is zero.

C. If the mark-to-market (MtM) value of a contract is negative, credit exposure is zero.

D. Credit exposure encompasses both current exposure and potential exposure.

Question CR-21 a.4 Bank Q905 • AnalystPrep Q.2902 (rebuilt) • re-filed from CR-19 a

Banks X and Y enter into a vanilla 5-year interest rate swap with a notional of USD 100 million and semiannual payments. Bank X pays six-month floating and Bank Y pays the fixed swap rate of 5%. Bank X calculates a 12-month peak exposure (potential future exposure) on the swap of USD 7.5 million at the 99% confidence level. Which statement is the correct interpretation?

A. There is a 1% probability that Bank X's exposure to Y in 12 months will be USD 7.5 million or less.

B. Bank X is 99% confident that its loss on the swap in 12 months will be no more than USD 7.5 million because of market moves alone.

C. There is at most a 1% probability that Bank X's positive mark-to-market value on the swap (its exposure to Bank Y) will exceed USD 7.5 million in 12 months.

D. Bank X is 99% confident that its exposure to Bank Y in 12 months will be at least USD 7.5 million.

Question CR-21 a.5 Bank Q925 • BT notes P2.T7.3.2 (rebuilt) • re-filed from CR-19 a

A bank wants to place its derivative counterparties in the same portfolio credit model as its loans. Which counterparty exposure measure is typically used as the basis for the loan-equivalent amount, even if a multiplier such as alpha is then applied?

A. Current exposure (the larger of current MtM and zero).

B. Expected exposure (EE) at a single future date.

C. Peak exposure (potential future exposure) at a high confidence level, such as 95%.

D. Expected positive exposure (EPE), the time-weighted average of the EE profile.

Question CR-21 a.6 New • built from GARP Ch.21, 21.1 (BCBS 2005 definitions)

A junior analyst at a Korean bank is preparing the expected positive exposure (EPE) of a 7-year netting set for the minimum capital calculation. The simulation engine reports expected exposures at 1, 3, 6 and 12 months, and then annually out to year 7. Which method is consistent with the definitions used in the chapter?

A. The time-weighted average of the expected exposures over the first year, each weighted by the share of the year that its interval represents.

B. The simple average of all the expected exposures out to year 7, with each reporting date counted once.

C. The highest 95th-percentile exposure reported over the first year.

D. The expected exposure at the one-year date, since capital is held for one year.

Solutions • CR-21 a

Solution CR-21 a.1 — Answer: C

Bank Q1083 • Schweser Module Quiz 39.1 Q1

Step 1 – Pick out the two features in the stem

"Mean of the distribution" and "at a specific future date".
⇒ The measure is a statistic of the exposure distribution at one date, and the statistic is the average.

Step 2 – Match against the four definitions

Peak exposure is a high percentile at a date; current exposure is today's $\max(0, \text{MtM})$ with no distribution; EPE averages the expected exposures across dates. Only expected exposure is the mean at a date.

Interpretation

EE and EPE are the pair GARP likes to swap: EE averages across scenarios at one date, and EPE then averages the EE profile across time.

Why the other options are wrong

A is incorrect. Peak exposure is the 95th or 99th percentile at the date, a tail statistic, not the mean.
B is incorrect. Current exposure is a value today; there is no future distribution to average.
D is incorrect. EPE is the time-weighted average of the expected exposures over many dates, not the mean at one date.

LO: CR-21 a: Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure. • **Reference:** Lynch, The Evolution of Stress Testing Counterparty Exposures (GARP CR Ch.21), 21.1; SchweserNotes 2026 Book 2, Reading 39, LO 39.a

Solution CR-21 a.2 — Answer: B

Bank Q1085 • BT answers P2.T6.707.1

Step 1 – Recall the four definitions

Current exposure $=$ $\max(0, \text{MtM today})$; peak exposure $=$ 95th or 99th percentile at a future date; EE $=$ mean at a future date; EPE $=$ time-weighted average of EE.

Step 2 – Test each statement

A gives EE's definition to current exposure; C turns EPE into a maximum; D gives current exposure's definition to EE. B matches the BCBS wording, including "generated for many future dates".
⇒ Peak exposure is a profile of tail values, one per future date.

Interpretation

Three of the four options are sibling swaps: a correct definition attached to the wrong name. Reading the noun (mean, percentile, maximum, average over time) before the measure's name avoids them.

Why the other options are wrong

A is incorrect. The mean at a future date is expected exposure; current exposure has no distribution.
C is incorrect. EPE is a weighted *average* of the EE profile, not its maximum.
D is incorrect. The larger of zero and today's value is current exposure (replacement cost), not expected exposure.

LO: CR-21 a: Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure. • **Reference:** Lynch (GARP CR Ch.21), 21.1

Solution CR-21 a.3 — Answer: C

Bank Q923 • BT notes P2.T6.407.2

Step 1 – Current exposure with a negative MtM

Current exposure is $\max(\text{MtM}, 0)$, so a negative MtM gives zero: B is true.

Step 2 – Credit exposure has a future part

Credit exposure is current exposure plus potential exposure. The MtM can turn positive before maturity, so a negative MtM today does not make credit exposure zero: C is false, D is true.
$\Rightarrow$ A negative value today only switches off the current part.

Step 3 – Conditionality

A bank loses its positive value only if the counterparty defaults, so exposure is conditional on default: A is true.

Interpretation

This is the logic of Figure 1 and of CR-21 h: current exposure can be zero while expected exposure and peak exposure are not. Treating current exposure as the whole exposure is the root pitfall.

Why the other options are wrong

A is incorrect. True: exposure is only lost if the counterparty defaults.
B is incorrect. True: current exposure is floored at zero.
D is incorrect. True: credit exposure combines what is at risk now and what may be at risk later.

LO: CR-21 a: Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure. • **Reference:** Lynch (GARP CR Ch.21), 21.1; Gregory, The xVA Challenge (GARP CR Ch.19), exposure definitions

Solution CR-21 a.4 — Answer: C

Bank Q905 • AnalystPrep Q.2902

Step 1 – What peak exposure measures

Exposure is the positive MtM that X would lose if Y defaulted. A 99% peak exposure is the 99th percentile of that exposure at the 12-month date.

Step 2 – Translate the percentile

The 99th percentile is exceeded with probability at most $100\% - 99\% = 1\%$.
$\Rightarrow$ At most a 1% chance that X's positive MtM (its exposure to Y) exceeds USD 7.5 million in 12 months.

Interpretation

Peak exposure is VaR-like, but in the opposite tail: it is a high percentile of a *gain* that is at risk, not of a loss. The pay and receive legs and the notional are not needed.

Why the other options are wrong

A is incorrect. Wrong side of the tail: the exposure is at or below USD 7.5 million 99% of the time, not 1%.
B is incorrect. Describes a market-risk VaR of losses; exposure is a positive value at risk of default.
D is incorrect. Reverses the inequality: USD 7.5 million is a ceiling exceeded 1% of the time, not a floor.

LO: CR-21 a: Differentiate among current exposure, peak exposure, expected exposure, and expected positive expo-

sure. • **Reference:** Lynch (GARP CR Ch.21), 21.1; Gregory (GARP CR Ch.19), potential future exposure

Solution CR-21 a.5 — Answer: D

Bank Q925 • BT notes P2.T7.3.2

Step 1 – What a loan equivalent needs

A loan has one exposure amount. To put a derivative beside it, the whole exposure profile must be collapsed into one representative amount.

Step 2 – Which measure does that

EPE is the time-weighted average of the EE profile, one number. Lynch: EPE models "allowed derivatives to be placed in portfolio credit risk models similarly to loans", and EPE times an alpha factor is the value that lets CCR exposures sit in a portfolio credit model with loans.
$\Rightarrow$ EPE, scaled by α, is the loan equivalent.

Interpretation

This is exactly how the credit-view stress test of CR-21 d is built: $\text{EL} = \sum p_i \, \alpha \, \text{EPE}_i \, \text{LGD}_i$.

Why the other options are wrong

A is incorrect. Current exposure ignores how exposure evolves; running it through a portfolio credit model is wrong (CR-21 c).
B is incorrect. EE at one date is a single point of the profile, not a summary of it.
C is incorrect. Peak exposure is a tail statistic used for limits, not an average amount lent.

LO: CR-21 a: Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure. • **Reference:** Lynch (GARP CR Ch.21), 21.1 and 21.4

Solution CR-21 a.6 — Answer: A

New • built from GARP Ch.21, 21.1

Step 1 – EPE is a time-weighted average

EPE is "the weighted average over time of expected exposures where the weights are the proportion that an individual expected exposure represents of the entire time interval".
$\Rightarrow$ With dates at 1, 3, 6 and 12 months the intervals are unequal (1, 2, 3 and 6 months), so weights of 1/12, 2/12, 3/12 and 6/12 apply, not equal weights.

Step 2 – Which interval for capital

For the minimum capital requirement the average is taken over the first year (or over the life of the longest contract if all mature within a year). The 7-year netting set therefore uses the first year only.

Interpretation

Two separate errors are on offer: the wrong weights (B) and the wrong statistic (C, D). EPE is neither a percentile nor a single date.

Why the other options are wrong

B is incorrect. Wrong horizon and wrong weights: the capital EPE uses the first year, and unequal intervals need time weights.
C is incorrect. That is the maximum peak exposure over the year, a tail statistic.
D is incorrect. That is a single point of the EE profile, not its average.

LO: CR-21 a: Differentiate among current exposure, peak exposure, expected exposure, and expected positive exposure. • **Reference:** Lynch (GARP CR Ch.21), 21.1, BCBS (2005) definitions

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 b

P2 CONCEPT

Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution.

Workbook: none homed here • revision notes in C9 Stress-Testing Practices and Scenario Design • Tier C • Vol 2 p.74

WHAT IS TESTED HERE

Why P2. Part of a reading Vardeez rates VERY IMPORTANT as a whole ("Full Important Chapter"); the map marks the LO VARDEEZ FOCUS. Revision notes in workbook C9 (Tier C).
Vardeez: one table. Treated as a credit risk, CCR leaves the institution exposed to changes in CVA, so CVA must be in the valuation; treated as a market risk, it can be hedged but leaves the institution exposed to credit deterioration and default; treating it as both is prudent but complex and difficult to interpret. "There is no clean answer, which is itself the examinable point."

Reported exam shapes: none listed in the map for this LO.

What the questions look like: two-part statements ("the market view does not allow hedging ..."), which blind spot each view leaves, how each view manages a deteriorating counterparty, how many stress results the dual view produces (at least $2N + 1$), and the history of the two views. All conceptual. One bank question filed here (Q1086, an EL versus stressed CVA calculation) is worked under CR-21 f.

1. Two lenses on one risk

In the 1990s, managing CCR meant evaluating the creditworthiness of derivatives counterparties and tracking their current exposure. Regulatory capital was an add-on to current exposure, set as a percentage of gross notional (Basel 1988). After the LTCM crisis the Counterparty Risk Management Policy Group (1999) cited deficiencies and called for better measures. Two families then grew side by side:

- **CCR as a credit risk.** First, potential exposure models to measure and limit CCR; then EPE models that let derivatives sit in portfolio credit models like loans (Canabarro, Picoult and Wilde, 2003). These two model types are the hallmark of the credit view. It was the predominant framework from 2000 to 2006 and became the basis for regulatory capital in Basel II (BCBS 2005), with netting and margining built into the models.
- **CCR as a market risk.** The price of CCR, the credit valuation adjustment (CVA), was first described for swaps in 1994 and 1996 and spread through the FAS 157 fair-value requirement (2006). Before the crisis credit spreads were stable and CVA was a small part of derivative valuations. In 2007-08 both spreads and exposures swung widely, and the combined effect produced *large losses and large, unusual gains*. Banks then built CVA risk management, and Basel III added a CVA capital charge (BCBS 2011). Figure 1 puts the milestones on one axis.

2. What each view implies

Both treatments are valid ways to manage the portfolio, but adopting one view alone leaves the institution blind to the risk from the other.

- **Credit view.** CCR is managed at inception or through collateral arranged in advance: potential future exposure (PFE) limits, netting and collateral, credit evaluation. Little can be done once the trades are in place. At default the bank must replace the defaulter's trades in the market *all at once* to rebalance its book, so the emphasis is on risk mitigants and credit evaluation. **Blind spot:** the bank is still exposed to changes in CVA and must include CVA in the valuation of its derivatives portfolio; inattention leads to balance-sheet surprises.
- **Market view.** CCR can be hedged. Instead of waiting for default, the bank replaces trades with the counterparty in the market *before* default, buying positions in proportion to the counterparty's probability of default. A low-PD counterparty has little replaced in advance; as its credit quality deteriorates, a larger share of its trades moves to other counterparties. At default the trades are already replaced and the default itself is a non-event. **Blind spot:** a bank that dynamically hedges CVA to limit market-risk losses remains exposed to large drops in creditworthiness and to sudden default.

A derivatives dealer is forced to consider both aspects. The choice also shapes the organisation: credit officers and limits on one side, a CVA hedging desk on the other.

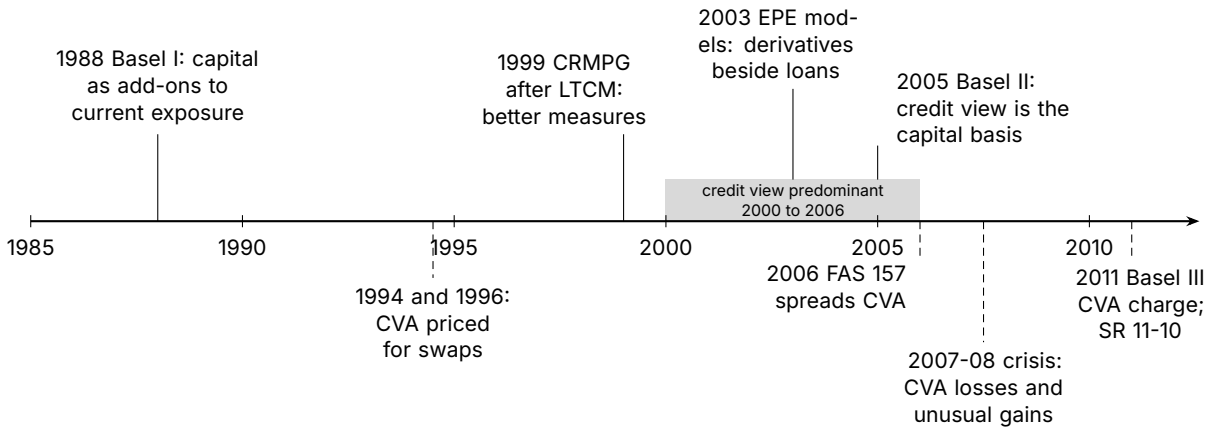


Figure 1. How the two views developed (GARP 21.1). Credit-view milestones above the axis (solid ticks), market-view milestones below it (dashed ticks). The grey band marks the years when the credit view was the dominant framework.

3. What the dual view does to stress testing

The dual nature of CCR produces many measures. On the credit side: current exposure, peak exposure and expected exposure. On the market side: CVA itself, and the risk of changes in CVA, measured for example by the VaR of CVA. This "dazzling array" is hard to interpret at both portfolio and counterparty level, and it raises a second question after "What is my counterparty credit risk?": "What CCR measures should I stress?"

Counting the results. In market risk, an equity crash stress yields one or two numbers: an instantaneous loss on the current portfolio and perhaps a stressed VaR. The CCR manager must run the same stress at the counterparty level and at the portfolio level, in both the credit and the market view:

$$\text{stress results} \;\geq\; 2N + 1 \qquad (N \text{ counterparties})$$

and the count at least doubles again if risk measures are stressed as well as instantaneous shocks; sub-portfolios by region or industry add more. With 50 counterparties that is at least 101 results, or at least 202. The volume "can bewilder even the most diligent risk manager, and overwhelm IT resources". Even so, a comprehensive view of the portfolio requires CCR stress tests, and supervisory guidance (SR 11-10) asks for them.

	CCR as a credit risk	CCR as a market risk
Measures	Current, peak and expected exposure; EPE	CVA; VaR of CVA
Period of dominance	2000 to 2006; basis of Basel II	Pricing only until 2007-08; FAS 157; Basel III CVA charge
How managed	At inception: PFE limits, netting, collateral, credit evaluation	Dynamically: hedge CVA; replace trades in proportion to PD
At default	Replace all the defaulter's trades at once	Trades already moved; default a non-event
Blind spot	Changes in CVA (must still value CVA)	Credit deterioration and sudden default
Stress output	Exposures and expected losses (CR-21 c, d)	Stressed CVA and BCVA (CR-21 e to g)

Exam traps

- **Polarity flip on hedging.** The market view *does* let the bank hedge market-risk losses on CVA; what it leaves open is credit deterioration and default.
- **Who replaces trades, and when.** Credit view: at default, all at once. Market view: before default, gradually, in proportion to the PD.
- **CVA is not optional under the credit view.** Even with limits, netting and collateral, the valuation must include CVA.
- **The count.** At least $2N + 1$ (both views at counterparty level, plus the portfolio), not $N + 1$ or $2N$; at least double that if risk measures are stressed too.
- **"Both" is not "easy".** Treating CCR as both is prudent, and the price is complexity and interpretability.
- **History.** The crisis produced large CVA losses *and* large, unusual gains; CVA was small and stable before it.

Practice questions: CR-21 b

Question CR-21 b.1

Bank Q1088 • Schweser Module Quiz 39.1 Q2

Is the following statement on the treatment of counterparty credit risk (CCR) correct?
"Treating CCR as a market risk does not allow an institution to hedge market risk losses, and it exposes the institution to declines in counterparty creditworthiness and default."

A. The statement is correct with regard to both hedging market risk losses and counterparty creditworthiness and default.

B. The statement is incorrect with regard to both hedging market risk losses and counterparty creditworthiness and default.

C. The statement is correct with regard to hedging market risk losses only.

D. The statement is correct with regard to counterparty creditworthiness and default only.

Question CR-21 b.2

New • built from GARP Ch.21, 21.2

A CCR manager is building the summary page of a stress-testing pack. Which of the following correctly assigns measures to the two ways of viewing counterparty credit risk described in the chapter?

A. Credit-risk view: current exposure, peak exposure and expected exposure; market-risk view: CVA and the variability of CVA, for example the VaR of CVA.

B. Credit-risk view: CVA and peak exposure; market-risk view: current exposure and expected exposure.

C. Credit-risk view: the VaR of CVA only; market-risk view: current, peak and expected exposure, since all three are marked to market.

D. Credit-risk view: CVA, because it is an expected loss; market-risk view: EPE, because it is a time average of market values.

Question CR-21 b.3

New • built from GARP Ch.21, 21.1

The risk committee of a regional bank treats CCR purely as a credit risk. It sets potential future exposure limits for each counterparty, requires collateral and netting agreements at inception, and does not include CVA in the valuation of its derivatives portfolio. According to the chapter, to which risk does this leave the bank most exposed?

A. Losses from counterparty defaults beyond its exposure limits, since the credit view provides no protection against default.

B. Wrong-way risk only, since collateral and netting remove every other source of counterparty risk.

C. Changes in the market value of counterparty risk (CVA) as credit spreads and exposures move, which can produce balance-sheet surprises.

D. None of significance, since limits, collateral and netting together capture all counterparty risk in a derivatives portfolio.

Question CR-21 b.4

New • built from GARP Ch.21, 21.1

Two dealers share a counterparty whose credit quality deteriorates over several months before it defaults. Dealer K treats CCR as a credit risk. Dealer M treats CCR as a market risk and manages its CVA dynamically. Which statement best describes the two dealers' experience?

A. K replaces the counterparty's trades gradually, in proportion to its probability of default, so the default is a non-event for K; M must replace all the trades at once at default.

B. K must replace all of the defaulting counterparty's trades in the market at once at default; M has already moved trades to other counterparties in proportion to the rising probability of default, so the default is close to a non-event for M.

C. Both dealers are fully protected: K by the collateral and netting set up at inception, and M by its CVA hedges.

D. Neither dealer can act before default, because under both views counterparty risk can be managed only at inception.

Question CR-21 b.5

New • built from GARP Ch.21, 21.1

Which of the following statements about the evolution of CCR measurement, as described in the chapter, is correct?

A. Stress testing of CCR evolved faster than the statistical measures of CCR, which is why the chapter builds the statistical measures on top of stress tests.

B. Treating CCR as a credit risk was the predominant framework from 2000 to 2006 and the basis of Basel II capital, while the market-risk treatment was largely confined to pricing CVA until the 2007-08 crisis.

C. Before the crisis CVA was a large and volatile part of bank derivative valuations, which prompted the FAS 157 accounting requirement.

D. During the crisis the combined swings in credit spreads and exposures produced large CVA losses but no gains, since credit spreads only widened.

Question CR-21 b.6

Bank Q1087 • BT answers P2.T6.707.2 • BT notes P2.T6.707.2

Rick is the bank's CCR manager, preparing a stress test report for a credit portfolio with positions with five major counterparties. In addition to default risk and credit deterioration, each position exposes the bank to fluctuations in its credit valuation adjustment (CVA). Rick wants to summarize all relevant values without showing more than necessary. Which of the following is the MOST LIKELY number of stress test results he will need to present?

A. One value for the aggregate portfolio.

B. Six values: one for the credit risk of each counterparty plus one for the credit risk of the portfolio.

C. Ten values: for each counterparty, both market and credit risk (the overall portfolio values being superfluous).

D. At least eleven values: both credit risk and market risk of each counterparty, plus the portfolio.

Solutions • CR-21 b

Solution CR-21 b.1 — Answer: D

Bank Q1088 • Schweser Module Quiz 39.1 Q2

Step 1 – First half: hedging

Under the market view CCR is priced as CVA and "can be hedged": the bank dynamically hedges its CVA to limit market-risk losses.

⇒ "Does not allow an institution to hedge market risk losses" is incorrect.

Step 2 – Second half: what is left open

A bank that hedges its CVA "remains exposed to large drops in creditworthiness or the sudden default of one of its counterparties".

⇒ This half is correct.

Interpretation

Each view covers one exposure and leaves the other open. The market view's weak point is the jump: a sudden default or a sharp downgrade outruns gradual hedging.

Why the other options are wrong

A is incorrect. Accepts the first half, but the market view does permit hedging.

B is incorrect. Rejects the second half, but the exposure to deterioration and default remains.

C is incorrect. Reverses which half is correct.

LO: CR-21 b: Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution. • **Reference:** Lynch (GARP CR Ch.21), 21.1; SchweserNotes 2026 Book 2, Reading 39, LO 39.b

Solution CR-21 b.2 — Answer: A

New • built from GARP Ch.21, 21.2

Step 1 – The credit side

"On the credit risk side, there are the important measures of exposure: current exposure; peak exposure; and expected exposure."

Step 2 – The market side

"On the market risk side, there is the valuation aspect coming from CVA, and there is the risk generated by changes in the CVA as measured by the value-at-risk (VaR) of CVA."

⇒ Exposures belong to the credit view; CVA and its risk belong to the market view.

Interpretation

This split is why the number of CCR stress results doubles: every scenario must be read through both sets of measures.

Why the other options are wrong

B is incorrect. Swaps the two lists: CVA is the market view's measure, and current and expected exposure belong to the credit view.

C is incorrect. Puts all the exposure measures in the market view; exposures are the credit view's statistics, whatever their inputs.

D is incorrect. CVA is a market price of CCR, not the credit view's expected loss; EPE is a credit-view exposure measure.

LO: CR-21 b: Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution. • **Reference:** Lynch (GARP CR Ch.21), 21.2

Solution CR-21 b.3 Answer: C

New • built from GARP Ch.21, 21.1

Step 1 – What the committee has covered

PFE limits, collateral and netting manage the default side of CCR: they are the credit view's tools.

Step 2 – What the credit view leaves open

"If CCR is treated as a credit risk, a bank can still be exposed to changes in CVA. ... it must still include CVA in the valuation of its derivatives portfolio. Inattention to this could lead to balance-sheet surprises."

$\Rightarrow$ The exposure is to swings in the fair value of counterparty risk.

Interpretation

The 2007-08 crisis made this concrete: spreads and exposures moved together and CVA swung by large amounts, in both directions, for banks that had treated CVA as a small pricing item.

Why the other options are wrong

A is incorrect. The credit view is built to manage default risk (limits, collateral, netting); its gap is the fair-value side.

B is incorrect. Absolute claim: collateral and netting do not remove CVA risk, and wrong-way risk is not the only residual.

D is incorrect. Absolute claim: the chapter says adopting one view alone leaves the bank blind to the other.

LO: CR-21 b: Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution. • **Reference:** Lynch (GARP CR Ch.21), 21.1

Solution CR-21 b.4 Answer: B

New • built from GARP Ch.21, 21.1

Step 1 – Credit view at default

"At default, the financial institution must replace the trades of the defaulting counterparty in the market all at once in order to rebalance its book."

Step 2 – Market view before default

The bank replaces trades "before it defaults by buying the positions in proportion to the counterparty's probability of default". As credit quality deteriorates, a larger share moves to other counterparties. "At default, the financial institution will have already replaced the trades and the default itself would be a non-event."

$\Rightarrow$ K faces a one-off replacement at default; M has replaced gradually.

Interpretation

This is the "implication for trading activities" in the LO wording: the market view turns default management into a continuous hedging activity.

Why the other options are wrong

A is incorrect. Swaps the two dealers' behaviour.

C is incorrect. Absolute claim: the credit-view dealer still faces CVA swings and replacement cost, and the market-view dealer is exposed to sudden default.

D is incorrect. True only of the credit view; the market view acts before default.

LO: CR-21 b: Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution. • **Reference:** Lynch (GARP CR Ch.21), 21.1

Solution CR-21 b.5 — Answer: B

New • built from GARP Ch.21, 21.1

Step 1 – The credit view

Potential exposure and EPE models "are the hallmark of treating CCR as a credit risk". This was "the predominant framework for measuring and managing CCR from 2000 to 2006, and was established as the basis for regulatory capital as part of Basel II".

Step 2 – The market view

The market treatment was "largely relegated to pricing in a credit valuation adjustment (CVA) prior to the financial crisis of 2007-08".
$\Rightarrow$ B is a direct statement of the chapter.

Interpretation

The chapter's premise runs the other way from A: the statistical measures moved fast and stress testing lagged, so stress tests are built on the statistical measures.

Why the other options are wrong

A is incorrect. Polarity flip: statistical measures progressed faster than stress testing.
C is incorrect. Before the crisis spreads were stable and CVA was a small part of valuations; FAS 157 is a fair-value rule, not a response to volatile CVA.
D is incorrect. The crisis produced "both large losses and large, unusual gains".

LO: CR-21 b: Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution. • **Reference:** Lynch (GARP CR Ch.21), 21.1

Solution CR-21 b.6 — Answer: D

Bank Q1087 • BT answers P2.T6.707.2

Step 1 – Both views are needed

Each position carries default and deterioration risk (credit view) and CVA risk (market view), so each counterparty needs a result in each view.

Step 2 – Count

At least twice the number of counterparties, plus one for the portfolio: $2 \times 5 + 1 = 11$.
$\Rightarrow$ At least eleven results; more if risk measures are stressed as well as instantaneous shocks.

Interpretation

GARP's point is that CCR stress testing multiplies results compared with market risk, where one stress gives one or two numbers. The "at least" matters: sub-portfolios and stressed risk measures push the count higher.

Why the other options are wrong

A is incorrect. Ignores the counterparty-level results and the dual view.
B is incorrect. Counts only the credit view: $5 + 1 = 6$.
C is incorrect. Counts both views for each counterparty ($2 \times 5 = 10$) but drops the portfolio result needed for aggregation.

LO: CR-21 b: Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution. • **Reference:** Lynch (GARP CR Ch.21), 21.2 and footnotes 1 and 2

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 c

P2 CONCEPT

Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio.

Workbook: none homed here • revision notes in C9 Stress-Testing Practices and Scenario Design • Tier C • C9.1 tags this LO (worked as Question CR-21 h.6)

WHAT IS TESTED HERE

Why P2. Part of a reading Vardeez rates VERY IMPORTANT throughout; the map marks the LO VARDEEZ FOCUS. Workbook C9.1, a best-practice audit, tags this LO and is worked under CR-21 h.

Vardeez: banks stress current exposures by repricing counterparty portfolios under scenarios such as an equity crash, and summarise the results in tables that show which counterparties are most vulnerable. These tests struggle with aggregation and ignore counterparty credit quality and wrong-way risk.

Reported exam shapes: none listed in the map for this LO.

What the questions look like: how a stressed current exposure is produced and what it is good for; its three problems (aggregation, credit quality, wrong-way risk); reading a Table 21.1-style report; how a loan-portfolio stress is built (PD stressed, EAD and LGD fixed) and how derivatives are fitted into it ($\alpha\times$EPE, PD and EPE stresses). All conceptual. The bank's Q1089, a current-exposure arithmetic question, is left out under the CONCEPT rule; Q1102 is re-filed here from CR-21 d.

1. Stressing current exposure

Section 1 of CR-21 a maps the chapter's four stress tests; this LO covers the first two. The most common CCR stress is a stress of **current exposure**. The bank assumes a scenario of changes in the underlying risk factors and *reprices* each counterparty's portfolio under it. The stressed current exposure is measured after the stress is applied but *before any collateral is collected*. Banks commonly report their top counterparties by current exposure in one table, then their top counterparties by stressed current exposure, one table per scenario.

Counterparty	Rating	MtM	Collateral	Current exposure	Stressed current exposure
A	A	0.5	0	0.5	303
B	AA	100	0	100	220
C	AA	35	0	35	119
D	BBB	20	20	0	76
E	BBB	600	600	0	75
F	A	−5	0	0	68
G	A	−10	0	0	50
H	BB	−50	0	0	24
I	A	35	20	15	17
J	BB	24	24	0	11

Current exposure stress test, equity market down 25% (US$ million), from GARP Table 21.1. Top 10 counterparties by stressed current exposure; no aggregate row.

A has almost no current exposure (0.5) and the largest stressed exposure (303); D and E are fully collateralised today yet show 76 and 75 before any new collateral arrives; F, G and H owe the bank

nothing today and still appear. Other scenarios (credit events, rate shocks) would rank *different* counterparties.

What it is good for. It shows which counterparties would be of concern in the stress event and how much each would owe: useful for monitoring individual counterparties.

Three problems.

(i) **Aggregation.** Summing the stressed exposures gives the loss if *every* counterparty defaulted in the scenario: "unless the scenario were the Apocalypse, this would clearly be an exaggeration". Feeding stressed current exposures into a portfolio credit model is also wrong, because expected exposures, not current exposures, belong there. Hence Table 21.1 has no aggregate.

(ii) **Credit quality.** The test counts only the value of the trades, not the counterparty's willingness or ability to pay: US$200 million owed by a start-up hedge fund is very different from US$200 million owed by a triple-A corporate. In Table 21.1 it is unclear whether C (AA, 119) or D (BBB, 76) is the greater worry. A limit structure for stressed exposure by rating and scenario is daunting, which is why limits are usually set on statistical measures such as peak exposure.

(iii) **Wrong-way risk.** A measure that omits credit quality cannot say anything about the correlation of exposure with credit quality.

2. Stressing a loan portfolio

The typical loan framework analyses how **expected loss** changes under a stress. For a pool of N loan counterparties,

$$\text{EL} = \sum_{i=1}^{N} p_i \cdot \text{ead}_i \cdot \text{lgd}_i.$$

A stress test takes EAD and LGD as **deterministic** and stresses the **PD**. Each p_i is a function of other variables, typically macroeconomic (an important exchange rate, the unemployment rate) or balance-sheet items; those variables are set to stressed values, giving stressed PDs p_i^S and

$$\text{EL}^S = \sum_{i=1}^{N} p_i^S \cdot \text{ead}_i \cdot \text{lgd}_i, \qquad \text{stress loss} = \text{EL}^S - \text{EL} \quad \text{(calculated in CR-21 d)}.$$

The bank can simply raise the PDs, or stress the variables behind them, and can report results per counterparty and in aggregate. PD stresses tend to push all counterparties in the same direction.

3. The same framework for a derivative portfolio

For CCR treated as a credit risk, PD and LGD are handled as for loans, but **EAD is now stochastic** and depends on market variables. EPE times an alpha factor is the exposure that lets CCR sit in a portfolio credit model beside loans, so

$$\text{EL} = \sum_{i=1}^{N} p_i \cdot \alpha \cdot \text{epe}_i \cdot \text{lgd}_i, \qquad \text{EL}^S = \sum_{i=1}^{N} p_i^S \cdot \alpha \cdot \text{epe}_i^S \cdot \text{lgd}_i.$$

Two kinds of stress are now available.

- **PD stresses**, as for loans: stress the PDs or their drivers (balance-sheet values, macroeconomic indicators, values of financial instruments). GARP's Table 21.2 parallels the rise in PDs by industry after the 2001-02 dotcom crash (CR-21 d). Loan and derivative stress losses can then be added together.

- **EPE stresses**: shock the market variables that drive exposure, such as equity prices and swap rates. The usual method is an instantaneous shock, often the same shock used for current exposure, applied to the initial market value *before* the EPE simulation is run. Shocks at future dates, or a series of shocks over time, are possible but not common. The effect on EPE depends on collateralisation and moneyness, and in aggregate an EPE stress may *raise or lower* expected loss, depending on the bank's directional bias and which counterparties are margined or hold excess margin (GARP Table 21.3). Because loan exposures are insensitive to market variables, EPE stresses need not be aggregated with the loan book.

Joint stresses of credit quality and market variables are straightforward in concept but hard in practice, because little connects macroeconomic or balance-sheet variables to market variables. Equity-based PDs (Merton, Kealhofer) come closest, but it is unclear how fast such PDs should react to an instantaneous exposure shock. **Wrong-way risk** is also hard to capture; the best existing approach in the credit framework is to stress current exposure, identify the counterparties most exposed to the stress, and then consider whether each is also subject to wrong-way risk.

The gain over current-exposure stresses: losses aggregate with the loan portfolio, and credit quality enters. The market-risk view goes further, with PDs inferred from market variables (CR-21 e).

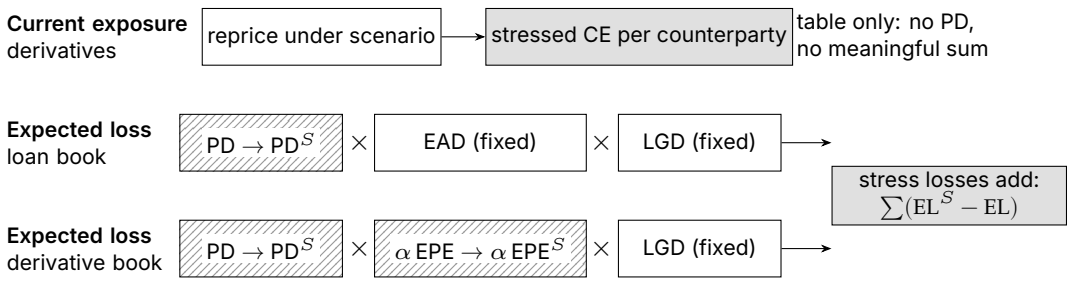


Figure 1. The two stress tests of this LO, with the current-exposure stress above them for contrast. Hatched boxes are the stressed inputs: only the PD for loans, the PD and the exposure for derivatives. Grey boxes are outputs. Expected-loss stresses on the two books can be added; stressed current exposures cannot.

	Stressed current exposure	EL stress: loans	EL stress: derivatives
What is shocked	Risk factors; portfolios repriced	PD (or its macro and balance-sheet drivers)	PD, and EPE through market variables
Exposure	Current exposure, before new collateral	EAD, deterministic	$\alpha \times$EPE, stochastic
Credit quality	Ignored	Physical PD	Physical PD
Aggregation	Not meaningful	Yes, by addition	Yes; adds to the loan stress
Direction	Per counterparty	Same way across names	EPE stresses may go either way
Wrong-way risk	No insight	Does not arise (fixed EAD)	Hard; screen via stressed CE

Exam traps

- **Summing stressed current exposures** gives the loss if every counterparty defaults at once; it is not a stress loss.
- **Stressed current exposure into a portfolio credit model**: wrong input; expected exposures

belong there.

- **"Easy to add credit quality"**: polarity flip. The test ignores credit quality, and building limits by rating and scenario is daunting.
- **Negative MtM or full collateral today is not safety under stress**: F, G, H and D, E in Table 21.1.
- **Loans: stress the PD, not the EAD.** EAD and LGD are deterministic in the loan framework.
- **Direction.** PD stresses push all names the same way; EPE stresses can raise or lower the aggregate.
- **EPE stresses and the loan book.** No need to aggregate them, because loan exposures do not move with market variables; PD-stress losses on the two books do add.

Practice questions: CR-21 c

Question CR-21 c.1

Bank Q1091 • AnalystPrep Q.2000

Financial institutions often stress test current exposure. What is done to create a stressed current exposure?

A. Assume a scenario of changes in underlying risk factors and reprice the portfolio under that scenario.

B. Select the major counterparties and perform detailed credit analysis of their recent financial information.

C. Estimate the worst-case loss at a given confidence level.

D. Randomly select a few counterparties and reprice their exposures for a given change in one market risk factor.

Question CR-21 c.2

Bank Q1093 • AnalystPrep Q.1997

A bank applies a 30% equity market crash stress test to each of its counterparties. How best would this stress test benefit the bank?

A. It allows the bank to set aside sufficient balance sheet provisions.

B. It ranks all counterparties from least to most risky.

C. It estimates when the next equity crash will occur.

D. It identifies the counterparties of concern in such a stress event and shows how much each would owe the bank.

Question CR-21 c.3

New • built from GARP Ch.21, 21.3 and Table 21.1

A CCR manager presents the following extract from the bank's current exposure stress report to the credit committee (scenario: equity market down 25%; US$ million).

Counterparty	Rating	MtM	Collateral	Current exposure	Stressed current exposure
A	A	0.5	0	0.5	303
C	AA	35	0	35	119
D	BBB	20	20	0	76
H	BB	−50	0	0	24
I	A	35	20	15	17

Which conclusion can the committee draw from this table alone?

A. The equity crash would cost the bank about US$539 million, the sum of the stressed current exposures.

B. Counterparty C is a greater concern than counterparty D in this scenario, because its stressed current exposure is larger.

C. The table shows which counterparties would owe the bank most if equities fell 25%, but it cannot say whether C or D is the greater concern, because stressed current exposure ignores credit quality.

D. Counterparty D needs no further attention in this scenario, because its trades are fully collateralised today.

Question CR-21 c.4

Bank Q1094 • AnalystPrep Q.2001

Stress tests of current exposure are useful for individual counterparties but create problems when aggregated. If a bank sums the stressed current exposures to obtain an aggregate stress exposure, the result represents the loss if every counterparty defaulted in the stress scenario. Is such an aggregate result a meaningful measure of stress loss?

A. No, the aggregate would clearly be an exaggeration of likely losses.

B. Yes, it provides a bird's-eye view of the maximum exposure and is therefore a sound stress loss.

C. Yes, it effectively simulates a global crisis like that of 2007-08.

D. No, because summing exposures ignores the individual counterparties' risks, which a portfolio model would capture exactly.

Question CR-21 c.5

Bank Q1092 • Schweser Module Quiz 39.2 Q1

An analyst notes that stress testing current exposure is problematic because aggregating results is typically not meaningful, although it is easy to account for the credit quality of the counterparty. Are the analyst's statements correct?

A. Correct with regard to both aggregating results and credit quality.

B. Correct with regard to aggregating results only.

C. Correct with regard to credit quality only.

D. Incorrect with regard to both aggregating results and credit quality.

Question CR-21 c.6 Bank Q1095 • Mock 2 Q62 (rebuilt, key corrected in the bank)

A risk manager stress tests the bank's current counterparty exposures under a scenario in which global equity markets fall 40%. The counterparties are ranked by stressed current exposure and the table is presented to the credit committee. Which of the following is the most significant shortcoming of this approach?

A. The 40% equity decline is implausible as a scenario.

B. It ranks counterparties on stressed current exposure alone, ignoring counterparty credit quality and giving no information on wrong-way risk.

C. It considers only a single risk factor rather than several.

D. It can be run only at the aggregate portfolio level and not for individual counterparties.

Question CR-21 c.7 New • built from GARP Ch.21, 21.4

The CRO of a mid-sized bank asks for a stress test of the corporate loan book in the expected-loss framework that the chapter starts from before adapting it to derivatives. Which design matches that framework?

A. Hold the PDs fixed and stress each EAD through instantaneous shocks to equity prices and swap rates, since the loans' exposures move with market variables.

B. Replace each EAD with alpha times EPE, and stress EPE by re-running the exposure simulation after a market shock.

C. Reprice each loan under the scenario and report the borrowers with the largest stressed current exposure, one table per scenario.

D. Hold EAD and LGD fixed, model each PD as a function of variables such as the unemployment rate or an important exchange rate, set those variables to stressed values, and recompute expected loss with the stressed PDs.

Question CR-21 c.8 Bank Q1102 • Schweser Module Quiz 39.2 Q2 (rebuilt) • re-filed from CR-21 d

Which of the following best explains why a financial institution that stresses EPE does not need to separately consider aggregating those stresses with its loan portfolio?

A. Loans are insensitive to market variables, so their exposures do not change when market variables are shocked.

B. EPE stresses are applied only to current exposure, which has no loan equivalent.

C. EPE and the loan portfolio are negatively correlated, so the stresses offset.

D. EPE and the loan portfolio are positively correlated, so the stresses are already captured in loan losses.

Solutions • CR-21 c

Solution CR-21 c.1

Answer: A

Bank Q1091 • AnalystPrep Q.2000

Step 1 – The procedure

"To create a stressed current value, the bank assumes a scenario of underlying risk factor changes and reprices the portfolio under that scenario."

Step 2 – Scope

The stress is applied to each counterparty, and the largest stressed exposures are reported per scenario.

⇒ Scenario plus full repricing, counterparty by counterparty.

Interpretation

Nothing in the procedure uses the counterparty's PD: that omission is the source of the credit-quality and wrong-way-risk problems.

Why the other options are wrong

B is incorrect. A credit review involves no scenario and no repricing.

C is incorrect. A worst loss at a confidence level is a statistical measure (VaR or peak exposure), not a scenario stress.

D is incorrect. The stress is applied to every counterparty, not a random sample.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.3

Solution CR-21 c.2

Answer: D

Bank Q1093 • AnalystPrep Q.1997

Step 1 – What the stress produces

A stressed current exposure for each counterparty under the equity crash, ranked in a table.

Step 2 – What that tells management

It "allows the bank to identify which counterparties would be of concern in such a stress event, and also how much the counterparty would owe the financial institution under the scenario".

⇒ A scenario-specific list of exposures, nothing more.

Interpretation

The ranking is valid only for this scenario and ignores credit quality, so it is neither a provisioning number nor a general risk ranking.

Why the other options are wrong

A is incorrect. Provisions follow expected losses; stressed current exposures carry no PD and cannot be summed into a loss.

B is incorrect. The ranking is by exposure in one scenario and ignores credit quality, so it is not a ranking of overall riskiness.

C is incorrect. A stress test measures the impact of a scenario; it does not forecast when it happens.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.3

Solution CR-21 c.3 — Answer: C

New • built from GARP Ch.21, 21.3 and Table 21.1

Step 1 – What the table measures

Stressed current exposure is the value of the trades after the shock and before new collateral: what each counterparty would owe the bank in the scenario.
⇒ It ranks exposure, not loss.

Step 2 – What it leaves out

It omits each counterparty's willingness or ability to pay. C has the larger stressed exposure (119) but is rated AA; D has 76 but is rated BBB. GARP: "it is unclear whether the CCR manager should be more concerned about counterparty C or counterparty D".

Interpretation

To compare C and D, exposure must be combined with credit quality, which is what the expected-loss stress of Sections 2 and 3 does.

Why the other options are wrong

A is incorrect. Absolute claim: $303 + 119 + 76 + 24 + 17 = 539$ is the loss only if every counterparty defaulted in the scenario.
B is incorrect. Ignores credit quality: a larger exposure to an AA name need not be the greater concern.
D is incorrect. D's collateral covers today's MtM of 20; the stressed exposure of 76 is measured before any further collateral is collected.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.3, Table 21.1

Solution CR-21 c.4 — Answer: A

Bank Q1094 • AnalystPrep Q.2001

Step 1 – What the sum assumes

Adding the stressed exposures counts every counterparty's exposure as lost, so it assumes all counterparties default together in the scenario.

Step 2 – GARP's verdict

"Unless the scenario were the Apocalypse, this would clearly be an exaggeration of the losses."
⇒ Not a meaningful stress loss.

Interpretation

A proper aggregate needs expected exposures and credit-quality information: the expected-loss framework of Sections 2 and 3.

Why the other options are wrong

B is incorrect. The worst-case sum overstates likely losses; it is not a sound stress loss.
C is incorrect. Even a severe crisis does not make every counterparty default at once.
D is incorrect. Right answer, wrong reason: a simple sum uses no portfolio model, and the problem is the joint-default assumption.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.3

Solution CR-21 c.5 Answer: B

Bank Q1092 • Schweser Module Quiz 39.2 Q1

Step 1 – Aggregation

Summing stressed exposures gives the loss if all counterparties defaulted at once, and a portfolio credit model needs expected exposures instead.

⇒ "Aggregating results is typically not meaningful" is correct.

Step 2 – Credit quality

Stressed current exposure reflects only trade values. Incorporating credit quality for each scenario and rating category is "daunting", and most banks have not done it.

⇒ "Easy to account for credit quality" is incorrect.

Interpretation

Two of the three named problems appear in one sentence; the second is phrased as a polarity flip.

Why the other options are wrong

A is incorrect. Accepts the false claim about credit quality.

C is incorrect. Reverses the two halves.

D is incorrect. Rejects the aggregation problem, one of the chapter's two main criticisms.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.3; SchweserNotes 2026 Book 2, Reading 39, LO 39.c

Solution CR-21 c.6 Answer: B

Bank Q1095 • Mock 2 Q62

Step 1 – The structural weaknesses

Stressed current exposure counts only the value of the trades, not the counterparty's ability to pay, and so says nothing about how exposure correlates with credit quality.

Step 2 – Compare with the options

B names both weaknesses. The other options name design choices or misstate the method.

⇒ Ranking by exposure alone can mislead the committee about where losses are likely.

Interpretation

GARP lists three problems: aggregation, credit quality, wrong-way risk. B covers the last two; the first is why the table carries no total.

Why the other options are wrong

A is incorrect. Scenario severity is a design choice, not a defect of the method.

C is incorrect. Single-factor scenarios are standard; the equity crash in Table 21.1 is one.

D is incorrect. Backwards: individual results are meaningful; it is the aggregation that fails.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.3

Solution CR-21 c.7 Answer: D

New • built from GARP Ch.21, 21.4

Step 1 – The loan framework

Expected loss $\text{EL} = \sum p_i \, \text{ead}_i \, \text{lgd}_i$ across the pool.

Step 2 – What is stressed

"A stress test could take EAD and LGD as deterministic, and focus on stresses where the probability of default is subject to a stress." The PD is a function of other variables, such as "an important

exchange rate or an unemployment rate", set to stressed values.
$\Rightarrow$ Stressed PDs give EL^S; the stress loss is $EL^S - EL$.

Interpretation

Each wrong option describes a real test, but for the derivative book: B is the credit-view derivative adaptation, C is the current-exposure stress, and A wrongly gives loans market-sensitive exposure.

Why the other options are wrong

A is incorrect. Loan exposures are insensitive to market variables; the loan stress works through the PD.
B is incorrect. $\alpha \times$EPE replaces EAD only for derivatives, whose exposure is stochastic.
C is incorrect. That is the current-exposure stress used for derivative counterparties.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4

Solution CR-21 c.8 Answer: A

Bank Q1102 • Schweser Module Quiz 39.2 Q2

Step 1 – What an EPE stress shocks

Market variables such as swap rates and equity prices, which drive derivative exposure.

Step 2 – Effect on loans

"Loans are insensitive to the market variables and thus will not have any change in exposure due to changes in market variables."
$\Rightarrow$ There is nothing on the loan side to aggregate with an EPE stress.

Interpretation

GARP's footnote adds two exceptions: loan commitments can have stochastic exposure, and loan expected losses can still rise if PDs are correlated with market variables. PD-stress losses on loans and derivatives are added together.

Why the other options are wrong

B is incorrect. EPE stresses shock the initial market value and then re-simulate EPE; the stated reason is the loans' insensitivity.
C is incorrect. Correlation is not the reason, and offsetting stresses would still need to be aggregated.
D is incorrect. Positive correlation would argue for aggregation, not against it.

LO: CR-21 c: Describe a stress test that can be performed on a loan portfolio, and on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4 and footnote 5; SchweserNotes 2026 Book 2, Reading 39, LO 39.d

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 d

P2 CALC

Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio.

Workbook: none homed here • revision notes in C9 Stress-Testing Practices and Scenario Design • Tier C • C9.1 tags this LO (worked as Question CR-21 h.6)

WHAT IS TESTED HERE

Why P2. One of the reading's three calculation LOs, in a chapter Vardeez rates VERY IMPORTANT throughout; the map marks it VARDEEZ FOCUS. Workbook C9.1 (worked as Question CR-21 h.6) tags it: derivative stress losses can be added to loan stress losses only when they are built from $\alpha \times$EPE in the expected-loss framework.

Vardeez: "Stressed expected loss is the expected loss recomputed under the stressed inputs. It is a level. Stress loss is the difference, $\text{EL}^S - \text{EL}$ ... It is a change." Loans: stress the PD only. Derivatives: stress the PD and the EPE, because exposure is itself stochastic.

Reported exam shapes: none listed in the map for this LO.

What the questions look like: a table of counterparties with PD, stressed PD, EAD or EPE, LGD (sometimes a recovery rate instead) and perhaps α; you are asked for the stress loss, a single name's contribution, or the largest contributor. The distractors are the base EL, the stressed EL, their sum, a dropped LGD, a dropped α and an unstressed EPE. Red herrings: a correlation, a current exposure, a notional.

1. A level and a change

The expected-loss framework of CR-21 c produces two numbers for each counterparty and for the portfolio:

- the **stressed expected loss** EL^S: the expected loss recomputed with the stressed inputs, a *level*;
- the **stress loss** $\text{EL}^S - \text{EL}$: how much worse the stressed world is than the current one, a *change*.

$$\text{Loans:} \quad \text{EL} = \sum_i p_i \, \text{ead}_i \, \text{lgd}_i, \qquad \text{EL}^S = \sum_i p_i^S \, \text{ead}_i \, \text{lgd}_i$$

$$\text{Derivatives:} \quad \text{EL} = \sum_i p_i \, \alpha \, \text{epe}_i \, \text{lgd}_i, \qquad \text{EL}^S = \sum_i p_i^S \, \alpha \, \text{epe}_i^S \, \text{lgd}_i$$

Only the PD carries the stress superscript for loans; for derivatives both the PD and the EPE do. Stress losses on the two books are added to give the bank's credit-portfolio stress loss.

2. Worked example: a loan portfolio

A recession scenario raises the PDs through the unemployment rate; EAD and LGD are fixed (USD million).

Loan	EAD	LGD	PD	Stressed PD	EL	Stressed EL	Stress loss
L1	200	45%	1.0%	2.5%	0.900	2.250	1.350
L2	120	60%	2.0%	4.0%	1.440	2.880	1.440
L3	80	30%	0.5%	0.8%	0.120	0.192	0.072
Portfolio					2.460	5.322	2.862

L1: EL $= 200 \times 0.45 \times 0.010 = 0.900$ and $\text{EL}^S = 200 \times 0.45 \times 0.025 = 2.250$, so the stress loss is 1.350. With only the PD stressed, each name's stress loss is simply ead $\times$ lgd $\times$ $(p^S - p)$: for L2, $120 \times 0.60 \times 0.020 = 1.440$. The portfolio stressed EL is 5.322, but the **stress loss is 2.862**.

3. Worked example: a derivative portfolio

Two derivative counterparties, with $\alpha = 1.2$ calculated at the expected-loss level. The scenario raises PDs *and* shocks equity markets, which changes the EPEs. D2's trades move against the bank's position, so its EPE falls.

Name	PD	PD^S	EPE	EPE^S	LGD	EL	EL^S	Stress loss
D1	0.40%	0.80%	50	65	60%	0.1440	0.3744	0.2304
D2	1.20%	1.80%	20	18	70%	0.2016	0.2722	0.0706
Portfolio						0.3456	0.6466	0.3010

D1: EL $= 0.004 \times 1.2 \times 50 \times 0.60 = 0.1440$ and $\text{EL}^S = 0.008 \times 1.2 \times 65 \times 0.60 = 0.3744$. For D2 the PD stress alone would add 0.1008, but the lower stressed EPE takes $1.2 \times 0.018 \times 2 \times 0.70 = 0.0302$ back. The bank's combined credit stress loss is $2.862 + 0.301 = 3.163$.

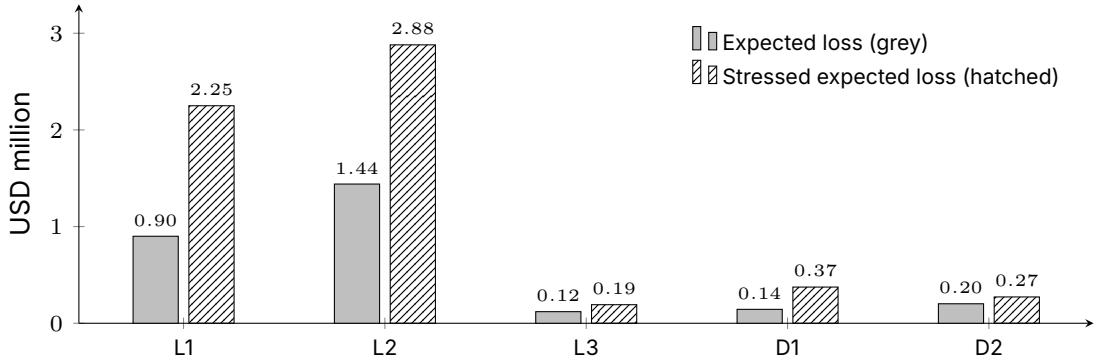


Figure 1. Worked examples of Sections 2 and 3. The stress loss is the gap between each hatched bar and the grey bar beside it, never the hatched bar itself: 2.862 for the loans and 0.301 for the derivatives, 3.163 in total.

4. GARP's two derivative tables

Table 21.2 (PD stress, dotcom crash). EPE and LGD are held fixed and the PDs are raised as they were by industry after 2001-02.

Name	PD (%)	EPE	LGD	EL	Stressed PD (%)	Stressed EL	Stress loss
AA	0.05	213.00	0.70	0.08	0.50	0.77	0.69
BB	0.03	202.50	0.60	0.04	0.30	0.38	0.34
CC	0.45	75.00	0.70	0.24	0.62	0.34	0.09
DD	0.90	30.00	0.65	0.18	1.20	0.24	0.06
EE	1.05	10.00	0.75	0.08	1.40	0.11	0.03
FF	0.09	157.00	0.50	0.07	0.12	0.10	0.02
GG	0.98	68.00	0.70	0.48	1.02	0.50	0.02
HH	2.17	3.00	0.34	0.02	3.00	0.03	0.01
II	0.03	150.00	0.20	0.01	0.05	0.02	0.01
JJ	0.50	50.00	0.60	0.15	0.50	0.15	0.00
Aggregate				1.36		2.63	1.27

PD stress, dotcom crash (US$ million), from GARP Table 21.2.

The largest stress losses come from the large, low-PD exposures whose PDs jump tenfold (AA, BB), not from the names with the highest PDs (HH, EE). All rows move the same way, as PD stresses do.

Table 21.3 (EPE stress, equity market down 25%). The PDs are held fixed and the current exposures of Table 21.1 are re-run as EPE stresses. Counterparty E, fully collateralised with zero current exposure, still has an EPE of 3.99 rising to 75.00; counterparty H, with a negative MtM and an EPE of only 0.02, has the largest stress loss (19.72 thousand) because its EPE jumps to 25.12 and its PD (1.2%) is the highest. With such small PDs the expected losses are measured in thousands: aggregate EL 3.62, stressed EL 52.32, stress loss 48.70 (US$ thousand), aggregated by simple addition. The table prints no LGD, so its EL columns cannot be rebuilt from the table alone; read it for the pattern.

Correction to the source notes. (1) GARP Table 21.2's EL columns run about 3% above PD × EPE × LGD: for AA, $0.0005 \times 213 \times 0.70 = 0.075$ (printed 0.08) and $0.005 \times 213 \times 0.70 = 0.746$ (printed 0.77). The table does not state α; with $\alpha = 1$ the aggregates are 1.32, 2.55 and 1.23 instead of 1.36, 2.63 and 1.27. The bank's version of the table (Question CR-21 d.3) uses $\alpha = 1$. In the exam, use the α the stem gives. (2) Vardeez calls α "the multiplier that converts effective EPE into an exposure at default", which is its regulatory-capital role. In this chapter α multiplies EPE and is "the alpha calculated at the expected loss", so it "may differ from the alpha used for economic or regulatory capital calculations" (GARP footnote 4).

Formula card: CR-21 d

Loan EL and stressed EL	$\text{EL} = \sum_i p_i \, \text{ead}_i \, \text{lgd}_i$, $\text{EL}^S = \sum_i p_i^S \, \text{ead}_i \, \text{lgd}_i$
Derivative EL and stressed EL	$\text{EL} = \sum_i p_i \, \alpha \, \text{epe}_i \, \text{lgd}_i$, $\text{EL}^S = \sum_i p_i^S \, \alpha \, \text{epe}_i^S \, \text{lgd}_i$
Stress loss	$\text{EL}^S - \text{EL}$, per name and in aggregate; add across the two books
PD-only stress, one name	$\text{ead} \times \text{lgd} \times (p^S - p)$ or $\alpha \, \text{epe} \times \text{lgd} \times (p^S - p)$
LGD from recovery	$\text{lgd} = 1 - R$
α	At the expected-loss level; may differ from the capital alpha (1 if not given)

Exam traps

- **Stressed EL reported as the stress loss** (intermediate result). The stress loss is the difference $\text{EL}^S - \text{EL}$.
- **Adding instead of subtracting.** $\text{EL} + \text{EL}^S$ is a classic distractor.
- **LGD slips.** Dropping LGD, or using the recovery rate as the LGD.
- **α slips.** Dropping α for derivatives, or applying it to loans.
- **Unstressed EPE.** When the scenario shocks market variables, the EPE changes too, and it can fall as well as rise.
- **Current exposure instead of EPE.** The derivative EL uses $\alpha \times$EPE; a current exposure in the table is a red herring.
- **Ranking by PD level.** The biggest contributor has the largest exposure × lgd × Δp, not the highest PD or stressed PD.
- **Red herrings.** Correlation, notional and current exposure do not enter the expected-loss stress.

Practice questions: CR-21 d

Question CR-21 d.1

Bank Q1101 • Mock 1 Q37

Bank Q's regulator requires all banks to calculate the stress loss on their loan portfolios. The CFO compiles the following data.

Parameter	Value
Loan portfolio	USD 250 million
Recovery rate	60%
PD, normal conditions	2.5%
PD, stressed conditions	7.0%
Correlation	0.25

What is the stress loss for Bank Q?

A. USD 2.5 million

B. USD 4.5 million

C. USD 7.0 million

D. USD 9.5 million

Question CR-21 d.2

Bank Q1100 • BT answers P2.T6.708.1 • BT notes P2.T6.708.1

A CCR manager analyses a portfolio of funded loans plus a sub-portfolio of unfunded, collateralised credit derivative positions. For each loan, expected loss is $p_i \times \text{EAD}_i \times \text{LGD}_i$, summed over the loans. The stress loss is $\text{SL} = \text{EL}^S - \text{EL}$, and expected and stressed expected losses are summed across the two sub-portfolios. Starting from the loan EL formula, he modifies it to produce a CCR-type stressed EL for the derivatives sub-portfolio. Each of the following modifications is likely EXCEPT which one, which is probably NOT appropriate?

A. For both sub-portfolios, retain the default probability p_i and stress it to a higher stressed probability p_i^S.

B. For the derivatives sub-portfolio, replace EAD_i with alpha multiplied by expected positive exposure, $\alpha \times \text{EPE}_i$.

C. For the derivatives sub-portfolio, stress exposure at default EAD_i to a higher stressed exposure at default EAD_i^S.

D. For the derivatives sub-portfolio, stress expected positive exposure EPE_i to a higher stressed EPE_i^S.

Question CR-21 d.3

Bank Q1099 • BT answers P2.T6.707.3 (rebuilt)

A bank re-examines its stress test of counterparty exposure within an expected-loss framework (PD stress, dotcom crash; $\alpha = 1$; USD million).

Name	PD	EPE	LGD	EL	Stressed PD	Stressed EL	Stress loss
AA	0.05%	213.00	70.0%	0.0746	0.50%	0.7455	0.6710
BB	0.03%	202.50	60.0%	0.0365	0.30%	0.3645	0.3281
CC	0.45%	75.00	70.0%	0.2363	0.62%	0.3255	0.0893
DD	1.00%	300.00	65.0%	1.9500	1.60%	?	?

What is the stress loss for counterparty DD?

A. USD 1,170,000

B. USD 1,800,000

C. USD 1,950,000

D. USD 3,120,000

Question CR-21 d.4

Bank Q1097 • BT answers P2.T6.25.2.2

A bank estimates expected loss (EL) and stressed expected loss for its top counterparties under normal and stressed conditions. LGD is unchanged under stress and EPE is unchanged ($\alpha = 1$).

Counterparty	PD (%)	EPE (US$m)	LGD	Stressed PD (%)
AA	0.05	190.0	55%	0.50
BB	0.03	202.5	60%	0.30
CC	0.45	75.0	70%	0.62
DD	0.90	30.0	65%	1.20

Which counterparty contributes the most to the stress loss?

A. AA

B. BB

C. CC

D. DD

Question CR-21 d.5

Bank Q1098 • BT answers P2.T6.25.2.3 (rebuilt)

A bank stresses its OTC derivatives portfolio with an equity market shock. For each counterparty, expected loss is EPE $\times$ LGD $\times$ PD ($\alpha = 1$), and the stress affects PD, LGD and EPE.

Item	Counterparty A	Counterparty B
PD	3%	18%
Stressed PD	4%	23%
LGD	5%	20%
Stressed LGD	10%	40%
EPE (USD million)	4.37	3.99
Stressed EPE (USD million)	5.46	4.99

What is the stress loss on the portfolio?

A. USD 42,000

B. USD 150,000

C. USD 331,000

D. USD 481,000

Question CR-21 d.6

New • built from GARP Ch.21, 21.4

A risk manager at a Canadian bank combines the stress losses on the loan book and the derivatives book under a recession scenario that raises PDs and shocks equity markets. Derivative exposures enter the credit framework as alpha times EPE, with alpha at the expected-loss level of 1.2. LGDs are unchanged by the stress. Amounts are in CAD million.

Loan segment	EAD	LGD	PD	Stressed PD
SME	400	40%	1.5%	3.0%
Corporate	250	50%	0.8%	1.6%

Counterparty	Current exposure	EPE	Stressed EPE	LGD	PD	Stressed PD
X	25	90	120	60%	0.5%	1.0%
Y	0	40	35	60%	2.0%	2.5%

Which of the following is closest to the combined stress loss on the two books?

A. CAD 3.40 million

B. CAD 3.90 million

C. CAD 3.99 million

D. CAD 8.29 million

Solutions • CR-21 d

Solution CR-21 d.1

Answer: B

Bank Q1101 • Mock 1 Q37

Step 1 – Loss given default

$\text{LGD} = 1 - 0.60 = 40\%$.

Step 2 – Expected loss and stressed expected loss

$$\text{EL} = 250 \times 0.025 \times 0.40 = 2.5, \qquad \text{EL}^S = 250 \times 0.070 \times 0.40 = 7.0$$

Step 3 – Stress loss

$$\text{EL}^S - \text{EL} = 7.0 - 2.5 = \text{USD } 4.5 \text{ million}$$

$\Rightarrow$ The scenario adds USD 4.5 million of expected loss.

Interpretation

The correlation is a red herring: it matters for unexpected loss and credit VaR, not for an expected-loss stress. This is the loan framework exactly: EAD and LGD fixed, PD stressed.

Why the other options are wrong

A is incorrect. The base expected loss, 2.5; the stress is ignored.
C is incorrect. The stressed expected loss, 7.0, a level rather than the change.
D is incorrect. Added the two ELs: $2.5 + 7.0 = 9.5$.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4

Solution CR-21 d.2

Answer: C

Bank Q1100 • BT answers P2.T6.708.1

Step 1 – What changes for derivatives

EAD is stochastic and depends on market variables, so it is *replaced* by $\alpha \times \text{EPE}_i$ (B appropriate).

Step 2 – What is then stressed

The PD is stressed in both books (A appropriate), and market-variable shocks flow through to a stressed EPE_i^S (D appropriate).
$\Rightarrow$ Once EAD has been replaced by $\alpha \times$EPE, there is no separate EAD left to stress.

Interpretation

GARP's derivative formula is $\text{EL}^S = \sum p_i^S \, \alpha \, \text{epe}_i^S \, \text{lgd}_i$: two stressed inputs, PD and EPE, against one (PD) for loans.

Why the other options are wrong

A is incorrect. Appropriate: stressing the PD is the standard stress for both books.
B is incorrect. Appropriate: this is how CCR exposure enters a portfolio credit model beside loans.
D is incorrect. Appropriate: shocks to market variables are applied to EPE.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4

Solution CR-21 d.3

Answer: A

Bank Q1099 • BT answers P2.T6.707.3

Step 1 – Base EL for DD (given)

$$\text{EL} = 0.0100 \times 300 \times 0.65 = 1.95$$

Step 2 – Stressed EL

$$\text{EL}^S = 0.0160 \times 300 \times 0.65 = 3.12$$

Step 3 – Stress loss

$$3.12 - 1.95 = 1.17 = \text{USD } 1{,}170{,}000$$

$\Rightarrow$ Equivalently $300 \times 0.65 \times (1.60\% - 1.00\%) = 1.17$.

Interpretation

DD dominates this extract because it pairs a large EPE with a sizeable PD increase. The other rows check out at $\alpha = 1$ (AA: $0.0005 \times 213 \times 0.70 = 0.0746$), which is why this version differs slightly from GARP's printed Table 21.2 (see the correction box).

Why the other options are wrong

B is incorrect. Dropped the LGD: $300 \times 0.006 = 1.80$.
C is incorrect. The base EL, already in the table.
D is incorrect. The stressed EL; the base EL was not subtracted.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4, Table 21.2

Solution CR-21 d.4

Answer: A

Bank Q1097 • BT answers P2.T6.25.2.2

Step 1 – Stress loss per name with only PD stressed

Stress loss = EPE $\times$ LGD $\times$ $(p^S - p)$ (USD million):

$$\text{AA: } 190 \times 0.55 \times 0.0045 = 0.470 \qquad \text{BB: } 202.5 \times 0.60 \times 0.0027 = 0.328$$
$$\text{CC: } 75 \times 0.70 \times 0.0017 = 0.089 \qquad \text{DD: } 30 \times 0.65 \times 0.0030 = 0.059$$

$\Rightarrow$ AA contributes most, USD 0.47 million of the 0.946 total.

Interpretation

The tenfold PD jump on a large exposure dominates, the same pattern as GARP's Table 21.2. High-PD names with small exposures contribute little.

Why the other options are wrong

B is incorrect. The largest EPE, but a PD rise of only 0.27 percentage points: 0.328.
C is incorrect. Confuses the PD level (0.45%) with the PD change (0.17 points): 0.089.
D is incorrect. Highest stressed PD, but the smallest exposure: 0.059.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4, Table 21.2

Solution CR-21 d.5

Answer: C

Bank Q1098 • BT answers P2.T6.25.2.3

Step 1 – Base expected loss

$$\text{EL}_A = 4.37\times0.05\times0.03 = 0.006555, \quad \text{EL}_B = 3.99\times0.20\times0.18 = 0.143640; \quad \text{total } 0.150195$$

Step 2 – Stressed expected loss (all three inputs stressed)

$$\text{EL}_A^S = 5.46\times0.10\times0.04 = 0.021840, \quad \text{EL}_B^S = 4.99\times0.40\times0.23 = 0.459080; \quad \text{total } 0.480920$$

Step 3 – Stress loss

$$0.480920 - 0.150195 = 0.330725 \approx \text{USD } 331{,}000$$

Interpretation

The EPEs are those of counterparties A and B in GARP's Table 21.3. Counterparty B, with the higher PD and LGD, produces almost all of the stress loss.

Why the other options are wrong

A is incorrect. Stressed only the PD, keeping the base EPE and LGD: $0.192280 - 0.150195 = 0.042$.

B is incorrect. The base expected loss, 0.150.

D is incorrect. The stressed expected loss, 0.481, before subtracting the base.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4, Table 21.3

Solution CR-21 d.6

Answer: C

New • built from GARP Ch.21, 21.4

Step 1 – Loan book (PD stressed, EAD and LGD fixed)

$$\text{EL} = 400(0.40)(0.015) + 250(0.50)(0.008) = 2.4 + 1.0 = 3.4, \qquad \text{EL}^S = 4.8 + 2.0 = 6.8$$

$\Rightarrow$ Loan stress loss 3.40.

Step 2 – Derivatives book ($\alpha\times$EPE, PD and EPE stressed)

$$\text{EL} = 1.2\,[0.005(90)(0.60) + 0.020(40)(0.60)] = 1.2(0.27 + 0.48) = 0.900$$
$$\text{EL}^S = 1.2\,[0.010(120)(0.60) + 0.025(35)(0.60)] = 1.2(0.72 + 0.525) = 1.494$$

$\Rightarrow$ Derivative stress loss $1.494 - 0.900 = 0.594$.

Step 3 – Combine the books

$$3.400 + 0.594 = 3.994 \approx \text{CAD } 3.99 \text{ million}$$

Interpretation

Both books are on the same expected-loss footing, so their stress losses add. Y's EPE falls under the equity shock, but its PD rise still increases its EL. The current exposures are a red herring: CCR enters the portfolio credit model through $\alpha \times$EPE.

Why the other options are wrong

A is incorrect. Dropped the derivatives book, perhaps misreading "EPE stresses need not be aggregated with the loan book": 3.40. That remark says loans have no EPE stress, not that derivative stress losses are left out.

B is incorrect. Omitted α: derivative stress loss $1.245 - 0.750 = 0.495$, total 3.90.

D is incorrect. Reported the total stressed EL, $6.8 + 1.494 = 8.29$, instead of the change.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. • **Reference:** Lynch (GARP CR Ch.21), 21.4, eqs. for EL and EL^S and footnote 4

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 e

P2 CONCEPT

Describe a stress test that can be performed on CVA.

Workbook: none homed here • revision notes in C9 Stress-Testing Practices and Scenario Design • Tier C • Vol 2 p.74

WHAT IS TESTED HERE

Why P2. Part of a reading Vardeez rates VERY IMPORTANT throughout; the map marks the LO VARDEEZ FOCUS. Revision notes in workbook C9 (Tier C).

Vardeez: four steps. Take a common simplified CVA formula that omits wrong-way risk; aggregate across N counterparties; calculate stressed CVA by applying an instantaneous shock to some of the market variables (the stress can affect EE and PD); compute the stress loss.

Reported exam shapes: none listed in the map for this LO.

What the questions look like: what each term of the CVA formula is (discounted risk-neutral EE, risk-neutral marginal PD, risk-neutral LGD); what the shock does (instantaneous, all later EEs recalculated, PDs from spread shocks); the sign of the stress loss; how the CVA stress differs from the expected-loss stress (risk-neutral against physical, life of the trades against a fixed horizon, market-based PD); and the extra stress a market-based PD allows (the exposure-PD correlation). Conceptual; the arithmetic is CR-21 f.

1. The market-risk view in a stress test

When CCR is treated as a market risk (CR-21 b), the stress is on the **market value** of counterparty risk: the losses that changes in market variables, including the counterparty's credit spread, would cause through CVA. Banks often stress their **unilateral** CVA, in which only the counterparty can default. Adding the bank's own default gives the bilateral CVA and the DVA, which is CR-21 g.

2. A CVA stress in four steps

Step 1: a simplified CVA per counterparty. The common formula that omits wrong-way risk (Gregory 2010) is

$$\text{CVA}_n = \text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j)$$

where $\text{EE}_n^*(t_j)$ is the **discounted expected exposure** during the jth period, calculated under the **risk-neutral** measure; $q_n^*(t_{j-1}, t_j)$ is the risk-neutral **marginal** default probability in that interval; LGD_n^* is the risk-neutral loss given default; and T is the final maturity.

Step 2: aggregate across the N counterparties, $\text{CVA} = \sum_{n=1}^{N} \text{CVA}_n$.

Step 3: shock the market variables. Every component depends on market variables: q^* is derived from the counterparty's credit spreads, LGD^* is set by convention or from market spreads, and EE^* depends on the values of the derivatives with the counterparty. A stressed CVA applies an **instantaneous shock** to some of these variables. Because the shock moves today's values, it changes the expected exposures of *all* later periods, so the whole EE profile must be recalculated, as with stressed current exposure. Stresses to the marginal default probabilities usually come from credit spread shocks. LGD could be stressed too, but CVA is linear in LGD, so the effect is obvious.

Step 4: the stress loss is $\text{CVA}^S - \text{CVA}$, a positive number when the stress raises CVA (calculated in

CR-21 f).

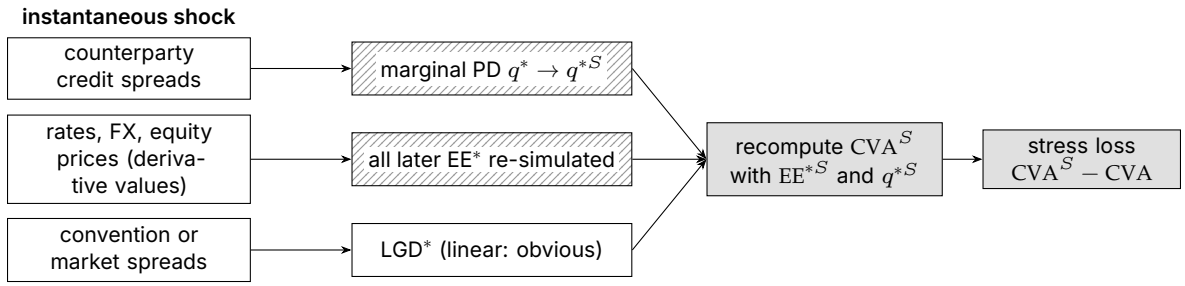


Figure 1. A CVA stress. The instantaneous shock hits market variables (left); the hatched inputs are the ones normally restated, the marginal default probabilities and the whole expected exposure profile; grey boxes are outputs.

3. How it differs from the expected-loss stress

Both stress tests treat expected loss as the product of LGD, exposure and PD, but the values are "quite different":

- **Risk-neutral versus physical** values: the most prominent difference.
- **Life versus horizon**: CVA uses expected losses over the life of the transactions; the expected-loss stress uses a specified time horizon.
- **Market-based PD**: in CVA the default probability comes from market prices.

The market-based PD brings a benefit: a **correlation between exposure and the probability of default** can be built in (Hull and White, 2012), which is how wrong-way risk enters CVA. It also makes possible an important additional stress test, a **stress of the correlation itself**. The correlation can have an important effect on CVA and is highly uncertain, so a bank should run stress tests to see the profit-and-loss impact if the correlation is wrong.

	Expected-loss stress (credit view)	CVA stress (market view)
PD	Physical, from a PD model	Risk-neutral marginal q^*, from credit spreads
Exposure	$\alpha \times$EPE	Discounted risk-neutral EE^* at each date
Horizon	A specified horizon	Life of the transactions
What is shocked	PD drivers; market variables behind EPE	Credit spreads and market variables, instantaneously
Wrong-way risk	Hard: screen via stressed current exposure	Exposure-PD correlation can be modelled and stressed
Combined with	The loan book	The trading book, via BCVA (CR-21 g)
Output	$\mathrm{EL}^S - \mathrm{EL}$	$\mathrm{CVA}^S - \mathrm{CVA}$

Exam traps

- **EE^* is a discounted, risk-neutral mean**, not a peak exposure and not a physical (real-world) expectation.

- q^* **is a marginal, risk-neutral PD** for the period, derived from spreads; not a cumulative PD and not a historical default rate.
- **Instantaneous, and the whole profile.** The shock is applied today, and every later EE must be recalculated, not only the first bucket.
- **Sign.** Stress loss $= \text{CVA}^S - \text{CVA}$: a rise in CVA is a loss. "Current minus stressed" reverses it.
- **DVA is not a wrong-way risk fix.** The simplified formula omits wrong-way risk; capturing it needs the exposure-PD correlation. DVA adds the bank's own default (CR-21 g).
- **LGD.** It can be stressed, but CVA is linear in LGD, so the result is mechanical.

Practice questions: CR-21 e

Question CR-21 e.1

Bank Q1103 • AnalystPrep Q.2007

A bank stress tests its unilateral CVA using the simplified formula, which omits wrong-way risk:

$$\text{CVA}_n = \text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j)$$

In this equation, what does $\text{EE}_n^*(t_j)$ represent?

A. The expected exposure at time j, calculated under a real-world measure for counterparty n.

B. The potential exposure at time j, calculated under a risk-neutral measure for counterparty n.

C. The discounted expected exposure during the jth time period, calculated under a real-world measure for counterparty n.

D. The discounted expected exposure during the jth time period, calculated under a risk-neutral measure for counterparty n.

Question CR-21 e.2

Bank Q1104 • AnalystPrep Q.2309 (rebuilt)

A bank stress tests the CVA on its derivatives portfolio. Which description of the stress test is most accurate?

A. Apply an instantaneous shock to market variables (credit spreads, market spreads, derivative values), recompute CVA, and take the stress loss as the stressed CVA minus the current CVA.

B. Re-estimate each counterparty's physical probability of default over a one-year horizon from historical default data and recompute expected loss.

C. Hold all market variables constant and increase only the bank's own default probability, then measure the change in CVA.

D. Apply a gradual shock to market variables over the life of each trade and take the stress loss as the current CVA minus the stressed CVA.

Question CR-21 e.3 New • built from GARP Ch.21, 21.5

A CVA desk applies an instantaneous 25% fall in equity prices to its equity derivatives with a hedge fund counterparty, together with the widening of the counterparty's CDS spread that accompanies the scenario. How does the scenario enter the stressed CVA?

A. Only the expected exposure in the first time bucket changes, because the shock is instantaneous; later buckets keep their base values.

B. The shock changes today's market values, so the expected exposures in all later periods must be recalculated, and the marginal default probabilities are restated from the shocked credit spread.

C. The expected exposures are left unchanged and only the LGD is stressed, since CVA is linear in LGD.

D. The desk replaces each period's expected exposure with the stressed current exposure, which gives the same answer at lower cost.

Question CR-21 e.4 New • built from GARP Ch.21, 21.5

A risk manager runs the same equity crash scenario through the expected-loss stress (CCR as a credit risk) and through the CVA stress (CCR as a market risk) on the same derivatives book. Which statement about the two exercises is correct?

A. Both use physical default probabilities over a one-year horizon; they differ only in the exposure measure.

B. The expected-loss stress uses default probabilities implied from CDS spreads, while the CVA stress uses historical default rates.

C. The CVA stress uses risk-neutral values, losses over the life of the transactions and a market-based model of default probability, while the expected-loss stress uses physical values over a specified horizon.

D. Because both are products of LGD, exposure and default probability, the two exercises give the same stress loss for the same scenario.

Question CR-21 e.5 New • built from GARP Ch.21, 21.5

A CVA desk computes CVA with market-implied default probabilities and a model in which exposure and the counterparty's default probability are correlated, as in Hull and White (2012). The correlation estimate is highly uncertain. Which additional stress test does the chapter recommend?

A. A stress of the alpha multiplier applied to EPE, since alpha converts exposure into a loan equivalent.

B. A stress of the bank's own default probability, since the DVA term captures wrong-way risk.

C. A stress of LGD alone, since LGD is the only CVA input that does not depend on market variables.

D. A stress of the correlation between exposure and the default probability, to see the profit-and-loss impact if the correlation is wrong.

Question CR-21 e.6 Bank Q1105 • BT answers P2.T6.708.2 • BT notes P2.T6.708.2 (rebuilt)

The simplified (unilateral) CVA for a counterparty is $\text{CVA}_n = \text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j)$. It is applied to a single derivative with a five-year life, with an interval of three months between dates. Each of the following statements is true EXCEPT which is false?

A. A stressed CVA is likely to shock EE to a stressed expected exposure.

B. To capture wrong-way risk, the formula should add a debt value adjustment (DVA) term.

C. A stressed CVA is likely to shock the marginal default probability q to a stressed risk-neutral marginal default probability.

D. The expected exposure and default probability will be estimated at twenty points in time.

Solutions • CR-21 e

Solution CR-21 e.1 — Answer: D

Bank Q1103 • AnalystPrep Q.2007

Step 1 – Which measure

CVA is a market (fair) value, so all its inputs are risk-neutral: "calculated under a risk-neutral measure".

Step 2 – Which exposure

GARP: $\text{EE}_n^*(t_j)$ "is the discounted expected exposure during the jth time period". It is the mean exposure, discounted, not a percentile.

$\Rightarrow$ Discounted, expected, risk-neutral, per period.

Interpretation

Real-world (physical) inputs belong to the expected-loss stress of CR-21 d; that is one of the differences the chapter lists between the two views.

Why the other options are wrong

A is incorrect. Wrong measure (real-world) and missing the discounting.

B is incorrect. Potential (peak) exposure is a tail percentile; CVA uses the mean exposure.

C is incorrect. Discounting is right, but the measure must be risk-neutral.

LO: CR-21 e: Describe a stress test that can be performed on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 e.2 — Answer: A

Bank Q1104 • AnalystPrep Q.2309

Step 1 – What CVA depends on

Credit spreads (through q^*), market spreads or convention (through LGD^*) and derivative values (through EE^*): all market variables.

Step 2 – The stress

"To calculate a stressed CVA, we would apply an instantaneous shock to some of these market variables", recompute CVA, and take $\text{CVA}^S - \text{CVA}$.

$\Rightarrow$ A rise in CVA is a stress loss.

Interpretation

This is Vardeez's four-step recipe in one sentence: simplified CVA, aggregate, instantaneous shock, stress loss.

Why the other options are wrong

B is incorrect. Describes the credit-view expected-loss stress (physical PD, fixed horizon), not a CVA stress.

C is incorrect. The bank's own default enters only through DVA in the bilateral version; a unilateral CVA stress shocks the counterparty side and exposures.

D is incorrect. The shock is instantaneous, and the sign is reversed: the stress loss is stressed minus current CVA.

LO: CR-21 e: Describe a stress test that can be performed on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 e.3 — Answer: B

New • built from GARP Ch.21, 21.5

Step 1 – Exposures

"An instantaneous shock will have some impact on the expected exposure calculated in later time periods, so all of the expected exposures will have to be recalculated."
⇒ The whole EE profile moves, not only the first bucket.

Step 2 – Default probabilities

"Stresses to the marginal probability of default are usually derived from credit spread shocks." The wider CDS spread gives new q^* values.

Interpretation

An equity crash against a hedge fund that is long equity is a natural wrong-way scenario: exposure and default probability rise together, so both inputs must be restated.

Why the other options are wrong

A is incorrect. The shock is applied today but affects every later expected exposure.
C is incorrect. LGD can be stressed, but the scenario's main effects are on exposure and default probability.
D is incorrect. Current exposure is not a substitute for expected exposure; that is one of the CR-21 h pitfalls.

LO: CR-21 e: Describe a stress test that can be performed on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 e.4 — Answer: C

New • built from GARP Ch.21, 21.5

Step 1 – What the two share

"There is a reliance in both cases on expected losses being the product of LGD, exposure and the probability of default."

Step 2 – Where they differ

"The use of risk-neutral values for CVA as opposed to physical values for expected losses" is the most prominent difference; "CVA uses expected losses over the life of the transactions, whereas expected losses use a specified time horizon"; and "the model for determining the probability of default is market-based in CVA".
⇒ C lists all three differences.

Interpretation

The two exercises can produce very different numbers for the same scenario, which is why CCR stress results multiply (CR-21 b) and why combining them is hard (CR-21 h).

Why the other options are wrong

A is incorrect. The CVA stress uses risk-neutral PDs over the life of the trades.
B is incorrect. Swaps the two views: spread-implied PDs belong to CVA.
D is incorrect. Same structure, different inputs: GARP says the values "will be quite different".

LO: CR-21 e: Describe a stress test that can be performed on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 e.5 — Answer: D

New • built from GARP Ch.21, 21.5

Step 1 – What a market-based PD allows

"It is possible in these circumstances to incorporate a correlation between the probability of de-

fault and the exposure." Hull and White (2012) "also demonstrate an important stress test that is available, a stress of the correlation between exposure and the probability of default".

Step 2 – Why run it

"Since there is likely to be a high degree of uncertainty around the correlation, a financial institution should run stress tests to determine the impact on profit and loss if the correlation is wrong."

Interpretation

This is the market view's answer to wrong-way risk, which the credit-view stresses can only screen for through stressed current exposure.

Why the other options are wrong

A is incorrect. Alpha belongs to the credit-view expected-loss framework, not to CVA.
B is incorrect. DVA reflects the bank's own default; it does not capture wrong-way risk.
C is incorrect. LGD is set by convention or from market spreads, and CVA is linear in it, so its stress is mechanical; it is not the stress the chapter highlights.

LO: CR-21 e: Describe a stress test that can be performed on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 e.6 Answer: B

Bank Q1105 • BT answers P2.T6.708.2

Step 1 – Count the dates

Five years at three-month intervals: $5/0.25 = 20$ points, so D is true.

Step 2 – What a stress shocks

Market shocks move the discounted EE and, through credit spreads, the risk-neutral marginal PD: A and C are true.

Step 3 – Wrong-way risk

The simplified formula omits wrong-way risk, the dependence between exposure and default. Capturing it needs a correlation between exposure and PD; a DVA term instead turns unilateral CVA into bilateral CVA.
$\Rightarrow$ B is false.

Interpretation

Two separate extensions of the simplified formula: wrong-way risk (exposure-PD correlation) and bilateral default (DVA). Mixing them up is the trap.

Why the other options are wrong

A is incorrect. True: the stress applies shocks to the exposure input.
C is incorrect. True: spread shocks restate the risk-neutral marginal PDs.
D is incorrect. True: $5 \times 4 = 20$ quarterly dates.

LO: CR-21 e: Describe a stress test that can be performed on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 f

P2 CALC

Calculate the stressed CVA and the stress loss on CVA.

Workbook: none homed here • revision notes in C9 Stress-Testing Practices and Scenario Design • Tier C • Vol 2 p.74

WHAT IS TESTED HERE

Why P2. One of the reading's three calculation LOs, in a chapter Vardeez rates VERY IMPORTANT throughout; the map marks it VARDEEZ FOCUS. Revision notes in workbook C9 (Tier C).

Vardeez: stressed CVA aggregated across counterparties, $\text{CVA}^S = \sum_n \text{LGD}_n^* \sum_j (\text{stressed exposure}) \times (\text{stressed marginal PD})$, and stress loss on CVA $= \text{CVA}^S - \text{CVA}$. Both exposure and PD carry the stress superscript, as in the derivative case of CR-21 d.

Reported exam shapes: none listed in the map for this LO.

What the questions look like: single-period (LGD $\times$ EE $\times$ q) or multi-period CVA tables; a spread shock (sometimes given as a spread and a recovery rate, so you build the marginal PDs), an exposure shock, or both; netted against gross exposures. You are asked for the stressed CVA or the stress loss, or the percentage change. Distractors: the level instead of the change, the base CVA, cumulative instead of marginal PDs, a spread not divided by $1 - R$, the sum of separate stresses for a joint shock. The bank's Q1086 is re-filed here from CR-21 b.

1. The calculation

From CR-21 e, with risk-neutral inputs and no wrong-way risk:

$$\text{CVA} = \sum_{n=1}^{N} \text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j), \qquad \text{CVA}^S = \sum_{n=1}^{N} \text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^{*S}(t_j)\, q_n^{*S}(t_{j-1}, t_j)$$

$$\boxed{\text{stress loss on CVA} = \text{CVA}^S - \text{CVA}}$$

When the stem gives a CDS spread s and a recovery rate R instead of PDs, build the marginal PDs from a constant hazard rate (CR-9 h and e):

$$\lambda \approx \frac{s}{1 - R}, \qquad q^*(t_{j-1}, t_j) = \mathrm{e}^{-\lambda t_{j-1}} - \mathrm{e}^{-\lambda t_j}.$$

A spread shock changes λ and therefore every q^*; a market shock changes the whole EE^* profile.

2. Worked example: a joint spread and exposure shock

One counterparty, three annual periods, $\text{LGD}^* = 60\%$. The scenario widens the counterparty's CDS spread from 120 bp to 300 bp ($R = 40\%$, so λ goes from 2% to 5%) and a rate shock raises the discounted expected exposures (USD million).

Year	EE^*	EE^{*S}	q^*	q^{*S}	Base term	Stressed term
1	8	11	1.980%	4.877%	0.0950	0.3219
2	10	13	1.941%	4.639%	0.1165	0.3619
3	6	8	1.902%	4.413%	0.0685	0.2118
CVA					0.2800	0.8956

Year 2: $q^* = \mathrm{e}^{-0.02} - \mathrm{e}^{-0.04} = 1.941\%$ and the base term is $0.60 \times 10 \times 0.01941 = 0.1165$; stressed, $q^{*S} = \mathrm{e}^{-0.05} - \mathrm{e}^{-0.10} = 4.639\%$ and $0.60 \times 13 \times 0.04639 = 0.3619$. The **stressed CVA** is 0.8956 and the **stress loss** is $0.8956 - 0.2800 = 0.6156$, a 220% increase.

Separate or joint? The spread shock alone gives a CVA of 0.6713 (stress loss 0.3913); the exposure shock alone gives 0.3734 (stress loss 0.0934). Their sum, 0.4847, falls short of the joint stress loss of 0.6156 by about 0.131, because CVA multiplies exposure by default probability. When a scenario moves both, run them together.

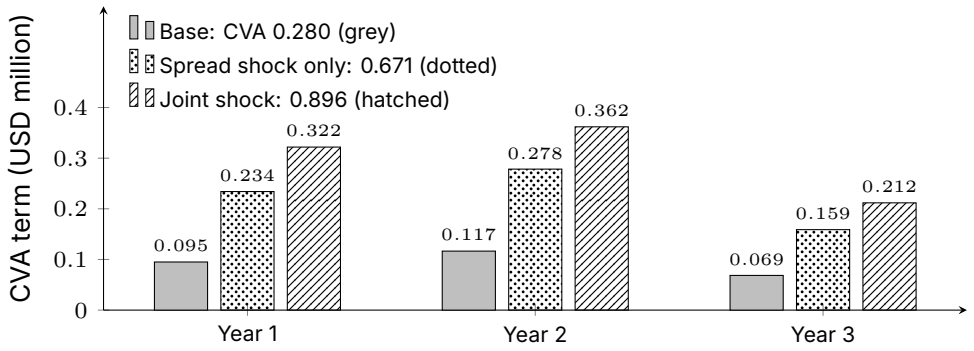


Figure 1. Base versus stressed CVA, term by term, for the worked example. The stress loss is the gap between the totals: 0.391 for the spread shock alone, 0.616 for the joint shock.

3. Portfolios, netting and sign conventions

Across counterparties, stressed CVAs and stress losses simply add. Netting lowers EE and so lowers CVA in the same proportion, since CVA is linear in EE; a netted stressed CVA is always below the gross one (Question CR-21 f.4). Under a pure spread shock with exposures unchanged, CVA is proportional to q^*: doubling q^* doubles CVA, a 100% increase.

Two sign conventions coexist in the GARP book. They describe the same economics:

	Lynch, this reading (Ch.21)	Gregory, CR-20 (Ch.20), and Vardeez
CVA	Positive charge: $\mathrm{LGD} \sum \mathrm{EE}\, q$	Negative adjustment: $-\mathrm{LGD} \sum \mathrm{EPE}(t)\, \mathrm{PD}$
DVA and BCVA	DVA positive, subtracted: $\mathrm{BCVA} = \mathrm{CVA} - \mathrm{DVA}$	DVA positive (ENE negative), added: $\mathrm{BCVA} = \mathrm{CVA} + \mathrm{DVA}$
Spreads widen	CVA rises	CVA becomes more negative
$\mathrm{CVA}^S - \mathrm{CVA}$ **for a loss**	Positive: a stress loss	Negative: a P&L loss

Correction to the source notes. Vardeez writes $\mathrm{CVA}^S = -\sum_n \mathrm{LGD}^*_n \sum_j \mathrm{EPE}^S_n(t_j) \times \mathrm{PD}^S_n(t_{j-1}, t_j)$ and then calls $\mathrm{CVA}^S - \mathrm{CVA}$ the stress loss. With the leading minus sign, a stress that raises counterparty risk makes that difference *negative*, so a "stress loss" would come out as a negative number. GARP's Lynch formula has no minus sign: CVA is a positive charge and the stress loss $\mathrm{CVA}^S - \mathrm{CVA}$ is positive (table above). The workbook and the bank questions follow Lynch, and so does this book. Two notation points: the exposure in each bucket is the discounted risk-neutral expected exposure $\mathrm{EE}^*_n(t_j)$ (Vardeez carries Gregory's "$\mathrm{EPE}(t_j)$", which in this chapter's BCBS terms is EE, since EPE here is a time average); and the default probability is the risk-neutral marginal q^*_n implied from spreads.

Formula card: CR-21 f

CVA per counterparty	$\text{CVA}_n = \text{LGD}_n^* \sum_j \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j)$; single period: LGD × EE × q
Stressed CVA	same formula with EE^{*S} and q^{*S}, summed over counterparties
Stress loss	$\text{CVA}^S - \text{CVA}$; percentage increase $(\text{CVA}^S - \text{CVA})/\text{CVA}$
Marginal PD from a spread	$\lambda \approx s/(1-R)$, $q^*(t_{j-1}, t_j) = \text{e}^{-\lambda t_{j-1}} - \text{e}^{-\lambda t_j}$
Linearity	CVA is linear in LGD, in EE and in q^* separately, but a joint shock multiplies: joint $\neq$ sum of separate stresses

Exam traps

- **Stressed CVA versus stress loss.** Read the question: the level CVA^S or the change $\text{CVA}^S - \text{CVA}$.
- **Marginal, not cumulative, PDs.** Each bucket uses the default probability for that period only.
- **Hazard from a spread.** Divide by $1 - R$; using s itself as the hazard understates the PDs.
- **Percentage change** is measured against the initial CVA, not the stressed one.
- **Joint shocks.** The stress loss of a joint spread and exposure shock exceeds the sum of the two separate stress losses.
- **Exposures can fall.** A market shock lowers EE for counterparties on the other side of the move.
- **Netting** lowers EE and CVA; a netted figure above the gross one is a reversal.
- **Sign convention.** Under Gregory's negative CVA, a loss shows as a negative $\text{CVA}^S - \text{CVA}$.

Practice questions: CR-21 f

Question CR-21 f.1

Bank Q1107 • AnalystPrep Q.3045 (rebuilt)

A risk analyst is calculating the stress loss on a portfolio's credit value adjustment (CVA). Under normal market conditions the portfolio CVA is USD 4.3 million. After an instantaneous 30% fall in markets, the stressed CVA is USD 76.0 million. What is the stress loss on the portfolio CVA?

A. USD 4.3 million

B. USD 71.7 million

C. USD 76.0 million

D. USD 80.3 million

Question CR-21 f.2

Bank Q1108 • BT answers P2.T6.25.3.1

A bank runs a CVA stress test. Counterparty 1 has an expected exposure (with netting) of USD 5 million, an LGD of 60% and a marginal default probability q^* of 1.5%. After a market shock that widens credit spreads, exposure is unchanged at USD 5 million but q^* rises to 3%. What is the stress loss on the CVA, and by what percentage does the stressed CVA exceed the initial CVA?

A. USD 45,000; 50%

B. USD 45,000; 100%

C. USD 90,000; 50%

D. USD 90,000; 100%

Question CR-21 f.3

Bank Q1086 • BT answers P2.T6.25.1.3 (rebuilt) • re-filed from CR-21 b

A bank has a 5-year interest rate swap with a counterparty. Assume a credit exposure of USD 60 million, a loss given default of 50%, and a one-period default probability of 1.50%. Treating counterparty credit risk as a credit risk, the bank measures its expected loss. Treating it as a market risk, the bank values the CVA after credit spreads widen so that the default probability rises to 1.70%, using the same exposure and LGD. Which option gives the expected loss (credit view) and the stressed CVA (market view)?

A. Expected loss USD 450,000; stressed CVA USD 60,000

B. Expected loss USD 450,000; stressed CVA USD 510,000

C. Expected loss USD 510,000; stressed CVA USD 450,000

D. Expected loss USD 900,000; stressed CVA USD 1,020,000

Question CR-21 f.4

Bank Q1109 • BT answers P2.T6.25.3.2 (rebuilt)

A bank stresses the CVA on its OTC derivative receivables, with and without netting agreements. LGD is 60% for all counterparties, and CVA for each counterparty = LGD × EE × q^*. The shock widens credit spreads (exposures unchanged) and raises q^* as shown.

Counterparty	EE, no netting (US$m)	EE, with netting (US$m)	Initial q^*	Stressed q^*
1	20	5	1.5%	3.0%
2	10	4	2.0%	4.0%
3	8	2	1.0%	2.5%

Which option gives the total stressed CVA across the three counterparties?

A. With netting USD 0.105m; without netting USD 0.348m

B. With netting USD 0.111m; without netting USD 0.372m

C. With netting USD 0.216m; without netting USD 0.720m

D. With netting USD 0.720m; without netting USD 0.216m

Question CR-21 f.5

New • built from GARP Ch.21, 21.5 and Ch.9 eq. 9.2, 9.3

A CVA desk at a Singapore bank stresses the CVA on a two-year swap with a single counterparty under a credit-spread shock that doubles the counterparty's flat CDS spread from 180 bp to 360 bp. The discounted risk-neutral expected exposures at the end of years 1 and 2 are USD 12 million and USD 15 million and are not affected by the shock. The recovery rate is 40%. Using a constant hazard rate $\lambda = s/(1-R)$ at each spread level and the marginal default probability in each year, which of the following is closest to the stressed CVA?

A. USD 0.44 million

B. USD 0.56 million

C. USD 0.91 million

D. USD 1.44 million

Question CR-21 f.6

New • built from GARP Ch.21, 21.5

A bank stresses the CVA on two counterparties under a scenario that widens credit spreads and moves market rates. The rate move raises the exposure to P and lowers the exposure to Q. Discounted risk-neutral expected exposures are in USD million; q^* is the marginal default probability for each year.

Counterparty (LGD*)	Year	EE*	Stressed EE*	q^*	Stressed q^*
P (60%)	1	20	26	1.0%	2.0%
	2	25	30	1.1%	2.2%
Q (75%)	1	10	9	2.5%	3.5%
	2	8	6	2.4%	3.4%

Which of the following is closest to the stress loss on the portfolio CVA?

A. USD 0.420 million

B. USD 0.434 million

C. USD 0.481 million

D. USD 0.558 million

Solutions • CR-21 f

Solution CR-21 f.1

Answer: B

Bank Q1107 • AnalystPrep Q.3045

Step 1 – Stress loss on CVA

$$\text{CVA}^S - \text{CVA} = 76.0 - 4.3 = \text{USD } 71.7 \text{ million}$$

$\Rightarrow$ The shock adds USD 71.7 million to the price of counterparty risk.

Interpretation

The rise from 4.3 to 76.0 shows how the combined effect of an exposure shock and a spread shock can multiply CVA, the crisis pattern that drew attention to CVA risk (CR-21 b).

Why the other options are wrong

A is incorrect. The unstressed CVA; the shock is ignored.
C is incorrect. The stressed CVA, a level rather than the change.
D is incorrect. Added the two CVAs instead of subtracting.

LO: CR-21 f: Calculate the stressed CVA and the stress loss on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 f.2

Answer: B

Bank Q1108 • BT answers P2.T6.25.3.1

Step 1 – Initial and stressed CVA

$$\text{CVA} = 0.60 \times 5{,}000{,}000 \times 0.015 = 45{,}000, \qquad \text{CVA}^S = 0.60 \times 5{,}000{,}000 \times 0.030 = 90{,}000$$

Step 2 – Stress loss and percentage increase

$$90{,}000 - 45{,}000 = 45{,}000, \qquad \frac{45{,}000}{45{,}000} = 100\%$$

$\Rightarrow$ With exposure and LGD fixed, CVA is proportional to q^*: doubling q^* doubles CVA.

Interpretation

The percentage is measured against the initial CVA. The loss equals the initial CVA exactly because the PD doubled.

Why the other options are wrong

A is incorrect. Right loss, but divided by the stressed CVA: $45{,}000/90{,}000 = 50\%$.
C is incorrect. Reported the stressed CVA as the loss and divided the change by the stressed level.
D is incorrect. Reported the stressed CVA (90,000) as the stress loss.

LO: CR-21 f: Calculate the stressed CVA and the stress loss on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 f.3

Answer: B

Bank Q1086 • BT answers P2.T6.25.1.3

Step 1 – Credit view: expected loss with the current PD

$$\text{EL} = 60{,}000{,}000 \times 0.50 \times 0.015 = 450{,}000$$

Step 2 – Market view: CVA repriced with the widened spread

$$\text{CVA}^S = 60{,}000{,}000 \times 0.50 \times 0.017 = 510{,}000$$

$\Rightarrow$ The CVA stress loss is $510{,}000 - 450{,}000 = 60{,}000$, a move the credit view does not see.

Interpretation

This toy example uses the same exposure, LGD and PD in both views to isolate one idea: CVA moves with credit spreads, an expected loss at a fixed PD does not. In practice the inputs would differ (risk-neutral against physical, life of the trade against a horizon; CR-21 e).

Why the other options are wrong

A is incorrect. Right EL, but reports the CVA change (60,000) instead of the stressed CVA level.
C is incorrect. Swaps the two measures.
D is incorrect. Omitted the 50% LGD: 900,000 and 1,020,000.

LO: CR-21 f: Calculate the stressed CVA and the stress loss on CVA. CR-21 b: Explain the treatment of counterparty credit risk (CCR) both as a credit risk and as a market risk and describe its implications for trading activities and risk management for a financial institution. • **Reference:** Lynch (GARP CR Ch.21), 21.1 and 21.5

Solution CR-21 f.4 **Answer: C**

Bank Q1109 • BT answers P2.T6.25.3.2

Step 1 – Stressed CVA without netting

$$0.60 \times (20 \times 0.030 + 10 \times 0.040 + 8 \times 0.025) = 0.60 \times 1.20 = 0.720$$

Step 2 – Stressed CVA with netting

$$0.60 \times (5 \times 0.030 + 4 \times 0.040 + 2 \times 0.025) = 0.60 \times 0.36 = 0.216$$

$\Rightarrow$ Netting cuts the stressed CVA by 70%, in proportion to the lower exposures.

Interpretation

Initial CVAs are 0.105 (netted) and 0.348 (gross), so the stress losses are 0.111 and 0.372. Netting reduces the stress loss as well as the level.

Why the other options are wrong

A is incorrect. Used the initial q^*: the unstressed CVAs, 0.105 and 0.348.
B is incorrect. Computed the stress losses (0.111 and 0.372), not the stressed CVAs.
D is incorrect. Reversed the netting effect; netting cannot raise exposure.

LO: CR-21 f: Calculate the stressed CVA and the stress loss on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 f.5 **Answer: C**

New • built from GARP Ch.21, 21.5

Step 1 – Hazard rates

$$\lambda = \frac{0.0180}{0.60} = 3\%, \qquad \lambda^S = \frac{0.0360}{0.60} = 6\%$$

Step 2 – Stressed marginal default probabilities

$$q_1^{*S} = 1 - \mathrm{e}^{-0.06} = 0.058235, \qquad q_2^{*S} = \mathrm{e}^{-0.06} - \mathrm{e}^{-0.12} = 0.054844$$

Step 3 – Stressed CVA

$$\text{CVA}^S = 0.60 \times (12 \times 0.058235 + 15 \times 0.054844) = 0.60 \times 1.52148 = 0.9129$$

$\Rightarrow$ About USD 0.91 million.

Step 4 – For reference, the base CVA and the stress loss

With $q_1^* = 0.029554$ and $q_2^* = 0.028681$: CVA $= 0.60 \times (0.35465 + 0.43021) = 0.4709$, so the stress loss is $0.9129 - 0.4709 = 0.4420$.

Interpretation

Doubling the spread slightly less than doubles CVA (0.471 to 0.913), because the marginal PDs of later years shrink as survival falls.

Why the other options are wrong

A is incorrect. Reported the stress loss, 0.44, when the stressed CVA was asked.

B is incorrect. Used the spread itself as the hazard rate (3.6%) without dividing by $1 - R$: $0.60 \times (12 \times 0.035360 + 15 \times 0.034109) = 0.56$.

D is incorrect. Used the cumulative PD for year 2, $1 - \text{e}^{-0.12} = 0.113080$: $0.60 \times (12 \times 0.058235 + 15 \times 0.113080) = 1.44$.

LO: CR-21 f: Calculate the stressed CVA and the stress loss on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5; Hull (GARP CR Ch.9), eqs. 9.2 and 9.3

Solution CR-21 f.6 **Answer: C**

New • built from GARP Ch.21, 21.5

Step 1 – Base CVA

$$\begin{aligned} \text{P}&: 0.60(20 \times 0.010 + 25 \times 0.011) = 0.2850 \\ \text{Q}&: 0.75(10 \times 0.025 + 8 \times 0.024) = 0.3315 \qquad \text{total } 0.6165 \end{aligned}$$

Step 2 – Stressed CVA (both inputs stressed)

$$\begin{aligned} \text{P}&: 0.60(26 \times 0.020 + 30 \times 0.022) = 0.7080 \\ \text{Q}&: 0.75(9 \times 0.035 + 6 \times 0.034) = 0.38925 \qquad \text{total } 1.09725 \end{aligned}$$

Step 3 – Stress loss

$$1.09725 - 0.6165 = 0.48075 \approx \text{USD } 0.481 \text{ million}$$

Interpretation

P's exposure and spread both rise, so its terms multiply up; Q's spread rise is partly offset by its lower exposure. Stressing spreads and exposures one at a time and adding the results misses the cross term.

Why the other options are wrong

A is incorrect. Stressed only the spreads, keeping base exposures: $1.0365 - 0.6165 = 0.420$.

B is incorrect. Added the spread-only (0.420) and exposure-only (0.014) stress losses: 0.434, missing the interaction.

D is incorrect. Kept Q's exposure at its base level, as if exposures could only rise: $0.708 + 0.4665 - 0.6165 = 0.558$.

LO: CR-21 f: Calculate the stressed CVA and the stress loss on CVA. • **Reference:** Lynch (GARP CR Ch.21), 21.5

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 g

P2 CALC

Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR.

Workbook: none homed here • revision notes in C9 Stress-Testing Practices and Scenario Design • Tier C • C9.1 tags this LO (worked as Question CR-21 h.6)

WHAT IS TESTED HERE

Why P2. One of the reading's three calculation LOs, in a chapter Vardeez rates VERY IMPORTANT throughout; the map marks it VARDEEZ FOCUS. Workbook C9.1 (worked as Question CR-21 h.6) tags it: leaving DVA out of a CVA stress is one of its wrong options.
Vardeez: the DVA calculation is the one set out at CR-20 e and g; the stress loss on BCVA is computed like the CVA stress loss, $\text{BCVA}^S - \text{BCVA}$; because CVA and DVA carry opposite signs, a stress that widens both the counterparty's and the bank's own spreads produces offsetting effects, so DVA cannot simply be ignored in the aggregation.

Reported exam shapes: none listed in the map for this LO.

What the questions look like: one-period or two-period tables with EE, NEE, both LGDs, both marginal PDs and both survival probabilities; you calculate the DVA, the BCVA, or the stress result of a widening in the bank's own spread (a counterintuitive gain). Conceptual items test what BCVA adds (NEE and survival probabilities), losses from an improvement in own credit, and adding BCVA stress results to market-risk stress results under common shocks.

1. From unilateral CVA to DVA

The CVA stress of CR-21 e and f assumes only the counterparty can default. "To capture the full impact of various scenarios on CVA profit and loss, a financial institution should include the liability side effects in the stress as well." This part of the bilateral CVA, the **debit valuation adjustment (DVA)**, is the value of the bank's option to default on its counterparties. Its formula differs from CVA in two ways:

(1) it uses the **negative expected exposure** NEE, the expected exposure calculated from the *counterparty's* point of view, priced with the bank's own LGD and marginal PD;

(2) the option to default has value only if the counterparty has survived, so the **counterparty's survival probability** $S_n^*(t_{j-1})$ enters. The CVA leg changes in the same way: a loss from the counterparty's default now requires that the bank has not defaulted first, so it is weighted by the bank's survival $S_I^*(t_{j-1})$.

$$\text{BCVA} = \sum_{n=1}^{N} \Bigg[\underbrace{\text{LGD}_n^* \sum_{j=1}^{T} \text{EE}_n^*(t_j)\, q_n^*(t_{j-1}, t_j)\, S_I^*(t_{j-1})}_{\text{CVA leg}} \underbrace{- \text{LGD}_I^* \sum_{j=1}^{T} \text{NEE}_n^*(t_j)\, q_I^*(t_{j-1}, t_j)\, S_n^*(t_{j-1})}_{\text{DVA}} \Bigg]$$

The subscript I is the bank. Without the survival weights the DVA is the unilateral DVA, $\text{UDVA} = \text{LGD}_I^* \sum_j \text{NEE}^*(t_j)\, q_I^*(t_{j-1}, t_j)$, the mirror image of the unilateral CVA. The survival probabilities

come from CDS spreads, and the DVA depends on the bank's *own* credit spread. In Gregory's signs (CR-20) the same thing is written BCVA $=$ CVA $+$ DVA with CVA negative; see the table in CR-21 f.

2. Worked example: calculating the DVA

One counterparty C, two annual periods, $\text{LGD}_C^* = \text{LGD}_I^* = 60\%$. Survival probabilities are at the start of each year (USD million).

Year	EE*	NEE*	q_C^*	q_I^*	$S_C^*(t_{j-1})$	$S_I^*(t_{j-1})$
1	40	20	3.0%	1.0%	1.000	1.000
2	50	25	2.9%	1.0%	0.970	0.990

$$\begin{aligned}\text{CVA leg} &= 0.60 \times (40 \times 0.030 \times 1.000 + 50 \times 0.029 \times 0.990) = 1.5813\\ \text{DVA} &= 0.60 \times (20 \times 0.010 \times 1.000 + 25 \times 0.010 \times 0.970) = 0.2655\\ \text{BCVA} &= 1.5813 - 0.2655 = 1.3158\end{aligned}$$

Unilaterally the CVA would be 1.59 and the DVA 0.27; the survival weights trim both.

3. Stressing BCVA

Stress A: a systemic widening. Both spreads widen together and exposures are unchanged: q_C^* becomes 6.0% and 5.6% with $S_C^*(1) = 0.940$; q_I^* becomes 3.0% and 2.9% with $S_I^*(1) = 0.970$.

$$\begin{aligned}\text{CVA leg}^S &= 0.60 \times (40 \times 0.060 + 50 \times 0.056 \times 0.970) = 3.0696\\ \text{DVA}^S &= 0.60 \times (20 \times 0.030 + 25 \times 0.029 \times 0.940) = 0.7689\\ \text{BCVA}^S - \text{BCVA} &= 2.3007 - 1.3158 = 0.9849\end{aligned}$$

A unilateral CVA stress would report $3.12 - 1.59 = 1.53$. The rise in DVA (0.50) and the lower own survival absorb about a third of it.

Stress B: the bank's own credit improves. Only the bank's spread tightens: q_I^* falls to 0.5% in both years and $S_I^*(1)$ rises to 0.995. The CVA leg edges up to 1.5857 and the DVA falls to 0.1328, so $\text{BCVA}^S = 1.4529$ and the stress loss is $+0.1371$: **a loss caused by an improvement in the bank's own credit quality**. A unilateral CVA stress shows nothing at all.

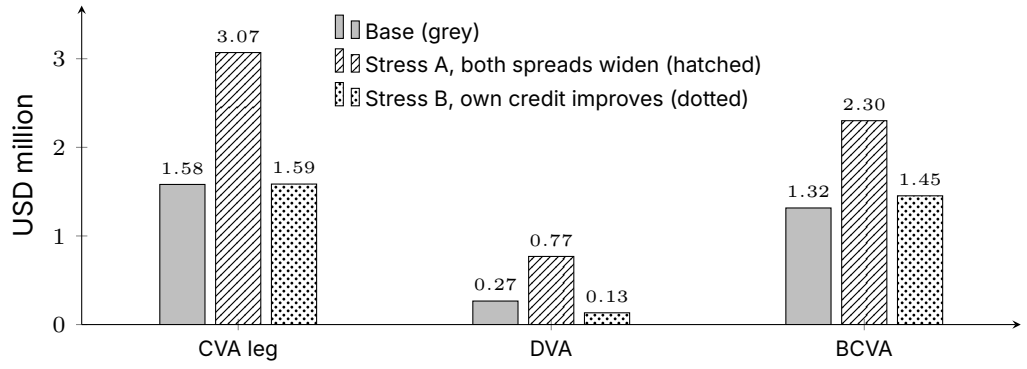


Figure 1. The worked example. In Stress A the DVA rises with the bank's own spread and offsets part of the CVA increase (stress loss 0.98 against 1.53 unilaterally). In Stress B the DVA shrinks and BCVA rises by 0.14 although no counterparty became riskier.

4. How stressed DVA enters the aggregation

- **The stress loss** is calculated as for CVA: a stressed BCVA minus the current BCVA. A negative value is a gain.
- **Counterintuitive results.** Because losses now depend on the bank's own credit spread, a bank can show losses when its own credit quality improves (Stress B) and gains when it deteriorates.
- **Correlation of spreads.** From a bilateral perspective the bank must consider how its own credit spread is correlated with its counterparties' spreads (Stress A widens them together).
- **Aggregation with market risk.** BCVA lets CCR be treated as a market risk, so it can be incorporated into market-risk stress testing coherently. The gains or losses from the BCVA stress can be added to the firm's market-risk stress results by simple addition, *as long as the same shocks to market variables are applied to the trading portfolio and to the BCVA*.

Correction to the source notes. (1) Vardeez concludes that dropping DVA "overstates the stress loss". That holds only for scenarios in which the bank's own spread widens (Stress A: 1.53 unilateral against 0.98 bilateral). When the bank's own credit improves, DVA falls and omitting it *understates* the loss, or misses it entirely (Stress B: 0 against 0.14). GARP's point is that the liability side must be included either way, because "losses occurring because the firm's own credit quality improves" are possible. (2) SchweserNotes' Key Concepts for LO 39.g call the bilateral CVA "also known as the debt value adjustment (DVA)". DVA is only the liability-side part of BCVA ("this part of the bilateral CVA, often called debit valuation adjustment"); BCVA is the CVA leg less the DVA.

Formula card: CR-21 g

Unilateral DVA	$\text{LGD}_I^* \sum_j \text{NEE}^*(t_j)\, q_I^*(t_{j-1}, t_j)$
DVA inside BCVA	$\text{LGD}_I^* \sum_j \text{NEE}^*(t_j)\, q_I^*(t_{j-1}, t_j)\, S_n^*(t_{j-1})$ (counterparty's survival)
CVA leg inside BCVA	$\text{LGD}_n^* \sum_j \text{EE}^*(t_j)\, q_n^*(t_{j-1}, t_j)\, S_I^*(t_{j-1})$ (bank's own survival)
BCVA	CVA leg $-$ DVA (Lynch); $=$ CVA $+$ DVA with Gregory's signs
Stress loss on BCVA	$\text{BCVA}^S - \text{BCVA}$; negative $=$ gain
Aggregation	total stress $=$ trading-book stress $+(\text{BCVA}^S - \text{BCVA})$, same market shocks

Exam traps

- **Pairings.** EE goes with the counterparty's PD and LGD; NEE goes with the bank's own PD and LGD.
- **Survival weights swapped.** The CVA leg carries the *bank's* survival; the DVA carries the *counterparty's*.
- **EPE in BCVA.** The new exposure term is NEE (negative expected exposure), not EPE.
- **Own spread widens ⇒ DVA up ⇒ BCVA down:** a stress gain, counterintuitive but correct. Own credit improves: a loss.
- **Ignoring DVA** misstates the stress in either direction; it does not always overstate it.
- **Sign of the result.** Stressed minus current BCVA; a fall in BCVA is a gain. "Current minus stressed" reverses it.
- **Common shocks.** BCVA stress results add to trading stress results only if the same market

shocks are used.

- **DVA is not wrong-way risk**, and BCVA is not "also called DVA".

Practice questions: CR-21 g

Question CR-21 g.1

Bank Q1110 • AnalystPrep Q.2005

When stress testing counterparty credit risk, a financial institution also considers scenarios in which the institution itself, as well as its counterparty, can default on its obligations. This approach effectively constitutes:

A. Unilateral CVA.

B. Bilateral CVA.

C. Negative exposure from the institution's point of view.

D. General wrong-way risk.

Question CR-21 g.2

Bank Q1116 • Schweser Module Quiz 39.2 Q3

Is the following statement on bilateral credit value adjustment (BCVA) correct?
"The formula for BCVA is similar to the formula for CVA, except that the BCVA formula uses expected positive exposure (EPE) and it incorporates the probability of the counterparty's survival."

A. The statement is correct with regard to both EPE and probability of survival.

B. The statement is correct with regard to EPE only.

C. The statement is correct with regard to probability of survival only.

D. The statement is incorrect with regard to both EPE and probability of survival.

Question CR-21 g.3

Bank Q1114 • BT answers P2.T6.708.3

The bilateral credit value adjustment (BCVA) formula is:

$$\text{BCVA} = \sum_{n=1}^{N} \text{LGD}_n \sum_{j=1}^{T} \text{EE}_n(t_j)\, q_n(t_{j-1}, t_j)\, S_I(t_{j-1}) - \sum_{n=1}^{N} \text{LGD}_I \sum_{j=1}^{T} \text{NEE}_n(t_j)\, q_I(t_{j-1}, t_j)\, S_n(t_{j-1})$$

Acme Bank applies this formula to measure its counterparty credit exposure and will stress test the BCVA. Each of the following statements is true EXCEPT which one?

A. If Acme's own credit quality improves, Acme can experience a loss on the BCVA.

B. For stress testing the BCVA, expected exposures should be replaced by observable current exposures.

C. Compared with unilateral CVA, BCVA adds the survival probabilities of Acme, S_I, and of its counterparty, S_n.

D. BCVA allows CCR to be treated as a market risk, so CCR can be incorporated into market risk stress testing coherently and additively.

Question CR-21 g.4

Bank Q1111 • BT answers P2.T6.25.3.3 (rebuilt, key corrected in the bank)

A bank (own LGD 60%, own marginal default probability 2%, own survival probability $S_I = 0.97$) stress tests three counterparties over one period, using unilateral CVA $= \text{LGD}_n \times \text{EE} \times q_n$ and bilateral CVA $= \text{LGD}_n \times \text{EE} \times q_n \times S_I - \text{LGD}_I \times \text{NEE} \times q_I \times S_n$. Amounts are in USD million.

Counterparty	EE	NEE	LGD_n	q_n	S_n
1	40	10	60%	2%	0.95
2	100	30	60%	4%	0.90
3	50	0	50%	1%	0.98

Which of the following statements about the CVA analysis is incorrect?

A. Counterparty 2 requires the largest charge under both the unilateral CVA and the BCVA.

B. In aggregate, the unilateral CVA (USD 3.13 million) exceeds the BCVA (USD 2.60 million).

C. The BCVA of counterparty 3 equals its unilateral CVA (USD 0.25 million) because its NEE is zero.

D. The DVA component for counterparty 1 is USD 0.114 million.

Question CR-21 g.5

Bank Q1112 • BT answers P2.T6.25.4.1 (rebuilt)

Xavier Bank has exposure to four counterparties, which also hold credit balances with the bank (negative exposure). For the bank, LGD is 50%, the marginal default probability is 1% and the survival probability is $S_I = 0.98$. Single-period data, in USD million:

Counterparty	EE	NEE	LGD_n	q_n	S_n
Alpha Pie	60	25	60%	2.0%	0.97
Bravo Ltd	90	70	60%	3.0%	0.96
Charlie Ltd	120	150	50%	3.0%	0.98
Delta Ltd	40	40	50%	5.0%	0.90

Using $\text{BCVA} = \text{LGD}_n\,\text{EE}\,q_n\,S_I - \text{LGD}_I\,\text{NEE}\,q_I\,S_n$, which counterparty has the highest BCVA adjustment?

A. Alpha Pie

B. Delta Ltd

C. Bravo Ltd

D. Charlie Ltd

Question CR-21 g.6

Bank Q1113 • BT answers P2.T6.25.4.2 (rebuilt)

A bank has a single-period exposure to a counterparty with EE = USD 150 million, NEE = USD 100 million, $\text{LGD}_n = 60\%$, $q_n = 2\%$ and $S_n = 0.97$. The bank's own LGD is 50%. Currently the bank's marginal default probability is $q_I = 1\%$ and its survival probability is $S_I = 0.98$. In a stress scenario the bank's credit spread widens, so that $q_I = 5\%$ and $S_I = 0.94$; all counterparty inputs and exposures are unchanged. Using $\text{BCVA} = \text{LGD}_n\,\text{EE}\,q_n\,S_I - \text{LGD}_I\,\text{NEE}\,q_I\,S_n$ and stress loss = stressed BCVA − current BCVA, what is the result of the stress?

A. A stress gain of USD 2.01 million

B. A stress gain of USD 1.94 million

C. A stress gain of USD 0.73 million

D. A stress loss of USD 2.01 million

Question CR-21 g.7

Bank Q1117 • BT answers P2.T6.25.4.3 (rebuilt, key corrected in the bank)

A bank's current CVA on a counterparty is USD 4.0 million and its DVA is USD 3.0 million. Under a stress scenario in which the bank's own credit spread widens sharply, the CVA rises to USD 6.5 million and the DVA rises to USD 9.0 million. Using BCVA = CVA − DVA, which of the following best describes the stress result?

A. A stress gain of USD 3.5 million, counterintuitive because it arises from the deterioration of the bank's own credit quality.

B. A stress loss of USD 2.5 million, being the increase in CVA.

C. A stress loss of USD 3.5 million, because the stressed BCVA is below the current BCVA.

D. A stress loss of USD 8.5 million, from a stressed BCVA of USD 15.5 million less a current BCVA of USD 7.0 million.

Question CR-21 g.8 Bank Q1115 • GARP PE2 Q75 (rebuilt)

A bank is designing the stress testing of its counterparty credit risk using the bilateral CVA (BCVA). Which of the following approaches is consistent with best practice?

A. Stress only the counterparty's credit spread, since the bank's own spread affects only the DVA, which is not relevant to stress losses.

B. Calculate the stress loss as the current BCVA minus the stressed BCVA, and keep BCVA stresses separate from market risk stress tests.

C. Assume that an improvement in the bank's own credit quality can never produce a loss on the BCVA.

D. Stress the bank's own credit spread together with counterparty spreads, consider their correlation, and apply the same market shocks as for the trading portfolio so that the BCVA stress loss can be added to market risk stress results.

Solutions • CR-21 g

Solution CR-21 g.1 — Answer: B

Bank Q1110 • AnalystPrep Q.2005

Step 1 – Unilateral against bilateral

Unilateral CVA admits only the counterparty's default. Letting the institution default too adds the value of its option to default, the DVA.

Step 2 – Name the measure

CVA (counterparty default) together with DVA (own default) is the bilateral CVA.
⇒ "We might consider not only that a financial institution's counterparty could default, but also that the financial institution in question could default to its counterparty. In such a case, the financial institution is considering its bilateral CVA."

Interpretation

The bilateral view brings the bank's own credit spread into the stress results, with the counterintuitive effects of Section 4.

Why the other options are wrong

A is incorrect. Unilateral CVA ignores the institution's own default.
C is incorrect. Negative exposure is an input to the DVA leg, not the overall approach.
D is incorrect. Wrong-way risk is the correlation between exposure and the counterparty's credit quality, not two-sided default.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 g.2 — Answer: C

Bank Q1116 • Schweser Module Quiz 39.2 Q3

Step 1 – The exposure term

The DVA leg uses the *negative* expected exposure (NEE), the exposure seen from the counterparty's side. EPE is already the CVA's exposure; BCVA does not add it.
⇒ The EPE part of the statement is wrong.

Step 2 – Survival

"The value of the option to default for the financial institution is dependent on the survival of the counterparty", so the counterparty's survival probability enters.
⇒ The survival part is correct.

Interpretation

BCVA adds two things: NEE and survival probabilities (the counterparty's in the DVA leg, the bank's own in the CVA leg).

Why the other options are wrong

A is incorrect. Accepts the EPE claim; the new exposure term is NEE.
B is incorrect. Reverses which half is right.
D is incorrect. The survival probability is part of BCVA.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5; SchweserNotes 2026 Book 2, Reading 39, LO 39.g

Solution CR-21 g.3 — Answer: B

Bank Q1114 • BT answers P2.T6.708.3

Step 1 – Check the true statements

A: losses depend on the bank's own spread, so an improvement in own credit (a smaller DVA) can cause a loss. C: S_I weights the CVA leg and S_n the DVA leg. D: with the same shocks, BCVA stress results add to market-risk stress results.

Step 2 – The false one

Using current exposure instead of expected exposure "can lead to substantial errors", especially at the money (CR-21 h).
$\Rightarrow$ B is false.

Interpretation

The question links CR-21 g to the pitfalls: whatever the framework, expected exposure is the right exposure input.

Why the other options are wrong

A is incorrect. True: the counterintuitive result that GARP flags.
C is incorrect. True: both survival probabilities enter the bilateral formula.
D is incorrect. True: coherent, additive aggregation with market-risk stress tests is BCVA's advantage.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5 and 21.6

Solution CR-21 g.4 — Answer: C

Bank Q1111 • BT answers P2.T6.25.3.3

Step 1 – Unilateral CVA

CP1 $0.6 \times 40 \times 0.02 = 0.480$; CP2 $0.6 \times 100 \times 0.04 = 2.400$; CP3 $0.5 \times 50 \times 0.01 = 0.250$; total 3.130.

Step 2 – CVA leg with the bank's survival $S_I = 0.97$

CP1 0.4656; CP2 2.3280; CP3 0.2425.

Step 3 – DVA leg, $0.6 \times \text{NEE} \times 0.02 \times S_n$

CP1 $0.6 \times 10 \times 0.02 \times 0.95 = 0.114$; CP2 $0.6 \times 30 \times 0.02 \times 0.90 = 0.324$; CP3 0.

Step 4 – BCVA

CP1 0.3516; CP2 2.0040; CP3 0.2425; total 2.598.
$\Rightarrow$ CP3's BCVA (0.2425) is below its unilateral CVA (0.25) even with zero NEE: C is the incorrect statement.

Interpretation

In the bilateral formula a loss from the counterparty's default counts only if the bank survives first, so the CVA leg is scaled by $S_I < 1$ even when there is no DVA.

Why the other options are wrong

A is incorrect. True: CP2 is the largest under both measures (2.400 and 2.004).
B is incorrect. True: 3.13 against 2.60.
D is incorrect. True: $0.6 \times 10 \times 0.02 \times 0.95 = 0.114$.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5, bilateral CVA formula

Solution CR-21 g.5 — Answer: C

Bank Q1112 • BT answers P2.T6.25.4.1

Step 1 – CVA legs, $\text{LGD}_n \, \text{EE} \, q_n \times 0.98$
Alpha 0.7056; Bravo 1.5876; Charlie 1.7640; Delta 0.9800.

Step 2 – DVA legs, $0.5 \times \text{NEE} \times 0.01 \times S_n$
Alpha 0.1213; Bravo 0.3360; Charlie 0.7350; Delta 0.1800.

Step 3 – BCVA
Alpha 0.5844; Bravo 1.2516; Charlie 1.0290; Delta 0.8000.
$\Rightarrow$ Bravo Ltd has the highest BCVA.

Interpretation
Charlie has the largest CVA leg, but its large NEE gives the largest DVA offset. Ranking on CVA alone, without the liability side, picks the wrong name: a small version of why DVA matters in aggregation.

Why the other options are wrong
A is incorrect. Alpha's BCVA, 0.584, is the lowest.
B is incorrect. Delta has the highest PD (5%), but its small EE keeps BCVA at 0.800.
D is incorrect. Charlie has the highest CVA leg (1.764) and unilateral CVA (1.80); ignoring its 0.735 DVA picks it.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 g.6 — Answer: A

Bank Q1113 • BT answers P2.T6.25.4.2

Step 1 – Current BCVA

$$0.6 \times 150 \times 0.02 \times 0.98 - 0.5 \times 100 \times 0.01 \times 0.97 = 1.7640 - 0.4850 = 1.2790$$

Step 2 – Stressed BCVA (own spread wider)

$$0.6 \times 150 \times 0.02 \times 0.94 - 0.5 \times 100 \times 0.05 \times 0.97 = 1.6920 - 2.4250 = -0.7330$$

Step 3 – Stress result

$$\text{BCVA}^S - \text{BCVA} = -0.7330 - 1.2790 = -2.012$$

$\Rightarrow$ A negative stress loss: a gain of about USD 2.01 million.

Interpretation
The wider own spread raises the DVA far more than the lower own survival trims the CVA leg. The bank gains from its own deterioration; the reverse move would produce a loss.

Why the other options are wrong
B is incorrect. Kept $S_I = 0.98$ in the CVA leg: $1.764 - 2.425 = -0.661$, and $-0.661 - 1.279 = -1.94$.
C is incorrect. Reported the stressed BCVA (-0.73) instead of the change.
D is incorrect. Sign error: a fall in the BCVA charge is a gain, not a loss.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 g.7 — Answer: A

Bank Q1117 • BT answers P2.T6.25.4.3

Step 1 – Current and stressed BCVA

$$\text{BCVA} = 4.0 - 3.0 = 1.0, \qquad \text{BCVA}^S = 6.5 - 9.0 = -2.5$$

Step 2 – Stress result

$$\text{BCVA}^S - \text{BCVA} = -2.5 - 1.0 = -3.5$$

$\Rightarrow$ A gain of USD 3.5 million, driven by the bank's own wider spread (DVA up by 6.0).

Interpretation

This is how stressed DVA enters the aggregation: a CVA-only stress would report a 2.5 loss, while the bilateral result is a 3.5 gain. The gain must be flagged as counterintuitive, not booked as comfort.

Why the other options are wrong

B is incorrect. Looked only at the CVA increase $(6.5 - 4.0)$ and ignored DVA.
C is incorrect. Right size, wrong sign: a fall in BCVA is a gain.
D is incorrect. Added DVA to CVA instead of subtracting: $15.5 - 7.0 = 8.5$.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5

Solution CR-21 g.8 — Answer: D

Bank Q1115 • GARP PE2 Q75

Step 1 – What must be stressed

BCVA depends on both parties' spreads (survival probabilities and DVA), so the bank's own spread is stressed with the counterparties', and "the financial institution will also have to consider how its own credit spread is correlated with its counterparties' credit spread".

Step 2 – How the result is used

Stress loss = stressed BCVA − current BCVA. "As long as the same shocks to market variables are applied to the trading portfolio and to the BCVA results, they can be aggregated by simple addition."
$\Rightarrow$ D states all three practices.

Interpretation

This is the full answer to the second half of the LO: DVA enters through the BCVA stress, and BCVA stress results join the market-risk stress results under common shocks.

Why the other options are wrong

A is incorrect. Ignores the DVA and the survival terms, which are exactly what the bilateral view adds.
B is incorrect. Reverses the stress-loss definition and gives up the main benefit, aggregation with market-risk stress tests.
C is incorrect. GARP warns that losses can occur because the firm's own credit quality improves.

LO: CR-21 g: Calculate the DVA and explain how stressing DVA enters into aggregating stress tests of CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.5

CR-21 • The Evolution of Stress Testing Counterparty Exposures • Vardeez: VERY IMPORTANT

CR-21 h

P2 CONCEPT

Describe the common pitfalls in stress testing CCR.

Workbook: C9 Stress-Testing Practices and Scenario Design • Tier C • Map #38 (3/16) • 2 Qs • Vol 2 p.74

WHAT IS TESTED HERE

Why P2. Map focus WORKBOOK C9 + VARDEEZ FOCUS. C9 is a Tier C section (Map #38, 3/16), reported as "2 to 3 questions on stress testing related: appropriate practices", that is, best-practice audits rather than computation; it sits in a chapter Vardeez rates VERY IMPORTANT throughout, hence P2.

Vardeez: four pitfalls. CCR is often not integrated with loan or trading stress tests; banks use current exposure instead of EE or EPE, which can cause significant errors; this is worst for at-the-money positions; and linearised delta sensitivities can be highly inaccurate because exposure is nonlinear.

Reported exam shapes

- **C9·S1** Appropriate stress-testing practices as a best-practice audit: what to stress, how to aggregate, what to avoid.

What the questions look like: a reviewer or auditor looks at a CCR stress programme and picks the sound practice, or the pitfall, from four. The wrong options are the pitfalls themselves: summing stressed current exposures, current exposure in a portfolio credit model, delta shortcuts for an at-the-money book, leaving DVA out. All conceptual. This block pulls the whole reading together, so it leans on CR-21 c, d and g.

1. The pitfalls

1. **No integration.** Stress testing beyond current exposure is recent, and it is rare for CCR to be aggregated with loan-portfolio or trading-position stress results in a consistent framework. Better models of CCR exposure and of CVA now make that aggregation possible (CR-21 d for loans, CR-21 g for trading).
2. **Current exposure where expected exposure belongs.** Since most banks already stress current exposure, it is tempting to reuse those stresses when combining CCR losses with loans or trading positions. Expected exposure or EPE should be used; current exposure "can lead to substantial errors".
3. **At-the-money positions are where the error is largest** (Section 2).
4. **Delta approximations.** Delta sensitivities save computation, but CCR exposure is highly nonlinear, so the linearisation can be highly misleading and its errors are not obvious. At-the-money portfolios hit by large price moves are especially prone to these errors.

From the chapter's conclusion, two more: **too many stress tests can hide the risk** of a portfolio, although a fair number is needed for a comprehensive view; and **combining default stresses with fair-value stresses** into a single comprehensive test "remains the true difficulty". Both kinds of loss can occur whichever way a bank manages CCR, so neither view can be dropped (CR-21 b).

2. Why at-the-money is the worst case

Gregory's normal approximation for a linear derivative with no intermediate payments gives $\text{EE} = m\,\Phi(m/\sigma) + \sigma\,\varphi(m/\sigma)$ for a starting MtM m and volatility σ. GARP's Figure 21.1 applies a US$1 million shock to the MtM and plots the change in MtM, in current exposure and in expected exposure against the starting MtM.

- **Deep in the money**: current and expected exposure both move almost one-for-one with the shock.
- **Deep out of the money**: both barely move.
- **At the money**: current exposure jumps from nothing to the full shock across a narrow band, while expected exposure moves smoothly by about half of it. "The difference between current exposure and expected exposure is almost half the value of the shock." Just above zero, current exposure overstates the change; just below it, current exposure shows no change at all and understates it.

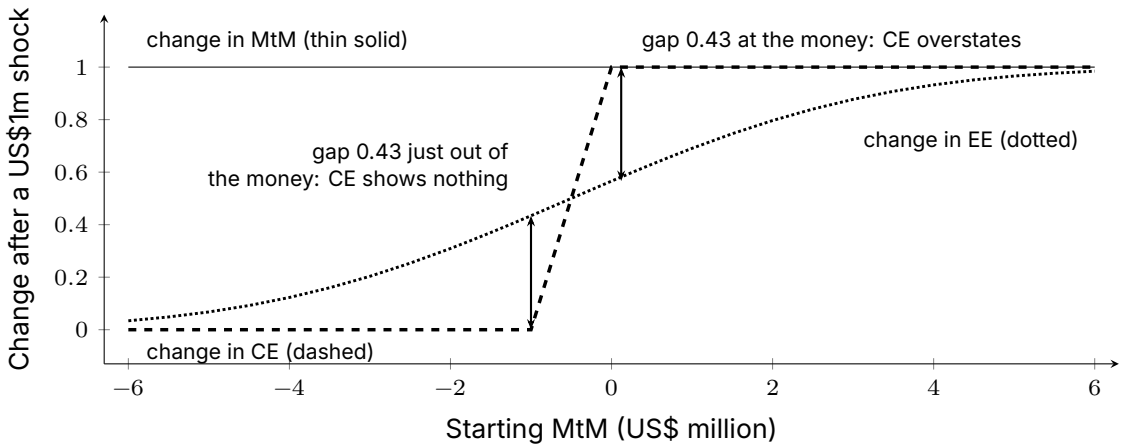


Figure 1. GARP Figure 21.1 redrawn (illustrative: normal approximation with $\sigma = 3$). Far from the money, stressed current exposure and expected exposure agree. Near the money they part by almost half the shock, in either direction.

3. A best-practice audit (shape S1)

	Sound practice	Pitfall
Exposure	EE or EPE ($\alpha \times$EPE in the credit framework)	Stressed current exposure reused for aggregation
Revaluation	Full revaluation; recalculate the whole EE profile after the shock	Delta (linear) shortcuts, worst for at-the-money books under large moves
Credit quality	PD in the stress (physical in EL, risk-neutral in CVA)	Exposure tables with no PD
Aggregation, credit view	EL^S — EL on derivatives added to the loan stress loss	Summing stressed current exposures; current exposure in a portfolio credit model
Aggregation, market view	BCVA^S — BCVA added to trading stresses under the same shocks	Different shocks for the trading book and the BCVA
Own credit	DVA included; own and counterparty spreads stressed together, with their correlation	DVA left out; "own credit can only help"
Wrong-way risk	Stress the exposure-PD correlation (CVA); screen stressed CE for wrong-way names	Assume independence untested
Volume	A fair number of well-chosen stresses	So many results that the risk is hidden

The chapter ends on an irony: regulators began moving derivatives to central clearing to reduce the CCR problem just as the ability to manage CCR was making major advances.

Exam traps

- **Polarity flip on delta.** Delta approximations are *least* reliable for at-the-money portfolios under large moves, and their errors are not obvious.
- **Direction of the at-the-money error.** Current exposure can overstate (just in the money) or show nothing (just out of the money); "almost half the shock" is the size of the gap.
- **Where the gap is small.** Deep in or deep out of the money, current and expected exposure respond alike.
- **Absolute claims.** "More stress tests always help"; "own credit can never cause a loss"; "managing CCR as a credit risk avoids fair-value losses".
- **The true difficulty** is combining default stresses and fair-value stresses, not running either one.

Practice questions: CR-21 h

Question CR-21 h.1

New • built from GARP Ch.21, 21.6 and Figure 21.1

A CCR analyst compares the change in current exposure with the change in expected exposure after a US$1 million shock to the market value of a linear derivative with no intermediate payments, using a normal approximation for expected exposure. For which starting position is the error from using stressed current exposure in place of expected exposure largest?

A. Deep in the money, because current exposure then takes the full shock while expected exposure takes almost none of it.

B. Deep out of the money, because current exposure stays at zero while expected exposure takes almost all of the shock.

C. Close to at the money, where the difference between the two changes is almost half the value of the shock.

D. The error is the same at every starting position, because current exposure and expected exposure are both linear in the market value.

Question CR-21 h.2

Bank Q1119 • New question written for this bank (GARP practice-exam style)

A risk analyst at a large bank is preparing the bank's counterparty credit risk (CCR) stress test for a portfolio of at-the-money interest rate swaps with a single counterparty. The current mark-to-market value of the portfolio is zero. Under the stress scenario, the mark-to-market value of the portfolio rises instantly by USD 40 million. The analyst plans to use the resulting change in current exposure to measure the increase in exposure that will be combined with the bank's loan portfolio stress losses. Using the expected exposure to the same shock instead would show an increase of approximately USD 20 million. Which of the following best describes the effect of using current exposure rather than expected exposure?

A. It overstates the increase in exposure by approximately USD 20 million, because for at-the-money exposures the difference between current exposure and expected exposure is almost half the size of the shock.

B. It has no material effect on the result, because for at-the-money exposures current exposure and expected exposure respond equally to a market shock.

C. It overstates the increase in exposure by approximately USD 40 million, because current exposure recognizes the full shock while expected exposure recognizes none of it.

D. It understates the increase in exposure by approximately USD 20 million, because current exposure ignores the potential for the portfolio value to rise further before maturity.

Question CR-21 h.3 Bank Q1120 • New question written for this bank (GARP practice-exam style)

A market risk manager at a bank wants to reduce the computational cost of stressing the bank's derivatives portfolio for counterparty credit risk by using delta sensitivities to approximate the change in exposure under large market moves. The portfolio is currently close to at-the-money with respect to its main risk factors. Which of the following statements best describes a pitfall of this approach?

A. Delta sensitivities are accurate for CCR because exposure is a linear function of the underlying risk factors, so the approximation is reliable even for large price moves.

B. Exposure is highly nonlinear in the risk factors, so linearizing with delta can introduce errors that are not obvious and can be highly misleading, especially for at-the-money portfolios subject to large price moves.

C. Delta approximations are unreliable only for deep out-of-the-money portfolios, where exposure is close to zero, and are acceptable for at-the-money portfolios.

D. The errors from delta approximations are large but easy to detect, because they always cause the stressed exposure to be understated relative to full revaluation.

Question CR-21 h.4 New • built from GARP Ch.21, 21.6 and Conclusion

The head of counterparty risk at a German bank summarises the chapter's conclusions for the risk committee. Which of the following statements is correct?

A. The more CCR stress tests the bank runs, the clearer its view of the risk, so the programme should report every available stress result.

B. A bank that manages CCR purely as a credit risk avoids fair-value losses, so it needs to stress only default losses.

C. More integrated stress tests come from combining the credit-risk view of CCR with the loan portfolio and the market-risk view with the trading book, but combining default stresses and fair-value stresses into one comprehensive test remains the true difficulty.

D. Central clearing has removed the need for CCR stress testing, which is why the chapter describes the recent advances in CCR management as ironic.

Question CR-21 h.5 Bank Q1118 • Mock 4 Q76 (rebuilt, key corrected in the bank)

An internal auditor at Bank Alpha is reviewing how the bank stress tests counterparty credit risk (CCR) and aggregates the results with its loan portfolio stress tests. Which of the following practices is consistent with sound practice?

A. Running stressed current exposures through a portfolio credit risk model to aggregate CCR with the loan portfolio.

B. Revaluing an at-the-money derivative portfolio under a large market shock using delta sensitivities only.

C. Summing the stressed current exposures across counterparties to obtain the aggregate stress loss.

D. Using expected exposure (or expected positive exposure), not current exposure, when combining CCR stress results with loan portfolio stress results.

Question CR-21 h.6 Workbook C9.1 • S1 • Map #38

The head of counterparty credit risk at a Japanese bank is reviewing the bank's CCR stress-testing program after supervisors asked for derivative stress results that can be combined with the stress results of the loan book. Which of the following changes should the head recommend?

A. Run the PD and market stresses through expected positive exposure scaled by alpha in the expected-loss framework, so that derivative stress losses can be added to the loan stress losses.

B. Sum the stressed current exposures across all counterparties in each scenario and report that total as the stress loss of the derivatives book for comparison with the loan book.

C. Reprice the derivatives under the large scenario shocks using delta sensitivities, since the linear approximation is most reliable for portfolios that are close to at-the-money.

D. Leave DVA out of the CVA stress test, since a bilateral view can only reduce reported stress losses and never creates a loss from changes in the bank's own credit quality.

Solutions • CR-21 h

Solution CR-21 h.1 — Answer: C

New • built from GARP Ch.21, 21.6 and Figure 21.1

Step 1 – Far from the money

Deep in the money, the exposure equals the MtM in almost every scenario, so current and expected exposure both rise by almost the full shock. Deep out of the money, both stay close to zero.

$\Rightarrow$ The two measures agree at both ends (Figure 1: gaps of about 0.02 and 0.03 at $+6$ and -6).

Step 2 – Near the money

Current exposure is a hinge at zero: its change jumps from 0 to the full shock over a narrow band. Expected exposure averages over future scenarios and moves smoothly, by about half the shock.

$\Rightarrow$ "For at-the-money exposures, the difference between current exposure and expected exposure is almost half the value of the shock."

Interpretation

This is why reusing current-exposure stresses for aggregation is a pitfall: the error peaks exactly where positions are most sensitive, at the money.

Why the other options are wrong

A is incorrect. Deep in the money expected exposure also takes almost the full shock; the gap is small.

B is incorrect. Deep out of the money expected exposure also barely moves; the gap is small.

D is incorrect. Current exposure is not linear in MtM (it is floored at zero), and expected exposure is a smooth nonlinear function of it.

LO: CR-21 h: Describe the common pitfalls in stress testing CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.6, Figure 21.1

Solution CR-21 h.2 — Answer: A

Bank Q1119 • New question written for this bank

Step 1 – Change in current exposure

Starting at zero, the USD 40 million rise in MtM raises current exposure, $\max(\text{MtM}, 0)$, by the full 40.

Step 2 – Change in expected exposure

Given as about 20: expected exposure responds only partly, because it already reflected the chance of the MtM rising.

Step 3 – The error

$40 - 20 = 20$, about half the shock.

$\Rightarrow$ Current exposure overstates the increase by about USD 20 million.

Interpretation

The result matches GARP's Figure 21.1: at the money the gap is almost half the shock. The overstatement would flow straight into the combined loan and CCR stress loss.

Why the other options are wrong

B is incorrect. The two measures diverge most at the money.

C is incorrect. Expected exposure does recognise part of the shock (about 20), so the overstatement is 20, not 40.

D is incorrect. Wrong direction: current exposure rises by more (40) than expected exposure

(20).

LO: CR-21 h: Describe the common pitfalls in stress testing CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.6, Figure 21.1

Solution CR-21 h.3 — Answer: B

Bank Q1120 • New question written for this bank

Step 1 – Shape of exposure

Exposure is the positive part of portfolio value, so it behaves like an option and is highly nonlinear in the risk factors.

Step 2 – What a delta approximation misses

A first-order (delta) approximation ignores that curvature, and the error grows with the size of the move. GARP: "the errors introduced are not obvious and the linearisation can be highly misleading. At-the-money portfolios with large price moves applied to the portfolio are especially prone to errors."

Interpretation

The cost saving is real, but it buys a stress number whose error is hidden and largest where it matters most.

Why the other options are wrong

A is incorrect. Treats exposure as linear; GARP says it is highly nonlinear.
C is incorrect. Reverses the risk: at-the-money portfolios are the most error-prone.
D is incorrect. The errors are "not obvious", and GARP does not say they always understate exposure.

LO: CR-21 h: Describe the common pitfalls in stress testing CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.6

Solution CR-21 h.4 — Answer: C

New • built from GARP Ch.21, Conclusion

Step 1 – Integration

"More integrated stress tests can be generated by combining the credit risk view with the loan portfolio, or the market risk view of CCR can be combined with the trading book."

Step 2 – What remains hard

"The true difficulty remains in combining the default stresses and the fair value stresses to get a single comprehensive stress test."
$\Rightarrow$ C quotes both points.

Interpretation

The two integrations are the end points of CR-21 d (EL^S — EL added to loans) and CR-21 g ($BCVA^S$ — BCVA added to trading). Joining the two families is still open.

Why the other options are wrong

A is incorrect. "Too many stress tests can hide the risk of a portfolio"; a fair number is needed, not every possible result.
B is incorrect. "Fair value losses and default losses can occur no matter how a financial institution manages its CCR."
D is incorrect. The irony is that regulators moved derivatives to central clearing just as CCR management was advancing; clearing does not make stress testing unnecessary.

LO: CR-21 h: Describe the common pitfalls in stress testing CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.6 and Conclusion

Solution CR-21 h.5 — Answer: D

Bank Q1118 • Mock 4 Q76

Step 1 – Recall the pitfalls

Stressed current exposures cannot be aggregated meaningfully; portfolio credit models need expected exposures; delta approximations fail for nonlinear exposure, especially at the money under large moves.

Step 2 – Test each practice

A, B and C are each one of those pitfalls. D uses the right exposure measure for aggregation.
⇒ "Expected exposure or EPE should be used as the exposure amount, and ... using current exposure instead would be a mistake."

Interpretation

An audit question is a list of pitfalls with one sound practice hidden among them: shape S1.

Why the other options are wrong

A is incorrect. Current exposures do not belong in a portfolio credit model; expected exposures do.

B is incorrect. Delta-only revaluation is highly misleading for an at-the-money book under a large shock.

C is incorrect. The sum is the loss if every counterparty defaulted together, a clear overstatement.

LO: CR-21 h: Describe the common pitfalls in stress testing CCR. • **Reference:** Lynch (GARP CR Ch.21), 21.3 and 21.6

Solution CR-21 h.6 — Answer: A

Workbook C9.1 • S1

Step 1 – What can be combined with a loan stress

Loan stress loss $= \text{EL}^S - \text{EL}$ with $\text{EL} = \sum p_i \, \text{ead}_i \, \text{lgd}_i$. For derivatives, $\alpha \times$ EPE "is the value that allows CCR exposures to be placed in a portfolio credit model along with loans"; then $\text{EL} = \sum p_i \, \alpha \, \text{epe}_i \, \text{lgd}_i$.
⇒ The derivative stress loss is measured in the same units and on the same physical-PD basis as the loan stress loss, so the bank "can also combine the stress losses on the loan portfolio and the stress losses on its derivatives portfolio by adding these stress losses together".

Step 2 – Which stresses to run

Stress PDs (or the macro and balance-sheet variables behind them) and stress EPE through market variables; note that EPE stresses can raise or lower the aggregate depending on directional bias and margining.

Interpretation

A is correct. It moves the program beyond simple current-exposure stresses: aggregation with loans becomes possible and counterparty credit quality enters the result.

Why the other options are wrong

B is incorrect. Absolute claim / scope error: the sum of stressed current exposures "would represent the loss that would occur if every counterparty defaulted in the stress scenario"; and current exposure ignores credit quality, so it cannot be compared with an expected-loss stress on loans.

C is incorrect. Polarity flip: CCR is highly nonlinear, and "at-the-money portfolios with large price moves applied to the portfolio are especially prone to errors from using delta approximations".

D is incorrect. Absolute claim: the bilateral formula "may lead to counterintuitive results such

as losses occurring because the firm's own credit quality improves"; to capture the full impact of a scenario on CVA P&L the liability side (DVA) should be included.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. CR-21 h: Describe the common pitfalls in stress testing CCR. • **Reference:** Lynch, The Evolution of Stress Testing Counterparty Exposures, Sections 21.3, 21.4, 21.5, 21.6

CR-22 • Structured Credit Risk • Vardeez: IMPORTANT

CR-22 i

P2 CONCEPT

Explain how the default probabilities and default correlations affect the credit risk in a securitization.

Workbook: B3 Default Correlation, Joint PD and Copulas • Tier B D15 Structured Credit and Implied Correlation • Tier D

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: B3 (Map #16, 6/16) carries the LO 40.i impacts table as shape S7, and D15 (Map #60) reports it again. Vardeez singles the LO out inside an IMPORTANT reading: "LO 40.i: This is Imp LO."

Vardeez: "Just remember the impacts via the table." His one possible calculation for the reading sits in CR-22 e and f, so this LO stays conceptual.

Reported exam shapes

- **B3·S7** The LO 40.i impacts table for the tranches (worked in D15.1).
- **D15·S1** The impacts table: how default probability and correlation move each tranche.

What the questions look like: direction questions. A protection seller on the senior or the junior tranche when correlation moves; which tranche is hurt by a rise in default probability *and* a fall in correlation; true-or-false statements on the mezzanine's two regimes and on how the equity's mean value and Credit VaR respond.

Worked elsewhere: D15.1 (an equity price rise read as a rise in implied correlation, with the grid of tranche values) under CR-22 l; B3.3 (the one-factor Gaussian copula) under CR-12 f. How the simulation grid is produced is CR-22 h.

1. Where the numbers come from

Malz simulates the CLO of CR-22 c ($85m senior, $10m mezzanine, $5m equity on 100 loans of $1m): 1,000 threads for each of 52 pairs of default probability ($\pi = 0.75\%$ to 9.75%) and constant pairwise correlation ($\rho = 0$, 0.3, 0.6, 0.9), each thread run through the waterfall (procedure: CR-22 h). The equity is valued at a 25% hurdle rate, about its $5m par at $\pi = 2.25\%$, $\rho = 0.3$; bond losses are writedowns in percent of par. This LO reads the mean results (GARP Table 22.3, Figure 1) and the 99% Credit VaRs (Table 22.4).

2. Default probability: bad for every tranche

GARP: "for any given correlation, an increase in the default rate will hurt all of the tranches." What differs is the *shape* of the response (Figure 1):

- **Equity: positively convex at low correlation.** Value falls fast and then flattens ("you can't beat a dead horse"): at $\rho = 0$, 6.59 to 1.06 by $\pi = 5.25\%$, then only to 0.22.
- **Senior: negatively convex.** Losses accelerate: at $\rho = 0$ nothing until $\pi = 6.75\%$, then 0.12%, 1.07%, 4.02% of par.
- **Mezzanine: both.** Negatively convex at low default rates, positively at high ones. It is thin (5% to 15%), so once pool losses reach its attachment point it moves quickly from par to a total loss (0% to 104% at $\rho = 0$; above 100% because the final coupon is lost too, up to $11.0m on $10m of par).
- **High correlation linearises:** at $\rho = 0.9$ all tranches respond more nearly linearly.

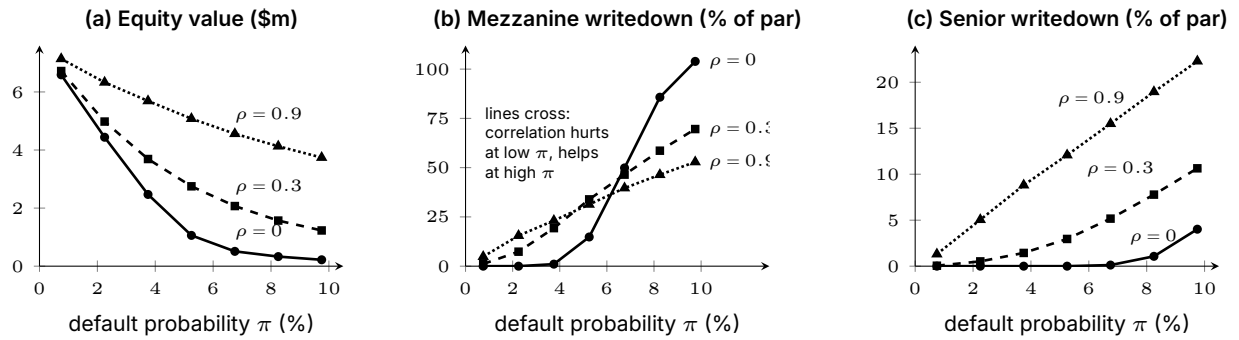


Figure 1. Mean tranche results over 1,000 threads (GARP Table 22.3): $\rho = 0$ solid with circles, $\rho = 0.3$ dashed with squares, $\rho = 0.9$ dotted with triangles. Every curve moves the wrong way as π rises. Correlation raises the equity value (a) and the senior writedown (c) at every π; the mezzanine lines (b) cross between $\pi = 5.25\%$ and 6.75%.

3. Correlation: a bet on dispersion

Correlation leaves the expected number of defaults unchanged; it changes how they spread across scenarios. With low correlation every thread has close to the expected number (Schweser: 1,000 coin tosses give close to 500 heads). With high correlation defaults arrive in **clusters**: some threads have very many, others very few.

- **Equity: long correlation.** Low correlation puts some defaults in almost every thread and the equity takes the first loss. High correlation raises the chance of a low-default state, the distribution turns U-shaped and the equity becomes "a lottery ticket": at $\pi = 2.25\%$, 4.44 at $\rho = 0$ to 6.33 at 0.9. "Correlation is more decisive for equity risk and value than default probability."
- **Senior: short correlation.** Only a cluster can breach its 15% attachment point: at $\pi = 2.25\%$ its writedown rises from 0% to 5.05%.
- **Mezzanine: depends on π.** Senior-like at low default rates, when a breach is unlikely (at $\pi = 0.75\%$, 0% to 4.84% as ρ rises); equity-like at high ones (at $\pi = 9.75\%$, 103.92% down to 52.87%).
- **Protection sellers** are long the tranche: a fall in correlation hurts the seller on the junior tranche and helps the seller on the senior.

4. Tail risk: Credit VaR (GARP Table 22.4, $m)

- **Equity.** VaR rises with correlation (1.62, 6.33, 6.85, 7.14 at $\pi = 0.75\%$): the mean rises, "but so also does its risk". It falls as π rises (6.33, 4.98, 1.23 at $\rho = 0.3$ for $\pi = 0.75\%$, 2.25%, 9.75%) because the expected loss is already so high.
- **Senior.** Virtually risk-free with uncorrelated defaults even at high π (8.33 on $85m at $\pi = 9.75\%$); of the order of half its par at high correlation, and at $\rho = 0.9$ almost insensitive to π (48 to 60). Correlation is the bigger risk factor.
- **Mezzanine.** Low and senior-like at low π and ρ; equity-like at high ones. Thin, so its 95% and 99% VaRs are close.

> **Correction to the source notes.** Vardeez's Credit VaR column (Schweser Figure 40.3: with ρ fixed, a higher π lowers the equity VaR, raises the senior VaR, and raises then lowers the mezzanine VaR) is a tendency, true at low and moderate correlation. In GARP Table 22.4 the equity VaR at $\rho = 0$ first rises (1.62 to 2.53 from $\pi = 0.75\%$ to 2.25%); at $\rho = 0.9$ the senior VaR peaks at $\pi = 2.25\%$ (59.99) and

drifts down to 47.79; at $\rho = 0.6$ and 0.9 the mezzanine VaR falls from the start. Exam answers follow the tendency (Schweser: "generally decreases").

	Equity	Mezzanine	Senior
π **up: value**	Falls	Falls	Falls
π **up: VaR**	Generally falls	Rises, then falls	Generally rises
ρ **up: value**	Rises	Falls at low π, rises at high π	Falls
ρ **up: VaR**	Rises	Senior-like at low π, ρ; equity-like at high	Rises (0.6 and 0.9 close)
Convexity in π	Positive	Negative, then positive	Negative
Correlation	Long	Depends on π	Short

The impacts table (shapes B3·S7 and D15·S1). Convexity at low correlation.

Exam traps

- **Polarity flip.** Correlation helps the equity and hurts the senior; a protection seller is long its tranche.
- **Uniform harm.** A higher π hurts every tranche; a higher ρ does not.
- **Mezzanine scope.** Senior-like at low default rates, equity-like at high ones; not "always like equity", and not "insensitive at $\rho = 0$" (its default sensitivity peaks there: 0.26% of par per bp at $\pi = 6.75\%$).
- **Expected loss versus allocation.** Correlation leaves the pool's expected loss unchanged and moves it between tranches.
- **Diversified is not safe.** High correlation exposes the senior of a diversified pool to large tail losses.
- **Mean versus risk.** The equity's value *and* its VaR rise with ρ; its VaR falls with π because the expected loss is already high.

Practice questions: CR-22 i

Question CR-22 i.1

Bank Q1169 • AnalystPrep Q.5520 (rebuilt)

A junior analyst is summarising, for the credit committee, a simulation of a three-tranche securitization run over a grid of default probabilities and pairwise default correlations. Which of the following statements about the effects on mean tranche values is correct?

A. A higher default probability lowers the value of all tranches, and a higher correlation lowers the value of the senior tranche but raises the value of the equity tranche.

B. A higher default probability lowers the value of all tranches, and a higher correlation lowers the value of every tranche.

C. A higher default probability lowers the value of the equity tranche but raises the value of the senior tranche, and a higher correlation lowers the value of the senior tranche.

D. A higher default probability lowers the value of all tranches, and a higher correlation raises the value of the senior tranche but lowers the value of the equity tranche.

Question CR-22 i.2

Bank Q1170 • AnalystPrep Q.5530 (rebuilt)

The CRO of an insurer asks how systematic risk, expressed as a high default correlation among the underlying loans, affects the senior bonds of a CDO the insurer holds. Which of the following is the most accurate answer?

A. It raises liquidity risk, making the senior bonds hard to trade.

B. It lowers the expected loss on the senior bonds as correlation rises.

C. It raises the likelihood of large losses on the senior bonds, shown by a higher Credit VaR.

D. It lowers the credit risk of the senior bonds, because the collateral is more diversified.

Question CR-22 i.3

Bank Q1166 • GARP PE1 Q46 • AP Mock 1 Q29 • Mock 3 Q34

A financial firm has sold default protection on the most senior tranche of a CDO. If the default correlation between the assets held in the CDO decreases sharply from the correlation used in pricing the tranches, assuming everything else is unchanged, how will the position of the financial firm be affected?

A. It will either increase or decrease, depending on the pricing model used and market conditions.

B. It will gain significant value, since the probability of the protection being exercised falls.

C. It will lose significant value, since the protection it sold will gain value.

D. It will neither gain nor lose value, since only expected default losses matter and correlation does not affect expected default losses.

Question CR-22 i.4

Bank Q1175 • Mock 1 Q63

Investor A has sold protection on the most junior tranche of a collateralized debt obligation (CDO). Investor B has sold default protection on the most senior tranche of the same CDO. If default correlations decrease, which of the following statements is incorrect?

I. Investor A's position will lose value.

II. Investor B's position will gain value.

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

Question CR-22 i.5

Bank Q1168 • AP Mock 1 Q34 (rebuilt)

A CDO analyst expects the default probability of the underlying assets to rise significantly while the default correlation falls, with all else equal. Which tranche is unambiguously reduced in value by BOTH changes?

A. The senior tranche

B. The equity tranche

C. The mezzanine tranche

D. None of the tranches, because the two effects always offset

Question CR-22 i.6

Bank Q1173 • BT P2.T6.314.3

In Malz's simulation, the mezzanine tranche has a width of only 10%. It is typical for the mezzanine tranche to be thinner than the senior bond. In regard to the mezzanine tranche, each of the following is true EXCEPT which is false?

A. At low default rates, an increase in correlation increases losses on the mezzanine bond (it behaves like a senior bond).

B. At high default rates, an increase in correlation decreases losses on the mezzanine bond (it behaves like equity).

C. The mezzanine tranche has negative convexity in default rates for low default rates (like a senior bond) but is positively convex for high default rates (like equity).

D. At low correlations, for example $\rho = 0$, the mezzanine tranche is essentially insensitive to changes in the default rate (like a senior bond).

Question CR-22 i.7

Bank Q1171 • BT P2.T6.314.1 (rebuilt)

A risk manager simulates a three-tranche securitization and measures the Credit VaR of the equity tranche as the shortfall of a low-percentile simulated value below the mean value. Regarding the equity tranche, each of the following statements is true EXCEPT which is false?

A. For a given correlation, an increase in the default rate lowers the expected equity tranche value.

B. For a given default rate, an increase in correlation raises the expected equity tranche value.

C. For a given default rate, an increase in correlation decreases the equity Credit VaR; for a given correlation, a higher default rate increases the equity Credit VaR.

D. At low correlations, the equity value is substantially positively convex in default rates.

Solutions • CR-22 i

Solution CR-22 i.1 — Answer: A

Bank Q1169 • AnalystPrep Q.5520

Step 1 – Default probability

For a given correlation, more defaults mean less collateral cash for everyone: equity values fall and both bonds' writedowns rise.
⇒ Default probability is bad for every tranche.

Step 2 – Correlation and the senior

Higher correlation makes defaults cluster, so some scenarios have enough defaults to breach the senior's subordination.
⇒ Senior value falls with correlation.

Step 3 – Correlation and the equity

Clustering also raises the chance of a scenario with very few defaults, the only one in which the first-loss tranche keeps most of its cash flow.
⇒ Equity value rises with correlation.

Interpretation

At $\pi = 2.25\%$ in GARP Table 22.3 the equity value rises from 4.44 to 6.33 as ρ goes from 0 to 0.9, while the senior writedown rises from 0% to 5.05% of par. The mezzanine is the mixed case and is left out of all four options.

Why the other options are wrong

B is incorrect. Treats correlation as uniformly harmful. The equity gains from the higher chance of few defaults.
C is incorrect. Has a higher default probability raising the senior's value. Every tranche is hurt.
D is incorrect. Reverses both correlation effects (polarity flip).

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. •
Reference: Malz, Ch.9 (GARP CR Ch.22), 22.3 Means of the Distributions, Table 22.3

Solution CR-22 i.2 — Answer: C

Bank Q1170 • AnalystPrep Q.5530

Step 1 – What high correlation means

GARP: high default correlation "means that there is a low but material probability of a state of the world in which an unusually large number of defaults occurs".
⇒ It fattens the tail of the pool loss distribution.

Step 2 – Who sits in the tail

The senior is reached only when losses exhaust the equity and the mezzanine, which needs such a cluster. At $\pi = 2.25\%$ its 99% Credit VaR is 0 at $\rho = 0$, 17.77 at 0.3, 41.43 at 0.6 and \$59.99m at 0.9.
⇒ Its probability of a large loss, and so its Credit VaR, rises with correlation.

Interpretation

This is GARP's "systematic risk" point: even if the collateral is well diversified, the senior bond has a risk of loss, potentially a large one, when correlation is high. Its expected loss may stay below the pool's, but its tail is fat.

Why the other options are wrong

A is incorrect. Liquidity is not the mechanism; correlation acts on the distribution of default

losses.
B is incorrect. Gets the direction wrong: the senior's expected writedown rises with correlation (0% to 5.05% at $\pi = 2.25\%$).
D is incorrect. High correlation means diversification works *less* well, not better.

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. •
Reference: Malz, Ch.9 (GARP CR Ch.22), 22.3 Credit VaR of the Tranches, Table 22.4; Systematic Risk

Solution CR-22 i.3 Answer: B

Bank Q1166 • GARP PE1 Q46 • AP Mock 1 Q29

Step 1 – The seller's exposure
Selling protection on the senior tranche is economically the same as holding it: the seller pays only if pool losses exceed all the subordination below the senior.
⇒ The seller is long the senior tranche, which is short correlation.

Step 2 – Lower correlation
With lower correlation, defaults are more independent and their number stays near its expectation, so breaches of the senior's attachment point become rarer.
⇒ The probability that the protection is called falls and the seller's position gains value.

Interpretation
The direction is not model-dependent: senior tranche value falls when correlation rises and rises when it falls. The mirror question (correlation rises, the senior protection seller loses) is the same idea.

Why the other options are wrong
A is incorrect. The sign is determined, not left to the pricing model or market conditions.
C is incorrect. Reverses the seller's side: the protection gains value only when it is more likely to be called, which happens when correlation rises.
D is incorrect. Correlation leaves the pool's expected loss unchanged but redistributes it across tranches, so the senior's expected loss does change.

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. •
Reference: Malz, Ch.9 (GARP CR Ch.22), 22.3 Means of the Distributions

Solution CR-22 i.4 Answer: D

Bank Q1175 • Mock 1 Q63

Step 1 – Both investors are long their tranches
A protection seller holds the tranche's credit risk.

Step 2 – Investor A, junior tranche
The junior (equity) tranche is long correlation: lower correlation removes the chance of a near-zero-loss scenario and puts some defaults in almost every one. A loses value.
⇒ Statement I is correct.

Step 3 – Investor B, senior tranche
The senior is short correlation: lower correlation makes the clusters that breach its subordination rarer. B gains value.
⇒ Statement II is correct.

Interpretation
Both statements are correct, so neither is incorrect. Read the stem twice: it asks for the *incorrect* statement.

Why the other options are wrong

A is incorrect. Treats I as incorrect, as if lower correlation helped the equity.

B is incorrect. Treats II as incorrect, as if the senior protection seller lost whenever correlation moved.

C is incorrect. Reverses the correlation exposure of both tranches.

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 Means of the Distributions

Solution CR-22 i.5 — Answer: B

Bank Q1168 • AP Mock 1 Q34

Step 1 – The default probability effect

A higher default probability lowers the value of every tranche.

Step 2 – The correlation effect, tranche by tranche

Lower correlation hurts the equity (fewer low-default scenarios), helps the senior (fewer clusters) and has a sign for the mezzanine that depends on the default rate.

Step 3 – Combine

Equity: both effects negative. Senior: negative and positive, net unclear. Mezzanine: negative and ambiguous.

$\Rightarrow$ Only the equity is unambiguously hurt.

Interpretation

Use the two rows of the impacts table and look for the tranche with the same sign in both. A rise in default probability with a *rise* in correlation would instead single out the senior.

Why the other options are wrong

A is incorrect. The senior is hurt by the higher default probability but helped by the lower correlation.

C is incorrect. The mezzanine's correlation effect switches sign with the default rate, so it is not hurt by both for certain.

D is incorrect. The effects offset for the senior, and possibly the mezzanine, but reinforce each other for the equity.

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 Means of the Distributions, Table 22.3

Solution CR-22 i.6 — Answer: D

Bank Q1173 • BT P2.T6.314.3

Step 1 – The two regimes

GARP: the mezzanine "behaves more like a senior bond at low default rates, when it is unlikely that losses will approach its attachment point" and "more like the equity tranche when default rates are high".

$\Rightarrow$ A and B are the two halves of the correlation effect and are true. At $\pi = 0.75\%$ its writedown rises from 0% to 4.84% with correlation; at $\pi = 9.75\%$ it falls from 103.92% to 52.87%.

Step 2 – Convexity

Negative convexity at low default rates, positive at high ones.

$\Rightarrow$ C is true.

Step 3 – Sensitivity at zero correlation

At $\rho = 0$ the mezzanine writedown goes 1.03%, 14.81%, 49.86%, 85.74% as π rises from 3.75% to

8.25%: once pool losses reach its 5% attachment point a thin tranche moves quickly from no loss to a total loss. Its default01 is the highest in GARP Table 22.5 at $\rho = 0$ (0.26% of par per bp at $\pi = 6.75\%$).
$\Rightarrow$ D is false.

Interpretation

Thinness is what makes the mezzanine so sensitive: conditional on any loss, the loss is large. With independent defaults the pool loss is close to its expectation, so the tranche's fate turns on whether that expectation has reached its attachment point.

Why the other options are wrong

A is incorrect. True: at low default rates higher correlation raises mezzanine losses, as for the senior.
B is incorrect. True: at high default rates higher correlation makes the few-default scenarios likelier, as for the equity.
C is incorrect. True: the convexity switches from senior-like (negative) to equity-like (positive).

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. •
Reference: Malz, Ch.9 (GARP CR Ch.22), 22.3 Means of the Distributions, Tables 22.3 and 22.5

Solution CR-22 i.7 Answer: C

Bank Q1171 • BT P2.T6.314.1

Step 1 – Mean values

A: more defaults reduce equity cash flows at any correlation. B: higher correlation raises the chance of very few defaults. Both true.

Step 2 – Convexity

D: at low correlation the equity loses value fast from a low default rate and then more slowly ("you can't beat a dead horse"). True.

Step 3 – Credit VaR

Higher correlation widens the distribution of equity values (more large losses and more near-zero losses), so the equity VaR *rises* with correlation: 1.62, 6.33, 6.85, 7.14 at $\pi = 0.75\%$. A higher default rate squeezes values towards zero, so the VaR generally *falls*: 6.33, 4.98, 1.23 at $\rho = 0.3$ for $\pi = 0.75\%$, 2.25%, 9.75%.
$\Rightarrow$ Both halves of C are reversed. C is false.

Interpretation

GARP: "Although the mean values of the equity tranche increase with correlation, so also does its risk." The fall in equity VaR at high default rates is not safety: the expected loss is already so high that little is left to lose.

Why the other options are wrong

A is incorrect. True: the default rate hurts the equity at every correlation.
B is incorrect. True: the equity is long correlation.
D is incorrect. True: positive convexity at low correlation.

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. •
Reference: Malz, Ch.9 (GARP CR Ch.22), 22.3 Credit VaR of the Tranches, Table 22.4

CR-22 • Structured Credit Risk • Vardeez: IMPORTANT

CR-22 I

P2 CONCEPT

Define implied correlation and describe how it can be measured.

Workbook: B3 Default Correlation, Joint PD and Copulas • Tier B D15 Structured Credit and Implied Correlation • Tier D

WHAT IS TESTED HERE

Why P2. Tier B workbook topic: B3 (Map #16, 6/16) reports "Gaussian copula, implied correlation" as shape S4, and D15 (Map #60) reports the concept again. Vardeez flags it inside an IMPORTANT reading: "Concept of LO 40.L, implied correlation, seems likely to be testable." **Vardeez:** implied correlation, likely testable. It works like implied volatility: the correlation that makes the model price equal the market price.

Reported exam shapes

- **B3·S4** and **D15·S4** Gaussian copula and implied correlation.
- **D15·S3** Implied correlation backed out of a tranche price.

What the questions look like: concept only. The direction of implied correlation when a tranche price moves and spreads do not; which inputs are observed and which one is solved for; risk-neutral versus physical inputs; why the values differ by tranche; the copula correlation against the other correlation concepts. Workbook D15.1 adds reading the grid of tranche values (D15·S1, S2).

Worked elsewhere: B3.3 (one-factor Gaussian copula calibrated to index tranches, conditional independence) under CR-12 f. Compound (tranche) versus base correlation and the skew they produce: CR-13 g.

1. What implied correlation is

CR-22 h and i run the model forward: assume a default probability and a correlation, simulate, read off tranche values. Implied correlation runs it **backwards**. GARP: "reverse engineer the modeling process, applying the model to observed market prices of structured credit products to estimate a default correlation." The result, the **implied credit** (or implied default) **correlation**, is a *risk-neutral* parameter, available wherever portfolio credit products are priced. Schweser's analogy: as one Black-Scholes-Merton volatility reproduces an option price, one copula correlation reproduces a tranche price.

2. Where the prices come from

CDX.NA.IG holds equal-notional CDS on 125 investment-grade names ($1m each per $125m); iTraxx covers Europe and Asia. Series start semiannually; the spread premium is fixed at initiation, so later trades exchange a present value. A default removes the name: notional and premium shrink by $1/125 = 0.8\%$. The **standard tranches** of a synthetic CDO on the index (equity 0 to 3%, junior mezzanine 3 to 7%, senior mezzanine 7 to 10%, senior 10 to 15%, super senior 15 to 30%) are liquid and quoted daily, unlike **bespoke** tranches issued "to order". So both inputs are observable: constituent CDS spreads and tranche prices.

Correction to the source notes. (1) GARP 22.4: with 40% recovery "9 defaults will leave just a small sliver of the equity tranche, and 10 defaults will entirely wipe it out". At 40% recovery a default costs

$0.6/125 = 0.48\%$ of notional, so the 0 to 3% equity ($3.75m on $125m) keeps $0.15m after 6 defaults and is wiped out by the 7th ($4.2m). GARP's 9 and 10 fit a 60% recovery ($3.6m and $4.0m). (2) GARP's illustration quotes a CDS-implied PD of "2 percent"; the Table 22.3 row it reads (4.98 at $\rho = 0.3$, 5.61 at $\rho = 0.6$) is $\pi = 2.25\%$, as in Workbook D15.1.

3. How it is measured

1. **Default probabilities:** *risk-neutral*, from each constituent's CDS spreads (along the term structure where available; recovery possibly from recovery swaps). Not physical PDs from ratings, as in the cash CLO.
2. **One correlation in a copula:** assume a constant pairwise ρ in a (usually Gaussian) copula, simulate the tranche cash flows and discount them at *risk-free* rates, not the tranche's own discount rate.
3. **Solve:** given the market tranche price (or points up front and running spread), find the unique ρ at which model value equals price.

Malz's example (Figure 2). Equity priced at about $5m with a CDS-implied PD of 2.25% matches $\rho = 0.3$; a rise to $5.6m with the PD unchanged means implied correlation has risen to 0.6, "an increase in the market's assessment of the systematic risk of the underlying loans". The equity is long correlation, so a higher price means a higher ρ.

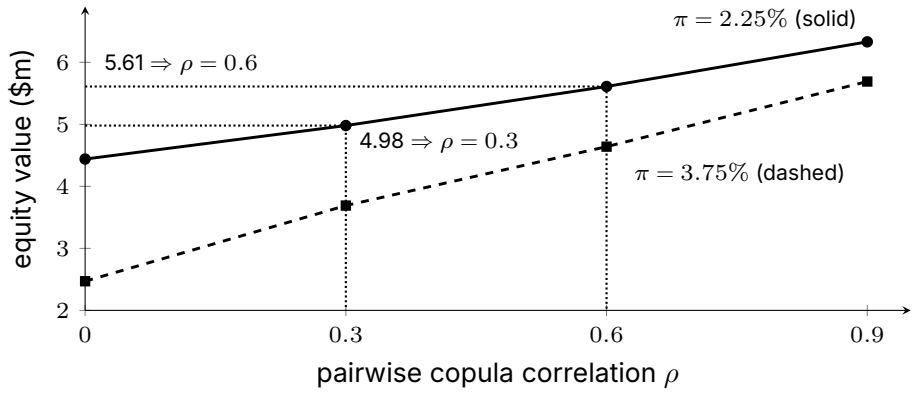


Figure 2. Backing implied correlation out of an equity price (GARP Table 22.3). With the CDS-implied PD fixed at 2.25% (solid), a price of 4.98 is matched only by $\rho = 0.3$ and a price of 5.61 only by $\rho = 0.6$ (dotted guides). On the dashed curve ($\pi = 3.75\%$) the same 4.98 would need $\rho \approx 0.7$: the PD must be pinned down from spreads before the correlation is solved for.

One value per tranche. Each tranche gives its own implied correlation. Malz calls it a **base correlation**, as it is tied to a tranche's attachment point; the values differ across tranches, the **correlation skew**. The constant pairwise correlation is an assumption *inside* each calibration, not a promise of one number for all tranches. Compound versus base correlation is CR-13 g.

4. Correlation concepts, and why this one is a market risk

	What it is	Nature
Default correlation	Correlation of two firms' default indicators	Credit; closest to portfolio credit risk
Asset return correlation	Correlation of log asset value changes (factor loadings)	Unobservable; proxied by equity
Equity return correlation	Correlation of log share price changes	Observable proxy
Copula correlation	Off-diagonal entries of the copula's correlation matrix	No direct economic meaning; Gaussian copula = single-factor correlation
Spread correlation	Correlation of spread changes (bp)	Market (mark-to-market)
Implied credit correlation	Copula correlation inferred from tranche prices	Market, risk-neutral

GARP: implied credit correlation "is as much a market-risk as a credit-risk concept". A standard tranche has a risk-neutral **partial spread01** to each of the 125 names rather than a default01 (CR-22 j), and implied correlation moves in its own right: both cause *mark-to-market* changes. In the cash CLO, default rates and correlation change expected cash flows: *fundamental* value.

Exam traps

- **Polarity flip.** Equity price up on unchanged spreads means implied correlation *up*; a senior tranche price falling on unchanged spreads means the same.
- **Physical for risk-neutral.** PDs from CDS spreads, not ratings or default history; discounting at risk-free rates.
- **Historical estimate.** Implied correlation is inferred from prices, not from joint defaults or equity returns.
- **Same for every tranche.** Values differ by tranche (skew); not a calibration error, and not averaged.
- **Wrong concept.** The copula correlation is not the default or the asset correlation. The Gaussian copula makes the transformed variables normal, not the default times (B3).

Practice questions: CR-22 I

Question CR-22 I.1

Bank Q1184 • Schweser Module Quiz 40.3 Q2 (rebuilt)

A risk manager who has just joined a correlation trading desk asks how the desk computes the implied correlation of a securitized tranche. Which of the following statements best describes the calculation?

A. The implied correlation is computed separately for each tranche using a constant pairwise correlation, so the resulting values are identical across all tranches.

B. Observed CDS prices are inserted directly into the pricing function as the tranche values, and no correlation is solved for.

C. The tranche pricing function is calibrated so that the model price matches the observed market price of the tranche, with the correlation as the unknown.

D. Historical (real-world) default probabilities are fed into the copula, and the correlation is chosen to match the market tranche price.

Question CR-22 I.2

Bank Q1183 • BT P2.T7.507.2 (rebuilt)

An analyst backs out the implied correlation of the CDX equity tranche using a Gaussian copula, constant pairwise correlation, and risk-neutral default probabilities extracted from the constituent CDS spreads. The equity tranche market price rises from $3.0 million to $3.2 million while the constituent CDS spreads (and hence the risk-neutral default probabilities) are unchanged. Which of the following is the most appropriate conclusion?

A. The implied correlation has increased, because equity holders benefit from higher correlation.

B. The implied correlation has decreased, because lower correlation reduces the probability of large clustered losses.

C. The implied correlation is unchanged, because the default probabilities are unchanged.

D. The risk-neutral default probabilities have fallen, so the model price no longer needs a correlation adjustment.

Question CR-22 I.3 New • built from GARP Ch.22, 22.4 Implied Correlation and Summary of Default Correlation Concepts • shape B3·S4, D15·S4

A model validation analyst at a Singapore bank is reviewing the note on the correlation parameter of the bank's tranche pricing model. The model joins the default-time distributions of the 125 CDX.NA.IG constituents with a Gaussian copula and a single constant pairwise correlation, which is recalibrated each day so that modelled standard tranche values match market prices. Which of the following statements in the note is correct?

A. The parameter is the constituents' default correlation, the correlation of their default indicators, so it can be checked against historical joint default frequencies.

B. The parameter is a copula correlation inferred from prices; it has no direct economic interpretation, but it equals the correlation of a Gaussian single-factor model.

C. The parameter is the constituents' asset return correlation, so it can be measured directly as the correlation of their equity returns.

D. The parameter is a physical (real-world) correlation, because it is estimated from the observed prices of liquid traded tranches.

Question CR-22 I.4 Bank Q1185 • New question written for this bank

An analyst backs out implied correlations for the CDX equity (0 to 3%), junior mezzanine (3 to 7%), and senior mezzanine (7 to 10%) tranches using a Gaussian copula with constant pairwise correlation and the same risk-neutral default probabilities extracted from the constituent CDS spreads. The calibrated values are different for each tranche. Which of the following is the most appropriate interpretation?

A. The differing values should be averaged, since the average is the true implied correlation of the index portfolio.

B. The differing values show a calibration error, because a constant pairwise correlation implies the same value must match every tranche.

C. The differing values arise because the analyst used historical default probabilities for some tranches and risk-neutral probabilities for others.

D. The differing values reflect a correlation skew; each implied (base) correlation is the value that makes the model price match the market price of that tranche.

Question CR-22 I.5 New • built from GARP Ch.22, 22.4 Credit Index Default Swaps and Standard Tranches; Implied Correlation

The head of a correlation trading desk explains to a new market risk manager why the desk's standard CDX tranche positions are risk-managed differently from the bank's holdings of cash CLO tranches. Which of the following statements is correct?

A. A standard tranche has a partial spread01 to each of the 125 constituents, and moves in spreads or in implied correlation cause mark-to-market changes; in a cash CLO, default rate and correlation changes alter fundamental value.

B. A standard tranche is valued with physical default probabilities taken from the constituents' ratings, so its main sensitivity is a default01, measured by shocking those probabilities by 10 bp.

C. Implied correlation is a pure credit-risk parameter that is fixed when a standard tranche is first traded, so it causes no mark-to-market changes in the desk's positions.

D. The CDX spread premium is reset every day to the fair spread of the constituents, so standard tranche positions carry no exposure to changes in constituent spreads.

Question CR-22 I.6 Workbook D15.1 • S1, S2, S3, S4 • Map #60

A structured credit analyst at an Australian superannuation fund monitors the fund's equity tranche in a CLO with the same structure as the one simulated in the reading. For loans with a one-year default probability of 2.25%, the simulation grid gives (equity value in AUD million; bond writedowns in percent of tranche par):

Copula correlation ρ	0.00	0.30	0.60	0.90
Equity value	4.44	4.98	5.61	6.33
Mezzanine writedown	0.00	7.35	12.82	15.49
Senior writedown	0.00	0.52	2.14	5.05

The market price of the equity tranche has risen from AUD 4.98 million to AUD 5.61 million, while the risk-neutral default probability implied by the loans' CDS spreads has stayed at 2.25%. Which of the following statements should the analyst include in the report to the investment committee?

A. The implied correlation has fallen from 0.60 to 0.30, because a higher equity price signals that defaults are now expected to be spread more evenly across scenarios.

B. The implied correlation has risen from 0.30 to 0.60, and the mezzanine bond's expected writedown has fallen, because the mezzanine tranche behaves like the equity tranche at any default rate.

C. The implied correlation has risen from 0.30 to 0.60, reflecting a higher market assessment of systematic risk, and the senior bond's expected writedown rises from 0.52% to 2.14% of par.

D. The implied correlation has risen from 0.30 to 0.60, and because it is estimated from rating-based physical default probabilities, it is a credit risk rather than a market risk parameter.

Solutions • CR-22 I

Solution CR-22 I.1 — Answer: C

Bank Q1184 • Schweser Module Quiz 40.3 Q2

Step 1 – What is observed

Two sets of prices: the constituent CDS spreads, which give risk-neutral default probabilities (and possibly recoveries), and the tranche prices.

Step 2 – What is solved for

Assume a constant pairwise correlation in the copula, feed in the default probabilities, and find the unique correlation at which the model tranche price equals the market price.

$\Rightarrow$ Calibration with the correlation as the unknown, exactly like backing an implied volatility out of an option price.

Interpretation

Schweser: "Starting with observed market prices and a pricing function for the tranches, it is possible to back out the unique implied correlation to calibrate the model price with the market price."

Why the other options are wrong

A is incorrect. The constant pairwise correlation is assumed within each calibration, but the solved values differ across tranches (correlation skew).

B is incorrect. CDS prices give the default probabilities, not the tranche values; tranche prices are observed separately and the correlation is solved for.

D is incorrect. The default probabilities are risk-neutral, from CDS spreads, not historical (sibling swap with the cash CLO simulation).

LO: CR-22 l: Define implied correlation and describe how it can be measured. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.4 Implied Correlation

Solution CR-22 I.2 — Answer: A

Bank Q1183 • BT P2.T7.507.2

Step 1 – Only one input can move

The CDS spreads, and so the risk-neutral default probabilities, are unchanged. The correlation is the only free input in the model.

Step 2 – Which way

The equity tranche is long correlation: higher correlation raises the chance of a scenario with very few defaults, which the first-loss holder values. A price rise from 3.0 to 3.2 therefore needs a higher correlation.

$\Rightarrow$ Implied correlation has increased.

Interpretation

This is Malz's own example in miniature (equity from \$5.0m to \$5.6m, correlation from 0.3 to 0.6). The precise value comes from the pricing model; the direction comes from the impacts table of CR-22 i.

Why the other options are wrong

B is incorrect. Polarity flip: lower correlation lowers the equity value, so it cannot explain a rise.

C is incorrect. Correlation is one of the two model inputs; with the other fixed, it must move when the price moves.

D is incorrect. Contradicts the stem: the spreads, and so the default probabilities, have not

changed.

LO: CR-22 l: Define implied correlation and describe how it can be measured. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.4 Implied Correlation

Solution CR-22 I.3 Answer: B

New • built from GARP Ch.22, 22.4

Step 1 – Name the parameter

A correlation entered in a copula and calibrated to market prices is an **implied credit correlation**: GARP calls it "an estimate of the copula correlation derived from market prices", not a distinct theoretical concept.

Step 2 – What a copula correlation is

Copula correlations are "the values entered into the off-diagonal cells of the correlation matrix" of the copula. They "have no direct economic interpretation", but "the correlation of a Gaussian copula is identical to the correlation of a Gaussian single-factor factor model".

$\Rightarrow$ B is correct.

Interpretation

Because it is backed out of prices with risk-neutral default probabilities, the parameter is a risk-neutral, market-based number. Like spread correlation it is a market rather than credit risk concept, and it moves every day with the tranche prices.

Why the other options are wrong

A is incorrect. Sibling swap: default correlation is the correlation of default indicators, a different concept, and is not what the calibration produces.

C is incorrect. Sibling swap: asset return correlation drives Merton-type models and is only proxied by equity correlation; neither is the calibrated copula parameter.

D is incorrect. Measure flip: a parameter calibrated to traded prices with CDS-implied default probabilities is risk-neutral, not physical.

LO: CR-22 l: Define implied correlation and describe how it can be measured. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.4 Implied Correlation; Summary of Default Correlation Concepts

Solution CR-22 I.4 Answer: D

Bank Q1185 • New question written for this bank

Step 1 – One calibration per tranche

Each tranche's implied correlation is the unique value that makes its model price equal its own market price.

Step 2 – Why they differ

The single-correlation Gaussian copula cannot fit all tranche prices with one number, so the solved values vary by tranche. GARP: "Correlations generally vary by tranche, a phenomenon called correlation skew."

$\Rightarrow$ D is correct.

Interpretation

Malz calls the tranche-by-tranche value a base correlation because it is associated with a tranche's attachment point. How compound (tranche) and base correlations differ, and the shapes they trace, is CR-13 g.

Why the other options are wrong

A is incorrect. There is no meaningful average: each value is the one that matches that tranche's

price.
B is incorrect. Constant pairwise correlation is an assumption inside each calibration, not a guarantee that every tranche returns the same value; the differences are the market's.
C is incorrect. The stem says all tranches use the same risk-neutral probabilities; the differences do not come from mixing measures.

LO: CR-22 l: Define implied correlation and describe how it can be measured. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.4 Implied Correlation

Solution CR-22 I.5 Answer: A

New • built from GARP Ch.22, 22.4

Step 1 – The standard tranche's sensitivities

GARP: "The value of each tranche has a distinct risk-neutral partial spread01, rather than a default01, that is, sensitivities to each of the constituents of the IG 125." Each spread01 is itself a function of the implied correlation, which "varies in its own right".
⇒ Spreads and implied correlation both move the mark-to-market value.

Step 2 – The cash CLO

In the cash CLO, "changes in default rates and correlation result in changes in expected cash flows and credit losses", that is, changes in fundamental value.
⇒ A is correct: implied credit correlation is as much a market-risk as a credit-risk concept.

Interpretation

This is why a correlation desk reports spread01s and correlation sensitivities, while a buy-and-hold CLO investor watches default rates and the waterfall.

Why the other options are wrong

B is incorrect. Standard tranches are valued with risk-neutral default probabilities from CDS spreads; the default01 is the cash CLO's measure (CR-22 j).
C is incorrect. Implied correlation moves with tranche prices and is a market concept; its changes are mark-to-market changes.
D is incorrect. The CDX spread premium is fixed on the initiation date, so later trading exchanges a present value that moves with the constituent spreads.

LO: CR-22 l: Define implied correlation and describe how it can be measured. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.4 Credit Index Default Swaps and Standard Tranches; Implied Correlation

Solution CR-22 I.6 Answer: C

Workbook D15.1 • S1, S2, S3, S4 • Map #60

Step 1 – Back out the implied correlation

With the CDS-implied default probability fixed at 2.25%, the only free input is the copula correlation. An equity value of 4.98 matches $\rho = 0.30$; a value of 5.61 matches $\rho = 0.60$.
⇒ The market now prices the equity as if defaults were more clustered: implied correlation has risen from 0.30 to 0.60.

Step 2 – Read the other tranches at the same default probability

Senior writedown: $0.52\% \rightarrow 2.14\%$ of par (up). Mezzanine writedown: $7.35\% \rightarrow 12.82\%$ (up).
⇒ At this low default rate the mezzanine sits with the senior: higher correlation hurts both bonds while it helps the equity.

Interpretation

C is correct. Higher correlation means a low but material probability of a state with unusually many defaults, the systematic risk that reaches the senior bond even on a diversified pool, while the equity, now a "lottery ticket", gains from the higher chance of a low-default state.

Why the other options are wrong

A is incorrect. Polarity flip. Equity value increases with correlation at this default rate (4.98 at 0.30, 5.61 at 0.60), so a higher price means a higher implied correlation. Evenly spread defaults describe low correlation, which is bad for the equity.

B is incorrect. Scope/condition error. The mezzanine behaves like the equity only at high default rates; at $\pi = 2.25\%$ its writedown rises from 7.35% to 12.82%.

D is incorrect. Sibling swap. Implied correlation is computed from risk-neutral default probabilities taken from single-name CDS spreads, discounting at risk-free rates; Malz calls implied credit correlation "as much a market-risk as a credit-risk concept", and each standard tranche's spread01s measure a market rather than a credit risk.

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. CR-22 l: Define implied correlation and describe how it can be measured. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 Means of the Distributions, Table 22.3; 22.4 Implied Correlation

CREDIT RISK • PART

P3

Third priority

Tier C and D topics, or LOs Vardeez flags in other chapters. 33 LOs from 15 readings.

CR-2 • Governance • Vardeez: NOT SO IMP

CR-2 c

P3 CONCEPT

Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight.

Workbook: none (no workbook section for CR-2) • Vardeez focus

WHAT IS TESTED HERE

Why P3. No workbook section and no reported shape; Vardeez flags the LO inside a reading he rates NOT SO IMP: "Just read this carefully (key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight)".
Vardeez: Guidelines, Skills, Limits, Oversight: read carefully.

Reported exam shapes: none listed in the map.

What the questions look like: "which arrangement is most consistent with best practice" statements, one per principle, with distractors that give risk management approval power, make guidelines legalistic or vague, or trade independence for isolation. One question reads a delegation-of-authority table (a lookup, no arithmetic). The three lines and the roles of the board, CEO and CRO are under CR-2 a.

1. Four principles, one purpose

GARP: best practice for the governance system rests on four principles, "critical to the quality of what gets originated" (Figure 1). All four aim to keep bad transactions out of the book, because no later portfolio management can repair a bad origination (CR-2 a).

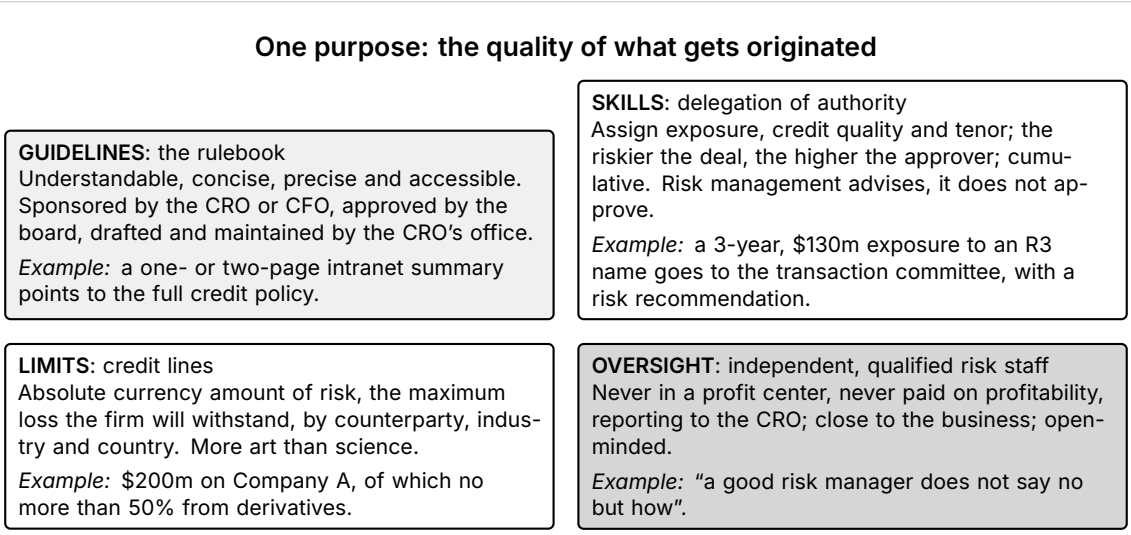


Figure 1. The four principles, each with its core requirement and one example from GARP Ch.2. Guidelines light grey, Oversight darker grey, Skills and Limits white.

2. Guidelines: the rulebook

Guidelines (credit policies, risk management standards) state the rules to be complied with *before* a transaction is concluded.

- **Four characteristics.** *Understandable*: plain language, an internal document that enables compliance, not a legal one. *Concise*: too long and no one reads them. *Precise*: brevity must not make them so general that bad deals slip through. *Accessible*: staff know where the current version is; a one- or two-page intranet summary helps.
- **Approval and ownership.** Sponsored by a senior executive (CRO or CFO) and **approved by the board**; a listed firm's audit committee reviews risk policies, and a board risk committee usually oversees and approves them. The CRO's office drafts them, seeks their approval, promulgates and maintains them, through an author senior enough for the knowledge, the politics and the board.
- **Review.** Periodically, and when the firm enters a new business, markets change, operations change (M&A), after any loss, even a minor one, and after events never contemplated (GARP: negative oil prices in April 2020, when sellers paid buyers $37 a barrel).
- **Content.** Purpose; how key parameters are defined; approval and delegation; new products and markets; review process; consequences of non-compliance.
- **Breach.** Rare and serious, often ending in termination. Carve-outs cover breaches outside the originator's control, such as an FX move that pushes an exposure over its limit. A central exposure database allows pre-trade checks and rejects, or documents, unauthorized trades.

> **Correction to the source notes.** Vardeez (CR-2 c, "Maintenance and content") has the guidelines "drafted, approved, and maintained" within the CRO's office, following SchweserNotes LO 20.c ("The CRO's office is responsible for drafting, approving, and maintaining the guidelines"). GARP says the CRO's office will "draft, seek approval for, promulgate, and maintain" them, and that approval "must be done by the Board of Directors". The CRO's office prepares and maintains; the board approves. Tested in Question CR-2 c.2.

3. Skills: delegating authority

The board cannot approve every transaction, so its authority is delegated down. Risk management's priorities are (1) not to lose money, (2) to make money; the business reverses the order, which is why the delegation rules cause friction.

- **Risk management advises.** Not a profit center, **no approval authority**. Guidelines say when its recommendation is required; a veto is rare; strong dissent goes into a written memo in the committee package, presented in person.
- **Two steps, one direction.** Assign the parameters (**amount of exposure, credit quality, tenor**), then delegate on them: the riskier the deal, the higher the approver, from originator to business head, transaction committee and credit committee. Deals too complex to parameterise go straight to a high-level body.
- **Cumulative.** Every intervening level approves; no jumping from the originator to the committee.
- **GARP's example** (Figure 2.2). Three years, R3 counterparty, $130 million exposure: the transaction committee's R3 authority runs to $150 million, so it approves; credit risk's recommendation is required above $100 million for R3, so that is needed too (Question CR-2 c.5).

4. Limits: credit lines

Limits are the absolute currency amount of risk the firm will take, its maximum loss, set by counterparty, industry and country. Where the loss on default depends on future markets (long-term commodity contracts, derivatives), combinations are used: $200 million on Company A, at most 50% from derivatives. Setting them is "more art than science": VaR models help, but experience often prevails. Management judges that a $150 million loss would be acceptable to shareholders, rating agencies and regulators, makes that the absolute limit for any counterparty, and accepts much less on weaker names (Schweser and Vardeez use $75 million). Originators prefer preapproved limits, allocated across business units in advance.

5. Oversight: independent and close

- **Independence: two rules.** Never located in a business unit with a profit center; pay never based on the business's profitability. The chain of command ends at the CRO, who usually reports to the CEO and has privileged access to the board's risk or audit committee.
- **Qualifications.** Risk managers must speak the business's language; a weak one is consulted late, only for the required recommendation.
- **Proximity.** They sit near operations and may join client meetings, but with a firm walk-away position (they are not seasoned negotiators), and without getting so close that independence suffers.
- **Open mind.** "A good risk manager does not say no but how."

	What best practice requires	Typical wrong option
Guidelines	Plain, short, precise, easy to find; CRO's office drafts and maintains, board approves; reviewed periodically and after losses	Legal language covering every case; vague to give flexibility; CRO alone approves; revised only when regulators ask
Skills	Parameters first (exposure, quality, tenor), then authority; higher approver for riskier deals; cumulative; risk advises	Risk management approves or vetoes; seniority of the originator decides; long tenor or low quality approved lower
Limits	Absolute maximum loss, overall and by counterparty, industry, country; analysis plus judgment	Models only; counterparty limits only; set by originators
Oversight	Outside profit centers, pay independent of profit, report to the CRO; close to the business; open mind	Inside the profit center; report to business heads; independence read as isolation

Exam traps

- **Who approves the guidelines.** The board; the CRO's office drafts, seeks approval for and maintains them.
- **Legal, exhaustive or vague guidelines.** The first two fail "understandable" and "concise"; vagueness fails "precise".
- **Risk management with approval power.** It advises and recommends; it rarely has a veto.
- **Skipping levels or reversing the ladder.** The lowest body whose limit covers the deal approves, *and* every level below it; long tenor and low quality go *higher*.
- **Independence read as isolation.** Risk managers must stay close to the business.

Practice questions: CR-2 c

Question CR-2 c.1

Bank Q31 • AnalystPrep Q.5804 (rebuilt)

A bank's CRO is redrafting the credit guidelines after a review found that originators rarely consult them. Which design is most consistent with best practice for guidelines?

A. Formal legal language for enforceability, with enough detail to cover every conceivable scenario.

B. Short, general principles, with the precise interpretation left to each business unit's judgment.

C. Plain language of reasonable length, specific enough to address real-life cases, with a short intranet summary.

D. Full text restricted to senior credit officers, with junior staff briefed verbally on what applies to them.

Question CR-2 c.2

Bank Q33 • AnalystPrep Q.5814 • AnalystPrep Q.5816 (rebuilt)

A bank listed on a major stock exchange is redesigning the governance process for its credit guidelines. Which arrangement is most consistent with best practice?

A. The head of origination drafts and approves the guidelines, since the business units know the products best, and revises them only when regulators require it.

B. The CRO's office drafts and maintains the guidelines, a senior executive sponsors them, the board approves them, and they are reviewed periodically and after losses.

C. Junior risk staff draft the guidelines to keep them simple, the CRO alone approves them, and they are reviewed once every ten years to keep them stable.

D. An external consultant drafts, approves and reviews the guidelines, so that the bank's own staff can concentrate on running day-to-day operations.

Question CR-2 c.3

Bank Q35 • Schweser Module Quiz 20.2 Q1

A firm's risk management unit is updating its guidelines relating to transaction approval and delegation. Risk managers are worried that even when transactions fall within available credit limits, foreign exchange volatility may cause breaches in these limits. Which of the following guideline parameters would be most appropriate to deal with this type of outcome?

A. The guidelines should include a carve-out for breaches that are due to foreign exchange rate volatility.

B. Breaches in the credit limit due to foreign exchange rate movements should be immediately escalated to the Chief Risk Officer (CRO).

C. The guidelines should treat each breach of the credit limit as an error by the originator, because currency movements should be anticipated.

D. Breaches of the credit limit due to foreign exchange rate movements should be immediately escalated to the risk committee of the board of directors.

Question CR-2 c.4

Bank Q38 • AnalystPrep Q.5810 (rebuilt)

A bank is designing the delegation of authority for its credit approvals. Which approach is most consistent with best practice?

A. Send every transaction, whatever its size, to the credit committee, so that all deals are treated uniformly.

B. Link approval authority to the originator's seniority, so that the most senior originators can approve any deal.

C. Allow lower approval levels for long-tenor and low-quality deals, because these transactions earn higher margins.

D. Assign exposure, credit quality and tenor to each deal, then set the approval level from them, higher for riskier deals.

Question CR-2 c.5

Bank Q30 • AnalystPrep Q.5950 (rebuilt) • re-filed from CR-2 b

A bank's approval process sets the following maximum exposure (USD millions) that each level can approve for a transaction with a tenor under five years. Delegation is cumulative, and the credit risk assessment unit must give a recommendation on every R3 transaction above USD 100 million.

Rating	Head of Trading	Transaction Committee	Executive Risk Committee
R3	75	150	250

A three-year transaction with an R3 counterparty generates an exposure of USD 130 million. Which approval path is required?

A. Head of Trading only.

B. Transaction Committee only, with no risk assessment recommendation.

C. Head of Trading and Transaction Committee, with a recommendation from the credit risk assessment unit.

D. Head of Trading, Transaction Committee and Executive Risk Committee, with a recommendation from the credit risk assessment unit.

Question CR-2 c.6

Bank Q34 • AnalystPrep Q.5817 (rebuilt)

A senior credit risk manager strongly disagrees with a large transaction that the originating unit will present to the credit committee next week. The bank's guidelines follow GARP's best practice. What should the risk manager do?

A. Attach a written memo setting out the objection to the committee package and present the opinion in person at the meeting.

B. Block the transaction by withholding sign-off, since risk management holds a veto over the transactions it opposes.

C. Approve the transaction under protest, since risk management carries approval authority for deals in its own area.

D. Report the transaction to the external auditors before the meeting, since dissent must be resolved outside management.

Question CR-2 c.7 Bank Q37 • AnalystPrep Q.5809 (rebuilt)

Which of the following statements about credit limits within a firm's governance framework is most accurate?

A. Limits are set solely from model outputs, with management judgment excluded so that they remain objective.

B. Limits cap the loss the firm will accept in absolute currency terms, with an overall limit and smaller sub-limits, set by analysis and judgment.

C. Limits apply only to individual counterparties, because an aggregate firm-wide limit cannot be defined.

D. Limits are set by each originator, who knows the clients best, and are never preapproved, so as not to slow deals.

Question CR-2 c.8 Bank Q32 • AnalystPrep Q.5812 (rebuilt)

An internal audit review of a bank's credit risk function lists four possible arrangements. Which one best reflects effective oversight of credit risk?

A. Risk managers sit inside the profit center to stay close to originators, and their bonuses depend on the unit's profitability.

B. Risk managers report to the head of the business unit that sponsors the transactions, since that person best understands the risks.

C. Risk managers avoid all contact with business units and clients, so that their judgment can never be influenced.

D. Risk managers sit outside profit centers, are paid independently of business profitability, and report to the CRO, who has board access.

Solutions • CR-2 c

Solution CR-2 c.1 — Answer: C

Bank Q31 • AnalystPrep Q.5804

Step 1 – The four characteristics

Guidelines must be understandable (plain language), concise (reasonable length), precise (specific enough to handle real-life situations) and accessible (staff know where to find them; a one- or two-page intranet summary helps).

Step 2 – Test each design

Only C delivers all four. A fails "understandable" and "concise"; B fails "precise"; D fails "accessible".

$\Rightarrow$ Plain, short, specific and easy to find.

Interpretation

Originators ignoring the guidelines is a symptom of documents that are too long, too legal or too hard to find. GARP's fix is a usable document, not a longer one: guidelines exist to enable compliance, not to build a legal case against an employee.

Why the other options are wrong

A is incorrect. Legal language and exhaustive length are exactly what makes guidelines unread; they are an internal document, not a contract.

B is incorrect. Brevity must not cost precision; vague guidelines let bad transactions "fall through the cracks".

D is incorrect. Everyone who needs the guidelines must be able to find the current version.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. • **Reference:** Bouteille and Coogan-Pushner, The Handbook of Credit Risk Management 2e, Ch.2 (GARP CR Ch.2), Guidelines

Solution CR-2 c.2 — Answer: B

Bank Q33 • AnalystPrep Q.5814 • AnalystPrep Q.5816

Step 1 – Ownership and drafting

The CRO's office owns the guidelines: it drafts them, seeks approval for them, promulgates and maintains them, using a senior author who knows the business.

Step 2 – Sponsorship and approval

A senior executive (CRO or CFO) sponsors them and the board approves them. For a listed firm the board's audit committee reviews risk policies; a board risk committee typically oversees and approves them.

Step 3 – Review

Periodic review, plus event-driven review: new business areas, market change, M&A, and any loss, even a minor one.

$\Rightarrow$ B matches every element.

Interpretation

The executive management is accountable for what the guidelines permit, which is why approval sits at board level. The CRO's office prepares and maintains; it does not approve (the correction in Section 2).

Why the other options are wrong

A is incorrect. Originators face a conflict of interest in writing their own rules, and review should

not wait for regulators.
C is incorrect. Drafters need seniority and expertise, the board approves, and ten years is far too infrequent.
D is incorrect. Management stays accountable; outsourcing drafting, approval and review is not best practice.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. • **Reference:** GARP CR Ch.2, Guidelines: Creation and Approval Process; Promulgating and Maintaining Guidelines

Solution CR-2 c.3 Answer: A

Bank Q35 • Schweser Module Quiz 20.2 Q1

Step 1 – Is the breach the originator's doing?

The transaction was within the limit when it was done; an exchange-rate move later pushed the exposure over it.

Step 2 – How guidelines handle it

GARP: guidelines can have carve-outs for variables such as foreign-exchange volatility, for example by expressing limits on a local currency basis, since FX management sits with the treasury function.
⇒ A carve-out written into the guidelines.

Interpretation

Breaches should be rare and are serious, often leading to termination. That is why guidelines must separate a deliberate breach from one caused by a market variable outside the originator's control.

Why the other options are wrong

B is incorrect. Escalating every FX-driven breach to the CRO is unnecessary once the guidelines carve it out.
C is incorrect. Treats an unintentional, external breach as misconduct.
D is incorrect. Escalating routine, tolerated breaches to the board's risk committee is excessive.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. • **Reference:** GARP CR Ch.2, Guidelines: Breach of Guidelines; Schweser-Notes 2026 Book 2, Reading 20, LO 20.c

Solution CR-2 c.4 Answer: D

Bank Q38 • AnalystPrep Q.5810

Step 1 – The two-step process

Step 1: assign each transaction its fundamental parameters (amount of exposure, credit quality, tenor). Step 2: delegate approval authority based on those parameters.

Step 2 – The direction of the rule

The riskier the transaction, the higher the approval level: high exposure, low quality or long tenor goes to senior committees; small, short, high-quality deals can be approved low down.
⇒ D.

Interpretation

Delegation puts each decision with people whose skills and seniority match its risk, and keeps the senior committees free for the deals that need them.

Why the other options are wrong

A is incorrect. Impractical: the board and its committees cannot approve everything, which is why authority is delegated.
B is incorrect. Authority follows transaction risk, not the originator's rank, and an originator approving his own deals is a conflict of interest.
C is incorrect. Reversed: long tenor and low credit quality require a higher approval level.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. • **Reference:** GARP CR Ch.2, Skills: Delegation of Authority

Solution CR-2 c.5 — Answer: C

Bank Q30 • AnalystPrep Q.5950 • re-filed from CR-2 b

Step 1 – Find the lowest level whose limit covers the exposure

USD 130 million exceeds the Head of Trading's R3 limit of 75 and is within the Transaction Committee's limit of 150.
⇒ The Transaction Committee is the approving body; the Executive Risk Committee (250) is not needed.

Step 2 – Apply cumulative delegation

Each intervening level must approve, so the Head of Trading approves before the Transaction Committee.

Step 3 – Check the recommendation trigger

USD 130 million is above the USD 100 million threshold for R3, so the credit risk assessment unit's recommendation is required.

Interpretation

This is GARP's own Figure 2.2 example (three-year tenor, R3, $130 million: transaction committee authority up to $150 million, risk recommendation above $100 million). The recommendation is advice the guidelines require, not an extra approval.

Why the other options are wrong

A is incorrect. USD 130 million exceeds the Head of Trading's limit of 75.
B is incorrect. Skips the cumulative lower level and the recommendation required above USD 100 million.
D is incorrect. Escalates unnecessarily: 130 is within the Transaction Committee's limit of 150.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. CR-2 b: Explain the processes that lead to risk taking including credit origination, credit risk assessment, and credit approval processes. • **Reference:** GARP CR Ch.2, Skills: Delegation of Authority, Figures 2.1 and 2.2

Solution CR-2 c.6 — Answer: A

Bank Q34 • AnalystPrep Q.5817

Step 1 – Does risk management decide?

No. It is not a profit center, has no approval authority and plays an advisory role; a veto right is rare.

Step 2 – How dissent is expressed

GARP: a risk manager who strongly disagrees must have a platform. A written memo is attached to the documentation package for the credit committee, and the risk manager sits on the committee and presents the opinion in person.
⇒ A.

Interpretation

The memo puts the dissent on the record that management and auditors will read first if the deal goes wrong, while the decision stays with the body that holds the authority.

Why the other options are wrong

B is incorrect. Risk management rarely has a veto; withholding sign-off to block a deal claims power it does not have.

C is incorrect. Risk management has no approval authority at all, under protest or otherwise.

D is incorrect. External auditors give independent assurance (third line); they are not a channel for resolving a credit decision.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. • **Reference:** GARP CR Ch.2, Skills

Solution CR-2 c.7 Answer: B

Bank Q37 • AnalystPrep Q.5809

Step 1 – What a limit is

The absolute currency amount of risk a company wants to take: the maximum loss it is prepared to withstand. Limits are frequently called credit lines.

Step 2 – How they are structured and set

An overall limit with smaller limits by counterparty, industry or country, and combinations for uncertain exposures. Setting them is "more art than science": models help, but experience and judgment often prevail.

$\Rightarrow$ B.

Interpretation

GARP's illustration: management judges a $150 million loss acceptable to shareholders, rating agencies and regulators, sets that as the absolute limit for any counterparty, and accepts much smaller exposures on weaker names.

Why the other options are wrong

A is incorrect. Management experience and judgment are explicitly part of limit setting.

C is incorrect. Firms have aggregate limits as well as counterparty, industry and country sub-limits.

D is incorrect. Originators do not set their own limits; they prefer preapproved limits so that they can market to clients.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. • **Reference:** GARP CR Ch.2, Setting Limits

Solution CR-2 c.8 Answer: D

Bank Q32 • AnalystPrep Q.5812

Step 1 – Independence

Two rules: risk management is never located in a business unit with a profit center, and a risk manager's pay is never based on the business's profitability. The chain of command ends at the CRO, who reports to the CEO and has privileged access to the board's risk or audit committee.

Step 2 – Closeness without capture

Risk managers must still understand the business, sit near operations and may join client meetings, with a firm walk-away position.

$\Rightarrow$ D satisfies independence without isolation.

Interpretation

Oversight needs independence, qualifications, proximity and an open mind together. A and B fail independence; C overcorrects into isolation and loses the business understanding GARP requires.

Why the other options are wrong

A is incorrect. A risk manager inside the profit center and paid on its profitability is not independent.

B is incorrect. Reporting to the sponsoring business head creates exactly the conflict of interest the CRO reporting line avoids.

C is incorrect. Independence does not mean isolation; risk managers need to understand the business, the clients and the transaction structures.

LO: CR-2 c: Discuss the following key principles underlying best practice for the governance system of credit risk: Guidelines, Skills, Limits, and Oversight. • **Reference:** GARP CR Ch.2, Oversight: Independence; Qualifications; Proximity to the Business Unit

CR-3 • Credit Risk Management • Vardeez: NOT SO IMP

CR-3 e

P3 CONCEPT

Describe and explain loan loss provisions and loan loss reserves.

Workbook: none (no workbook section for CR-3) • Vardeez focus

WHAT IS TESTED HERE

Why P3. No workbook section and no reported shape; Vardeez singles out three LOs in a reading he rates NOT SO IMP: "read these 3 LOs carefully. LO 19.e & LO 19.g; and LO 19.h". His "19" is this reading (GARP Ch.3); the letters match CR-3 e, g and h.
Vardeez: loan loss provisions and reserves.

Reported exam shapes: none listed in the map.

What the questions look like: statement questions. Which is the flow and which the stock, what an adequate reserve must take into account, and where impairments, provisions and write-offs land in the financial statements. No calculations. This block also carries the background for CR-3 g and CR-3 h.

1. Where provisions come from

GARP (van Greuning and Brajovic Bratanovic, Ch.3): "asset classification provides a basis for determining an adequate level of provisions for possible loan losses."

Category	Meaning	Delinquency rule
Standard (pass)	Debt service beyond doubt; fully cash-secured loans qualify whatever the arrears	none
Specially mentioned	Potential weaknesses that could later jeopardise repayment	none
Substandard	Well-defined weaknesses; the bank looks to collateral or other secondary sources	nonperforming, typically 90 days or more
Doubtful	Collection in full questionable; loss possible	180 days or more, unless sufficiently secured
Loss	Uncollectible; not worth deferring the write-off, though some recovery may come	one year or more, unless very well secured

- **Expected loss sets the provision.** EL $=$ PD $\times$ LGD $\times$ EAD (CR-3 f): who you lent to, what product, how long since the loan began. GARP: "the EL calculation is the basis for performing the book provisions."
- **Unexpected loss does not.** UL is a point in the tail of the loss distribution (for example the 99th percentile). It is covered by capital (CR-4), not by provisions.

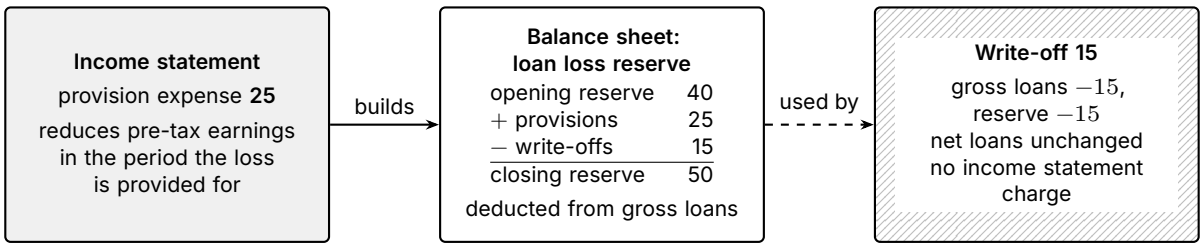
2. Provisions versus reserves

- **Loan loss provision** (a *flow*). The expense set aside in each period, charged to the income statement, to cover potential future losses on loans. It comes out of the classification process.
- **Loan loss reserve** (a *stock*; also called the allowance). The loan loss provisions accumulated over time, carried on the balance sheet as a deduction from gross loans.
- **Write-off.** A loss asset removed from the books "against the reserve": gross loans and the reserve fall by the same amount, so net loans do not change. The earnings hit came when the provision

was made. So the reserve is accumulated provisions *less* write-offs:

$$\text{closing reserve} = \text{opening reserve} + \text{provisions for the period} - \text{write-offs}$$

- **Impairment recognition** (GARP 3.3). When a bank is unlikely to collect the amounts due, it reduces the loan's carrying amount to its estimated realizable value, either through an existing allowance or by charging the income statement in the period of impairment. A restructured (renegotiated) loan is written down to net realizable value, counting the cost of all concessions, with the charge in the period of restructuring (Question CR-3 e.6).
- **General versus specific.** Specific provisions sit against identified problem assets. GARP: provisions, "together with general loss reserves that are normally counted as Tier 2 capital and are not assigned to specific assets, form the basis for establishing a bank's capacity to absorb losses". International loans may carry loan-by-loan provisions or special reserves set country by country for country and transfer risk.



Net loans: before the write-off $1{,}000 - 65 = 935$; after it $985 - 50 = 935$.

Figure 1. From provision to write-off, with illustrative numbers. The solid arrow is the provision building the reserve; the dashed arrow is a write-off drawing on it. Income statement box light grey, reserve white, write-off hatched. The write-off moves gross loans and the reserve together and leaves net loans at 935.

3. What an adequate reserve must reflect

GARP and the notes list the factors to weigh: the quality of credit policies and procedures; prior loss experience; loan growth; the quality of management in the lending area; loan collection and recovery practices; and changes in national and local economic and business conditions and general trends. An analysis of the overall allowance should also cover the provisioning policy (especially the value attributed to collateral and whether it can be enforced), the classification and review process, the *current factors likely to cause losses that differ from historical experience*, a long-run trend of overdue loans, and an opinion on adequacy with the additional provisions needed to bring the total in line with IFRS. Provisioning has always involved judgment, but that discretion must follow established policies; IFRS 9 now requires a forward-looking approach (CR-3 g). Whether a bank retains or writes off loss assets changes how large its reserve *looks* (CR-3 h).

Correction to the source notes. SchweserNotes (LO 21.e) states that "loan loss provisions and reserves are both normally counted as Tier 2 capital", and the bank's explanation to Q70 repeats it. GARP attaches the Tier 2 treatment only to *general loss reserves that are not assigned to specific assets*. This matches Basel: general provisions qualify for Tier 2 (up to 1.25% of credit risk-weighted assets under the standardised approach), while provisions against identified deterioration of particular assets are excluded. Tested in Question CR-3 e.3.

	Provision	Reserve (allowance)	Write-off
What it is	Expense of the period	Accumulated provisions less write-offs	Removal of a loss asset from the books
Where	Income statement	Balance sheet, deducted from gross loans	Gross loans and reserve, together
Earnings effect	Reduces them	None by itself	None if already provided for
Net loans	Falls (reserve rises)	Net of the reserve	Unchanged
Driven by	Classification, EL, current conditions	Six factors above	Retain or write off policy (CR-3 h)

Exam traps

- **Flow and stock swapped.** The provision is the period's charge; the reserve is the accumulated balance.
- **Reserves for unexpected loss.** Provisions and reserves are built from expected losses; the tail (UL) is the job of capital.
- **Double counting the loss.** A write-off of a fully provided loan does not hit earnings again.
- **"Both are Tier 2".** Only general reserves not assigned to specific assets normally count as Tier 2.
- **History alone.** Low past losses do not justify a low reserve when growth, staff quality, collection practices or the economy have changed.
- **Circular inputs.** The current reserve balance or a list of overdue loans is not a determinant of the adequate reserve.

Practice questions: CR-3 e

Question CR-3 e.1

Bank Q70 • AnalystPrep Q.5873 (rebuilt)

A bank analyst is reviewing the balance sheet during an economic downturn. She wants to confirm that she correctly understands how the bank's loan loss provisions differ from its loan loss reserves. Which statement is correct?

A. Reserves are the loan loss provisions accumulated over time and carried on the bank's balance sheet.

B. Provisions are the cumulative balance on the balance sheet, while reserves are the amount set aside in one period.

C. Reserves are held to absorb unexpected losses in the extreme tail, while provisions cover the expected losses.

D. Provisions are counted as Tier 2 capital, while reserves are excluded from regulatory capital altogether.

Question CR-3 e.2 New • built from GARP Ch.3, 3.3 (Impairment recognition) and 3.5

Last year a bank classified a USD 4 million loan to a manufacturer as loss and set aside a specific provision for the full balance. This year, after collection efforts end with no recovery, the bank writes the loan off against the loan loss reserve. Which statement about the effect on the bank's pre-tax earnings is correct?

A. Both last year's provision and this year's write-off reduce pre-tax earnings, each by USD 4 million.

B. Last year's provision reduced pre-tax earnings by USD 4 million; this year's write-off leaves them unchanged.

C. Last year's provision left pre-tax earnings unchanged; this year's write-off reduces them by USD 4 million.

D. Neither entry affects pre-tax earnings, because both only move amounts between balance sheet accounts.

Question CR-3 e.3 New • built from GARP Ch.3, 3.5 Loan Loss Provisioning

A risk analyst is assessing a bank's capacity to absorb credit losses. The bank holds specific provisions against identified problem loans and a general loss reserve that is not assigned to any specific asset. Which statement is most consistent with GARP's description of these balances?

A. Only the specific provisions add to the capacity to absorb losses, because the general reserve is not linked to any loan.

B. The general reserve is the expense of the period, while the specific provisions are the balance carried on the balance sheet.

C. Both the specific provisions and the general reserve are normally counted as Tier 2 capital, since both are loss allowances.

D. The provisions, together with the general reserve that normally counts as Tier 2 capital, form the basis of the capacity to absorb losses.

Question CR-3 e.4 Bank Q71 • Schweser Module Quiz 21.2 Q1

Two senior credit managers are reviewing their bank's loan loss reserve procedures. Which of the following elements is most important when determining the appropriate level of reserves?

A. Current loan loss reserves.

B. The bank's experience with previous losses.

C. The current number of nonperforming loans in aggregate.

D. A list of all loans that are delinquent for more than 180 days.

Question CR-3 e.5 Bank Q72 • New question written for this bank (rebuilt)

A risk analyst at a regional bank is reviewing the adequacy of the bank's loan loss allowance. Over the past three years the bank's realized losses have been very low. Over the same period the loan book grew 40% through a new lending segment, the bank hired many inexperienced lending officers, and collection and recovery procedures for the segment were relaxed. Which of the following is the most appropriate conclusion for the review?

A. The allowance should reflect current risk factors that differ from historical experience, so low past losses alone cannot support it.

B. The allowance can be held at its current level, because low realized losses show that the existing provisioning policy is adequate.

C. The allowance should be set from the aggregate number of nonperforming loans, because that is the only objective measure of losses.

D. The allowance can be reduced, because general loss reserves count as Tier 2 capital and already provide a buffer against losses.

Question CR-3 e.6 Bank Q58 • MK-AP Mock 3 Q47 • AnalystPrep Q.5860 • re-filed from CR-3 a

A community bank has restructured several loans because of borrowers' financial difficulties. Management is considering the appropriate accounting treatment. According to sound financial accounting practice, how should the bank reflect these restructured loans in its financial statements?

A. Keep the recorded investment of restructured loans unchanged to maintain consistency in the loan portfolio.

B. Increase the recorded investment of restructured loans to cover potential future losses.

C. Measure the restructured loan by reducing its recorded investment to net realizable value, charging the reduction to the income statement.

D. Record the reduction in loan value as a deferred charge, to be recognized in future financial periods.

Solutions • CR-3 e

Solution CR-3 e.1 — Answer: A

Bank Q70 • AnalystPrep Q.5873

Step 1 – Provisions

Provisions are the amounts set aside, out of the asset classification process, to offset potential future losses: an expense of each period.

Step 2 – Reserves

Reserves are those provisions accumulated over multiple years and carried on the balance sheet (strictly, net of the write-offs charged against them).
$\Rightarrow$ A: the reserve is the stock, the provision the flow.

Interpretation

The bank keys this question D in its original order; rebuilt here with the same correct statement as A. Its explanation adds that "both provisions and reserves are normally counted as Tier 2 capital", which overstates GARP (correction box, Section 3).

Why the other options are wrong

B is incorrect. Reverses the two terms: the cumulative balance is the reserve, the period's charge is the provision.
C is incorrect. Both are built from expected losses; losses in the extreme tail are unexpected losses, covered by capital.
D is incorrect. GARP says general loss reserves not assigned to specific assets are normally counted as Tier 2 capital, so reserves are not excluded.

LO: CR-3 e: Describe and explain loan loss provisions and loan loss reserves. • **Reference:** van Greuning and Brajovic Bratanovic, Analyzing Banking Risk 4e, Ch.7 (GARP CR Ch.3), 3.5 Loan Loss Provisioning; SchweserNotes 2026 Book 2, LO 21.e

Solution CR-3 e.2 — Answer: B

New • built from GARP Ch.3, 3.3 and 3.5

Step 1 – Last year: the provision

The provision is an expense charged to the income statement. It cut last year's pre-tax earnings by USD 4 million and raised the loan loss reserve by the same amount, reducing net loans.

Step 2 – This year: the write-off

Writing the loan off against the reserve removes USD 4 million from gross loans and USD 4 million from the reserve. Net loans do not change and nothing passes through the income statement.
$\Rightarrow$ The loss hit earnings once, when it was provided for.

Interpretation

GARP recognises impairment either through an existing allowance or by charging the income statement in the period of impairment. Here the allowance already holds the full amount, so the write-off is a balance sheet clean-up. Had the loan not been provided for, the write-off would need a charge in the year it happens.

Why the other options are wrong

A is incorrect. Double counts the loss: the write-off uses the reserve built by last year's charge.
C is incorrect. Reverses the timing: the provision is the expense; the write-off of a fully provided loan is not.
D is incorrect. The provision is an income statement expense, not a transfer between balance sheet accounts.

LO: CR-3 e: Describe and explain loan loss provisions and loan loss reserves. • **Reference:** GARP CR Ch.3, 3.3 Bank-Specific Policies (Impairment recognition); 3.5 Workout Procedures for Loss Assets

Solution CR-3 e.3 Answer: D

New • built from GARP Ch.3, 3.5

Step 1 – GARP's sentence

"Such provisions, together with general loss reserves that are normally counted as Tier 2 capital and are not assigned to specific assets, form the basis for establishing a bank's capacity to absorb losses."

Step 2 – Which balance carries the Tier 2 treatment

The general reserve, the one not tied to specific assets. Basel agrees: general provisions held against presently unidentified losses qualify for Tier 2 (within a cap), provisions against identified deterioration of particular assets do not.
$\Rightarrow$ D.

Interpretation

The general reserve is a cushion for losses not yet identified, which is why regulators treat it as capital-like. Specific provisions are already earmarked for losses the bank has recognised.

Why the other options are wrong

A is incorrect. GARP counts both; the general reserve is exactly the cushion for losses not yet tied to a loan.
B is incorrect. Confuses general versus specific with flow versus stock; both are balances built by provisions.
C is incorrect. SchweserNotes' "both" (see the correction box): only the general reserve normally counts as Tier 2.

LO: CR-3 e: Describe and explain loan loss provisions and loan loss reserves. • **Reference:** GARP CR Ch.3, 3.5 Loan Loss Provisioning; Basel III capital definition (general provisions in Tier 2)

Solution CR-3 e.4 Answer: B

Bank Q71 • Schweser Module Quiz 21.2 Q1

Step 1 – The factors in an adequate reserve

Credit policies and procedures; prior loss experience; loan growth; quality of lending management; collection and recovery practices; economic and business conditions.

Step 2 – Match the options

Only the bank's experience with previous losses is on the list.
$\Rightarrow$ B.

Interpretation

The other three options are outputs of the process (the existing reserve) or classification statistics (counts and delinquency lists). They describe the book; they do not tell you how much loss it will produce.

Why the other options are wrong

A is incorrect. Circular: the reserve should not be set by what is already on the books.
C is incorrect. A count of nonperforming loans ignores severity, history and conditions; it is a classification statistic.
D is incorrect. A delinquency cut-off list is a classification output, and 180 days is too late a trigger for reserving.

LO: CR-3 e: Describe and explain loan loss provisions and loan loss reserves. • **Reference:** GARP CR Ch.3, 3.5 Loan Loss Provisioning; SchweserNotes 2026 Book 2, LO 21.e

Solution CR-3 e.5 Answer: A

Bank Q72 • New question written for this bank

Step 1 – What has changed

Three of the six reserve factors have moved against the bank: loan growth (40% in a new segment), quality of lending staff (many inexperienced officers) and collection practices (relaxed).

Step 2 – What the allowance review must do

GARP's analysis of the allowance includes "determination of the current factors that are likely to cause losses ... and that differ from the historical experience of loss".
⇒ Low past losses are one input; on their own they cannot justify the current allowance.

Interpretation

Fast growth into a new segment hides losses for a while, because new loans have not had time to default. That is exactly when history understates the reserve needed.

Why the other options are wrong

B is incorrect. Relies solely on past losses and ignores the changed factors.
C is incorrect. One classification statistic is not the basis of the allowance; the six factors are.
D is incorrect. Capital treatment does not reduce the need for adequate provisions.

LO: CR-3 e: Describe and explain loan loss provisions and loan loss reserves. • **Reference:** GARP CR Ch.3, 3.3 (analysis of adequacy of the overall allowance) and 3.5

Solution CR-3 e.6 Answer: C

Bank Q58 • MK-AP Mock 3 Q47 • AnalystPrep Q.5860 • re-filed from CR-3 a

Step 1 – What restructuring means

Renegotiated debt is a loan restructured to reduce interest or principal because the borrower's position has deteriorated. The concessions lower what the bank can collect.

Step 2 – GARP's treatment

Measure the loan by reducing its recorded investment to net realizable value, counting the cost of all concessions at the restructuring date, and record the reduction as a charge to the income statement for that period.
⇒ C.

Interpretation

This is impairment recognition in practice: the loss reaches earnings when the concession is made, not later. Much renegotiated debt is normally a sign that a bank has problems.

Why the other options are wrong

A is incorrect. Leaving the carrying amount unchanged ignores the concessions and overstates assets and income.
B is incorrect. Backwards: the concessions lower the value; increasing the recorded investment inflates assets.
D is incorrect. Deferring the charge delays loss recognition; it belongs to the period of restructuring.

LO: CR-3 e: Describe and explain loan loss provisions and loan loss reserves. • **Reference:** GARP CR Ch.3, 3.3 Bank-Specific Policies (Impairment recognition; Renegotiated debt treatment)

CR-3 • Credit Risk Management • Vardeez: NOT SO IMP

CR-3 g

P3 CONCEPT

Explain the requirements for estimating expected loss under IFRS 9.

Workbook: none (no workbook section for CR-3) • Vardeez focus

WHAT IS TESTED HERE

Why P3. Vardeez's second flagged LO in CR-3 (his "LO 19.g"); no workbook section.
Vardeez: IFRS 9 expected loss.

Reported exam shapes: none listed in the map.

What the questions look like: place a loan in its stage and pick the horizon and the interest base that go with it; explain the move from incurred to expected loss. The requirements are the staging rules, not a computation, so the bank's lifetime-ECL discounting question (Q79) is not used. Provisions and reserves are under CR-3 e.

1. From incurred loss to expected loss

Until 2017, IAS 39 required an **incurred loss** model: a loan was impaired only once a loss event had occurred, so provisions came late. For financial years beginning on or after **1 January 2018**, IFRS 9 replaced it with an **expected loss** model. GARP: IFRS 9 "has moved accounting standards closer to sound credit risk management practices". Two requirements follow:

- **Forward-looking estimates.** IFRS 9 "requires a forward-looking approach to estimated loan losses and consequent provisioning" (GARP 3.3). Expected loss is still built from PD, LGD and EAD (CR-3 f), but the inputs must reflect current and expected conditions, not only past averages.
- **Three stages.** The horizon of the expected loss, and the base for interest income, depend on the asset's credit stage.

2. The three stages

- **Stage 1: performing assets** (not in arrears). Provisions on a **12-month** expected loss; effective interest on the **gross** amount.
- **Stage 2: assets in arrears, or where a significant change in the credit environment has occurred**. GARP's example: the underwriting decision would have been significantly different on price or other terms given changes in the macroeconomy, such as higher interest rates affecting the client's ability to service its debt. Provisions on **lifetime** expected loss; effective interest still on the **gross** amount.
- **Stage 3: nonperforming assets** (typically 90 days or more past due, CR-3 d). **Lifetime** expected loss; effective interest on the **net (carrying)** amount, that is, gross less the allowance.

The two attributes switch at *different* boundaries: the horizon changes between stages 1 and 2, the interest base between stages 2 and 3. Stage 2 is the only stage that pairs a lifetime horizon with a gross interest base, the cell examiners like to swap. A 12-month expected loss counts only defaults that could occur in the next 12 months; lifetime counts defaults over the remaining life, so for a multi-year loan the provision jumps when it moves from stage 1 to stage 2. Interest on the net amount is smaller: on a gross balance of 100 with an allowance of 30 and an 8% effective rate, interest is 8.0 on the gross basis but $0.08 \times 70 = 5.6$ on the net basis.

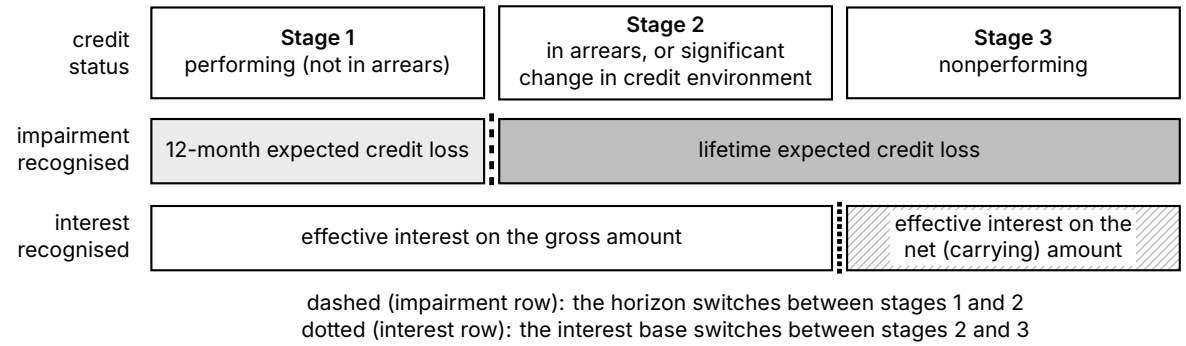


Figure 1. GARP Table 3.3 as a grid. The dashed boundary between stages 1 and 2 changes the expected loss horizon (12-month light grey, lifetime darker grey); the dotted boundary between stages 2 and 3 changes the interest base (gross white, net hatched). Stage 2 pairs lifetime with gross.

Correction to the source notes. Vardeez (CR-3 g) and SchweserNotes (LO 21.g) define stage 2 as "assets with any level of delinquency". GARP's stage 2 is wider: "assets in arrears, *or where a significant change in the credit environment has occurred*". A loan that is fully current can therefore be in stage 2, with lifetime expected loss, if conditions have changed enough that the bank would now underwrite it on materially different terms. Tested in Question CR-3 g.4.

	IAS 39 (to 2017)	IFRS 9 (from 1 January 2018)
Model	Incurred loss	Expected loss, three stages
When losses are provided	After a loss event has occurred	Before default: 12-month expected loss from day one
Outlook	Backward-looking	Forward-looking (current and expected conditions)
Horizon	Not staged	12-month (stage 1) or lifetime (stages 2 and 3)
Interest income	Not staged	Gross (stages 1 and 2) or net (stage 3)

Exam traps

- **Stage 2 swapped.** Stage 2 is lifetime *and* gross. "Lifetime and net" is stage 3; "12-month and gross" is stage 1.
- **Delinquency as the only trigger.** A significant change in the credit environment moves a current loan to stage 2.
- **Lifetime for everything.** Performing (stage 1) assets carry only a 12-month expected loss.
- **Backward-looking estimates.** A flat historical loss rate ignores IFRS 9's forward-looking requirement.
- **Date.** IFRS 9 applies from financial years beginning 1 January 2018; it replaced IAS 39's incurred loss model.

Practice questions: CR-3 g

Question CR-3 g.1

Bank Q76 • AP Mock 2 Q70 • AnalystPrep Q.5875

A bank's audit committee is discussing IFRS 9 and its expected credit loss model. Which of the following scenarios exemplifies compliance with IFRS 9's expected loss estimation model?

A. Allocating a uniform loss provision percentage across the entire loan portfolio based on average historical loss data.

B. Estimating expected losses on a performing loan based on potential losses over its entire lifetime, regardless of its current performance status.

C. Assessing expected losses on underperforming assets by evaluating the 12-month loss outlook and adjusting provisions annually.

D. Applying a differentiated approach, where performing assets are assessed for 12-month expected losses and underperforming or non-performing assets for lifetime expected losses.

Question CR-3 g.2

New • built from GARP Ch.3, 3.5 (IFRS 9 Implications)

A bank's audit committee asks why the bank's loan loss allowance rose when it adopted IFRS 9 in 2018, even though no additional loans had defaulted. Which explanation is most accurate?

A. IFRS 9 replaced the incurred loss model of IAS 39 with an expected loss model, so losses are provided for before any default occurs.

B. IFRS 9 kept the incurred loss model of IAS 39 but required the provision on every impaired loan to cover its lifetime losses.

C. IFRS 9 required every loan to carry a provision equal to its lifetime expected losses from the day it is granted.

D. IFRS 9 required provisions to be counted as Tier 1 capital, so the bank built them up to protect its capital ratios.

Question CR-3 g.3

Bank Q78 • Schweser Module Quiz 21.2 Q3 (rebuilt)

A bank's loan is delinquent but not yet nonperforming, because a short-term macroeconomic event is temporarily preventing repayment. Under IFRS 9, which treatment is required for this loan?

A. Stage 1: 12-month expected loss with effective interest on the gross amount.

B. Stage 2: lifetime expected loss with effective interest on the gross amount.

C. Stage 3: lifetime expected loss with effective interest on the net (carrying) amount.

D. Stage 2: 12-month expected loss with effective interest on the net (carrying) amount.

Question CR-3 g.4 New • built from GARP Ch.3, 3.5 (IFRS 9 Implications)

A corporate borrower of a European bank is current on all its payments. Since the loan was granted, interest rates have risen sharply, and the credit officer judges that the original underwriting decision would have been significantly different on price and terms if it were made today. A junior analyst, working from study notes that define stage 2 as "assets with any level of delinquency", proposes stage 1. Under IFRS 9 as described by GARP, how should the loan be staged?

A. Stage 1, because only a missed payment can move a performing loan out of stage 1.

B. Stage 3, because a loan that would now be priced on different terms is treated as nonperforming.

C. Stage 2, because a significant change in the credit environment moves a loan to stage 2 even without arrears.

D. Stage 2 for the interest calculation but stage 1 for the provision, until the borrower falls into arrears.

Question CR-3 g.5 Bank Q77 • AnalystPrep Q.5876

A long-standing corporate client's loan has deteriorated significantly in credit quality and is now nonperforming. Which provisioning approach correctly aligns with IFRS 9 Stage 3 requirements?

A. Adopting a steady-state provisioning method based on the long-standing relationship with the client.

B. Continuation of provisioning based on initial PD and LGD at loan origination.

C. Provisions based on lifetime expected losses, with effective interest on the net (carrying) amount.

D. Relying on external credit ratings to determine the necessary provisions.

Question CR-3 g.6 New • built from GARP Ch.3, 3.3 (loan portfolio review) and 3.5

A bank's IFRS 9 model sets the provision on each stage 1 loan by applying the bank's average annual loss rate over the past ten years to the gross balance. Economic forecasts now point to a sharp downturn, but the model's output has not changed. Which criticism of the model is most valid?

A. IFRS 9 requires a forward-looking estimate, so expected losses should reflect current and expected conditions, not past averages alone.

B. IFRS 9 prohibits any use of historical loss data, so the ten-year average must be replaced entirely by external credit ratings.

C. Stage 1 loans require lifetime expected losses, so the horizon of the model is too short whatever data it uses.

D. IFRS 9 provisions must be a uniform percentage of the whole portfolio, so applying the rate loan by loan is wrong.

Solutions • CR-3 g

Solution CR-3 g.1 — Answer: D

Bank Q76 • AP Mock 2 Q70 • AnalystPrep Q.5875

Step 1 – The structure of IFRS 9

Three stages: performing assets (stage 1) carry a 12-month expected loss; assets in arrears or with a significant change in credit environment (stage 2) and nonperforming assets (stage 3) carry lifetime expected loss.

Step 2 – Find the option with the differentiation

Only D ties the horizon to credit quality.
$\Rightarrow$ D.

Interpretation

The defining feature is that the provisioning horizon depends on the asset's stage, and every estimate is forward-looking.

Why the other options are wrong

A is incorrect. A flat historical-average rate is neither staged nor forward-looking; it resembles the old approach.
B is incorrect. A performing loan sits in stage 1, which uses a 12-month horizon, not lifetime.
C is incorrect. Reversed: underperforming and nonperforming assets need lifetime, not 12-month, expected loss.

LO: CR-3 g: Explain the requirements for estimating expected loss under IFRS 9. • **Reference:** van Greuning and Brajovic Bratanovic, Analyzing Banking Risk 4e, Ch.7 (GARP CR Ch.3), 3.5 IFRS 9 Implications, Table 3.3

Solution CR-3 g.2 — Answer: A

New • built from GARP Ch.3, 3.5

Step 1 – What IAS 39 did

Under the incurred loss model a loan was impaired only after a loss event had occurred, so performing loans carried little or no provision.

Step 2 – What IFRS 9 changed

For financial years beginning on or after 1 January 2018, IFRS 9 abandoned the incurred loss model in favour of an expected loss model. Even a performing loan now carries a 12-month expected loss, and deteriorated loans a lifetime one.
$\Rightarrow$ Provisions rise on adoption without any new defaults.

Interpretation

GARP's phrase is that IFRS 9 moved accounting "closer to sound credit risk management practices": losses are recognised when they are expected, not when they have happened.

Why the other options are wrong

B is incorrect. IFRS 9 did not keep the incurred loss model; it replaced it.
C is incorrect. Lifetime losses from day one apply only after deterioration (stages 2 and 3); performing loans take 12 months. (Lifetime-from-origination is the route of the US CECL standard, not IFRS 9.)
D is incorrect. An accounting standard does not decide regulatory capital, and provisions are not Tier 1 capital.

LO: CR-3 g: Explain the requirements for estimating expected loss under IFRS 9. • **Reference:** GARP CR Ch.3, 3.5 IFRS 9 Implications

Solution CR-3 g.3 — Answer: B

Bank Q78 • Schweser Module Quiz 21.2 Q3

Step 1 – Rule out stages 1 and 3

The loan is in arrears, so it is not stage 1 (performing, not in arrears). It is not yet nonperforming, so it is not stage 3.

Step 2 – Stage 2 attributes

Lifetime expected loss, with effective interest still on the gross amount.
$\Rightarrow$ B.

Interpretation

A temporary cause does not change the staging: being in arrears is a stage 2 trigger in itself. The interest base moves to the net amount only when the loan becomes nonperforming.

Why the other options are wrong

A is incorrect. Stage 1 is only for performing assets with no arrears.
C is incorrect. Stage 3 is for nonperforming assets; this loan is not yet nonperforming.
D is incorrect. Mixes the attributes: stage 2 uses lifetime (not 12-month) expected loss and the gross (not net) amount.

LO: CR-3 g: Explain the requirements for estimating expected loss under IFRS 9. • **Reference:** GARP CR Ch.3, 3.5 IFRS 9 Implications, Table 3.3; SchweserNotes 2026 Book 2, LO 21.g

Solution CR-3 g.4 — Answer: C

New • built from GARP Ch.3, 3.5

Step 1 – GARP's stage 2 trigger

"Assets in arrears, or where a significant change in the credit environment has occurred", for example where the underwriting decision would have been significantly different on price or other terms given macroeconomic changes such as higher interest rates.

Step 2 – Apply it

The borrower is current, but the credit officer's judgment matches GARP's example exactly.
$\Rightarrow$ Stage 2: lifetime expected loss, effective interest on the gross amount.

Interpretation

The study notes' rule ("any level of delinquency") is the narrower Vardeez and Schweser wording; GARP adds the significant-change trigger (correction box). Waiting for arrears would recognise the deterioration late, which is what IFRS 9 was designed to stop.

Why the other options are wrong

A is incorrect. Missed payments are one trigger, not the only one.
B is incorrect. Stage 3 is for nonperforming assets; this loan is fully current.
D is incorrect. Stage determines both attributes together; there is no split staging.

LO: CR-3 g: Explain the requirements for estimating expected loss under IFRS 9. • **Reference:** GARP CR Ch.3, 3.5 IFRS 9 Implications (stage 2)

Solution CR-3 g.5 — Answer: C

Bank Q77 • AnalystPrep Q.5876

Step 1 – Identify the stage

The loan is nonperforming: stage 3.

Step 2 – Stage 3 requirements

Lifetime expected loss, with effective interest on the net (carrying) amount, that is, after deducting the allowance.
⇒ C.

Interpretation

Moving from stage 2 to stage 3 does not change the horizon (lifetime in both); it changes the interest base from gross to net, so interest income falls.

Why the other options are wrong

A is incorrect. The length of the relationship does not change the required method.
B is incorrect. IFRS 9 requires estimates updated for the current, deteriorated credit quality, not origination parameters.
D is incorrect. External ratings can be an input but they are not the requirement.

LO: CR-3 g: Explain the requirements for estimating expected loss under IFRS 9. • **Reference:** GARP CR Ch.3, 3.5 IFRS 9 Implications, Table 3.3

Solution CR-3 g.6 **Answer: A**

New • built from GARP Ch.3, 3.3 and 3.5

Step 1 – The requirement being missed

GARP: IFRS 9 "requires a forward-looking approach to estimated loan losses and consequent provisioning". An estimate that does not move when the economic outlook worsens is not forward-looking.

Step 2 – Check the horizon

Stage 1 loans correctly use a 12-month horizon, so the flaw is the inputs, not the horizon.
⇒ A.

Interpretation

Historical data remain a legitimate input; the error is relying on them alone. GARP's stage 2 example (a macroeconomic change such as higher rates) shows the same forward-looking logic at work in staging.

Why the other options are wrong

B is incorrect. Absolute claim: history is allowed as an input, and external ratings are not required.
C is incorrect. Stage 1 uses 12-month expected loss; lifetime applies from stage 2.
D is incorrect. A uniform portfolio-wide percentage is exactly what IFRS 9's staged, loan-level approach moved away from.

LO: CR-3 g: Explain the requirements for estimating expected loss under IFRS 9. • **Reference:** GARP CR Ch.3, 3.3 (loan portfolio review) and 3.5 IFRS 9 Implications

CR-3 • Credit Risk Management • Vardeez: NOT SO IMP

CR-3 h

P3 CONCEPT

Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets.

Workbook: none (no workbook section for CR-3) • Vardeez focus

WHAT IS TESTED HERE

Why P3. Vardeez's third flagged LO in CR-3 (his "LO 19.h"); no workbook section.
Vardeez: workout: retain versus write off loss assets.

Reported exam shapes: none listed in the map.

What the questions look like: name the workout strategy behind an action, judge how a workout function should be organised and assessed, and read a reserve level through the retain or write-off choice. No calculations. Provisions, reserves and write-offs are defined under CR-3 e.

1. Workout procedures

GARP: if timely action is not taken on problem loans, chances to strengthen or collect them are missed and losses can grow until they threaten the bank's solvency. During a workout each loan and borrower is considered **on its own merits**. Typical strategies (Vardeez: they can be used in any order):

1. **Reduce the bank's exposure:** the borrower provides additional capital, funds, collateral or guarantees.
2. **Work with the borrower:** assess the problems and find ways to raise repayment capacity: advice, a programme to cut operating costs and raise earnings, selling assets, a debt restructuring programme or changed loan terms. Vardeez: this "borders on consulting".
3. **Bring in a third party:** arrange for the borrower to be bought or taken over by a more creditworthy party, or a joint-venture partnership.
4. **Liquidate the exposure:** out-of-court settlement or other legal action, calling on guarantees, foreclosing or liquidating collateral.

Assessing the workout function. Review its organization (departments and responsible staff), its performance (attempted and successful recoveries, by number and by volume, and the average time to recovery), the workout methods used and the involvement of senior management. Vardeez: review historical attempts to collect and their success rate.

2. Retaining versus writing off loss assets

	Retain loss assets	Write off loss assets
Tradition	British	United States
What happens	Loss assets stay on the books until all remedies for collection are exhausted	Loss assets are promptly written off against the reserve, that is, removed from the books
Reserve looks	Unusually large	Smaller relative to the loan portfolio
Character	Keeps time for recovery efforts	More conservative; assets are nonbankable but not necessarily nonrecoverable
Net loans and earnings	Same as write-off once the loss is fully provided	Same as retention once the loss is fully provided

GARP's warning for analysts: when evaluating a bank's provisions, "an analyst must clearly understand whether the bank is aggressively writing off its losses or is simply providing for them". The approach used in a country often depends on how the tax authorities treat provisions. To compare a retaining bank with a writing-off bank, put both on the same basis: remove the retained loss assets from gross loans *and* from the reserve (Question CR-3 h.5).

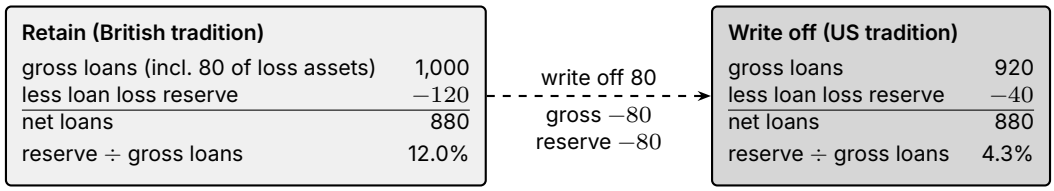


Figure 1. The same fully provided loss assets (80) under the two approaches, illustrative numbers. Writing them off (dashed arrow) removes 80 from gross loans and 80 from the reserve: net loans stay at 880, but the reserve falls from 12.0% to 4.3% of gross loans. Retain panel light grey, write-off panel darker grey.

Correction to the source notes. Vardeez's CR-3 h diagram labels the retention effect "loss reserve increases", and its caption says "retaining the asset raises the loss reserve on the balance sheet". Retention adds nothing to the reserve: the provision was made when the loss was recognised, and retention simply keeps that reserve, and the loss asset, on the books. GARP's wording is that the reserve "may appear unusually large"; writing off makes it "appear smaller in relation to the outstanding loan portfolio". The comparison is about how large the reserve *looks*, not about a new provision. Tested in Questions CR-3 h.4 and h.5.

Exam traps

- **Traditions swapped.** Retain is British, write-off is US; a small reserve ratio points to write-off.
- **Which is conservative.** GARP calls the US write-off approach more conservative, not the larger-reserve British one.
- **Write-off read as giving up.** Written-off assets are nonbankable but not necessarily nonrecoverable; collection can continue.
- **Wrong comparability fix.** Adjust a retaining bank by reducing *both* gross loans and the reserve; never increase the reserve.
- **Strategy labels.** New collateral or guarantees reduce exposure; calling on a guarantee liquidates it; selling assets or restructuring is working with the borrower.

- **One rule for all.** Workouts treat each borrower on its merits; blanket legal action or outsourcing is not a workout procedure.

Practice questions: CR-3 h

Question CR-3 h.1

Bank Q80 • AP Mock 3 Q46 • AnalystPrep Q.5834

A regional bank is reassessing how it manages credit risk on high-risk borrowers and problem loans. It needs more effective handling of accounts that have exceeded credit limits, become delinquent, or entered default. Which policy revision would best enhance the bank's ability to manage these situations?

A. Outsourcing the management of all high-risk accounts to external consultants for immediate action.

B. Introducing a dynamic credit limit system that automatically adjusts based on the borrower's repayment behavior.

C. Establishing a specialized internal team dedicated to the workout of delinquent or defaulted accounts with structured and independent oversight.

D. Implementing a policy of immediate legal action against all borrowers whose accounts become delinquent.

Question CR-3 h.2

New • built from GARP Ch.3, 3.5 (Workout Procedures for Loss Assets)

A bank's workout unit records one action on each of four problem loans:

(i) The owners inject fresh capital and pledge additional collateral.

(ii) The bank advises the borrower to sell a non-core division and restructures the remaining debt.

(iii) The bank arranges a joint-venture partner that takes a controlling stake in the borrower.

(iv) The bank calls on the parent company's guarantee and liquidates the collateral.

Which of the following correctly matches the actions to GARP's typical workout strategies?

A. (i) reducing exposure; (ii) working with the borrower; (iii) third-party partner or buyer; (iv) liquidating exposure.

B. (i) working with the borrower; (ii) working with the borrower; (iii) third-party partner or buyer; (iv) liquidating exposure.

C. (i) reducing exposure; (ii) liquidating exposure; (iii) third-party partner or buyer; (iv) liquidating exposure.

D. (i) reducing exposure; (ii) working with the borrower; (iii) third-party partner or buyer; (iv) reducing exposure.

Question CR-3 h.3 New • built from GARP Ch.3, 3.5 (Workout Procedures for Loss Assets)

A bank supervisor is assessing the effectiveness of a bank's workout procedures for problem loans. Which evidence is most relevant to the assessment, according to GARP?

A. The volume of new loans originated by relationship managers who previously worked in the workout unit.

B. The share of problem loans handled under one standard restructuring template applied to every borrower.

C. The size of the loan loss reserve relative to the loan book, compared with peer banks in the same country.

D. The unit's organization and staff, its attempted and successful recoveries by number and volume, and its average time to recovery.

Question CR-3 h.4 Bank Q82 • Schweser Module Quiz 21.2 Q4 (rebuilt)

A financial analyst for ABC Bank notices that loan loss reserves are small relative to the size of the bank's loan portfolio. He tells a coworker this is a warning sign. The coworker replies that there may be a simple explanation that management could clear up. The coworker is suggesting that ABC Bank could be following the:

A. British model and retaining loss assets on the balance sheet.

B. U.S. model and promptly writing off loss assets against the reserve.

C. U.S. model and providing internal consulting to help the borrower improve its ability to pay.

D. U.S. model and allowing more time for collection efforts or collateral enhancement to work out.

Question CR-3 h.5 Bank Q81 • BT answers P2.T6.24.5.2 • BT notes P2.T6.24.5.2

Converse Bank and Old Crow Bank each made a $1,000,000 loan to a technology startup, Janzee. Janzee defaulted and is unlikely to repay. Old Crow retains the loss asset on its balance sheet (carrying a matching loan loss reserve) in the hope of some later recovery, while Converse writes the loan off against its reserve. All else is equal. An analyst wants the two banks' financial statements to be comparable and believes the Janzee loan will never be recovered. What adjustment should he make?

A. Reduce Old Crow's asset values only.

B. Increase Old Crow's loan loss reserves only.

C. Reduce Old Crow's asset values and its loan loss reserves.

D. Reduce Old Crow's asset values and increase its loan loss reserves.

Question CR-3 h.6 New • built from GARP Ch.3, 3.5 (Workout Procedures for Loss Assets)

Two banks in different countries hold similar loan books. Bank R follows the British tradition and retains loss assets on its books until all collection remedies are exhausted; Bank W follows the US tradition and promptly writes loss assets off against its reserve. Which statement comparing the two approaches is correct?

A. Bank W's approach is the more conservative, and writing a loss asset off does not mean that recovery on it is abandoned.

B. Bank R's approach is the more conservative, because its larger loan loss reserve shows it provides more for each loss asset.

C. Bank W gives up all collection efforts on written-off assets, since a write-off declares the asset to be unrecoverable.

D. The choice between the approaches is set by IFRS 9 staging, so it does not vary with the national tax treatment of provisions.

Solutions • CR-3 h

Solution CR-3 h.1 — Answer: C

Bank Q80 • AP Mock 3 Q46 • AnalystPrep Q.5834

Step 1 – Name the problem

Accounts already over limit, delinquent or in default: that is the domain of the workout function.

Step 2 – What sound practice looks like

A dedicated workout unit with responsible staff, senior management involvement and an independent review of its recoveries and time to recovery, handling each borrower on its merits (more collateral, restructuring, a third-party partner or liquidation).

$\Rightarrow$ C.

Interpretation

A specialised team can choose the right strategy case by case and can be measured on its results, which blanket rules and outsourcing cannot deliver.

Why the other options are wrong

A is incorrect. Wholesale outsourcing gives up control and the bank's own accountability for recoveries.

B is incorrect. Dynamic limits are a preventive tool for performing borrowers; they do not resolve accounts already in trouble.

D is incorrect. Immediate legal action for every delinquency ignores case-by-case treatment; liquidation is only one of four strategies.

LO: CR-3 h: Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets. • **Reference:** van Greuning and Brajovic Bratanovic, Analyzing Banking Risk 4e, Ch.7 (GARP CR Ch.3), 3.5 Workout Procedures for Loss Assets

Solution CR-3 h.2 — Answer: A

New • built from GARP Ch.3, 3.5

Step 1 – Reducing exposure

GARP: "having the borrower provide additional capital, funds, collateral, or guarantees". Action (i) adds capital and collateral.

Step 2 – Working with the borrower

Advice, cost programmes, selling assets, debt restructuring or changed terms. Action (ii) combines an asset sale with a restructuring.

Step 3 – Third party, then liquidation

A takeover or joint venture is the third-party strategy (iii). Calling on guarantees, foreclosing and liquidating collateral are the liquidation strategy (iv).

$\Rightarrow$ A.

Interpretation

The two guarantee items look alike but sit in different strategies: *obtaining* a new guarantee reduces exposure; *calling* on one liquidates it. Selling assets to restructure is working with the borrower, not liquidation.

Why the other options are wrong

B is incorrect. Treats a capital injection and new collateral as consulting; they reduce the bank's exposure.

C is incorrect. Treats the asset sale and restructuring as liquidation; GARP lists them under

working with the borrower.
D is incorrect. Treats calling on a guarantee as adding security; it is part of liquidating the exposure.

LO: CR-3 h: Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets. • **Reference:** GARP CR Ch.3, 3.5 Workout Procedures for Loss Assets

Solution CR-3 h.3 Answer: D

New • built from GARP Ch.3, 3.5

Step 1 – GARP's assessment criteria

Organization of the function (departments and responsible staff); performance of the workout units, through attempted and successful recoveries by number and volume and the average time for recovery; the methods used; senior management involvement.

Step 2 – Match

Only D measures the unit's organization and recovery performance.
$\Rightarrow$ D.

Interpretation

Recovery rates and speed are the outcome measures; organization and senior involvement explain them. Vardeez puts the same idea as reviewing historical collection attempts and their success rate.

Why the other options are wrong

A is incorrect. Origination by former workout staff says nothing about the workout unit's results.
B is incorrect. A single template for every borrower contradicts the rule that each loan is considered on its own merits.
C is incorrect. The reserve ratio depends heavily on whether the bank retains or writes off loss assets (and on tax rules), so it is not a measure of workout effectiveness.

LO: CR-3 h: Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets. • **Reference:** GARP CR Ch.3, 3.5 Workout Procedures for Loss Assets

Solution CR-3 h.4 Answer: B

Bank Q82 • Schweser Module Quiz 21.2 Q4

Step 1 – The two approaches

Retention (British) keeps loss assets on the books until collection remedies are exhausted, so the reserve looks unusually large. Prompt write-off (US) removes them against the reserve, so the reserve looks smaller relative to the portfolio.

Step 2 – Read the observation

A small reserve relative to loans fits the write-off approach.
$\Rightarrow$ B.

Interpretation

The small ratio need not mean under-provisioning: the loss assets and the reserve held against them have both left the balance sheet (Figure 1). Management can clear this up by explaining its write-off policy.

Why the other options are wrong

A is incorrect. Reversed: retention makes the reserve look large, not small.

C is incorrect. Consulting the borrower is a workout strategy, not a balance sheet treatment, and does not explain the ratio.
D is incorrect. Allowing time for collection is the rationale for retention, which enlarges the reserve; it is not the US approach.

LO: CR-3 h: Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets. • **Reference:** GARP CR Ch.3, 3.5 Workout Procedures for Loss Assets; SchweserNotes 2026 Book 2, LO 21.h

Solution CR-3 h.5 Answer: C

Bank Q81 • BT answers P2.T6.24.5.2

Step 1 – Converse's balance sheet

Writing off removed the $1,000,000 loan from assets and the matching reserve from the books.

Step 2 – Old Crow's balance sheet

Retention keeps the loan in gross assets and the matching reserve on the books, so both are $1,000,000 larger than at Converse.

Step 3 – The adjustment

Put Old Crow on the write-off basis: reduce its assets by $1,000,000 and its loan loss reserve by $1,000,000. Net assets do not change.
$\Rightarrow$ C.

Interpretation

Retention inflates both sides of the same pair of accounts; writing off shrinks both. The reserve is not "too small" at either bank: the difference is presentation.

Why the other options are wrong

A is incorrect. Removing the asset but leaving its reserve would understate Old Crow's net assets by $1,000,000.
B is incorrect. Wrong direction: Old Crow's reserve is already larger, because it still holds the reserve against the retained loss.
D is incorrect. Removing the asset is right, but the reserve must fall with it, not rise.

LO: CR-3 h: Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets. • **Reference:** GARP CR Ch.3, 3.5 Workout Procedures for Loss Assets

Solution CR-3 h.6 Answer: A

New • built from GARP Ch.3, 3.5

Step 1 – GARP on the US approach

Prompt write-off against the reserve "is typical of the U.S. tradition and is more conservative; loss assets are considered to be nonbankable but not necessarily nonrecoverable".

Step 2 – GARP on what drives the choice

"The approach used in a particular country often depends on the taxation applied to provisions by the fiscal authorities."
$\Rightarrow$ A.

Interpretation

The larger reserve under retention reflects loss assets still on the books, not more generous provisioning, which is why the analyst must know whether a bank writes off aggressively or simply provides.

Why the other options are wrong

B is incorrect. Polarity flip: GARP calls the write-off approach more conservative; the British reserve only looks large.

C is incorrect. Absolute claim: written-off assets are nonbankable but not necessarily nonrecoverable.

D is incorrect. IFRS 9 staging governs expected loss provisioning, not this choice; GARP links the choice to national tax treatment.

LO: CR-3 h: Describe a workout procedure for loss assets and compare the following two approaches used to manage loss assets: retaining loss assets and writing off loss assets. • **Reference:** GARP CR Ch.3, 3.5 Workout Procedures for Loss Assets

CR-5 • Introduction to Credit Risk Modeling and Assessment • Vardeez: VERY IMPORTANT

CR-5 c

P3 CALC

Estimate risk-weighted assets and capital adequacy ratio of a financial institution.

Workbook: D3 Regulatory vs Economic Capital and Capital Adequacy • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D3 (Map #61, 3/16) reports the risk-weighted assets and capital adequacy ratio as the computational partner of "regulatory versus economic capital". Vardeez's flags in this reading go to LOs d to g, not c.

Vardeez: the CAR against 8% (Basel II) or 10.5% (Basel III), the standardized weights and the IRB route through the ASRF model.

Reported exam shapes

- **D3·S3** Risk-weighted assets (standardized weights plus IRB $K \times 12.5 \times \text{EAD}$) and the capital adequacy ratio against a minimum.

What the questions look like: numerical. Exposures with their risk weights (and sometimes an IRB book with its K), Tier 1 and Tier 2 capital, and a minimum; you build RWA, divide, and compare. Red herrings: economic capital and unweighted total assets. A few concept items test the IRB building blocks (ASRF, asset correlation, maturity adjustment).

Worked elsewhere: D3.2 (regulatory versus economic capital) under CR-4 g.

1. The capital adequacy ratio

Financial institutions should hold enough capital to meet the risk of their assets (GARP 5.5):

$$\boxed{\text{CAR} = \frac{\text{Tier 1} + \text{Tier 2 capital}}{\text{Risk-weighted assets}} \geq \alpha} \qquad \alpha = 8\% \text{ (Basel II)}, \quad \alpha = 10.5\% \text{ (Basel III)}$$

Tier 1 is core capital (equity and reserves), the primary source for covering unexpected losses; **Tier 2** is supplementary and not as secure. Basel III's 10.5% is the 8% total capital minimum plus the 2.5% conservation buffer (ORR-22). Dividing Tier 1 alone by RWA gives the Tier 1 ratio; dividing by unweighted assets gives a leverage-style ratio. Neither is the CAR.

2. Risk-weighted assets: two routes

RWA $= \sum_j w_j \times \text{Exposure}_j$, with the weight w_j set by one of the two Basel II approaches.

Standardized approach. Prescribed risk weights, set by the supervisor and common to all cases; the risk assessment relies on external models (ratings). Easy to deploy but costlier, because capital requirements are set higher to allow for its ambiguity. Vardeez lists the Basel I buckets: 0% cash, gold and OECD sovereigns; 20% OECD banks and agencies; 50% uninsured residential mortgages; 100% corporates and non-OECD banks. Basel II's standardized weights are finer (A-rated corporates 50%, residential mortgages 35%). Exam stems give the weights.

Internal ratings-based (IRB) approach. PD comes from the bank's internal rating or scoring systems; LGD is set by the supervisor (**foundation** IRB) or modelled internally (**advanced** IRB). Capital comes from the **asymptotic single risk factor** (ASRF) model, which assumes a completely diversified portfolio: idiosyncratic risks cancel and only systematic risk matters, so capital can be computed loan by loan and added up. For a corporate exposure:

$$\boxed{\text{RWA} = K \times 12.5 \times \text{EAD}} \qquad K = \left[\text{LGD} \cdot N\left(\frac{N^{-1}(\text{PD}) + \sqrt{R}\,N^{-1}(0.999)}{\sqrt{1-R}}\right) - \text{PD} \cdot \text{LGD}\right] \times \text{MA}$$

with the maturity adjustment $\text{MA} = (1+(M-2.5)b)/(1-1.5b)$ and $b = (0.11852-0.05478\ln \text{PD})^2$. K is a capital charge per unit of EAD covering unexpected loss only (the $-\text{PD}\cdot\text{LGD}$ term removes expected loss); $12.5 = 1/8\%$ turns it back into RWA. The $N(\cdot)$ term is the worst-case default rate (WCDR) at 99.9%. The asset correlation R **falls as PD rises** (riskier firms default for idiosyncratic reasons) and **rises with firm size**. The maturity adjustment **rises with maturity** and is **larger for low-PD borrowers**, which have more room to be downgraded. Granularity adjustments (positive or negative) handle the concentration the ASRF ignores.

Worked example. A bank's books (USD million): corporates under IRB, EAD 1,000 with PD 1%, LGD 45%, $M = 2.5$, $R = 0.19$; residential mortgages 2,000 at 35%; OECD sovereigns 800 at 0%; claims on banks 500 at 20%; unrated corporates 400 at 100%. Tier 1 is 170 and Tier 2 is 50.

- WCDR $= N((-2.3263 + 0.4359 \times 3.0902)/0.9) = N(-1.088) = 13.83\%$; $b = 0.1375$, maturity factor $1/(1-1.5b) = 1.2598$.
- $K = 0.45 \times (0.1383 - 0.01) \times 1.2598 = 7.27\%$, so $\text{RWA}_{\text{IRB}} = 0.0727 \times 12.5 \times 1{,}000 = 909$ (an effective weight of 90.9%).
- Standardized: $700 + 0 + 100 + 400 = 1{,}200$. Total RWA $= 2{,}109$.
- CAR $= 220/2{,}109 = 10.43\%$: above 8% (Basel II), but 1.4 short of the $0.105 \times 2{,}109 = 221.4$ Basel III needs. The Tier 1 ratio is $170/2{,}109 = 8.06\%$.

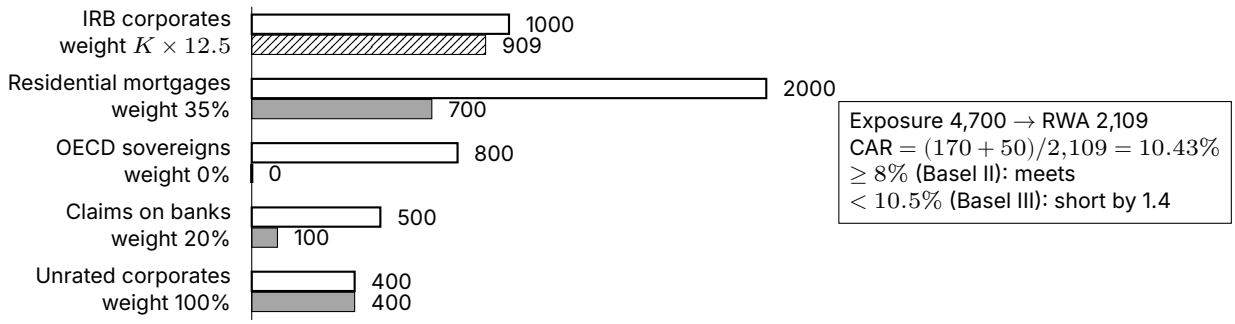


Figure 1. The worked example, book by book (USD million). The upper bar of each pair (white, outlined) is the exposure; the lower bar is its RWA, hatched for the IRB book ($K \times 12.5 \times \text{EAD}$) and grey for the standardized books. Weighting shrinks 4,700 of exposure to 2,109 of RWA.

Correction to the source notes. GARP (5.5) and SchweserNotes (LO 23.c) describe RWA as "a weighted average" of the assets. It is a weighted *sum*, $\sum_j w_j \times \text{Exposure}_j$, with no division by the weights; every worked example and every bank question computes it that way. GARP and Schweser also print "Asymptotic Risk Factor (ASFR) model"; the Basel name is the asymptotic single risk factor (ASRF) model. GARP's "log(PD)" in the maturity adjustment is the natural logarithm.

Formula card: CR-5 c

Capital adequacy ratio	$\text{CAR} = (\text{Tier 1} + \text{Tier 2})/\text{RWA} \geq \alpha; \quad \alpha = 8\%$ (Basel II), 10.5% (Basel III)
Standardized RWA	$\sum_j w_j \times \text{Exposure}_j$, prescribed weights w_j
IRB RWA	$K \times 12.5 \times \text{EAD} \quad (12.5 = 1/8\%)$
IRB capital charge	$K = \text{LGD}\,(\text{WCDR} - \text{PD}) \times \text{MA}$
Worst-case default rate	$\text{WCDR} = N((N^{-1}(\text{PD}) + \sqrt{R}\,N^{-1}(0.999))/\sqrt{1-R})$
Maturity adjustment	$(1 + (M - 2.5)b)/(1 - 1.5b), \quad b = (0.11852 - 0.05478 \ln \text{PD})^2$

Exam traps

- **IRB book mis-entered.** $K \times$ EAD is capital, not RWA (dropped 12.5 inflates the CAR); nor does the book enter at a 100% weight on EAD.
- **Wrong numerator or denominator.** Tier 1 alone gives the Tier 1 ratio; unweighted assets give a leverage-style ratio.
- **WCDR without −PD.** Capital covers unexpected loss only; using the WCDR alone double counts expected loss.
- **Wrong minimum.** 8% is Basel II, 10.5% Basel III. A higher weight on one class re-weights that class only.
- **Economic capital.** The bank's own UL-based estimate (CR-4 g) plays no part in the regulatory ratio.

Practice questions: CR-5 c

Question CR-5 c.1

Bank Q162 • AnalystPrep Q.3102 (rebuilt)

A bank examiner is reviewing Bank M, which holds USD 60 million of Tier 1 capital and USD 20 million of Tier 2 capital. Its assets are USD 400 million of corporate loans (risk weight 100%), USD 300 million of residential mortgages (risk weight 50%) and USD 200 million of loans to highly rated banks (risk weight 20%). Using the standardized approach, which of the following is closest to Bank M's capital adequacy ratio?

A. 8.89%

B. 10.17%

C. 13.56%

D. 15.69%

Question CR-5 c.2

Bank Q165 • BT P2.T7.24.10.1 (rebuilt)

An analyst at Bank DeNiro is computing risk-weighted assets under Basel I and under a Basel II standardized approach. The balance sheet shows:

Asset	Book value	Basel I weight	Basel II weight
Corporate loans (A rated)	USD 100 million	100%	50%
OECD government loans (AAA)	USD 10 million	0%	0%
Residential mortgages (uninsured)	USD 50 million	50%	35%

Which of the following is closest to the bank's RWA under Basel I and under Basel II, respectively?

A. USD 67.5 million and USD 125.0 million

B. USD 125.0 million and USD 67.5 million

C. USD 135.0 million and USD 67.5 million

D. USD 160.0 million and USD 67.5 million

Question CR-5 c.3

Bank Q164 • AnalystPrep Q.5961 (rebuilt)

A bank has total capital (Tier 1 plus Tier 2) of USD 120 million and risk-weighted assets of USD 1,000 million, of which USD 200 million of exposures carry a 100% risk weight. The regulator raises the risk weight on that asset class to 150%, with no change in capital or exposures. Which of the following is closest to the bank's capital adequacy ratio after the change?

A. 8.00%

B. 10.91%

C. 12.00%

D. 13.33%

Question CR-5 c.4

Bank Q166 • AnalystPrep Q.5938 (rebuilt)

A supervisor applying the Basel III minimum capital adequacy ratio of 10.5% reviews two banks (USD million):

	Tier 1 capital	Tier 2 capital	RWA
Bank A	80	20	1,000
Bank B	100	30	1,100

Which of the following statements is correct?

A. Only Bank A meets the minimum.

B. Only Bank B meets the minimum.

C. Both banks meet the minimum.

D. Neither bank meets the minimum.

Question CR-5 c.5

Bank Q160 • BT P2.T6.24.6.3 (rebuilt)

A credit analyst assesses Company XYZ's capital adequacy over the last three years using the three key capital ratios: CET1, Tier 1 and total capital, each divided by total RWAs. Excerpt from the annual report (USD million):

	2017	2016	2015
Common Equity Tier 1 capital	146,424	142,367	137,100
Additional Tier 1 capital	22,639	20,443	17,600
Tier 2 capital	22,456	27,564	38,200
Total regulatory capital	191,519	190,374	192,900
Credit risk RWAs	960,763	989,639	968,600
Market risk RWAs	44,100	36,910	49,600
Operational risk RWAs	293,825	256,300	224,300
Total RWAs	1,298,688	1,282,849	1,242,500

Over the three years, what do the three ratios signal about the company's capital adequacy?

A. Mixed

B. Declining

C. Improving

D. Not enough data to determine

Question CR-5 c.6

Bank Q161 • Schweser Module Quiz 23.1 Q3 (rebuilt)

A bank's model validation team is summarising the internal ratings-based (IRB) approach it uses to compute risk-weighted assets for its capital adequacy ratio. Which of the following statements relates correctly to the IRB approach?

A. Idiosyncratic risk is explicitly factored into the model.

B. Estimates of the probability of default are made by regulators.

C. Maturity adjustments should be higher for borrowers with high probabilities of default.

D. The maturity parameter reflects that capital requirements should increase with maturity.

Question CR-5 c.7

Bank Q170 • AnalystPrep Q.5970

A bank using the advanced IRB approach under Basel II is reviewing the asset correlation parameter in the RWA formula. Which statement correctly describes how the asset correlation depends on borrower characteristics?

A. Asset correlation decreases with firm size, because larger firms have higher idiosyncratic risk.

B. Asset correlation is constant regardless of PD and firm size.

C. Asset correlation increases with PD, because higher-PD firms are more exposed to the systematic factor.

D. Asset correlation decreases as PD increases and increases with firm size, because larger firms are more closely tied to the general economy.

Question CR-5 c.8

Bank Q171 • BT P2.T7.24.10.3 (rebuilt)

Bank Fortuna is estimating risk-weighted assets for a retail loan portfolio under the Basel II IRB approach, using $\text{RWA} = 12.5 \times \text{EAD} \times \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA}$:

Exposure at default (EAD)	USD 400,000,000
Loss given default (LGD)	35%
Worst-case default rate (WCDR)	10%
Probability of default (PD)	2%
Maturity adjustment (MA)	1 (none applies to retail)

Which of the following is closest to the portfolio's RWA?

A. USD 11.2 million

B. USD 35.0 million

C. USD 140.0 million

D. USD 175.0 million

Question CR-5 c.9

Workbook D3.1 • S2, S3 • Map #61

A regulatory reporting analyst at a Singapore bank is preparing the year-end capital adequacy return. The bank's banking book and capital are summarised below (SGD million):

- Corporate loans under foundation IRB: EAD 2,000; IRB capital requirement $K = 7.5\%$ of EAD
- Residential mortgages under the standardized approach: exposure 3,000; prescribed risk weight 35%
- Claims on the domestic sovereign: exposure 1,500; prescribed risk weight 0%
- Claims on other banks: exposure 800; prescribed risk weight 20%
- Tier 1 capital: 262; Tier 2 capital: 70
- Internal economic capital for credit risk ($\text{UL}_p \times$ CM at 99.97%): 300

The supervisor applies the Basel III minimum capital adequacy ratio of 10.5%. Which of the following is closest to the bank's risk-weighted assets and capital adequacy ratio, together with the correct conclusion?

A. RWA 7,300; CAR 4.5%; the bank falls short of the 10.5% minimum

B. RWA 3,085; CAR 8.5%; the bank falls short of the 10.5% minimum

C. RWA 3,210; CAR 10.3%; the bank falls short of the 10.5% minimum

D. RWA 3,085; CAR 10.8%; the bank meets the 10.5% minimum

Solutions • CR-5 c

Solution CR-5 c.1

Answer: C

Bank Q162 • AnalystPrep Q.3102

Step 1 – Risk-weighted assets

$$\text{RWA} = 400(100\%) + 300(50\%) + 200(20\%) = 400 + 150 + 40 = \text{USD } 590 \text{ million}$$

Step 2 – Capital

Tier 1 plus Tier 2: $60 + 20 = 80$.

Step 3 – Capital adequacy ratio

$$\text{CAR} = 80/590 = 13.56\%$$

$\Rightarrow$ Comfortably above the 8% Basel II minimum.

Interpretation

The weights shrink USD 900 million of assets to USD 590 million of RWA; the high-quality bank loans count for only one fifth of their size.

Why the other options are wrong

A is incorrect. Divided by unweighted total assets: $80/900 = 8.89\%$.

B is incorrect. Used Tier 1 only: $60/590 = 10.17\%$, the Tier 1 ratio.

D is incorrect. Applied the simple average weight, $(100 + 50 + 20)/3 = 56.67\%$, to all assets: $80/510 = 15.69\%$, ignoring the sizes of the books.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.2

Answer: B

Bank Q165 • BT P2.T7.24.10.1

Step 1 – Basel I

$$100(100\%) + 10(0\%) + 50(50\%) = 100 + 0 + 25 = \text{USD } 125.0 \text{ million}$$

Step 2 – Basel II standardized

$$100(50\%) + 10(0\%) + 50(35\%) = 50 + 0 + 17.5 = \text{USD } 67.5 \text{ million}$$

$\Rightarrow$ The finer Basel II weights reward the A rating and the mortgage collateral.

Interpretation

The same balance sheet needs much less capital under Basel II's rating-sensitive weights: at 8%, 10.0 against 5.4.

Why the other options are wrong

A is incorrect. Swapped the two regimes' results.

C is incorrect. Put a 100% weight on the OECD government loan under Basel I: $100 + 10 + 25 = 135$.

D is incorrect. Used the unweighted book value for Basel I: $100 + 10 + 50 = 160$.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.3 — Answer: B

Bank Q164 • AnalystPrep Q.5961

Step 1 – Change in RWA

The USD 200 million class moves from $200 \times 100\% = 200$ to $200 \times 150\% = 300$ of RWA: an increase of 100.

Step 2 – New RWA and CAR

$$\text{RWA} = 1{,}000 + 100 = 1{,}100, \qquad \text{CAR} = 120/1{,}100 = 10.91\%$$

$\Rightarrow$ Down from $120/1{,}000 = 12.00\%$ before the change.

Interpretation

With capital fixed, any rise in a risk weight lowers the CAR; only the re-weighted class changes.

Why the other options are wrong

A is incorrect. Scaled all RWA by 1.5: $120/1{,}500 = 8.00\%$.
C is incorrect. Ignored the change: $120/1{,}000 = 12.00\%$.
D is incorrect. Moved RWA the wrong way, to 900: $120/900 = 13.33\%$.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.4 — Answer: B

Bank Q166 • AnalystPrep Q.5938

Step 1 – Bank A

$$\text{CAR}_A = (80 + 20)/1{,}000 = 10.00\% < 10.5\%$$

Step 2 – Bank B

$$\text{CAR}_B = (100 + 30)/1{,}100 = 11.82\% \geq 10.5\%$$

$\Rightarrow$ Only Bank B meets the Basel III minimum.

Interpretation

Bank A would pass the 8% Basel II test; the 2.5% conservation buffer in Basel III's 10.5% is what it fails.

Why the other options are wrong

A is incorrect. Reverses the result: Bank A's 10.00% is below 10.5%.
C is incorrect. True only against the 8% Basel II minimum; the stem applies 10.5%.
D is incorrect. Uses Tier 1 only: $80/1{,}000 = 8.00\%$ and $100/1{,}100 = 9.09\%$, both below 10.5%.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.5 — Answer: A

Bank Q160 • BT P2.T6.24.6.3

Step 1 – CET1 ratio (CET1 / total RWAs)

2015: $137{,}100/1{,}242{,}500 = 11.03\%$; 2016: 11.10%; 2017: $146{,}424/1{,}298{,}688 = 11.27\%$. Rising.

Step 2 – Tier 1 ratio (CET1 + AT1)

2015: $154{,}700/1{,}242{,}500 = 12.45\%$; 2016: 12.69%; 2017: $169{,}063/1{,}298{,}688 = 13.02\%$. Rising.

Step 3 – Total capital ratio
2015: $192{,}900/1{,}242{,}500 = 15.53\%$; 2016: 14.84%; 2017: $191{,}519/1{,}298{,}688 = 14.75\%$. Falling.
$\Rightarrow$ Two ratios improve and one declines: mixed.

Interpretation
High-quality capital grew faster than RWAs, but Tier 2 fell from 38,200 to 22,456, pulling the total ratio down. Every ratio uses total RWAs (credit, market and operational).

Why the other options are wrong
B is incorrect. Looks only at the total capital ratio and ignores the rising CET1 and Tier 1 ratios.
C is incorrect. Looks only at CET1 and Tier 1 and ignores the fall in the total capital ratio.
D is incorrect. Capital by tier and total RWAs are all given; each ratio can be computed.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.6 Answer: D

Bank Q161 • Schweser Module Quiz 23.1 Q3

Step 1 – The maturity adjustment
GARP: long-term borrowers are riskier, since downgrades are more likely over a longer life, so the capital requirement should increase with maturity; and low-PD borrowers have more room to be downgraded, so their adjustment is larger.
$\Rightarrow$ D states the first consideration.

Step 2 – The other building blocks
The ASRF model assumes a completely diversified portfolio, so idiosyncratic risk cancels out; PD comes from the bank's own internal rating systems.

Interpretation
In $b = (0.11852 - 0.05478 \ln \text{PD})^2$, a lower PD makes $-\ln \text{PD}$ larger, so b and the adjustment are larger: the formula encodes the low-PD point directly.

Why the other options are wrong
A is incorrect. The ASRF model assumes idiosyncratic risk is diversified away; only systematic risk matters.
B is incorrect. Under IRB the bank estimates PD with its internal systems; supervisors set LGD only in foundation IRB.
C is incorrect. Reversed: the adjustment is higher for *low*-PD borrowers.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.7 Answer: D

Bank Q170 • AnalystPrep Q.5970

Step 1 – Correlation and PD
GARP: asset correlations decrease with increasing PD, because the higher the PD, the larger the idiosyncratic component of the obligor's risk.

Step 2 – Correlation and size
Asset correlations increase with firm size: larger firms are more closely tied to the general economy, smaller firms default more for idiosyncratic reasons.
$\Rightarrow$ D states both effects with the right direction.

Interpretation

R is the borrower's dependence on the single systematic factor; it is set by a Basel Committee formula, not estimated by the bank.

Why the other options are wrong

A is incorrect. Reversed: correlation rises with firm size, and it is smaller firms whose risk is more idiosyncratic.

B is incorrect. The Basel formula makes correlation vary with PD and size.

C is incorrect. Reversed: correlation falls as PD rises.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.8

Answer: C

Bank Q171 • BT P2.T7.24.10.3

Step 1 – Capital charge per unit of EAD

$$K = \text{LGD} \times (\text{WCDR} - \text{PD}) \times \text{MA} = 0.35 \times (0.10 - 0.02) \times 1 = 0.028$$

$\Rightarrow$ Capital covers unexpected loss only: the 99.9% default rate less the expected default rate.

Step 2 – RWA

$$\text{RWA} = 12.5 \times 400 \times 0.028 = \text{USD } 140.0 \text{ million}$$

Interpretation

The capital requirement is $0.028 \times 400 = 11.2$ million; RWA is 12.5 times that, so that 8% of RWA returns the capital charge.

Why the other options are wrong

A is incorrect. Dropped the 12.5: $400 \times 0.35 \times 0.08 = 11.2$ is the capital, not RWA.

B is incorrect. Used PD instead of WCDR − PD: $12.5 \times 400 \times 0.35 \times 0.02 = 35.0$.

D is incorrect. Used the WCDR alone, double counting expected loss: $12.5 \times 400 \times 0.35 \times 0.10 = 175.0$.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. • **Reference:** Doumpos et al. 2019, Ch.1 (GARP CR Ch.5), 5.5

Solution CR-5 c.9

Answer: D

Workbook D3.1 • S2, S3

Step 1 – RWA of the IRB book

$$\text{RWA}_{\text{IRB}} = K \times 12.5 \times \text{EAD} = 0.075 \times 12.5 \times 2{,}000 = 1{,}875$$

$\Rightarrow$ The corporate book carries a 93.75% effective risk weight: K is a capital charge, and 12.5 $(= 1/8\%)$ converts it back into risk-weighted assets.

Step 2 – RWA of the standardized books

$$\text{RWA}_{\text{SA}} = 0.35(3{,}000) + 0(1{,}500) + 0.20(800) = 1{,}050 + 0 + 160 = 1{,}210$$

$\Rightarrow$ Prescribed weights set by the supervisor: the sovereign book adds nothing, the interbank book only one fifth of its face value.

Step 3 – Total RWA

$$\text{RWA} = 1{,}875 + 1{,}210 = 3{,}085$$

Step 4 – Capital adequacy ratio

$$\text{CAR} = \frac{\text{Tier 1} + \text{Tier 2}}{\text{RWA}} = \frac{262 + 70}{3{,}085} = \frac{332}{3{,}085} = 10.76\%$$

$\Rightarrow$ The bank holds SGD 10.76 of regulatory capital for every SGD 100 of risk-weighted assets, about 0.26 percentage points above the 10.5% minimum.

Interpretation

The bank meets the Basel III level (8% total capital plus the 2.5% conservation buffer), but with a thin cushion of about SGD 8 million ($332 - 0.105 \times 3{,}085 = 8.1$). The SGD 300 million of economic capital is a red herring: it is the bank's own UL-based estimate at its chosen confidence level and plays no part in the regulatory ratio, which is a formula-driven minimum.

Why the other options are wrong

A is incorrect. Divided capital by unweighted total exposures, $2{,}000 + 3{,}000 + 1{,}500 + 800 = 7{,}300$, giving $332/7{,}300 = 4.5\%$: a leverage-style ratio, not the risk-based CAR.

B is incorrect. Correct RWA, but used Tier 1 only in the numerator: $262/3{,}085 = 8.5\%$. That is the Tier 1 ratio; the CAR counts Tier 1 plus Tier 2.

C is incorrect. Entered the IRB loans at their EAD (an implicit 100% weight) instead of $K \times 12.5 \times \text{EAD}$: $\text{RWA} = 2{,}000 + 1{,}210 = 3{,}210$, $\text{CAR} = 332/3{,}210 = 10.3\%$, which wrongly flips the conclusion.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. CR-4 g: Describe how economic capital is derived. • **Reference:** Doumpos et al., Ch.1, 5.5; Schroeck, Ch.5, Box 4.1 and eq. 4.17

CR-6 • Credit Scoring and Rating • Vardeez: MODERATE

CR-6 b

P3 CONCEPT

Differentiate between through-the-cycle credit rating system and point-in-time credit rating system.

Workbook: D7 Early Warning Indicators and PIT vs TTC • Tier D • Map #45 (2/16)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: Map #45 reports "point in time & through the cycle" twice in the same recall list (D7). The rest of CR-6 is a read-only chapter.
Vardeez: credit rating agency methodologies, "just give a good read". His pairing to hold: the system that is slower to move is the agencies' and suits long-term loans; the one that reacts fastest is the banks' internal system for short-term loans.

Reported exam shapes

- **D7·S3** PIT versus TTC: definitions and users.
- **D7·S4** The procyclicality polarity: PIT is the volatile, procyclical one.

What the questions look like: all conceptual. Two philosophies and five properties (horizon, sensitivity, update frequency, user, loan type), plus one consequence: what each does to capital over the cycle. The distractors swap the labels, swap the users, or claim that one philosophy is simply more accurate.

Note: D7.2 also tags ORR-18 d and appears in the ORR book as well.

1. Scoring and rating: the shared background for CR-6

Both quantify a borrower's credit risk, but GARP (Doumpos et al.) separates them by output and use:

- **Credit scoring**: a *numerical* score from automated analytical models, for consumers and small businesses, *solely for internal use* by lenders and corporate clients.
- **Credit rating**: a grade on an *ordinal* scale, each grade tied to an empirically estimated PD; used for corporate loans, bond issues and countries, often *publicly available*, and may combine analytical and judgmental assessment.

Either can be built internally (banks, on their own historical data) or bought externally (agencies, bureaus, consultants). The four benefits: less subjectivity in loan evaluation; analytical credit risk management (scenario analysis, stress testing); consistency and transparency through a common framework; lower time and cost of appraisal. How the agencies assign their ratings, and the criticisms of them, are CR-6 d.

2. Two rating philosophies

The Basel Committee distinguishes two philosophies:

- **Through the cycle (TTC)**: "primarily of interest for major credit rating agencies". A long-term orientation covering *at least one business cycle*, so the assessment is robust to short-term changes and updated less often.
- **Point in time (PIT)**: "usually employed by internal systems developed by credit institutions and banks". Focuses on the current and short-term condition of the borrower, in most cases over no

more than one year. PIT ratings are more volatile and updated promptly as new information arrives, so they are used for short-term transactions and loans.

Two GARP qualifications are easy to miss:

- **Neither is simply better.** On predictive power "the findings from existing research are mixed"; the choice depends on the scope and requirements of the analysis.
- **Default is easier to define PIT** (for example, default within one year). Extending to TTC is hard because the *time until default* must be defined and modelled explicitly.

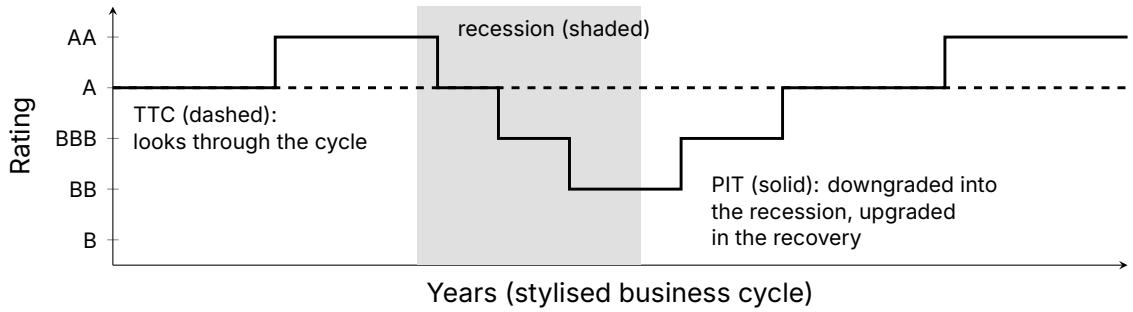


Figure 1. One borrower rated under the two philosophies (stylised, not data). The PIT rating (solid) tracks current conditions: it migrates down in the recession (shaded) and back up in the recovery, which is why PIT PDs and the capital built on them are procyclical. The TTC rating (dashed) is anchored to the whole cycle and barely moves.

3. The procyclicality polarity (shape S4)

Because a PIT rating follows the cycle, a borrower is *less likely to keep the same rating* under PIT than under TTC. Feed PIT PDs into economic capital and capital falls when the credit environment improves and rises when it deteriorates: it is **procyclical**. Crouhy, Galai and Mark (ORR-18 d) therefore keep PIT PDs (structural, KMV-type) for near-term expected loss and pricing, and TTC PDs (largely the agencies' approach) for economic capital, profitability and strategic decisions: "the TTC approach therefore reduces the volatility of economic capital".

Do not confuse this with the criticism of the agencies (CR-6 d). The agencies *claim* TTC ratings, yet critics found them too optimistic in growth and too pessimistic in recessions. Both statements hold: TTC is the stable philosophy by design, and the complaint is that agency practice moved with the cycle despite the label.

	Through the cycle (TTC)	Point in time (PIT)
Main users	Major rating agencies	Banks' internal systems
Horizon	At least one full business cycle	Current condition, usually up to one year
Updates, volatility	Robust to short-term changes; updated less often; stable	Reacts to new information; updated promptly; more volatile
Suited to	Long-term loans; economic capital, strategy	Short-term loans; near-term EL, pricing
Capital over the cycle	Smooths economic capital	Procyclical capital
Defining default	Hard: time to default must be modelled	Easier (default within one year)

Exam traps

- **Polarity flip.** PIT is the volatile, promptly updated, procyclical philosophy; TTC is the stable one. Options that call TTC "updated promptly" or "more volatile" have swapped the labels.
- **Role misassignment.** TTC belongs to the major agencies, PIT to banks' internal systems. "TTC relies on internal data" is a swap.
- **Loan horizon.** TTC suits long-term loans, PIT short-term ones and short-horizon investors.
- **Absolute claim on accuracy.** Research on predictive power is mixed; "PIT is unambiguously more accurate" (or TTC) is false.
- **TTC as a forecast.** A TTC rating does not anticipate the recession and downgrade early; it deliberately looks through it.
- **Identical borrowers, identical ratings.** The rating philosophy, not just the fundamentals, decides how much current conditions feed through.
- **Agency procyclicality.** The CR-6 d criticism (agency ratings behaved procyclically) does not make TTC the volatile philosophy.

Practice questions: CR-6 b

Question CR-6 b.1

Bank Q232 • AnalystPrep Q.5886

Which of the following best characterizes a through-the-cycle (TTC) credit rating approach when compared to a point-in-time (PIT) approach?

A. TTC assessments react swiftly to short-term economic fluctuations and are frequently updated to reflect the current credit condition of the borrower.

B. PIT ratings take a long-term perspective, often encompassing a full business cycle, and aim to remain stable against economic ups and downs.

C. TTC ratings have a long-term view and tend to be less sensitive to short-term economic conditions than PIT ratings.

D. Both TTC and PIT ratings prioritize near-term economic indicators and are subject to frequent revisions.

Question CR-6 b.2

Bank Q235 • AnalystPrep Q.5889

Which of the following statements is true regarding the frequency of updates in credit rating systems?

A. TTC rating systems are updated more frequently than PIT systems to give a timely reflection of economic conditions.

B. PIT rating systems are typically updated less frequently, coinciding with significant shifts in the business cycle.

C. PIT rating systems are updated more frequently than TTC systems to incorporate current conditions and short-term trends.

D. Both TTC and PIT systems are updated on a fixed annual schedule regardless of economic or borrower changes.

Question CR-6 b.3 Bank Q238 • Schweser Module Quiz 24.1 Q2

A lender is considering using a through-the-cycle approach when assessing the creditworthiness of a potential borrower. Which of the following statements regarding this approach is correct? This method:

A. will capture default risk in the best way possible.

B. should be used if the loan is short term in nature.

C. should be used if the loan is long term in nature.

D. is a robust option because it relies on internal data only.

Question CR-6 b.4 Bank Q236 • AnalystPrep Q.5890

Which of the following might be a consequence of relying on a TTC credit rating system for external reporting compared to a PIT system?

A. Stakeholders might perceive an overstatement of credit risk during downturns because of the system's sensitivity to short-term fluctuations.

B. External reports would likely show less volatility in credit risk metrics over time, giving a more consistent long-term perspective.

C. Financial reports might display rapid changes in credit risk assessments that mirror the market's perception during economic instability.

D. Reported credit risk may change frequently, requiring frequent updates to external communication and possibly confusing stakeholders.

Question CR-6 b.5 Bank Q231 • AnalystPrep Q.5388 (rebuilt)

Bank A uses a through-the-cycle (TTC) rating system and Bank B uses a point-in-time (PIT) rating system. Both banks lend to identical borrowers, and all borrowers were rated at the start of the year when economic conditions were benign. During the year the country enters a severe recession. Which of the following is the most likely observation about rating changes at year-end?

A. Bank A's ratings show substantially more downgrades than Bank B's, because TTC ratings react most quickly to the recession.

B. Bank B's ratings show substantially more downgrades than Bank A's, because PIT ratings reflect current conditions while TTC ratings are robust to short-term changes.

C. Both banks' ratings show the same downgrades, because the borrowers and their fundamentals are identical.

D. Bank A's ratings show more downgrades than Bank B's, because TTC ratings look ahead across a full cycle and anticipate the recession.

Question CR-6 b.6 Workbook D7.2 • S3, S4 • Map #45 • also ORR-18 d

A credit risk manager at a German bank is presenting to a group of newly hired analysts on the rating philosophies used across the bank. Its internal rating system for the short-term trade finance book is point-in-time (PIT), its economic capital model currently uses through-the-cycle (TTC) PDs, and its bond portfolio relies on agency ratings. Which of the following statements is correct for the manager to make?

A. Because TTC ratings are updated promptly to reflect any new information about a borrower, they are the more volatile of the two philosophies and the more procyclical.

B. TTC assessments are mainly employed in banks' internal systems for short-term loans, while PIT assessments are primarily of interest for the major credit rating agencies.

C. Research shows that PIT assessments have unambiguously higher predictive power than TTC assessments, so the bank should use PIT PDs for every purpose, including economic capital.

D. Switching the economic capital model from TTC to PIT PDs would make capital more volatile and procyclical, because a borrower is less likely to keep the same rating under a PIT approach.

Solutions • CR-6 b

Solution CR-6 b.1

Answer: C

Bank Q232 • AnalystPrep Q.5886

Step 1 – Recall the TTC profile

GARP: TTC assessments have "a long-term orientation covering at least one business cycle", so they are "robust to short-term changes" and updated less often.

Step 2 – Recall the PIT profile

PIT focuses on the current and short-term condition (usually up to one year), is more volatile and is updated promptly.

⇒ Only C gives TTC the long-term, less sensitive profile.

Interpretation

Every distractor on this LO is built from the same two columns: describe one philosophy and attach the other's label. Fix the column first, then read the label.

Why the other options are wrong

A is incorrect. Describes PIT (swift reaction, frequent updates) and calls it TTC.

B is incorrect. Swaps the labels the other way: the long-term, stable profile belongs to TTC, not PIT.

D is incorrect. Wrong for TTC: only PIT prioritises near-term conditions and frequent revision.

LO: CR-6 b: Differentiate between through-the-cycle credit rating system and point-in-time credit rating system. • **Reference:** Doumpos et al., Analytical Techniques in the Assessment of Credit Risk, Ch.2 (GARP CR Ch.6), 6.2.1 Through the Cycle and Point in Time Assessments

Solution CR-6 b.2

Answer: C

Bank Q235 • AnalystPrep Q.5889

Step 1 – Update frequency of each philosophy

TTC assessments "are subject to less frequent updates". PIT ratings "are updated in a timely manner to reflect any new information regarding the status of the borrower".

⇒ PIT is updated more often than TTC.

Interpretation

Update frequency follows from the horizon: a rating meant to describe the whole cycle has no reason to move on short-term news, while a rating meant to describe the next year must.

Why the other options are wrong

A is incorrect. Reverses the frequencies.

B is incorrect. Describes TTC behaviour (infrequent updates tied to the cycle) and attributes it to PIT.

D is incorrect. Neither philosophy is defined by a fixed calendar; PIT updates as information arrives. (Basel's separate minimum is that every borrower is re-rated at least annually, CR-6 d.)

LO: CR-6 b: Differentiate between through-the-cycle credit rating system and point-in-time credit rating system. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.2.1

Solution CR-6 b.3 Answer: C

Bank Q238 • Schweser Module Quiz 24.1 Q2

Step 1 – Match the horizon to the loan

A TTC assessment covers at least one full business cycle, a long-term view.
⇒ It matches the long maturity of a long-term loan; PIT, focused on the next year or so, suits short-term loans and transactions.

Interpretation

Vardeez's pairing: slow-moving TTC, used by the agencies, for long-term loans; fast-moving PIT, run internally by banks, for short-term loans.

Why the other options are wrong

A is incorrect. Absolute claim: research on the predictive power of the two approaches is mixed, and PIT captures current default risk more quickly.
B is incorrect. Short-term loans call for PIT.
D is incorrect. Role misassignment: TTC is primarily the agencies' philosophy; banks' internal systems are typically PIT.

LO: CR-6 b: Differentiate between through-the-cycle credit rating system and point-in-time credit rating system. •
Reference: Doumpos et al., Ch.2 (GARP CR Ch.6), 6.2.1; SchweserNotes 2026 Book 2, Reading 24, LO 24.b

Solution CR-6 b.4 Answer: B

Bank Q236 • AnalystPrep Q.5890

Step 1 – What TTC does to a time series of risk metrics

TTC ratings are robust to short-term changes and updated less often, so ratings, and the risk metrics built on them, change little from period to period.
⇒ Externally reported credit risk is smoother and more consistent over time.

Interpretation

The price of that stability is timeliness: a TTC report may not show a short-term deterioration that a PIT report would pick up at once.

Why the other options are wrong

A is incorrect. Sensitivity to short-term fluctuations is the PIT trait, not TTC.
C is incorrect. Rapid, market-mirroring changes describe PIT.
D is incorrect. Frequently changing reported risk is the PIT consequence.

LO: CR-6 b: Differentiate between through-the-cycle credit rating system and point-in-time credit rating system. •
Reference: Doumpos et al., Ch.2 (GARP CR Ch.6), 6.2.1

Solution CR-6 b.5 Answer: B

Bank Q231 • AnalystPrep Q.5388

Step 1 – Classify the shock

A recession is a change in current conditions within the cycle.

Step 2 – Apply each philosophy

Bank B's PIT system reflects the current condition of each borrower, so its ratings migrate down sharply. Bank A's TTC system is designed to be robust to short-term changes, so its ratings move little.
⇒ The recession shows up mainly in Bank B's downgrades (Figure 1).

Interpretation

This is the mechanism behind shape S4: PIT ratings migrate with the cycle, so capital built on them is procyclical.

Why the other options are wrong

A is incorrect. Reverses the philosophies: TTC is the stable, slow-moving one.

C is incorrect. Identical borrowers do not imply identical ratings; the philosophy decides how much of the recession feeds through.

D is incorrect. Confuses the long TTC horizon with forecasting a downturn. TTC looks through the cycle and deliberately does not move with the recession.

LO: CR-6 b: Differentiate between through-the-cycle credit rating system and point-in-time credit rating system. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.2.1

Solution CR-6 b.6 Answer: D

Workbook D7.2 • S3, S4

Step 1 – Which philosophy is volatile

Doumpos: PIT ratings "are more volatile, and they are updated in a timely manner"; TTC assessments "are robust to short-term changes and they are subject to less frequent updates".

Step 2 – What that does to economic capital

Crouhy, Galai and Mark: "The probability of a firm's staying in the same rating when it is assessed using a PIT approach is smaller than when it is assessed using a TTC approach. The TTC approach therefore reduces the volatility of economic capital". Capital calibrated to current conditions is "more procyclical (capital goes down when the credit environment improves and goes up when it deteriorates)".

$\Rightarrow$ D is correct. The bank's current split (PIT for short-term trade finance, TTC for economic capital) is exactly Crouhy's.

Interpretation

Crouhy keeps PIT PDs for near-term expected loss and pricing, and TTC PDs for economic capital, profitability and strategic decisions; comparing both in RAROC at a normal point and at the worst point of the cycle shows the effect.

Why the other options are wrong

A is incorrect. Polarity flip: prompt updating and volatility describe PIT, not TTC.

B is incorrect. Role misassignment: TTC is "primarily of interest for major credit rating agencies"; PIT is "usually employed by internal systems developed by credit institutions and banks".

C is incorrect. Absolute claim: "the findings from existing research are mixed", so the choice depends on the scope of the analysis, and Crouhy reserves TTC for economic capital.

LO: CR-6 b: Differentiate between through-the-cycle credit rating system and point-in-time credit rating system. ORR-18 d: Explain challenges that arise when using RAROC for performance measurement, including choosing a time horizon, measuring default probability, and choosing a confidence level. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.2.1; Crouhy, Galai and Mark, The Essentials of Risk Management 2e, Ch.17, Default Probabilities: Point-in-Time vs. Through-the-Cycle

CR-6 • Credit Scoring and Rating • Vardeez: MODERATE

CR-6 d

P3 CONCEPT

Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings.

Workbook: none homed here • Vardeez focus (D7 notes carry the procyclicality criticism)

WHAT IS TESTED HERE

Why P3. Vardeez flags it in a Moderate chapter: "credit rating agency methodologies: give it a good read". No workbook question.
Reported exam shapes: none listed in the map.

What the questions look like: all conceptual, in two halves. *Methodology*: issuer versus issue-specific ratings, the rating scale and the investment-grade line, the inputs that strengthen a rating, how the agencies combine analysis and judgment, and Basel's requirements for rating systems. *Criticisms*: GARP's four (transparency and conflict of interest, debt explosion, poor predictive ability, procyclicality), usually as "which is (not) a criticism".

Re-filed: Bank Q243 (Basel's specification requirements) from CR-6 c, a P4 LO; Vardeez and Schweser both teach the requirements under this LO.
Recap: scoring versus rating and the TTC claim are in CR-6 b.

1. Issuer ratings and issue-specific ratings

- **Issuer credit rating**: "a forward-looking opinion about an obligor's creditworthiness". It can be long- or short-term, and comes as counterparty, corporate or sovereign ratings. The usual reason to obtain one is *access to capital markets or other funding*: a top rating means lower interest rates, easier external financing and better terms with suppliers and investors.
- **Issue-specific rating**: an opinion on the obligor's creditworthiness *with respect to a specific obligation* (for example, a specific loan). It considers the terms of that debt: its duration, the collateral, the use of the loan (for a mortgage, the value of the house) and anything else in the risk profile of the issue.

So one issuer can have several issue ratings that differ from its issuer rating and from each other (Figure 1).

2. How the agencies assign ratings

- **The market.** Moody's, S&P and Fitch hold more than 90% between them (Moody's and S&P about 40% each, Fitch about 15% with about 350,000 ratings). They rate government, corporate and municipal bonds, certificates of deposit, preferred stock, and MBS and CDOs.
- **The scale.** Ordinal, multi-grade: investment grades Aaa/AAA, Aa/AA, A/A, Baa/BBB; speculative grades from Ba/BB down; defaults at the bottom. Historical one-year default rates are below 0.2% on average for investment grades and range from about 0.7% (Ba/BB) to more than 25% (C and D) for speculative grades (GARP Table 6.2).
- **Horizons.** Short-term ratings measure risk over a short period (typically one year); medium- and long-term ratings cover longer periods, sometimes on different scales.
- **The method.** Agency ratings "combine analytical and judgmental approaches" over a wide range of inputs: financial ratios (profitability, solvency, liquidity, management efficiency), behavioural

and transaction data, size and age, market conditions and competitive advantages, stock market data for listed firms, corporate governance, and news. The data come from proprietary databases, industry and market reports, and *on-site meetings* between agency analysts and the rated company's executives. A credit *score*, by contrast, is the output of an automated model.

- **Basel's five requirements** for any scoring or rating system (GARP 6.3; Vardeez and Schweser put them here): (1) meaningful differentiation of risk, with grades designed to distinguish risk rather than to minimise regulatory capital, borrowers in the same grade treated differently by transaction, and no excessive concentration in one grade; (2) continuous evaluation, with every borrower re-rated at least annually; (3) oversight, with monitoring, controls such as stress tests, and feedback; (4) correct selection of risk attributes; (5) a substantial, representative database.

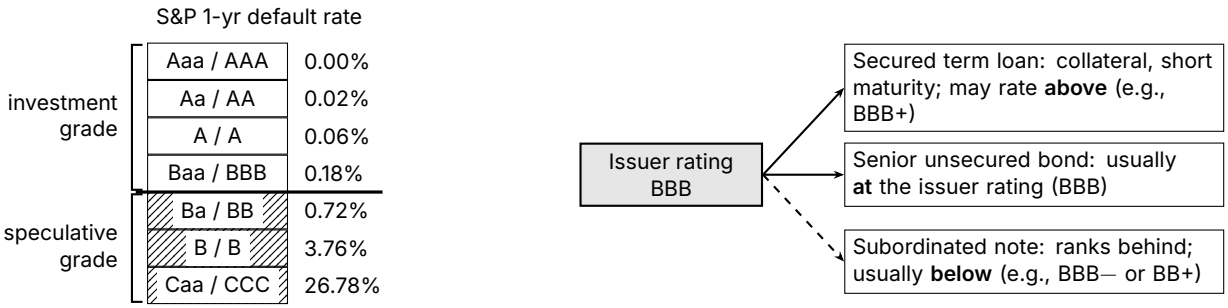


Figure 1. Left: the ordinal scale (investment grades unshaded, speculative grades hatched; the thick line is the investment-grade boundary) with S&P's historical average one-year default rates from GARP Table 6.2. Right: one issuer, three issue-specific ratings. GARP names the factors an issue rating weighs (duration, collateral, use of the loan); the notch sizes shown are illustrative.

3. The main criticisms

GARP lists four (Vardeez and Schweser count five by splitting the first):

Criticism	What it says
Lack of transparency and accountability; conflict of interest	How agencies prepare ratings is generally not publicly disclosed. Their revenues depend on fees paid by the entities being rated (issuer pays).
Promoting debt explosion	Better risk estimates let lenders cut risk premiums and the cost of debt. Debt became more accessible, which may raise systemic risk through credit bubbles and crunches.
Poor predictive ability	The agencies missed Lehman Brothers, Enron, WorldCom and the 2007 to 2008 crisis; studies find ratings weaker predictors of default than accounting- and market-based models and banks' internal systems.
Procyclicality	Agencies claim TTC ratings, yet ratings were too optimistic in growth and too pessimistic in recessions, which may exacerbate crises.

The balancing view. Ratings remain integral to risk management. They are good indicators of *systematic* risk, are used to benchmark internal models, and work best as a complement to other approaches: one measure is rarely enough.

Exam traps

- **Issuer versus issue swap.** The issue rating is the one that weighs collateral, maturity and use of funds; the issuer rating is the entity-wide view.
- **Investment-grade boundary.** The lowest investment grade is Baa/BBB; Ba/BB is the top of speculative grade.
- **Ratio direction.** Higher margins and lower leverage support a stronger rating.
- **Debt explosion polarity.** The criticism is that ratings made debt *too accessible*, not that they were too conservative and restricted lending.
- **Who pays.** The conflict of interest arises because the *rated entities* pay the fees, not investors and not shareholders.
- **Transparency versus timeliness.** Undisclosed methodology is the transparency criticism; downgrading after the trouble is visible is poor predictive ability.
- **Basel requirements.** Grades must distinguish risk, not manage capital; borrowers in one grade may be treated differently; re-rate at least annually, not only after an event.

Practice questions: CR-6 d

Question CR-6 d.1

Bank Q245 • AnalystPrep Q.5896 (rebuilt)

Which statement correctly distinguishes an issuer credit rating from an issue-specific rating?

A. An issuer rating reflects the terms, collateral and use of a specific loan, whereas an issue-specific rating is a general view of the entity.

B. An issuer rating is a forward-looking opinion on an obligor's overall creditworthiness and can help the entity access capital markets or funding, whereas an issue-specific rating also considers the terms, collateral and use of a specific obligation.

C. An issue-specific rating ignores collateral and loan terms and always equals the issuer rating.

D. An issuer rating is a backward-looking summary of historical default statistics used only for regulatory capital.

Question CR-6 d.2

Bank Q246 • AnalystPrep Q.5897 (rebuilt)

On the Moody's / S&P scale, which rating is the lowest category still classified as investment grade?

A. Baa / BBB

B. Ba / BB

C. A / A

D. Caa / CCC

Question CR-6 d.3 New • built from GARP Ch.6, 6.4.1 and 6.5

The treasurer of a mid-sized Spanish manufacturer is preparing for the company's first rated bond issue and asks the bank's credit analyst how the agency will arrive at its rating. Which of the following statements is correct for the analyst to make?

A. The rating will be the output of an automated scoring model applied to the latest financial ratios, so no meeting with management is involved and qualitative factors are excluded.

B. The agency will publish one rating on a single scale for all horizons, since short-term and long-term credit risk are measured identically.

C. The rating will be updated in real time as the company's share price moves, in the same way as a bank's point-in-time internal score.

D. The rating will combine analytical and judgmental assessment of a wide range of inputs, gathered from proprietary databases, industry and market reports, and on-site meetings between the agency's analysts and the company's executives.

Question CR-6 d.4 Bank Q244 • AnalystPrep Q.1778 (rebuilt)

A rating analyst compares four otherwise similar firms using financial ratios. Which firm's ratios would, other things equal, normally support the strongest credit rating?

A. Higher operating profit margins and a lower debt-to-equity ratio

B. Higher operating profit margins and a higher debt-to-equity ratio

C. Lower operating profit margins and a lower debt-to-equity ratio

D. Lower operating profit margins and a higher debt-to-equity ratio

Question CR-6 d.5 Bank Q243 • BT answers P2.T6.702.2 (rebuilt) • re-filed from CR-6 c

Which of the following statements about the Basel Committee's specification requirements for a credit scoring or rating system is correct?

A. Credit grades should be designed so as to manage regulatory capital requirements.

B. Borrowers in the same credit grade must be treated identically regardless of transaction-level differences.

C. Borrower ratings should be updated on a reasonable basis, in practice no less frequently than an annual review.

D. Borrower ratings need be reviewed only when a default or downgrade event occurs.

Question CR-6 d.6

Bank Q247 • AnalystPrep Q.5898

Credit rating agencies face various criticisms regarding their methodologies and practices. Which of the following criticisms points out a potential concern regarding the transparency of the rating process?

A. Ratings may not be as proactive as desired and are often downgraded after a financial downturn has become apparent.

B. CRAs are compensated by the entities they rate, leading to potential biases in the ratings provided.

C. Methodologies and factors influencing the final rating are not always made clear to investors or the entities being rated.

D. Ratings can lead to herding behavior among investors, exacerbating financial market volatility.

Question CR-6 d.7

Bank Q248 • AnalystPrep Q.5900

In relation to credit rating agencies, the aspect of "conflict of interest" has been a persistent concern. What does this criticism entail?

A. CRAs face inherent conflicts because they are publicly traded entities, with obligations to maximize shareholder value over rating accuracy.

B. CRAs have a conflict of interest due to the fact that they are typically compensated by the entities they rate, which may lead to biased ratings.

C. CRAs are paid by the investors who use the ratings, which gives them an incentive to publish overly pessimistic ratings.

D. CRAs are conflicted as they have to balance the interests of rated entities with those of investors using the ratings for investment decisions.

Question CR-6 d.8

Bank Q249 • BT answers P2.T6.24.8.3 (rebuilt)

Credit rating agency (CRA) ratings have attracted several well-known criticisms. Which of the following is NOT one of the main criticisms of CRA ratings?

A. Rating processes are not publicly disclosed, and fees paid by rated entities create conflicts of interest.

B. Ratings were too optimistic during economic expansions and too pessimistic during recessions.

C. Ratings are overly conservative, restricting debt availability and discouraging lending.

D. Ratings have been weaker predictors of corporate default than accounting- and market-based models.

Solutions • CR-6 d

Solution CR-6 d.1 — Answer: B

Bank Q245 • AnalystPrep Q.5896

Step 1 – Issuer rating

GARP: "a forward-looking opinion about an obligor's creditworthiness"; a strong one improves access to financing and relationships with stakeholders.

Step 2 – Issue-specific rating

An opinion about the obligor's creditworthiness "with respect to a specific financial obligation", considering its duration, collateral, the uses of the loan and other features of that debt.

⇒ Only B states both correctly.

Interpretation

Because the issue rating adds the terms of the debt, two issues of the same company can be rated differently (Figure 1).

Why the other options are wrong

A is incorrect. Swaps the two definitions.

C is incorrect. Issue ratings do weigh collateral and terms, so they need not equal the issuer rating.

D is incorrect. Issuer ratings are forward-looking, and their uses go well beyond regulatory capital.

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.2.2 Issuer Ratings and Issue-Specific Ratings

Solution CR-6 d.2 — Answer: A

Bank Q246 • AnalystPrep Q.5897

Step 1 – Recall the two sets of grades (GARP Table 6.1)

Investment grades: Aaa/AAA, Aa/AA, A/A, Baa/BBB. Speculative grades: Ba/BB, B/B, Caa/CCC, Ca/CC, C and D.

⇒ The investment-grade boundary sits between Baa/BBB and Ba/BB, so Baa/BBB is the lowest investment grade.

Interpretation

The boundary matters because default rates jump across it: S&P's average one-year default rate is 0.18% for BBB but 0.72% for BB, four times as high (Table 6.2).

Why the other options are wrong

B is incorrect. Ba/BB is the highest speculative grade, one step below the boundary.

C is incorrect. A/A is investment grade but two categories above the lowest.

D is incorrect. Caa/CCC is deep in speculative territory, with a high likelihood of default.

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.5 Credit Rating Agencies, Tables 6.1 and 6.2

Solution CR-6 d.3 — Answer: D

New • built from GARP Ch.6, 6.4.1 and 6.5

Step 1 – How the agencies work

GARP: ratings issued by the major agencies "combine analytical and judgmental approaches and consider a wide range of different input data" (the financial, behavioural, market, governance and news inputs of Sect. 6.4.1). The data come "through proprietary databases and industry/-market reports, as well as through onsite meetings of expert rating analysts with executives of the rated companies."

⇒ D describes the agency process.

Step 2 – Check the distractors against the scoring and PIT columns

An automated model with no judgment is credit *scoring*; real-time updating is a PIT trait; and agencies may use different scales for different horizons.

Interpretation

The judgmental element is what lets a rating weigh management quality, competitive position and governance; it is also one root of the transparency criticism, because the judgment is not publicly disclosed.

Why the other options are wrong

A is incorrect. Describes a credit score (automated, internal), not an agency rating.

B is incorrect. Agencies issue short-term ratings (typically one year) and medium- and long-term ratings, and "may adopt different rating scales for ratings of different horizons".

C is incorrect. Agency ratings are TTC by claim and updated infrequently (CR-6 b); real-time movement is the PIT profile.

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.4.1 Data Collection and Pre-processing; 6.5 Credit Rating Agencies

Solution CR-6 d.4 — Answer: A

Bank Q244 • AnalystPrep Q.1778

Step 1 – Profitability

Profitability ratios (profit margins, ROA, ROE) measure the ability to generate profits from assets and capital: higher is better.

Step 2 – Solvency

Solvency ratios (debt to assets, debt to equity, interest coverage) measure the debt burden: lower leverage is better.

⇒ The firm that is better on both, higher margins and lower debt-to-equity, supports the strongest rating.

Interpretation

Financial ratios are only one block of the agency's inputs, but they set the direction: stronger profitability and a lighter debt load both point to a lower probability of default.

Why the other options are wrong

B is incorrect. Treats higher leverage as a strength.

C is incorrect. Low leverage helps, but weaker margins mean less capacity to service debt.

D is incorrect. Worst on both dimensions, so the weakest rating.

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.4.1.1 Financial Data

Solution CR-6 d.5 — Answer: C

Bank Q243 • BT answers P2.T6.702.2

Step 1 – The five Basel requirements

Meaningful differentiation of risk; continuous evaluation of borrowers; oversight of the system's operation; correct selection of risk assessment attributes; a substantial database.

Step 2 – Continuous evaluation

"All borrowers and loans in a credit portfolio should be, at a minimum, annually re-rated", taking all new information into account.
⇒ C is correct.

Interpretation

The first requirement is the one examiners invert: grades must distinguish risk, "rather than to minimize regulatory capital requirements", and borrowers within a grade "must be treated differently according to their transaction's characteristics".

Why the other options are wrong

A is incorrect. Reverses meaningful differentiation: grades are designed to distinguish risk, not to manage capital.
B is incorrect. Reverses the rule: borrowers in the same grade are treated differently according to transaction characteristics.
D is incorrect. Event-only review contradicts continuous evaluation and the annual minimum.

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. CR-6 c: Describe the process for developing credit risk scoring and rating models. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.3 Modeling Requirements; SchweserNotes 2026 Book 2, Reading 24, LO 24.d

Solution CR-6 d.6 — Answer: C

Bank Q247 • AnalystPrep Q.5898

Step 1 – The transparency criticism

GARP: "the way CRAs prepare their risk ratings is generally not publicly disclosed, thus raising questions on the practices being followed."
⇒ Undisclosed methodology and rating drivers, option C, is the transparency concern.

Interpretation

GARP bundles transparency, accountability and conflict of interest in one bullet, which is why B is tempting. The question asks for the transparency strand specifically.

Why the other options are wrong

A is incorrect. Late downgrades are the poor-predictive-ability (timeliness) criticism.
B is incorrect. Issuer-pays fees are the conflict-of-interest strand, related but distinct.
D is incorrect. Herding by investors is not one of GARP's criticisms of agency ratings.

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.5 Credit Rating Agencies

Solution CR-6 d.7 — Answer: B

Bank Q248 • AnalystPrep Q.5900

Step 1 – Source of the conflict

GARP: "the fact that CRAs revenues depend on the fees paid by the entities being rated, raises further concerns about possible conflicts of interest between the raters and those rated."
⇒ The bias incentive runs towards favourable ratings for the fee payer: B.

Interpretation

The same issuer-pays logic resurfaces for sovereign ratings in CR-8 f, where GARP argues it is weaker because sovereign fees are small; there the upward bias comes mainly from sub-sovereign revenue and government-supplied data.

Why the other options are wrong

A is incorrect. Ownership structure is not the conflict GARP identifies.
C is incorrect. Reverses the fee flow and the direction of bias: issuers pay, and the incentive is towards favourable ratings.
D is incorrect. Too vague: it does not identify the revenue source that creates the conflict.

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.5 Credit Rating Agencies

Solution CR-6 d.8 — Answer: C

Bank Q249 • BT answers P2.T6.24.8.3

Step 1 – Map each option to GARP's list

A is transparency and conflict of interest; B is procyclicality; D is poor predictive ability.

Step 2 – Test C against the debt-explosion criticism

GARP: better risk estimates let lenders lower risk premiums, so "debt is now more accessible for firms and consumers", which may add systemic risk through credit bubbles and crunches.
⇒ C states the opposite of that criticism, so it is the one that is NOT a criticism.

Interpretation

The procyclicality criticism (B) is the bridge to CR-6 b: the agencies claim TTC ratings, yet in practice their ratings moved with the cycle.

Why the other options are wrong

A is incorrect. A genuine criticism: undisclosed processes and issuer-pays fees.
B is incorrect. A genuine criticism: procyclical behaviour despite the TTC claim.
D is incorrect. A genuine criticism: studies find ratings weaker predictors than accounting- and market-based models (Lehman, Enron, WorldCom).

LO: CR-6 d: Describe rating agencies' assignment methodologies for issue and issuer ratings, and identify the main criticisms of the credit rating agencies' ratings. • **Reference:** Doumpos et al., Ch.2 (GARP CR Ch.6), 6.5 Credit Rating Agencies

CR-7 • Credit Scoring and Retail Credit Risk Management • Vardeez: MODERATE

CR-7 b

P3 CONCEPT

Explain the differences between retail credit risk and corporate credit risk.

Workbook: D13 Retail Credit Risk • Tier D • Map #57 (2/16)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: Map #57 reports "retail credit". The examinable retail-specific material is the dark-side argument and risk-based pricing (D13); this LO is the base both stand on.

Vardeez: for CR-7, LOs f and h "can be important". His summary of this LO: what makes retail easier to model is the sheer number of small, independent borrowers; what makes corporate harder is that a few large exposures carry the portfolio.

Reported exam shapes

- **D13·S3** Retail versus corporate credit risk: predictable EL, and small but non-zero capital.

What the questions look like: conceptual. Bite-sized exposures against single-name concentrations, expected loss priced in against unexpected loss, early warnings against the "supertanker", and the regulatory capital treatment. Workbook D13.1 also covers shapes S1 (CR-7 h) and S2 (CR-7 c) and is worked here.

1. What counts as retail: the shared background for CR-7

Basel defines retail exposures as **homogeneous portfolios** of a large number of small, low-value loans, with a consumer or business focus, *where the incremental risk of any single exposure is small*. Examples: credit cards, installment loans (personal, education, auto, leasing), revolving credits (overdrafts, home equity lines) and residential mortgages. Small business loans qualify if the total exposure to the borrower is **below EUR 1 million**.

Mortgages and home equity loans are secured by the property (the loan-to-value ratio is the key risk variable); small business loans by business assets or the owners' guarantees; **credit card revolving loans are unsecured**. Credit risk is the main risk of retail banking; the others (interest rate, asset valuation, operational, business, reputation) are CR-7 a.

2. Five differences

- **Size of each exposure.** Retail exposures "arrive in bite-sized pieces, so that default by a single customer is never expensive enough to threaten a bank". Corporate portfolios hold large single-name exposures and concentrations in firms that are economically intertwined by region or sector.
- **Loss behaviour.** In normal markets a retail book behaves like a well-diversified portfolio, so the expected loss can be estimated, treated like any other cost of doing business (branches, cheque processing) and **built into the price**. Corporate risk "is dominated by the risk that credit losses will rise to some unexpected level".
- **Early warning.** A rise in retail defaults is often signalled by customer behaviour, such as a missed minimum card payment. The bank can then cut lending to existing customers, change its marketing and acceptance rules to attract less risky customers, or price in the risk. A commercial portfolio is "something of a supertanker": by the time trouble is obvious it is often too late.
- **Capital.** Regulators accept that retail risk is relatively predictable and mortgages safer, so retail banks set aside **a relatively small amount of capital** under Basel II and III: small, not zero. In re-

turn they must supply PD, LGD and EAD for clearly differentiated segments, segmented by credit score (or an equivalent) and by *vintage*. Basel uses three formulas for three subsectors: residential mortgages, revolving credit and other retail.

- **The caveat.** Crouhy warns the distinction "can be overstated": diversification can be "a fickle friend". In 2007 to 2009 house prices fell at about the same time across the US, and a systematic change in lending behaviour (lending without checking incomes) planted a hidden common risk in whole portfolios. That is the dark side, CR-7 c (dotted curve in Figure 1).

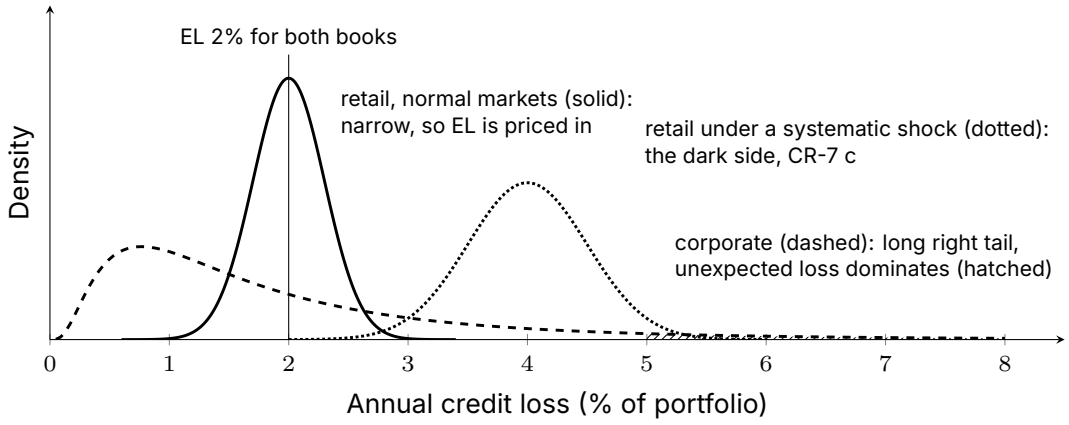


Figure 1. Stylised loss distributions with the same 2% expected loss. The diversified retail book (solid) is narrow, so its loss rate is predictable and can be charged to customers. The concentrated corporate book (dashed) has a long right tail (hatched beyond 5%, about 6% probability here). A common risk factor shifts the whole retail distribution to the right (dotted): diversification across many small loans does not protect against it.

	Retail credit risk	Corporate credit risk
Exposures	Many small, homogeneous loans	Fewer, large single names; sector and regional concentrations
One default	Never threatens the bank	Can cause a severe loss
Loss profile	Predictable EL, priced in as a cost of business	Dominated by unexpected loss
Warning signs	Customer behaviour signals trouble early	Often visible only when too late (supertanker)
Bank's response	Change lending, acceptance, marketing, pricing	Little room to act in time
Regulatory capital	Relatively small; PD, LGD, EAD by score and vintage segment	Larger for the same exposure
Weak spot	A common systematic factor (dark side)	Concentration

Exam traps

- **Size polarity.** Retail exposures are bite-sized; corporate ones are large. Any option giving retail "large pieces" or saying a single corporate default "rarely threatens" the bank is reversed.
- **Size of the book is not the risk.** Two books of equal total size can have very different risk profiles: granularity and concentration matter, not the total.

- **Absolute claim on capital.** Predictable EL does not mean zero capital: retail carries "a relatively small amount".
- **Early-warning polarity.** Retail gives behavioural warnings; commercial portfolios usually do not.
- **Sibling swap.** A large loss on one big customer is the corporate risk profile, not the retail dark side.
- **Diversification is not a guarantee.** It works in normal markets and fails against a systematic factor.

Practice questions: CR-7 b

Question CR-7 b.1

Bank Q256 • AnalystPrep Q.2010

Why is the credit exposure of retail banking different from that of corporate banking?

A. Credit exposures of retail banking arrive in large pieces, but individual exposures by a single consumer are not expensive enough to threaten the bank.

B. Credit exposures of retail banking arrive in bite-sized pieces; hence, an individual consumer's exposure is not big enough to threaten the bank.

C. Corporate exposures arrive in bite-sized pieces which must be repaid before a new facility can be offered; hence, individual corporate exposures rarely threaten the bank.

D. Credit exposures of corporates arrive in large pieces, each of which must be repaid before a new facility can be offered; hence, individual corporate exposures rarely threaten the bank.

Question CR-7 b.2

Bank Q257 • BT P2.T6.603.1 • BT Mock Q37

Retail exposures include credit cards, installment loans (e.g., personal finance, educational loans, auto loans, leasing), revolving credits (e.g., overdrafts, home equity lines of credit) and residential mortgages. Each of the following tends to be a stronger feature of retail credit risk rather than corporate credit risk EXCEPT which is more typical of corporate credit risk?

A. The portfolio tends to behave like a well-diversified portfolio in normal markets.

B. Default by a single customer is never expensive enough to threaten a bank.

C. A rise in defaults is often signaled in advance by a change in customer behavior.

D. Portfolio risk is dominated by the risk that credit losses will rise to unexpected levels due to concentrations of exposures.

Question CR-7 b.3 Bank Q258 • BT P2.T6.25.5.2 (rebuilt)

Silver Lining Bank has a retail loan portfolio and a corporate loan portfolio of equal total size. Which of the following statements comparing retail and corporate credit risk is LEAST accurate?

A. A default by a single retail customer is unlikely to threaten the bank, whereas a default by a large corporate borrower can have a significant impact.

B. Expected losses on the retail portfolio can be estimated reliably and priced into customer charges as a cost of doing business, whereas corporate portfolio risk is dominated by the risk of unexpected losses.

C. Changes in customer behavior often signal a rise in retail defaults early enough for preemptive action, whereas corporate problems often become known too late to correct.

D. Because the two portfolios are equal in size, they have comparable risk profiles, so a loss rate above expectations would hurt each portfolio equally.

Question CR-7 b.4 Bank Q259 • Mock 1 Q41

Which of the following is most accurate about early warning signals of retail and commercial exposures?

A. Neither retail nor commercial exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

B. Both retail and commercial exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

C. Commercial exposures but not retail exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

D. Retail exposures but not commercial exposures provide early warning signals of default that allow banks time to take action to minimize PD and LGD.

Question CR-7 b.5 New • built from GARP Ch.7, Box 7.1, 7.1 and 7.6 • shape D13·S3

The capital planning team at a Dutch bank is comparing the regulatory treatment of its retail book (mortgages, credit cards and loans to small businesses) with that of its corporate loan book under Basel II and III. Which of the following statements is correct?

A. Loans to small businesses can never be managed as retail exposures, because Basel's retail definition covers consumer loans only.

B. Retail exposures attract more capital than corporate loans of the same size, because the dark side makes retail losses harder to predict.

C. Retail banks set aside a relatively small amount of capital, but must provide PD, LGD and EAD for clearly differentiated segments, segmented by credit score or an equivalent measure and by vintage.

D. Basel applies one formula to all retail exposures, so a mortgage and a credit card with the same PD carry the same capital requirement.

Question CR-7 b.6 Workbook D13.1 • S1, S2, S3 • Map #57 • also CR-7 c, CR-7 h

The head of retail credit risk at a UK bank is preparing a report for the board risk committee. The credit card unit proposes to replace its single advertised interest rate with pricing by score band, and the mortgage unit plans to launch a near-prime product for which the bank has little loss history. Which of the following statements is correct for the report to include?

A. Keeping a single interest rate for all card applicants is likely to cause adverse selection, attracting mainly high-risk customers, and the extent of this may only become apparent when the economic climate deteriorates.

B. Because retail credit losses are relatively predictable, the dark side of retail credit is mainly the risk that a loss on a single large customer exceeds its expected level in a downturn.

C. Because a retail portfolio behaves like a well-diversified portfolio, its expected loss can be priced in and the bank need not hold any capital against retail unexpected losses.

D. Under tiered pricing above the cutoff score, the higher a customer's score band, the higher the interest rate charged, since high-scoring customers tend to use their credit lines more.

Solutions • CR-7 b

Solution CR-7 b.1

Answer: B

Bank Q256 • AnalystPrep Q.2010

Step 1 – The defining feature of retail exposure

GARP: retail exposures "arrive in bite-sized pieces, so that default by a single customer is never expensive enough to threaten a bank".

Step 2 – The corporate contrast

Corporate portfolios "often contain large exposures to single names" and concentrations, so one default can matter.

⇒ Only B states the retail feature and draws the right conclusion.

Interpretation

Small size and large numbers are the root of every other difference: diversification, predictable expected loss and early warnings all follow from them.

Why the other options are wrong

A is incorrect. Self-contradictory: retail exposures are small, which is why a single default does not threaten the bank.

C is incorrect. Reverses the roles: corporate exposures are large; the repay-before-a-new-facility clause is irrelevant.

D is incorrect. Right that corporate exposures are large, wrong conclusion: large single-name exposures can threaten the bank.

LO: CR-7 b: Explain the differences between retail credit risk and corporate credit risk. • **Reference:** Crouhy, Galai and Mark, The Essentials of Risk Management 2e, Ch.9 (GARP CR Ch.7), 7.1 The Nature of Retail Credit Risk

Solution CR-7 b.2

Answer: D

Bank Q257 • BT P2.T6.603.1

Step 1 – Classify each statement

A: diversified behaviour in normal markets, retail. B: no single default threatens the bank, retail. C: behavioural early warnings, retail.

Step 2 – The corporate feature

D: "the risk of loss from many commercial credit portfolios is dominated by the risk that credit losses will rise to some unexpected level", driven by concentrations.

⇒ D is the EXCEPT.

Interpretation

Concentration is what turns a corporate book's risk into an unexpected-loss problem; granularity is what turns a retail book's risk into an expected-loss cost.

Why the other options are wrong

A is incorrect. A retail feature: it is why retail EL can be estimated.

B is incorrect. A retail feature: exposures are small.

C is incorrect. A retail feature: corporate problems usually surface too late.

LO: CR-7 b: Explain the differences between retail credit risk and corporate credit risk. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.1

Solution CR-7 b.3 — Answer: D

Bank Q258 • BT P2.T6.25.5.2

Step 1 – Test A, B and C

All three restate GARP: small retail exposures against large corporate ones (A), predictable and priced-in retail EL against unexpected-loss-dominated corporate risk (B), and early retail warnings against the supertanker (C).

Step 2 – Test D

Risk depends on granularity, concentration and predictability, not on total size. A loss rate above expectations is far more damaging for a concentrated corporate book.
⇒ D is the least accurate statement.

Interpretation

Equal size is the red herring in the stem: it invites the reader to equate the two books, which is exactly the error D makes.

Why the other options are wrong

A is incorrect. Accurate: the exposure-size difference.
B is incorrect. Accurate: EL predictability against unexpected-loss dominance.
C is incorrect. Accurate: behavioural warnings allow preemptive action in retail, rarely in commercial lending.

LO: CR-7 b: Explain the differences between retail credit risk and corporate credit risk. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.1

Solution CR-7 b.4 — Answer: D

Bank Q259 • Mock 1 Q41

Step 1 – Retail

"A rise in defaults is often signaled in advance by a change in customer behavior", for example customers failing to make a minimum card payment. The bank can then cut exposures, change acceptance rules or raise rates for riskier customers.

Step 2 – Commercial

"A commercial credit portfolio is something of a supertanker. By the time it is obvious that something is going wrong, it's often too late to do much about it."
⇒ Retail but not commercial: D.

Interpretation

GARP adds a warning of its own: the retail signals are not always heeded, because banks are tempted to keep chasing fast-growing business lines (Box 7.4, CR-7 c).

Why the other options are wrong

A is incorrect. Retail portfolios do give early behavioural signals.
B is incorrect. Commercial portfolios typically do not give timely warnings.
C is incorrect. Reverses the two.

LO: CR-7 b: Explain the differences between retail credit risk and corporate credit risk. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.1

Solution CR-7 b.5 Answer: C

New • built from GARP Ch.7, Box 7.1, 7.1 and 7.6

Step 1 – How regulators view retail risk

"Regulators accept the idea that retail credit risk is relatively predictable, and also that mortgage loans are safer", so retail banks "are asked to set aside a relatively small amount of risk capital under Basel II and III compared with regulatory capital for corporate loans".

Step 2 – What they ask for in return

Banks "will have to provide regulators with PD, LGD and EAD statistics for clearly differentiated segments", with segmentation "based on credit scores or some equivalent measure and on vintage".

$\Rightarrow$ C is correct.

Interpretation

Shape S3: predictable EL means small capital, not zero capital.

Why the other options are wrong

A is incorrect. Small business loans can be managed as retail if the total exposure to the borrower is below EUR 1 million (Box 7.1).

B is incorrect. Polarity flip: retail attracts relatively less capital than corporate lending.

D is incorrect. Basel treats three retail subsectors (residential mortgages, revolving credit, other retail) with three different formulas.

LO: CR-7 b: Explain the differences between retail credit risk and corporate credit risk. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), Box 7.1; 7.1; 7.6 The Basel Regulatory Approach

Solution CR-7 b.6 Answer: A

Workbook D13.1 • S1, S2, S3

Step 1 – The card unit's single rate (S1)

Crouhy, Galai and Mark: "a 'one price fits all' policy in a competitive market leads to adverse selection, i.e., the bank will primarily attract high-risk customers, to whom the product is attractive, and discourage low-risk customers (for the opposite reason). The degree of adverse selection suffered by a bank may only become apparent when the economic climate deteriorates."

$\Rightarrow$ A is correct, and it is the case for the card unit's move to risk-based pricing.

Step 2 – The mortgage unit's new product (S2)

Little loss history is the first prime cause of the dark side: keep that exposure limited and stress test it. One stem, three shapes (CR-7 h, CR-7 c and this LO).

Why the other options are wrong

B is incorrect. Sibling swap: large single-name exposures define corporate portfolios. The dark side is "some unforeseen but systematic risk factor that influences the behavior of many of the credits".

C is incorrect. Absolute claim: predictable EL is priced in, but regulators ask for "a relatively small amount of risk capital", not none, and the dark side is exactly the risk of losses rising to unexpected levels.

D is incorrect. Polarity flip: tiered pricing means "the higher the score, the lower the price".

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. CR-7 b: Explain the differences between retail credit risk and corporate credit risk. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.1, Box 7.2, 7.5 and 7.8

CR-7 • Credit Scoring and Retail Credit Risk Management • Vardeez: MODERATE

CR-7 c

P3 CONCEPT

Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem.

Workbook: D13 Retail Credit Risk • Tier D • Map #57 (2/16) • D13.1 worked as Question CR-7 b.6

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: Map #57 names "the dark-side-of-retail-credit argument" as one of the two examinable retail items (D13).
Vardeez: four causes and the CFPA underwriting rules; the CFPA itself is gap-filled from GARP.

Reported exam shapes

- **D13·S2** The dark side of retail credit: a systematic factor that drives many credits at once.

What the questions look like: conceptual. Why diversification fails, the four prime causes, which responses actually address the problem (limited exposure to new risks, stress tests, the CFPA's ability-to-repay rule and qualified mortgages), and the pre-crisis slide in standards.

Re-filed: Bank Q268 (qualified mortgages) from CR-7 e, because GARP, Vardeez and Schweser all present the CFPA as a dark-side measure; Bank Q255 (pre-crisis lending conditions, Box 7.4) from CR-7 a. Both are P4 LOs.
Worked elsewhere: D13.1 (dark side, risk-based pricing and capital in one stem) under CR-7 b.

1. What the dark side is

Recap from CR-7 b: a retail book's expected loss is predictable *because* it is diversified across many small, independent loans. The dark side is what happens when they stop being independent. GARP's Box 7.2: "the danger that losses will suddenly rise to unexpected levels because of some unforeseen but systematic risk factor that influences the behavior of many of the credits in a bank's retail portfolio." The result is a sudden lurch upward in default rates together with unexpected falls in collateral values (house prices), the dotted curve in Figure 1 of CR-7 b. It "played a significant role in sparking the 2007 to 2009 crisis" and was already in the pre-crisis 2006 edition of Crouhy's book.

2. The four prime causes

1. **New products without loss history.** Not all innovative retail products have enough historical loss data to make risk assessments reliable (subprime, for most banks).
2. **The perfect storm.** Even well-understood products can behave unexpectedly when risk factors all worsen at once, for example a deep recession with higher interest rates while house prices fall sharply.
3. **A changing social and legal system.** The tendency to default reflects social and legal attitudes; the rising acceptability of personal bankruptcy in the US in the 1990s is GARP's example.
4. **Operational issues with a systematic effect.** Consumer credit is a semiautomated decision process, so a flaw in credit assessment hits the whole portfolio at once.

3. The measures

- **Limit, do not avoid, exposure to new risks.** "It's difficult to put a risk number to these kinds of wild-card risk." Banks should make sure only a limited number of retail portfolios are especially vulnerable to new kinds of risk: "a little exposure" can open a lucrative line and generate the data to measure the risk; "a lot makes the bank a hostage to fortune".
- **Stress test the big conventional portfolios.** Where large portfolios (mortgages) are vulnerable to sharp changes in several risk factors, banks "must use stress tests to gauge how devastating each plausible worst-case scenario might be".
- **Post-crisis regulation.** The Consumer Financial Protection Bureau (CFPB) grew out of the Dodd-Frank Act. The CFPA requires originators to determine whether the consumer has the **ability to repay** the mortgage; a **qualified mortgage** (QM) lets the creditor assume the requirement is met.

Qualified mortgage features (GARP Box 7.3)	"Ability to repay": the eight underwriting standards
No excess upfront points and fees	Current employment status; current income or assets
No toxic features: negative amortization, terms over 30 years, interest-only periods	Credit history; monthly payment on the mortgage
A cap on income going to debt, e.g., DTI below 43%	Monthly payments on other loans on the property and on mortgage-related obligations (property taxes)
No balloon payments	Other debt obligations; the monthly DTI ratio (or residual income) with the new mortgage

Correction to the source notes. Vardeez lists "the underwriting standards a lender is asked to consider" as the debt-to-income ratio, upfront fees and points, loan type (balloon, interest-only) and loan term, and his gap-fill then says the "ability to repay" rule asks the lender to consider "the underwriting standards listed above". Those four items are the *qualified mortgage features* of GARP Box 7.3. The ability-to-repay rule asks for the eight underwriting standards in the right-hand column, of which only the DTI ratio overlaps. Separately, SchweserNotes (LO 25.c) gives the QM debt-to-income cap as 45%; GARP Box 7.3 gives 43%, the figure used here (and in Question CR-7 c.6).

4. A perfect storm the industry built

Box 7.4: between 2002 and 2007 lenders and borrowers believed house prices would keep rising. Low interest rates, poorly structured broker incentives and an increasingly competitive market lowered underwriting standards: less creditworthy borrowers, high cumulative loan-to-value ratios and limited or no income verification. Many loans were 2/28 or 3/27 hybrid ARMs whose teaser rates made them viable only if refinanced on rising house prices, and because many were written at about the same time they set up "a systemic wave of either refinancing or default". Retail banks had ignored early warnings to chase volume; it was a perfect storm "created by the banking industry itself".

Exam traps

- **Sibling swap with concentration.** The dark side is a systematic factor moving many small credits together, not a large loss on one big borrower.
- **Diversification as the cure.** Splitting loans further or spreading them wider does not remove a common factor.

- **Expected versus unexpected.** The dark side is an unexpected loss; pricing in the expected loss does not address it.
- **Cause versus response.** Capital, stress tests and limits are responses; new products, the perfect storm, social and legal change and operational flaws are the causes.
- **Absolute claims about measures.** Not "put a precise number on it" (it is a wild card), not "exclude every new product" and not "grow fast to gather data": *limit* exposure, and stress test.
- **QM polarity.** DTI *below* about 43%; 30 years is a *maximum* term; interest-only and balloon features disqualify a loan.
- **QM features versus ability to repay.** Two different lists; only DTI is on both.

Practice questions: CR-7 c

Question CR-7 c.1

Bank Q264 • BT P2.T6.25.5.3 (rebuilt)

A retail lender holds small consumer loans spread across many regions and income groups. During a severe recession defaults rise simultaneously across all segments, and losses far exceed the level the credit model projected using historical data. Which statement best describes the structural risk exposed?

A. A common macroeconomic risk factor made borrower defaults move together, so diversification across many small loans gave only imperfect protection.

B. The losses were mainly idiosyncratic, so the portfolio should be split into even smaller loans to remove them.

C. The loss reflects expected loss, which should simply have been built into loan pricing.

D. The model was sound; the excess losses show only that borrowers' credit scores were too high at origination.

Question CR-7 c.2

Bank Q260 • AnalystPrep Q.2012 (rebuilt)

Retail lending entails offering facilities to a large group of borrowers in relatively small amounts, thereby naturally bringing about diversification. However, certain lending behaviors can erode the benefits of diversification and result in market-wide financial turmoil, as the 2007 to 2009 financial crisis showed. Which of the following behaviors is MOST likely to result in less than perfect protection from systematic risks?

A. International lending

B. Relaxation of lending requirements, such as advancing money without checking incomes

C. Multi-currency lending

D. Charging borrowers a risk-based interest rate

Question CR-7 c.3

Bank Q263 • Schweser Module Quiz 25.1 Q2

The dark side of retail credit risk is perpetuated by all of the following factors EXCEPT:

A. capital set aside to protect a bank in the event of default.

B. process flaws resulting in high-risk applicants receiving credit.

C. new products that do not have sufficient historical loss data.

D. a social acceptance of bankruptcy and borrowers "walking away" from their obligations.

Question CR-7 c.4

New • built from GARP Ch.7, Box 7.2

The chief risk officer of a UK card issuer learns that for six months a parameter error in the bank's semiautomated application engine approved applicants who scored below the cutoff, across every card product and region. Which of the prime causes of the "dark side" of retail credit risk does this illustrate?

A. Innovative products that lack enough historical loss data for their risk assessments to be reliable.

B. A perfect storm, in which well-understood products behave unexpectedly because several risk factors worsen at the same time.

C. A change in the social and legal acceptability of default, such as personal bankruptcy.

D. An operational issue in credit assessment that has a systematic effect on the whole consumer portfolio.

Question CR-7 c.5

Bank Q262 • BT P2.T6.603.2 (rebuilt)

The dark side of retail risk management has four prime causes: innovative products lack historical loss data; sensitivity to the macroeconomic environment; consumer defaults are the product of a social and legal system that changes; and operational issues that can have a systematic effect. Which of the following measures attempts to address the problem?

A. Banks should assign a precise risk number to these wild-card risks and price them into the loans.

B. Banks should exclude every new type of retail product from their portfolios.

C. Banks should use stress tests to gauge how devastating each plausible worst-case scenario might be.

D. Banks should expand quickly into new products such as subprime lending to accumulate loss data.

Question CR-7 c.6 Bank Q268 • AnalystPrep Q.2011 • re-filed from CR-7 e

The Consumer Financial Protection Act (CFPA) requires lenders to assess a borrower's ability to repay a mortgage, and a loan labeled a qualified mortgage (QM) is presumed to satisfy this requirement. Which of the following characterizes a qualified mortgage?

A. The debt-to-income (DTI) ratio must be greater than 43%.

B. It must not have excess upfront points or fees.

C. It must have a minimum term of 30 years.

D. It must have interest-only features.

Question CR-7 c.7 Bank Q255 • AnalystPrep Q.2017 • re-filed from CR-7 a

What characterized the retail lending market in the run-up to the 2007/2008 financial crisis?

A. Enforcement of strict underwriting standards, elevated interest rates to discourage speculative borrowing, and controlled competition among a few dominant lenders.

B. Somewhat relaxed underwriting criteria combined with cyclically adjusted interest rates and a competitive landscape dominated by traditional banking institutions.

C. Loosely applied underwriting standards, historically low interest rates fueled by expansive monetary policies, and fierce competition among a broad array of lending institutions.

D. Conservatively modeled underwriting processes that inadequately captured borrower risk, variable interest rates influenced by short-term economic indicators, and a fragmented competitive environment with aggressive non-bank entrants.

Solutions • CR-7 c

Solution CR-7 c.1 — Answer: A

Bank Q264 • BT P2.T6.25.5.3

Step 1 – Read the symptom

Defaults rose *simultaneously* across regions and segments: the signature of a common factor, not of borrower-specific events.

Step 2 – Name the risk

GARP: in 2007 to 2009 house prices fell at about the same time across a large economy, so "diversification turned out to offer less than perfect protection". Losses rising to unexpected levels because of a systematic factor is the dark side.

$\Rightarrow$ A.

Interpretation

The credit model was calibrated on history in which defaults were largely independent; the recession made them move together, which is exactly what the history could not show.

Why the other options are wrong

B is incorrect. Splitting loans further does not remove systematic risk; the losses were correlated, not idiosyncratic.

C is incorrect. Expected loss is the predictable part that is priced in; the problem is unexpected loss far above the projection.

D is incorrect. Unsupported: the failure was ignoring correlation under stress, not score levels at origination.

LO: CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.1 and Box 7.2

Solution CR-7 c.2 — Answer: B

Bank Q260 • AnalystPrep Q.2012

Step 1 – How diversification fails

It fails when many credits become exposed to the same hidden factor.

Step 2 – GARP's example

"A systematic change in behavior in consumer lending industries, e.g., advancing money to consumers without checking their incomes, can introduce a hidden systematic risk into credit portfolios, and even whole credit industries."

$\Rightarrow$ B.

Interpretation

In a downturn that hidden factor produces "sudden lurches upward in the default rate" and unexpected falls in collateral values: the dark side in action.

Why the other options are wrong

A is incorrect. International lending spreads exposures across economies; if anything it adds diversification.

C is incorrect. Multi-currency lending also spreads exposures; it is not the behaviour GARP identifies.

D is incorrect. Risk-based pricing aligns price with risk (CR-7 h); it does not create a common exposure.

LO: CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. •
Reference: Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.1

Solution CR-7 c.3 — Answer: A

Bank Q263 • Schweser Module Quiz 25.1 Q2

Step 1 – List the four prime causes

New products without loss data; well-understood products behaving unexpectedly when risk factors worsen together; a changing social and legal system; operational flaws in a semiautomated credit process.

Step 2 – Match the options

B is an operational flaw, C is missing loss data, D is the social acceptability of default.
⇒ Capital (A) is a response that absorbs losses, not a cause.

Interpretation

Separating causes from responses is the usual trap on this LO: capital, limits and stress tests sit on the response side.

Why the other options are wrong

B is incorrect. A listed cause: an operational flaw that grants credit to higher-risk applicants.
C is incorrect. A listed cause: innovative products without historical loss data.
D is incorrect. A listed cause: the social and legal acceptability of default.

LO: CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. •
Reference: Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), Box 7.2; SchweserNotes 2026 Book 2, Reading 25, LO 25.c

Solution CR-7 c.4 — Answer: D

New • built from GARP Ch.7, Box 7.2

Step 1 – What went wrong

A process error in the application engine let below-cutoff applicants through, in every product and region at once.

Step 2 – Match it to the causes

GARP: "any operational issue that affects the credit assessment of customers can have a systematic effect on the whole consumer portfolio. Because consumer credit is run as a semiautomated decision-making process rather than as a series of tailored decisions, it's vital that the credit process be designed and operated correctly."
⇒ D.

Interpretation

The same automation that makes retail credit cheap and consistent (CR-7 d) turns a single coding error into a portfolio-wide exposure: one common factor, many credits.

Why the other options are wrong

A is incorrect. No new product is involved; the products are existing cards.
B is incorrect. No economic shock is described; the problem is internal to the credit process.
C is incorrect. Borrowers' attitudes to default have not changed; the bank approved the wrong applicants.

LO: CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. •
Reference: Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), Box 7.2

Solution CR-7 c.5

Answer: C

Bank Q262 • BT P2.T6.603.2

Step 1 – Can the risk be measured directly?

"Almost by definition, it's difficult to put a risk number to these kinds of wild-card risk." That rules out A.

Step 2 – Limit, do not avoid

Banks should make sure only a limited number of portfolios are vulnerable to new kinds of risk: a little exposure is informative, "a lot makes the bank a hostage to fortune". That rules out B (avoid everything) and D (expand fast).

Step 3 – Large conventional portfolios

"Banks must use stress tests to gauge how devastating each plausible worst-case scenario might be."

$\Rightarrow$ C.

Interpretation

The two measures fit the two kinds of portfolio: limits for exposure to new risks, stress tests for big established books exposed to several risk factors at once.

Why the other options are wrong

A is incorrect. Wild-card risks cannot be given a reliable risk number.

B is incorrect. Over-corrects: GARP asks banks to limit, not eliminate, exposure to new products.

D is incorrect. Large, fast exposure to unproven products makes the bank a hostage to fortune.

LO: CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), Box 7.2

Solution CR-7 c.6

Answer: B

Bank Q268 • AnalystPrep Q.2011

Step 1 – Recall the QM features (Box 7.3)

No excess upfront points and fees; no toxic features (negative amortization, terms over 30 years, interest-only periods); a cap on income going to debt (DTI below 43%); no balloon payments.

$\Rightarrow$ Only B is a QM feature.

Interpretation

The QM is a plain-vanilla mortgage: meeting the features lets the creditor presume the borrower can repay. It is the CFPA's direct response to the slipping standards of Box 7.4.

Why the other options are wrong

A is incorrect. Reverses the DTI rule: a QM caps DTI below about 43%.

C is incorrect. Thirty years is a maximum, not a minimum; longer terms are toxic features.

D is incorrect. Interest-only periods disqualify a loan from QM status.

LO: CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. CR-7 e: Discuss the key variables in a mortgage credit assessment and describe the use of cutoff scores, default rates, and loss rates in a credit scoring model. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.1 and Box 7.3

Solution CR-7 c.7

Answer: C

Bank Q255 • AnalystPrep Q.2017

Step 1 – GARP's account (Box 7.4)

"Combined with low interest rates, poorly structured incentives for brokers, and an increasingly competitive environment, this led to a lowering of underwriting standards."

Step 2 – The loans that resulted

Less creditworthy borrowers, high cumulative LTVs, limited or no income verification, and teaser-rate hybrid ARMs that worked only if house prices rose.
$\Rightarrow$ Loose standards, low rates and fierce competition: C.

Interpretation

This is the dark side made by the industry itself: a systematic change in lending behaviour put the same hidden risk into whole portfolios at once.

Why the other options are wrong

A is incorrect. Describes the opposite: standards were lax and rates low.
B is incorrect. Understates the laxity and wrongly implies cyclically adjusted rates and dominance by traditional banks.
D is incorrect. Underwriting was loose, not conservatively modelled, and rates were low.

LO: CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. CR-7 a: Analyze the credit risks and other risks generated by retail banking. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), Box 7.4 Slipping Standards in Subprime Lending

CR-7 • Credit Scoring and Retail Credit Risk Management • Vardeez: MODERATE

CR-7 f

P3 CONCEPT

Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques.

Workbook: none homed here • Vardeez focus (D13 notes summarise CAP and AR)

WHAT IS TESTED HERE

Why P3. Vardeez flags LO 25.f as one that "can be important" in a Moderate chapter. No workbook question.
Vardeez: CAP and accuracy ratio.

Reported exam shapes: none listed in the map.

What the questions look like: conceptual. Reading a CAP (perfect, random and actual curves; which way the axis runs), what the accuracy ratio measures and its bounds, when a scorecard must be rebuilt, and the cutoff trade-off between false goods and false bads.

Not used: Bank Q275 asks for two accuracy ratios from given areas; it is a calculation under a CONCEPT LO (TYPE rule).
Re-filed: Bank Q273 (good and bad score distributions, false goods and false bads) and Q269 (re-setting the cutoff from loss and profit experience) from CR-7 e, a P4 LO; GARP opens its scorecard-performance section with exactly that trade-off.

1. What a scorecard must do

A scorecard predicts which applicants will prove good or bad, so it must assign high scores to good credits and low scores to bad ones. GARP: "the goal of the scorecard is to minimize the overlapping area of the distribution of the good and bad credits." Because the two distributions overlap, any cutoff makes two errors: **false bads** (good accounts rejected, to the left of the cutoff) and **false goods** (bad accounts accepted, to the right). Raising the cutoff cuts off more bads but forgoes more goods; the two errors move in opposite directions. Given the cutoff, the bank measures the product's loss rate and profitability from experience, usually updated quarterly, and long series are needed to capture a full economic cycle (CR-7 e).

2. The cumulative accuracy profile

- **Axes.** Horizontal: the customer population *sorted by score from the highest risk to the lowest*. Vertical: the cumulative percentage of actual defaults, from the bank's records.
- **Perfect model.** If the model predicts that 10% of accounts will default in the next 12 months and is perfect, every default falls in the riskiest 10%: the line rises to 100% at 10% and is flat after.
- **Random model.** The 45-degree line: the riskiest $x\%$ of accounts holds $x\%$ of the defaults. It cannot tell good from bad.
- **Actual model.** Lies between the two; the bank hopes it sits close to the perfect line, climbing steeply at the left.

The **accuracy ratio** summarises the curve:

$$\boxed{\text{AR} = \frac{A_R}{A_P}}$$

where A_R is the area between the actual curve and the random line, and A_P the area between the perfect curve and the random line. AR runs from 0 for a random model to 1 for a perfect one; the closer to 1, the more accurate the model. (Vardeez's appendix already corrects his notes' "0.5 is random": 0.5 is the random benchmark for AUROC, a different statistic, with $\text{AR} = 2\,\text{AUROC} - 1$.)

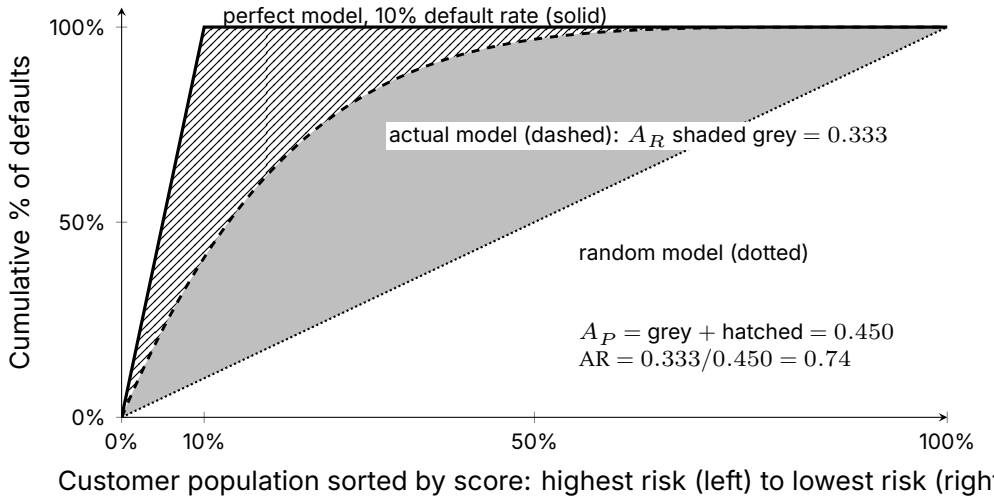


Figure 1. Cumulative accuracy profile (after GARP Figure 7.3). The perfect model (solid) captures every default in the riskiest 10%; the random model (dotted) captures defaults in proportion to the population; the actual model (dashed) lies between. A_R is the grey region between the actual curve and the diagonal; A_P is the whole region between the perfect curve and the diagonal (grey plus hatched). Here the curve is $1-(1-x)^5$, giving $\text{AR} = 0.74$.

Correction to the source notes. GARP's text says "the area under the perfect model is denoted A_P, while the area under the actual rating model is denoted A_R". Read literally, as areas down to the horizontal axis, a random model would score $0.5/0.95 = 0.53$ (with a 10% default rate) instead of zero, which is the origin of the "0.5 means random" error. The ratio is meaningful only when both areas are measured from the 45-degree line, as Schweser states and as used here.

	Perfect model	Actual model	Random model
Shape of the CAP	Rises to 100% at the default rate, then flat	Concave, between the two	45-degree line
Riskiest 10% (10% default rate)	100% of defaults	Between 10% and 100%	10% of defaults
Area from the diagonal	A_P	A_R	0
Accuracy ratio	1	A_R/A_P, between 0 and 1	0

3. Monitoring and replacing a scorecard

- **Monitor regularly.** Performance can be tracked, say every quarter, with a CAP curve, and the model replaced when it deteriorates. Retail banks have many customers, so data to assess a scorecard accumulate quickly.
- **Deterioration is gradual.** Scorecard performance "tends not to change abruptly", but it decays because the characteristics of the population change over time, or its behaviour evolves so that different variables become associated with default. On a CAP the curve drifts towards the diagonal and AR falls.

- **A product change is a rebuild.** If a lender sells its auto-loan business and issues credit cards instead, the target population will probably differ enough to justify a new custom scorecard.

Exam traps

- **Axis direction.** Population sorted from *highest* risk to lowest; that is why a good model climbs steeply at the left.
- **Random is 0, not 0.5.** 0.5 is AUROC's random benchmark; AR's is 0. A perfect model has AR $= 1$, the maximum.
- **Benchmark swap.** At the default rate the perfect line is at 100%; the random line is at the default rate itself.
- **Areas from the diagonal.** Both A_R and A_P are measured from the random line, not from the horizontal axis.
- **The cutoff does not change AR.** AR measures how well the scores rank the whole population; moving the cutoff changes the mix of false goods and false bads, not the scorecard's discriminating power.
- **Cutoff polarity.** A higher cutoff means fewer false goods and more false bads; no cutoff reduces both.
- **Monitoring.** Quarterly, gradual decay, and a new custom scorecard when the product (and so the population) changes.

Practice questions: CR-7 f

Question CR-7 f.1

Bank Q273 • BT P2.T6.604.2 (rebuilt) • re-filed from CR-7 e

Suppose the population of credit applicants, which is ordinarily unknowable, divides into either "good" or "bad" accounts, to use Crouhy's terminology. Good accounts have a mean credit score of 730 with a standard deviation of 25. Bad accounts have a mean credit score of 600 with a standard deviation of 40. Both distributions are approximately normal. A bank is evaluating whether to set its cutoff score at 680; that is, applicants with a score of 680 or greater will be approved. There are two types of underwriting error: a "false bad" is the rejection of a good account; a "false good" is the acceptance of a bad account. Which of the following is a TRUE statement?

A. Raising the cutoff score improves the scorecard's accuracy ratio, because fewer bad accounts are accepted.

B. The bank can reduce the fraction of false bads by increasing the cutoff score.

C. The bank can reduce the fraction of false goods by increasing the cutoff score.

D. The bank can simultaneously reduce the fraction of errors (both false bads and false goods) by increasing the cutoff score.

Question CR-7 f.2

Bank Q277 • BT P2.T7.20.4.3 (rebuilt)

A bank's scorecard predicts that 10% of accounts will default over the next 12 months. Accounts are ranked from riskiest to safest by score. Actual results show that the riskiest 10% of accounts contain 70% of all defaults that occurred. Which statement about the cumulative accuracy profile (CAP) at the riskiest 10% of the population is correct?

A. The perfect model line shows 10% of defaults and the random model line shows 100%, so the scorecard is between them.

B. The perfect model line shows 100% of defaults and the random model line shows 10%; the scorecard's 70% lies between them, so it is better than random but not perfect.

C. The perfect model line shows 100% of defaults and the random model line shows 50%; the scorecard is better than random.

D. The perfect model line shows 70% of defaults, which equals the scorecard, so the accuracy ratio is 1.

Question CR-7 f.3

New • built from GARP Ch.7, 7.4 and Figure 7.3

A model validation team at a consumer lender compares two application scorecards on the same 12-month cohort, in which 8% of accounts defaulted. Accounts are sorted from the highest to the lowest risk score, and the team records the share of all defaults captured:

Riskiest share of the population	8%	20%	50%
Scorecard X: share of defaults captured	52%	78%	95%
Scorecard Y: share of defaults captured	30%	55%	80%

Which of the following statements is correct?

A. X's CAP lies above Y's at each point shown, so X sits closer to the perfect-model line and should have the higher accuracy ratio.

B. Y is the better scorecard, because its CAP lies closer to the 45-degree line, which is the benchmark a good model should track.

C. A perfect scorecard would capture 8% of defaults in the riskiest 8% of accounts, so both scorecards beat the perfect model at that point.

D. The accuracy ratios of X and Y cannot be compared, because the bank's cutoff score has not been given.

Question CR-7 f.4 Bank Q276 • BT P2.T6.604.3

About the cumulative accuracy profile (CAP), each of the following statements is true EXCEPT, which is not?

A. A perfect credit scoring model generates an accuracy ratio (AR) of 1.0, which is the upper bound on the AR.

B. A purely random model that cannot differentiate between good and bad customers is likely to generate an accuracy ratio (AR) of 0.40 to 0.60; i.e., $50\% \pm 10\%$.

C. The CAP curve, which plots the actual rating model as a cumulative percentage of defaults, is monotonically increasing (also known as nondecreasing or weakly increasing).

D. The CAP curve plots the fraction of defaulted customers (y-axis) against the fraction of the entire customer population sorted by score from highest risk (left) to lowest risk (right).

Question CR-7 f.5 New • built from GARP Ch.7, 7.4

The head of retail analytics at a Canadian bank reviews the quarterly CAP report for the bank's auto-loan application scorecard. Over the last four quarters the curve has drifted slowly towards the 45-degree line. The bank has also agreed to sell its auto-loan business and to launch a credit card. Which of the following statements is most appropriate?

A. Scorecard performance usually changes abruptly, so the CAP should be recomputed daily and the scorecard replaced the first time its accuracy ratio falls.

B. The auto-loan scorecard can be reused for the credit card, because a scorecard's accuracy ratio depends only on the bank's own default rate.

C. Monitoring is needed only once the economic cycle turns, because the characteristics and behaviour of a scored population are stable in normal times.

D. The drift signals deterioration, which can come from changes in the population's characteristics or behaviour, and the switch to credit cards changes the target population enough to justify a new custom scorecard.

Question CR-7 f.6 Bank Q269 • AnalystPrep Q.5469 (rebuilt) • re-filed from CR-7 e

A bank has set the cutoff score for its mortgage scorecard. Over time it accumulates data on the loss rate and profitability of each score band. According to the standard approach to cutoff scores, which of the following is the most appropriate way for the bank to adjust the cutoff?

A. Estimate loss rates and profitability from a long history that spans a full economic cycle, and move the cutoff to the level that best balances risk and profitability.

B. Raise the cutoff to the highest feasible level so that no defaults occur, since a higher score means lower risk.

C. Base the cutoff on the last twelve months of experience only, because recent data best reflect current conditions.

D. Lower the cutoff whenever losses are low, because a lower cutoff always increases profit without affecting the loss rate.

Solutions • CR-7 f

Solution CR-7 f.1

Answer: C

Bank Q273 • BT P2.T6.604.2

Step 1 – Where the errors sit

False bads are good accounts below the cutoff; false goods are bad accounts at or above it.

Step 2 – Move the cutoff up

Fewer bad accounts lie above a higher cutoff, so false goods fall; more good accounts lie below it, so false bads rise.

$\Rightarrow$ Only C is true.

Interpretation

At 680 the two error rates happen to be equal: $\Pr(\text{good} < 680) = \Phi((680-730)/25) = \Phi(-2) = 2.28\%$ and $\Pr(\text{bad} \geq 680) = 1 - \Phi((680-600)/40) = 2.28\%$. Moving the cutoff trades one error for the other; only a better scorecard (less overlap) reduces both.

Why the other options are wrong

A is incorrect. The accuracy ratio measures how well the scores rank the whole population; it does not depend on where the cutoff is set.

B is incorrect. Reversed: raising the cutoff increases false bads; lowering it reduces them.

D is incorrect. With overlapping distributions the two errors move in opposite directions; no cutoff reduces both.

LO: CR-7 f: Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques. CR-7 e: Discuss the key variables in a mortgage credit assessment and describe the use of cutoff scores, default rates, and loss rates in a credit scoring model. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.3 From Cutoff Scores to Default Rates and Loss Rates (Figure 7.2); 7.4

Solution CR-7 f.2

Answer: B

Bank Q277 • BT P2.T7.20.4.3

Step 1 – Perfect model at the riskiest 10%

With a 10% default rate, a perfect model puts every default in the riskiest 10%: 100%.

Step 2 – Random model at the riskiest 10%

A random model captures defaults in proportion to the population: 10%.

Step 3 – Place the scorecard

70% lies between 10% and 100%.

$\Rightarrow$ Better than random, not perfect, so $0 < \text{AR} < 1$: B.

Interpretation

One point does not give the accuracy ratio, which needs the whole curve, but it shows where the scorecard sits between the two benchmarks.

Why the other options are wrong

A is incorrect. Swaps the benchmarks: the perfect line is the higher one (100%), the random line the lower (10%).

C is incorrect. The random model captures 10% at the riskiest 10%, not 50%.

D is incorrect. Treats the observed 70% as the perfect line; the perfect line is at 100%, so AR is below 1.

LO: CR-7 f: Discuss the measurement and monitoring of a scorecard performance including the use of cumulative

accuracy profile (CAP) and the accuracy ratio (AR) techniques. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.4 Measuring and Monitoring the Performance of a Scorecard, Figure 7.3

Solution CR-7 f.3 Answer: A

New • built from GARP Ch.7, 7.4 and Figure 7.3

Step 1 – The benchmarks at each point

Perfect model: 100% of defaults by the riskiest 8% (the default rate) and at every later point. Random model: 8%, 20% and 50%.

Step 2 – Compare the curves

X captures more defaults than Y at every point (52% against 30%, 78% against 55%, 95% against 80%), and both lie above the random line.

$\Rightarrow$ X's curve is closer to the perfect line and encloses more area above the diagonal, so its accuracy ratio is higher: A.

Interpretation

No arithmetic is needed: a CAP lying everywhere above another has the larger A_R, and both share the same A_P (same cohort, same default rate).

Why the other options are wrong

B is incorrect. Polarity flip: the 45-degree line is the random model; the closer a curve sits to it, the worse the scorecard.

C is incorrect. Benchmark swap: 8% of defaults in the riskiest 8% is the random model; the perfect model captures 100% there.

D is incorrect. The CAP is built from the whole population sorted by score; no cutoff is involved, so the two ratios are comparable.

LO: CR-7 f: Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.4, Figure 7.3

Solution CR-7 f.4 Answer: B

Bank Q276 • BT P2.T6.604.3

Step 1 – The random model

It is the 45-degree line, so the area between it and the random line is zero and $\text{AR} = 0/A_P = 0$, not 0.40 to 0.60.

Step 2 – Check the other three

A: a perfect model has $\text{AR} = 1$, the maximum. C: a cumulative share of defaults can only stay flat or rise. D: the standard construction, riskiest accounts first.

$\Rightarrow$ B is the false statement.

Interpretation

The "50% $\pm$ 10%" figure comes from mixing up AR with the area under the ROC curve (random 0.5) or from reading GARP's "area under" literally (see the correction box).

Why the other options are wrong

A is incorrect. True: AR cannot exceed 1.

C is incorrect. True: a cumulative curve never decreases.

D is incorrect. True: this is how the CAP is drawn.

LO: CR-7 f: Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP

CR Ch.7), 7.4, Figure 7.3

Solution CR-7 f.5 Answer: D

New • built from GARP Ch.7, 7.4

Step 1 – Read the drift

A CAP moving towards the diagonal means the scorecard ranks defaulters less well: performance is deteriorating. GARP: it "tends not to change abruptly" but deteriorates because "the characteristics of the underlying population may change over time, and/or the behavior of the population may evolve".

Step 2 – The product change

"If a financial institution that offers auto loans decides to sell this business and issue credit cards instead, it is highly probable that the target customer population will be different enough to justify the development of a new custom scorecard."
$\Rightarrow$ D.

Interpretation

GARP's cycle is quarterly monitoring, with replacement when performance deteriorates or the product changes.

Why the other options are wrong

A is incorrect. Performance tends to decay gradually, not abruptly; GARP suggests quarterly monitoring.
B is incorrect. A scorecard is built for a population; a new product brings a different population and calls for a new scorecard.
C is incorrect. Populations and their behaviour change over time in any phase of the cycle, which is why monitoring is continuous.

LO: CR-7 f: Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.4

Solution CR-7 f.6 Answer: A

Bank Q269 • AnalystPrep Q.5469

Step 1 – What the cutoff trades off

A higher score means lower risk. Too low a cutoff raises default risk; too high a cutoff rejects profitable low-risk borrowers.

Step 2 – How to recalibrate

The bank measures loss rate and profitability from experience ("only by using longer time series can the bank hope to capture behavior through a normal economic cycle") and adjusts the cutoff to optimise the margin.
$\Rightarrow$ A.

Interpretation

Statistics are updated quarterly, but the calibration window must be long: a benign year would set the cutoff too low.

Why the other options are wrong

B is incorrect. A very high cutoff rejects many good, profitable borrowers (false bads); the aim is balance, not zero defaults.
C is incorrect. Twelve months may cover only a benign or only a stressed phase; GARP wants a

full cycle.

D is incorrect. A lower cutoff admits riskier applicants and raises the loss rate.

LO: CR-7 f: Discuss the measurement and monitoring of a scorecard performance including the use of cumulative accuracy profile (CAP) and the accuracy ratio (AR) techniques. CR-7 e: Discuss the key variables in a mortgage credit assessment and describe the use of cutoff scores, default rates, and loss rates in a credit scoring model. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.3

CR-7 • Credit Scoring and Retail Credit Risk Management • Vardeez: MODERATE

CR-7 h

P3 CONCEPT

Discuss the benefits of risk-based pricing of financial services.

Workbook: D13 Retail Credit Risk • Tier D • Map #57 (2/16) • D13.1 worked as Question CR-7 b.6

WHAT IS TESTED HERE

Why P3. Tier D workbook topic (Map #57: "the risk-based-pricing / adverse-selection dynamic"), and Vardeez flags LO 25.h as one that "can be important".
Vardeez: risk-based pricing: different prices for the same product, set from risk factors and judged against performance metrics band by band.

Reported exam shapes

- **D13·S1** Risk-based pricing and adverse selection under a "one price fits all" policy; tiered pricing direction.

What the questions look like: conceptual. Which way adverse selection runs, which way tiered prices run, what goes into a risk-based price, and what the benefits are.

Not used: Bank Q283 and Q284 (incremental risk-adjusted income and segment margins) are calculations under a CONCEPT LO (TYPE rule).
Worked elsewhere: D13.1 at Question CR-7 b.6 (its option A is the adverse-selection shape).

1. What risk-based pricing is

GARP: "explicitly incorporating risk-driven account economics into the annualized interest rate that is charged to the customer at the account level". Customers with different risk profiles pay *different amounts for the same product*. The key economic factors:

- operating expenses; the **probability of take-up** (that the customer accepts the offer);
- the probability of default, the loss given default and the exposure at default;
- the capital allocated to the transaction and the institution's cost of equity capital.

RBP combines **external market data** (the probability of take-up, itself a function of price and credit limit) with **internal data** (the cost of capital). Many leading banks use some form of it for auto loans, credit cards and mortgages.

2. Why one price fails

"A 'one price fits all' policy in a competitive market leads to adverse selection, i.e., the bank will primarily attract high-risk customers, to whom the product is attractive, and discourage low-risk customers (for the opposite reason). The degree of adverse selection suffered by a bank may only become apparent when the economic climate deteriorates." The single price is cheap for risky customers relative to their expected loss and dear for safe ones, who take their business to competitors that price them lower (Figure 1).

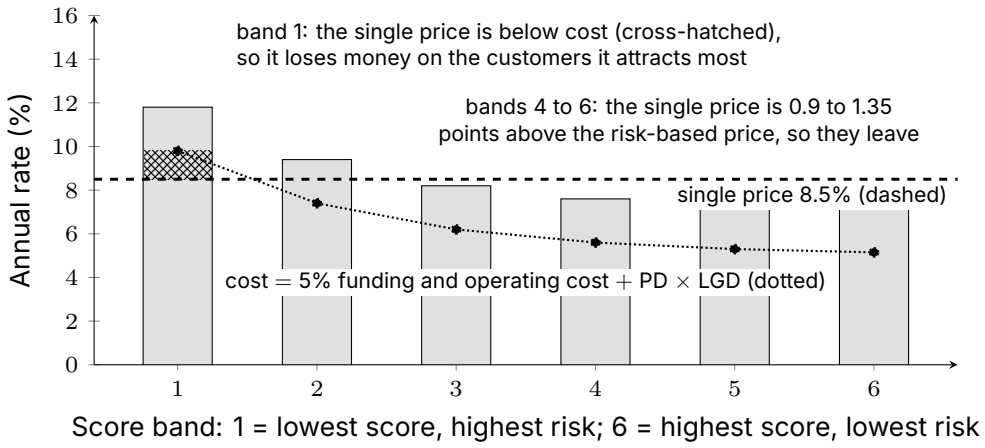


Figure 1. Illustrative tiered pricing. Bars: a risk-based price per score band, here cost plus a 2-point margin (11.80%, 9.40%, 8.20%, 7.60%, 7.30%, 7.15%), falling as the score rises. Dotted: cost per band, 5% plus expected loss with PDs of 8%, 4%, 2%, 1%, 0.5% and 0.25% and a 60% LGD. Dashed: a single 8.5% price for everyone. It loses money in band 1 (cross-hatched) and overcharges bands 4 to 6, so the pool drifts towards the riskiest bands: adverse selection.

3. The benefits, and how far banks have got

- **Tiered pricing.** Pricing differentiated by score band above the cutoff: "the higher the score, the lower the price". Setting price as an increasing function of riskier score bands "can make risk-based pricing easier and more effective".
- **Management information.** A well-designed RBP strategy maps alternative pricing strategies at the credit-score level to key corporate metrics: revenue, profit, loss, risk-adjusted return, market share and portfolio value.
- **Shareholder value.** RBP lets managers meet their objectives under several constraints, trading off profit, market share and risk (mathematical programming, such as integer programming, finds the best mix). Pricing balances growing market share against the rate of bad accounts, as on a *strategy curve* of the bad rate against the acceptance rate.
- **Money on the table.** Traditional pricing leaves a lot of it: "better pricing can improve key corporate performance metrics by 10 to 20 percent or more."
- **Still in its infancy.** Banks' objectives are seldom fully reflected in pricing; low-balance accounts and high-balance revolvers are often mispriced; cutoffs and tiers are often set by ad hoc heuristics. Since the crisis banks also build longer-term considerations into RBP. Nonbanks lending to consumers should use it too, but many lack the infrastructure.

	One price fits all	Risk-based pricing
Price	The same for every customer	Differs by risk for the same product
High-risk customers	Cheap relative to risk: attracted	Charged for their risk
Low-risk customers	Overcharged: leave for competitors	Lower price: retained
Portfolio drift	Towards higher risk; visible in a downturn	Pool and price stay aligned
Inputs	Little beyond average cost	Take-up probability, PD, LGD, EAD, capital, cost of equity, operating costs
Management use	None by band	Map pricing to revenue, profit, loss, market share, risk-adjusted return

Exam traps

- **Adverse-selection polarity.** One price attracts the *high*-risk customers and repels the low-risk ones; the portfolio's average PD rises, it does not fall.
- **Tiered-pricing polarity.** Higher score band, lower price; higher PD, higher price.
- **Adverse selection is not fraud.** A borrower hiding information, a job loss or a borrower misjudging capacity are not selection effects of pricing.
- **RBP still needs the risk parameters.** It is built on account-level PD, LGD, EAD, capital and costs; it does not remove the need to estimate them.
- **Timing.** The damage from adverse selection may only show when the economy turns.
- **Absolute claims.** RBP does not remove the trade-off between market share and bad accounts, and it is still in its infancy.

Practice questions: CR-7 h

Question CR-7 h.1

Bank Q282 • AnalystPrep Q.5941

A regional bank charges one interest rate to all applicants regardless of risk. Which situation correctly illustrates adverse selection in this lending process?

A. A borrower deliberately conceals a poor financial history to obtain a loan.

B. Low-risk borrowers are deterred by the high rate, leaving mostly high-risk borrowers applying.

C. A borrower cannot repay because of an unexpected job loss.

D. A borrower overestimates their ability to meet repayments and defaults.

Question CR-7 h.2 Bank Q285 • BT P2.T6.605.1 (rebuilt)

A lender charges the same price for a retail product to all customers regardless of risk. Which outcome is most likely?

A. Low-risk customers are drawn in by the price while high-risk customers are driven away.

B. High-risk customers are attracted by a price that is low relative to their risk, while low-risk customers are pushed away by a price that is high relative to their risk.

C. The portfolio's average probability of default falls because the single price screens out risky applicants.

D. Revenue is maximised because the lender avoids differentiating among score bands.

Question CR-7 h.3 Bank Q287 • Schweser Module Quiz 25.2 Q2

By implementing risk-based pricing on its mortgage products, a bank will likely charge a:

A. higher interest rate to a customer with a higher FICO score.

B. lower interest rate to a customer with a lower credit bureau score.

C. higher interest rate to a customer with a higher probability of default.

D. lower interest rate to a customer positioned on a lower relative score band.

Question CR-7 h.4 New • built from GARP Ch.7, 7.8

A pricing team at an Australian bank is building a risk-based pricing engine for its auto loans. Which of the following statements about the inputs to the price is correct?

A. The probability of take-up is an external market input that itself depends on the price and credit limit offered, while the cost of equity capital is an internal input.

B. Once the PD of a score band is known, the price can be set from the PD alone, since LGD, EAD and allocated capital are the same for every account.

C. Operating expenses are left out, because risk-based pricing prices credit risk only.

D. The probability of take-up is an internal input, estimated from the bank's own default history in each score band.

Question CR-7 h.5 Bank Q286 • BT P2.T6.605.3 (rebuilt)

Which of the following is NOT a benefit of risk-based pricing of retail financial services?

A. It lets managers map pricing strategies to metrics such as profit/loss, revenue, market share and risk-adjusted return at each score band.

B. It removes the need to estimate PD, LGD and EAD at the account level because prices are set by score band alone.

C. It lets managers evaluate trade-offs among profitability, market share and risk in order to increase shareholder value.

D. For the same product, it lets the bank charge higher prices to higher-risk customers and lower prices to lower-risk customers.

Question CR-7 h.6 New • built from GARP Ch.7, 7.8

The head of cards at a Singapore bank is making the case to the executive committee for replacing the card's single annual percentage rate with risk-based pricing. Which of the following statements is most accurate according to Crouhy, Galai and Mark?

A. Risk-based pricing is now mature in retail banking, and banks' key business objectives are generally well reflected in their pricing strategies.

B. Under risk-based pricing the bank no longer faces a trade-off between growing market share and limiting the rate of bad accounts.

C. Risk-based pricing suits banks only; nonbanks that extend credit to consumers should avoid it because it does not apply outside banking.

D. Traditional retail pricing leaves considerable money on the table: better pricing can improve key corporate performance metrics by 10 to 20 percent or more, with price used to balance market share against the rate of bad accounts.

Solutions • CR-7 h

Solution CR-7 h.1 — Answer: B

Bank Q282 • AnalystPrep Q.5941

Step 1 – What adverse selection is here

A single price that ignores risk is high relative to low-risk borrowers' risk and low relative to high-risk borrowers' risk, so it skews *who applies*.
⇒ Low-risk borrowers deterred, high-risk borrowers remaining: B.

Interpretation

Adverse selection is a property of the applicant pool that the pricing creates. Risk-based pricing exists to stop it.

Why the other options are wrong

A is incorrect. Concealment by one borrower is misrepresentation, not a selection effect of the price.
C is incorrect. An unexpected job loss is a shock to one borrower, not a selection effect.
D is incorrect. Misjudging one's own capacity is a borrower error, not a selection effect.

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.5 From Default Risk to Customer Value

Solution CR-7 h.2 — Answer: B

Bank Q285 • BT P2.T6.605.1

Step 1 – Price against risk

The single price is cheap for high-risk customers relative to their PD and LGD, and expensive for low-risk customers.

Step 2 – Who takes it up

High-risk customers take up the product; low-risk customers go to competitors with lower prices (Figure 1).
⇒ B.

Interpretation

GARP's warning is about timing: "the degree of adverse selection suffered by a bank may only become apparent when the economic climate deteriorates", because in good times the riskier pool still pays.

Why the other options are wrong

A is incorrect. Reverses the effect: it is the high-risk customers who find the flat price attractive.
C is incorrect. Adverse selection raises the portfolio's average PD; it does not lower it.
D is incorrect. The flat price loses low-risk business and undercharges high-risk business; nothing about it maximises risk-adjusted revenue.

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.5 and 7.8

Solution CR-7 h.3 — Answer: C

Bank Q287 • Schweser Module Quiz 25.2 Q2

Step 1 – The direction of risk-based pricing

The riskier the customer, the higher the price. Tiered pricing: "the higher the score, the lower

the price".

Step 2 – Apply it

A higher PD means a higher rate; a higher FICO or bureau score, or a higher score band, means a lower rate.
⇒ C.

Interpretation

Every distractor reverses the score-to-price direction once; check each against "higher risk, higher price".

Why the other options are wrong

A is incorrect. A higher FICO score signals lower risk, so a lower rate.
B is incorrect. A lower bureau score signals higher risk, so a higher rate.
D is incorrect. A lower score band means higher risk, so a higher rate.

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.8 Risk-Based Pricing; SchweserNotes 2026 Book 2, Reading 25, LO 25.h

Solution CR-7 h.4 Answer: A

New • built from GARP Ch.7, 7.8

Step 1 – The inputs

GARP lists "operating expenses, the probability of take-up (i.e., the probability that the customer will accept a product offering), the probability of default, the loss given default, the exposure at default, the amount of capital allocated to the transaction, and the cost of equity capital".

Step 2 – External versus internal

RBP "incorporates key factors from both the external market data (such as the probability of take-up, which in turn is a function of price and credit limit) and internal data (such as the cost of capital)".
⇒ A.

Interpretation

Take-up is what makes pricing a two-sided problem: raise the price to cover risk and fewer customers accept, which is why the price has to be optimised rather than simply marked up.

Why the other options are wrong

B is incorrect. PD is one of seven inputs; LGD, EAD and capital differ across accounts and are priced in.
C is incorrect. Operating expenses are an explicit input.
D is incorrect. Swaps external for internal, and confuses take-up (will the customer accept the offer?) with default.

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.8 Risk-Based Pricing

Solution CR-7 h.5 Answer: B

Bank Q286 • BT P2.T6.605.3

Step 1 – The true benefits

Different prices for the same product by risk (D); pricing strategies mapped to profit, revenue, market share and risk-adjusted return by score band (A); trade-offs among profitability, market share and risk weighed to raise shareholder value (C).

Step 2 – Test B

The price rests on account-level economics: take-up probability, PD, LGD, EAD, allocated capital, cost of equity and operating expenses. These must still be estimated.
⇒ B is not a benefit.

Interpretation

Risk-based pricing needs more risk measurement, not less; that is one reason GARP says it requires infrastructure many nonbank lenders lack.

Why the other options are wrong

A is incorrect. A true benefit: the mapping of pricing strategies to corporate metrics.
C is incorrect. A true benefit: trade-offs among profit, market share and risk.
D is incorrect. A true benefit: the definition of risk-based pricing.

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.8

Solution CR-7 h.6 **Answer: D**

New • built from GARP Ch.7, 7.8

Step 1 – The benefit claim

GARP: "Traditional retail pricing leaves a considerable amount of money on the table; better pricing can improve key corporate performance metrics by 10 to 20 percent or more." "Pricing is a key tool for retail bankers as they balance the goal of increasing market share against the goal of reducing the rate of bad accounts."
⇒ D.

Step 2 – The state of practice

"RBP in the financial retail area remains in its infancy. A bank's key business objectives are seldom adequately reflected in its pricing strategy."

Interpretation

The strategy curve (the bad rate against the acceptance rate) is the tool for this balance: RBP manages the trade-off, it does not remove it.

Why the other options are wrong

A is incorrect. The opposite of GARP: RBP is in its infancy and objectives are seldom adequately reflected in pricing.
B is incorrect. Absolute claim: pricing is used to balance market share against bad accounts, so the trade-off remains.
C is incorrect. GARP says RBP "should also be used" when nonbanks offer credit; the obstacle is that many lack the logistical and operational infrastructure.

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. • **Reference:** Crouhy, Galai and Mark, Ch.9 (GARP CR Ch.7), 7.8

CR-8 • Country Risk: Determinants, Measures, and Implications • Vardeez: MODERATE

CR-8 g

P3 CONCEPT

Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default.

Workbook: D23 Sovereign Default Risk and CDS • Tier D • Map #56 (1/16)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic (D23), and Vardeez flags it in a Moderate chapter: "same as Level 1, but the last LO is very important on CDS". Map #56: sovereign CDS as a measure of country default risk "is the one item to actually secure".
Vardeez: the three measures trade the same thing against each other: moving from ratings to default spreads to CDS buys responsiveness and pays for it with contamination.

Reported exam shapes

- **D23·S1** Sovereign CDS spreads as a measure of country default risk, compared with ratings and bond default spreads.

What the questions look like: conceptual "which statement is correct". Who leads whom, what each measure gets right and wrong, what a sovereign CDS does and does not protect against, and how far the evidence on CDS really goes. The numbers in the notes (spreads, CDS premiums) are illustrations, not exam calculations.

Not used: Bank Q313 repeats three of the four claims in workbook D23.1 (near-duplicate).

1. Ratings against market default spreads

The sovereign default spread is the yield on a government's bond in a foreign currency, usually US dollars, minus the riskless rate in that currency. GARP Table 8.10 (July 2022, 10-year dollar bonds): Brazil $6.19\% - 3.02\% = 3.17\%$, Chile 1.31%, Colombia 2.56%, Indonesia 1.36%, Mexico 2.05%, Peru 1.74%, Turkey 7.90%. Spreads and ratings are positively correlated, but spreads have three advantages:

- **More granular.** Colombia and Indonesia are both Baa2, yet the market charges Colombia 120 bp more.
- **More dynamic.** In December 2005 Brazil's spread was 3.18% and Venezuela's 3.09%; by December 2009 Brazil's had fallen to 1.32% while Venezuela's had widened to 10.26%, with no matching change in ratings.
- **Earlier.** Spreads adjust quickly to information, so they give earlier signals of imminent danger.

The costs: spreads are **far more volatile** than ratings and move with liquidity and investor demand for reasons that have nothing to do with default. They also need a **default-free security in the same currency**: local currency bonds of different governments cannot be compared, because rate differences can reflect expected inflation, and even a dollar spread assumes the US Treasury is default free.

What the studies agree on. (1) Default spreads are correlated with both ratings and actual default. (2) "The sovereign bond market leads ratings agencies": spreads usually climb *before* a downgrade and drop before an upgrade. (3) A rating change is still an informational event with a price impact when it happens. So ratings are not useless: bond markets draw on ratings, just as the agencies draw on market data.

2. Sovereign CDS spreads

How the contract works. The buyer pays the spread, a percentage of the notional, each period. If the reference entity fails to pay, restructures or goes bankrupt (a credit event), the seller either takes the bond and pays par (physical settlement) or pays par minus the bond's market price (cash settlement). GARP's example: protection on USD 10 million of 5-year Colombian bonds at 250 bp costs USD 250,000 a year. Two gaps remain: *no credit event, no payment*, even if the bond price collapses; and the protection is only as good as the seller's credit standing (the buyer may also default on the spread payments).

The market. Sovereign CDS are a small part of a market dominated by corporate and then bank CDS. It is narrow: a few players account for most trading, so the failure of a big one can throw it into turmoil, as Lehman's did in 2008.

What the evidence says. If counterparty and liquidity risk are assumed away, CDS prices give updated measures of default risk, where ratings are updated infrequently. In Greece from 2006 to 2010, ratings stayed stagnant for most of the period while CDS and default spreads changed every month (Figure 1). The findings:

1. Changes in CDS spreads **lead** changes in sovereign bond yields and in sovereign ratings.
2. "While the debate still continues", evidence is emerging that CDS spread changes are better predictors of sovereign default than ratings.
3. CDS spreads rise with economic policy uncertainty, together with currency depreciation.
4. CDS prices **cluster**: within GARP's emerging-market cluster 1 the average correlation is 0.516, against 0.210 with the rest of the market.

The limits. Exposure to *counterparty and liquidity risk*, endemic to the market, moves CDS prices for reasons unrelated to default: much of the surge in the last quarter of 2008 followed Lehman's failure. The narrowness of the market makes single names prone to liquidity squeezes. GARP's verdict: CDS changes provide information, "albeit noisy", about changes in default risk, but "there is little to indicate that it is superior to market default spreads (obtained from government bonds)".

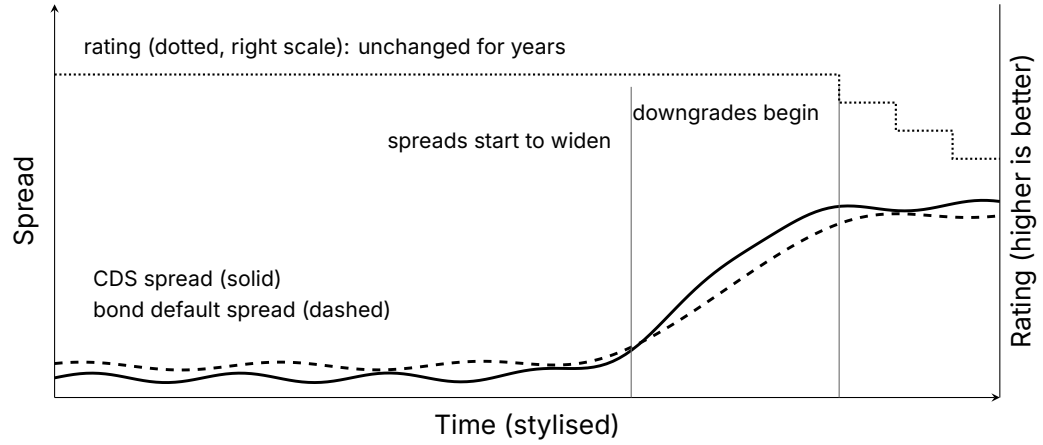


Figure 1. Stylised, not data: the pattern of GARP Figure 8.8 (Greece, 2006 to 2010). The rating (dotted, right scale) is unchanged for most of the period, then cut several times in quick succession; the CDS spread (solid) and the bond default spread (dashed) move every month and widen well before the first downgrade. The market measures lead, but they also carry noise (the small wiggles) unrelated to default.

Corrections to the source notes. (1) GARP's text says Brazil's 10-year dollar bond yielded 6% against 3.02% on Treasuries, "the difference of 1.90% between the two rates (6%−3.02% = 2.98%)". The sentence contradicts itself (1.90% against 2.98%) and its own Table 8.10, which gives 6.19%−3.02% = 3.17%; SchweserNotes (LO 26.g) repeats 6.00% and 2.98%. The table figure, 3.17%, is used here. (2) SchweserNotes dates the Brazil and Venezuela spreads of 1.32% and 10.26% to December 2010; GARP gives December 2009. (3) Vardeez and SchweserNotes say CDS spreads carry "market and liquidity" risk on top of default risk; GARP names *counterparty* and liquidity risk. Schweser also states flatly that CDS spreads "are better predictors of sovereign risk than credit ratings"; GARP says only that, "while the debate still continues", evidence is emerging for it.

	Sovereign rating	Bond default spread	CDS spread
What it is	Agency's ordinal grade	Dollar bond yield minus Treasury yield	Annual premium for default protection
Updated	Infrequently, after committee review	Continuously, as bonds trade	Continuously, as CDS trade
Granularity	Coarse (Colombia = Indonesia = Baa2)	Fine (2.56% against 1.36%)	Fine
Timing	Lags: moves after spreads	Leads ratings	Leads bond yields and ratings
Noise	Stable, but stale; may overreact in crises (CR-8 f)	Volatile; liquidity and investor demand	Counterparty and liquidity risk; narrow market; clustering
Needs	Analysts and data	A default-free bond in the same currency	A functioning CDS market and a credit event to pay
Verdict	Still informative: changes move prices	Correlated with ratings and default	Noisy; little evidence it beats default spreads

Exam traps

- **Lead-lag polarity.** Markets lead the agencies: spreads widen before downgrades and narrow before upgrades; CDS changes lead bond yields and ratings.
- **Absolute claim.** "CDS spreads are superior to every other measure" is false: there is "little to indicate" they beat bond default spreads. "Better predictor than ratings" is the supported (and still debated) claim.
- **Pure default risk.** A CDS spread is not a clean default probability; counterparty and liquidity risk are inside it, so reading it as pure default risk can overstate that risk.
- **Liquidity polarity.** The sovereign CDS market is narrow, not deep: illiquidity is a weakness.
- **Scope error.** Default spreads need a default-free bond in the same currency; comparing local currency yields across countries measures inflation as well.
- **Ratings are not useless.** A rating change is still an informational event with a price impact.
- **Protection gaps.** No credit event, no payout; and the seller can fail.

Practice questions: CR-8 g

Question CR-8 g.1

Bank Q311 • AnalystPrep Q.6202 (rebuilt)

An analyst compares sovereign credit ratings with sovereign CDS spreads as indicators of default risk. Which of the following statements is most accurate?

A. Ratings adjust more quickly to new information than CDS spreads, so ratings are the better predictor of default.

B. CDS spreads adjust more quickly to market information than ratings, and changes in CDS spreads have been found to lead changes in ratings.

C. A CDS spread isolates pure sovereign credit risk, so it can be read directly as a default probability estimate.

D. CDS spreads are preferable to ratings because the sovereign CDS market is deep and highly liquid.

Question CR-8 g.2

New • built from GARP Ch.8, Market Interest Rates, Table 8.10

A fixed-income analyst at a Swiss private bank compares Moody's ratings with default spreads on 10-year US dollar government bonds (July 2022):

	Colombia	Indonesia	Brazil
Moody's rating	Baa2	Baa2	Ba2
Default spread over US Treasuries	2.56%	1.36%	3.17%

Which of the following statements about using default spreads rather than ratings to gauge sovereign default risk is correct?

A. Because Colombia and Indonesia share a rating, the gap between their spreads must reflect liquidity rather than default risk, so the spreads add nothing to the ratings.

B. The same comparison could be made with the three countries' local currency bond yields, provided each is measured against the US Treasury yield.

C. Default spreads are less volatile than ratings, because they average the views of many investors.

D. Spreads give a more granular and more dynamic view of default risk than ratings, but they are more volatile and can move with liquidity and investor demand for reasons unrelated to default.

Question CR-8 g.3 New • built from GARP Ch.8, Credit Default Swaps

A portfolio manager holds USD 10 million of 5-year Colombian government bonds and buys 5-year CDS protection on Colombia at 250 basis points to hedge default risk. Which of the following statements about the protection is correct?

A. If the bonds' price falls sharply in a liquidity squeeze while Colombia keeps paying, the protection seller must compensate the manager for the fall.

B. Because the reference entity is a sovereign, the manager carries no exposure to the protection seller.

C. After a failure to pay, a restructuring or a bankruptcy, the seller either buys the bonds at par or pays par minus their market price, and the protection is only as good as the seller's own credit standing.

D. Under cash settlement the seller pays the full par value of USD 10 million and also takes delivery of the bonds.

Question CR-8 g.4 Bank Q312 • AnalystPrep Q.5917 (rebuilt)

An analyst plans to use sovereign credit default swap (CDS) spreads as a stand-alone gauge of a country's default risk. Which of the following is a key limitation of this approach?

A. Sovereign CDS spreads are not observable because CDS contracts are privately negotiated.

B. CDS spreads are affected by counterparty and liquidity risk as well as default risk, so using them to measure only default risk could overstate that risk.

C. CDS spreads adjust more slowly to new information than sovereign credit ratings do.

D. CDS spreads cannot be compared across countries because there is no risk-free security to serve as a benchmark.

Question CR-8 g.5 New • built from GARP Ch.8, Market Interest Rates and Credit Default Swaps

At an investment committee meeting of a Japanese pension fund, one member argues: "Market measures lead the agencies, so sovereign ratings are useless and we should stop using them." Which of the following responses is best supported by the evidence in Damodaran?

A. A rating change is still an informational event with a price impact when it occurs, and bond markets draw on ratings just as agencies draw on market data, so ratings remain useful alongside spreads.

B. Ratings are better predictors of default than either market measure, because each rating is set only after a committee review.

C. Default spreads are uncorrelated with ratings, so the two measures carry unrelated information and both should be kept.

D. Ratings can be replaced by a portfolio of emerging-market sovereign CDS, because those spreads move independently of each other and so diversify country risk.

Question CR-8 g.6 Workbook D23.1 • S1 • Map #56 • also CR-8 b

A sovereign risk analyst at a Singapore asset manager is presenting to the investment committee on how to monitor default risk in a portfolio of emerging market government bonds. The analyst compares agency ratings, default spreads on US dollar sovereign bonds, and sovereign CDS spreads. Which of the following statements is correct for the analyst to make?

A. Changes in sovereign CDS spreads tend to lead changes in bond yields and ratings, but a spike may reflect counterparty and liquidity risk rather than higher default risk.

B. Sovereign CDS spreads update continuously, and the evidence shows them to be clearly superior to default spreads on government bonds as a measure of default risk.

C. Default spreads on sovereign bonds tend to widen only after a rating downgrade, so the bond market confirms agency assessments rather than anticipating them.

D. Default spreads can be measured from local currency government bonds by comparing their yields across countries, without needing a default-free bond in each currency.

Solutions • CR-8 g

Solution CR-8 g.1 — Answer: B

Bank Q311 • AnalystPrep Q.6202

Step 1 – Timeliness

Ratings "get updated infrequently"; CDS prices adjust to reflect current information on default risk.

Step 2 – The evidence

"Changes in CDS spreads lead changes in the sovereign bond yields and in sovereign ratings."
⇒ B.

Interpretation

Speed is the CDS spread's strength; its weakness is what else is in the price (Question CR-8 g.4).

Why the other options are wrong

A is incorrect. Reverses the timeliness: ratings are the slow, infrequently updated measure.
C is incorrect. CDS prices also reflect counterparty and liquidity risk, so a spread is not a pure default-risk measure or a direct default probability.
D is incorrect. The CDS market is narrow and prone to liquidity problems; liquidity is a limitation, not an advantage.

LO: CR-8 g: Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default. • **Reference:** Damodaran, Country Risk 2022 (GARP CR Ch.8), Sovereign Default Risk: Credit Default Swaps, CDS and Default Risk

Solution CR-8 g.2 — Answer: D

New • built from GARP Ch.8, Market Interest Rates, Table 8.10

Step 1 – What the table shows

Colombia and Indonesia carry the same Baa2 rating, but the market charges Colombia 2.56% and Indonesia 1.36%: "the market differentiation for risk is more granular than the ratings agencies".

Step 2 – The costs of spreads

Market measures "tend to be far more volatile than ratings and can be affected by variables that have nothing to do with default. Liquidity and investor demand can sometimes cause shifts in spreads".
⇒ D states both sides.

Interpretation

Spreads are also more dynamic: Brazil and Venezuela had almost equal spreads in December 2005 (3.18% and 3.09%) but 1.32% and 10.26% by December 2009.

Why the other options are wrong

A is incorrect. The gap is the granularity advantage: the market sees slightly more default risk in Colombia.
B is incorrect. Scope error: local currency yields embed expected inflation, so they cannot be compared across countries; a default-free bond in the same currency is needed.
C is incorrect. Polarity flip: spreads are far more volatile than ratings.

LO: CR-8 g: Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default. • **Reference:** Damodaran, Country Risk 2022 (GARP CR Ch.8), Market Interest Rates: The Sovereign Default Spread, Table 8.10; The Spread as a Predictor of Default

Solution CR-8 g.3 Answer: C

New • built from GARP Ch.8, Credit Default Swaps

Step 1 – What triggers payment

The seller makes the buyer whole "if the issuer of the bond (reference entity) fails to pay, restructures or goes bankrupt (credit event)", by physical settlement (buy the bond at par) or cash settlement (pay par minus the market price).

Step 2 – The two gaps

"If there is no credit event, and the market price of the bond collapses, you as the buyer will not be compensated"; and "the guarantee is only as good as the credit standing of the seller of the CDS".

$\Rightarrow$ C.

Interpretation

The premium here is 2.5% of USD 10 million, USD 250,000 a year for five years. The two gaps are why a CDS spread is not a clean default measure: it prices the seller's risk and the market's liquidity too.

Why the other options are wrong

A is incorrect. No credit event, no payout: a price fall without default is not covered.

B is incorrect. The buyer is exposed to the seller's ability to pay whatever the reference entity.

D is incorrect. Mixes the two settlements: cash settlement pays par minus market price, with no delivery; physical settlement pays par against delivery.

LO: CR-8 g: Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default. • **Reference:** Damodaran, Country Risk 2022 (GARP CR Ch.8), Credit Default Swaps: How Does a CDS Work?

Solution CR-8 g.4 Answer: B

Bank Q312 • AnalystPrep Q.5917

Step 1 – GARP's first limitation

"The exposure to counterparty and liquidity risk, endemic to the CDS market, can cause changes in CDS prices that have little to do with default risk": much of the late-2008 surge followed Lehman's failure.

Step 2 – GARP's second limitation

"The narrowness of the CDS market can make an individual CDS susceptible to liquidity problems".

$\Rightarrow$ A spread that mixes default, counterparty and liquidity effects, read as pure default risk, can overstate that risk: B.

Interpretation

Vardeez and Schweser call the extra components "market and liquidity" risk; GARP's word is counterparty (see the correction box).

Why the other options are wrong

A is incorrect. CDS spreads are quoted and observable, even though the contracts trade over the counter.

C is incorrect. Reverses the evidence: CDS spreads react faster than ratings, which is an advantage.

D is incorrect. The missing-benchmark problem belongs to local currency bond default spreads; a CDS is quoted directly as a spread on the notional.

LO: CR-8 g: Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting

default. • **Reference:** Damodaran, Country Risk 2022 (GARP CR Ch.8), Credit Default Swaps: CDS and Default Risk

Solution CR-8 g.5 Answer: A

New • built from GARP Ch.8, Market Interest Rates and Credit Default Swaps

Step 1 – What the default-spread studies conclude

"Notwithstanding the lead-lag relationship, a change in sovereign ratings is still an informational event that creates a price impact at the time that it occurs. In summary, it would be a mistake to conclude that sovereign ratings are useless, since sovereign bond markets seem to draw on ratings (and changes in these ratings) when pricing bonds, just as ratings agencies draw on market data". ⇒ A.

Step 2 – Check the CDS claim in D

GARP's fourth CDS finding is clustering: CDS prices across groups of countries move together (cluster 1: average internal correlation 0.516 against 0.210 external).

Interpretation

The three measures are complements: ratings for stability and their information events, default spreads for granular pricing, CDS spreads as the earliest but noisiest warning.

Why the other options are wrong

B is incorrect. Polarity flip: market measures adjust faster, and CDS changes are emerging as better predictors than ratings.

C is incorrect. Default spreads are correlated with ratings and with actual default.

D is incorrect. CDS spreads cluster, so a basket of correlated sovereign CDS does not diversify country risk, and nothing in the evidence says CDS should replace ratings.

LO: CR-8 g: Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default. • **Reference:** Damodaran, Country Risk 2022 (GARP CR Ch.8), The Spread as a Predictor of Default; CDS and Default Risk, Table 8.11

Solution CR-8 g.6 Answer: A

Workbook D23.1 • S1

Step 1 – What the evidence says about sovereign CDS

Damodaran: "changes in CDS spreads lead changes in the sovereign bond yields and in sovereign ratings". But "the exposure to counterparty and liquidity risk, endemic to the CDS market, can cause changes in CDS prices that have little to do with default risk": much of the late-2008 surge followed Lehman's failure, and the narrowness of the market adds liquidity squeezes.

⇒ A is correct: CDS spreads are the most timely signal, and their known noise has a named source.

Interpretation

The committee should use all three measures: ratings for stability and their information events, bond default spreads for granular and dynamic pricing in a currency with a default-free benchmark, and CDS spreads as the earliest (but noisy) warning, discounting moves that coincide with dealer stress or thin trading.

Why the other options are wrong

B is incorrect. Absolute claim: CDS changes "provide information, albeit noisy", but "there is little to indicate that it is superior to market default spreads (obtained from government bonds)". The supported claim is only that CDS changes may predict default better than ratings.

C is incorrect. Polarity flip on the lead-lag: "the sovereign bond market leads ratings agencies,

with default spreads usually climbing ahead of a rating downgrade and dropping before an upgrade".

D is incorrect. Scope error: a market default spread needs "a default free security in the currency in which the bonds are issued". Local currency yields across countries "cannot be compared to each other, since the differences in rates can be due to differences in expected inflation".

LO: CR-8 g: Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default. CR-8 b: Evaluate the methods for measuring country risk and discuss the limitations of using those methods.

• **Reference:** Damodaran, Country Risk 2022 (GARP CR Ch.8), Sovereign Default Risk: Market Interest Rates (The Sovereign Default Spread; The Spread as a Predictor of Default) and Credit Default Swaps (CDS and Default Risk)

CR-9 • Estimating Default Probabilities • Vardeez: VERY IMPORTANT

CR-9 g

P3 CONCEPT

Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller.

Workbook: D14 Credit Derivatives and Credit-Linked Notes • Tier D (D14.1, shape S3)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic (D14, Map #58, 2/16), and Vardeez says skip the CDS here because it is covered later (CR-13).
Vardeez: the definition and the two legs, physical against cash settlement, the FI Advisors and ELF example, credit indices, and the fixed-coupon table for the upfront payment.

Reported exam shapes

- **D14·S3** CDS mechanics: obligations of buyer and seller, settlement (Question CR-9 g.7).

What the questions look like: who pays what, and when. A credit event triggers the payoff, a price fall does not; premiums run quarterly in arrears with a final accrual; physical settlement exchanges bonds for face value, cash settlement pays face value minus the cheapest-to-deliver price; under fixed coupons the side that gets the better of the coupon pays upfront.

Kept outside the TYPE rule: GARP Practice Question 9.5 (the seller's cash flows) is a GARP-practice numerical; it is Question CR-9 g.4.

Elsewhere: hazard rates from CDS spreads under CR-9 h; CDS valuation, indices and synthetic CDOs under CR-13.

1. Definition and the two legs

A credit default swap insures against default by a particular company, the **reference entity**; its default is a **credit event**, usually defined as a failure to pay as payments fall due, a restructuring of debt, or a bankruptcy (restructuring is sometimes excluded in North American contracts).

- **Buyer of protection.** Pays the **CDS spread** (the total paid per year as a percentage of notional) in periodic payments, usually *quarterly in arrears*, until maturity or a credit event. Because the payments are in arrears, a **final accrual payment** is due for the part-period up to the credit event. In return the buyer has the right to sell bonds of the reference entity for their face value after a credit event.
- **Seller of protection.** Receives the payments and, after a credit event, agrees to buy the bonds for their face value. The total face value that can be sold is the CDS's **notional principal**.
- **The reference entity is not a party**, and the buyer need not own its bonds: protection bought without an exposure is a "naked" CDS (banned on European sovereign debt in 2013). Without a credit event the buyer receives nothing, however far the bonds' price falls.

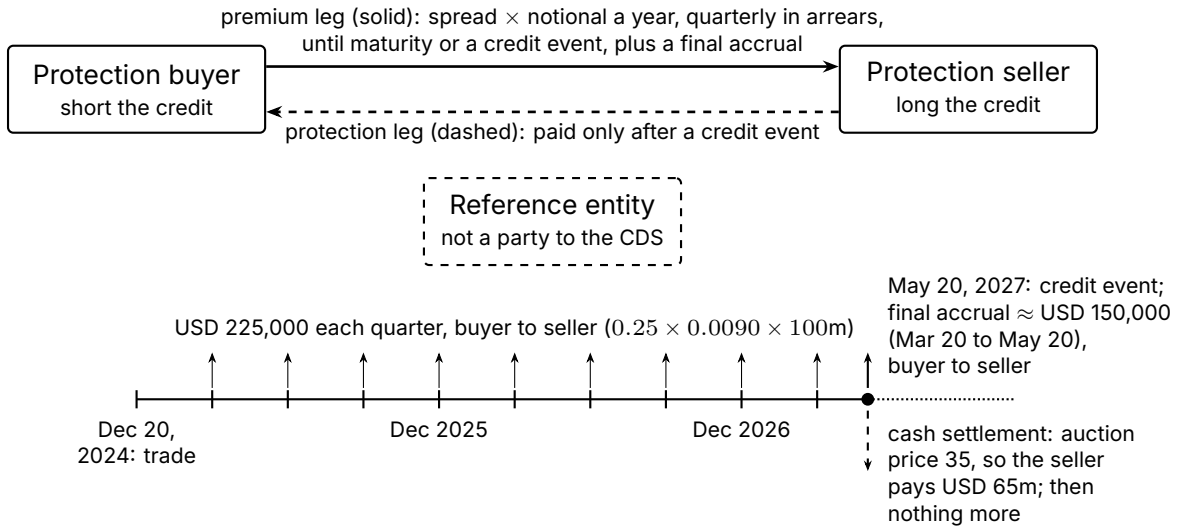


Figure 1. The two legs (top) and GARP's example on a timeline (bottom): a five-year CDS from December 20, 2024 on USD 100 million at 90 bp a year, quarterly in arrears. Solid arrows run from buyer to seller, the dashed arrow from seller to buyer. The contract would have run to December 20, 2029; the credit event ends it.

2. Settlement after a credit event

- **Physical settlement.** The buyer delivers bonds with a total face value equal to the notional; the seller pays their face value (USD 100 million in the example).
- **Cash settlement** (now usual). A two-stage auction, held several days after the credit event, sets the mid-market value of the **cheapest deliverable bond**. The seller pays notional $\times(1-$auction price): at 35 per 100, USD 65 million. In general the payoff is $L(1-R)$, with the recovery R set by the auction.
- **Cheapest-to-deliver option.** Several bonds of the same seniority can usually be delivered, and they need not trade at the same percentage of face value after default. The buyer benefits from the choice; the auction values the cheapest one.
- **Lehman, 2008.** The auction recovery was about 8 cents per dollar, so buyers received about 92% of notional; settlement went smoothly.

3. Quotes, dates and fixed coupons

- **Bid and ask.** A dealer quoting 250/260 bp will *buy* protection paying 250 bp a year and *sell* it for 260 bp. Five years is the most popular maturity; contracts mature on March 20, June 20, September 20 or December 20, so the first payment covers a stub period.
- **Fixed coupons.** Standard contracts trade with a fixed coupon; the market quotes the spread. At the trade the present value of the difference between spread and coupon payments is settled upfront: if the quoted spread is *below* the coupon, the **seller** pays the buyer; if it is *above*, the **buyer** pays the seller. Afterwards the buyer pays the coupon.
- **Who uses CDSs.** Banks tend to be net buyers of protection, insurance companies net sellers (AIG's sales of protection on AAA mortgage tranches led to its 2008 bailout). Indices such as CDX NA IG and iTraxx Europe (125 names each) trade the same way (CR-13).
- **Two flaws.** Asymmetric information: a lender close to the company may know more about its default risk than other participants. Manufactured credit events: in 2018 Hovnanian agreed with

GSO Capital Partners, which had bought protection on it, to skip an interest payment and trigger a technical default.

Correction to the source notes. (1) SchweserNotes LO 27.g (body and summary) says the buyer makes "fixed quarterly payments (in advance)". GARP: the payments "are usually made in arrears every quarter", which is why a final accrual payment is due after a credit event; Schweser's own example then uses payment in arrears. (2) Vardeez's cash-settlement line says the seller pays the buyer "the fair market value of the defaulted bonds". The seller pays the face value *minus* the auction value of the cheapest deliverable bond: USD 65 million, not 35 million, in GARP's example. (3) Vardeez gives June 20 and December 20 as index "settlement dates"; they are the usual index *maturity* dates.

	Protection buyer	Protection seller
Credit view	Short the credit: gains if it worsens	Long the credit: gains if it holds up
Upfront (fixed coupon)	Pays if the quoted spread is above the coupon	Pays if the quoted spread is below the coupon
Before a credit event	Pays the spread (or coupon), quarterly in arrears	Receives the payments; pays nothing
Price fall, no credit event	Keeps paying; receives nothing	Keeps receiving; pays nothing
At a credit event	Pays the final accrual; payments stop	Receives the final accrual
Physical settlement	Delivers bonds with face value = notional; receives face value	Pays face value; receives the bonds
Cash settlement	Receives notional $\times(1 -$ CTD auction price$)$	Pays notional $\times(1 -$ CTD auction price$)$

Exam traps

- **Price fall is not a credit event.** No failure to pay, restructuring or bankruptcy, no payoff: the buyer keeps paying.
- **Arrears, not advance.** Payments are made in arrears, so the buyer owes a final accrual after the credit event.
- **Payoff is the loss, not the recovery.** Cash settlement pays $1-$ auction price (65), not the auction price (35).
- **Who delivers.** In physical settlement the buyer delivers the bonds and the seller pays face value.
- **Cheapest to deliver.** The buyer's option to choose the bond raises the payoff; the auction prices the cheapest.
- **Upfront direction.** Quoted spread below the coupon: the seller pays upfront. Above: the buyer pays.
- **Not insurance.** The buyer need not own the bonds.
- **Bid and ask.** The dealer buys protection at the bid and sells at the ask.

Practice questions: CR-9 g

Question CR-9 g.1

GARP book • Ch.9 Practice Question 9.6 (rewritten as multiple choice)

A junior trader at a Paris bank asks how a credit default swap can be settled after a credit event. Which statement is correct?

A. Physical settlement: the seller delivers the reference entity's bonds to the buyer at their post-default price; cash settlement: the seller pays the buyer the market value of the defaulted bonds.

B. Physical settlement: the buyer delivers bonds with a face value equal to the notional and receives their face value; cash settlement: the seller pays the notional times one minus the auction value of the cheapest deliverable bond per unit of face value.

C. Physical settlement: the buyer delivers bonds and receives their market value just before the default; cash settlement: the buyer pays the seller the accrued premium only.

D. Physical settlement: the reference entity buys back its bonds from the buyer at face value; cash settlement: the seller pays the buyer the full notional.

Question CR-9 g.2

Bank Q344 • BT notes P2.T6.24.10.1 (rebuilt)

Chicago Firefighters Pension (CFP) owns Belize government bonds with a par value of $10,000,000 and has bought 10-year credit default swap protection on them from Old Crow Bank at a CDS spread of 300 basis points per year (3% of par). In year 8, the market value of the bonds falls to $8,000,000 because of a decrease in market demand for the bonds. Belize continues to make all of its scheduled payments and no credit event is declared. What is the cash flow between CFP and Old Crow Bank for year 8?

A. CFP receives a net payment of $1.7M ($2M payout less $300,000 premium).

B. CFP receives a payment of $2M from Old Crow Bank.

C. CFP pays $300,000 to Old Crow Bank and receives no payout.

D. No payments are made by either party because the swap is marked to market and terminated when the bond price falls.

Question CR-9 g.3

New • built from GARP Ch.9, 9.4 and 9.4.1

A CDS on Delta Corp allows any of three senior unsecured Delta bonds to be delivered. Several days after a credit event, the three bonds trade at 38, 35 and 41 per 100 of face value, and the contract is cash settled through the standard auction. Which statement about the payoff is correct?

A. The auction values the cheapest deliverable bond, the one trading at 35, so the seller pays 65% of notional; the cheapest-to-deliver option benefits the protection buyer.

B. The payoff is based on the average of the three prices, so neither party gains from the choice of bond.

C. The payoff is based on the bond trading at 41, so the seller pays 59% of notional; the choice of bond benefits the protection seller.

D. The payoff is based on the bond trading at 35, and the buyer pays 35% of notional to the seller.

Question CR-9 g.4 GARP book • Ch.9 Practice Question 9.5 (rewritten as multiple choice) • GARP-practice numerical

A credit default swap requires a semiannual payment at the rate of 60 basis points per year. The principal is $300 million and the swap is settled in cash. A default occurs after four years and two months, and the calculation agent estimates that the price of the cheapest deliverable bond is 40% of its face value shortly after the default. Which of the following correctly describes the cash flows for the seller of the credit default swap?

A. Receives $900,000 every six months for four years, then pays $120 million at four years and two months.

B. Receives $900,000 every six months for four years, then pays $180 million at four years and two months.

C. Receives $900,000 every six months for four years; at four years and two months receives a final accrual of $300,000 and pays $180 million.

D. Receives $1.8 million every six months for four years; at four years and two months receives a final accrual of $600,000 and pays $180 million.

Question CR-9 g.5 New • built from GARP Ch.9, 9.4.3

A five-year CDS on an investment-grade name trades with a standard fixed coupon of 100 basis points a year. The quoted CDS spread is 60 basis points. Which statement correctly describes the cash flows at the trade and afterwards?

A. The buyer pays the seller the present value of the 40 bp a year difference upfront, then pays 60 bp a year.

B. The seller pays the buyer the present value of the 40 bp a year difference upfront; afterwards the buyer pays the 100 bp coupon until maturity or a credit event.

C. No upfront payment is made because the quoted spread is below the coupon; the buyer pays 60 bp a year.

D. The buyer pays the seller the present value of the 40 bp a year difference upfront; afterwards the buyer pays the 100 bp coupon.

Question CR-9 g.6 GARP book • Ch.9 Practice Question 9.20 (rewritten as multiple choice)

A credit trainer tells a class: "The position of a buyer of a credit default swap is similar to the position of someone who is long a risk-free bond and short a corporate bond." Which explanation of the statement is correct?

A. The buyer gains when the reference entity's credit improves, as a holder of its bonds would.

B. The buyer pays the risk-free rate and receives the corporate yield, so the position earns the credit spread.

C. The statement is false: a CDS buyer has no exposure to the reference entity unless it also owns the entity's bonds.

D. A corporate bond plus bought protection is approximately a risk-free bond, so bought protection is approximately a long risk-free bond plus a short corporate bond: the buyer gains as the issuer's credit worsens and pays about the credit spread to hold the position.

Question CR-9 g.7 Workbook D14.1 • S1, S2, S3, S4 • Map #58

A structured credit specialist at a German bank is training newly hired analysts on four instruments the bank uses to transfer or take credit risk. Which of the following statements should the specialist present as correct?

A. In a total return swap used to finance a bond purchase, the total return receiver retains ownership of the bond, and the spread over the floating rate compensates the receiver for the risk that the payer defaults.

B. In a credit default swap, the protection buyer must own bonds of the reference entity, as with an insurance contract, and receives the notional times one minus the auction price after a credit event.

C. In a synthetic CDO priced with the one-factor Gaussian copula, base correlation is the correlation that prices each individual tranche, while compound correlation prices the tranche from zero to its detachment point.

D. In a credit-linked note issued through an SPV, the investor pays the principal upfront and bears the credit risk of the assets sold to the SPV, with no credit linkage to the sponsoring entity.

Solutions • CR-9 g

Solution CR-9 g.1

Answer: B

GARP book • Ch.9 Practice Question 9.6

Step 1 – Physical settlement

The buyer has the right to sell bonds of the reference entity for their face value; the seller must buy them. Bonds with a face value equal to the notional change hands for that face value.

Step 2 – Cash settlement

An auction sets the value of the cheapest deliverable bond; the seller pays the notional times one minus that value (CTD at 35 gives 65% of notional).

$\Rightarrow$ B.

Interpretation

Both routes leave the buyer with about the face value: either cash for the bonds, or cash equal to the loss on bonds it can keep or sell. Cash settlement is now usual.

Why the other options are wrong

A is incorrect. Reverses who delivers, and the cash leg pays the recovery value instead of the loss (the slip in Vardeez's notes).

C is incorrect. The buyer receives face value, not a pre-default market value; the accrual is owed in addition to the payoff, not instead of it.

D is incorrect. The reference entity is not a party to the CDS, and cash settlement pays the loss, not the full notional.

LO: CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.4 Credit Default Swaps; Practice Question 9.6

Solution CR-9 g.2

Answer: C

Bank Q344 • BT notes P2.T6.24.10.1

Step 1 – Has a credit event occurred

A credit event is a failure to pay, a restructuring or a bankruptcy. Belize is paying in full; the price fell because demand fell.

$\Rightarrow$ No credit event, so the seller owes nothing.

Step 2 – The buyer's obligation

The buyer pays the spread until maturity or a credit event: $0.03 \times 10{,}000{,}000 = \$300{,}000$ a year.

Interpretation

A CDS protects against default, not against market price moves. CFP's \$2 million mark-to-market loss is unhedged by this contract.

Why the other options are wrong

A is incorrect. Treats the price fall as a credit event and nets the premium: $2.0 - 0.3 = 1.7$.

B is incorrect. Same payout error, and it drops the premium that is still due.

D is incorrect. A CDS is not terminated by a price fall; it runs to maturity or a credit event.

LO: CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.4

Solution CR-9 g.3 Answer: A

New • built from GARP Ch.9, 9.4 and 9.4.1

Step 1 – Which bond sets the payoff

The buyer may deliver any of the eligible bonds and will choose the cheapest; the auction therefore determines the value of the cheapest deliverable bond, here 35.

Step 2 – Who pays and how much

The seller pays notional $\times(1-0.35)=65\%$ of notional.
$\Rightarrow$ The choice of bond is an option held by the buyer: A.

Interpretation

The cheapest-to-deliver option makes protection more valuable to the buyer, one reason it pushes the CDS-bond basis in a positive direction (CR-9 i).

Why the other options are wrong

B is incorrect. The payoff is not an average; the cheapest bond sets it.
C is incorrect. The most expensive bond would favour the seller; the option belongs to the buyer.
D is incorrect. Reverses the direction and pays the recovery value instead of the loss.

LO: CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.4 and 9.4.1 The Cheapest-to-Deliver Bond

Solution CR-9 g.4 Answer: C

GARP book • Ch.9 Practice Question 9.5

Step 1 – Regular payments to the seller

$\frac{1}{2}\times 0.0060\times 300{,}000{,}000=\$900{,}000$ every six months, at $0.5, 1.0, \ldots, 4.0$ years (8 payments)

Step 2 – Final accrual

Payments are made in arrears, so the buyer owes the premium for the two months since the last payment date:

$$0.0060\times 300{,}000{,}000\times\tfrac{2}{12}=\$300{,}000$$

Step 3 – Cash settlement

$300{,}000{,}000\times(1-0.40)=\$180{,}000{,}000$ paid by the seller at 4 years and 2 months

Interpretation

The seller collects \$7.5 million in total $(8\times 0.9+0.3)$ and pays \$180 million. A GARP-practice numerical under a CONCEPT LO: the numbers only record the mechanics (who pays, how often, in arrears, and the loss rather than the recovery).

Why the other options are wrong

A is incorrect. Paid the bond's recovery value instead of the loss $(300\times 0.40=120)$, and omitted the accrual.
B is incorrect. Omitted the final accrual that payment in arrears requires.
D is incorrect. Paid the full annual rate every six months: $0.0060\times 300\text{m}=\$1.8$ million, with an accrual of \$600,000.

LO: CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.4; Practice Question 9.5

Solution CR-9 g.5 Answer: B

New • built from GARP Ch.9, 9.4.3

Step 1 – Who gets the better of the coupon

Paying 100 bp a year for protection worth 60 bp overpays the seller by 40 bp a year.

Step 2 – The upfront

When the quoted spread is below the coupon, the seller pays the buyer the present value of the difference at the trade. After that the buyer pays the fixed coupon.
$\Rightarrow$ B.

Interpretation

The upfront payment converts a standard coupon into a trade at the market spread: the side that gets the better of the coupon compensates the other.

Why the other options are wrong

A is incorrect. Wrong payer, and after a fixed-coupon trade the running payment is the coupon, not the quoted spread.
C is incorrect. An upfront is needed whenever spread and coupon differ.
D is incorrect. Wrong direction: the buyer pays upfront only when the spread is above the coupon.

LO: CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.4.3 The Use of Fixed Coupons

Solution CR-9 g.6 Answer: D

GARP book • Ch.9 Practice Question 9.20

Step 1 – The hedge

Buying a corporate bond and buying protection on its issuer converts the bond, approximately, into a risk-free bond (GARP: a 7% bond with 200 bp protection earns about 5%).

Step 2 – Rearrange

Corporate bond + protection $\approx$ risk-free bond, so protection $\approx$ risk-free bond $-$ corporate bond.
$\Rightarrow$ A protection buyer is like someone long a risk-free bond and short a corporate bond: D.

Interpretation

Both positions gain when the issuer's credit worsens and both cost roughly the credit spread to carry. This is also why the CDS spread should be close to the bond yield spread.

Why the other options are wrong

A is incorrect. Describes the seller (or a bondholder), who is long the credit.
B is incorrect. Earning the spread is the seller's side; the buyer pays it.
C is incorrect. The buyer's exposure comes from the contract itself; owning the bonds is not required.

LO: CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. • **Reference:** Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.4 and 9.5.1; Practice Question 9.20

Solution CR-9 g.7 **Answer: D**

Workbook D14.1 • S1, S2, S3, S4

Step 1 – The CLN statement

Choudhry (CR-23): "Unlike CLNs issued by originators direct, CLNs issued by SPVs do not have any credit-linkage to the sponsoring entity. The note is linked instead to assets that have been sold to the SPV, and its performance is dependent on the performance of these assets." The note is funded: the investor's principal is paid upfront and is repaid less credit losses, so the investor is the protection seller (outside-source for the funded mechanics).
$\Rightarrow$ D is correct: funded, investor bears the reference credit risk, and the SPV removes the sponsor's own credit from the note.

Interpretation

The four instruments split cleanly by who funds and who holds the asset: a CLN investor funds and bears credit risk; a CDS seller bears credit risk unfunded; a TRS payer holds the bond and lays off its total return; a synthetic CDO tranche holder sells protection on a slice of a CDS portfolio. Each wrong option misplaces one of those attributions.

Why the other options are wrong

A is incorrect. Role misassignment: in Hull's financing structure the payer buys and "retains ownership of the bond for the life of the swap", and "the spread over the floating rate received by the payer is compensation for bearing the risk that the receiver will default".
B is incorrect. Scope/condition error: the settlement clause is right, but ownership is not a condition. Hull: unlike an insurance contract, "in the case of a CDS, the underlying asset does not have to be owned"; CDS volume on a company can exceed its debt (Lehman).
C is incorrect. Sibling swap: compound (tranche) correlation is the ρ that prices an individual tranche $\{a_{q-1}, a_q\}$; base correlation is the ρ that prices the $\{0, a_q\}$ tranche. The definitions are reversed.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation. CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. • **Reference:** Hull, OFOD 11e, Ch.25, 13.1, 13.7, 13.10; Choudhry, Ch.12 (CR-23), SPV applications

CR-12 • Credit Risk • Vardeez: IMPORTANT

CR-12 c

P3 CALC

Calculate the probability of default using credit spreads.

Workbook: D5 Credit Spread and PD from Spreads • Tier D • Map #42 (2/16)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D5 (Map #42, 2 of 16 sources) reports "deriving PD from credit spreads (LO 30.c)" and the one-period PD formula, and warns that the formula card's credit spread formula has its sign inverted.

Vardeez: a repeat of CR-9 ("just skim through"); his note for this objective is the hazard rate approximation $\bar{\lambda} = s(T)/(1-R)$, worked in full at CR-9 e and CR-9 h.

Reported exam shapes

- **D5·S1** Credit spread from the market value of debt, with the card's inverted sign as the distractor.
- **D5·S2** Deriving PD from credit spreads.
- **D5·S3** One-period PD $= (1/\text{LGD}) \times [1 - (1+r_f)/(1+y)]$, where recovery decides the answer.

What the questions look like: all numerical. A spread, a recovery rate and a maturity give a hazard rate and then a cumulative or interval PD; or yields and recoveries on one-year zeros give a one-period PD. Risk-free rates, notionals and old recovery assumptions appear as red herrings.

Worked elsewhere: D5.2 (three issuers ranked by one-period PD) at Question CR-9 j.7; spread-implied hazard rates from CDS quotes (A15.5, A15.6) at Questions CR-9 h.1 and CR-9 h.7; constant-hazard probability functions under CR-9 e.

1. From a spread to a hazard rate and a PD (shape S2)

A credit spread is, roughly, the expected loss rate per year: the hazard rate times the loss given default. Inverting it (GARP eq. 12.2, Hull eq. 9.3):

$$\bar{\lambda}(T) = \frac{s(T)}{1-R} \qquad Q(T) = 1 - \mathrm{e}^{-\bar{\lambda}(T)\,T}$$

Schweser's example: a one-year bond 125 bp over the risk-free rate with 55% recovery gives $\bar{\lambda} = 0.0125/0.45 = 2.78\%$. The hazard rate is already per year; the maturity enters only when it is turned into a cumulative PD.

PDs for each interval of a CVA calculation (GARP 12.1). Split the life of the portfolio at $t_1, t_2, \ldots$, interpolate a spread $s(t_i)$ for each end point, and take the drop in survival across the interval:

$$q_i = \exp\left(-\frac{s(t_{i-1})\,t_{i-1}}{1-R}\right) - \exp\left(-\frac{s(t_i)\,t_i}{1-R}\right)$$

Example. Spreads of 100, 120 and 140 bp at one, two and three years, $R = 40\%$: average hazard rates 1.667%, 2.000% and 2.333%; survival 0.9835, 0.9608 and 0.9324; so $q_1 = 1.65\%$, $q_2 = 2.27\%$ and $q_3 = 2.84\%$. Each q_i uses its own maturity's average hazard rate. These are **risk-neutral** probabilities, because CVA values future cash flows (CR-9 k); they sit well above real-world PDs from default data (CR-9 j).

2. Getting the spread itself (shape S1)

For zero-coupon debt with face value F, current market value D_t and maturity T (continuous compounding), $D_t = F\mathrm{e}^{-yT}$, so

$$y = -\frac{1}{T}\ln\frac{D_t}{F} \qquad \text{CS} = y - r = -\frac{1}{T}\ln\frac{D_t}{F} - r$$

Because $D_t < F$, the logarithm is negative and the leading minus sign makes the yield positive. A face value of 100 trading at 85 with three years to run and $r = 3\%$: $y = -\ln(0.85)/3 = 5.42\%$, so CS $= 242$ bp. Under Merton's model the debt value is $V_0 - E_0$ (CR-9 l), which is how Workbook D5.1 gets D_t.

The same idea values a derivative with credit risk in GARP's special case of a single uncollateralised payoff: discount at the risk-free rate *plus* the counterparty's T-year spread, $f = f_{nd}\,\mathrm{e}^{-(y-y_{nd})T}$. Example 12.2: a \$3 option with a 1.5% two-year spread is worth $3\mathrm{e}^{-0.015\times 2} = \$2.91$.

3. The one-period version (shape S3)

Price a one-year zero-coupon bond two ways: at its yield, and as the risk-neutral expected payoff discounted at the risk-free rate (annual compounding, default only at maturity, recovery R of face value):

$$\frac{(1-\text{PD}) + \text{PD}\cdot R}{1+r_f} = \frac{1}{1+y} \quad\Longrightarrow\quad \boxed{\text{PD} = \frac{1}{\text{LGD}}\left(1 - \frac{1+r_f}{1+y}\right)}$$

$1-(1+r_f)/(1+y) = (y-r_f)/(1+y)$ is the expected loss rate; dividing by LGD turns it into a PD. **Example:** $y = 5.5\%$, $r_f = 3\%$, $R = 45\%$: expected loss rate $1 - 1.03/1.055 = 2.370\%$, so PD $= 2.370\%/0.55 = 4.31\%$. The shortcut $(y-r_f)/(1-R) = 4.55\%$ ignores the $1/(1+y)$ term. **Recovery decides:** at the same spread, a higher recovery means a higher implied PD, so ranking issuers by spread can reverse their PD ranking (Workbook D5.2). This exact discrete form is an outside-source formula; it is the one-period cousin of $\bar{\lambda} = s/(1-R)$.

Formula card: CR-12 c

Average hazard rate	$\bar{\lambda}(T) = s(T)/(1-R)$ (per year; do not divide by T)
Cumulative PD	$Q(T) = 1 - \mathrm{e}^{-\bar{\lambda}(T)\,T}$
Interval PD (CVA)	$q_i = \mathrm{e}^{-s(t_{i-1})t_{i-1}/(1-R)} - \mathrm{e}^{-s(t_i)t_i/(1-R)}$
Spread from debt value	$\text{CS} = -\frac{1}{T}\ln(D_t/F) - r$ (not $+\frac{1}{T}\ln(D_t/F) - r$)
Credit-risky discounting	$f = f_{nd}\,\mathrm{e}^{-(y-y_{nd})T}$
One-period PD	$\text{PD} = \frac{1}{1-R}\left(1 - \frac{1+r_f}{1+y}\right) = \frac{y-r_f}{(1+y)(1-R)}$

Exam traps

- **Wrong-input twin.** Dividing the spread by R instead of $1-R$, or multiplying by $1-R$ instead of dividing.
- **Hazard rate divided by T.** $s/(1-R)$ is already an annual rate; a two-year bond does not halve it.
- **Hazard read as a PD.** $\bar{\lambda}$ is a rate; the T-year PD is $1 - \mathrm{e}^{-\bar{\lambda}T}$, and $\bar{\lambda}T$ is the linear shortcut.
- **Inverted sign** in the spread formula (shape S1): a negative "spread" on risky debt is the card's

error. Forgetting $1/T$ reports a T-year log loss as an annual spread; forgetting $-r$ reports the yield.

- **Unconditional versus conditional** interval PDs: q_i is seen from today; dividing by survival to the start of the interval gives the conditional PD.
- **Spread is not PD.** The spread prices expected loss, PD $\times$ LGD; bonds of one issuer with different recoveries must offer different yields.

Practice questions: CR-12 c

Question CR-12 c.1

Bank Q538 • Schweser Module Quiz 30.1 Q2 (rebuilt)

A credit analyst observes that a 2-year corporate bond yields 190 basis points above the risk-free rate. With a recovery rate of 35%, the average hazard rate for Years 1 and 2 is closest to:

A. 0.67%

B. 1.46%

C. 2.92%

D. 5.43%

Question CR-12 c.2

Bank Q536 • BT P2.T6.24.20.2 (rebuilt)

A risk analyst at an investment bank is assessing the credit risk of a derivatives counterparty. The counterparty's 5-year credit spread is 200 basis points and the risk-free rate is 5%. The recovery rate is now assumed to be 40%; the analyst who last reviewed the counterparty used 50%. Assuming a constant hazard rate equal to the average hazard rate implied by the spread, which of the following is closest to the counterparty's 5-year cumulative risk-neutral probability of default?

A. 5.82%

B. 15.35%

C. 16.67%

D. 18.13%

Question CR-12 c.3

New • built from GARP Ch.12, 12.1 (CVA and DVA)

A CVA analyst at a Swiss bank divides the next three years into annual intervals. The counterparty's credit spreads, interpolated to the interval end points, and other inputs are:

Maturity	1 year	2 years	3 years
Credit spread (bp)	90	110	130

The expected recovery rate is 40% and the risk-free rate is 2.5% for all maturities. Following the method GARP sets out for CVA, which of the following is closest to q_3, the risk-neutral probability that the counterparty defaults during the third year?

A. 2.05%

B. 2.69%

C. 2.79%

D. 6.29%

Question CR-12 c.4

Workbook D5.1 • S1 • Map #42

A credit analyst at a Japanese life insurer is checking a colleague's formula card, which gives the credit spread on zero-coupon debt as $\text{CS} = (1/T)\ln(D/F) - R_f$. The analyst applies the standard definition to an issuer whose only debt is a zero-coupon bond. Data (JPY billion, rates continuously compounded):

- Market value of the firm's assets from a calibrated Merton model: 68.0
- Market value of equity: 24.5; equity volatility: 55%
- Face value of the zero-coupon bond: 50.0, maturing in 4 years
- Risk-free rate: 1.5% per annum

Which of the following is closest to the credit spread on the issuer's debt?

A. -498 bps

B. 198 bps

C. 348 bps

D. 1,243 bps

Question CR-12 c.5

Bank Q539 • BT notes P2.T6.301.3 (rebuilt)

A single-period model gives the minimum coupon spread z an investor requires as compensation for default risk by setting the expected payoff equal to the risk-free payoff: $(1-\text{PD})(1+r+z)+\text{PD}\times R = 1+r$. The risk-free rate is 2.0%, the default probability is 3.0%, and the loss given default is 60.0% (so $R = 40\%$). The minimum spread z is closest to:

A. 124 basis points

B. 180 basis points

C. 186 basis points

D. 192 basis points

Question CR-12 c.6 New • built from GARP Ch.12, 12.1 and Workbook D5 (one-period formula) • shape D5·S3

A fixed-income analyst at a Korean asset manager prices two one-year zero-coupon bonds issued by the same company. Each bond either repays its face value at maturity or defaults then, recovering a fixed fraction of face value. Valuation is risk-neutral with annual compounding, and the one-year risk-free rate is 3.0%.

Bond	One-year yield	Expected recovery rate
Senior unsecured	5.5%	45%
Subordinated	?	15%

Both bonds carry the same issuer default probability. Which of the following is closest to the yield at which the subordinated bond is priced consistently with the senior bond?

A. 5.50%

B. 6.86%

C. 6.92%

D. 7.83%

Solutions • CR-12 c

Solution CR-12 c.1

Answer: C

Bank Q538 • Schweser Module Quiz 30.1 Q2

Step 1 – Loss given default

$1 - R = 1 - 0.35 = 0.65.$

Step 2 – Average hazard rate

$$\bar{\lambda} = \frac{s}{1 - R} = \frac{0.0190}{0.65} = 0.02923 = 2.92\% \text{ per year}$$

$\Rightarrow$ The spread is the annual expected loss rate, so dividing by LGD gives the annual default intensity.

Interpretation

The average hazard rate applies to each of the two years; the maturity enters only if a cumulative PD is wanted: $1 - \mathrm{e}^{-2 \times 0.02923} = 5.68\%$ over two years.

Why the other options are wrong

A is incorrect. Multiplied the spread by the recovery rate: $1.90\% \times 0.35 = 0.67\%$.

B is incorrect. Divided the correct 2.92% by the 2-year maturity; the hazard rate is already per year.

D is incorrect. Divided by the recovery rate instead of LGD: $1.90\%/0.35 = 5.43\%$.

LO: CR-12 c: Calculate the probability of default using credit spreads. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.1, eq. 12.2; Hull, RMFI 6e, Ch.17 (GARP CR Ch.9), 9.6.1 eq. 9.3

Solution CR-12 c.2

Answer: B

Bank Q536 • BT P2.T6.24.20.2

Step 1 – Average hazard rate with the current recovery

$$\bar{\lambda} = \frac{0.0200}{1 - 0.40} = 0.03333$$

Step 2 – Five-year cumulative PD

$$Q(5) = 1 - \mathrm{e}^{-0.03333 \times 5} = 1 - \mathrm{e}^{-0.16667} = 1 - 0.84648 = 15.35\%$$

$\Rightarrow$ A 15.35% risk-neutral probability that the counterparty defaults within five years.

Interpretation

The risk-free rate is a red herring: it discounts losses in a CVA but does not enter the PD. The old 50% recovery is a trap: the implied PD depends on the recovery assumed, and the current one must be used. (CVA itself is fairly insensitive to the recovery assumption when the PD is implied from the spread with the same R, because the PD scales with $1/(1-R)$ and the loss with $1-R$.)

Why the other options are wrong

A is incorrect. Multiplied the spread by LGD instead of dividing: $1 - \mathrm{e}^{-0.02 \times 0.6 \times 5} = 5.82\%$.

C is incorrect. Linear shortcut: $\bar{\lambda}T = 0.03333 \times 5 = 16.67\%$ treats the hazard rate as a simple annual probability.

D is incorrect. Used the stale 50% recovery: $\bar{\lambda} = 0.04$, $1 - \mathrm{e}^{-0.20} = 18.13\%$.

LO: CR-12 c: Calculate the probability of default using credit spreads. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.1, eqs. 12.1 and 12.2

Solution CR-12 c.3

Answer: B

New • built from GARP Ch.12, 12.1

Step 1 – Average hazard rates to each end point

$$\bar{\lambda}(2) = \frac{0.0110}{0.60} = 1.8333\%, \qquad \bar{\lambda}(3) = \frac{0.0130}{0.60} = 2.1667\%$$

Step 2 – Survival to the start and end of year 3

$$V(2) = \mathrm{e}^{-0.018333\times 2} = 0.96400, \qquad V(3) = \mathrm{e}^{-0.021667\times 3} = 0.93707$$

Step 3 – Interval PD

$$q_3 = V(2) - V(3) = 0.96400 - 0.93707 = 0.02693 = 2.69\%$$

$\Rightarrow$ Seen from today, a 2.69% chance that the counterparty defaults during year 3.

Interpretation

Each survival probability uses the average hazard rate for its own maturity, so a rising spread curve gives a forward hazard rate for year 3 of $3 \times 2.1667\% - 2 \times 1.8333\% = 2.83\%$, well above any of the average rates. The risk-free rate discounts the losses in the CVA sum; it does not enter q_3.

Why the other options are wrong

A is incorrect. Used the three-year average hazard for both end points: $\mathrm{e}^{-0.021667\times 2} - \mathrm{e}^{-0.065} = 2.05\%$.

C is incorrect. Divided by survival to the start of year 3: $0.02693/0.96400 = 2.79\%$, the conditional PD (sibling swap).

D is incorrect. Reported the cumulative three-year PD: $1 - 0.93707 = 6.29\%$.

LO: CR-12 c: Calculate the probability of default using credit spreads. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.1 CVA and DVA

Solution CR-12 c.4

Answer: B

Workbook D5.1 • S1

Step 1 – Market value of debt

$$D_t = V_0 - E_0 = 68.0 - 24.5 = 43.5$$

$\Rightarrow$ Debt with a JPY 50.0 billion face value is worth JPY 43.5 billion today.

Step 2 – Yield on the debt

$$D_t = F\mathrm{e}^{-yT} \Rightarrow y = -\frac{1}{T}\ln\frac{D_t}{F} = -\frac{1}{4}\ln\frac{43.5}{50.0} = -\frac{1}{4}(-0.13926) = 0.03482$$

$\Rightarrow$ The debt yields 3.48% per annum. The logarithm is negative because $D_t < F$; the leading minus sign turns it into a positive yield.

Step 3 – Credit spread

$$\mathrm{CS} = y - r = 0.03482 - 0.015 = 0.01982 \approx 198 \text{ bps}$$

$\Rightarrow$ Investors demand 1.98% a year over the risk-free rate for this issuer's default risk.

Interpretation

The promised payment is worth $50e^{-0.06} = 47.09$ risk-free, so the market charges $(47.09 - 43.5)/47.09 = 7.6\%$ of its value for expected default losses. The 198 bps spread is the rate that, added to r, discounts the promised JPY 50.0 billion back to its market value; it is the input a risk-neutral PD or CVA would use. Equity volatility is a red herring here: it calibrates V_0 and σ_V but plays no role once the debt value is known.

Why the other options are wrong

A is incorrect. Used the card's inverted sign: $\frac{1}{4}\ln(0.87) - 0.015 = -0.0348 - 0.015 = -0.0498$. A negative spread on risky debt is the polarity flip.

C is incorrect. Stopped at the intermediate result: 348 bps is the yield y, not the spread; the risk-free rate was not subtracted.

D is incorrect. Did not annualise: $-\ln(0.87) - 0.015 = 0.1393 - 0.015 = 0.1243$. The 13.9% is a four-year cumulative log loss, not a per-annum yield.

LO: CR-12 c: Calculate the probability of default using credit spreads. CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. • **Reference:** Hull, OFOD 11e, Ch.24, 12.1 Special Cases (eqs. 12.5, 12.6, Example 12.2); Hull, RMFI 6e, Ch.17, 9.5, 9.8, Problem 9.17

Solution CR-12 c.5

Answer: D

Bank Q539 • BT notes P2.T6.301.3

Step 1 – Solve the single-period equation for z

$$(1 - \text{PD})(1 + r + z) + \text{PD} \cdot R = 1 + r \quad \Longrightarrow \quad z = \frac{\text{PD}\,(1 + r - R)}{1 - \text{PD}}$$

Step 2 – Substitute

$$z = \frac{0.03 \times (1.02 - 0.40)}{0.97} = \frac{0.0186}{0.97} = 0.019175 \approx 192\ \text{bp}$$

$\Rightarrow$ The investor needs about 1.92% over the risk-free rate to break even on a 3% PD.

Interpretation

Run backwards, the same equation turns a spread into a PD: $\text{PD} = z/(1 + r + z - R) = 0.019175/0.639175 = 3.0\%$. The spread must cover the expected loss of principal, the lost interest r in the default state, and the fact that it is earned only in the 97% survival state. Here recovery applies to principal only; in the zero-coupon formula of Section 3 it applies to the face value.

Why the other options are wrong

A is incorrect. Used the recovery rate instead of LGD: $\text{PD} \times R/(1 - \text{PD}) = 0.012/0.97 = 1.24\%$.

B is incorrect. Expected loss alone, $\text{PD} \times \text{LGD} = 1.80\%$, ignoring that the spread is earned only on survival.

C is incorrect. $\text{PD} \times \text{LGD}/(1 - \text{PD}) = 1.856\%$: adjusts for survival but ignores the lost interest r.

LO: CR-12 c: Calculate the probability of default using credit spreads. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.1; single-period default model (outside source)

Solution CR-12 c.6 **Answer: C**

New • built from GARP Ch.12, 12.1 and Workbook D5

Step 1 – Expected loss rate on the senior bond

$$\text{PD} \times \text{LGD}_{sen} = 1 - \frac{1.03}{1.055} = 0.023697$$

Step 2 – Issuer PD

$$\text{PD} = \frac{0.023697}{0.55} = 0.043085$$

$\Rightarrow$ A 4.31% risk-neutral probability that the issuer defaults within the year.

Step 3 – Expected loss rate on the subordinated bond

$$\text{PD} \times \text{LGD}_{sub} = 0.043085 \times 0.85 = 0.036622$$

Step 4 – Consistent yield

$$\frac{1.03}{1 + y_{sub}} = 1 - 0.036622 \quad \Longrightarrow \quad y_{sub} = \frac{1.03}{0.963378} - 1 = 6.92\%$$

Interpretation

One default probability, two recoveries: the subordinated bond loses 85% instead of 55% in default, so its expected loss rate is $0.85/0.55 = 1.55$ times the senior one, and its yield must rise to match. Spreads price expected loss, not PD.

Why the other options are wrong

A is incorrect. Ignored recovery and gave the subordinated bond the senior yield because the PD is the same.

B is incorrect. Used the shortcut PD $\approx (y - r_f)/(1 - R) = 4.545\%$ and a spread of $4.545\% \times 0.85 = 3.864\%$: $3\% + 3.864\% = 6.86\%$. The question specifies the exact one-period pricing.

D is incorrect. Divided by the recovery rate instead of LGD: PD $= 0.023697/0.45 = 5.266\%$, then $1.03/(1 - 0.05266 \times 0.85) - 1 = 7.83\%$.

LO: CR-12 c: Calculate the probability of default using credit spreads. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.1; Workbook D5 one-period formula (outside source)

CR-12 • Credit Risk • Vardeez: IMPORTANT

CR-12 e

P3 CONCEPT

Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models.

Workbook: none homed here B3 revision notes cover reduced form versus structural

WHAT IS TESTED HERE

Why P3. No workbook shape, but Vardeez flags the objective in a chapter he rates IMPORTANT. **Vardeez:** "LO 30.e here we can consider it relevant for the exam. So read it carefully."

Reported exam shapes: none listed in the map.

What the questions look like: "which statement is correct" items on two things: why default correlation matters (loss distribution, diversification, risk-neutral versus real-world PDs), and the trade-off between reduced form and structural models (mechanism, achievable correlation, speed). Distractors swap the two model families or reverse a property. One GARP-tagged item (AP Mock 1).

1. Why default correlation matters

Default correlation is the tendency of two companies to default at about the same time. GARP gives three reasons it exists: companies in the same industry or region are hit by the same external events; economic conditions make average default rates higher in some years than in others; and a default by one company can cause another to default (**credit contagion**). Its significance:

- **It shapes the portfolio loss distribution.** The expected loss is the sum of the individual expected losses and does not depend on correlation; the tails do (Figure 1). Credit VaR is a tail measure (CR-12 g).
- **Credit risk cannot be fully diversified away.** Adding names reduces idiosyncratic risk only; the common component remains.
- **It is the major reason risk-neutral PDs exceed real-world PDs** (CR-9 j, k): correlated defaults are systematic risk, and bond investors demand a premium for bearing it.
- **It prices multi-name products.** The value of a kth-to-default CDS or a CDO tranche depends critically on default correlation (CR-13 f).

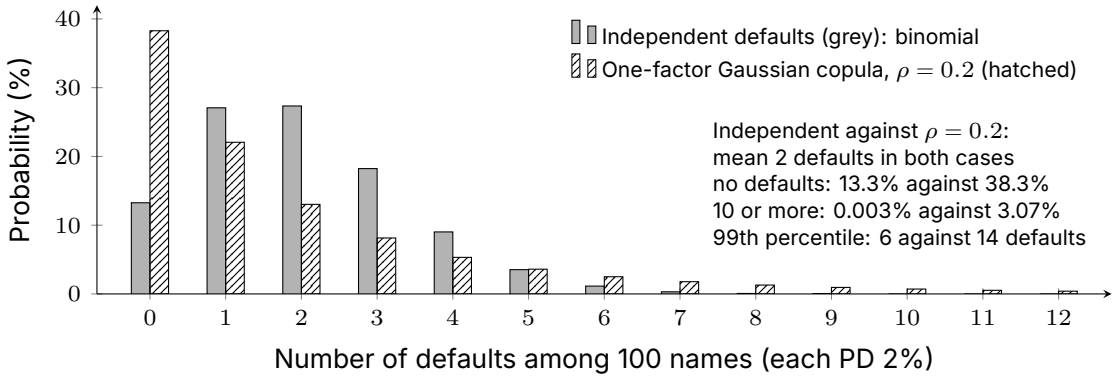


Figure 1. Same expected number of defaults, very different risk. Correlation moves probability into both ends: "no defaults" and "many defaults" both become more likely. The independent case reproduces GARP's 13.9 figures: 86.74% chance of one or more defaults, 0.0034% of ten or more.

2. Reduced form versus structural models

	Reduced form models	Structural models
Mechanism	Hazard rates follow stochastic processes correlated with macroeconomic variables; when A's hazard rate is high, B's tends to be high	Merton-type: a company defaults when its asset value falls below a level; the asset processes of A and B are correlated
Achievable correlation	Limited: even with perfectly correlated hazard rates, the chance of both defaulting in the same short period is usually very low	Can be made as high as desired
Strengths	Mathematically attractive; reflect how economic cycles generate default correlation	Realistic correlation for firms in the same industry and country, or financially dependent firms
Weakness and fix	Range too low; fix by letting hazard rates jump	Computationally slow (Monte Carlo); hard to calibrate as dynamic models

The trade-off runs one way down the table: the model that is easy to compute struggles to produce high correlation; the model that can produce it is slow. Two links to other objectives:

- The practical tool, the **Gaussian copula model of time to default**, "can be shown to be similar to Merton's structural model" (CR-12 f).
- As **dynamic models** of portfolio loss (GARP 13.11, CR-13 h), structural models correlate many asset processes and default each firm at a barrier, so they need Monte Carlo and are hard to calibrate; reduced form models need jumps in the hazard rates to build in a realistic amount of correlation.

Exam traps

- **Mechanism swap.** Correlated hazard rates belong to reduced form models; correlated asset values to structural models.
- **Property reversal.** Structural models allow high correlation but are slow; reduced form models are fast but limited. Vardeez's table says structural models produce "high" correlations: more precisely they *allow* correlation to be set as high as needed.

- **Perfect hazard correlation is not joint default.** Two perfectly correlated hazard rates still rarely produce defaults in the same short period.
- **Expected loss.** Correlation changes the spread and tail of portfolio losses, not their mean.
- **Direction of the risk premium.** Default correlation makes risk-neutral PDs *higher* than real-world PDs, not lower.

Practice questions: CR-12 e

Question CR-12 e.1

Bank Q546 • AnalystPrep Q.6052 (rebuilt)

An investment bank wants to quantify the benefit of diversification in its credit portfolio. Why is estimating default correlation among the obligors important?

A. Because with positive default correlation, defaults of different obligors offset each other, so portfolio credit risk can be diversified away by adding more obligors.

B. Because default correlation matters only for pricing single-name exposures; it has no bearing on the loss distribution of a portfolio.

C. Because default correlation changes the expected loss of each obligor, while the variability of portfolio losses is unaffected.

D. Because default correlation drives the portfolio's loss distribution: positive correlation makes defaults cluster, so credit risk cannot be fully diversified away and tail losses are larger than under independence.

Question CR-12 e.2

New • built from GARP Ch.12, 12.2 (Default Correlation)

A credit strategist at a pension fund's fixed-income team is preparing a briefing on default correlation for the investment committee. Which of the following statements should the strategist include?

A. Credit contagion, where one company's default triggers another's, reduces default correlation, because the surviving companies take over the defaulted company's business.

B. In a reduced form model, perfectly correlated hazard rates for two companies make it highly likely that both default within the same short period.

C. Default correlation is the major reason that risk-neutral default probabilities are greater than real-world default probabilities, because correlated defaults create systematic risk that investors require compensation for.

D. Because default is rare, default correlation can be ignored in a well-diversified portfolio of investment-grade bonds without affecting the tail of the loss distribution.

Question CR-12 e.3

Bank Q543 • AP Mock 1 Q26 (rebuilt)

A risk manager is explaining how reduced-form and structural models generate default correlation between two firms. Which statement is most accurate?

A. Reduced-form models induce default correlation through correlated asset value processes, whereas structural models assume the firms' hazard rates are correlated with macroeconomic variables.

B. Structural models ignore systematic factors, whereas reduced-form models capture common market and sector influences through correlated hazard rates.

C. Reduced-form models assume the hazard rates of the firms follow stochastic processes that are correlated with macroeconomic variables, whereas structural models, similar to Merton's model, induce correlation by assuming the firms' asset values follow correlated stochastic processes.

D. Reduced-form models can be calibrated to achieve any desired level of default correlation, whereas structural models can only produce low default correlations.

Question CR-12 e.4

Bank Q545 • Schweser Module Quiz 30.3 Q1

Which of the following statements is most accurate regarding reduced-form versus structural models used to estimate default correlation?

A. Structural models tend to take a long time to process.

B. Reduced-form models are more computationally intensive.

C. Reduced-form models allow for very high default correlations.

D. Economic cycle impacts on default correlation trends are best reflected in structural models.

Question CR-12 e.5

Bank Q544 • BT P2.T6.24.21.1 (rebuilt)

A regional bank's loan portfolio is exposed to a downturn in the local mining industry. Two borrowers, Alpha Inc. (machinery supplier to miners) and Delta Inc. (logistics provider to miners), are likely to be hit by the same shock at nearly the same time. The risk manager wants a default correlation model that can represent a high probability of near-simultaneous default of these two firms. Which model is most suitable, and why?

A. A reduced-form model, because hazard rates correlated with macroeconomic variables can be calibrated to give as high a default correlation as needed over short horizons.

B. A structural model, because correlation can be introduced through correlated stochastic processes for the firms' assets and can be set as high as needed, at the cost of longer computation time.

C. A reduced-form model, because it is mathematically simple and the range of default correlations it can produce is wide.

D. A structural model, because it best reflects how default correlation trends move with the economic cycle and is quick to compute.

Question CR-12 e.6 Bank Q597 • Schweser Module Quiz 31.3 Q3 (rebuilt; re-filed from CR-13 h)

A structured credit quant is reviewing alternatives to the one-factor Gaussian copula for modelling default correlation in a CDO. Which of the following correctly classifies the structural model of credit risk and states its main drawback?

A. A heterogeneous model: it allows each reference entity its own time-to-default distribution, so the binomial formula for the number of defaults no longer applies.

B. A dynamic model: the asset values of many companies follow correlated processes and each company defaults when its asset value reaches a barrier, so it must be run by Monte Carlo simulation and is difficult to calibrate.

C. An implied copula model: a probability distribution for the average hazard rate of the portfolio is implied from tranche prices, so it cannot be used without market quotes.

D. A random recovery model: recovery rates are negatively related to default rates, so the copula correlation rises as the common factor falls.

Solutions • CR-12 e

Solution CR-12 e.1 — Answer: D

Bank Q546 • AnalystPrep Q.6052

Step 1 – What default correlation does

Default correlation is the tendency of companies to default at about the same time (common industry or regional shocks, economic conditions, contagion).

Step 2 – Effect on the portfolio

Higher correlation makes defaults cluster: some years see many, others few. The loss distribution widens and its tail thickens, while the expected loss is unchanged (Figure 1).

$\Rightarrow$ Credit risk cannot be fully diversified away, so the correlation must be estimated to measure portfolio risk.

Interpretation

This is why the same 100-name portfolio can have a 99th percentile of 6 defaults or 14, depending only on correlation. Diversification benefit is exactly what the correlation estimate measures.

Why the other options are wrong

A is incorrect. Polarity flip: positive correlation makes defaults occur together rather than offset, so diversification is limited.

B is incorrect. Scope error: correlation is irrelevant for one name but central to the loss distribution of a portfolio.

C is incorrect. Reversed: the expected loss is the sum of individual expected losses and does not depend on correlation; it is the variability and tail of losses that do.

LO: CR-12 e: Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2 Default Correlation

Solution CR-12 e.2 — Answer: C

New • built from GARP Ch.12, 12.2

Step 1 – Default correlation and risk-neutral PDs

GARP: default correlation "means that credit risk cannot be completely diversified away and is the major reason why risk-neutral default probabilities are greater than real-world default probabilities".

$\Rightarrow$ C is correct: correlated defaults are systematic risk, priced into spreads.

Step 2 – Check the other claims

Contagion is one of GARP's three *sources* of default correlation. In reduced form models, "even when there is perfect correlation between the hazard rates of the two companies, the probability that they will both default during the same short period of time is usually very low".

Interpretation

The gap between spread-implied and historical PDs (CR-9 j) is largest in relative terms for investment-grade names, exactly where correlated, systematic default risk dominates idiosyncratic risk.

Why the other options are wrong

A is incorrect. Polarity flip: contagion, one default causing another, is a reason default correlation exists; it increases it.

B is incorrect. True fact reversed: perfect hazard rate correlation still leaves joint default in a

short period unlikely, which is the main weakness of reduced form models.
D is incorrect. Absolute claim: rarity does not make correlation negligible. Correlation drives the tail of the loss distribution, and diversification cannot remove the common component.

LO: CR-12 e: Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2 Default Correlation

Solution CR-12 e.3 — Answer: C

Bank Q543 • AP Mock 1 Q26

Step 1 – Reduced form mechanism

Hazard rates of different companies follow stochastic processes correlated with macroeconomic variables; a high hazard rate for A tends to coincide with a high hazard rate for B.

Step 2 – Structural mechanism

Merton-type: default occurs when asset value falls below a threshold; correlation comes from assuming correlated stochastic processes for the firms' assets.
$\Rightarrow$ Option C states both mechanisms correctly.

Interpretation

Each family generates correlation through a different channel: correlated default intensities (reduced form) or correlated asset values (structural). The channel explains the properties: intensities rarely produce simultaneous defaults; asset values can be made as correlated as needed.

Why the other options are wrong

A is incorrect. Swaps the mechanisms: asset-value correlation belongs to structural models, macro-correlated hazard rates to reduced form models.
B is incorrect. Structural models do include systematic factors, through the correlated asset processes.
D is incorrect. Reversed: structural models allow correlation to be set as high as needed; reduced form models achieve only a low range.

LO: CR-12 e: Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2 Default Correlation

Solution CR-12 e.4 — Answer: A

Bank Q545 • Schweser Module Quiz 30.3 Q1

Step 1 – Computation

Structural models simulate correlated asset processes, so they are "liable to be computationally quite slow". Reduced form models are mathematically attractive and fast.

Step 2 – Correlation range and the cycle

Reduced form models give only a low range of default correlations but reflect the tendency of economic cycles to generate correlation; structural models can reach any correlation.
$\Rightarrow$ Only A is accurate.

Interpretation

Speed against realism is the whole comparison: memorise one strength and one weakness for each family.

Why the other options are wrong

B is incorrect. Reversed: reduced form models are the simpler, faster ones.

C is incorrect. Reversed: reduced form models achieve only low default correlations.
D is incorrect. Reversed: reflecting economic-cycle effects is a strength of reduced form models, whose hazard rates are tied to macroeconomic variables.

LO: CR-12 e: Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2; SchweserNotes 2026 Book 2, Reading 30, LO 30.e

Solution CR-12 e.5 Answer: B

Bank Q544 • BT P2.T6.24.21.1

Step 1 – What the case needs

Two firms exposed to the same industry shock need a model that can deliver a high probability of near-simultaneous default, that is, high default correlation over a short horizon.

Step 2 – Match the model

Reduced form models link hazard rates to macroeconomic variables, but joint default in a short period stays unlikely even with perfectly correlated hazard rates. Structural models correlate the asset processes and can set the correlation as high as needed; the price is computation time.
$\Rightarrow$ A structural model fits.

Interpretation

GARP names this exact situation, companies in the same industry and country, or financially dependent on each other, as the case where "a relatively high default correlation may be warranted".

Why the other options are wrong

A is incorrect. Reduced form models achieve only a low range of default correlations over short periods, whatever the macro link.
C is incorrect. Right that they are mathematically simple, wrong that the correlation range is wide.
D is incorrect. Structural models are slow, and the economic-cycle effect is a strength of reduced form models.

LO: CR-12 e: Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.2 Default Correlation

Solution CR-12 e.6 Answer: B

Bank Q597 • Schweser Module Quiz 31.3 Q3

Step 1 – Where the structural model sits

GARP 13.11 lists dynamic models, which model the evolution of portfolio loss through time, in three types: structural, reduced form and top down.

Step 2 – How it works and its drawback

In the structural version the asset prices of many companies follow correlated processes and a company defaults when its asset price reaches a barrier. These models "have to be implemented with Monte Carlo simulation and calibration is therefore difficult".
$\Rightarrow$ The structural model is a dynamic model, with the computational burden that CR-12 e attributes to structural models.

Interpretation

The same split runs through the dynamic models: the reduced form version must add jumps in

hazard rates to reach realistic correlation, the structural version reaches it but is slow and hard to calibrate.

Why the other options are wrong

A is incorrect. True description of the heterogeneous model, the wrong model: it relaxes the homogeneity of the Gaussian copula and says nothing about asset values.

C is incorrect. True description of the implied copula model, the wrong model.

D is incorrect. Describes the Andersen and Sidenius random recovery and random factor loading model, the wrong model.

LO: CR-12 e: Describe the significance of estimating default correlation for credit portfolios and distinguish between reduced form and structural default correlation models. CR-13 h: Discuss alternative approaches used to estimate default correlation. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.11 Dynamic Models; Ch.24 (GARP CR Ch.12), 12.2

CR-12 • Credit Risk • Vardeez: IMPORTANT

CR-12 g

P3 CALC

Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach.

Workbook: D20 Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D20 (Map #41, 2 of 16 sources, notes only) asks you to know CreditMetrics as the ratings-migration, mark-to-market model. Vardeez flags credit VaR as important. TYPE is CALC by an approved judgment call: the Gaussian copula worst case default rate gives a short calculation.
Vardeez: "LO30.g: This LO which talks abt Credit Var is imp for exam."

Reported exam shapes

- **D20·S2** CreditMetrics as the ratings-migration, mark-to-market model (Question CR-12 g.6).

What the questions look like: credit VaR $= L(1-R)\times$WCDR, with or without the WCDR computed for you; what moves the WCDR; CreditMetrics thresholds from a transition matrix; and statements contrasting the two approaches. One GARP-tagged item (GARP PE2).

Worked elsewhere: B1.1 (credit VaR as a quantile minus EL, Malz) at Question CR-10 b.8; A31.1 and A31.2 (WCDR and IRB capital) at Questions CR-10 d.8 and CR-10 d.9; the four-model comparison (Vasicek, CreditRisk+, CreditMetrics) under CR-10 e. The conditional PD behind the WCDR is derived under CR-12 f.

1. Credit VaR from the one-factor Gaussian copula

Credit VaR is defined like market VaR: the credit loss over a horizon that will not be exceeded at a confidence level (typically one year and 99.9%). For a very large portfolio of similar loans with a common PD $Q(T)$ and copula correlation ρ, the fraction defaulting given the factor is $Q(T \mid F)$ (CR-12 f). We are $X\%$ certain that $F > -N^{-1}(X)$, so we are $X\%$ certain the default rate will be below Vasicek's **worst case default rate** (GARP eq. 12.10):

$$V(X,T) = N\left(\frac{N^{-1}[Q(T)] + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right) \qquad \boxed{\text{Credit VaR} \approx L\,(1-R)\,V(X,T)}$$

L is the size of the portfolio and R the recovery rate. A loan of size L_i contributes $L_i(1-R)V(X,T)$.

GARP Example 12.5 (Vardeez's worked example). \$100 million of retail exposures, one-year PD 2%, recovery 60%, $\rho = 0.1$:

$$V(0.999,1) = N\left(\frac{-2.0537 + 0.3162 \times 3.0902}{\sqrt{0.9}}\right) = N\left(\frac{-1.0766}{0.9487}\right) = N(-1.1348) = 12.82\%$$

$$\text{Credit VaR} = 100 \times 0.1282 \times (1-0.6) = \$5.13 \text{ million}$$

(Rounding the WCDR to 0.128 first gives \$5.12 million.)

- **What moves the WCDR:** a higher ρ (0.2 gives 22.6%), a higher confidence level (99% gives only 8.2%) or a higher PD raise it. With $\rho = 0$ the large portfolio's default rate is simply the PD, 2%. Recovery changes the credit VaR, not the WCDR.
- **Which definition.** Hull's $L(1-R)V(X,T)$ is a loss quantile with no expected loss deducted. Malz's portfolio credit VaR (CR-11) and Basel capital (CR-10 d) subtract the expected loss, here $100 \times 0.02 \times 0.4 = \$0.8$ million. Follow the stem.

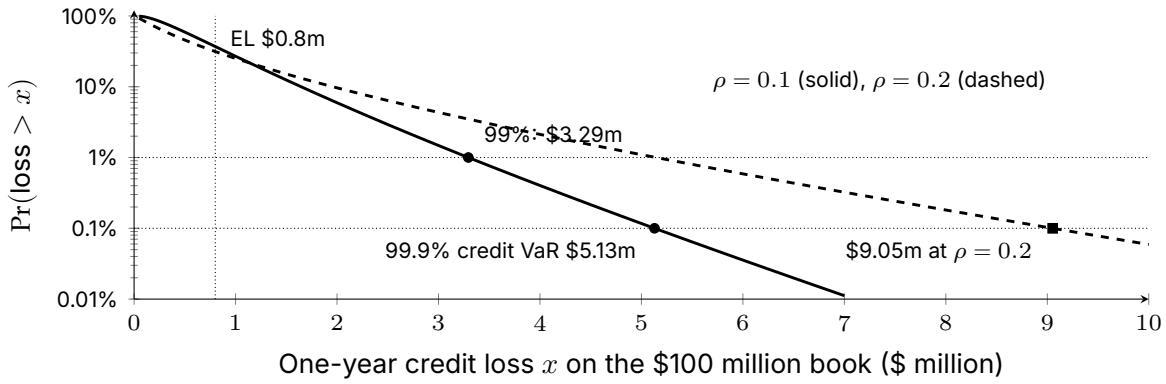


Figure 1. Tail of the loss distribution for Example 12.5 ($L = \$100$m, PD 2%, $R = 60\%$), log scale. Credit VaR at X is where the curve crosses $1 - X$ (dotted lines). Doubling the copula correlation leaves the expected loss at \$0.8 million but moves the 99.9% loss from \$5.13 million to \$9.05 million.

2. Credit VaR with CreditMetrics

CreditMetrics (JPMorgan) estimates the loss distribution by **Monte Carlo simulation of the rating changes of all counterparties**. On each trial: sample each counterparty's end-of-year rating, including default; revalue the outstanding contracts; total the credit losses. Many trials give a loss distribution, from which the VaR is read.

- **Losses from downgrades as well as defaults**, with upgrades counted as negative losses: a mark-to-market model, not a default-mode one. Credit mitigation clauses can be approximately incorporated. The cost: it is computationally time intensive.
- **Correlated rating changes.** Rating changes are not independent: a Gaussian copula joins them, with the copula correlation usually set to the correlation between the companies' equity returns.
- **Thresholds from the transition matrix.** With GARP Table 12.4 and equity correlation 0.2, sample correlated x_A, x_B. The AAA company stays AAA if $x_A < N^{-1}(0.8983) = 1.2719$, becomes AA if $1.2719 \le x_A < N^{-1}(0.9920) = 2.4089$, A if $2.4089 \le x_A < N^{-1}(0.9975) = 2.8070$, and so on. The BBB company becomes AAA if $x_B < N^{-1}(0.0001) = -3.7190$, AA if $-3.7190 \le x_B < N^{-1}(0.0011) = -3.0618$, A if $-3.0618 \le x_B < N^{-1}(0.0370) = -1.7866$, ..., and defaults if $x_B > N^{-1}(0.9983) = 2.9290$.

Correction to the source notes. GARP's CreditMetrics illustration (Ch.12, 12.3) prints the A band for the BBB company as "$-1.7866 \le x_B < -3.0618$". The bounds are reversed: no number satisfies that. From the cumulative probabilities $N^{-1}(0.0011) = -3.0618$ and $N^{-1}(0.0370) = -1.7866$, the BBB company moves to A when $-3.0618 \le x_B < -1.7866$ (as Question CR-12 g.5 uses).

Formula card: CR-12 g

Worst case default rate	$V(X,T) = N\left(\frac{N^{-1}[Q(T)] + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right)$
Key quantiles	$N^{-1}(0.999) = 3.0902, \quad N^{-1}(0.99) = 2.3263$
Credit VaR (Hull)	$L(1-R)V(X,T)$; loan contribution $L_i(1-R)V(X,T)$
Expected loss	$L(1-R)Q(T)$ (subtract only if the stem defines credit VaR as unexpected loss)
CreditMetrics thresholds	N^{-1} of cumulative transition probabilities along the rating row

	Gaussian copula (Vasicek)	CreditMetrics
Method	Closed form for a large homogeneous portfolio	Monte Carlo simulation of rating changes and revaluation
Losses counted	Defaults only	Defaults and downgrades (upgrades as gains)
Correlation	One copula correlation ρ	Gaussian copula on rating changes, ρ from equity returns
Inputs	PD, ρ, R, L	Transition matrix, spread curves by rating, recoveries, correlations
Cost	A one-line calculation	Computationally intensive

Exam traps

- **Recovery for LGD.** Credit VaR multiplies by $1 - R$, not R.
- **Average PD for the worst case.** $L(1-R)Q(T)$ is the expected loss; the VaR needs $V(X,T)$.
- **Expected loss deducted (or not).** Hull's formula is a loss quantile; subtract EL only when the question defines credit VaR as unexpected loss.
- **Wrong quantile.** $N^{-1}(0.99) = 2.326$ for a 99.9% VaR, or a missing square root on ρ.
- **Sibling swap of models.** CreditMetrics captures downgrades; the copula formula captures defaults only. CreditMetrics correlates rating changes; it does not sample them independently.
- **Threshold bands.** Read cumulative probabilities along the row; a sample falls in the band between successive N^{-1} values.

Practice questions: CR-12 g

Question CR-12 g.1

Bank Q555 • Schweser Module Quiz 30.3 Q3

Assuming a loan portfolio of L, a recovery rate of RR, and the percentage of losses on a portfolio less than $V(T, X)$, which of the following formulas is used to estimate credit VaR?

A. $L \times RR \times V(T, X)$

B. $L \times (1 - RR)/V(T, X)$

C. $V(T, X)/[L \times (1 - RR)]$

D. $L \times (1 - RR) \times V(T, X)$

Question CR-12 g.2

Bank Q554 • GARP PE2 Q76 (rebuilt)

A bank has a retail loan exposure of GBP 85 million. The 1-year average probability of default is 2.50% and the average recovery rate is 72%. Using the one-factor Gaussian copula approach with a copula correlation of 0.06, the 99.9% worst-case default rate over one year is approximately 10.73% (10.7337% unrounded). What is the 99.9% credit VaR of the exposure?

A. GBP 0.595 million

B. GBP 1.960 million

C. GBP 2.555 million

D. GBP 6.569 million

Question CR-12 g.3

New • built from GARP Ch.12, 12.3 (eq. 12.10, Example 12.5)

A risk manager at a South African bank holds a ZAR 2,400 million portfolio of 8,000 similar small-business loans. The one-year probability of default averages 1.5%, the recovery rate averages 35%, and the copula correlation, estimated from the equity returns of listed peers, is 0.12. The CRO asks for the one-year 99.9% credit VaR, defined as the credit loss that the bank is 99.9% confident will not be exceeded, using the one-factor Gaussian copula. Which of the following is closest?

A. ZAR 101.3 million

B. ZAR 113.8 million

C. ZAR 164.7 million

D. ZAR 188.1 million

Question CR-12 g.4

New • built from GARP Ch.12, 12.3 (Example 12.5)

A credit risk analyst reproduces GARP's Example 12.5: a USD 100 million retail portfolio, one-year PD 2%, recovery 60% and copula correlation 0.1, which gives a 99.9% worst case default rate (WCDR) of 12.8% and a credit VaR of USD 5.13 million. She then tests the sensitivity of the WCDR, changing one input at a time. Which of the following statements is correct?

A. Raising the copula correlation from 0.1 to 0.2 lowers the WCDR, because more correlated loans behave like fewer independent loans.

B. Lowering the confidence level from 99.9% to 99% lowers the WCDR to about 8.2%.

C. Raising the recovery rate from 60% to 70% lowers the WCDR, because each default then costs less.

D. With a copula correlation of zero, the WCDR would still exceed the 2% PD, because the 99.9% confidence level always adds a tail term.

Question CR-12 g.5

Bank Q553 • BT P2.T6.24.21.3 (rebuilt)

In a CreditMetrics simulation, a AAA company (A) and a BBB company (B) have an equity return correlation of 0.2, which is used as the Gaussian copula correlation between their standard normal samples x_A and x_B. The 1-year thresholds implied by the transition matrix are:

Company	Sample range	Rating after 1 year
A (AAA)	$x_A < 1.2719$	AAA
A (AAA)	$1.2719 \le x_A < 2.4089$	AA
A (AAA)	$2.4089 \le x_A < 2.8070$	A
B (BBB)	$x_B < -3.7190$	AAA
B (BBB)	$-3.7190 \le x_B < -3.0618$	AA
B (BBB)	$-3.0618 \le x_B < -1.7866$	A

On one simulation trial the correlated samples are $x_A = 2.50$ and $x_B = -3.40$. What are the simulated ratings of the two companies?

A. A: AA; B: AA

B. A: A; B: AAA

C. A: A; B: AA

D. A: A; B: A

Question CR-12 g.6 Bank Q552 • AnalystPrep Q.6057 (rebuilt) • shape D20·S2

A risk analyst compares two ways to estimate the 1-year, 99.9% credit VaR of a bank's loan book: (i) the one-factor Gaussian copula approach applied to a large portfolio of similar loans, and (ii) the CreditMetrics approach. Which statement is most accurate?

A. Approach (i) converts the probability of default and the copula correlation into a 99.9% worst-case default rate, which is multiplied by $L(1 - RR)$; approach (ii) simulates rating changes, so its losses include downgrades as well as defaults.

B. Approach (ii) counts losses only from defaults, whereas approach (i) also captures losses from rating downgrades.

C. Approach (i) multiplies the unconditional default probability by $L(1 - RR)$, so the copula correlation does not affect the credit VaR.

D. Approach (ii) samples the rating changes of different counterparties independently, which is why it is computationally quick.

Solutions • CR-12 g

Solution CR-12 g.1 — Answer: D

Bank Q555 • Schweser Module Quiz 30.3 Q3

Step 1 – Worst-case fraction defaulting

$V(T, X)$ is the default rate on a large portfolio that is not exceeded with confidence X over T years.

Step 2 – Loss per defaulted unit

Each defaulted unit of exposure loses $1 - RR$.

$\Rightarrow$ Credit VaR $= L \times (1 - RR) \times V(T, X)$.

Interpretation

Exposure, times the stressed default rate, times loss given default: the same three factors as expected loss, with the average PD replaced by the worst case default rate.

Why the other options are wrong

A is incorrect. Uses the recovery rate instead of loss given default, so it measures amounts recovered rather than lost.

B is incorrect. Divides by $V(T, X)$; the default rate must multiply the exposure.

C is incorrect. Inverts the formula; the result is a ratio, not a monetary loss.

LO: CR-12 g: Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.3 Credit VaR

Solution CR-12 g.2 — Answer: C

Bank Q554 • GARP PE2 Q76

Step 1 – Check the worst case default rate

$$V = N\left(\frac{N^{-1}(0.025) + \sqrt{0.06}\, N^{-1}(0.999)}{\sqrt{0.94}}\right) = N\left(\frac{-1.9600 + 0.2449 \times 3.0902}{0.9695}\right)$$

$$V = N(-1.2408) = 10.7337\%$$

Step 2 – Loss given default

$1 - RR = 1 - 0.72 = 0.28$.

Step 3 – Credit VaR

$$85 \times 0.28 \times 0.107337 = \text{GBP } 2.555 \text{ million}$$

$\Rightarrow$ The bank is 99.9% confident that one-year credit losses will not exceed about GBP 2.56 million.

Interpretation

The worst case default rate already embeds the correlation; the high 72% recovery shrinks the loss per default to 28% of exposure.

Why the other options are wrong

A is incorrect. Expected loss: $85 \times 0.025 \times 0.28 = 0.595$, the average PD in place of the worst case.

B is incorrect. Subtracted the expected default rate: $85 \times 0.28 \times (0.107337 - 0.025) = 1.960$. GARP's estimate is the loss quantile, with no expected loss deducted.

D is incorrect. Used the recovery rate instead of LGD: $85 \times 0.107337 \times 0.72 = 6.569$.

LO: CR-12 g: Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach. • **Reference:** Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.3, eq. 12.10

Solution CR-12 g.3 — Answer: D

New • built from GARP Ch.12, 12.3

Step 1 – Inputs to the worst case default rate

$N^{-1}(0.015) = -2.1701$, $\sqrt{0.12} = 0.3464$, $N^{-1}(0.999) = 3.0902$, $\sqrt{0.88} = 0.9381$.

Step 2 – Worst case default rate

$$V(0.999, 1) = N\left(\frac{-2.1701 + 0.3464 \times 3.0902}{0.9381}\right) = N\left(\frac{-1.0996}{0.9381}\right) = N(-1.1722) = 12.06\%$$

$\Rightarrow$ In the 1-in-1,000 year, about 12% of the loans default, eight times the average.

Step 3 – Credit VaR

$$2{,}400 \times (1 - 0.35) \times 0.12056 = \text{ZAR } 188.1 \text{ million}$$

Interpretation

The number of loans is a red herring beyond confirming that the portfolio is large enough for the Vasicek approximation. The stem defines credit VaR as a loss quantile, so the expected loss of ZAR 23.4 million is not deducted.

Why the other options are wrong

A is incorrect. Used the recovery rate instead of LGD: $2{,}400 \times 0.35 \times 0.12056 = 101.3$.
B is incorrect. Used the 99% quantile 2.326: $V = N(-1.4543) = 7.29\%$, so $2{,}400 \times 0.65 \times 0.0729 = 113.8$.
C is incorrect. Deducted the expected loss: $2{,}400 \times 0.65 \times (0.12056 - 0.015) = 164.7$, which is unexpected loss, not the quantile the CRO defined.

LO: CR-12 g: Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach. • Reference: Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.3, eq. 12.10, Example 12.5

Solution CR-12 g.4 — Answer: B

New • built from GARP Ch.12, 12.3

Step 1 – Confidence level

At 99%, $N^{-1}(0.99) = 2.3263$: $V = N((-2.0537 + 0.3162 \times 2.3263)/0.9487) = N(-1.3893) = 8.24\%$.

$\Rightarrow$ Lower confidence, smaller worst case: B is correct.

Step 2 – Correlation and the zero-correlation limit

With $\rho = 0.2$: $V = N((-2.0537 + 0.4472 \times 3.0902)/0.8944) = N(-0.7510) = 22.6\%$. With $\rho = 0$: $V = N(N^{-1}(0.02)) = 2\%$.

Interpretation

Only Q, ρ and X enter the WCDR. Recovery and portfolio size scale the credit VaR afterwards. Correlation is what turns a 2% average into a double-digit worst case: with independent defaults a large portfolio defaults at its average rate almost surely.

Why the other options are wrong

A is incorrect. Polarity flip: higher correlation raises the WCDR, to 22.6%; clustering makes the bad years worse.

C is incorrect. Sibling swap: recovery lowers the credit VaR (to USD 3.85 million) but leaves the WCDR at 12.8%.
D is incorrect. With $\rho = 0$ the tail term vanishes and the WCDR equals the PD, 2%: the law of large numbers removes all uncertainty about the default rate.

LO: CR-12 g: Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach. •
Reference: Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.3, eq. 12.10, Example 12.5

Solution CR-12 g.5 — Answer: C

Bank Q553 • BT P2.T6.24.21.3

Step 1 – Where the thresholds come from

They are N^{-1} of cumulative transition probabilities, e.g. AAA row: $N^{-1}(0.8983) = 1.2719$, $N^{-1}(0.8983 + 0.0937) = 2.4089$, $N^{-1}(0.8983 + 0.0937 + 0.0055) = 2.8070$.

Step 2 – Company A

$2.4089 \leq 2.50 < 2.8070$: downgraded from AAA to A.

Step 3 – Company B

$-3.7190 \leq -3.40 < -3.0618$: upgraded from BBB to AA.
$\Rightarrow$ A becomes A; B becomes AA.

Interpretation

The 0.2 correlation affects how the pair (x_A, x_B) is drawn; once drawn, each rating follows from its own thresholds. Revaluing all positions on each trial and repeating builds the loss distribution; B's upgrade counts as a negative loss.

Why the other options are wrong

A is incorrect. Placed 2.50 in the AA band; 2.50 exceeds 2.4089.
B is incorrect. AAA for company B needs $x_B < -3.7190$; -3.40 is above it.
D is incorrect. The A band for B starts at -3.0618; -3.40 is lower, so B is in the AA band.

LO: CR-12 g: Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach. •
Reference: Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.3 CreditMetrics, Table 12.4

Solution CR-12 g.6 — Answer: A

Bank Q552 • AnalystPrep Q.6057

Step 1 – The Gaussian copula approach

For a large homogeneous portfolio, $V(T, X)$ converts the PD and the copula correlation into a worst case default rate; credit VaR $= L(1 - RR)V(T, X)$. Both inputs drive the answer.

Step 2 – CreditMetrics

Monte Carlo simulation of rating changes for all counterparties, correlated through a Gaussian copula; revaluing after each trial gives a loss distribution that includes downgrades (upgrades count as negative losses).
$\Rightarrow$ Only A describes both correctly.

Interpretation

This is the D20 distinction: CreditMetrics is the ratings-migration, mark-to-market model; the copula formula (Vasicek) counts defaults only. The price of CreditMetrics' richer loss definition is heavy computation.

Why the other options are wrong

B is incorrect. Reverses the approaches: CreditMetrics captures downgrades; the copula formula counts defaults only.

C is incorrect. Ignores correlation: $V(T, X)$ rises with ρ and exceeds $Q(T)$ at high confidence, so the unconditional PD understates the VaR.
D is incorrect. CreditMetrics correlates rating changes with a Gaussian copula (correlation from equity returns) and is computationally intensive.

LO: CR-12 g: Describe how to estimate credit VaR using the Gaussian copula and the CreditMetrics approach. •
Reference: Hull, OFOD 11e, Ch.24 (GARP CR Ch.12), 12.3 Credit VaR and CreditMetrics

CR-13 • Credit Derivatives • Vardeez: MODERATE

CR-13 a

P3 CONCEPT

Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO).

Workbook: D14 Credit Derivatives and Credit-Linked Notes • Tier D • Map #58 (2/16)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D14 (Map #58, 2 of 16 sources) asks you to know the credit-linked note structure, total return swaps and the CDS mechanics. This objective is the shared background for the whole reading.
Vardeez: a MODERATE chapter; for this objective his notes describe only the CDS (mechanics at CR-9 g). The total return swap and the CDO are gap-filled from GARP.

Reported exam shapes

- **D14·S1** CLN structure: funded, the note investor bears the credit risk.
- **D14·S2** Total return swaps: who owns the bond, who earns the spread.
- **D14·S3** CDS mechanics: obligations of buyer and seller, settlement.

What the questions look like: "which statement is correct" items on who pays whom, who bears the credit risk and what happens at a credit event or at maturity; and which position gains from a credit view. Amounts appear inside statements (a CDS payoff, a TRS settlement), never as a free calculation.

Worked elsewhere: D14.1 (four instruments, including the CLN issued through an SPV) at Question CR-9 g.7, which also covers CDS mechanics in full; synthetic CDO valuation under CR-13 f; compound and base correlation under CR-13 g.

1. Credit default swaps (shape S3)

A **credit derivative** is a contract whose payoff depends on the creditworthiness of one or more companies or countries. The most popular is the **credit default swap**: protection against a **credit event** (bankruptcy, failure to pay, restructuring) by a **reference entity**.

- **Buyer's obligation.** Pay the **CDS spread** (a percentage of notional per year), usually **quarterly in arrears**, until maturity or a credit event. GARP: 90 bp on $100 million is $225,000 a quarter.
- **Seller's obligation.** On a credit event, pay $L(1-R)$. Under **physical settlement** the buyer sells bonds with face value L to the seller for L. Under **cash settlement**, now usual, an ISDA auction sets the mid-market value of the cheapest deliverable bond: at an auction price of 35 per 100 the payoff is $65 million.
- **Final accrual.** Because premiums are paid in arrears, the buyer owes the premium accrued since the last payment date: in GARP's example, a credit event on 20 May after a quarterly payment on 20 March costs the buyer two months of the annual 90 bp, about $150,000.
- **Not insurance.** The buyer need not own the reference entity's bonds, and the payoff comes only on a credit event, not on a fall in price.
- **Positions.** Buying protection is a short credit position (it gains when spreads widen). Selling protection is like owning the bond: GARP notes that a long bond has a similar risk to a short CDS position on its issuer.

Correction to the source notes. SchweserNotes 2026 Book 2, LO 27.g summary, says the CDS buyer makes "fixed quarterly payments (in advance)". GARP (Ch.13, 13.1) says payments are "usually made quarterly in arrears", which is why a final accrual payment is due after a credit event; Schweser's own Reading 31 also says arrears. Use **in arrears**.

2. Total return swaps (shape S2)

A **total return swap** exchanges the *total* return on a bond (or any portfolio of assets), meaning coupons, interest and the gain or loss in value, for a floating rate plus a spread. GARP's example: 5 years, $100 million notional, floating plus 25 bp.

- The **total return payer** pays the coupons on $100 million of the bond; the **receiver** pays floating plus 25 bp on $100 million.
- At the end, a 10% rise in the bond means the payer pays $10 million; a 15% fall means the receiver pays $15 million. On a default the swap usually ends and the receiver pays $100 million minus the bond's market value.
- **Financing use.** The receiver wants to invest in the bond; the payer buys it and **keeps ownership**. The receiver is in the position of having borrowed at floating plus 25 bp; the bond acts as collateral, as in a repo.
- **Who earns the spread.** The payer, as compensation for the risk that the receiver defaults when the bond has fallen. It depends on the credit quality of the receiver, of the bond issuer, and their correlation.

If the payer owns the bond, the swap passes the bond's credit risk to the receiver; if it does not, the swap is a short position in the bond.

3. Credit-linked notes (shape S1)

A **credit-linked note** is a **funded** credit derivative. The investor pays the note's principal at issue and earns a coupon that includes a credit premium; at maturity the principal comes back less any credit loss on the reference entity. The investor is therefore the **protection seller**, and the issuer, the protection buyer, already holds the cash (mechanics from the workbook; outside the GARP excerpts). Who else's credit the investor carries depends on the issuer (GARP Ch.23, Choudhry): a CLN issued directly by a bank also carries that bank's credit risk; a CLN issued through an SPV has no credit linkage to the sponsor, only to the assets sold to the SPV.

4. Collateralized debt obligations

An asset-backed security whose underlying assets are bonds is a **CDO**. A waterfall allocates interest and principal so that a more senior tranche is more likely to be paid. Built from bonds that the tranche holders fund, it is a **cash CDO**.

A **synthetic CDO** replaces the bonds by CDSs (a long bond resembles a short CDS). The originator sells CDS protection on a portfolio of companies; the CDO principal is the total CDS notional. Inflows are the CDS spreads, outflows the default payouts, and both are passed to tranches. GARP's example on $100 million:

Tranche	Losses absorbed	Principal	Spread on outstanding principal
Equity	0 to 5%	$5 million	1,000 bp
Mezzanine	5 to 20%	$15 million	100 bp
Senior	above 20%	$80 million	10 bp

Payouts of \$2 million cut the equity principal to \$3 million, on which it then earns 1,000 bp. A further \$4 million wipes out the equity tranche and cuts the mezzanine principal to \$14 million. Synthetic tranche holders make no initial investment but post the tranche principal as collateral. Tranches on the index portfolios trade on their own (**single-tranche trading**): iTraxx Europe 0 to 3, 3 to 6, 6 to 9, 9 to 12, 12 to 22 and 22 to 100%; CDX NA IG 0 to 3, 3 to 7, 7 to 10, 10 to 15, 15 to 30 and 30 to 100%.

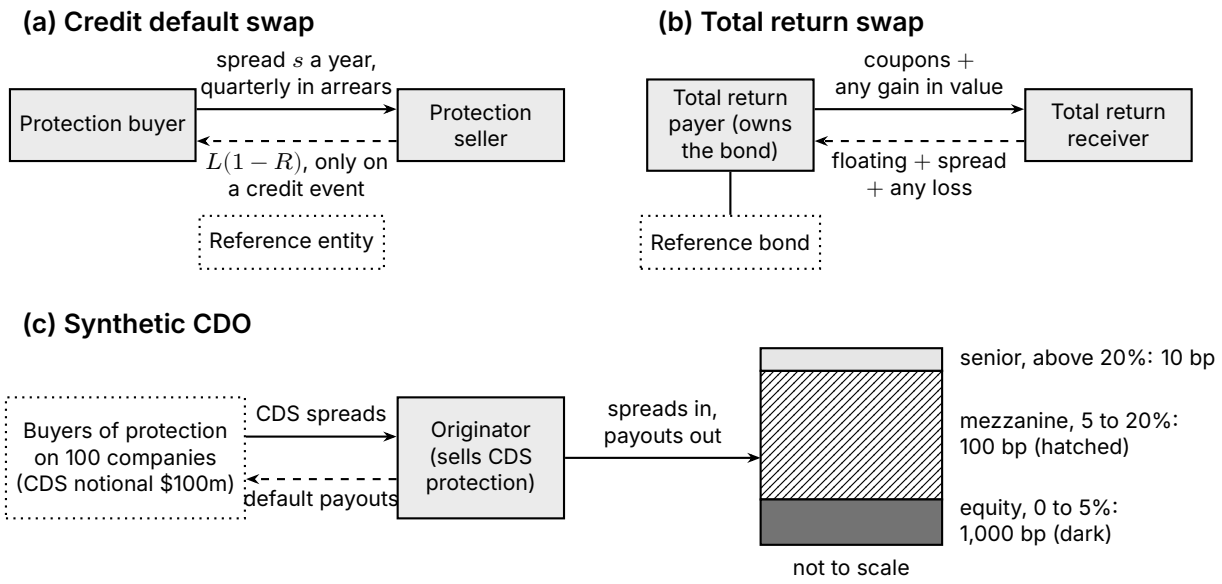


Figure 1. Solid arrows are regular payments, dashed arrows payments that happen only on a credit event or a loss. (a) The CDS seller pays only after a credit event of the reference entity (dotted box), which is not a party to the contract. (b) The TRS payer keeps the bond and passes its total return; the receiver pays floating plus spread and bears any loss. (c) GARP's synthetic CDO: the originator's CDS income and payouts are passed to three tranches, and losses are absorbed from the equity tranche (dark) upwards through the mezzanine (hatched) to the senior tranche (light).

	Funded?	Who bears the reference credit risk	Payments, and what happens on a credit event
CDS	No	The protection seller	Buyer pays the spread quarterly in arrears; seller pays $L(1-R)$, physically or in cash
Total return swap	No	The receiver (with the market risk)	Payer passes coupons and gains; receiver pays floating plus spread and losses, and on default the notional minus market value
Credit-linked note	Yes	The note investor (also the issuer, if issued directly)	Investor pays principal upfront, earns a coupon with a credit premium, is repaid less the credit loss
Synthetic CDO tranche	No (collateral posted)	The tranche holder, in order of seniority	Tranche earns its spread on outstanding principal; payouts in its range write the principal down

Exam traps

- **CDS treated as insurance.** The buyer need not own the bonds, and a price fall without a credit event pays nothing.
- **Payoff.** $L(1-R)$, not $L \times R$; physical settlement pays L against delivery of bonds worth LR.
- **Timing.** Premiums are paid in arrears (not in advance), so a final accrual payment follows a credit event.
- **TRS roles.** The payer owns the bond and receives floating plus spread; the receiver bears the whole return, losses included.
- **Funded versus unfunded.** The CLN investor pays cash upfront and is the protection seller; calling the investor the protection buyer is a role flip.
- **Hedge direction.** Buying a CDO tranche, selling protection and receiving total return all *add* credit risk.

Practice questions: CR-13 a

Question CR-13 a.1

Bank Q557 • AnalystPrep Q.6061 (rebuilt)

A risk manager wants a credit derivative in which one party pays the total return on a reference asset (coupons plus any gain or loss in value) and receives a floating rate plus a spread, so that the other party gets the economic exposure of owning the asset without owning it. Which instrument fits this description?

A. Credit default swap (CDS)

B. Total return swap (TRS)

C. Physically settled CDS

D. Collateralized debt obligation (CDO)

Question CR-13 a.2

Bank Q559 • AnalystPrep Q.6065 (rebuilt)

An insurance company's investment team wants a credit derivative that arranges the credit risk of a reference portfolio into tiers of priority, so that it can choose a distinct risk-return profile by buying one tier. Which instrument is the most appropriate?

A. Single-name credit default swap (CDS)

B. Total return swap (TRS)

C. Synthetic collateralized debt obligation (CDO)

D. Add-up basket CDS

Question CR-13 a.3

Bank Q564 • Mock 2 Q40 (rebuilt)

An investor wishes to reduce the credit risk of his bond portfolio without selling the bonds. The best approach for this investor is to:

A. buy a collateralized debt obligation (CDO) tranche.

B. sell protection in an index credit default swap (CDS).

C. enter into a total return swap (TRS) as the total return receiver.

D. buy protection in a credit default swap (CDS) on the portfolio constituents.

Question CR-13 a.4

Bank Q558 • AnalystPrep Q.6064 (rebuilt)

A portfolio manager expects the credit quality of a reference entity to improve, so its CDS spread should narrow. All else equal, which position should gain from this view?

A. Buying protection on the entity through a single-name CDS

B. Selling protection on the entity through a single-name CDS

C. Buying protection on a first-to-default basket CDS that includes the entity

D. Being the total return payer on a swap referencing a bond of the entity

Question CR-13 a.5

Bank Q562 • BT P2.T6.607.3 (rebuilt)

Financial Bank holds bonds issued by Konkcore Corporation. To hedge their credit risk it buys protection in an over-the-counter CDS on Konkcore from Counterparty Finservices, LLC. Which of the following is true about the bank's CDS position?

A. If the price of Konkcore's bond falls because interest rates rise, the value of the CDS will increase.

B. If the yield on Konkcore's bond rises because of a rating downgrade, the value of the CDS will increase.

C. If the default correlation between Konkcore and Finservices increases, the value of the CDS will increase.

D. If Konkcore defaults and the CDS is cash settled, Finservices pays the bank approximately the mid-market value of the defaulted bonds determined by the auction.

Question CR-13 a.6

Bank Q560 • BT P2.T6.24.14.3 (rebuilt)

A bank holds $10 million par of Konkcore bonds and buys 5-year CDS protection on $10 million notional at a spread of 200 bp a year, paid quarterly. Konkcore later defaults, and the post-default auction values the bonds at 40% of par. Which of the following statements is NOT correct?

A. Under cash settlement the protection seller pays the bank $6.0 million.

B. Under physical settlement the bank delivers the bonds and receives $10 million.

C. Whichever settlement is used, the bond-plus-CDS position leaves the bank with $10 million of value (ignoring premiums paid and the accrued premium).

D. The CDS payoff to the bank is $4.0 million, because the payoff equals the notional times the recovery rate.

Question CR-13 a.7 Bank Q561 • BT P2.T6.24.22.1 (rebuilt)

Sesame Street Bank owns a $200 million bond portfolio and enters a 5-year total return swap as the total return payer with a hedge fund. During the swap the bank pays the coupons earned on the bonds and receives SOFR plus 25 bp on $200 million. At maturity the portfolio's value has fallen by 15% and there has been no default. Which statement about the end-of-swap payment is correct?

A. The bank pays the hedge fund $30 million, since the bank bears the price decline.

B. The bank receives $30 million from the hedge fund, since the receiver pays the decline in value.

C. No payment is made, because the coupons paid by the bank offset the price decline.

D. The bank receives $170 million, the portfolio's value at maturity.

Question CR-13 a.8 New • built from Workbook D14 revision notes and GARP Ch.23, 23.1 (Choudhry, CLNs) • shape D14·S1

A treasury analyst at a Malaysian insurer is offered a 5-year credit-linked note issued directly by Bank X and referencing Company Y. The insurer would pay USD 20 million at issue and receive a coupon of SOFR plus 180 basis points. Which of the following statements about the insurer's position is correct?

A. The insurer is the protection buyer: if Y suffers a credit event, Bank X compensates the insurer for the loss on Y.

B. Like a seller of CDS protection, the insurer pays nothing at inception and pays cash only if Y suffers a credit event.

C. The insurer has sold protection on Y in funded form: after a credit event its principal is repaid less the loss on Y, and because Bank X issued the note directly the insurer is also exposed to Bank X's own credit risk.

D. Because Bank X holds the insurer's cash, the coupon spread compensates the insurer for Bank X's credit risk only, and a credit event at Y would not reduce the redemption amount.

Solutions • CR-13 a

Solution CR-13 a.1 — Answer: B

Bank Q557 • AnalystPrep Q.6061

Step 1 – What is exchanged

The total return (coupons plus the gain or loss in value) against floating plus a spread: that is GARP's definition of a total return swap.

Step 2 – Who gets the exposure

The receiver gets the economics of owning the asset while the payer keeps ownership.
$\Rightarrow$ A total return swap.

Interpretation

A CDS transfers only default risk, and only on a credit event; a TRS transfers the whole return, market moves included, which is why it works as synthetic financing of the bond.

Why the other options are wrong

A is incorrect. A CDS pays $L(1-R)$ on a credit event against a spread; there is no exchange of coupons or of gains and losses in value.
C is incorrect. Physical settlement is a way of settling a CDS after a credit event, not an exchange of total return.
D is incorrect. A CDO tranches the credit risk of a portfolio; it is not a bilateral swap of returns.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.7 Total Return Swaps

Solution CR-13 a.2 — Answer: C

Bank Q559 • AnalystPrep Q.6065

Step 1 – Tiers of priority

Tranching allocates losses by seniority: the equity tranche absorbs the first losses, then the mezzanine, then the senior tranche.

Step 2 – Match the instrument

A synthetic CDO sells CDS protection on a portfolio and passes the income and payouts to tranches, each with its own spread and risk.
$\Rightarrow$ A synthetic CDO.

Interpretation

Investors choose a tranche to choose a risk-return profile: 1,000 bp for the first 5% of losses in GARP's example, 10 bp for losses above 20%.

Why the other options are wrong

A is incorrect. A single-name CDS covers one reference entity with no tranching.
B is incorrect. A TRS passes the whole return of one asset or portfolio to one counterparty; it has no tiers.
D is incorrect. An add-up basket CDS pays on every default in the basket; it is not divided into tiers of priority.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.8 Collateralized Debt Obligations, Synthetic CDOs

Solution CR-13 a.3 — Answer: D

Bank Q564 • Mock 2 Q40

Step 1 – What reduces credit risk

A position that gains when credit quality deteriorates, offsetting losses on the bonds.

Step 2 – Check each instrument

A CDS protection buyer pays a spread and receives $L(1-R)$ on a credit event of the reference entity. Buying a CDO tranche, selling protection and receiving total return each make the investor a taker of credit risk.

$\Rightarrow$ Buying CDS protection on the constituents is the hedge.

Interpretation

CDS protection transfers default losses to the seller without selling the bonds; the investor swaps issuer default risk for (much smaller) counterparty risk on the seller.

Why the other options are wrong

A is incorrect. A CDO tranche holder absorbs losses on the reference portfolio: more credit exposure, not less.

B is incorrect. A protection seller pays out on defaults in the index, which adds credit risk.

C is incorrect. The total return receiver bears the asset's credit and market losses, which replicates a leveraged long position.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.1, 13.7 and 13.8

Solution CR-13 a.4 — Answer: B

Bank Q558 • AnalystPrep Q.6064

Step 1 – What a narrowing spread does

New protection becomes cheaper, so an existing contract that *receives* the old, higher spread gains value, and one that pays it loses.

Step 2 – Who receives the spread

The protection seller. Selling protection is a long credit position, like owning the bond.

$\Rightarrow$ Selling protection gains from the view.

Interpretation

GARP's equivalence (a long bond has a similar risk to a short CDS position) is the quickest test: whatever helps a bondholder helps a protection seller.

Why the other options are wrong

A is incorrect. A protection buyer is short credit and loses when spreads narrow.

C is incorrect. A first-to-default buyer is also short credit; improving credit makes the protection worth less.

D is incorrect. The total return payer passes on the bond's gain in value to the receiver, so it loses when the bond rallies.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.1 and 13.7

Solution CR-13 a.5 — Answer: B

Bank Q562 • BT P2.T6.607.3

Step 1 – What the CDS is linked to

The credit quality of Konkcore. A downgrade raises the market's default probability, so new protection costs more and the bank's existing protection gains value.

Step 2 – Check the other drivers

A rise in rates moves the bond price without any change in credit risk. Correlation between Konkcore and the seller is wrong-way risk: it makes the protection less reliable. At cash settlement the auction sets the bond's mid-market value, and the seller pays face value minus that value.

$\Rightarrow$ Only B is true.

Interpretation

The bank holds a credit hedge, not an interest-rate hedge; and its protection is only as good as the seller's ability to pay when Konkcore defaults.

Why the other options are wrong

A is incorrect. A rate-driven fall in the bond price is market risk, not credit risk; the CDS does not pay for it.

C is incorrect. Polarity flip: higher default correlation between the reference entity and the protection seller reduces the value of the protection.

D is incorrect. The payoff is the face value *minus* the auction value, $L(1-R)$, not the value of the defaulted bonds.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.1 Credit Default Swaps

Solution CR-13 a.6 — Answer: D

Bank Q560 • BT P2.T6.24.14.3

Step 1 – Cash settlement

Payoff $= L(1-R) = 10 \times (1-0.40) = \$6.0$ million: A is correct.

Step 2 – Physical settlement

The bank delivers bonds worth \$4.0 million and receives their \$10 million face value: B is correct.

Step 3 – The hedged position

Bonds worth \$4.0 million plus a \$6.0 million payoff, or \$10 million in cash: the bank ends with \$10 million either way, so C is correct.

$\Rightarrow$ D is the incorrect statement.

Interpretation

Both settlement methods restore par; they differ only in whether the bonds change hands. The premium ceases at the credit event, apart from the final accrual.

Why the other options are wrong

A is incorrect. Correct: $10 \times 0.60 = \$6.0$ million.

B is incorrect. Correct: physical settlement exchanges the bonds for their face value.

C is incorrect. Correct: the protection makes the bondholder whole at par.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.1 Credit Default Swaps

Solution CR-13 a.7 — Answer: B

Bank Q561 • BT P2.T6.24.22.1

Step 1 – Size of the change in value

$15\% \times \$200 \text{ million} = \$30 \text{ million}$.

Step 2 – Direction

The payer pays gains in value and receives losses; the receiver pays floating plus spread and any loss. The value fell, so the hedge fund (receiver) pays the bank $30 million.

$\Rightarrow$ The bank receives $30 million.

Interpretation

The bank kept the bonds but passed their whole return, price risk included, to the hedge fund; its 25 bp spread compensates it for the risk that the fund cannot pay when the bonds have fallen.

Why the other options are wrong

A is incorrect. Reverses the direction: the payer receives the decline.

C is incorrect. Coupons are exchanged during the swap; the change in value is settled in addition at the end.

D is incorrect. Confuses the change in value, $30 million, with the portfolio's value at maturity, $200 \times 0.85 = \$170$ million.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.7 Total Return Swaps

Solution CR-13 a.8 — Answer: C

New • built from Workbook D14 and GARP Ch.23

Step 1 – Funded protection

The insurer pays USD 20 million at issue and is repaid less any credit loss on Y: it has sold protection on Y, and the spread in the coupon is its premium.

Step 2 – The issuer's credit

Bank X holds the cash, so Bank X faces no counterparty risk on the protection. The insurer, by contrast, is repaid by Bank X: a note issued directly carries Bank X's credit risk as well (an SPV issue would remove that link).

$\Rightarrow$ C is correct.

Interpretation

The CLN is the funded mirror of a CDS sale. The funding is what moves counterparty risk from the protection buyer to the investor.

Why the other options are wrong

A is incorrect. Role flip: the note investor is the protection seller and loses on a credit event at Y.

B is incorrect. That describes an unfunded CDS sale; a CLN is funded at issue.

D is incorrect. The redemption amount is reduced by credit losses on the reference entity; that is the credit link.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). • **Reference:** Workbook D14 revision notes (CLN mechanics, outside source); Choudhry (GARP CR Ch.23), 23.1 SPV applications

CR-13 • Credit Derivatives • Vardeez: MODERATE

CR-13 b

P3 CONCEPT

Explain how to account for credit risk exposure in valuing a CDS.

Workbook: none homed here • Vardeez focus (the D14 revision notes summarise GARP's valuation example)

WHAT IS TESTED HERE

Why P3. No workbook shape, but Vardeez flags the objective: know the process, because a full valuation is too long for an exam calculation.
Vardeez: "LO 31.b - Just know the Process here - This can be very lengthy on the exam especially calculations. So just KNOW the process."

Reported exam shapes: none listed in the map.

What the questions look like: which leg is weighted by survival and which by default probabilities, the mid-period default convention and the accrual term it creates, which default probabilities are used, what the breakeven spread equates, how a recovery assumption enters, and the sign of a mark-to-market. The worked example below shows the process; the bank's numerical items on single legs (Q570, Q571) are not used, under the TYPE rule.

Worked elsewhere: CDS mechanics under CR-13 a and CR-9 g; implying hazard rates from CDS spreads (A15.5, A15.6) under CR-9 h.

1. The valuation in five steps

A CDS is valued by equating two present values: what the protection buyer expects to pay and what it expects to receive. Credit risk enters every step through the reference entity's default probabilities.

1. **Default probabilities.** Use **risk-neutral** probabilities, implied from bond prices or from CDS quotes (CR-9 k: valuation needs risk-neutral PDs). With a constant hazard rate λ, survival to t is $V(t) = \mathrm{e}^{-\lambda t}$ and the probability of default in year t is $V(t-1) - V(t)$.
2. **Premium leg.** A payment is made only if the entity survives to the payment date, so each payment is weighted by **survival** and discounted from the payment date: $A = \sum V(t_i)\,\Delta t\, v(t_i)$ per unit of spread.
3. **Accrual.** Defaults are assumed to happen **halfway** through a period. The buyer then owes the premium accrued for half a period, weighted by the default probability: $B = \sum \mathrm{PD}_i \,\frac{1}{2}\Delta t\, v(\mathrm{mid})$ per unit of spread.
4. **Protection leg.** The payoff $1 - R$ per unit of notional is weighted by the **default probability** and discounted from mid-period: $C = \sum \mathrm{PD}_i (1-R)\, v(\mathrm{mid})$.
5. **Breakeven spread.** $s(A + B) = C$, so $s = C/(A + B)$.

Vardeez's example. Three years, hazard rate 3%, risk-free rate 4% (continuous), recovery 35%, annual payments, per \$1 of notional:

Year	$V(t)$	PD	$v(t)$	$v(t-0.5)$	Premium ($\times s$)	Accrual ($\times s$)	Payoff
1	0.97045	0.0296	0.9608	0.9802	0.9324	0.0145	0.0188
2	0.94176	0.0287	0.9231	0.9418	0.8693	0.0135	0.0176
3	0.91393	0.0278	0.8869	0.9048	0.8106	0.0126	0.0164
Total					$A = 2.6123$	$B = 0.0406$	$C = 0.0528$

$$s = \frac{C}{A+B} = \frac{0.0528}{2.6529} = 1.99\% \ (199 \text{ bp})$$

Dropping the accrual term gives $0.0528/2.6123 = 2.02\%$: forgetting B *raises* the spread, because the buyer is credited with paying less. GARP's own example (5 years, $\lambda = 2\%$, $r = 5\%$, $R = 40\%$) gives $A = 4.0728$, $B = 0.0422$, $C = 0.0506$ and $s = 0.0506/4.1150 = 123$ bp.

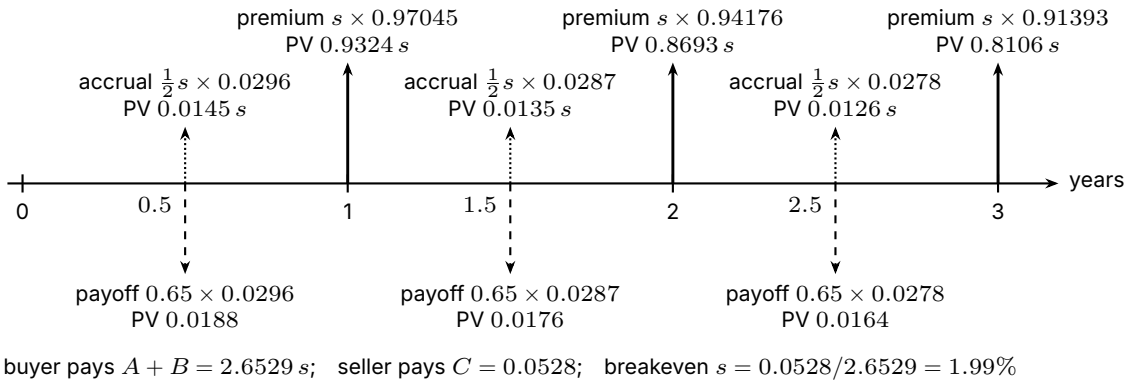


Figure 1. Vardeez's three-year CDS on a timeline. Arrows above the axis are paid by the protection buyer: premiums on the anniversary dates, weighted by survival (thick solid), and the accrued half-year premium if default occurs mid-year (dotted). Arrows below the axis are paid by the seller: the payoff $1 - R = 0.65$ at the assumed mid-year default dates, weighted by the probability of default in that year (dashed).

2. Marking to market, recovery and binary CDS

- **Mark-to-market.** Revalue an existing CDS with today's risk-neutral probabilities. Since today's spread satisfies $C = s_{\text{mkt}}(A+B)$, the value to the **seller** is $s_{\text{deal}}(A+B) - C = (s_{\text{deal}} - s_{\text{mkt}})(A+B)$; the buyer's value is the negative. GARP: a deal at 150 bp, now quoted at 123 bp, is worth $4.1150 \times 0.015 - 0.0506 = 0.0111$ per \$1 to the seller.
- **Implied hazard rates.** In practice the hazard rate is found by searching for the λ that reproduces the market spread (GARP: 100 bp implies 1.63% a year), and a term structure of hazard rates from a term structure of spreads.
- **Recovery barely matters for a regular CDS** when the same R is used to imply the PDs and to value the trade: implied PDs scale roughly with $1/(1-R)$ and the payoff with $1 - R$.
- **Binary CDS.** The payoff is a fixed amount (the notional), whatever the recovery. C becomes $\sum \text{PD}_i \, v(\text{mid})$, so the spread is higher and depends on R: 205 bp against 123 bp in GARP's example, and $0.0528/0.65/2.6529 = 3.06\%$ in Vardeez's.

Leg	Paid by	Weighted by, and when	Amount per \$1 notional
Premium	Buyer	Survival to each payment date; discounted from the payment date	$s\,\Delta t$
Accrual	Buyer	Default probability in the period; discounted from mid-period	$\frac{1}{2}s\,\Delta t$
Protection	Seller	Default probability in the period; discounted from mid-period	$1 - R$ (binary CDS: 1)

Exam traps

- **Weights swapped.** Premiums are weighted by survival, the payoff and the accrual by default probabilities.
- **Real-world PDs.** Valuation uses risk-neutral PDs from CDS or bond prices, not historical default rates.
- **Accrual forgotten.** Leaving out B raises the computed spread.
- **Timing.** Mid-period defaults are discounted over a shorter time than end-of-period ones, so their PV is *higher*.
- **Recovery.** The payoff uses $1 - R$, not R; a binary CDS ignores R in the payoff and costs more.
- **Sign of the mark-to-market.** Spreads widen: protection buyers gain. Spreads narrow: sellers gain.

Practice questions: CR-13 b

Question CR-13 b.1

Bank Q566 • Schweser Module Quiz 31.1 Q3

CDS spreads are calculated such that the PV of:

A. accrual payments is zero.

B. expected payoff is less than the PV of expected payments during the life of the CDS.

C. expected payoff is equal to the PV of expected payments during the life of the CDS.

D. expected payoff is greater than the PV of expected payments during the life of the CDS.

Question CR-13 b.2

Bank Q569 • AnalystPrep Q.6074

A credit analyst calculates the present value of expected payments and payoff for a CDS. What default timing convention is used within each payment period?

A. Defaults occur at the start of each period.

B. Defaults occur at random times, averaged over the life of the contract.

C. Defaults occur at the end of each period.

D. Defaults occur at the midpoint of each period.

Question CR-13 b.3

Bank Q567 • AnalystPrep Q.6067

When valuing a CDS, the risk manager assumes that a default, if it occurs within a year, happens at the middle of that year rather than at the end. The risk-free rate is 3% per annum with continuous compounding. Holding the default probability and the payoff amount constant, how does the present value of the expected payoff under the mid-year assumption compare with the end-of-year assumption?

A. It is higher, because the payoff is discounted over a shorter period.

B. It is lower, because the payoff is discounted over a longer period.

C. It is unchanged, because the assumption does not affect the discount factor.

D. It cannot be determined without knowing the spread payment rate.

Question CR-13 b.4 Bank Q568 • AnalystPrep Q.6068

In valuing a CDS, the recovery rate is 40%. How is the recovery rate used to compute the expected payoff on a default event?

A. The probability of default in the period is multiplied by (1 − recovery rate) and by the notional.

B. The recovery rate is added to the probability of default to reflect the higher potential return.

C. The recovery rate is used as a direct multiplier of the probability of default and the notional.

D. The recovery rate is applied to the expected spread payments to estimate the actual cash flows.

Question CR-13 b.5 Bank Q573 • AnalystPrep Q.6073 (rebuilt; re-filed from CR-13 c)

An analyst at a dealer is valuing a 5-year CDS on a corporate issuer for the trading book. Which default probabilities should be used, and how are they typically obtained?

A. Real-world default probabilities, estimated from historical default rates for the issuer's rating class.

B. Real-world default probabilities, estimated from the issuer's historical accounting ratios.

C. Risk-neutral default probabilities, implied from market CDS spreads or bond prices.

D. Risk-neutral default probabilities, set equal to the historical average annual default rate.

Question CR-13 b.6 New • built from GARP Ch.13, 13.2 (Recovery Rate, Binary CDS)

A CDS desk at a Brazilian bank values protection on a corporate. It implies a constant hazard rate from the corporate's CDS quotes using a 40% recovery rate, then uses the same 40% to value its trades. A junior analyst lists four claims about the valuation. Which claim is correct?

A. Leaving out the accrual payment on default lowers the breakeven spread, because the buyer is then assumed to pay less.

B. Re-running the valuation of a regular CDS with a 20% recovery rate, re-implying the hazard rate from the same quotes, changes its value very little; a binary CDS on the same name is far more sensitive to the assumption.

C. A binary CDS on the same name should trade at a lower spread than the regular CDS, because its payoff is fixed in advance.

D. The expected premium payments are weighted by the probability of default in each year, and the expected payoff by the probability of survival to each payment date.

Question CR-13 b.7 New • built from GARP Ch.13, 13.2 (Marking to Market a CDS)

Two years ago a hedge fund bought 5-year protection on a retailer at 180 basis points. The retailer's 3-year CDS spread is now 240 basis points. The fund's risk system values the position by the standard method of GARP Section 13.2. Which of the following describes the value of the position to the fund?

A. Negative: the fund pays 180 bp for protection now worth 240 bp, so it would have to pay the seller to close out.

B. Zero: a CDS is revalued at zero after inception because its spread is reset to the market level.

C. Positive, equal to the present value of the expected payoff, C, computed with today's hazard rate.

D. Positive, equal to 60 bp times $A+B$, the present value of the premium and accrual payments per unit of spread, computed with risk-neutral probabilities implied from today's spreads.

Solutions • CR-13 b

Solution CR-13 b.1

Answer: C

Bank Q566 • Schweser Module Quiz 31.1 Q3

Step 1 – The two sides

The buyer's side is the PV of expected premium and accrual payments, $s(A + B)$; the seller's side is the PV of the expected payoff, C.

Step 2 – Breakeven

A new CDS has zero value, so the spread is set where $s(A + B) = C$.
$\Rightarrow$ The PV of the expected payoff equals the PV of the expected payments.

Interpretation

This is the whole valuation in one line, $s = C/(A+B)$. Every credit input (hazard rate, recovery) works through A, B or C.

Why the other options are wrong

A is incorrect. The accrual payments have a positive PV whenever default is possible; they are part of the buyer's side.
B is incorrect. If the payoff were worth less than the payments, the protection buyer would be overpaying; the contract would not be at breakeven.
D is incorrect. The reverse imbalance: the seller would be underpaid.

LO: CR-13 b: Explain how to account for credit risk exposure in valuing a CDS. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.2 Valuation of Credit Default Swaps

Solution CR-13 b.2

Answer: D

Bank Q569 • AnalystPrep Q.6074

Step 1 – The convention

GARP assumes defaults happen halfway through each period, so the payoff and the accrual are discounted from the mid-point (0.5, 1.5, 2.5, ... years with annual payments).

Step 2 – Why

Default can happen at any time within the period; the mid-point is a simple average, and it creates the accrual payment the buyer owes for the half period.
$\Rightarrow$ Defaults occur at the midpoint of each period.

Interpretation

The premium leg is discounted from the payment dates; the protection and accrual legs from the mid-points. That timing difference is why the accrual term exists at all.

Why the other options are wrong

A is incorrect. Defaults at the start would leave no accrued premium and discount the payoff over too short a time.
B is incorrect. The model assigns a specific time within each period, not an average over the life of the contract.
C is incorrect. End-of-period defaults would also remove the accrual, and understate the PV of the payoff.

LO: CR-13 b: Explain how to account for credit risk exposure in valuing a CDS. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.2, Tables 13.2 to 13.4

Solution CR-13 b.3 Answer: A

Bank Q567 • AnalystPrep Q.6067

Step 1 – Compare the discount factors

A payoff in year t is discounted by $e^{-0.03(t-0.5)}$ under the mid-year assumption and by $e^{-0.03t}$ at year end. The first is larger by the factor $e^{0.015}$.

Step 2 – Effect on the PV

Same probability, same payoff, earlier payment: a higher present value.

$\Rightarrow$ The PV is higher.

Interpretation

With a positive interest rate, earlier cash is worth more. The difference is small (about 1.5% of each term here) but it raises C and so the breakeven spread.

Why the other options are wrong

B is incorrect. Reversed: the mid-year payoff is discounted over a shorter period, not a longer one.

C is incorrect. The timing assumption changes the discount factor directly.

D is incorrect. The spread enters the premium and accrual legs, not the PV of the payoff.

LO: CR-13 b: Explain how to account for credit risk exposure in valuing a CDS. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.2, Table 13.4

Solution CR-13 b.4 Answer: A

Bank Q568 • AnalystPrep Q.6068

Step 1 – The payoff on default

The seller pays the notional less what the bonds recover: $L(1-R)$.

Step 2 – The expected payoff

Weight it by the probability of default in the period: $\text{PD}_i \times (1-R) \times L$, here $0.60 \times \text{PD}_i \times L$, then discount from mid-period.

$\Rightarrow$ Multiply the PD by one minus the recovery rate and by the notional.

Interpretation

Recovery affects only the protection leg. The premium and accrual legs do not depend on R directly, only through the PDs implied from the market.

Why the other options are wrong

B is incorrect. Adding a rate to a probability has no meaning, and a higher recovery lowers the payoff.

C is incorrect. Multiplying by R gives the amount recovered, not the amount lost.

D is incorrect. The recovery rate does not enter the premium leg.

LO: CR-13 b: Explain how to account for credit risk exposure in valuing a CDS. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.2, Table 13.4

Solution CR-13 b.5 Answer: C

Bank Q573 • AnalystPrep Q.6073

Step 1 – Valuation needs risk-neutral probabilities

A CDS value is the PV of expected cash flows discounted at the risk-free rate, so the expectations must use risk-neutral probabilities (CR-9 k).

Step 2 – Where they come from

GARP: the default probabilities used to value a CDS should be risk-neutral, implied from bond prices or from CDS quotes (searching for the hazard rate that reproduces the quoted spread).
$\Rightarrow$ Risk-neutral PDs implied from market prices.

Interpretation

Real-world PDs from default data are for scenario analysis and capital; they are lower, and would undervalue protection.

Why the other options are wrong

A is incorrect. Real-world, historical PDs are for scenario analysis, not valuation.
B is incorrect. Accounting ratios estimate real-world default risk (for example a credit score); they are not market-implied.
D is incorrect. Right label, wrong source: a historical average default rate is a real-world measure, whatever it is called.

LO: CR-13 b: Explain how to account for credit risk exposure in valuing a CDS. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.2, Estimating Default Probabilities (CR-13 c)

Solution CR-13 b.6 — Answer: B

New • built from GARP Ch.13, 13.2

Step 1 – Recovery and a regular CDS

If the hazard rate is implied from market spreads with recovery R, the implied PDs scale roughly with $1/(1-R)$ while the payoff scales with $1-R$. In GARP's example, re-implying with $R = 20\%$ instead of 40% moves the value of a 150 bp deal from 0.0111 to 0.0112 per \$1.

Step 2 – Recovery and a binary CDS

A binary CDS pays the full notional, so its payoff does not shrink as R falls while the implied PDs do: its spread falls from 205 bp to about 154 bp.
$\Rightarrow$ B is correct.

Interpretation

GARP's point: the value of a regular CDS is not very sensitive to the recovery rate as long as the same R is used to imply the default probabilities; a binary CDS is.

Why the other options are wrong

A is incorrect. Polarity flip: without B the denominator of $s = C/(A+B)$ shrinks, so the spread rises (2.02% against 1.99% in Vardeez's example).
C is incorrect. A binary CDS pays more on default (all of the notional rather than $1-R$ of it), so it costs more: 205 bp against 123 bp.
D is incorrect. Weights swapped: premiums are weighted by survival, the payoff by default probabilities.

LO: CR-13 b: Explain how to account for credit risk exposure in valuing a CDS. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.2, Recovery Rate and Binary Credit Default Swaps

Solution CR-13 b.7 — Answer: D

New • built from GARP Ch.13, 13.2

Step 1 – Value to the protection buyer

The buyer receives the protection leg and pays the deal spread: value $= C - s_{\text{deal}}(A+B)$, with A, B and C computed from today's risk-neutral probabilities for the remaining three years.

Step 2 – Use the market spread

Today's 3-year spread satisfies $C = s_{\text{mkt}}(A+B)$, so the value is $(s_{\text{mkt}} - s_{\text{deal}})(A+B) =$

$(0.0240 - 0.0180)(A + B)$.
$\Rightarrow$ Positive: spreads have widened, so the fund's old protection is cheap.

Interpretation

The mark-to-market is the spread difference times a risky annuity (the PV of the premium and accrual payments per unit of spread). Schweser's quiz version computes it for a given C; the exam usually asks only for the sign or the structure.

Why the other options are wrong

A is incorrect. Polarity flip: paying 180 bp for protection now worth 240 bp is a gain for the buyer; it is the seller who would pay to close out.
B is incorrect. A CDS is worth zero only at inception; its spread is fixed for the life of the contract.
C is incorrect. Ignores the premiums still to be paid: the value is C minus $s_{\text{deal}}(A + B)$, not C alone.

LO: CR-13 b: Explain how to account for credit risk exposure in valuing a CDS. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.2, Marking to Market a CDS

CR-13 • Credit Derivatives • Vardeez: MODERATE

CR-13 f

P3 CONCEPT

Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach.

Workbook: D14 Credit Derivatives and Credit-Linked Notes • Tier D • Map #58 (2/16)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D14 (Map #58, 2 of 16 sources) reports synthetic CDO valuation as shape S4 and flags this objective as "worth knowing".
Vardeez: "LO 31.f: Know this as well."

Reported exam shapes

- **D14·S4** Synthetic CDO valuation and the two implied correlations, compound and base (the correlations are CR-13 g).

What the questions look like: process questions. What the spread payments approach equates, why the tranche principal declines, why conditioning on the common factor gives a binomial distribution, what is integrated over the factor, how tranche spreads relate at inception, and how correlation moves them.

Worked elsewhere: D14.1 (option C swaps compound and base correlation) at Question CR-9 g.7; the conditional PD $Q(t \mid F)$ under CR-12 f; the three legs of a CDS valuation under CR-13 b; the synthetic CDO structure under CR-13 a.

1. Tranche principal and attachment points

A tranche covers portfolio losses between the **attachment point** a_L and the **detachment point** a_H (recap of the structure: CR-13 a). With n equally weighted names and recovery R, each default costs $(1-R)/n$ of the portfolio. With an initial tranche principal of 1, after k defaults the tranche principal is

$$\begin{cases} 1 & k < m(n_L) \\ \dfrac{a_H - k(1-R)/n}{a_H - a_L} & m(n_L) \le k < m(n_H) \\ 0 & k \ge m(n_H) \end{cases} \qquad n_L = \frac{a_L n}{1-R}, \qquad n_H = \frac{a_H n}{1-R}$$

where $m(x)$ is the smallest integer greater than x. For the iTraxx 3 to 6% tranche ($n = 125$, $R = 40\%$) each default costs 0.48% of the portfolio, $n_L = 6.25$ and $n_H = 12.5$: the seventh default is the first to hit the tranche, ten defaults leave 40% of its principal, and the thirteenth wipes it out (Figure 1).

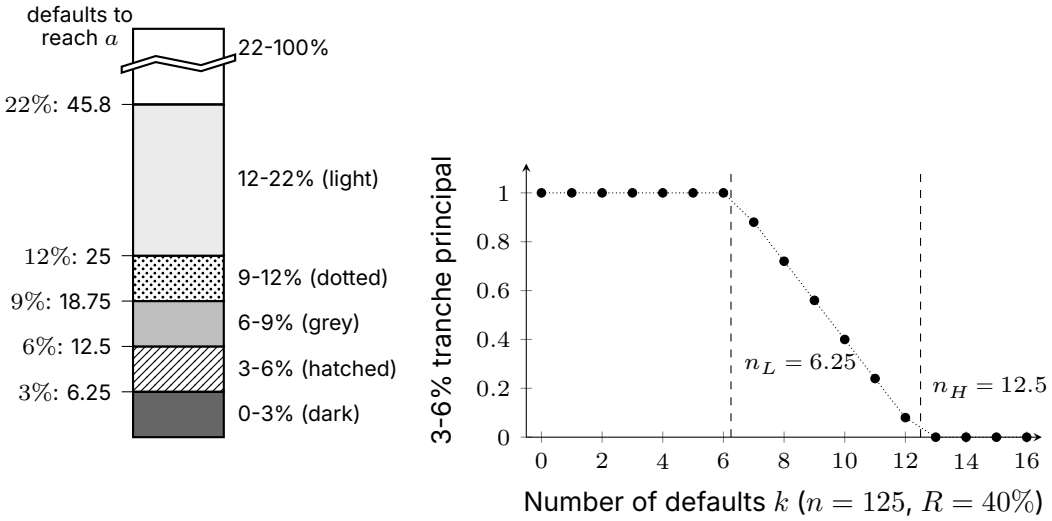


Figure 1. Left: the six standard iTraxx Europe tranches, each shaded differently (scale broken above 22%), with the number of defaults at 40% recovery needed to reach each attachment point. Right: principal of the 3 to 6% tranche after k defaults (dots). It is untouched up to six defaults, falls to 0.88 at the seventh and by 0.16 per default after that (0.40 after ten), and is gone at the thirteenth (dashed lines at n_L and n_H). The spread payments approach needs the *expected* value of this principal on each payment date.

2. The spread payments approach

Vardeez's three steps: compute the PV of expected inflows (spread payments) and outflows (losses); adjust for accrued payments as in a CDS (CR-13 b); set the spread that equates them. The difference from a CDS is that the spread is paid on the **outstanding tranche principal**, which falls as losses arrive. With payment dates $\tau_1, \dots, \tau_m$, expected tranche principal E_j at τ_j ($E_0 = 1$) and discount factor $v(\tau)$ (GARP eqs. 13.1 to 13.4):

$$A = \sum_j (\tau_j - \tau_{j-1}) E_j \, v(\tau_j), \qquad C = \sum_j (E_{j-1} - E_j) \, v(\tau_{j-\frac{1}{2}}),$$

$$B = \sum_j \tfrac{1}{2} (\tau_j - \tau_{j-1})(E_{j-1} - E_j) \, v(\tau_{j-\frac{1}{2}}), \qquad \text{where } \tau_{j-\frac{1}{2}} = \tfrac{1}{2}(\tau_{j-1} + \tau_j)$$

sA is the PV of the regular spread payments, sB the accrual on losses assumed mid-period, and C the PV of the expected losses. The value to the protection buyer is $C - sA - sB$, so the breakeven spread is $s = C/(A + B)$. A tranche quoted as an upfront payment plus a fixed running spread s^* (the 0 to 3% tranche, with $s^* = 500$ bp) has upfront $C - s^*(A + B)$. Everything hinges on the expected tranche principals E_j, and the copula supplies them.

3. The Gaussian copula approach (the standard market model)

1. **Unconditional PD.** Assume one constant hazard rate λ for all names, consistent with the index spread (found by the CDS search of CR-13 b): $Q(t) = 1 - \mathrm{e}^{-\lambda t}$.
2. **Conditional PD.** One-factor Gaussian copula with copula correlation ρ (CR-12 f): $Q(t \mid F) = N\big((N^{-1}[Q(t)] - \sqrt{\rho}\, F)/\sqrt{1-\rho}\big)$.
3. **Number of defaults.** Given F, defaults are independent, so the probability of exactly k defaults by t is binomial: $P(k, t \mid F) = \binom{n}{k} Q(t \mid F)^k \big[1 - Q(t \mid F)\big]^{n-k}$.
4. **Conditional values.** Weight the tranche principal after k defaults (Section 1) by $P(k, \tau_j \mid F)$ to get $E_j(F)$, and from it $A(F)$, $B(F)$ and $C(F)$.

5. **Integrate over F.** Weight by the standard normal density of F (in practice by Gaussian quadrature) to get the unconditional A, B, C, then $s = C/(A + B)$.

GARP Example 13.2. iTraxx 3 to 6% tranche, $\rho = 0.15$, $R = 40\%$, flat 3.5% rates, quarterly payments, index spread 50 bp (hazard rate 0.83%): $A = 4.2846$, $B = 0.0187$, $C = 0.1496$, so $s = 0.1496/4.3033 = 348$ bp.

Role of correlation (GARP 13.9). With low correlation the equity tranche is very risky and the senior tranches very safe. As correlation rises, defaults cluster: scenarios with few defaults and scenarios with many both become more likely, so the junior tranches become less risky and the senior tranches more risky. With perfect correlation and zero recovery all tranches are equally risky.

	Spread payments approach	Gaussian copula approach
Role	Turns expected tranche principals into a breakeven spread	Produces the expected tranche principals
Key idea	PV of spread on the declining principal, plus accrual, equals PV of expected losses	Conditional on F, defaults are independent and binomial; then integrate over F
Inputs	E_j on each payment date, discount factors	Hazard rate from the index spread, ρ, R (usually 40%), n, attachment points
Output	$s = C/(A + B)$, or upfront $C - s^*(A + B)$	A, B, C for each tranche

Exam traps

- **Fixed principal.** The tranche spread is paid on the outstanding, declining principal, not on the original notional.
- **Unconditional binomial.** Defaults are independent only *given* F; the binomial is applied conditionally and the result integrated over F.
- **Correlation direction.** Higher correlation makes the equity tranche safer and the senior tranches riskier; it does not change each name's PD.
- **Single spread for all tranches.** Tranche spreads reflect seniority; the average CDS spread is the index level, not a tranche price.
- **Attachment arithmetic.** Losses are $k(1 - R)/n$ of the portfolio, not k/n: at 40% recovery the 3% attachment point needs 6.25 defaults, not 3.75.

Practice questions: CR-13 f

Question CR-13 f.1

Bank Q587 • AnalystPrep Q.6083 (rebuilt)

In the one-factor Gaussian copula model of time to default used to value a synthetic CDO, the model's key role is to:

A. assume defaults are independent so that tranche losses follow a simple binomial distribution unconditionally.

B. link the marginal default-time distributions of the reference entities through a correlation parameter, so that the distribution of the number of defaults, and hence tranche losses, can be derived.

C. assign a different hazard rate to each entity so that the model is heterogeneous.

D. predict which reference entity will default first and when.

Question CR-13 f.2

Bank Q588 • AnalystPrep Q.6084 (rebuilt)

A structured credit analyst values iTraxx tranches with the one-factor Gaussian copula model. She works with the conditional default probability $Q(t \mid F)$ rather than the unconditional $Q(t)$ because, given the common factor F:

A. the default time of each entity becomes exactly predictable.

B. hazard rates are replaced by historical default frequencies.

C. defaults of different entities become independent, so the probability of exactly k defaults follows a binomial distribution; the results are then integrated over F.

D. correlation is eliminated from the valuation, so the tranche price no longer depends on ρ.

Question CR-13 f.3

Bank Q586 • AnalystPrep Q.6082 (rebuilt)

A risk manager values a tranche of a synthetic CDO using the spread payments approach. Which description of the method is most accurate?

A. Find the tranche spread that sets the PV of expected spread payments (on the declining tranche principal, including accrual) equal to the PV of expected default-related payoffs.

B. Take the average CDS spread of the reference entities and apply it unchanged to every tranche.

C. Set the tranche spread equal to the observed bid-ask midpoint of the standard index tranche, with no model of defaults.

D. Set the tranche spread so that the PV of expected losses on the senior tranche equals that on the equity tranche.

Question CR-13 f.4 Bank Q589 • BT P2.T6.24.23.2 (rebuilt)

Omega Capital is valuing a synthetic CDO referencing 100 corporate entities, with equity (first 5% of losses), mezzanine and senior tranches, using the one-factor Gaussian copula model of time to default. Which step is the core of computing each tranche's expected loss and hence its breakeven spread?

A. Compute the unconditional default probability $Q(t)$ only and apply it directly to the tranche attachment points.

B. Set the tranche spread equal to the average CDS spread of the 100 entities.

C. Assume each entity defaults independently, so no integration is needed.

D. Compute conditional default probabilities and expected tranche principals given the common factor, then integrate these over the standard normal distribution of the factor.

Question CR-13 f.5 Bank Q590 • Schweser Module Quiz 31.3 Q1 (rebuilt)

A bank originates a $100 million synthetic CDO by selling CDS protection on 25 names, $4 million each, at an average CDS spread of 2%. It creates equity, mezzanine and senior tranches. At inception, the tranche spreads are set so that:

A. each tranche earns the same spread rate.

B. each tranche's spread rate reflects its seniority (risk), and the spread payments across the tranches add up to the CDS premiums received on the reference portfolio.

C. the senior tranche earns the highest spread rate because it has the largest notional.

D. the equity tranche earns the lowest spread rate because it is the smallest.

Question CR-13 f.6 New • built from GARP Ch.13, 13.9 and Practice Question 13.26

A structured credit analyst at a French bank prices the standard iTraxx Europe tranches with the one-factor Gaussian copula, keeping the index spread, recovery rate and hazard rate fixed. She raises the copula correlation from 0.10 to 0.30. Which of the following describes the effect on the tranches?

A. All tranche spreads rise, because higher correlation raises each company's probability of default.

B. The equity tranche becomes riskier and the senior tranches safer, because defaults become more frequent in every scenario.

C. The equity tranche becomes less risky and the senior tranches riskier, so the equity tranche's price for protection falls and the senior tranche spreads rise; the index spread is unchanged.

D. No tranche price changes, because the expected loss on the whole portfolio does not depend on correlation.

Solutions • CR-13 f

Solution CR-13 f.1

Answer: B

Bank Q587 • AnalystPrep Q.6083

Step 1 – Two ingredients

Each name's marginal default-time distribution $Q(t)$ comes from the market (a hazard rate consistent with the index spread). The copula joins these marginals with one correlation parameter ρ.

Step 2 – What it delivers

The joint behaviour gives the probability distribution of the number of defaults on each date, hence the expected tranche principals and losses.

$\Rightarrow$ B describes the copula's role.

Interpretation

The copula is the dependence engine; the spread payments approach then prices each tranche from the losses the copula generates.

Why the other options are wrong

A is incorrect. Defaults are independent only conditional on F; unconditionally they are correlated, which is the point of the model.

C is incorrect. The standard market model is homogeneous: one hazard rate for all names. Different hazard rates define the heterogeneous model (CR-13 h).

D is incorrect. The model values tranches from the distribution of defaults; it does not forecast which name defaults first.

LO: CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, Using the Gaussian Copula Model of Time to Default

Solution CR-13 f.2

Answer: C

Bank Q588 • AnalystPrep Q.6084

Step 1 – Conditional independence

In $x_i = \sqrt{\rho}\,F + \sqrt{1-\rho}\,Z_i$ the Z_i are independent. Fixing F leaves only the Z_i, so defaults are independent with common probability $Q(t \mid F)$.

Step 2 – The binomial and the integration

$P(k, t \mid F) = \binom{n}{k} Q(t \mid F)^k [1 - Q(t \mid F)]^{n-k}$ gives the conditional distribution of defaults; the tranche values $A(F)$, $B(F)$, $C(F)$ are then integrated over the normal density of F.

$\Rightarrow$ C is correct.

Interpretation

Conditioning turns an intractable joint distribution of 125 default times into a one-dimensional integral of binomial probabilities.

Why the other options are wrong

A is incorrect. Given F, each default time is still random (through Z_i); only its probability is fixed.

B is incorrect. The hazard rates are risk-neutral, implied from the index spread; conditioning does not replace them with historical data.

D is incorrect. Correlation enters through $Q(t \mid F)$, which depends on ρ; after integration the tranche price depends strongly on ρ.

LO: CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, eqs. 13.5 to 13.7

Solution CR-13 f.3 Answer: A

Bank Q586 • AnalystPrep Q.6082

Step 1 – Inflows to the protection seller

The spread s is paid on the outstanding tranche principal, which declines as losses arrive: PV sA, plus the accrual sB on losses assumed mid-period.

Step 2 – Outflows

The PV of expected losses absorbed by the tranche, C.

Step 3 – Breakeven

$sA + sB = C$, so $s = C/(A + B)$.
$\Rightarrow$ A describes the approach.

Interpretation

It is the CDS valuation of CR-13 b with the survival-weighted notional replaced by the expected tranche principal E_j.

Why the other options are wrong

B is incorrect. The average CDS spread is the index level; tranches are priced by seniority, not at the average.
C is incorrect. The approach values a tranche from a model of expected losses; it does not copy a market quote.
D is incorrect. Each tranche is priced on its own expected losses; there is no condition linking senior and equity losses.

LO: CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, eqs. 13.1 to 13.4

Solution CR-13 f.4 Answer: D

Bank Q589 • BT P2.T6.24.23.2

Step 1 – Conditional quantities

For each value of F: $Q(t \mid F)$, the binomial probabilities of k defaults, and from them the expected principal of each tranche on each payment date.

Step 2 – Unconditional quantities

Integrate $A(F)$, $B(F)$ and $C(F)$ over the standard normal distribution of F; then $s = C/(A + B)$.
$\Rightarrow$ D is the core step.

Interpretation

Example 13.2 does exactly this for the iTraxx 3 to 6% tranche with 60 quadrature points, giving $A = 4.2846$, $B = 0.0187$, $C = 0.1496$ and a 348 bp spread.

Why the other options are wrong

A is incorrect. The unconditional PD alone ignores correlation, which is what determines how losses are spread across tranches.
B is incorrect. The average CDS spread is the index level; it cannot price tranches of different seniority.

C is incorrect. Unconditional independence ignores correlation; independence holds only given F, which is why the integration is needed.

LO: CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, eqs. 13.5 to 13.12, Example 13.2

Solution CR-13 f.5 Answer: B

Bank Q590 • Schweser Module Quiz 31.3 Q1

Step 1 – Where the income comes from

The originator receives the CDS premiums on the reference portfolio: 2% of \$100 million, \$2 million a year.

Step 2 – How it is shared

The income is allocated to the tranches, with spread rates set by risk. In Schweser's example equity (0 to 5%) earns 10%, mezzanine (5 to 25%) 5.625% and senior 0.5%: $0.5+1.125+0.375 =$ \$2 million.

$\Rightarrow$ B is correct.

Interpretation

The junior tranche earns the highest rate because it absorbs the first losses; the senior tranche earns the least because losses must exceed 25% before it pays anything.

Why the other options are wrong

A is incorrect. Equal rates ignore the different risks of the tranches.

C is incorrect. The senior tranche earns the lowest rate; its size does not make it riskier.

D is incorrect. Reversed: the equity tranche earns the highest rate because it bears the first losses.

LO: CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. • **Reference:** SchweserNotes 2026 Book 2, Reading 31, LO 31.f; Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.8 Synthetic CDOs

Solution CR-13 f.6 Answer: C

New • built from GARP Ch.13, 13.9 and Practice Question 13.26

Step 1 – What correlation changes

Each name keeps the same hazard rate, so the portfolio's expected loss and the index spread are unchanged. Correlation changes how defaults cluster: few defaults and many defaults both become more likely.

Step 2 – Effect by tranche

More scenarios with few defaults protect the equity tranche; more scenarios with many defaults reach the senior tranches. With the January 2007 inputs (index 23 bp), the standard market model gives an equity upfront of 13.3% at $\rho = 0.10$ and 5.5% at $\rho = 0.30$, while the 12 to 22% spread rises from about 0 to 5 bp.

$\Rightarrow$ The equity tranche becomes less risky, the senior tranches riskier.

Interpretation

This is why tranche quotes carry information about correlation, and why a single implied correlation can be backed out of each tranche price (CR-13 g).

Why the other options are wrong

A is incorrect. Correlation does not change each name's PD; it redistributes losses between tranches.

B is incorrect. Polarity flip: the equity tranche gains from clustering and the senior tranches lose.

D is incorrect. The expected portfolio loss is unchanged, but its distribution across the tranches changes.

LO: CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. • **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.9 Role of Correlation in a Basket CDS and CDO; Practice Question 13.26

CR-13 • Credit Derivatives • Vardeez: MODERATE

CR-13 g

P3 CONCEPT

Define the two measures of implied correlation: compound (tranche) correlation and base correlation.

Workbook: D14 Credit Derivatives and Credit-Linked Notes • Tier D • Map #58 (2/16)

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D14 reports the two implied correlations as part of shape S4 and flags this objective as "worth knowing".
Vardeez: "LO 31.g: This can be imp too."

Reported exam shapes

- **D14·S4** Synthetic CDO valuation and the two implied correlations, compound and base (the valuation is CR-13 f).

What the questions look like: definitions (which tranche each correlation prices and how it is found), smile versus skew, what flat implied correlations would mean, how base correlations are built from compound ones, and how a nonstandard tranche is priced. The classic distractor swaps compound and base.

Worked elsewhere: D14.1 (option C swaps the two definitions) at Question CR-9 g.7; D15.1 (implied correlation from tranche prices) at Question CR-22 l.6; the tranche valuation behind every implied correlation under CR-13 f.

1. Compound and base correlation

In the standard market model (CR-13 f) the recovery rate is usually set at 40% and the hazard rate comes from the index spread, which leaves the copula correlation ρ as the **only unknown**. Just as traders imply a volatility from an option price in Black-Scholes-Merton, they imply a correlation from a tranche quote. For iTraxx Europe the tranches are $\{a_0, a_1\}, \{a_1, a_2\}, \ldots$ with $a_0 = 0$, $a_1 = 3\%$, $a_2 = 6\%$, $a_3 = 9\%$, $a_4 = 12\%$, $a_5 = 22\%$.

- **Compound (tranche) correlation** of the tranche $\{a_{q-1}, a_q\}$: the ρ for which the model spread equals the market spread of *that tranche*, found by iterative search.
- **Base correlation** for detachment point a_q: the ρ that prices the $\{0, a_q\}$ tranche consistently with the market. GARP's four steps: (1) find the compound correlation of each tranche; (2) use it to compute C_p, the PV of expected loss on tranche p as a share of its principal; (3) sum $\sum_{p \le q} C_p(a_p - a_{p-1})$, the PV of expected loss on $\{0, a_q\}$ as a share of the whole portfolio; (4) divide by a_q and search for the single ρ that gives this C-value for the $\{0, a_q\}$ tranche.

Vardeez's wording (base correlation "adjusts the tranche correlations ... for each tranche") is made precise by step 4: one base correlation prices the *whole* equity-like tranche from zero to a_q. For the first tranche the two measures coincide, because $\{0, a_1\}$ is both.

Detachment point a_q	3%	6%	9%	12%	22%
Compound correlation of $\{a_{q-1}, a_q\}$	17.7%	7.8%	14.0%	18.2%	23.3%
Base correlation of $\{0, a_q\}$	17.7%	28.4%	36.5%	43.2%	60.5%

GARP Table 13.8: 5-year iTraxx Europe, 31 January 2007 (index 23 bp, hazard rate 0.382%, rates 3%, $R = 40\%$).

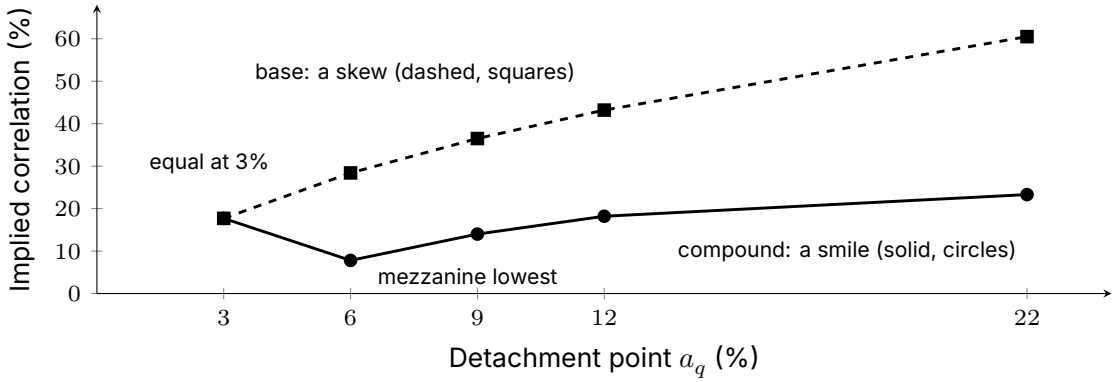


Figure 1. GARP Table 13.8. Compound correlations (solid line, circles) first fall and then rise with seniority, a correlation smile. Base correlations (dashed line, squares) rise steadily with the detachment point, a correlation skew. If market prices were consistent with the one-factor Gaussian copula, both lines would be flat at a single correlation.

2. What the patterns mean, and nonstandard tranches

- **The model is wrong, but useful as a quoting device.** Pronounced smiles and skews show that market prices are not consistent with the one-factor Gaussian copula. Implied correlations are a way of quoting prices, like implied volatilities.
- **Nonstandard tranches.** To quote a 4 to 8% tranche, one approach interpolates base correlations for 0 to 4% and 0 to 8%, computes the two expected losses and takes the difference. GARP prefers to interpolate the **expected losses** themselves: for no arbitrage, the PV of expected loss on the $\{0, X\}$ tranche must increase with X at a decreasing rate, and interpolated base correlations often violate this. The difference of expected losses then gives a compound correlation and a breakeven spread for the new tranche.

	Compound (tranche) correlation	Base correlation
Prices	One tranche, $\{a_{q-1}, a_q\}$	The tranche from zero, $\{0, a_q\}$
Found from	That tranche's market spread, by iterative search	Expected losses of all tranches up to a_q (from their compound correlations), then iterative search
Typical pattern	Smile: high for equity, lowest for mezzanine, rising for senior	Skew: rises with the detachment point
First tranche	17.7%	17.7% (the same tranche)

Exam traps

- **Definitions swapped (sibling swap).** Compound prices a single tranche; base prices the tranche from zero to a_q.
- **Shapes swapped.** Compound shows the smile, base the skew.
- **Historical estimate.** Both are implied from market prices, with $R = 40\%$ assumed; neither is a historical correlation of defaults or spreads.
- **Flat means consistent.** A consistent model would give the same correlation for every tranche; the patterns are evidence against the model.
- **Interpolation.** Interpolate expected losses, not base correlations, for nonstandard tranches.

Practice questions: CR-13 g

Question CR-13 g.1

Bank Q594 • Schweser Module Quiz 31.3 Q2

Base correlation and compound correlation are both:

A. tranche correlations.

B. implied correlations.

C. not relevant for synthetic CDOs.

D. exhibiting a skew with correlations rising with tranche seniority.

Question CR-13 g.2

Bank Q591 • AnalystPrep Q.6085 (rebuilt)

In pricing a synthetic CDO with the one-factor Gaussian copula model and a 40% recovery rate, the copula correlation is the only unknown input. The compound (tranche) correlation for a given tranche is best described as the correlation that:

A. makes the model-calculated spread for that tranche equal to the spread quoted in the market, found by an iterative search.

B. makes the model price of the tranche $\{0, a\}$ consistent with the market, using the cumulative expected loss up to the tranche's detachment point.

C. is estimated directly from historical co-movements in the credit spreads of the reference entities.

D. is set equal across all tranches so that the model spreads of all tranches reproduce the market spreads simultaneously.

Question CR-13 g.3

Bank Q592 • AnalystPrep Q.6086 (rebuilt)

A trader implies the base correlation for the detachment point a_q $(q > 1)$ of a synthetic CDO. Which statement best describes base correlation?

A. It is the correlation that makes the model spread of the $\{a_{q-1}, a_q\}$ tranche equal to its quoted spread.

B. It is the correlation that makes the $\{0, a_q\}$ tranche price consistently with the market, with that tranche's expected loss built from the compound correlations of the tranches up to a_q.

C. It is the correlation of the most senior tranche, applied to every junior tranche beneath it.

D. It is the average of the compound correlations of all the tranches in the capital structure.

Question CR-13 g.4 Bank Q593 • BT P2.T6.24.23.3 (rebuilt)

Global Credit Solutions is reviewing implied correlations for its synthetic CDO tranches. Which of the following statements are correct?

i. The compound correlation of a tranche is the correlation at which the model spread equals the market-quoted spread for that tranche, found by an iterative search.

ii. The base correlation for detachment point a_q is the correlation that prices the $\{0, a_q\}$ tranche consistently with the market, using expected losses obtained from the compound correlations of the tranches.

iii. If market prices were consistent with the one-factor Gaussian copula model, the implied correlations would be the same for all tranches.

iv. Compound correlations exhibit a skew rising with seniority, while base correlations exhibit a smile.

A. i only

B. i and ii only

C. i, ii and iii only

D. ii, iii and iv only

Question CR-13 g.5 New • built from GARP Ch.13, 13.10 (Implied Correlation, Table 13.8)

A correlation trader at a London bank reviews GARP's implied correlations for 5-year iTraxx Europe tranches on 31 January 2007:

Detachment point	3%	6%	9%	12%	22%
Compound correlation	17.7%	7.8%	14.0%	18.2%	23.3%
Base correlation	17.7%	28.4%	36.5%	43.2%	60.5%

Which statement is correct?

A. The base correlations form a smile: they fall from the equity tranche to the 0 to 6% tranche and then rise.

B. The 0 to 6% tranche has a base correlation of 7.8%, the lowest in the capital structure.

C. The market is pricing every tranche with the one-factor Gaussian copula and a single correlation of about 18%, the equity tranche's correlation.

D. The compound correlations form a smile, lowest for the 3 to 6% tranche, while the base correlations rise steadily with the detachment point; the two coincide at 3%, where the tranche is both a single tranche and a base tranche.

Question CR-13 g.6 New • built from GARP Ch.13, 13.10 (Valuing Nonstandard Tranches)

A dealer is asked to quote the 4 to 8% tranche of iTraxx Europe, which is not one of the standard tranches. Market quotes exist for all the standard tranches. Which approach does GARP describe as the better one?

A. Interpolate the PV of expected loss on the 0 to X% tranche across the standard detachment points to estimate it for 0 to 4% and 0 to 8%; take the difference as the expected loss on the 4 to 8% tranche, then imply its compound correlation and breakeven spread.

B. Average the compound correlations of the 3 to 6% and 6 to 9% tranches and use the result as the 4 to 8% tranche's correlation.

C. Interpolate base correlations at 4% and 8% and use them directly, because interpolated base correlations always satisfy the no-arbitrage condition on expected losses.

D. Quote the index spread for the 4 to 8% tranche, because a tranche straddling two standard tranches has the same risk as the index.

Solutions • CR-13 g

Solution CR-13 g.1

Answer: B

Bank Q594 • Schweser Module Quiz 31.3 Q2

Step 1 – How each is obtained

Both are found by searching for the copula correlation that makes the model reproduce market prices: compound for a single tranche, base for the $\{0, a_q\}$ tranche.
$\Rightarrow$ Both are implied correlations.

Interpretation

Implied correlations are to tranche quotes what implied volatilities are to option prices: a way of quoting a price through the one unknown parameter of a standard model.

Why the other options are wrong

A is incorrect. Only the compound correlation is a tranche correlation; base correlation prices the tranche from zero to a_q.
C is incorrect. Both exist only because synthetic CDO tranches are quoted.
D is incorrect. Only base correlations show a skew rising with seniority; compound correlations show a smile.

LO: CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation.
• **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, Implied Correlation

Solution CR-13 g.2

Answer: A

Bank Q591 • AnalystPrep Q.6085

Step 1 – The one unknown

With $R = 40\%$ and the hazard rate from the index spread, ρ is the only free input.

Step 2 – Compound correlation

For the tranche $\{a_{q-1}, a_q\}$, the value of ρ at which the model spread equals the market spread, found by iterative search.
$\Rightarrow$ A is the definition.

Interpretation

Each tranche gets its own compound correlation; together they trace the smile (17.7%, 7.8%, 14.0%, 18.2%, 23.3% for iTraxx in January 2007).

Why the other options are wrong

B is incorrect. Sibling swap: that is the definition of base correlation.
C is incorrect. Implied correlations come from current market prices, not from historical spread co-movements.
D is incorrect. If one correlation priced every tranche there would be no smile; in practice each tranche needs its own.

LO: CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation.
• **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, Implied Correlation

Solution CR-13 g.3

Answer: B

Bank Q592 • AnalystPrep Q.6086

Step 1 – Build the expected loss on $\{0, a_q\}$

Use each tranche's compound correlation to get its expected loss C_p, then sum $\sum_{p \le q} C_p(a_p -$

a_{p-1}) and divide by a_q.

Step 2 – Imply one correlation

Search for the single ρ that gives this expected loss for the $\{0, a_q\}$ tranche.
$\Rightarrow$ B is the definition.

Interpretation

Because each base tranche starts at zero, base correlations rise monotonically with a_q (17.7% to 60.5%), which makes them easier to interpolate than compound correlations, although GARP warns that interpolating them can break no-arbitrage.

Why the other options are wrong

A is incorrect. Sibling swap: that is the compound correlation of a single tranche.
C is incorrect. No correlation is carried down from the senior tranche; each base correlation is implied separately.
D is incorrect. Base correlation is implied by a search, not averaged; and it uses only the tranches up to a_q.

LO: CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation.
• **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, Implied Correlation, steps 1 to 4

Solution CR-13 g.4 Answer: C

Bank Q593 • BT P2.T6.24.23.3

Step 1 – Statements i and ii

Both restate GARP's definitions: compound for one tranche by iterative search; base for $\{0, a_q\}$ using expected losses from the compound correlations.

Step 2 – Statement iii

GARP: if market prices were consistent with the one-factor Gaussian copula model, the implied correlations (compound and base) would be the same for all tranches. True.

Step 3 – Statement iv

Reversed: compound correlations show the smile, base correlations the skew.
$\Rightarrow$ i, ii and iii only.

Interpretation

The smile and skew are the market's verdict on the model, in the same way that the volatility smile is the verdict on Black-Scholes-Merton.

Why the other options are wrong

A is incorrect. Omits ii and iii, which are both true.
B is incorrect. Omits iii, the consistency statement.
D is incorrect. Excludes i, a correct definition, and includes iv, which swaps smile and skew.

LO: CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation.
• **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, Implied Correlation, Table 13.8

Solution CR-13 g.5 Answer: D

New • built from GARP Ch.13, 13.10

Step 1 – Read the compound row

17.7%, then 7.8% for 3 to 6%, then rising to 23.3%: down then up, a smile with its low point at the mezzanine tranche.

Step 2 – Read the base row

17.7%, 28.4%, 36.5%, 43.2%, 60.5%: rising with the detachment point, a skew. At 3% the tranche $\{0, 3\%\}$ is both the first tranche and a base tranche, so both measures are 17.7%.
$\Rightarrow$ D is correct.

Interpretation

GARP: the compound correlations exhibit a "correlation smile" and the base correlations a "correlation skew". Both would be flat if the market priced with the model.

Why the other options are wrong

A is incorrect. Shapes swapped: the base correlations rise throughout; the smile belongs to the compound row.
B is incorrect. 7.8% is the compound correlation of the 3 to 6% tranche; the base correlation for 0 to 6% is 28.4%.
C is incorrect. A single correlation would make both rows flat; the patterns show the market is not pricing with one correlation.

LO: CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation.
• **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, Implied Correlation, Table 13.8

Solution CR-13 g.6 Answer: A

New • built from GARP Ch.13, 13.10

Step 1 – The traditional approach and its flaw

Interpolating base correlations to 0 to 4% and 0 to 8% and computing expected losses from them often violates the no-arbitrage condition: the PV of expected loss on $\{0, X\}$ must increase with X at a decreasing rate. The base correlation is a nonlinear function of that expected loss.

Step 2 – The better approach

Compute the expected loss on each standard $\{0, X\}$ tranche, interpolate the expected losses directly (which can be done so as to respect the condition), take the 0 to 8% minus the 0 to 4% value, then imply a compound correlation and breakeven spread.
$\Rightarrow$ A is the approach GARP prefers.

Interpretation

Interpolate the quantity that carries the economics (expected loss), not the model parameter that maps to it nonlinearly.

Why the other options are wrong

B is incorrect. Averaging compound correlations of tranches with different attachment points has no pricing basis, and compound correlations are not monotonic.
C is incorrect. Absolute qualifier reversed: GARP says interpolated base correlations often fail the no-arbitrage condition.
D is incorrect. The index spread prices protection on the whole portfolio; a mezzanine tranche has a very different risk.

LO: CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation.
• **Reference:** Hull, OFOD 11e, Ch.25 (GARP CR Ch.13), 13.10, Valuing Nonstandard Tranches

CR-14 • Derivatives • Vardeez: NOT SO IMP

CR-14 i

P3 CONCEPT

Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods.

Workbook: none homed here (the reading's only section, B2, covers CR-14 j)

WHAT IS TESTED HERE

Why P3. Vardeez flags this one LO in a NOT SO IMP chapter: "Counterparty risk intermediaries: these can be important from the exam point of view". No workbook question is homed here and the map lists no shapes.
Vardeez: counterparty risk intermediaries.

What the questions look like: conceptual. Match a description (asset isolation, separate capital, credit wraps, unhedged protection selling) to the right entity; pick the true statement about a limitation, or the false one in an EXCEPT list; say what a CCP does that these structures did not.

1. Four ways to manufacture a better counterparty

Before central clearing, the market tried to create counterparties of higher quality than the institutions behind them. Each structure moved risk somewhere else, and that is where each one failed.

Special purpose vehicle (SPV, or SPE). A legal entity (a company or limited partnership) created to isolate a firm from financial risk: assets are transferred to it, or it finances a large project, without putting the whole firm or a counterparty at risk; jurisdictions may require that it is not owned by the firm for which it is set up. It aims to change bankruptcy rules so that, if a derivative counterparty is insolvent, the client still receives its full investment before other claims are paid. Typical use: structured notes, protecting the principal to a triple-A level, better than the issuer's own rating. *Limitation:* it **transforms counterparty risk into legal risk**. A bankruptcy court may **consolidate** the SPV's assets with the originator's (US courts have a history of this; UK courts less so). Lehman's SPV "flip" clauses, which put note investors first, were ruled unenforceable in the US and enforceable in the UK. And priority for one party means less for others: risk is moved, not removed.

Derivatives product company (DPC). A triple-A-rated, bankruptcy-remote subsidiary of a major dealer which, unlike an SPV, is **separately capitalised** to earn its rating, protecting its counterparties against failure of the parent (e.g. Merrill Lynch Derivative Products, Salomon Swapco). The rating rests on minimal market risk (mirror trades with the parent, a matched book), parent support (on the parent's default the DPC passes to another institution or is terminated at mid-market, a "pre-packaged bankruptcy"), and operating rules (counterparty limits, daily MtM and margining). *Limitation:* is a triple-A subsidiary of a double-A bank really a better counterparty than the bank? In the crisis the Bear Stearns DPCs were wound down by J.P. Morgan and two Lehman DPCs filed for Chapter 11: the DPC's fate was tied to its parent, and counterparty risk had become legal, market and operational risk.

Monolines. Financial guarantee insurers with triple-A ratings that sold "credit wraps" on bonds, then moved into single-name CDS and structured finance. Because of their ratings they did not post collateral in normal times. *Limitation:* downgrade clauses **triggered collateral posting** exactly as MtM losses on the protection sold mounted. Once downgrades began they were swift, and the collateral calls essentially sent them into default (from November 2007 XL Financial Assurance, AMBAC and MBIA essentially failed); AIG, the same kind of protection seller, was bailed out for about USD 182bn.

Protection buyers carried strong **wrong-way risk**: the protection was worth most when the seller was weakest.

Credit derivative product company (CDPC). The DPC structure with a monoline's business model: it sells credit protection, mainly CDS, for premium income. *Limitation:* unlike a DPC it does not hedge with offsetting trades, so it is not market neutral; it is **highly leveraged**; like a monoline it posts little or no collateral and relies on its rating. CDPCs fared somewhat better than monolines only for timing reasons (many were not operational until after July 2007), and in October 2008 Fitch withdrew its ratings on the five CDPCs it rated.

Correction to the source notes. Vardeez writes that "during and after the crisis many CDPCs experienced large losses and failed" (Schweser: "many CDPCs failed during or shortly after the global financial crisis"). GARP says something different: CDPCs "fared somewhat better during the GFC, but only for timing reasons", because many were not fully operational until after July 2007; the warning sign it cites is Fitch withdrawing ratings on the five CDPCs it rated (October 2008). The limitation to learn is the business model (leverage, no hedge, no collateral, reliance on ratings), not a record of failures.

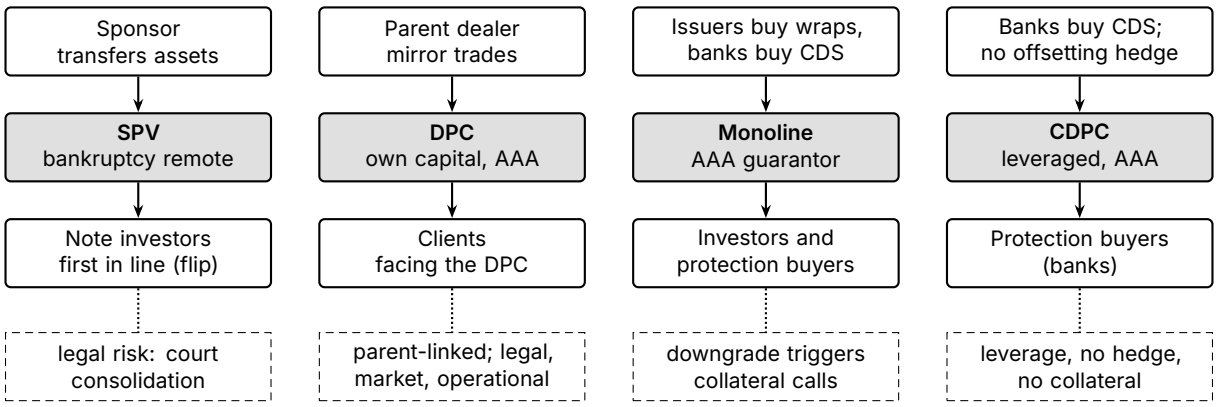


Figure 1. The four structures (entity in the grey box) and, in the dashed boxes, where the counterparty risk went. None removed the risk: the SPV and DPC turned it into legal and parent risk; the monoline and CDPC concentrated one-way credit risk in a leveraged entity whose protection depended on a rating.

2. The common flaw, and how a CCP differs

GARP argues that, as risk mitigants, all four "have fundamental flaws", which is why there is little evidence of them today. A CCP shares some of their features, so the lessons carry over. Like an SPV or DPC, it gives derivative counterparties priority at the expense of other creditors and depends on a legal framework that will be challenged. Unlike monolines and CDPCs, it runs a **matched book** and requires **variation and initial margin in all situations**, and it mutualises a member's failure across the membership instead of concentrating it in one entity. But it centralises risk in a large, possibly too-big-to-fail entity, sharp margin increases in a crisis could recreate the monoline dynamic, and "systemic risk insurance" is a misnomer: systemic risk cannot be diversified.

	SPV	DPC	Monoline	CDPC
What it is	Bankruptcy-remote legal entity	Separately capitalised AAA dealer subsidiary	AAA financial guarantor	DPC structure, monoline business
Hedged?	n/a	Yes (matched book)	No	No
Collateral?	n/a	Daily margining	No	No
Key limitation	Legal risk: consolidation	Fate tied to parent	Downgrade triggers collateral	Leverage, no hedge
Crisis	Lehman flip clauses disputed	Bear wound down, Lehman DPCs in Chapter 11	AMBAC, MBIA, XL failed; AIG bailed out	Fitch withdrew ratings

Exam traps

- **SPV versus DPC (sibling swap).** Asset isolation, project finance and flip clauses are SPV features; separate capital and mirror trades are DPC features.
- **DPC versus CDPC.** The DPC hedges (matched book); the CDPC does not. Monolines and CDPCs differ in structure, not in their lack of hedging or collateral.
- **Purpose of DPCs.** They were created so that institutions below triple-A could trade long-dated derivatives through a triple-A vehicle; that is not the CDPC's purpose.
- **AAA means no risk (absolute claim).** A triple-A rating reduced, never removed, counterparty risk, and the rating could disappear quickly.
- **Monolines posted collateral.** Not in normal times; downgrade clauses forced it at the worst moment.

Practice questions: CR-14 i

Question CR-14 i.1

Bank Q649 • AnalystPrep Q.6174

A legal entity, typically a company or limited partnership, is created to isolate a firm from financial risk. A firm may transfer assets to it for management or use it to finance a large project. Jurisdictions may require that it is not owned by the firm on whose behalf it is set up. It aims to change bankruptcy rules so that, if a counterparty is insolvent, a client can receive its full investment before other claims are paid, which necessarily imposes a less favourable position on other parties. It is commonly used in structured notes to guarantee counterparty risk on the note's principal to a very high (typically triple-A) level, and rating agencies examine its mechanics and legal specifics before granting a rating. Which entity is described?

A. Credit derivative product company (CDPC)

B. Monoline insurance company

C. Derivatives product company (DPC)

D. Special purpose vehicle (SPV)

Question CR-14 i.2 Bank Q647 • AnalystPrep Q.6172 (rebuilt)

Derivatives product companies (DPCs) were developed to mitigate counterparty risk in OTC derivatives markets. Which of the following statements about DPCs is correct?

A. DPCs primarily change bankruptcy rules so that clients receive their full investment ahead of other claims if a counterparty becomes insolvent.

B. DPCs hold no capital of their own; their triple-A rating is always taken directly from the rating of the sponsoring parent.

C. DPCs are legal entities created to manage assets and finance large projects, isolating the entire firm from financial risk.

D. DPCs are typically triple-A-rated, separately capitalised, bankruptcy-remote subsidiaries of a major dealer, which give counterparties a degree of protection against failure of the DPC parent.

Question CR-14 i.3 Bank Q648 • AnalystPrep Q.6173

Monoline insurers were established as financial guarantee companies with strong credit ratings and provided "credit wraps" to enhance the credit quality of obligations. They later entered single-name credit default swaps (CDS) and structured finance, and ran into major difficulties during the global financial crisis. Which of the following statements about monolines is most accurate?

A. Monolines were legal entities created to manage assets and finance large projects, isolating the entire firm from financial risk.

B. The primary function of monolines was to act as structured entities designed to mitigate counterparty risk in OTC derivatives markets.

C. Monolines were financial guarantee companies that provided "credit wraps", and later also sold protection in the CDS and structured finance markets.

D. Monolines evolved to allow counterparties with less than triple-A credit quality to trade long-dated derivatives.

Question CR-14 i.4 Bank Q646 • AnalystPrep Q.6171

Within the OTC derivatives market, firms have used various structures to mitigate counterparty risk and manage credit exposure, including special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs). Which of the following statements is correct?

A. DPCs are legal entities created primarily for asset management and to finance large projects without exposing the entire firm to financial risk.

B. Monolines and CDPCs are primarily used to change bankruptcy rules so that clients receive their full investment if a counterparty becomes insolvent.

C. SPVs are designed to isolate a firm from financial risk, typically by transferring assets to the SPV for management or by using it to finance a large project without risking the entire firm.

D. CDPCs evolved to allow counterparties with less than triple-A credit quality to trade long-dated derivatives.

Question CR-14 i.5 Bank Q652 • BT P2.T6.319.3 (rebuilt)

Which of the following statements about triple-A-rated monoline insurers is correct?

A. The triple-A rating effectively eliminates counterparty risk to a monoline.

B. The triple-A rating implies that the monoline must post collateral against declines in the market value of its contracts.

C. In normal times monolines did not have to post collateral because of their triple-A ratings, but rating-downgrade clauses could force collateral posting just when their losses were rising, so counterparties depended on the rating.

D. CDPCs are safer than monolines because they exhibit right-way risk and do not create systemic risk.

Question CR-14 i.6 Bank Q653 • BT P2.T6.323.2 (rebuilt)

Each of the following is a limitation of using special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines or credit derivative product companies (CDPCs) as counterparty risk mitigants, EXCEPT:

A. An SPV transforms counterparty risk into legal risk, since a bankruptcy court may consolidate the SPV's assets with those of the sponsor.

B. A DPC's protection proved weaker than expected because its fate was linked to its parent: the Bear Stearns DPCs were wound down by JPMorgan after the parent's rescue, two Lehman DPCs filed for Chapter 11, and DPCs also carry market and operational risk.

C. Monolines and CDPCs did not post collateral in normal times, but downgrade clauses forced posting just as their insured losses rose.

D. DPCs reliably achieved an orderly, pre-packaged wind-down when their parents failed in the global financial crisis, removing counterparty risk for their clients.

Question CR-14 i.7 Bank Q654 • Schweser Module Quiz 32.2 Q3

Which of the following statements is an enhancement offered by the central counterparty (CCP) structure relative to the special purpose vehicle (SPV), the derivative product company (DPC), and the monoline insurance models? The CCP structure:

A. enables financial institutions to remove assets from their balance sheets.

B. enables counterparty risk to be outsourced, but in a nondiversified format.

C. spreads losses over a group of counterparties to minimize potential systemic risk.

D. enables a counterparty transaction originator to fail and not affect the other member firms.

Solutions • CR-14 i

Solution CR-14 i.1 Answer: D

Bank Q649 • AnalystPrep Q.6174

Step 1 – Pick out the clues

Asset transfer or project finance; not owned by the sponsor; a change to bankruptcy priority so that a client is paid first; use in structured notes to reach triple-A; agencies rating the legal mechanics.

Step 2 – Match the entity

These are the defining features of a special purpose vehicle (or special purpose entity).
$\Rightarrow$ D.

Interpretation

The phrase "necessarily imposes a less favourable position on other parties" is GARP's point that an SPV moves risk rather than removing it. Its own limitation follows from the same feature: because its value rests on the legal structure, it converts counterparty risk into legal (consolidation) risk.

Why the other options are wrong

A is incorrect. A CDPC sells credit protection with a monoline-like model; it does not rearrange bankruptcy priorities.

B is incorrect. A monoline sells financial guarantees (credit wraps); it is an insurer, not an asset-isolating vehicle.

C is incorrect. A DPC is a separately capitalised dealer subsidiary used as a triple-A derivatives counterparty, not a priority-shifting structure for notes.

LO: CR-14 i: Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.4 Special Purpose Vehicles

Solution CR-14 i.2 Answer: D

Bank Q647 • AnalystPrep Q.6172

Step 1 – What a DPC is

A bankruptcy-remote subsidiary of a major dealer, set up by one or more banks, which, unlike an SPV, is separately capitalised to obtain a triple-A rating.

Step 2 – What it protects against

Its counterparties are partly protected from the failure of the parent, with the rating maintained by capital, mirror trades, margin and activity restrictions.
$\Rightarrow$ D.

Interpretation

The limitation sits inside the definition: protection against the parent's failure depends on the DPC really being independent of it. In the crisis two Lehman DPCs filed for Chapter 11 alongside their parent, and the Bear Stearns DPCs were wound down.

Why the other options are wrong

A is incorrect. Rearranging bankruptcy priority for investors (flip clauses) is the SPV mechanism.

B is incorrect. DPCs are separately capitalised; GARP notes only that "some" derived their quality simply from the sponsor's rating, so "always" and "no capital" are wrong.

C is incorrect. Asset management and project finance with risk isolation describe an SPV.

LO: CR-14 i: Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.4.3 Derivatives Product Companies

Solution CR-14 i.3 Answer: C

Bank Q648 • AnalystPrep Q.6173

Step 1 – Monolines' business

Single-line financial guarantee companies with strong (typically triple-A) ratings that sold credit wraps, then entered single-name CDS and structured finance in search of diversification and better returns.

Step 2 – Eliminate the borrowed descriptions

A describes an SPV, B a DPC-style structure built to reduce counterparty risk, D the original purpose of DPCs.
$\Rightarrow$ C.

Interpretation

Moving from wrapping bonds to selling CDS on structured credit gave monolines a large one-way exposure to exactly the assets that collapsed in 2007, with protection buyers relying on their rating rather than on collateral.

Why the other options are wrong

A is incorrect. The SPV description (asset transfer, project finance, risk isolation).
B is incorrect. Monolines are guarantors that take on credit risk for a premium, not bankruptcy-remote structures designed to reduce counterparty risk.
D is incorrect. Letting sub-triple-A institutions trade long-dated derivatives was the motivation for DPCs.

LO: CR-14 i: Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.4.4 Monolines and CDPCs

Solution CR-14 i.4 Answer: C

Bank Q646 • AnalystPrep Q.6171

Step 1 – Attach each description to its owner

Asset management and project finance with risk isolation: SPV. Changing bankruptcy rules so that clients are paid in full: SPV. Allowing sub-triple-A institutions to trade long-dated derivatives: DPC.

Step 2 – Find the statement that attaches it correctly

Only C gives the isolation role to the SPV.
$\Rightarrow$ C.

Interpretation

This question is a matching exercise across all four entities, and each wrong option attaches a true feature to the wrong entity. Learn one defining phrase each: SPV, legal isolation; DPC, separate capital and mirror trades; monoline, credit wraps; CDPC, unhedged protection selling.

Why the other options are wrong

A is incorrect. Gives the SPV's role to a DPC; a DPC is a separately capitalised triple-A dealer subsidiary.

B is incorrect. Shifting bankruptcy priority is an SPV feature; monolines and CDPCs are leveraged protection sellers.
D is incorrect. That was the purpose of DPCs; CDPCs extended the DPC structure to a monoline-like business of selling credit protection.

LO: CR-14 i: Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.4, 14.4.3 and 14.4.4

Solution CR-14 i.5 Answer: C

Bank Q652 • BT P2.T6.319.3

Step 1 – Collateral in normal times

Monolines and CDPCs typically did not post collateral against falls in the market value of their contracts, because of their excellent ratings.

Step 2 – What happened in the crisis

MtM losses on protection sold raised doubts about the ratings; downgrade clauses then demanded collateral the monolines could not meet, and once agencies began downgrading, the decline was swift.
⇒ Protection buyers were relying on a rating that could vanish: C.

Interpretation

This is wrong-way risk at its starkest: the protection was worth most to the buyer exactly when the seller was weakest, and the rating trigger turned that weakness into a liquidity crisis. AIG followed the same path and was bailed out.

Why the other options are wrong

A is incorrect. A triple-A rating lowers but does not eliminate counterparty risk; it collapsed for the monolines.
B is incorrect. The rating is the reason monolines did *not* post collateral in normal times; posting came only after a downgrade.
D is incorrect. CDPCs had a monoline-like model, highly leveraged and unhedged, carrying wrong-way risk for their protection buyers, not right-way risk; Fitch withdrew its ratings on the five it rated in October 2008.

LO: CR-14 i: Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.4.4 Monolines and CDPCs

Solution CR-14 i.6 Answer: D

Bank Q653 • BT P2.T6.323.2

Step 1 – Test A

An SPV turns counterparty risk into legal risk through possible consolidation; US courts have a history of consolidation rulings. A true limitation.

Step 2 – Test B

The Bear Stearns DPCs were wound down by J.P. Morgan and two Lehman DPCs filed for Chapter 11; DPCs carry legal, market and operational risk. A true limitation.

Step 3 – Test C

Monolines and CDPCs posted no collateral in normal times; downgrade clauses forced posting as losses rose. A true limitation.

Step 4 – Test D

The orderly, pre-packaged workout was what DPCs promised; it did not happen when their parents failed.

⇒ D is the statement that is not a limitation (and is false).

Interpretation

Each true statement names the specific risk the structure converted counterparty risk into: legal risk (SPV), parent and operational risk (DPC), and rating-triggered liquidity risk (monoline and CDPC).

Why the other options are wrong

A is incorrect. A genuine limitation of SPVs.

B is incorrect. A genuine limitation of DPCs, borne out in the crisis.

C is incorrect. A genuine limitation of monolines and CDPCs.

LO: CR-14 i: Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.4, 14.4.3 and 14.4.4

Solution CR-14 i.7

Answer: C

Bank Q654 • Schweser Module Quiz 32.2 Q3

Step 1 – What each older structure does

The SPV isolates assets off the originator's balance sheet; the DPC is a separately capitalised subsidiary that survives the parent's failure; the monoline takes counterparty risk onto one insurer, undiversified.

Step 2 – What the CCP adds

Margin, a default fund and loss mutualisation: a member's default is absorbed across all clearing members instead of resting on one entity.

⇒ C.

Interpretation

GARP's comparison adds two structural differences from monolines and CDPCs: a CCP runs a matched book, and it collects variation and initial margin in all situations. The caveat is that the CCP itself becomes a concentration of risk, and its margin calls can be procyclical.

Why the other options are wrong

A is incorrect. Removing assets from the balance sheet describes the SPV.

B is incorrect. Outsourcing counterparty risk in a non-diversified form describes the monoline; the CCP does the opposite.

D is incorrect. Surviving the originator's failure describes the DPC; in a CCP a member's default does reach the other members, through the default fund.

LO: CR-14 i: Define special purpose vehicles (SPVs), derivatives product companies (DPCs), monolines, and credit derivatives product companies (CDPCs) and describe the limitations of using them as risk mitigating methods. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), CCPs in Context; SchweserNotes 2026 Book 2, Reading 32, LO 32.i

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 a

P3 CONCEPT

Describe counterparty risk and differentiate it from lending risk.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No recall workbook section covers CR-15, but Vardeez flags all eight LOs: this is "the foundational chapter for the counterparty risk chapters, so give it a good read and understand the points". The background for the whole reading sits here; later LOs recap it and point back. **Vardeez:** the rows of the comparison table that decide most questions are the amount at risk (known for a loan, uncertain for a derivative) and who bears it (unilateral against bilateral).

Reported exam shapes: none listed in the Master Map. The bank questions test three things: the definition (bilateral, uncertain, a credit risk whose size is set by market variables), the loan-against-swap contrast, and pre-settlement against settlement risk.

What the questions look like: all qualitative: "which statement is correct" or "true EXCEPT", often built on a firm that has both a loan and a swap with the same bank. Distractors swap the two columns of the comparison table or add an absolute qualifier ("only", "no relationship", "eliminated").

1. What counterparty risk is

Counterparty credit risk, usually just counterparty risk, is the risk that the entity with whom one has entered into a financial contract will fail to fulfil its side of the agreement, for example by defaulting (GARP 15.1). As with all credit risk, the loss comes from an obligor that is unable or unwilling to pay. What is new is the size of the claim: it is the *future value of a contract*, which can turn out positive or negative.

GARP places counterparty risk mainly in **OTC derivatives that are bilaterally cleared and uncollateralised**. Four other cases carry *less* of it, because they combine margining, central clearing and short maturities: exchange-traded derivatives (always centrally cleared, typically margined daily), securities financing transactions such as repos (margined daily, sometimes cleared), centrally cleared OTC derivatives (margined daily) and collateralised bilateral OTC derivatives (bespoke collateral agreement). Less is not none. A small risk per unit can sit on a very large position: a bank's notional facing a central counterparty (CCP) can be extremely large (see the correction box in CR-15 b).

2. Counterparty risk versus lending risk

Traditional credit risk is **lending risk**: one party owes an amount to another and may fail to pay some or all of it because of insolvency. GARP names two features of lending risk, and counterparty risk differs on both (GARP 15.1.1):

	Lending risk	Counterparty risk
Products	Loans, bonds, mortgages, credit cards	OTC derivatives (swaps, FX forwards, CDSs) and securities financing (repos, securities lending)
Amount at risk	Known with a degree of certainty: a bond is close to par; an amortising mortgage's balance is predictable; a card line can be assumed fully drawn	Uncertain: the net value of all future cash flows at the default date, positive or negative, usually a fraction of notional
Who bears it	Unilateral: only the lender (a bond issuer loses nothing if the buyer defaults)	Bilateral: whichever party is in the money at default is exposed; the two sides can be unequal
Market variables	Moderate effect on the amount owed	Set the exposure: market risk sizes it, credit risk triggers it
Measurement	Balance or notional	Simulation of future values (PFE, EE: CR-19)

Bilateral, but not always symmetric. Each party has risk to the other, yet the two risks can differ: a bought option with the premium paid upfront creates counterparty risk only for the buyer (CR-15 b), and parties with different credit quality or different collateral terms face different risks.

One client, two products. A firm borrows from a bank at a floating rate and swaps the loan to fixed with the same bank (GARP 14.2.5). On the loan the bank is owed roughly the principal and only the bank can lose. On the swap only net coupons change hands, and the value moves with rates, so it can become an asset to either side. The two trades also sit in different books for capital: the loan in the banking book, the swap in the trading book.

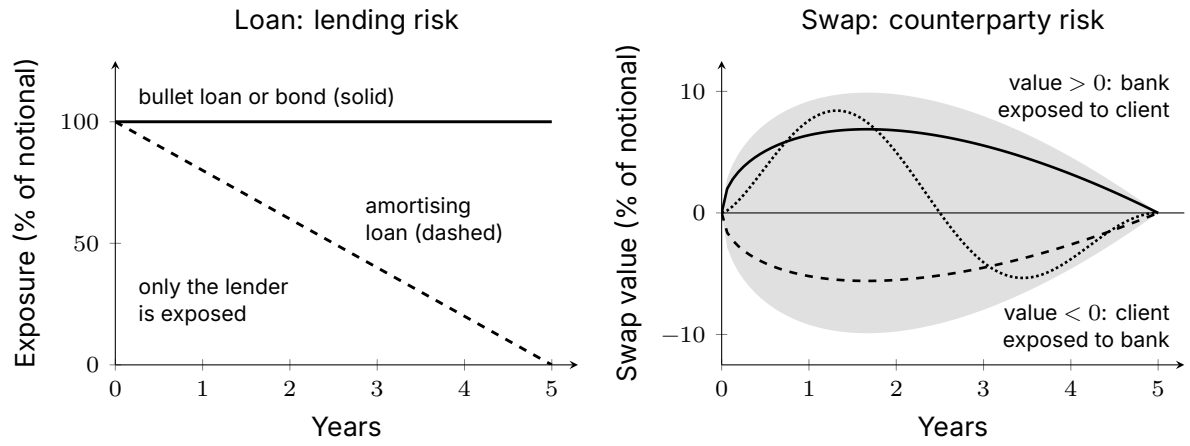


Figure 1. Left: the lender's exposure is known in advance and one-way. Right: three illustrative value paths of a five-year swap (solid, dashed, dotted) inside a grey band of likely values. The value starts near zero, can move to either side of the zero line, returns to zero at maturity, and stays a small fraction of notional; note the different vertical scales.

Market risk and credit risk together. The default is the credit event; the market decides how much is owed at that moment. Counterparty risk is therefore a credit risk sized by market variables, not one or the other (Questions CR-15 a.2 and a.3).

3. Pre-settlement and settlement risk

- **Pre-settlement risk:** the counterparty defaults before the final settlement of the transaction. This is what "counterparty risk" normally means.
- **Settlement risk:** arises at settlement times, especially maturity, from timing differences between when each party performs its obligations.

	Pre-settlement risk	Settlement risk
When	Any time over the life of the trade	At settlement, for hours or days
Exposure	The value difference, a fraction of notional	Up to 100% of the notional paid
Chance of default in the window	Substantially higher	Low, because the window is short
Dominant in	Long-dated OTC derivatives	Spot and short-dated trades
Controls	Counterparty limits, CVA	Separate settlement limits; CLS (payment versus payment) for FX; delivery versus payment for securities

GARP's FX example. A forward exchanges EUR 1m for USD 1.1m. Settlement risk is the full USD 1.1m if the euros are paid and the dollars never arrive, but only on the settlement day. Pre-settlement risk is only the value difference between the two legs, and it can bite at any time. **Herstatt** (26 June 1974) is the classic settlement failure: the bank was closed after the German interbank payment system shut at 3:30 pm local time, when it was 10:30 am in New York, so counterparties that had paid Deutschmarks never received their dollars.

Margin period of risk (MPoR). Margined positions still carry risk while collateral is awaited: 10 business days is standard for collateralised bilateral OTC derivatives, 5 for centrally cleared ones (CR-17).

> **Correction to the source notes.** In the FX example GARP (15.1.2) says a move in the rate from 1.10 to 1.15 "would translate into a loss of about $45,000". The value difference on EUR 1m is $1{,}000{,}000 \times (1.15 - 1.10) = \text{USD } 50{,}000$ (before any discounting to the settlement date). GARP's point is unaffected: tens of thousands of dollars of pre-settlement exposure against USD 1.1m of settlement exposure.

Exam traps

- **"Counterparty risk is market risk."** It is a credit risk: a default must occur. Market variables only size the claim.
- **Column swap.** Known amount and one-way belong to lending; uncertain amount and two-way belong to counterparty risk.
- **Exposure equals notional.** A swap exchanges no principal; its exposure is the value of the remaining net flows, a fraction of notional.
- **Bilateral means equal.** Each side can be exposed, but not necessarily by the same amount (upfront premium, different collateral terms).

- **A negative value today means no risk.** A contract that is a liability today can become an asset before the counterparty defaults.
- **Settlement against pre-settlement.** Settlement risk is large and brief; pre-settlement risk is smaller and lasts the whole life, so default before settlement is far more likely.
- **"Eliminated".** Choosing a strong counterparty or a prime broker does not remove counterparty risk; clearing and margin reduce it.

Practice questions: CR-15 a

Question CR-15 a.1

Bank Q659 • AnalystPrep Q.1860 (rebuilt)

A junior analyst is drafting a glossary entry on counterparty risk for an OTC derivatives desk. Which of the following statements is correct?

A. Counterparty risk is unilateral: only the party that is owed money at inception bears it.

B. Counterparty risk is bilateral: the contract value can become positive or negative, so either party can end up exposed to the other.

C. Counterparty risk is symmetric by construction: both parties always carry the same exposure to each other.

D. The future value of the contract is known at inception, so only the default probability is uncertain.

Question CR-15 a.2

Bank Q658 • BT P2.T6.302.3 (rebuilt)

A risk manager is explaining counterparty risk to a new member of the credit committee. Which of the following statements is correct?

A. A party can lose from its counterparty's default only if the contract has a positive value to it at the default date, and that value is driven by market variables.

B. A party can lose from its counterparty's default only if the contract has a negative value to it at the default date.

C. Because counterparty risk is a credit risk, it has no relationship to market risk.

D. Because counterparty risk is a market risk, it has no relationship to default risk.

Question CR-15 a.3

Bank Q661 • BT P2.T6.316.1

Consider the following assertions about the differences between lending risk and counterparty risk:

I. Lending risk is a type of credit risk, but counterparty risk is a type of market risk.

II. Lending exposure is somewhat predictable, but counterparty credit exposure is highly uncertain.

III. Lending risk is unilateral, but counterparty risk is bilateral.

IV. Lending risk arises from an obligor's inability to meet a contractual obligation, but counterparty risk arises due to market variables.

V. Lending risk cannot be easily quantified, but counterparty risk is amenable to straightforward conventional quantification.

Which of the above are true differences between lending and counterparty risk?

A. None are true

B. Only II and III are true

C. I, II and V are true

D. All are true

Question CR-15 a.4

Bank Q660 • BT P2.T6.900.1 • BT Mock Q47

Julia is analysing the credit risk of a contract her firm is considering entering. Which feature of her work most likely indicates that this is counterparty credit risk rather than traditional lending risk?

A. Interest rate fluctuations have little impact on the amount owed.

B. The exposure at default is 100% of the notional over most of the life of the contract.

C. Her firm is the only party assuming credit risk: if her firm defaults, the other party faces no losses.

D. She is using Monte Carlo simulation because the future value of the contract is highly uncertain.

Question CR-15 a.5

Bank Q664 • AnalystPrep Q.6142 (rebuilt)

A firm borrows from a bank at a floating rate and hedges the interest rate risk by entering into a pay-fixed, receive-floating swap with the same bank. Which statement best describes the difference in the bank's credit risk on the two transactions?

A. Both exposures are about equal to the notional amount throughout, and both are unilateral.

B. The loan exposure is approximately known and unilateral, while the swap exposure is uncertain and bilateral, since either party may owe the other depending on rates.

C. The loan exposure is uncertain and bilateral, while the swap exposure is known and unilateral.

D. Both exposures are uncertain to the same degree, because both depend on the level of interest rates.

Question CR-15 a.6 Bank Q663 • Schweser Module Quiz 33.1 Q3

Which of the following statements regarding counterparty credit risk is most accurate?

A. Counterparty risk is unilateral.

B. Over-the-counter (OTC) derivatives contain less counterparty risk than exchange-traded derivatives because the counterparty is known.

C. The precise future value of the contract is uncertain, but the counterparties are aware of whether the future value will be positive or negative.

D. Counterparty risk is typically associated with counterparty default prior to settlement rather than default during the settlement process.

Question CR-15 a.7 Bank Q662 • BT P2.T6.302.2 (rebuilt)

Each of the following statements about counterparty risk is true EXCEPT:

A. Counterparty risk arises when a party stands to gain on a contract and another party must fulfil a contractual obligation.

B. Counterparty risk is a special form of credit risk, typically relevant where credit risk is two-way rather than one-way.

C. Exchange-traded derivatives carry less counterparty risk than over-the-counter (OTC) derivatives.

D. Counterparty risk can be essentially eliminated if the trading partner is a prime broker.

Solutions • CR-15 a

Solution CR-15 a.1

Answer: B

Bank Q659 • AnalystPrep Q.1860

Step 1 – What sets the direction of the exposure

A derivative's value is the net value of all its future cash flows. Market moves can push it above or below zero over the life of the trade.

⇒ Whoever is in the money when the other side defaults is the one that loses.

Step 2 – Apply it to both parties

Either party can end up in the money, so each has risk to the other: the risk is bilateral.

Interpretation

This is the line that separates counterparty risk from lending risk, where only the lender can lose. GARP adds that bilateral risk can be asymmetric, so "bilateral" does not mean "equal".

Why the other options are wrong

A is incorrect. Unilateral risk is the lending case. The party owed money at inception may be the debtor later.

C is incorrect. Absolute qualifier: the two exposures differ with the trade's profile, credit quality and collateral terms (an option bought with an upfront premium creates risk for the buyer only).

D is incorrect. The future value is the uncertain part; it is typically highly uncertain as seen from today.

LO: CR-15 a: Describe counterparty risk and differentiate it from lending risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.1 Counterparty Risk versus Lending Risk

Solution CR-15 a.2

Answer: A

Bank Q658 • BT P2.T6.302.3

Step 1 – When a default causes a loss

If the contract is an asset to the surviving party (positive value), the defaulter owes it money and the claim may not be paid. If the contract is a liability (negative value), the survivor still has to pay and cannot gain from the default.

⇒ The loss needs a positive value at the default date.

Step 2 – What sets that value

The value is the present value of the remaining cash flows, which moves with rates, FX and other market variables.

⇒ Counterparty risk is a credit risk whose size is set by market risk.

Interpretation

The risk exists all the time, because today's negative value can turn positive before a default. The *loss* only appears if the value is positive when the default happens.

Why the other options are wrong

B is incorrect. Reverses the sign: a negative value is an amount owed to the defaulter.

C is incorrect. Absolute qualifier: the exposure is driven by market variables, so the link to market risk is strong.

D is incorrect. Also absolute: without a default there is no loss, so default risk is the core of it.

LO: CR-15 a: Describe counterparty risk and differentiate it from lending risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1, 15.1.1 and 15.3.1

Solution CR-15 a.3 — Answer: B

Bank Q661 • BT P2.T6.316.1

Step 1 – Check each assertion against GARP's two features

I false: both are credit risk; counterparty risk is simply more sensitive to market variables.
II true: the lending amount is roughly known; derivative exposure at default is uncertain.
III true: only the lender is at risk; in a derivative either party can be the creditor.

Step 2 – The remaining assertions

IV false: in both cases the loss is caused by an obligor unable or unwilling to pay.
V false: reversed; counterparty risk needs simulation of future values, which is harder than measuring a loan balance.
$\Rightarrow$ Only II and III survive.

Interpretation

GARP's two differentiating aspects are exactly II and III: an uncertain future value and a bilateral risk.

Why the other options are wrong

A is incorrect. II and III are genuine differences.
C is incorrect. Accepts I and V, both false, and leaves out III.
D is incorrect. Accepts I, IV and V, which are false.

LO: CR-15 a: Describe counterparty risk and differentiate it from lending risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.1

Solution CR-15 a.4 — Answer: D

Bank Q660 • BT P2.T6.900.1

Step 1 – Sort the features into the two columns

Little sensitivity to rates (A), exposure at full notional (B) and one-way risk (C) are all lending features.

Step 2 – Find the counterparty feature

A future value so uncertain that it has to be simulated is the hallmark of a derivative exposure.
$\Rightarrow$ D points to counterparty risk.

Interpretation

Monte Carlo simulation of future values is how PFE and expected exposure are produced (CR-19); a loan needs no such machinery because its balance is roughly known.

Why the other options are wrong

A is incorrect. A predictable amount owed is lending-like.
B is incorrect. Exposure fixed at about full notional is lending-like; derivative exposure is usually a variable fraction of notional.
C is incorrect. Unilateral risk is lending risk; counterparty risk is bilateral.

LO: CR-15 a: Describe counterparty risk and differentiate it from lending risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.1

Solution CR-15 a.5 — Answer: B

Bank Q664 • AnalystPrep Q.6142

Step 1 – The loan

The bank's amount at risk is about the principal outstanding, known with reasonable certainty.

A floating rate creates only moderate uncertainty over the amount owed, and only the bank can lose.

Step 2 – The swap

Only net coupons are exchanged and no principal. The swap's value depends on where rates go and can be an asset to the bank or to the firm.
⇒ Uncertain and bilateral: the firm is also exposed to the bank.

Interpretation

This is GARP's own example of a client hedging a floating-rate loan with a swap from the same bank (14.2.5). Even with one client and one bank, the two trades carry different risks and sit in different books for capital.

Why the other options are wrong

A is incorrect. Swap exposure is a variable fraction of notional and is bilateral.
C is incorrect. Reverses the two transactions.
D is incorrect. The loan principal is still approximately known; the degree of uncertainty is very different.

LO: CR-15 a: Describe counterparty risk and differentiate it from lending risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.1; Ch.2 (GARP CR Ch.14), 14.2.5 Banks and End Users

Solution CR-15 a.6

Answer: D

Bank Q663 • Schweser Module Quiz 33.1 Q3

Step 1 – Pre-settlement against settlement

GARP: pre-settlement risk "is what counterparty risk usually refers to"; settlement risk is the separate risk at settlement times.
⇒ D is accurate.

Step 2 – Test the others

Counterparty risk is bilateral (A false). Exchange-traded derivatives are centrally cleared and margined daily, so they carry *less* counterparty risk (B reversed). Neither the size nor the sign of the future value is known (C false).

Interpretation

Unless a question says otherwise, read "counterparty risk" as default before final settlement.

Why the other options are wrong

A is incorrect. That is lending risk.
B is incorrect. Reversed, and knowing who the counterparty is does not reduce the risk.
C is incorrect. The sign of the future value is unknown too, which is why either party may be exposed.

LO: CR-15 a: Describe counterparty risk and differentiate it from lending risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1 and 15.1.2 Settlement, Pre-settlement, and Margin Period of Risk

Solution CR-15 a.7

Answer: D

Bank Q662 • BT P2.T6.302.2

Step 1 – Check the true statements

A: a loss needs a gain on the contract plus a counterparty that owes performance. B: the two-way nature is what distinguishes counterparty risk from lending risk. C: exchange-traded contracts are centrally cleared and margined daily.

Step 2 – The exception

The identity of the counterparty does not remove the risk. A prime broker can default too; margining, clearing and netting reduce the risk but nothing "essentially eliminates" it.

Interpretation

GARP: no risk mitigant is perfect and some residual counterparty risk always remains. Lehman Brothers was a prime broker.

Why the other options are wrong

A is incorrect. True, so not the exception.
B is incorrect. True: counterparty risk is credit risk with a two-way exposure.
C is incorrect. True: less, though not zero.

LO: CR-15 a: Describe counterparty risk and differentiate it from lending risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1, 15.1.1 and 15.1.3

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 b

P3 CONCEPT

Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No workbook section; Vardeez flags the LO in his foundation chapter for counterparty risk.

Vardeez: OTC derivatives (interest rate swaps, FX, credit default swaps) and securities financing transactions, with the two legs of a repo; "exchange transactions do not carry counterparty risk" (corrected below: they carry less, not none).

Reported exam shapes: none listed in the Master Map. The bank questions ask which product has the largest notional or the least risk, who is exposed in a repo or a CDS, and how settlement risk differs from pre-settlement risk.

What the questions look like: qualitative "true EXCEPT" items on one product at a time. **GARP practice item kept:** Question CR-15 b.7 (GARP PE1 Q45) gives option inputs and numerical options, but the answer is an exposure of zero; it is kept as a GARP practice question.

1. Two families of transactions

Counterparty risk lives in **OTC derivatives** and **securities financing transactions** (SFTs). Exchange-traded and centrally cleared trades carry much less (CR-15 a, Section 1). Customisation reduces basis risk, which is why a much greater notional trades OTC than on exchanges; interest rate swaps make up the bulk of OTC notional.

Transaction	How counterparty risk arises	Who bears it
Interest rate swap	Fixed against floating; only net coupons, no principal; value moves with rates	Either party, whichever is in the money; a fraction of notional
FX forward, cross-currency swap	Notionals are exchanged; FX volatility is large and cross-currency swaps can be long-dated	Either party; plus settlement risk on the full notional at the exchange
Credit default swap	Value moves with the reference entity's spread and jumps at its default; wrong-way risk	Mainly the protection buyer (larger potential exposure)
Option, premium paid upfront	The writer has received everything it is owed	Buyer only; the writer's exposure is zero
Repo, reverse repo	Borrower fails to repurchase while the collateral falls; lender fails to return excess collateral	Mainly the cash lender; also the borrower, for the haircut
Securities lending	As a repo, with securities rather than cash on one side	Both, as for a repo
Exchange-traded, cleared	Default of the CCP or a clearing member; margined daily	Small per unit, but on large positions

In GARP's survey split of CVA by asset class (Figure 15.3) interest rate products are a large share, with important contributions from FX (volatile rates, long-dated cross-currency swaps) and credit derivatives (wrong-way risk).

2. Repos and securities lending: the gap between two legs

In a repo the seller (the cash borrower) sells securities for cash and buys them back later at a higher price; the difference is the repo interest. The cash lender receives a risk-free charge plus a counterparty charge, and protects itself with a **haircut**: to borrow USD 100m against a 2% haircut, the borrower delivers securities worth $100/(1-0.02) = \text{USD } 102.04\text{m}$. A reverse repo is the same trade seen from the other side; securities lending is a repo with securities in place of cash.

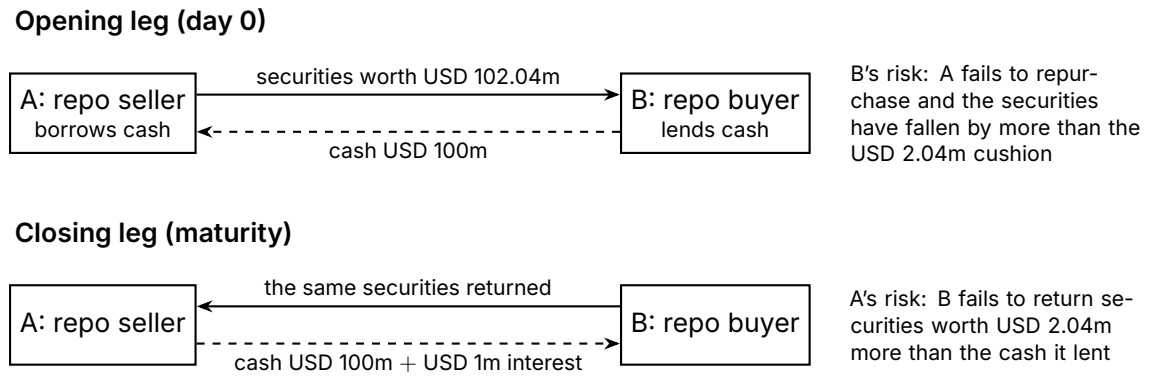


Figure 2. A repo with a 2% haircut. Solid arrows carry securities, dashed arrows carry cash. Neither leg on its own is the risk; the risk is the gap between them. The lender is exposed if the borrower fails to repurchase, the borrower is exposed for the over-collateralisation, and both are exposed to moves in the collateral value in between.

The lender also carries market, liquidity, operational and legal risk on the collateral. Daily margining and short maturities are why GARP rates SFTs as relatively low in counterparty risk.

3. Two products that need care

Credit default swaps. The protection buyer is exposed to the protection seller, and the exposure grows exactly when the reference entity deteriorates. If the seller's credit is linked to the reference entity (Schweser's example: protection on Greek sovereign debt bought from a Greek bank), the buyer is "winning" just as the seller's ability to pay weakens: **wrong-way risk**. Keep the roles apart: the *reference entity's* spread drives the CDS value and hence the exposure; the *counterparty's* spread drives its default probability and hence the CVA. Exposure is measured gross of recovery, and a derivative claim ranks pari passu with senior unsecured bonds.

Options with an upfront premium. Once the buyer has paid the premium it owes nothing more, so the writer can never be owed money: the writer's counterparty exposure is zero whatever the strike, spot or volatility, and only the buyer carries counterparty risk (GARP's example of asymmetric bilateral risk).

Settlement risk by product (CR-15 a, Section 3). Spot trades carry almost only settlement risk, long-dated OTC derivatives mainly pre-settlement risk; FX exchanges full notionals, mostly through CLS, with cross-currency swaps an exception.

> **Correction to the source notes.** Vardeez writes that "exchange transactions do not carry counterparty risk", and Schweser (LO 33.b) that exchange-traded derivatives "do not carry counterparty risk because the exchange is usually the counterparty". GARP (15.1) lists exchange-traded derivatives among the cases that carry *less* counterparty risk, because they are always centrally cleared and typically margined daily, and warns that a small risk can relate to a very large position (a bank's notional facing a CCP can be extremely large). Read "none" in an option as wrong and "least" or "less" as right (Question CR-15 b.4).

Exam traps

- **Option writer.** With the premium received upfront, the writer's exposure is zero; strike, spot, volatility and rate are red herrings.
- **Only the lender in a repo.** The repo seller is exposed too, for the haircut it has handed over.
- **CDS roles swapped.** The reference entity's spread sets the exposure; the counterparty's spread sets the default probability used in CVA.
- **Largest notional against largest risk.** Interest rate swaps dominate notional; FX and CDS carry more risk per unit of notional.
- **"None" for exchange-traded.** Less, not none.

Practice questions: CR-15 b

Question CR-15 b.1

Bank Q673 • Schweser Module Quiz 33.1 Q1

When considering counterparty credit risk, which of the following financial products has the largest outstanding notional amount in the marketplace?

A. Credit default swaps.

B. Foreign exchange forwards.

C. Interest rate swaps.

D. Repos and reverse repos.

Question CR-15 b.2

Bank Q668 • AnalystPrep Q.1861 (rebuilt)

Which of the following best defines settlement risk?

A. The risk that a counterparty defaults at some point between trade inception and the final settlement date, causing loss of the remaining contractual payments.

B. The risk that arises at final settlement because the two parties perform their obligations at different times, so one may pay while the other fails to deliver.

C. The risk that a borrower fails to repay the principal amount of a loan.

D. The risk that the market value of collateral falls between margin calls.

Question CR-15 b.3

Bank Q669 • AnalystPrep Q.1862 (rebuilt)

Counterparty risk is often divided into settlement risk and pre-settlement risk. Which of the following statements about the two is correct?

A. Settlement risk generates much smaller exposures than pre-settlement risk, but default at settlement is much more likely.

B. Settlement risk generates potentially much larger exposures, but default before the final settlement date is substantially more likely than default at settlement.

C. Settlement risk generates much larger exposures and is also much more likely to occur than pre-settlement default.

D. The two risks are similar in both size of exposure and likelihood, so they should be limited jointly.

Question CR-15 b.4

Bank Q670 • BT P2.T6.316.2

Which of the following transactions is likely to carry the LEAST amount of counterparty risk?

A. An over-the-counter (OTC) derivatives contract where the exposure is large but the default probability is small

B. An over-the-counter (OTC) derivatives contract where the exposure is small but the default probability is large

C. An exchange-traded derivative contract where the exchange clears the trade via an attached central counterparty

D. A repurchase agreement (repo) with zero haircut applied to illiquid collateral

Question CR-15 b.5

Bank Q671 • BT P2.T6.406.2 (rebuilt)

In a classic repurchase agreement (repo), Bank A is the repo seller (borrows cash and delivers securities) and Counterparty B is the repo buyer (lends cash and receives securities). A haircut applies. Each of the following statements is true EXCEPT:

A. A higher haircut (over-collateralisation) reduces the counterparty risk faced by the lender, Counterparty B.

B. The securities sold by Bank A become collateral legally owned by Counterparty B, who, if Bank A defaults, can liquidate them to recover the cash lent.

C. Because Bank A borrows cash, it bears no counterparty risk; only Counterparty B is exposed.

D. Counterparty B is exposed to credit risk (default of Bank A), market risk (collateral price volatility), liquidity risk (illiquid collateral), operational risk and legal risk.

Question CR-15 b.6

Bank Q672 • BT P2.T6.407.3

Income Partners, LLC, the protection buyer, purchases a credit default swap (CDS) from Institutional Traders, LLC, which sells protection by writing the CDS. The CDS is marked to market, and the reference entity of the CDS is Acme Corporation. Each of the following statements is true EXCEPT:

A. Income Partners has more exposure than its counterparty, Institutional Traders, due to potential future exposure at high confidence levels.

B. For the exposure faced by Income Partners due to the CDS, the credit spread of Acme Corporation does NOT have an impact, but the credit spread of Institutional Traders DOES.

C. In the event of a bankruptcy by Institutional Traders, Income Partners will likely be pari passu with the firm's senior, unsecured bondholders.

D. Exposure faced by Income Partners to Institutional Traders is measured gross of recovery, but the estimated recovery rate affects the quantification of counterparty risk through the credit value adjustment (CVA).

Question CR-15 b.7

Bank Q666 • GARP PE1 Q45 (rebuilt)

A derivative trading firm sells a European-style call option on stock JKJ with a time to expiration of 9 months, a strike price of EUR 45, an underlying asset price of EUR 67 and an implied annual volatility of 27%. The option premium is paid in full upfront. The annual risk-free interest rate is 2.5%. What is the trading firm's counterparty credit exposure from this transaction?

A. EUR 0

B. EUR 22.00

C. EUR 22.84

D. EUR 23.02

Solutions • CR-15 b

Solution CR-15 b.1

Answer: C

Bank Q673 • Schweser Module Quiz 33.1 Q1

Step 1 – Where counterparty risk exists

Two families: OTC derivatives and securities financing transactions. The OTC derivatives market is far larger.

Step 2 – Within OTC derivatives

Interest rate swaps make up the bulk of OTC notional.
⇒ Interest rate swaps have the largest notional outstanding.

Interpretation

Largest notional is not largest risk per unit: a swap exchanges no principal, so its exposure is a small fraction of notional, while FX products and CDSs contribute more risk per unit of notional (GARP Figure 15.3).

Why the other options are wrong

A is incorrect. CDS notional is far smaller than swap notional.
B is incorrect. FX forwards carry large risk per trade, but their total notional is much smaller.
D is incorrect. Securities financing is smaller than the OTC derivatives market.

LO: CR-15 b: Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1 and 15.1.4; SchweserNotes 2026 Book 2, Reading 33, LO 33.b

Solution CR-15 b.2

Answer: B

Bank Q668 • AnalystPrep Q.1861

Step 1 – GARP's definition

Settlement risk "arises at settlement times ... due to timing differences between when each party performs its obligations under the contract".
⇒ One side may have paid while the other fails to deliver, so the loss can be the full amount paid.

Step 2 – Separate it from the neighbours

Default at any point before final settlement is pre-settlement risk, the usual meaning of counterparty risk.

Interpretation

The Herstatt failure is the standard case: Deutschmarks paid in Germany, dollars never received in New York.

Why the other options are wrong

A is incorrect. That is pre-settlement risk.
C is incorrect. That is lending risk.
D is incorrect. That is the residual market risk of collateral over the margin period of risk, not settlement risk.

LO: CR-15 b: Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.2

Solution CR-15 b.3 Answer: B

Bank Q669 • AnalystPrep Q.1862

Step 1 – Size of the exposure

Settlement risk can reach 100% of the notional paid. Pre-settlement exposure is the value difference between the legs, a fraction of notional.

Step 2 – Likelihood

Settlement risk lasts hours or days; pre-settlement risk lasts the whole life of the trade.
⇒ GARP: "default prior to the expiration of the contract is substantially more likely than default at the settlement date".

Interpretation

Large but brief against smaller but long: that is why institutions set separate settlement limits rather than folding settlement risk into counterparty limits.

Why the other options are wrong

A is incorrect. Reverses both size and likelihood.
C is incorrect. Right on size, wrong on likelihood: the settlement window is short.
D is incorrect. Ignores the clear differences, and GARP says the two are limited separately.

LO: CR-15 b: Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.2

Solution CR-15 b.4 Answer: C

Bank Q670 • BT P2.T6.316.2

Step 1 – Rank the mitigants

An exchange-traded contract is cleared through a central counterparty and margined daily, so the residual counterparty risk is small.

Step 2 – The other three

Both OTC contracts are bilateral and unmitigated: counterparty risk needs both exposure and default probability, and each still has plenty of one. A repo with no haircut on illiquid collateral has no cushion against a fall in the collateral value.
⇒ C carries the least.

Interpretation

"Least", not "none": GARP lists exchange-traded derivatives among the cases with less counterparty risk, and the exposure to a CCP can be very large (correction box).

Why the other options are wrong

A is incorrect. A large uncollateralised exposure remains, even with a small default probability.
B is incorrect. A small exposure helps, but a large default probability leaves material risk.
D is incorrect. A zero haircut on illiquid collateral removes the main mitigant; this is among the riskiest choices.

LO: CR-15 b: Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1

Solution CR-15 b.5 Answer: C

Bank Q671 • BT P2.T6.406.2

Step 1 – The lender's side

A haircut gives B a cushion (A true); title to the securities passes to B, who can sell them if A

defaults (B true); B faces credit, market, liquidity, operational and legal risk (D true).

Step 2 – The borrower's side

With a haircut, A has handed over securities worth more than the cash received (USD 102.04m against USD 100m at 2%). If B defaults and does not return them, A loses the excess.
⇒ C is false: both sides are exposed.

Interpretation

The repo is a two-sided exposure that is deliberately unequal: the haircut protects the lender at the borrower's expense (Figure 2).

Why the other options are wrong

A is incorrect. True, so not the exception.
B is incorrect. True, so not the exception.
D is incorrect. True, so not the exception.

LO: CR-15 b: Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. • **Reference:** SchweserNotes 2026 Book 2, Reading 33, LO 33.b; Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1

Solution CR-15 b.6 Answer: B

Bank Q672 • BT P2.T6.407.3

Step 1 – What drives the CDS value

The CDS is worth more to the buyer as Acme's spread widens, and much more if Acme defaults. Acme's spread therefore drives Income Partners' exposure.

Step 2 – What the seller's spread does

Institutional Traders' own spread measures its default probability. That enters the CVA, not the exposure.
⇒ B reverses the roles, so it is the false statement.

Step 3 – Confirm the others

A: the buyer's gain can be large (a jump at Acme's default) while the seller's gain is limited to premiums. C: derivative claims rank with senior unsecured debt. D: exposure is gross of recovery; recovery enters through LGD in CVA.

Interpretation

If the seller's credit is tied to Acme, the buyer also faces wrong-way risk: the protection is worth most when the seller is least able to pay.

Why the other options are wrong

A is incorrect. True: the protection buyer has the larger potential future exposure.
C is incorrect. True: GARP says holders of derivatives are generally pari passu with senior bondholders.
D is incorrect. True: exposure ignores recovery by convention; LGD enters CVA.

LO: CR-15 b: Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.4, 15.3.3 and 15.3.5

Solution CR-15 b.7

Answer: A

Bank Q666 • GARP PE1 Q45

Step 1 – Who owes whom

The firm wrote the call and has received the premium in full. The buyer has no further payment obligation.

$\Rightarrow$ The firm can never be owed money by the buyer.

Step 2 – Exposure

Exposure is $\max(V, 0)$, with V the contract value to the firm. For a written option $V \leq 0$ always, so the exposure is zero.

Interpretation

All the inputs are red herrings for the writer's credit exposure. The writer bears market risk on the short call; the counterparty risk sits with the buyer, which is exposed to the firm. This is GARP's example of asymmetric bilateral risk.

Why the other options are wrong

B is incorrect. Intrinsic value, $67 - 45 = 22.00$: what the firm owes the buyer, not a claim on the buyer.

C is incorrect. $67 - 45\mathrm{e}^{-0.025\times 0.75} = 22.84$, the forward-style value: again the buyer's gain, the firm's liability.

D is incorrect. The Black-Scholes premium, about 23.02: already received, and the option is a liability to the writer.

LO: CR-15 b: Describe transactions that carry counterparty risk and explain how counterparty risk can arise in each transaction. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.1 (footnote 4: an option with an upfront premium creates counterparty risk for the buyer only)

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 c

P3 CONCEPT

Identify and describe institutions that take on significant counterparty risk.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No workbook section; Vardeez flags the LO in his foundation chapter for counterparty risk.

Vardeez: the three tiers of derivatives player (large, medium, small) and their characteristics; "the collateral column is the discriminating one".

Reported exam shapes: none listed in the Master Map. The questions ask you to place an institution in its tier from its activity profile, to say who posts collateral and who does not, to separate risk-takers from third-party service providers, and to read where counterparty risk ends up in a bank's book.

What the questions look like: short profiles ("a government agency with a few cross-currency swaps and two dealers") and "which statement is correct". Two questions are re-filed from CR-14 d (a P4 LO, not built), and one is new from GARP's CVA breakdown.

1. Three groups of derivatives users

The institutions that take on counterparty risk through trading differ in size, volume, coverage of asset classes, and willingness and ability to post collateral. GARP describes them in its derivatives chapter (14.2.4); Vardeez and Schweser call the third group "small players", GARP calls it "end users".

Group: who	Trading profile	Collateral
Large player (dealer): large global bank	Vast number of trades; many clients; trade with each other; all asset classes; member of most or all exchanges and CCPs; the largest 14 dealers hold about four-fifths of notional and provide most liquidity	Collateralise, as long as the counterparty does the same (sometimes even if not)
Medium-sized player: smaller bank or other financial institution, often making markets in some products	Several asset classes but not all (often no credit, commodities or exotics); may cover one market, such as a regional bank in local currencies; fewer clients	Generally willing to collateralise
End user (small player): large corporate, sovereign, or smaller financial institution with specific needs	Few trades, few counterparties; often one asset class (a corporate only FX, a miner only commodity forwards, a pension fund only rates and inflation); directional positions	Often unable or unwilling to post; or posts illiquid collateral, or infrequently

The collateral column decides. Vardeez: it "inverts the intuition: the smallest players, with the least capacity to absorb a loss, are also the ones least able to post collateral against it". GARP adds that

many OTC players have neither strong credit quality nor the ability to post collateral, so counterparty risk is "arguably an unavoidable consequence of the derivatives market".

One-way collateral. A special group of end users with exceptionally high credit quality (sovereigns, supranational entities, multilateral development banks) has historically received collateral from banks without posting any, sometimes with rating triggers that require posting after a downgrade. Such arrangements are now seen as much more costly because of the funding and capital they consume (Workbook A21.4, worked as Question CR-17 f.7).

Third parties. Firms offering settlement, margining and collateral management, software, trade compression and clearing services help participants reduce counterparty risk and related risks (such as legal risk) and improve efficiency. They do not take on counterparty risk by trading, so they are not derivatives players in this sense.

2. Where the counterparty risk ends up

Large global banks have exposure to all asset classes; smaller banks have more limited exposure (mainly interest rate and some FX products); end users are limited too: a corporate may use only interest rate and cross-currency swaps (GARP 15.1.4). GARP's breakdown of one bank's CVA by rating and by counterparty type (Figure 15.4, Royal Bank of Scotland, 2016; the full table is in Question CR-15 c.6) shows where the risk sits:

- **High-quality credits** contribute little because their default probability is low (AAA: GBP 4m of 618m in 2016).
- **Banks and other financial institutions** contribute little: relatively good credit quality, and most of their trades are probably collateralised (banks GBP 22m, 3.6% of the 2016 total).
- **The bulk** sits with medium ratings (BBB− to A−: GBP 388m, 62.8% in 2016) and with corporates and governments (GBP 526m, 85.1%), most of which is likely to be uncollateralised.
- **Non-investment grade** is smaller than its default probabilities would suggest, probably because banks are partly reluctant to trade with such names.

Coverage has widened: banks once quantified only poor credits and uncollateralised names; counterparty risk to CCPs is the next area to get attention.

> **Correction to the source notes.** Vardeez's tier table gives "a hedge fund or pension fund" as typical medium derivatives players. GARP (14.2.4) uses a pension fund active only in interest rate and inflation products as an example of an *end user*, and defines the medium-sized player as a smaller bank or other financial institution with significant OTC activity, including market making. Classify by activity profile (number of trades and counterparties, asset-class coverage, collateral), not by type of institution: a hedge fund or pension fund can fall in either group.

Exam traps

- **Sovereigns are big, so they are large players.** A sovereign with a few hedging trades is an end user; tiers follow activity, not size.
- **End users post liquid collateral.** They often post none, or illiquid collateral, or post infrequently.
- **Medium players "definitely" post collateral.** They are generally willing to; only dealers routinely collateralise.
- **One-way CSAs are risk-free for the bank.** Receiving nothing while posting is extra risk for

the bank, and costly in funding and capital.

- **Service providers as risk-takers.** Clearing, compression and collateral-management firms reduce risk for others; they are not derivatives players.
- **Highest default probability means highest CVA.** Exposure and collateral matter too: in 2016 medium-rated, uncollateralised corporates and governments dominated.

Practice questions: CR-15 c

Question CR-15 c.1

Bank Q676 • New question written for this bank

A senior risk manager is comparing the characteristics of three types of institutions that take on counterparty risk through OTC derivatives trading. Which of the following statements is correct?

A. Small players hold the largest number of OTC derivatives trades because they spread their activity across many asset classes.

B. Medium players are definitely required to post collateral, but they trade in a narrower range of asset classes than small players.

C. Large players trade with each other and with many clients, cover a very wide range of asset classes, and post collateral against their positions.

D. Large players are typically sovereigns and corporations whose derivatives needs are specific and tied to a single asset class.

Question CR-15 c.2

Bank Q675 • New question written for this bank

A risk analyst at a bank is classifying the firm's OTC derivatives counterparties by activity profile. One counterparty is a national government agency that enters into a small number of cross-currency swaps with only two dealers to hedge its foreign-currency debt. If it posts collateral, the collateral is usually illiquid. Which of the following best describes this counterparty?

A. A large derivatives player, because sovereign entities typically trade across commodities, equity, foreign exchange, interest rate and credit derivatives.

B. A small derivatives player, because it trades with few counterparties, specialises in one asset class, and posts illiquid collateral if any.

C. A medium derivatives player, because it has a large number of clients but is less active than dealers in some asset classes.

D. A small derivatives player, because small players are always required to post liquid collateral to dealers to offset their lower trading volume.

Question CR-15 c.3 Bank Q625 • AnalystPrep Q.2694 • re-filed from CR-14 d

Which of the following OTC counterparties is most likely to post collateral regularly to its trading counterparties?

A. Private equity funds

B. Sovereigns

C. Corporates

D. Dealer banks

Question CR-15 c.4 Bank Q621 • AnalystPrep Q.6133 • re-filed from CR-14 d

Which type of OTC derivatives end user is especially known for receiving, but not posting, collateral in transactions with banks, subject to possible rating triggers?

A. High-frequency trading firms specialising in algorithmic derivatives trading.

B. Corporate clients engaged in large-scale hedging of operational risks.

C. Sovereigns, supranational entities or multilateral development banks with high credit ratings.

D. Investment funds using derivatives for portfolio diversification and risk mitigation.

Question CR-15 c.5 Bank Q677 • New question written for this bank

A consultant presents a firm that provides central clearing services, trade compression and collateral management software to derivatives market participants. A junior analyst states that the firm should be grouped with large derivatives players because it is critical to counterparty risk in the OTC market. Which of the following is the most accurate assessment?

A. The analyst is correct, because any firm involved in clearing must be a large dealer that takes on significant trading counterparty risk.

B. The analyst is incorrect, because the firm is a medium derivatives player that has a large number of clients and a high volume of OTC trades.

C. The analyst is correct, because trade compression and collateral management only exist inside large dealer banks.

D. The analyst is incorrect, because the firm is a third-party provider of products and services that reduce counterparty risk and improve efficiency, not a derivatives player taking on risk through trading.

Question CR-15 c.6 New • built from GARP Ch.15, 15.1.4 and Figure 15.4

The head of counterparty risk at a UK bank is reviewing the following CVA breakdown, taken from GARP's example (Royal Bank of Scotland annual report, GBP millions):

Rating	2016	2015	Counterparty type	2016	2015
AAA	4	37	Banks	22	18
AA to AA+	22	66	Other financial institutions	70	126
A to AA−	52	49	Corporate	337	470
BBB− to A−	388	293	Government	189	160
Non-investment grade and unrated	152	329			
Total	618	774	Total	618	774

Which of the following conclusions about the 2016 figures is most consistent with the breakdown and with GARP's explanation of it?

A. The largest share of CVA comes from non-investment-grade and unrated names, because their default probabilities are the highest.

B. Bank counterparties contribute little CVA because interbank derivatives are exempt from collateral agreements.

C. The largest share of CVA comes from BBB− to A− names and from corporates and governments, most of which trade uncollateralised.

D. AAA names contribute a material share of CVA because their low default probability allows large uncollateralised exposures.

Solutions • CR-15 c

Solution CR-15 c.1

Answer: C

Bank Q676 • New question written for this bank

Step 1 – The large player

Large global dealer banks: a vast number of trades, many clients and each other as counterparties, all major asset classes, and collateral posted against positions.
$\Rightarrow$ C matches.

Step 2 – The other tiers

Medium players are generally willing (not required) to collateralise and cover several asset classes, more than end users. End users have few trades and often a single asset class.

Interpretation

The tiers differ on four dimensions at once: number of trades, number of counterparties, asset-class coverage and collateral. A question usually changes one of them.

Why the other options are wrong

A is incorrect. Small players have the fewest trades and often one asset class.
B is incorrect. "Definitely" is too strong, and medium players cover a wider range than small players, not a narrower one.
D is incorrect. Describes end users, not dealers.

LO: CR-15 c: Identify and describe institutions that take on significant counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.2.4 Market Participants and Collateralisation; SchweserNotes 2026 Book 2, Reading 33, LO 33.c

Solution CR-15 c.2

Answer: B

Bank Q675 • New question written for this bank

Step 1 – Read the activity profile

A small number of cross-currency swaps, two dealers, one asset class, a specific hedging need, and illiquid collateral if any.

Step 2 – Match it to a tier

Every feature is an end-user (small-player) feature. Sovereigns are explicitly listed among end users.

Interpretation

Size of the institution does not decide the tier: a national government agency is large, but its derivatives activity is narrow and directional.

Why the other options are wrong

A is incorrect. A sovereign hedging its own debt does not trade across all asset classes.
C is incorrect. Medium players have many clients and significant OTC activity, often including market making.
D is incorrect. Right tier, wrong reason: end users often post no collateral or illiquid collateral.

LO: CR-15 c: Identify and describe institutions that take on significant counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.2.4

Solution CR-15 c.3 — Answer: D

Bank Q625 • AnalystPrep Q.2694

Step 1 – Collateral behaviour by group

Dealers collateralise as long as the counterparty makes the same commitment. Corporates and many funds cannot or will not commit, or post illiquid collateral infrequently. Highly rated sovereigns often receive collateral without posting it.
⇒ Dealer banks post the most collateral, and most regularly.

Interpretation

This is the collateral column of the tier table, and the reason the counterparty risk in a bank's book is concentrated in end users rather than in other banks.

Why the other options are wrong

A is incorrect. Private equity funds behave like end users and often do not commit to posting.
B is incorrect. Sovereigns have historically received collateral but not posted it.
C is incorrect. Corporates are typically unable or unwilling to post, or post illiquid collateral infrequently.

LO: CR-15 c: Identify and describe institutions that take on significant counterparty risk. CR-14 d: Identify the participants and describe the use of collateralization in the derivatives market. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.2.4

Solution CR-15 c.4 — Answer: C

Bank Q621 • AnalystPrep Q.6133

Step 1 – The special case in GARP

Counterparties of exceptionally high credit quality (sovereigns, supranationals, multilateral development banks) have had a favourable arrangement in which they receive but do not post collateral.

Step 2 – The rating trigger

They may have to start posting if they are downgraded.
⇒ C.

Interpretation

For the bank this one-way arrangement is worse than no collateral agreement at all: it posts when it owes, receives nothing when it is owed, and must fund the collateral and hold capital against the exposure. GARP notes such terms are now seen as much more costly.

Why the other options are wrong

A is incorrect. Trading firms have no exceptional credit quality and no such arrangement.
B is incorrect. Corporates often do not post, but they do not receive one-way terms either; they are simply uncollateralised.
D is incorrect. Investment funds are ordinary end users.

LO: CR-15 c: Identify and describe institutions that take on significant counterparty risk. CR-14 d: Identify the participants and describe the use of collateralization in the derivatives market. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.2.4

Solution CR-15 c.5 — Answer: D

Bank Q677 • New question written for this bank

Step 1 – Who takes counterparty risk

Large, medium and small players take on counterparty risk through their own trading.

Step 2 – Who sells services

Clearing services, software, trade compression and collateral management are offered by third parties so that participants can reduce counterparty risk and improve efficiency.
$\Rightarrow$ The firm is a third-party provider, not a derivatives player.

Interpretation

Being important to counterparty risk is not the same as bearing it. (A CCP that clears trades does take on counterparty risk; that is CR-18 territory. Here the firm only provides services.)

Why the other options are wrong

A is incorrect. Offering clearing services does not make a firm a dealer.
B is incorrect. Medium players trade a high volume of OTC derivatives themselves; this firm provides services.
C is incorrect. These services are offered by third parties, not only inside dealers.

LO: CR-15 c: Identify and describe institutions that take on significant counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.2 (GARP CR Ch.14), 14.2.4; SchweserNotes 2026 Book 2, Reading 33, LO 33.c

Solution CR-15 c.6 — Answer: C

New • built from GARP Ch.15, 15.1.4 and Figure 15.4

Step 1 – Shares by rating (2016)

BBB− to A−: $388/618 = 62.8\%$. Non-investment grade and unrated: $152/618 = 24.6\%$. AAA: $4/618 = 0.6\%$.
$\Rightarrow$ Medium ratings carry the majority.

Step 2 – Shares by counterparty type (2016)

Corporates and governments: $(337 + 189)/618 = 526/618 = 85.1\%$. Banks: $22/618 = 3.6\%$.

Step 3 – GARP's explanation

High-quality credits have low default probabilities; banks and other financial institutions have relatively good credit quality and are mostly collateralised; corporates and governments are mostly uncollateralised; banks are partly reluctant to trade with non-investment-grade names.

Interpretation

CVA needs exposure as well as default probability. The names a bank trades most with, without collateral, dominate. The 2015 column is a red herring for the 2016 question, and a warning: in 2015 the non-investment-grade bucket (329 of 774, 42.5%) was the largest rating bucket.

Why the other options are wrong

A is incorrect. In 2016 non-investment grade is 152, well below the 388 for BBB− to A−; GARP attributes the gap to reluctance to trade with such names.
B is incorrect. Banks contribute little, but because of good credit quality and collateralisation, not an exemption from collateral.
D is incorrect. AAA contributes GBP 4m, under 1%: a low default probability means a small CVA.

LO: CR-15 c: Identify and describe institutions that take on significant counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.4 Product Type, Figure 15.4

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 d

P3 CONCEPT

Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No workbook section; Vardeez flags the LO in his foundation chapter for counterparty risk.

Vardeez: the eight definitions; "credit migration cuts both ways": improvement is more likely for weak ratings, deterioration over long contracts, and the conditions attached to each are swappable.

Reported exam shapes: none listed in the Master Map. The questions test the asymmetry of exposure (a claim if positive, still owed if negative), mark-to-market against replacement cost, current against potential exposure, LGD against recovery, and how migration shapes the term structure of default probability.

What the questions look like: definitional "true EXCEPT" items with small numbers as decoration. **GARP practice item kept:** Question CR-15 d.6 (AP Mock 2 Q63) is a numerical current-exposure question on a describe LO; it is kept as a GARP practice question, with its distractors rebuilt from real errors.

1. From valuation to exposure

Mark-to-market (MTM) is the present value of all the payments an institution expects to receive less those it is obliged to make (GARP 15.3.2). It is not an amount due today, and it can be positive or negative. It drives every valuation adjustment (CR-15 h) and is needed today and at every future date at which the counterparty could default. As a measure of what could be lost, a plain sum of MTMs ignores netting, collateral and hedging (Vardeez).

Replacement cost is the cost of entering into an equivalent transaction with another counterparty (GARP 15.3.3). It is close to MTM but includes real costs such as bid-offer spreads, large for illiquid trades (and even a standard contract may be illiquid at the default time). Default documentation refers to replacement costs, so these costs can enter the amount owed at close-out.

Credit exposure is the loss if the counterparty defaults. A positive value is a claim on the defaulter; a negative value must still be paid, and the survivor cannot gain by being released from it (except in special, non-standard cases). Hence exposure $= \max(\text{value}, 0)$. By convention exposure ignores recovery. It is conditional on default, although market practice computes it independently of default, which implicitly assumes no wrong-way risk (CR-20). Schweser and Vardeez split it into current exposure (today's claim), future exposure and contingent claims; PFE and expected exposure are CR-19.

2. Default probability and credit migration

Counterparty risk must be assessed over the whole life of the trades, which can be very long. Two questions matter: the probability of default over a horizon, and the probability of a decline in credit quality (a downgrade or spread widening). **Credit migration** matters because it shapes the term structure of default probability and can cause problems before any default. GARP's three effects on the default probability for a future period, such as year 4 to 5:

- it **tends to fall** with the horizon, because default may already have happened earlier: a firm is unlikely to default between years 20 and 21 because it is unlikely to survive that long;
- an **expected deterioration** raises it over time (until the first effect takes over);
- an **expected improvement** lowers it over time, and the first effect accelerates the fall.

Credit quality is empirically **mean-reverting**: good credits tend to deteriorate, poor ones to improve (Figure 4). Default probabilities can be real-world (historical) or risk-neutral (implied from spreads such as CDS); the risk-neutral kind is now standard for CVA (CR-15 e). Estimation is CR-9.

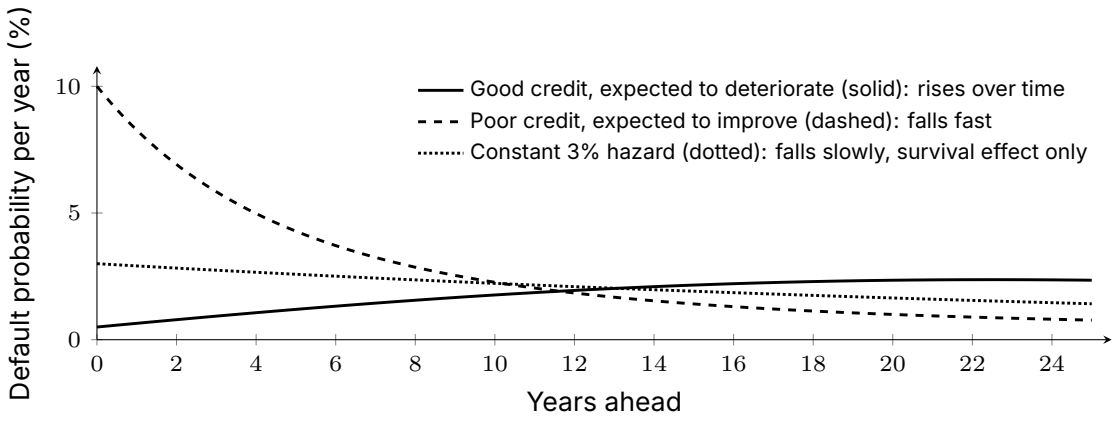


Figure 4. Illustrative term structures of the default probability per year, seen today. Even with a constant conditional default rate (dotted) the unconditional probability falls, because fewer firms survive to default later. Mean reversion bends the good credit upwards and the poor credit downwards.

3. Recovery and loss given default

The **recovery rate** is the percentage of the outstanding claim recovered when the counterparty defaults; **loss given default** is LGD $= 1 -$ recovery rate (70% recovery means a 30% loss). LGD is highly uncertain. Exposure is measured before recovery, and LGD enters CVA. Derivative claims rank **pari passu** with senior bondholders, so OTC derivatives, bonds and CDSs may appear to share one LGD, but not always. Timing differs too: a bond's LGD is realised at once, since the bond can be sold; a CDS settles within days through the auction; a derivative with a defaulted counterparty cannot be freely sold, so its LGD can differ (Lehman, 2008).

> **Correction to the source notes.** Schweser (LO 33.d) says that "recovery is not usually considered when pricing credit risk". Recovery is ignored when *exposure* is measured, but GARP (15.3.5) is explicit that LGD is relevant in the quantification of CVA, which is the price of counterparty risk. Schweser also calls MTM "an accrual accounting measure"; it is a fair-value measure, the present value of future cash flows at current market rates.

Term	What it is	Point to watch
Mark-to-market	PV of receipts less PV of payments	Not the next net payment; can be negative
Replacement cost	Cost to enter the same trade with a new counterparty	MTM plus bid-offer and other real costs
Credit exposure	Loss if the counterparty defaults: $\max(\text{value}, 0)$	Gross of recovery; conditional on default
Credit migration	Change in credit quality before default	Shapes the term structure of PD
Default probability	Real-world or risk-neutral	Per-period PD tends to fall with horizon
Recovery rate	% of the claim recovered	Derivatives pari passu with senior bonds
LGD	$1-$ recovery rate	A severity, not a probability

Exam traps

- **Default as a release.** A survivor with a negative value still pays; it cannot gain from the default.
- **Negative MTM today, so no exposure ever.** Current exposure is zero, but future exposure can be positive.
- **MTM equals the next payment.** MTM is the PV of all remaining net flows.
- **Recovery in replacement cost.** Replacement cost is about re-entering the trade; recovery affects the loss, not the replacement price.
- **Per-year PD rises with horizon.** For distant periods it tends to fall, because the firm may not survive that long.
- **Risk-neutral PD needs the real-world PD.** The two are separate estimates: one from spreads, one from history.

Practice questions: CR-15 d

Question CR-15 d.1

Bank Q691 • BT P2.T6.317.1

Each of the following statements is true about credit exposure EXCEPT:

A. Credit exposure defines the loss in the event of a counterparty default.

B. Credit exposure does not assume counterparty default; that is, it is not conditional on default.

C. Credit exposure includes the risk from current (actual) claims.

D. Credit exposure includes the risk from possible future (potential) claims.

Question CR-15 d.2

Bank Q687 • BT P2.T6.407.2 (rebuilt)

Which of the following statements about credit exposure, current exposure and potential exposure is FALSE?

A. Credit exposure is the loss conditional on the counterparty defaulting.

B. If the mark-to-market (MTM) of a contract is negative today, current exposure is zero.

C. If the MTM of a contract is negative today, the contract's credit exposure over its remaining life is necessarily zero.

D. Credit exposure encompasses both current exposure and potential future exposure.

Question CR-15 d.3

Bank Q686 • BT P2.T6.320.3 (rebuilt)

Financial institution XYZ has an interest rate swap with Counterparty A and hedges it with an identical offsetting swap with Counterparty B. At a future date the MTM of the swap with A is $+V$ (in XYZ's favour, $V > 0$), so the swap with B is $-V$. Assume zero recovery, no collateral and no netting across counterparties. Which of the following statements is correct?

A. If either counterparty defaults, XYZ's loss is zero because the two swaps hedge each other.

B. If only Counterparty B defaults, XYZ gains V because it is released from its obligation.

C. If only Counterparty A defaults, XYZ loses V.

D. If only Counterparty B defaults, XYZ loses V.

Question CR-15 d.4

Bank Q684 • BT P2.T6.317.3

A bank has an interest rate swap with a counterparty with a remaining term of three years. The next settlement, which nets both parties' payments, is a USD 300,000 payment from the bank to the counterparty. The present value of all payments the bank expects to receive over the next three years is USD 5.6 million; the present value of all payments it expects to make, including the USD 300,000, is USD 4.3 million. The payments will be netted. Each of the following is true EXCEPT:

A. The current MTM is negative, −USD 300,000.

B. The MTM is USD 5.6 million − USD 4.3 million = +USD 1.3 million.

C. The current exposure is USD 1.3 million.

D. Replacement cost is unlikely to equal the MTM exactly, but is often approximated by it.

Question CR-15 d.5

Bank Q685 • BT P2.T6.318.1

A bank holds an OTC derivatives position with a counterparty. The current MTM of the bank's position is +USD 4.0 million, but the position is highly illiquid. There is no collateral or margin, and the net counterparty risk adjustment is zero. If the counterparty defaults, the bank expects a recovery rate of 35%. Which of the following is the best estimate of the position's replacement cost?

A. About USD 1.4 million

B. About USD 2.6 million

C. Less than USD 4.0 million

D. More than USD 4.0 million

Question CR-15 d.6

Bank Q679 • AP Mock 2 Q63 (rebuilt)

Prime Bank enters into a GBP 200 million, five-year, annual-pay interest rate swap, receiving 8% fixed against 12-month LIBOR. Over the first year the swap rate for the remaining maturity declines by 150 basis points. Immediately after the first annual exchange, which of the following is closest to Prime Bank's current exposure on the swap?

A. GBP 9.94 million

B. GBP 10.28 million

C. GBP 12.00 million

D. GBP 12.47 million

Question CR-15 d.7

Bank Q682 • AnalystPrep Q.1967 (rebuilt)

An analyst is documenting the inputs to a bank's credit value adjustment (CVA). What does loss given default (LGD) represent?

A. The fraction of the exposure expected to be recovered in the event of counterparty default.

B. The fraction of the exposure expected to be lost in the event of counterparty default.

C. The probability of counterparty default given that it defaulted on a previous contract.

D. The probability that the loss following a counterparty default exceeds a given threshold.

Question CR-15 d.8

Bank Q683 • BT P2.T6.317.2

Why is credit migration crucial to measures of counterparty risk (choose the best answer)?

A. Because it affects the current mark-to-market (MTM).

B. Because it affects the recovery.

C. Because it affects the term structure of default probability.

D. None of the above: credit migration has no impact on counterparty risk.

Question CR-15 d.9 Bank Q690 • Schweser Module Quiz 33.1 Q2

Liz Parker is a junior quantitative analyst who is preparing a report on credit migration. An excerpt of her report contains the following statements:

I. Future default probability will likely increase over time, especially for periods far into the future.

II. When computing the default probability of a counterparty under a risk-neutral measure, we need first to determine the actual default probability.

Which of Parker's statements is (are) correct?

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

Solutions • CR-15 d

Solution CR-15 d.1

Answer: B

Bank Q691 • BT P2.T6.317.1

Step 1 – The definition

Exposure "defines the loss in the event of a counterparty defaulting" (A true) and covers current and potential future claims (C, D true).

Step 2 – Conditionality

Exposure matters only if the counterparty defaults, so its quantification is conditional on default.
$\Rightarrow$ B is false.

Interpretation

GARP's nuance: market practice often computes exposure independently of the default event, which implicitly assumes no wrong-way risk. That is a modelling shortcut, not the definition.

Why the other options are wrong

A is incorrect. True: the definition of credit exposure.
C is incorrect. True: current claims are part of exposure.
D is incorrect. True: potential future claims are part of exposure.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.3 Replacement Cost and Credit Exposure

Solution CR-15 d.2

Answer: C

Bank Q687 • BT P2.T6.407.2

Step 1 – Current exposure

Current exposure is today's claim, $\max(\text{MTM}, 0)$. A negative MTM gives zero (B true).

Step 2 – Future exposure

The MTM can turn positive before the counterparty defaults, so potential future exposure can be positive even when today's MTM is negative.
$\Rightarrow$ C is false.

Interpretation

This is why an out-of-the-money swap still uses credit limit and attracts CVA: the exposure is about the future value, not today's.

Why the other options are wrong

A is incorrect. True: exposure matters only on default.
B is incorrect. True: current exposure is floored at zero.
D is incorrect. True: credit exposure is the general term covering both.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.3; SchweserNotes 2026 Book 2, Reading 33, LO 33.d

Solution CR-15 d.3

Answer: C

Bank Q686 • BT P2.T6.320.3

Step 1 – Default of A

XYZ's MTM with A is $+V$: a claim. With zero recovery, A's default costs XYZ V.

Step 2 – Default of B

XYZ's MTM with B is $-V$: XYZ owes B. B's default creates no loss, and XYZ must still honour the liability, so there is no gain either.

$\Rightarrow$ Only C is correct.

Interpretation

The offsetting swap hedges market risk, but the counterparty risk is per counterparty: exposure is $\max(\text{MTM}, 0)$ counterparty by counterparty, and the loss and the liability do not offset because they face different names.

Why the other options are wrong

A is incorrect. Confuses a market hedge with credit protection.

B is incorrect. A survivor cannot gain by being released from its liability.

D is incorrect. XYZ has no claim on B, so B's default costs nothing.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.3

Solution CR-15 d.4 — Answer: A

Bank Q684 • BT P2.T6.317.3

Step 1 – MTM

MTM is the PV of receipts less the PV of payments: $5.6 - 4.3 = +\text{USD } 1.3$ million (B true).

Step 2 – Exposure and replacement cost

Current exposure $= \max(1.3, 0) = 1.3$ (C true). Replacement cost adds bid-offer and other costs, so it is close to but not exactly the MTM (D true).

Step 3 – The next payment

The USD 300,000 is only the next net cash flow, already inside the USD 4.3 million.

$\Rightarrow$ A is false.

Interpretation

A swap can be worth a lot to the bank even when its next net payment goes out: value depends on all remaining flows.

Why the other options are wrong

B is incorrect. True: the GARP definition of MTM.

C is incorrect. True: positive MTM with no collateral is the current exposure.

D is incorrect. True: MTM is the usual approximation of replacement cost.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.2 Valuation and Mark-to-Market and 15.3.3

Solution CR-15 d.5 — Answer: D

Bank Q685 • BT P2.T6.318.1

Step 1 – Start from MTM

MTM of $+$USD 4.0 million is the usual approximation of replacement cost.

Step 2 – Add the real costs of replacing

Replacing the trade means paying bid-offer and transaction costs, which are large for a highly illiquid position.

$\Rightarrow$ Replacement cost is above USD 4.0 million.

Interpretation

The 35% recovery rate is a red herring: it affects the loss on default, not the price of re-entering the trade. With no collateral and a zero net counterparty adjustment, nothing pulls the cost below MTM.

Why the other options are wrong

A is incorrect. Applied the recovery rate to MTM: $4.0 \times 35\% = 1.4$.
B is incorrect. Applied LGD to MTM: $4.0 \times 65\% = 2.6$.
C is incorrect. Transaction costs push replacement cost up, not down.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.3

Solution CR-15 d.6

Answer: B

Bank Q679 • AP Mock 2 Q63

Step 1 – The rate advantage

After one year four annual payments remain. A replacement swap would receive $8\% - 1.5\% = 6.5\%$, so Prime Bank's swap is worth an extra $200 \times 1.5\% =$ GBP 3m a year.

Step 2 – Value the advantage

$$\text{MTM} = 3 \times \frac{1 - 1.065^{-4}}{0.065} = 3 \times 3.4258 = \text{GBP } 10.28 \text{ million}$$

Step 3 – Current exposure

The MTM is positive, so current exposure $= \max(10.28, 0) =$ GBP 10.28 million.

Interpretation

Falling rates help the fixed receiver, which is now the creditor: this is the replacement cost it would face if the counterparty defaulted today.

Why the other options are wrong

A is incorrect. Discounted at the old 8% rate: $3 \times 3.3121 = 9.94$.
C is incorrect. Left the four GBP 3m advantages undiscounted: 12.00.
D is incorrect. Counted five remaining payments instead of four: $3 \times 4.1557 = 12.47$.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.2 and 15.3.3

Solution CR-15 d.7

Answer: B

Bank Q682 • AnalystPrep Q.1967

Step 1 – GARP's definition

LGD in percentage terms is 100% minus the recovery rate: the fraction of the exposure lost on default.

Step 2 – Its role

LGD is a severity that multiplies exposure and default probability in CVA.

Interpretation

Recovery 40% means LGD 60%. Higher recovery, lower LGD, one for one.

Why the other options are wrong

A is incorrect. That is the recovery rate, the complement of LGD.

C is incorrect. LGD is not a probability.
D is incorrect. Also a probability; LGD is the expected fraction lost.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.5 Recovery and Loss Given Default

Solution CR-15 d.8 Answer: C

Bank Q683 • BT P2.T6.317.2

Step 1 – What migration is

Discrete changes in credit quality, such as rating changes.

Step 2 – Why it matters

GARP: credit migrations "are crucial since they influence the term structure of default probability", for example the probability of default between years 4 and 5 rather than just over the next year.

Interpretation

Counterparty risk runs over the whole life of the trades, so the evolution of default probability over that life matters, and mean reversion shapes it.

Why the other options are wrong

A is incorrect. Current MTM depends on market variables and contract terms.
B is incorrect. Recovery is a separate input.
D is incorrect. Migration matters, and can cause problems even before default.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.4 Default Probability, Credit Migration, and Credit Spreads

Solution CR-15 d.9 Answer: D

Bank Q690 • Schweser Module Quiz 33.1 Q2

Step 1 – Statement I

The default probability for a distant period tends to decrease, because default may already have occurred before the period starts. I is false.

Step 2 – Statement II

A risk-neutral probability is implied from market prices such as credit spreads; it does not need the real-world probability, which is estimated from historical data. II is false.

Interpretation

Statement I is Vardeez's swappable condition: rising default probability over time belongs to a good credit expected to deteriorate, not to every counterparty.

Why the other options are wrong

A is incorrect. I reverses the tendency.
B is incorrect. II mixes the two measures.
C is incorrect. Both are false.

LO: CR-15 d: Describe credit exposure, credit migration, recovery, mark-to-market, replacement cost, default probability, loss given default, and the recovery rate. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.4

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 e

P3 CONCEPT

Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No workbook section; Vardeez flags the LO in his foundation chapter for counterparty risk.

Vardeez: CVA and credit limits "answer different questions": CVA *prices* the risk (and helps pick counterparties and optimise netting), credit limits *cap* it. The full CVA definition and the credit-adjusted value identity are at CR-12 b; the calculation is CR-20.

Reported exam shapes: none listed in the Master Map. The questions test what CVA is, its key advantage over a limit, which level each tool works at and what each does to the number of counterparties, how limits are run, and why CVA became risk-neutral.

What the questions look like: "which statement is correct / true EXCEPT". Two questions are new from GARP 15.1.5 and 15.1.7; Question CR-15 e.2 is a bank calculation rebuilt as a qualitative question to respect the CONCEPT tag.

1. Credit limits: the traditional control

Credit limits (credit lines) control the amount of counterparty risk taken over time, and diversify it by limiting the exposure to any one counterparty, sector or region, with regard to the underlying credit quality (GARP 15.1.5). Where diversification is impractical because of specialisation or relationships, exposures can become too large and need other mitigants.

- **PFE against a limit.** The limit is set subjectively from risk appetite and compared with the potential future exposure, a bad-scenario exposure at a confidence level (about the loan size for a loan; modelled for derivatives, CR-19).
- **Limits step down with the horizon,** because credit quality may deteriorate and investment-grade default probabilities tend to rise over time.
- **Incremental PFE.** A trade that would breach the limit at *any* future date is refused unless specifically approved. What counts is PFE after minus PFE before, smaller than the standalone PFE; a risk-reducing trade can lower it.
- **Soft and hard limits.** A new trade that would breach is simply refused. Market moves may breach a soft limit (say USD 10m) without action; a breach of the hard limit (say USD 15m) requires remedial action: unwind, restructure or hedge.
- **Binary.** A full limit blocks even very profitable trades; penalties near a limit tend to be ad hoc.

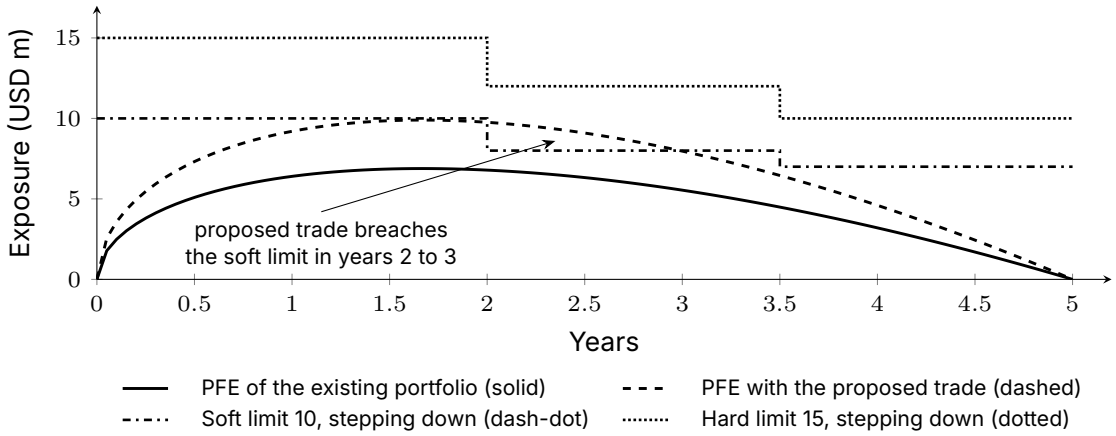


Figure 5. Illustrative PFE profiles against limits that step down with the horizon. The existing portfolio fits everywhere; the proposed trade fits today but breaches the soft limit in years 2 to 3, so it is refused unless specifically approved. A later rise in PFE from market moves alone could run up to the hard limit before remedial action is required.

2. CVA: the price of counterparty risk

CVA is the difference between the value of a derivative assuming the counterparty cannot default and its value allowing for default: the market value of counterparty credit risk (GARP 15.1.6; CR-12 b). It turns the approval question into "is the trade still profitable once the counterparty risk is priced in?", so it sets a *minimum revenue*. With credit limits, by contrast, the implied price is either zero (trade accepted) or infinite (trade rejected). Credit limits can therefore send the wrong signal: a low-profit trade accepted because the limit has room, a high-profit one rejected because it has not.

CVA is a counterparty-level (strictly, netting-set-level) calculation, done incrementally: the change in CVA from adding the trade to the existing netting set. Because of this, CVA is **additive across counterparties** and does not see concentration: one very large exposure, or several highly correlated names in the same region or sector.

3. Using the two together

	Credit limits	CVA
Question answered	How much exposure may we take?	At what price should we take it?
Level	Portfolio: limits concentrations	Trade (all trade features) and counterparty (netting, collateral)
Output	Binary: accept or refuse	A price: the minimum revenue for the trade
Concentration	Seen and limited	Not seen: additive across counterparties
Number of counterparties	Encourages more (smaller exposures, diversification)	Encourages fewer (bigger netting benefits)
Who runs it	Credit risk department approves the trade	xVA desk prices the CVA before the trade is done

The two act on different levels and pull in opposite directions on the number of counterparties, so banks use them as complements (GARP Figure 15.6): the credit department approves or rejects, then

the xVA desk prices in the CVA.

4. What CVA represents: actuarial or risk-neutral

A price can be an **actuarial** expected value with a risk premium (the insurance view), or the cost of a **hedging strategy** (the risk-neutral, market-implied view). Banks once treated CVA actuarially, as a reserve for expected counterparty losses, much like a loan loss reserve. The risk-neutral approach now dominates, for four reasons (GARP 15.1.7):

- **Market practice:** large US banks adopted it early, and novation prices spread it to others.
- **Accounting:** the exit-price concept of FAS 157 and IFRS 13 implies risk-neutral default probabilities; US and Canadian banks under FAS 157 adopted early, while many European banks under IAS 39 stayed actuarial until IFRS 13 (from 2013). Exit price also requires DVA, which is seen as problematic.
- **Basel III:** capital rules define CVA with respect to credit spreads, but do not permit DVA, a conflict with accounting.
- **Regulators:** models "based on expected loss or historical calculations ignoring risk premia" are not acceptable.

Historical default probabilities are now uncommon in CVA.

Exam traps

- **Levels swapped.** CVA works at trade and counterparty level; limits at portfolio level.
- **Incentives swapped.** CVA favours fewer counterparties (netting); limits favour more (diversification).
- **"Limits cannot use PFE".** Limits are normally set on PFE; their weakness is that they are binary.
- **"CVA captures concentration".** It is additive across counterparties; limits handle concentration.
- **Standalone PFE against the limit.** The incremental PFE counts, and it is usually smaller.
- **Actuarial and risk-neutral swapped.** The hedging-cost view is risk-neutral; the reserve view is actuarial.
- **DVA everywhere.** Accounting (IFRS 13) recognises DVA; Basel III capital does not.

Practice questions: CR-15 e

Question CR-15 e.1

Bank Q692 • AP Mock 2 Q10 • AnalystPrep Q.1965 (rebuilt)

A bank's derivatives desk adjusts the value of an OTC derivative to reflect the possibility that the counterparty defaults before meeting its obligations. The adjustment is the difference between the value of the contract assuming no counterparty default and its true (risky) value. Which of the following best describes this credit value adjustment (CVA)?

A. The market value of counterparty market risk.

B. The market value of counterparty credit risk.

C. The cost of posting initial margin over the life of the transaction.

D. The cost of holding regulatory capital against the transaction over its life.

Question CR-15 e.2

Bank Q695 • BT P2.T6.901.3 (rebuilt as a qualitative question)

HFQ LLC, a market maker, holds an OTC derivatives position for which it has estimated the following, each stated as a positive amount: a risk-free value of USD 32.0 million, a credit value adjustment (CVA) of USD 5.0 million, and a debt value adjustment (DVA) of USD 8.0 million. Based only on this information, which statement about the risky (credit-adjusted) value of the position from HFQ's perspective is correct?

A. It is below the risk-free value, because both CVA and DVA reduce the value to the party that holds the position.

B. It equals the risk-free value less CVA, because DVA never enters the valuation of a derivative.

C. It is above the risk-free value, because the DVA benefit from HFQ's own default risk exceeds the CVA cost.

D. It cannot be determined without estimates of the counterparty's default probability and loss given default.

Question CR-15 e.3

Bank Q693 • BT P2.T6.320.1

Traditional counterparty credit risk management depends on credit limits, but a more sophisticated approach uses a credit value adjustment (CVA). According to Gregory, which is the key advantage of CVA over a credit line?

A. CVA can include potential future exposure (PFE), but credit limits cannot incorporate PFE.

B. CVA incorporates a portfolio perspective, but credit limits are not portfolio credit risk management.

C. CVA enables transaction decisions based on profitability at the trade level, whereas limits only control exposure at the portfolio level.

D. CVA includes default probability, recovery and correlation between counterparties, but credit limits cannot.

Question CR-15 e.4

Bank Q694 • BT P2.T6.902.1

Patricia is a risk analyst building a counterparty risk management system for her international financial services firm. The Risk Committee wants the system to assess counterparty risk at three levels: trade, counterparty and portfolio. Patricia is weighing credit limits against the credit value adjustment (CVA). Each of the following is true EXCEPT:

A. CVA and credit limits are complements in this three-level system.

B. Credit limits act at the portfolio level by limiting exposures to avoid concentrations.

C. CVA focuses on evaluating counterparty risk at the trade level and the counterparty level.

D. CVA encourages maximising the number of trading counterparties in order to maximise the benefits of netting, but credit limits encourage minimising this number to encourage smaller exposures.

Question CR-15 e.5

New • built from GARP Ch.15, 15.1.5

The credit risk department of a bank controls its exposure to a corporate counterparty with a PFE-based limit structure: a soft limit of USD 10 million and a hard limit of USD 15 million, both stepping down for later time horizons. Which of the following practices is consistent with GARP's description of credit limits?

A. A rise in PFE to USD 12 million caused purely by market moves breaches the soft limit but does not by itself force trades to be unwound, whereas a breach of the hard limit requires remedial action.

B. A proposed trade is compared with the remaining headroom using its standalone PFE, since netting with existing trades is captured only in CVA.

C. The limits should step up at longer horizons, because exposures on most derivatives grow with maturity.

D. Because the soft limit can be breached by market moves, new trades that take PFE above USD 10 million at a future date can be booked without approval.

Question CR-15 e.6

New • built from GARP Ch.15, 15.1.7

A bank's valuation committee is reviewing why its CVA methodology moved from historical default probabilities to market-implied ones. Which of the following statements is most accurate?

A. Under the actuarial approach, CVA is the cost of a hedging strategy built from CDS spreads, so it already uses market-implied default probabilities.

B. European banks reporting under IAS 39 were the early adopters of risk-neutral CVA, while US banks under FAS 157 kept actuarial reserves until IFRS 13 forced convergence.

C. Regulators generally accept CVA models based on expected loss from historical default rates, provided the risk premium is excluded.

D. The exit-price concept in FAS 157 and IFRS 13 points to risk-neutral default probabilities, and Basel III also defines CVA with respect to credit spreads, although Basel III does not permit DVA.

Solutions • CR-15 e

Solution CR-15 e.1

Answer: B

Bank Q692 • AP Mock 2 Q10

Step 1 – Read the definition in the stem

Risk-free value minus risky value: the reduction in value caused by the possibility of counterparty default.

Step 2 – Name it

That reduction is what the market would charge for bearing the counterparty's default risk: the market value of counterparty *credit* risk.

Interpretation

CVA prices the risk that a credit limit only caps. The other two options are sibling xVA terms (CR-15 h).

Why the other options are wrong

A is incorrect. Counterparty risk is a credit risk; market variables only size the exposure.
C is incorrect. That is MVA, the margin value adjustment.
D is incorrect. That is KVA, the capital value adjustment.

LO: CR-15 e: Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.6 Credit Value Adjustment and 15.2.3

Solution CR-15 e.2

Answer: C

Bank Q695 • BT P2.T6.901.3

Step 1 – The value identity

Credit-adjusted value = risk-free value − CVA + DVA (CR-12 b). CVA is the cost of the counterparty's default; DVA is the benefit from HFQ's own default risk.

Step 2 – Direction

The net adjustment is $-5.0 + 8.0 = +3.0$, so the risky value ($32.0 - 5.0 + 8.0 =$ USD 35.0 million) is above the risk-free value.

Interpretation

DVA is recognised for accounting under the exit-price concept of IFRS 13, but not in Basel III capital; "never" in option B is the trap. When DVA exceeds CVA, the institution's own credit is worse than the counterparty's after weighting by exposure.

Why the other options are wrong

A is incorrect. Treats DVA as a cost: $32.0 - 5.0 - 8.0 = 19.0$.
B is incorrect. Drops DVA altogether, $32.0 - 5.0 = 27.0$; DVA does enter fair value.
D is incorrect. PD and LGD are inputs to CVA and DVA, which are already given.

LO: CR-15 e: Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk. CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.7 and 15.2.3

Solution CR-15 e.3 — Answer: C

Bank Q693 • BT P2.T6.320.1

Step 1 – How a limit decides

Limits are set on PFE at portfolio level; a trade is accepted if it fits and refused if it breaches. The implied price of counterparty risk is zero or infinite.

Step 2 – How CVA decides

CVA is the actual price of counterparty risk at trade and counterparty level, so the question becomes whether the trade is profitable once CVA is priced in.

$\Rightarrow$ The key advantage is profitability-based decisions.

Interpretation

Limits can accept a low-profit trade because the line has room and refuse a high-profit one because it has not; CVA corrects that incentive.

Why the other options are wrong

A is incorrect. Limits are normally expressed in PFE.

B is incorrect. Reversed: limits are the portfolio tool; CVA is counterparty-level and misses concentration.

D is incorrect. Limits are set with regard to credit quality; the distinguishing advantage is pricing, not the inputs.

LO: CR-15 e: Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.5 and 15.1.6

Solution CR-15 e.4 — Answer: D

Bank Q694 • BT P2.T6.902.1

Step 1 – Levels

CVA covers trade level (trade features) and counterparty level (netting, collateral); limits act at portfolio level (B and C true). Different levels make them complements (A true).

Step 2 – Incentives

Netting benefits are largest when trades are concentrated with few counterparties, so CVA encourages *fewer*; limits cap each name, so they encourage *more*, smaller exposures.

$\Rightarrow$ D reverses both, so it is the false statement.

Interpretation

GARP Figure 15.6: trade and counterparty information feed CVA, portfolio information feeds limits.

Why the other options are wrong

A is incorrect. True: they are used together.

B is incorrect. True: limits avoid concentrations.

C is incorrect. True: CVA captures trade features and counterparty-level mitigants.

LO: CR-15 e: Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.6, Figure 15.6

Solution CR-15 e.5 Answer: A

New • built from GARP Ch.15, 15.1.5

Step 1 – Two ways to breach

A new trade that would breach a limit is refused unless escalated. A breach caused by market moves is handled through soft and hard limits.

Step 2 – Soft against hard

GARP: a soft limit "may be breached through market movements (but not new transactions)", while a breach of the hard limit requires remedial action (unwind, restructure or hedge); its example is USD 10m soft and USD 15m hard.
$\Rightarrow$ A describes exactly this.

Interpretation

Figure 5 shows the logic: the proposed trade breaches the soft limit in years 2 to 3 and is refused, while the existing portfolio could drift up to the hard limit on market moves.

Why the other options are wrong

B is incorrect. The incremental PFE (after minus before) is what counts, and it is smaller than the standalone PFE because of portfolio effects.
C is incorrect. Polarity flip: limits are often reduced over time, because credit quality may deteriorate over long horizons.
D is incorrect. New trades that would breach the limit at any future point are refused unless specifically approved.

LO: CR-15 e: Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.5 Credit Limits, Figure 15.5

Solution CR-15 e.6 Answer: D

New • built from GARP Ch.15, 15.1.7

Step 1 – Accounting

FAS 157 and IFRS 13 reference an exit price, which implies risk-neutral default probabilities (and recognition of DVA).

Step 2 – Regulation

Basel III defines CVA with respect to credit spreads, so it also points to risk-neutral CVA, but it does not permit DVA: a conflict with accounting.
$\Rightarrow$ D is accurate.

Interpretation

Market practice and local regulators pushed the same way; historical default probabilities are now uncommon in CVA.

Why the other options are wrong

A is incorrect. Reverses the definitions: the hedging-cost view is risk-neutral; the actuarial view is an expected value with a risk premium.
B is incorrect. Reversed: US and Canadian banks under FAS 157 adopted early; many European banks under IAS 39 used actuarial CVA.
C is incorrect. Regulators say models "based on expected loss or historical calculations ignoring risk premia" are not acceptable.

LO: CR-15 e: Describe credit value adjustment (CVA) and compare the use of CVA and credit limits in evaluating and mitigating counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.7 What Does CVA Represent?

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 f

P3 CONCEPT

Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No workbook section; Vardeez flags the LO in his foundation chapter for counterparty risk.

Vardeez: seven ways to *manage* counterparty risk (high-quality counterparties, cross-product netting, close-out, collateral, walkaway features, diversification, clearinghouses) and five ways to *mitigate* it (netting, collateral, hedging, CCPs, other clauses), each with its side effect; quantification by trade, counterparty and portfolio level (CR-15 e).

Reported exam shapes: none listed in the Master Map. The questions ask for the best tool in a situation (a concentrated market, a weakening counterparty, a negative MTM at default) and for the new risk each mitigant creates.

What the questions look like: short scenarios with four tools as options, and "true EXCEPT" lists pairing mitigants with risks. **GARP practice item kept:** Question CR-15 f.3 (AP Mock 1 Q7) is a netting calculation on a describe LO; it is kept as a GARP practice question.

1. Quantify, manage, mitigate

Quantify at three levels (CR-15 e): the *trade* on its own (cash flows, risk factors), the *counterparty* or netting set (netting and collateral, giving the incremental effect of a trade), and the *portfolio* of all counterparties (knowing only a few will default in a period). CVA covers the first two; credit limits the third. **Manage** and **mitigate** with the tools below.

Tool	How it works	The catch
High-quality counterparties	Trade only with strong credits, which may not even be asked for collateral	Concentrates on few names; quality can deteriorate
Cross-product netting	Aggregate positive and negative values with one counterparty; only the net is at risk	Legal risk if the agreement is not enforceable
Close-out	On default, terminate all trades at once and net the values: pay if the net is negative, claim if positive	Relies on enforceable netting; losses crystallise
Collateral (margin)	Counterparty posts cash or securities against the net exposure; in theory exposure falls to zero	Residual risk over the MPoR; operational, liquidity, legal and FX risk
Walkaway feature	Non-defaulting party may cancel the trades if it owes the defaulter	Does not reduce exposure; a gain that offsets losses; uncommon since the 1992 ISDA (CR-16)
Diversification	Limit exposure to any one name, sector or region	Impractical in concentrated or specialised markets
Exchanges and CCPs	The CCP becomes the counterparty to every trade and guarantees performance	Centralises the risk: operational, liquidity and systemic risk
Other clauses	Resets or additional termination events reset the MTM or end trades early	Operational and liquidity risk
Hedging	Buy CDS protection on the counterparty	Cost; market risk from the hedge's MTM; operational risk

Netting in one line. Values of $+30$, -12 and $+5$ with one counterparty: the gross exposure without netting is $30+5=35$; with an enforceable netting agreement the exposure is the net, $30-12+5=23$.

2. Mitigation converts risk rather than removing it

GARP (15.1.3): no mitigant is perfect, residual counterparty risk always remains, and quantifying it can be harder than quantifying the original risk. Each mitigant turns counterparty risk into other risks:

- **Netting** creates legal risk where netting cannot be enforced in a jurisdiction.
- **Collateral** leaves market risk over the time it takes to receive it, creates operational risk (logistics) and liquidity risk (posting must be funded; collateral has price and FX volatility). Rehypothecation, segregation and even cash collateral can create wrong-way risk.
- **Other clauses** (resets, additional termination events) create operational and liquidity risk.
- **Hedging** with CDSs creates operational risk and market risk through the hedge's MTM volatility, and can feed systemic risk.
- **CCPs** require centralisation, heavy collateralisation and mutualisation of losses: operational and liquidity risk, and systemic risk because a CCP failure would be a major disturbance.

Mitigation is a double-edged sword: it improves stability, but it can loosen constraints such as capital and credit limits and so let volumes grow. Without netting and collateral the OTC market would never have reached its size. Think of mitigation as risk transfer.

Hedging and the CVA desk (GARP 15.1.8). The simplest hedge is a "jump-to-default" hedge: buy single-name CDS protection on the counterparty to increase its credit limit. Contingent CDSs (CCDSs) and risk participation agreements (RPAs) index the protection to the exposure on a specific derivative, but that market has never become liquid, and even single-name CDS covers few names. In practice CVA is hedged dynamically through credit spreads (often CDS indices, which are more liquid) and market variables. A CVA desk, now usually an **xVA desk**, prices and may take over the counterparty risk from the originating desk by transfer pricing; in the largest banks this goes back to the late 1990s.

> **Correction to the source notes.** Schweser (LO 33.f) says that "taking collateral equal to or greater than the notional amount of principal should theoretically eliminate all counterparty risk". Collateral is posted against MTM, the net exposure, not against notional (GARP 15.1.3; Vardeez: "to cover net exposures"); a swap's exposure is a small fraction of its notional. Even full collateral leaves residual risk over the margin period of risk.

Exam traps

- **Walkaway direction.** It helps the non-defaulting party only when that party owes the defaulter (negative MTM).
- **Netting takes exposure to zero.** Netting leaves the net amount; only sufficient collateral takes it towards zero.
- **Diversify in a concentrated market.** With only a handful of counterparties there is nobody to diversify into.
- **CDS from the same dealers.** Protection bought from the concentrated group of counterparties adds wrong-way risk.
- **Netting creates basis risk or extra exposure.** Its side effect is legal risk; it cannot raise exposure.
- **Collateral reduces every risk.** It reduces counterparty risk and adds operational, liquidity, market and legal risk.
- **CCPs eliminate counterparty risk.** They centralise it, and can create systemic risk.

Practice questions: CR-15 f

Question CR-15 f.1

Bank Q698 • BT P2.T6.24.24.1 (rebuilt)

G Corporation has a 3-year OTC interest rate swap with notional USD 50 million with Sunset Financial Bank. The swap currently has a positive mark-to-market value to G, so Sunset owes G. Concerns about Sunset's creditworthiness have grown. Which action would best mitigate G's counterparty credit risk on this swap?

A. Terminate the swap immediately and leave the interest rate risk unhedged.

B. Require Sunset to post collateral under a collateral (margin) agreement covering the current mark-to-market exposure.

C. Negotiate a walkaway feature in the swap contract.

D. Increase the frequency of mark-to-market valuations without changing the contract terms.

Question CR-15 f.2 Bank Q704 • Schweser Module Quiz 33.2 Q1

Ondine Financial, Inc. (Ondine) uses a variety of techniques to manage counterparty risk. It has entered into an interest rate swap with Scarbo, Inc. (Scarbo). Currently, Ondine's position in the swap has a mark-to-market value of −USD 1 million. Based on the information provided, which of the following credit risk mitigation techniques would be most advantageous to Ondine if Scarbo defaults?

A. Close-out.

B. Collateralisation.

C. Netting.

D. Walkaway.

Question CR-15 f.3 Bank Q706 • AP Mock 1 Q7 (rebuilt)

A hedge fund has the following mark-to-market positions with a single counterparty, AB&B, all covered by a legally enforceable cross-product netting agreement (positive means AB&B owes the fund):

Contract	A	B	C	D	E
MTM (USD millions)	+44.0	+88.0	−35.2	−3.3	+20.0

Assuming AB&B defaults with zero recovery, what is the fund's counterparty exposure to AB&B after netting?

A. USD 38.5 million

B. USD 113.5 million

C. USD 152.0 million

D. USD 190.5 million

Question CR-15 f.4 Bank Q703 • GARP PE1 Q28 • Mock 3 Q52

A derivative trading firm only trades derivatives on rare commodities. The company and a handful of other firms, all of whom have large notional outstanding contracts with the company, dominate the market for such derivatives. The company's management would like to mitigate its overall counterparty exposure, with the goal of reducing it to almost zero. Which of the following methods, if implemented, could best achieve this goal?

A. Ensuring that sufficient collateral is posted by counterparties

B. Diversifying among counterparties

C. Cross-product netting on a single-counterparty basis

D. Purchasing credit derivatives, such as credit default swaps

Question CR-15 f.5 Bank Q702 • BT P2.T6.900.2

Acme Financial Incorporated has an outstanding derivatives contract with its counterparty. Currently neither party posts collateral, but they agree to modify the contract to include a bilateral collateral arrangement. Which risk has Acme reduced?

A. Counterparty risk

B. Operational risk

C. Market and liquidity risk

D. All of the above (counterparty, operational, and market and liquidity risk)

Question CR-15 f.6 Bank Q701 • BT P2.T6.406.3

According to Gregory, risk mitigation is imperfect: it cannot perfectly eliminate counterparty risk, and it creates new risks. Each of the following pairings of a mitigation method with the risk it creates is true EXCEPT:

A. Collateral leads to operational and liquidity risks through the need to run a complex collateral management function.

B. Central counterparties create systemic risk, since the failure of a CCP could be a significant systemic disturbance.

C. Hedging may lead to additional market risk through the mark-to-market volatility of the hedging instruments.

D. Netting creates additional exposure in the form of basis risk, because the mark-to-market of a position can become positive for both counterparties.

Question CR-15 f.7 Bank Q700 • BT P2.T6.406.1

Which of the following is TRUE about counterparty risk?

A. Counterparty risk excludes pre-settlement and settlement risks.

B. In contrast to counterparty risk, settlement risk is characterised by small exposures for long periods of time.

C. Counterparty risk mitigation does not eliminate risk but instead converts counterparty risk into other types of financial risk.

D. In the credit derivatives market, notional outstanding is one of the more accurate measures of counterparty risk in the system.

Solutions • CR-15 f

Solution CR-15 f.1 — Answer: B

Bank Q698 • BT P2.T6.24.24.1

Step 1 – Who bears the risk

G is owed money (positive MTM), so G is exposed to Sunset.

Step 2 – Match the tool

Collateral posted by Sunset against the MTM offsets G's loss if Sunset defaults, and keeps the hedge in place.

Interpretation

The three distractors fail in different ways: one swaps counterparty risk for market risk, one helps the wrong party, and one only measures.

Why the other options are wrong

A is incorrect. Leaves G unhedged against rates: it converts risk rather than mitigating it, and may realise a claim on a weak name.

C is incorrect. A walkaway clause benefits the party with negative MTM if the other defaults; that is Sunset, not G.

D is incorrect. Monitoring measures exposure but does not reduce it.

LO: CR-15 f: Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.3 Mitigating Counterparty Risk

Solution CR-15 f.2 — Answer: D

Bank Q704 • Schweser Module Quiz 33.2 Q1

Step 1 – Ondine's position

MTM −USD 1 million: Ondine owes Scarbo. Scarbo's default costs Ondine nothing in lost receivables.

Step 2 – Which tool helps

A walkaway feature lets the non-defaulting party cancel the trades, so Ondine need not pay the USD 1 million.

Interpretation

GARP (CR-16): walkaway features do not reduce exposure; they produce a gain that offsets other losses, were common before the 1992 ISDA Master Agreement and are not part of standard documentation now. In the Drexel Burnham Lambert default, counterparties chose not to walk away.

Why the other options are wrong

A is incorrect. Close-out crystallises the −USD 1 million, which Ondine pays.

B is incorrect. Collateral protects the party with positive exposure; Ondine has none.

C is incorrect. Netting offsets other trades; with this single swap Ondine still owes USD 1 million.

LO: CR-15 f: Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. • **Reference:** SchweserNotes 2026 Book 2, Reading 33, LO 33.e and 33.f; Gregory, The xVA Challenge 4e, Ch.6 (GARP CR Ch.16), 16.3.3 Payment Under Close-Out

Solution CR-15 f.3 — Answer: B

Bank Q706 • AP Mock 1 Q7

Step 1 – Sum the claims

$44.0 + 88.0 + 20.0 = 152.0.$

Step 2 – Sum the liabilities

$-35.2 - 3.3 = -38.5.$

Step 3 – Net

$152.0 - 38.5 = 113.5$, a net claim, so the exposure is USD 113.5 million.

Interpretation

Netting cuts the exposure from USD 152.0 million gross to USD 113.5 million. Without enforceable netting the fund would lose USD 152.0 million and still owe USD 38.5 million.

Why the other options are wrong

A is incorrect. The liabilities alone, $35.2 + 3.3 = 38.5$.
C is incorrect. Gross positive exposure, ignoring the netting agreement.
D is incorrect. Added absolute values of all five contracts: $44.0+88.0+35.2+3.3+20.0 = 190.5$.

LO: CR-15 f: Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.3; Ch.6 (GARP CR Ch.16), 16.3.2 Close-Out Netting

Solution CR-15 f.4 — Answer: A

Bank Q703 • GARP PE1 Q28

Step 1 – Read the market structure

A handful of firms, all already counterparties with large notionals.

Step 2 – Test each tool against "almost zero"

Diversification needs other counterparties, and there are none. Netting reduces gross to net but leaves the net exposure. CDS protection would be bought from the same few dealers. Sufficient collateral posted against the net exposure can in theory take it to zero with each existing counterparty.

Interpretation

Only collateral targets the level of exposure itself; the others reduce or move it. The residual risk over the margin period of risk still remains.

Why the other options are wrong

B is incorrect. Needs a larger pool of counterparties than the market offers.
C is incorrect. Leaves the net exposure, usually well above zero.
D is incorrect. The sellers are the same concentrated group (wrong-way risk), and hedging adds market risk.

LO: CR-15 f: Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.3 and 15.1.5

Solution CR-15 f.5 — Answer: A

Bank Q702 • BT P2.T6.900.2

Step 1 – What collateral reduces

Collateral posted against the net exposure reduces the loss on the counterparty's default.

Step 2 – What it adds

Margin calls, valuation and transfers add operational risk; collateral must be funded and its value moves with prices and FX, adding liquidity and market risk; enforceability adds legal risk.
⇒ Only counterparty risk is reduced.

Interpretation

This is GARP's "conversion" point: mitigation transforms counterparty risk into other risks.

Why the other options are wrong

B is incorrect. Operational risk rises.
C is incorrect. Market and liquidity risk rise.
D is incorrect. Only counterparty risk falls.

LO: CR-15 f: Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.3

Solution CR-15 f.6 — Answer: D

Bank Q701 • BT P2.T6.406.3

Step 1 – Check the documented side effects

Collateral: operational and liquidity risk (A true). CCPs: systemic risk (B true). Hedging: market risk from the hedge's MTM (C true).

Step 2 – Netting

Netting offsets values within a netting set; it cannot raise exposure. Its side effect is legal risk where the agreement is not enforceable.
⇒ D is false.

Interpretation

Learn the pairs: netting and legal risk; collateral and operational and liquidity risk; hedging and market risk; CCPs and systemic risk.

Why the other options are wrong

A is incorrect. True, so not the exception.
B is incorrect. True, so not the exception.
C is incorrect. True, so not the exception.

LO: CR-15 f: Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.3

Solution CR-15 f.7 — Answer: C

Bank Q700 • BT P2.T6.406.1

Step 1 – GARP's statement

Risk mitigants do not remove counterparty risk per se but convert it into other forms of financial risk.

Step 2 – Eliminate the rest

Counterparty risk includes pre-settlement and settlement risk (A false); settlement risk is large and brief (B reversed); notional overstates risk and is a poor measure (D false).

Interpretation

Each mitigant leaves residual counterparty risk and adds legal, operational, liquidity, market or systemic risk.

Why the other options are wrong

A is incorrect. The two are the components of counterparty risk.

B is incorrect. Reversed: large exposures for short periods.
D is incorrect. Notional is not an accurate measure of counterparty risk.

LO: CR-15 f: Identify and describe the different ways institutions can quantify, manage, and mitigate counterparty risk. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.2 and 15.1.3

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 g

P3 CONCEPT

Identify and explain the costs of an OTC derivative.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No workbook section; Vardeez flags the LO in his foundation chapter for counterparty risk.

Vardeez: "the costs depend on which side of the trade you are on": in the money, the counterparty risk is theirs to create and yours to bear; out of the money it is yours to create. Funding runs the other way: a cost in the money, a benefit out of the money. (His placement of the funding cost is corrected below.)

Reported exam shapes: none listed in the Master Map. The questions give the sign of the MTM and the collateral terms and ask which costs and benefits arise, and test that capital and initial margin cost money whatever the sign.

What the questions look like: four parallel option bundles ("counterparty risk from ... and a funding ..., plus ..."). One question is re-filed from CR-15 h and two are new from GARP 15.2.2 and 15.3.6.

1. Costs beyond counterparty risk

After the global financial crisis four costs linked to counterparty risk gained attention alongside CVA (GARP 15.2.1): **funding** (the more risk on the balance sheet, the dearer the borrowing), **collateral** (a cost or a benefit depending on its type), **initial margin** (over-collateralisation, required for example in central clearing, and so a funding cost) and **regulatory capital** against counterparty risk and CVA.

2. The lifetime costs of a derivative

Which costs arise depends on the sign of the portfolio value and on the collateral thresholds: the counterparty posts above *its* threshold, you post below *yours* (GARP 15.2.2, Figure 15.9). In practice thresholds are often zero or infinity.

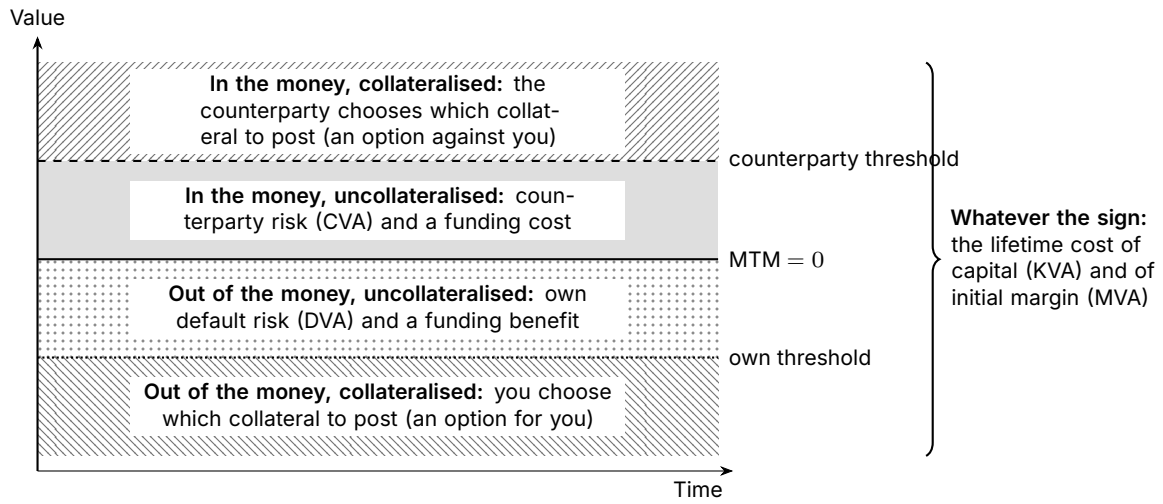


Figure 6. The lifetime costs of a derivative by region of value, after GARP Figure 15.9. Grey: in the money and uncollateralised; dotted: out of the money and uncollateralised; hatched: the collateralised regions beyond each threshold. The brace marks the costs that apply in every region.

	Positive MTM (in the money)	Negative MTM (out of the money)
Counterparty risk	The counterparty's default (CVA), on the uncollateralised part	Your own default (DVA)
Funding	A cost on the uncollateralised part: an asset you must fund	A benefit to the extent the counterparty is uncollateralised: you do not post
Collateral choice	The counterparty chooses what to post	You choose what to post
Capital and initial margin	A cost	A cost

Symmetry, and where it breaks. One party's positive value is the other's negative value, so one party's CVA cost is the other's DVA benefit. Capital and initial margin break the symmetry: they are costs to both parties with no equal and opposite benefit.

3. Where the costs come from

GARP 15.3.6 gives one example of each:

- **Funding.** An uncollateralised trade that requires an upfront cash payment must be funded, for example by issuing a bond, because the cash flows that repay it come later; lenders require compensation. In a collateralised trade the margin offsets the payment. Initial margin offsets nothing and always needs funding.
- **Collateral.** Funding is defined in base-currency cash. Collateral in another currency or in securities

earns a different rate: receiving bonds may be a funding benefit, but a repo rate defines the cost of turning them into cash. The poster may also have a contractual choice of currency or security.

- **Capital.** A trade that raises capital requirements has a cost, because shareholders require a return (dividends) on the capital.

Some inputs are objective (yields on the bank's own bonds); others subjective (the funding maturity a trade needs, future dividends).

The bank's classic set-up (GARP 14.2.5). A bank hedges an uncollateralised client trade with a collateralised trade in the interbank market. When the client trade moves in the bank's favour, the hedge moves against it: the bank posts collateral on the hedge but receives none from the client, and must fund the difference. This asymmetry is the source of most funding costs.

> **Correction to the source notes.** Vardeez's CR-15 g says that in the money "you might incur funding costs for the collateralized part, depending on who chooses the collateral", and that out of the money "you might still incur funding costs on the collateralized part". GARP (15.2.2) places the funding cost on the *uncollateralised* part of a positive value (together with the counterparty risk), and the funding benefit on the uncollateralised part of a negative value. On the collateralised part the issue is who chooses the collateral: the counterparty when you are in the money, you when you are out of the money.

Exam traps

- **Sign swap.** Counterparty risk from the counterparty's default and a funding cost go with a positive MTM; own-default risk and a funding benefit go with a negative MTM.
- **No costs at zero MTM.** Capital and initial margin cost money whatever the MTM.
- **Initial margin as a benefit.** It is over-collateralisation and a funding cost.
- **Every cost has a mirror benefit.** CVA and DVA mirror each other; capital and initial margin do not.
- **A hedged book has no funding cost.** An uncollateralised client trade hedged with a collateralised trade creates one.
- **Generic costs.** Brokerage, documentation and an option premium are not the lifetime economic costs in this LO.

Practice questions: CR-15 g

Question CR-15 g.1

Bank Q708 • AP Mock 1 Q19

Global Finance Bank (GFB) is training its derivatives team on the economic costs of OTC derivatives over their lifetime, and on how the mark-to-market position affects those costs. The case study is an interest rate swap currently in a positive MTM position: the transaction is in the money to GFB. The swap is partially collateralised. Which of the following accurately describes the economic costs GFB would incur in this scenario?

A. Counterparty risk costs due to GFB's potential for default, and a funding benefit from the uncollateralised component of the MTM position.

B. Counterparty risk and funding costs arising from the uncollateralised component of the MTM position, and funding costs for the capital needed for the transaction and for initial margin.

C. Legal and compliance costs for contract documentation, and market risk costs due to potential unfavourable movements in interest rates.

D. Brokerage and transaction fees for executing the swap, and premium costs for buying the derivative.

Question CR-15 g.2

Bank Q709 • New question written for this bank

A derivatives risk manager at a bank reviews an interest rate swap whose mark-to-market (MTM) is currently negative (out of the money) for the bank. The counterparty does not post collateral. Which of the following best describes the economic costs and benefits the bank faces over the life of the swap?

A. Counterparty risk from the bank's own default and a funding benefit on the uncollateralised MTM, plus a funding cost for the capital and any initial margin.

B. Counterparty risk from the counterparty's default and a funding cost on the uncollateralised MTM, plus the funding cost of the capital held.

C. No counterparty risk and no funding impact, because a negative MTM means the bank cannot lose on the counterparty's default.

D. Counterparty risk from the bank's own default and a funding cost on the uncollateralised MTM, but no cost for capital or initial margin.

Question CR-15 g.3 Bank Q710 • New question written for this bank

A bank enters a centrally cleared swap that requires it to post initial margin, and the swap's MTM is subsequently close to zero. A trainee argues that the bank has no economic costs on the position because the MTM is neither positive nor negative. Which of the following best evaluates the trainee's claim?

A. The claim is correct, because economic costs of a derivative arise only when MTM is positive.

B. The claim is incorrect, because initial margin represents over-collateralisation that carries a funding cost, and funding the capital held against the transaction is a cost regardless of the MTM sign.

C. The claim is incorrect, because posting initial margin creates a funding benefit that offsets the capital cost to zero.

D. The claim is correct, because initial margin is returned immediately and so carries no funding or capital implications.

Question CR-15 g.4 Bank Q715 • BT P2.T6.902.2 • BT Mock Q57 (rebuilt) • re-filed from CR-15 h

A bank's CVA desk (market maker) enters an uncollateralised trade with a client and hedges it with a collateralised trade in the market. The client trade has a positive MTM to the bank, so the collateralised hedge has a negative MTM. Market moves now make the client trade's positive MTM even larger. Using the convention that a cost is a positive adjustment, which of the following implications is most likely true?

A. The funding cost (FVA) is positive.

B. From the trader's (market maker's) perspective, CVA decreases.

C. Because the position is hedged, there is no associated regulatory cost of capital to the bank.

D. The trader (market maker) writes a contingent credit default swap (CCDS) to transfer its counterparty risk.

Question CR-15 g.5 New • built from GARP Ch.15, 15.3.6

A treasury analyst at a bank is explaining to the derivatives desk how funding and collateral costs arise. Ignoring initial margin and capital, which of the following statements is correct?

A. A collateralised trade on which the bank makes an upfront payment needs funding just like an uncollateralised one, because margin never offsets an upfront payment.

B. Receiving government bonds instead of cash as collateral removes all funding and collateral costs, because bonds can always be converted to cash at no cost.

C. Funding costs are fully objective inputs, because the bank can read them from its bond spreads for any maturity it chooses.

D. An uncollateralised trade on which the bank makes an upfront payment must be funded, for example by issuing a bond, because the cash flows that repay it arrive later.

Question CR-15 g.6 New • built from GARP Ch.15, 15.2.2

Two banks trade a swap on which both must hold regulatory capital and post initial margin. A trainee claims that every xVA cost borne by one bank is an equal and opposite benefit to the other. Which assessment is most accurate?

A. The trainee is right for every term, because one party's positive value is always the other's negative value.

B. The trainee is wrong for every term, because CVA and DVA are costs to both parties.

C. The trainee is right for CVA and DVA, since one party's CVA cost is the other's DVA benefit, but wrong for capital and initial margin, which are costs to both parties with no offsetting benefit.

D. The trainee is right for capital, because one bank's regulatory capital charge is released as capital relief at its counterparty.

Solutions • CR-15 g

Solution CR-15 g.1

Answer: B

Bank Q708 • AP Mock 1 Q19

Step 1 – In the money

The counterparty owes GFB. On the uncollateralised part GFB bears counterparty risk and must fund an asset for which it holds no collateral: a funding cost.

Step 2 – The collateralised part

The counterparty chooses which collateral to post; no further exposure cost.

Step 3 – Whatever the sign

Capital and initial margin must be funded.
$\Rightarrow$ B combines the three.

Interpretation

GARP's Figure 15.9 above the centre line: counterparty risk and funding cost, then the collateral option, with capital and initial margin spanning the whole picture.

Why the other options are wrong

A is incorrect. Describes the out-of-the-money case: own-default risk and a funding benefit.
C is incorrect. Documentation and generic market risk are not the lifetime economic costs that depend on MTM and collateral.
D is incorrect. A swap has no premium, and brokerage is not one of these costs.

LO: CR-15 g: Identify and explain the costs of an OTC derivative. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.2 Economic Costs of a Derivative, Figure 15.9

Solution CR-15 g.2

Answer: A

Bank Q709 • New question written for this bank

Step 1 – Out of the money

The bank owes the counterparty, so the counterparty risk arises from the bank's own default (DVA).

Step 2 – Funding

The counterparty is uncollateralised: the bank posts nothing against its liability, which works as a funding benefit.

Step 3 – Whatever the sign

Capital and any initial margin cost money.

Interpretation

The mirror image of Question CR-15 g.1: the counterparty risk and the funding effect both flip, the capital and margin costs do not.

Why the other options are wrong

B is incorrect. Describes the in-the-money case.
C is incorrect. Own-default risk and funding effects remain, and capital and margin still cost money.
D is incorrect. Funding effect reversed, and the capital and margin costs omitted.

LO: CR-15 g: Identify and explain the costs of an OTC derivative. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.2

Solution CR-15 g.3 Answer: B

Bank Q710 • New question written for this bank

Step 1 – Initial margin

Initial margin is over-collateralisation: the bank must fund it for as long as the trade lives.

Step 2 – Capital

The capital held against the trade must also be funded, whether MTM is positive, negative or zero.

Interpretation

GARP: "whether or not the portfolio has a positive or negative value, there are costs from funding the capital … and any initial margin".

Why the other options are wrong

A is incorrect. Costs arise in both MTM states, and capital and margin costs apply always.
C is incorrect. Initial margin is a cost, not a benefit.
D is incorrect. Initial margin is held for the life of the trade.

LO: CR-15 g: Identify and explain the costs of an OTC derivative. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.1 and 15.2.2

Solution CR-15 g.4 Answer: A

Bank Q715 • BT P2.T6.902.2

Step 1 – The hedge

The collateralised hedge moves further against the bank, so the bank posts more collateral on it.

Step 2 – The client trade

The client trade is uncollateralised: the bank receives nothing against its larger claim, so the collateral it posts must be funded.
$\Rightarrow$ A funding cost: positive FVA in the cost-positive convention.

Interpretation

This is GARP's classic bank set-up (14.2.5): uncollateralised client, collateralised hedge. The market risk is hedged; the funding and the counterparty risk are not.

Why the other options are wrong

B is incorrect. The exposure to the client has grown, so CVA increases.
C is incorrect. The hedge removes market risk, not the counterparty risk or the capital held against the client leg.
D is incorrect. To transfer counterparty risk the desk would buy a CCDS; writing one adds risk.

LO: CR-15 g: Identify and explain the costs of an OTC derivative. CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.1.8 and 15.2.2; Ch.2 (GARP CR Ch.14), 14.2.5

Solution CR-15 g.5 Answer: D

New • built from GARP Ch.15, 15.3.6

Step 1 – GARP's funding example

An uncollateralised trade with an upfront cash payment must be funded, perhaps by issuing a bond, because the offsetting cash flows arrive later; lenders require compensation for that.

Step 2 – The collateralised version

With collateral, the margin offsets the upfront payment, so the funding need disappears (initial

margin excepted).

Interpretation

Funding, collateral and capital costs are, like credit spreads, hard to assess, and they are key inputs to xVA (CR-15 h).

Why the other options are wrong

A is incorrect. Polarity flip: in a collateralised trade the margin offsets the payment.
B is incorrect. A repo rate defines the cost of converting bonds into cash; the collateral type changes the cost, it does not remove it.
C is incorrect. Some inputs are subjective, such as the maturity of funding a trade needs and the dividends shareholders will require.

LO: CR-15 g: Identify and explain the costs of an OTC derivative. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.6 Funding, Collateral, and Capital Costs

Solution CR-15 g.6 Answer: C

New • built from GARP Ch.15, 15.2.2

Step 1 – Where the symmetry holds

One party's positive value is the other's negative value, so one party's CVA cost is the other's DVA benefit.

Step 2 – Where it breaks

GARP: capital and initial margin costs "are always present and do not provide an equal and opposite benefit to the counterparties". Both banks pay them.

Interpretation

That is why KVA and MVA are always costs in the xVA sign table, while CVA and DVA, and the funding terms, can mirror each other (CR-15 h).

Why the other options are wrong

A is incorrect. Ignores the asymmetric terms, capital and initial margin.
B is incorrect. DVA is a benefit to the party whose own default it reflects.
D is incorrect. Each bank's capital is set by its own regulator on its own exposure; there is no mirror relief.

LO: CR-15 g: Identify and explain the costs of an OTC derivative. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.2

CR-15 • Counterparty Risk and Beyond • Vardeez: MODERATE

CR-15 h

P3 CONCEPT

Explain the components of the X-Value Adjustment (xVA) term.

Workbook: none (no workbook section for CR-15)

WHAT IS TESTED HERE

Why P3. No workbook section; Vardeez flags the LO in his foundation chapter for counterparty risk.

Vardeez: "more than the definitions themselves, the examinable content is the sign column": CVA a cost, DVA a benefit, FVA and ColVA either, KVA and MVA always costs.

Reported exam shapes: none listed in the Master Map. The questions name the term from a description, test which terms can be benefits, map each term to its market and cost components (GARP Table 15.1), and test the overlaps between terms.

What the questions look like: "which xVA term is described" and "which statement is true". Question CR-15 h.4 is a bank calculation rebuilt as a qualitative question to respect the CONCEPT tag; two questions are new from GARP 15.3.

1. What an xVA term is

An xVA term quantifies the cost, or benefit, of one component (counterparty risk, collateral, funding or capital) over the lifetime of a transaction or portfolio (GARP 15.2.3). It is computed by integrating the relevant profile against a cost curve: a credit spread, a funding curve, a collateral rate or a cost of capital. The adjustments correct a **base valuation** that ignores these effects, giving the actual valuation (GARP Figure 15.11).

Term	Defines	Cost or benefit	Market component	Cost component
CVA	Counterparty risk: the cost of the counterparty's default	Cost	Credit exposure	Default probability
DVA	Counterparty risk from the point of view of one's own default	Benefit	Credit exposure (negative side)	Own default probability
FVA	Cost and benefit of funding the transaction: FVA = FCA + FBA	Either	Valuation (funding position)	Funding cost
ColVA	Embedded options in the collateral agreement (choice of currency or security) and non-standard collateral terms	Either	Collateral amount	Collateral cost
KVA	Cost of holding capital, typically regulatory, over the life of the trade	Always a cost	Capital amount	Capital cost
MVA	Cost of posting initial margin over the life of the trade	Always a cost	Initial margin amount	Funding cost

2. Every term is a market component times a cost component

GARP 15.3.1: counterparty risk combines **market risk**, which defines the exposure, with **credit risk**, which defines the counterparty's credit quality. Every xVA term has the same structure: a market component, directly or indirectly related to the base portfolio value, and a cost (or benefit) component, the cost of bearing that market component (Table 15.1, last two columns above). The valuation itself defines, directly or indirectly, every market component: the credit exposure, the funding position (an asset if positive, a liability if negative), the initial margin (from the variability of the value), the collateral amount and the capital. Market components are needed today and in the future; for margin, the future value of required margin and of the securities posted must also be modelled.

3. Signs, overlaps and recursion

- **Two sign conventions.** In a valuation (the adjustment added to the base value, as in Vardeez's table and in CR-20) costs are negative: CVA negative, DVA positive. In GARP's xVA profile charts (Figure 15.10) costs are plotted as *positive* values and benefits as negative. Read the stem's convention before choosing a sign.
- **Always costs.** KVA and MVA can never be benefits: capital and initial margin cost both parties and have no mirror benefit (CR-15 g).
- **Overlaps.** The definitions are standard but not unique. Collateral must be funded, so some costs could sit in FVA or ColVA; and own-default risk is widely seen as a funding benefit, so DVA and FBA overlap. Adding every term blindly can double count.
- **Recursion.** The valuation is an input to the adjustments, yet the true valuation should include them. Practice adds the adjustments linearly to a base value, which is only an approximation, since the real problem is non-linear and recursive.

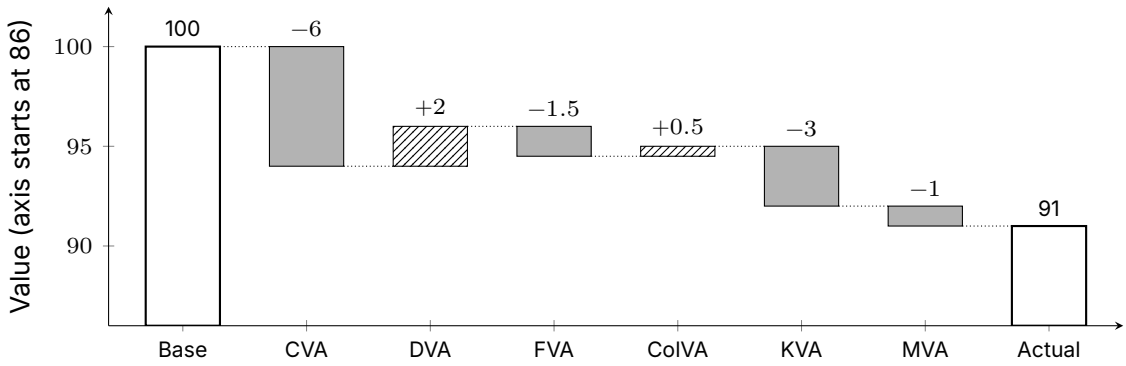


Figure 7. An illustrative xVA waterfall in the valuation convention: costs (grey) reduce the base valuation, benefits (hatched) add to it. Here FVA is a net funding cost and ColVA a small benefit from a collateral choice; either could have the other sign. KVA and MVA are always reductions. $100 - 6 + 2 - 1.5 + 0.5 - 3 - 1 = 91$.

Exam traps

- **MVA and KVA swapped.** MVA is initial margin; KVA is capital.
- **ColVA as initial margin.** ColVA is collateral optionality and non-standard terms; initial margin is MVA.
- **DVA as the counterparty's default.** DVA is one's own default; CVA is the counterparty's.
- **FVA always a cost.** FVA = FCA + FBA: it can be a benefit.
- **KVA or MVA as a benefit.** Never.

- **Wrong cost component.** CVA pairs exposure with default probability; funding, capital and collateral costs belong to FVA and MVA, KVA and ColVA.
- **Sign convention.** Valuation: costs negative. GARP's profile charts: costs positive.
- **Double counting.** DVA and FBA, and FVA and ColVA, can overlap.

Practice questions: CR-15 h

Question CR-15 h.1

Bank Q711 • AnalystPrep Q.4868 (rebuilt)

A bank's valuation of a derivative includes a funding value adjustment (FVA). Which of the following best describes FVA?

A. It is the sum of a funding cost adjustment (FCA) and a funding benefit adjustment (FBA), so it can be either a cost or a benefit.

B. It is the cost of posting initial margin over the lifetime of the transaction.

C. It is always a cost, because funding a derivative can never generate a benefit.

D. It is the cost of holding regulatory capital over the lifetime of the transaction.

Question CR-15 h.2

Bank Q713 • AnalystPrep Q.5516 (rebuilt)

New regulations require Global Finance Bank to post initial margin on its OTC derivatives, which over-collateralises the positions. The Chief Trader wants a valuation adjustment that captures the cost of posting this initial margin over the lifetime of the transactions. Which xVA component is being referred to?

A. Margin value adjustment (MVA)

B. Credit value adjustment (CVA)

C. Funding value adjustment (FVA)

D. Capital value adjustment (KVA)

Question CR-15 h.3

Bank Q716 • BT P2.T6.900.3 (rebuilt)

Which of the following statements about xVA terms is TRUE?

A. ColVA is the cost of posting initial margin over the lifetime of a transaction.

B. DVA represents counterparty risk from the point of view of the counterparty's default.

C. KVA is the cost of holding capital, typically regulatory capital, over the lifetime of a transaction.

D. CVA and DVA on the same trade are always equal in size, so they cancel in a bilateral valuation.

Question CR-15 h.4 Bank Q717 • BT P2.T6.902.3 (rebuilt as a qualitative question)

Peter, a risk analyst, has a basic valuation (no counterparty risk, funding or capital effects) of USD 50.0 million for his firm's uncollateralised derivative position. He must now adjust for the following xVA terms, each stated as a positive amount: CVA USD 5.0 million (cost); DVA USD 2.0 million (benefit); FVA USD 1.0 million (unsecured funding cost); ColVA zero (the contract is uncollateralised); KVA USD 4.0 million (cost); MVA USD 3.0 million (cost). Which of the following correctly describes how the terms move the final valuation relative to the basic valuation?

A. All five non-zero terms are deducted, because every valuation adjustment is a cost by definition.

B. CVA and DVA are netted into a bilateral charge, while FVA, KVA and MVA are excluded because they are internal management costs rather than valuation adjustments.

C. CVA, FVA, KVA and MVA are deducted and DVA is added, so the final valuation is below the basic valuation by the excess of these costs over the DVA benefit.

D. CVA, KVA and MVA are deducted, while FVA and DVA are added, because funding adjustments are always benefits to the party holding the position.

Question CR-15 h.5 New • built from GARP Ch.15, 15.3.1 and Table 15.1

An xVA analyst is documenting, for each valuation adjustment, its market component and its cost component, as in GARP Table 15.1. Which of the following pairings is correct?

A. CVA: credit exposure as the market component and the institution's funding cost as the cost component.

B. MVA: the initial margin amount as the market component and the cost of capital as the cost component.

C. KVA: the valuation as the market component and the counterparty's default probability as the cost component.

D. FVA: the valuation (funding position) as the market component and the funding cost as the cost component.

Question CR-15 h.6 New • built from GARP Ch.15, 15.2.3 and 15.3.2

The head of an xVA desk is reviewing a junior quant's summary of the framework. Which of the following statements is most accurate?

A. Every cost belongs to exactly one xVA term, so the terms can always be added without double counting.

B. Since own-default risk is widely seen as a funding benefit, DVA and the funding benefit adjustment (FBA) can overlap, so adding both may double count.

C. xVA terms are computed from the final, fully adjusted valuation, so adding them linearly to a base value involves no approximation.

D. In GARP's convention for plotting xVA profiles, benefits are shown as positive values and costs as negative values.

Solutions • CR-15 h

Solution CR-15 h.1

Answer: A

Bank Q711 • AnalystPrep Q.4868

Step 1 – GARP's definition

FVA "defines the cost and benefit arising from the funding of the transaction. It is divided into two terms: funding cost adjustment (FCA) and funding benefit adjustment (FBA)."

Step 2 – Sign

A positive-value uncollateralised position must be funded (a cost); a negative-value position facing an uncollateralised counterparty gives a funding benefit.
$\Rightarrow$ FVA can be either.

Interpretation

FVA and ColVA are the two "either" terms in the sign table.

Why the other options are wrong

B is incorrect. That is MVA.
C is incorrect. Ignores FBA, the benefit leg.
D is incorrect. That is KVA.

LO: CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.3 xVA Terms

Solution CR-15 h.2

Answer: A

Bank Q713 • AnalystPrep Q.5516

Step 1 – What is being priced

The cost of posting initial margin over the life of the trades.

Step 2 – Name it

GARP: MVA "defines the cost of posting initial margin over the lifetime of the transaction".

Interpretation

Initial margin is over-collateralisation that offsets nothing, so it must be funded: MVA is always a cost.

Why the other options are wrong

B is incorrect. CVA prices the counterparty's default.
C is incorrect. FVA covers funding of the transaction in general; initial margin is carved out as MVA.
D is incorrect. KVA is the cost of holding capital.

LO: CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.3

Solution CR-15 h.3

Answer: C

Bank Q716 • BT P2.T6.900.3

Step 1 – Recall the definitions

CVA and DVA: counterparty risk (the counterparty's default, one's own default). FVA: funding. ColVA: collateral optionality and non-standard terms. KVA: holding capital. MVA: posting initial margin.

Step 2 – Test each statement

Only C matches: KVA is the cost of holding capital, typically regulatory, over the lifetime of the transaction.

Interpretation

The distractors are the classic sibling swaps: ColVA for MVA, and whose default DVA reflects.

Why the other options are wrong

A is incorrect. That is MVA; ColVA covers collateral optionality and non-standard terms.
B is incorrect. DVA reflects the party's own default; the counterparty's default is CVA.
D is incorrect. CVA uses the counterparty's default probability and the positive exposure, DVA one's own default probability and the negative exposure; they are not equal in general.

LO: CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.3

Solution CR-15 h.4 — Answer: C

Bank Q717 • BT P2.T6.902.3

Step 1 – Classify each term

Costs: CVA (counterparty default), FVA (stated as a funding cost), KVA (capital), MVA (initial margin). Benefit: DVA (own default). ColVA is zero.

Step 2 – Direction

Costs reduce the basic valuation and benefits add to it: $50.0 - (5.0 + 1.0 + 4.0 + 3.0) + 2.0 =$ USD 39.0 million, below the basic valuation.

Interpretation

FVA could have been a benefit for a different position; the stem says it is a cost here. KVA and MVA are always costs.

Why the other options are wrong

A is incorrect. Treats DVA as a cost: $50.0 - 15.0 = 35.0$.
B is incorrect. Ignores FVA, KVA and MVA, which are valuation adjustments: $50.0 - 5.0 + 2.0 = 47.0$.
D is incorrect. Adds FVA although it is stated as a cost: $50.0 - 12.0 + 3.0 = 41.0$.

LO: CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.3, Figure 15.11

Solution CR-15 h.5 — Answer: D

New • built from GARP Ch.15, 15.3.1 and Table 15.1

Step 1 – The structure

Each xVA term is a market component (from the valuation) combined with a cost component (the cost of bearing it).

Step 2 – Table 15.1

CVA/DVA: credit exposure and default probability. FVA: valuation and funding cost. MVA: initial margin amount and funding cost. ColVA: collateral amount and collateral cost. KVA: capital amount and capital cost.
$\Rightarrow$ D is correct.

Interpretation

The valuation is the funding position: an asset if positive, a liability if negative, which is why FVA can be a cost or a benefit.

Why the other options are wrong

A is incorrect. CVA's cost component is the default probability, not funding.
B is incorrect. MVA's cost component is the funding cost of the margin, not the cost of capital.
C is incorrect. KVA pairs the capital amount with the cost of capital.

LO: CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.3.1 Overview, Table 15.1

Solution CR-15 h.6 Answer: B

New • built from GARP Ch.15, 15.2.3 and 15.3.2

Step 1 – Overlaps

GARP: there are potential overlaps between the terms, "for example, between DVA and FBA, where own-default risk is widely seen as a funding benefit"; and since collateral must be funded, some costs could be defined as either FVA or ColVA.

Step 2 – The other statements

The valuation is an input to the adjustments while the correct valuation should include them: the linear approach is an approximation. GARP's profile charts put costs on the positive axis.

Interpretation

Before adding up an xVA stack, check that DVA and FBA are not both claiming the same own-credit benefit.

Why the other options are wrong

A is incorrect. The definitions overlap, so blind addition can double count.
C is incorrect. The problem is non-linear and recursive; the linear approach is only an approximation.
D is incorrect. Reversed: costs are positive and benefits negative in GARP's profile convention.

LO: CR-15 h: Explain the components of the X-Value Adjustment (xVA) term. • **Reference:** Gregory, The xVA Challenge 4e, Ch.3 (GARP CR Ch.15), 15.2.3 and 15.3.2

CR-18 • Central Clearing • Vardeez: MODERATE

CR-18 f

P3 CONCEPT

Provide examples of a loss waterfall.

Workbook: D24 Central Clearing and the CCP Loss Waterfall • Tier D • Map #59

WHAT IS TESTED HERE

Why P3. Tier D workbook topic (D24, Map #59). Vardeez and the map see central clearing as Part I material, "not likely testable; just give a good read", with one carve-out: "know the loss waterfall".
Vardeez: know the loss waterfall.

Reported exam shapes

- **D24·S1** The order of the loss waterfall, with sequence errors as distractors: initial margin and default fund contribution of the defaulter, CCP equity, mutualised default fund, assessments.

What the questions look like: conceptual ordering questions (what comes next, where skin in the game sits, which three events come in which order), plus one workbook question that walks a loss down the waterfall in numbers. Workbook D24.1 is numerical and is kept under this CONCEPT LO, as workbook questions always are.

1. What a loss waterfall is

When a clearing member defaults, the CCP neutralises its portfolio by macro-hedging and then auctioning it to the surviving members (CR-18 e). The defaulter's initial margin and default fund contribution are there to pay for that. If they are not enough, or the auction fails, the CCP has further resources and ways of allocating losses. A **loss waterfall** sets the order in which they are used (GARP Figure 18.9):

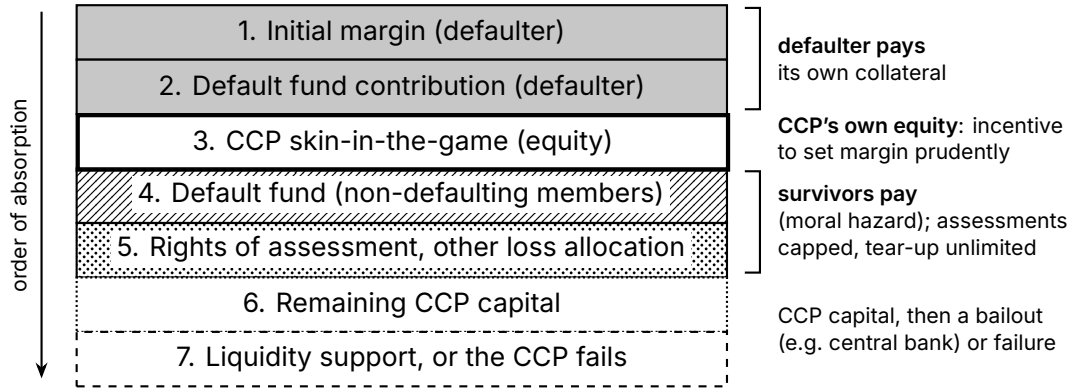


Figure 1. A typical CCP loss waterfall (after GARP Figure 18.9): grey for the defaulter's resources, a heavy frame for the CCP's skin-in-the-game, hatching and dots for what the surviving members bear, then dotted and dashed frames for the CCP's remaining capital and the last resort. The initial margin of non-defaulting members appears nowhere.

2. The layers

- **Defaulter pays (1, 2).** The defaulter's initial margin (sized to cover close-out losses at 99% or more) and then its own default fund contribution. Any extra variation margin the CCP holds also counts as defaulter-pays.

- **Skin-in-the-game (3).** A slice of the CCP's own equity, placed after the defaulter's resources so the CCP has an incentive to avoid losses beyond them. It is not in every rule book: in 2014 a futures broker on the Korea Exchange (KRX) defaulted after an algorithmic trading error, and KRX mutualised a USD 4.3m loss to its members' default fund without losing anything itself, because its rules put its own loss after the roughly USD 200m fund.
- **Survivors' default fund (4).** Now the members who did nothing wrong pay. The loss may be shared pro rata, or by rules that put winning auction bidders last, to encourage proactive bidding. A member can lose even if it never traded with the defaulter or has no net position with the CCP.
- **Rights of assessment and other loss allocation (5).** *Rights of assessment* are an unfunded obligation to top up a depleted default fund, typically invoked once a significant fraction (for example 25%) is used; they are capped, since unlimited calls would create moral hazard and hit members at the worst time. *Variation margin gains haircutting* (VMGH), probably the most common alternative, pays members with gains less than they are owed while those owing pay in full; it is guaranteed to work if the portfolio can be auctioned at mid-market. *Tear-up* cancels opposite trades to restore a matched book; *forced allocation* makes members accept positions at CCP-set prices. Tear-up and forced allocation are potentially unlimited.
- **Remaining CCP capital (6),** if loss allocation is finite; then **liquidity support or failure (7)**. Some CCPs add costly lines of credit or insurance guarantees.
- **Not in the waterfall:** the initial margin of non-defaulting members. EMIR Article 45(4) bars its use; initial margin haircutting has been suggested but not used. So survivors' initial margin is in theory risk free, though unlimited tear-up makes that small comfort.

Schweser's figure shows six stages, without the remaining CCP capital; the order of the others is the same.

3. Examples

One CCP, four losses. An illustrative CCP holds defaulter initial margin of 400, a defaulter default fund contribution of 60, skin-in-the-game of 40, survivors' default fund of 900, assessments capped at 450 and remaining capital of 150.

Layer	Loss 250	Loss 800	Loss 1,800	Loss 2,200
1. Initial margin (defaulter), 400	250	400	400	400
2. Default fund (defaulter), 60	0	60	60	60
3. Skin-in-the-game, 40	0	40	40	40
4. Survivors' default fund, 900	0	300	900	900
5. Rights of assessment, cap 450	0	0	400	450
6. Remaining CCP capital, 150	0	0	0	150
7. Uncovered: support or failure	0	0	0	200

At 250 the defaulter's own margin covers everything and 150 goes back to its administrator. At 800 the survivors lose a third of their fund (300 of 900), above the 25% that typically triggers a recapitalisation. At 1,800 the fund is gone and assessments are called. At 2,200 every funded and capped layer is exhausted and 200 is left: support or failure.

What has happened in practice. Lehman at SwapClear (2008): 90% of the risk was macro-hedged within a week and all 66,000 trades auctioned within three weeks, using only about a third of Lehman's initial margin (layer 1); the rest went back to the administrators. All seven defaults in LCH's history were managed within the defaulter's initial margin. KRX (2014) reached layer 4 with no skin-in-the-game ahead of it (above). The more recent Nasdaq default was less successful: a substantial part of

the default fund was needed (layer 4). The Hong Kong Futures Exchange (1987) made losses that dwarfed its default fund.

	Whose money	Funded in advance?	Limit
1, 2. Defaulter's IM and DF	The defaulter (defaulter pays)	Yes	Its own deposits
3. Skin-in-the-game	CCP shareholders	Yes	Set in the rule book (may be absent)
4. Survivors' DF	Surviving members (mutualised)	Yes	Size of the fund
5. Assessments, VMGH, tear-up	Surviving members	No	Assessments capped; tear-up and forced allocation potentially unlimited
6, 7. CCP capital, support	CCP, then outside support	Partly	CCP fails if exhausted

Exam traps

- **Sequence swap.** Skin-in-the-game sits after the defaulter's initial margin and default fund contribution (used before anyone else's money, not pro rata with the survivors') and before the mutualised fund. Moving it below the fund overstates the survivors' loss.
- **Survivors' initial margin.** Never in the waterfall (EMIR 45(4)); survivors pay through the default fund and loss allocation.
- **Scope error.** Mutualised losses do not depend on having traded with the defaulter.
- **Absolute claims.** Rights of assessment are capped; only tear-up and forced allocation are potentially unlimited, so "the CCP can never fail" and "a member's loss is limited to its default fund contribution" are both false in general.
- **VMGH versus a collateral haircut.** VMGH cuts the variation margin paid to members with gains; it is not a haircut on securities posted as margin.

Practice questions: CR-18 f

Question CR-18 f.1

Bank Q867 • AnalystPrep Q.6102 (rebuilt)

In a CCP loss waterfall, the first resource applied to losses from a clearing member's default is the defaulting member's initial margin. Which statement best describes its role?

A. It is the defaulter's own collateral, sized to cover potential losses on its portfolio over the close-out period, and is consumed before any resources of the CCP or surviving members.

B. It is a pooled buffer contributed by all members, consumed pro rata with the defaulter's share first.

C. It is consumed only after the CCP has contributed its own equity (skin in the game).

D. It is drawn from non-defaulting members' margin in proportion to their positions, to spread losses evenly.

Question CR-18 f.2 Bank Q868 • AnalystPrep Q.6103 (rebuilt)

After the defaulting member's initial margin is exhausted, which resource is applied next in the CCP loss waterfall?

A. The CCP's own equity (skin in the game).

B. The defaulting member's own default fund contribution.

C. The default fund contributions of surviving members.

D. Rights of assessment on surviving members.

Question CR-18 f.3 Bank Q869 • AnalystPrep Q.6104 (rebuilt)

In the CCP loss waterfall, the CCP's "skin in the game" refers to:

A. A contribution of the CCP's own equity, applied after the defaulter's margin and default fund contribution but before surviving members' default fund contributions.

B. A contribution of the CCP's own equity, applied before the defaulter's initial margin to give members confidence.

C. The additional contributions called from surviving members after the default fund is depleted.

D. A compensation payment made by the CCP to non-defaulting members whose positions gained value.

Question CR-18 f.4 Bank Q872 • Schweser Module Quiz 36.2 Q2

Given the following three events, what is the proper order of the CCP loss waterfall?

I. Nondefaulting members' default fund contributions are exhausted.

II. Defaulting member's collateral and default fund contributions are exhausted.

III. The CCP taps an amount of its equity that enables it to function normally.

A. I, II, III.

B. II, I, III.

C. II, III, I.

D. III, II, I.

Question CR-18 f.5 Bank Q876 • AnalystPrep Q.6170 (rebuilt) • re-filed from CR-18 g

Which of the following best describes variation margin gains haircutting (VMGH) as a loss allocation method of a CCP?

A. The CCP reduces the initial margin of non-defaulting members to cover the shortfall left by the defaulter.

B. The CCP applies a haircut to the value of securities posted as margin to reflect their price volatility.

C. The CCP reduces the variation margin it pays to members whose positions have gained in value, while members who owe variation margin must still pay in full.

D. The CCP reduces the variation margin owed by all members, both those with gains and those with losses, by the same percentage.

Question CR-18 f.6 Workbook D24.1 • S1 • Map #59

A risk consultant is reviewing the default rules of a European CCP on behalf of a clearing member. In a fire drill, one clearing member defaults, and the macro-hedging and auction of its portfolio cost EUR 900 million. The resources in the CCP's rule book are:

- Initial margin posted by the defaulter: EUR 500 million
- Default fund contribution of the defaulter: EUR 80 million
- CCP skin-in-the-game: EUR 50 million
- Default fund contributions of the surviving members: EUR 920 million
- Initial margin posted by the surviving members: EUR 6,400 million

Which of the following statements is correct for the consultant to include in the report?

A. The surviving members' default fund contributions absorb EUR 320 million, because the CCP's skin-in-the-game ranks after the mutualised default fund.

B. The surviving members' default fund contributions absorb nothing, because the surviving members' initial margin is applied before the mutualised default fund.

C. The surviving members' default fund contributions absorb EUR 270 million, shared only among the members that held trades with the defaulter.

D. The surviving members' default fund contributions absorb EUR 270 million, and none of the surviving members' initial margin is used.

Solutions • CR-18 f

Solution CR-18 f.1 — Answer: A

Bank Q867 • AnalystPrep Q.6102

Step 1 – Where initial margin sits

The waterfall starts with the defaulter-pays resources: the defaulter's initial margin, then its default fund contribution.

Step 2 – What it is sized for

Initial margin is the member's own collateral, calibrated to cover losses on its portfolio during close-out (the margin period of risk) at a high confidence level, 99% or more.

$\Rightarrow$ It is the first line of defence, used before the CCP's or anyone else's money: A.

Interpretation

Because initial margin covers most defaults on its own (all seven of LCH's; the Lehman default used about a third of it), the deeper layers are rarely reached. That is the design: the defaulter pays first and in full.

Why the other options are wrong

B is incorrect. Describes the default fund, which is pooled; initial margin is member-specific.

C is incorrect. Reverses the order: skin-in-the-game comes after the defaulter's margin and default fund contribution.

D is incorrect. Non-defaulting members' initial margin is never used in the waterfall (EMIR Article 45(4)); mutualisation runs through default fund contributions.

LO: CR-18 f: Provide examples of a loss waterfall. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8 (GARP CR Ch.18), 18.3 The Loss Waterfall, Figure 18.9

Solution CR-18 f.2 — Answer: B

Bank Q868 • AnalystPrep Q.6103

Step 1 – Finish the defaulter-pays block

Both the initial margin and the default fund contribution of the defaulter are defaulter-pays resources, and both are used before anyone else's money.

Step 2 – Then the rest

Only after that come the CCP's skin-in-the-game, the surviving members' default fund contributions and then rights of assessment.

$\Rightarrow$ B.

Interpretation

The defaulter's default fund contribution is part of a mutualised pool, but in a default the defaulter's own share is taken first, not pro rata with the survivors'. This is one of the sequence traps the D24 shape is built on.

Why the other options are wrong

A is incorrect. Skips the defaulter's default fund contribution: skin-in-the-game comes after both defaulter-pays resources.

C is incorrect. Survivors' contributions come after the CCP's own equity.

D is incorrect. Rights of assessment come only after the default fund is depleted.

LO: CR-18 f: Provide examples of a loss waterfall. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8 (GARP CR Ch.18), 18.3 The Loss Waterfall, Figure 18.9

Solution CR-18 f.3 Answer: A

Bank Q869 • AnalystPrep Q.6104

Step 1 – What skin-in-the-game is

A contribution of the CCP's own equity to absorbing default losses.

Step 2 – Where it sits and why

After the defaulter-pays resources and before the surviving members' default fund, so that the CCP bears a loss before mutualisation starts and has an incentive to set margin and default fund requirements prudently.
⇒ A.

Interpretation

It is not universal: under KRX's 2014 rules the exchange's own loss came only after the default fund, so a USD 4.3m loss was mutualised to members while KRX lost nothing. Where skin-in-the-game exists, its position is what the D24 distractors move.

Why the other options are wrong

B is incorrect. Wrong order: the CCP's equity is never used before the defaulter's own resources.
C is incorrect. Describes rights of assessment.
D is incorrect. Not a feature of the waterfall; if anything, members with gains may have their variation margin haircut (VMGH).

LO: CR-18 f: Provide examples of a loss waterfall. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8 (GARP CR Ch.18), 18.3 The Loss Waterfall (Skin in the Game box)

Solution CR-18 f.4 Answer: C

Bank Q872 • Schweser Module Quiz 36.2 Q2

Step 1 – Defaulter first

The defaulting member's collateral (initial margin) and default fund contribution are used first: event II.

Step 2 – Then the CCP

The CCP taps its own equity, its skin-in-the-game, to a level at which it can still function normally: event III.

Step 3 – Then the survivors

Mutualisation: the non-defaulting members' default fund contributions, followed by rights of assessment: event I.
⇒ II, III, I.

Interpretation

Defaulter, CCP, survivors: the CCP's own slice is the boundary between the defaulter-pays and the survivors-pay parts of the waterfall.

Why the other options are wrong

A is incorrect. Puts the mutualised fund before the defaulter's own resources.
B is incorrect. Uses the survivors' fund before the CCP's equity.
D is incorrect. Uses the CCP's equity before the defaulter's resources.

LO: CR-18 f: Provide examples of a loss waterfall. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8 (GARP CR Ch.18), 18.3, Figure 18.9; SchweserNotes 2026 Book 2, Reading 36, LO 36.f

Solution CR-18 f.5 — Answer: C

Bank Q876 • AnalystPrep Q.6170

Step 1 – The matched book

A CCP runs a matched book, so the losses on the defaulter's positions must be matched by gains to other members.

Step 2 – The haircut

VMGH "haircuts" those gains: members whose positions have gained receive only part of the variation margin owed to them, while members who owe variation margin pay in full, so the CCP stays funded.

$\Rightarrow$ C.

Interpretation

VMGH sits in layer 5 of the waterfall, alongside rights of assessment. GARP calls it probably the most common alternative loss-allocation method; it mimics the economics of a bilateral market and is guaranteed to work if the defaulter's portfolio can be auctioned at mid-market, but fails if the CCP must close out at a large premium. A member hit by VMGH may not have made an overall profit, since its cleared gains may hedge losses elsewhere.

Why the other options are wrong

A is incorrect. Describes initial margin haircutting, which has been suggested but not used, and which EMIR Article 45(4) bars.

B is incorrect. Describes a collateral haircut on posted securities, a valuation control, not loss allocation.

D is incorrect. Haircutting the members who owe would leave the CCP unfunded; only payments to members with gains are cut.

LO: CR-18 f: Provide examples of a loss waterfall. CR-18 g: Explain the different methods of absorbing losses and managing the default of one or more members of a CCP. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8 (GARP CR Ch.18), 18.3 The Loss Waterfall; SchweserNotes 2026 Book 2, Reading 36, LO 36.g

Solution CR-18 f.6 — Answer: D

Workbook D24.1 • S1 • Map #59

Step 1 – Defaulter pays: initial margin, then its own default fund contribution

$$900 - 500 = 400, \qquad 400 - 80 = 320$$

$\Rightarrow$ After the defaulter's own resources, EUR 320 million of loss remains.

Step 2 – CCP skin-in-the-game

$$320 - 50 = 270$$

$\Rightarrow$ The CCP's equity takes the next EUR 50 million, which gives it an incentive to manage defaults well.

Step 3 – Survivors pay: mutualised default fund

$270 \leq 920 \Rightarrow$ survivors' default fund absorbs EUR 270 million; nothing reaches assessments

$\Rightarrow$ About 29% of the survivors' contributions are consumed, so the CCP may invoke a capped recapitalisation, but no further layer is needed to absorb this default.

Interpretation

The order is defaulter's initial margin, defaulter's default fund contribution, CCP skin-in-the-game, survivors' default fund, then assessments and other loss allocation. The EUR 6,400 million of surviving members' initial margin is a red herring: Gregory notes that none of the loss allocation methods lets a CCP use the initial margin of non-defaulting members (EMIR Article 45(4)), which is why initial margin is "in theory" risk free to the poster. The clearing member's real exposure is its share of the EUR 270 million plus any assessment or other allocation.

Why the other options are wrong

A is incorrect. Sequence swap: moved skin-in-the-game below the mutualised fund, so the survivors absorb $900 - 500 - 80 = 320$. In Figure 18.9 skin-in-the-game ranks immediately after the defaulter-pays resources.

B is incorrect. Role misassignment: survivors' initial margin is not in the waterfall; only the defaulter's initial margin is. Survivors pay through their default fund contributions.

C is incorrect. Scope error: correct amount, but mutualisation does not depend on trading with the defaulter. A member "can suffer default losses even if it never traded with the defaulted counterparty", has no net position with the CCP, or has a position in the same direction.

LO: CR-18 f: Provide examples of a loss waterfall. CR-18 g: Explain the different methods of absorbing losses and managing the default of one or more members of a CCP. • **Reference:** Gregory, The xVA Challenge 4e, Ch.8, 18.3 The Loss Waterfall, Figure 18.9; 18.2 Margin and Default Funds

CR-19 • Future Value and Exposure • Vardeez: VERY IMPORTANT

CR-19 a

P3 CALC

Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure.

Workbook: C8 Counterparty Exposure Profiles: EE, PFE, EPE • Tier C

WHAT IS TESTED HERE

Why P3. Tier C workbook topic (C8, Map #37); Vardeez's focus in this reading is on the profiles and the netting factor (CR-19 d, e, f), not on the metrics themselves. They are the vocabulary for CVA (CR-20) and CR-21, so they are worth one clean pass.

Reported exam shapes

- **C8·S1** The six-way metric family: expected MtM (EFV), expected exposure / EPE, PFE, ENE, effective EE / EPE and average EPE, maximum PFE.

What the questions look like: a scenario table or a normal distribution at one date (EFV, EPE, ENE, PFE), or an EE or PFE profile over time (average EPE, effective EE and EPE, maximum PFE). The traps are the denominator, the names and what is additive.

Worked elsewhere: C8.2 (the glossary of metrics: additivity, average EPE, ENE and DVA) under CR-19 f. Positive and negative exposure, and the sold-option red herring, are set out under CR-19 d and h.

1. One date: four metrics on one distribution

Exposure is the positive part of the future value of the netting set, $\max(\text{value}, 0)$; negative exposure is $\min(\text{value}, 0)$, the counterparty's exposure to the bank (GARP eqs. 19.1 and 19.2). Recovery is not deducted. At a given future date, the distribution of value gives four metrics (GARP 19.1.5):

- **Expected future value (EFV)**, also *expected mark-to-market*: the average of all values. It equals the current value at time zero and drifts away from it through cash flow differentials, forward rates and asymmetric margin terms. It can be negative, and it is **additive** across trades.
- **Potential future exposure (PFE)**: the exposure exceeded with probability no more than $1-\alpha$, "an equivalent metric to VaR". It is a quantile, not the worst case.
- **Expected positive exposure (EPE)**, also called *expected exposure* (EE): the average of the exposures with negative values counted as *zero*, not dropped.
- **Expected negative exposure (ENE)**: the same for negative values, the bank's own default (DVA). Since $\max(v, 0) + \min(v, 0) = v$, EFV = EPE + ENE.

GARP example (five equally likely scenarios: 70, 50, 30, -10, -30):

$$\text{EFV} = \tfrac{110}{5} = 22 \qquad \text{PFE} = 70\ (\alpha > 80\%) \qquad \text{EPE} = \tfrac{70+50+30}{5} = 30 \qquad \text{ENE} = \tfrac{-10-30}{5} = -8$$

Check: $30 - 8 = 22$. Averaging only the positive scenarios would give 50, not 30.

Normal future value with mean μ and standard deviation σ:

$$\text{EPE} = \mu\,\Phi\left(\tfrac{\mu}{\sigma}\right) + \sigma\,\phi\left(\tfrac{\mu}{\sigma}\right), \qquad \text{ENE} = \mu - \text{EPE}, \qquad \text{PFE}_\alpha = \mu + z_\alpha \sigma$$

GARP: $\mu = 2$, $\sigma = 2$ gives EPE 2.17, ENE -0.17, 99% PFE $2 + 2.326 \times 2 = 6.65$. With $\sigma = 4$: EPE 2.79, ENE -0.79, PFE 11.30 (GARP prints 11.31, from $z = 2.3263$). EFV stays at 2. Higher volatility

raises EPE, |ENE| and PFE, because exposure is option-like; ENE rises almost fivefold because it is out of the money. At $\mu = 0$, EPE $= \sigma/\sqrt{2\pi} \approx 0.4\sigma$ (the MPoR formula of CR-19 h).

2. Across time: profiles and their summaries

Repeating the calculation at each future date gives the EE and PFE *profiles*. Three summaries:

- **Average EPE** (often just "EPE"): the time-weighted average of the EE profile, the *loan equivalent*. With unequal intervals, weight each EE by its interval: $\sum_j \mathrm{EE}(t_j)\Delta t_j / \sum_j \Delta t_j$. It is crude (it averages out both market randomness and time), but it has a strong theoretical basis for capital and xVA.
- **Maximum PFE** (maximum exposure, Hull's "maximum peak exposure"): the peak of the PFE profile over the life. A single number.
- **Effective EE and effective EPE** (Basel, outside the GARP excerpt): effective EE is the EE profile made *non-decreasing*, $\mathrm{EffEE}_k = \max(\mathrm{EffEE}_{k-1}, \mathrm{EE}_k)$; effective EPE is its time-weighted average (over the first year for capital). They capture **rollover risk**: a short-dated trade whose EE decays to zero is in practice replaced, so its exposure returns. EffEE $\geq$ EE at every date, so effective EPE $\geq$ EPE, with equality only when EE never falls. Neither involves a confidence level.

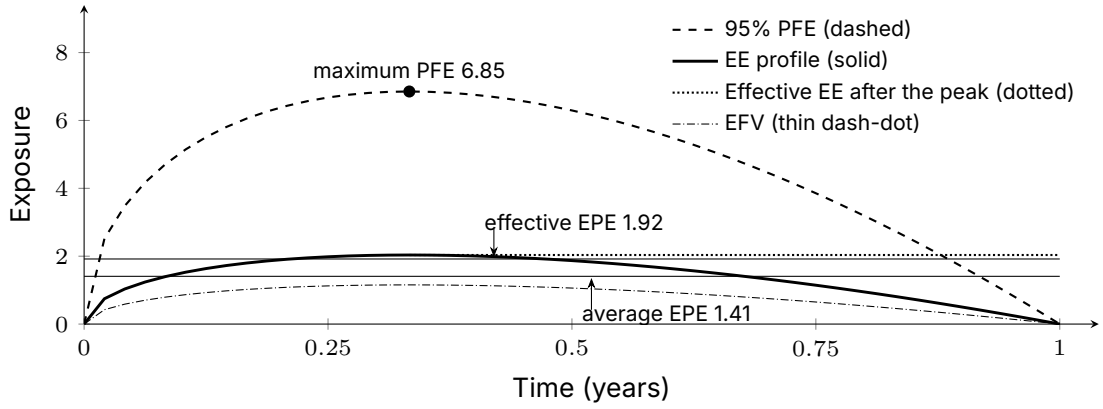


Figure 1. A stylised one-year profile with a normal future value (mean $3\sqrt{t}(1-t)$, standard deviation $9\sqrt{t}(1-t)$). The PFE (dashed) sits above the EE (solid); its peak is the maximum PFE (dot). The EE peaks at 2.04 and falls; effective EE (dotted) holds that peak, so effective EPE (1.92) exceeds the average EPE (1.41), both shown as thin horizontal lines.

Correction to the source notes.

- Vardeez and SchweserNotes: ENE is "the opposite" (Schweser: "exact opposite") of EPE. It is not $-$EPE: EFV $=$ EPE $+$ ENE. In GARP's example EPE is 30 and ENE -8.
- Vardeez: PFE is "the worst-case MtM scenario at a specific future point". PFE is a quantile at a stated confidence level, exceeded with probability up to $1-\alpha$ (it equals the highest value only when $\alpha > 1 - 1/n$ in a table of n scenarios).
- Vardeez and Schweser: EE "is larger than" expected MtM. It is at least as large, $\max(v, 0) \geq v$, with equality when no outcome is negative.
- GARP's normal-distribution example prints ENE as $+0.17$ and $+0.79$. On GARP's own definition, $\min(\text{value}, 0)$, used in its scenario table (-8), they are -0.17 and -0.79.

Formula card: CR-19 a

Exposure	$\text{PE} = \max(V, 0)$; $\text{NE} = \min(V, 0)$; recovery not deducted
n equally likely scenarios	$\text{EFV} = \frac{1}{n}\sum V_k$; $\text{EPE} = \frac{1}{n}\sum \max(V_k, 0)$; $\text{ENE} = \frac{1}{n}\sum \min(V_k, 0)$ (divide by n)
PFE at α	α-quantile of exposure; normal: $\mu + z_\alpha \sigma$
Normal EPE	$\mu\Phi(\mu/\sigma) + \sigma\phi(\mu/\sigma)$; $\mu = 0$: $0.3989\,\sigma$; $\text{ENE} = \mu - \text{EPE}$
Average EPE	$\sum_j \text{EE}(t_j)\Delta t_j / \sum_j \Delta t_j$ (simple average if equally spaced)
Effective EE	$\text{EffEE}_k = \max(\text{EffEE}_{k-1}, \text{EE}_k)$ (running maximum)
Effective EPE	time-weighted average of effective EE; $\geq$ EPE
Maximum PFE	$\max_j \text{PFE}(t_j)$
Aggregation	EFV additive; EPE, ENE sub-additive (CR-19 f)

Exam traps

- **Wrong denominator.** EPE averages over all scenarios with negatives set to zero; dividing by the number of positive scenarios overstates it.
- **Name collisions.** EFV is also called expected MtM and even "EE"; EPE at one date is also called EE; average EPE is also called EPE. Read the definition in the stem.
- **Confidence level in the wrong metric.** Only PFE (and maximum PFE) depends on α; EE, EPE and effective EPE do not.
- **Maximum PFE averaged.** It is the peak of the PFE profile, not its average, and not the peak of the EE profile.
- **Effective EE allowed to fall.** It carries the running maximum to the end of the period.
- **Unequal intervals ignored.** Weight each EE by its interval when dates are unequally spaced.
- **Zero exposure read as zero value.** A sold option adds nothing to EPE but its negative value still enters EFV.
- **Netting assumed.** Without an enforceable netting agreement, add standalone EPEs.

Practice questions: CR-19 a

Question CR-19 a.1

Bank Q899 • BT P2.T6.321.1 (rebuilt, key corrected)

A counterparty risk analyst is comparing expected mark-to-market (expected future value) with expected exposure (EE) for a derivatives position at a future date. Which of the following statements is FALSE?

A. Expected MtM can be negative.

B. EE always equals expected MtM.

C. EE is greater than or equal to expected MtM.

D. Expected MtM can differ significantly from the current MtM.

Question CR-19 a.2

Bank Q928 • AnalystPrep Q.1995 (rebuilt)

At a single future date, a netting set has five equally likely MtM outcomes to the bank (in USD millions): 40, 20, 0, -20 and -30. Which of the following is closest to the expected exposure (EE) at that date?

A. USD 2 million

B. USD 12 million

C. USD 30 million

D. USD 40 million

Question CR-19 a.3

Bank Q916 • BT P2.T6.411.3

A position has a future mark-to-market value that is normally distributed. Acme Bank has the following exposure metrics at the 99% confidence level (values in millions):

Mean (μ)	2.0
Standard deviation (σ)	4.0
Confidence level	99%
Expected exposure (EE)	2.79
Potential future exposure (PFE)	11.31

Each of the following statements about Acme Bank's PFE is true EXCEPT:

A. There is a 1.0% probability that the future (mark-to-market) value will be greater than +USD 11.31 million.

B. If the mean is zero, there is no impact on PFE, as PFE is conditional on gains.

C. At a 95% confidence level, the PFE is less than USD 11.31 million.

D. At a 50% confidence level, the PFE is 2.0.

Question CR-19 a.4

Bank Q920 • BT P2.T6.908.3 (rebuilt)

A counterparty risk analyst has the following probability distribution for the future value (FV) of a derivative position. The mean is USD 3.0 and the standard deviation is USD 12.56; the distribution is symmetric about the mean in shape but not exactly normal.

Future value	Probability	Future value	Probability
USD 33.0	2.5%	(−USD 1.0)	20.0%
USD 25.0	5.0%	(−USD 6.0)	15.0%
USD 18.0	7.5%	(−USD 12.0)	7.5%
USD 12.0	15.0%	(−USD 19.0)	5.0%
USD 7.0	20.0%	(−USD 27.0)	2.5%

Which of the following is the 95% potential future exposure (PFE) of this position?

A. USD 18.00

B. USD 23.66

C. USD 25.00

D. USD 33.00

Question CR-19 a.5

Workbook C8.1 • S1, S2, S3 • Map #37

A counterparty risk analyst at a Canadian bank is estimating exposure metrics to a pension fund counterparty at the six-month horizon. The bank has two trades with the fund:

- A receive-fixed interest rate swap.
- A one-year European put on an equity index that the bank sold to the fund on 20,000 units, premium received in full at inception. Inputs: index level 4,000; strike 3,800; implied volatility 20%; risk-free rate 3%. Black-Scholes value today: CAD 3.5 million.

The two trades were booked under different master agreements, and legal counsel has confirmed that close-out netting between them is not enforceable. The bank's simulation produces five equally likely scenarios for the value of each trade to the bank in six months (CAD million):

Scenario	1	2	3	4	5
Swap	18.0	9.0	3.0	−6.0	−14.0
Sold put	−0.5	−1.0	−2.0	−4.5	−9.0

Which of the following is closest to the bank's expected positive exposure (EPE) to the fund and the expected future value (EFV) of the two trades at the six-month horizon?

A. EPE CAD 2.5 million; EFV CAD −1.4 million

B. EPE CAD 5.3 million; EFV CAD −1.4 million

C. EPE CAD 6.0 million; EFV CAD 2.0 million

D. EPE CAD 6.0 million; EFV CAD −1.4 million

Question CR-19 a.6

Bank Q924 • BT P2.T6.909.1 (rebuilt)

A credit risk analyst has the expected exposure (EE) profile of a position below, with EE treated as constant over the period ending at each observation date.

Observation date (years)	0.5	1	2	3	5
EE (USD millions)	3	5	8	6	2

Which of the following is closest to the expected positive exposure (EPE), computed as the time-weighted average of the EE profile?

A. USD 4.40 million

B. USD 4.80 million

C. USD 5.50 million

D. USD 8.00 million

Question CR-19 a.7

Bank Q906 • BT P2.T6.25.1.2

Over a period of five months, a bank has had the following counterparty exposures (USD millions):

Period	EE	95% PFE	Effective (non-decreasing) EE
1	5	9.80	5
2	10	19.60	10
3	9	17.64	10
4	5	9.80	10
5	4	7.84	10

Which row correctly gives the expected positive exposure (EPE), the effective expected positive exposure (EEPE) and the maximum potential future exposure?

A. EPE $= 6.6$, EEPE $= 9.0$ and maximum PFE $= 19.6$

B. EPE $= 9.0$, EEPE $= 6.6$ and maximum PFE $= 7.84$

C. EPE $= 6.6$, EEPE $= 10.0$ and maximum PFE $= 19.6$

D. EPE $= 6.6$, EEPE $= 12.9$ and maximum PFE $= 19.6$

Question CR-19 a.8

Bank Q922 • BT P2.T6.909.3 (rebuilt)

A credit risk manager has the expected exposure (EE) of a short-dated position, measured at the end of each month for twelve months (USD millions):

Month	1	2	3	4	5	6	7	8	9	10	11	12
EE	2	4	6	8	10	9	8	7	6	5	4	3

Which of the following is closest to the effective expected positive exposure (effective EPE) of this position?

A. USD 6.00 million

B. USD 6.67 million

C. USD 8.33 million

D. USD 10.00 million

Question CR-19 a.9

Bank Q918 • BT P2.T6.412.2 (key corrected)

To determine the counterparty credit risk capital charge, the Basel III framework requires an effective expected positive exposure (effective EPE, or EEPE) calculated with stressed parameters. Each of the following statements about the EEPE is true EXCEPT:

A. If expected exposure (EE) increases over a few months to a peak and then declines toward zero by the end of the year, EEPE must be greater than EPE.

B. If expected exposure (EE) is monotonically increasing, EEPE is identical to EPE.

C. The intent of the non-decreasing constraint (in effective EE and EEPE) is to capture roll-off or rollover risk, for transactions close to maturity that are likely to be replaced in practice.

D. Because EEPE is a weighted average over time, it is a single number, but it depends on a confidence level, typically 99%, which can be set at more conservative levels.

Solutions • CR-19 a

Solution CR-19 a.1

Answer: B

Bank Q899 • BT P2.T6.321.1

Step 1 – Compare the two averages

Expected MtM averages all outcomes; EE averages $\max(V, 0)$, with negatives set to zero. Since $\max(V, 0) \geq V$ in every scenario, EE $\geq$ expected MtM, with equality only if no outcome is negative.

$\Rightarrow$ C is true and B ("always equals") is false.

Step 2 – The other statements

Expected MtM can be negative (a position expected to lose value), and it drifts away from the current MtM through cash flow differentials, forward rates and asymmetric margin terms.

$\Rightarrow$ A and D are true.

Interpretation

EE is the option-like quantity; expected MtM is the linear one. In GARP's example they are 30 and 22.

Why the other options are wrong

A is incorrect. True: EFV can be negative; exposure cannot.

C is incorrect. True: setting negatives to zero can only raise the average.

D is incorrect. True: over long horizons EFV can differ significantly from today's value.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5

Solution CR-19 a.2

Answer: B

Bank Q928 • AnalystPrep Q.1995

Step 1 – Exposure in each scenario

$\max(V, 0) = 40,\ 20,\ 0,\ 0,\ 0.$

Step 2 – Average over all five scenarios

$$\text{EE} = \frac{40 + 20 + 0 + 0 + 0}{5} = \frac{60}{5} = \text{USD 12 million}$$

$\Rightarrow$ Negative scenarios contribute zero but stay in the denominator.

Interpretation

For completeness: EFV $= (40 + 20 + 0 - 20 - 30)/5 = 2$ and ENE $= -50/5 = -10$, and $12 - 10 = 2$.

Why the other options are wrong

A is incorrect. Reported the expected MtM, 2, letting negatives offset positives.

C is incorrect. Averaged only the positive outcomes, $60/2 = 30$.

D is incorrect. Took the largest outcome, a PFE at a very high confidence level.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5, Figure 19.3 example

Solution CR-19 a.3

Answer: B

Bank Q916 • BT P2.T6.411.3

Step 1 – PFE is a quantile of the whole distribution

$$\text{PFE}_{99} = \mu + z_{0.99}\sigma = 2.0 + 2.326 \times 4.0 = 11.30 \ (11.31 \text{ with } z = 2.3263)$$

$\Rightarrow$ By definition only 1% of outcomes lie above it (A true).

Step 2 – Change the confidence level

95%: $2.0 + 1.645 \times 4.0 = 8.58 < 11.31$ (C true). 50%: $z = 0$, so PFE $= \mu = 2.0$ (D true).

Step 3 – Change the mean

With $\mu = 0$: $\text{PFE}_{99} = 2.326 \times 4.0 = 9.30$, a fall of 2.0.

$\Rightarrow$ PFE moves one-for-one with the mean; B is false.

Interpretation

These are GARP's own numbers (EPE 2.79 and PFE 11.31 for $\mu = 2$, $\sigma = 4$). PFE is not conditional on gains; that description fits EPE.

Why the other options are wrong

A is incorrect. True: the 99th percentile is exceeded with 1% probability.

C is incorrect. True: a lower confidence level gives a lower quantile.

D is incorrect. True: the median of a normal is its mean.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5

Solution CR-19 a.4

Answer: C

Bank Q920 • BT P2.T6.908.3

Step 1 – Accumulate probability from the top

$\Pr(\text{FV} > 25) = 2.5\%$; $\Pr(\text{FV} > 18) = 2.5\% + 5.0\% = 7.5\%$.

$\Rightarrow$ 18 is exceeded 7.5% of the time, more than the 5% a 95% PFE allows; 25 is exceeded only 2.5% of the time.

Step 2 – Cross-check from the bottom

Cumulative probability up to 18 is 92.5%, up to 25 is 97.5%. The 95th percentile is the first value reaching 95%: USD 25.00.

Interpretation

When the stem gives the actual distribution, read the quantile from it; the normal approximation is only for when the distribution is normal.

Why the other options are wrong

A is incorrect. USD 18 is the 90% PFE (7.5% of outcomes exceed it).

B is incorrect. Normal approximation: $3 + 1.645 \times 12.56 = 23.66$; the stem says the distribution is not normal.

D is incorrect. USD 33 is the maximum outcome, a 100% level.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5

Solution CR-19 a.5 **Answer: D**

Workbook C8.1 • S1, S2, S3

Step 1 – Decide the aggregation level

Netting is not enforceable, so each trade is its own netting set. Exposure is computed trade by trade with $\max(\text{value}, 0)$ and then summed.

$\Rightarrow$ The two trades cannot offset one another in a default; the bank would have to pay the put's value in full while claiming the swap's value as an unsecured creditor.

Step 2 – Swap EPE

$$\text{EPE}_{\text{swap}} = \tfrac{1}{5}(18 + 9 + 3 + 0 + 0) = \tfrac{30}{5} = 6.0$$

$\Rightarrow$ Negative scenarios count as zero but stay in the denominator: on average CAD 6.0 million is at risk from the swap.

Step 3 – Sold put EPE

$$\text{EPE}_{\text{put}} = \tfrac{1}{5}(0 + 0 + 0 + 0 + 0) = 0$$

$\Rightarrow$ Every value to the bank is negative or zero: having received the premium, the bank is owed nothing by the fund, so the put adds no exposure. The index level, strike, volatility and the CAD 3.5 million Black-Scholes value price the option but are irrelevant to the seller's exposure.

Step 4 – Total EPE

$$\text{EPE} = 6.0 + 0 = 6.0$$

Step 5 – EFV (additive, includes every value)

$$\text{EFV}_{\text{swap}} = \frac{18 + 9 + 3 - 6 - 14}{5} = 2.0, \qquad \text{EFV}_{\text{put}} = \frac{-0.5 - 1.0 - 2.0 - 4.5 - 9.0}{5} = -3.4$$

$$\text{EFV} = 2.0 - 3.4 = -1.4$$

$\Rightarrow$ The two trades are expected to be worth CAD 1.4 million against the bank in six months, yet the bank still has CAD 6.0 million of expected positive exposure.

Interpretation

EFV and EPE answer different questions. EFV is the average of all values and is additive, so the sold put pulls it negative. EPE keeps only what the counterparty would owe the bank, so the sold put contributes nothing to it and, without enforceable netting, cannot reduce the swap's exposure either. The CVA desk should price CAD 6.0 million of EPE, and the legal team's view on netting is worth CAD 0.7 million of EPE here.

Why the other options are wrong

A is incorrect. Red herring used as an offset: subtracted the sold put's Black-Scholes value from the swap EPE, $6.0 - 3.5 = 2.5$. Today's option value is neither an exposure nor a netting offset.

B is incorrect. Assumed netting where none is enforceable: summed values scenario by scenario, 17.5, 8.0, 1.0, -10.5, -23.0, giving EPE $(17.5 + 8.0 + 1.0)/5 = 5.3$. This is the netted EPE (a netting factor below 1), not the exposure the bank actually has.

C is incorrect. Correct EPE, but dropped the sold put from EFV because it has zero exposure, reporting only the swap's EFV of 2.0. Zero exposure does not mean zero value; EFV averages all values.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive

exposure, and maximum exposure. CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. CR-12 a: Assess the credit risks of derivatives. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11, 19.1.1, 19.1.5, 19.3.1, Tables 19.1 and 19.2; Hull, OFOD 11e, Ch.24, 12.1

Solution CR-19 a.6

Answer: A

Bank Q924 • BT P2.T6.909.1

Step 1 – Interval lengths

0 to 0.5: 0.5; 0.5 to 1: 0.5; 1 to 2: 1; 2 to 3: 1; 3 to 5: 2. Total 5 years.

Step 2 – Time-weighted average

$$\text{EPE} = \frac{3(0.5) + 5(0.5) + 8(1) + 6(1) + 2(2)}{5} = \frac{22}{5} = \text{USD } 4.40 \text{ million}$$

$\Rightarrow$ The long, low final period pulls the average down.

Interpretation

GARP: average EPE is "the weighted average (the weights being the time intervals) of the EPE profile"; a simple average applies only when the points are equally spaced.

Why the other options are wrong

B is incorrect. Unweighted average, $24/5 = 4.80$, ignoring unequal intervals.
C is incorrect. Dropped the final, longest period: $(3 + 5 + 8 + 6)/4 = 5.50$.
D is incorrect. The peak EE, 8.00, not an average.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5, Figure 19.4

Solution CR-19 a.7

Answer: A

Bank Q906 • BT P2.T6.25.1.2

Step 1 – EPE

$$\frac{5 + 10 + 9 + 5 + 4}{5} = \frac{33}{5} = 6.6$$

Step 2 – Effective EPE

$$\frac{5 + 10 + 10 + 10 + 10}{5} = \frac{45}{5} = 9.0$$

Step 3 – Maximum PFE

$$\max(9.80,\ 19.60,\ 17.64,\ 9.80,\ 7.84) = 19.6$$

Interpretation

Three different operations on two profiles: average the EE, average the running maximum of the EE, and take the maximum of the PFE. Effective EPE (9.0) exceeds EPE (6.6) because EE falls after period 2.

Why the other options are wrong

B is incorrect. Swapped EPE and EEPE (EEPE cannot be below EPE) and took the smallest PFE.

C is incorrect. Used the peak effective EE (10) instead of its average.
D is incorrect. Averaged the PFE profile: $(9.8+19.6+17.64+9.8+7.84)/5 = 12.94$, a different quantity.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** Gregory, The xVA Challenge 4e, Ch.11 (GARP CR Ch.19), 19.1.5; SchweserNotes 2026 Book 2, Reading 37, LO 37.a

Solution CR-19 a.8 Answer: C

Bank Q922 • BT P2.T6.909.3

Step 1 – Effective EE (running maximum)

2, 4, 6, 8, 10, then 10 for months 6 to 12.
$\Rightarrow$ After the month-5 peak, effective EE stays at 10.

Step 2 – Effective EPE

$$\frac{2+4+6+8+10\times 8}{12} = \frac{100}{12} = 8.33$$

Interpretation

The plain EPE is $72/12 = 6.00$. The 2.33 gap is the rollover allowance: a short-dated trade whose EE decays is assumed to be replaced, so its exposure does not fall back.

Why the other options are wrong

A is incorrect. The plain EPE, $72/12 = 6.00$, with no non-decreasing adjustment.
B is incorrect. Averaged only the first six effective EE values, $40/6 = 6.67$.
D is incorrect. The peak EE, 10.00, not the average of the effective profile.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** SchweserNotes 2026 Book 2, Reading 37, LO 37.a (effective EE and effective EPE); Basel II (BCBS 2005)

Solution CR-19 a.9 Answer: D

Bank Q918 • BT P2.T6.412.2

Step 1 – Peaked EE (A) and rising EE (B)

After a peak, effective EE stays at the peak while EE falls, so EEPE $>$ EPE (A true). If EE never falls, effective EE equals EE at every date and EEPE $=$ EPE (B true).

Step 2 – Purpose (C)

The non-decreasing constraint captures rollover risk for short-dated trades that would be replaced (C true).

Step 3 – Confidence level (D)

EEPE is an average of expected exposures; it involves no quantile and no confidence level. That belongs to PFE.
$\Rightarrow$ D is false.

Interpretation

D is the only statement that mixes an average measure with a quantile. Under Basel, conservatism enters EEPE through stressed calibration and the alpha multiplier, not through a confidence level.

Why the other options are wrong

A is incorrect. True: the floor binds once EE falls.
B is incorrect. True: the floor never binds when EE keeps rising.
C is incorrect. True: rollover risk is the stated purpose.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. • **Reference:** SchweserNotes 2026 Book 2, Reading 37, LO 37.a; Basel III counterparty credit risk framework (EEPE)

CR-22 • Structured Credit Risk • Vardeez: IMPORTANT

CR-22 h

P3 CONCEPT

Describe a simulation approach to calculating credit losses for different tranches in a securitization.

Workbook: D15 Structured Credit and Implied Correlation • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: D15 (Map #60, 2/16) reports reading tranche values and write-downs off the simulation grid as "the one possible calculation" for the reading's simulation material.

Vardeez: three steps (estimate the parameters, generate default-time simulations, compute the tranche losses); the tail of each tranche's distribution gives its Credit VaR.

Reported exam shape

- **D15·S2** Reading tranche values and writedowns off the simulation grid.

What the questions look like: concept only. The order of the steps; what to do when correlation cannot be estimated; what needs a fresh simulation and what only reprocessing; how a Credit VaR is read off 1,000 threads; which statements a grid of simulated results supports.

Worked elsewhere: D15.1 (reading the grid at $\pi = 2.25\%$ and backing out implied correlation) under CR-22 l. What the grid shows about default probability and correlation is CR-22 i.

1. Why simulate

The scenario analysis of CR-22 e and f runs the waterfall at one constant default rate. That ignores **default correlation**, the propensity of defaults to coincide, and with it the timing and clustering of defaults. A copula simulation of the collateral pool (the approach of CR-11) gives the whole distribution of losses for each bond and of IRRs or values for the equity, and so valuation and risk statistics such as Credit VaR, not just particular outcomes.

2. The three steps

1. **Estimate parameters.** A default probability π_n for each of the $N = 100$ loans, physical (from a rating) or risk-neutral (from spreads, where comparable spread data exist); Malz takes a physical one-year PD, the same for every loan, and converts it to a hazard rate $\lambda = -\ln(1-\pi)$: $\pi = 2.25\%$ gives $\lambda = 0.022757$. Then the pairwise copula correlations. There is little market or financial data to estimate them and the copula correlation is not intuitive, so Malz assumes one constant $\rho \geq 0$ and runs a **grid**: 13 default probabilities (0.75% to 9.75%) $\times$ 4 correlations (0, 0.3, 0.6, 0.9) $= 52$ pairs.

2. **Generate default times.** For each correlation, draw $I = 1{,}000$ threads of a 100-dimensional standard normal with pairwise correlation ρ, and map each draw to a default time

$$t_{n,i} = -\frac{\ln\left[1 - \Phi(z_{n,i})\right]}{\lambda}.$$

   GARP's first draw, $z = -1.2625$, gives $t = 4.80$ years: a default within the CLO's five-year term. Count, in each thread, the defaults in each year: the first thread at $(\pi, \rho) = (2.25\%, 0.3)$ has $(11, 5, 7, 7, 7)$, cumulatively 37 by year 5, "one of the grimmer simulation threads".

3. **Compute the credit losses.** Pass each thread's default counts through the waterfall of CR-22 c, e

and f. Each thread gives a loss for each bond and a cash-flow stream for the equity, discounted at a 25% hurdle rate; 1,000 threads give the distributions.

What needs a new simulation. One matrix of correlated normals is needed per *correlation*, not per default probability. A new recovery rate or loan coupon, or a different liability structure, only changes how the simulations are processed. Fresh simulations are needed only for more threads, a different number of loans, or a new correlation setting. The default sensitivities of CR-22 j reuse the same normals for the same reason: a new seed would add simulation noise.

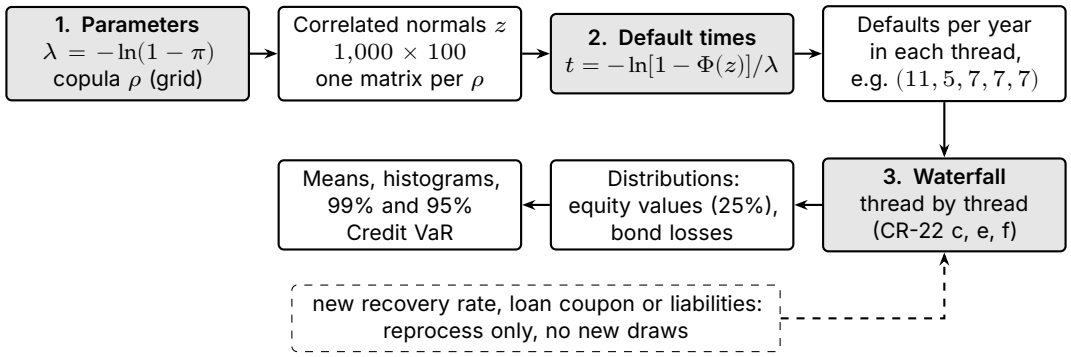


Figure 1. Malz's simulation, with the three steps in grey boxes. Solid arrows: one pass for each of the 52 (π, ρ) pairs. Dashed arrow: changing the recovery rate, the loan coupon or the liability structure re-enters at the waterfall and needs no new draws; a new ρ, more threads or more loans means new normals.

3. Reading the output

- **Means** (GARP Table 22.3): the grid of mean equity values and mean bond writedowns, one cell per (π, ρ) pair; CR-22 i reads it. Means of the two bond writedowns "don't add up" even though each thread does: at $(\pi, \rho) = (0.75\%, 0.3)$ the senior's 0.05% of par comes from a handful of threads in which the mezzanine is wiped out, while only 19 threads show any mezzanine loss.
- **Distributions** (Figures 22.2 to 22.4). Equity: a bell curve at low correlation, squeezed against zero at high π; U-shaped at high correlation. Bonds: losses bunched at zero with a long tail; the thin mezzanine tends to a U shape at high π and ρ.
- **Credit VaR** from 1,000 sorted threads: the 99% level uses the 10th ordered result and the 95% level the 50th. Equity: mean value minus the 10th (50th) lowest value. Bonds: the 10th (50th) highest loss minus the expected loss.
- **A simulation artifact.** At $(0.75\%, 0.3)$ only seven threads show a senior loss, so the 10th highest loss is zero, below the expected loss: the senior's 99% VaR is negative (-0.04). It would disappear at 99.5% or 99.9%, but higher confidence needs more threads to keep simulation error down.

Correction to the source notes. (1) GARP 22.3 first gives the default probabilities as "from $\pi = 0.075$ to $\pi = 0.975$, in increments of 0.075"; the grid is 0.0075 to 0.0975 in steps of 0.0075, as GARP lists it later (13 values, hence 52 pairs). (2) GARP illustrates the bond means that "don't add up" with "the parameter pair $\pi = 0.0225$ and $\rho = 0.30$" and a senior loss of 0.05% of par. In Table 22.3 that pair shows 0.52% (senior) and 7.35% (mezzanine); the 0.05% and 1.11% belong to $\pi = 0.0075$, $\rho = 0.30$. (3) GARP's text and Workbook D15's formula card measure the equity VaR from the \$5m par. Table 22.4 measures it from the mean: at $\pi = 0.75\%$ the equity VaRs for $\rho = 0.6$ and 0.9 (6.85 and 7.14) exceed the \$5m par, which no par-based VaR can do, and equal the mean values in Table 22.3 (the 1% quantile is zero). Use mean minus quantile, as for the bonds.

	Per thread	99% (95%) Credit VaR, 1,000 threads	Shape of the distribution
Equity	PV of cash flows at a 25% hurdle	Mean minus 10th (50th) lowest value	Bell curve at low ρ; U-shaped, a lottery ticket, at high ρ
Mezzanine	Loss, % of par	10th (50th) highest loss minus EL	Bunched at zero, long tail; thin, so 95% and 99% VaR close
Senior	Loss, % of par	10th (50th) highest loss minus EL	Almost all at zero; tail near half of par at high ρ

Exam traps

- **Step order.** Parameters first, then default times for every *loan* (not for a tranche), then the waterfall applied thread by thread, not once to the average default count.
- **Correlation is not set to zero** for want of data: zero correlation removes the clustering that threatens the senior. Run a grid of pairs.
- **What triggers new draws.** A new correlation, more threads or more loans; not a new recovery rate, coupon or default probability.
- **Quantile position.** With 1,000 threads, 99% is the 10th ordered result and 95% the 50th.
- **Subtract the mean.** Credit VaR is the quantile loss minus the expected loss; the raw quantile overstates it.
- **Negative VaR.** A sign of too few tail threads, not of a riskless tranche.

Practice questions: CR-22 h

Question CR-22 h.1

Bank Q1163 • New question written for this bank

A risk analyst at an asset manager is documenting how credit losses for the tranches of a CLO are calculated using a simulation approach. Which of the following correctly describes the sequence of the procedure?

A. Estimate the default probabilities and the correlation parameter, generate correlated default times for each loan in the collateral pool, and then run the resulting default counts through the waterfall to compute losses for each tranche in every simulation thread.

B. Estimate the default probabilities, run the waterfall once at the expected default rate to compute tranche losses, and then adjust each tranche loss upward to reflect the correlation parameter.

C. Estimate the correlation parameter, generate default times for the equity tranche only, and then infer the losses of the mezzanine and senior tranches from the equity losses.

D. Generate default times for each loan, estimate the default probabilities and correlation from the simulated default times, and then run the average number of defaults through the waterfall to compute tranche losses.

Question CR-22 h.2 Bank Q1165 • New question written for this bank

A structured credit analyst must simulate losses on the tranches of a CLO. Default probabilities for the loans can be inferred from market CDS spreads, but there are almost no market data from which to estimate the pairwise default correlation. Which of the following is the most appropriate way to proceed?

A. Use the observed equity return correlation as the copula correlation, because the copula correlation has a direct economic interpretation as an equity relationship.

B. Infer the correlation from the tranche ratings and estimate the default intensities from a single assumed correlation value.

C. Infer the default intensities from spread data and run the simulation across a range of default probability and correlation pairs to examine how tranche results change.

D. Set the correlation to zero, since correlation cannot be observed, and rely on the diversification of the pool to produce the tranche loss distribution.

Question CR-22 h.3 Bank Q1179 • New question written for this bank (rebuilt) • re-filed from CR-22 j

An analyst who has simulated a CLO with 1,000 threads of correlated normal draws now computes the default01 of each tranche. Which of the following procedures is consistent with how the simulation is reused to measure default sensitivities?

A. Shock the default probability by 1 bp, redraw all random numbers for the revaluation, and report the result as a negative number for bonds because their values fall.

B. Shock the correlation parameter up and down by 10 bps, redraw the normal variables for each shock, and divide the difference in mean losses by 20.

C. Shock the default probability up and down by 10 bps, keep the original default counts fixed, and revalue each tranche from the existing simulated cash flows.

D. Shock the default probability up and down by 10 bps, reuse the same correlated normal draws, recompute the default counts and subsequent cash flows, and divide the difference in mean values or losses by 20.

Question CR-22 h.4 Bank Q1164 • New question written for this bank (rebuilt as a concept question)

A risk manager simulates 1,000 threads of a CLO and passes each through the waterfall, following the approach in the reading: equity tranche outcomes are recorded as present values and bond outcomes as credit losses. Which pair of measures gives the 99% Credit VaR of the equity tranche and of the mezzanine bond?

A. Equity: mean value minus the 50th lowest value. Mezzanine: 50th highest loss minus the mean loss.

B. Equity: mean value minus the 10th lowest value. Mezzanine: 10th highest loss minus the mean loss.

C. Equity: mean value minus the 10th highest value. Mezzanine: 10th lowest loss minus the mean loss.

D. Equity: the 10th lowest value. Mezzanine: the 10th highest loss, with no deduction of the mean.

Question CR-22 h.5 Bank Q1174 • BT P2.T6.315.2 • re-filed from CR-22 i

In the three-tiered (senior bond, mezzanine, and equity tranche) securitization structure, Malz tends to measure the risk of a tranche by its 95% and/or 99% Credit VaR. In regard to tranche risks, each of the following is true EXCEPT which is false?

A. Credit VaR is the worst-case loss at the 0.05 or 0.01 quantile minus the expected loss (EL).

B. High systemic risk can be expressed as high default correlation, which tends to increase the risk (Credit VaR) of senior bonds.

C. Because the mezzanine tranche is thick, at high correlations and high default rates the mezzanine tranche behaves like the senior bond, with a bell-curve-shaped loss distribution.

D. An increase in granularity can significantly diminish securitization risk.

Question CR-22 h.6 Bank Q1172 • BT P2.T6.314.2 (rebuilt; bank key corrected) • re-filed from CR-22 i

A simulated three-tranche securitization produces the following 99% Credit VaR for the senior bond tranche ($ million; VaR = the 10th highest simulated loss minus the expected loss, so a larger number means more risk). Rows are the default probability π and columns the default correlation ρ.

π	$\rho = 0.00$	$\rho = 0.30$	$\rho = 0.90$
0.0225	0.00	17.77	59.99
0.0525	0.00	35.74	56.23
0.0975	8.33	42.60	47.79

With respect to the senior bond tranche, each of the following is true EXCEPT which is false?

A. The low-default, high-correlation case ($\pi = 0.0225$, $\rho = 0.90$) has a lower senior 99% Credit VaR than the high-default, zero-correlation case ($\pi = 0.0975$, $\rho = 0$).

B. For a given default rate, an increase in default correlation increases senior bond losses.

C. For a given positive default correlation, an increase in the default rate generally increases expected senior bond losses.

D. At $\rho = 0.90$ the senior 99% Credit VaR is very large (roughly $48 to $60 million in the exhibit) and not highly sensitive to the default rate.

Solutions • CR-22 h

Solution CR-22 h.1 — Answer: A

Bank Q1163 • New question written for this bank

Step 1 – Estimate the parameters

Default probabilities (converted to hazard rates) and the copula correlation come first: they drive everything else.

Step 2 – Simulate default times for every loan

Correlated standard normals, one 100-dimensional draw per thread, are mapped to a default time for each loan.

Step 3 – Pass each thread through the waterfall

Count the defaults in each year of each thread and run them through the waterfall to get a loss for every tranche in every thread.

$\Rightarrow$ A gives the steps in the right order and keeps the whole distribution.

Interpretation

Vardeez's three steps: estimate the parameters, generate the default-time simulations, compute the credit losses. The distribution across threads, not one average, is what the simulation buys.

Why the other options are wrong

B is incorrect. Describes one scenario analysis with a correlation add-on: no distribution of tranche losses.

C is incorrect. Default times are simulated for each loan in the pool, not for a tranche; tranche losses come out of the waterfall.

D is incorrect. Reverses steps 1 and 2 (parameters cannot be estimated from the draws they generate) and averages over threads, which throws away the distribution.

LO: CR-22 h: Describe a simulation approach to calculating credit losses for different tranches in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 The Simulation Procedure and the Role of Correlation

Solution CR-22 h.2 — Answer: C

Bank Q1165 • New question written for this bank

Step 1 – The default probabilities

Spread data give risk-neutral default intensities (hazard rates); ratings would give physical ones.

Step 2 – The correlation

GARP: the correlations "are more difficult to obtain, since the copula correlation … is not a natural or intuitive quantity, and there is not much market or financial data with which to estimate it."

$\Rightarrow$ Run the simulation over a grid of (π, ρ) pairs and study how the tranche results change, as Malz does with 52 pairs.

Interpretation

Schweser: the copula correlation "could be useful in theory but suffers from empirical imprecision in practice. Instead, a sensitivity analysis is performed for various default and correlation pairs."

Why the other options are wrong

A is incorrect. The copula correlation has no direct economic interpretation; equity correlation is at best a proxy for asset correlation, a different concept.

B is incorrect. Reverses the logic: spreads inform the default intensities, not the correlation, and one assumed value skips the needed sensitivity analysis.

D is incorrect. Zero correlation removes the clustering of defaults that threatens the senior tranche, so tail losses would be badly understated.

LO: CR-22 h: Describe a simulation approach to calculating credit losses for different tranches in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 The Simulation Procedure and the Role of Correlation

Solution CR-22 h.3 — Answer: D

Bank Q1179 • New question written for this bank

Step 1 – What a change in π touches

The correlated normals z depend only on ρ. A new default probability changes the hazard rate, hence the default times $t = -\ln[1 - \Phi(z)]/\lambda$, the default counts and every cash flow after them.
$\Rightarrow$ Reuse z, but recompute default times, counts and the waterfall.

Step 2 – The default01

Shock π by ± 10 bp, revalue, and divide the change by 20 to express it per basis point; it is reported as a positive number (a fall in value or a rise in loss).
$\Rightarrow$ D is correct.

Interpretation

GARP: "We can reuse our correlated normal simulations z. In fact, we should, in order to avoid a change of random seed and the attendant introduction of additional simulation noise." The default01 itself is CR-22 j; the point here is which parts of the simulation have to be rerun.

Why the other options are wrong

A is incorrect. Redrawing the random numbers adds simulation noise to the sensitivity, and default01 is quoted as a positive number.
B is incorrect. Shocks the correlation, which measures a different sensitivity; a new ρ would indeed need new draws.
C is incorrect. Keeping the default counts fixed ignores the shock: the counts must change for the revaluation to change.

LO: CR-22 h: Describe a simulation approach to calculating credit losses for different tranches in a securitization. CR-22 j: Explain how default sensitivities for tranches are measured. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 The Simulation Procedure; Default Sensitivities of the Tranches

Solution CR-22 h.4 — Answer: B

Bank Q1164 • New question written for this bank

Step 1 – The quantile's position

With 1,000 threads, the worst 1% is the 10 worst outcomes: the 10th ordered result. The 50th is the 95% level.

Step 2 – The equity, recorded as values

Bad outcomes are *low* values: the 99% VaR is the mean value minus the 10th lowest value.

Step 3 – The bonds, recorded as losses

Bad outcomes are *high* losses: the 99% VaR is the 10th highest loss minus the expected (mean) loss.
$\Rightarrow$ B is correct.

Interpretation

Credit VaR is an unexpected loss: the distance between the tail outcome and the mean. On GARP's histograms it is the horizontal distance between the solid line (mean) and the dashed line (1% quantile). GARP's text says the equity VaR is measured from the \$5m par; its Table 22.4

measures it from the mean (see the correction box in Section 3).

Why the other options are wrong

A is incorrect. Uses the 50th ordered result: that is the 95% Credit VaR.
C is incorrect. Reverses the tails: high equity values and low bond losses are good outcomes.
D is incorrect. Stops at the quantile: the raw tail outcome is not the Credit VaR until the mean is deducted.

LO: CR-22 h: Describe a simulation approach to calculating credit losses for different tranches in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 Credit VaR of the Tranches

Solution CR-22 h.5 Answer: C

Bank Q1174 • BT P2.T6.315.2

Step 1 – Check each true statement

A is the definition used for every tranche. B is GARP's systematic risk point: high correlation raises the senior's tail losses. D: GARP, "Granularity can significantly diminish securitization risks"; a lumpy pool of a few large loans has more extreme outcomes.

Step 2 – The mezzanine's thickness and shape

The mezzanine (5% to 15%) is thin, not thick. At high correlations and default rates its simulated losses tend to a U shape, all or nothing, like the equity; GARP: the U-shape tendency "is much stronger for the mezzanine, which ... behaves like the equity at high correlations and default probabilities".
$\Rightarrow$ C is false on both counts.

Interpretation

Thinness means that, conditional on any loss, the loss is likely to be large, which is also why the mezzanine's 95% and 99% Credit VaRs are close for most (π, ρ) pairs.

Why the other options are wrong

A is incorrect. True: the quantile loss minus the expected loss.
B is incorrect. True: correlation is the senior's main risk factor.
D is incorrect. True: more, smaller loans reduce the probability of extreme outcomes, so a granular pool can support a larger senior tranche with no rise in Credit VaR.

LO: CR-22 h: Describe a simulation approach to calculating credit losses for different tranches in a securitization. CR-22 k: Describe risk factors that impact structured products. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 Characteristics of the Distributions; Credit VaR of the Tranches; Granularity

Solution CR-22 h.6 Answer: A

Bank Q1172 • BT P2.T6.314.2

Step 1 – Read the two cells in A

$(\pi, \rho) = (0.0225, 0.90)$: \$59.99m. $(0.0975, 0)$: \$8.33m. The first is far *higher*.
$\Rightarrow$ A is false.

Step 2 – Check the others against the grid

B: along each row VaR rises with correlation (0.00, 17.77, 59.99 at $\pi = 0.0225$). C: more defaults hurt every tranche, and the senior's expected writedown rises with π at every positive ρ (GARP Table 22.3). D: the $\rho = 0.90$ column stays between 47.79 and 59.99.
$\Rightarrow$ B, C and D are true.

Interpretation

Correlation is a more important driver of the senior's tail than the default probability: indepen-

dent defaults stay near their expected level and rarely reach the 15% attachment point, even at a 9.75% default rate, while clusters reach it even at 2.25%.

Why the other options are wrong

B is incorrect. True: higher correlation clusters defaults and raises the senior's tail loss.
C is incorrect. True: every tranche is hurt by a higher default rate; the senior default01 is positive.
D is incorrect. True: at high correlation the VaRs are similar whatever the default rate; GARP says the senior's risk "varies surprisingly little with default probability".

LO: CR-22 h: Describe a simulation approach to calculating credit losses for different tranches in a securitization. CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. • **Reference:** Malz, Ch.9 (GARP CR Ch.22), 22.3 Credit VaR of the Tranches, Table 22.4

CR-23 • An Introduction to Securitisation • Vardeez: IMPORTANT

CR-23 c

P3 CONCEPT

Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust.

Workbook: none homed here; the structures are summarised in the A9 revision notes D14 CLN issued through an SPV (cross-reference)

WHAT IS TESTED HERE

Why P3. No workbook question is homed here, but Vardeez flags the LO: "LO 39.c: This can be important" (the map matches his "39.c" to CR-23 c) in a chapter he rates IMPORTANT rather than Very Important.
Vardeez: Trust vs SPV.

Reported exam shapes: none listed in the map.

What the questions look like: concept only, no calculations. Which structure fits a given asset and issuer; what happens to principal during a revolving period and when a trigger trips; what the trust form adds; why the vehicle must be bankruptcy remote.

Worked elsewhere: D14.1 (a credit-linked note issued through an SPV) at Question CR-9 g.7; the trigger ratios (MPR, delinquency, default) and the WAL under CR-23 g.

1. The SPV and its two legal forms

The SPV (also SPE or SPC) buys the assets from the originator in a **true sale** and funds the purchase by issuing notes. GARP: "An SPV is a legal trust or company that is not, for legal purposes, linked in any way to the originator of the securitisation. As such it is **bankruptcy-remote** from the sponsor." Investors are exposed only to the SPV's assets, so the notes are rated on the pool, not on the originator. SPVs are often incorporated offshore (Jersey, the Cayman Islands) or in SPV-friendly centres (Dublin, the Netherlands), mainly for tax, including withholding tax on the note coupons.

"For legal purposes an SPV is categorised as either a Company or a Trust." The LO's "trust versus SPV" is therefore **trust versus company**: both are SPVs (Vardeez's table labels the company form "SPV").

- **Company** (usual in the Euromarkets, where accounting rules allow it): one entity buys the assets for cash and issues claims directly against them.
- **Trust** (more common in the US): interests are represented by a **trustee**, usually a bank's agency services department or a specialist trust company. Schweser: a company "may not distance the originator from the assets enough for accounting purposes in the United States", so two entities are used. The originator sells the assets for cash to a master trust (SPV 1), which deposits them in a grantor trust (SPV 2) and receives a beneficial interest in it. The notes rest on that beneficial claim, not on the assets directly: the same economics as one SPV, with more distance between originator and assets.

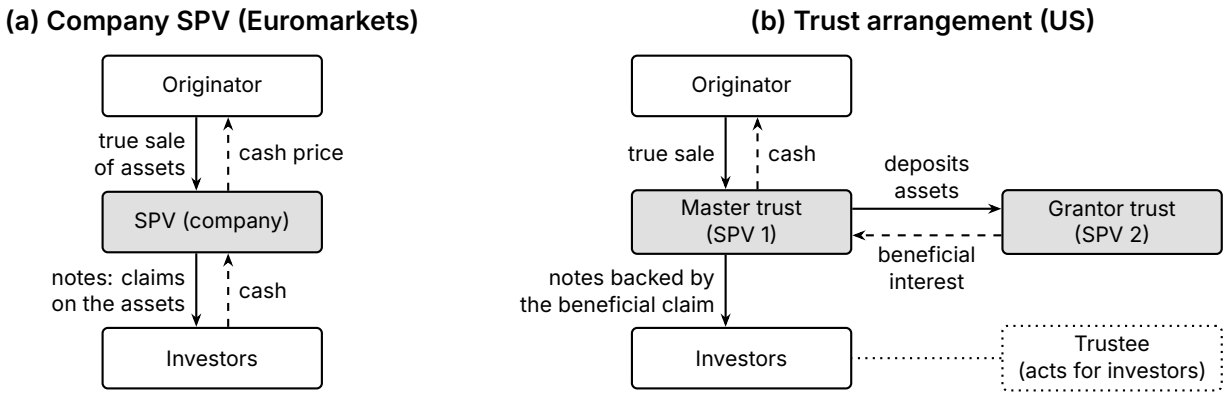


Figure 1. The two legal forms of an SPV. Solid arrows carry assets and claims, dashed arrows carry cash or an interest flowing back; shaded boxes are SPVs. The trust form puts a second entity between the originator and the assets (GARP 23.1; Schweser, Figure 41.3).

2. The three SPV structures

GARP: "There are essentially two main securitisation structures, amortising (pass-through) and revolving. A third type, the master trust, is used by frequent issuers."

- **Amortising (pass-through).** Principal and interest go to investors coupon by coupon through the life of the security. The notes are priced and traded on expected maturity and WAL, which incorporates prepayment assumptions (CR-23 g, h). Used for residential and commercial MBS and consumer loan ABS.
- **Revolving.** During the **revolving period**, principal collections buy new receivables that meet the eligibility criteria. It suits short-dated assets with a relatively high prepayment speed: credit card debt and auto loans. In the **amortisation period** investors are repaid in equal instalments (**controlled amortisation**) or principal is trapped in a separate account and paid in one lump sum at the expected maturity (**soft bullet**). Early amortisation triggers (an MPR below its floor, delinquency or default above theirs) end the revolving period early.
- **Master trust.** Frequent issuers under US and UK law issue multiple securitisations from the same SPV: the originator transfers assets to the master trust and notes are issued out of the pool as investors demand. Used by MBS and credit card ABS originators; since 1991 the preferred vehicle for credit cards. Each series draws on the whole, changing pool pro rata to its invested amount; excess spread released from the trust can be shared with series suffering interest shortfalls, and a financial guaranty is often added because card debt is unsecured with low recoveries.

Vardeez's discriminator: an asset that pays down on a schedule goes into an amortising structure, one that pays back quickly and repeatedly into a revolving one; an *issuer* that comes to market repeatedly uses a master trust. The master trust is defined by the issuer's behaviour, not the asset.

	Amortising	Revolving	Master trust
Principal	Passed through each coupon	Buys new receivables, then repaid	Shared by every series from one pool
Typical assets	Mortgages (RMBS, CMBS), consumer loans	Credit cards, auto loans	MBS and credit card ABS
Repayment	From the start; priced on expected maturity and WAL	Controlled amortisation or soft bullet	Per series, from the common pool
Defined by	The asset's amortisation schedule	The asset's short life and repayment speed	The issuer's frequency of issue

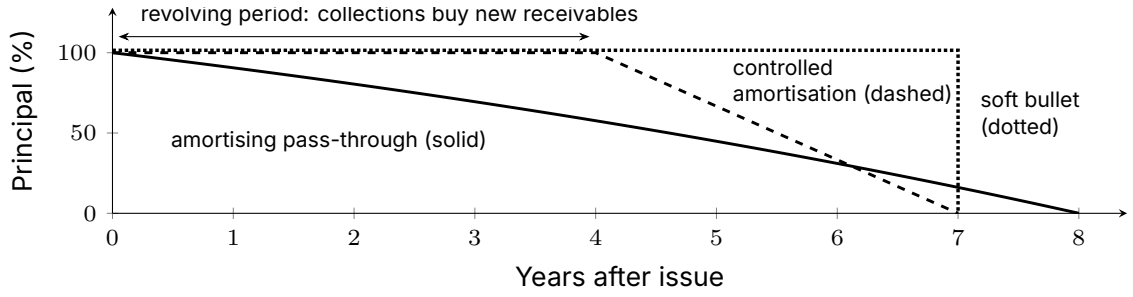


Figure 2. How principal comes back (stylised). A pass-through repays from the first coupon; a revolving structure reinvests collections, then repays in instalments or in one lump sum. A tripped early amortisation trigger brings the decline forward.

Correction to the source notes. SchweserNotes (LO 41.c) and the bank's key to Q1212 say a revolving structure pays principal "in large lump sums rather than a pre-specified amortization schedule". In GARP, collections are *reinvested* during the revolving period and investors are then repaid by controlled amortisation or a soft bullet: the lump sum is one of two options, and it comes after the revolving period.

Exam traps

- **Geography swap.** Trusts are more common in the US, companies in the Euromarkets. **Trust is not the opposite of SPV:** both forms are SPVs, both need a true sale and bankruptcy remoteness; the trust adds a second entity and a beneficial interest.
- **Sibling swap.** Passing principal through and pricing on WAL describe the amortising structure; reinvesting principal describes the revolving one.
- **Master trust as an asset type.** It answers "how often does the issuer come to market?". An infrequent issuer of card debt uses a plain revolving structure.
- **Trigger polarity.** An early amortisation trigger ends the revolving period sooner and shortens the notes' life; it does not extend the revolving period or leave the WAL unchanged.
- **Whose credit?** SPV notes carry the pool's credit, de-linked from the originator's rating, which can make funding cheaper. Over-collateralisation means notes issued are *less* than the assets.

Practice questions: CR-23 c

Question CR-23 c.1

Bank Q1197 • Mock 2 Q29 • re-filed from CR-23 a

A bank is establishing a special purpose vehicle (SPV) to securitize a portfolio of receivables. The bank's structuring team is explaining to the treasury department why the SPV is necessary and what characteristics it must possess. Which of the following is the most important feature the SPV must have for the securitization to achieve its intended purpose?

A. The SPV must be domiciled in the same jurisdiction as the originating bank so that a single legal regime governs the transaction.

B. The SPV must be consolidated onto the originating bank's balance sheet so that investors have recourse to the bank's capital.

C. The SPV must be bankruptcy remote, so that the insolvency of the originating bank does not give its creditors a claim on the transferred assets.

D. The SPV must retain the originating bank as servicer so that the credit quality of the pool is actively managed.

Question CR-23 c.2

New • built from GARP Ch.23, 23.1 Example 23.1; Schweser LO 41.c

A UK card issuer that brings a credit card ABS to market several times a year is choosing the legal form of its issuing vehicle. Its US counsel recommends a trust arrangement; its London counsel says a company incorporated in Jersey would serve. Which of the following statements about the two forms is correct?

A. A company SPV is the usual form in the US market, because US accounting rules are satisfied by a single entity that issues notes directly against its assets.

B. Choosing the trust form removes the need for a true sale, because the trustee then holds the receivables on behalf of the originator.

C. A company SPV cannot be bankruptcy remote, so investors in its notes keep a claim on the originator if the receivables underperform.

D. In the trust arrangement the receivables are sold to a master trust that deposits them in a grantor trust; the notes rest on a beneficial interest, which further distances the originator.

Question CR-23 c.3

Bank Q1208 • AnalystPrep Q.2929 (rebuilt)

Besides cash flow securitization, a special purpose vehicle (SPV) can also be used to issue credit-linked notes (CLNs). Which of the following correctly describes a CLN issued by an SPV?

A. The note's performance depends on the credit of the sponsoring entity that set up the SPV.

B. The note's performance depends on the performance of assets that have been sold to the SPV, not on the sponsoring entity.

C. The note is issued directly by the originator, which keeps the reference assets on its own balance sheet.

D. The note can only be issued if the SPV is not legally separate from the sponsoring entity.

Question CR-23 c.4

Bank Q1209 • BT P2.T6.610.3

Which of the following structures is most likely to be used by an infrequent issuer of securities whose asset pool consists of credit card debt, that is, short-dated assets with a relatively high prepayment speed?

A. Master trust

B. Amortizing structure

C. Revolving structure

D. None of the above

Question CR-23 c.5

Bank Q1211 • Mock 3 Q59 (rebuilt)

A bank is planning to securitize a portfolio of automobile loans and is considering an amortizing or a revolving structure for the special purpose vehicle. Which of the following statements correctly describes a revolving securitization structure?

A. Principal received from the underlying loans is passed through to investors as it is collected.

B. The pool of underlying assets is fixed at closing and cannot be replenished.

C. Investors are exposed to the credit risk of the originating bank rather than of the underlying loan pool.

D. During the revolving period, principal collections are used to purchase new receivables, and investors are later repaid through controlled amortization or a single soft bullet payment.

Question CR-23 c.6

Bank Q1220 • Mock 4 Q61 • re-filed from CR-23 f

A revolving securitization of credit card receivables includes an early amortization trigger that is activated if the monthly payment rate (MPR) falls below a minimum threshold. A risk analyst at an investor is explaining the purpose and effect of this feature to a client. Which of the following statements is correct?

A. The trigger allows the originator to extend the revolving period if the MPR deteriorates, giving the pool time to recover.

B. The trigger ends the revolving period early and begins returning principal to investors, protecting them from further deterioration in the pool.

C. The trigger requires the originator to substitute higher-quality receivables into the pool until the MPR returns above the threshold.

D. The trigger converts the transaction from a revolving to an amortizing structure while leaving the expected weighted-average life of the notes unchanged.

Question CR-23 c.7 Bank Q1210 • GARP PE1 Q21 • GARP PE2 Q60 (rebuilt)

A bank plans to securitize car loans using an SPV. The bank would sell the loans to the SPV through a "true sale" and the SPV would issue securities under a revolving securitization structure. Which of the following statements is correct?

A. The credit quality of the securitized car loan assets is enhanced if the principal value of the securities issued is higher than the principal value of the assets.

B. The SPV can give the bank access to cheaper funding if the credit quality of the securitized car loan assets is higher than the credit quality of the bank's balance sheet.

C. Under a revolving structure, prepayment assumptions are not incorporated, which typically results in principal being paid to investors coupon by coupon throughout the life of the security.

D. Under a revolving structure, principal collected on the loans is passed through to investors as it is received, so the notes are priced and traded on the weighted-average life of the loans.

Solutions • CR-23 c

Solution CR-23 c.1

Answer: C

Bank Q1197 • Mock 2 Q29

Step 1 – What the SPV is for

Securitisation works only if investors' exposure depends on the asset pool, not on the originator's solvency. That needs a true sale to a separate legal entity with no recourse to the originator.

Step 2 – The feature that delivers it

The SPV must be bankruptcy remote: if the originator fails, its creditors have no claim on the transferred assets.

⇒ C is the defining requirement.

Interpretation

GARP: an SPV "is not, for legal purposes, linked in any way to the originator", so the sponsor's bankruptcy "will have no impact on the SPV". Everything else in the options is either optional or would undo the isolation.

Why the other options are wrong

A is incorrect. SPVs are often domiciled offshore (Jersey, Cayman) or in SPV-friendly centres for tax reasons; the same jurisdiction is not required.

B is incorrect. Consolidation with recourse to the bank would undo the true sale and tie investors back to the originator's credit.

D is incorrect. Servicing is an operational choice (the originator often services the pool) and gives no legal isolation.

LO: CR-23 c: Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust. CR-23 a: Define securitization, describe the securitization process, and explain the role of participants in the process. • **Reference:** Choudhry, Structured Credit Products 2e, Ch.12 (GARP CR Ch.23), 23.1 Example 23.1 Special Purpose Vehicles; 23.3 The Process of Securitisation

Solution CR-23 c.2

Answer: D

New • built from GARP Ch.23, 23.1; Schweser LO 41.c

Step 1 – Both forms are SPVs

GARP: "An SPV is a legal trust or company" that is not linked to the originator and is bankruptcy remote. A company SPV can be bankruptcy remote just as a trust can.

⇒ C fails.

Step 2 – Where each form is used

The trust "is more common in the US market, and its interests are represented by a Trustee"; in the Euromarkets SPVs "are often incorporated as companies". Schweser: the company form may not distance the originator enough for US accounting purposes.

⇒ A swaps the geography.

Step 3 – What the trust arrangement adds

The originator sells the receivables to a master trust (SPV 1), which deposits them in a grantor trust (SPV 2) and receives a beneficial interest. The notes are backed by that beneficial claim, which further distances the originator from the assets (Figure 1).

⇒ D is correct. A true sale is still needed (B fails), and the trustee acts for the investors, not the originator.

Interpretation

For a card issuer that comes to market several times a year, the trust form also carries the master trust structure, which issues series after series from one pool of receivables.

Why the other options are wrong

A is incorrect. Geography swap: the single-entity company form is the Euromarkets choice; US practice favours the trust because the company may not give enough distance for accounting purposes.

B is incorrect. A trust is still an SPV that must buy the assets in a true sale; the trustee represents the investors' interests.

C is incorrect. Absolute qualifier: GARP defines an SPV, company or trust, as bankruptcy remote from the sponsor.

LO: CR-23 c: Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), 23.1 Example 23.1 Special Purpose Vehicles; SchweserNotes 2026 Book 2, Reading 41, LO 41.a and 41.c, Figure 41.3

Solution CR-23 c.3 Answer: B

Bank Q1208 • AnalystPrep Q.2929

Step 1 – The SPV owns the assets

The SPV is legally separate from the sponsor and holds the assets sold to it.

Step 2 – What the CLN is linked to

GARP: "Unlike CLNs issued by originators direct, CLNs issued by SPVs do not have any credit-linkage to the sponsoring entity. The note is linked instead to assets that have been sold to the SPV, and its performance is dependent on the performance of these assets."

$\Rightarrow$ B is correct.

Interpretation

Issuing through an SPV strips the sponsor's own credit out of the note: the investor takes the credit risk of the assets in the vehicle only. The same separation is what makes any SPV note rate on its pool (Workbook D14.1, worked as Question CR-9 g.7).

Why the other options are wrong

A is incorrect. Describes a CLN issued by the originator directly; an SPV-issued CLN has no credit linkage to the sponsor.

C is incorrect. Confuses the SPV with the originator: the SPV issues the note and owns the reference assets.

D is incorrect. Reverses the logic: the SPV's legal separateness is what de-links the note from the sponsor.

LO: CR-23 c: Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), 23.1 Example 23.1 Special Purpose Vehicles

Solution CR-23 c.4 Answer: C

Bank Q1209 • BT P2.T6.610.3

Step 1 – Match the asset

Short-dated receivables with a high prepayment speed and no amortisation schedule suit a revolving structure, in which principal collections buy new receivables during the revolving period.

GARP names credit card debt and auto loans.

Step 2 – Match the issuer

The master trust is for frequent issuers that issue multiple series from one pool. This issuer is infrequent, so a stand-alone revolving structure is enough.
⇒ C.

Interpretation

Two questions pick the structure: how does the asset pay back (schedule or revolving), and how often does the issuer come to market (once or repeatedly)? A frequent card issuer would use a master trust; this one does not need it.

Why the other options are wrong

A is incorrect. The master trust is the frequent issuer's vehicle; the stem says infrequent.
B is incorrect. Amortising (pass-through) structures suit assets with amortisation schedules, such as mortgages and consumer loans.
D is incorrect. The revolving structure fits the description.

LO: CR-23 c: Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), 23.3 SPV Structures: Revolving Structures, Master Trust

Solution CR-23 c.5 Answer: D

Bank Q1211 • Mock 3 Q59

Step 1 – What revolves

In a revolving structure, principal collections are not passed through: during the revolving period they buy new receivables that meet the eligibility criteria.

Step 2 – How investors are repaid

In the amortisation period principal is paid in equal instalments (controlled amortisation) or trapped in a separate account until the expected maturity and paid as one lump sum (soft bullet).
⇒ D.

Interpretation

The replenishing pool is what lets a vehicle funded by medium-term notes hold receivables that each last only months, such as auto loans or card balances.

Why the other options are wrong

A is incorrect. Passing principal through as collected is the amortising (pass-through) structure.
B is incorrect. A fixed, non-replenished pool is also the amortising case; a revolving pool is replenished.
C is incorrect. Contradicts the true sale: investors bear the pool's credit risk, not the originator's.

LO: CR-23 c: Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), 23.3 SPV Structures: Revolving Structures

Solution CR-23 c.6 Answer: B

Bank Q1220 • Mock 4 Q61

Step 1 – What the trigger watches

The rating agencies require every non-amortising ABS to set a minimum MPR as an early amortisation trigger; delinquency and default ratios are tied to triggers too (CR-23 g).

Step 2 – What happens when it trips

Reinvestment of principal collections stops and principal starts flowing to the noteholders: the revolving period ends early.
⇒ Investors are repaid sooner from a pool that is deteriorating: B.

Interpretation

The trigger is a protection for investors, not a lifeline for the originator. It shortens the notes' expected life, which is why the WAL cannot stay the same.

Why the other options are wrong

A is incorrect. Reverses the effect: the trigger ends the revolving period; it does not extend it.
C is incorrect. Substituting receivables is not the trigger's mechanism; the response is to stop revolving and amortise.
D is incorrect. Early amortisation returns principal sooner, so the expected weighted-average life shortens.

LO: CR-23 c: Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master trust. CR-23 f: Explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), 23.3 Revolving Structures; 23.5 Performance Analysis for Credit Card ABS

Solution CR-23 c.7 Answer: B

Bank Q1210 • GARP PE1 Q21 • GARP PE2 Q60

Step 1 – The funding statement

A true sale places the loans in a separate SPV, so the notes are rated on the pool, de-linked from the bank's own rating. If the car loans are better credit than the bank's balance sheet, the notes can fund more cheaply than the bank's unsecured debt.
⇒ B is correct.

Step 2 – The overcollateralisation statement

Over-collateralisation means the notes issued are *less* than the assets; issuing more notes than assets weakens credit quality.
⇒ A is backwards.

Step 3 – The two structure statements

Paying principal coupon by coupon and pricing on WAL with prepayment assumptions describe the amortising structure. In a revolving structure, principal collections buy new receivables first.
⇒ C and D both describe the wrong structure.

Interpretation

GARP's funding argument: the ABS market is less liquid, which adds to the yield, but the higher rating of the SPV's notes frequently makes the originator's cost of funds lower, a large benefit for lower-rated institutions.

Why the other options are wrong

A is incorrect. Reverses over-collateralisation: enhancement comes from issuing notes below the asset value.
C is incorrect. Coupon-by-coupon principal describes the amortising structure, which does incorporate prepayment assumptions through the WAL.
D is incorrect. Passing principal through and pricing on WAL is again the amortising (pass-through) structure.

LO: CR-23 c: Analyze the differences in the mechanics of issuing securitized products using a trust versus a special purpose vehicle (SPV) and distinguish between the three main SPV structures: amortizing, revolving, and master

trust. • **Reference:** Choudhry, Ch.12 (GARP CR Ch.23), 23.2 Funding; 23.3 Mechanics of Securitisation, SPV Structures